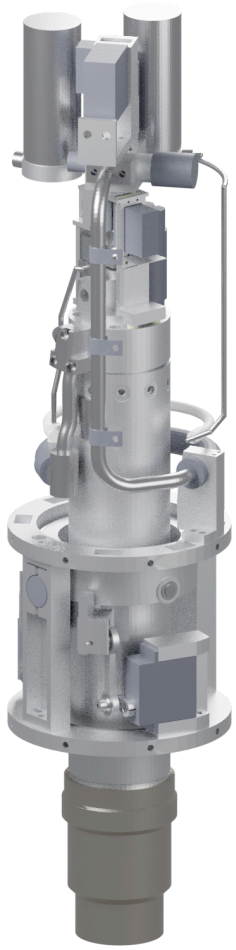
ARGOS H1 · ENGINE PROGRAMME · ARGOS-H1-ENGINE-DATASHEET

Engine technical datasheet

N_2O / printed-ABS hybrid rocket engine · servo-pintle throttle · two-axis gimbal · for a self-landing rocket



FINAL CAD v3.5.2 assembly, neutral pose. Visualisation of tessellated display geometry (GLB 4bcc264f683e128f), not a drawing. Not in the GLB display mesh, so shown in no render: all 57 of the 57 standard-hardware lines of BOM /hardware (ENG-HW-101 to ENG-HW-170; hardware_v352 lists 57 lines), so no screw, spring, pin or insert, among them ENG-HW-101 and ENG-HW-170 (the J-FWD radial screw and the disc spring under its head); ENG-GM-12 (Transport / neutral lock, 2 half rings (REMOVE BEFORE FLIGHT)). DERIVED T974

PRELIMINARY · rev1

SIM / CALC / DERIVED; not physical approval; no hardware has been tested. Nothing in this datasheet is physical approval or acceptance. No value is MEASURED. Every number carries one of the ten evidence tags (SIM, CALC, DERIVED, ASSUMED, DATASHEET, LITERATURE, AS RECALLED, TO-VERIFY, MISSING, ESTIMATE) or the source-class pill DECIDED (a design value, decided limit or requirement: chosen, not measured or computed), and a trace reference (T###, Appendix A). Missing data are shown as MISSING, TO-DEFINE or PENDING, never estimated by the generator.

RATED THRUST (SL), PC AT THE SOFTWARE CAP
500 N SIM T033

CHAMBER PRESSURE, RATED
2.50 MPa (25.0 bar) SIM T036

THROTTLE FLOOR TARGET (20 %)
100 N SIM T037

ISP, RATED (SL)
193.1 s SIM T034

CAD MASS, WET, ALL GROUPS (HARDWARE CALC)
5.34303 kg DERIVED T038

ENVELOPE (CAD BOUNDING BOX)
139.5 × 140.2 × 593.3 mm DERIVED
T043 T044 T040

FOUND NEGATIVES IN THE MARGIN MATRIX (§9.3)
33 DERIVED T697

INPUTS STILL PENDING (§1.4)
2 DERIVED PENDING T705

Configuration	Version	sha256/16
CAD (FINAL)	v3.5.2	`manifest a4713d45b3b24a0f`
Combustion	v2.4d	`7dbf425c12300658`
Mechanical	v2.4	`66bc53d88d444679`
Electronics	v4.1	`c862375163393adc`
Space/GNC	v3.2	`bde317782574cbd9`
3D FEA report	FINAL handback of the v3.5.2 re-check	`dc53f4cd45d75f4e`
Margin matrix (rows v8)	v352	`6c694fa7147fb511`
Requirements	rev 2	`012b14cbaaddfdc6`
Decision log	266 lines	`4cde8acb297d195c`
Review register	18 reviews	`076172727cddd074`
Decision overrides	ovr-7	`6746adf53921b1ef`

Generated 2026-10-21T16:02:27Z by gen/build_datasheet.py 0.3.1. Quotations: delivery Sønderborg, Denmark, DK-6400 (every price TO-QUOTE). Units SI; pressures in bar abs unless marked.

Contents

1 Identification 4
- 1.1 Document and configuration 4
- 1.2 What changed since rev0 4
- 1.3 Evidence classes 4
- 1.4 At a glance: found negatives, open rulings and pending inputs 5
- 1.5 Recent lead decisions 5

2 Key performance 6
- 2.1 Combustion datasheet inputs 6
- 2.2 Throttle points (20 C tank, initial port) 6
- 2.3 Pressure ladder 7
- 2.4 FA-1 hard start: equivalent static pressure spectrum 8

3 Throttle and thrust vector control 9
- 3.1 Throttle 9
- 3.2 Gimbal and TVC requirements (GNC v3.2 datasheet inputs) 9
- 3.3 Axis stiffness band and rocking modes (FEA3D, R3 poses) 9
- 3.4 TVC actuation record (mechanical.json v2.1, not re-derived for v3.5.2) 9

4 Geometry and mass 10
- 4.1 Envelope and principal dimensions 10
- 4.2 Assembly views 10
- 4.3 Part families 12
- 4.4 Mass 13
- 4.5 Airframe interface and engine mounting (current sources) 14
- 4.6 CAD build, fit and interference checks 14

5 Parts list 16
- 5.1 BL — Head-end bleed (provision) 16
- 5.2 EL — Sensors and encoders 16
- 5.3 FD — Oxidiser feed 16
- 5.4 FS — Main valve module (fixed) 17
- 5.5 GM — Gimbal and thrust frame 17
- 5.6 GR — Fuel grain 18
- 5.7 HW — Bearings and hardware 18
- 5.8 IG — Igniter 18
- 5.9 NZ — Nozzle and joint 18
- 5.10 PD — Pintle drive 18
- 5.11 TC — Thrust chamber 19
- 5.12 Standard hardware (fasteners, springs, bearings; BOM /hardware, CALC masses) 19
- 5.13 Ground support tools, provisions and contingency (not flight mass) 21

6 Materials and thermal protection 22
- 6.1 Materials 22
- 6.2 Short-time exposure of the 6061-T6 case and head (T-F1) 22
- 6.3 Thermal protection and thermal loads 22

7 Combustion, sequences and the abort set 24
- 7.1 Design point 24
- 7.2 Fuel grain 24
- 7.3 Injector, feed and igniter 24
- 7.4 Start and relight sequence 24
- 7.5 Relight purge (decided schedule and conditions) 26
- 7.6 Abort set (Combustion matrix; firmware causes in §10.5) 26
- 7.7 FA-1 hard start: rulings of record 27
- 7.8 Stability 27
- 7.9 Test specifications: CF-4 and T1-T4 27

8 Plume CFD 29
- 8.0 Operating points 29
- 8.1 Key results at the v1 point (O/F 7.76) 29
- 8.2 Fields 29
- 8.3 Afterburning and model limits 31
- 8.4 Illustration 32
- 8.5 Review status 32

9 Structural analysis: 3D FEA of the assembled engine (CAD v3.5.2) 33
- 9.1 Model, loads and status 33
- 9.2 Margin matrix: every part × every load case 33
- 9.3 Found negatives: every negative cell, its class and its closure 35
- 9.4 Named findings stated plainly 37
- 9.5 J-FWD joint rows (TC-01 / TC-02 / HW-101 screw) 38
- 9.6 Mounting data: loads at the 16 airframe holes 43
- 9.7 Gimbal axis stiffness (R3 poses) and rocking modes 44
- 9.8 Secondary joints J-FD01 / J-FD09 (F4) 44
- 9.9 Mechanical follow-ups after FEA3D-2: tap wall v3, LC-TB70 v2, C1 seat 45
- 9.10 FEA3D-2 open items and owners 45
- 9.11 Not run, superseded, and runs that wait for the lead 46

10 Electronics and control 47
- 10.0 Status of electronics.json v4.1 under line 266 47
- 10.1 Engine control unit and firmware 47
- 10.2 Sensors 48
- 10.3 Limits of firmware v2.12 48
- 10.4 Light rule and start monitors 49
- 10.5 Abort and refusal causes (firmware v2.12) 49
- 10.6 Power, head-end bleed provision and fail-safe states 50
- 10.7 Lead rulings on v2.11 and items routed to Electronics v2.12 51
- 10.8 Lead rulings on the Electronics v2.12 handback (CHANGELOG 264) and on its review (CHANGELOG 266) 52
- 10.9 Electronics open issues 52

11 Mission and landing (GNC v3.2) 55
- 11.1 Mission headline numbers 55
- 11.2 GNC open items 55

12 Verification status 57
- 12.1 Review status of the deliverables used here 57
- 12.2 Review register 57
- 12.3 Founder actions (traced to their decision) 58
- 12.4 Open items with owners 62
- 12.5 Requirement verification methods 63
- 12.6 Founder test list 63

13 Decision overrides and the contradiction register 65
13.1 Decision overrides 65
13.2 Superseded values withheld 65
13.3 Contradiction register 65
14 Unknowns register 69
15 Revision history and sources 71
15.1 Revision history 71
15.2 Sources 71
Appendix A Value trace 73

1 Identification

SIM / CALC / DERIVED; not physical approval; no hardware has been tested. This datasheet is generated from the engine programme hub by gen/build_datasheet.py 0.3.1. Every value in a data column is a traced value (T###, Appendix A) with its evidence class. No value is MEASURED: the engine has not been built or fired. Missing data are shown as MISSING, TO-DEFINE or PENDING and never estimated here.

1.1 Document and configuration

ARGOS-H1-ENGINE-DATASHEET rev1, CAD v3.5.2. Full source list with paths in §15.2.

Item	Version · sha256/16 (this build)
CAD set (FINAL v3.5.2)	v3.5.2 final · sha256/16 621481ddeccc228b T707
CAD set status	FINAL v3.5.2 CAD set (lead release 19:09:50Z, chain E ended 19:06:26Z; publication approved 19:22:55Z), built 2026-10-20 19:15:43-19:16:54Z from src/ (rev 5) with hardware v1.7; supersedes the INTERIM r4h6 set (its set file, with the rev 2 - rev 4 texts, is history/set_v3.5.2.json.31bb2b1e8b985e41). DESIGN MODEL: not released for fabrication; the geometric checks are software checks, not physical validation; nothing here is physical approval. TC-09 thinning (206 (2a)) stays conditional on its re-run, as in the … DECIDED *Design value, set_v3.5.2.json v3.5.2 final* T725
CAD manifest	456 rows · sha256/16 a4713d45b3b24a0f T708
Assembly GLB (renders)	v3.5.2 · sha256/16 4bcc264f683e128f T709
Bill of materials	v3.5.2 · sha256/16 9234c953c5b6beb8 T710
Mass properties	v3.5.2 · sha256/16 c1ff170be0ac8cc5 T711
Combustion interface	v2.4d · sha256/16 7dbf425c12300658 T712
Combustion datasheet inputs	v2.4d · sha256/16 8dc583578748b06f T713
Mechanical interface	v2.4 · sha256/16 66bc53d88d444679 T714
Mechanical 04-fea/v352 manifest	226 rows · sha256/16 4f7e7655ccc25f70 T715
GNC interface	v3.2 · sha256/16 bde317782574cbd9 T716
GNC datasheet inputs	v3.2 · sha256/16 b8b0dcaf45a5d915 T717
Electronics interface	v4.1 · sha256/16 c862375163393adc T718
FEA3D v3.5.2 report	FINAL handback of the v3.5.2 re-check · sha256/16 dc53f4cd45d75f4e T719
Margin matrix of record (rows v8)	v352 · sha256/16 6c694fa7147fb511 T720
Decision log	266 lines · sha256/16 4cde8acb297d195c T721
Review register	18 reviews · sha256/16 076172727cddd074 T722
Requirements	rev 2 · sha256/16 012b14cbaaddfdc6 T723
Airframe ICD (out of the finish scope, CHANGELOG 222; no value is taken from it)	draft 0.2 · sha256/16 706455abf4098182 T724

1.2 What changed since rev0

Rev0 values from the rev0 sources.json (07-datasheet/out/sources.json, kept); rev1 values from this build.

Item	rev0 (2026-10-13)	rev1 (this build)
CAD	v3.4.1 DERIVED T726	v3.5.2 DECIDED *Design value, render_manifest.json v3.5.2* T727
Combustion interface	v2.4 DERIVED T728	v2.4d DERIVED T729
Mechanical interface	v2 DERIVED T730	v2.4 DERIVED T731
GNC interface	v3 DERIVED T732	v3.2 DERIVED T733
Electronics interface	v2 DERIVED T734	v4.1 DERIVED T735
Decision log	176 lines DERIVED T736	266 lines DERIVED T737
Parts changed vs v3.5.1	–	ENG-FD-01; ENG-FD-02; ENG-FD-04; ENG-GM-01; ENG-GM-02; ENG-NZ-01; ENG-NZ-06; ENG-TC-01; ENG-TC-02; ENG-TC-03; ENG-TC-09 DECIDED *Design value, set_v3.5.2.json v3.5.2 final* T738
Mass change and the GNC trigger	–	BELOW the trigger with and without the conditional TC-09 thinning: no re-fly by mass DERIVED T739
GNC row 8: servo output stiffness K_s (filed under ENG-11): upper end of the design-to window	**120** N m/rad DERIVED T741 SUPERSEDED	design-to **100** DERIVED T743–**110** N m/rad DERIVED T744 at the ASSUMED servo J **1.6e-03** kg m² ASSUMED T745, not robust across J: It is not robust: it moves with J (R3 pose table, cons: 0.76e-3 130→= 160; 2.7e-3 120→= 160; no K_s passes at all three J), the inertia convention (n48 at J 1.6e-3, CAD v3.5.1: none) and the rotor model (two-mass, CAD v3.5.1: 150→= 160), and it is empty at 0.9 x first bending (CAD v3.5.1). DERIVED T746
Case-TC limits	–	Lead consistency fold found while applying: the case-TC in-burn abort 423.15 K (243 (3), 249 (1)) and the RELIGHT refusal 403.15 K (249 (4)) replace the stale 448.15 K and 433.15 K in the combustion texts. DECIDED *Lead decision, line 257* T747
3D FEA of the assembled engine	–	2026-10-21, FINAL handback of the v3.5.2 re-check (seat FEA3D-2 from 2026-10-21T00:41Z; seat FEA3D-1 before). The assembled-engine model on CAD v3.5.2 (asm v1.3): LC1 and LC2 on the S-AX gap-conductance bounds GVLO / GVHI, FA-1 (FA1Q / D / U / S / T) on the NF2b relight state, TB70 / TBS, the LC6 … DERIVED T748

1.3 Evidence classes

Class	Meaning
SIM	model run (CFD, FE, Monte Carlo, quasi-1D, CEA, SIL)
CALC	hand calculation on model loads or standard geometry
DERIVED	computed from other traced values (by a specialist, a reviewer or this generator)
ASSUMED	an input adopted without a measurement or a source
DATASHEET	maker data
LITERATURE	published data, cited
AS RECALLED	a value quoted from memory by its author; to be sourced
TO-VERIFY	needed from, or to be confirmed by, the named owner
MISSING	not available; never estimated by the generator
ESTIMATE	an estimate named as such by its source
DECIDED	a design value, decided limit or requirement: chosen, not measured or computed; where the source gives a basis, that basis and its tag are shown beside it (source class: design value, lead decision line N, or requirement)
REF	a reference or provenance text (a source, a review id, a "no number changed" statement): it carries no value and no pill
PENDING	an input that has not landed (never assumed)
SUPERSEDED	withdrawn by a newer decision, shown only as history
TO-DEFINE	not yet defined by its owner (the value is MISSING)

Every number carries one of the ten evidence tags (SIM, CALC, DERIVED, ASSUMED, DATASHEET, LITERATURE, AS RECALLED, TO-VERIFY, MISSING, ESTIMATE) or the source-class pill DECIDED. The tag comes from the source tag or the quoted text; where neither gives one, the trace names the class set at the call site. This generator sets no default class: a value without a class is counted as UNTAGGED in qa_report. A lead line that quotes a SIM, DERIVED or CALC number keeps that number's tag; DECIDED applies only to the chosen value itself. PENDING, SUPERSEDED and TO-DEFINE are statuses shown beside the tag. A review verdict, a matching number or a passing check is not physical approval.

1.4 At a glance: found negatives, open rulings and pending inputs

Found negatives. 33 DERIVED T697 margin-matrix cells are negative (BR **7** T698, BT **2** T699, CH **3** T700, DC **1** T701, MF **9** T702, TS-183 **9** T703, TS-212 **2** T704); every one is listed with its class and closure in §9.3. Named: TC-04 (BT), the TC-01 J-FWD seat-line P+Q (DC; the R2-B select fit closes it only in the CALC band model), the J-FWD screw LCF (C6 OPEN: single-use screws), the screw head fillet from zero (§9.4). The mesh convergence of the seat-line P+Q is not shown (§9.1.1).

Inputs still pending at this build, each quoted in full (REV-DS-1 D1-06). Where a value waits for one of them it is marked at the value: SUSPENDED UNTIL v2.13 (266 (6)) for the option (b) values; OPEN for the T4 relight length (§7.9).

Pending input	Status (quoted)
Electronics v2.13 delta (the E4-01 fix) and its review: QUEUED, NOT STARTED; the founder says when (266 (8)). Until it is reviewed, option (b) is not used for any start with ignition (266 (6))	QUEUED, NOT STARTED (the founder says when): the Electronics v2.13 delta and its review: the light-window hold (latched at IGNITION entry until E2 + 0.45 s, bumpless resume), the TC4 step check (1.5 K per conversion) and rate check (5 K/s slew, 3 K trip) with a latched f3 SEAT_STEP, the E4-03 fix, unit tests UT-1 to UT-4, the S13T TC-fault set, the FMEA row, the B-TC-1 TC4 row with the spark on, the v4.2 folds, and notes E4-N01 (T_cal 10-30 C or kPcSeatArmDevMax_K 40 K), E4-N02 (the 264 (2) pad rule and the 264 (7) inhibit implemented), E4-N03 (47.5 against 47.75 bar abs, TO-VERIFY), E4-N04 (the P_torch re-base needs compensation or a photodiode-guarded gate; torch local heat MISSING) and E4-N11 (a SIL MANIFEST_v2). DECIDED *Lead decision, line 266 (8)* T030
T4 re-run: the length of the shorter T4 relight (OPEN, 261 (2) and 264 (5))	Combustion sets the length in the T4 re-run of 261 (2) with Mechanical's soak-back. DECIDED *Lead decision, line 264 (5)* T494

1.5 Recent lead decisions

The last 10 lead decision lines of the CHANGELOG (CL 266 lines).

Line	Date	Decision text (first 900 characters; full line in the CHANGELOG)
L253	2026-10-21	Lead decisions and verifications after line 252 (2026-10-21 05:24Z to 07:09:15Z, times from date -u; SIM / CALC / DERIVED unless tagged; none of it is physical approval). (1) cD in Mechanical's v7 inputs, ruled 05:27Z (sent between 05:26:33Z and 05:27:43Z; the message said 05:28Z, a lead stamp slip): the SOAK52 under-head band strain MS is a local refined-mesh value, so (C) applies: (b2) 0.296 stands and cD 0.267 is shown beside it (strain ratio 1.023, inside the 5 % band; the MF note applies); resultant-type inputs take cD as an instance, worst-of as in v6: FA1S N_head_min 2717.2 N, FA1S seat window 455.0 N, FA1S spread 0.1549, SOAK70 M_max 3438.9 N mm at x' 30.5; the record results/v352/jfwd_v7inputs_cD_v352.json 5a41c8d74897bdda (md 424146c89c8c9f01) is Mechanical's re-map input with v6; the datasheet J-FWD UH row reads '0.296 (s6), cD 0.267 beside'. (2) Line order: line 251 is the ... DECIDED *Lead decision, line 253* T749
L255	2026-10-21	Lead line covering 07:13Z-07:52Z (times from date -u; SIM / CALC / DERIVED as tagged; none of it is physical approval). (1) MECHANICAL RE-MAP GO AND RULING: Mechanical handed back at 07:33:56Z without publishing (its run ended before the go; hub unchanged). The lead verified every work-tree and hub sha16 of its 07:28:05Z pre-publication post and its publish scripts (07:30:26-07:37:34Z) and resumed it at 07:37:34Z with the GO and LEAD RULING R2-B, quoted verbatim in line 254 (1); R2-B governs the ENG-TC-02 plug OD: SELECT FIT to the measured TC-01 bore, radial gap g in [0.040, 0.0631] mm, D_meas - 0.1262 <= OD <= D_meas - 0.080 (window 0.0462 mm less 2U, U TO-DEFINE), measured at 20 +/- 1 C with paired serials; drawn option A (57.20 -0.030/-0.040, MS +0.005 FLAG) not adopted; data strings unedited in v3.5.2 and superseded by the drawing note. Answers to its three questions: the texts ... DECIDED *Lead decision, line 255* T750
L256	2026-10-21	Lead line covering 07:53Z-08:45Z (times from date -u; SIM / CALC / DERIVED as tagged; none of it is physical approval). (1) REV-COMB-4, the independent delta re-review of Combustion v2.4b: interim 07:52:49Z, full review 08:32:27Z, saved verbatim by the lead as reviews/REVIEW-COMBUSTION-R4.md (2908321d7f75d439; artefacts reviews/combustion/r4/, MANIFEST_REV-COMB-4.sha256 996475830cae731f, 23/23 verified 08:37:28Z; SUPERVISION.json entry added). VERDICT ACCEPT WITH CORRECTIONS: 0 blocking, 2 major, 11 minor, 5 notes; line 230 (1)-(4) CONFIRMED and (5) CONFIRMED as a decision (its supporting sentence is refuted for 1 of the 3 low corners, which crosses 37.16 N at E2 + 0.857 s); all 17 REV-COMB-3 items closed (10 closed, 7 with residual). The lead ACCEPTS the review. (2) PURGE, 253 (7)(ii), now FINAL: the 2.43 s scheduled flow with the 2.39 s cap is ADOPTED [DERIVED by REV-COMB-4 on the ... DECIDED *Lead decision, line 256* T751
L257	2026-10-21	08:46-09:39Z (date -u). (1) COMBUSTION v2.4c PUBLISHED by the lead at 09:38:09Z (text only; no model re-run): the REV-COMB-4 corrections (R4-01 to R4-08, R4-12, R4-14 to R4-18, the section 6 texts) with the decisions of 227 (5), 230, 243 (3), 249 (4), 253 (7), 256 and (2) below. Files: interfaces/combustion.json v2.4c 661db0a593462382; 02-combustion/REPORT.md 7a9acd3df5937303 (new section 17; lines 1-3161 keep their numbers); CF-4-SPEC.md v2.2 4e36be703a9e7d44; T1-T4-REGRESSION-SPEC.md v3.1 b00b2d852ed3475c; REVISION-RESPONSE-R3.md 370875c3645eb49c (annotated only); DATASHEET-INPUTS-COMBUSTION.md v2.4c 29ed6d45c55ff234; writer 02-combustion/src/apply_v2p4c_rev_comb_4.py 063195b1b3a12bed (run 09:35:44Z). The v2.4b set is archived in 02-combustion/superseded/v2p4b/ with HASHES.txt (json fd3a7da6d27c7c66). 29 JSON leaves were replaced, each kept in superseded_v2p4c with its finding; all ... DECIDED *Lead decision, line 257* T752
L258	2026-10-21	09:40-10:47Z (date -u). (1) REV-GNC-7 ACCEPTED: its final (reviews/REVIEW-GNC-R5.md 6b86fcd326568646, saved verbatim by the lead at 10:39:38Z; reviewer manifest reviews/gnc/rev4b/MANIFEST_REV-GNC-7.sha256 e04257acba555551 verified 94/94 OK at 10:38:50Z; SUPERVISION.json entry added) is PUBLISHABLE WITH CORRECTIONS (0 blocking; 4 major RV7-01 to RV7-04; 5 minor; 3 notes). gnc.json v3.2 = the candidate 01-systems-gnc/staging_v3.1/rev4b/v32/gnc_v3.2_rev4b.json 43d58c4aff164b86 with the 34 ops of reviews/gnc/rev4b/data/b/rv7_v32_corrections.json f7c0c6051efd5a08 (reviewer trial fea33eacdd8a7d56), and DATASHEET-INPUTS-GNC.md with DS ops 1-7 plus rows for the new open items 53-57; the lead publishes it after an independent apply that matches the trial and t10_postcheck v3 --publication --rv7 exits 0. ENG-11 (RV7-01): design-to 100 <= K_s <= 110 N m/rad (two-sided) at the ASSUMED servo J ... DECIDED *Lead decision, line 258* T753
L260	2026-10-21	10:48-10:57Z (date -u). (1) GNC.JSON v3.2 PUBLISHED by the lead (built and stamped 10:56:48Z, promoted 10:57:09Z): interfaces/gnc.json v3.2 bde317782574cbd9 (1,415,700 B) = the candidate 01-systems-gnc/staging_v3.1/rev4b/v32/gnc_v3.2_rev4b.json 43d58c4aff164b86 with the 34 REV-GNC-7 operations of reviews/gnc/rev4b/data/b/rv7_v32_corrections.json f7c0c6051efd5a08, applied independently by the lead and byte-equal to the reviewer trial fea33eacdd8a7d56 without its trial marker (236107a89ca4a240), plus lead folds L1-L4: L1 the status header (published) and the status item (4) sentences replaced by the texts of ops 6 and 7; L2 flight_software.rev4b_entries.mode3_filter_reinit.v with the text of op 13; L3 status_after_rev_gnc_7 notes on v32_items_rev4b items 6 and 14 (S17 v2.11 CONFIRMED 0.8 s for W10 < 6 m/s with conditions); L4 date 2026-10-21, supersedes and ... DECIDED *Lead decision, line 260* T754
L261	2026-10-21	10:58-11:26Z (date -u). (1) COMBUSTION v2.4d PUBLISHED by the lead at 11:24:52Z (text only; no model re-run; no SIM, CEA or FE result changed): the fold of 258 (2)-(4), 259 and 260. Files: interfaces/combustion.json v2.4d 7dbf425c12300658; 02-combustion/REPORT.md 0f2af843861176c8 (new section 18; lines 1-3314 keep their numbers); CF-4-SPEC.md v2.3 6f0d4afd7fb3a798; T1-T4-REGRESSION-SPEC.md v3.2 edde58cf7e1aadf7; DATASHEET-INPUTS-COMBUSTION.md v2.4d 8dc583578748b06f (row added: FA-1 class Z3 v2 and the J-FWD replace rule); REVISION-RESPONSE-R3.md unchanged (370875c3645eb49c); writer 02-combustion/src/apply_v2p4d_lead.py 6c43216fc7b22306 (run 11:24:28Z; two builds with one stamp byte-identical). The v2.4c set is archived in 02-combustion/superseded/v2p4c/ with HASHES.txt (json 661db0a593462382). 18 JSON leaves replaced, each kept in superseded_v2p4d with its decision; all v2.4c keys kept; ... DECIDED *Lead decision, line 261* T755
L262	2026-10-21	Lead decisions and verifications after line 261 (2026-10-21 11:29Z to 12:14Z, times from date -u; SIM / CALC / DERIVED unless tagged; none of it is physical approval). (1) FOUNDER DECISION: Gaspar tapped 'Three agents' on the lead's 04:00Z decision card at 11:44:54Z: at most three agents at a time from now (two since his 2026-10-14 restart card). The lead told him at 11:49Z the new estimate for the full package in both repos, about 17:00-18:30Z (19:00-20:30 Danish); the long pole is the Electronics v2.12 round and its review REV-ELEC-4. (2) REV-DS-1, the independent review of datasheet rev1 (a fresh agent, effort max), started 11:48:03Z in the third slot. Phase A returned 12:08:02Z with no findings, because no draft PDF exists yet. Its re-trace harness is reviews/datasheet/r1/scripts/retrace.py 2684ef2459023d5f (rev0 test, 111 ids: 101 match; the 4 mismatches all come from sources ... DECIDED *Lead decision, line 262* T756
L264	2026-10-21	Lead rulings on the Electronics v2.12 handback (line 263), 2026-10-21 12:43:15Z to 12:49:16Z, times from date -u or the transcript; SIL / CALC / DERIVED unless tagged, none of it physical approval. (1) RECEIVED AND VERIFIED: the handback arrived at 12:43:15Z (author stamp 12:42:14Z) and is saved verbatim as [working copy; path not printed] (cd6981c0ba07b670); staging_v2.12/SHA256SUMS-v2.12.txt f6fd3a63817dfb34: 84 of 84 entries verify (lead check from absolute paths right after the 12:44:46Z date -u reading), 85 files in the folder; CHANGELOG 263 lines / 1157b92de515db02 (line 263 sha16 33fe3e2d40e2cb28, ASCII, four fields). interfaces/electronics.json stays v4 11f7896ad0d0613b; results/electronics_v4.1.json c862375163393adc stays PENDING REVIEW until REV-ELEC-4's verdict is recorded, and the datasheet shows v4.1 values as PENDING REVIEW. (2) R3-01 ... DECIDED *Lead decision, line 264* T757
L266	2026-10-21	Lead decisions and verifications after line 264 (2026-10-21 12:50Z to 14:46Z, times from date -u or the transcript; SIM / CALC / DERIVED unless tagged; none of it is physical approval). (1) REV-ELEC-4 brief: [working copy; path not printed] 1e659eff4bceecde (the sha16 that 264 (8) promised); the reviewer was resumed 12:51Z at effort max. PUSH 13:00:10Z: p1 PR #5 cf92b16..4c852a7 (96 paths, all under argos-engine/: lines 262-264, staging_v2.12, REV-DS-1 phase A; scans clean); reviews/datasheet/r1/scripts/form_check.py is held back from the public repo because its string list names the private benchmark. (2) DATASHEET rev1-draft (author post 13:19:27Z): the lead re-hashed all 15 files at 13:21:00Z, all match (PDF 8e7d007d128dfb58, 151 pages). Rulings 13:24:27Z: 166.14 bar abs is DERIVED (1.1 x FA1S 151.04) everywhere and the requirement cites 256 (R4-12) and 264 (4); the nine ... DECIDED *Lead decision, line 266* T758

2 Key performance

2.1 Combustion datasheet inputs

Every row of 02-combustion/DATASHEET-INPUTS-COMBUSTION.md (DSIC v2.4d), quoted; the pill is the row's tag.

Item	Value	Condition / basis	Source
Rated thrust (maximum commanded)	500 N; Pc 25.01 bar abs; O/F 10.44; Isp (sea level) 193.1 s; c* 1401 m/s delivered; propellant flow 264.1 g/s (N2O 241.0, ABS 23.1) SIM T760	20 C saturated tank, initial port 25 mm, nominal regression law, eta_c* 0.92, eta_cf 0.975, pintle Cd 0.70 SIM T759	combustion.json operating_points.max_thrust T761
Chamber pressure limits	software cap 25 bar; ECU abort 30 bar; MEOP 45 bar; proof 67.5 bar; MEOP basis 44.33 bar (margin 0.67 bar to MEOP) SIM *Lead decision, per DATASHEET-INPUTS-COMBUSTION.md v2.4d* T763	MEOP basis: late light at the 24 C interlock with the designed line loss and the conservative bands SIM *Lead decision, per DATASHEET-INPUTS-COMBUSTION.md v2.4d* T762	combustion.json operating_points.pressure_limits, meop_basis_v2p4 T764
Thrust ceiling (available, not commanded)	ceiling 603-747 N for Cd 0.5-0.8 at 20 C (Dyer); worst model case HEM at 15 C, Cd 0.6: 499 N (Cd_eff 0.407) SIM T766	ceiling = F at min(full stroke 4 mm, injector dP/Pc >= 0.25) SIM T765	combustion.json operating_points.ceiling_vs_Cd_20C_and_models T767
Worst-case steady thrust and Pc (pintle full-stroke failure)	619-822 N, Pc 30.7-40.3 bar abs at 15-25 C (Cd 0.6-0.8); 876 N, [withheld: superseded value, SUP-01] bar abs at a 30 C state (SUPERSEDED in combustion.json: 30 C is not credible, the ARM interlock is at 24 C; REV-COMB-4 R4-17) SIM T769	initial port, steady; the ECU abort (30 bar) acts on it; the hard-start transient is the FA-1 row SIM T768	combustion.json operating_points.pressure_limits.full_stroke_failure_S230 T770
Throttle range	62.5-500 N (12.5-100 %): Pc 4.37-25.01 bar abs; O/F 4.06-10.44; Isp (sea level) 140.2-193.1 s SIM T772	25 % floor (125 N): met (dP/Pc 5.8); 20 % target (100 N): met (dP/Pc 7.1); 12.5 % stretch: model meets it, low-flux combustion UNVERIFIED (static fire T3) SIM T771	combustion.json operating_points.throttle_points_20C_initial_port, floor_* T773
Burn time and total impulse at a 500 N hold (2.0 kg tank)	500 N held 4.06-7.01 s; liquid end 6.77-7.50 s; impulse to liquid end 3387-3616 N s; N2O 1.630-1.737 kg SIM T775	15-25 C tank (the 25 C end is a bound: ARM 15-24 C), equilibrium and non-equilibrium tank SIM T774	combustion.json operating_points.max_thrust_hold_baseline_tank T776
Mission burns (2.0 kg tank, 3.167 L)	A1 dispersed p99: 8.26 s, 2968.2 N s, O/F 9.05, Isp 190.7 s (T0 294.12 K); A1 p99 throat-growth record (tg711, T0 297.56 K): 8.14 s, 2932.0 N s; B1 85 m p99: 21.78 s, 2984.5 N s, O/F 5.57, Isp 172.3 s; B1 max: 23.63 s, 3008.5 N s, Isp 170.1 s; B1 p99 throat-growth record (tg711, T0 289.99 K): 21.66 s, 3333.2 N s, Isp 174.8 s SIM T778	gnc.json v3 records flown on the S230 grain (non-equilibrium tank); the A1 rows are the 300 m record (comparator). A1 is decided at 210 m (lines 214, 227 (1)): GNC rev4 burn p99 8.443 / 8.270 s (RV6-05: not shorter); its impulse is MISSING until gnc.json v3.2; B1 governs sizing SIM T777	02-combustion/data/v2/dc_gncv3/summary_v2p4.json (7f4f181b80d55331); CHANGELOG 227 (9) T779
N2O tank	2.00 kg (founder, line 219). Option 2.20 kg: A1 burn p99 8.92 / 8.71 s (max 9.24 s) over the 8.258 s A1 sizing counter; first-order A1 throat-area growth 1.96-2.20 % per flight (nominal, n 0.5), residual web >= 2.54 mm DERIVED *Lead decision, per DATASHEET-INPUTS-COMBUSTION.md v2.4d* T781	2.20 kg: B1, thermal and erosion TO-VERIFY (line 227 (9)); no B1 record with 2.20 kg (OI-50) DERIVED *Lead decision, per DATASHEET-INPUTS-COMBUSTION.md v2.4d* T780	data/v2p4b/tank_2p20kg_option_v2p4b.json (59995244124f7c9a) T782
Nozzle and grain	throat 13.72 mm, exit 19.41 mm, expansion ratio 2.0 (graphite insert, one per flight); ABS grain 53.5 mm OD x 230 mm, one straight port 25 mm initial DECIDED *Design value, DATASHEET-INPUTS-COMBUSTION.md v2.4d* T784	graphite grade TO-QUOTE (Sønderborg DK-6400) DECIDED *Design value, DATASHEET-INPUTS-COMBUSTION.md v2.4d* T783	combustion.json gas_path T785
Start level (relight)	start-level lift 0.21776 mm at the A1 relight state (286.70 K), 0.2039 mm at 20 C (the ECU ignition-lift row at the tank temperature); lit thrust from E2 to E2 + 0.8 s >= 33.14 N (3 of 72 corners never reach 37.16 N); dead-engine cold flow <= 12.86 N (CONDITIONAL on CF-3); one of the 3 low corners crosses 37.16 N at E2 + 0.857 s, the other two stay below it to the end of the 1.0 s record (R4-03); with the SPI flux the cold-flow bound is 19.0-20.1 N over 283.15-289.5 K (19.5 N at 286.70 K), above the 16.31 N margin, so the CF-3 start-lift acceptance is not a formality (R4-17) SIM / DERIVED *Design value, DATASHEET-INPUTS-COMBUSTION.md v2.4d* T787	lit thrust: grain_q1d over 72 corners of the ASSUMED bands; lift: ECU table (Electronics) SIM / DERIVED *Design value, DATASHEET-INPUTS-COMBUSTION.md v2.4d* T786	data/v2p4b/p4c_lit_side_v2p4b.json (12f7699ddbeeecda); data/v2p4b/cold_flow_start_lift_v2p4b.json (28ce48092f33da1b) T788
Relight purge flow (decided)	2.43 s scheduled flow (both-open overlap 2.38 s + 0.05 s) with the 2.39 s both-open cap (LEAD 253 (7)(ii), final in 256 (2)); needed flow 2.2 / 1.9 / 1.8 s at 283.15 / 289.5 / 297.15 K unlined; vented N2O at 2.43 s <= 66.7 / 70.8 / 75.1 g at 283.15 / 289.5 / 297.15 K (Cd 0.70), 75.75 g at the 2.44 s of flow the cap allows; basis 29.25 g/s = 76 g / 2.5982 s of flow at 289.5 K (27.58-31.06 g/s over 283.15-297.15 K) DERIVED / ASSUMED *Lead decision, per DATASHEET-INPUTS-COMBUSTION.md v2.4d; basis SIM* T790	conditions (256 (2)): (a) 2.2 s at 283.15 K stated wherever 1.9 s is quoted; (b) measured Cd x A of the two-tap path >= 0.91 x the Cd 0.70 value (>= 0.95 x with the FD-01 element bound), else re-size; (c) the line stays unlined (lined, 283.15 K needs 2.5 s); (257 (2)): (d) after CF-4 the flow and cap are re-set from the measured Cd x A by the measured-time form (mass flow at 297.15 K), never above 2.43 / 2.39 s, so the 297.15 K vent stays <= 76 g; (258 (2)): (d1) the re-set takes the upper 95 % bound of the measured N2O Cd x A; (d2) CF-4 shows the 283.15 K need at the re-set flow; at Cd 0.80 the fixed schedule would vent about 85.5 g at 297.15 K [ESTIMATE] and fail GNC's A1 reserve test ... DERIVED / ASSUMED *Lead decision, per DATASHEET-INPUTS-COMBUSTION.md v2.4d; basis SIM* T789	data/v2p4b/purge_basis_line216_v2p4b.json (ca2ee1f0ea0f0194); reviews/combustion/r4/r4_purge_corners.json (9249ed093c21a3a3); CHANGELOG 257 (2), 258 (2); reviews/REVIEW-GNC-R5.md (6b86fcd326568646) T791
FA-1 hard start: structural load	head equivalent static pressure, all 896 cases, undamped: 93.6 bar abs (T <= 0.1 ms), 120.3 (0.1-1 ms; 115.4 at zeta 0.02, 109.6 at zeta 0.05), 146.9 (1-20 ms), 93.6 (20-50 ms, at T = 21.1 ms near its lower edge); peak head pressure 88.0 bar abs; FA1S 151.04 bar abs bounds every period and case (max 146.88, margin 4.16 bar) SIM T793	band values 94 / 121 / 147 / 94 bar abs DECIDED for structural SDOF paths (LEAD 230 (2); the line-194 (1) values 90 / 93 / 102 are superseded); FA1S is the design case for pressure-filled passages (next row); per-mode values are an analysis path only; damping ASSUMED (no modal test) SIM T792	data/v2p4b/fa1_spectra_damped_v2p4b.json (3a84a905052d5364) T794
FA-1 in gas-filled passages (P_torch passage, Pc port)	design case FA1S 151.04 bar abs uniform over the passage; linear lossless passage peaks <= 129.9 bar abs at the J-FWD tap and <= 148.7 bar abs at the torch seat (c 150-1200 m/s); Pc-port face <= 130.2 / 131.3 / 130.5 bar abs (CC-1 6 mm3 / CC-1 20 mm3 / CC-2) over c 181-1200 m/s; the band values do not bound gas-filled passages DERIVED T796	linear lossless duct (ASSUMED); not modelled: wave steepening and shock reflection, losses, torch-chamber compliance, N2O decomposition, a blocked-outlet deflagration; Mechanical's tap-wall row v3 at FA1S passes: governing SOAK52 MS_y +0.670 / MS_u +0.785; screw-temperature sensitivity MS_y +0.275 at SOAK70 [CALC on SIM]; ligament lowest local natural frequency >= 26 kHz [CALC, edges and span ASSUMED]; R4-01 closed (CHANGELOG 259 (1); wording LEAD 258 (4)) DERIVED T795	reviews/combustion/r4/r4_torch_passage.json (8cb4b0c80173b88d), r4_pc_port_face.json (94f5a39166a8f778); 04-fea/v352/jfwd_tapwall_v352_v3.json (ef70572ec92b05b0) T797
FA-1 class Z3 v2 and the J-FWD replace rule	Z3 v2: 63.1 < head <= 88.1 bar abs inside the light window, with the undamped SDOF ESP of the head history <= 151.04 bar abs (FA1S) at each of the 406 periods 0.02-50 ms, first-to-last crossing of 63.1 bar abs <= 3.0 ms and 10-90 % first rise >= 0.8 ms: 896 / 896 FA-1 grid cases inside (ESP max 146.881 bar abs at 3.5 ms, margin 4.159 bar); replace rule: Q_JFWD = max(head peak, head ESP at 0.2-0.5 ms) > 60.0 bar abs at any time in a run means the line-231 J-FWD inspection and the TC-02 taps gauged before the next run (298 / 896 grid cases; 185 for the peak alone) DERIVED *Lead decision, line 258 (3); basis SIM* T799	grid counts on an ASSUMED construct, not rates; Z3 v2 NOT EVALUATED in flight (1 kHz record); not meant for normal lights (a light to MEOP trips Q_JFWD only below a 0.288 ms 10-90 % rise); a noise-floor rule for measured stand traces TO-VERIFY; the 0.2-0.5 ms band ASSUMED (TO-VERIFY on a v3.5.2 modal); damped variant not adopted (damping MISSING) DERIVED *Lead decision, line 258 (3); basis SIM* T798	interfaces/mechanical.json v2.4 (66bc53d88d444679) lc_tb70_envelope_v352; 04-fea/v352/lc_tb70_envelope_v2.json (7ebd2fd63dc878ef); combustion.json z3_jfwd_rulings_v2p4d; CHANGELOG 258 (3), 259 (2)-(3) T800
Pc transducer proof	flight unit >= 166.14 bar abs (1.1 x FA1S 151.04); the C1 Kulite XTL-190SM-2000A pick, 206.8 bar abs (DATASHEET 1.5x rule), meets it; stand dynamic channel: flush >= 103.4 bar abs (1.1 × 94), 0.1-1 ms path >= 133.1 bar abs (1.1 × 121), range >= 100 bar abs flush and >= 132 bar abs with a sense line DATASHEET *Lead decision, line 249 (4)* T803	path period > 1 ms not acceptable for the dynamic channel; the P_torch 070BG over-ranges during FA-1 (its proof and recovery are Electronics v2.12 items) DATASHEET *Lead decision, line 249 (4)* T801 [P_torch: lead 264 (4) T802 selects the C1-family Kulite XTL-190SM-2000A; this 070BG text is superseded; the threshold re-base is OPEN: contradiction register C-14]	combustion.json v2.4c pc_dynamic_channel T804
FA-1 at TC-01 J-FWD (per-mode)	time-domain modal bound 114.6 bar abs undamped (108.9 at zeta 0.02), E rule 127.0 (121.4); spectral sums at zeta 0: signed 101.5-109.4, absolute 137.3-147.9, SRSS 55.7-58.1 bar abs; break-even 116.3 yield / 125.8 ultimate; MS at the bound +0.015 / +0.097; on the restated 0.1-1 ms band value 121 bar abs -0.039 / +0.040 (line 228 lists +0.25 / +0.23 on the 93 / 102 band reading) DERIVED T806	worst of the FE frequency factors x0.85 / 0.90 / 1.00; intervals because FEA3D publishes 9 of 60 modal shares (OI-51); the bound takes the residual at its published signed share 0.382 and stays below the yield break-even while the residual absolute share is <= 0.487; TC-01 LC1 node: 111.8 (bound), 122.7 (E rule) DERIVED T805	data/v2p4b/fa1_modal_th_bound_v2p4b.json (639f5fbf3db77ee8), fa1_per_mode_peq_v2p4b.json (7ade3ee7ef544f7a) T807

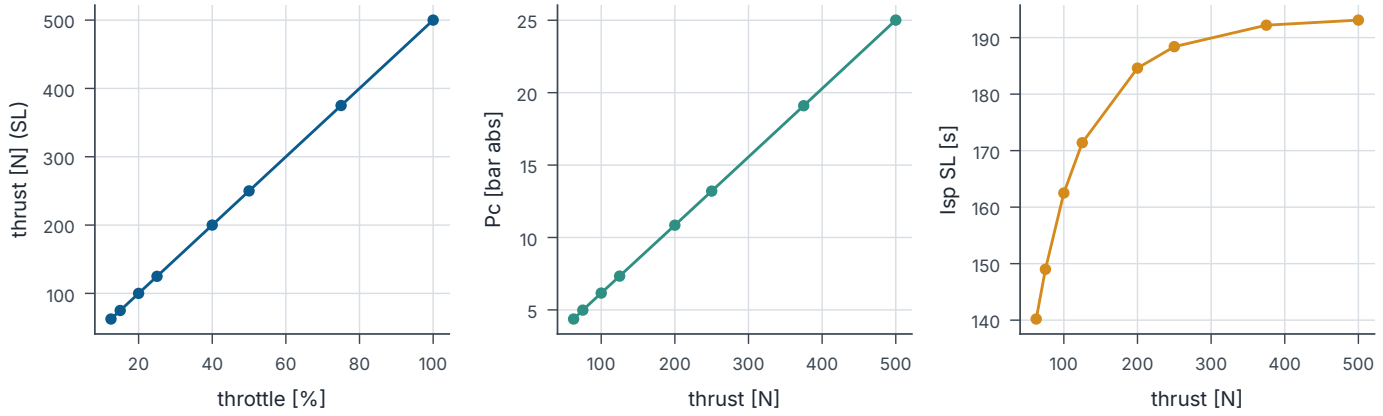
Notes of the combustion datasheet inputs (quoted)
A1 apogee target 210 m is decided (lines 214, 227 (1)); the A1 records above are the 300 m comparators until gnc.json v3.2 (OI-49). DECIDED *Design value, DATASHEET-INPUTS-COMBUSTION.md v2.4d* T808
Prices: none quoted here; any quote is TO-QUOTE, delivered to Sønderborg DK-6400. DECIDED *Design value, DATASHEET-INPUTS-COMBUSTION.md v2.4d* T809
OI-41 is DECIDED (LEAD 230 (2) band values; 249 (4) flight Pc proof); the per-mode row is an analysis path only and still depends on the FEA3D share vector (OI-51) and on damping (OI-52). Z3 v2 and the J-FWD replace rule on Q_JFWD are ADOPTED (LEAD 258 (3); table row 'FA-1 class Z3 v2 and the J-FWD replace rule'). Open: the PURGE command time (Electronics v2.12); before flight acceptance (LEAD): the flight consequence of a flagged first light, the damped ESP variant and the 0.2-0.5 ms J-FWD band. DECIDED *Design value, DATASHEET-INPUTS-COMBUSTION.md v2.4d* T810
pcf_tmax 0.8 s (LEAD 227 (5)) holds on the dead side only with the photodiode main-chamber view (FLAMEOUT_OPT armed) or the GNC reference re-taken at the ECU E2 / light; photodiode-blind families release 42 / 98 of 1000 dead runs at 0.8 s (OI-PHOTODIODE-VIEW, open for IGN-1). DECIDED *Design value, DATASHEET-INPUTS-COMBUSTION.md v2.4d* T811

2.2 Throttle points (20 C tank, initial port)

combustion.json /operating_points/throttle_points_20C_initial_port; basis: 20 C saturated tank (P_tank 50 bar row of the map for lift and dP/Pc), initial port 25 mm, nominal law. SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* 0.92, eta_cf 0.975)

Fraction	Thrust	Pc	O/F	Isp SL	N2O flow	ABS flow	Pintle lift	dP_inj / Pc

Fraction	Thrust	Pc	O/F	Isp SL	N2O flow	ABS flow	Pintle lift	dP_inj / Pc
1.000 SIM T817	500.0 N SIM T033	25.01 bar SIM T812	10.44 SIM T035	193.1 s SIM T034	241.0 g/s SIM T813	23.09 g/s SIM T814	1.916 mm SIM T815	0.91 SIM T816
0.750 SIM T826	375.0 N SIM T818	19.10 bar SIM T819	8.99 SIM T820	192.2 s SIM T821	179.0 g/s SIM T822	19.90 g/s SIM T823	1.252 mm SIM T824	1.55 SIM T825
0.500 SIM T835	250.0 N SIM T827	13.20 bar SIM T828	7.34 SIM T829	188.4 s SIM T830	119.1 g/s SIM T831	16.24 g/s SIM T832	0.755 mm SIM T833	2.74 SIM T834
0.400 SIM T844	200.0 N SIM T836	10.85 bar SIM T837	6.58 SIM T838	184.6 s SIM T839	95.9 g/s SIM T840	14.57 g/s SIM T841	0.588 mm SIM T842	3.58 SIM T843
0.250 SIM T853	125.0 N SIM T845	7.33 bar SIM T846	5.32 SIM T847	171.4 s SIM T848	62.6 g/s SIM T849	11.77 g/s SIM T850	0.367 mm SIM T851	5.80 SIM T852
0.200 SIM T861	100.0 N SIM T037	6.16 bar SIM T854	4.85 SIM T855	162.5 s SIM T856	52.0 g/s SIM T857	10.73 g/s SIM T858	0.301 mm SIM T859	7.10 SIM T860
0.150 SIM T870	75.0 N SIM T862	4.98 bar SIM T863	4.34 SIM T864	149.0 s SIM T865	41.7 g/s SIM T866	9.61 g/s SIM T867	0.238 mm SIM T868	9.02 SIM T869
0.125 SIM T879	62.5 N SIM T871	4.37 bar SIM T872	4.06 SIM T873	140.2 s SIM T874	36.5 g/s SIM T875	8.98 g/s SIM T876	0.207 mm SIM T877	10.43 SIM T878

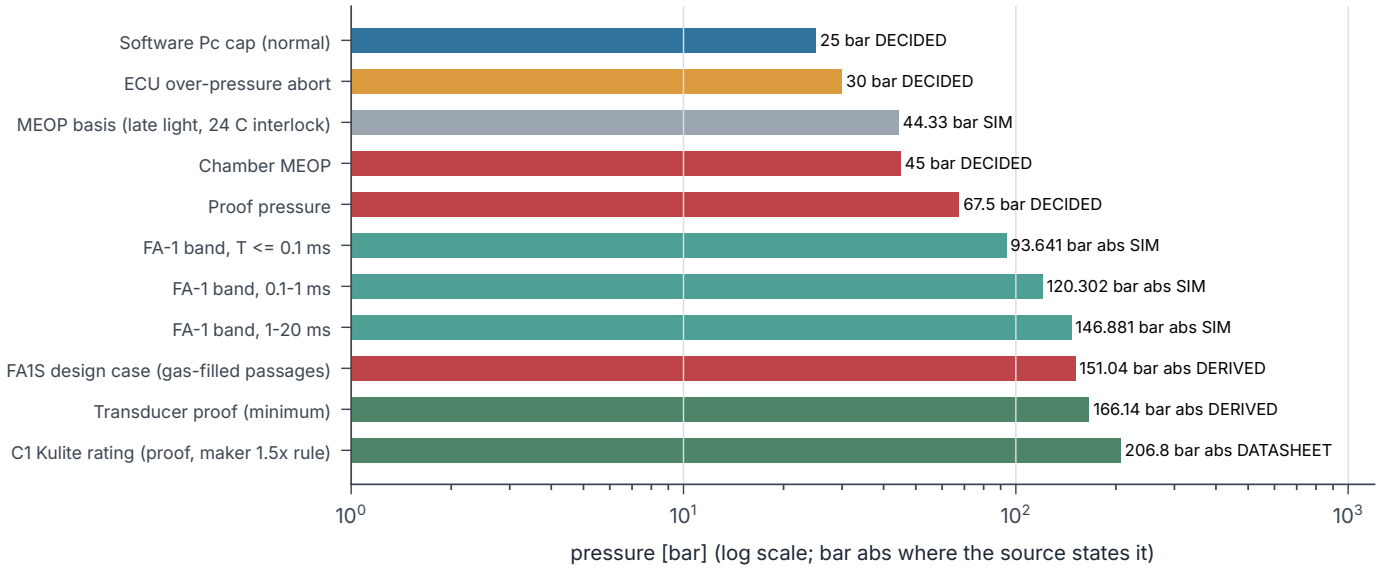


Throttle points: thrust vs throttle fraction, chamber pressure and sea-level Isp vs thrust (SIM). Data: T033, T034, T037, T812, T818, T819, T821, T827, T828, T830, T836, T837, T839, T845, T846, T848, T854, T856, T862, T863, T865, T871, T872, T874

2.3 Pressure ladder

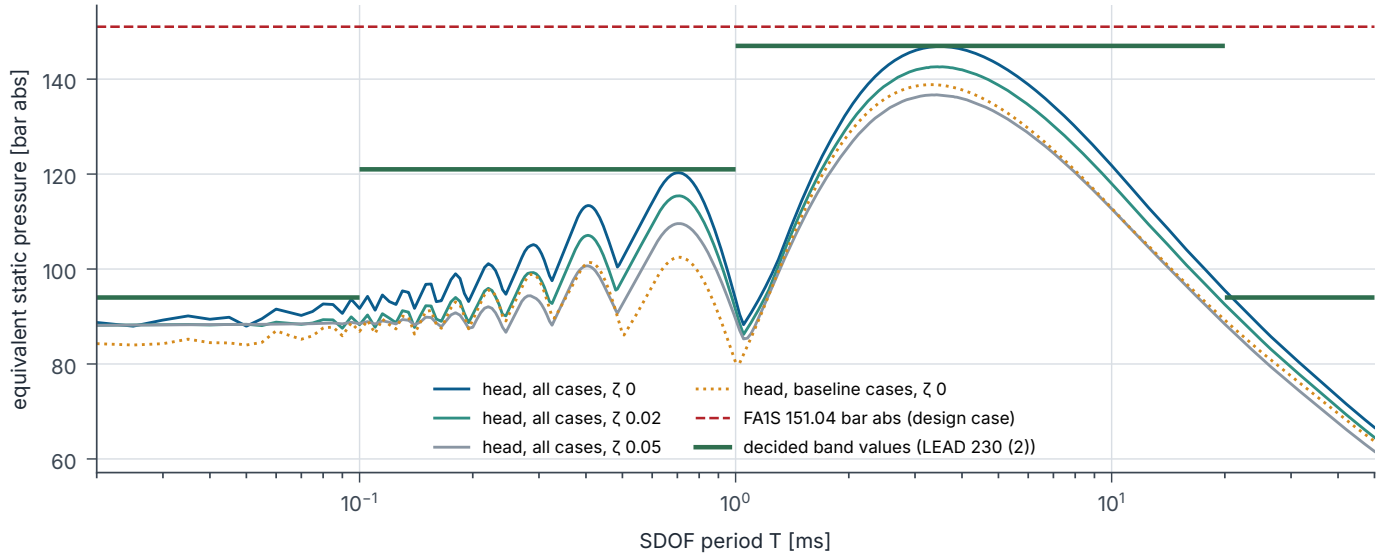
Each value at the precision of its source, with the unit its source states: bar abs where the source says so; otherwise bar, with "not stated in the source" (REV-DS-1 D1-08).

Pressure level	Value	Basis of the unit (bar or bar abs)
Software Pc cap (normal)	25 bar DECIDED *Lead decision of 2026-10-12* T880	not stated in the source (bar); combustion.json pressure_limits in Pa
ECU over-pressure abort	30 bar DECIDED *Lead decision of 2026-10-12* T881	not stated in the source (bar); combustion.json pressure_limits in Pa
MEOP basis (late light, 24 C interlock)	44.33 bar SIM T884	not stated in the source (bar); combustion.json meop_basis_v2p4 basis_bar
Chamber MEOP	45 bar DECIDED *Lead decision of 2026-10-12* T882	not stated in the source (bar); combustion.json pressure_limits in Pa
Proof pressure	67.5 bar DECIDED *Lead decision of 2026-10-12* T883	not stated in the source (bar); combustion.json pressure_limits in Pa
FA-1 band, T <= 0.1 ms	93.641 bar abs SIM T885	bar abs: combustion.json key head_all_band_maxima_bar_abs
FA-1 band, 0.1-1 ms	120.302 bar abs SIM T886	bar abs: combustion.json key head_all_band_maxima_bar_abs
FA-1 band, 1-20 ms	146.881 bar abs SIM T887	bar abs: combustion.json key head_all_band_maxima_bar_abs
FA1S design case (gas-filled passages)	151.04 bar abs DERIVED T890	bar abs: combustion.json fa1_p_torch_v2p4c ("FA1S ... bar abs")
Transducer proof (minimum)	166.14 bar abs DERIVED T892 = 1.1 DECIDED *Lead decision, line 249 (4)* T893 × FA1S 151.04 bar abs DERIVED T890 Required of the flight Pc transducer and of P_torch: CHANGELOG 256 (5), R4-12 DECIDED *Lead decision, line 256 (5)* T894, and 264 (4) DECIDED *Lead decision, line 264 (4)* T802	bar abs: DATASHEET-INPUTS-COMBUSTION.md row "Pc transducer proof" (">= ... bar abs")
C1 Kulite rating (proof, maker 1.5x rule)	206.8 bar abs DATASHEET T896	bar abs: DATASHEET-INPUTS-COMBUSTION.md row "Pc transducer proof" (C1 rating "... bar abs")



Pressure ladder (bar, log scale; bar abs where the source states it, see the table), each level with its class: limits (DECIDED, lead decisions), the MEOP basis (SIM), FA-1 equivalent-static band maxima (SIM on ASSUMED inputs; the decided band values of line 230 (2) are rounded up from these), the FA1S design case (DERIVED) and the transducer ratings. Data: T880, T881, T882, T883, T884, T885, T886, T887, T888, T890, T892, T896

2.4 FA-1 hard start: equivalent static pressure spectrum



FA-1 equivalent static pressure at the head vs SDOF period (406 periods, 0.02-50 ms, counted from the file), undamped and damped (damping ASSUMED); red: the FA1S design case, 151.04 bar abs; green: the decided band values for structural SDOF paths (line 230 (2)). SIM histories on ASSUMED inputs, DERIVED spectra (Combustion v2.4b data, unchanged by the text-only v2.4c and v2.4d folds: table below). Data: T890, T897, T899

Currency check of the FA-1 spectrum data	Record
File named by combustion.json	same file: combustion.json v2.4d names 02-combustion/data/v2p4b/fa1_pressure_spectra_published_v2p4b.csv with sha256/16 a2c380515cada638; the file read here has a2c380515cada638 DERIVED T901
v2.4c fold (REPORT §17)	Applied by the lead with `src/apply_v2p4c_rev_comb_4.py` (sha256/16 `063195b1b3a12bed`, run at 2026-10-21T09:35:44Z from date -u). Text only: no model was re-run and no SIM, CEA or FE result changed. T902
v2.4d fold (REPORT §18)	Applied by the lead with `src/apply_v2p4d_lead.py` (sha256/16 `6c43216fc7b22306`, run at 2026-10-21T11:24:28Z from date -u). Text only: no model was re-run and no SIM, CEA or FE result changed; one planning value changed (the T4 N2O, §18.1). T903
FA-1 data after v2.4d (REPORT §18.3)	- No Combustion number changed except the T4 planning value; the FA-1 grid histories are the v2.4b set and the reviewer's statistics in `fa1_grid_statistics_v2p4c` are kept. T904

3 Throttle and thrust vector control

3.1 Throttle

Item	Value (combustion.json)
floor requirement 25pct	F_N: v: 125.0; tag: REQ; Pc_Pa: v: 733390.0; tag: SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* 0.92, eta_cf 0.975); OF: v: 5.32; tag: SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* 0.92, eta_cf 0.975); status: met (dP/Pc 5.8) SIM T905
floor target 20pct	F_N: v: 100.0; tag: REQ; Pc_Pa: v: 616171.0; tag: SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* 0.92, eta_cf 0.975); OF: v: 4.85; tag: SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* 0.92, eta_cf 0.975); status: met (dP/Pc 7.1) SIM T906
floor stretch 12p5pct	F_N: v: 62.5; tag: REQ stretch; Pc_Pa: v: 436985.0; tag: SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* 0.92, eta_cf 0.975); OF: v: 4.06; tag: SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* 0.92, eta_cf 0.975); status: model meets it: dP/Pc 10.4, chug gain margin 38.6 at 10 ms delay, attached nozzle except incipient lip separation (Schmucker); G_ox 74 → ~25 kg/m2s as the port opens: low-flux combustion ... SIM T907
thrust ceiling rule	F_ceiling = F at min(stroke 4 mm, injector dP/Pc >= 0.25) SIM T908
chamber fill time constant s	0.0034 SIM T909
throttle architecture	Servo-driven variable-area pintle (needle-in-seat) at the head end + separate NC main isolation valve upstream (tank side, off-gimbal) DECIDED *Lead decision of 2026-10-11* T910

3.2 Gimbal and TVC requirements (GNC v3.2 datasheet inputs)

01-systems-gnc/DATASHEET-INPUTS-GNC.md rows 6-8 (DSIG v3.2).

Item	Value for the datasheet	Basis
Gimbal range	±7.0 deg commanded limit, on the vector norm; 8.0 deg conical mechanical stop. Flown engine angle: p99 7.32 deg, max 7.36 deg (flexible-mount overshoot, inside the stop). SIM T911	7 deg gives P 0.984. 6 deg gives P 0.982 (3 lost / 2 gained, p 1). 5 deg gives P 0.924 (30 lost / 0 gained, p 1.9e-9). Recorded at the v3 basis. SIM T912
Gimbal slew and dynamics	Slew >= 100 deg/s in the worst direction at 7 deg, at the 7.4 V nominal plug (ENG-07). Other limits: angular acceleration >= 34.9 rad/s2; bandwidth >= 8 Hz; resolution <= 0.1 deg; backlash <= 0.1 deg; command-to-motion latency <= 10 ms. DECIDED *Requirement, DATASHEET-INPUTS-GNC.md v3.2; basis SIM* T913	Comparators: 80 deg/s loses (11 lost / 2 gained, p 0.022); 60 deg/s gives P 0.716. Delivered slew at 7 deg / 7.4 V (DERIVED extrapolation): 111.2 deg/s at the mechanical.json v2 table load; 99.8 deg/s at the v2 term sum; 97.8 deg/s with the hose at its routing limit. TO-VERIFY with the exact kinematics (row 11k). SIM T914
GNC row 8: servo output stiffness K_s (filed under ENG-11; ENG-11 (REQ) is the gimbal axis stiffness, §3.3 and §12.5)	design-to **100** DERIVED T743–**110** N m/rad DERIVED T744 at the ASSUMED servo J **1.6e-03** kg m² ASSUMED T745, not robust across J: It is not robust: it moves with J (R3 pose table, cons: 0.76e-3 130→= 160; 2.7e-3 120→= 160; no K_s passes at all three J), the inertia convention (n48 at J 1.6e-3, CAD v3.5.1: none) and the rotor model (two-mass, CAD v3.5.1: 150→= 160), and it is empty at 0.9 x first bending (CAD v3.5.1). DERIVED T746	24 / 24 variants (RV6-14 (a)-(e)); check run reproduces REV-GNC-6 v05 part B on V5c. Groups: J_1.6_cons_FE_structure <= 90-120; J_1.6_cons_FE_structure_and_bending NONE; J_0.76_cons_FE 130→= 160; J_2.7_cons_FE 125→= 160 (set by A10, compliant + hot E -10 %; the row's 120->= 160 is the rigid T_REF variant A03); J_all_cons_FE_rigid_TREF NONE; J_all_cons_FE_worst_structure 130-140, 160; J_all_n48_FE_rigid_TREF NONE; V5c_reference_and_bending NONE. Summary findings (verbatim): V5c plant (v3.2 basis, CAD v3.3), J 1.6e-3 cons: both axes pass 100-130 N m/rad (reproduces the v3.2 window). FE plant ... DERIVED T915

GNC row 8 (servo output stiffness K_s, filed under ENG-11) changed since rev0. REV-GNC-7 ACCEPTED: its final (reviews/REVIEW-GNC-R5.md 6b86fcd326568646, saved verbatim by the lead at 10:39:38Z; reviewer manifest reviews/gnc/rev4b/MANIFEST_REV-GNC-7.sha256 e04257acba555551 verified 94/94 OK at 10:38:50Z; SUPERVISION.json entry added) is PUBLISHABLE WITH CORRECTIONS (0 blocking; 4 major RV7-01 to RV7-04; 5 minor; 3 notes). gnc.json v3.2 = the candidate 01-systems-gnc/staging_v3.1/rev4b/v32/gnc_v3.2_rev4b.json 43d58c4aff164b86 with the 34 ops of reviews/gnc/rev4b/data/b/rv7_v32_corrections.json f7c0c6051efd5a08 (reviewer trial fea33eacdd8a7d56), and DATASHEET-INPUTS-GNC.md with DS ops 1-7 plus rows for the new open items 53-57; the lead publishes it after an independent apply that matches the trial and t10_postcheck v3 --publication --rv7 exits 0. ENG-11 (RV7-01): design-to 100 <= K_s <= 110 N m/rad (two-sided) at the ASSUMED servo J 1.6e-3 on the CAD v3.5.2 R3 plant (pose-wise <= 90-110; band maximum <= 90-100; band minimum <= 90-115 and 150→=160; only 100 passes at both band ends; no K_s passes at all three J, so not robust across J) [DERIVED]; the rev0 datasheet row 8 window 100-120 is withdrawn. DECIDED *Lead decision, line 258 (1)* T740 The rev0 window is SUPERSEDED and not used. ENG-11 (REQ), the gimbal axis stiffness of REQUIREMENTS.md line 62, is in §3.3 and §12.5.

3.3 Axis stiffness band and rocking modes (FEA3D, R3 poses)

Per-pose values in §9.7. Servo k_s ASSUMED (TO-VERIFY at B-TVC-1). The FE rows are ENG-11 (REQ), the gimbal axis stiffness, per axis.

Axis stiffness K over the 17 R3 poses	min (pose) – max (pose)
FE, α	**7294** N m/rad SIM T916 (stop000 T918) – **9337** N m/rad SIM T917 (stop180 T919)
FE, β	**3972** N m/rad SIM T920 (stop180 T922) – **7172** N m/rad SIM T921 (stop000 T923)
with servo k_s 58, α	**1150** N m/rad DERIVED T924 (neutral T926) – **1426** N m/rad DERIVED T925 (stop000 T927)
with servo k_s 58, β	**791** N m/rad DERIVED T928 (stop180 T930) – **1799** N m/rad DERIVED T929 (stop000 T931)
with servo k_s 174, α	**2697** N m/rad DERIVED T932 (neutral T934) – **3076** N m/rad DERIVED T933 (stop000 T935)
with servo k_s 174, β	**1697** N m/rad DERIVED T936 (stop180 T938) – **3594** N m/rad DERIVED T937 (stop000 T939)

ENG-11 (REQ) status on these FE values (REQUIREMENTS.md line 62): the hard minimum ≥ 3,100 N·m/rad per axis DERIVED T942 holds (beta minimum **3971.84** N m/rad SIM T920); the spread element (alpha / beta spread across poses ≤ ±**20** % DERIVED T941) is not met on beta: ±**28.7** % DERIVED T944 half-range over mid-range (alpha ±**12.3** % DERIVED T943); OPEN under 188 (7) T945: GNC robustness assessment, then the lead amends the requirement or Mechanical stiffens (§12.5; open item 11b, §11.2).

3.4 TVC actuation record (mechanical.json v2.1, not re-derived for v3.5.2)

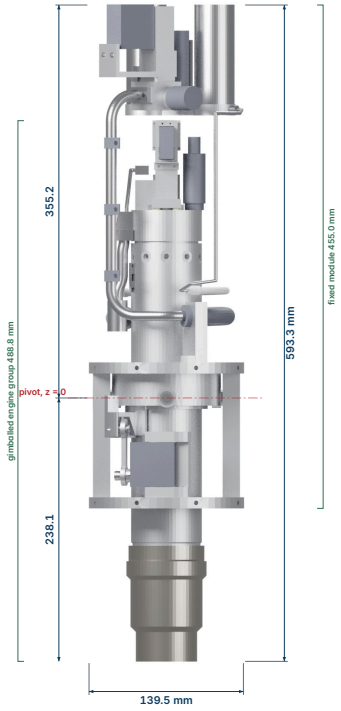
These v2.1 sections were not re-derived for v3.5.2 (mechanical.json not_re_derived_v352); where they conflict with a _v352 section, the _v352 section governs.

Item	Value (v2.1 record)
concept	2 x Savox SB-2292SG on the fixed lower ring GM-10 below the gimbal ring, shafts tangential; 7075 clamp horns r_h; vertical dog-bone pushrods (GE 5 E spherical plain bearings both ends) to two levers on the engine collar DERIVED *Design value, mechanical.json v2.4* T946
gear ratio n	4.8 DECIDED *Lead decision, per mechanical.json v2.4* T947
servo angle range deg	cmd_7deg: amp_deg: 7.0; psi_min_deg: [-32.37, -32.37]; psi_max_deg: [37.39, 37.39]; rod_tilt_max_deg: [7.37, 7.37]; stop_8deg: amp_deg: 8.0; psi_min_deg: [-36.82, -36.82]; psi_max_deg: [44.22, 44.22]; rod_tilt_max_deg: [8.83, 8.83] DERIVED *Design value, mechanical.json v2.4* T948
rate engine dps under 3.19Nm	6.6: worst_dps: 95.3; worst_dir_deg: 315.0; alpha_dps: 153.3; beta_dps: 152.2; centre_worst_dps: 117.5; 7.4: worst_dps: 111.2; worst_dir_deg: 315.0; alpha_dps: 178.0; beta_dps: 175.5; centre_worst_dps: 139.5; 7.6: worst_dps: 118.3; worst_dir_deg: 315.0; alpha_dps: 189.0; beta_dps: 185.9; centre_worst_dps: 149.4; 7.8: worst_dps: 125.4; worst_dir_deg: 315.0; alpha_dps: 200.0; beta_dps: 196.2; ... DERIVED *Design value, mechanical.json v2.4* T949
ENG-07 rate status	as built: worst-direction slew at 7 deg 111.2 deg/s at 7.4 V, 118.3 at 7.6 V (>= 100); 95.3 deg/s at the 6.6 V degraded plug voltage, the case GNC validated for n 4.8 (lead FINAL: P(landing) 0.982 at 6.6 V) CALC T950 [class: CALC / DERIVED, with SIM for the P value it quotes]
hinge moment design Nm	37.95 DERIVED *Design value, mechanical.json v2.4* T951
ENG-11 status	first rocking mode 16.0-27.8 Hz (excl. J_int) vs the 25 Hz minimum: below it whenever the servo is at the low end of its ASSUMED stiffness; the servo caps the total (servo alone through a rigid structure 16.2-28.7 Hz incl. pose); lead route: ENG-07d closes by control + B-TVC-1 measurement; GNC to state the minimum K / f for a 6 dB flex gain margin, Mechanical stiffens to it (stiffened concept: structure-only 3400/1975 vs 2442/1608 N m/rad at the centre) SIM T952 [class: SIM / DERIVED]
gimbal architecture	two-axis Cardan gimbal of the whole thrust chamber; split engine collar (GM-01/02) clamped on the case at the pivot plane carries the inner pins (+/-Y) in 608-2RS bearings in the ring GM-03; the ring carries the outer pins (+/-X) in 608-2RS bearings in two yoke posts GM-08 on the thrust rings GM-09/GM-10 (fixed module); 8 deg conical PU stop GM-11 DERIVED *Design value, mechanical.json v2.4* T953
gimbal range deg	7.0 DECIDED *Requirement, mechanical.json v2.4* T954
gimbal mechanical stop deg	8.0 DERIVED *Design value, mechanical.json v2.4* T955
gimbal bearings	4 × 608-2RS (8×22×7); pins 8 h6 17-4PH H1150 DECIDED *Design value, mechanical.json v2.4* T956

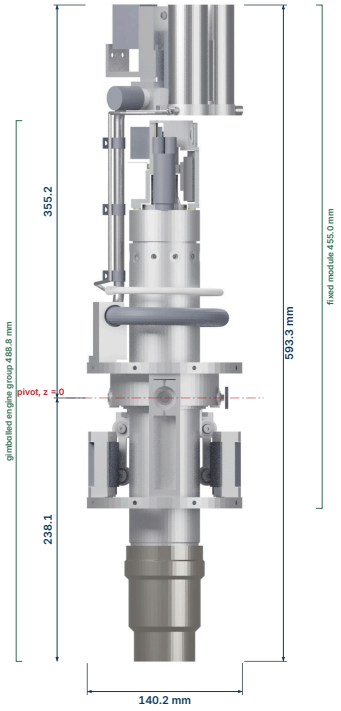
4 Geometry and mass

Renders are visualisation only. They are produced by gen/render_views.sh from the tessellated GLB display mesh of the FINAL CAD v3.5.2 (GLB sha256/16 4bcc264f683e128f; render manifest 9e3d1d0e8156ad4a). They are not drawings and carry no tolerances. Dimensions printed on them are bounding-box extents from the assembly manifest; the STEP solids are the CAD of record.

4.1 Envelope and principal dimensions



Front view (looking along +Y)



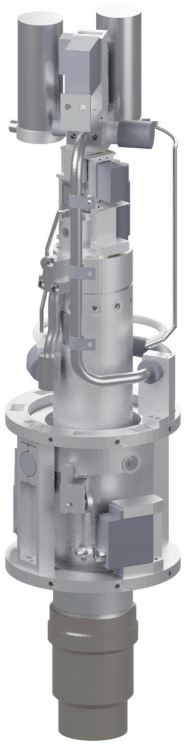
Side view (looking along −X)

Envelope from the union of the 149 instance bounding boxes in the assembly manifest (engine frame, mm, origin at the gimbal pivot, +z forward). An axis-aligned box does not bound the radial extent; the pose clearance check is listed in §4.6. The bay requirement is REQUIREMENTS.md §1 (the airframe ICD draft 0.2 is not used, see §4.5).

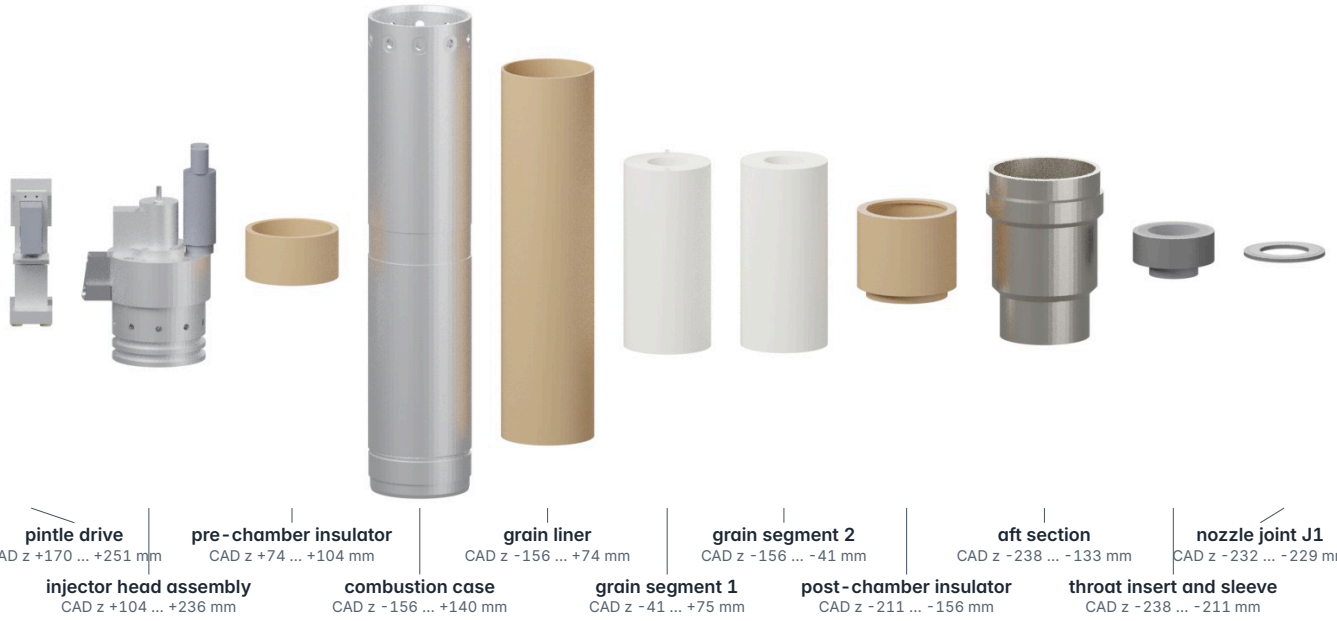
Dimension	Value
Overall length along the engine axis (z)	**593.3** mm DERIVED T040
Overall width X / Y (axis-aligned bounding box)	**139.50** mm DERIVED T043 / **140.23** mm DERIVED T044
Pivot to forward end / pivot to aft end	**355.2** mm DERIVED T041 / **238.1** mm DERIVED T042
Gimballed engine group: z range / length	**−238.1–250.7** mm DERIVED T046 / **488.8** mm DERIVED T045
Fixed module (thrust frame, valve module, servos): z range / length	**−99.8–355.2** mm DERIVED T048 / **455.0** mm DERIVED T047
GLB read-back: meshes / bounds error / result	**149** DERIVED T957 / **0.087** mm DERIVED T958 / **yes** DERIVED T959
Clearance at 18 poses: not-whitelisted interferences / smallest moving clearance	**0** DERIVED T960 / **0.525** mm DERIVED T961
Engine-bay inner diameter (requirement)	≥ 140 mm DECIDED *Requirement, REQUIREMENTS.md rev 2* T962
Thrust rings GM-09 / GM-10: outer diameter (CAD bounds) / radial gap to the bay bore	**139.50** mm DERIVED T964 / **0.250** mm DERIVED T965
Instances reaching beyond the ring radius (CAD bounds)	ENG-BL-06#0: 70.48 mm from the axis (V2 vent bulkhead fitting (316L) through the bay wall at az 112; outside: non-propulsive opposed tee (airframe ICD-14)) DERIVED T966

4.2 Assembly views

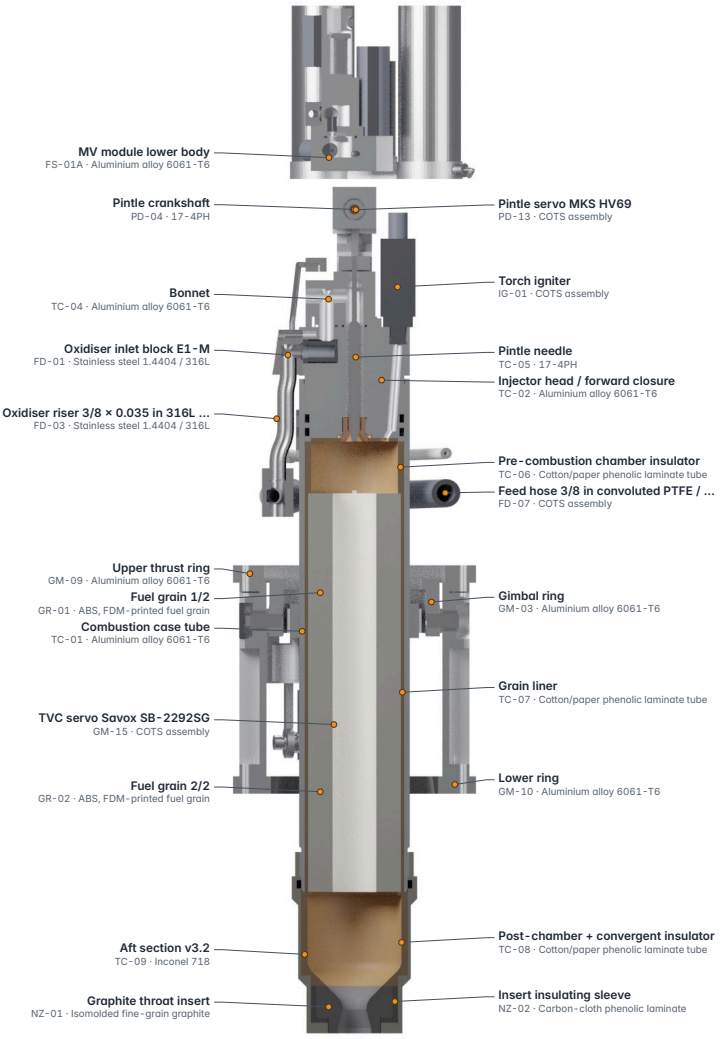
What the renders do not show. Not in the GLB display mesh, so shown in no render: all 57 of the 57 standard-hardware lines of BOM /hardware (ENG-HW-101 to ENG-HW-170; hardware_v352 lists 57 lines), so no screw, spring, pin or insert, among them ENG-HW-101 and ENG-HW-170 (the J-FWD radial screw and the disc spring under its head); ENG-GM-12 (Transport / neutral lock, 2 half rings (REMOVE BEFORE FLIGHT)). DERIVED T974



Isometric view, FINAL CAD v3.5.2, neutral pose (render 0e2b467ecc117872). Visualisation only; parts not in the GLB are listed above the figure.



Thrust chamber and pintle stack laid out side by side along the axis order (pintle drive at left, nozzle joint at right), true scale. Axial positions are not preserved; the CAD z range of each group is printed under it. Visualisation only.



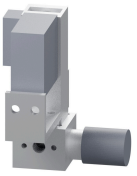
Half section in the plane y = 0 (the y < 0 half removed; cut faces darkened). Labels: BOM name, part number and material, placed at the area-weighted centroid of each part's cut face. Visualisation of display geometry, not a drawing. The section shows no screw, spring or other standard hardware (not in the GLB; see the note in §4.2). Parts without a cut face in this plane: ENG-FS-10. BOM rows: T105, T207, T247, T083, T087, T241, T151, T175, T177, T257, T191, T163, T225, T187, T249, T243, T251, T139, T093, T253, T153, T255, T193

4.3 Part families

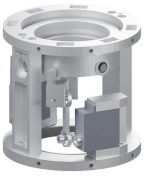
BL · Head-end bleed (provision)
11 DERIVED T387 lines · **0.783** kg DERIVED
T388

EL · Sensors and encoders
6 DERIVED T389 lines · **0.074** kg DERIVED
T390

FD · Oxidiser feed
11 DERIVED T391 lines · **0.276** kg DERIVED
T392

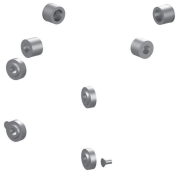


FS · Main valve module (fixed)
15 DERIVED T393 lines · **0.336** kg DERIVED
T394



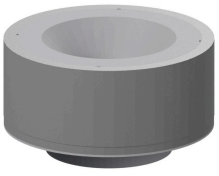
GM · Gimbal and thrust frame
19 DERIVED T395 lines · **1.153** kg DERIVED
T396

GR · Fuel grain
2 DERIVED T397 lines · **0.424** kg DERIVED
T398



HW · Bearings and hardware
4 DERIVED T399 lines · **0.024** kg DERIVED
T400

IG · Igniter
2 DERIVED T401 lines · **0.074** kg DERIVED
T402



NZ · Nozzle and joint
5 DERIVED T403 lines · **0.063** kg DERIVED
T404

PD · Pintle drive
20 DERIVED T405 lines · **0.175** kg DERIVED
T406

TC · Thrust chamber
11 DERIVED T407 lines · **1.664** kg DERIVED T408

Each family is rendered at its assembly position (not exploded). Line counts and masses are sums over the BOM lines of the family (qty × unit mass, DERIVED by this generator; the transport lock and the standard hardware lines are excluded). The HW family render shows the modelled BOM lines ENG-HW-20, ENG-HW-21, ENG-HW-22, ENG-HW-22-IR, ENG-HW-25 (render metadata) and no fastener or spring: the standard hardware of BOM /hardware and the transport lock are not in the GLB (§4.2).

4.4 Mass

Mass by subassembly and state, massprops MP v3.5.2. The states include the standard hardware carried as max(0.20 kg allowance, CALC sum) at the host parts (lead 188 (11)). Group "all" is the CAD model total. Not weighed.

Group	Wet (full grain)	Mid-burn	Burn-out	No grain
gimballed engine	**2.8210** kg DERIVED T975	**2.7380** kg DERIVED T976	**2.6550** kg DERIVED T977	**2.3967** kg DERIVED T978
gimbal ring	**0.2166** kg DERIVED T979	**0.2166** kg DERIVED T980	**0.2166** kg DERIVED T981	**0.2166** kg DERIVED T982
TVC horns	**0.0200** kg DERIVED T983	**0.0200** kg DERIVED T984	**0.0200** kg DERIVED T985	**0.0200** kg DERIVED T986
TVC pushrods	**0.0184** kg DERIVED T987	**0.0184** kg DERIVED T988	**0.0184** kg DERIVED T989	**0.0184** kg DERIVED T990
fixed module (thrust frame, MV module, servos)	**2.2669** kg DERIVED T991	**2.2669** kg DERIVED T992	**2.2669** kg DERIVED T993	**2.2669** kg DERIVED T994
total	**5.3430** kg DERIVED T995	**5.2600** kg DERIVED T996	**5.1770** kg DERIVED T997	**4.9187** kg DERIVED T998

Gimballed body (group "engine": thrust chamber, grain, collar and levers, inner pins, pintle stack, engine shares of hose and lines, J-FWD hardware), mechanical.json /gimballed_mass_properties_v352 (GNC input). Full inertia tensors in the source.

State	Gimballed engine mass	CG (x, y, z) from the pivot	I about the pivot: Ixx, Iyy, Izz
wet	**2.82102** kg DERIVED T999	**-2.543, 0.892, 4.539** mm DERIVED T1000	**0.05752, 0.05775, 0.00218** kg·m² DERIVED T1002
mid	**2.73802** kg DERIVED T1003	**-2.620, 0.919, 5.934** mm DERIVED T1004	**0.05699, 0.05722, 0.00214** kg·m² DERIVED T1006
burnout	**2.65502** kg DERIVED T1007	**-2.702, 0.948, 7.417** mm DERIVED T1008	**0.05646, 0.05670, 0.00211** kg·m² DERIVED T1010
nograin	**2.39666** kg DERIVED T1011	**-2.993, 1.050, 12.688** mm DERIVED T1012	**0.05482, 0.05506, 0.00200** kg·m² DERIVED T1014

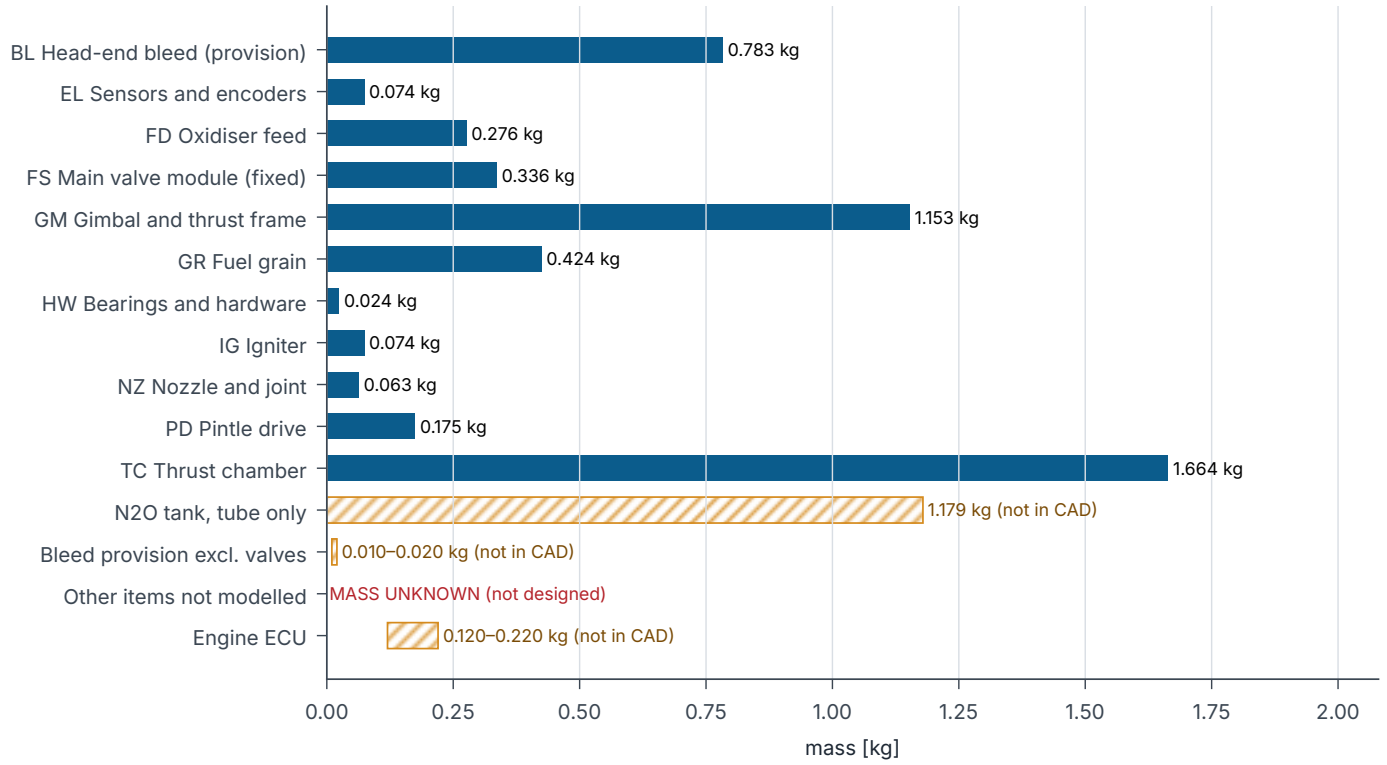
Item	Value
Hardware carried: allowance / CALC sum / carried	**0.20** kg CALC *Lead decision, line 188 (11)* T1025 / **0.22314** kg CALC T1026 / **0.22314** kg CALC T1027
Hardware lines without a mass	HW-128; HW-158; HW-159; HW-160; HW-161; HW-163; HW-164; HW-165; HW-166; HW-168 MISSING T1028
Landing-mass change vs v3.5.1: as built / without the conditional TC-09 thinning / the thinning	**−0.0662** kg CALC T1019 / **+0.0118** kg CALC T1020 / **−0.0780** kg CALC T1021
GNC re-fly trigger and verdict	**0.10** kg DECIDED *Lead decision, per massdelta_v3.5.2_vs_v3.5.1.json v3.5.2* T1022 BELOW the trigger with and without the conditional TC-09 thinning: no re-fly by mass CALC T1023 CALC (CAD volume x engmat density + hardware CALC); not weighed, so not a 'confirmed' mass; COTS masses unchanged CALC T1024
CAD set mass statement	massprops_v3.5.2.json (engine group mid 2.73802 kg, CG [-2.62, 0.919, 5.934] mm; hardware carried 0.22314 kg CALC) and massdelta_v3.5.2_vs_v3.5.1.json v9: landing -0.0662 kg vs v3.5.1 (+0.0118 kg without the conditional TC-09 thinning), below the 0.10 kg GNC trigger with and without it; CALC, not weighed CALC T1029
Electronics mass statement vs the trigger	Electronics net -0.061 kg; worst combination: Mechanical v3.5.2 +0.0118 kg (without the TC-09 thinning) + the C1 Pc addition with no credit for the P_feed / P_mid reduction = +0.025 kg; below the 0.10 kg re-fly trigger in every combination. Not CONFIRMED (nothing weighed): information for GNC, not a re-fly input DERIVED T1030

Mass by BOM part family (DERIVED: Σ qty × unit mass over the BOM lines).

BOM family	Lines	Mass
BL · Head-end bleed (provision)	**11** DERIVED T387	**0.7832** kg DERIVED T388
EL · Sensors and encoders	**6** DERIVED T389	**0.0740** kg DERIVED T390
FD · Oxidiser feed	**11** DERIVED T391	**0.2759** kg DERIVED T392
FS · Main valve module (fixed)	**15** DERIVED T393	**0.3364** kg DERIVED T394
GM · Gimbal and thrust frame	**19** DERIVED T395	**1.1528** kg DERIVED T396
GR · Fuel grain	**2** DERIVED T397	**0.4244** kg DERIVED T398
HW · Bearings and hardware	**4** DERIVED T399	**0.0242** kg DERIVED T400
IG · Igniter	**2** DERIVED T401	**0.0740** kg DERIVED T402
NZ · Nozzle and joint	**5** DERIVED T403	**0.0632** kg DERIVED T404
PD · Pintle drive	**20** DERIVED T405	**0.1746** kg DERIVED T406
TC · Thrust chamber	**11** DERIVED T407	**1.6640** kg DERIVED T408
Sum of BOM part lines (transport lock and standard hardware excluded)		**5.0466** kg DERIVED T409
BOM mass_sum_kg as published (BOM header)		**5.0466** kg DERIVED T412
Standard hardware lines (BOM /hardware, CALC)		**0.22314** kg CALC T411

Items excluded from the CAD mass. They are not zero: each is carried at its source value or as UNKNOWN. Rows from mechanical.json v2.1 sections are labelled: they were not re-derived for v3.5.2.

Not in the CAD mass	Mass	Status / source
N2O tank: tube only (bulkheads, valves, fittings not designed)	**1.179** kg DERIVED T483	6061-T6 tube 101.6 × 3.175 + 2 bolted bulkheads (radial M6, 2 FKM O-rings each) DERIVED T484 mechanical.json v2.1 record (not re-derived for v3.5.2: not_re_derived_v352)
Head-end bleed provision, excluding valves (BOM provision lines)	**0.010–0.020** kg ASSUMED T485	v3.5: bleed MODELLED (ENG-BL-01..10 in the BOM lines); provision = opposed vent tee outside the wall only ASSUMED T486
Other items not modelled (mechanical.json list)	MISSING MISSING T488	thrust-ring radial screws (16 x M5); isolation valve + actuator (mass UNKNOWN: Electronics/Combustion); tee, filter (<= 15 um, OI-7), relief, fill/vent; hose and lines fixed shares (~0.07 kg ASSUMED); tank MISSING T487
Engine ECU (on the fixed structure, not in CAD)	**0.16** kg ASSUMED T489 (**0.12–0.22** kg T490)	separate engine ECU (Teensy 4.1) on the fixed thrust structure, CAN link to flight computer DERIVED T491
N2O load (propellant)	**2.00** kg DECIDED *Lead decision of 2026-10-12* T492	LEAD decision 2026-10-12 (CHANGELOG): 2.00 kg / 3.17 L (85 % fill) DECIDED *Lead decision, per combustion.json v2.4d* T493



CAD mass by BOM family (solid) and items outside the CAD mass (hatched; ranges where the source gives one). Data: T388, T390, T392, T394, T396, T398, T400, T402, T404, T406, T408, T483, T485, T488, T489, T492

4.5 Airframe interface and engine mounting (current sources)

Airframe-facing engine data from the current sources: REQUIREMENTS.md rev 2, the CAD v3.5.2 BOM and assembly manifest, combustion.json v2.4d, electronics.json v4.1, gnc.json v3.2 and massdelta v3.5.2. Engine mounting hole loads and ring resultants of the v3.5.2 FEA: §9.6 (they govern for the airframe).

Item	Value
Bay, attachment and gimbal	
Engine-bay inner diameter (REQUIREMENTS.md §1)	≥ 140 mm DECIDED *Requirement, REQUIREMENTS.md rev 2* T962
Thrust ring outer diameter / radial gap to the bay bore (CAD bounds)	**139.50** mm DERIVED T964 / **0.250** mm DERIVED T965
Airframe attachment: ENG-GM-09 (BOM)	ENG-GM-09: Upper thrust ring (engine module to airframe, thrust path) DECIDED *Design value, BOM_v3.5.2.json v3.5.2* T151
Airframe attachment: ENG-GM-10 (BOM)	ENG-GM-10: Lower ring (stop ring, TVC servo base, frame tie) DECIDED *Design value, BOM_v3.5.2.json v3.5.2* T153
Airframe screws: ENG-HW-125 (BOM /hardware)	**8 × M5** Airframe screw into the upper thrust ring GM-09 (radial, 8x) - ICD-AIRFRAME DECIDED *Design value, BOM_v3.5.2.json v3.5.2* T315
Airframe screws: ENG-HW-126 (BOM /hardware)	**8 × M5** Airframe screw into the lower ring GM-10 (radial, 8x) - ICD-AIRFRAME DECIDED *Design value, BOM_v3.5.2.json v3.5.2* T317
Airframe screws in total	**16** DERIVED T1031 (the mounting holes of §9.6)
Gimbal range / mechanical stop (REQ ENG-07)	**7 °** DECIDED *Requirement, ENG-07* T968 / **8 °** DECIDED *Requirement, ENG-07* T969
Head-end bleed vent through the bay wall (BOM)	ENG-BL-06: V2 vent bulkhead fitting (316L) through the bay wall at az 112; outside: non-propulsive opposed tee (airframe ICD-14) DECIDED *Design value, BOM_v3.5.2.json v3.5.2* T963
Feed across the gimbal (mechanical.json v2.1 record, not re-derived for v3.5.2)	-6 PTFE-lined stainless-braided hose (oxygen-cleaned, PFPE only), JIC-37 ends: fixed end at the isolation-valve outlet above the engine, engine end on a JIC 90 deg elbow at the bonnet -6 ORB inlet (az 90 deg) DERIVED T1032
Loads into the airframe	
Max thrust (requirement ENG-01)	**500** N DECIDED *Requirement, ENG-01* T971
Rated thrust (sea level), throttle point at the software Pc cap	**500.0** N SIM T033
Over-thrust abort threshold (firmware v2.12 config)	**650** N DECIDED *Lead decision of 2026-10-13* T1033
Chamber MEOP / proof pressure	**4.50** MPa (45.0 bar) DECIDED *Lead decision of 2026-10-12* T1034 / **6.75** MPa (67.5 bar) DECIDED *Lead decision of 2026-10-12* T1035
Lateral thrust at the gimbal range (max thrust × sin range)	**60.9** N DERIVED T972
Axial acceleration in flight, max of the dispersed set (gnc.json v3.2)	**4.04** g SIM T1036; p99 **3.77** g SIM T1037
Axial acceleration after first contact (touchdown), max (gnc.json v3.2)	**5.43** g SIM T1038; p99 **5.00** g SIM T1039
Lateral acceleration at the pivot, touchdown, max (gnc.json v3.2)	**6.29** g SIM T1040; p99 **5.22** g SIM T1041
Landing shock envelope (gnc.json v3.2)	**10.00** g DERIVED T1042
Mass	
CAD mass, wet, all groups (hardware CALC carried; not weighed)	**5.3430** kg DERIVED T038
GNC vehicle basis (massprops v3.5.1): liftoff mass / liftoff T/W (DATASHEET-INPUTS-GNC row 4)	**17.44** kg DERIVED T1016 / **2.92** DERIVED T1018
Landing-mass change v3.5.2 vs v3.5.1 / without the conditional TC-09 thinning; GNC re-fly trigger	**−0.0662** kg CALC T1019 / **+0.0118** kg CALC T1020; trigger **0.10** kg DECIDED *Lead decision, per massdelta_v3.5.2_vs_v3.5.1.json v3.5.2* T1022
Verdict (massdelta v3.5.2)	BELOW the trigger with and without the conditional TC-09 thinning: no re-fly by mass CALC T1023

ICD-AIRFRAME. ICD-AIRFRAME v2.1 is REMOVED from the finish scope and is not written now; 00-requirements/ICD-AIRFRAME.md draft 0.2 stays as it is (superseded in places, not updated). DECIDED *Lead decision, line 222* T1043 No value in this datasheet is taken from ICD-AIRFRAME draft 0.2 (2026-10-13): every row above has a current source. The items that were queued for it are kept as engine data or open items in the engine datasheet and this log, for when the airframe is tackled: bay bore >= 140 mm down to the nozzle and the v3.5.1 envelope; mounting loads at the 16 M5 holes and ring resultants (FEA3D lines 217/220); engine-mount modes (line 221); BL-06 vent hole; tank station 0.58-1.08 m; first-flight apogee 210 m; the ENG-11 requirement to MEASURE the airframe first bending mode before K_s is frozen (line 218); the 2.20 kg tank option (line 219). DECIDED *Lead decision, line 222* T1044 In this datasheet: bay bore and envelope §4.1 and above; mounting loads §9.6; engine-mount modes and axis stiffness §9.7; the BL-06 vent above; tank station and the 210 m apogee §11.1; the airframe first bending mode and the 2.20 kg tank option §11.2 (rows 11d, 11g).

4.6 CAD build, fit and interference checks

Check	Record (quoted)
Build record	checks/make_all_log_v3.5.2.json, checks/build_log_v3.5.2.json, checks/roundtrip_parts_v3.5.2.json (112 / 112 STEP round-trips), checks/roundtrip_assembly_v3.5.2.json (neutral and stop045 PASS), checks/glb_readback_v3.5.2.json (149 meshes, PASS) DERIVED T1045

Check	Record (quoted)
STEP round trips (parts)	**112** DERIVED T1046 of **112** DERIVED T1047
Final set vs the interim set	checks/test_final352_v1.json: of the 112 part numbers only ENG-TC-01 differs from the interim (material removal only: 104.037 mm3 removed = the d11.2 flats against the d10.5 flats, dV -103.54 mm3; final-minus-interim 0.03968 mm3 = coplanar-face slivers < 0.001 mm thick on the r 31.313 planes, STEP-precision boolean residue); placements byte-identical; BOM lines added gse_tool/ENG-TC-12, gse_tool/ENG-TC-13, hw/ENG-HW-170, changed part/ENG-TC-01, part/ENG-TC-02 (notes), hw/ENG-HW-101 DERIVED T1048
Parts changed vs v3.5.1 (STL sha256)	checks/test_final352_v1.json parts_changed_vs_v351: STL sha256 of every final part against 03-cad/v3.5.1/parts: exactly these 11 part numbers differ (ENG-NZ-06 new); the other 101 STL files are byte-identical DERIVED T1049
Interference checks of record	interim checks (unchanged geometry): checks/clear_interim352_v2.json, checks/test_interim352b_v1.json, checks/fillet352/ (rev 2 / rev 3), checks/test_r4_v1.json (rev 4), checks/test_r4s_v1.json (head seat (a)); rev 5: 04-fea/v352/checks_rev5/test_rev5_jfwd_v2.json (lead 17:55:55Z: CAD check ACCEPTED; flats, pocket vs forward features and ENG-FD-04, raised head and spring vs every part at the poses, spring rim vs pocket wall, supported area, TC-02 tap walls vs clocking, datum B) and checks/test_final352_v1.json (the final build against the interim: TC-01 material removal only) DERIVED T1050
Rebuild proof	the v3.5.2 set rebuilds from the hub copy of the archive with the five named inputs (shas checked): STL, part STEP DATA, BOM, GLB, placements and manifests byte-identical; massprops identical except the run-time field /cg_iteration[0]/t_s (massdelta follows); the two assembly STEP files identical up to entity order and numbering (OCCT colour/style writer order) DERIVED T1051

Software geometric check at 18 poses (CLR v2; run on the interim set, whose geometry the final set keeps except the ENG-TC-01 material removal, see the CAD set record above). Pairs closer than 3 mm include expected contacts; a clean software check is not a physical fit check.

Pose	Interference records (whitelisted)	Not whitelisted	Pairs closer than 3 mm	Smallest of those
neutral T1052	**4** DERIVED T1056	**0** DERIVED T1057	**8** DERIVED T1058	**1.188** mm DERIVED T1059
alpha+ T1060	**0** DERIVED T1064	**0** DERIVED T1065	**10** DERIVED T1066	**1.188** mm DERIVED T1067
alpha- T1068	**0** DERIVED T1072	**0** DERIVED T1073	**9** DERIVED T1074	**1.188** mm DERIVED T1075
beta+ T1076	**0** DERIVED T1080	**0** DERIVED T1081	**13** DERIVED T1082	**0.816** mm DERIVED T1083
beta- T1084	**0** DERIVED T1088	**0** DERIVED T1089	**13** DERIVED T1090	**0.948** mm DERIVED T1091
combined045 T1092	**0** DERIVED T1096	**0** DERIVED T1097	**13** DERIVED T1098	**1.052** mm DERIVED T1099
combined135 T1100	**0** DERIVED T1104	**0** DERIVED T1105	**13** DERIVED T1106	**1.188** mm DERIVED T1107
combined225 T1108	**0** DERIVED T1112	**0** DERIVED T1113	**13** DERIVED T1114	**1.188** mm DERIVED T1115
combined315 T1116	**0** DERIVED T1120	**0** DERIVED T1121	**15** DERIVED T1122	**1.052** mm DERIVED T1123
stop000 T1124	**0** DERIVED T1128	**0** DERIVED T1129	**14** DERIVED T1130	**0.000** mm DERIVED T1131
stop045 T1132	**0** DERIVED T1136	**0** DERIVED T1137	**14** DERIVED T1138	**0.000** mm DERIVED T1139
stop090 T1140	**0** DERIVED T1144	**0** DERIVED T1145	**12** DERIVED T1146	**0.000** mm DERIVED T1147
stop135 T1148	**0** DERIVED T1152	**0** DERIVED T1153	**14** DERIVED T1154	**0.000** mm DERIVED T1155
stop180 T1156	**0** DERIVED T1160	**0** DERIVED T1161	**14** DERIVED T1162	**0.000** mm DERIVED T1163
stop225 T1164	**0** DERIVED T1168	**0** DERIVED T1169	**14** DERIVED T1170	**0.000** mm DERIVED T1171
stop270 T1172	**0** DERIVED T1176	**0** DERIVED T1177	**14** DERIVED T1178	**0.000** mm DERIVED T1179
stop315 T1180	**0** DERIVED T1184	**0** DERIVED T1185	**16** DERIVED T1186	**0.000** mm DERIVED T1187
stopFD07B T1188	**0** DERIVED T1192	**0** DERIVED T1193	**19** DERIVED T1194	**0.000** mm DERIVED T1195

5 Parts list

BOM BOM v3.5.2. Price rule: every line TO-QUOTE delivered to Sønderborg DK-6400 unless dated; no zero and no "from" prices.

Item	Value
Delivery address for quotations	Sonderborg, Denmark, DK-6400 DECIDED *Requirement, BOM_v3.5.2.json v3.5.2* T1196
Price status	prices not given: every line TO-QUOTE unless dated DECIDED *Design value, BOM_v3.5.2.json v3.5.2* T1197
BOM part lines / placed instances	**107** DERIVED T1198 / **142** DERIVED T1199
Standard hardware: lines / items	**57** DERIVED T1200 / **174** DERIVED T1201
Hardware mass rule	CALC masses (standard nominal geometry), NOT datasheet: the hardware is carried in massprops as max(0.20 kg allowance, CALC sum) at the host parts (lead 188 (11)) CALC T1202
Total cost	**TO-QUOTE (not summed)** MISSING T413

Columns: part number (with its BOM-row trace), name and supplier as written in the BOM, quantity, material, process, unit mass with its evidence class, and price status. COTS = commercial part. Wetted oxygen-clean parts are marked O_2.

5.1 BL — Head-end bleed (provision)

BL family: 11 lines, 0.7832 kg (Σ qty × unit mass).

PN	Part and supplier	Qty	Material	Process	Unit mass [kg]	Price
ENG-BL-01 T049	Bleed tap block T4 crown (316L) with R1 0.71 mm restrictor insert, O-ring face seal to a Ø1.5 TC-04 … O_2 in-house / workshop TO-QUOTE	1	Stainless steel 1.4404 / 316L	CNC-mill	**0.0083** DERIVED T050	TO-QUOTE
ENG-BL-02 T051	Bleed tap adapter on FD-01 (316L): M8×1 port, <= 15 um sintered 316L disc, R2 0.71 mm, integral tee (T4 leg … O_2 in-house / workshop TO-QUOTE	1	Stainless steel 1.4404 / 316L	CNC-lathe	**0.0061** DERIVED T052	TO-QUOTE
ENG-BL-03A T053	Bleed line T4 leg BL-01 → BL-02 tee: 1/8 × 0.028 in 316L, 49 mm developed (routing polyline; bends R >= 10 … O_2 in-house / workshop TO-QUOTE	1	Stainless steel 1.4404 / 316L	cut-stock	**0.0021** DERIVED T054	TO-QUOTE
ENG-BL-03E T055	Bleed line common line BL-02 → BL-04A: 1/8 × 0.028 in 316L, 53 mm developed (routing polyline; bends R >= … O_2 in-house / workshop TO-QUOTE	1	Stainless steel 1.4404 / 316L	cut-stock	**0.0023** DERIVED T056	TO-QUOTE
ENG-BL-03F T057	Bleed line fixed side: 1/8 × 0.028 in 316L, 266 mm developed (wall clamps to the airframe TO-DEFINE) O_2 in-house / workshop TO-QUOTE	1	Stainless steel 1.4404 / 316L	cut-stock	**0.0117** DERIVED T058	TO-QUOTE
ENG-BL-04 T059	Bleed crossing hose -2 class PTFE / 316 braid, bundled 23 mm above the FD-07 coil (same sweep), 180 mm O_2 TO-SELECT	1	PTFE COTS	COTS	**0.0126** ASSUMED T060	TO-QUOTE
ENG-BL-05 T061	Bleed valve 2/2 NC solenoid, 12 V, seat >= 2 mm, sub-base mount (ENVELOPE Ø36 × 90) O_2 TO-SELECT	2	Stainless steel 1.4404 / 316L COTS	COTS	**0.3000** ASSUMED T062	TO-QUOTE
ENG-BL-06 T063	V2 vent bulkhead fitting (316L) through the bay wall at az 112; outside: non-propulsive opposed tee … O_2 TO-SELECT	1	Stainless steel 1.4404 / 316L COTS	COTS	**0.0100** ASSUMED T064	TO-QUOTE
ENG-BL-07 T065	Bleed valve manifold (6061-T6): V1/V2 pads, Ø1.5 mid passage with P_mid port, Ø3 V2 vent passage O_2 in-house / workshop TO-QUOTE	1	Aluminium alloy 6061-T6	CNC-mill	**0.0749** DERIVED T066	TO-QUOTE
ENG-BL-08 T067	P_mid pressure transducer (ENVELOPE Ø22 × 55, Electronics selects) on the manifold boss O_2 Electronics (TO-SELECT)	1	Stainless steel 1.4404 / 316L COTS	COTS	**0.0500** ASSUMED T068	TO-QUOTE
ENG-BL-10 T069	V1 inlet fitting (316L): fixed line BL-03F into the manifold O_2 TO-SELECT	1	Stainless steel 1.4404 / 316L COTS	COTS	**0.0050** ASSUMED T070	TO-QUOTE

5.2 EL — Sensors and encoders

EL family: 6 lines, 0.0740 kg (Σ qty × unit mass).

PN	Part and supplier	Qty	Material	Process	Unit mass [kg]	Price
ENG-EL-03 T071	Gimbal encoder, inner axis (beta): AS5048A on a round PCB 20 × 2.8 (REQUEST to Electronics) ams AS5048A on custom 20 mm PCB (Electronics)	1	COTS assembly COTS	COTS/custom PCB	**0.0030** ASSUMED T072	TO-QUOTE
ENG-EL-04 T073	Gimbal encoder, outer axis (alpha): AS5048A on a round PCB 20 × 2.8 recessed in the -X post bore (REQUEST to … ams AS5048A on custom 20 mm PCB (Electronics)	1	COTS assembly COTS	COTS/custom PCB	**0.0030** ASSUMED T074	TO-QUOTE
ENG-EL-05 T075	Pintle-position encoder AS5048A on a 14 mm round PCB (chip + PCB envelope 14 × 2.4) in the PD-03 cover ams AS5048A on a 14 mm round PCB (Electronics request 2026-10-12)	1	COTS assembly COTS	COTS/custom PCB	**0.0020** ASSUMED T076	TO-QUOTE
ENG-EL-06 T077	Diametral NdFeB magnet 6 × 2.5 (AS5048A), bonded on the pin end ams AS5048 evaluation magnet / Magnet-Shop DK	2	NdFeB magnet N35 COTS	COTS	**0.0005** DERIVED T078	TO-QUOTE
ENG-EL-11 T079	P_feed transducer J602 TE M3234-000005 class, 1/4-18 NPT male, vertical on the FS-01A boss (Electronics … O_2 TO-QUOTE (delivery Sonderborg DK-6400)	1	COTS assembly COTS	COTS (Electronics)	**0.0500** ASSUMED T080	TO-QUOTE
ENG-EL-12 T081	MSS-2 Type T MI thermocouple 1/16 in in a 1/16 × 1/8-27 NPT male connector (Electronics part; tip 9 mm into … O_2 TO-QUOTE (delivery Sonderborg DK-6400)	1	COTS assembly COTS	COTS (Electronics)	**0.0150** ASSUMED T082	TO-QUOTE

5.3 FD — Oxidiser feed

FD family: 11 lines, 0.2759 kg (Σ qty × unit mass).

PN	Part and supplier	Qty	Material	Process	Unit mass [kg]	Price
ENG-FD-01 T083	Oxidiser inlet block E1-M (316L): riser socket, 8 mm passages, P_feed tap, face seal to the filter cartridge O_2 in-house / workshop TO-QUOTE	1	Stainless steel 1.4404 / 316L	CNC mill (2 setups) + socket weld of the riser (TIG, back-purged)	**0.0434** DERIVED T084	TO-QUOTE
ENG-FD-02 T085	Filter cartridge 15 um absolute, pleated 316L Dutch-twill mesh, area >= 1450 mm2 (custom, in the TC-02 cap) O_2 custom element: Purolator Advanced Filtration / GKN Sinter Metals / Mott …	1	COTS assembly COTS	COTS custom (filter maker)	**0.0040** ASSUMED T086	TO-QUOTE
ENG-FD-03 T087	Oxidiser riser 3/8 × 0.035 in 316L seamless, 71 mm developed, 2 S-bends (R >= 24 mm) O_2 316L seamless 3/8 × 0.035 in (ASTM A269): Swagelok SS-T6-S-035-6ME / Sandvik …	1	Stainless steel 1.4404 / 316L	tube bender + socket weld into FD-01 (TIG, back-purged) + coil-end …	**0.0137** DERIVED T088	TO-QUOTE
ENG-FD-04 T089	Riser radiation shield: 0.2 mm polished 316 foil + 2 band-clip stand-offs (case OD → riser) in-house / workshop TO-QUOTE	1	Stainless steel 1.4404 / 316L	cut + roll + spot weld (or 3D-printed PEEK clips, TO-DECIDE by the …	**0.0054** DERIVED T090	TO-QUOTE
ENG-FD-05 T091	Riser-foot TEE (316L): riser socket (top), hose socket (side, 25.9 deg), MSS-1 1/8-27 NPT (bottom) O_2 in-house / workshop TO-QUOTE	1	Stainless steel 1.4404 / 316L	CNC lathe + mill (2 setups); socket welds of FD-03 and FD-07A (TIG …	**0.0407** DERIVED T092	TO-QUOTE

PN	Part and supplier	Qty	Material	Process	Unit mass [kg]	Price
ENG-FD-07 T093	Feed hose 3/8 in convoluted PTFE / 316 braid, rising coil 203 deg, flexible length 180 mm O2 convoluted PTFE 3/8 in, 1× 316 braid, O2-clean, pressure-test certificate ...	1	COTS assembly COTS	COTS hose assembly (both ends crimped; FD-07A/B are its end fittings)	0.0379 TO-VERIFY T094	TO-QUOTE
ENG-FD-08 T095	Fixed feed line 3/8 × 0.035 in 316L, 279 mm developed: 3D bend, vertical on the bay wall at az 219.5, R24 ... O2 316L seamless 3/8 × 0.035 in (ASTM A269): Swagelok SS-T6-S-035-6ME / Sandvik ...	1	Stainless steel 1.4404 / 316L	tube bender (R 24 die) + orbital/TIG butt weld to FD-07B + socket ...	0.0538 DERIVED T096	TO-QUOTE
ENG-FD-08F T097	Line flange (316L): socket for FD-08, AS568-012 FKM face seal to FS-01A, 2x M4 O2 in-house / workshop TO-QUOTE	1	Stainless steel 1.4404 / 316L	CNC mill	0.0135 DERIVED T098	TO-QUOTE
ENG-FD-09 T099	Fixed hose-end clamp (6061-T6, base + cap, 2x M4 into GM-09 top face, bare aluminium on the sleeve), v3.4.1 ... in-house / workshop TO-QUOTE	1	Aluminium alloy 6061-T6	CNC mill	0.0380 DERIVED T100	TO-QUOTE
ENG-FD-11 T101	P-clip 3/8 in, 316 band + silicone cushion, M4 to the airframe wall (insert/rivet nut in the airframe = ICD) Essentra / RS Components DK P-clip 3/8 in SS (TO-QUOTE)	3	COTS assembly COTS	COTS	0.0040 ASSUMED T102	TO-QUOTE
ENG-FD-12F T103	Tank-line flange (316L) on FS-01B: socket for the tank line (route TO-DEFINE with the tank, ICD-5) ... O2 in-house / workshop TO-QUOTE	1	Stainless steel 1.4404 / 316L	CNC mill	0.0135 DERIVED T104	TO-QUOTE

5.4 FS — Main valve module (fixed)

FS family: 15 lines, 0.3364 kg (Σ qty × unit mass).

PN	Part and supplier	Qty	Material	Process	Unit mass [kg]	Price
ENG-FS-01A T105	MV module lower body (6061-T6): outlet chamber 18 × 12.3 (segment side), line flange face, J602 / MSS-2 / ... O2 in-house / workshop TO-QUOTE	1	Aluminium alloy 6061-T6	CNC mill 3 setups	0.0521 DERIVED T106	TO-QUOTE
ENG-FS-01B T107	MV module upper body (6061-T6): seat web + PCTFE seat counterbore, inlet chamber 16, piston bore 11.9 H8 ... O2 in-house / workshop TO-QUOTE	1	Aluminium alloy 6061-T6	CNC lathe + mill	0.0444 DERIVED T108	TO-QUOTE
ENG-FS-02 T109	MV-P poppet: disc 14 × 3 + stem 3 h7, hollow 1.2 (balance hole), M3 into FS-04 O2 in-house / workshop TO-QUOTE	1	17-4PH	CNC lathe, stem ground	0.0046 DERIVED T110	TO-QUOTE
ENG-FS-03 T111	MV-P soft seat insert PCTFE: land 11.5-12.5 × 0.3, OD 16 × 2.3 (retention lip TO-DESIGN) O2 in-house / workshop TO-QUOTE	1	PCTFE	lathe from rod	0.0002 DERIVED T112	TO-QUOTE
ENG-FS-04 T113	MV-P balance piston 11.9 f7 + rod 4 h7 (one piece), seal groove, cross hole 1.2 into the balance chamber O2 in-house / workshop TO-QUOTE	1	17-4PH	CNC lathe, rod ground	0.0077 DERIVED T114	TO-QUOTE
ENG-FS-06 T115	MV actuator housing (dry side, 6061-T6): spring pocket, cam slot, 2x MR85 bearing bores (servo-side bore ... in-house / workshop TO-QUOTE	1	Aluminium alloy 6061-T6	CNC mill	0.0292 DERIVED T116	TO-QUOTE
ENG-FS-07 T117	MV closing spring EN 10270-3 1.4310, d 1.8 / Dm 12 / n 6.6 active, L0 31.3, 85 N closed / 107.6 N open ... TO-QUOTE (delivery Sonderborg DK-6400)	1	Stainless spring wire 1.4310 COTS	COTS custom spring (Gutekunst / Lee Spring, TO-QUOTE)	0.0080 ASSUMED T118	TO-QUOTE
ENG-FS-08 T119	MV follower (17-4PH H900, flat hardened face) on the rod top, M3 in-house / workshop TO-QUOTE	1	17-4PH	CNC lathe + H900	0.0045 DERIVED T120	TO-QUOTE
ENG-FS-09 T121	MV cam (r 6.0 → 8.8 over 45 deg, 15 deg dwell, width 5; closed 25 deg before the rise) + shaft 5 in 2x ... in-house / workshop TO-QUOTE	1	17-4PH	wire EDM + lathe + broach	0.0081 DERIVED T122	TO-QUOTE
ENG-FS-10 T123	MV opener servo Savox SB-2292SG (Electronics, commonality) on the housing +u face, spline coupled to the cam ... TO-QUOTE (delivery Sonderborg DK-6400)	1	COTS assembly COTS	COTS	0.0810 DATASHEET T124	TO-QUOTE
ENG-FS-11 T125	MV latch + release + dashpot + OPEN/CLOSED microswitches: RESERVED ENVELOPE 16 × 12 × 30 (TO-DESIGN with ... TO-QUOTE (delivery Sonderborg DK-6400)	1	COTS assembly COTS	TO-DESIGN	0.0250 ASSUMED T126	TO-QUOTE
ENG-FS-12 T127	S1 back-relief check, insert cartridge 5.5 mm class, crack 0.3-0.7 bar (outlet → inlet chamber) ... O2 TO-QUOTE (delivery Sonderborg DK-6400)	1	COTS assembly COTS	COTS (TO-SELECT: The Lee Company Chek insert class)	0.0020 ASSUMED T128	TO-QUOTE
ENG-FS-13 T129	Secondary rupture disc 9.5 MPa +/-5 %, reverse-acting, 1/4-18 NPT holder with 1/4 NPT outlet to the ... O2 TO-QUOTE (delivery Sonderborg DK-6400)	1	COTS assembly COTS	COTS (TO-SELECT: Fike / BS&B / Oseco mini disc assemblies)	0.0450 ASSUMED T130	TO-QUOTE
ENG-FS-14 T131	Drain port plug 1/8-27 NPT flush socket plug 316 (ground drain/purge; segment low point of the module) O2 TO-QUOTE (delivery Sonderborg DK-6400)	1	COTS assembly COTS	COTS	0.0040 ASSUMED T132	TO-QUOTE
ENG-FS-15 T133	MV module mounting bracket (6061-T6): 2x M4 to FS-01A/B, 2x M5 radial into the airframe wall at az 219.5 ... in-house / workshop TO-QUOTE	1	Aluminium alloy 6061-T6	CNC mill	0.0206 DERIVED T134	TO-QUOTE

5.5 GM — Gimbal and thrust frame

GM family: 19 lines, 1.1528 kg (Σ qty × unit mass).

PN	Part and supplier	Qty	Material	Process	Unit mass [kg]	Price
ENG-GM-01 T135	Engine collar half +Y (servo 1 lever, 135 deg) in-house / workshop TO-QUOTE	1	Aluminium alloy 7075-T6	CNC mill (pair machined as one ring, then split)	0.0678 DERIVED T136	TO-QUOTE
ENG-GM-02 T137	Engine collar half -Y (servo 2 lever, 225 deg), mirror of GM-01 in-house / workshop TO-QUOTE	1	Aluminium alloy 7075-T6	CNC mill (pair machined as one ring, then split)	0.0691 DERIVED T138	TO-QUOTE
ENG-GM-03 T139	Gimbal ring (Cardan) in-house / workshop TO-QUOTE	1	Aluminium alloy 6061-T6	CNC mill + 4th-axis boring (from 100 mm bar or 16 mm plate)	0.1899 DERIVED T140	TO-QUOTE
ENG-GM-04 T141	Inner pivot pin 8 h5 × 18, needle raceway (AISI 440C, 58-62 HRC) in-house / workshop TO-QUOTE	2	Martensitic stainless 440C	CNC lathe, harden, ground h5	0.0068 DERIVED T142	TO-QUOTE
ENG-GM-05 T143	Outer pivot pin 8 h5 × 18.5, needle raceway (AISI 440C, 58-62 HRC) in-house / workshop TO-QUOTE	2	Martensitic stainless 440C	CNC lathe, harden, ground h5	0.0069 DERIVED T144	TO-QUOTE
ENG-GM-06 T145	Inner-axis thrust washer 8.1/14 × 1.5 (PTFE composite) TO-SELECT	2	Stainless steel 1.4301 / AISI 304 COTS	COTS (TO-SELECT)	0.0012 ASSUMED T146	TO-QUOTE
ENG-GM-07 T147	Outer-axis thrust washer 8.1/14 × 1.4 (PTFE composite) TO-SELECT	2	Stainless steel 1.4301 / AISI 304 COTS	COTS (TO-SELECT)	0.0011 ASSUMED T148	TO-QUOTE
ENG-GM-08 T149	Yoke post (outer-axis bearing housing + frame post), x2 in-house / workshop TO-QUOTE	2	Aluminium alloy 6061-T6	CNC mill from 20 × 30 bar (3 setups)	0.0774 DERIVED T150	TO-QUOTE
ENG-GM-09 T151	Upper thrust ring (engine module to airframe, thrust path) in-house / workshop TO-QUOTE	1	Aluminium alloy 6061-T6	CNC mill from 12 mm plate	0.1520 DERIVED T152	TO-QUOTE
ENG-GM-10 T153	Lower ring (stop ring, TVC servo base, frame tie) in-house / workshop TO-QUOTE	1	Aluminium alloy 6061-T6	CNC mill from 12 mm plate	0.1927 DERIVED T154	TO-QUOTE
ENG-GM-11 T155	Conical stop bumper, PU 90A, 8 deg line contact PU 90A tube 100 × 80: Kubo / RS Components DK (TO-QUOTE)	1	Polyurethane bumper 90A	cast or turned from PU 90A tube	0.0092 DERIVED T156	TO-QUOTE
ENG-GM-12 T157	Transport / neutral lock, 2 half rings (REMOVE BEFORE FLIGHT) transport lock, not placed, excluded from mass in-house / workshop TO-QUOTE	1	PETG	FDM print PETG	0.0622 DERIVED T158	TO-QUOTE

PN	Part and supplier	Qty	Material	Process	Unit mass [kg]	Price
ENG-GM-13 T159	TVC servo 1 bracket (C-frame: top/bottom ear blocks + back rail) in-house / workshop TO-QUOTE	1	Aluminium alloy 6061-T6	CNC mill from 40 × 50 bar (3 setups)	0.0489 DERIVED T160	TO-QUOTE
ENG-GM-14 T161	TVC servo 2 bracket (mirror of GM-13) in-house / workshop TO-QUOTE	1	Aluminium alloy 6061-T6	CNC mill from 40 × 50 bar (3 setups)	0.0489 DERIVED T162	TO-QUOTE
ENG-GM-15 T163	TVC servo Savox SB-2292SG (electronics.json) Lindinger (AT) item 9738546 per electronics.json v1	2	COTS assembly COTS	COTS	0.0810 DATASHEET T164	119.99 EUR incl. VAT (electronics.json v1, 2026-10-11; DK delivery TO-QUOTE) DK delivery TO-QUOTE
ENG-GM-16 T165	TVC horn r 11.1 mm, 25T clamp (7075-T6) in-house / workshop TO-QUOTE	2	Aluminium alloy 7075-T6	CNC mill + wire-EDM 25T spline (or re-machined COTS 25T clamp horn)	0.0022 DERIVED T166	TO-QUOTE
ENG-GM-17 T167	TVC pushrod dog-bone 42.3 mm between eyes (7075-T6), fixed length in-house / workshop TO-QUOTE	2	Aluminium alloy 7075-T6	CNC mill	0.0042 DERIVED T168	TO-QUOTE
ENG-GM-18 T169	Lever pivot stud: flange 7.5 × 1.5 (= washer), 5 h7 shoulder x 7.5, M4 x 4.5, 2.5 hex (17-4PH) in-house / workshop TO-QUOTE	2	17-4PH	CNC lathe	0.0020 DERIVED T170	TO-QUOTE
ENG-GM-19 T171	Joint spacer washer 5.1 × 7.5 × 1.5 (17-4PH H1150), ground faces in-house / workshop TO-QUOTE	6	17-4PH	CNC lathe from bar, faces ground	0.0003 DERIVED T172	TO-QUOTE
ENG-GM-20 T173	Horn-joint shoulder screw: head 8.5 × 4, 5 h7 × 10 shoulder, M4 × 3.5 (17-4PH) in-house / workshop TO-QUOTE	2	17-4PH	CNC lathe	0.0035 DERIVED T174	TO-QUOTE

5.6 GR — Fuel grain

GR family: 2 lines, 0.4244 kg (Σ qty × unit mass).

PN	Part and supplier	Qty	Material	Process	Unit mass [kg]	Price
ENG-GR-01 T175	Fuel grain 1/2: ABS 53.5 OD x 115.0, straight centred port 25 mm (grain v2 S230) in-house / workshop TO-QUOTE	1	ABS, FDM-printed fuel grain	FDM print ABS 100 % infill, concentric perimeters	0.2122 DERIVED T176	TO-QUOTE
ENG-GR-02 T177	Fuel grain 2/2: ABS 53.5 OD x 115.0, straight centred port 25 mm (grain v2 S230) in-house / workshop TO-QUOTE	1	ABS, FDM-printed fuel grain	FDM print ABS 100 % infill, concentric perimeters	0.2122 DERIVED T178	TO-QUOTE

5.7 HW — Bearings and hardware

HW family: 4 lines, 0.0242 kg (Σ qty × unit mass).

PN	Part and supplier	Qty	Material	Process	Unit mass [kg]	Price
ENG-HW-20 T179	Drawn-cup needle roller bearing HK0808 (Fw 8 x D 12 x C 8), open ends INA HK0808-B: RS Components (RS 137-1723, GBP 5.16 ex VAT each, 2026-10-13) ...	4	Carbon/bearing steel COTS	COTS	0.0030 DATASHEET T180	TO-QUOTE
ENG-HW-21 T181	Retaining washer DIN 6799-6 (shaft 7-9 mm), spring steel Bossard DK / RS Components DK	4	Carbon/bearing steel COTS	COTS	0.0003 ASSUMED T182	TO-QUOTE
ENG-HW-22 T183	Spherical plain bearing GE 5 E (steel/steel) - outer ring (pressed in the pushrod eye) SKF / INA / Hepco GE 5 E class: RS Components DK (TO-QUOTE)	4	Carbon/bearing steel COTS	COTS	0.0025 DERIVED T184	TO-QUOTE
ENG-HW-25 T185	Horn axial screw ISO 10642 M3 × 6 (countersunk, 10.9), Loctite 243 Bossard DK / RS Components DK	2	Carbon/bearing steel COTS	COTS	0.0005 ASSUMED T186	TO-QUOTE

5.8 IG — Igniter

IG family: 2 lines, 0.0740 kg (Σ qty × unit mass).

PN	Part and supplier	Qty	Material	Process	Unit mass [kg]	Price
ENG-IG-01 T187	Torch igniter (N2O/propane, spark) - ENVELOPE 20 × 45 mm + M10×1 nose + 2 inlet fittings (Combustion ... Combustion design: body CNC 316L, micro spark plug (model-engine class ...	1	COTS assembly COTS	Combustion design (316L body, micro spark plug)	0.0700 ASSUMED T188	TO-QUOTE
ENG-IG-02 T189	Torch inlet check valve, cartridge insert in the torch inlet bore (N2O and propane, x2) O2 Lee Company Chek cartridge (stainless, 5.5 mm class, cracking <= 0.3 bar ...	2	COTS assembly COTS	COTS	0.0020 ASSUMED T190	TO-QUOTE

5.9 NZ — Nozzle and joint

NZ family: 5 lines, 0.0632 kg (Σ qty × unit mass).

PN	Part and supplier	Qty	Material	Process	Unit mass [kg]	Price
ENG-NZ-01 T191	Graphite throat insert (Dt 13.72, De 19.41, OD 44, aft spigot 30) isomolded graphite rod 50 mm (Toyo Tanso IG-110 / SGL R6510 class): Schunk ...	1	Isomolded fine-grain graphite	CNC lathe from isomolded graphite rod (dust extraction, sharp carbide)	0.0469 DERIVED T192	TO-QUOTE
ENG-NZ-02 T193	Insert insulating sleeve (carbon-phenolic 50/44, 3 mm) in-house / workshop TO-QUOTE	1	Carbon-cloth phenolic laminate	lathe from carbon-fabric/phenolic tube (tape-wrapped or moulded ...	0.0120 DERIVED T194	TO-QUOTE
ENG-NZ-03 T195	Nozzle-joint hot layer J1: carbon-phenolic washer 30.6/49.8 × 2.0 carbon-fabric / phenolic laminate plate >= 2.5 mm (aerospace ablative or ...	1	Carbon-cloth phenolic laminate	water-jet + face-grind from carbon-fabric/phenolic laminate plate ...	0.0035 DERIVED T196	TO-QUOTE
ENG-NZ-04 T197	Nozzle-joint lip-face seal J1: flexible graphite ring 30.6/49.8 × 0.5, oxidation-inhibited flexible graphite sheet 0.5 mm, oxidation-inhibited grade, >= 98 % C (SGL ...	1	Flexible graphite sheet	cut from sheet (steel-rule die / water-jet)	0.0007 DERIVED T198	TO-QUOTE
ENG-NZ-06 T199	Nozzle-joint load-spreading foil (v3.5.2 N-F2): flexible graphite ring 30.6/43.9 × 0.25 flexible graphite roll foil 0.25 mm, >= 99.85 % C, 1.0 g/cm3 (SGL SIGRAFLEX TH ...	1	Flexible graphite roll foil 0.25 mm	cut from roll foil (steel-rule die / water-jet)	0.0002 DERIVED T200	TO-QUOTE

5.10 PD — Pintle drive

PD family: 20 lines, 0.1746 kg (Σ qty × unit mass).

PN	Part and supplier	Qty	Material	Process	Unit mass [kg]	Price
ENG-PD-01 T201	Pintle drive frame, 2 pieces: A (-Y foot + bridge + leg + servo plate + cap) and B (+Y foot + rail plate + ... in-house / workshop TO-QUOTE	1	Aluminium alloy 6061-T6	CNC mill from 6061-T6 block (3 setups per piece) or welded/bolted ...	0.0389 DERIVED T202	TO-QUOTE
ENG-PD-02 T203	Pintle crank housing (bearing seats A 18 and B 19, crank cavity, spring cavity 22) in-house / workshop TO-QUOTE	1	Aluminium alloy 6061-T6	CNC mill + lathe (bores in one setup)	0.0198 DERIVED T204	TO-QUOTE
ENG-PD-03 T205	Crank-housing cover with encoder PCB pocket in-house / workshop TO-QUOTE	1	Aluminium alloy 6061-T6	CNC mill	0.0026 DERIVED T206	TO-QUOTE
ENG-PD-04 T207	Pintle crankshaft (throw r 3.0, pin 3, journals 12 / 10, dog end, spring hub, magnet pocket) in-house / workshop TO-QUOTE	1	17-4PH	CNC lathe + mill (eccentric pin turned in a second setup) + grind ...	0.0125 DERIVED T208	TO-QUOTE
ENG-PD-05 T209	Dog hub on the HV69 6 mm 25T spline (one-sided drive tooth 60 deg) in-house / workshop TO-QUOTE	1	Aluminium alloy 7075-T6	CNC lathe + broach 25T (or re-machined COTS aluminium horn)	0.0006 DERIVED T210	TO-QUOTE
ENG-PD-06 T211	Pintle connecting link 28 mm centres, eyes OD 8 × 3.8 with 4.5 bores, shank 4 × 3 in-house / workshop TO-QUOTE	1	17-4PH	wire EDM from plate + ream	0.0039 DERIVED T212	TO-QUOTE
ENG-PD-07 T213	Pintle carriage: needle clamp arm (M3 stem thread) + link clevis + plate on the MGN09H block in-house / workshop TO-QUOTE	1	Aluminium alloy 6061-T6	CNC mill, hard anodise	0.0100 DERIVED T214	TO-QUOTE
ENG-PD-08 T215	Miniature linear guide block HIWIN MGN09H (C0 4020 N, M0Y 18.6 N m) HIWIN MGN09H-ZF (preload ZF, end seals) via HIWIN GmbH / Misumi (TO-QUOTE ...	1	Carbon/bearing steel COTS	COTS	0.0300 TO-VERIFY T216	TO-QUOTE

PN	Part and supplier	Qty	Material	Process	Unit mass [kg]	Price
ENG-PD-09 T217	Spiral (clock) return spring, 0.25 N m at closed + 0.12 N m/rad (+/-10 %), envelope 21/8 × 4.5 custom flat spiral spring 1.4310 strip (Gutekunst / Lesjofors DK, TO-QUOTE ...	1	Stainless spring wire 1.4310 COTS	COTS custom (spring maker)	0.0055 ASSUMED T218	TO-QUOTE
ENG-PD-10 T219	Radiation shield under the HV69: 0.5 mm polished aluminium L-sheet on 2 G11 stand-offs in-house / workshop TO-QUOTE	1	Aluminium alloy 6061-T6	cut + bend from 0.5 mm 1050A-H14 sheet, polished both faces	0.0007 DERIVED T220	TO-QUOTE
ENG-PD-11 T221	Isolator washer G11 7/3.4 × 2 (frame feet to head top) NEMA G11 sheet 2 mm: Rochling / Isola / RS Components DK (TO-QUOTE, delivery ...	4	Glass-epoxy laminate NEMA LI-1 G11	punched / turned from G11 sheet	0.0001 DERIVED T222	TO-QUOTE
ENG-PD-12 T223	G11 pad 1.5 mm under the HV69 ears (window for the case) NEMA G11 sheet 1.5 mm (TO-QUOTE)	1	Glass-epoxy laminate NEMA LI-1 G11	CNC from G11 sheet	0.0009 DERIVED T224	TO-QUOTE
ENG-PD-13 T225	Pintle servo MKS HV69 (29 × 13 × 28.7, 21 g, 6 mm 25T spline), case below the shaft MKS HV69 (MKS S0022012); Lindinger AT item 9788343	1	COTS assembly COTS	COTS	0.0210 DATASHEET T226	TO-QUOTE ref. Lindinger AT catalogue 62.99 EUR incl. 20 % AT VAT, 2026-10-12, Electronics pintle_mini_servo_v1; not a DK quote (not a DK quote)
ENG-PD-14 T227	Thin-section ball bearing 12×18×4 (NSK/NMB 6701 class, 'NB701' in pintle_v2p3), C0r 655 N NSK / NMB / SKF 61701 class (TO-QUOTE, Sonderborg DK-6400)	1	Carbon/bearing steel COTS	COTS	0.0031 ASSUMED T228	TO-QUOTE
ENG-PD-15 T229	Deep-groove ball bearing 6800-2Z 10×19×5, C0r 840 N SKF 61800-2Z / NSK 6800ZZ (TO-QUOTE, Sonderborg DK-6400)	1	Carbon/bearing steel COTS	COTS	0.0055 ASSUMED T230	TO-QUOTE
ENG-PD-16 T231	Pin 3 g6 × 12 (17-4PH H900) with e-clip grooves (crank pin bush / carriage link pin) in-house / workshop TO-QUOTE	1	17-4PH	turned + ground (or ISO 2338 3 m6 × 12 hardened)	0.0007 DERIVED T232	TO-QUOTE
ENG-PD-17 T233	PTFE-lined steel-backed bush 3 × 4.5 × 3.8 (DU class) GGB DU / SKF PCM class 3 × 4.5 (maker part TO-SELECT, TO-QUOTE)	2	Carbon/bearing steel COTS	COTS	0.0003 ASSUMED T234	TO-QUOTE
ENG-PD-18 T235	Encoder magnet NdFeB N35 diametral 6 × 1 diametral 6 × 1 mm N35 (Supermagnete / Magnetladen, TO-QUOTE)	1	NdFeB magnet N35	COTS	0.0002 DERIVED T236	TO-QUOTE
ENG-PD-19 T237	Miniature rail HIWIN MGNR09R, length 46.1 mm (cut to length) HIWIN MGNR09R cut to 46 mm (TO-QUOTE)	1	Carbon/bearing steel COTS	COTS	0.0175 DATASHEET T238	TO-QUOTE
ENG-PD-20 T239	G11 stand-off 5/2.2 × 2 (shield to servo plate) G11 rod 6 mm (TO-QUOTE)	2	Glass-epoxy laminate NEMA LI-1 G11	turned from G11 rod	0.0001 DERIVED T240	TO-QUOTE

5.11 TC — Thrust chamber

TC family: 11 lines, 1.6640 kg (Σ qty × unit mass).

PN	Part and supplier	Qty	Material	Process	Unit mass [kg]	Price
ENG-TC-01 T241	Combustion case tube 6061-T6 tube with turning stock on the OD (rev 5: OD 63.5 h9 with runout 0.04 ...	1	Aluminium alloy 6061-T6	cut-stock + CNC lathe (bore fwd seal land, turn aft land and thread) ...	0.4613 DERIVED T242	TO-QUOTE
ENG-TC-02 T243	Injector head / forward closure (pintle seat bore, manifold, Pc and torch ports, E1-M filter bore and inlet ... O2 in-house / workshop TO-QUOTE	1	Aluminium alloy 6061-T6	CNC lathe + 3-axis mill (2 setups)	0.4372 DERIVED T244	TO-QUOTE
ENG-TC-03 T245	Pintle seat insert (5.5 mm sharp-edged seat, 37.5 deg diffuser) O2 in-house / workshop TO-QUOTE	1	Dispersion-strengthened copper GlidCop	CNC lathe	0.0149 DERIVED T246	TO-QUOTE
ENG-TC-04 T247	Bonnet: stem seal + guide housing, E1-M inlet boss (8 mm transfer, cross-drilling at the bore top ... O2 in-house / workshop TO-QUOTE	1	Aluminium alloy 6061-T6	CNC lathe + mill	0.0535 DERIVED T248	TO-QUOTE
ENG-TC-05 T249	Pintle needle (15 deg metering cone, 6 mm body, 3 mm stem) O2 in-house / workshop TO-QUOTE	1	17-4PH	CNC lathe, stem ground h7 Ra 0.2	0.0166 DERIVED T250	TO-QUOTE
ENG-TC-06 T251	Pre-combustion chamber insulator (phenolic, bore 49.9, wall 3.60 mm, 30 mm long) in-house / workshop TO-QUOTE	1	Cotton/paper phenolic laminate tube	lathe from phenolic tube	0.0244 DERIVED T252	TO-QUOTE
ENG-TC-07 T253	Grain liner (phenolic tube 57.10/53.95) paper- or glass-phenolic tube 2.25 × 0.0625 in class: Aerocon / Kinetic / RS ...	1	Cotton/paper phenolic laminate tube	cut from stock tube, OD sanded to slip fit	0.0853 DERIVED T254	TO-QUOTE
ENG-TC-08 T255	Post-chamber + convergent insulator (phenolic, 3.50 mm over the post-chamber, contour per combustion v2) in-house / workshop TO-QUOTE	1	Cotton/paper phenolic laminate tube	lathe from glass- or silica-phenolic rod/tube	0.0515 DERIVED T256	TO-QUOTE
ENG-TC-09 T257	Aft section v3.2: post-chamber barrel + nozzle holder + retaining lip (IN718, M63×1.5 socket on the case) in-house / workshop TO-QUOTE	1	Inconel 718	CNC lathe (bore from the forward end) + flats	0.5143 DERIVED T258	TO-QUOTE
ENG-TC-10 T259	Injector-face insulator disc, phenolic 2 mm glass- or paper-phenolic sheet 2 mm: RS Components DK / Rochling	1	Cotton/paper phenolic laminate tube	water-jet or lathe from sheet	0.0047 DERIVED T260	TO-QUOTE
ENG-TC-11 T261	Stem guide bushing 5/3 × 4 (PTFE + 25 % glass) O2 PTFE-glass rod 10 mm: RS Components DK / Polyfluor	1	PTFE	lathe from rod	0.0001 DERIVED T262	TO-QUOTE

5.12 Standard hardware (fasteners, springs, bearings; BOM /hardware, CALC masses)

Torque as written in the line notes (DESIGN; J-FWD: set per flight lot by J-F1, see M-T1J). Unit masses are CALC on standard nominal geometry, not datasheet.

PN	Item and standard	Qty	Group	Material	Torque (notes)	Unit mass [kg]	Price
ENG-HW-101 T273	J-FWD forward-closure radial screw ENG-HW-101-IN718 (case TC-01 to head plug TC-02), with the disc spring HW-170 under ... ENG-HW-101-IN718 (Mechanical drawing, TO-DRAW): ISO 7380-1 button-head form with the F3 ...	10	engine	IN718	2.747 N m	0.00218 CALC T274	TO-QUOTE
ENG-HW-102 T275	J-BON bonnet screw (TC-04 on TC-02) ISO 4762 M4 × 12	4	engine	A4-70	2.05 N m	0.00205 CALC T276	TO-QUOTE
ENG-HW-103 T277	FD-01 inlet block screw, lower pair (FD-01 ears into the TC-02 flat r 29.5) DIN 7984 M4 × 20	2	engine	A4-80	1.8 N m	0.00237 CALC T278	TO-QUOTE
ENG-HW-103B T279	FD-01 inlet block screw, upper (FD-01 body into the TC-02 flat r 29.5, above the passages) DIN 7984 M4 × 16	1	engine	A4-80	1.8 N m	0.00206 CALC T280	TO-QUOTE
ENG-HW-104 T281	BL-01 bleed tap block screw (into the TC-04 boss) ISO 4762 M2.5 × 12	2	engine	A4-70	0.5 N m	0.00065 CALC T282	TO-QUOTE
ENG-HW-105 T283	PD-01 frame foot screw through the G11 isolator washer PD-11 (into TC-02) ISO 4762 M3 × 12	4	engine	A2-70	0.6 N m	0.00104 CALC T284	TO-QUOTE
ENG-HW-106 T285	PD-02 crank housing to the PD-01 shelf and cap ISO 4762 M3 × 8	4	engine	A2-70	0.88 N m	0.00086 CALC T286	TO-QUOTE
ENG-HW-107 T287	HV69 pintle servo ear screw through the G11 pad PD-12 (into PD-01) ISO 4762 M2 × 8	4	engine	A2-70	0.24 N m	0.00031 CALC T288	TO-QUOTE
ENG-HW-108 T289	MGNR09R rail screw (into the PD-01 rail plate) ISO 4762 M3 × 8	3	engine	A2-70	0.88 N m	0.00086 CALC T290	TO-QUOTE
ENG-HW-109 T291	PD-07 carriage plate to the MGN09H block ISO 4762 M3 × 6	4	engine	A2-70	0.88 N m	0.00077 CALC T292	TO-QUOTE
ENG-HW-110 T293	PD-03 cover screw (into PD-02 end face) ISO 4762 M2 × 6	3	engine	A2-70	0.24 N m	0.00028 CALC T294	TO-QUOTE
ENG-HW-111 T295	PD-05 dog-hub clamp screw ISO 4762 M2 × 6	1	engine	A2-70	0.24 N m	0.00028 CALC T296	TO-QUOTE
ENG-HW-112 T297	PD-10 radiation shield screw through the G11 stand-offs ISO 4762 M2 × 8	2	engine	A2-70	0.24 N m	0.00031 CALC T298	TO-QUOTE

PN	Item and standard	Qty	Group	Material	Torque (notes)	Unit mass [kg]	Price
ENG-HW-113 T299	Collar clamp screw (GM-01 clearance + counterbore into GM-02 tapped) DIN 7984 M3 × 10	4	engine	A4-70	1.0 N m	0.00080 CALC T300	TO-QUOTE
ENG-HW-114 T301	Horn radial pinch-clamp screw ISO 4762 M3 × 10	2	horn	12.9	2.16 N m	0.00095 CALC T302	TO-QUOTE
ENG-HW-115 T303	EL-03 encoder holder screw (into the GM-03 ring) ISO 4762 M2 × 6	2	ring	A2-70	0.24 N m	0.00028 CALC T304	TO-QUOTE
ENG-HW-120 T305	Yoke post to the upper thrust ring GM-09 (counterbored from the top) ISO 4762 M5 × 12	4	fixed	A4-70	4.15 N m	0.00351 CALC T306	TO-QUOTE
ENG-HW-121 T307	Yoke post to the lower ring GM-10 (counterbored from below) ISO 4762 M5 × 12	4	fixed	Ti-6Al-4V	4.15 N m	0.00194 CALC T308	TO-QUOTE
ENG-HW-122 T309	Dowel pin 4 m6 × 16, hardened (posts to rings) ISO 8734 4 m6 × 16	4	fixed			0.00158 CALC T310	TO-QUOTE
ENG-HW-123 T311	TVC servo bracket to the lower ring GM-10 (counterbored from below) ISO 4762 M4 × 12	8	fixed	Ti-6Al-4V	2.05 N m	0.00114 CALC T312	TO-QUOTE
ENG-HW-124 T313	TVC servo ear screw (servo to bracket GM-13/14) ISO 4762 M4 × 10	8	fixed	A4-70	2.05 N m	0.00190 CALC T314	TO-QUOTE
ENG-HW-125 T315	Airframe screw into the upper thrust ring GM-09 (radial, 8x) - ICD-AIRFRAME ISO 10642 M5 × 10	8	fixed	A4-70	4.15 N m	0.00209 CALC T316	TO-QUOTE
ENG-HW-126 T317	Airframe screw into the lower ring GM-10 (radial, 8x) - ICD-AIRFRAME ISO 10642 M5 × 10	8	fixed	A4-70	4.15 N m	0.00209 CALC T318	TO-QUOTE
ENG-HW-127 T319	FD-09 hose-end clamp foot screw into GM-09 (with helicoil) ISO 4762 M4 × 12	2	fixed	A4-80	1.8 N m	0.00205 CALC T320	TO-QUOTE
ENG-HW-128 T321	Wire thread insert M4 × 1.5 D free-running, 304 (Helicoil class) DIN 8140 M4 × 6	2	fixed			MISSING MISSING T322	TO-QUOTE
ENG-HW-129 T323	FD-09 clamp cap screw ISO 4762 M3 × 20	2	fixed	A4-70	0.88 N m	0.00141 CALC T324	TO-QUOTE
ENG-HW-130 T325	P-clip FD-11 screw into the airframe insert / rivet nut (ICD) ISO 7380-1 M4 × 8	3	fixed	A4-70	2.05 N m	0.00102 CALC T326	TO-QUOTE
ENG-HW-131 T327	FD-08F line flange screw (into FS-01A) ISO 4762 M4 × 12	2	fixed	A4-70	2.05 N m	0.00205 CALC T328	TO-QUOTE
ENG-HW-132 T329	FD-12F tank-line flange screw (into FS-01B) ISO 4762 M4 × 12	2	fixed	A4-70	2.05 N m	0.00205 CALC T330	TO-QUOTE
ENG-HW-133 T331	MV-P body screw FS-01A to FS-01B ISO 4762 M4 × 25	4	fixed	A4-70	2.05 N m	0.00308 CALC T332	TO-QUOTE
ENG-HW-134 T333	MV-P cover FS-06 screw (into FS-01B) ISO 4762 M3 × 12	4	fixed	A4-70	0.88 N m	0.00105 CALC T334	TO-QUOTE
ENG-HW-135 T335	MV follower FS-08 screw on the FS-04 rod top ISO 10642 M3 × 8	1	fixed	A4-70	0.88 N m	0.00053 CALC T336	TO-QUOTE
ENG-HW-136 T337	FS-15 bracket to FS-01A/B ISO 4762 M4 × 12	2	fixed	A4-70	2.05 N m	0.00205 CALC T338	TO-QUOTE
ENG-HW-137 T339	FS-15 bracket to the airframe wall (radial, ICD-2) ISO 4762 M5 × 10	2	fixed	A4-70	4.15 N m	0.00325 CALC T340	TO-QUOTE
ENG-HW-138 T341	Bleed valve to manifold BL-07 (sub-base screws, per valve) ISO 4762 M3 × 25	8	fixed	A4-70	0.88 N m	0.00164 CALC T342	TO-QUOTE
ENG-HW-139 T343	Bleed manifold BL-07 to the airframe (ICD-14) ISO 4762 M4 × 12	2	fixed	A4-70	2.05 N m	0.00205 CALC T344	TO-QUOTE
ENG-HW-150 T345	O-ring AS568-226 FKM SAE AS568 -226 (ID 50.39 +/- 0.46 x CS 3.53 +/- 0.10 mm, DATASHEET: Freudenberg-NOK ...	2	engine	FKM		0.00315 CALC T346	TO-QUOTE
ENG-HW-151 T347	O-ring AS568-016 FFKM SAE AS568 -016 (ID 15.6 x CS 1.78 mm, AS RECALLED)	1	engine	FFKM		0.00027 CALC T348	TO-QUOTE
ENG-HW-152 T349	O-ring AS568-013 FKM SAE AS568 -013 (ID 10.82 x CS 1.78 mm, AS RECALLED)	1	engine	FKM		0.00019 CALC T350	TO-QUOTE
ENG-HW-153 T351	O-ring AS568-012 FKM SAE AS568 -012 (ID 9.25 x CS 1.78 mm, AS RECALLED)	1	engine	FKM		0.00016 CALC T352	TO-QUOTE
ENG-HW-154 T353	O-ring AS568-010 FKM SAE AS568 -010 (ID 6.07 x CS 1.78 mm, AS RECALLED)	1	engine	FKM		0.00012 CALC T354	TO-QUOTE
ENG-HW-155 T355	O-ring AS568-013 FKM SAE AS568 -013 (ID 10.82 x CS 1.78 mm, AS RECALLED)	1	engine	FKM		0.00019 CALC T356	TO-QUOTE
ENG-HW-156 T357	O-ring AS568-229 FKM SAE AS568 -229 (ID 56.74 x CS 3.53 mm, AS RECALLED)	1	engine	FKM		0.00352 CALC T358	TO-QUOTE
ENG-HW-157 T359	O-ring AS568-012 FKM SAE AS568 -012 (ID 9.25 x CS 1.78 mm, AS RECALLED)	2	fixed	FKM		0.00016 CALC T360	TO-QUOTE
ENG-HW-158 T361	Spring-energised PTFE rod seal 4 mm (pintle stem, TC-04) maker catalogue rod 4 mm	1	engine			MISSING MISSING T362	TO-QUOTE
ENG-HW-159 T363	PTFE / 25 % glass guide bushing (pintle stem, TC-04) maker catalogue 4 mm	1	engine			MISSING MISSING T364	TO-QUOTE
ENG-HW-160 T365	BL-01 face O-ring (FFKM/FKM, N2O) SAE AS568 TO-SELECT	1	engine			MISSING MISSING T366	TO-QUOTE
ENG-HW-161 T367	BL-02 M8×1 port seal (bonded seal or O-ring boss) maker catalogue M8×1	1	engine			MISSING MISSING T368	TO-QUOTE
ENG-HW-162 T369	Crush washer M10 copper (torch IG-01 nose) DIN 7603 A 10 × 14 × 1	1	engine	Cu		0.00067 CALC T370	TO-QUOTE
ENG-HW-163 T371	Port seals of the M10×1 transducer ports (P_feed FD-01, P_manifold TC-04) and the Pc port maker catalogue M10×1 / Pc port	3	engine			MISSING MISSING T372	TO-QUOTE
ENG-HW-164 T373	MSS-1 thermocouple compression fitting 1/16 tube x 1/8-27 NPT male (Swagelok SS-100-1-2 class) maker catalogue 1/16 × 1/8 NPT	1	engine			MISSING MISSING T374	TO-QUOTE
ENG-HW-165 T375	MV-P balance-piston seal (FS-04 groove) and poppet seat seal maker catalogue 11.9 piston / 14 seat	2	fixed			MISSING MISSING T376	TO-QUOTE
ENG-HW-166 T377	Bleed valve sub-base O-rings (2 per valve) SAE AS568 TO-SELECT	4	fixed			MISSING MISSING T378	TO-QUOTE
ENG-HW-167 T379	G11 isolator washers under the HV69 ears (not modelled) in-house M2	4	engine			0.00003 CALC T380	TO-QUOTE
ENG-HW-168 T381	Bundle clip BL-04 bleed hose to the FD-07 coil TO-DESIGN	2	hose			MISSING MISSING T382	TO-QUOTE
ENG-HW-169 T383	Thread locker / retaining compounds: Loctite 243 (GM-18, GM-20, HW-25, TC-05 stem), 648 (GM-04/05 pins), 638 (PD-18) ... maker	1	engine			0.00000 ASSUMED T384	TO-QUOTE
ENG-HW-170 T385	J-FWD disc spring under each ENG-HW-101-IN718 head (route (i), lead 16:53Z R2): spring A baseline DIN EN 16983 (previously DIN 2093) group 1, series A A10: De 10.0 x Di 5.2 x t 0.5, l0 ...	10	engine	1.4568		0.00023 CALC T386	TO-QUOTE

Method	Tightening method (BOM)
M-T1	torque-controlled: calibrated setting torque wrench (ISO 6789-1 Type II; tool tolerance per its class, TO-VERIFY at purchase), screws snugged then taken to the final torque in one pass, opposite-pair / cross order on multi-screw patterns; threads clean, lubrication state as on the drawing (the K basis); N2O-system joints: oxygen-compatible PFPE grease only DECIDED *Design value, BOM_v3.5.2.json v3.5.2* T1203

Method	Tightening method (BOM)
M-T2	as M-T1 with a calibrated setting torque screwdriver (ISO 6789-1 Type II) for final torques below 1 N m DECIDED *Design value, BOM_v3.5.2.json v3.5.2* T1204
M-T1L	as M-T1 (M-T2 tool below 1 N m) on threads coated with the thread locker of the line (cure per the maker sheet before load); torque applied on the wet locker DECIDED *Design value, BOM_v3.5.2.json v3.5.2* T1205
M-T1Ti	as M-T1; torque from a torque-tension test of the anodised (AMS 2488 Type 2) + PFPE-lubricated Ti screw (K TO-VERIFY); the v3.5.1 A4-70 torque shown until that test DECIDED *Design value, BOM_v3.5.2.json v3.5.2* T1206
M-NONE	not a torque joint (dowel, insert, seal, washer, clip, consumable) DECIDED *Design value, BOM_v3.5.2.json v3.5.2* T1207
M-T1J	J-FWD (HW-101, lead 17:55:55Z F1 / F2, 16:53Z R3): as M-T1L (wet thread locker of the line, cure per the maker sheet before load) with the final torque SET PER FLIGHT LOT by J-F1: T_lot = 3750 N x K_min(lot) x 5 mm, so that the lot's K_min gives <= 3750 N (today's K window 0.1465-0.267: 2.747 N m, minimum 2057.6 N at K 0.267); the disc spring HW-170 under each head; C4 registration (lead 17:55:55Z F2): two GSE shoulder screws ENG-TC-13 (not flight), selected to the pin-gauged TC-01 holes, go into two opposite holes before any flight screw; the flight screws are fitted one at a time, each taken to the lot torque and recorded (hole number, screw and spring lot, torque, wrench); the two shoulder screws come out last and are replaced by flight screws, recorded; never re-torqued (single-use, R3) DECIDED *Design value, BOM_v3.5.2.json v3.5.2* T1208
Preload scatter rule	+/-25 % about F = T / (K d), K 0.20 (both ASSUMED; the FEA3D F4 basis, alpha_A = 1.25 / 0.75 = 1.67): the joint margins of 04-fea/assembly3d F4 and 04-fea/v352/joints_v352_v1.json use it; a torque-tension test on the original parts that shows a wider scatter re-opens those joints ASSUMED T1209

5.13 Ground support tools, provisions and contingency (not flight mass)

PN	Item	Qty	Status	Unit mass [kg]	Price
ENG-TC-12 T269	J-FWD hole plug set, 10 plugs + band, O-ring passage aid in-house / workshop TO-QUOTE	1	gse (not flight)	MISSING MISSING T270	TO-QUOTE
ENG-TC-13 T271	J-FWD registration shoulder screw pair (C4 registration) in-house / workshop TO-QUOTE	1	gse (not flight)	MISSING MISSING T272	TO-QUOTE
ENG-BL-06N T263	bleed vent non-propulsive opposed tee outside the airframe wall (on ENG-BL-06) TO-SELECT	1	provision (not modelled)	MISSING MISSING T264	TO-QUOTE
ENG-TK-OUT T265	N2O tank outlet on the aft bulkhead (OI-32): position, bore, sump / dip tube / anti-vortex - MISSING TO-SELECT	1	provision (not modelled: MISSING)	MISSING MISSING T266	TO-QUOTE
ENG-NZ-05 T267	J1b contingency: insulating layer 0.5 mm phlogopite mica between the NZ-03 hot layer and the NZ-04 seal phlogopite mica sheet rated >= 1002 C (J1b FE mica-face peak 901.8 C + 100 K ...	0	contingency (not fitted)	TO-DEFINE MISSING T268	TO-QUOTE

Bleed provision mass estimate (excluding the valves): **0.010–0.020** kg ASSUMED T1210. Valve mass: TO-VERIFY (datasheet) TO-VERIFY T1211

6 Materials and thermal protection

6.1 Materials

Material usage over the BOM part lines (DERIVED by this generator; transport lock and standard hardware excluded).

Material (BOM material_name)	Lines	Mass	Oxygen-clean lines
Aluminium alloy 6061-T6 (EN AW-6061 T6), bar/tube	20 DERIVED T420	2.0705 kg DERIVED T421	5 DERIVED T422
Stainless steel 1.4404 / 316L, annealed bar (EN 10088-3), oxidiser-wetted parts	16 DERIVED T414	0.8795 kg DERIVED T415	15 DERIVED T416
COTS assembly (mixed materials), mass from datasheet or assumed	17 DERIVED T423	0.5409 kg DERIVED T424	8 DERIVED T425
Inconel 718 (UNS N07718, AMS 5662 solution + AMS 5663 aged bar): TC-03 seat-insert alternative	1 DERIVED T480	0.5143 kg DERIVED T481	0 DERIVED T482
ABS, FDM-printed fuel grain (100 % infill)	2 DERIVED T453	0.4244 kg DERIVED T454	0 DERIVED T455
Cotton/paper phenolic laminate tube (NEMA LI-1 grade CE/XX class)	4 DERIVED T477	0.1659 kg DERIVED T478	0 DERIVED T479
Aluminium alloy 7075-T6 (EN AW-7075 T6), bar	5 DERIVED T441	0.1503 kg DERIVED T442	0 DERIVED T443
Carbon/bearing steel (generic COTS)	9 DERIVED T456	0.0809 kg DERIVED T457	0 DERIVED T458
Isomolded fine-grain graphite (reference grade Toyo Tanso IG-110 or equivalent)	1 DERIVED T459	0.0469 kg DERIVED T460	0 DERIVED T461
17-4PH (1.4542) condition H1150, pins	7 DERIVED T429	0.0456 kg DERIVED T430	3 DERIVED T431
Martensitic stainless 440C (UNS S44004, 1.4125), hardened + low temper (raceway pins for drawn-cup needle bearings, v3.5.1)	2 DERIVED T444	0.0274 kg DERIVED T445	0 DERIVED T446
17-4PH (1.4542) condition H900: pintle crankshaft, crank/link pins, MV-P poppet	4 DERIVED T438	0.0258 kg DERIVED T439	0 DERIVED T440
Carbon-cloth phenolic laminate (MX-4926N / FM-5055 class), tube or plate: NZ-02 sleeve, J1 hot-layer washer NZ-03	2 DERIVED T462	0.0155 kg DERIVED T463	0 DERIVED T464
Dispersion-strengthened copper GlidCop AL-15 (UNS C15715), as-consolidated rod (v3.5.1 TC-03 pintle seat insert)	1 DERIVED T474	0.0149 kg DERIVED T475	1 DERIVED T476
Stainless spring wire 1.4310 (302/301) EN 10270-3 or 17-7PH (springs: MV-P closing, pintle spiral spring)	2 DERIVED T435	0.0135 kg DERIVED T436	0 DERIVED T437
PTFE	2 DERIVED T417	0.0127 kg DERIVED T418	2 DERIVED T419
Polyurethane bumper 90A	1 DERIVED T450	0.0092 kg DERIVED T451	0 DERIVED T452
Stainless steel 1.4301 / AISI 304, annealed bar (EN 10088-3)	2 DERIVED T447	0.0046 kg DERIVED T448	0 DERIVED T449
Glass-epoxy laminate NEMA LI-1 G11 (EN 60893-3-2 EP GC 203 class), high-temperature epoxy	3 DERIVED T471	0.0014 kg DERIVED T472	0 DERIVED T473
NdFeB magnet N35 (diametral, Ni-plated)	2 DERIVED T426	0.0012 kg DERIVED T427	0 DERIVED T428
Flexible graphite sheet, oxidation-inhibited grade (GrafTech Grafoil GTJ / SGL Sigraflex inhibited class), 0.5 mm: J1 lip-face seal NZ-04	1 DERIVED T465	0.0007 kg DERIVED T466	0 DERIVED T467
PCTFE (Kel-F 81 / Neoflon M-300 class) soft seat, oxygen-compatible	1 DERIVED T432	0.0002 kg DERIVED T433	1 DERIVED T434
Flexible graphite roll foil 0.25 mm, >= 99.85 % C, 1.0 g/cm3 (SGL SIGRAFLEX TH class): v3.5.2 N-F2 load-spreading foil NZ-06	1 DERIVED T468	0.0002 kg DERIVED T469	0 DERIVED T470

6.2 Short-time exposure of the 6061-T6 case and head (T-F1)

04-fea/v352/texts/item_c_tf1_texts_v352_v5.json. The 10 000 h values are conservative for a short exposure; (d) stays OPEN until a sourced short-time allowable exists.

Item	Value
T-F1 founder action (text v5)	T-F1. Short-time exposure allowables for the 6061-T6 parts TC-01 (case tube) and TC-02 (injector head / forward closure). GATE: before the first static fire. WHY: criterion 183 (1)(d) / 188 (5) needs each part's temperature within a SOURCED time-at-temperature limit. The usable limit on file for 6061-T6 is 150 C (long time). The strength cells use the engmat 10,000 h allowables (conservative for strength), so (d) is reported OPEN, never as passed, for TC-01 and TC-02; it gates nominal operation as well as the abort value. PREDICTED EXPOSURE [SIM, v3.5.2 rev-4 thermal, B1 p99 tg711, gap-conductance bounds GVLO | GVHI]: TC-02 part maximum 188.4 | 181.9 C at the end of burn, above 150 C for 12.3 | 9.9 s per firing. TC-01 part maximum 191.7 | 187.0 C at the chamber bore about 14 mm aft of the injector face, in the soak (51.65 | 43.65 s), above 150 C for 488.8 | 479.5 s per firing; at the J-FWD screw row 159.3 | 129.6 C, 48.5 | 0.0 s above 150 C; while pressurised (burn) TC-01 stays at or below 109.6 | 118.6 C. Over the 50-cycle design life (183 (1)(c)) the cumulative time above 150 C is TC-02 10.2 | 8.2 min and TC-01 6.8 | 6.7 h (part maxima). EXPOSURE BASIS, TB STATE (RESTATED, lead 2026-10-21T05:50:45Z R9; CHANGELOG 249 (3)) [ASSUMED surrogate, sax.py: the end-of-burn field shifted uniformly so that the TC-01 maximum equals A; the shifted-history times apply the same shift to ... CALC T1212 [continues; printed in full in §12.3 with routes (i) and (ii), the coupon test and its pass rules]
Criteria status	a_to_c: PASS on both bounds (lowest TC-02 TBS +0.409 / +0.745; lead, CHANGELOG 249 (3)); d: OPEN (T-F1): no sourced short-time exposure allowable; the cells use the engmat 10,000 h basis CALC T1214 [CALC on SIM; accepted, 249 (3) T1215]
Abort value	A = 150 C (CHANGELOG 249 (1) ACCEPTED) DECIDED *Lead decision, per item_c_tf1_texts_v352_v5.json v5* T1213
6061-T6 allowables at 150 C (10 000 h basis): Fty / Ftu	184.3 MPa LITERATURE T1216 / 194.7 MPa LITERATURE T1217
6061-T6 allowables at 175 C (10 000 h basis): Fty / Ftu	140.5 MPa LITERATURE T1218 / 155.4 MPa LITERATURE T1219
6061-T6 allowables at 178.4 C (10 000 h basis): Fty / Ftu	134.5 MPa LITERATURE T1220 / 150.1 MPa LITERATURE T1221
6061-T6 allowables at 188.4 C (10 000 h basis): Fty / Ftu	116.9 MPa LITERATURE T1222 / 134.4 MPa LITERATURE T1223
6061-T6 allowables at 191.7 C (10 000 h basis): Fty / Ftu	111.2 MPa LITERATURE T1224 / 129.2 MPa LITERATURE T1225
6061-T6 allowables at 193.4 C (10 000 h basis): Fty / Ftu	108.2 MPa LITERATURE T1226 / 126.5 MPa LITERATURE T1227
6061-T6 allowables at 200 C (10 000 h basis): Fty / Ftu	96.6 MPa LITERATURE T1228 / 116.2 MPa LITERATURE T1229
6061-T6 allowables at 208.4 C (10 000 h basis): Fty / Ftu	84.9 MPa LITERATURE T1230 / 104.6 MPa LITERATURE T1231
6061-T6 allowables at 223.4 C (10 000 h basis): Fty / Ftu	68.8 MPa LITERATURE T1232 / 86.6 MPa LITERATURE T1233
6061-T6 allowables at 225 C (10 000 h basis): Fty / Ftu	67.1 MPa LITERATURE T1234 / 84.7 MPa LITERATURE T1235
6061-T6 allowables at 238.4 C (10 000 h basis): Fty / Ftu	52.7 MPa LITERATURE T1236 / 68.7 MPa LITERATURE T1237
6061-T6 allowables at 248.4 C (10 000 h basis): Fty / Ftu	42.0 MPa LITERATURE T1238 / 56.7 MPa LITERATURE T1239
6061-T6 allowables at 250 C (10 000 h basis): Fty / Ftu	40.3 MPa LITERATURE T1240 / 54.8 MPa LITERATURE T1241

6.3 Thermal protection and thermal loads

Item	Value
Grain-zone liner thickness	1.6 mm DECIDED *Design value, combustion.json v2.4d* T1249
Post-chamber insulator	phenolic, thickness set by thermal FEA; 1-D bound needs >= 5 mm and still exceeds the 192 C aft-land allowable (open issue OI-4) DERIVED T1250
Peak convective heat flux at the throat (Bartz, 600 K wall)	21.92 MW/m² SIM T1242
Peak gas-side heat-transfer coefficient	10259 W/(m²·K) SIM T1243
Adiabatic wall temperature at the throat	2737 K (2464 °C) SIM T1244
Incident flux on a cold wall at start (nominal–conservative)	27.12–31.55 MW/m² SIM T1245
Throat insert replacement rule	replace the insert when dA_t/A_t reaches 5 % (measured pin gauge / bore scan after each firing) DERIVED T1251
Throat insert acceptance (NZ-01 criterion)	LEAD 186 (3) as worded by 194 (5): refutation is any of: a throat-area change more than 2 % beyond the erosion predicted for that firing, the prediction frozen before the firing and updated only with earlier firings; a non-axisymmetric step; a missing fragment SIM *Lead decision, per combustion.json v2.4d* T1252
Nozzle gasket NZ-03 finding / joint J1	mica/vermiculite 900 C rating vs 1595 C FE: fails by 695 K DERIVED T1253 2.0 mm carbon-phenolic flange + 0.5 mm flexible-graphite seal at the lip face DERIVED T1254
Case-TC abort A / start and relight limit / nozzle-holder limit (firmware v2.12)	423.15 K (150 °C) DECIDED *Lead decision, line 243 (3)* T1246 / 403.15 K (130 °C) DECIDED *Lead decision of 2026-10-20* T1247 / 523.15 K (250 °C) ASSUMED T1248

Item	Value
Case-TC decisions (CHANGELOG 257)	Lead consistency fold found while applying: the case-TC in-burn abort 423.15 K (243 (3), 249 (1)) and the RELIGHT refusal 403.15 K (249 (4)) replace the stale 448.15 K and 433.15 K in the combustion texts. DECIDED *Lead decision, line 257* T1255

Thermal margins of the parts: 3D FEA thermal and thermo-mechanical cases, §9 (margin matrix columns LC3, TD, LC6, TB70, TBS).

7 Combustion, sequences and the abort set

Reading this section. combustion.json v2.4d is the source of every value here unless a row names another source. Where a quoted text still carries wording that a newer lead decision replaced, the old words are ~~struck through~~ and the decision is inserted with its override id (§13.1).

7.1 Design point

Item	Value
Fuel	ABS (FDM printed, 100 % infill); PETG backup DECIDED *Requirement, combustion.json v2.4d* T1256
Oxidiser load and tank volume	**2.00** kg DECIDED *Lead decision of 2026-10-12* T492 in **3.17** L DECIDED *Lead decision of 2026-10-12* T1257
Tank conditioning	Self-pressurised N2O conditioned 15-24 C at ARM (ENG-12 as amended; was 15-25 C) with the heater hard-disabled at ARM and the T_tank <= 24 C interlock at ARM and ignition (guard band: interlock_guard_band); heater blanket removed at liftoff; N2 supercharge NOT baselined DERIVED T1258
Ignition interlock guard band	T_tank <= 24 C and P_tank <= 55.28 bar abs, each lowered by the total channel uncertainty (Electronics states it) MISSING *Lead decision, line 172 (4)* T1259
Model basis	20 C saturated tank (P_tank 50 bar row of the map for lift and dP/Pc), initial port 25 mm, nominal law. SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* 0.92, eta_cf 0.975) SIM T1260
Combustion and nozzle efficiencies η_{c*} / η_{cf}	**0.920** ASSUMED T1261 / **0.975** ASSUMED T1262
Oxidiser flux at the head, rated, initial port	**491** kg/(m²·s) SIM T1263
Regression law (form)	r = a * G_ox^n (oxidiser-flux law, local G_ox along the port) ASSUMED T1264
Regression coefficient a / exponent n	**5.770e-05** m/s·(kg/m²s)^-n ASSUMED T1265 / **0.50** ASSUMED T1266
Regression law note	No verified N2O/ABS law in accessible literature; every result is checked at x0.6 and x1.5 on a. Static fires T1-T2 measure it. ASSUMED T1267
MEOP basis (SIM)	{"basis_bar": 44.33, "margin_to_MEOP_bar": 0.67, "proof_margin_min_bar": 21.9, "supersedes": "v2.3 basis [withheld: superseded value, SUP-01] bar (RC-08)"} SIM T1268
Stored energy (flight tank)	decomposition_TNT_kg: 0.7737; R_6p9kPa_m: 16.4 DERIVED T1269

7.2 Fuel grain

Item	Value
Segments (BOM)	Fuel grain 1/2: ABS 53.5 OD x 115.0, straight centred port 25 mm (grain v2 S230) T175 Fuel grain 2/2: ABS 53.5 OD x 115.0, straight centred port 25 mm (grain v2 S230) T177
Segment length from the CAD manifest (GR-01 / GR-02)	**116.8** mm DERIVED T1270 / **115.0** mm DERIVED T1271
Grain outer diameter / total length	**53.5** mm DECIDED *Design value, combustion.json v2.4d* T1272 / **230.0** mm DERIVED T1273
Port	single centred straight round port DECIDED *Lead decision of 2026-10-12* T1274, initial Ø **25.0** mm DECIDED *Design value, combustion.json v2.4d* T1275
Initial web	**14.25** mm DERIVED T1276
Port entrance (decided)	Rounded port entrance r >= 4 mm is now REQUIRED on every grain, flight and stand (decomposition peak 40.6 vs 43.7 bar abs, MEOP 45). The fuel-assisted fast part at relight stays controlled by the torch-first gate alone: FMEA hazard with a single control. DECIDED *Lead decision, line 160 (4)* T1277
Density (printed ABS)	**1000** kg/m³ ASSUMED T1278
Grain mass: Combustion / CAD	**0.404** kg DERIVED T1279 / **0.4244** kg DERIVED T1280
Residual web rule	web >= 1 mm at the x1.5 regression bound for every flight duty cycle DERIVED T1281
Retention	forward face on the pre-chamber insulator shoulder, aft face on the post-chamber insulator; anti-rotation key (Mechanical) DECIDED *Design value, combustion.json v2.4d* T1282

7.3 Injector, feed and igniter

Item	Value
Feed architecture	tank → NC main isolation valve (off-gimbal) → -6 PTFE-lined braided hose across the gimbal → <= 15 um filter → pintle injector (on gimbal) DECIDED *Design value, combustion.json v2.4d* T1284
Injector flow model	Dyer et al. AIAA 2007-5702 (SPI/HEM blend); HEM and SPI bound it (G ratio 0.55-0.68 and 1.34-1.45 at 15 C) LITERATURE T1285
Filter	sintered <= 15 um, Cv >= 2.2 DERIVED T1286
Isolation valve: minimum Cv / MEOP / closing time	**3.0** DERIVED T1287 / **7.00** MPa (70.0 bar) DECIDED *Lead decision, per combustion.json v2.4d* T1288 / **0.15** s DERIVED T1289
Relief: trapped liquid, back-relief and tank	SUPERSEDED (v2.4a): v1 "required (<= 7.5 MPa) between the isolation valve and the pintle" is replaced by the back-relief check MV segment → tank (Cv >= 0.1) and the secondary rupture disc 9.5 MPa +/- 5 % on FS-01 (LEAD line 172 (6)); see relief_v2p4a DECIDED *Lead decision, line 172 (6)* T1290
Tank relief (v2.4a)	tank_relief_set_MPa: 6.9; tank_relief_basis: below the 7.0 MPa tank MEOP (ENG-09); fire case relieving at 8.35 MPa; fire_relieving_MPa: 8.35; MAWP_MPa_min: 8.42; proof_MPa_min: 10.5; back_relief_check: MV segment → tank, Cv >= 0.1; secondary_disc: 9.5 MPa +/- 5 % on FS-01, discharge overboard; supersedes: trapped-liquid relief <= 7.5 MPa (v1; ENG-09 text superseded by LEAD line 172 (6)); ... DERIVED *Lead decision, line 172 (6)* T1291
Hose across the gimbal	-6 accepted: 6.3 m/s at 0.241 kg/s nominal, 6.5-7.1 m/s at 0.26 kg/s (15-25 C computed; ENG-12 as amended allows 15-24 C at ARM, so 25 C is a bound); hose friction 8-17 kPa; slow-closure surge (40-150 ms, 0.4-0.8 m line) 14-106 kPa DERIVED T1292
Pc tap and sense line	head top (Mechanical 2026-10-12) via sense line to the fixed side DECIDED *Design value, combustion.json v2.4d* T1293 1/8 in OD (1.6 mm ID) SS, length <= 0.3 m, transducer dead volume <= 0.3 cm3 DERIVED T1294
Igniter type	spark-ignited N2O/propane torch, radial into the pre-chamber DECIDED *Design value, combustion.json v2.4d* T1295
Torch chamber pressure / O/F / throat diameter	**3.0** bar DERIVED T1296 / **5.5** DERIVED T1297 / **2.0** mm DERIVED T1298
Torch flows N2O / propane	**0.503** g/s DERIVED T1299 / **0.091** g/s DERIVED T1300
Spark: supply / energy / rate	**12** V DERIVED T1301 / **0.05** J DERIVED T1302 / **50** Hz DERIVED T1303
Igniter system mass: Combustion / BOM ENG-IG-01	**0.20–0.30** kg ASSUMED T1283 / **0.070** kg ASSUMED T188
Starts per flight	>= 3; limited by propane (0.055 g/start) and N2O torch flow (0.30 g/start) DERIVED T1304
Light windows W from the lift command	**0.40** s DECIDED *Lead decision, per combustion.json v2.4d* T1305 first start · **0.65** s DECIDED *Lead decision, per combustion.json v2.4d* T1306 relight · **0.35** s DECIDED *Lead decision, per combustion.json v2.4d* T1307 stand

7.4 Start and relight sequence

Start and relight sequence (combustion.json v2.4d /igniter/sequence; times relative to E0 where the source states one).

Epoch	t [s]	Action
PURGE (A1 relight only)	**−2.900** s DERIVED *Lead decision, line 153* T1309	PURGE commits iff the lead to the plain relight trigger is >= L* = 1.3 s (LEAD 186 (1), ADOPTED for both wall heat-transfer brackets; replaces the 1.7 s interim of 181 (2) and the PROVISIONAL rule of 175 (3)); below L* no purge and no unpurged relight on that pass (drogue descent; vehicle loss in the FMEA); the ECU unpurged RELIGHT guard enforces it (LEAD 203 (7), v2.11). A commit lead below about 2.1 s is not counted as a landing (LEAD 204 (1), GNC). When committed: MV latch and opener ramp (0.35 s) with the pintle seated; V1 and V2 open; both taps flow 2.43 s (the decided scheduled flow, LEAD 253 (7)(ii), final in 256 (2); the v2.4b records were run with 2.5 s of flow); the close command leads by the measured valve closing time so the flow stops by E0 - 0.05 s (scheduled flow end, LEAD line 164). Closure (V1 first, on P_mid) is confirmed by the closure deadline <= the torch gate (Electronics v4: E0 + 0.16 s = the torch gate, 160 ms, kTorchGate_s), otherwise BLEED_FAULT (see E1). ECU both-open cap <= 2.39 s (LEAD 253 (7)(ii), final in 256 (2); it replaces the 2.55 s of 203 (3)); vented N2O at 2.43 s, unlined, Cd 0.70 <= 66.7 / 70.8 / 75.1 g at 283.15 / 289.5 / 297.15 K and 75.75 g at the 2.44 s of flow the cap allows; at Cd 0.80 the fixed schedule would vent about 80.5 g at 289.5 K and 85.5 g at 297.15 K [ESTIMATE], so after CF-4 the flow and cap are re-set from the measured Cd x A, never above 2.43 / 2.39 s, keeping the 297.15 K vent <= 76 g (LEAD 257 (2); REV-COMB-4 R4-07; REV-GNC-7; purge_decision_v2p4c) DERIVED *Lead decision, line 153* T1310

Epoch	t [s]	Action
E0	0.000 s DERIVED *Lead decision, line 153* T1312	START or RELIGHT at T_fire: MV opens with the pintle seated (already open after PURGE); SOL1 (N2O) and SOL2 (propane) open; spark 50 Hz, 50 mJ from E0 + 0.02 s DERIVED *Lead decision, line 153* T1313
torch gate	0.160 s DERIVED *Lead decision, line 153* T1315	no E1 by E0 + 0.16 s → TORCH_FAULT (A1); no main N2O admitted DERIVED *Lead decision, line 153* T1316
E1	0.090 s DERIVED *Lead decision, line 153* T1318	E1 = torch flame confirmed on P_torch AND photodiode (2oo2, FA-1 (a)), 10 ms persistence, AND, on an A1 relight with the bleed fitted, [bleed closure confirmed OR the closure deadline reached] (deadline <= torch gate E0 + 0.16 s; LEAD line 166 (3)). At the deadline without confirmation: flight - BLEED_FAULT to the FC (log-only, line 168), V2 commanded closed, the lift proceeds; stand - abort. Then the pintle is commanded to THE start-level lift h_s(T_tank at E0), the row of the ignition-lift table: 0.2039 mm at 20 C, 0.21776 mm at the A1 relight state 286.70 K (t_lift 28.7-29.5 ms); CF-3 calibrates the table to <= 37.1 g/s at 20 C and <= 37.4 g/s at the A1 state and measures at the commanded lift with the encoder recorded (start_level_lift_v2p4b; LEAD 203 (7)). From E1 to E2 the torch is supervised on both sensors: loss on BOTH for >= 5 ms closes the pintle and the MV (FA-1 (b), A16) DERIVED *Lead decision, line 153* T1319
W	0.810 s DERIVED *Lead decision, line 153* T1321	abort deadline W from the lift command: 0.40 s flight first start, 0.65 s flight relight, 0.35 s stand (A2) DERIVED *Lead decision, line 153* T1322
E2	0.442 s DERIVED *Lead decision, line 153* T1324	light: Pc >= 0.8 bar above the pre-lift baseline on filtered AND raw Pc for 10 ms, above 1.8 bar abs; SOL1 off first, SOL2 within 20 ms (torch ends propane-only), spark off, ramp released. LIGHT_SPIKE event logged (replaces A5) DERIVED *Lead decision, line 153* T1325
E3	—	thrust >= 90 % of the command (ENG-04 latency, combustion REPORT §7.7) DERIVED *Lead decision, line 153* T1328

One field per value (combustion.json /igniter_sequence_fields_v2p4c, the generated-table input for Electronics); the pill is each field's own tag.

Field	Value	Source (as written)
purge.commit_lead_L_star	1.3 s DECIDED *Lead decision, per combustion.json v2.4d* T1329	igniter.sequence.v.start[0]; LEAD 186 (1) DECIDED *Design value, combustion.json v2.4d* T1330
purge.PURGE_command_analysed	-2.9 s DECIDED *Design value, combustion.json v2.4d* T1331	relight_purge.v.timing_s; Electronics v2.12 states the flight value DECIDED *Design value, combustion.json v2.4d* T1332
purge.MV_ramp	0.35 s DECIDED *Design value, combustion.json v2.4d* T1333	relight_purge.v.timing_s DECIDED *Design value, combustion.json v2.4d* T1334
purge.flow_scheduled	2.43 s DECIDED *Lead decision, per combustion.json v2.4d* T1335	purge_decision_v2p4c; LEAD 253 (7)(ii), 256 (2) DECIDED *Design value, combustion.json v2.4d* T1336
purge.flow_end	-0.05 s DECIDED *Lead decision, per combustion.json v2.4d* T1337	LEAD line 164 DECIDED *Design value, combustion.json v2.4d* T1338
purge.both_open_cap	2.39 s DECIDED *Lead decision, per combustion.json v2.4d* T1339	purge_decision_v2p4c; LEAD 256 (2) DECIDED *Design value, combustion.json v2.4d* T1340
purge.needed_flow_unlined_283p15K	2.2 s DERIVED T1341	purge_decision_v2p4c; REV-COMB-4 R4-07 DECIDED *Design value, combustion.json v2.4d* T1342
purge.needed_flow_unlined_289p5K	1.9 s SIM T1343	relight_purge.v.timing_s.needed_flow_unlined DECIDED *Design value, combustion.json v2.4d* T1344
purge.valve_open_close_times	0.03, 0.08 s ASSUMED T1345	purge_basis_line216_v2p4b.v.cap_rule DECIDED *Design value, combustion.json v2.4d* T1346
purge.closure_deadline	0.16 s DECIDED *Design value, combustion.json v2.4d* T1347	igniter.sequence.v.start[3]; Electronics v4 kTorchGate_s (= the torch gate) DECIDED *Design value, combustion.json v2.4d* T1348
spark.rate	50 Hz DECIDED *Design value, combustion.json v2.4d* T1349	igniter.sequence.v.start[1] DECIDED *Design value, combustion.json v2.4d* T1350
spark.energy	0.05 J DECIDED *Design value, combustion.json v2.4d* T1351	igniter.sequence.v.start[1] (50 mJ) DECIDED *Design value, combustion.json v2.4d* T1352
spark.start	0.02 s DECIDED *Design value, combustion.json v2.4d* T1353	igniter.sequence.v.start[1] DECIDED *Design value, combustion.json v2.4d* T1354
spark.train_max	0.5 s DECIDED *Design value, combustion.json v2.4d* T1355	firmware_constant_sources_v2p4b kSparkTrain_s DECIDED *Design value, combustion.json v2.4d* T1356
torch_gate.latest_E1	0.16 s DECIDED *Design value, combustion.json v2.4d* T1357	igniter.sequence.v.start[2] DECIDED *Design value, combustion.json v2.4d* T1358
E1.median	0.09 s SIM T1359	igniter.sequence.v.start[3] (torch_gate_v2p4) DECIDED *Design value, combustion.json v2.4d* T1360
E1.persistence	0.01 s DECIDED *Design value, combustion.json v2.4d* T1361	igniter.sequence.v.start[3] DECIDED *Design value, combustion.json v2.4d* T1362
E1_to_E2.torch_loss_both_sensors	0.005 s DECIDED *Design value, combustion.json v2.4d* T1363	igniter.sequence.v.start[3] (FA-1 (b), A16) DECIDED *Design value, combustion.json v2.4d* T1364
start_lift.h_s_20C	0.2039 mm DECIDED *Design value, combustion.json v2.4d* T1365	igniter.sequence.v.start[3]; start_level_lift_v2p4b DECIDED *Design value, combustion.json v2.4d* T1366
start_lift.h_s_A1_286p70K	0.21776 mm DECIDED *Design value, combustion.json v2.4d* T1367	start_level_lift_v2p4b.v.line216_9 (ECU row) DECIDED *Design value, combustion.json v2.4d* T1368
start_lift.t_lift	0.0287, 0.0295 s DERIVED T1369	igniter.sequence.v.start[3] DECIDED *Design value, combustion.json v2.4d* T1370
start_lift.flow_cap_20C	37.1 g/s DECIDED *Lead decision, per combustion.json v2.4d* T1371	igniter.sequence.v.start[3]; CF-3 DECIDED *Design value, combustion.json v2.4d* T1372
start_lift.flow_cap_A1	37.4 g/s DECIDED *Lead decision, per combustion.json v2.4d* T1373	igniter.sequence.v.start[3]; CF-3 DECIDED *Design value, combustion.json v2.4d* T1374
W.flight_first_start_from_lift_command	0.4 s DECIDED *Lead decision, per combustion.json v2.4d* T1375	igniter.sequence.v.light_windows_W_s DECIDED *Design value, combustion.json v2.4d* T1376
W.flight_relight_from_lift_command	0.65 s DECIDED *Lead decision, per combustion.json v2.4d* T1377	igniter.sequence.v.light_windows_W_s DECIDED *Design value, combustion.json v2.4d* T1378
W.stand_from_lift_command	0.35 s DECIDED *Lead decision, per combustion.json v2.4d* T1379	igniter.sequence.v.light_windows_W_s DECIDED *Design value, combustion.json v2.4d* T1380
W.relight_plan_limit_E2	0.81 s DECIDED *Lead decision, per combustion.json v2.4d* T1381	igniter.sequence.v.start[4] (torch gate 0.16 + W 0.65) DECIDED *Design value, combustion.json v2.4d* T1382
E2.rise_over_prelift_baseline	0.8 bar DECIDED *Design value, combustion.json v2.4d* T1383	igniter.sequence.v.start[5] (filtered AND raw Pc) DECIDED *Design value, combustion.json v2.4d* T1384
E2.persistence	0.01 s DECIDED *Design value, combustion.json v2.4d* T1385	igniter.sequence.v.start[5] DECIDED *Design value, combustion.json v2.4d* T1386
E2.floor	1.8 bar abs DECIDED *Design value, combustion.json v2.4d* T1387	igniter.sequence.v.start[5] DECIDED *Design value, combustion.json v2.4d* T1388
E2.SOL2_after_SOL1_max	0.02 s DECIDED *Design value, combustion.json v2.4d* T1389	igniter.sequence.v.start[5] DECIDED *Design value, combustion.json v2.4d* T1390
E2.ECU_before_model_max	0.0147 s DERIVED T1391	E2_rule_v2p4b.v.shift_bound_s DECIDED *Design value, combustion.json v2.4d* T1392
E2.worst_purged_median_p99_max	0.442, 0.589, 0.626 s SIM T1393	igniter.sequence.v.start[5] (rg01_line_states_v2p4) DECIDED *Design value, combustion.json v2.4d* T1394
E3.thrust_fraction_of_command	0.9 - DECIDED *Requirement, combustion.json v2.4d* T1395	igniter.sequence.v.start[6] (ENG-04) DECIDED *Design value, combustion.json v2.4d* T1396
propane.SOL2_close_backstop_time	0.4 s DECIDED *Design value, combustion.json v2.4d* T1397	firmware_constant_sources_v2p4b kPropaneOffPcAbs_Pa DECIDED *Design value, combustion.json v2.4d* T1398
propane.SOL2_close_backstop_Pc	5 bar abs DERIVED T1399	firmware_constant_sources_v2p4b kPropaneOffPcAbs_Pa DECIDED *Design value, combustion.json v2.4d* T1400
ramp.start_ramp_max	2500 N/s DECIDED *Design value, combustion.json v2.4d* T1401	firmware_constant_sources_v2p4b kStartRamp_Nps DECIDED *Design value, combustion.json v2.4d* T1402
ramp.pintle_to_ignition_lift_legacy	0.12 s DECIDED *Design value, combustion.json v2.4d* T1403	firmware_constant_sources_v2p4b kPintleToIgnitionLift_s DECIDED *Design value, combustion.json v2.4d* T1404
shutdown.throttle_down_before_MECO_max	0.75 - DECIDED *Lead decision, per combustion.json v2.4d* T1405	igniter.sequence.v.shutdown[0] (ENG-05) DECIDED *Design value, combustion.json v2.4d* T1406
shutdown.MV_seat_confirm_Pc_below	2 bar abs DECIDED *Design value, combustion.json v2.4d* T1407	igniter.sequence.v.shutdown[2] DECIDED *Design value, combustion.json v2.4d* T1408
shutdown.MV_deenergise_after_command_max	0.3 s DECIDED *Design value, combustion.json v2.4d* T1409	igniter.sequence.v.shutdown[2] DECIDED *Design value, combustion.json v2.4d* T1410
shutdown.ground_hold_before_safing	30 s DECIDED *Design value, combustion.json v2.4d* T1411	igniter.sequence.v.shutdown[3] DECIDED *Design value, combustion.json v2.4d* T1412
arm.tank_T_range	15, 24 C DECIDED *Requirement, combustion.json v2.4d* T1413	igniter.sequence.v.arm_preconditions[0] (ENG-12; interlock_guard_band) DECIDED *Design value, combustion.json v2.4d* T1414
arm.P_torch_refuse_above	1.05 bar abs DECIDED *Design value, combustion.json v2.4d* T1415	igniter.sequence.v.arm_preconditions[2] DECIDED *Design value, combustion.json v2.4d* T1416
arm.relight_case_TC_refuse_above	403.15 K DECIDED *Lead decision, per combustion.json v2.4d* T1417	igniter.sequence.v.arm_preconditions[5]; LEAD 249 (4) kCaseRelightMax_K DECIDED *Design value, combustion.json v2.4d* T1418
release.accel_thrust_threshold	37.16 N DECIDED *Lead decision, per combustion.json v2.4d* T1419	drogue_release_hold.v.line175_5_note_v2p4a (LEAD 175 (5)) DECIDED *Design value, combustion.json v2.4d* T1420
release.frames	3 - DECIDED *Lead decision, per combustion.json v2.4d* T1421	drogue_release_hold.v.line175_5_note_v2p4a DECIDED *Design value, combustion.json v2.4d* T1422
release.frame_rate	100 Hz DECIDED *Lead decision, per combustion.json v2.4d* T1423	drogue_release_hold.v.line175_5_note_v2p4a DECIDED *Design value, combustion.json v2.4d* T1424
release.pcf_tmax	0.8 s DECIDED *Lead decision, per combustion.json v2.4d* T1425	drogue_release_hold.v.line175_5_note_v2p4a; LEAD 227 (5) DECIDED *Design value, combustion.json v2.4d* T1426
stand_abort.N2_purge_min	60 s DECIDED *Lead decision, per combustion.json v2.4d* T1427	igniter.sequence.v.abort_if_no_ignition[1] DECIDED *Design value, combustion.json v2.4d* T1428
stand_abort.N2_purge_rate_min	10 g/s DECIDED *Lead decision, per combustion.json v2.4d* T1429	igniter.sequence.v.abort_if_no_ignition[1] DECIDED *Design value, combustion.json v2.4d* T1430
stand_abort.hold_before_approach_min	15 min DECIDED *Lead decision, per combustion.json v2.4d* T1431	igniter.sequence.v.abort_if_no_ignition[1] DECIDED *Design value, combustion.json v2.4d* T1432

Phase	Rule
ARM preconditions	tank 15-24 C at ARM (ENG-12 as amended); T_tank <= 24 C at ARM and at ignition (liquid-line interlock, combustion REPORT §2.8) with the thresholds lowered by the channel uncertainty (interlock_guard_band); tank heater hard-disabled at ARM DERIVED *Lead decision, line 153* T1433
	pintle seated, encoder healthy; MV-P closed; bleed valves V1 and V2 closed, drivers inhibited DERIVED *Lead decision, line 153* T1434
	P_torch auto-zero; ARM refused if P_torch > 1.05 bar abs DERIVED *Lead decision, line 153* T1435

Phase	Rule
	Pc transducer: flight unit RESELECTED by Electronics (LEAD 205): miniature seat-mounted, body within 12 mm x 30 mm, front dead volume <= 20 mm3, proof >= 166.14 bar abs (1.1 x FA1S 151.04; LEAD 249 (4)); C1 Kulite XTL-190SM-2000A 206.8 bar abs (DATASHEET 1.5x rule) meets it; OI-41 DECIDED by LEAD 230 (2), range covering the 45 bar MEOP with the 30 bar abort resolvable, >= 120 C; over-range bit and above-full-scale flag (LEAD 203 (1)(a)); P_torch 0-70 bar class (ENG-09; it sees the FA-1 passage pressure, design case FA1S 151.04 bar abs: its proof (>= 166.14 bar abs, or isolation) and its saturation and recovery are Electronics v2.12 items, REV-COMB-4 R4-12); P_feed, P_tank healthy; Pc reads ambient DERIVED *Lead decision, line 153* T1436 [P_torch: lead 264 (4) T802 selects the C1-family Kulite XTL-190SM-2000A; the "0-70 bar class" and "or isolation" wording is superseded (isolation is rejected); the threshold re-base is OPEN: contradiction register C-14]
	igniter continuity (spark coil, SOL1, SOL2); photodiode healthy (needed for the 2oo2 E1) DERIVED *Lead decision, line 153* T1437
	relight only: chamber vented; RELIGHT refused if a case thermocouple reads > 403.15 K (130 C; kCaseRelightMax_K, LEAD 249 (4); the start and ARM limit on TC1 / TC2 is the same value; the 433.15 K of v2.4b is superseded) DERIVED *Lead decision, line 153* T1438
Relight	MV closed during coasts (LEAD); the v2.3 text "the isolation valve may stay open between burns" is withdrawn DERIVED *Lead decision, line 153* T1439
	A1 relight: PURGE at E0 - 2.90 s (2.5 s of flow as analysed in the v2.4b records; the decided scheduled flow is 2.43 s with the 2.39 s cap, LEAD 253 (7)(ii) and 256 (2); Electronics v2.12 states the command time, E0 - 2.90 s or E0 - 2.83 s; theta <= 2 deg, balanced or axial vent), then the start sequence from E0 DERIVED *Lead decision, line 153* T1440
	settling: liquid at the tank outlet from E0 - 2.90 s - t_settle (combustion REPORT §2.4 note); RELIGHT refused if a case thermocouple > 403.15 K (LEAD 249 (4)) DERIVED *Lead decision, line 153* T1441
	light window W 0.65 s; E0 → E2 max 0.626 s on the worst purged line state (rg01_line_states_v2p4); the 0.81 s plan holds on the predictor path (PURGE committed with lead >= L* = 1.3 s; LEAD 181 (2) scope); after a commit lead below L* there is no relight on that pass DERIVED *Lead decision, line 153* T1442
Shutdown	pre-MECO throttle-down to <= 75 % (ENG-05) DERIVED *Lead decision, line 153* T1443
	pintle closes at full drive speed DERIVED *Lead decision, line 153* T1444
	MV de-energised on seat confirmation (encoder at seat and Pc < 2 bar abs) or 0.30 s after the command DERIVED *Lead decision, line 153* T1445
	trapped liquid relieves to the tank through S1 (combustion REPORT §2.7); ground: 30 s hold then safing DERIVED *Lead decision, line 153* T1446
Abort if no ignition	pintle closed; torch solenoids closed (SOL1 not after SOL2); MV closed (abort is MV-first) DERIVED *Lead decision, line 153* T1447
	no retry into a pooled chamber; stand: N2 purge >= 60 s at >= 10 g/s, then a hold of >= 15 min before approach with Pc at ambient and the head thermocouples falling, then safing (LEAD line 172 (1); stand_safing_v2p4a). SUPERSEDED (v2.4a): "wait >= 5 s" withdrawn DERIVED *Lead decision, line 153* T1448
	flight: no retry; vehicle-level response per GNC DERIVED *Lead decision, line 153* T1449
	A5 (Pc > 8 bar for 5 ms) is DELETED (LEAD line 160): it tripped on 98 % of the late-light grid; LIGHT_SPIKE records the event instead DERIVED *Lead decision, line 153* T1450

Stand safing after a no-light. no retry into a pooled chamber; N2 purge of the chamber >= 60 s at >= 10 g/s; hold >= 15 min before approach, with Pc at ambient and the head thermocouples falling; then the safing procedure (REPORT §10.2) DECIDED *Lead decision, line 172 (1)* T1451

7.5 Relight purge (decided schedule and conditions)

combustion.json /purge_decision_v2p4c (decided by LEAD 253 (7)(ii), final in 256 (2); conditions 256 (2), 257 (2), 258 (2)). The v2.4b records were run with 2.5 s of flow (the analysed basis, more flow than decided).

Item	Value
Commit rule	PURGE commits iff the lead to the plain relight trigger is >= L* = 1.3 s; below L* no purge and no unpurged relight on that pass SIM / DERIVED *Lead decision, per combustion.json v2.4d* T1452
Decision	2.43 s scheduled purge flow (both-open overlap 2.38 s + 0.05 s) with the 2.39 s both-open cap, ADOPTED by LEAD 253 (7)(ii) and FINAL in 256 (2) on REV-COMB-4; it replaces the 2.5 s schedule and the 2.55 s cap DERIVED / ASSUMED / ESTIMATE *Lead decision, per combustion.json v2.4d; basis SIM* T1453
Scheduled flow / both-open overlap / both-open cap / flow at the cap	**2.43** s DERIVED *Lead decision, per combustion.json v2.4d; basis SIM* T1454 / **2.38** s DERIVED *Lead decision, per combustion.json v2.4d; basis SIM* T1455 / **2.39** s DERIVED *Lead decision, per combustion.json v2.4d; basis SIM* T1456 / **2.44** s DERIVED *Lead decision, per combustion.json v2.4d; basis SIM* T1457
Needed flow, unlined line (283.15 / 289.5 / 297.15 K)	283.15 K: 2.2; 289.5 K: 1.9; 297.15 K: 1.8 DERIVED T1458
Margins at 2.43 s, unlined	283.15 K: 0.23; 289.5 K: 0.53; 297.15 K: 0.63 DERIVED T1459
Vented N2O bound (law) at 2.43 s	283.15 K: 66.4; 289.5 K: 70.7; 297.15 K: 75.4; at_cap_2p44s_of_flow: 75.75 DERIVED T1460
Vented N2O at Cd 0.80, fixed schedule	289.5 K: 80.5; 297.15 K: 85.5; 297.15 K, 2.44 s of flow at the cap: 85.8 ESTIMATE T1461 ESTIMATE: 2.43 s x 33.13 g/s at 289.5 K (average rate); at 297.15 K the Cd 0.80 / 0.70 ratio 1.1326 on 31.06 g/s (REV-GNC-7 interim 2); with the fixed 2.43 / 2.39 s schedule, which (d) re-sets after CF-4 ESTIMATE T1462
A1 reserve test (REV-GNC-7)	test: CHANGELOG line 208 against the v3d reference (A1 210 m, v351, N = 500 per base, dip onset 80 g); 76 g: UB95 0.854 pp liquid / 0.617 pp powered: PASS; 80.5 g: UB95 0.972 / 0.617 pp: PASS (margin 0.028 pp); first_fail_g: 82.1; 85.5-85.8 g: UB95 1.206 pp liquid: FAIL (powered 0.736 pp passes); tag: DERIVED from SIM by REV-GNC-7 (interim 2, 09:25:51Z); above 80 g a linear per-run shift, ESTIMATE DERIVED T1463
Condition (a), LEAD 256 (2)	wherever 1.9 s is quoted as the needed flow, 2.2 s at the 283.15 K cold corner is stated with it DECIDED *Lead decision, per combustion.json v2.4d* T1464
Condition (b), LEAD 256 (2)	CF-4 / B-BLEED-1 shows a measured Cd x A of the two-tap path >= 0.91 x the Cd 0.70 value (>= 0.95 x with the FD-01 element bound), else the flow is re-sized DECIDED *Lead decision, per combustion.json v2.4d* T1465
Condition (c), LEAD 256 (2)	the line stays unlined (lined, 283.15 K needs 2.5 s) DECIDED *Lead decision, per combustion.json v2.4d* T1466
Condition (d), LEAD 257 (2)	after CF-4 / B-BLEED-1, kPurgeFlow_s and the both-open cap are re-set from the measured Cd x A by the measured-time form of purge_basis_line216_v2p4b, with the mass flow taken at 297.15 K: both-open overlap <= 76 g / mdot(297.15 K) - (t_close - t_open), never above the decided 2.43 s / 2.39 s; the 297.15 K vent then stays <= 76 g at any measured Cd x A, and the 283.15 K need stays met because need and limit both scale with 1 / (Cd x A) (margin on the 10 ms steps 0.23 s at a Cd x A ratio of 1, 0.21 s at 1.08 and 1.13 (Cd 0.80), 0.19 s at 1.2) [DERIVED; the scaling is ASSUMED, as in (b)]; if the constants are not re-set, the measured Cd x A must also be <= 1.08 x the Cd 0.70 value (vent <= 82 g at the cap, REV-GNC-7's bound) or GNC re-runs the A1 reserve test at the measured vented mass; a temperature-scheduled flow (1.8 s needed at 297.15 K) is not adopted DECIDED *Lead decision, per combustion.json v2.4d* T1467
Condition (d1), LEAD 258 (2)	the (d) re-set takes the upper 95 % bound of the measured N2O Cd x A of the two-tap path, not its mean: at the Cd 0.80 ratio r 1.1326 a ratio measured 7.9 % low gives a true vent of 81.99 g, just under the 82.1 g first fail of the A1 reserve test DECIDED *Lead decision, per combustion.json v2.4d* T1468
Condition (d2), LEAD 258 (2)	CF-4 shows the 283.15 K need at the re-set flow (need and limit both scale with 1 / (Cd x A), ASSUMED); below a ratio r of 1 condition (b) governs DECIDED *Lead decision, per combustion.json v2.4d* T1469
PURGE command time (open)	open: Electronics v2.12 states E0 - 2.90 s (the flow ends at E0 - 0.12 s) or E0 - 2.83 s (the flow ends at E0 - 0.05 s) [DERIVED] DERIVED T1470
CF-4	CF-4 v2.3 (v2.4d) runs the decided flow, adds a run at or below 283.15 K or states the cold corner model-only, checks the Cd x A acceptance in P5, records the measured Cd x A that re-sets the flow and cap (d), takes the re-set from its upper 95 % bound (d1) and shows the 283.15 K need at the re-set flow (d2) (LEAD 258 (2)) DECIDED *Lead decision, per combustion.json v2.4d* T1471
Electronics constants	kPurgeFlow_s 2.43 s; command span 2.38 s inside the 2.39 s cap; flow at the cap 2.44 s (Electronics v4 head_end_bleed_provision.cap_decision_230_3.option_not_made, now decided); after CF-4 both are calibration constants re-set by (d) with (d1) / (d2) (LEAD 258 (2)), never above these values DECIDED *Design value, combustion.json v2.4d* T1472 [implemented in electronics.json v4.1 as the purge FINAL calibration constants: ACCEPTED (REV-ELEC-4, line 266) T027; §10.6]
Vented-mass reserve	vented N2O is judged against the 76 g basis (G73 decided, G80 sensitivity) and the A1 N2O reserve (a1_reserve_test_REV_GNC_7); with (d) the vent stays <= 76 g, which passes; the 253 (7)(ii) re-fly-trigger wording is withdrawn (REV-COMB-4 R4-11) DECIDED *Lead decision, per combustion.json v2.4d* T1473

7.6 Abort set (Combustion matrix; firmware causes in §10.5)

combustion.json /abort_matrix_v2p4 (LEAD lines 160, 164, 166, 168, 175; REPORT v2.4a §7.4). The F4 PC_SENSOR HIGH abort (TB-state) and the flight J-FWD indication (JFWD_LS; Q_JFWD on the stand at B-DPC-1), routed to Electronics v2.12 by LEAD 258 (4), are implemented in electronics.json v4.1: ACCEPTED (REV-ELEC-4, line 266) T027; §10.

ID	Combustion abort matrix (as published)
A2	no E2 within W of the lift command (0.40 / 0.65 / 0.35 s) DECIDED *Lead decision, line 160* T1474
A4	Pc > 30 bar abs on the FILTERED channel for 5 ms (LEAD line 166 (4); amends line 160 "raw"); the hardware OVP comparator (31.47 bar g) stays the fast path for a blocked throat. Decomposition spike filtered peak <= 21.4 bar: no trip DECIDED *Lead decision, line 160* T1475
A5	DELETED → LIGHT_SPIKE log DECIDED *Lead decision, line 160* T1476
A7	Pc loss or implausibility, bands and the flight START high-side rule (Electronics sets values) DECIDED *Lead decision, line 160* T1477
A14	bleed valve fault, or closure not confirmed by the closure deadline (closure deadline = torch gate, E0 + 0.16 s; Electronics v4) → BLEED_FAULT to the FC (FC action LOG-ONLY, line 168); flight: V2 commanded closed and the lift proceeds (E1 term, line 166 (3)); stand: abort; fail-closed → no purge → no relight on that pass; closure from P_mid is a pad go/no-go (line 168). SUPERSEDED (v2.4a): "closure not confirmed by E0 - 0.05 s" (E0 - 0.05 s stays the scheduled flow end only) DECIDED *Lead decision, line 160* T1478
A15	pintle or chamber anomaly during PURGE: inhibit torch, close MV DECIDED *Lead decision, line 160* T1479
A16	torch loss on BOTH sensors >= 5 ms between E1 and E2: close pintle and MV DECIDED *Lead decision, line 160* T1480
A3_unpurged_note_v2p4a	A3 trips on the post-light vapour gap of an unpurged start (ratio <= 0.24 against 0.40 for >= 22 ms), after the drogue-release term is already true: see purge_commit_rule_v2p4a; unpurged relight excluded by design (LEAD 186 (1); ECU guard 203 (7)) DECIDED *Lead decision, line 160* T1481

Abort action (firmware v2.12, SIL only). every abort de-energises the NC valve at once and commands the pintle closed in parallel; no in-flight relight retry after a no-light; normal shutdown: pintle first, MV on seat confirmation or the timeout DECIDED *Lead decision, per electronics.json v4.1* T1482

7.7 FA-1 hard start: rulings of record

combustion.json /z3_jfwd_rulings_v2p4d and /fa1_p_torch_v2p4c. Grid counts are on an ASSUMED construct, not probabilities. The FA-1 pressure spectrum is in §2.4; the J-FWD criterion rows in §9.5.

Item	Value (quoted)
Z3 v2 (FA-1 class; LEAD 258 (3))	
decision	ADOPTED by LEAD 258 (3) at the go 2026-10-21T10:47:17Z: REV-COMB-4 Z3' as written (head peak <= 88.1 bar abs AND undamped SDOF ESP <= 151.04 bar abs at each of the 406 published periods 0.02-50 ms, S from rest at the record start) plus the v1 FA-1 class conditions kept (first-to-last crossing of 63.1 bar abs <= 3.0 ms; 10-90 % first rise >= 0.8 ms); Z3 allowed only inside the light window (record start to 150 ms after the first 8 bar abs crossing) DERIVED / CALC *Lead decision, line 258 (3); basis SIM / ASSUMED* T1483
grid	896 / 896 grid cases inside (850 inside v1); ESP max 146.881 bar abs at 3.5 ms, margin 4.159 bar to FA1S [DERIVED on SIM; grid counts on an ASSUMED construct, not probabilities] DERIVED / CALC *Lead decision, line 258 (3); basis SIM / ASSUMED* T1484
flight	NOT EVALUATED in flight (1 kHz record; the LIGHT_SPIKE raw peak is a lower bound) DERIVED / CALC *Lead decision, line 258 (3); basis SIM / ASSUMED* T1485
not adopted	shape limits (option 2: the smallest box that admits every grid case admits a trapezoid with ESP 163.71 bar abs, above FA1S), a raised FA1S (option 3; needs FE) and the damped ESP variant (structural damping MISSING) DERIVED / CALC *Lead decision, line 258 (3); basis SIM / ASSUMED* T1486
J-FWD replace rule on Q_JFWD (LEAD 258 (3))	
decision	ADOPTED by LEAD 258 (3): Q_JFWD = max(head peak over the record, max of S(T) over T 0.2-0.5 ms on the light window and each excursion window), not the peak alone; Q_JFWD > 60.0 bar abs at any time in a run, no minimum duration, means the line-231 J-FWD inspection (C6 (2) list) and the TC-02 taps gauged (M5-6H GO / NO-GO) before the next run DERIVED / CALC *Lead decision, line 258 (3); basis SIM / ASSUMED* T1487
grid	298 / 896 grid cases against 185 for the peak alone; the 113 added cases have a head peak of 49.781-60.0 bar abs (Q / peak in gauge up to 1.2989) [SIM grid counts on an ASSUMED construct, not rates] DERIVED / CALC *Lead decision, line 258 (3); basis SIM / ASSUMED* T1488
why	Q_JFWD >= the peak by construction, so it keeps every flag of the 256 (4) peak rule; a false trigger costs an inspection, a missed one is an unanalysed joint load DERIVED / CALC *Lead decision, line 258 (3); basis SIM / ASSUMED* T1489
normal lights	not meant for normal lights (REV-COMB-4 R4-10): a light to MEOP 45 bar abs trips it only below a 0.288 ms 10-90 % rise [DERIVED] DERIVED / CALC *Lead decision, line 258 (3); basis SIM / ASSUMED* T1490
measured traces	on measured stand traces a noise-driven trigger still means the inspection until a noise-floor rule is derived from a no-event B-DPC-1 record and reviewed (TO-VERIFY) DERIVED / CALC *Lead decision, line 258 (3); basis SIM / ASSUMED* T1491
basis	Mechanical's rows at 60 bar abs, nominal factors: MS_y +0.328 (TC-01 ligament, LC2U) / MS_u +0.312 (TC-01 ligament trend, 78.6 C) [CALC on SIM] DERIVED / CALC *Lead decision, line 258 (3); basis SIM / ASSUMED* T1492
band	0.2-0.5 ms J-FWD band ASSUMED (no v3.5.2 J-FWD modal in this package: TO-VERIFY) DERIVED / CALC *Lead decision, line 258 (3); basis SIM / ASSUMED* T1493
replaces	the v1 'any head excursion above 60.0 bar abs: replace all J-FWD screws' rule and the HW-101 '60 bar' action (drawing texts v2 58683aa9a66a65dd; no hardware-data, BOM or geometry edit) DERIVED / CALC *Lead decision, line 258 (3); basis SIM / ASSUMED* T1494
electronics	Q_JFWD logging and the inside_v2 re-classification of the S6R runs and the N-04 burn-on cases: Electronics v2.12 (LEAD 258 (4)) DERIVED / CALC *Lead decision, line 258 (3); basis SIM / ASSUMED* T1495
Open before flight acceptance (LEAD 258 (3))	
open item 1	the flight consequence of a flagged first light (relight on a used joint; line 231 covers one Z3 event per assembly) (LEAD) TO-VERIFY T1496
open item 2	the damped ESP variant (structural damping MISSING) TO-VERIFY T1497
open item 3	the 0.2-0.5 ms J-FWD band (ASSUMED; TO-VERIFY on a v3.5.2 modal) TO-VERIFY T1498
open item 4	a shutdown from a case hotter than the rows' soak basis (F4 observation; Mechanical, no limit derived) TO-VERIFY T1499
Torch passage pressure and the J-FWD rows	
P_torch design case (FA1S)	P_torch during FA-1: design case FA1S 151.04 bar abs uniform over the passage (REV-COMB-4 R4-01); linear lossless passage peaks <= 129.9 bar abs at the J-FWD tap and <= 148.7 at the torch seat for c 150-1200 m/s; the band values 94 / 121 / 147 / 94 do not bound gas-filled passages DECIDED *Lead decision, per combustion.json v2.4d* T1500
Mechanical row (tap wall v3)	tap-wall v3, published (04-fea/v352/jfwd_tapwall_v352_v3.json ef70572ec92b05b0; CHANGELOG 259 (1)), at P_torch = FA1S 151.04 bar abs uniform over the passage, DLF 1.0, 3750 N J-F1 cap, wall 1.6355 mm net at +/-0.25 deg: governing row SOAK52 (TC-02 thread 94.9 C): sigma 136.9 MPa, MS_y +0.670 / MS_u +0.785; screw-temperature sensitivity MS_y +0.294 at SOAK52 and +0.275 at SOAK70 (158.6 C); 0 flags; Lame factor 2.2094 [CALC on SIM]; ligament lowest local natural frequency >= 26 kHz against <= 3.45 kHz in the passage [CALC; edges and span ASSUMED; not an FE modal]; R4-01 CLOSED by the ... CALC T1501
Grid statistics status	CLOSED (LEAD 258 (3), go 2026-10-21T10:47:17Z): Z3 v2 ADOPTED = Z3' above as written plus the v1 window (first-to-last crossing of 63.1 bar abs <= 3.0 ms) and rise (10-90 % first rise >= 0.8 ms) kept, Z3 allowed only inside the light window; 896 / 896 grid cases inside [DERIVED on SIM; grid counts on an ASSUMED construct]; NOT EVALUATED in flight; 253 (7)(i) stays adopted, its 256 (3) condition ... DECIDED *Lead decision, per combustion.json v2.4d* T1502
Combustion side	no Combustion number changes: the FA-1 grid histories are the v2.4b set and fa1_grid_statistics_v2p4c keeps the reviewer's statistics; T1-T4 adds no series-stop item (T-R2 already halts the series at any Pc sample above 45 bar abs; the inspection after a Q_JFWD trigger is Mechanical's rule for every run of a J-FWD joint) DERIVED / CALC *Lead decision, line 258 (3); basis SIM / ASSUMED* T1503

7.8 Stability

Stability indicators at the initial port (SIM).

Throttle	dP/Pc	Chug gain margin, 5 ms	Chug gain margin, 10 ms	1L acoustic	Helmholtz	Thermal lag
100.0 %	**0.93** SIM T1504	**6.5** SIM T1505	**4.7** SIM T1506	**1460** Hz SIM T1507	**1125** Hz SIM T1508	**2.17** Hz SIM T1509
75.0 %	**1.58** SIM T1510	**10.2** SIM T1511	**7.5** SIM T1512	**1477** Hz SIM T1513	**1138** Hz SIM T1514	**1.61** Hz SIM T1515
50.0 %	**2.78** SIM T1516	**17.0** SIM T1517	**12.5** SIM T1518	**1499** Hz SIM T1519	**1155** Hz SIM T1520	**1.07** Hz SIM T1521
25.0 %	**5.86** SIM T1522	**32.1** SIM T1523	**23.7** SIM T1524	**1532** Hz SIM T1525	**1180** Hz SIM T1526	**0.56** Hz SIM T1527
20.0 %	**7.18** SIM T1528	**36.8** SIM T1529	**27.2** SIM T1530	**1539** Hz SIM T1531	**1186** Hz SIM T1532	**0.47** Hz SIM T1533
12.5 %	**10.54** SIM T1534	**52.2** SIM T1535	**38.6** SIM T1536	**1544** Hz SIM T1537	**1190** Hz SIM T1538	**0.33** Hz SIM T1539

injector dP/Pc >= 0.25 everywhere (ceiling rule); chug gain margin from the lumped feed/chamber model; acoustic modes far above the actuator band DERIVED T1540
classical hybrid low-frequency (thermal-lag/boundary-layer) instability 0.3-2 Hz band is identified but its amplitude is NOT predicted; static fire T2/T3 with the 1 kHz control channel for the 0.3-2 Hz band and the stand dynamic channel (>= 10-20 kHz, pc_dynamic_channel) for the 1L / Helmholtz modes at 1.1-1.8 kHz, which 1 kHz logging aliases TO-VERIFY T1541

7.9 Test specifications: CF-4 and T1-T4

Physical tests are run by the founders; none has been run. The T4 rows of T1-T4 v3.2 stay the DC-D computation [SIM] until Combustion re-runs them (OPEN).

Item	Value (quoted)
CF-4 claim	A1 relight lights within W 0.65 s with P_abort <= 1e-3, venting <= 76 g, theta <= 2 deg DECIDED *Lead decision, line 164* T1542
CF-4 specification	CF-4-SPEC v2.3 (v2.4d, 6f0d4afd7fb3a798) DECIDED *Design value, combustion.json v2.4d* T1543
CF-4 pass criterion 1	P1 liquid at the pintle inlet within 18.86 cm3 of lift-flow volume after reaching the start-level lift: 0.275 s at 0.21776mm (A1 state), 0.293 s at 0.20390mm (v2.8 basis) (CF-4 v2.1; volume form) DECIDED *Lead decision, line 164* T1544
CF-4 pass criterion 2	P2 >= 0.35 s continuous liquid or inside the void budget DECIDED *Lead decision, line 164* T1545
CF-4 pass criterion 3	P3 cap clears FD-01 crown <= 2.3 s after flow start DECIDED *Lead decision, line 164* T1546
CF-4 pass criterion 4	P4 per tap and per phase: vapour >= 3.4 g/s per tap (80 % of HEM 4.25), liquid >= 12.5 g/s per tap (80 % of Dyer 15.62), Cd_eff >= 0.56, dP V1+V2 <= 11 %; judged against the open-element reading; the barrier-element T4 stall is a model construct (bubble point ~0.7 kPa vs 45.5 bar) and a measured stall is a finding DECIDED *Lead decision, line 164* T1547
CF-4 pass criterion 5	P5 <= 76 g vented (IS-1, IS-2) DECIDED *Lead decision, line 164* T1548
CF-4 pass criterion 6	P6 flow stopped by E0 - 0.05 s (scheduled); closure confirmed (V1 first, P_mid) by the deadline <= torch gate (Electronics v4: E0 + 0.16 s); one valve held open → BLEED_FAULT at the deadline; flight build: V2 commanded closed and the lift proceeds; stand build: abort DECIDED *Lead decision, line 164* T1549
CF-4 pass criterion 7	P7 no unfiltered path to the seat DECIDED *Lead decision, line 164* T1550
T1-T4 law rule	default gate "current rule OR a_meas >= 1.25 → change grain" unless the oxidiser-flux law is positively shown at sigma(S) <= 1.5 % on the T3 holds and a T1/T2 firing (LEAD line 172 (3)) DERIVED T1551
T1-T4 specification	T1-T4 regression-rate measurement and law form (RG-03), v3.2 DECIDED *Design value, T1-T4-REGRESSION-SPEC.md v3.2* T1552
T4 duration (N-07) and re-size	CONFIRMED by GNC in gnc.json v3.2 (bde317782574cbd9; duty_cycles.note_n07_rev_gnc_7; published in CHANGELOG 260): no GNC objective needs the DC-D duty as one firing; GNC's static-fire objectives need DC-D segments, each flown as its own segment or firing inside the A = 150 C case limit (REV-ELEC-3 N-07: the DC-D first burn would cross it at about 23.5 s, merged bound [DERIVED]); the A1-like relight objective sits with the shortest T4 firing. T4 is re-sized to that firing (LEAD 261): planning value about 2.0 kg of N2O [ESTIMATE]; the T4 rows of T1-T4 v3.2 lines 61, 143-145 and 171 stay the DC-D computation [SIM] until Combustion re-runs them on the re-sized profile (OPEN) DECIDED *Lead decision, per combustion.json v2.4d* T1553

Item	Value (quoted)
T4 decision (CHANGELOG 261 (2))	T4 RE-SIZED (DECIDED): GNC's N-07 answer in gnc.json v3.2 (duty_cycles.note_n07_rev_gnc_7) confirms that no GNC objective needs the DC-D duty as one firing and that GNC's DC-D segments fly as their own segments or firings inside the A = 150 C case limit (REV-ELEC-3 N-07: the DC-D first burn would cross it at about 23.5 s, merged bound [DERIVED]). T4 is re-sized to REV-COMB-4's shortest firing (B1 p99 burn 21.76 s, a coast of about 19 s, one relight of about 4.6 s at about 0.47 throttle): planning value about 2.0 kg of N2O [ESTIMATE] against 7.02 kg. The T4 rows of T1-T4 lines 61, 143-145 and 171 stay the DC-D computation [SIM] until Combustion re-runs them (OPEN, Combustion; not in the finish run). Why: a T4 that trips the case abort at about 23.5 s of its first burn cannot be flown as written, and REV-COMB-4, REV-ELEC-3 and GNC all point to the shortest firing. DECIDED *Lead decision, line 261 (2)* T1554 [the relight length of 261 (2) is superseded by 264 (5), next row; contradiction register C-15]
T4 relight (CHANGELOG 264 (5), N-07, OI-T4-RESIZE)	N-07 (OI-T4-RESIZE): SHORTER RELIGHT ADOPTED: the T4 relight after the B1 p99 burn (case about 137.8 C at its end [DERIVED]) is cut so that no planned segment crosses A = 150 C, keeping the light, the latency measurement and a steady-Pc segment; Combustion sets the length in the T4 re-run of 261 (2) with Mechanical's soak-back. The coast is not lengthened to make margin (soak-back can raise the case after shutdown and is not modelled). T4 then demonstrates relight latency from a soaked state, not a 4.59 s landing-length relight from the B1-soaked state; bench plan v5 and the datasheet say so. DECIDED *Lead decision, line 264 (5)* T1555 **What T4 demonstrates (264 (5)):** T4 then demonstrates relight latency from a soaked state, not a 4.59 s landing-length relight from the B1-soaked state; bench plan v5 and the datasheet say so. DECIDED *Lead decision, line 264 (5)* T1556 [the length is OPEN: Combustion sets it in the T4 re-run with Mechanical's soak-back; T4 is a stand relight and waits for v2.13 (266 (6)) T025; contradiction register C-15]
Test budgets	T1_100pct_2t_8s_kg: 1.928; share_of_flight_load: 0.964; T1_max_2t_s_on_flight_tank: 6.04; stand_tank: MISSING (founders): the T1 2t = 8 s firing uses 96 % of a 2.0 kg flight load; with a flight-size tank 2t <= 6.0 s (T1-T4 v3 states both); CF4: runs_lower: 59; runs_base_5_per_state_and_attitude: 75; runs_upper_with_4deg_repeats: 105; N2O_per_run_g: 169.5642883766552; N2O_per_run_terms_g: vent_max: 76.0; lift_flow: 55.76428837665523; segment_state: 37.8; N2O_per_run_g_at_0p2039mm: 166.015; … DERIVED T1557

8 Plume CFD

Simulation, not measurement. Version v2.1a for validation and v2.2 for the time-matched afterburn, 2026-10-12. Owner: Combustion and Propulsion. Status: digital simulation only, to be reviewed. A simulation is a prediction, not a measurement. Text revision v2.2b, 2026-10-13: wording changes for review findings RC-10, RC-11 and RC-19 (`reviews/REVIEW-COMBUSTION.md`, ACCEPT WITH COMMENTS); no case was re-run. SIM T1559

8.0 Operating points

Table 8.0: operating points of the plume CFD (RPL §0) and the current max-thrust point (combustion.json /operating_points). Every table and figure of this section names its point.

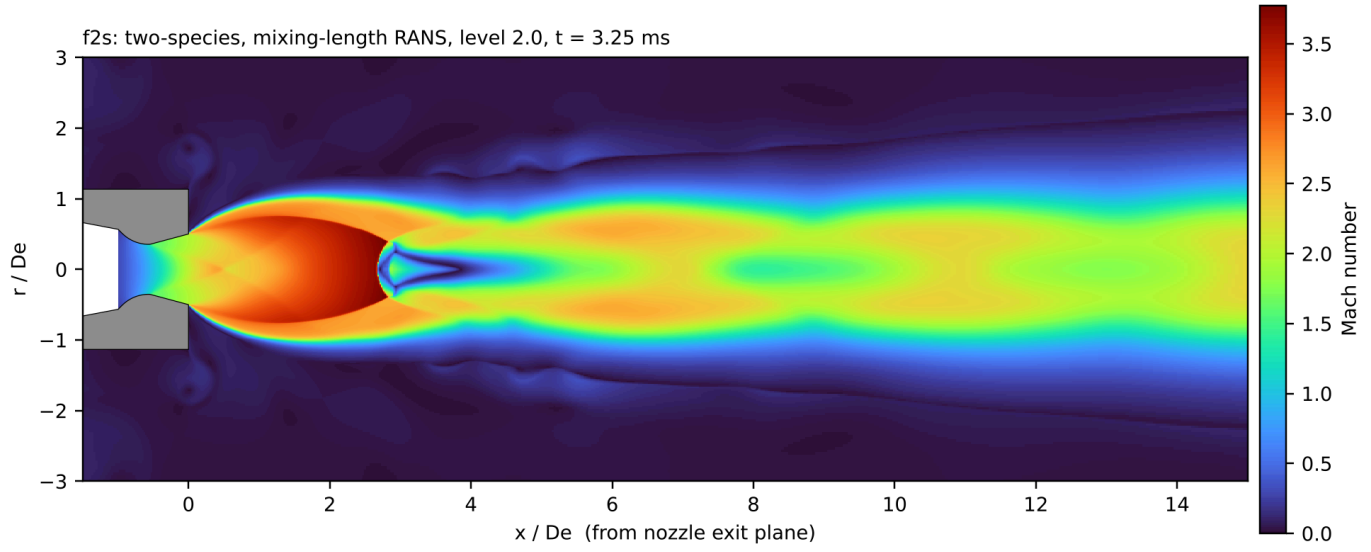
Item	Value
Plume v1 point: every grid level, sensitivity and OpenFOAM run	v1 (combustion.json v1): Pc 24.99 bar, O/F 7.76, T0 = eta_c*^2 Tc = 2844.9 K. All grid levels, sensitivities and OpenFOAM runs use it. SIM T1560
Plume v2 point: coarse grid only	v2 (combustion.json v2.1 S230 point): Pc 25.01 bar, O/F 10.435, T0 2757 K. SIM T1561
Current max-thrust point (combustion.json): thrust / Pc / O/F / delivered T0	**500** N SIM T1566 / **25.01** bar SIM T1567 / **10.435** SIM T035 / **2757** K SIM T1568
Relation of the plume points to the current max-thrust point	The current max-thrust point (combustion.json v2.4d: 500 N, Pc 25.01 bar, O/F 10.435) is the plume v2 point (same Pc and O/F; coarse grid only). The fine-grid results and the field figures of §8.1-8.2 are at the v1 point (O/F 7.76, Pc 24.99 bar). On the coarse grid the v2 point differs from the v1 point by +0.015 % in thrust, +2.46 % in the Mach disk position, +3.22 % in mass flow and −5.8 % in the axis temperature at 10 De (validation_v2.json /v2op_vs_v1). DERIVED T1573
Currency of the plume report and results	RPL (05-plume-cfd/REPORT.md, text revision v2.2b, 140b3c59f8323bba) and PVAL (validation_v2.json v2.1a, cbfbce89c41689f3) are the current plume files: the newest file in 05-plume-cfd/ is 05-plume-cfd/REPORT.md (2026-10-13T11:24Z); REPORT.md equals REPORT_v2p2b.txt. DERIVED T1574

8.1 Key results at the v1 point (O/F 7.76)

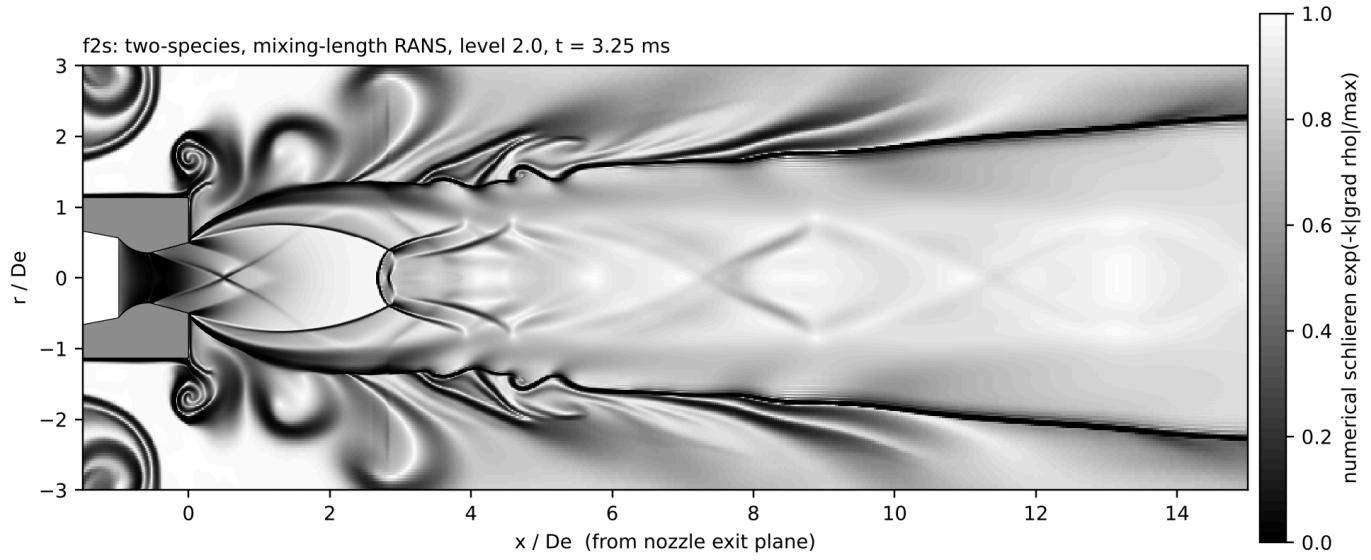
Plume CFD results (PVAL v2.1a), fine grid at the v1 point, O/F 7.76 (current max-thrust point O/F 10.435: Table 8.0) unless a row says otherwise; lengths in exit diameters De.

Quantity	Value
Exit Mach number / pe/pa / fully expanded Mach number (1-D)	**2.012** DERIVED T1589 / **3.236** DERIVED T1590 / **2.623** DERIVED T1591
Mach disk position, fine grid (in mm) and Lewis–Carlson correlation	**2.676** De SIM T1575 (**51.9** mm DERIVED T1592) — correlation **2.663** De DERIVED T1576
Mach disk resolvability	grid differences within 2 sigma of the temporal scatter: GCI not meaningful; report the band SIM T1593
Shock-cell spacing, first / mean of three (in mm)	**3.96** De SIM T1577 (**76.8** mm DERIVED T1594) / **4.10** De SIM T1578 (**79.6** mm DERIVED T1595)
Shock-cell spacing, Prandtl–Pack–Tam correlation	**4.82** De DERIVED T1579
Shock-cell spacing grid sensitivity	**6.54** % SIM T1596 — oscillatory: 3 x max-deviation bound SIM T1597
Peak Mach number before the disk	**3.69** SIM T1598
Potential core length, axis Z ≥ 0.95 (in mm)	**14.9** De SIM T1580 (**289** mm DERIVED T1599)
Axis temperature at 10 De / 20 De (frozen, no afterburning)	**2222** K (1949 °C) SIM T1581 / **2058** K (1785 °C) SIM T1582
Thrust, exit integral (fine) vs 1-D with divergence factor; deviation	**500.8** N SIM T1583 vs **505.8** N DERIVED T1584; **−0.98** % SIM T1585
Thrust grid sensitivity	**0.15** % SIM T1600 — monotone without observed convergence (p < 0.5): 3 x max-deviation bound SIM T1601
Mass flow, CFD vs 1-D; deviation	**0.2542** kg/s SIM T1586 vs **0.2558** kg/s DERIVED T1587; **−0.65** % SIM T1588
Starting-jet front at the end of the run	**44.9** De SIM T1602
v2 point (= the current max-thrust point when Table 8.0 says so), coarse grid: Mach disk CFD / correlation; thrust deviation	**2.709** De SIM T1603 / **2.666** De DERIVED T1604; **−1.09** % SIM T1605
v2 point remark	same coarse grid and models as own_c_2s_rans (v1 point O/F 7.76); v2 - v1 differences in v2op_vs_v1 carry no grid error but do carry the temporal band of the coarse level SIM T1606

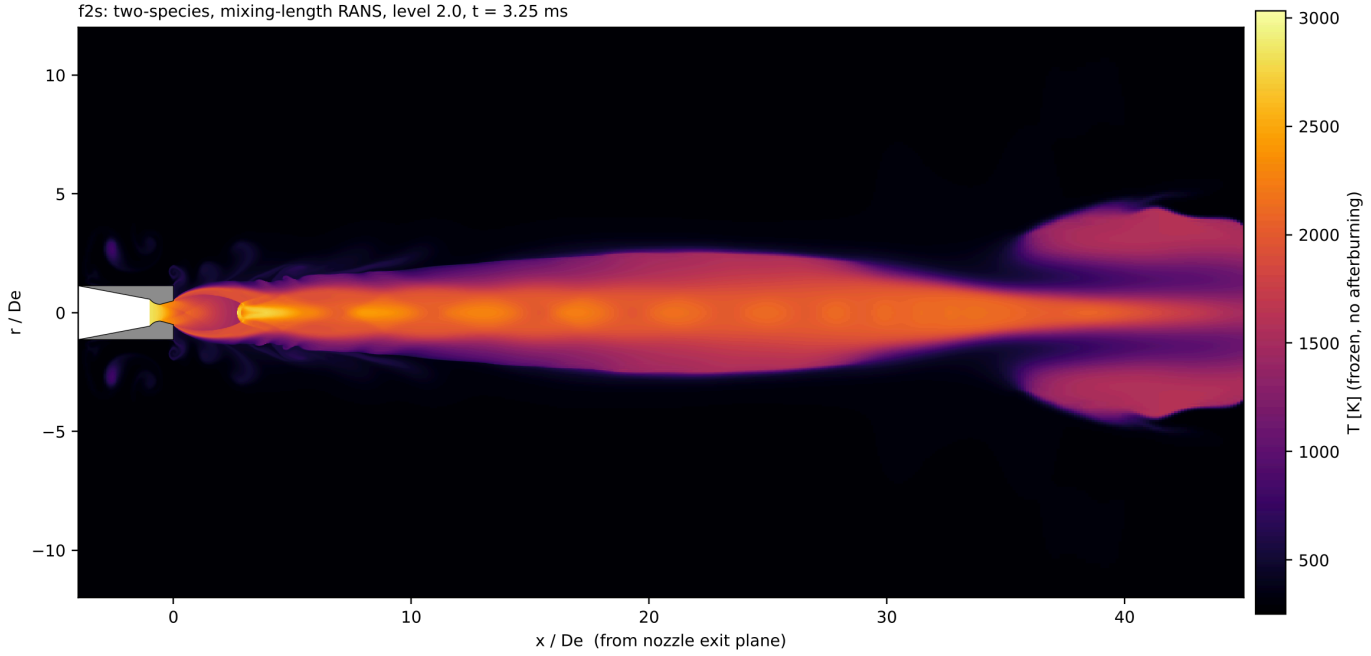
8.2 Fields



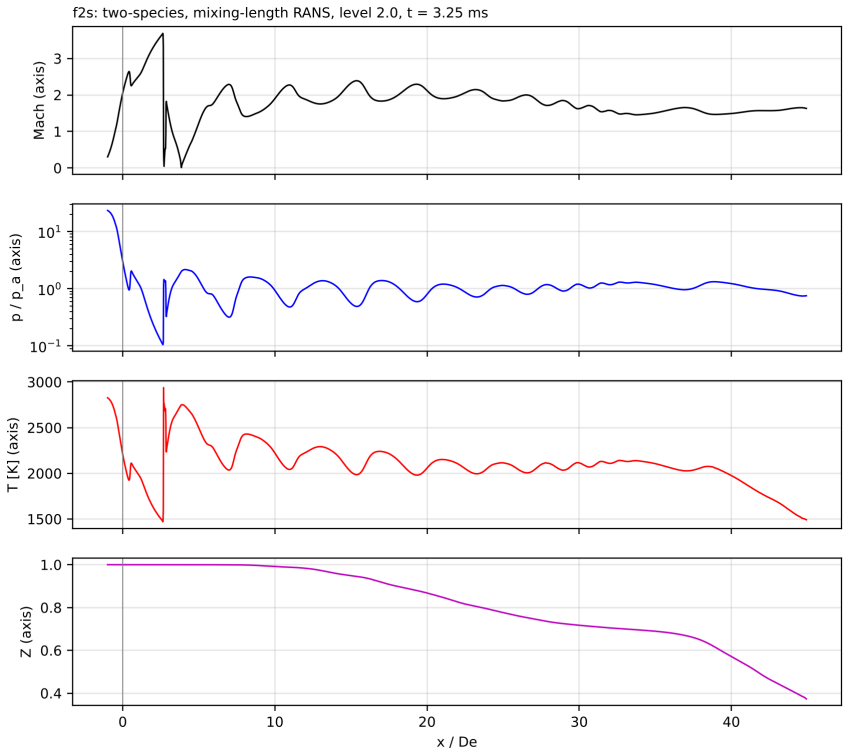
Mach number, near field (SIM, fine grid, v1 point, O/F 7.76 (current max-thrust point O/F 10.435: Table 8.0)). Source PF-MACH: 05-plume-cfd/ownruns/f_2s_100/fig/f2s_mach_near.png (sha256/16 39cfd050d53b9f68).



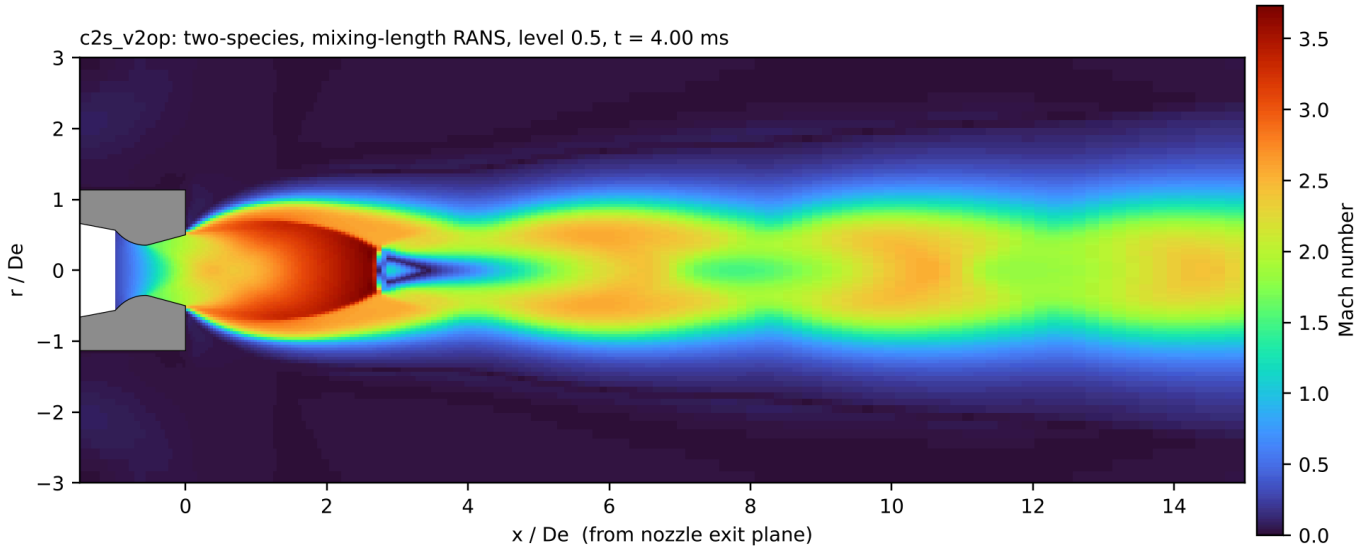
Numerical schlieren |∇ρ| (SIM, fine grid, v1 point, O/F 7.76 (current max-thrust point O/F 10.435: Table 8.0)): Mach disk, reflected shocks and the shock-cell train. Source PF-SCHL: 05-plume-cfd/ownruns/f_2s_100/fig/f2s_schlieren_near.png (sha256/16 5a559717ba7c3a76).



Static temperature, full domain (SIM, fine grid, v1 point, O/F 7.76 (current max-thrust point O/F 10.435: Table 8.0); frozen chemistry, no afterburning). The far field is not statistically stationary. Source PF-TEMP: 05-plume-cfd/ownruns/f_2s_100/fig/f2s_T_full.png (sha256/16 b7fe094ff216873c).



Centreline profiles (SIM, fine grid, v1 point, O/F 7.76 (current max-thrust point O/F 10.435: Table 8.0)). Source PF-AXIS: 05-plume-cfd/ownruns/f_2s_100/fig/f2s_centreline.png (sha256/16 de88abf544299723).



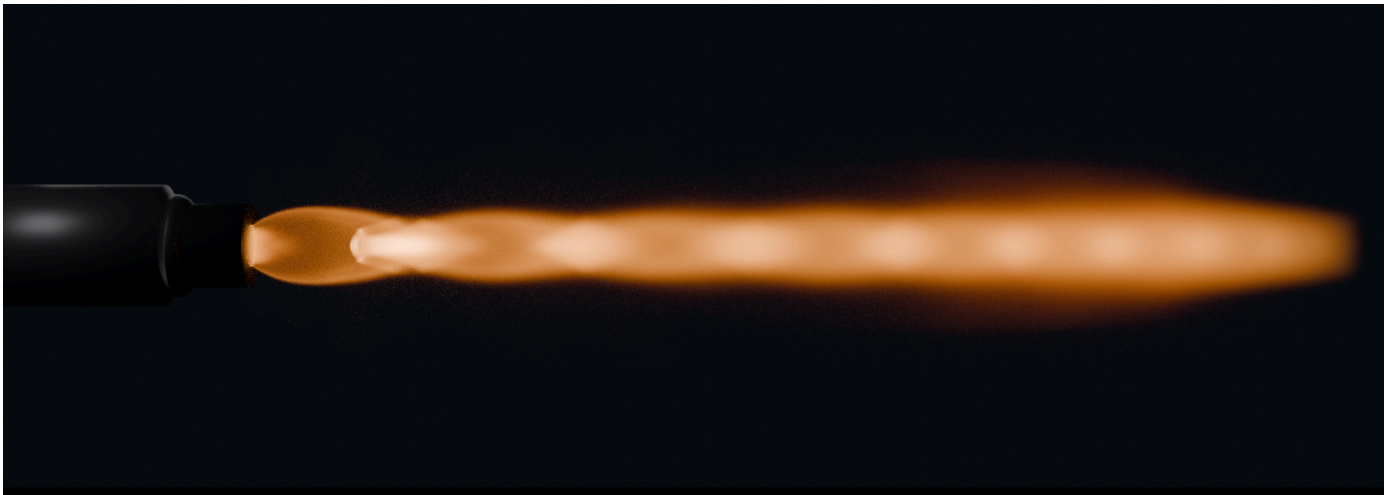
Mach number at the v2 point (SIM, coarse grid; O/F 10.435, the current max-thrust point, Table 8.0). v2-v1 differences carry the grid error of the coarse level. Source PF-V2MACH: 05-plume-cfd/ownruns/c_2s_100_v2op/fig/c2s_v2op_mach_near.png (sha256/16 034d7f18ff3075f3).

8.3 Afterburning and model limits

Afterburning by operating point (RPL §0) and the model limits (RPL §10); operating points in Table 8.0.

Item	Report statement
Afterburning at the v1 point (O/F 7.76)	adds 172-182 K at O/F 7.76 (temporal band 172-178 K; time-matched fine maximum 179.2 K; Richardson extrapolation of the three levels 181.8 K); SIM T1607
Afterburning at the v2 point (O/F 10.435; the current max-thrust point when Table 8.0 says so)	adds nothing at the v2 point for an equilibrium exhaust (O/F 10.4, oxidiser-rich; 0.44 K). The stream-B upper bound, which releases the c* deficit by fast chemistry, raises the plume maximum from 2705 K to 3011 K at that point (difference of maxima, +306 K; ownruns/c_2s_100_v2op/fig/c2s_v2op_afterburn_stats.json); SIM T1608
Afterburning at the stretch floor (12.5 %)	adds up to 1476 K at the fuel-rich 12.5 % floor. SIM T1609
Limit: Soot and radiation	**Soot and radiation.** There is no soot model and no radiative transfer. The synthetic photo uses an ASSUMED soot proxy. SIM T1610
Limit: Finite-rate chemistry	**Finite-rate chemistry and turbulence-chemistry interaction.** Afterburning is a fast-chemistry, mean-Z equilibrium bound. Real afterburning is lower, and possibly zero if the mixture quenches. SIM T1611
Limit: Exhaust gas model	**Exhaust gas model.** The exhaust is a calorically perfect gas with frozen calibrated gamma, which gives core temperature errors up to about -500/+30 K in the strongly expanded region (§4). SIM T1612
Limit: Geometry and setting	**Geometry and setting.** The model is axisymmetric only (no gimbal angle, no 3-D structure) and has no ground plane, so no landing impingement or recirculation is modelled. SIM T1613
Limit: Duration	**Duration.** Simulated time is 3-4 ms. The far field (> 15 De) is not statistically stationary, and the fine level covers only 0.25 ms. SIM T1614
Limit: No acoustics	**No acoustics or plume loads on the vehicle.** SIM T1615

8.4 Illustration



ARGOS H1 plume at 500 N, v1 design point (Pc 25 bar, O/F 7.76); fields: medium-grid 2-D axisymmetric CFD (t = 3 ms) with the fast-chemistry afterburning increment, revolved

ARTISTIC RENDER (Blender Cycles): appearance only. Colours, brightness, exposure and the soot-proxy emitter are chosen for illustration;

NOT a measurement or a validated radiation result. Traceable synthetic image: synth_photo (forward Abel, qualitative).

Illustration, not simulation. Blender render of revolved CFD fields with exposure, colour and emitter chosen for appearance (ARTISTIC); it is not a radiation prediction, carries no data and is not tied to an operating point for any value. Source PF-ART: 05-plume-cfd/blender/final/argos_plume_artistic_side.png (sha256/16 7c722b7f001b2ae4).

8.5 Review status

Independent reviews of the plume CFD in register order, newest last (reviews/SUPERVISION.json). A review verdict is not physical approval.

Review	Verdict	Comments (or criteria)	Reviewed file hashes vs current files
REV-PLUME-1 T1617	ACCEPT WITH COMMENTS DERIVED T1618	RC-10 thrust GCI R=1.09 diverges: reword; RC-11 correlation claim wording; RC-19 presentation DERIVED T1619	0 unchanged, 1 changed since review (05-plume-cfd/REPORT.md), 0 not found DERIVED T1616
REV-PLUME-2 T1621	ACCEPT DERIVED T1622	R2-12 summary wording (lead file) corrected: HANDBACK-SUMMARY d12c8862bf5524a4 DERIVED T1623	1 unchanged, 0 changed since review (none), 0 not found DERIVED T1620

Item	Value
Report text revision answering the comments	Version v2.1a for validation and v2.2 for the time-matched afterburn, 2026-10-12. Owner: Combustion and Propulsion. Status: digital simulation only, to be reviewed. A simulation is a prediction, not a measurement. Text revision v2.2b, 2026-10-13: wording changes for review findings RC-10, RC-11 and RC-19 (`reviews/REVIEW-COMBUSTION.md`, ACCEPT WITH COMMENTS); no case was re-run. SIM T1559

9 Structural analysis: 3D FEA of the assembled engine (CAD v3.5.2)

SIM / CALC / DERIVED; not physical approval; no hardware has been tested. Every margin below is a model result (SIM), a hand calculation on model loads (CALC) or a value derived from them (DERIVED). Negatives are shown as found. Delivered by / reviewed by / date: fea3d (FEA3D-2) / not reviewed: status to be reviewed until the independent review / 2026-10-21. DERIVED T1624

9.1 Model, loads and status

FEA3D v3.5.2 report header (F3D F3D FINAL handback of the v3.5.2 re-check) and the rows-v8 matrix of record.

Item	Value
Status	2026-10-21, FINAL handback of the v3.5.2 re-check (seat FEA3D-2 from 2026-10-21T00:41Z; seat FEA3D-1 before). The assembled-engine model on CAD v3.5.2 (asm v1.3): LC1 and LC2 on the S-AX gap-conductance bounds GVLO / GVHI, FA-1 (FA1Q / D / U / S / T) on the NF2b relight state, TB70 / TBS, the LC6 soak column, LC3 at the 8 stop poses and F2 (GS / TD / LC6 / LC2F) at the 9 range-cone poses with both lever roots at R3, the LC3 / F2 hot bounds, F3 stiffness, the modal corrections (RM2-06 / RM2-07), the mounting data, the F4 joints, the thermal-joint re-posts (RM2-13) and the J-FWD one-sector contact sub-model with its seat patches, envelope and convergence check. Margin matrix of every part x case (3A) re-run on the J-FWD envelope v5 and, the second and last time, on the final envelope v6; neither re-run moved a cell. Supersedes the v3.5.1 REPORT for v3.5.2; the v3.5.1 record stays as ... DERIVED T1625
Author	fea3d (FEA3D-1, then FEA3D-2), lead briefs and rulings as cited per section (CHANGELOG 231-248; lead rulings 2026-10-20 / 21 by UTC time) DERIVED T1626
Configuration	CAD v3.5.2 INTERIM set as relayed (manifests in 2.1; rev 1 `05a48a8efb89ea8f` ... hardware v1.6 `e25b45808a8f9cf6`). The FINAL v3.5.2 set (CHANGELOG 247: `MANIFEST_v3.5.2.sha256` `a94851ae4dc22cc4`, 441 rows; re-issued with the archive inputs as `0658b176e918f91f`, 448 rows, CHANGELOG 248 (6)) differs in ENG-TC-01 only (J-FWD spot-face flats d11.2 instead of d10.5, coplanar): disclosed in 2.1 and 10A.2; the J-FWD convergence check cD is on the final TC-01. DERIVED T1627 [manifest of record: 03-cad/v3.5.2/MANIFEST_v3.5.2.sha256 a4713d45b3b24a0f T708, 456 rows T1628; the issues quoted in this row are its earlier issues]
Tool	CalculiX 2.21 (SPOOLES direct), C3D10 tets (gmsh 4.15.2), own Python pre / post (`scripts/`), assembly model fea3d-asm v1.3 ('tiex' tie formulation, 7.1) DERIVED T1629
Units	mm, N, MPa, degC. MS = allowable / (SF x stress) - 1 DERIVED T1630
Tags	SIM (FE output), DERIVED (computed from SIM / data), CALC (Mechanical's hand calculation), ASSUMED, DATASHEET, LITERATURE, TO-VERIFY, AS RECALLED, ESTIMATE DERIVED T1631
Independent review	DECIDED: no independent review of the FEA3D v3.5.2 run (04-fea/assembly3d_v352/, FINAL in line 252) is held in this finish run. The datasheet review-status table says so plainly and cites this item. DECIDED *Lead decision, line 262 (3)* T1632
Margin matrix of record	Cell = MS_y / MS_u after the 234 (4) all-node rule. Flags: r = regular value stands (the all-node minimum sits at an idealisation a named sub-model / contact FE replaces), a = all-node value is the cell, c = includes v3.5.1 carried values (unchanged part), p = some sources pending, s = hot-bound all-node value lowered to the node-wise bound at its node (hb_singular_check_v352), w = hot-bound all-node value replaced by the rigorous part-wide bound over every node of the part, stored value beside it (hb_partwide_v352; lead 21:47:29Z: lc3_bound does not sum the increments of parts outside SAME_MESH_ENGINE node by node). MF (gov. ...) = model-fidelity class with the governing sub-model value (lead 21:15:29Z (3)); BT = bad-tet class (lead 21:15:29Z (4)). '-' = case not evaluated for the part; n/a = no allowable in that slot (brittle parts: one value). DERIVED T1633
Matrix delta vs the FEA3D-2 table	Mechanical's single J-FWD re-map on the FINAL envelope v6 was published under the lead go 2026-10-21T07:37:34Z (CHANGELOG line 254, sha16 c7d52385af9ce594): rows v8 jfwd_in718_rows_v352_v8.json af072cc2e70711ca, with HW-101 5020A Eq. 22 on the R6 basis (V 4079.2 N FE (b1), M 4354.6 N mm CALC). DERIVED T1634

9.1.1 Mesh convergence status

The seat-line P+Q is NOT shown mesh-converged; s6 + C1 stand in every state (CHANGELOG 250 (10), 252 (1)).

Point	Status (quoted)
Seat-line P+Q (J-FWD)	MF note (lead 01:30:18Z (5b)). The seat-line P+Q is not shown converged: s9 → s6 raises it by 12-18 % on the governing line. Under 234 (4) the s6 value is quoted as found. SIM T1635
Check cD (finer mesh, FINAL TC-01)	cD END [SIM]: seat_sDL12_mu0 (s6 + two Ball fields h 0.04 mm at the d+6.051 z'0 ends, FINAL TC-01 d11.2, mu 0) rc 0, 6639 s, peak RSS 6.88 GB; result jfwd_cD_result_v352.json 8209e80f09f28af7 (script 6dfd6d331b946b48, runs/post_cD.log 5f696dded48b0c3c): governing-line PmPb cD / s6 PRELOAD 0.927, SOAK52 0.947, SOAK 0.966, SOAK70 0.959, FA1S 0.985, so by the (C) rule s6 + C1 stand in every state (lead 05:17:36Z (1)); MF note: s9 → s6 +12 to +18 %, s6 → cD -1.5 to -7.3 %, not monotonic, mesh and geometry not split, NOT shown converged; the patch-boundary extrapolation (9 sD nodes over 0.01 mm, none on the flats) is information. SIM T1636
Ruling on cD	cD RULING (05:17:36Z): s6 + C1 stand in every state; band anchors unchanged (PRELOAD 235.5 / SOAK52 370.3 / SOAK 366.8 / SOAK70 362.1 / FA1S 448.3 MPa); cD lines recorded beside, not entered; MF note kept with the numbers: s9 → s6 +12 to +18 %, s6 → cD -1.5 to -7.3 %, not monotonic, mesh and geometry change not split, so the seat-line P+Q is NOT shown converged; the extrapolation and the 9 sD nodes are information. ENVELOPE v6 ADOPTED as FINAL by the rule fixed at 02:01:34Z before the run: cD's resultants enter as an instance as solved (SOAK70 window max 3123.4 → 3151.1 N, +0.9 %; FA1S N min 2734.6 → 2728.5 N, -0.2 %; FA1S window min 462.5 → 455.0 N, -1.6 %); per-instance SOAK70 threaded-length M max 3013.4 → 3095.9 N mm and M at x' 30.5 3415.2 → 3438.9 N mm recorded for Mechanical; cD's result script first checked one side of N and window, now two-sided (recorded in the delta). ... DECIDED *Lead decision, line 250 (10)* T1637

9.2 Margin matrix: every part × every load case

Cell = MS_y / MS_u after the 234 (4) all-node rule. Flags: r = regular value stands (the all-node minimum sits at an idealisation a named sub-model / contact FE replaces), a = all-node value is the cell, c = includes v3.5.1 carried values (unchanged part), p = some sources pending, s = hot-bound all-node value lowered to the node-wise bound at its node (hb_singular_check_v352), w = hot-bound all-node value replaced by the rigorous part-wide bound over every node of the part, stored value beside it (hb_partwide_v352; lead 21:47:29Z: lc3_bound does not sum the increments of parts outside SAME_MESH_ENGINE node by node). MF (gov. ...) = model-fidelity class with the governing sub-model value (lead 21:15:29Z (3)); BT = bad-tet class (lead 21:15:29Z (4)). '-' = case not evaluated for the part; n/a = no allowable in that slot (brittle parts: one value). DERIVED T1633

Read the matrix as found: a negative cell is a FOUND NEGATIVE of the model and is never shown as passing. Cells are MS_y / MS_u (yield / ultimate; one value for brittle parts). Red cells are negative; amber cells are below +0.05. "–" means the case is not evaluated for that part (§9.2.4). Classes and closures of every negative cell are in §9.3.

Table 9.1: Static, pressure and FA-1 cases (MS_y / MS_u; flags r, a, c, p, s, w as defined in the matrix text above; MF / BT classes; source MMXMD MMXMD FEA3D margin matrix v3.5.2).

Part	LC1	LC2	LC2F	FA1Q	FA1D	FA1U	FA1S	FA1T
TC-01	+0.955 / +0.580 r T1638	+0.440 / +0.164 r T1639	–	+0.412 / +0.522 r T1641	+0.107 / +0.194 r T1642	**−0.083 / −0.012 r MF[1]** T1643	**−0.107 / −0.037 r MF[2]** T1644	–
TC-02	+0.496 / +0.223 r T1646	+0.469 / +0.202 r T1647	–	+1.705 / +1.918 r T1649	+1.098 / +1.263 r T1650	+0.725 / +0.861 r T1651	+0.679 / +0.811 r T1652	–
TC-03	**−0.088 / −0.170** T1654	**−0.092 / −0.175** T1655	–	+9.382 / +11.587 T1657	+8.834 / +10.922 T1658	+8.247 / +10.211 T1659	+8.153 / +10.097 T1660	–
TC-04	+0.459 / +0.184 a T1662	+0.459 / +0.183 a T1663	–	+1.803 / +2.030 a T1665	+1.802 / +2.029 a T1666	+1.800 / +2.027 a T1667	+1.800 / +2.027 a T1668	–
TC-09	+1.062 / +0.796 a T1670	+1.003 / +0.744 a T1671	–	+2.053 / +2.636 T1673	+1.548 / +2.034 T1674	+1.197 / +1.617 T1675	+1.151 / +1.563 T1676	–
NZ-01	**−0.520 / n/a a** T1678	**−0.489 / n/a a** T1679	–	+3.234 / n/a T1681	+2.134 / n/a a T1682	+1.504 / n/a a T1683	+1.476 / n/a r T1684	–
NZ-02	+0.595 / n/a a T1686	+0.596 / n/a a T1687	–	+4.107 / n/a a T1689	+2.967 / n/a a T1690	+2.265 / n/a a T1691	+2.178 / n/a a T1692	–
NZ-03	+0.445 / n/a r T1694	**−0.088 / n/a r** T1695	–	+1.615 / n/a r T1697	+1.032 / n/a r T1698	+0.672 / n/a r T1699	+0.628 / n/a r T1700	–
NZ-04	+3.584 / n/a a T1702	+2.897 / n/a a T1703	–	+6.534 / n/a a T1705	+5.282 / n/a a T1706	+4.415 / n/a a T1707	+4.301 / n/a a T1708	–
FD-01	+1.227 / +3.422 a T1710	+1.197 / +3.362 a T1711	–	+3.170 / +9.743 a T1713	+3.088 / +9.531 a T1714	+3.011 / +9.334 a T1715	+3.000 / +9.304 a T1716	–
GM-01	+3.352 / +2.750 a T1718	+1.439 / +1.102 a T1719	+2.551 / +1.720 a T1720	+0.767 / +1.030 a T1721	+0.373 / +0.578 a T1722	+0.130 / +0.299 a T1723	+0.100 / +0.264 a T1724	+0.777 / +1.042 a T1725
GM-02	+6.556 / +5.511 a T1726	+3.255 / +2.667 a T1727	+3.585 / +2.512 a T1728	+2.091 / +2.551 a T1729	+1.404 / +1.762 a T1730	+0.980 / +1.275 a T1731	+0.927 / +1.215 a T1732	+2.105 / +2.568 a T1733
GM-03	+13.611 / +10.787 a T1734	+6.254 / +4.852 a T1735	+8.795 / +6.024 a T1736	+3.585 / +3.932 a T1737	+2.564 / +2.833 a T1738	+1.934 / +2.156 a T1739	+1.856 / +2.072 a T1740	+1.353 / +1.531 a T1741
GM-08a	+61.651 / +49.544 a T1742	+32.135 / +25.732 a T1743	+32.080 / +22.722 a T1744	+21.291 / +22.978 a T1745	+16.337 / +17.649 a T1746	+13.281 / +14.362 a T1747	+12.900 / +13.952 a T1748	+10.455 / +11.322 a T1749
GM-08b	+56.638 / +45.500 a T1750	+28.473 / +22.777 a T1751	+35.973 / +25.514 a T1752	+18.126 / +19.573 a T1753	+13.877 / +15.003 a T1754	+11.256 / +12.183 a T1755	+10.929 / +11.832 a T1756	+8.839 / +9.584 a T1757
GM-09	+64.003 / +51.441 a T1758	+31.350 / +25.099 a T1759	+35.853 / +25.428 a T1760	+19.350 / +20.890 a T1761	+14.799 / +15.994 a T1762	+11.999 / +12.983 a T1763	+11.651 / +12.608 a T1764	+9.410 / +10.198 a T1765
GM-10	+89.997 / +72.412 T1766	+46.871 / +37.620 T1767	+22.208 / +15.643 T1768	+30.505 / +32.889 T1769	+23.597 / +25.458 T1770	+19.309 / +20.846 T1771	+18.773 / +20.270 T1772	+19.238 / +20.769 a T1773
GM-13	+67.053 / +53.901 a T1774	+34.896 / +27.959 a T1775	+15.399 / +10.760 a T1776	+22.696 / +24.490 a T1777	+17.500 / +18.900 a T1778	+14.274 / +15.430 a T1779	+13.872 / +14.997 a T1780	+22.434 / +24.208 a T1781
GM-14	>+99 / >+99 a T1782	>+99 / >+99 a T1783	+18.968 / +13.319 a T1784	>+99 / >+99 a T1785	>+99 / >+99 a T1786	>+99 / >+99 a T1787	>+99 / >+99 a T1788	>+99 / >+99 a T1789
GM-16a	+29.165 / +24.993 a T1790	+15.354 / +13.092 a T1791	+5.523 / +3.996 a T1792	+10.140 / +11.799 a T1793	+7.697 / +8.993 a T1794	+6.181 / +7.251 a T1795	+5.992 / +7.033 a T1796	+10.140 / +11.799 a T1797
GM-16b	>+99 / +89.126 a T1798	+89.402 / +76.900 a T1799	+8.123 / +5.988 a T1800	>+99 / >+99 a T1801	>+99 / >+99 a T1802	>+99 / >+99 a T1803	>+99 / >+99 a T1804	>+99 / >+99 a T1805
PD-01R	+1.886 / +1.338 r T1806	+1.862 / +1.318 r T1807	–	+4.444 / +4.883 r T1809	+4.207 / +4.628 r T1810	+2.827 / +3.136 T1811	+2.670 / +2.966 T1812	–
PD-01S	+1.590 / +1.097 r T1814	+1.590 / +1.097 r T1815	–	+4.273 / +4.699 r T1817	+4.008 / +4.412 r T1818	+2.850 / +3.161 r T1819	+2.698 / +2.997 r T1820	–
HW-101 (J-FWD)	–	–	–	–	–	–	–	–
HW-103 (J-FD01)	–	–	–	–	–	–	–	–
HW-127 (J-FD09)	–	–	–	–	–	–	–	–

[1] TC-01 FA1U: -0.083 / -0.012 r MF (gov. sub-model +0.005 / +0.239)

[2] TC-01 FA1S: -0.107 / -0.037 r MF (gov. sub-model +0.005 / +0.239)

Table 9.2: Landing, gimbal-stop and thermal cases (MS_y / MS_u; flags r, a, c, p, s, w as defined in the matrix text above; MF / BT classes; source MMXMD MMXMD FEA3D margin matrix v3.5.2).

Part	LC3 cold	LC3 hot bound	GS	TD	LC6 F2	F2 hot bound	LC6 soak
TC-01	+1.044 / +0.657 a T1846	+0.890 / +0.528 r T1847	+1.080 / +0.686 a T1848	+1.118 / +0.717 a T1849	+30.584 / +24.649 a T1850	+0.877 / +0.518 r T1851	**−0.356 / −0.482 r** T1852

Part	LC3 cold	LC3 hot bound	GS	TD	LC6 F2	F2 hot bound	LC6 soak
TC-02	+7.561 / +5.883 r T1853	+0.504 / +0.190 r T1854	+7.254 / +5.637 r T1855	+7.689 / +5.986 r T1856	>+99 / >+99 r T1857	+0.493 / +0.181 r T1858	+1.068 / +0.656 a T1859
TC-03	+25.346 / +22.957 a T1860	**−0.107 / −0.188 r** T1861	+25.274 / +22.892 a T1862	+25.529 / +23.124 a T1863	>+99 / >+99 a T1864	**−0.107 / −0.188 r** T1865	+0.992 / +0.811 r T1866
TC-04	>+99 / >+99 a T1867	**+0.228 / −0.004 as BT** T1868	>+99 / +92.532 a T1869	>+99 / >+99 a T1870	>+99 / >+99 a T1871	**+0.223 / −0.008 as BT** T1872	+6.477 / +5.017 a T1873
TC-09	+10.348 / +9.106 a T1874	+1.031 / +0.774 a T1875	+10.456 / +9.202 a T1876	+10.545 / +9.281 a T1877	>+99 / >+99 a T1878	+1.091 / +0.826 aw[1] T1879	+1.075 / +0.705 T1880
NZ-01	+4.804 / n/a r T1881	**−0.403 / n/a aw[2]** T1882	+4.860 / n/a r T1883	+4.866 / n/a r T1884	>+99 / n/a r T1885	**−0.401 / n/a** T1886	**−0.426 / n/a r** T1887
NZ-02	+1.559 / n/a a T1888	+0.781 / n/a a T1889	+1.581 / n/a a T1890	+1.581 / n/a a T1891	>+99 / n/a a T1892	+0.784 / n/a as T1893	>+99 / n/a a T1894
NZ-03	+0.179 / n/a r T1895	+0.243 / n/a r T1896	+0.187 / n/a r T1897	+0.190 / n/a r T1898	>+99 / n/a T1899	+0.252 / n/a r T1900	>+99 / n/a r T1901
NZ-04	+13.606 / n/a a T1902	+3.004 / n/a a T1903	+13.685 / n/a a T1904	+13.702 / n/a a T1905	>+99 / n/a T1906	+3.007 / n/a as T1907	+6.093 / n/a a T1908
FD-01	+33.700 / +66.045 a T1909	+0.837 / +2.620 aw[3] T1910	+35.220 / +68.982 a T1911	+32.090 / +62.934 a T1912	+85.170 / >+99 a T1913	+0.830 / +2.606 aw[4] T1914	**−0.187 / +0.780 a** T1915
GM-01	+0.534 / +0.322 T1916	+0.220 / +0.052 T1917	+0.648 / +0.420 T1918	+2.660 / +2.154 a T1919	+42.346 / +36.351 T1920	+0.289 / +0.111 T1921	+0.166 / +0.005 a T1922
GM-02	+0.400 / +0.206 T1923	+0.202 / +0.036 T1924	+0.591 / +0.371 T1925	+3.643 / +3.001 a T1926	+57.337 / +49.269 T1927	+0.337 / +0.152 T1928	+0.200 / +0.034 a T1929
GM-03	+1.227 / +0.796 a T1930	+1.225 / +0.795 a T1931	+1.893 / +1.334 a T1932	+4.793 / +3.674 a T1933	+11.646 / +9.202 a T1934	+1.881 / +1.324 a T1935	+13.300 / +10.536 a T1936
GM-08a	+5.528 / +4.266 a T1937	+5.523 / +4.262 a T1938	+12.101 / +9.569 a T1939	+15.110 / +11.997 a T1940	+37.240 / +29.850 a T1941	+12.079 / +9.551 a T1942	+37.360 / +29.947 a T1943
GM-08b	+10.694 / +8.434 a T1944	+10.667 / +8.412 a T1945	+10.525 / +8.298 a T1946	+22.858 / +18.247 a T1947	+49.715 / +39.914 a T1948	+10.499 / +8.277 a T1949	+49.877 / +40.045 a T1950
GM-09	+5.134 / +3.948 a T1951	+5.131 / +3.946 a T1952	+12.056 / +9.533 a T1953	+14.319 / +11.359 a T1954	+42.664 / +34.226 a T1955	+11.963 / +9.458 a T1956	+43.439 / +34.851 a T1957
GM-10	+3.708 / +2.798 T1958	+3.646 / +2.748 T1959	+3.618 / +2.726 T1960	+14.128 / +11.205 T1961	+76.758 / +61.732 a T1962	+3.596 / +2.708 T1963	+77.024 / +61.946 a T1964
GM-13	+2.418 / +1.757 a T1965	+2.375 / +1.723 a T1966	+2.578 / +1.887 a T1967	+9.510 / +7.479 a T1968	+61.967 / +49.799 a T1969	+2.531 / +1.849 a T1970	+76.436 / +61.471 a T1971
GM-14	+2.434 / +1.771 a T1972	+2.418 / +1.758 a T1973	+2.509 / +1.831 a T1974	+10.565 / +8.330 a T1975	+77.419 / +62.264 a T1976	+2.492 / +1.817 a T1977	>+99 / +81.872 a T1978
GM-16a	+0.372 / +0.183 a T1979	+0.369 / +0.179 a T1980	+0.568 / +0.351 a T1981	+3.197 / +2.616 a T1982	+29.491 / +25.274 a T1983	+0.548 / +0.334 a T1984	+47.292 / +40.613 a T1985
GM-16b	+0.450 / +0.250 a T1986	+0.449 / +0.248 a T1987	+0.531 / +0.319 a T1988	+4.299 / +3.566 a T1989	+45.191 / +38.802 a T1990	+0.524 / +0.313 a T1991	+71.675 / +61.624 a T1992
PD-01R	+4.237 / +3.243 T1993	+1.477 / +1.006 r T1994	+2.484 / +1.823 T1995	+3.956 / +3.015 T1996	+20.413 / +16.349 T1997	+0.981 / +0.604 r T1998	**−0.315 / −0.451 r** T1999
PD-01S	+5.864 / +4.562 T2000	+1.323 / +0.881 r T2001	+3.513 / +2.657 T2002	+5.515 / +4.279 T2003	+21.487 / +17.219 T2004	+0.971 / +0.596 r T2005	**−0.378 / −0.502 r** T2006
HW-101 (J-FWD)	–	–	–	–	–	–	–
HW-103 (J-FD01)	–	–	–	–	–	–	–
HW-127 (J-FD09)	–	–	–	–	–	–	–

[1] TC-09 F2 hot bound: +1.091 / +0.826 aw (part-wide; stored +1.092 / +0.827)

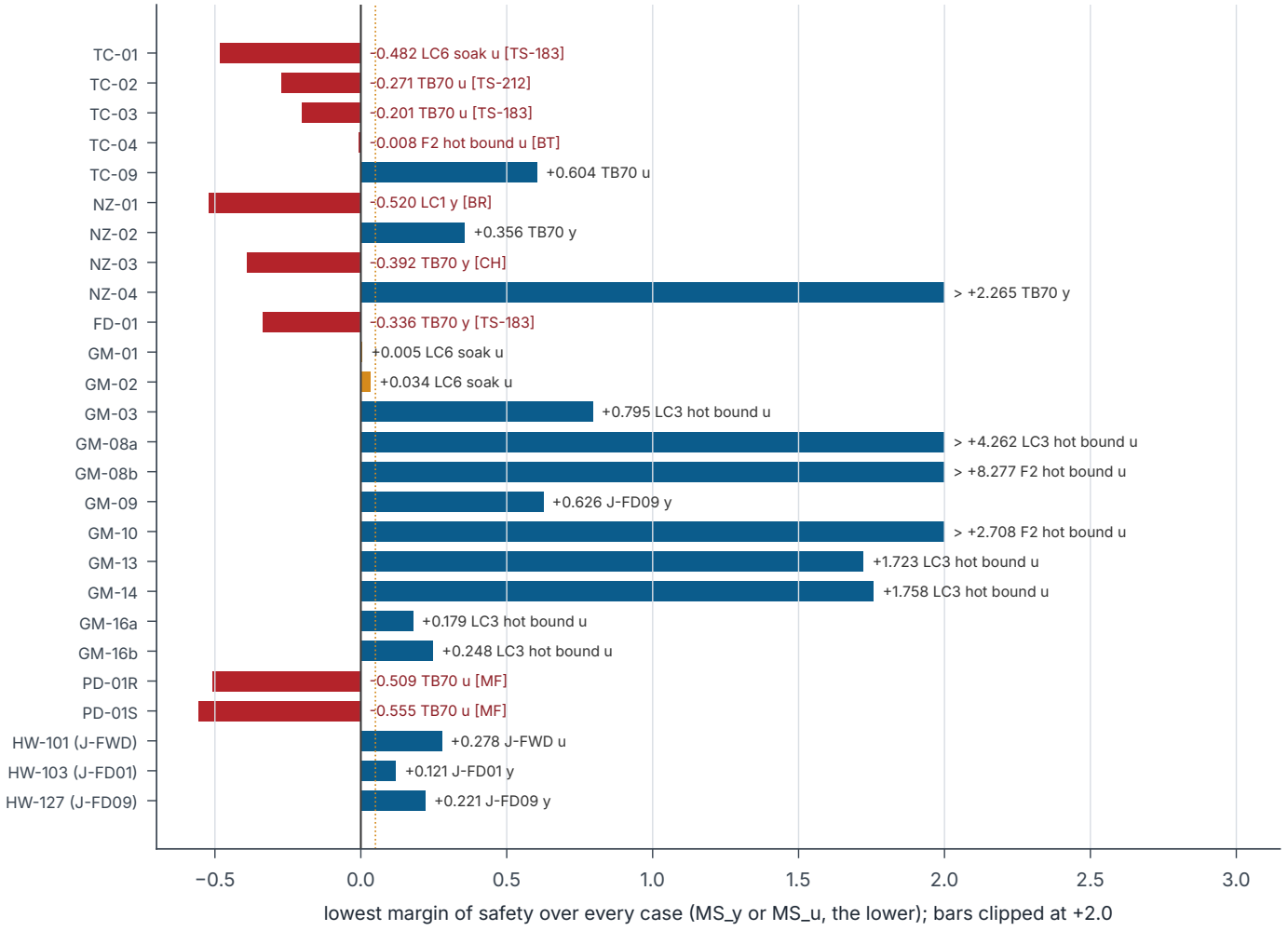
[2] NZ-01 LC3 hot bound: -0.403 / n/a aw (part-wide; stored -0.402 / n/a)

[3] FD-01 LC3 hot bound: +0.837 / +2.620 aw (part-wide; stored +0.848 / +2.640)

[4] FD-01 F2 hot bound: +0.830 / +2.606 aw (part-wide; stored +0.840 / +2.625)

Table 9.3: Thermal-abort states, joints and the part minimum (MS_y / MS_u; flags r, a, c, p, s, w as defined in the matrix text above; MF / BT classes; source MMXMD MMXMD FEA3D margin matrix v3.5.2).

Part	TB70	TBS	J-FWD	J-FD01	J-FD09	min
TC-01	**+0.120 / −0.110 r** T2028	+0.570 / +0.109 r T2029	**−0.075 / +0.239** T2030	–	–	**−0.482** (LC6 soak u) T2033
TC-02	**−0.236 / −0.271 r** T2034	**+0.138 / −0.035 r** T2035	+0.045 / +0.162 T2036	+0.404 / n/a T2037	–	**−0.271** (TB70 u) T2039
TC-03	**−0.121 / −0.201 r** T2040	+0.325 / +0.071 r T2041	–	–	–	**−0.201** (TB70 u) T2045
TC-04	+0.401 / +0.122 a T2046	+1.102 / +0.496 a T2047	–	–	–	**−0.008** (F2 hot bound u) T2051
TC-09	+0.861 / +0.604 a T2052	+1.763 / +1.117 a T2053	–	–	–	+0.604 (TB70 u) T2057
NZ-01	**−0.315 / n/a a** T2058	**−0.054 / n/a a** T2059	–	–	–	**−0.520** (LC1 y) T2063
NZ-02	+0.356 / n/a a T2064	+0.665 / n/a a T2065	–	–	–	+0.356 (TB70 y) T2069
NZ-03	**−0.392 / n/a r** T2070	**−0.253 / n/a r** T2071	–	–	–	**−0.392** (TB70 y) T2075
NZ-04	+2.265 / n/a a T2076	+3.195 / n/a a T2077	–	–	–	+2.265 (TB70 y) T2081
FD-01	**−0.336 / +0.484 a** T2082	**−0.005 / +0.976 a** T2083	–	–	–	**−0.336** (TB70 y) T2087
GM-01	+0.291 / +0.113 a T2088	+0.936 / +0.483 a T2089	–	–	–	+0.005 (LC6 soak u) T2093
GM-02	+0.241 / +0.069 a T2094	+0.861 / +0.425 a T2095	–	–	–	+0.034 (LC6 soak u) T2099
GM-03	+72.008 / +57.900 a T2100	>+99 / +71.545 a T2101	–	–	–	+0.795 (LC3 hot bound u) T2105
GM-08a	>+99 / >+99 a T2106	>+99 / >+99 a T2107	–	–	–	+4.262 (LC3 hot bound u) T2111
GM-08b	>+99 / >+99 a T2112	>+99 / >+99 a T2113	–	–	–	+8.277 (F2 hot bound u) T2117
GM-09	>+99 / >+99 a T2118	>+99 / >+99 a T2119	–	–	+0.626 / n/a T2122	+0.626 (J-FD09 y) T2123
GM-10	+32.616 / +26.119 T2124	+45.559 / +32.388 T2125	–	–	–	+2.708 (F2 hot bound u) T2129
GM-13	+24.422 / +19.509 a T2130	+34.211 / +24.250 a T2131	–	–	–	+1.723 (LC3 hot bound u) T2135
GM-14	>+99 / >+99 a T2136	>+99 / >+99 a T2137	–	–	–	+1.758 (LC3 hot bound u) T2141
GM-16a	+10.802 / +9.170 a T2142	+15.346 / +11.521 a T2143	–	–	–	+0.179 (LC3 hot bound u) T2147
GM-16b	>+99 / >+99 a T2148	>+99 / >+99 a T2149	–	–	–	+0.248 (LC3 hot bound u) T2153
PD-01R	**−0.387 / −0.509 r** T2154	**−0.081 / −0.346 r** T2155	–	–	–	**−0.509** (TB70 u) T2159
PD-01S	**−0.444 / −0.555 r** T2160	**−0.166 / −0.407 r** T2161	–	–	–	**−0.555** (TB70 u) T2165
HW-101 (J-FWD)	–	–	+0.650 / +0.278 T2168	–	–	+0.278 (J-FWD u) T2171
HW-103 (J-FD01)	–	–	–	+0.121 / +0.379 T2175	–	+0.121 (J-FD01 y) T2177
HW-127 (J-FD09)	–	–	–	–	+0.221 / +0.485 T2182	+0.221 (J-FD09 y) T2183



Lowest margin of safety per part over every load case (the matrix "min" field; red = negative, amber = below +0.05; class of the governing cell in brackets where the matrix gives one). Bars are clipped at +2.0. DERIVED from SIM / CALC cells. Data: T2184, T2185, T2186, T2187, T2188, T2189, T2190, T2191, T2192, T2193, T2194, T2195, T2196, T2197, T2198, T2199, T2200, T2201, T2202, T2203, T2204, T2205, T2206, T2207, T2208, T2209, T2210, T2211, T2212, T2213, T2214, T2215, T2216, T2217, T2218, T2219, T2220, T2221, T2222, T2223 and 12 more

9.2.4 Cells shown "–" and why

Column	Why
FA1T	**FA1T**: evaluated on the gimbal / mount parts only (post_f1 GIMBAL_PARTS: GM-01, GM-02, GM-03, GM-08a/b, GM-09, GM-10, GM-13, GM-14, GM-16a/b): the thrust spike counts only for parts with T <= 20 ms (lead 194); the pressure parts are bounded by FA1Q / D / U / S DERIVED T2236
LC2F	**LC2F**: evaluated on the gimbal / mount parts only (post_s3d GIMBAL_PARTS; the pressure parts are not at a defined Pc in this case), as published for v3.5.1 (234 (2)) DERIVED T2237
J-FWD	**J-FWD**: only TC-01, TC-02 and the HW-101 screw are in the J-FWD sub-model; the global TC-01 / TC-02 cells at the J-FWD connectors (z >= 97.5) defer to this column (NAMED rules) DERIVED T2238
J-FD01	**J-FD01**: F4 joint: the HW-103 / 103B screws and the TC-02 host thread only DERIVED T2239
J-FD09	**J-FD09**: F4 joint: the HW-127 screws and the GM-09 host thread only DERIVED T2240
joint rows	**joint rows**: HW-101 / HW-103 / HW-127 are connectors in the global model (no FE body): their rows sit in the joint columns only DERIVED T2241

9.3 Found negatives: every negative cell, its class and its closure

33 DERIVED T697 cells of the matrix are negative. By class (REV-MECH-2 section 10 classes; class rules of CHANGELOG 234 (4)): **BR 7** T698 · **BT 2** T699 · **CH 3** T700 · **DC 1** T701 · **MF 9** T702 · **TS-183 9** T703 · **TS-212 2** T704.

Every negative matrix cell (MMXMD, section "Negative cells and their classes"). The class decides the closure route; no class turns a negative into a pass. Owners: see §9.10.

Part	Column	Cell y / u	Governing record	Class	Closure (what closes it)	Ref
TC-01	FA1U SIM T495	**−0.083 / −0.012** T496	n9090, A1_22.2, regular , None, 45.7, -0.058, +0.833 SIM T497	MF T498	MF (n9090 regular, 1.0451 mm from the J-FWD RBE3 in the J-FWD region; the all-node minimum n614 -0.241 / -0.182 is the RBE3 itself, d 0.0): (a) global values kept as found; (b) governing = the J-FWD sub-model's TC-01 FA1S criterion rows, minimum +0.005 / … DERIVED T499	lead 21:15:29Z (3) under 234 (4); basis as the 239 TB70 ruling (lines 231 (3), 239) T500
TC-01	FA1S SIM T501	**−0.107 / −0.037** T502	n9090, A1_22.2, regular , None, 45.7, -0.083, +0.785 SIM T503	MF T504	MF (n9090 regular, 1.0451 mm from the J-FWD RBE3 in the J-FWD region; the all-node minimum n614 -0.261 / -0.203 is the RBE3 itself, d 0.0): (a) global values kept as found; (b) governing = the J-FWD sub-model's TC-01 FA1S criterion rows, minimum +0.005 / … DERIVED T505	lead 21:15:29Z (3) under 234 (4); basis as the 239 TB70 ruling (lines 231 (3), 239) T506
TC-01	LC6 soak SIM T507	**−0.356 / −0.482** T508	n13829, B1_70.7, regular , None, 159.2, +42.139, +0.932 SIM T509	TS-183 T510	thermal-secondary (provisional, lead 08:04Z); in the J-FWD region (z >= 97.5) the J-FWD sub-model governs: js5 soak P+Q <= 2 Fty(T) rows (J-FWD column) DERIVED T511	lead 07:51Z (d), 08:04Z, 08:40Z T512
TC-01	TB70 SIM T513	**+0.120 / −0.110** T514	n9090, TB, regular , None, 132.0, +0.353, +2.360 SIM T515	MF T516	n9090 1.05 mm from the J-FWD RBE3: connector artefact, closed by the J-FWD sub-model (FA1S / LC2 bound the primary) DERIVED T517	lead 08:58Z T518

Part	Column	Cell y / u	Governing record	Class	Closure (what closes it)	Ref
TC-01	J-FWD SIM T519	−0.075 / +0.239 T520	CALC T521	DC T522	TC-01 seat / ring line P+Q (secondary, shakedown, P+Q <= 2 Fty(T)) FOUND NEGATIVE on the worst computed instance of the FINAL envelope v6 (lead 02:01:34Z (E), 03:30:46Z (3), 05:17:36Z (3): sharp js5 / (a) / (a') / (a'') + (b2) R0.2 incl. its repeats + (b1) s9 … DERIVED T523	flags (0 to +0.05): FA1S (ii) hole bearing, CALC V 3888.1 N, L 2.365 (governing, 240 (5)) +0.005/+0.239 [CALC]; SOAK52 IN718 under-head band Neuber (local strain) [ASSUMED (sigma_el x A_ANN 47.12 / A_supp; edge loading at the tangential overhang not represented)] at 3750 N +0.039/n/a [CALC / DERIVED; ASSUMED scaling]. R0.2 bore-side break adopted (240 (5)); route (i) adopted (HW-170 disc spring under the head, lead 16:53Z R2; rev-5 CAD check accepted 17:55:55Z); J-F1 maximum capped at 3750 N (lead 17:56:12Z). (b1) at C1 is negative (SOAK52 -0.030), so the 240 (10) seat-line P+Q select-fit band APPLIES (IN718 [0.040, 0.0761] mm radial, as a narrower TC-02 spigot OD tolerance); it is re-sized ONCE by Mechanical on the final envelope from the FEA3D P+Q-vs-radial-gap sensitivity [DERIVED, existing runs, no new run]; if no band inside [0.040, 0.0761] closes it, the seat-line P+Q stays a found negative in the datasheet; the governing minimum on the final envelope is SOAK -0.071 ((b2) rep); owner Mechanical T524 [superseded: R2-B [0.040 T526, 0.0631 mm DECIDED Lead ruling R2-B (2026-10-21T07:37:34Z), quoted verbatim in Mechanical line 254 (1) T527], 254 (1) T525; band re-size done, band v1 · sha256/16 1ba3147b42edfeed DERIVED T528]
TC-02	TB70 SIM T529	−0.236 / −0.271 T530	n30266, TB, regular , None, 228.7, +1.267, +1.292 SIM T531	TS-212 T532	TC-02 under 212 (1): (a)-(c) pass; (d) time at temperature open with T-F1 (Mechanical's (b) package) DERIVED T533	REV-MECH-2 rows 2-4; lead 08:58Z, 09:15Z (3) T534
TC-02	TBS SIM T535	+0.138 / −0.035 T536	n30266, TB, regular , None, 228.7, +2.131, +1.276 SIM T537	TS-212 T538	TC-02 under 212 (1): (a)-(c) pass; (d) time at temperature open with T-F1 (Mechanical's (b) package) DERIVED T539	REV-MECH-2 rows 2-4; lead 08:58Z, 09:15Z (3) T540
TC-03	LC1 SIM T541	−0.088 / −0.170 T542	n46040, B1_1, regular , None, 89.6, +8.043, +1.737 SIM T543	TS-183 T544	thermal-secondary 183 (1): (a) primary and (b) P+Q pass; (c) LCF and (d) GlidCop time at temperature with Mechanical; MF on the value (gap-variant field, RM2-13) DERIVED T545	REV-MECH-2 rows 5-7; lead 07:51Z (a), 08:58Z T546
TC-03	LC2 SIM T547	−0.092 / −0.175 T548	n46040, B1_1, regular , None, 89.6, +7.684, +1.723 SIM T549	TS-183 T550	thermal-secondary 183 (1): (a) primary and (b) P+Q pass; (c) LCF and (d) GlidCop time at temperature with Mechanical; MF on the value (gap-variant field, RM2-13) DERIVED T551	REV-MECH-2 rows 5-7; lead 07:51Z (a), 08:58Z T552
TC-03	LC3 hot bound SIM T553	−0.107 / −0.188 T554	n38065, B1_1, regular , None, 89.6, n/a, n/a SIM T555	TS-183 T556	thermal-secondary 183 (1): (a) primary and (b) P+Q pass; (c) LCF and (d) GlidCop time at temperature with Mechanical; MF on the value (gap-variant field, RM2-13) DERIVED T557	REV-MECH-2 rows 5-7; lead 07:51Z (a), 08:58Z T558
TC-03	F2 hot bound SIM T559	−0.107 / −0.188 T560	n38065, B1_1, regular , None, 89.6, n/a, n/a SIM T561	TS-183 T562	thermal-secondary 183 (1): (a) primary and (b) P+Q pass; (c) LCF and (d) GlidCop time at temperature with Mechanical; MF on the value (gap-variant field, RM2-13) DERIVED T563	REV-MECH-2 rows 5-7; lead 07:51Z (a), 08:58Z T564
TC-03	TB70 SIM T565	−0.121 / −0.201 T566	n48254, TB, regular , None, 181.3, +6.297, +1.638 SIM T567	TS-183 T568	thermal-secondary 183 (1): (a) primary and (b) P+Q pass; (c) LCF and (d) GlidCop time at temperature with Mechanical; MF on the value (gap-variant field, RM2-13) DERIVED T569	REV-MECH-2 rows 5-7; lead 07:51Z (a), 08:58Z T570
TC-04	LC3 hot bound SIM T571	+0.228 / −0.004 T572	n42993, B1_19, BT bad_tets, 0.0, 30.0, n/a, n/a SIM T573	BT T574	found negative: DERIVED lower bound at a mesh-defect node (not an exception under 234 (4): a bad tet is a discretisation defect, refinement convergence not shown; no refinement or sub-model replaces it); n42993 sits on 2 TC-04 tets with q 0.040 / 0.027 (BT … DERIVED T575	lead 21:15:29Z (4) T576
TC-04	F2 hot bound SIM T577	+0.223 / −0.008 T578	n42993, B1_19, BT bad_tets, 0.0, 30.0, n/a, n/a SIM T579	BT T580	found negative: DERIVED lower bound at a mesh-defect node (not an exception under 234 (4): a bad tet is a discretisation defect, refinement convergence not shown; no refinement or sub-model replaces it); n42993 sits on 2 TC-04 tets with q 0.040 / 0.027 (BT … DERIVED T581	lead 21:15:29Z (4) T582
NZ-01	LC1 SIM T583	−0.520 / n/a T584	n73561, B1_19, RE reentrant_edges, 0.5, 1311.6, n/a, n/a SIM T585	BR T586	brittle and thermal: 183 (2) capture / containment shown + test basis, else design change DERIVED T587	REV-MECH-2 rows 8-10 T588
NZ-01	LC2 SIM T589	−0.489 / n/a T590	n74175, B1_1, BT bad_tets, 0.5389, 1332.4, n/a, n/a SIM T591	BR T592	brittle and thermal: 183 (2) capture / containment shown + test basis, else design change DERIVED T593	REV-MECH-2 rows 8-10 T594
NZ-01	LC3 hot bound SIM T595	−0.403 / n/a T596	part-wide bound (every node; hb_partwide_v352), stored minimum at: n51953, B1_1, regular , None, 610.6, n/a, n/a SIM T597	BR T598	brittle and thermal: 183 (2) capture / containment shown + test basis, else design change DERIVED T599	REV-MECH-2 rows 8-10 T600
NZ-01	F2 hot bound SIM T601	−0.401 / n/a T602	n51953, B1_1, regular , None, 610.6, n/a, n/a SIM T603	BR T604	brittle and thermal: 183 (2) capture / containment shown + test basis, else design change DERIVED T605	REV-MECH-2 rows 8-10 T606
NZ-01	LC6 soak SIM T607	−0.426 / n/a T608	n75257, B1_70.7, regular , None, 911.8, n/a, n/a SIM T609	BR T610	brittle and thermal: 183 (2) capture / containment shown + test basis, else design change DERIVED T611	REV-MECH-2 rows 8-10 T612
NZ-01	TB70 SIM T613	−0.315 / n/a T614	n75213, TB, RE reentrant_edges, 0.2789, 1374.0, n/a, n/a SIM T615	BR T616	brittle and thermal: 183 (2) capture / containment shown + test basis, else design change DERIVED T617	REV-MECH-2 rows 8-10 T618
NZ-01	TBS SIM T619	−0.054 / n/a T620	n75213, TB, RE reentrant_edges, 0.2789, 1374.0, n/a, n/a SIM T621	BR T622	brittle and thermal: 183 (2) capture / containment shown + test basis, else design change DERIVED T623	REV-MECH-2 rows 8-10 T624
NZ-03	LC2 SIM T625	−0.088 / n/a T626	n81357, B1_21.65, regular , None, 1294.1, n/a, n/a SIM T627	CH T628	char primary vs the ASSUMED 20 MPa: N-F1 (>= 21.96 MPa inside the 23.7 MPa acceptance); tie edge 188 (3) contact FE + N-F1 DERIVED T629	REV-MECH-2 LC2 table T630
NZ-03	TB70 SIM T631	−0.392 / n/a T632	n81357, TB, regular , None, 1359.5, n/a, n/a SIM T633	CH T634	char primary vs the ASSUMED 20 MPa: N-F1 (>= 21.96 MPa inside the 23.7 MPa acceptance); tie edge 188 (3) contact FE + N-F1 DERIVED T635	REV-MECH-2 LC2 table T636
NZ-03	TBS SIM T637	−0.253 / n/a T638	n81357, TB, regular , None, 1359.5, n/a, n/a SIM T639	CH T640	char primary vs the ASSUMED 20 MPa: N-F1 (>= 21.96 MPa inside the 23.7 MPa acceptance); tie edge 188 (3) contact FE + N-F1 DERIVED T641	REV-MECH-2 LC2 table T642
FD-01	LC6 soak SIM T643	−0.187 / +0.780 T644	n124943, B1_70.7, ID tie_patch_edges, 0.0, 99.2, +80.663, +1.440 SIM T645	TS-183 T646	thermal-secondary 183 (1), all-node value at the FD01_TC02_screws bonded-patch edge kept in the cell (234 (4): no named sub-model): (a) primary and (b) P+Q <= 2 Fty(T) pass; (c) LCF with Mechanical; (d) passes on the sourced long-time allowable where T <= 150 … DERIVED T647	lead 08:58Z, 09:47Z (6) T648

Part	Column	Cell y / u	Governing record	Class	Closure (what closes it)	Ref
FD-01	TB70 SIM T649	−0.336 / +0.484 T650	n124901, TB, ID tie_patch_edges, 0.0, 113.8, +1.939, +0.991 SIM T651	TS-183 T652	thermal-secondary 183 (1), all-node value at the FD01_TC02_screws bonded-patch edge kept in the cell (234 (4): no named sub-model): (a) primary and (b) P+Q <= 2 Fty(T) pass; (c) LCF with Mechanical; (d) passes on the sourced long-time allowable where T <= 150 ... DERIVED T653	lead 08:58Z, 09:47Z (6) T654
FD-01	TBS SIM T655	−0.005 / +0.976 T656	n124901, TB, ID tie_patch_edges, 0.0, 113.8, +3.427, +0.989 SIM T657	TS-183 T658	thermal-secondary 183 (1), all-node value at the FD01_TC02_screws bonded-patch edge kept in the cell (234 (4): no named sub-model): (a) primary and (b) P+Q <= 2 Fty(T) pass; (c) LCF with Mechanical; (d) passes on the sourced long-time allowable where T <= 150 ... DERIVED T659	lead 08:58Z, 09:47Z (6) T660
PD-01R	LC6 soak SIM T661	−0.315 / −0.451 T662	n115356, B1_70.7, regular , None, 96.4, +24.533, +1.055 SIM T663	MF T664	PD-02 rigid in the model (same alloy): the PD-01 / PD-02 mismatch is a model artefact; primary-only > +20 DERIVED T665	lead 07:51Z (e), 08:04Z T666
PD-01R	TB70 SIM T667	−0.387 / −0.509 T668	n115356, TB, regular , None, 104.1, +22.563, +0.840 SIM T669	MF T670	PD-02 rigid in the model (same alloy): the PD-01 / PD-02 mismatch is a model artefact; primary-only > +20 DERIVED T671	lead 07:51Z (e), 08:04Z T672
PD-01R	TBS SIM T673	−0.081 / −0.346 T674	n115356, TB, regular , None, 104.1, +22.266, +0.839 SIM T675	MF T676	PD-02 rigid in the model (same alloy): the PD-01 / PD-02 mismatch is a model artefact; primary-only > +20 DERIVED T677	lead 07:51Z (e), 08:04Z T678
PD-01S	LC6 soak SIM T679	−0.378 / −0.502 T680	n118875, B1_70.7, regular , None, 96.4, +26.014, +0.865 SIM T681	MF T682	PD-02 rigid in the model (same alloy): the PD-01 / PD-02 mismatch is a model artefact; primary-only > +20 DERIVED T683	lead 07:51Z (e), 08:04Z T684
PD-01S	TB70 SIM T685	−0.444 / −0.555 T686	n118875, TB, regular , None, 104.1, +23.533, +0.669 SIM T687	MF T688	PD-02 rigid in the model (same alloy): the PD-01 / PD-02 mismatch is a model artefact; primary-only > +20 DERIVED T689	lead 07:51Z (e), 08:04Z T690
PD-01S	TBS SIM T691	−0.166 / −0.407 T692	n118875, TB, regular , None, 104.1, +23.225, +0.669 SIM T693	MF T694	PD-02 rigid in the model (same alloy): the PD-01 / PD-02 mismatch is a model artefact; primary-only > +20 DERIVED T695	lead 07:51Z (e), 08:04Z T696

Class key (matrix class rules): **BT** bad-tet node (mesh defect; lead 21:15:29Z (4)); **MF** model fidelity (a named sub-model governs; lead 21:15:29Z (3)); **BR** brittle part (graphite), thermal; **DC** design change route (J-FWD seat line); **TS-183** thermal stress, 183 (1) (c) LCF / (d) time at temperature; **TS-212** thermal stress, 212 (1) (d) time at temperature; **CH** char layer against an ASSUMED strength. The key is a reading aid written for this datasheet; the class of each cell is the traced value in the table.

9.3.1 Negative cells in the information columns (information, not governing)

11 DERIVED T2276 more cells are negative in the information columns (TS2 (info), TB70 SF1.0 (info), LC1 merged v351 map (info)). They are shown as information: they do not govern and are not in the count above. **Flagged:** TC-01 TS2 (info) **−0.034** SIM T2242 has no class (UNCLASSIFIED; closure: to the lead).

margin_matrix_v352.json (rows v8), information columns /info_columns; cell values and classes as published.

Part	Column	Cell y / u	Class	Closure (quoted)
TC-01	TS2 (info)	−0.034 / n/a SIM T2242	UNCLASSIFIED DERIVED T2243	to the lead DERIVED T2244
TC-03	LC1 merged v351 map (info)	−0.332 / −0.393 SIM T2245 T2246	TS-183 DERIVED T2247	thermal-secondary 183 (1): (a) primary and (b) P+Q pass; (c) LCF and (d) GlidCop time at temperature with Mechanical; MF on the value (gap-variant field, RM2-13) DERIVED T2248
NZ-01	TS2 (info)	−0.257 / n/a SIM T2249	BR DERIVED T2250	brittle and thermal: 183 (2) capture / containment shown + test basis, else design change DERIVED T2251
NZ-01	LC1 merged v351 map (info)	−0.522 / n/a SIM T2252	BR DERIVED T2253	brittle and thermal: 183 (2) capture / containment shown + test basis, else design change DERIVED T2254
NZ-03	TB70 SF1.0 (info)	−0.087 / n/a SIM T2255	CH DERIVED T2256	char primary vs the ASSUMED 20 MPa: N-F1 (>= 21.96 MPa inside the 23.7 MPa acceptance); tie edge 188 (3) contact FE + N-F1 DERIVED T2257
FD-01	TS2 (info)	−0.022 / n/a SIM T2258	TS-183 DERIVED T2259	thermal-secondary 183 (1), all-node value at the FD01_TC02_screws bonded-patch edge kept in the cell (234 (4): no named sub-model): (a) primary and (b) P+Q <= 2 Fty(T) pass; (c) LCF with Mechanical; (d) passes on the ... DERIVED T2260
FD-01	TB70 SF1.0 (info)	−0.004 / n/a SIM T2261	TS-183 DERIVED T2262	thermal-secondary 183 (1), all-node value at the FD01_TC02_screws bonded-patch edge kept in the cell (234 (4): no named sub-model): (a) primary and (b) P+Q <= 2 Fty(T) pass; (c) LCF with Mechanical; (d) passes on the ... DERIVED T2263
PD-01R	TS2 (info)	−0.204 / n/a SIM T2264	MF DERIVED T2265	PD-02 rigid in the model (same alloy): the PD-01 / PD-02 mismatch is a model artefact; primary-only > +20 DERIVED T2266
PD-01R	TB70 SF1.0 (info)	−0.080 / n/a SIM T2267	MF DERIVED T2268	PD-02 rigid in the model (same alloy): the PD-01 / PD-02 mismatch is a model artefact; primary-only > +20 DERIVED T2269
PD-01S	TS2 (info)	−0.279 / n/a SIM T2270	MF DERIVED T2271	PD-02 rigid in the model (same alloy): the PD-01 / PD-02 mismatch is a model artefact; primary-only > +20 DERIVED T2272
PD-01S	TB70 SF1.0 (info)	−0.166 / n/a SIM T2273	MF DERIVED T2274	PD-02 rigid in the model (same alloy): the PD-01 / PD-02 mismatch is a model artefact; primary-only > +20 DERIVED T2275

9.4 Named findings stated plainly

Named findings of the brief, each quoted from its source. The TC-01 seat-line P+Q is a FOUND NEGATIVE at the drawn fit; the R2-B select fit closes it only in the CALC-on-SIM band model and only if the select-fit process is carried out.

Finding	What the sources say (quoted)	Owner / route
TC-04 hot-bound cells (class BT)	LC3 hot bound +0.228 / -0.004, F2 hot bound +0.223 / -0.008 [DERIVED node-wise lower bound]; regular nodes within 2 mm >= +0.643; a found negative in the datasheet DERIVED T2277	lead (ruled 2026-10-21T03:30:46Z (2): the settling route NOT RUN, final, for disk) DECIDED *Lead decision, per REPORT-FEA3D-v352.md FINAL handback of the v3.5.2 re-check* T2278
TC-01 J-FWD seat-line P+Q (class DC)	Matrix cell (drawn fit): **-0.075 / +0.239** T2030 R2-B select fit and the band (Mechanical, CALC on SIM): ENG-TC-02 PLUG OD, J-FWD SEAT-LINE BAND (lead 2026-10-21T05:50:45Z R2, restated to the lead ruling R2-B; jfwd_pq_band_v352_v1.json on relax v8, the final anchors s6 + C1; the TC-01 bore 57.25 H8, 57.250-57.296): LEAD RULING R2-B (2026-10-21T07:37:34Z): the ENG-TC-02 plug OD is a SELECT FIT to its own measured TC-01 bore, flag-free band: radial gap g in [0.040, 0.0631] mm, i.e. D_meas - 0.1262 <= OD <= D_meas - 0.080 mm (window 0.0462 mm on the diameter, less 2U, U the expanded (k=2) uncertainty of the bore and OD measurements, TO-DEFINE); bore and OD measured at 20 +/- 1 C and recorded with the paired TC-01 / TC-02 serials; the TC-02 OD is finished last, from a blank with finishing stock, to the measured bore. Basis: jfwd_pq_band_v352_v1.json 1ba3147b42edfeed [CALC on SIM]: seat-line MS >= +0.05 at 3750 N at the band edge, and positive in the ASSUMED +4 K stand T0 24 C information case (MS 0 there at g 0.0638). Option A (drawn 57.20 -0.030/-0.040, MS +0.005, FLAG) is not adopted (information only). The parts / BOM / hardware data strings stay unedited in v3.5.2 and are superseded by this drawing note. The seat-line P+Q stays NOT shown mesh-converged (R4 MF note); this is CALC on SIM, not physical approval. At the ruled edge g 0.0631 mm [CALC on SIM, jfwd_pq_band_v352_v1.json]: seat-line MS +0.0500 (k_b hi) / +0.0534 (k_b lo) at 3750 N (SOAK); 4096 N information +0.0479 / +0.0513; ASSUMED +4 K information (stand T0 24 C, k_b hi, 3750 N) +0.0067 (SOAK). INFORMATION: MS 0 at g 0.06862 mm (k_b hi) / 0.06892 mm (k_b lo) at 3750 N. The drawn 57.20 f7 (radial 0.040-0.078) leaves the seat line a FOUND NEGATIVE (rows v8 at 3750 N: SOAK -0.075 (b2) rep, SOAK70 -0.059, SOAK52 ... CALC T2279 [a build procedure: printed in full in §12.3 (R2-B select fit)]	Founder / shop: select-fit process with U TO-DEFINE (R2-B, CHANGELOG 254 (1)); Mechanical / FEA3D: convergence (not shown)
J-FWD screw LCF at the thread root (C6 OPEN, single-use screws)	FOUND NEGATIVE (lead 14:14Z): quoted as negative, never as passing DECIDED *Lead decision, per jfwd_screw_lcf_v352_v3.json v3* T2282 Kf 3.0, per firing: N_allow **5.6** CALC T2284, MS **−0.888** CALC T2283; assembly from zero: N_allow **0.6** CALC T2286, MS **−0.988** CALC T2285 (1) S01 and S02 are replaced after any FA-1-class event, together with the line-231 J-FWD inspection (lead 10:10Z rule). (2) Line-231 post-FA-1 inspection list: the ten J-FWD screws and their HW-170 disc springs replaced (new, single-use, (3)); the ten holes gauged for elongation and bearing set with the HW-101 pins GO 5.50 / NO-GO 5.62 (C1 (g)); a hole that takes the 5.62 pin is rejected; hole edges and the bore within 3 mm of a hole at 10x for nicks or raised metal; S01 / S02 replaced; the removed screws inspected per (3). (3) J-FWD screw life: SINGLE-USE founder rule until C6 closes (lead 16:53Z R3): the J-FWD screws are new at every assembly, never reused or re-torqued; removed screws are kept per hole and inspected at 10x plus dye penetrant at the thread root AND the head-to-shank fillet (LEAD ADDITION 18:36:37Z), as data. C6 is OPEN: the screw LCF at the thread root is a FOUND NEGATIVE (183 (1)(c); jfwd_screw_lcf_v352_v3.json on rows v8, SOAK (b1) as solved, Kf 3.0: per firing N_allow 5.6 (MS -0.888), assembly from zero N_allow 0.6 (MS -0.988); the Kf 2.2 sensitivity row per firing N_allow 26.9 (MS -0.462; v2 +0.360: SIGN CHANGE in v3); with (b1)'s moment x 1.4764 (R5, information) per firing N_allow 1.3 (MS -0.974)); the head-to-shank fillet from zero is a FOUND NEGATIVE that does not govern (jfwd_head_fillet_v352_v2.json: N_allow 29.3 (MS -0.414); per firing 1092.0 (MS +20.840); Kt LITERATURE, secondary via AMESweb citing Pilkey 2005, the Pilkey page and the T-head Kt TO-SOURCE; the no-spring head-end resultants carried over ASSUMED); the thread root governs. Both are quoted as negative, never as passing. Closure paths 2 (FE levers: (b1) is solved and entered, rows v8 under C1 / R5, and the row stays negative; the P+Q select-fit band, C9 (6), is sized for the seat line and takes no screw credit) and 3 (qualification test, TO-DEFINE, founder) remain; path 1 (MJ thread) closed NOT SOURCED. HW-170 disc spring SINGLE-USE (lead 19:53:27Z, CHANGELOG 248 (7)): a new spring with each new screw at every J-FWD assembly; removed springs are scrapped (C9 (2)). (4) POST-ABORT RULE (founder rule, lead 248 (7)): after any case-TC or TB-state abort the J-FWD joint counts as used: the ten ENG-HW-101-IN718 screws and their HW-170 springs are replaced (new, single-use) ... CALC T2280 [a founder action: printed in full in §12.3 (founder rule C6, mechanical.json /jfwd_v352/founder_rules/C6, the same text)]	Mechanical (closure path 2) / founder (path 3, qualification test TO-DEFINE); until then every J-FWD screw is single-use

Finding	What the sources say (quoted)	Owner / route
J-FWD screw head-to-shank fillet from zero	At the J-F1 cap 3750 N the fillet root (d 5.120, A 20.59 mm2, Z 13.18 mm3; head-end FE moment, IN718-scaled) carries a nominal fibre of 743.9 MPa at SOAK70|(cD) (thread root of the same case 1274.8 MPa): static net-section MS_y +0.339, MS_u +0.587 (the thread-root screw-fibre MS_y is -0.343, rows v8). LCF on the quoted stepped-bar Kt (r 0.2: Kt_t 2.583, Kt_b 2.459; Kf = Kt): (i) per firing N_allow 1092.0 (MS +20.840) against the thread root 5.6 (MS -0.888); (ii) assembly from zero N_allow 29.3 (MS -0.414) against 0.6 (MS -0.988). The fillet does NOT govern over the thread root: it would govern only at Kt x 2.61 ((i); Kt_t / Kt_b [6.742, 6.417]) or x 2.327 ((ii)) of the quoted values; it reaches N_allow 50 at Kt x 1.725 ((i)) / x 0.917 ((ii)). Kt: stepped-bar fit, web-sourced (AMESweb citing Pilkey 2005), page TO-SOURCE; the bolt-head (T-head) loading is NOT SOURCED. CALC T2281	Mechanical: Kt page and T-head Kt TO-SOURCE
HW-101 IN718 screw, J-FWD cell (positive; rows v8)	+0.650 / +0.278 CALC T2168	Mechanical (rows v8, R6 basis)

TC-04 negative cells as listed in §9.3: LC3 hot bound +0.228 / -0.004, F2 hot bound +0.223 / -0.008.

9.5 J-FWD joint rows (TC-01 / TC-02 / HW-101 screw)

J-FWD column rows (cell = minimum over the criterion rows; information rows listed after them) DERIVED T2287. The criterion rows set the matrix cell; the hand-form LC2N / LC2M rows are shown beside them as information (lead 03:30:46Z, CHANGELOG 250 (2)); every other information row follows in a third table. Values are from the matrix JSON (MMX), row by row.

9.5.1 TC-01: J-FWD cell −0.075 / +0.239

TC-01: criterion rows (54); cell **−0.075–0.239** T2288.

State	Row	MS_y	MS_u	Preload	Tag	Criterion / note
FA1S T2289	(i) ligament Pm+Pb, half-ligament average SIM T2290	+0.171 T2291	+0.263 T2292	–	SIM / DERIVED SIM	231 (1)(i) T2293
FA1S T2294	(i) ligament Pm, half-ligament average SIM T2295	+1.268 T2296	+1.446 T2297	–	SIM / DERIVED SIM	231 (1)(i) T2298
FA1S T2299	(i) shear-out flank+ Pm SIM T2300	+0.991 T2301	+1.147 T2302	–	SIM / DERIVED SIM	231 (1)(i) T2303
FA1S T2304	(i) shear-out flank+ Pm+Pb SIM T2305	+0.809 T2306	+0.951 T2307	–	SIM / DERIVED SIM	231 (1)(i) T2308
FA1S T2309	(i) shear-out flank- Pm SIM T2310	+0.952 T2311	+1.105 T2312	–	SIM / DERIVED SIM	231 (1)(i) T2313
FA1S T2314	(i) shear-out flank- Pm+Pb SIM T2315	+0.769 T2316	+0.907 T2317	–	SIM / DERIVED SIM	231 (1)(i) T2318
FA1S T2319	(ii) hole bearing, exact V 3603.6 N on t 2.688 (sharp) SIM T2320	+0.233 T2321	+0.519 T2322	–	SIM / DERIVED SIM	231 (1)(ii) T2323
FA1S T2324	(iii) band Neuber n30 (local strain) SIM T2325	+0.518 T2326	–	–	SIM / DERIVED SIM	231 (1)(iii) T2328
FA1S T2329	(ii) hole bearing, CALC V 3888.1 N, L 2.365 (governing, 240 (5)) CALC T2330	+0.005 T2331	+0.239 T2332	–	CALC CALC	231 (1)(ii) T2333
FA1S T2334	(i) shear-out CALC T2335	+0.727 T2336	+0.937 T2337	–	CALC CALC	231 (1)(i) T2338
FA1S T2339	(i) net section Pm CALC T2340	+1.483 T2341	+1.678 T2342	–	CALC CALC	231 (1)(i) T2343
FA1S T2344	(i) ligament Pm+Pb CALC T2345	+0.161 T2346	+0.252 T2347	–	CALC CALC	231 (1)(i) T2348
SOAK70 T2349	band Neuber n30 (under-head band, local strain) SIM T2350	+0.788 T2351	–	–	SIM / DERIVED SIM	lead 06:36Z (DC head seat) T2353
SOAK70 T2354	54 seat lines P+Q <= 2 Fty(T), envelope minimum (line d+6.051 z'+0.0, (b2); A4-80 FE) SIM T2355	**−0.055** T2356	–	–	SIM / DERIVED SIM	240 (9)(i) T2358
SOAK T2359	band Neuber n30 (under-head band, local strain) SIM T2360	+0.832 T2361	–	–	SIM / DERIVED SIM	lead 06:36Z (DC head seat) T2363
SOAK T2364	54 seat lines P+Q <= 2 Fty(T), envelope minimum (line d+6.051 z'+0.0, (b2) rep; A4-80 FE) SIM T2365	**−0.071** T2366	–	–	SIM / DERIVED SIM	240 (9)(i) T2368
SOAK52 T2369	band Neuber n30 (under-head band, local strain) SIM T2370	+0.479 T2371	–	–	SIM / DERIVED SIM	lead 06:36Z (DC head seat) T2373
SOAK52 T2374	54 seat lines P+Q <= 2 Fty(T), envelope minimum (line d+6.051 z'+0.0, (b1); A4-80 FE) SIM T2375	**−0.030** T2376	–	–	SIM / DERIVED SIM	240 (9)(i) T2378
FA1U T2379	js5b FA1U (FA1, SF [1.0, 1.0]): (i) ligament Pm+Pb, half-ligament average SIM T2380	+0.222 T2381	+0.318 T2382	–	SIM / DERIVED SIM	231 (1)(i) T2383
FA1U T2384	js5b FA1U (FA1, SF [1.0, 1.0]): (i) ligament Pm, half-ligament average SIM T2385	+1.319 T2386	+1.501 T2387	–	SIM / DERIVED SIM	231 (1)(i) T2388
FA1U T2389	js5b FA1U (FA1, SF [1.0, 1.0]): (i) shear-out flank+ Pm+Pb SIM T2390	+0.769 T2391	+0.908 T2392	–	SIM / DERIVED SIM	231 (1)(i) T2393
FA1U T2394	js5b FA1U (FA1, SF [1.0, 1.0]): (i) shear-out flank- Pm+Pb SIM T2395	+0.733 T2396	+0.869 T2397	–	SIM / DERIVED SIM	231 (1)(i) T2398
FA1U T2399	js5b FA1U (FA1, SF [1.0, 1.0]): (ii) hole bearing, exact V 3562.7 N on t 2.688 SIM T2400	+0.247 T2401	+0.537 T2402	–	SIM / DERIVED SIM	231 (1)(ii) T2403
FA1U T2404	js5b FA1U (FA1, SF [1.0, 1.0]): (iii) band Neuber n30 (local strain) SIM T2405	+0.403 T2406	–	–	SIM / DERIVED SIM	231 (1)(iii) T2408
FA1D T2409	js5b FA1D (FA1, SF [1.0, 1.0]): (i) ligament Pm+Pb, half-ligament average SIM T2410	+0.257 T2411	+0.355 T2412	–	SIM / DERIVED SIM	231 (1)(i) T2413
FA1D T2414	js5b FA1D (FA1, SF [1.0, 1.0]): (i) ligament Pm, half-ligament average SIM T2415	+1.641 T2416	+1.848 T2417	–	SIM / DERIVED SIM	231 (1)(i) T2418
FA1D T2419	js5b FA1D (FA1, SF [1.0, 1.0]): (i) shear-out flank+ Pm+Pb SIM T2420	+1.121 T2421	+1.288 T2422	–	SIM / DERIVED SIM	231 (1)(i) T2423
FA1D T2424	js5b FA1D (FA1, SF [1.0, 1.0]): (i) shear-out flank- Pm+Pb SIM T2425	+1.080 T2426	+1.244 T2427	–	SIM / DERIVED SIM	231 (1)(i) T2428
FA1D T2429	js5b FA1D (FA1, SF [1.0, 1.0]): (ii) hole bearing, exact V 2825.1 N on t 2.688 SIM T2430	+0.573 T2431	+0.938 T2432	–	SIM / DERIVED SIM	231 (1)(ii) T2433
FA1D T2434	js5b FA1D (FA1, SF [1.0, 1.0]): (iii) band Neuber n30 (local strain) SIM T2435	+1.231 T2436	–	–	SIM / DERIVED SIM	231 (1)(iii) T2438
LC2U T2439	js5b LC2U (FE-factored, SF [1.0, 1.0]): (i) ligament Pm+Pb, half-ligament average SIM T2440	–	+0.424 T2442	–	SIM / DERIVED SIM	231 (1)(i) T2443 FE-factored: the x1.5 / x2.0 state solved with contact at the 3750 N maximum preload (J-F1 cap), SF 1.0; LC2Y = yield slot, LC2U = ... T2444
LC2U T2445	js5b LC2U (FE-factored, SF [1.0, 1.0]): (i) ligament Pm, half-ligament average SIM T2446	–	+2.277 T2448	–	SIM / DERIVED SIM	231 (1)(i) T2449 FE-factored: the x1.5 / x2.0 state solved with contact at the 3750 N maximum preload (J-F1 cap), SF 1.0; LC2Y = yield slot, LC2U = ... T2450
LC2U T2451	js5b LC2U (FE-factored, SF [1.0, 1.0]): (i) shear-out flank+ Pm+Pb SIM T2452	–	+1.544 T2454	–	SIM / DERIVED SIM	231 (1)(i) T2455 FE-factored: the x1.5 / x2.0 state solved with contact at the 3750 N maximum preload (J-F1 cap), SF 1.0; LC2Y = yield slot, LC2U = ... T2456
LC2U T2457	js5b LC2U (FE-factored, SF [1.0, 1.0]): (i) shear-out flank- Pm+Pb SIM T2458	–	+1.512 T2460	–	SIM / DERIVED SIM	231 (1)(i) T2461 FE-factored: the x1.5 / x2.0 state solved with contact at the 3750 N maximum preload (J-F1 cap), SF 1.0; LC2Y = yield slot, LC2U = ... T2462
LC2U T2463	js5b LC2U (FE-factored, SF [1.0, 1.0]): (ii) hole bearing, exact V 2044.7 N on t 2.688 SIM T2464	–	+1.605 T2466	–	SIM / DERIVED SIM	231 (1)(ii) T2467 FE-factored: the x1.5 / x2.0 state solved with contact at the 3750 N maximum preload (J-F1 cap), SF 1.0; LC2Y = yield slot, LC2U = ... T2468
LC2Y T2469	js5b LC2Y (FE-factored, SF [1.0, 1.0]): (i) ligament Pm+Pb, half-ligament average SIM T2470	+0.421 T2471	–	–	SIM / DERIVED SIM	231 (1)(i) T2473 FE-factored: the x1.5 / x2.0 state solved with contact at the 3750 N maximum preload (J-F1 cap), SF 1.0; LC2Y = yield slot, LC2U = ... T2474
LC2Y T2475	js5b LC2Y (FE-factored, SF [1.0, 1.0]): (i) ligament Pm, half-ligament average SIM T2476	+2.539 T2477	–	–	SIM / DERIVED SIM	231 (1)(i) T2479 FE-factored: the x1.5 / x2.0 state solved with contact at the 3750 N maximum preload (J-F1 cap), SF 1.0; LC2Y = yield slot, LC2U = ... T2480
LC2Y T2481	js5b LC2Y (FE-factored, SF [1.0, 1.0]): (i) shear-out flank+ Pm+Pb SIM T2482	+1.631 T2483	–	–	SIM / DERIVED SIM	231 (1)(i) T2485 FE-factored: the x1.5 / x2.0 state solved with contact at the 3750 N maximum preload (J-F1 cap), SF 1.0; LC2Y = yield slot, LC2U = ... T2486
LC2Y T2487	js5b LC2Y (FE-factored, SF [1.0, 1.0]): (i) shear-out flank- Pm+Pb SIM T2488	+1.615 T2489	–	–	SIM / DERIVED SIM	231 (1)(i) T2491 FE-factored: the x1.5 / x2.0 state solved with contact at the 3750 N maximum preload (J-F1 cap), SF 1.0; LC2Y = yield slot, LC2U = ... T2492
LC2Y T2493	js5b LC2Y (FE-factored, SF [1.0, 1.0]): (ii) hole bearing, exact V 1514.6 N on t 2.688 SIM T2494	+1.871 T2495	–	–	SIM / DERIVED SIM	231 (1)(ii) T2497 FE-factored: the x1.5 / x2.0 state solved with contact at the 3750 N maximum preload (J-F1 cap), SF 1.0; LC2Y = yield slot, LC2U = ... T2498
LC2Y T2499	js5b LC2Y (FE-factored, SF [1.0, 1.0]): (iii) band Neuber n30 (local strain) SIM T2500	+3.070 T2501	–	–	SIM / DERIVED SIM	231 (1)(iii) T2503 FE-factored: the x1.5 / x2.0 state solved with contact at the 3750 N maximum preload (J-F1 cap), SF 1.0; LC2Y = yield slot, LC2U = ... T2504
SOAK70 T2505	IN718 flat bearing vs Fbry on the route (i) spring footprint (A 46.06 mm2, registered stack) CALC T2506	+0.355 T2507	–	3750 T2509	CALC / DERIVED CALC	no-spring records (information): union 36.401 mm2 0.071, A_ROW 38.171 mm2 0.123 T2510
SOAK70 T2511	IN718 under-head band Neuber (local strain) [ASSUMED (sigma_el x A_ANN 47.12 / A_supp; edge loading at the tangential overhang not represented)] CALC T2512	+0.092 T2513	–	3750 T2515	CALC / DERIVED; ASSUMED scaling CALC	(b2); full_float_0.557 (lead 17:55:55Z F1), mean-pressure scaling ASSUMED T2516

State	Row	MS_y	MS_u	Preload	Tag	Criterion / note
SOAK70 T2517	IN718 seat line d6.051 z'0 P+Q <= 2 Fty(T), F_sh form CALC T2518	−0.059 T2519	–	3750 T2521	CALC / DERIVED CALC	240 (9)(i), (10) T2522 (b2) T2523
SOAK70 T2524	IN718 ring line P+Q <= 2 Fty(T), F_sh form CALC T2525	−0.059 T2526	–	3750 T2528	CALC / DERIVED CALC	240 (9)(i), (10) T2529 (b2) T2530
SOAK T2531	IN718 flat bearing vs Fbry on the route (i) spring footprint (A 46.06 mm2, registered stack) CALC T2532	+0.354 T2533	–	3750 T2535	CALC / DERIVED CALC	no-spring records (information): union 36.401 mm2 0.07, A_ROW 38.171 mm2 0.122 T2536
SOAK T2537	IN718 under-head band Neuber (local strain) [ASSUMED (sigma_el x A_ANN 47.12 / A_supp; edge loading at the tangential overhang not represented)] CALC T2538	+0.191 T2539	–	3750 T2541	CALC / DERIVED; ASSUMED scaling CALC	(b1); full_float_0.557 (lead 17:55:55Z F1), mean-pressure scaling ASSUMED T2542
SOAK T2543	IN718 seat line d6.051 z'0 P+Q <= 2 Fty(T), F_sh form CALC T2544	−0.075 T2545	–	3750 T2547	CALC / DERIVED CALC	240 (9)(i), (10) T2548 (b2) rep T2549
SOAK T2550	IN718 ring line P+Q <= 2 Fty(T), F_sh form CALC T2551	−0.075 T2552	–	3750 T2554	CALC / DERIVED CALC	240 (9)(i), (10) T2555 (b2) rep T2556
SOAK52 T2557	IN718 flat bearing vs Fbry on the route (i) spring footprint (A 46.06 mm2, registered stack) CALC T2558	+0.373 T2559	–	3750 T2561	CALC / DERIVED CALC	no-spring records (information): union 36.401 mm2 0.085, A_ROW 38.171 mm2 0.138 T2562
SOAK52 T2563	IN718 under-head band Neuber (local strain) [ASSUMED (sigma_el x A_ANN 47.12 / A_supp; edge loading at the tangential overhang not represented)] CALC T2564	+0.039 T2565	–	3750 T2567	CALC / DERIVED; ASSUMED scaling CALC	(b2); full_float_0.557 (lead 17:55:55Z F1), mean-pressure scaling ASSUMED T2568
SOAK52 T2569	IN718 seat line d6.051 z'0 P+Q <= 2 Fty(T), F_sh form CALC T2570	−0.033 T2571	–	3750 T2573	CALC / DERIVED CALC	240 (9)(i), (10) T2574 (b1) T2575
SOAK52 T2576	IN718 ring line P+Q <= 2 Fty(T), F_sh form CALC T2577	−0.033 T2578	–	3750 T2580	CALC / DERIVED CALC	240 (9)(i), (10) T2581 (b1) T2582
FA1S T2583	IN718 under-head band Neuber (local strain) [ASSUMED (sigma_el x A_ANN 47.12 / A_supp; edge loading at the tangential overhang not represented)] CALC T2584	+0.259 T2585	–	3750 T2587	CALC / DERIVED; ASSUMED scaling CALC	(b2); full_float_0.557 (lead 17:55:55Z F1), mean-pressure scaling ASSUMED T2588

TC-01: hand-form LC2N / LC2M rows, INFORMATION beside the criterion (rule (A), lead 03:30:46Z) (12).

State	Row	MS_y	MS_u	Preload	Tag	Criterion / note
LC2N T2589	js5b LC2N (LC2, SF [1.5, 2.0]): (i) ligament Pm+Pb, half-ligament average SIM T2590	+0.016 T2591	−0.183 T2592	–	SIM / DERIVED SIM	231 (1)(i) T2593 hand form: SF on the total stress including preload, linear superposition ASSUMED (information; lead 2026-10-21T03:30:46Z (1)) T2594
LC2N T2595	js5b LC2N (LC2, SF [1.5, 2.0]): (i) ligament Pm, half-ligament average SIM T2596	+1.751 T2597	+1.212 T2598	–	SIM / DERIVED SIM	231 (1)(i) T2599 hand form: SF on the total stress including preload, linear superposition ASSUMED (information; lead 2026-10-21T03:30:46Z (1)) T2600
LC2N T2601	js5b LC2N (LC2, SF [1.5, 2.0]): (i) shear-out flank+ Pm+Pb SIM T2602	+0.944 T2603	+0.563 T2604	–	SIM / DERIVED SIM	231 (1)(i) T2605 hand form: SF on the total stress including preload, linear superposition ASSUMED (information; lead 2026-10-21T03:30:46Z (1)) T2606
LC2N T2607	js5b LC2N (LC2, SF [1.5, 2.0]): (i) shear-out flank- Pm+Pb SIM T2608	+0.948 T2609	+0.566 T2610	–	SIM / DERIVED SIM	231 (1)(i) T2611 hand form: SF on the total stress including preload, linear superposition ASSUMED (information; lead 2026-10-21T03:30:46Z (1)) T2612
LC2N T2613	js5b LC2N (LC2, SF [1.5, 2.0]): (ii) hole bearing, exact V 988.2 N on t 2.688 SIM T2614	+1.933 T2615	+1.695 T2616	–	SIM / DERIVED SIM	231 (1)(ii) T2617 hand form: SF on the total stress including preload, linear superposition ASSUMED (information; lead 2026-10-21T03:30:46Z (1)) T2618
LC2N T2619	js5b LC2N (LC2, SF [1.5, 2.0]): (iii) band Neuber n30 (local strain) SIM T2620	+5.146 T2621	–	–	SIM / DERIVED SIM	231 (1)(iii) T2623 hand form: SF on the total stress including preload, linear superposition ASSUMED (information; lead 2026-10-21T03:30:46Z (1)) T2624
LC2M T2625	js5b LC2M (LC2, SF [1.5, 2.0]): (i) ligament Pm+Pb, half-ligament average SIM T2626	+0.266 T2627	+0.021 T2628	–	SIM / DERIVED SIM	231 (1)(i) T2629 hand form: SF on the total stress including preload, linear superposition ASSUMED (information; lead 2026-10-21T03:30:46Z (1)) T2630
LC2M T2631	js5b LC2M (LC2, SF [1.5, 2.0]): (i) ligament Pm, half-ligament average SIM T2632	+2.220 T2633	+1.599 T2634	–	SIM / DERIVED SIM	231 (1)(i) T2635 hand form: SF on the total stress including preload, linear superposition ASSUMED (information; lead 2026-10-21T03:30:46Z (1)) T2636
LC2M T2637	js5b LC2M (LC2, SF [1.5, 2.0]): (i) shear-out flank+ Pm+Pb SIM T2638	+1.279 T2639	+0.840 T2640	–	SIM / DERIVED SIM	231 (1)(i) T2641 hand form: SF on the total stress including preload, linear superposition ASSUMED (information; lead 2026-10-21T03:30:46Z (1)) T2642
LC2M T2643	js5b LC2M (LC2, SF [1.5, 2.0]): (i) shear-out flank- Pm+Pb SIM T2644	+1.276 T2645	+0.837 T2646	–	SIM / DERIVED SIM	231 (1)(i) T2647 hand form: SF on the total stress including preload, linear superposition ASSUMED (information; lead 2026-10-21T03:30:46Z (1)) T2648
LC2M T2649	js5b LC2M (LC2, SF [1.5, 2.0]): (ii) hole bearing, exact V 995.4 N on t 2.688 SIM T2650	+1.952 T2651	+1.723 T2652	–	SIM / DERIVED SIM	231 (1)(ii) T2653 hand form: SF on the total stress including preload, linear superposition ASSUMED (information; lead 2026-10-21T03:30:46Z (1)) T2654
LC2M T2655	js5b LC2M (LC2, SF [1.5, 2.0]): (iii) band Neuber n30 (local strain) SIM T2656	+6.441 T2657	–	–	SIM / DERIVED SIM	231 (1)(iii) T2659 hand form: SF on the total stress including preload, linear superposition ASSUMED (information; lead 2026-10-21T03:30:46Z (1)) T2660

TC-01: other information rows (68); they do not set the cell.

State	Row	MS_y	MS_u	Preload	Tag	Criterion / note
FA1S T2661	elastic band (contact-edge field) SIM T2662	−0.576 T2663	−0.543 T2664	–	SIM / DERIVED SIM	–
FA1S T2665	all-node (hole / head contact edge) SIM T2666	−0.946 T2667	−0.942 T2668	–	SIM / DERIVED SIM	n388 4390.4 MPa T2669
FA1S T2670	(iii) hole-edge Neuber (Mechanical's information row) CALC T2671	+1.384 T2672	–	–	CALC CALC	–
PRELOAD T2674	band Neuber n30 (under-head band, local strain) SIM T2675	+9.300 T2676	–	–	SIM / DERIVED SIM	lead 06:36Z (DC head seat) T2678
PRELOAD T2679	flat-seat head bearing vs Fbry (A4-80 FE head force) SIM T2680	+3.222 T2681	–	–	SIM / DERIVED SIM	flight row = IN718 below T2683
PRELOAD T2684	ligament station P+Q (0.087 mm from the hole wall; contact-edge field) SIM T2685	−0.096 T2686	–	–	SIM / DERIVED SIM	240 (9): information T2688
PRELOAD T2689	all-node (hole / head contact edge) SIM T2690	−0.941 T2691	−0.936 T2692	–	SIM / DERIVED SIM	n388 4073.3 MPa T2693
PRELOAD T2694	54 seat lines P+Q <= 2 Fty(T), envelope minimum (line d+6.051 z'+0.0, (b2); A4-80 FE) SIM T2695	+1.038 T2696	–	–	SIM / DERIVED SIM	240 (9)(i) T2698
SOAK70 T2699	flat-seat head bearing vs Fbry (A4-80 FE head force) SIM T2700	+1.003 T2701	–	–	SIM / DERIVED SIM	flight row = IN718 below T2703
SOAK70 T2704	ligament station P+Q (0.087 mm from the hole wall; contact-edge field) SIM T2705	−0.395 T2706	–	–	SIM / DERIVED SIM	240 (9): information T2708
SOAK70 T2709	all-node (hole / head contact edge) SIM T2710	−0.950 T2711	−0.946 T2712	–	SIM / DERIVED SIM	n398 3371.4 MPa T2713
SOAK T2714	flat-seat head bearing vs Fbry (A4-80 FE head force) SIM T2715	+0.522 T2716	–	–	SIM / DERIVED SIM	flight row = IN718 below T2718
SOAK T2719	ligament station P+Q (0.087 mm from the hole wall; contact-edge field) SIM T2720	−0.200 T2721	–	–	SIM / DERIVED SIM	240 (9): information T2723
SOAK T2724	all-node (hole / head contact edge) SIM T2725	−0.956 T2726	−0.953 T2727	–	SIM / DERIVED SIM	n338 3842.7 MPa T2728
SOAK52 T2729	flat-seat head bearing vs Fbry (A4-80 FE head force) SIM T2730	+0.836 T2731	–	–	SIM / DERIVED SIM	flight row = IN718 below T2733
SOAK52 T2734	ligament station P+Q (0.087 mm from the hole wall; contact-edge field) SIM T2735	−0.208 T2736	–	–	SIM / DERIVED SIM	240 (9): information T2738
SOAK52 T2739	all-node (hole / head contact edge) SIM T2740	−0.964 T2741	−0.962 T2742	–	SIM / DERIVED SIM	n395 5034.6 MPa T2743
PRELOAD T2744	IN718 flat bearing vs Fbry on the route (i) spring footprint (A 46.06 mm2, registered stack) CALC T2745	+3.127 T2746	–	3750 T2748	CALC / DERIVED CALC	no-spring records (information): union 36.401 mm2 2.261, A_ROW 38.171 mm2 2.42 T2749
PRELOAD T2750	IN718 under-head band Neuber (local strain) [ASSUMED (sigma_el x A_ANN 47.12 / A_supp; edge loading at the tangential overhang not represented)] CALC T2751	+5.910 T2752	–	3750 T2754	CALC / DERIVED; ASSUMED scaling CALC	(b1); full_float_0.557 (lead 17:55:55Z F1), mean-pressure scaling ASSUMED T2755
PRELOAD T2756	IN718 ligament station P+Q (0.08 mm from the hole wall; contact-edge field) CALC T2757	−0.096 T2758	–	3750 T2760	CALC / DERIVED CALC	240 (9): information T2761
PRELOAD T2762	IN718 seat line d6.051 z'0 P+Q <= 2 Fty(T), F_sh form CALC T2763	+1.038 T2764	–	3750 T2766	CALC / DERIVED CALC	240 (9)(i), (10) T2767 (b2) T2768

State	Row	MS_y	MS_u	Preload	Tag	Criterion / note
PRELOAD T2769	IN718 seat line d6.051 z'0 P+Q, head-force bound CALC T2770	+1.038 T2771	–	3750 T2773	CALC / DERIVED CALC	(b2) T2774
PRELOAD T2775	IN718 ring line P+Q <= 2 Fty(T), F_sh form CALC T2776	+1.038 T2777	–	3750 T2779	CALC / DERIVED CALC	240 (9)(i), (10) T2780 (b2) T2781
PRELOAD T2782	IN718 ring line P+Q, head-force bound CALC T2783	+1.038 T2784	–	3750 T2786	CALC / DERIVED CALC	(b2) T2787
PRELOAD T2788	IN718 flat bearing vs Fbry on the route (i) spring footprint (A 46.06 mm2, registered stack) CALC T2789	+2.778 T2790	–	4096 T2792	CALC / DERIVED CALC	no-spring records (information): union 36.401 mm2 1.986, A_ROW 38.171 mm2 2.131; 4096 N column = information (J-F1 maximum capped at 3750 ... T2793
PRELOAD T2794	IN718 under-head band Neuber (local strain) [ASSUMED (sigma_el x A_ANN 47.12 / A_supp; edge loading at the tangential overhang not represented)] CALC T2795	+4.973 T2796	–	4096 T2798	CALC / DERIVED; ASSUMED scaling CALC	(b1); full_float_0.557 (lead 17:55:55Z F1), mean-pressure scaling ASSUMED; 4096 N column = information (J-F1 maximum capped at 3750 N ... T2799
PRELOAD T2800	IN718 ligament station P+Q (0.08 mm from the hole wall; contact-edge field) CALC T2801	**−0.173** T2802	–	4096 T2804	CALC / DERIVED CALC	240 (9): information T2805 4096 N column = information (J-F1 maximum capped at 3750 N, lead 17:56:12Z) T2806
PRELOAD T2807	IN718 seat line d6.051 z'0 P+Q <= 2 Fty(T), F_sh form CALC T2808	+1.033 T2809	–	4096 T2811	CALC / DERIVED CALC	240 (9)(i), (10) T2812 (b2); 4096 N column = information (J-F1 maximum capped at 3750 N, lead 17:56:12Z) T2813
PRELOAD T2814	IN718 seat line d6.051 z'0 P+Q, head-force bound CALC T2815	+0.866 T2816	–	4096 T2818	CALC / DERIVED CALC	(b2); 4096 N column = information (J-F1 maximum capped at 3750 N, lead 17:56:12Z) T2819
PRELOAD T2820	IN718 ring line P+Q <= 2 Fty(T), F_sh form CALC T2821	+1.033 T2822	–	4096 T2824	CALC / DERIVED CALC	240 (9)(i), (10) T2825 (b2); 4096 N column = information (J-F1 maximum capped at 3750 N, lead 17:56:12Z) T2826
PRELOAD T2827	IN718 ring line P+Q, head-force bound CALC T2828	+0.866 T2829	–	4096 T2831	CALC / DERIVED CALC	(b2); 4096 N column = information (J-F1 maximum capped at 3750 N, lead 17:56:12Z) T2832
SOAK70 T2833	IN718 ligament station P+Q (0.08 mm from the hole wall; contact-edge field) CALC T2834	**−0.440** T2835	–	3750 T2837	CALC / DERIVED CALC	240 (9): information T2838
SOAK70 T2839	IN718 seat line d6.051 z'0 P+Q, head-force bound CALC T2840	**−0.142** T2841	–	3750 T2843	CALC / DERIVED CALC	(b2) T2844
SOAK70 T2845	IN718 ring line P+Q, head-force bound CALC T2846	**−0.142** T2847	–	3750 T2849	CALC / DERIVED CALC	(b2) T2850
SOAK70 T2851	IN718 TC-01 lines with Pm > Fty (Mechanical) CALC T2852	–	–	3750 T2855	CALC / DERIVED CALC	{"per_history": {"js5": {"seat": {"A4-80": 12, "IN718_Fsh": 13, "IN718_headforce_bound": 16, "max_Pm_over_Fty_bound": 1.138, "n_lines" ... T2856
SOAK70 T2857	IN718 flat bearing vs Fbry on the route (i) spring footprint (A 46.06 mm2, registered stack) CALC T2858	+0.300 T2859	–	4096 T2861	CALC / DERIVED CALC	no-spring records (information): union 36.401 mm2 0.028, A_ROW 38.171 mm2 0.078; 4096 N column = information (J-F1 maximum capped at 3750 ... T2862
SOAK70 T2863	IN718 under-head band Neuber (local strain) [ASSUMED (sigma_el x A_ANN 47.12 / A_supp; edge loading at the tangential overhang not represented)] CALC T2864	+0.009 T2865	–	4096 T2867	CALC / DERIVED; ASSUMED scaling CALC	(b2); full_float_0.557 (lead 17:55:55Z F1), mean-pressure scaling ASSUMED; 4096 N column = information (J-F1 maximum capped at 3750 N ... T2868
SOAK70 T2869	IN718 ligament station P+Q (0.08 mm from the hole wall; contact-edge field) CALC T2870	**−0.471** T2871	–	4096 T2873	CALC / DERIVED CALC	240 (9): information T2874 4096 N column = information (J-F1 maximum capped at 3750 N, lead 17:56:12Z) T2875
SOAK70 T2876	IN718 seat line d6.051 z'0 P+Q <= 2 Fty(T), F_sh form CALC T2877	**−0.061** T2878	–	4096 T2880	CALC / DERIVED CALC	240 (9)(i), (10) T2881 (b2); 4096 N column = information (J-F1 maximum capped at 3750 N, lead 17:56:12Z) T2882
SOAK70 T2883	IN718 seat line d6.051 z'0 P+Q, head-force bound CALC T2884	**−0.177** T2885	–	4096 T2887	CALC / DERIVED CALC	(b2); 4096 N column = information (J-F1 maximum capped at 3750 N, lead 17:56:12Z) T2888
SOAK70 T2889	IN718 ring line P+Q <= 2 Fty(T), F_sh form CALC T2890	**−0.061** T2891	–	4096 T2893	CALC / DERIVED CALC	240 (9)(i), (10) T2894 (b2); 4096 N column = information (J-F1 maximum capped at 3750 N, lead 17:56:12Z) T2895
SOAK70 T2896	IN718 ring line P+Q, head-force bound CALC T2897	**−0.177** T2898	–	4096 T2900	CALC / DERIVED CALC	(b2); 4096 N column = information (J-F1 maximum capped at 3750 N, lead 17:56:12Z) T2901
SOAK70 T2902	IN718 TC-01 lines with Pm > Fty (Mechanical) CALC T2903	–	–	4096 T2906	CALC / DERIVED CALC	{"per_history": {"js5": {"seat": {"A4-80": 12, "IN718_Fsh": 13, "IN718_headforce_bound": 16, "max_Pm_over_Fty_bound": 1.202, "n_lines" ... T2907
SOAK T2908	IN718 ligament station P+Q (0.08 mm from the hole wall; contact-edge field) CALC T2909	**−0.272** T2910	–	3750 T2912	CALC / DERIVED CALC	240 (9): information T2913
SOAK T2914	IN718 seat line d6.051 z'0 P+Q, head-force bound CALC T2915	**−0.142** T2916	–	3750 T2918	CALC / DERIVED CALC	(b2) rep T2919
SOAK T2920	IN718 ring line P+Q, head-force bound CALC T2921	**−0.142** T2922	–	3750 T2924	CALC / DERIVED CALC	(b2) rep T2925
SOAK T2926	IN718 TC-01 lines with Pm > Fty (Mechanical) CALC T2927	–	–	3750 T2930	CALC / DERIVED CALC	{"per_history": {"js5": {"seat": {"A4-80": 12, "IN718_Fsh": 13, "IN718_headforce_bound": 18, "max_Pm_over_Fty_bound": 1.206, "n_lines" ... T2931
SOAK T2932	IN718 flat bearing vs Fbry on the route (i) spring footprint (A 46.06 mm2, registered stack) CALC T2933	+0.299 T2934	–	4096 T2936	CALC / DERIVED CALC	no-spring records (information): union 36.401 mm2 0.026, A_ROW 38.171 mm2 0.076; 4096 N column = information (J-F1 maximum capped at 3750 ... T2937
SOAK T2938	IN718 under-head band Neuber (local strain) [ASSUMED (sigma_el x A_ANN 47.12 / A_supp; edge loading at the tangential overhang not represented)] CALC T2939	+0.087 T2940	–	4096 T2942	CALC / DERIVED; ASSUMED scaling CALC	(b1); full_float_0.557 (lead 17:55:55Z F1), mean-pressure scaling ASSUMED; 4096 N column = information (J-F1 maximum capped at 3750 N ... T2943
SOAK T2944	IN718 ligament station P+Q (0.08 mm from the hole wall; contact-edge field) CALC T2945	**−0.302** T2946	–	4096 T2948	CALC / DERIVED CALC	240 (9): information T2949 4096 N column = information (J-F1 maximum capped at 3750 N, lead 17:56:12Z) T2950
SOAK T2951	IN718 seat line d6.051 z'0 P+Q <= 2 Fty(T), F_sh form CALC T2952	**−0.076** T2953	–	4096 T2955	CALC / DERIVED CALC	240 (9)(i), (10) T2956 (b2) rep; 4096 N column = information (J-F1 maximum capped at 3750 N, lead 17:56:12Z) T2957
SOAK T2958	IN718 seat line d6.051 z'0 P+Q, head-force bound CALC T2959	**−0.183** T2960	–	4096 T2962	CALC / DERIVED CALC	(b2) rep; 4096 N column = information (J-F1 maximum capped at 3750 N, lead 17:56:12Z) T2963
SOAK T2964	IN718 ring line P+Q <= 2 Fty(T), F_sh form CALC T2965	**−0.076** T2966	–	4096 T2968	CALC / DERIVED CALC	240 (9)(i), (10) T2969 (b2) rep; 4096 N column = information (J-F1 maximum capped at 3750 N, lead 17:56:12Z) T2970
SOAK T2971	IN718 ring line P+Q, head-force bound CALC T2972	**−0.183** T2973	–	4096 T2975	CALC / DERIVED CALC	(b2) rep; 4096 N column = information (J-F1 maximum capped at 3750 N, lead 17:56:12Z) T2976
SOAK T2977	IN718 TC-01 lines with Pm > Fty (Mechanical) CALC T2978	–	–	4096 T2981	CALC / DERIVED CALC	{"per_history": {"js5": {"seat": {"A4-80": 12, "IN718_Fsh": 13, "IN718_headforce_bound": 20, "max_Pm_over_Fty_bound": 1.257, "n_lines" ... T2982
SOAK52 T2983	IN718 ligament station P+Q (0.08 mm from the hole wall; contact-edge field) CALC T2984	**−0.283** T2985	–	3750 T2987	CALC / DERIVED CALC	240 (9): information T2988
SOAK52 T2989	IN718 seat line d6.051 z'0 P+Q, head-force bound CALC T2990	**−0.085** T2991	–	3750 T2993	CALC / DERIVED CALC	(b2) T2994
SOAK52 T2995	IN718 ring line P+Q, head-force bound CALC T2996	**−0.085** T2997	–	3750 T2999	CALC / DERIVED CALC	(b2) T3000
SOAK52 T3001	IN718 TC-01 lines with Pm > Fty (Mechanical) CALC T3002	–	–	3750 T3005	CALC / DERIVED CALC	{"per_history": {"js5": {"seat": {"A4-80": 0, "IN718_Fsh": 0, "IN718_headforce_bound": 16, "max_Pm_over_Fty_bound": 1.075, "n_lines": 54} ... T3006
SOAK52 T3007	IN718 flat bearing vs Fbry on the route (i) spring footprint (A 46.06 mm2, registered stack) CALC T3008	+0.320 T3009	–	4096 T3011	CALC / DERIVED CALC	no-spring records (information): union 36.401 mm2 0.043, A_ROW 38.171 mm2 0.094; 4096 N column = information (J-F1 maximum capped at 3750 ... T3012
SOAK52 T3013	IN718 under-head band Neuber (local strain) [ASSUMED (sigma_el x A_ANN 47.12 / A_supp; edge loading at the tangential overhang not represented)] CALC T3014	**−0.057** T3015	–	4096 T3017	CALC / DERIVED; ASSUMED scaling CALC	(b2); full_float_0.557 (lead 17:55:55Z F1), mean-pressure scaling ASSUMED; 4096 N column = information (J-F1 maximum capped at 3750 N ... T3018
SOAK52 T3019	IN718 ligament station P+Q (0.08 mm from the hole wall; contact-edge field) CALC T3020	**−0.316** T3021	–	4096 T3023	CALC / DERIVED CALC	240 (9): information T3024 4096 N column = information (J-F1 maximum capped at 3750 N, lead 17:56:12Z) T3025

State	Row	MS_y	MS_u	Preload	Tag	Criterion / note
SOAK52 T3026	IN718 seat line d6.051 z'0 P+Q <= 2 Fty(T), F_sh form CALC T3027	−0.035 T3028	–	4096 T3030	CALC / DERIVED CALC	240 (9)(i), (10) T3031 (b1); 4096 N column = information (J-F1 maximum capped at 3750 N, lead 17:56:12Z) T3032
SOAK52 T3033	IN718 seat line d6.051 z'0 P+Q, head-force bound CALC T3034	−0.131 T3035	–	4096 T3037	CALC / DERIVED CALC	(b1); 4096 N column = information (J-F1 maximum capped at 3750 N, lead 17:56:12Z) T3038
SOAK52 T3039	IN718 ring line P+Q <= 2 Fty(T), F_sh form CALC T3040	−0.035 T3041	–	4096 T3043	CALC / DERIVED CALC	240 (9)(i), (10) T3044 (b1); 4096 N column = information (J-F1 maximum capped at 3750 N, lead 17:56:12Z) T3045
SOAK52 T3046	IN718 ring line P+Q, head-force bound CALC T3047	−0.131 T3048	–	4096 T3050	CALC / DERIVED CALC	(b1); 4096 N column = information (J-F1 maximum capped at 3750 N, lead 17:56:12Z) T3051
SOAK52 T3052	IN718 TC-01 lines with Pm > Fty (Mechanical) CALC T3053	–	–	4096 T3056	CALC / DERIVED CALC	{"per_history": {"js5": {"seat": {"A4-80": 0, "IN718_Fsh": 0, "IN718_headforce_bound": 16, "max_Pm_over_Fty_bound": 1.127, "n_lines": 54} ... T3057
FA1S T3058	IN718 under-head band Neuber (local strain) [ASSUMED (sigma_el x A_ANN 47.12 / A_supp; edge loading at the tangential overhang not represented)] CALC T3059	+0.012 T3060	–	4096 T3062	CALC / DERIVED; ASSUMED scaling CALC	(b2); full_float_0.557 (lead 17:55:55Z F1), mean-pressure scaling ASSUMED; 4096 N column = information (J-F1 maximum capped at 3750 N ... T3063

9.5.2 TC-02: J-FWD cell +0.045 / +0.162

TC-02: criterion rows (24); cell **0.045–0.162** T3064.

State	Row	MS_y	MS_u	Preload	Tag	Criterion / note
FA1S T3065	(i) shear-out to g2 at x' 28.0 Pm+Pb SIM T3066	+0.981 T3067	+1.136 T3068	–	SIM / DERIVED SIM	231 (1)(i) T3069
FA1S T3070	(i) shear-out to g2 at x' 28.0 Pm SIM T3071	+3.593 T3072	+3.954 T3073	–	SIM / DERIVED SIM	231 (1)(i) T3074
FA1S T3075	(ii) thread shear on A_n 68.4 SIM T3076	+1.969 T3077	+2.330 T3078	–	SIM / DERIVED SIM	231 (1)(ii) T3079
FA1S T3080	(ii) tap bearing on L_e 6.84 SIM T3081	+2.115 T3082	+2.839 T3083	–	SIM / DERIVED SIM	231 (1)(ii) T3084
FA1S T3085	(iii) band Neuber n15 SIM T3086	+7.893 T3087	–	–	SIM / DERIVED SIM	231 (1)(iii) T3089
FA1S T3090	(ii) tapped-thread bearing CALC T3091	+1.385 T3092	+1.940 T3093	–	CALC CALC	231 (1)(ii) T3094
FA1S T3095	(ii) internal thread shear at 3965.0 N on A_n 67.9 CALC T3096	+1.332 T3097	+1.615 T3098	–	CALC CALC	231 (1)(ii) T3099
FA1U T3100	js5b FA1U (FA1, SF [1.0, 1.0]): (i) shear-out to g2 at x' 28.0 Pm+Pb SIM T3101	+1.449 T3102	+1.641 T3103	–	SIM / DERIVED SIM	231 (1)(i) T3104
FA1U T3105	js5b FA1U (FA1, SF [1.0, 1.0]): (ii) thread shear on A_n 68.4 SIM T3106	+1.650 T3107	+1.971 T3108	–	SIM / DERIVED SIM	231 (1)(ii) T3109
FA1U T3110	js5b FA1U (FA1, SF [1.0, 1.0]): (ii) tap bearing on L_e 6.84 SIM T3111	+2.112 T3112	+2.835 T3113	–	SIM / DERIVED SIM	231 (1)(ii) T3114
FA1D T3115	js5b FA1D (FA1, SF [1.0, 1.0]): (i) shear-out to g2 at x' 28.0 Pm+Pb SIM T3116	+1.645 T3117	+1.853 T3118	–	SIM / DERIVED SIM	231 (1)(i) T3119
FA1D T3120	js5b FA1D (FA1, SF [1.0, 1.0]): (ii) thread shear on A_n 68.4 SIM T3121	+1.597 T3122	+1.913 T3123	–	SIM / DERIVED SIM	231 (1)(ii) T3124
FA1D T3125	js5b FA1D (FA1, SF [1.0, 1.0]): (ii) tap bearing on L_e 6.84 SIM T3126	+2.970 T3127	+3.894 T3128	–	SIM / DERIVED SIM	231 (1)(ii) T3129
LC2U T3130	js5b LC2U (FE-factored, SF [1.0, 1.0]): (i) shear-out to g2 at x' 28.0 Pm+Pb SIM T3131	–	+3.239 T3133	–	SIM / DERIVED SIM	231 (1)(i) T3134 FE-factored: the x1.5 / x2.0 state solved with contact at the 3750 N maximum preload (J-F1 cap), SF 1.0; LC2Y = yield slot, LC2U = ... T3135
LC2U T3136	js5b LC2U (FE-factored, SF [1.0, 1.0]): (ii) thread shear on A_n 68.4 SIM T3137	–	+1.633 T3139	–	SIM / DERIVED SIM	231 (1)(ii) T3140 FE-factored: the x1.5 / x2.0 state solved with contact at the 3750 N maximum preload (J-F1 cap), SF 1.0; LC2Y = yield slot, LC2U = ... T3141
LC2U T3142	js5b LC2U (FE-factored, SF [1.0, 1.0]): (ii) tap bearing on L_e 6.84 SIM T3143	–	+5.493 T3145	–	SIM / DERIVED SIM	231 (1)(ii) T3146 FE-factored: the x1.5 / x2.0 state solved with contact at the 3750 N maximum preload (J-F1 cap), SF 1.0; LC2Y = yield slot, LC2U = ... T3147
LC2Y T3148	js5b LC2Y (FE-factored, SF [1.0, 1.0]): (i) shear-out to g2 at x' 28.0 Pm+Pb SIM T3149	+5.228 T3150	–	–	SIM / DERIVED SIM	231 (1)(i) T3152 FE-factored: the x1.5 / x2.0 state solved with contact at the 3750 N maximum preload (J-F1 cap), SF 1.0; LC2Y = yield slot, LC2U = ... T3153
LC2Y T3154	js5b LC2Y (FE-factored, SF [1.0, 1.0]): (ii) thread shear on A_n 68.4 SIM T3155	+1.330 T3156	–	–	SIM / DERIVED SIM	231 (1)(ii) T3158 FE-factored: the x1.5 / x2.0 state solved with contact at the 3750 N maximum preload (J-F1 cap), SF 1.0; LC2Y = yield slot, LC2U = ... T3159
LC2Y T3160	js5b LC2Y (FE-factored, SF [1.0, 1.0]): (ii) tap bearing on L_e 6.84 SIM T3161	+6.085 T3162	–	–	SIM / DERIVED SIM	231 (1)(ii) T3164 FE-factored: the x1.5 / x2.0 state solved with contact at the 3750 N maximum preload (J-F1 cap), SF 1.0; LC2Y = yield slot, LC2U = ... T3165
SOAK70 T3166	IN718 thread shear on THE ROW A_n 67.93 (route (i): M5 × 10.7 + HW-170 spring, L_e min 7.294) CALC T3167	+0.087 T3168	+0.207 T3169	3750 T3170	CALC / DERIVED CALC	234 (1)(b) T3171 N 8227.4 ((b2)), T 97.7 C T3172
SOAK T3173	IN718 thread shear on THE ROW A_n 67.93 (route (i): M5 × 10.7 + HW-170 spring, L_e min 7.294) CALC T3174	+0.104 T3175	+0.227 T3176	3750 T3177	CALC / DERIVED CALC	234 (1)(b) T3178 N 8110.3 (js5), T 96.0 C T3179
SOAK52 T3180	IN718 thread shear on THE ROW A_n 67.93 (route (i): M5 × 10.7 + HW-170 spring, L_e min 7.294) CALC T3181	+0.045 T3182	+0.162 T3183	3750 T3184	CALC / DERIVED CALC	234 (1)(b) T3185 N 8572.4 ((b2) rep), T 94.9 C; within 0.05: FE confirmation T3186
FA1S T3187	IN718 thread shear on THE ROW A_n 67.93 (route (i): M5 × 10.7 + HW-170 spring, L_e min 7.294) CALC T3188	+1.802 T3189	+2.143 T3190	3750 T3191	CALC / DERIVED CALC	234 (1)(b) T3192 N 3302.3 (js5), T 45.6 C T3193
FA1S T3194	(ii) tap bearing, CALC V 3888.1 N on FEA3D's d x L_e basis (Mechanical rows) CALC T3195	+1.891 T3196	+2.564 T3197	–	CALC CALC	231 (1)(ii) T3198 FEA3D tap_bearing_derived sigma 106.0 at V 3603.6 (d x L_e 34.00 mm2, L_e 6.85), allowables back-computed (330.6 / 407.6 MPa); V-driven ... T3199

TC-02: hand-form LC2N / LC2M rows, INFORMATION beside the criterion (rule (A), lead 03:30:46Z) (6).

State	Row	MS_y	MS_u	Preload	Tag	Criterion / note
LC2N T3200	js5b LC2N (LC2, SF [1.5, 2.0]): (i) shear-out to g2 at x' 28.0 Pm+Pb SIM T3201	+6.251 T3202	+4.829 T3203	–	SIM / DERIVED SIM	231 (1)(i) T3204 hand form: SF on the total stress including preload, linear superposition ASSUMED (information; lead 2026-10-21T03:30:46Z (1)) T3205
LC2N T3206	js5b LC2N (LC2, SF [1.5, 2.0]): (ii) thread shear on A_n 68.4 SIM T3207	+0.526 T3208	+0.276 T3209	–	SIM / DERIVED SIM	231 (1)(ii) T3210 hand form: SF on the total stress including preload, linear superposition ASSUMED (information; lead 2026-10-21T03:30:46Z (1)) T3211
LC2N T3212	js5b LC2N (LC2, SF [1.5, 2.0]): (ii) tap bearing on L_e 6.84 SIM T3213	+6.137 T3214	+5.558 T3215	–	SIM / DERIVED SIM	231 (1)(ii) T3216 hand form: SF on the total stress including preload, linear superposition ASSUMED (information; lead 2026-10-21T03:30:46Z (1)) T3217
LC2M T3218	js5b LC2M (LC2, SF [1.5, 2.0]): (i) shear-out to g2 at x' 28.0 Pm+Pb SIM T3219	+2.851 T3220	+2.089 T3221	–	SIM / DERIVED SIM	231 (1)(i) T3222 hand form: SF on the total stress including preload, linear superposition ASSUMED (information; lead 2026-10-21T03:30:46Z (1)) T3223
LC2M T3224	js5b LC2M (LC2, SF [1.5, 2.0]): (ii) thread shear on A_n 68.4 SIM T3225	+0.793 T3226	+0.496 T3227	–	SIM / DERIVED SIM	231 (1)(ii) T3228 hand form: SF on the total stress including preload, linear superposition ASSUMED (information; lead 2026-10-21T03:30:46Z (1)) T3229
LC2M T3230	js5b LC2M (LC2, SF [1.5, 2.0]): (ii) tap bearing on L_e 6.84 SIM T3231	+6.096 T3232	+5.506 T3233	–	SIM / DERIVED SIM	231 (1)(ii) T3234 hand form: SF on the total stress including preload, linear superposition ASSUMED (information; lead 2026-10-21T03:30:46Z (1)) T3235

TC-02: other information rows (30); they do not set the cell.

State	Row	MS_y	MS_u	Preload	Tag	Criterion / note
FA1S T3236	condition: through-wall SCL at the band node (must pass) SIM T3237	+0.346 T3238	+0.452 T3239	–	SIM / DERIVED SIM	lead 07:21Z (4) condition T3240 passes T3241
FA1S T3242	elastic band (contact-edge field, 0.99 mm from the TC-01 hole-edge footprint) SIM T3243	+0.155 T3244	+0.245 T3245	–	SIM / DERIVED SIM	–
FA1S T3246	all-node (bonded-thread tie edge) SIM T3247	−0.890 T3248	−0.881 T3249	–	SIM / DERIVED SIM	n1660 2148.3 MPa T3250
PRELOAD T3251	thread shear on A_n 68.4, js5 N (A4-80 FE) SIM T3252	+1.526 T3253	+1.846 T3254	–	SIM / DERIVED SIM	A4-80 FE force (A4-80 withdrawn, 240 (4)): the flight row is Mechanical's IN718 THE ROW T3255

State	Row	MS_y	MS_u	Preload	Tag	Criterion / note
PRELOAD T3256	all-node (bonded-thread tie edge) SIM T3257	−0.865 T3258	−0.854 T3259	–	SIM / DERIVED SIM	n1660 1779.6 MPa T3260
SOAK70 T3261	thread shear on A_n 68.4, js5 N (A4-80 FE) SIM T3262	+0.581 T3263	+0.756 T3264	–	SIM / DERIVED SIM	A4-80 FE force (A4-80 withdrawn, 240 (4)): the flight row is Mechanical's IN718 THE ROW T3265
SOAK70 T3266	thread shear on A_n 68.4 at the envelope N 7471.8 ((b2); A4-80 FE) DERIVED T3267	+0.206 T3268	+0.339 T3269	–	DERIVED DERIVED	A4-80 FE force (A4-80 withdrawn, 240 (4)): the flight row is Mechanical's IN718 THE ROW T3270
SOAK70 T3271	all-node (bonded-thread tie edge) SIM T3272	−0.779 T3273	−0.764 T3274	–	SIM / DERIVED SIM	n1660 1034.8 MPa T3275
SOAK T3276	thread shear on A_n 68.4, js5 N (A4-80 FE) SIM T3277	+0.221 T3278	+0.357 T3279	–	SIM / DERIVED SIM	A4-80 FE force (A4-80 withdrawn, 240 (4)): the flight row is Mechanical's IN718 THE ROW T3280
SOAK T3281	thread shear on A_n 68.4 at the envelope N 7382.9 (js5; A4-80 FE) DERIVED T3282	+0.221 T3283	+0.357 T3284	–	DERIVED DERIVED	A4-80 FE force (A4-80 withdrawn, 240 (4)): the flight row is Mechanical's IN718 THE ROW T3285
SOAK T3286	all-node (bonded-thread tie edge) SIM T3287	−0.891 T3288	−0.884 T3289	–	SIM / DERIVED SIM	n1642 2100.4 MPa T3290
SOAK52 T3291	thread shear on A_n 68.4, js5 N (A4-80 FE) SIM T3292	+0.375 T3293	+0.528 T3294	–	SIM / DERIVED SIM	A4-80 FE force (A4-80 withdrawn, 240 (4)): the flight row is Mechanical's IN718 THE ROW T3295
SOAK52 T3296	thread shear on A_n 68.4 at the envelope N 7764.0 ((b2) rep; A4-80 FE) DERIVED T3297	+0.162 T3298	+0.291 T3299	–	DERIVED DERIVED	A4-80 FE force (A4-80 withdrawn, 240 (4)): the flight row is Mechanical's IN718 THE ROW T3300
SOAK52 T3301	all-node (bonded-thread tie edge) SIM T3302	−0.824 T3303	−0.812 T3304	–	SIM / DERIVED SIM	n17845 1298.7 MPa T3305
PRELOAD T3306	IN718 thread shear on THE ROW A_n 67.93 (route (i): M5 × 10.7 + HW-170 spring, L_e min 7.294) CALC T3307	+1.509 T3308	+1.826 T3309	3750 T3310	CALC / DERIVED CALC	234 (1)(b) T3311 N 3750.0 (js5), T 20.0 C T3312
PRELOAD T3313	IN718 thread shear on FEA3D's drawn-basis A_n 68.4 (M5 × 10.2, lead 13:14Z; superseded by THE ROW) CALC T3314	+1.526 T3315	+1.845 T3316	3750 T3317	CALC / DERIVED CALC	N 3750.0 (js5), T 20.0 C T3318
PRELOAD T3319	IN718 thread shear on THE ROW A_n 67.93 (route (i): M5 × 10.7 + HW-170 spring, L_e min 7.294) CALC T3320	+1.297 T3321	+1.587 T3322	4096 T3323	CALC / DERIVED CALC	234 (1)(b) T3324 N 4096.0 (js5), T 20.0 C; 4096 N column = information (J-F1 maximum capped at 3750 N, lead 17:56:12Z) T3325
PRELOAD T3326	IN718 thread shear on FEA3D's drawn-basis A_n 68.4 (M5 × 10.2, lead 13:14Z; superseded by THE ROW) CALC T3327	+1.313 T3328	+1.605 T3329	4096 T3330	CALC / DERIVED CALC	N 4096.0 (js5), T 20.0 C; 4096 N column = information (J-F1 maximum capped at 3750 N, lead 17:56:12Z) T3331
SOAK70 T3332	IN718 thread shear on FEA3D's drawn-basis A_n 68.4 (M5 × 10.2, lead 13:14Z; superseded by THE ROW) CALC T3333	+0.095 T3334	+0.215 T3335	3750 T3336	CALC / DERIVED CALC	N 8227.4 ((b2)), T 97.7 C T3337
SOAK70 T3338	IN718 thread shear on THE ROW A_n 67.93 (route (i): M5 × 10.7 + HW-170 spring, L_e min 7.294) CALC T3339	+0.043 T3340	+0.159 T3341	4096 T3342	CALC / DERIVED CALC	234 (1)(b) T3343 N 8573.4 ((b2)), T 97.7 C; within 0.05: FE confirmation; 4096 N column = information (J-F1 maximum capped at 3750 N, lead 17:56:12Z) T3344
SOAK70 T3345	IN718 thread shear on FEA3D's drawn-basis A_n 68.4 (M5 × 10.2, lead 13:14Z; superseded by THE ROW) CALC T3346	+0.051 T3347	+0.166 T3348	4096 T3349	CALC / DERIVED CALC	N 8573.4 ((b2)), T 97.7 C; 4096 N column = information (J-F1 maximum capped at 3750 N, lead 17:56:12Z) T3350
SOAK T3351	IN718 thread shear on FEA3D's drawn-basis A_n 68.4 (M5 × 10.2, lead 13:14Z; superseded by THE ROW) CALC T3352	+0.112 T3353	+0.235 T3354	3750 T3355	CALC / DERIVED CALC	N 8110.3 (js5), T 96.0 C T3356
SOAK T3357	IN718 thread shear on THE ROW A_n 67.93 (route (i): M5 × 10.7 + HW-170 spring, L_e min 7.294) CALC T3358	+0.059 T3359	+0.176 T3360	4096 T3361	CALC / DERIVED CALC	234 (1)(b) T3362 N 8456.3 (js5), T 96.0 C; 4096 N column = information (J-F1 maximum capped at 3750 N, lead 17:56:12Z) T3363
SOAK T3364	IN718 thread shear on FEA3D's drawn-basis A_n 68.4 (M5 × 10.2, lead 13:14Z; superseded by THE ROW) CALC T3365	+0.066 T3366	+0.184 T3367	4096 T3368	CALC / DERIVED CALC	N 8456.3 (js5), T 96.0 C; 4096 N column = information (J-F1 maximum capped at 3750 N, lead 17:56:12Z) T3369
SOAK52 T3370	IN718 thread shear on FEA3D's drawn-basis A_n 68.4 (M5 × 10.2, lead 13:14Z; superseded by THE ROW) CALC T3371	+0.052 T3372	+0.170 T3373	3750 T3374	CALC / DERIVED CALC	N 8572.4 ((b2) rep), T 94.9 C T3375
SOAK52 T3376	IN718 thread shear on THE ROW A_n 67.93 (route (i): M5 × 10.7 + HW-170 spring, L_e min 7.294) CALC T3377	+0.005 T3378	+0.116 T3379	4096 T3380	CALC / DERIVED CALC	234 (1)(b) T3381 N 8918.4 ((b2) rep), T 94.9 C; within 0.05: FE confirmation; 4096 N column = information (J-F1 maximum capped at 3750 N, lead 17:56:12Z) T3382
SOAK52 T3383	IN718 thread shear on FEA3D's drawn-basis A_n 68.4 (M5 × 10.2, lead 13:14Z; superseded by THE ROW) CALC T3384	+0.012 T3385	+0.124 T3386	4096 T3387	CALC / DERIVED CALC	N 8918.4 ((b2) rep), T 94.9 C; 4096 N column = information (J-F1 maximum capped at 3750 N, lead 17:56:12Z) T3388
FA1S T3389	IN718 thread shear on FEA3D's drawn-basis A_n 68.4 (M5 × 10.2, lead 13:14Z; superseded by THE ROW) CALC T3390	+1.821 T3391	+2.165 T3392	3750 T3393	CALC / DERIVED CALC	N 3302.3 (js5), T 45.6 C T3394
FA1S T3395	IN718 thread shear on THE ROW A_n 67.93 (route (i): M5 × 10.7 + HW-170 spring, L_e min 7.294) CALC T3396	+1.537 T3397	+1.845 T3398	4096 T3399	CALC / DERIVED CALC	234 (1)(b) T3400 N 3648.3 (js5), T 45.6 C; 4096 N column = information (J-F1 maximum capped at 3750 N, lead 17:56:12Z) T3401
FA1S T3402	IN718 thread shear on FEA3D's drawn-basis A_n 68.4 (M5 × 10.2, lead 13:14Z; superseded by THE ROW) CALC T3403	+1.554 T3404	+1.865 T3405	4096 T3406	CALC / DERIVED CALC	N 3648.3 (js5), T 45.6 C; 4096 N column = information (J-F1 maximum capped at 3750 N, lead 17:56:12Z) T3407

9.5.3 HW-101 (J-FWD): J-FWD cell +0.650 / +0.278

HW-101 (J-FWD): criterion rows (11); cell **0.65–0.278** T3408.

State	Row	MS_y	MS_u	Preload	Tag	Criterion / note
FA1S T3409	IN718 5020A Eq. 22, preload omitted (criterion, 240 (3)); R6 basis: V 4079.2 N FE (b1), M 4354.6 N mm CALC (arm 1.120 mm RETAINED, lead 15:37Z (R2)) CALC T3410	–	+0.278 T3412	–	CALC CALC	NASA-STD-5020A Eq. 22 T3413 preload included 0.065 (information); fix v2 CALC record (V 3888.1 N) 0.301 / included 0.082 superseded by R6 T3414
FA1S T3415	IN718 direct shear on the minor area (231 (ii)) CALC T3416	+0.933 T3417	+1.408 T3418	–	CALC CALC	231 (ii) T3419
LC2N T3420	js5b LC2N (LC2, SF [1.5, 2.0]): IN718 FE screw, LC2: combined vM (preload on A_s + factored shear on A_3) / (engmat AMS 5662/5663 bar minimums) SIM T3421	+1.894 T3422	+2.110 T3423	–	SIM / DERIVED SIM	234 (1)(b) T3424
LC2M T3425	js5b LC2M (LC2, SF [1.5, 2.0]): IN718 FE screw, LC2: combined vM (preload on A_s + factored shear on A_3) / (engmat AMS 5662/5663 bar minimums) SIM T3426	+2.218 T3427	+2.394 T3428	–	SIM / DERIVED SIM	234 (1)(b) T3429
SOAK70 T3430	IN718 tension on A_s (234 (1)(b)) CALC T3431	+0.717 T3432	+1.034 T3433	3750 T3434	CALC / DERIVED CALC	N 8227.4 ((b2)), T 158.6 C T3435
SOAK70 T3436	IN718 head on the HW-170 spring, A 39.19 mm2 (IN718 Fty / Ftu, no bearing credit) CALC T3437	+3.745 T3438	+4.622 T3439	3750 T3440	CALC / DERIVED CALC	N 8227.4 ((b2)), T 158.6 C; spring side TO-SOURCE (Mechanical) T3441
SOAK T3442	IN718 tension on A_s (234 (1)(b)) CALC T3443	+0.757 T3444	+1.088 T3445	3750 T3446	CALC / DERIVED CALC	N 8110.3 (js5), T 127.0 C T3447
SOAK T3448	IN718 head on the HW-170 spring, A 39.19 mm2 (IN718 Fty / Ftu, no bearing credit) CALC T3449	+3.855 T3450	+4.770 T3451	3750 T3452	CALC / DERIVED CALC	N 8110.3 (js5), T 127.0 C; spring side TO-SOURCE (Mechanical) T3453
SOAK52 T3454	IN718 tension on A_s (234 (1)(b)) CALC T3455	+0.650 T3456	+0.955 T3457	3750 T3458	CALC / DERIVED CALC	N 8572.4 ((b2) rep), T 154.1 C T3459
SOAK52 T3460	IN718 head on the HW-170 spring, A 39.19 mm2 (IN718 Fty / Ftu, no bearing credit) CALC T3461	+3.560 T3462	+4.405 T3463	3750 T3464	CALC / DERIVED CALC	N 8572.4 ((b2) rep), T 154.1 C; spring side TO-SOURCE (Mechanical) T3465
FA1S T3466	IN718 head on the HW-170 spring, A 39.19 mm2 (IN718 Fty / Ftu, no bearing credit) CALC T3467	+11.188 T3468	+13.595 T3469	3750 T3470	CALC / DERIVED CALC	N 3302.3 (js5), T 45.6 C; spring side TO-SOURCE (Mechanical) T3471

HW-101 (J-FWD): other information rows (29); they do not set the cell.

State	Row	MS_y	MS_u	Preload	Tag	Criterion / note
FA1S T3472	A4-80 FE: interaction (234 (1)(b)) / shear-only yield on A_3 SIM T3473	+0.145 T3474	+0.508 T3475	–	SIM / DERIVED SIM	A4-80 withdrawn (240 (4)) T3476
PRELOAD T3477	A4-80 FE: tension on A_s (234 (1)(b)) SIM T3478	+1.269 T3479	+2.025 T3480	–	SIM / DERIVED SIM	A4-80 withdrawn (240 (4)) T3481
SOAK70 T3482	A4-80 FE: tension on A_s (234 (1)(b)) SIM T3483	+0.113 T3484	+0.890 T3485	–	SIM / DERIVED SIM	A4-80 withdrawn (240 (4)) T3486
SOAK T3487	A4-80 FE: tension on A_s (234 (1)(b)) SIM T3488	−0.089 T3489	+0.476 T3490	–	SIM / DERIVED SIM	A4-80 withdrawn (240 (4)) T3491
SOAK52 T3492	A4-80 FE: tension on A_s (234 (1)(b)) SIM T3493	−0.025 T3494	+0.644 T3495	–	SIM / DERIVED SIM	A4-80 withdrawn (240 (4)) T3496
FA1U T3497	js5b FA1U (FA1, SF [1.0, 1.0]): IN718 FE screw, FA1 (234 (1)(b)): ultimate interaction = criterion; yield sh (engmat AMS 5662/5663 bar minimums) SIM T3498	+1.070 T3499	+1.287 T3500	–	SIM / DERIVED SIM	234 (1)(b) T3501
FA1D T3502	js5b FA1D (FA1, SF [1.0, 1.0]): IN718 FE screw, FA1 (234 (1)(b)): ultimate interaction = criterion; yield sh (engmat AMS 5662/5663 bar minimums) SIM T3503	+1.641 T3504	+1.724 T3505	–	SIM / DERIVED SIM	234 (1)(b) T3506
LC2U T3507	js5b LC2U (FE-factored, SF [1.0, 1.0]): IN718 FE screw, FE-factored: combined vM (preload on A_s + factored shear on (engmat AMS 5662/5663 bar ... SIM T3508	–	+2.145 T3510	–	SIM / DERIVED SIM	234 (1)(b) T3511

State	Row	MS_y	MS_u	Preload	Tag	Criterion / note
LC2Y T3512	js5b LC2Y (FE-factored, SF [1.0, 1.0]): IN718 FE screw, FE-factored: combined vM (preload on A_s + factored shear on (engmat AMS 5662/5663 bar ... SIM T3513	+1.919 T3514	–	–	SIM / DERIVED SIM	234 (1)(b) T3516
PRELOAD T3517	IN718 tension on A_s (234 (1)(b)) CALC T3518	+2.910 T3519	+3.693 T3520	3750 T3521	CALC / DERIVED CALC	N 3750.0 (js5), T 20.0 C T3522
PRELOAD T3523	IN718 head on the HW-170 spring, A 39.19 mm2 (IN718 Fty / Ftu, no bearing credit) CALC T3524	+9.806 T3525	+11.970 T3526	3750 T3527	CALC / DERIVED CALC	N 3750.0 (js5), T 20.0 C; spring side TO-SOURCE (Mechanical) T3528
PRELOAD T3529	IN718 tension on A_s (234 (1)(b)) CALC T3530	+2.580 T3531	+3.296 T3532	4096 T3533	CALC / DERIVED CALC	N 4096.0 (js5), T 20.0 C; 4096 N column = information (J-F1 maximum capped at 3750 N, lead 17:56:12Z) T3534
PRELOAD T3535	IN718 head on the HW-170 spring, A 39.19 mm2 (IN718 Fty / Ftu, no bearing credit) CALC T3536	+8.893 T3537	+10.874 T3538	4096 T3539	CALC / DERIVED CALC	N 4096.0 (js5), T 20.0 C; spring side TO-SOURCE (Mechanical); 4096 N column = information (J-F1 maximum capped at 3750 N, lead 17:56:12Z) T3540
SOAK70 T3541	IN718 soak fibre yield (secondary: relaxation / shakedown / LCF, lead 10:14Z) CALC T3542	**−0.219** T3543	–	3750 T3545	CALC / DERIVED CALC	(cD); shakedown 1.078, range 958.8 MPa T3546
SOAK70 T3547	IN718 tension on A_s (234 (1)(b)) CALC T3548	+0.648 T3549	+0.952 T3550	4096 T3551	CALC / DERIVED CALC	N 8573.4 ((b2)), T 158.6 C; 4096 N column = information (J-F1 maximum capped at 3750 N, lead 17:56:12Z) T3552
SOAK70 T3553	IN718 head on the HW-170 spring, A 39.19 mm2 (IN718 Fty / Ftu, no bearing credit) CALC T3554	+3.553 T3555	+4.395 T3556	4096 T3557	CALC / DERIVED CALC	N 8573.4 ((b2)), T 158.6 C; spring side TO-SOURCE (Mechanical); 4096 N column = information (J-F1 maximum capped at 3750 N, lead 17:56:12Z) T3558
SOAK70 T3559	IN718 soak fibre yield (secondary: relaxation / shakedown / LCF, lead 10:14Z) CALC T3560	**−0.267** T3561	–	4096 T3563	CALC / DERIVED CALC	(cD); shakedown 0.964, range 1014.4 MPa; 4096 N column = information (J-F1 maximum capped at 3750 N, lead 17:56:12Z) T3564
SOAK T3565	IN718 soak fibre yield (secondary: relaxation / shakedown / LCF, lead 10:14Z) CALC T3566	**−0.343** T3567	–	3750 T3569	CALC / DERIVED CALC	(b1); shakedown 0.657, range 1212.5 MPa T3570
SOAK T3571	IN718 tension on A_s (234 (1)(b)) CALC T3572	+0.685 T3573	+1.002 T3574	4096 T3575	CALC / DERIVED CALC	N 8456.3 (js5), T 127.0 C; 4096 N column = information (J-F1 maximum capped at 3750 N, lead 17:56:12Z) T3576
SOAK T3577	IN718 head on the HW-170 spring, A 39.19 mm2 (IN718 Fty / Ftu, no bearing credit) CALC T3578	+3.657 T3579	+4.534 T3580	4096 T3581	CALC / DERIVED CALC	N 8456.3 (js5), T 127.0 C; spring side TO-SOURCE (Mechanical); 4096 N column = information (J-F1 maximum capped at 3750 N, lead 17:56:12Z) T3582
SOAK T3583	IN718 soak fibre yield (secondary: relaxation / shakedown / LCF, lead 10:14Z) CALC T3584	**−0.398** T3585	–	4096 T3587	CALC / DERIVED CALC	(b1); shakedown 0.518, range 1323.9 MPa; 4096 N column = information (J-F1 maximum capped at 3750 N, lead 17:56:12Z) T3588
SOAK52 T3589	IN718 soak fibre yield (secondary: relaxation / shakedown / LCF, lead 10:14Z) CALC T3590	**−0.201** T3591	–	3750 T3593	CALC / DERIVED CALC	(b2) rep; shakedown 1.14, range 932.1 MPa T3594
SOAK52 T3595	IN718 tension on A_s (234 (1)(b)) CALC T3596	+0.586 T3597	+0.880 T3598	4096 T3599	CALC / DERIVED CALC	N 8918.4 ((b2) rep), T 154.1 C; 4096 N column = information (J-F1 maximum capped at 3750 N, lead 17:56:12Z) T3600
SOAK52 T3601	IN718 head on the HW-170 spring, A 39.19 mm2 (IN718 Fty / Ftu, no bearing credit) CALC T3602	+3.383 T3603	+4.195 T3604	4096 T3605	CALC / DERIVED CALC	N 8918.4 ((b2) rep), T 154.1 C; spring side TO-SOURCE (Mechanical); 4096 N column = information (J-F1 maximum capped at 3750 N, lead ... T3606
SOAK52 T3607	IN718 soak fibre yield (secondary: relaxation / shakedown / LCF, lead 10:14Z) CALC T3608	**−0.247** T3609	–	4096 T3611	CALC / DERIVED CALC	(b2) rep; shakedown 1.037, range 979.1 MPa; 4096 N column = information (J-F1 maximum capped at 3750 N, lead 17:56:12Z) T3612
FA1S T3613	IN718 tension on A_s at FA1S (information beside Eq. 22) CALC T3614	+3.410 T3615	+4.281 T3616	3750 T3617	CALC / DERIVED CALC	N 3302.3 (js5), T 45.6 C T3618
FA1S T3619	IN718 tension on A_s at FA1S (information beside Eq. 22) CALC T3620	+2.992 T3621	+3.780 T3622	4096 T3623	CALC / DERIVED CALC	N 3648.3 (js5), T 45.6 C; 4096 N column = information (J-F1 maximum capped at 3750 N, lead 17:56:12Z) T3624
FA1S T3625	IN718 head on the HW-170 spring, A 39.19 mm2 (IN718 Fty / Ftu, no bearing credit) CALC T3626	+10.032 T3627	+12.211 T3628	4096 T3629	CALC / DERIVED CALC	N 3648.3 (js5), T 45.6 C; spring side TO-SOURCE (Mechanical); 4096 N column = information (J-F1 maximum capped at 3750 N, lead 17:56:12Z) T3630
FA1S T3631	IN718 5020A Eq. 22 at arm +0.4 mm (sensitivity), preload omitted CALC T3632	–	+0.165 T3634	–	CALC CALC	NASA-STD-5020A Eq. 22 T3635 preload included −0.035 (information) T3636

9.6 Mounting data: loads at the 16 airframe holes

Generated 2026-10-20T23:06:59Z by scripts/icd_export_v352.py from the f1.dat reaction totals of the listed runs (SIM unless tagged). 16 radial M5 holes: GM-09 forward ring z +27 mm, GM-10 aft ring z -94.8 mm. Forces of the engine module ON the airframe; radial + = pull-out; shear = hypot(tangential, axial) in the radial screw. SIM T3638

Governing rule: lead 22:54:03Z: governing = the larger of the as-found value and the moment-free estimate (L1 global: OP moment zero through pose_neutral_v352r3R3's EA steps, ASSUMED; R3 pose: the two-face estimate and OP moment zero), per hole and component, then enveloped with the carried v3.5.1 compliant rows; DERIVED DERIVED T3637 The CHANGELOG ruling: OP-MOMENT ARTEFACT AND HOLE LOADS (22:24:09Z, 22:54:03Z, 23:12:52Z): FOUND (FEA3D): the v3.5.2 OP rod forces carry a non-physical OP couple from two TC-09 aft-boundary pressure faces (elements 24664 / 23610 at p/Pc 0.75 where the station carries 1.0; about 1.56 N m/MPa; the physical OP moment is 0); ACCEPTED as an MF observation: the as-found cells stand, with the moment-free statics estimate beside each affected row (op_moment_check_v352.json bf849ad982fb8bb1, op_moment_global_v352.json a7c92f815fb0f55d). RULED: every datasheet hole-load row governs on the larger of as found and the estimate, tagged DERIVED, both shown, '!' where the estimate governs: L1 = max(as found, OP-zero via pose_neutral's EA steps, transfer ASSUMED); R3 = max(as found, two-face, OP-zero); then enveloped with the carried v3.5.1 compliant rows. Governing case maxima, max screw shear [DERIVED]: the estimate governs at AF_GM09_0 for LC1 76, LC2 150, FA1Q 348, FA1D 448, FA1U 544, FA1S 559 and FA1T 657 N (the as-found L1 rows were 3.5-7.8 % low there); as found governs LC3 890, GS 808, TD Pc 0 169, TD Pc max 306, LC6 80 and LC2F 288 N; pull-out FA1T 120 N (icd_momentfree_v352.json 785b4e48f407ddfb; engine_mount_data_v352.json e5f41bde43a00ba4, md ec986ff940cfed33). GM-01: the rod-force path governs its DERIVED bound (ROD_1 with and without the residue), replacing the hole-ratio scaling; the as-found cells stand with the sign-free bound beside them (FA1S +0.077 / +0.238); the hot-bound base residue (about 86 / 66 N rod force at Pc_max) goes beside the cells as worst-sign bounds (LC3 GM-01 +0.052 → +0.031, GM-02 +0.036 → +0.015), ACCEPTED, no sign change. Datasheet line: 'OP rod forces include a non-physical OP couple from two TC-09 boundary faces (model artefact, MF); moment-free statics estimate beside each affected row'. DECIDED *Lead decision, line 249 (6)* T3639

Cell = max shear / max |axial| / max pull-out, N; ! = an estimate governs that value by 0.5 N or more; * = a compliant-airframe value exceeds the rigid governing one by more than 1 N and 5 %. DERIVED T3640

Per-hole governing loads, part 1 of 2: max shear / max |axial| / max pull-out [N] per case; "!" = the moment-free estimate governs that value by 0.5 N or more; "*" = a compliant-airframe value exceeds the rigid one (MOUNTMD MOUNTMD v3.5.2 re-check with RM2-11 folded).

Hole (x, y, z mm)	LC1	LC2	FA1Q	FA1D	FA1U	FA1S	FA1T
AF_GM09_0 (59.4, 24.6, 27.0)	76! / 68! / 20! T3641	150! / 136! / 33! T3642	348! / 320! / 64! T3643	448! / 412! / 82! T3644	544! / 500! / 99! T3645	559! / 514! / 102! T3646	657! / 604! / 120! T3647
AF_GM09_1 (24.6, 59.4, 27.0)	5 / 3 / 0 * T3648	5 / 5 / 0 T3649	10! / 9! / 0 T3650	13! / 12! / 0 T3651	16! / 14! / 0 T3652	16! / 14! / 0 T3653	19! / 17! / 0 T3654
AF_GM09_2 (-24.6, 59.4, 27.0)	3 / 3 / 0 T3655	5 / 5 / 0 T3656	10! / 9! / 0 T3657	14! / 12! / 0 T3658	16! / 14! / 0 T3659	17! / 15! / 0 T3660	20! / 18! / 0 T3661
AF_GM09_3 (-59.4, 24.6, 27.0)	71! / 65! / 14! T3662	143! / 131! / 27! T3663	336! / 310! / 61! T3664	434! / 400! / 78! T3665	527! / 486! / 95! T3666	542! / 499! / 98! T3667	638! / 588! / 115! T3668
AF_GM09_4 (-59.4, -24.6, 27.0)	76! / 69! / 15! T3669	149! / 136! / 28! T3670	346! / 320! / 62! T3671	446! / 411! / 79! T3672	541! / 499! / 96! T3673	556! / 513! / 99! T3674	654! / 603! / 116! T3675
AF_GM09_5 (-24.6, -59.4, 27.0)	3 / 3 / 1 * T3676	5 / 5 / 0 T3677	10! / 9! / 0 T3678	13! / 12! / 0 T3679	16! / 14! / 0 T3680	17! / 15! / 0 T3681	20! / 18! / 0 T3682
AF_GM09_6 (24.6, -59.4, 27.0)	6 / 6 / 0 * T3683	8 / 8 / 0 T3684	10! / 9! / 0 T3685	13! / 11! / 0 T3686	15! / 14! / 0 T3687	16! / 14! / 0 T3688	19! / 17! / 0 T3689
AF_GM09_7 (59.4, -24.6, 27.0)	72! / 66! / 18! T3690	145! / 133! / 32! T3691	343! / 316! / 62! T3692	442! / 408! / 80! T3693	538! / 495! / 97! T3694	553! / 509! / 100! T3695	651! / 600! / 118! T3696
AF_GM10_0 (53.1, 22.0, -94.8)	47 / 47 / 0 * T3697	76! / 72! / 0 T3698	183! / 174! / 0 T3699	236! / 224! / 0 T3700	286! / 272! / 0 T3701	294! / 279! / 0 T3702	347! / 329! / 0 T3703
AF_GM10_1 (22.0, 53.1, -94.8)	19 / 19 / 4 T3704	33 / 32 / 6 T3705	64 / 63 / 12 T3706	81 / 80 / 16 T3707	99 / 97 / 19 T3708	101 / 100 / 19 T3709	68 / 68 / 13 T3710
AF_GM10_2 (-22.0, 53.1, -94.8)	39 / 39 / 1 T3711	76 / 75 / 0 T3712	176 / 175 / 0 T3713	225 / 224 / 0 T3714	273 / 272 / 0 T3715	280 / 279 / 0 T3716	170 / 169 / 0 T3717
AF_GM10_3 (-53.1, 22.0, -94.8)	48 / 48 / 0 * T3718	76! / 73! / 0 T3719	183! / 176! / 0 T3720	236! / 227! / 0 T3721	286! / 276! / 0 T3722	294! / 283! / 0 T3723	346! / 333! / 0 T3724
AF_GM10_4 (-53.1, -22.0, -94.8)	45 / 45 / 0 * T3725	76! / 73! / 0 T3726	184! / 177! / 0 T3727	237! / 228! / 0 T3728	288! / 277! / 0 T3729	296! / 285! / 0 T3730	349! / 336! / 0 T3731
AF_GM10_5 (-22.0, -53.1, -94.8)	20 / 20 / 1 T3732	23 / 23 / 0 T3733	21 / 21 / 0 T3734	25 / 25 / 0 T3735	30 / 30 / 0 T3736	30 / 30 / 0 T3737	27 / 27 / 0 T3738
AF_GM10_6 (22.0, -53.1, -94.8)	3 / 3 / 1 * T3739	4! / 4! / 1 T3740	6! / 4! / 1! T3741	8! / 6! / 1! T3742	10! / 8! / 2! T3743	10! / 8! / 2! T3744	12! / 10! / 2! T3745
AF_GM10_7 (53.1, -22.0, -94.8)	47 / 47 / 0 * T3746	75! / 71! / 0 T3747	180! / 171! / 0 T3748	232! / 220! / 0 T3749	282! / 267! / 0 T3750	290! / 274! / 0 T3751	341! / 323! / 0 T3752

Per-hole governing loads, part 2 of 2: max shear / max |axial| / max pull-out [N] per case; "!" = the moment-free estimate governs that value by 0.5 N or more; "*" = a compliant-airframe value exceeds the rigid one (MOUNTMD MOUNTMD v3.5.2 re-check with RM2-11 folded).

Hole (x, y, z mm)	LC3	GS	TD0	TDP	LC6	LC2F
AF_GM09_0 (59.4, 24.6, 27.0)	306! / 303 / 295 T3753	333 / 298 / 86 T3754	113 / 91 / 88 T3755	186! / 157! / 120! T3756	79 / 69 / 44 T3757	197! / 175! / 75! T3758
AF_GM09_1 (24.6, 59.4, 27.0)	102 / 16 / 20! * T3759	10 / 9 / 5 T3760	6 / 4 / 6 T3761	9 / 6! / 7 T3762	3 / 2 / 2 T3763	7! / 6! / 4 T3764
AF_GM09_2 (-24.6, 59.4, 27.0)	72 / 7 / 5 * T3765	10 / 9 / 1 T3766	2 / 2 / 2 T3767	5! / 4! / 2 T3768	1 / 1 / 1 T3769	5! / 5! / 0 T3770
AF_GM09_3 (-59.4, 24.6, 27.0)	323! / 269 / 94! T3771	338 / 301 / 74 T3772	118 / 95 / 31 T3773	176! / 149! / 48! T3774	78 / 68 / 20 T3775	191! / 170! / 42! T3776
AF_GM09_4 (-59.4, -24.6, 27.0)	332! / 276! / 97! T3777	349 / 311 / 76 T3778	121 / 97 / 32 T3779	181! / 153! / 48! T3780	79 / 69 / 20 T3781	196! / 175! / 43! T3782

Hole (x, y, z mm)	LC3	GS	TD0	TDP	LC6	LC2F
AF_GM09_5 (-24.6, -59.4, 27.0)	36 / 9 / 4 * T3783	10 / 9 / 1 T3784	4 / 4 / 2 T3785	6! / 6! / 2 T3786	2 / 2 / 1 T3787	5! / 5! / 0 T3788
AF_GM09_6 (24.6, -59.4, 27.0)	140 / 20 / 22! * T3789	14 / 12 / 6 T3790	14 / 8 / 8 T3791	17 / 11 / 10 T3792	10 / 9 / 4 T3793	10! / 9! / 4 T3794
AF_GM09_7 (59.4, -24.6, 27.0)	304! / 302 / 296 T3795	329 / 295 / 85 T3796	113 / 92 / 89 T3797	185! / 157! / 121 T3798	80 / 70 / 44 T3799	195! / 174! / 75! T3800
AF_GM10_0 (53.1, 22.0, -94.8)	130! / 116! / 162! T3801	161 / 148 / 23 T3802	50 / 48 / 27 T3803	87! / 83! / 26 T3804	41 / 40 / 15 T3805	98! / 94! / 2 T3806
AF_GM10_1 (22.0, 53.1, -94.8)	329! / 311 / 164! T3807	284 / 274 / 83 T3808	56 / 52 / 17 T3809	109 / 104 / 34 T3810	10 / 8 / 4 T3811	100 / 98 / 31 T3812
AF_GM10_2 (-22.0, 53.1, -94.8)	890 / 889 / 214 T3813	808 / 806 / 98 T3814	169 / 169 / 28 T3815	306 / 304 / 28! T3816	55 / 55 / 10 T3817	288 / 287 / 6 T3818
AF_GM10_3 (-53.1, 22.0, -94.8)	195 / 160 / 217 T3819	162 / 156 / 29 T3820	48 / 44 / 13 T3821	78! / 73! / 11 T3822	39 / 37 / 8 T3823	96! / 92! / 0 T3824
AF_GM10_4 (-53.1, -22.0, -94.8)	196 / 161 / 211 T3825	162 / 156 / 34 T3826	49 / 45 / 14 T3827	80! / 75! / 12 T3828	40 / 38 / 8 T3829	97! / 93! / 0 T3830
AF_GM10_5 (-22.0, -53.1, -94.8)	889! / 889! / 221 T3831	807 / 807 / 108 T3832	163 / 163 / 30 T3833	275 / 275 / 29! T3834	45 / 45 / 9 T3835	233 / 233 / 2 T3836
AF_GM10_6 (22.0, -53.1, -94.8)	322! / 311! / 164! T3837	281 / 275 / 81 T3838	54 / 50 / 16 T3839	98 / 95 / 30 T3840	12 / 10 / 3 T3841	82 / 81 / 25 T3842
AF_GM10_7 (53.1, -22.0, -94.8)	162 / 160 / 163! * T3843	160 / 146 / 22 T3844	52 / 49 / 27 T3845	88! / 84! / 25 T3846	42 / 40 / 15 T3847	98! / 94! / 2 T3848

Resultant of the 16 hole loads on the airframe, rigid airframe, per case (DERIVED).

Case	Total force on the airframe, max |Fx|, |Fy|, |Fz| [N]	Total moment, max |Mx|, |My|, |Mz| [N mm]
LC1	24.4, 24.4, 346.8 T3849	2432, 995, 496 T3850
LC2	24.6, 24.7, 751.6 T3851	3915, 2057, 568 T3852
FA1Q	1.0, 1.3, 1903.5 T3853	7295, 4979, 697 T3854
FA1D	1.3, 1.6, 2450.3 T3855	9299, 6412, 793 T3856
FA1U	1.5, 2.0, 2976.8 T3857	11229, 7793, 886 T3858
FA1S	1.6, 2.0, 3058.5 T3859	11529, 8007, 900 T3860
FA1T	1.0, 1.3, 3600.0 T3861	7341, 5066, 723 T3862
LC3	321.6, 320.7, 506.5 T3863	22897, 11991, 2577 T3864
GS	140.3, 140.5, 514.4 T3865	56100, 50983, 7415 T3866
TD0	294.0, 294.1, 215.2 T3867	8899, 6024, 1514 T3868
TDP	357.2, 356.4, 511.3 T3869	10100, 11757, 1967 T3870
LC6	118.5, 118.5, 395.1 T3871	1436, 1312, 312 T3872
LC2F	137.6, 135.7, 907.7 T3873	2669, 11645, 1427 T3874

Mounting data header	Value
Included FE model mass, v3.5.2 (L1)	**4.033** kg DERIVED T3875
Engine group in the model	**2.829** kg DERIVED T3876
Excluded airframe-wall equipment (loads go to the wall directly)	**1.310** kg DERIVED T3877

9.7 Gimbal axis stiffness (R3 poses) and rocking modes

SIM (FE unit steps on the v3.5.2 R3 pose runs) + DERIVED (servo in series; k_s 58 / 174 N m/rad ASSUMED, TO-VERIFY bench B-TVC) SIM T3878

Axis stiffness K at the 17 R3 poses [N m/rad], coupled values: FE alone (SIM) and with the servo in series at k_s 58 and 174 N m/rad (ASSUMED, TO-VERIFY at B-TVC-1). Fraction = the pose change over the stop045 change (KTAB).

Pose	FE α	FE β	k_s 58: α	k_s 58: β	k_s 174: α	k_s 174: β	Fraction max
stop000 stop (8 deg)	7294 SIM T3879	7172 SIM T3880	1426 DERIVED T3881	1799 DERIVED T3882	3076 DERIVED T3883	3594 DERIVED T3884	1.208 DERIVED T3885
stop045 stop (8 deg)	7403 SIM T3886	6881 SIM T3887	1362 DERIVED T3888	1559 DERIVED T3889	2987 DERIVED T3890	3219 DERIVED T3891	1.000 DERIVED T3892
stop090 stop (8 deg)	7756 SIM T3893	5808 SIM T3894	1190 DERIVED T3895	1124 DERIVED T3896	2732 DERIVED T3897	2431 DERIVED T3898	0.872 DERIVED T3899
stop135 stop (8 deg)	8595 SIM T3900	4569 SIM T3901	1184 DERIVED T3902	871 DERIVED T3903	2785 DERIVED T3904	1892 DERIVED T3905	0.725 DERIVED T3906
stop180 stop (8 deg)	9337 SIM T3907	3972 SIM T3908	1291 DERIVED T3909	791 DERIVED T3910	3034 DERIVED T3911	1697 DERIVED T3912	0.664 DERIVED T3913
stop225 stop (8 deg)	8606 SIM T3914	4268 SIM T3915	1184 DERIVED T3916	860 DERIVED T3917	2786 DERIVED T3918	1839 DERIVED T3919	0.725 DERIVED T3920
stop270 stop (8 deg)	7764 SIM T3921	5443 SIM T3922	1190 DERIVED T3923	1109 DERIVED T3924	2733 DERIVED T3925	2364 DERIVED T3926	0.871 DERIVED T3927
stop315 stop (8 deg)	7402 SIM T3928	6678 SIM T3929	1362 DERIVED T3930	1549 DERIVED T3931	2987 DERIVED T3932	3174 DERIVED T3933	0.998 DERIVED T3934
neutral neutral	8246 SIM T3935	5764 SIM T3936	1150 DERIVED T3937	1085 DERIVED T3938	2697 DERIVED T3939	2364 DERIVED T3940	0.990 DERIVED T3941
alpha+ range cone, single axis (7 deg)	7880 SIM T3942	5501 SIM T3943	1179 DERIVED T3944	1106 DERIVED T3945	2722 DERIVED T3946	2366 DERIVED T3947	0.958 DERIVED T3948
alpha- range cone, single axis (7 deg)	7873 SIM T3949	5823 SIM T3950	1178 DERIVED T3951	1118 DERIVED T3952	2721 DERIVED T3953	2423 DERIVED T3954	0.958 DERIVED T3955
beta+ range cone, single axis (7 deg)	7393 SIM T3956	7005 SIM T3957	1334 DERIVED T3958	1633 DERIVED T3959	2941 DERIVED T3960	3342 DERIVED T3961	1.201 DERIVED T3962
beta- range cone, single axis (7 deg)	9216 SIM T3963	4223 SIM T3964	1263 DERIVED T3965	826 DERIVED T3966	2974 DERIVED T3967	1782 DERIVED T3968	0.764 DERIVED T3969
combined045 cone pose, alpha and beta both non-zero (7 deg)	7508 SIM T3970	6744 SIM T3971	1290 DERIVED T3972	1456 DERIVED T3973	2880 DERIVED T3974	3051 DERIVED T3975	1.081 DERIVED T3976
combined135 cone pose, alpha and beta both non-zero (7 deg)	8627 SIM T3977	4754 SIM T3978	1185 DERIVED T3979	899 DERIVED T3980	2788 DERIVED T3981	1958 DERIVED T3982	0.820 DERIVED T3983
combined225 cone pose, alpha and beta both non-zero (7 deg)	8636 SIM T3984	4492 SIM T3985	1185 DERIVED T3986	890 DERIVED T3987	2789 DERIVED T3988	1912 DERIVED T3989	0.821 DERIVED T3990
combined315 cone pose, alpha and beta both non-zero (7 deg)	7508 SIM T3991	6560 SIM T3992	1290 DERIVED T3993	1448 DERIVED T3994	2880 DERIVED T3995	3013 DERIVED T3996	1.077 DERIVED T3997

Lumped 2-DOF rocking frequencies (ROCK; rigid airframe). Servo J values: J_low: 0.00076; J_nom: 0.0016; J_high: 0.0027 T4006.

Servo stiffness and inertia case	Rocking-mode range over the 17 poses
k_s 58 N m/rad, J free	18.3 – 29.8 Hz DERIVED T3998
k_s 58 N m/rad, J low	16.6 – 23.0 Hz DERIVED T3999
k_s 58 N m/rad, J nom	15.2 – 19.1 Hz DERIVED T4000
k_s 58 N m/rad, J high	13.8 – 16.2 Hz DERIVED T4001
k_s 174 N m/rad, J free	26.8 – 40.3 Hz DERIVED T4002
k_s 174 N m/rad, J low	24.2 – 31.1 Hz DERIVED T4003
k_s 174 N m/rad, J nom	22.2 – 27.2 Hz DERIVED T4004
k_s 174 N m/rad, J high	20.1 – 24.0 Hz DERIVED T4005

9.8 Secondary joints J-FD01 / J-FD09 (F4)

F4 secondary joints on the v3.5.2 hardware (F4J; preload scatter ±25 % and K 0.20 ASSUMED). The matrix cells J-FD01 / J-FD09 are in Table 9.3.

Joint	Hardware	Torque / nominal preload	Variant / area	Worst bolt MS_y (bounding)	MS_y with Φ 0.2
J-FD01	HW-103 2x M4×20 DIN 7984 A4-80 (lower, y +/-8.5, z 146.51, engagement 8.5) + HW-103B 1x M4×16 (upper, y 0, z 165.51, engagement 5.5), 1.8 N m M-T1, into the TC-02 flat x -29.5 (hardware_v352.json) DECIDED *Design value, f4_v352.json v1* T4007	**1.8** N m DECIDED *Design value, f4_v352.json v1* T4008 / **2250** N DERIVED T4009	α_F4_bounding: R7 bounding T4012	**+0.121** SIM T4010 LC3 hard-landing envelope (10 g) T4011	**+0.403** SIM T4013
J-FD01	HW-103 2x M4×20 DIN 7984 A4-80 (lower, y +/-8.5, z 146.51, engagement 8.5) + HW-103B 1x M4×16 (upper, y 0, z 165.51, engagement 5.5), 1.8 N m M-T1, into the TC-02 flat x -29.5 (hardware_v352.json) DECIDED *Design value, f4_v352.json v1* T4007	**1.8** N m DECIDED *Design value, f4_v352.json v1* T4008 / **2250** N DERIVED T4009	b_bolt_group_vector: R7 bounding T4016	**+0.128** SIM T4014 LC3 touchdown (5.43 g SIM) T4015	**+0.405** SIM T4017
J-FD09	HW-127 2x M4×12 ISO 4762 A4-80, 1.8 N m M-T1, into GM-09 with helicoil 1.5 D (thread check as a direct M4 tap in 6061, 6.0 mm: conservative) DECIDED *Design value, f4_v352.json v1* T4018	**1.8** N m DECIDED *Design value, f4_v352.json v1* T4019 / **2250** N DERIVED T4020	0.184: EI 0.184 T4023	**+0.476** SIM T4021 LC3 hard-landing envelope (10 g) T4022	**+0.660** SIM T4024
J-FD09	HW-127 2x M4×12 ISO 4762 A4-80, 1.8 N m M-T1, into GM-09 with helicoil 1.5 D (thread check as a direct M4 tap in 6061, 6.0 mm: conservative) DECIDED *Design value, f4_v352.json v1* T4018	**1.8** N m DECIDED *Design value, f4_v352.json v1* T4019 / **2250** N DERIVED T4020	0.5: EI 0.5 T4027	**+0.221** SIM T4025 LC3 hard-landing envelope (10 g) T4026	**+0.586** SIM T4028

9.9 Mechanical follow-ups after FEA3D-2: tap wall v3, LC-TB70 v2, C1 seat

Tap wall v3 (TAPW TAPW v3); R4-01 CLOSED in CHANGELOG 259 (1).

TC-02 torch-tap wall at FA1S (R4-01)	Value
Governing (TC-02 thread, SOAK52)	**+0.670** CALC T4031 / **+0.785** CALC T4032
Screw-temperature sensitivity (SOAK70)	**+0.275** CALC T4033 / **+0.365** CALC T4034
Verdict	Tap wall to the torch passage (az 18 / 342), wall 1.6355 mm net at +/-0.25 deg, CALC on SIM at the 3750 N J-F1 cap, P_torch = FA1S 151.04 bar abs uniform over the passage at every state: governing SOAK52 (N 8572.4 N from (b2) rep, p_thread 48.21 MPa, passage 15.003 MPa gauge): sigma 136.9 MPa, MS_y +0.670 / MS_u +0.785 at the TC-02 thread temperature 94.9 C; with the screw temperature as a sensitivity MS_y +0.275 (SOAK70, 158.6 C). v1 (P_torch 102 bar abs, rows v7): MS_y +0.814 / screw T +0.389. Flags within 0.05 or negative (3750 column, all walls): 0; 4096 information column: 0. R4-01: ligament lowest local natural frequency >= 26 kHz on the extreme bounding strip (CALC, ASSUMED edges), far above the passage (<= 3.45 kHz): DLF 1.0 on the passage peak; NOT MODELLED: wave steepening and shock reflection, losses, torch-chamber compliance, N2O decomposition, a blocked-outlet ... CALC T4030
Status	CALC on SIM; ASSUMED conservative ligament model; negatives SHOWN AS FOUND; not physical approval CALC T4029

LC-TB70 envelope v2 (TB70 TB70 v2); ruled in CHANGELOG 258 (3).

LC-TB70 envelope v2	Text (quoted)
Z3 v2 (ruled)	Z3 (LC-TB70 v2): 63.1 < P_head <= 88.1 bar abs inside the light window (record start before the light to 150 ms after the first crossing of 8 bar abs), with the undamped SDOF ESP of the head-end pressure on that window <= 151.04 bar abs (FA1S) at every period 0.02-50 ms (the published 406-period grid; from rest at the record start, g = P - 1.01325 bar piecewise linear and held after the window, the larger of g clipped at 0 and unclipped; sample step <= 0.05 ms), the window above 63.1 bar abs <= 3.0 ms and the 10-90 % rise >= 0.8 ms; REV-COMB-4 Z3' adopted as written with the two v1 shape conditions kept; exact on SIM traces, conservative on measured traces (the undamped test integrates noise: a no-event stand record is checked first, TO-VERIFY); 896 of 896 FA-1 grid cases inside (ESP max 146.881 at 3.5 ms, margin 4.159 bar; peak max 88.02; window max 1.6693 ms; rise min 0.8419 ms) [DERIVED on SIM] DECIDED *Lead decision, per lc_tb70_envelope_v2.json v2* T4035
J-FWD replace rule on Q_JFWD	J-FWD replace rule (LC-TB70 v2): Q_JFWD > 60.0 bar abs at any time in a run (Q_JFWD = the larger of the head peak and the head ESP at the J-FWD periods 0.2-0.5 ms; no minimum duration): the joint counts as used; before the next run the line-231 J-FWD inspection (C6 (2) list: the ten ENG-HW-101-IN718 screws and their HW-170 springs replaced, the TC-01 holes pin-gauged GO 5.50 / NO-GO 5.62, hole edges and bore at 10x, S01 / S02 replaced, removed screws per C6 (3)) and the TC-02 taps gauged (M5-6H GO / NO-GO). Basis: at 60.0 bar abs every J-FWD row meets the nominal factors at 78.6 C (yield slot 9.0 MPa = js5b LC2U, minimum MS_y +0.328 TC-01 ligament (LC2U); ultimate slot 12.0 MPa minimum MS_u +0.312 TC-01 ligament trend 78.6 C); above it only the SF 1.0 FA-1 rows (line-231 one-off). Not meant for normal lights (Z0, <= 45 bar abs: a rise to 45 bar abs trips it only below a 0.2879 ms 10-90 % rise); 298 of the 896 FA-1 hard-start bracket cases exceed it (185 by the peak alone) [SIM grid count on an ASSUMED construct, not a rate of normal lights]. Signal: SIL plant head pressure (exact); stand B-DPC-1 DC-restored with C1 (after a no-event noise check, TO-VERIFY); flight C1 LIGHT_SPIKE raw peak (lower bound) against Electronics' chain-model threshold (TO-VERIFY; ideal sampler 40.381 bar abs). DECIDED *Lead decision, per lc_tb70_envelope_v2.json v2* T4036
F4 TB-state abort set	CONFIRMED, with one addition: PC_SENSOR raised on the HIGH side (transducer output >= 4.75 V = 146.5 bar abs: the stand latches and aborts; in flight the non-sticky over-range bit, log v12 ovr_flags b0) CALC T4037
Status	DERIVED from the analysed structural cases and the Combustion v2.4b FA-1 grid (SIM on an ASSUMED construct; grid counts are not probabilities); margins are the published tables, negatives SHOWN AS FOUND; not physical approval CALC T4038

9.10 FEA3D-2 open items and owners

FEA3D-2 §13 as published (05:45Z). **Item 1** quotes the select-fit band of line 240 (10) and its one re-size by Mechanical [superseded: R2-B [**0.040** T526, **0.0631** mm DECIDED *Lead ruling R2-B (2026-10-21T07:37:34Z), quoted verbatim in Mechanical line 254 (1)* T527], 254 (1) T525; band re-size done, band v1 · sha256/16 1ba3147b42edfeed DERIVED T528] Closure: R2-B select fit (254 (1) T525); stays a found negative at the drawn fit (contradiction register C-11, §13.3). **Item 2** quotes **+0.301** CALC T4111 for the HW-101 J-FWD u cell: the rows-v8 matrix of record gives **+0.278** CALC T4112 (§9.5.3; contradiction register C-01, §13.3).

#	Item	Status / numbers	Owner	Closure
1	TC-01 J-FWD seat / ring line P+Q (secondary, shakedown; class DC) DERIVED T4039	found negative on the final envelope v6 `43a5d78b67c615fc` (lead 05:17:36Z (3); the seat lines are those of v5): SOAK -0.071 ((b2) rep), SOAK70 -0.055 ((b2)), SOAK52 -0.030 ((b1) at C1) (P+Q <= 2 Fty(T), A4-80 FE; the worst computed instance, not a proven bound; the contact solution depends on the predecessor state, 10A.5). Not shown converged with the mesh (MF note with cD: s9 → s6 +12 to +18 %, s6 → cD -1.5 to ... SIM T4040	Mechanical (IN718 re-map once on the final envelope v6 with `jfwd_v7inputs_cD_v352.json`, lead 05:27Z (3); select-fit band re-size from the P+Q-vs-gap sensitivity, lead 00:39:30Z (1), (4)) DECIDED *Lead decision, per REPORT-FEA3D-v352.md FINAL handback of the v3.5.2 re-check* T4041	the band inside [0.040, 0.0761] mm that closes it, or the seat-line P+Q stays a found negative in the datasheet; the HW-170 disc-spring route (i) and the preload-retention audit (240 (10)) DERIVED T4042 [superseded: R2-B [**0.040** T526, **0.0631** mm DECIDED *Lead ruling R2-B (2026-10-21T07:37:34Z), quoted verbatim in Mechanical line 254 (1)* T527], 254 (1) T525; band re-size done, band v1 · sha256/16 1ba3147b42edfeed DERIVED T528]
2	HW-101 IN718 screw rows on the final envelope DERIVED T4043	Eq. 22 inputs V 4079.2 N (FE, (b1)) and M 4354.7 N mm (CALC) (lead 01:30:18Z (6)); the cell quotes +0.301 at V 3888.1 N until the re-map; Mechanical's soak fibre rows negative (SOAK -0.257 at 3750 N; 10A.8); (b1) moments as solved, M x 1.404 / x 1.476 as information (lead 02:01:34Z (B)) SIM T4044	Mechanical DECIDED *Lead decision, per REPORT-FEA3D-v352.md FINAL handback of the v3.5.2 re-check* T4045	the re-map on the final envelope v6; the relaxation / shakedown / LCF rows DERIVED T4046
3	TC-02 TB70 / TBS (n30266, 228.7 C; TS-212) DERIVED T4047	-0.236 / -0.271; +0.138 / -0.035 [SIM] SIM T4048	Mechanical / founder DECIDED *Lead decision, per REPORT-FEA3D-v352.md FINAL handback of the v3.5.2 re-check* T4049	212 (1) (d) time at temperature with T-F1; Mechanical's (b) package DERIVED T4050
4	The seat region at TB (4.3) DERIVED T4051	Mechanical's S-AX v352r4 fails line 106 at SF 1.0 at TB70 (TC-02 -0.326 / -0.371; TC-03 undercut root -0.191 / -0.264) [CALC / SIM, Mechanical] SIM T4052	Mechanical DECIDED *Lead decision, per REPORT-FEA3D-v352.md FINAL handback of the v3.5.2 re-check* T4053	T-F2 contact runs and the abort search DERIVED T4054
5	TC-03 LC1 / LC2 / LC3 / F2 hot bound / TB70 (TS-183) DERIVED T4055	-0.088 / -0.170 ... -0.121 / -0.201 [SIM / DERIVED]; S-AX governs +2.564 / +2.241 (lead 07:51Z (a)) SIM T4056	Mechanical DECIDED *Lead decision, per REPORT-FEA3D-v352.md FINAL handback of the v3.5.2 re-check* T4057	183 (1) (c) LCF and (d) GlidCop time at temperature DERIVED T4058
6	FD-01 bonded-patch edge (TS-183) DERIVED T4059	LC6 soak -0.187 / +0.780, TB70 -0.336 / +0.484, TBS -0.005 / +0.976 (A 175 C, hotter than the adopted 150 C) [SIM] SIM T4060	Mechanical DECIDED *Lead decision, per REPORT-FEA3D-v352.md FINAL handback of the v3.5.2 re-check* T4061	183 (1) (c) LCF; (d) on the sourced long-time allowable where T <= 150 C, else T-F1 DERIVED T4062
7	NZ-01 graphite (BR) DERIVED T4063	-0.520 ... -0.054 (yield only, brittle, thermal) [SIM]; S-AX v352_NF2b is the named sub-model SIM T4064	Mechanical / Combustion DECIDED *Lead decision, per REPORT-FEA3D-v352.md FINAL handback of the v3.5.2 re-check* T4065	183 (2) capture / containment plus a test basis, else a design change DERIVED T4066
8	NZ-03 char (CH) DERIVED T4067	LC2 -0.088, TB70 -0.392, TBS -0.253 (n81357) against the ASSUMED 20 MPa SIM T4068	founder / Mechanical DECIDED *Lead decision, per REPORT-FEA3D-v352.md FINAL handback of the v3.5.2 re-check* T4069	N-F1 (>= 21.96 MPa inside the 23.7 MPa acceptance); 188 (3) contact FE at the tie edge DERIVED T4070
9	PD-01R / PD-01S (MF) DERIVED T4071	-0.315 / -0.451 ... -0.444 / -0.555; primary-only above +20 SIM T4072	fea3d (model) DECIDED *Lead decision, per REPORT-FEA3D-v352.md FINAL handback of the v3.5.2 re-check* T4073	PD-02 is rigid in the model; a deformable PD-02 if the lead asks for it (lead 07:51Z (e), 08:04Z) DERIVED T4074
10	TC-04 BT node n42993 (q 0.040 / 0.027) DERIVED T4075	LC3 hot bound +0.228 / -0.004, F2 hot bound +0.223 / -0.008 [DERIVED node-wise lower bound]; regular nodes within 2 mm >= +0.643; a found negative in the datasheet SIM T4076	lead (ruled 2026-10-21T03:30:46Z (2): the settling route NOT RUN, final, for disk) DECIDED *Lead decision, per REPORT-FEA3D-v352.md FINAL handback of the v3.5.2 re-check* T4077	the settling route (remesh TC-04 to q >= 0.1, re-solve the hot-bound bases and the 17 R3 poses, shared with item 11; about 2.7 h ccx, about 2.50 GiB free at each start) if disk is ever granted; until then the cell stays negative, class BT DERIVED T4078
11	OP-moment residue (MF observation, 4.5, 9B.3) DERIVED T4079	L1 Q [539, 484] N mm/MPa; R3 poses [304-307, 2298]; cells stand as found with the moment-free estimates beside them; datasheet hole loads take the larger value (8) SIM T4080	lead (ruled 03:30:46Z (2): NOT RUN, final, for disk) DECIDED *Lead decision, per REPORT-FEA3D-v352.md FINAL handback of the v3.5.2 re-check* T4081	a moment-free OP step on the shared route of item 10, if disk is ever granted DERIVED T4082
12	J-FWD contact history dependence (10A.4, 10A.5) DERIVED T4083	the soak states depend on the predecessor state at mu 0 and mu 0.2; carried as the envelope over the computed histories SIM T4084	fea3d / lead DECIDED *Lead decision, per REPORT-FEA3D-v352.md FINAL handback of the v3.5.2 re-check* T4085	a history-robust contact model (mortar, or a softened penalty at the ring) is not run and not proposed for this schedule DERIVED T4086

#	Item	Status / numbers	Owner	Closure
13	J-FWD on the FINAL TC-01 (2.1, 10A.2) DERIVED T4087	all J-FWD FE except cD on the interim TC-01 (d10.5 flats; the final d11.2 flats are coplanar); cD ran on the FINAL TC-01: cD / s6 0.927-0.985 on the governing line, s6 + C1 stand; mesh and geometry not split (10A.6) SIM T4088	fea3d / Mechanical DECIDED *Lead decision, per REPORT-FEA3D-v352.md FINAL handback of the v3.5.2 re-check* T4089	effect on the other J-FWD rows ASSUMED small; Mechanical cites the same basis in its re-map DERIVED T4090
13a	LC2 hand form, TC-01 ligament (information, 10A.13) DERIVED T4091	LC2N Pm+Pb +0.016 / -0.183, LC2M +0.266 / +0.021 with SF 1.5 / 2.0 on the total stress including the preload (about 129 of 151.6 MPa) [SIM / DERIVED; linear superposition ASSUMED]; criterion = FE-factored LC2Y / LC2U +0.421 / +0.424 (rule (A), lead 03:30:46Z (1)) SIM T4092	lead (ruled) / datasheet owner DECIDED *Lead decision, per REPORT-FEA3D-v352.md FINAL handback of the v3.5.2 re-check* T4093	shown as information beside the cells (10A, the datasheet J-FWD table); no closure needed under the rule DERIVED T4094
14	Bearing, washer and screw datasheets DERIVED T4095	HK0808 C0 / C, PTFE washers: TO-VERIFY; loads in 6.1 and 8 SIM T4096	Mechanical DECIDED *Lead decision, per REPORT-FEA3D-v352.md FINAL handback of the v3.5.2 re-check* T4097	datasheet values into the hub (none invented here) DERIVED T4098
15	7075-T6 temperature table (RM2-10, RM1-06) DERIVED T4099	allowables temperature-independent 470 / 540 MPa [LITERATURE] SIM T4100	Mechanical DECIDED *Lead decision, per REPORT-FEA3D-v352.md FINAL handback of the v3.5.2 re-check* T4101	the 188 (5) table and the mill certificate DERIVED T4102
16	Servo output stiffness and inertia DERIVED T4103	k_s 58-174 N m/rad and J 0.76-2.7e-3 kg m2 ASSUMED; rocking 14.9-29.0 Hz at neutral (9A) SIM T4104	Electronics / bench B-TVC-1; REV-GNC DECIDED *Lead decision, per REPORT-FEA3D-v352.md FINAL handback of the v3.5.2 re-check* T4105	bench measurement DERIVED T4106
17	Runs not made (section 11) DERIVED T4107	c71x tie-impact runs and the J-FWD spring-footprint run (NOT RUN: no space after js5b, lead 18:50:09Z (5)); the TC-04 / OP shared route (NOT RUN, final, 03:30:46Z (2)); a compliant-airframe v3.5.2 run; a solved 21.5 s state; an R4 lever-root mesh; the modal on the production mesh (8 GB cap) SIM T4108	fea3d (if disk space or memory is granted) DECIDED *Lead decision, per REPORT-FEA3D-v352.md FINAL handback of the v3.5.2 re-check* T4109	as section 11 DERIVED T4110

9.11 Not run, superseded, and runs that wait for the lead

FEA3D-2 §11: nothing in this list is a margin.

Item (quoted)
c71x (the tie-impact runs `pose_stop045_c71x`, `pose_neutral_c71x`, `lc1v351_c71x`; CAD v3.5.1 with tiex): NOT RUN (lead 16:20Z (3), 18:50:09Z (5)). It could run only if disk space appeared after js5b; none did, and the tail F ended 2026-10-21T03:18:21Z with 'spring run and c71x NOT RUN'. Consequence: the K change between the v3.5.1 and v3.5.2 models (section 9) is not split between the asm v1.3 interfaces, the TC-09 mesh level and the CAD change. DERIVED T4113
The J-FWD spring-footprint run (route (i) for the flat bearing under the screw head): NOT RUN (lead 18:50:09Z (5)). It was ranked after (b1) and js5b and could run only if space appeared after js5b; none did (tail F end line 03:18:21Z). The flat-bearing row stays Mechanical's IN718 row on A_ROW (10A.11). DERIVED T4114
The shared re-solve route for the TC-04 BT cells (hot bounds -0.004 / -0.008 u at n42993, section 4) and the OP-moment residue of the F2 cells (section 9B): remesh TC-04 to tet quality q >= 0.1, re-solve the 2 hot-bound bases x 4 states and the 17 R3 poses with a moment-free OP step, then re-post, lc3_bound and the matrix. Cost [DERIVED from the logged runs]: about 2.7 h ccx, 3.1 h wall, peak RSS 5.64 GB, about 0.46 GB transient per run and 0.49 GB held. **NOT RUN, final** (lead 2026-10-21T03:30:46Z (2); earlier 22:24:34Z, 21:47:29Z (4)). Each run needs about 2.50 GiB free at its start and would end under the 1.86 GiB floor; the route leaves free space about 0.95 GB lower, and nothing ... DERIVED T4115
js5c (the TC-01 / TC-02 fit contact): dropped. The fit closure is CLOSED on the Lame bound (lead 10:36Z, 240 (7); section 10B.7). DERIVED T4116
No solved 21.5 s thermal state: the 21.5 s TC-02 rows are constant-stress [DERIVED] (10B.6). A solved state needs a 21.5 s S-AX L1 state from Mechanical, or a direct thermal-frame map plus one LC1 solve. DERIVED T4117
No LC1 on the v3.5.2 NF2b merged field: the v3.5.1 merged map stands for TC-02 / TC-03 as the perfect-contact information bound (10B), not on NZ-01. DERIVED T4118
No R4 lever-root mesh: where R2 → R3 changes by more than 5 %, the cell quotes the p = 1 bound (5.2). DERIVED T4119
No compliant-airframe v3.5.2 run: the datasheet rows carry the v3.5.1 compliant rows, marked c351 (8.1). DERIVED T4120
The runs that would isolate the stop045 K change (`pose_stop045_v352L1`, `v352L1x`, `v352`) no longer hold fields (9). DERIVED T4121
lc1v352hb (rev-1 lever, R3 hot-bound base, default S-AX map; DONE 04:01:54Z): **superseded, never posted.** GVLO / GVHI (about 04:15Z) and the rev-3 bases hbg3 / hbh3 (about 05:55Z) replaced it before any post. Its fields were pruned at 08:53:19Z as uncited (`runs/PRUNED_v352.txt` lines 23-30). Nothing uses it. ACCEPTED as found, not re-run (lead 21:47:29Z (2)). DERIVED T4122
The R2 / R3 stop045 runs without a check_fbd JSON are ACCEPTED as found (lead 21:47:29Z (2)). stop045 is the K band reference; its mirror pose stop315 corroborates it (fractions 0.955-0.998, fbd 0.0087 N; section 9). DERIVED T4123
`modal_v351ca2b_c71` is superseded by `modal_v351ca2c_c71` for RM2-06 only. The 9A mode list stands. DERIVED T4124
The v3.5.1 K files (`k_stop_c71`, `k_cone_c71`, `k_range_v351`) are superseded for every with-servo value (9). DERIVED T4125

10 Electronics and control

electronics.json v4.1 (firmware v2.12, SIL only) is the published interface and the source of record of this section (CHANGELOG 266 (7)). Line 266 governs v4.1 where they differ. The labels at the values: SUSPENDED UNTIL v2.13 (266 (6)) and ACCEPTED (REV-ELEC-4, line 266), as listed in 266 (6); every other value keeps its own evidence tag. Where line 266 (7) corrects a v4.1 item, the corrected text is printed at the item, quoted from the REV-ELEC-4 review as 266 (7) applies it. SIL and CALC evidence only: no hardware exists.

10.0 Status of electronics.json v4.1 under line 266

CHANGELOG line 266 (lead). Each item is a lead decision; the numbers it quotes keep their own tags. Not physical approval.

Item	Text of CHANGELOG line 266 (quoted in full)
Promotion (266 (7))	PROMOTED 14:45:39Z: interfaces/electronics.json = staging_v2.12 results/electronics_v4.1.json c862375163393adc, byte for byte (v4 11f7896ad0d0613b stays as staging_v2.11/results/electronics_v4.json). Firmware v2.12 and its SIL are the design baseline under the carve-out of (6). DECIDED *Lead decision, line 266 (7)* T023
REV-ELEC-4 verdict (266 (4))	REV-ELEC-4 (independent delta review of Electronics v2.12; interim E4-01 / E4-02 13:57Z; closed 14:34:56Z and returned 14:35:54Z by its stamps, delivered 14:38:25Z): ACCEPT WITH COMMENTS, with option (b) carved out of every start with ignition until a reviewed v2.13 delta; 0 BLOCKING, 1 MAJOR (E4-01), 5 MINOR (E4-02 to E4-06), 11 NOTES DECIDED *Lead decision, line 266 (4)* T029
Carve-out (266 (6), verbatim)	Option (b) (seat coefficient loaded, setPcSeatCoef != 0) is not used for any start with ignition (pad first start, stand starts, relights) until a v2.13 delta with the light-window hold and the TC4 step / rate checks is reviewed. The coefficient may be loaded for B-PC-1 / B-TC-1 bench work without ignition. DECIDED *Lead decision, line 266 (6)* T024
Consequences (266 (6))	Consequences under the rulings in force: A1 FLIGHT IS NO-GO until v2.13 is reviewed (264 (2): the A1 pad is no-go unless the coefficient is loaded and TC4 is valid at ARM); stand first starts may run on the fallback only within 264 (2) (k <= 1 default build, k <= 1 B-PC-1-shown build, with k measured at B-PC-1 first; F-10 MISSING); no stand relight (the fallback refuses it NOT_VENTED [SIM]), so the T4 relight waits for v2.13. Option (b) stays the design: it is the only configuration that keeps the A1 relight at k = 3 [SIM S13T]. DECIDED *Lead decision, line 266 (6)* T025
SUSPENDED UNTIL v2.13 (266 (6))	Suspended until v2.13, and marked so wherever shown: the option (b) firmware rows, residuals and S13T outcomes; the klgnBaseDrift 0.06 bar basis for any relight; the evidence (not the decision) of 264 (3), to be re-shown on v2.13; the A1 pad seat row; the stand relight after B1; the B-PC-1 / B-TC-1 TC4 rows. DECIDED *Lead decision, line 266 (6)* T026
ACCEPTED (REV-ELEC-4, line 266)	Accepted now: purge FINAL 2.43 / 2.39 s with rule (d), (d1), (d2); TB-state; JFWD_LS; kPtorchRezeroMax_Pa 0.60 bar (re-base OPEN, 264 (4)); log v13; klgnBaseCheck_Pa 0.76 / 0.46 bar; the U330 band rows; R4-02, R4-09, R4-13; F-13, F-14, B-MV-1; the T1-T4 planning rule. DECIDED *Lead decision, line 266 (6)* T027
Corrections that line 266 (7) applies to v4.1	Until v4.2 this line governs v4.1 where they differ: its STAGED status text is superseded; TC4 is the C1 seat TC (MAX31856 U904, J904, BOM row 205), not bore top (the 264 (2) rename is made by this line; the file keeps the old label until v4.2); the reviewer's corrections apply as written in the review: E4-02 (the coefficient-within-5-percent-at-k-3 acceptance, also in the 264 (2) wording, is replaced by the k-dependent limit of its Q3 table, eps_max 11.2 / 8.1 / 6.0 / 3.4 / 1.8 / 0.8 % at k = 1 / 1.25 / 1.5 / 2 / 2.5 / 3 with the 0.1012 bar adders [DERIVED]; beyond it the stand relight after B1 is planned REFUSED NOT_VENTED, fail-safe, with no relight objective claimed), E4-03 (canArm subtracts the static term twice after an earlier ARM; fixed in v2.13), E4-04 (the U330 ordering gains the stand relight light: the U330 low edge falls below the 30 bar abs firmware abort from k = 0.725 at T0 24 C and from k = 1.21 at T0 0 C [DERIVED]), E4-05 (1)-(9) (the stale limit bases, the missing limit rows, the pad go / no-go seat and JFWD_LS rows, labels, BOM v2.4 eb6e77dcd4610383, the 264 (1)-(8) folds such as P_TORCH XTL-190SM-2000A, and the purge mass basis DERIVED 71.1-75.8 g with the SIL 76.38 g an ESTIMATE) and E4-06 (bench plan v5: A1 pad NO-GO until v2.13; T4 restated as a relight-latency demonstration from a soaked state, its length from Combustion's T4 re-run, waiting for v2.13; BOM row 187 relabelled; the C7 stand-DAQ TC added as a stand-only row). Bench plan v5 does not go to the founders before these corrections. DECIDED *Lead decision, line 266 (7)* T028
E4-01, the MAJOR finding (266 (5))	E4-01 (MAJOR) [SIM reviewer SIL variant; CALC probes]: with option (b) the seat compensation acts in every state, is not frozen in the light window and has no TC4 step or rate check, so a TC4 reading error after E1 gives a false or premature E2 at every start. The unlit cold-flow rise is about 0.42 bar, about 0.38 bar short of the 0.80 bar E2 rise, which an error of about 15 K at k = 1 or 5 K at k = 3 closes [DERIVED]; 13 of 44 unlit relight runs reach FALSE_E2 and 33 of 64 lit runs lose the relight; at A1 k = 3 with a -30 K error the gnc v3.2 release predicate holds from the false E2 and run-lift flow enters the unlit chamber for 280 ms (a drogue release with no thrust; the FC is not modelled). Credible causes: spark-coupled interference on the grounded-junction TC at the C1 boss, a lead short, a MAX31856 conversion error. DECIDED *Lead decision, line 266 (5)* T032
Queued and routed (266 (8))	QUEUED, NOT STARTED (the founder says when): the Electronics v2.13 delta and its review: the light-window hold (latched at IGNITION entry until E2 + 0.45 s, bumpless resume), the TC4 step check (1.5 K per conversion) and rate check (5 K/s slew, 3 K trip) with a latched f3 SEAT_STEP, the E4-03 fix, unit tests UT-1 to UT-4, the S13T TC-fault set, the FMEA row, the B-TC-1 TC4 row with the spark on, the v4.2 folds, and notes E4-N01 (T_cal 10-30 C or kPcSeatArmDevMax_K 40 K), E4-N02 (the 264 (2) pad rule and the 264 (7) inhibit implemented), E4-N03 (47.5 against 47.75 bar abs, TO-VERIFY), E4-N04 (the P_torch re-base needs compensation or a photodiode-guarded gate; torch local heat MISSING) and E4-N11 (a SIL MANIFEST_v2). DECIDED *Lead decision, line 266 (8)* T030 Routed: E4-N08 (the hardware both-open backstop allows up to 2.741 s against Combustion's 2.39 s never-above value) to Combustion and the lead at B-BLEED-1; the FC rule for SEAT_STEP before a relight and the compensated-Pc release predicate are GNC and lead items with v2.13. DECIDED *Lead decision, line 266 (8)* T031

10.1 Engine control unit and firmware

Item	Value
Interface status	DESIGN + SIL + schematic, ~~STAGED for the lead (06-electronics/staging_v2.12/; interfaces/electronics.json v4 stays the published file until the lead promotes this one)~~ PROMOTED 14:45:39Z (v4.1 is the published file); the STAGED status text is superseded until v4.2 **OVR-10**. Firmware v2.12 runs only in SIL (no Teensy HAL): the REV-ELEC-3 R3-01 option (b) seat compensation of the C1 Pc reading on the working basis given at 11:25Z (LEAD 261 (3); the lead confirms or changes it, the fallback and decision (3) at this handback), the TB-state of the LC-TB70 v2 F4 set with PC_SENSOR HIGH, the flight J-FWD indication, the purge FINAL as calibration constants with rule (d), log v13. The ECU v2.5 schematic does NOT show the v2.11 / v2.12 board items (schematic v2.6 NOT drawn); no board has been built and nothing here is hardware-validated or physical approval. Values carry tags; open items are in open_issues. DECIDED *Design value, electronics.json v4.1* T4126 [266 (7) T4137: the STAGED status text is superseded; v4.1 is the source of record; contradiction register C-02]
Architecture	separate engine ECU (Teensy 4.1) on the fixed thrust structure, CAN link to flight computer DERIVED T491
Microcontroller	PJRC Teensy 4.1 (i.MX RT1062 600 MHz, native SDIO microSD, 3 CAN incl. 1 CAN-FD) DECIDED *Design value, electronics.json v4.1* T4127
PCB outline / enclosure reservation	**120, 60, 1.6** mm DECIDED *Design value, electronics.json v4.1* T4128 / **128, 68, 32** mm DECIDED *Design value, electronics.json v4.1* T4129
ECU mass	**0.16** kg ASSUMED T489
Schematic	version: v2.5 (KiCad 7.0.11, generated from hardware/kicad_gen/ecu_design.py); erc: v: 0 errors, 0 warnings, 9 sheets (Tue Oct 13 13:58:00 2026); tag: KiCad eeschema ERC (GUI on Xvfb); self-consistency only (RE-15); netlist_crosscheck: v: components 471, nets 288, pins 1314 of 1301 intended, errors 0, warnings 0; tag: check_netlist.py v1; negative_control: v2.3: PREG PWR_FLAG removed → 1 error power_pin_not_driven U802 pin 1 (erc/negative_control_v2p3); not repeated for v2.5; files: ... DECIDED *Design value, electronics.json v4.1* T4130
Firmware version	v2.12 DECIDED *Design value, electronics.json v4.1* T4131
Firmware target	PJRC Teensy 4.1 (i.MX RT1062). HAL NOT written (RE-06; LEAD 166 (9): Electronics open item, required before B-PC-1). The core runs in SIL and test_core only TO-VERIFY T4132
Firmware unit tests (the v2.11 record, as carried in v4.1; v2.12 below)	hooks_build: 326 passed, 0 failed (results/sil_v2p11/test_core_log.txt 3d9c40a8cd5d417d); no_hooks_build: 325 passed, 0 failed (results/sil_v2p11/test_core_nohooks_log.txt 226a5bb1e36b1954; R2-08: the core built without ECU_SIL_TEST_HOOKS, the one hook-only case skipped); v2p11_cases: PASS LogRecord v12 is 180 bytes (v2.11: + ovr_flags; 1 kHz x 180 B = 180 kB/s to SD; v11 179 B, v10 150 B, v8/v9 ... SIM T4133
SIL record v2.11	comparison: results/sil_v2p11/compare_v2p10_v2p11.json 7bb97d757dbdcc46 (+ .txt), calc/v2p11/sil_compare_v1.py: every v2.10 result file case by case or leaf by leaf; per file below; per_file_vs_v2p10: S1_mission.json: leaves 83 / 83, 10 changed; S2_steps.json: leaves 176 / 176, 66 changed; S3_deep.json: leaves 524 / 524, 205 changed; S4_hover.json: leaves 5 / 5, 4 changed; S5_mc.json: leaves 48 / ... SIM T4134
CAN link	ISO 11898-2 CAN, classic 2.0B 1 Mbit/s, 11-bit IDs; CAN-FD-capable transceivers so FD is a firmware upgrade DECIDED *Design value, electronics.json v4.1* T4135
Change summary v3 to v4	v3 (firmware v2.10) → v4 (firmware v2.11, SIL record 8c1a534c3af1b0ba, MANIFEST eaa9efc5112a8178); STAGED, the hub v3 file is unchanged DECIDED *Design value, electronics.json v4.1* T4136
Firmware v2.12 unit tests (SIL)	332 passed, 0 failed SIM T4139 / no hooks: 331 passed, 0 failed SIM T4140
REV-ELEC-4 reproduction (266 (4))	Reproduction [CALC / SIM, a repeat]: rebuild byte-identical (4 binaries); test_core 332 / 0 and 331 / 0 with byte-identical logs; every author v2.12 run byte-identical; the v2.11 SIL record re-run on v2.12 changes only S12 (1 outcome leaf) and S13B (82 / 70 outcome leaves, PC_SENSOR to RUN or to NO_IGNITION; 0 unlit runs reach RUN), as predicted. DECIDED *Lead decision, line 266 (4)* T4138

10.1.1 Firmware v2.12 changes and their status under line 266

SIL only (firmware v2.12 has no Teensy HAL); no hardware exists.

Change (electronics.json v4.1 /firmware/v2p12/changes, quoted)	Label (266 (6)) and correction (266 (7))
seat compensation (R3-01 option (b)): Pc_comp = Pc_in - coef x (T_seat - T_seat,ARM) before every use; coef per serial (setPcSeatCoef), T_seat = Inputs::tc_bore_K (TC4, MAX31856 U904); SEAT_NOREF / SEAT_FROZEN fallbacks; U330 not compensated DECIDED *Design value, electronics.json v4.1* T4141	SUSPENDED UNTIL v2.13 (266 (6)) T026 option (b) firmware row
ARM sanity static term: |gauge - coef x (T_seat - T_cal)| <= kPcArmZeroMax_Pa (T_cal per serial, setPcSeatTcal; v2.11 check when coef 0) DECIDED *Design value, electronics.json v4.1* T4142	SUSPENDED UNTIL v2.13 (266 (6)) T026 option (b) firmware row **Corrected by 266 (7), E4-03:** E4-03 (canArm subtracts the static term twice after an earlier ARM; fixed in v2.13) DECIDED *Lead decision, line 266 (7)* T4143
TB-state (f3 b0/b1): HW_OVERPRESSURE, PC_OVERPRESSURE, HARD_START, OVER_TEMP, PC_SENSOR HIGH (>= 4.75 V) in any non-SAFE state; stand: the HIGH over-range aborts PC_SENSOR as before and now sets TB_STATE; flight: TB_STATE latched, no abort DECIDED *Design value, electronics.json v4.1* T4144	ACCEPTED (REV-ELEC-4, line 266) T027 TB-state
flight J-FWD indication (f3 b5): LIGHT_SPIKE raw >= kJfwdLsRawAbs_Pa 47.5 bar abs or raw >= 4.75 V DECIDED *Design value, electronics.json v4.1* T4145	ACCEPTED (REV-ELEC-4, line 266) T027 JFWD_LS [E4-N03, TO-VERIFY (266 (8))]

Change (electronics.json v4.1 /firmware/v2p12/changes, quoted)	Label (266 (6)) and correction (266 (7))
torch re-zero limit kPtorchRezeroMax_Pa 0.60 bar (070BG basis; f3 b6 on a rejection) DECIDED *Design value, electronics.json v4.1* T4147	ACCEPTED (REV-ELEC-4, line 266) T027 re-base OPEN, 264 (4) [E4-N04 (266 (8))]
kIgnBaseCheck_Pa = kIgnResidual_Pa + kIgnBaseDrift_Pa 0.06 bar + 0.10 bar DECIDED *Design value, electronics.json v4.1* T4149	ACCEPTED (REV-ELEC-4, line 266) T027 kIgnBaseCheck_Pa 0.76 / 0.46 bar SUSPENDED UNTIL v2.13 (266 (6)) T026 the kIgnBaseDrift 0.06 bar basis for any relight
purge FINAL: kPurgeFlow_s 2.43 s, kHbleedCap_s 2.39 s, both calibration constants (setPurgeCal, SAFE only, rule (d) bounds) DECIDED *Design value, electronics.json v4.1* T4150	ACCEPTED (REV-ELEC-4, line 266) T027 purge FINAL
log v13: + seat_comp_hPa, seat_ref_dK, f3_flags (185 B) DECIDED *Design value, electronics.json v4.1* T4151	ACCEPTED (REV-ELEC-4, line 266) T027 log v13

10.2 Sensors

Item	Value
Pressure transducers (selection, electronics.json v4.1)	PC: Kulite XTL-190SM-2000A (C1, seat-mounted on the head, J601); P_FEED: Kulite XTL-190SM-2000A (C1, J602); P_MID: Kulite XTL-190SM-2000A (C1, J511); P_TORCH: TE M3234-000005-070BG (J901); P_TANK: TE M3234-000005-170BG (J603); stand_static_reference: TE M3234-000005-070BG (B-PC-1, BOM 199) DECIDED *Design value, electronics.json v4.1* T4156 [266 (7), E4-05 (7) T4157: the 264 (4) fold applies: P_TORCH is the XTL-190SM-2000A; the file keeps the 070BG entry until v4.2; contradiction register C-10]
P_torch transducer: lead 264 (4) supersedes the v4 selection	P_torch (J901, on the torch body, not in the head; torch_port_bl08_v352_v1.json 20220647d32cbbbc) moves from the TE M3234-000005-070BG (proof 141.0 bar abs) to a C1-family Kulite XTL-190SM-2000A (proof 206.8 bar abs [DATASHEET]) against the 166.14 bar abs requirement (1.1 x FA1S 151.04 bar abs) [DERIVED]; it is the same part as P_mid (ENG-BL-08) and C1. DECIDED *Lead decision, line 264 (4)* T802 New unit: proof **206.8** bar abs DATASHEET T4158 against the required **166.14** bar abs DERIVED T4159. Its thresholds are re-based in a later Electronics round (OPEN, §12.4; contradiction register C-10, C-14). [266 (8) T4148: E4-N04, queued with v2.13]
Hardware over-pressure comparator	U330 on the C1 chain: trip 32.59 bar abs nominal, band 31.08-34.11 bar abs; firmware abort 30 bar abs filtered 5 ms DERIVED T4160
Coast re-zero	**span (kPcRezeroSpan_Pa)** = 5000 Pa DECIDED *Design value, electronics.json v4.1* T4152 **max (kPcRezeroMax_Pa)** = 30000 Pa DECIDED *Lead decision, line 264 (2) (kPcRezeroMax_Pa stays 0.30 bar on the compensated residual); v4.1 is the source of record (266 (7))* T4153 **second_rezero_max_age (kPcRezero2MaxAge_s)** = 0.5 s DECIDED *Design value, electronics.json v4.1* T4154 **rule:** second re-zero ending before PURGE/RELIGHT (RE-07); flight NOT_VENTED = flag, stand = refusal DECIDED *Design value, electronics.json v4.1* T4155 [**Carve-out, 266 (6)**: Option (b) (seat coefficient loaded, setPcSeatCoef != 0) is not used for any start with ignition (pad first start, stand starts, relights) until a v2.13 delta with the light-window hold and the TC4 step / rate checks is reviewed. The coefficient may be loaded for B-PC-1 / B-TC-1 bench work without ignition. DECIDED *Lead decision, line 266 (6)* T824]
Thermocouple channels	TC1 type K case az 90 (U604); TC2 type K case az 270 (U605); TC3 type K nozzle holder (U606); TC4 type K bore top (U904); TC5 type T MSS-1 (U905); TC6 type T MSS-2 (U906) DECIDED *Design value, electronics.json v4.1* T4161 **Corrected by 266 (7), TC4 label:** TC4 is the C1 seat TC (MAX31856 U904, J904, BOM row 205), not bore top (the 264 (2) rename is made by this line; the file keeps the old label until v4.2) DECIDED *Lead decision, line 266 (7)* T4162
Case abort / relight refusal	**423.15** K (150 °C) DECIDED *Lead decision, line 243 (3)* T4163 / **403.15** K (130 °C) DECIDED *Lead decision of 2026-10-20* T4164
Case abort rule	in-burn abort on TC1 / TC2 above the case abort for the persistence (abort_matrix OVER_TEMP); channel budget: case_tc_channel DECIDED *Lead decision, per electronics.json v4.1* T4165
Persistence	**0.2** s DECIDED *Design value, electronics.json v4.1* T4166
Start and ARM limit / stand ARM maximum	**403.15** K (130 °C) DECIDED *Lead decision of 2026-10-20* T4167 / **297.15** K (24 °C) DECIDED *Lead decision, per electronics.json v4.1* T4168
Nozzle-holder limit	**523.15** K (250 °C) ASSUMED T4169
Photodiode	FA-1 head on J902 (Si photodiode + log TIA, 0-5 V), part TO-SELECT (BOM row); valid 0.02 V to 4.8 V DECIDED *Design value, electronics.json v4.1* T4170
Encoders	AS5048A x3 (gimbal A/B J610/J611, pintle crank J612), SPI1 1 MHz DECIDED *Design value, electronics.json v4.1* T4171
Stand dynamic Pc channel	B-DPC-1: PCB 113B22 (5 kpsi = 344.75 bar at +/-5 V; maximum pressure 15 kpsi = 1034 bar) + NI 9234 at 51.2 kS/s (AC coupled, -3 dB 0.5 Hz), flat to >= 10 kHz incl. the mount; head flush at the C1 seat (stand only, IGN-1 and the T series); fallback IEPE conditioner BOM v2.3 line 265; see pc_overrange_fa1.stand_dynamic_channel DATASHEET *Lead decision, line 172 (5)* T4172

10.3 Limits of firmware v2.12

electronics.json /limits_v2p11. Values are firmware constants (config.h), SIL only; each with its unit, at the precision of the source.

Limit	Value	Basis (quoted)
case-TC abort A (TC1 / TC2)	**kCaseOverTemp_K** = **423.15** K DECIDED *Lead decision, line 243 (3)* T4173	A = 150 C (LEAD 243 (3)); false-trip side 7.78 K worst case / 3.55 K RSS inside +12.3 K at the 137.74 C B1 station (case_tc_channel); persistence 200 ms (2-3 MAX31856 conversions) DERIVED T4174
case-TC start / relight limit and ARM limit (TC1 / TC2)	**kCaseRelightMax_K** = **403.15** K DECIDED *Lead decision of 2026-10-20* T4175	A1 purge-to-end-of-relight rise 11.9 K + 1 s longer relight 3.0 K + 1 K relative channel = 15.9 K below A (130 C + 15.9 K = 145.9 C <= 150 C); A1 nominal start reading <= 99.5 C DERIVED T4176
nozzle-holder TC3 abort and ARM limit	**kNozzleHolderOverTemp_K** = **523.15** K ASSUMED T4177	250 C ASSUMED (unchanged); the ARM refusal on it is accepted DERIVED T4178
stand T0 ARM limit (stand mode, TC1 / TC2)	**kCaseStandArmMax_K** = **297.15** K DECIDED *Lead decision, per electronics.json v4.1* T4179	24 C: the 12.26 K false-trip margin is a 20 C-start margin (dT_station / dT0 about 0.96); a backstop, the founder checklist row F-13 is the control; re-derived after B-TC-1 as T0_max = 20 + (12.26 - e_hi,cal) / 0.96 C DERIVED T4180
purge both-open command cap	**kHbleedCap_s** = **2.39** s DECIDED *Lead decision, line 256 (2)* T4181 ACCEPTED (REV-ELEC-4, line 266) T027 purge FINAL 2.43 / 2.39 s with rule (d), (d1), (d2)	LEAD 203 (3) <= 2.55 s; 230 (3): the scheduled command span 2.45 s does not fit a 2.39 s cap, so 2.55 s is kept; physical flow at the cap 2.60 s = 76.05 g on the one basis 29.25 g/s, 80.76 g at the warm end 31.06 g/s DERIVED T4182 **Corrected by 266 (7), E4-05 (1):** 1. limits_v2p11[4]: the basis text is stale (2.55 s / 76.05 g / 80.76 g). Replace it with the v2.12 FINAL basis: kPurgeFlow_s 2.43 s and kHbleedCap_s 2.39 s (defaults and maxima, setPurgeCal SAFE only), cap 2.397 s at the defaults, rule (d) with (d1) / (d2). DERIVED T4183
hardware both-commanded-open timer (U576-U579, DESIGN, not drawn)	**kHbleedHwBothNom_s** = **2.659** s DECIDED *Design value, electronics.json v4.1* T4184	LTC6993-2 RSET 63.4k, DIVCODE 8 (AS-RECALLED, TO-VERIFY): 2.577-2.741 s with the 3.1 % tolerance; drops V1 only; the firmware cap acts first (minimum 2.577 s > cap + 1 tick) DERIVED T4185
vented-mass limit and flow basis	**kPurgeVentLimit_kg** = **0.076** kg DECIDED *Lead decision, line 216 (5)* T4186	LEAD 216 (5): the limit is the vented mass; one flow basis 29.25 g/s (LEAD 230 (3)); measured-time form after CF-4 / B-BLEED-1: overlap <= 76 g / mdot_meas - (t_close - t_open)_meas DERIVED T4187
ride-through persistence / window / T45 / over-range N / fixed RC (CANDIDATE, default OFF)	**kRideOvpTime_s** = 0.015 s DERIVED T4188 **kRideWinPostE2_s** = 0.015 s DERIVED T4189 **kRideT45_s** = 0.007 s DERIVED T4190 **kRideOverrangeN** = 2 DECIDED *Lead decision, line 216 (3)* T4191 **kRideCmpTauNom_s** = 0.012825 s DECIDED *Design value, electronics.json v4.1* T4192 **kRideCmpTauMin_s** = 0.0121 s DERIVED T4193 **kRideCmpTauMax_s** = 0.013552 s DERIVED T4194	see ride_through.constants (each with its basis) DERIVED T4195 **Corrected by 266 (7), E4-05 (4):** 4. limits_v2p11[7] still says "(CANDIDATE, default OFF)". Restate it as ADOPTED on the 253 (7)(i) / 256 (3) conditions. DERIVED T4196
Pc transducer proof	C1 206.8 bar abs DATASHEET T4197	requirement >= 166.1 bar abs (1.1 x FA1S 151.04, 230 (2)), replacing 112 bar abs DERIVED T4198 [requirement **166.14** bar abs DERIVED T4159, 256 (R4-12) T894, 264 (4) T802]
Pc over-range / above-FS levels (C1)	**kPcSensorMax_V** = 4.75 V DECIDED *Design value, electronics.json v4.1* T4199 **kPcAboveFs_V** = 4.56 V DECIDED *Design value, electronics.json v4.1* T4200	146.5 / 139.96 bar abs on the C1 transfer (pc_transducer_v3 E) DERIVED T4201
firmware Pc over-pressure abort	**kPcOverPressureAbs_Pa** = 3000000 Pa DECIDED *Lead decision, per electronics.json v4.1* T4202 **kPcOverPressureTime_s** = 0.005 s DECIDED *Lead decision of 2026-10-13* T4203	filtered absolute Pc (RE-20); hardware U330 on C1 32.59 bar abs (31.08-34.11) DERIVED T4204
R2 window / ECU backstop / hold lift / FC re-send bound	**kR2WindowMax_s** = 0.8 s DECIDED *Lead decision, line 227 (5)* T4205 **kR2HoldBackstop_s** = 0.85 s DECIDED *Design value, electronics.json v4.1* T4206 **kR2HoldLift_m** = 0.00021776 m DECIDED *Lead decision, line 230 (4)* T4207 **kFcDrogueResendMax** = 10 DECIDED *Design value, electronics.json v4.1* T4208	0.8 s for W10 < 6 m/s (227 (5), confirmed with conditions by S17 v2.11); backstop = window + 2 FC frames + link latency (216 (2)(b)); Combustion v2.4b start-level lift (230 (4)); 10 re-sends = 0.2 s (216 (2)(a)) DERIVED T4209
cold rule	**kTankIgnMin_K** = 288.15 K DECIDED *Requirement, ENG-12* T4210 **kTankColdArmMargin_K** = 0.42 K DERIVED T4211	ARM margin = 1.5 x the PREPRESS Tsat drop 0.23 K + repeatability 0.07 K, rounded up (SIL S10v211 drop max 0.2219 K) DERIVED T4212
A7 high-side blank	**kA7StartHiBlank_s** = **0.45** s DECIDED *Lead decision, line 166 (7)* T4213	E2 .. E2 + 0.45 s on the stand and in flight (four conditions, start_monitors.A7_flight_blank) DERIVED T4214
light-rule baseline check	**kIgnBaseCheck_Pa** = **76000** Pa DECIDED *Lead decision, line 264 (2) (kIgnBaseCheck_Pa 0.76 / 0.46 bar, default / B-PC-1-shown build)* T4215 ACCEPTED (REV-ELEC-4, line 266) T027 kIgnBaseCheck_Pa 0.76 / 0.46 bar SUSPENDED UNTIL v2.13 (266 (6)) T026 the kIgnBaseDrift 0.06 bar basis for any relight	residual + 0.1 bar; per location (pad / stand refuse, A1 keeps / zeroes P0) DERIVED T4216 **Corrected by 266 (7), E4-05 (2):** 2. limits_v2p11[14]: the basis "residual + 0.1 bar" lacks the drift term. Write "residual + kIgnBaseDrift 0.06 + 0.10 bar = 0.76 / 0.46 bar". DERIVED T4217

3. Missing limit rows, each with its value, tag and source: DERIVED T4218 The values are firmware v2.12 config.h constants as the review lists them; v4.2 adds the rows.

Limit row that v4.1 lacks (REV-ELEC-4 E4-05 (3), applied by 266 (7))	Label
the seat constants (kPcSeatCoefMax_PaPerK 1.0e4, kPcSeatDevMax_K 150, kPcSeatTcMin_K / kPcSeatTcMax_K 223.15 / 473.15, the T_cal minimum / default / maximum, kPcSeatArmDevMax_K 30); DERIVED T4219	SUSPENDED UNTIL v2.13 (266 (6)) T026 option (b) firmware rows

Limit row that v4.1 lacks (REV-ELEC-4 E4-05 (3), applied by 266 (7))	Label
kIgnBaseDrift 0.06 bar; DERIVED T4220	SUSPENDED UNTIL v2.13 (266 (6)) T026 the basis for any relight ACCEPTED (REV-ELEC-4, line 266) T027 as a part of kIgnBaseCheck_Pa 0.76 / 0.46 bar
kPtorchRezeroMax_Pa 0.60 bar; DERIVED T4221	ACCEPTED (REV-ELEC-4, line 266) T027 re-base OPEN, 264 (4)
kJfwdLsRawAbs 47.5 bar abs. DERIVED T4222	ACCEPTED (REV-ELEC-4, line 266) T027 JFWD_LS [E4-N03: 47.5 against 47.75 bar abs, TO-VERIFY (266 (8))]

10.4 Light rule and start monitors

Item	Value
Light rule	P0 = mean of filtered / raw gauge Pc over [E1 - 20 ms, E1]. Check against the auto-zero ambient, per location (LEAD CHANGELOG 166 (1), 216 (4)): pad first start and stand: P0 above +check (BASE_HIGH) or below -check (BASE_AMB) → the lift is refused (PC_SENSOR abort before any pintle command, MV closed by the abort); A1 relight: BASE_HIGH → P0 kept, flagged to the FC, RUN on the measured P0; BASE_AMB → P0 := 0. Light (E2) = filtered AND raw rise over P0 >= kIgnRise_Pa with filtered AND raw Pc >= the floor, kIgnConfirmRelN consecutive ticks DECIDED *Lead decision, line 216 (4)* T4223 [**Carve-out, 266 (6)**: Option (b) (seat coefficient loaded, setPcSeatCoef != 0) is not used for any start with ignition (pad first start, stand starts, relights) until a v2.13 delta with the light-window hold and the TC4 step / rate checks is reviewed. The coefficient may be loaded for B-PC-1 / B-TC-1 bench work without ignition. DECIDED *Lead decision, line 266 (6)* T024]
kIgnRise_Pa	**80000** Pa DECIDED *Lead decision, per electronics.json v4.1* T4224
kIgnConfirmRelN	**10** DECIDED *Lead decision, per electronics.json v4.1* T4225
kIgnPreN	**20** DECIDED *Lead decision, per electronics.json v4.1* T4226
kIgnResidual_Pa	**60000** Pa DECIDED *Lead decision, per electronics.json v4.1* T4227
kIgnBaseCheck_Pa	**76000** Pa DECIDED *Lead decision, line 264 (2) (kIgnBaseCheck_Pa 0.76 / 0.46 bar, default / B-PC-1-shown build)* T4228 ACCEPTED (REV-ELEC-4, line 266) T027 kIgnBaseCheck_Pa 0.76 / 0.46 bar SUSPENDED UNTIL v2.13 (266 (6)) T026 the kIgnBaseDrift 0.06 bar basis for any relight
kIgnFloorAbs_Pa	**180000** Pa DECIDED *Lead decision, per electronics.json v4.1* T4229
Start monitor A3	(Pc - P0)/(Pc_map_lo - P0) < 0.4 for 20 ms in RUN; denominator >= 0.5 bar DECIDED *Lead decision, per electronics.json v4.1* T4230
Start monitor A7	HOLD band from E1 (low edge P0 - 0.3 bar), START band E2 → E3, RUN_MAP after E3; 50 ms; IGNITION / stand → PC_SENSOR abort, flight RUN → flag once per burn; HIGH side blanked from E2 to E2 + 450 ms on the stand and in flight (RE-12; flight extension ACCEPTED as DESIGN, LEAD 216 (8)) DECIDED *Lead decision, per electronics.json v4.1* T4231
Start monitor E3	first RUN tick with the ramp done and Pc >= P0 + 0.9 (setpoint - P0), or E2 + 600 ms DECIDED *Design value, electronics.json v4.1* T4232
Start monitor A7_flight_blank	**decision:** CHANGELOG line 216 (lead, 2026-10-13): ...ed time, never a wait with the MV open. (8) RE-12: the flight extension of the A7 high-side blank E2..E2 + 0.45 s is ACCEPTED as DESIGN with the reviewer's four conditions (config.h l.369-372 corrected; electronics.json states the blind window and that the 30 bar abort, comparator, A3 and the A7 low side stay armed; the FC owner confirms no decision uses the A7 high flag before E2 + 0.45 s; test v210-2 extended on d6... DECIDED *Lead decision, line 216 (8), as quoted in electronics.json v4.1* T4233 **blind window:** E2 .. E2 + 450 ms, A7 HIGH side only (stand and flight). Still ARMED in it: the filtered 30 bar abort (5 ms; ride-through build 15 ms to E2 + 15 ms), the hardware comparator U330, A3 (flame-out, low side) and the A7 LOW side DECIDED *Design value, electronics.json v4.1* T4234 **condition 1:** n: 1; condition: config.h comment corrected (REV-ELEC-2 cited v2.10 l.369-372); status: DONE: config.h l.413-416 (v2.11 numbering) DECIDED *Design value, electronics.json v4.1* T4235 **condition 2:** n: 2; condition: electronics.json states the blind window and what stays armed; status: DONE: blind_window above DECIDED *Design value, electronics.json v4.1* T4236 **condition 3:** n: 3; condition: the FC owner confirms no decision uses the A7 high flag (d6 b7 via PC_SENSOR) before E2 + 0.45 s; status: OPEN: FC owner (requests) TO-VERIFY T4237 **condition 4:** n: 4; condition: test v210-2 extended on d6 b7; status: DONE: PASS v2.11 LEAD 216 (8) condition 4 (v210-2 extended on TLM_RELIGHT d6 b7): b7 = 1 on every tick of the 7.6 bar excursion inside the E2..E2 + 0.45 s A7 high-side blank (blind window: no A7 high flag, b7 untouched); b7 = 0 from the A7 flag tick after the blank and stays 0 (latched by fault::PC_SENSOR) with Pc back in band [b7 bad in blank 0, b7 set after the flag 0, b7 first 0 at tick 516 (A7 flag tick 516), ticks 660] SIM T4238
Start monitor light_spike_record	no late-light abort (the v2.9 pressure-hold abort is deleted, RE-01, LEAD CHANGELOG 160): LIGHT_SPIKE record from 8 bar abs (raw and filtered peak, time above 8 bar and above the 30 bar limit, time from light) in the log and EVENT 0x20; v2.11 (LEAD 203 (1)(a)): a sample at or above 4.56 V (above the C1 full scale) sets the above-full-scale flag (LIGHT_SPIKE ls_flags b7, log v12 ovr_flags) with PC_VALID untouched DECIDED *Lead decision, per electronics.json v4.1* T4239
Stand relight after B1 (v4.1 r3_01_c1_drift)	option (b): RUN (all comp / comp_over runs; comp_under refused NOT_VENTED at k >= 2-3: a coefficient 10 % low leaves 0.27-0.40 bar on the 0.30 bar limit); fallback (coefficient 0): REFUSED NOT_VENTED (refuse 8) in every run that arms - the uncompensated drift 0.78-4.10 bar exceeds the re-zero limit, both re-zeros are rejected and the stand vent check refuses (safe: no light attempted). RULE: a stand relight after a B1-length burn needs option (b) with the per-serial coefficient within about 5 % at k = 3 (B-PC-1 row) SIM T4240 SUSPENDED UNTIL v2.13 (266 (6)) T026 the stand relight after B1 **Corrected by 266 (7), E4-02:** E4-02 (the coefficient-within-5-percent-at-k-3 acceptance, also in the 264 (2) wording, is replaced by the k-dependent limit of its Q3 table, eps_max 11.2 / 8.1 / 6.0 / 3.4 / 1.8 / 0.8 % at k = 1 / 1.25 / 1.5 / 2 / 2.5 / 3 with the 0.1012 bar adders [DERIVED]; beyond it the stand relight after B1 is planned REFUSED NOT_VENTED, fail-safe, with no relight objective claimed) DECIDED *Lead decision, line 266 (7)* T4241
U330 band rows (v4.1 r3_01_c1_drift A8)	by_k: 1.0: A1_relight_burn_end_T0_0C_shift_bar: -0.114; A1_relight_burn_end_T0_0C_band_bar_abs: [30.96, 34.22]; A1_relight_burn_end_T0_20C_shift_bar: 0.382; A1_relight_burn_end_T0_20C_band_bar_abs: [30.7, 34.48]; A1_relight_burn_end_T0_40C_shift_bar: 0.878; A1_relight_burn_end_T0_40C_band_bar_abs: [30.2, 34.98]; B1_burn_end_stand_T0_0C_shift_bar: 0.027; B1_burn_end_stand_T0_0C_band_bar_abs: [31.05, 34.13]; B1_burn_end_stand_T0_24C_shift_bar: 0.622; B1_burn_end_stand_T0_24C_band_bar_abs: [30.45, 34.73]; stand_relight_light_T0_0C_shift_bar: 0.89; stand_relight_light_T0_0C_band_bar_abs: [30.19, 34.99]; stand_relight_light_T0_24C_shift_bar: 1.485; stand_relight_light_T0_24C_band_bar_abs: [29.59, 35.59]; 1.5: A1_relight_burn_end_T0_0C_shift_bar: -0.171; A1_relight_burn_end_T0_0C_band_bar_abs: [30.91, 34.27]; A1_relight_burn_end_T0_20C_shift_bar: 0.573; ... DERIVED T4242 ACCEPTED (REV-ELEC-4, line 266) T027 the U330 band rows **Corrected by 266 (7), E4-04:** E4-04 (the U330 ordering gains the stand relight light: the U330 low edge falls below the 30 bar abs firmware abort from k = 0.725 at T0 24 C and from k = 1.21 at T0 0 C [DERIVED]) DECIDED *Lead decision, line 266 (7)* T4243

10.5 Abort and refusal causes (firmware v2.12)

31 causes in electronics.json /abort_matrix/causes (SIL only; no hardware); thresholds with their units, at the precision of the source.

Code	Cause	Trigger	Thresholds
1 T4244	**CMD** T4245	FC ABORT frame 0x010 (magic 0xA5 0x5A, code, seq, CRC-8; no sequence check) or CMD_STATE ABORT: acted on at the next tick ahead of every other command; NC valve holding magnet released and pintle commanded closed on that tick; EVENT STATE with cause CMD sent three times; log abort_cause CMD. Works with the Pc channel invalid (test v210-15) DECIDED *Design value, electronics.json v4.1* T4246	no threshold
2 T4247	**HARD_START** T4248	IGNITION, or RUN within 300 ms of entry: filtered absolute Pc above 30 bar abs for 5 ms (RE-20, filtered channel); in IGNITION a Pc transducer output at or above 4.75 V (a high over-range is taken as real pressure; flight: the over-range bit is non-sticky and PC_SENSOR is not latched, stand as built). Ride-through BUILD OPTION (kRideThroughBuild = 0 in the flight build: OFF; flight relight only, from E1 to E2 + 15 ms): the filtered persistence becomes 15 ms, the IGNITION over-range rule needs 2 consecutive over-range readings, and a raw reading above the MEOP 45 bar abs held for 7 ms aborts at once (T45 guard); the hardware comparator RC of that BOM variant is FIXED at 12.82 ms nominal. No late-light abort (the v2.9 pressure hold is deleted, RE-01): a LIGHT_SPIKE record is logged from 8 bar abs (EVENT 0x20) with the above-full-scale flag at 4.56 V DECIDED *Design value, electronics.json v4.1* T4249	**kRunSettle_s** = 0.3 s DECIDED *Design value, electronics.json v4.1* T4250 **kPcOverPressureAbs_Pa** = 3000000 Pa DECIDED *Lead decision, per electronics.json v4.1* T4251 **kPcOverPressureTime_s** = 0.005 s DECIDED *Lead decision of 2026-10-13* T4252 **kPcSensorMax_V** = 4.75 V DECIDED *Design value, electronics.json v4.1* T4253 **kRideThroughBuild** = 0 DECIDED *Lead decision, line 203 (1)* T4254 **kRideWinPostE2_s** = 0.015 s DERIVED T4255 **kRideOvpTime_s** = 0.015 s DERIVED T4256 **kRideOverrangeN** = 2 DECIDED *Lead decision, line 216 (3)* T4257 **kPcMeopLeadAbs_Pa** = 4500000 Pa DECIDED *Lead decision, per electronics.json v4.1* T4258 **kRideT45_s** = 0.007 s DERIVED T4259 **kRideCmpTauNom_s** = 0.012825 s DECIDED *Design value, electronics.json v4.1* T4260 **kLightSpikeLoAbs_Pa** = 800000 Pa DECIDED *Lead decision, line 160* T4261 **kPcAboveFs_V** = 4.56 V DECIDED *Design value, electronics.json v4.1* T4262
3 T4263	**NO_IGNITION** T4264	no light within W from the lift by the relative rule: filtered AND raw rise over P0 at least 0.8 bar with filtered AND raw Pc at least the floor 1.8 bar abs for 10 consecutive ticks; W = 400 ms flight first start, 650 ms flight relight, 350 ms stand. Also the fault bit of TORCH_FAULT and TORCH_LOST DECIDED *Design value, electronics.json v4.1* T4265	**kIgnRise_Pa** = 80000 Pa DECIDED *Lead decision, per electronics.json v4.1* T4266 **kIgnFloorAbs_Pa** = 180000 Pa DECIDED *Lead decision, per electronics.json v4.1* T4267 **kIgnConfirmRelN** = 10 DECIDED *Lead decision, per electronics.json v4.1* T4268 **kIgnWindowFirst_s** = 0.4 s DECIDED *Lead decision of 2026-10-13* T4269 **kIgnWindowRelightLong_s** = 0.65 s DECIDED *Lead decision of 2026-10-13* T4270 **kIgnWindowFirstStand_s** = 0.35 s DECIDED *Lead decision of 2026-10-13* T4271
4 T4272	**FLAMEOUT** T4273	RUN, Pc valid: A3 ratio (Pc - P0)/(Pc_map_lo - P0) below 0.4 for 20 ms (Pc_map lagged by 40 ms). RUN, Pc invalid: FLAMEOUT_OPT, photodiode dark for 20 ms once the main-chamber view was proven (20 lit ticks after the torch closed). The legacy gauge rule (1.5 bar, 0.35) is not used DECIDED *Design value, electronics.json v4.1* T4274	**kFlameoutRel** = 0.4 DECIDED *Lead decision of 2026-10-13* T4275 **kFlameoutTime_s** = 0.02 s DECIDED *Lead decision of 2026-10-13* T4276 **kFlameoutMapTau_s** = 0.04 s DECIDED *Design value, electronics.json v4.1* T4277 **kPhotoFlameoutTime_s** = 0.02 s DECIDED *Design value, electronics.json v4.1* T4278 **kPhotoViewProveN** = 20 DECIDED *Design value, electronics.json v4.1* T4279 **kFlameoutPc_Pa** = 150000 Pa DECIDED *Design value, electronics.json v4.1* T4280 **kFlameoutFrac** = 0.35 DECIDED *Design value, electronics.json v4.1* T4281
5 T4282	**PC_BAND** T4283	RUN after 300 ms: |Pc - Pc_ref| above 0.3 of Pc_ref and above 2 bar for 150 ms DECIDED *Design value, electronics.json v4.1* T4284	**kRunSettle_s** = 0.3 s DECIDED *Design value, electronics.json v4.1* T4285 **kPcBandFrac** = 0.3 DECIDED *Design value, electronics.json v4.1* T4286 **kPcBandMin_Pa** = 200000 Pa DECIDED *Design value, electronics.json v4.1* T4287 **kPcBandTime_s** = 0.15 s DECIDED *Design value, electronics.json v4.1* T4288
6 T4289	**OVER_TEMP** T4290	case TC (TC1 / TC2) above 423.15 K (150.0 C) or nozzle holder (TC3) above 523.15 K (250.0 C) for 200 ms in a burn, PREPRESS or PURGE (the persistence spans two to three MAX31856 conversions); ARM refused with this cause when TC1 or TC2 reads above the start limit 403.15 K (130.0 C) or TC3 above 523.15 K (250.0 C), and unless BOTH case TCs are valid (RE-08); a start command (IGNITE / RELIGHT) in COAST above 403.15 K (130.0 C) is refused CASE_HOT; one TC lost: continue on the other; both lost: stand treated hot, flight burn continues and the relight is refused (CASE_HOT). Servo-bracket NTC above 333.15 K (60.0 C) only flags DECIDED *Design value, electronics.json v4.1* T4291	**kCaseOverTemp_K** = 423.15 K DECIDED *Lead decision, line 243 (3)* T1246 **kNozzleHolderOverTemp_K** = 523.15 K ASSUMED T1248 **kOverTempTime_s** = 0.2 s DECIDED *Design value, electronics.json v4.1* T4292 **kCaseRelightMax_K** = 403.15 K DECIDED *Lead decision of 2026-10-20* T1247 **kServoBracketOverTemp_K** = 333.15 K DERIVED T4293
7 T4294	**TANK_OVERPRESSURE** T4295	P_tank above 65 bar for 200 ms in ARMED or PREPRESS (never aborts a burn) DECIDED *Design value, electronics.json v4.1* T4296	**kTankOverPressure_Pa** = 6500000 Pa DECIDED *Design value, electronics.json v4.1* T4297 **kTankOverPressureTime_s** = 0.2 s DECIDED *Design value, electronics.json v4.1* T4298

Code	Cause	Trigger	Thresholds
8 T4299	**GIMBAL_FAULT** T4300	command-model vs encoder mismatch above 1.5 deg for 50 ms while burning; the allowance grows by 40 ms times the model rate; NOT inhibited near touchdown (see decision). Detection bound on the as-built linkage: see sil_fail_dispositions (RE-24) DECIDED *Design value, electronics.json v4.1* T4301	**kGimbalFaultErr_rad** = 0.02618 rad DECIDED *Design value, electronics.json v4.1* T4302 **kGimbalFaultTime_s** = 0.05 s DECIDED *Design value, electronics.json v4.1* T4303 **kGimbalFaultRateWin_s** = 0.04 s DECIDED *Design value, electronics.json v4.1* T4304
9 T4305	**PINTLE_FAULT** T4306	pintle rail below 5.5 V for 2 ms (LTC4365 trip or regulator failure); jam: no progress for 100 ms (time path) or 50 ms (current path); encoder invalid at ARM; needle off seat or encoder invalid during a purge on the stand DECIDED *Design value, electronics.json v4.1* T4307	**kPrailLost_V** = 5.5 V DECIDED *Design value, electronics.json v4.1* T4308 **kPrailLostTime_s** = 0.002 s DECIDED *Design value, electronics.json v4.1* T4309 **kPintleStallTime_s** = 0.1 s DECIDED *Design value, electronics.json v4.1* T4310 **kPintleStallTimel_s** = 0.05 s DECIDED *Design value, electronics.json v4.1* T4311
10 T4312	**LINK_LOST** T4313	no valid CMD_FAST: degraded after 20 ms, lost after 100 ms; SAFE/ARMED/PREPRESS → SAFE, RUN → SHUTDOWN by default (link policy) DECIDED *Design value, electronics.json v4.1* T4314	**kLinkDegraded_s** = 0.02 s DECIDED *Design value, electronics.json v4.1* T4315 **kLinkLost_s** = 0.1 s DECIDED *Design value, electronics.json v4.1* T4316
11 T4317	**PREPRESS_FAIL** T4318	feed not primed (P_feed above 0.8 of P_tank) within 600 ms, or the NC valve OPEN switch missing DECIDED *Design value, electronics.json v4.1* T4319	**kFeedPrimedFrac** = 0.8 DECIDED *Design value, electronics.json v4.1* T4320 **kPrepressTimeout_s** = 0.6 s DECIDED *Design value, electronics.json v4.1* T4321
12 T4322	**PC_SENSOR** T4323	transducer output outside 0.25 V to 4.75 V: IGNITION abort (low side), stand RUN abort after 5 ms, flight RUN continues open loop; flight high over-range is non-sticky (kPcOverrangeNonSticky = true: PC_VALID cleared on the over-range readings only, no PC_SENSOR latch), stand as built (latched); A7 plausibility band outside for 50 ms (HOLD band low edge P0 minus 0.3 bar; START high side blanked from E2 for 450 ms on the stand and in flight): IGNITION and stand abort, flight RUN flag once per burn; RE-02 baseline per location: on the pad first start and on the stand the lift is refused when P0 is above the auto-zero ambient by more than 0.7 bar (BASE_HIGH) or below it by more than that (BASE_AMB); the A1 relight keeps the measured P0 on BASE_HIGH and sets P0 to zero on BASE_AMB; ARM: |Pc| above 0.5 bar before the auto-zero DECIDED *Design value, electronics.json v4.1* T4324 [0.7 bar in this trigger text is the v2.11 klgnBaseCheck_Pa; v4.1 holds klgnBaseCheck_Pa at the 264 (2) values **0.76** T4327 / **0.46** bar DECIDED *Lead decision, line 264 (2)* T4328 (§10.4; ACCEPTED, REV-ELEC-4, line 266): contradiction register C-13]	**kPcSensorMin_V** = 0.25 V DECIDED *Design value, electronics.json v4.1* T4331 **kPcSensorMax_V** = 4.75 V DECIDED *Design value, electronics.json v4.1* T4332 **kPcSensorLossStand_s** = 0.005 s DECIDED *Design value, electronics.json v4.1* T4333 **kPcOverrangeNonSticky** = yes DECIDED *Lead decision, line 203 (1)* T4334 **kA7Time_s** = 0.05 s ASSUMED T4335 **kA7HoldLow_Pa** = 30000 Pa DECIDED *Design value, electronics.json v4.1* T4336 **kA7StartHiBlank_s** = 0.45 s DECIDED *Lead decision, line 166 (7)* T4337 **klgnBaseCheck_Pa** = 76000 Pa DECIDED *Lead decision, line 264 (2) (klgnBaseCheck_Pa 0.76 / 0.46 bar, default / B-PC-1-shown build)* T4338 ACCEPTED (REV-ELEC-4, line 266) T027 klgnBaseCheck_Pa 0.76 / 0.46 bar SUSPENDED UNTIL v2.13 (266 (6)) T026 the klgnBaseDrift 0.06 bar basis for any relight **kPcArmZeroMax_Pa** = 50000 Pa DECIDED *Design value, electronics.json v4.1* T4339
13 T4340	**BATT_LOW** T4341	VBAT below 6.6 V for 200 ms in pad states before the first start (flag in flight); ARM refused below 8.1 V rest voltage DECIDED *Design value, electronics.json v4.1* T4342	**kBattLow_V** = 6.6 V DECIDED *Design value, electronics.json v4.1* T4343 **kBattLowTime_s** = 0.2 s DECIDED *Design value, electronics.json v4.1* T4344 **kBattArmMin_V** = 8.1 V DECIDED *Lead decision of 2026-10-12* T4345
14 T4346	**MV_FAULT** T4347	NC valve CLOSED switch missing at ARM, ladder OPEN+CLOSED or short (RE-19); direct relight OPEN switch missing by 950 ms from E0: stand abort, flight flag; CLOSED expected within 500 ms of the latch drop (flag) DECIDED *Design value, electronics.json v4.1* T4348	**kMvOpenBy_s** = 0.95 s DECIDED *Design value, electronics.json v4.1* T4349 **kMvCloseConfirm_s** = 0.5 s DECIDED *Design value, electronics.json v4.1* T4350
15 T4351	**ARM_LOST** T4352	arming plug removed, or ARM bus not on after the request DECIDED *Design value, electronics.json v4.1* T4353	no threshold
16 T4354	**WATCHDOG** T4355	hardware watchdog timeout 50 ms: reboot lands in SAFE, MV and bleed closed, RUN never reported after the reset (no in-flight recovery path, see decision); a lit-session marker becomes a relight lockout DECIDED *Design value, electronics.json v4.1* T4356	**kHwWatchdog_s** = 0.05 s DECIDED *Design value, electronics.json v4.1* T4357
17 T4358	**PC_OVERPRESSURE** T4359	outside the ignition phase: filtered absolute Pc above 30 bar for 5 ms DECIDED *Design value, electronics.json v4.1* T4360	**kPcOverPressureAbs_Pa** = 3000000 Pa DECIDED *Lead decision, per electronics.json v4.1* T4361 **kPcOverPressureTime_s** = 0.005 s DECIDED *Lead decision of 2026-10-13* T4362
18 T4363	**HW_OVERPRESSURE** T4364	hardware comparator U330 on the C1 Pc chain (INA333 output before the ADC; tap RC, trip and band in thresholds_hardware) sets the U331 latch: ARM bus off in hardware, independent of the firmware; the firmware reports HW_OVERPRESSURE after OVP_OK low for 2 ms (OVP_OK ignored for 5 ms after ARM_REQ rises) DECIDED *Design value, electronics.json v4.1* T4365	**kHwOvpTime_s** = 0.002 s DECIDED *Design value, electronics.json v4.1* T4366 **kHwOvpArmSettle_s** = 0.005 s DECIDED *Design value, electronics.json v4.1* T4367
19 T4368	**TVC_SUPPLY** T4369	pad only: predicted both-stall TVC plug voltage below 6.6 V for 100 ms DECIDED *Design value, electronics.json v4.1* T4370	**kTvcSupplyMin_V** = 6.6 V DECIDED *Lead decision of 2026-10-12* T4371 **kTvcSupplyTime_s** = 0.1 s DECIDED *Design value, electronics.json v4.1* T4372
20 T4373	**OVERCURRENT** T4374	firmware e-fuse before the first start: battery current above 22 A for 1 s DECIDED *Design value, electronics.json v4.1* T4375	**kEfuse_A** = 22 A DECIDED *Design value, electronics.json v4.1* T4376 **kEfuseTime_s** = 1 s DECIDED *Design value, electronics.json v4.1* T4377
21 T4378	**MAINT_JAM** T4379	ARM refused: a pintle jam recorded in NV memory and not cleared (CLEAR_JAM, bearing A rule) DECIDED *Design value, electronics.json v4.1* T4380	no threshold
22 T4381	**TORCH_FAULT** T4382	torch not confirmed by 160 ms from E0: P_torch above 2 bar abs AND photodiode lit (2oo2, FA-1) for 3 ticks with the main chamber below 1 bar gauge; abort before the pintle step DECIDED *Design value, electronics.json v4.1* T4383	**kTorchGate_s** = 0.16 s DECIDED *Design value, electronics.json v4.1* T4384 **kTorchLitAbs_Pa** = 200000 Pa ESTIMATE *Design value, electronics.json v4.1* T4385 **kTorchConfirmN** = 3 DECIDED *Design value, electronics.json v4.1* T4386 **kTorchGateMaxPcG_Pa** = 100000 Pa DECIDED *Design value, electronics.json v4.1* T4387
23 T4388	**RELIGHT_LOCKOUT** T4389	ARM, IGNITE and RELIGHT refused after an abort following a lit burn or on a relight, until CLEAR_LOCKOUT (ground); after a ground no-light CLEAR_LOCKOUT is refused for 960 s (safing hold, see decision) DECIDED *Design value, electronics.json v4.1* T4390	**kNoLightSafingMin_s** = 960 s DERIVED T4391
24 T4392	**OVER_THRUST** T4393	flight RUN: FC_THRUST above 650 N on 3 consecutive fresh valid frames (rate 100 Hz, stale after 50 ms); never on the stand DECIDED *Design value, electronics.json v4.1* T4394	**kOverThrust_N** = 650 N DECIDED *Lead decision of 2026-10-13* T1033 **kOverThrustFrames** = 3 DECIDED *Lead decision of 2026-10-13* T4395 **kFcThrustRate_Hz** = 100 Hz DECIDED *Lead decision, per electronics.json v4.1* T4396 **kFcThrustStale_s** = 0.05 s DECIDED *Design value, electronics.json v4.1* T4397
25 T4398	**MODE_MISMATCH** T4399	ARM refused: mode strap invalid, or strap and load-cell reading disagree DECIDED *Design value, electronics.json v4.1* T4400	no threshold
26 T4401	**BAY_HOT** T4402	ARM refused: bay temperature above 318.15 K (45.0 C) or no valid reading DECIDED *Design value, electronics.json v4.1* T4403	**kBayArmMax_K** = 318.15 K TO-VERIFY TO-DEFINE T4404
27 T4405	**TANK_HOT** T4406	ARM refused: tank wall above 296.59 K (23.4 C) (limit 297.15 K (24.0 C) minus the wall-channel uncertainty 0.56 K), or Tsat(P_tank) above the limit with P_tank above 50.34 bar abs uncalibrated / 54.3 bar abs after B-PT-1 (guard band, see decision), or no valid tank temperature DECIDED *Design value, electronics.json v4.1* T4407	**kTanklgnMaxWall_K** = 296.587 K DERIVED T4408 **kTanklgnMax_K** = 297.15 K DECIDED *Lead decision of 2026-10-13* T4409 **kTankWallUnc_K** = 0.5625 K DERIVED T4410 **kTanklgnMaxP_Pa** = 5033700 Pa DECIDED *Design value, electronics.json v4.1* T4411 **kTanklgnMaxPCal_Pa** = 5429700 Pa DECIDED *Design value, electronics.json v4.1* T4412
28 T4413	**BLEED_FAULT** T4414	head-end bleed: pad BIT step failed (any mode); STAND: closure not confirmed, purge path not proven, or P_mid high with both valves commanded closed for 200 ms. FLIGHT: an unconfirmed closure is a FLAG to the FC and the relight proceeds (closure step bound 210 ms; deadline 160 ms from E0) DECIDED *Design value, electronics.json v4.1* T4415	**kHbleedLeakTime_s** = 0.2 s DECIDED *Design value, electronics.json v4.1* T4416 **kHbleedCloseConfirm_s** = 0.21 s DERIVED T4417 **kTorchGate_s** = 0.16 s DECIDED *Design value, electronics.json v4.1* T4418
29 T4419	**TORCH_LOST** T4420	torch confirmed (E1, 2oo2) but BOTH flame sensors dark for 5 ms before E2: pintle and NC valve close DECIDED *Design value, electronics.json v4.1* T4421	**kTorchLossTime_s** = 0.005 s DECIDED *Lead decision, line 164 (8b)* T4422
30 T4423	**TANK_COLD** T4424	ARM refused when no valid tank temperature exists or the coldest tank estimate (lower of the wall DS18B20 readings and Tsat(P_tank)) is below 288.15 K (15.0 C) plus the ARM margin 0.42 K (sized from the PREPRESS Tsat drop 0.23 K with a factor and the repeatability 0.07 K); IGNITE in PREPRESS refused below 288.15 K (15.0 C) on the pad first start and on the stand, with the safe exit: back to ARMED on the refusal tick, MV latch dropped, pintle seated, CLOSED switch expected within 500 ms; a flight relight is flagged only (RE-17) DECIDED *Design value, electronics.json v4.1* T4425	**kTanklgnMin_K** = 288.15 K DECIDED *Requirement, ENG-12* T4426 **kTankColdArmMargin_K** = 0.42 K DERIVED T4427 **kTankColdPrepressDrop_K** = 0.23 K TO-VERIFY T4428 **kTankColdRepeat_K** = 0.07 K DATASHEET T4429 **kMvCloseConfirm_s** = 0.5 s DECIDED *Design value, electronics.json v4.1* T4430
31 T4431	**CASE_WARM_STAND** T4432	stand mode only: ARM refused when TC1 or TC2 reads above 297.15 K (24.0 C) (stand T0 rule; a backstop for gross cases, the founder pre-fire checklist row is the control); flight has no T0 rule DECIDED *Design value, electronics.json v4.1* T4433	**kCaseStandArmMax_K** = 297.15 K DECIDED *Lead decision, per electronics.json v4.1* T4434

10.6 Power, head-end bleed provision and fail-safe states

Item	Value
Battery	LiPo 2S 2.2 Ah (16.3 Wh), usable 13 Wh DECIDED *Design value, electronics.json v4.1* T4435
Logic power, typical / peak	**1.25** T4436 / **2.6** W DERIVED T4437
Energy per flight day	**1.73** Wh DERIVED T4438
Energy per static-fire session	**2.79** Wh DERIVED T4439
Peak current	**35.7** A DERIVED T4440
Minimum voltage at the ECU	**6.11** V DERIVED T4441
Fusing	F101: 25 A ATM main; F102: 2 A 1206 logic; F301: 15 A ATM ARM bus (feeds the 12 V boost); F501: 3 A 1206 VBAT vent-branch fuse DECIDED *Design value, electronics.json v4.1* T4442
bleed: architecture	2 NC valves in series (V1, V2), restrictors 2 × 0.707 mm upstream of V1 (1.0 mm equivalent), P_mid Kulite XTL-190SM-2000A (C1, J511, LEAD 206 (3)) between them on the U570 comparator into the pin 23 ladder; V1 = pin 15 AND PURGE_WIN AND NOT (both-open timer), V2 = pin 24 AND PURGE_WIN, low-side on +12V_ARM DECIDED *Design value, electronics.json v4.1* T4443

Item	Value
bleed: schedule	purge_to_E0: v: 2.9; unit: s; tag: LEAD CHANGELOG 164 (2); src: config.h kPurgeRelight_s; flow_start: v: 0.42; unit: s; tag: DERIVED; src: config.h kPurgeFlowStart_s; flow: v: 2.43; unit: s; tag: LEAD 256 (2) / 257 (2); src: config.h kPurgeFlow_s; flow_min_fc_planning: v: 1.9; unit: s; tag: COMB CHANGELOG 163, LEAD 164 (2); src: config.h kPurgeFlowMin_s; flow_end_lead: v: 0.05; unit: s; tag: LEAD CHANGELOG 164 (1); src: config.h kPurgeFlowEndLead_s; path_proof_extra_wait: v: 0.1; unit: s; tag: ... DECIDED *Design value, electronics.json v4.1* T4444 [v2.11 text, still carried in v4.1; the both-open cap is superseded by 256 (2) T4445 (**2.43** s T4446 flow, **2.39** s T4447 cap), implemented in v2.12 (v4.1): ACCEPTED (REV-ELEC-4, line 266) T023 T027: contradiction register C-12] ACCEPTED (REV-ELEC-4, line 266) T027 purge FINAL 2.43 / 2.39 s with rule (d), (d1), (d2)
bleed: closure rule	E1 = 2oo2 torch AND [closure confirmed OR deadline 160 ms from E0]; flight: not confirmed → BLEED_FAULT flag to the FC, V2 commanded closed, relight proceeds; stand: abort DECIDED *Lead decision, line 166 (3)* T4452
bleed: cap decision 230 3	question: CHANGELOG line 230 (lead, 2026-10-19): ...ve on FA1S, so no other margin changes. (3) Purge (OI-42; line 216 (5)): the ONE flow basis 29.25 g/s at 289.5 K is ADOPTED. It gives 76.05 g at the 2.55 s cap, needs <= 2.548 s for 76 g, and needs 2.397 s at the warm end of Combustion's band (27.58-31.06 g/s over 283.15-297.15 K; Cd ASSUMED, 25.30-33.13 g/s over Cd 0.60-0.80). Electronics v2.11 sets the provisional fixed cap to 2.39 s if the scheduled purge flow fits inside it; if it does ... DECIDED *Design value, electronics.json v4.1* T4453 [v2.11 text, still carried in v4.1; the both-open cap is superseded by 256 (2) T4445 (**2.43** s T4446 flow, **2.39** s T4447 cap), implemented in v2.12 (v4.1): ACCEPTED (REV-ELEC-4, line 266) T023 T027: contradiction register C-12]
bleed: hardware both open timer	decision: CHANGELOG line 216 (lead, 2026-10-13): ...TORCH_FAULT; all unlit runs 0 false E2. (5) R2-05, 203 (3) RESTATED: the purge limit is the vented mass, so the ECU cap takes measured-time form after CF-4/B-BLEED-1: both-open overlap <= 76 g / mdot_measured - (t_close - t_open)_measured. Until then one flow basis is carried, the one Combustion v2.4b states, with GNC's G80 arm as the high-basis sensitivity; the 2.55 s commanded cap stays as a provisional software limit and the 'not truncated' ... DECIDED *Design value, electronics.json v4.1* T4454 [v2.11 text, still carried in v4.1; the both-open cap is superseded by 256 (2) T4445 (**2.43** s T4446 flow, **2.39** s T4447 cap), implemented in v2.12 (v4.1): ACCEPTED (REV-ELEC-4, line 266) T023 T027: contradiction register C-12] [routed (266 (8)) T031: E4-N08, the backstop allows up to 2.741 s against the 2.39 s never-above value; Combustion and the lead at B-BLEED-1]
bleed: pad bit	PREPRESS steps S0-S4 (valve pattern vs P_mid); any failure → BLEED_FAULT; a missing purge window fails S2 (V1 not opening) DECIDED *Design value, electronics.json v4.1* T4455
purge FINAL: kPurgeFlow_s	**kPurgeFlow_s = 2.43** s DECIDED *Lead decision, line 256 (2)* T4456 ACCEPTED (REV-ELEC-4, line 266) T027 purge FINAL 2.43 / 2.39 s with rule (d), (d1), (d2)
purge FINAL: kHbleedCap_s	**kHbleedCap_s = 2.39** s DECIDED *Lead decision, line 256 (2)* T4457 ACCEPTED (REV-ELEC-4, line 266) T027 purge FINAL 2.43 / 2.39 s with rule (d), (d1), (d2)
purge FINAL: calibration	both constants are calibration values: the config values are the DEFAULTS and the MAXIMA; EngineEcu::setPurgeCal(flow, cap) in SAFE only, 1.0 s <= flow <= 2.43 s, cap <= 2.39 s, flow + t_o - t_c + 2 ticks < cap DECIDED *Design value, electronics.json v4.1* T4458
purge FINAL: rule d	after CF-4 / B-BLEED-1: cap <= 76 g / mdot(297.15 K) - (t_close - t_open) with mdot from (d1) the upper 95 % bound of the measured N2O Cd x A; flow <= cap + t_close - t_open - 2 ticks; never above 2.43 / 2.39 s; (d2) CF-4 shows the 283.15 K need at the re-set flow; 2.397 s at the defaults DECIDED *Lead decision, line 257 (2)* T4459
purge FINAL: purge command time	PURGE at E0 - 2.90 s; flow start PURGE + 0.42 s; flow end E0 - 0.05 s; scheduled both-open command span 2.38 s DERIVED T4460
purge FINAL: vented mass DERIVED	scheduled 2.43 s: 71.1 g at 29.25 g/s (289.5 K), 75.5 g at 31.06 g/s (297.15 K); at the cap (flow 2.44 s) 71.4 / 75.8 g DERIVED T4461
purge mass basis (266 (7), E4-05 (9))	**Corrected by 266 (7), E4-05 (9):** 9. The stale "<= 73.0 g at 2.5 s" at v4.1 l.9974, sil_main.cpp l.2505 / 2668 and S12_bleed.json is replaced by the FINAL basis: DERIVED 71.1-75.8 g; SIL 76.38 g, an ESTIMATE (E4-N05). DERIVED T4462
Fail-safe: ECU power loss	MV latch drops → MV-P spring + pressure bias close it; the pintle spring closes the unpowered HV69; NC bleed valves close; igniter dead DECIDED *Design value, electronics.json v4.1* T4463
Fail-safe: MCU hang	hardware watchdog (50 ms) resets MCU and drops ARM bus → same as above; reboot lands in SAFE, never resumes RUN DECIDED *Design value, electronics.json v4.1* T4464
Fail-safe: link loss	see link_loss DECIDED *Design value, electronics.json v4.1* T4465
Fail-safe: Pc sensor fault	RUN: open-loop feed-forward map on P_feed + fault flag; IGNITION: abort; flight HIGH over-range: PC_VALID 0 on that tick only (non-sticky, LEAD 203 (1)(a)), LOW still latches DECIDED *Design value, electronics.json v4.1* T4466
Fail-safe: gimbal encoder fault	flag; RUN: continue on servo commands (default) unless mismatch abort criterion met DECIDED *Design value, electronics.json v4.1* T4467
Fail-safe: arming plug removed	ARM bus dead: igniter, pintle and main valve power removed regardless of firmware DECIDED *Design value, electronics.json v4.1* T4468
Fail-safe: ECU reset in flight	reboot to SAFE, MV and bleed closed, RUN never reported after the reset, the drogue is not released; no in-flight recovery path (LEAD 166 (8)); HIL-RST-1 proves it DECIDED *Design value, electronics.json v4.1* T4469
Fail-safe: pintle rail fail-high	LTC4365 OV disconnect 8.06-8.34 V: servo unpowered, spring closes; PRAIL_OK low drops the MV latch in hardware (U306) DECIDED *Design value, electronics.json v4.1* T4470
Fail-safe: Pc over-pressure	hardware comparator U330 on the C1 chain (trip 32.59 bar abs nominal, band 31.08-34.11) latches and removes the ARM bus independent of firmware; firmware abort 30 bar abs filtered for 5 ms DECIDED *Design value, electronics.json v4.1* T4471
Fail-safe: purge window	FC PURGE_EN low or the one-shot expired: both bleed drives forced off in hardware; both drives high for longer than the both-commanded-open timer (2.577-2.741 s, DESIGN, not drawn): V1 forced off DECIDED *Design value, electronics.json v4.1* T4472
Fail-safe: case TC	both case TCs lost: stand treated hot (abort), flight burn continues and the relight is refused CASE_HOT (RE-08) DECIDED *Design value, electronics.json v4.1* T4473

10.7 Lead rulings on v2.11 and items routed to Electronics v2.12

electronics.json /lead_rulings_v2p11 and the CHANGELOG items that route work to Electronics v2.12.

Item	Value (quoted)
case tc abort channel	CHANGELOG line 249 (lead, 2026-10-21): Lead decisions and verifications after line 248 (2026-10-20 21:17Z to 2026-10-21 03:19Z, times from date -u; SIM / CALC / DERIVED unless tagged, none of it physical approval; the owners' records are Electronics' work tree electronics/results/ (case_tc_channel_v1.json 6f29c93eff893adf, tf1_rate_v1.json 0c1bb777739dcf8f (5b995ead2f99d0e7 when ruled; updated since by its author), fa1_chain_v2p11.json eea3b964e41f0afe) with v2p11_notes/PROGRESS.md, and FEA3D's / FEA3D-2's fea3d/results/v352/, fea3d/runs/tail_F.log and fea3d/HANDOVER-FEA3D.md f46bbb008fc76329). (1) CASE-TC ABORT CHANNEL (Electronics v2.11 Part A, ruled 21:46:49Z): the A = 150 C (423.15 K) abort on the case TC station (TC1 / TC2, type K, MAX31856 U604 / U605) is ACCEPTED: high-reading error 7.78 K worst case, 3.55 K RSS, inside the +12.3 K false-trip margin (9.08 K over a -40..105 C board) [DATASHEET MAX31856 19-7534 Rev 0 / LITERATURE IEC 60584-1, AS RECALLED / DERIVED; EMI / ground 0.5 K and CJ gradient 1.0 K ASSUMED, TO-VERIFY B-TC-1]; the 0.20 s persistence already existed in v2.10 (engine_ecu.cpp 6e62a7e4e37930bd l.1043 / 1050, kOverTempTime_s; the brief's 'no persis... DECIDED *Lead decision, per electronics.json v4.1* T4474
stand T0 rule	CHANGELOG line 249 (lead, 2026-10-21): ...erms, else the tolerance is re-derived. (2) STAND T0 RULE (founder action): B1-length stand burns only with the engine soaked at a stand ambient <= 24 C (facility thermometer) [DERIVED: 20 + 4.48 / 0.96 = 24.67 C less 0.5 K thermometer, rounded down; dT_station / dT0 about 0.96, DERIVED]; in stand mode v2.11 refuses ARM if TC1 or TC2 reads > 24 C (297.15 K, own refuse code, test_core case) as a backstop for gross cases; the founder pre-fire checklist row is the control; no hot-day FE run; re-derive after B-TC-1 as T0_max = 20 + (12.26 - e_hi,cal) / 0.96; the A1 flight has no T0 rule (> 45 K margin). (3) T-F1 EXPOSURE BASIS (21:51:07Z, SUPERSEDING the 21:46:49Z 'A 165 rows' wording): the A 175 rows (TB, tb352g / h, both bounds; they cover r <= about 2... DECIDED *Lead decision, per electronics.json v4.1* T4475
T F1 basis	CHANGELOG line 249 (lead, 2026-10-21): ... flight has no T0 rule (> 45 K margin). (3) T-F1 EXPOSURE BASIS (21:51:07Z, SUPERSEDING the 21:46:49Z 'A 165 rows' wording): the A 175 rows (TB, tb352g / h, both bounds; they cover r <= about 22.8 K/s), because the GVHI rev-4 B1 end-of-burn station rate is 7.03 K/s [SIM; GVLO 6.74] above the 7.00 K/s limit of the A 165 rows (one method lands 0.004 K inside 165 C, below its resolution); the surrogate sets the TC-01 maximum to A and the station reads A - 1.43 / 1.54 K (GVLO / GVHI). A 175 rows [SIM / ASSUMED surrogate]: GVLO TC-02 248.4 C, above 150 C for 1192.0 s, TC-01 257.1 C (1435.8 s above 150 C); GVHI TC-02 233.0 C, 929.7 s, TC-01 243.4 C; (a)-(c) PASS on both bounds (lowest TC-02 TBS +0.409 / +0.745), no leak; (d) OPEN (T-F1). The TB rows quoted for the abort are the A 175 rows, with the A 150 rows beside them as the reading-equals-truth case; A stays 150 C (a lower A would trip a nominal merged-bound burn with the uncalibrated channel: 137.74 + 7.78 = 145.52 C); re-basing route: if B-TC-1 calibrates the channel (Electronics: tol_lo 10.28 K at 7.03 K/s, TC-01 maximum 161.82 C [DERIVED]), the lead may move T-F1 to the A 165 rows before T-F1 is run; Mechanical's re-map restates tf1_exposure_v352 (d23e72f089cca33a) and the T-F1 ... DECIDED *Lead decision, per electronics.json v4.1* T4476
relight and transducer	CHANGELOG line 249 (lead, 2026-10-21): ...he T-F1 founder text on the A 175 rows. (4) RELIGHT AND TRANSDUCER (21:46:49Z): kCaseRelightMax_K = 403.15 K (130 C) on Electronics' basis (A1 merged purge-to-end-of-relight rise 11.9 K + 1 s longer relight 3.0 K + 1 K relative channel); ARM refusals on the start limit (TC1 / TC2) and the TC3 nozzle-holder limit accepted, each limit with value and basis in electronics_v4; line 205's Pc-transducer proof requirement becomes >= 166 bar abs (1.1 x FA1S 151.04, 230 (2)), replacing 112 bar; the C1 pick meets it (206.8 bar abs, DATASHEET 1.5x rule); the head-face peaks 104.5 / 121.6 / 125.1 bar abs (CC-1 at c 222 / 400 m/s, CC-2) are SIM on the ASSUMED 1-D lumped passage model (fa1_chain_v2p11.json), all within FA1S 151.04. (5) FEA3D (21:47:29Z, 21:59:05Z): lc1v352hb accepted as found (superseded, pruned uncited, used by no cell, not re-run; run table 'superseded, never posted'); pose_stop045 R... DECIDED *Lead decision, per electronics.json v4.1* T4477
disk	CHANGELOG line 249 (lead, 2026-10-21): ...nges an envelope quantity, v6 is final. (12) DISK (01:30:39Z, 01:52:16Z): while js5b ran, Electronics' stop / ask level was lowered from 2.25 to 2.05 GiB (js5b projected lowest free 2.119 GiB; lead worst case about 2.02 GiB, above the 1.86 GiB floor [CALC]); its 60 MB net-new budget is unchanged (+6.3 MB used); the 01:49:04Z projection of 1.890 GiB was FEA3D-2's 44 MB costing scratch (deleted); transient scratch over... DECIDED *Lead decision, per electronics.json v4.1* T4478
runs	CHANGELOG line 249 (lead, 2026-10-21): ...tarted 03:18:46Z (start checks passed). Electronics v2.11's final test record 2 is done (MANIFEST_v1.json 620b3d566fe0c8b9, record_utc 02:15:54Z; all binaries identical after comment-only batches); its Part B handback is pending. (14) FOUNDER (2026-10-20 23:38:31Z) asked to see progress as step coun... DECIDED *Lead decision, per electronics.json v4.1* T4479
runs note	the MANIFEST cited in 249 (13) is superseded by two re-records with byte-identical binaries (comment-only batch, Makefile ovp target fix): firmware.record.re_records_after_line_249_13 DECIDED *Lead decision, per electronics.json v4.1* T4480
open	CHANGELOG line 249 (lead, 2026-10-21): ...torch, the face-peak inputs of (4) and Electronics' S6R input finding: 10 S6R runs have their input outside LC-TB70, the Combustion v2.4a replay head-face peak 88.02 bar being above the Z3 level 75.519 bar, and 46 of 896 grid cases peak above 75.519 bar [SIM]); the datasheet and its review; the final push (p1-propulsion PR #5, ... DECIDED *Lead decision, per electronics.json v4.1* T4481
Routed to v2.12 (LEAD 258 (4))	ROUTING: the F4 PC_SENSOR HIGH abort (>= 4.75 V = 146.5 bar abs; LOW side not added), the inside_v2 re-classification of the S6R runs (mechanical requests_v352[4]) and Q_JFWD logging go to Electronics v2.12; the combustion.json fa1_p_torch_v2p4c mechanical_row wording (governing row SOAK52 +0.670 / +0.785; +0.275 at SOAK70 is the screw-temperature sensitivity, same numbers) is corrected in combustion v2.4d together with the Z3 ruling and (d1) / (d2). DECIDED *Lead decision, line 258 (4)* T4482
Electronics v2.12 (LEAD 261 (3))	ELECTRONICS v2.12: the Part A interim (R3-01) arrived at 11:19Z and is saved verbatim in [working copy; path not printed]. It recommends option (b), a temperature-compensated zero on a new seat TC; option (a), a widened limit, would defeat OI-12 [DERIVED]. Working basis given at 11:25Z: option (b) with the seat TC on the MAX31856 #4 channel and the flagged v2.11 fallback; the lead rules on both at the handback. DECIDED *Lead decision, line 261 (3)* T4483

10.8 Lead rulings on the Electronics v2.12 handback (CHANGELOG 264) and on its review (CHANGELOG 266)

CHANGELOG line 264 (lead). Each item is a lead decision (DECIDED, Lead decision, line 264 (k)); numbers quoted in an item keep their own tags. electronics.json v4.1 implements them; it is the source of record under line 266 (§10.0), with option (b) carved out of every start with ignition until a reviewed v2.13 delta (266 (6)).

Ruling	Text (quoted) and values
Electronics v2.12 staged (Electronics, line 263)	Hub 06-electronics/staging_v2.12/: 84 files + SHA256SUMS-v2.12.txt f6fd3a63817dfb34 (firmware/ = changed files only, overlay on staging_v2.11/firmware); interfaces/electronics.json v4 and staging_v2.11/ unchanged; candidate results/electronics_v4.1.json c862375163393adc, docs/bench_plan_v5.md 360c1d6695cc305c, results/bom_v2p4.csv eb6e77dcd4610383. T4484
Source of record and the v4.1 candidate (264 (1))	interfaces/electronics.json stays v4 11f7896ad0d0613b; results/electronics_v4.1.json c862375163393adc stays PENDING REVIEW until REV-ELEC-4's verdict is recorded, and the datasheet shows v4.1 values as PENDING REVIEW. DECIDED *Lead decision, line 264 (1)* T022 [superseded by the promotion, 266 (7) T023]
R3-01 Pc zero, OI-C1-SEAT-DRIFT (264 (2))	R3-01 (OI-C1-SEAT-DRIFT): option (b) ADOPTED for flight and stand: Pc_comp = Pc - coef x (T_seat - T_seat,ARM) with the per-serial coefficient and T_cal from B-PC-1 (0-85 C) and the bench plan v5 acceptance (coefficient within 5 % at k = 3); kPcRezeroMax_Pa stays 0.30 bar on the compensated residual; kIgnBaseCheck_Pa 0.76 / 0.46 bar (default / B-PC-1-shown build). The seat TC is MAX31856 #4 (TC4, U904, J904; BOM v2.4 row 205): CONFIRMED; the v4.1 channel list still calls TC4 'bore top' and it is renamed 'C1 seat' at promotion. The fallback (coefficient 0, the flagged v2.11 behaviour) is NOT a flight configuration: it loses the A1 relight at k = 3 (default build) and from k = 1.5 on the minus sign (B-PC-1-shown build) [SIM S13T]; it is allowed on the stand only with k <= 2 (default build) or k <= 1 (B-PC-1-shown build) by F-10 and k x 0.0248 bar/K x (T0 - T_cal) clearly inside 0.5 bar at ARM [DERIVED]. NEW RULE (lead; not implemented yet): A1 pad no-go unless the B-PC-1 coefficient is loaded (non-zero) and the seat TC is valid at ARM; Electronics adds the pad go/no-go row after REV-ELEC-4. Residuals stay OPEN with owners: the per-serial coefficient and T_cal (founders, B-PC-1), the thermal-zero maximum (F-10), the seat-TC mount at the C1 boss and the P_mid / U570 temperature (Mechanical, MISSING), and the ASSUMED error-budget terms (coefficient 10 %, gradient, 0.25 s lag, 0.3 K). DECIDED *Lead decision, line 264 (2)* T4325 [**Carve-out, 266 (6)**: Option (b) (seat coefficient loaded, setPcSeatCoef != 0) is not used for any start with ignition (pad first start, stand starts, relights) until a v2.13 delta with the light-window hold and the TC4 step / rate checks is reviewed. The coefficient may be loaded for B-PC-1 / B-TC-1 bench work without ignition. DECIDED *Lead decision, line 266 (6)* T024] **Corrected by 266 (7), E4-02:** E4-02 (the coefficient-within-5-percent-at-k-3 acceptance, also in the 264 (2) wording, is replaced by the k-dependent limit of its Q3 table, eps_max 11.2 / 8.1 / 6.0 / 3.4 / 1.8 / 0.8 % at k = 1 / 1.25 / 1.5 / 2 / 2.5 / 3 with the 0.1012 bar adders [DERIVED]; beyond it the stand relight after B1 is planned REFUSED NOT_VENTED, fail-safe, with no relight objective claimed) DECIDED *Lead decision, line 266 (7)* T4241 **Corrected by 266 (7), TC4 label:** TC4 is the C1 seat TC (MAX31856 U904, J904, BOM row 205), not bore top (the 264 (2) rename is made by this line; the file keeps the old label until v4.2) DECIDED *Lead decision, line 266 (7)* T4162
Pc re-zero limit on the compensated residual; BASE check, default / B-PC-1-shown build (264 (2))	**0.30** bar DECIDED *Lead decision, line 264 (2)* T4326; **0.76** T4327 / **0.46** bar DECIDED *Lead decision, line 264 (2)* T4328 ACCEPTED (REV-ELEC-4, line 266) T027 kIgnBaseCheck_Pa 0.76 / 0.46 bar [the v2.11 values these replace (electronics.json v4): contradiction register C-13] [**Carve-out, 266 (6)**: Option (b) (seat coefficient loaded, setPcSeatCoef != 0) is not used for any start with ignition (pad first start, stand starts, relights) until a v2.13 delta with the light-window hold and the TC4 step / rate checks is reviewed. The coefficient may be loaded for B-PC-1 / B-TC-1 bench work without ignition. DECIDED *Lead decision, line 266 (6)* T024]
A1 BASE_AMB outcome (264 (3))	DECISION (3), A1 BASE_AMB: ADOPTED as built: a BASE_AMB at the A1 relight aborts PC_SENSOR (fail-safe, no false E2, relight lost: the lost-relight outcome of gnc.json v3.2, drogue descent and vehicle loss); with option (b) it occurs in 0 of the S13T A1 runs (T0 0 / 20 / 40 C, k 1-3, both signs, coefficient +/-10 %, TC +/-0.3 K) [SIM]. DECIDED *Lead decision, line 264 (3)* T4485 [closes OI-A1-BASE-AMB (§10.9): contradiction register C-16] SUSPENDED UNTIL v2.13 (266 (6)) T026 the evidence (not the decision) of 264 (3); re-shown on v2.13 [**Carve-out, 266 (6)**: Option (b) (seat coefficient loaded, setPcSeatCoef != 0) is not used for any start with ignition (pad first start, stand starts, relights) until a v2.13 delta with the light-window hold and the TC4 step / rate checks is reviewed. The coefficient may be loaded for B-PC-1 / B-TC-1 bench work without ignition. DECIDED *Lead decision, line 266 (6)* T024]
P_torch transducer, OI-PTORCH-PROOF (264 (4))	R4-12 (OI-PTORCH-PROOF): option 2 ADOPTED: P_torch (J901, on the torch body, not in the head; torch_port_bl08_v352_v1.json 20220647d32cbbbc) moves from the TE M3234-000005-070BG (proof 141.0 bar abs) to a C1-family Kulite XTL-190SM-2000A (proof 206.8 bar abs [DATASHEET]) against the 166.14 bar abs requirement (1.1 x FA1S 151.04 bar abs) [DERIVED]; it is the same part as P_mid (ENG-BL-08) and C1. Isolating the 070BG (option 1) is rejected: it changes the torch gate reading and is not designed. OPEN for the next Electronics round after REV-ELEC-4: re-base kTorchArmZeroMax_Pa and kPtorchRezeroMax_Pa (0.60 bar now, on the 070BG basis) on the C1 zero including the torch-body thermal zero (the local torch heat is MISSING: Combustion / Mechanical); the output saturates at the INA333 rail above 146.5 bar abs during FA-1 and its recovery is TO-VERIFY at B-PC-1 / F-12; the BOM gains one C1-family unit, TO-QUOTE delivered to Sonderborg DK-6400. No CAD v3.5.2 geometry changes (P_torch is not modelled; ENG-IG-01 is an envelope). DECIDED *Lead decision, line 264 (4)* T4486 [contradiction register C-10 and C-14]
T4 relight, OI-T4-RESIZE (264 (5))	N-07 (OI-T4-RESIZE): SHORTER RELIGHT ADOPTED: the T4 relight after the B1 p99 burn (case about 137.8 C at its end [DERIVED]) is cut so that no planned segment crosses A = 150 C, keeping the light, the latency measurement and a steady-Pc segment; Combustion sets the length in the T4 re-run of 261 (2) with Mechanical's soak-back. The coast is not lengthened to make margin (soak-back can raise the case after shutdown and is not modelled). T4 then demonstrates relight latency from a soaked state, not a 4.59 s landing-length relight from the B1-soaked state; bench plan v5 and the datasheet say so. DECIDED *Lead decision, line 264 (5)* T1555 [§7.9; contradiction register C-15] [T4 is a stand relight: it waits for v2.13 (266 (6)) T025] [**Carve-out, 266 (6)**: Option (b) (seat coefficient loaded, setPcSeatCoef != 0) is not used for any start with ignition (pad first start, stand starts, relights) until a v2.13 delta with the light-window hold and the TC4 step / rate checks is reviewed. The coefficient may be loaded for B-PC-1 / B-TC-1 bench work without ignition. DECIDED *Lead decision, line 266 (6)* T024]
A1 flight T0 range (264 (6))	A1 FLIGHT T0 0-40 C: ADOPTED as the ASSUMED design range (DECIDED) of the A1 relight analyses; the founders confirm it against the launch-site climate for 1 Sept 2027 before flight; the stand keeps T0 <= 24 C. DECIDED *Lead decision, line 264 (6)* T4487 Range **0** °C DECIDED *Lead decision, line 264 (6); an ASSUMED design range (264 (6))* T4488 to **40** °C DECIDED *Lead decision, line 264 (6); an ASSUMED design range (264 (6))* T4489
OI-JFWD-FLIGHT, PROVISIONAL (264 (7))	OI-JFWD-FLIGHT: PROVISIONAL lead rule until flight acceptance: a flagged first light (f3 JFWD_LS) inhibits the A1 relight and the vehicle takes the lost-relight outcome of gnc.json v3.2 (a relight on a joint that has used its one analysed Z3 event, 231 / 258 (4), is not flown); the founders confirm or change it at flight acceptance, with T-F1. DECIDED *Lead decision, line 264 (7)* T4490 [264 (7) provisional inhibit; not implemented yet: 266 (8) T4491, with v2.13]
REV-ELEC-4 and work not done (264 (8))	v4.1 is promoted only after its verdict. The author did NOT produce the v2.12 SIL MANIFEST or re-run S1-S11 / S14-S17 on v2.12: REV-ELEC-4 judges which of them the v2.12 changes can affect. DECIDED *Lead decision, line 264 (8)* T4492
REV-ELEC-4 verdict (266 (4))	REV-ELEC-4 (independent delta review of Electronics v2.12; interim E4-01 / E4-02 13:57Z; closed 14:34:56Z and returned 14:35:54Z by its stamps, delivered 14:38:25Z): ACCEPT WITH COMMENTS, with option (b) carved out of every start with ignition until a reviewed v2.13 delta; 0 BLOCKING, 1 MAJOR (E4-01), 5 MINOR (E4-02 to E4-06), 11 NOTES DECIDED *Lead decision, line 266 (4)* T029
E4-01, the MAJOR finding (266 (5))	E4-01 (MAJOR) [SIM reviewer SIL variant; CALC probes]: with option (b) the seat compensation acts in every state, is not frozen in the light window and has no TC4 step or rate check, so a TC4 reading error after E1 gives a false or premature E2 at every start. The unlit cold-flow rise is about 0.42 bar, about 0.38 bar short of the 0.80 bar E2 rise, which an error of about 15 K at k = 1 or 5 K at k = 3 closes [DERIVED]; 13 of 44 unlit relight runs reach FALSE_E2 and 33 of 64 lit runs lose the relight; at A1 k = 3 with a -30 K error the gnc v3.2 release predicate holds from the false E2 and run-lift flow enters the unlit chamber for 280 ms (a drogue release with no thrust; the FC is not modelled). Credible causes: spark-coupled interference on the grounded-junction TC at the C1 boss, a lead short, a MAX31856 conversion error. DECIDED *Lead decision, line 266 (5)* T032
Carve-out and promotion (266 (6), (7))	quoted in full in §10.0
Queued and routed (266 (8))	QUEUED, NOT STARTED (the founder says when): the Electronics v2.13 delta and its review: the light-window hold (latched at IGNITION entry until E2 + 0.45 s, bumpless resume), the TC4 step check (1.5 K per conversion) and rate check (5 K/s slew, 3 K trip) with a latched f3 SEAT_STEP, the E4-03 fix, unit tests UT-1 to UT-4, the S13T TC-fault set, the FMEA row, the B-TC-1 TC4 row with the spark on, the v4.2 folds, and notes E4-N01 (T_cal 10-30 C or kPcSeatArmDevMax_K 40 K), E4-N02 (the 264 (2) pad rule and the 264 (7) inhibit implemented), E4-N03 (47.5 against 47.75 bar abs, TO-VERIFY), E4-N04 (the P_torch re-base needs compensation or a photodiode-guarded gate; torch local heat MISSING) and E4-N11 (a SIL MANIFEST_v2). DECIDED *Lead decision, line 266 (8)* T030 Routed: E4-N08 (the hardware both-open backstop allows up to 2.741 s against Combustion's 2.39 s never-above value) to Combustion and the lead at B-BLEED-1; the FC rule for SEAT_STEP before a relight and the compensated-Pc release predicate are GNC and lead items with v2.13. DECIDED *Lead decision, line 266 (8)* T031

10.9 Electronics open issues

electronics.json /open_issues (34 items as published). An item a later lead line closes is listed under the closed group, with the line that closes it; only a residual that a lead line names stays open.

ID	Severity	Issue	Owner	Closure
Open: 31 T4666				
OI-HAL	HIGH DECIDED *Design value, electronics.json v4.1* T4496	Teensy HAL not written (RE-06); TLM_SLOW undefined; 96 ARM -Wdouble-promotion warnings (v2.10: 93); objects only, not linked TO-VERIFY T4495	Electronics DECIDED *Design value, electronics.json v4.1* T4497	HAL-1, B-ECU-1 DECIDED *Design value, electronics.json v4.1* T4494
OI-POWER	HIGH DECIDED *Design value, electronics.json v4.1* T4501	RE-09: +12V_ARM and the ARM bus over rating in the purge and E0 overlaps; the E0 row CANNOT PASS as designed (OI-E0-SUPPLY, bench plan v4 B-PWR-2); time-resolved budget, sequencing and peak-and-hold not done; v2.11 adds about 15 mA of C1 bridge current on +5V_SNS (power_budget.v2p11_sense_rail; the re-sum is not redone) TO-VERIFY T4500	Electronics DECIDED *Design value, electronics.json v4.1* T4502	power sequencing / capacity decided before the layout; B-PWR-2 DECIDED *Design value, electronics.json v4.1* T4499
OI-BLEED-VALVE	HIGH DECIDED *Design value, electronics.json v4.1* T4506	bleed valve not selected (LEAD 206 (4) search continues); peak-and-hold not drawn; valve times ASSUMED (30 / 80 ms) in the cap, the masses and the timer margins TO-VERIFY T4505	Electronics / Mechanical DECIDED *Design value, electronics.json v4.1* T4507	selection; B-BLEED-1 DECIDED *Design value, electronics.json v4.1* T4504
OI-WINDOW-LATENT	MEDIUM DECIDED *Design value, electronics.json v4.1* T4516	stuck-high purge-window chain and the both-open timer are invisible to the ECU (no free pin); a pre-flight check with PURGE_EN low is not in firmware v2.11 TO-VERIFY T4515	Electronics / FC owner DECIDED *Design value, electronics.json v4.1* T4517	CHK-PW, CHK-BT; firmware BIT step DECIDED *Design value, electronics.json v4.1* T4514
OI-BLEED-DETECT	MEDIUM DECIDED *Design value, electronics.json v4.1* T4521	S12 (SIM): both bleed valves stuck open (the 203 (4) label is corrected in head_end_bleed_provision.oi_bleed_detect) are not detected on the stand and are flagged 5.05 s after injection in the flight first burn (P_mid below the U570 band on the decided path); SIL vented 28.36 / 50.37 / 78.60 g (d 0.6 / 0.8 / 1.0 mm, after the lift), 104.17 g (first burn); no detection design in v2.11 TO-VERIFY T4520	Electronics / LEAD DECIDED *Design value, electronics.json v4.1* T4522	LEAD: accept the FMEA residual (FMEA-A1-BLEED-BOTH-OPEN) or ask for a detection design; B-BLEED-1 DECIDED *Design value, electronics.json v4.1* T4519

ID	Severity	Issue	Owner	Closure
OI-TEB-FALSE-LIGHT	MEDIUM DECIDED *Design value, electronics.json v4.1* T4533	S9 +/-1.034 bar TEB cases fail as in v2.9 / v2.10 (the v2.11 run uses the C1 chain with the same injected biases) TO-VERIFY T4532	LEAD / Electronics DECIDED *Design value, electronics.json v4.1* T4534	B-PC-1 residual <= 0.3 bar, then kBpc1Shown DECIDED *Design value, electronics.json v4.1* T4531
OI-GUARD-BAND	MEDIUM DECIDED *Design value, electronics.json v4.1* T4538	cold side CLOSED by LEAD 216 (7) (ARM margin 0.42 K, PREPRESS safe exit; S5: TANK_COLD 26, safe exits 0). Open: the uncalibrated P_tank hot interlock needs the tank <= about 19.8 C; wall-vs-bulk offset MISSING; countdown margin above 15 C MISSING TO-VERIFY T4537	Electronics / Mechanical DECIDED *Design value, electronics.json v4.1* T4539	B-PT-1 DECIDED *Design value, electronics.json v4.1* T4536
OI-E2-LATENCY	LOW DECIDED *Design value, electronics.json v4.1* T4543	E2 - light (SIL): p50 12 / max 13 ms in S5, 12 / 13 ms in S15 (flight), 12 / 13 ms (stand); A1 relight 13-20 ms over the S13 B / C / D lit cases, 18 ms at the offset corners within the 0.3 bar B-PC-1 residual, 20 ms at the -0.6 bar corners: the IGN-1 criterion (<= 20 ms) has no margin at those corners; GNC models thrust onset at E2 (GNC to confirm 20 ms) TO-VERIFY T4542	Combustion / Electronics / GNC DECIDED *Design value, electronics.json v4.1* T4544	IGN-1 DECIDED *Design value, electronics.json v4.1* T4541
OI-RE15	HIGH DECIDED *Design value, electronics.json v4.1* T4548	no layout; pin-function check, footprints, derating table not done TO-VERIFY T4547	Electronics DECIDED *Design value, electronics.json v4.1* T4549	RE-15 checks before fabrication DECIDED *Design value, electronics.json v4.1* T4546
OI-SCHEMATIC-V2P6	HIGH DECIDED *Design value, electronics.json v4.1* T4553	schematic v2.6 NOT drawn: three C1 bridge channels, U330 / U570 values on C1, the ride-through C330B variant, the both-open timer U576-U579, the C1 harness breaks; ERC and the netlist cross-check cover v2.5 only (ecu.schematic.v2p11_board_items_not_drawn) TO-VERIFY T4552	Electronics DECIDED *Design value, electronics.json v4.1* T4554	schematic v2.6 + ERC + netlist cross-check before RE-15 DECIDED *Design value, electronics.json v4.1* T4551
OI-GIMBAL-JAM	MEDIUM DECIDED *Design value, electronics.json v4.1* T4558	S6 gimbal_jam_moving_cmd: small-amplitude jam undetected (RE-24 test-based, LEAD 203 (6)); the abort is not inhibited near touchdown (LEAD 227 (7)) TO-VERIFY T4557	Electronics + GNC DECIDED *Design value, electronics.json v4.1* T4559	B-TVC-1 DECIDED *Design value, electronics.json v4.1* T4556
OI-TB-MEOP	ACCEPTED DECIDED *Design value, electronics.json v4.1* T4563	S6 throat block: plant Pc 53.42 bar abs, 33 ms above MEOP (R2-15: not 72 ms); inside LC-TB70 (S6C) TO-VERIFY T4562	Mechanical FEA DECIDED *Design value, electronics.json v4.1* T4564	LC-TB70 (04-fea/v352/lc_tb70_envelope_v1.json) DECIDED *Design value, electronics.json v4.1* T4561
OI-FC	HIGH DECIDED *Design value, electronics.json v4.1* T4568	FC: PURGE_EN discrete; R2 v2 at W 0.8 s with the mode-3 flush at the Pc-invalid onset (227 (6)), the reference choice and the frame-noise budget (<= 2.51 N rms iid at 0.8 s); DROGUE reports with the re-send rule (<= 10 re-sends, 0.2 s) and the DROGUE_ECHO log (216 (2)(a)); the d6 b6 echo check before a RELIGHT; no decision on the A7 high flag before E2 + 0.45 s (216 (8) condition 3); the FC Pc branch margin at E2; the PURGE commit L* 1.3 s with no unpurged RELIGHT (the ECU refuses it: UNPURGED 16); FC_THRUST valid flag clear on A1; PC_VALID launch commit; IMU filter delay TO-VERIFY T4567	FC owner + GNC DECIDED *Design value, electronics.json v4.1* T4569	B-FC-1, HIL-OT-1 DECIDED *Design value, electronics.json v4.1* T4566
OI-38-R2-COMMAND-HOLD	HIGH DECIDED *Design value, electronics.json v4.1* T4573	ECU side in SIL: lift-commanded hold at 0.21776 mm, backstop E2 + 850 ms, per-tick Pc validity (S17 v2.11: 0 dead releases with the hold and the flush at either window, W10 < 6 m/s, photodiode view). Open: the FC side; GNC R2 v2 re-run with the flush / reference / noise budget; the FC Pc branch margin; the lit F_min 33.14 N corners (230 (5)); CF-3 at 0.21776 mm; injector / feed soak-back heat TO-VERIFY TO-VERIFY T4572	GNC (FC) / Combustion / Electronics DECIDED *Design value, electronics.json v4.1* T4574	GNC R2 v2 re-run; B-FC-1; CF-3 start-lift point DECIDED *Design value, electronics.json v4.1* T4571
OI-FA1-PC-OVERRANGE	HIGH DECIDED *Design value, electronics.json v4.1* T4578	C1 chain (v2.11): 0 of 896 FA-1 grid cases reach the over-range or the above-FS flag; as built the A1 relight aborts 374-379 of 896 (filtered 30 bar / 5 ms and the comparator); the ride-through option (default OFF) aborts 0 and S6R aborts 105 of 105 required in-window over-pressure runs inside LC-TB70; adoption is the lead's; C1 output above full scale and recovery MISSING (F-10) TO-VERIFY T4577	LEAD (adopt the ride-through or not) / Electronics / founders (F-10, B-PC-1, B-DPC-1) DECIDED *Design value, electronics.json v4.1* T4579	LEAD decision; B-PC-1; B-DPC-1 at IGN-1 / T series; HIL-TR-1 DECIDED *Design value, electronics.json v4.1* T4576
OI-A1-PCLOSS-OVERTHRUST	ACCEPTED (FMEA record) DECIDED *Design value, electronics.json v4.1* T4583	A1: no over-thrust detector after an in-flight Pc loss (fmea_rows_a1) TO-VERIFY T4582	LEAD / GNC DECIDED *Design value, electronics.json v4.1* T4584	none on A1; RG-16 budget before FC_THRUST is armed on any mission DECIDED *Design value, electronics.json v4.1* T4581
OI-STAND-LATE-LIGHT	LOW DECIDED *Design value, electronics.json v4.1* T4588	S15 v2.11 stand: 4 of 230 lit plants end NO_IGNITION (light later than the stand W); flight 0 of 230; fail-safe by design (R2-19) TO-VERIFY T4587	Combustion / LEAD DECIDED *Design value, electronics.json v4.1* T4589	IGN-1 (record E1 → light on every stand start) DECIDED *Design value, electronics.json v4.1* T4586
OI-UNPURGED-RELIGHT	MEDIUM DECIDED *Design value, electronics.json v4.1* T4593	the ECU guard is DONE (LEAD 203 (7): flight build with the bleed fitted refuses a RELIGHT in COAST, UNPURGED 16). Remaining: (a) with an UNPRIMED / UNKNOWN line in flight with the bleed fitted there is no relight path (PURGE needs the direct path: refused PURGE_PATH; RELIGHT in COAST refused UNPURGED); (b) the torch-kept ride-through of the gap is not implemented; (c) Combustion's latency model takes E2 as Pc > 3 bar abs while the ECU flight E2 is the relative rule TO-VERIFY T4592	LEAD / Combustion / GNC (FC) DECIDED *Design value, electronics.json v4.1* T4594	LEAD on (a); FC rule in the FSW record (B-FC-1); CF-4 O-3, IGN-1 DECIDED *Design value, electronics.json v4.1* T4591
OI-GEN-TABLES	MEDIUM DECIDED *Design value, electronics.json v4.1* T4598	gen_tables.py v3.3 cannot parse the combustion.json v2.4b igniter sequence; engine_params.h / plant_tables.h / gnc_tables.h / valve_map_table.h stay generated from combustion.json v2.3 (HISTORICAL); v2.3 → v2.4b leaf diff of the read keys: 0 numeric changes outside the igniter sequence, 6 inside (results/gen_tables_v2p4b_leafdiff_v1.json); the four values with no source text are now sourced by Combustion v2.4b (firmware_constant_sources_v2p4b; mapping confirmed in requests_answered) TO-VERIFY T4597	Electronics (parser) + Combustion (machine-readable sequence) DECIDED *Design value, electronics.json v4.1* T4599	gen_tables v3.4, regenerate, diff, re-run the SIL record DECIDED *Design value, electronics.json v4.1* T4596
OI-FMIN-LIT	MEDIUM DECIDED *Design value, electronics.json v4.1* T4603	lit F_min 33.14 N (3 of 72 Combustion corners never reach 37.16 N at 0.21776 mm, LEAD 230 (5)): S17 v2.11 with the lit thrust capped there: no lit run released at either window with the flush (relight lost, drogue kept); without the flush, reference none, 274 of 1000 normal lit runs released TO-VERIFY T4602	GNC (R2 v2 re-run) / LEAD DECIDED *Design value, electronics.json v4.1* T4604	230 (5) open item DECIDED *Design value, electronics.json v4.1* T4601
OI-PHOTODIODE-VIEW	HIGH DECIDED *Design value, electronics.json v4.1* T4608	the dead-side result with reference none needs the photodiode main-chamber view (FLAMEOUT_OPT armed): photodiode-blind families with the flush release 20 / 42 (normal_noopt) and 42 / 98 (starved040_noopt) of 1000 dead runs at 0.5 / 0.8 s; with the reference re-taken at the ECU E2 or at the light: none TO-VERIFY T4607	Combustion / Mechanical (view) / GNC (reference) DECIDED *Design value, electronics.json v4.1* T4609	IGN-1 photodiode row (view proven with the torch closed) or the GNC reference choice DECIDED *Design value, electronics.json v4.1* T4606
OI-FC-PC-BRANCH	MEDIUM DECIDED *Design value, electronics.json v4.1* T4613	with the FC Pc branch (release on Pc valid and filtered Pc >= 3.0 bar abs) every normal-family run is released at the E2 frame: Pc_f at E2 3.04 bar abs, 0.04 bar above the branch threshold, so R2 never decides those runs (drogue_release_support.r2v2_sweep.verdict_227_5.fc_pc_branch_finding) TO-VERIFY T4612	GNC / FC owner DECIDED *Design value, electronics.json v4.1* T4614	state the branch margin or move the threshold; GNC R2 v2 re-run DECIDED *Design value, electronics.json v4.1* T4611
OI-R2-23-GLITCH	LOW DECIDED *Design value, electronics.json v4.1* T4618	a HIGH over-range glitch later in the burn (E2 + 350 ms) is also seen by the filtered channel and the U330 tap: as built HW_OVERPRESSURE at every command, the ride-through build PC_OVERPRESSURE / HW_OVERPRESSURE except at 200 N (fail-safe, relight lost); the real glitch rate is MISSING TO-VERIFY T4617	Electronics / founders DECIDED *Design value, electronics.json v4.1* T4619	B-PC-1, HIL-TR-1 DECIDED *Design value, electronics.json v4.1* T4616
OI-FACE-PEAK	ROUTED DECIDED *Design value, electronics.json v4.1* T4623	C1 seat-face peaks 104.5-125.1 bar abs (SIM on the ASSUMED 1-D lumped passage model) against the 102 bar abs T <= 1 ms ultimate of line 205; LEAD 249 (4) restated the proof requirement (>= 166 bar abs; C1 206.8 bar abs) and the peaks lie within FA1S 151.04 TO-VERIFY T4622	REV-COMB-4 / Combustion / Mechanical (CHANGELOG 249 (15)) DECIDED *Design value, electronics.json v4.1* T4624	REV-COMB-4 DECIDED *Design value, electronics.json v4.1* T4621
OI-S6R-INPUT	ROUTED DECIDED *Design value, electronics.json v4.1* T4628	10 S6R runs have their INPUT outside LC-TB70: the Combustion v2.4a FA-1 replay max-face case peaks 88.02 bar abs > Z3 75.519 bar abs; 46 of 896 grid cases peak above 75.519 (141 above 63.1); no v2.11 claim depends on them TO-VERIFY T4627	REV-COMB-4 / Mechanical (CHANGELOG 249 (15)) DECIDED *Design value, electronics.json v4.1* T4629	REV-COMB-4; LC-TB70 DECIDED *Design value, electronics.json v4.1* T4626
OI-SEAT-RINGING	MEDIUM DECIDED *Design value, electronics.json v4.1* T4633	ideal 25 us steps ring the C1 seat passage in the SIL model (laminar-only damping ASSUMED): the T45 guard does not fire on them (record: the ride-build ideal-step aborts come at the window end or the comparator, ride_through.seat_ringing_finding); the 1 kHz raw swing about 38-61 bar abs is from a pre-record diagnostic (not in the record); whether a real passage rings like this is MISSING; the main step cases use a 1.0 ms rise (LC-TB70 pulse convention) TO-VERIFY T4632	Electronics / Combustion / founders DECIDED *Design value, electronics.json v4.1* T4634	B-DPC-1, HIL-TR-1 DECIDED *Design value, electronics.json v4.1* T4631
OI-C1-SEAT-DRIFT	HIGH DECIDED *Design value, electronics.json v4.1* T4644	R3-01: option (b) implemented in SIL; needs the lead ruling, the per-serial coefficient and T_cal (B-PC-1 over 0-85 C), the thermal-zero maximum (F-10), the seat TC mounting at the C1 boss and the P_mid / U570 temperature (MISSING) TO-VERIFY T4643	LEAD (ruling) / Electronics / founders (B-PC-1, F-10) / Mechanical (seat TC mount) DECIDED *Design value, electronics.json v4.1* T4645	lead ruling; B-PC-1 zero-TC row; F-10 answer DECIDED *Design value, electronics.json v4.1* T4641 [264 (2) ruled option (b); 266 (6) T024: option (b) is not used for any start with ignition until a reviewed v2.13 delta (carve-out, §10.0); residuals OPEN (264 (2)) T4642] SUSPENDED UNTIL v2.13 (266 (6)) T026 option (b) rows and residuals

ID	Severity	Issue	Owner	Closure
OI-STAND-RELIGHT-COEF	MEDIUM DECIDED *Design value, electronics.json v4.1* T4649	S13T: a stand relight after a B1-length burn is refused NOT_VENTED when the seat coefficient is 10 % low at k >= 2 (0.27-0.40 bar residual on the 0.30 bar limit): B-PC-1 must hold the coefficient to about 5 % at k = 3 (safe refusal) TO-VERIFY T4648	Electronics / founders DECIDED *Design value, electronics.json v4.1* T4650	B-PC-1 per-serial residual over the 50.4 K span DECIDED *Design value, electronics.json v4.1* T4647 **Corrected by 266 (7), E4-02:** E4-02 (the coefficient-within-5-percent-at-k-3 acceptance, also in the 264 (2) wording, is replaced by the k-dependent limit of its Q3 table, eps_max 11.2 / 8.1 / 6.0 / 3.4 / 1.8 / 0.8 % at k = 1 / 1.25 / 1.5 / 2 / 2.5 / 3 with the 0.1012 bar adders [DERIVED]; beyond it the stand relight after B1 is planned REFUSED NOT_VENTED, fail-safe, with no relight objective claimed) DECIDED *Lead decision, line 266 (7)* T4241 SUSPENDED UNTIL v2.13 (266 (6)) T026 the stand relight after B1
OI-PTORCH-PROOF	HIGH DECIDED *Design value, electronics.json v4.1* T4654	R4-12: 070BG proof 141.0 bar abs < 166.14 bar abs required at FA1S; isolate or move P_torch to a C1-family unit (rev_comb_4_items_v2p12.R4-12) TO-VERIFY T4653	LEAD / Electronics / Combustion / Mechanical DECIDED *Design value, electronics.json v4.1* T4655	lead choice; F-7 / F-12 data DECIDED *Design value, electronics.json v4.1* T4652 [264 (4): the C1-family Kulite; threshold re-base OPEN; 266 (8) T4148: E4-N04 with v2.13; contradiction register C-10, C-14]
OI-T4-RESIZE	MEDIUM DECIDED *Design value, electronics.json v4.1* T4659	N-07: at the B1 end-of-burn rate the re-sized T4 relight (4.59 s) needs the case TCs <= 112.7 C at the relight; the B1 p99 burn ends at about 137.8 C; a coast longer than about 19 s or a shorter relight (bench plan v5 T1-T4) TO-VERIFY T4658	Combustion / Mechanical / LEAD DECIDED *Design value, electronics.json v4.1* T4660	Combustion re-run of the T4 rows on the re-sized profile; Mechanical soak-back DECIDED *Design value, electronics.json v4.1* T4657 [264 (5): the shorter relight; T4 is a stand relight and waits for v2.13 (266 (6)) T025; contradiction register C-15]
OI-JFWD-FLIGHT	MEDIUM DECIDED *Design value, electronics.json v4.1* T4664	a flagged first light (f3 JFWD_LS) leaves the relight on a used joint; line 231 covers one Z3 event per assembly (LEAD 258 (4) OPEN) TO-VERIFY T4663	LEAD DECIDED *Design value, electronics.json v4.1* T4665	lead ruling before flight acceptance DECIDED *Design value, electronics.json v4.1* T4662 [264 (7) provisional inhibit; not implemented yet: 266 (8) T4491, with v2.13]
Closed by a lead line or marked CLOSED in the file (kept for the record): 3 T4667				
OI-PURGE-CAP	CLOSED (LEAD 256 (2), 257 (2), 258 (2)) DECIDED *Design value, electronics.json v4.1* T4511	FINAL: 2.43 s flow / 2.39 s cap as calibration constants, rule (d) (d1) (d2) (purge_final_v2p12); residual: CF-4 / B-BLEED-1 re-set TO-VERIFY T4510	LEAD DECIDED *Design value, electronics.json v4.1* T4512	lead decision on the cap; measured-time form after CF-4 / B-BLEED-1 DECIDED *Design value, electronics.json v4.1* T4509 [lead decision made, 256 (2) T4445; implemented in v4.1; open: measured-time form after CF-4 / B-BLEED-1 T4448; E4-N08 routed (266 (8)) T031; contradiction register C-12] ACCEPTED (REV-ELEC-4, line 266) T027 purge FINAL
OI-A1-BASE-AMB	MEDIUM DECIDED *Design value, electronics.json v4.1* T4526	S13T with the drift model: option (b) 0 BASE_AMB in the A1 sweep (both builds); fallback BASE_AMB → PC_SENSOR at k = 3 (default build) and at k >= 1.5 on the - sign (B-PC-1-shown build) (r3_01_c1_drift.decision_3); LEAD decision (3) at this handback TO-VERIFY T4525	LEAD DECIDED *Design value, electronics.json v4.1* T4527	accept the outcome or rule the alternative (keep the measured P0 below ambient; not implemented) DECIDED *Design value, electronics.json v4.1* T4524 [closed by 264 (3) T4485 (BASE_AMB at the A1 relight ADOPTED as built); the evidence (not the decision) of 264 (3) is suspended until v2.13 and is re-shown on v2.13 (266 (6)); open only for the residual a lead line names: the A1 pad no-go rule of 264 (2) T4528 (NEW RULE; NO-GO until v2.13, 266 (6); §12.4). 253 (5) T4529: the "0 of 500" evidence is structural, not an occurrence rate. Contradiction register C-16] SUSPENDED UNTIL v2.13 (266 (6)) T026 the evidence of 264 (3)
OI-STALE-LABELS	CLOSED (N-09, v2.12) DECIDED *Design value, electronics.json v4.1* T4638	config.h comments corrected in v2.12 (rev_elec_3_author_status.N-09) TO-VERIFY T4637	Electronics DECIDED *Design value, electronics.json v4.1* T4639	v2.12 text pass (comments, labels) DECIDED *Design value, electronics.json v4.1* T4636

11 Mission and landing (GNC v3.2)

Basis. The basis is the published gnc.json v3.2, `interfaces/gnc.json` (bde317782574cbd9, published by the lead 2026-10-21, CHANGELOG 260). It is the candidate `01-systems-gnc/staging_v3.1/rev4b/v32/gnc_v3.2_rev4b.json` (43d58c4aff164b86) with the REV-GNC-7 corrections (reviews/REVIEW-GNC-R5.md) and the lead folds L1-L4 (publication_note.published_v3p2). It folds the lead decisions of CHANGELOG lines 204-227, the REV-GNC-6 corrections and the REV-GNC-7 folds of lines 250 (6), 253 (7), 256 (2), 257 (2) and 258. This file is the rev4b file (9aacdefaaa0df250) with the REV-GNC-7 datasheet operations DS 1-7 and rows 11m-11q added by the lead. The ENG-11 FE-plant re-run (CAD v3.5.1, `rev4b/eng11_fe/`) and REV-GNC-7's check on the CAD v3.5.2 R3 pose table and K band are folded into the v3.2 text (ENG-11_servo_output_stiffness, k_band_r3_v352). DECIDED *Design value, DATASHEET-INPUTS-GNC.md v3.2* T4668

11.1 Mission headline numbers

01-systems-gnc/DATASHEET-INPUTS-GNC.md (DSIG v3.2); rows 6-8 (gimbal, slew, servo stiffness) are in §3.2. **Vehicle basis of rows 1-5 and 9: massprops v3.5.1** (the basis column names it); the CAD v3.5.2 mass change is in the table below.

Item	Value for the datasheet	Basis	Source
Mission set: A1 apogee target (baseline flight)	210 m. Profile: TVC ascent, nose drogue at apogee, relight, throttled TVC landing. DECIDED *Requirement, DATASHEET-INPUTS-GNC.md v3.2; basis SIM* T4669	The line-208 band rule, judged against the v3d 285 m reference with the dip onset at 80 g. liq|G73: 210 m gives -25/+20, p 0.551, UB95 0.802 pp. pw|G73: the band is 220 m. The decision takes the lower state, 210 m. N = 1000 per main base (A1_none, A1J_pc); massprops v3.5.1 (488b47f9d55ff91b); sim6 v2.9-dev. Four-base pooled (informational): liq 210 m UB95 0.595 pp. DECIDED *Requirement, DATASHEET-INPUTS-GNC.md v3.2; basis SIM* T4670	v3.2 `v32_items_rev4b[3]`; `01-systems-gnc/staging_v3.1/rev4/out_rev4/p2/p2_rev4.json` (68cf1f643effd5b9) T4671
Mission set: B1 development hop cap	62 m, PROVISIONAL until the T1-T4 regression data. The 85 m dispersed cycles stay the thermal and erosion sizing cases. ASSUMED *Requirement, DATASHEET-INPUTS-GNC.md v3.2; basis SIM* T4672	At 62 m (regression band plus 7-11 % throat growth): P 0.996 (SIM, N = 500, ledger v3d / iface_v3c). The GNC cap is 62.9 m, rounded down. B1 is not re-flown on massprops v3.5.1 (TO-VERIFY, GNC). ASSUMED *Requirement, DATASHEET-INPUTS-GNC.md v3.2; basis SIM* T4673	v3.2 `b1_regression_band.lead_cap`, `flight_profile.mission_plan_v3.v.B1`, `open_items[25]` T4674
A1 landing P at 210 m	P = 0.9835. This is P(sim6 landing AND >= 80 g liquid at touchdown), pooled A1_none 0.984 / A1J_pc 0.983. Plain sim6 landing is 0.986 / 0.985. The pw state gives 0.988, and the v3d reference 0.986. SIM T4675	N = 1000 per main base (2000 flights), state liq|G73, v3.5.1, sim6 v2.9-dev. This P is not used for flight acceptance yet: that waits for the 210 m gate-delay arm and the canopy-delay bound (227 (8); row 11a). SIM T4676	`p2_rev4.json` (68cf1f643effd5b9) `variants.v351.states.liq.bands.210` T4677
Liftoff T/W	2.92 DERIVED T4678	Rated 500 N; liftoff mass 17.44 kg; nominal deterministic A1 flight; 0 kg payload; 2.00 kg N2O tank at 0.58 m; v3.5.1. DERIVED T4679	v3.2 `v32_items_rev4b[15]` (c); `01-systems-gnc/staging_v3.1/rev4/out_rev4/payload_n500/payload_rev4.json` (6d62104dac3f4fcd) T4680
Touchdown T/W; landing floor ratio	3.16; 0.633. The floor ratio is the 20 % floor thrust over the touchdown weight. DERIVED T4681	Nominal touchdown mass 16.11 kg (as row 4). The landing ceiling is >= 237.6 N, from 1.5 × 16.15 kg weight (DERIVED). The floor is <= 20 % of F_max (DERIVED; 25 % flies, 40 % fails without ballast). DERIVED T4682	as row 4; v3.2 `requirements.throttle` T4683
Landing-mass basis	Nominal planning touchdown mass <= 16.15 kg. A planning mass above 16.15 kg needs its own paired arm. Lower bounds (unchanged): 7.08 / 11.33 / 14.16 kg at the 12.5 / 20 / 25 % floors. Record only, not a window: the dispersed 14.18-18.13 kg. DERIVED *Requirement, DATASHEET-INPUTS-GNC.md v3.2; basis SIM* T4684	16.15 kg is the v3.5.1 estimate covered by the 210 m decision MC (SIM, N = 1000 per main base). The GNC nominal touchdown is 16.11 kg (DERIVED). In that MC, P(80 g) is 0.976 at >= 16.15 kg against 0.989 below (Fisher p 0.049). DERIVED *Requirement, DATASHEET-INPUTS-GNC.md v3.2; basis SIM* T4685	v3.2 `icd_inputs_for_airframe.landing_mass_window_kg`; `01-systems-gnc/staging_v3.1/rev4/out_rev4/p2/landing_mass_coverage_h210.json` (d40472c8d92eaac9) T4686
Payload record (informational)	Baseline: no payload, 2.00 kg N2O tank (founder, line 219). +0.5 kg: 190 m band. +1 kg: no band with the 2.00 kg tank; it reaches 210 m only with a 2.20 kg tank, whose A1 engine-on p99 8.92 / 8.71 s (max 9.24 s) exceeds the 8.258 s A1 sizing counter, so it is TO-VERIFY (B1, thermal and erosion check) before it is quoted. +2 kg: not met; it fails on landing control, not propellant. SIM T4687	Criterion: the line-208 rule with the sweep floor at 180 m, N = 500 per base. +0.5 kg: 190 m PASS, UB95 0.18 pp; 200 m fails the bound. +1 kg, 2.20 kg tank at 210 m: UB95 0.54 pp; paired against the 0 kg vehicle -8/+8, UB95 0.66 pp. SIM T4688	v3.2 `v32_items_rev4b[15]`; `payload_rev4.json` (6d62104dac3f4fcd) T4689

The GNC figures above (rows 1-5 and 9: the A1 apogee, the landing P, the liftoff mass and T/W) are on massprops v3.5.1. The CAD v3.5.2 landing-mass change is below GNC's re-fly trigger with and without the conditional TC-09 thinning: no re-fly by mass. The change is CALC, not weighed. gnc.json /reference_vehicle/liftoff_mass_kg is the ledger-v3d reference vehicle and is not quoted as current (lead 262 (5)).

Item	Value
GNC vehicle basis (row 4): liftoff mass / liftoff T/W (massprops v3.5.1)	**17.44** kg DERIVED T1016 / **2.92** DERIVED T1018
Landing-mass change v3.5.2 vs v3.5.1: as built / without the conditional TC-09 thinning	**−0.0662** kg CALC T1019 / **+0.0118** kg CALC T1020
GNC's re-fly trigger (landing-mass growth)	**0.10** kg DECIDED *Lead decision, per massdelta_v3.5.2_vs_v3.5.1.json v3.5.2* T1022
Verdict and evidence (03-cad/v3.5.2/massdelta_v3.5.2_vs_v3.5.1.json /gnc_trigger)	BELOW the trigger with and without the conditional TC-09 thinning: no re-fly by mass CALC T1023 CALC (CAD volume x engmat density + hardware CALC); not weighed, so not a 'confirmed' mass; COTS masses unchanged CALC T1024

11.2 GNC open items

Open item	What is needed (quoted)	Owner / source
11a Open item: 210 m gate-delay arm	D = 0.25 / 0.5 / 0.75 / 1.0 s, both states, N = 1000 per main base (about 16,000 flights), plus a bound on the drogue deploy-to-inflation delay. Needed before the 210 m relight P is used for flight acceptance; not needed for v3.2 or the datasheet. TO-VERIFY T4690	Owner: GNC (arm); recovery / airframe with GNC (canopy bound); v3.2 `open_items[42]` T4691
11b Open item: ENG-11 on the FE plant	The CAD v3.5.1 re-run (`rev4b/eng11_fe/`) and REV-GNC-7's CAD v3.5.2 check are done (row 8; design-to 100-110 N m/rad on CAD v3.5.2). Remaining: B-TVC-1 and the bending mode. DERIVED T4692	Owner: REV-GNC reviewer (check); lead (datasheet sentence); v3.2 `open_items[43]` T4693
11c Open item: B-TVC-1 bench	Measure K_s with its uncertainty, J at the servo output shaft, the reflected-inertia convention, and the rocking modes at the poses. K_s is frozen only on an ENG-07d PASS. TO-VERIFY T4694	Owner: founders / Electronics (bench); GNC (re-run); v3.2 `open_items[46]` T4695
11d Open item: airframe first bending mode	The full and landing modes, measured or from the airframe FE. airframe_first_bending_min_Hz 10 Hz is NOT CONSISTENT with ENG-07d at the window with one fixed notch. TO-VERIFY T4696	Owner: airframe work (deferred by the founder, line 222); GNC re-runs ENG-07d; v3.2 `open_items[44]` T4697
11e Open item: second ECU notch	A notch on the measured bending mode (RESEARCH PROPOSAL). MISSING T4698	Owner: GNC; v3.2 `open_items[45]` T4699
11f Open item: R2 with Pc invalid	Electronics v2.11 S17 CONFIRMED pcf_tmax 0.8 s for W10 < 6 m/s with conditions: the ECU R2 hold, the mode-3 flush, the photodiode main-chamber view (or the reference re-taken), the FC Pc branch off or moved with margin (OI-FC-PC-BRANCH), FC frame noise <= 2.51 N rms iid at 0.8 s, cold flow x 1.3 margin +1.242 N. Lit side: 3 of 72 P4c corners stay below 37.16 N (OI-FMIN-LIT). GNC re-runs R2 v2 with the flush. TO-VERIFY T4700	Owner: Electronics v2.11 (S17, budget); Combustion v2.4b (lit side); then GNC re-runs R2 v2; v3.2 `open_items[47]`, `[48]` T4701
11g Open item: 2.20 kg tank option	Its B1, thermal and erosion check (A1 engine-on p99 8.92 / 8.71 s, max 9.24 s, > 8.258 s). TO-VERIFY T4702	Owner: Combustion / Mechanical; v3.2 `open_items[49]` T4703
11h Open item: 210 m coverage and duty cycle	The coverage set at 210 m (e19, L400, wind sweep) and a 210 m A1 duty-cycle record. B1 governs thermal and erosion sizing. MISSING T4704	Owner: GNC; v3.2 `open_items[50]` T4705
11i Open item: sim relight start level	100 N in the sim vs Combustion's 62.5 N start lift (open model item). SIM T4706	Owner: GNC; v3.2 `open_items[51]` T4707
11j Open item: GIMBAL_FAULT near touchdown	The abort is NOT inhibited. Revisit it if B-TVC-1 gives a jam rate. Encoder noise, vibration and supply sag are not modelled. DECIDED *Requirement, DATASHEET-INPUTS-GNC.md v3.2; basis SIM* T4708	Owner: lead (revisit); Electronics / founders (jam rate); GNC (re-run); v3.2 `open_items[52]` T4709
11k Open item: delivered slew at 7 deg / 7.4 V	Recompute with the exact linkage kinematics (GNC extrapolation 97.8-111.2 deg/s). TO-VERIFY T4710	Owner: Mechanical / Electronics T4711
11l Open item: B1 cap	The final cap after T1-T4 (regression law measured). B1 is not re-flown on massprops v3.5.1 (row 2). TO-VERIFY T4712	Owner: lead with the Combustion data (cap); GNC (re-fly); v3.2 `open_items[25]` T4713
11m Open item: FC Pc branch (OI-FC-PC-BRANCH)	The FC Pc branch of the release predicate (Pc valid AND Pc >= 300 kPa abs) releases every normal-family run at the E2 frame in S17 (Pc_f at E2 3.04 bar abs), including runs that flame out 1-50 ms later outside R2. State the branch margin or move it, then re-fly the release statistics. SIM T4714	Owner: GNC / FC owner; v3.2 `open_items[53]` T4715

Open item	What is needed (quoted)	Owner / source
11n Open item: photodiode view (OI-PHOTODIODE-VIEW, HIGH)	pcf_tmax 0.8 s (W10 < 6 m/s, 227 (5)) holds on the dead side only with the photodiode main-chamber view (FLAMEOUT_OPT armed) or the GNC reference re-taken at the ECU E2 / light; photodiode-blind families release 42 / 98 of 1000 dead runs at 0.8 s. Closes at IGN-1. SIM T4716	Owner: Combustion / Mechanical (view); GNC (reference option); v3.2 `open_items[54]` T4717
11o Open item: lit-side F_min (OI-FMIN-LIT)	At the 0.21776 mm start-level lift 3 of 72 P4c corners never reach 37.16 N within 0.8 s (F_min 33.14 N): a lost relight in the Pc-invalid branch. 37.16 N and 0.8 s are kept. The start-level lift is TO-VERIFY at CF-3. SIM T4718	Owner: GNC / lead; Combustion (CF-3); v3.2 `open_items[55]` T4719
11p Open item: C1 thermal zero (R3-01, REV-ELEC-3 MAJOR)	The C1 transducer thermal zero is not in the re-zero / BASE limits: a negative drift loses the A1 relight (PC_SENSOR), with drogue descent and vehicle loss. GNC's A1 P does not include it. The maximum coefficient is MISSING (F-10). DERIVED T4720	Owner: Electronics v2.12; Mechanical (seat node); founders (F-10); GNC (re-fly); v3.2 `open_items[56]` T4721
11q Open item: purge condition (d)	kPurgeFlow_s and the both-open cap are re-set from the CF-4 / B-BLEED-1 measured N2O Cd x A (upper 95 % bound) with the mass flow at 297.15 K, never above 2.43 / 2.39 s, and CF-4 shows the 283.15 K need at the re-set flow. Without the re-set: measured Cd x A <= 1.08 x the Cd 0.70 value, or GNC re-runs the A1 reserve test at the measured vent. DECIDED *Requirement, DATASHEET-INPUTS-GNC.md v3.2; basis DERIVED* T4722	Owner: Combustion / Electronics (CF-4, constants); GNC (reserve test if needed); v3.2 `open_items[57]` T4723

12 Verification status

SIM / CALC / DERIVED; not physical approval; no hardware has been tested. Independent review verdicts for the engine team's digital deliverables. A verdict is not physical approval: no hardware has been tested. DECIDED *Design value, SUPERVISION.json 18 reviews* T4724 No value here is a measurement. Review verdicts, matching file hashes, clean CAD checks, passing unit tests and this document's own trace check are digital evidence only.

12.1 Review status of the deliverables used here

Verdicts from reviews/SUPERVISION.json and, for REV-ELEC-4 and REV-DS-1 (not in the register yet), from their review files; dispositions from the CHANGELOG. A verdict covers the hashed files only (§12.2). A verdict is not physical approval.

Review	Deliverable	Verdict	Lead disposition / plan (quoted)
REV-COMB-4	Combustion v2.4b (delta re-review); folded in v2.4c / v2.4d	**ACCEPT WITH CORRECTIONS** DERIVED T4725	(1) REV-COMB-4, the independent delta re-review of Combustion v2.4b: interim 07:52:49Z, full review 08:32:27Z, saved verbatim by the lead as reviews/REVIEW-COMBUSTION-R4.md (2908321d7f75d439; artefacts reviews/combustion/r4/, MANIFEST_REV-COMB-4.sha256 996475830cae731f, 23/23 verified 08:37:28Z; (2) PURGE, 253 (7)(ii), now FINAL: the 2.43 s scheduled flow with the 2.39 s cap is ADOPTED [DERIVED by REV-COMB-4 on the author's SIM model: needed flow 2.2 / 1.9 / 1.8 s at 283.15 / 289.5 / 297.15 K ... DECIDED *Lead decision, line 256* T4726
REV-ELEC-3	Electronics v2.11 (firmware v2.11, electronics.json v4)	**ACCEPT WITH COMMENTS** DERIVED T4727	(4) REV-ELEC-3 (independent delta review of Electronics v2.11; started 05:45:38Z, returned 06:59:45Z): ACCEPT WITH COMMENTS; 0 BLOCKING, 1 MAJOR (R3-01), 5 MINOR (R3-02 to R3-06), 10 NOTE; Artefacts reviews/electronics/r3/ (MANIFEST_REV-ELEC-3.sha256 44fdbc2ebb7e355c, lead 31 of 31 OK 07:02:55Z); the review is saved by the lead as reviews/REVIEW-ELECTRONICS-R3.md fcf05f8bc7382cd1; (7) Decisions on Electronics' six items, with REV-ELEC-3's recommendations: (i) ADOPTED: the R2-03 ride-through is ... DECIDED *Lead decision, line 253* T4728
REV-GNC-7	GNC rev4b / gnc.json v3.2	**PUBLISHABLE WITH CORRECTIONS** DERIVED T4729	(1) REV-GNC-7 ACCEPTED: its final (reviews/REVIEW-GNC-R5.md 6b86fcd326568646, saved verbatim by the lead at 10:39:38Z; reviewer manifest reviews/gnc/rev4b/MANIFEST_REV-GNC-7.sha256 e04257acba555551 verified 94/94 OK at 10:38:50Z; DECIDED: the two REV-GNC-7 additions are adopted as (d1) the re-set uses the upper 95 % bound of the measured N2O Cd x A (at r 1.1326 a ratio measured 7.9 % low gives a true vent of 81.99 g, just under the 82.1 g first fail) and (d2) CF-4 shows the 283.15 K need at the ... DECIDED *Lead decision, line 258* T4730
REV-MECH-2	Mechanical FEA and CAD (v3.5 era)	**REVISE** DERIVED T4731	CHANGELOG line 234 DECIDED *Lead decision, per SUPERVISION.json 18 reviews* T4732
REV-ELEC-4	Electronics v2.12 (firmware v2.12 and SIL; electronics.json v4.1; bench plan v5; BOM v2.4)	Overall: ACCEPT WITH COMMENTS, with option (b) carved out of every start with ignition until a reviewed v2.13 delta. Findings: 0 BLOCKING, 1 MAJOR, 5 MINOR, 11 NOTES. DERIVED T4733 Review file: reviews/REVIEW-ELECTRONICS-R4.md, sha256/16 70b7103472eebab4 T4734; line 266	**CHANGELOG 266 (4):** REV-ELEC-4 (independent delta review of Electronics v2.12; interim E4-01 / E4-02 13:57Z; closed 14:34:56Z and returned 14:35:54Z by its stamps, delivered 14:38:25Z): ACCEPT WITH COMMENTS, with option (b) carved out of every start with ignition until a reviewed v2.13 delta; 0 BLOCKING, 1 MAJOR (E4-01), 5 MINOR (E4-02 to E4-06), 11 NOTES DECIDED *Lead decision, line 266 (4)* T029 **Carve-out, 266 (6), verbatim:** Option (b) (seat coefficient loaded, setPcSeatCoef != 0) is not used for any start with ignition (pad first start, stand starts, relights) until a v2.13 delta with the light-window hold and the TC4 step / rate checks is reviewed. The coefficient may be loaded for B-PC-1 / B-TC-1 bench work without ignition. DECIDED *Lead decision, line 266 (6)* T024 **266 (7):** PROMOTED 14:45:39Z: interfaces/electronics.json = staging_v2.12 results/electronics_v4.1.json c862375163393adc, byte for byte (v4 11f7896ad0d0613b stays as staging_v2.11/results/electronics_v4.json). Firmware v2.12 and its SIL are the design baseline under the carve-out of (6). DECIDED *Lead decision, line 266 (7)* T023
FEA3D-2	3D FEA of the v3.5.2 assembly (REPORT-FEA3D-v352.md) and the rows-v8 matrix	**no independent review held in this finish run**	DECIDED: no independent review of the FEA3D v3.5.2 run (04-fea/assembly3d_v352/, FINAL in line 252) is held in this finish run. The datasheet review-status table says so plainly and cites this item. DECIDED *Lead decision, line 262 (3)* T1632
REV-DS-1	This datasheet: the rev1-draft (PDF 8e7d007d128dfb58, not published)	**ACCEPT WITH CORRECTIONS** DERIVED T4735 No finding is BLOCKING. DERIVED T4736 There are 5 MAJOR findings (D1-01, -02, -03, -04, -06) and 8 MINOR (D1-05, D1-07 to D1-13). DERIVED T4737	REV-DS-1 (independent review of the rev1-draft): ACCEPT WITH CORRECTIONS, 0 BLOCKING, 5 MAJOR, 8 MINOR; corrections applied in this build; the check of this final build is recorded in the CHANGELOG after it. DERIVED T4738 **CHANGELOG 266 (3):** REV-DS-1 (independent review of the rev1-draft; phase B from 13:23Z; batches 13:41:04Z and 14:05Z; full review returned 14:28:20Z, stamp 14:25:25Z): ACCEPT WITH CORRECTIONS; 0 BLOCKING, 5 MAJOR (D1-01 to D1-04, D1-06), 8 MINOR (D1-05, D1-07 to D1-13); 100 headline values re-traced, all MATCH; 86 of 86 sources match; the renders match the CAD (mean abs difference at most 2.9 / 255). Conditions: the final applies every finding and lead ruling; an independent check of the final passes against reviews/datasheet/r1/data/final_build_checklist_rev1draft.json e2cc78f1a0b50551; the draft PDF is not published. Every finding ACCEPTED and sent to the author (13:47:38Z, 14:05:35Z, 14:30:54Z). DECIDED *Lead decision, line 266 (3)* T4739

12.2 Review register

reviews/SUPERVISION.json (076172727cddd074), 18 entries in register order. Not physical approval.

Review	Deliverable	Verdict	Reviewed files now (generator check)
REV-COMB-1 T4740 2026-10-12/13	02-combustion (REPORT.md v2.3, combustion.json v2.3, HANDBACK-SUMMARY.md) DERIVED T4743	**REVISE** DERIVED T4742	0 of 3 reviewed files unchanged; changed since review: 02-combustion/REPORT.md, interfaces/combustion.json, 02-combustion/HANDBACK-SUMMARY.md T4744
REV-PLUME-1 T4745 2026-10-12/13	05-plume-cfd (REPORT.md) DERIVED T4747	**ACCEPT WITH COMMENTS** DERIVED T1618	0 of 1 reviewed files unchanged; changed since review: 05-plume-cfd/REPORT.md T4748
REV-GNC-1 T4749 2026-10-13	01-systems-gnc v3 (REPORT-v3.md, gnc.json v3) with addendum on v3.1 staging (FC_THRUST) DERIVED T4752	**REVISE** DERIVED T4751	4 of 5 reviewed files unchanged; changed since review: interfaces/gnc.json T4753
REV-ELEC-1 T4754 2026-10-13	06-electronics v2.9 (firmware v2.9, SIL sil_v2p9, KiCad ECU v2.4, BOM v2.1, bench_plan_v2.md, REPORT-v2.9-HANDBACK.md, interfaces/electronics.json v2) DERIVED T4757	**REVISE** DERIVED T4756	3 of 10 reviewed files unchanged; changed since review: 06-electronics/firmware/src/core/config.h, 06-electronics/firmware/src/core/engine_ecu.cpp, 06-electronics/firmware/src/core/engine_ecu.h, 06-electronics/firmware/src/core/control.cpp, 06-electronics/hardware/kicad_gen/ecu_design.py, interfaces/electronics.json; not in the hub: 06-electronics/REPORT-v2.9-HANDBACK.md T4758
REV-GNC-2 T4759 2026-10-13	01-systems-gnc/staging_v3.1/rev1 (REPORT-GNC-rev1.md, rev1_items.json, FSW record v1.4, sim6 v2.8-dev MWFPRS) responding to REV-GNC-1 DERIVED T4762	**CONDICIONADO** DERIVED T4761	6 of 6 reviewed files unchanged T4763
REV-COMB-2 T4764 2026-10-13	02-combustion v2.4 (REPORT.md, REVISION-RESPONSE.md, CF-4-SPEC.md, T1-T4-REGRESSION-SPEC.md, combustion.json v2.4) DERIVED T4767	**REVISE (targeted)** DERIVED T4766	0 of 5 reviewed files unchanged; changed since review: 02-combustion/REPORT.md, 02-combustion/REVISION-RESPONSE.md, interfaces/combustion.json, 02-combustion/CF-4-SPEC.md, 02-combustion/T1-T4-REGRESSION-SPEC.md T4768
REV-PLUME-2 T4769 2026-10-13	05-plume-cfd v2.2b (REPORT.md) DERIVED T4771	**ACCEPT** DERIVED T1622	1 of 1 reviewed files unchanged T4772
REV-GNC-3 T4773 2026-10-13	01-systems-gnc/staging_v3.1/rev2 (REPORT-GNC-rev2.md, staging JSON, sim6 v2.9-dev amendments G, K, V) and the lead's provisional decisions in CHANGELOG line 175 DERIVED T4776	**CONDICIONADO** DERIVED T4775	6 of 7 reviewed files unchanged; not in the hub: interfaces/CHANGELOG.md (lines 155-175 read at) T4777
REV-MECH-1 T4778 2026-10-13	04-fea (S-AX L1_TC09IN718_thJ1revB_TC09IN, S-3D v1.1, joints v1p1, stiffness V5c) and 03-cad v3.5 (BOM, massprops, assembly, checks run 6) DERIVED T4781	**REVISE (FEA); REVISE, not freezable (CAD v3.5)** DERIVED T4780	5 of 12 reviewed files unchanged; not in the hub: interfaces/mechanical.json v2.1, results/sax_results_L1_TC09IN718_thJ1revB_TC09IN_v1.json, results/sax_margin_table_L1_TC09IN718_thJ1revB_TC09IN_v1.csv, results/s3d_results_v1.1.json, results/joints_v1p1.json, assembly/ENGINE_v3.5_neutral.step, assembly/ENGINE_v3.5_neutral.glb T4782
REV-GNC-4 T4783 2026-10-13	01-systems-gnc/staging_v3.1/rev2p1 (gnc.json v3.1 publication candidate) DERIVED T4786	**PUBLISHABLE WITH CORRECTIONS (not publishable as staged)** DERIVED T4785	1 of 5 reviewed files unchanged; not in the hub: results/gnc_json/gnc_v3.1_staging_rev2p1.json, results/comparators/comparators_rev2p1.json, results/mass_v35/mass_v35_vs_v3d.json, interfaces/CHANGELOG.md (186 lines) T4787

Review	Deliverable	Verdict	Reviewed files now (generator check)
REV-COMB-3 T4788 2026-10-13	02-combustion v2.4a (REPORT.md, REVISION-RESPONSE-R2.md, CF-4-SPEC.md, T1-T4-REGRESSION-SPEC.md, combustion.json v2.4a) and lead line 186 DERIVED T4791	**REVISE (targeted)** DERIVED T4790	2 of 6 reviewed files unchanged; changed since review: 02-combustion/REPORT.md, 02-combustion/CF-4-SPEC.md, 02-combustion/T1-T4-REGRESSION-SPEC.md, interfaces/combustion.json T4792
REV-GNC-5 T4793 2026-10-13	01-systems-gnc/staging_v3.1/rev3 (GNC rev3: P2-P6, P8-P9; 11 v3.2 proposals) and lead line 204 DERIVED T4796	**PUBLISHABLE WITH CORRECTIONS** DERIVED T4795	3 of 4 reviewed files unchanged; changed since review: interfaces/gnc.json T4797
REV-ELEC-2 T4798 2026-10-13	06-electronics v2.10 hub copy (firmware, SIL sil_v2p10, FA-1 harness, bench_plan_v3, BOM v2.2, REPORT-v2.10-HANDBACK) and interfaces/electronics.json v3; lead line 203 DERIVED T4801	**REVISE** DERIVED T4800	10 of 11 reviewed files unchanged; changed since review: interfaces/electronics.json T4802
REV-GNC-6 T4803 2026-10-19	01-systems-gnc/staging_v3.1/rev4 (GNC rev4: A1 band on v3.5.1, landing mass, ENG-11, RE-24, R2 v2 answers, payload record; v3.2 proposals [0]-[15]) and lead lines 213-223 as they bear on it DERIVED T4806	**PUBLISHABLE WITH CORRECTIONS** DERIVED T4805	4 of 6 reviewed files unchanged; changed since review: interfaces/gnc.json; not in the hub: 04-fea/assembly3d/REPORT-FEA3D.md (sections 9/9A, read for [12]) T4807
REV-MECH-2 T4808 2026-10-20	04-fea/assembly3d (FEA3D final on CAD v3.5.1, asm v1.2, CalculiX 2.21 C3D10, L1 171,123 nodes) and 04-fea/s3d_v351, with lead lines 217-231 as they bear on it DERIVED T4810	**REVISE** DERIVED T4731	5 of 6 reviewed files unchanged; not in the hub: interfaces/CHANGELOG.md (230 lines at the brief) T4811
REV-ELEC-3 T4812 2026-10-21	06-electronics/staging_v2.11 (Electronics v2.11: firmware v2.11 core, SIL record sil_v2p11, electronics_v4.json, bench_plan_v4, BOM v2.3, pc_transducer_v3 (C1 reselection), case_tc_channel_v1, ... DERIVED T4814	**ACCEPT WITH COMMENTS** DERIVED T4727	4 of 6 reviewed files unchanged; not in the hub: interfaces/combustion.json (v2.4b, read), 04-fea/v352/lc_tb70_envelope_v1.json (read) T4815
REV-COMB-4 T4816 2026-10-21	02-combustion v2.4b (REPORT.md, REVISION-RESPONSE-R3.md, DATASHEET-INPUTS-COMBUSTION.md, HANDBACK-SUMMARY.md, CF-4-SPEC.md v2.1, T1-T4-REGRESSION-SPEC.md v3, interfaces/combustion.json v2.4b) with ... DERIVED T4818	**ACCEPT WITH CORRECTIONS** DERIVED T4725	1 of 8 reviewed files unchanged; changed since review: 02-combustion/REPORT.md, interfaces/combustion.json, 02-combustion/REVISION-RESPONSE-R3.md, 02-combustion/DATASHEET-INPUTS-COMBUSTION.md, 02-combustion/CF-4-SPEC.md, 02-combustion/T1-T4-REGRESSION-SPEC.md; not in the hub: interfaces/CHANGELOG.md (255 lines at the review) T4819
REV-GNC-7 T4820 2026-10-21	01-systems-gnc/staging_v3.1/rev4b (GNC rev4b: gnc.json v3.2 candidate, ENG-11 FE-plant re-run, CAD v3.5.2 R3 stiffness band, DATASHEET-INPUTS-GNC.md) and lead lines 227-257 as they bear on it, with ... DERIVED T4822	**PUBLISHABLE WITH CORRECTIONS** DERIVED T4729	3 of 6 reviewed files unchanged; not in the hub: interfaces/gnc.json (v3.1 at the review), interfaces/combustion.json (v2.4c), interfaces/electronics.json (v4) T4823

12.3 Founder actions (traced to their decision)

Every founder action, gate and build procedure prints in full, with its acceptance limit and its source lines, so that it can be carried out from the printed page (lead 2026-10-21 14:05Z). The J-FWD build procedure C4 is printed step by step in Annex C4, after this table.

The founders run every physical step; none has been run. Purchases: TO-QUOTE delivered to Sønderborg DK-6400. Bench plan v5 does not go to the founders before the corrections of CHANGELOG 266 (7).

Founder action	Decision and text (quoted in full)
IN718 M5 J-FWD screw: quote, and the M6 fallback	RULED: the J-FWD screw is IN718 M5 × 10, ENG-HW-101-IN718: AMS 5662 bar aged to AMS 5663, ISO 7380-1 form dk 9.5 / k 2.75, threads rolled after ageing preferred (TO-DEFINE), maker TO-SELECT, TO-QUOTE delivered to DK-6400. FALLBACK only if IN718 cannot be procured: stock BUMAX 88-MKT M6 × 10 A4-80 (TO-QUOTE), which needs a CAD change (d6.6 holes, d11.5 spot faces, wall 2.600, M6 taps, a further FD-04 trim), a tap-wall probe toward the torch passage (about 1.41 mm left) and an FE re-run; recorded as a fallback, not a passed design. DECIDED *Lead decision, line 240* T4825
HW-170 disc spring: quote with relaxation data at about 160 C	Spring A (Schnorr 001 500 A in 1.4568) is the baseline and spring B (Lesjofors 7909) the alternate, TO-QUOTE delivered to DK-6400 with relaxation data at about 160 C. DECIDED *Lead decision, line 243* T4826 HW-170 disc spring SINGLE-USE (lead 19:53:27Z, CHANGELOG 248 (7)): a new spring with each new ENG-HW-101-IN718 screw at every J-FWD assembly; removed springs are scrapped, never refitted or measured for reuse; the rows assume an as-new spring; the maker's relaxation data at about 160 C is TO-QUOTE with the spring (delivered to Sonderborg DK-6400). HW-170 drawing / BOM note: 'SINGLE-USE: scrap on removal'. DECIDED *Lead decision, per mechanical.json v2.4* T4827
J-F1: torque cap 3750 N per flight lot and the Prescale footprint check (TO-VERIFY)	F1: the J-F1 MAXIMUM PRELOAD IS CAPPED AT 3750 N (J-F1 sets the flight-lot torque so K_min gives <= 3750 N); the 3750 N column (FE-anchored: the patch runs ran at 3750 N) governs every J-FWD row and 4096 N is information; the under-head row is quoted at full float at 3750 N as a FLAG with the ASSUMED tag; TO-VERIFY: a pressure-film (Prescale) footprint check at J-F1. DECIDED *Lead decision, line 245* T4828
J-F1b: 175 C peak with the dwell (method TO-DEFINE)	J-F1b (the optional J-F1 extension of hardware v1.7 HW-101; rule lead 19:53:27Z, PEAK 175 C ACCEPTED 21:17:03Z, dwell CORRECTED, CHANGELOG 248 (7); work-tree record jf1b_loctite_v352_v1.json 2ba9d10745f3f651, read-only [SIM]): 5 of the J-F1 joints (with the spring, the load washer kept in; J-F1 coupons, never flight hardware), the cycle 20 → 175 C → 20 C, at least 10 min at the peak from when the spring flat reaches the peak, 10 cycles as in hardware v1.7, residual preload and breakaway torque recorded after each cycle as data for the preload-retention audit (the check on the relaxation model). Peak 175 C = max(160 C, the governing SOAK70 zone maximum 160.29 C + 10 C) rounded up to 5 C (the js5 screw set alone gives 170 C, information); the 10 min is more than ten times the governing state's 53.0 s above (zone maximum - 10 C). METHOD: a uniform oven cycle puts the tap LOCTITE 243 at 175 C against 97.7 C in service, so it is allowed only once Henkel's service range for 243 is verified to cover 175 C (AS RECALLED -55 to +180 C, TO-VERIFY); otherwise the spring and head zone are heated locally and the tap temperature is recorded; TO-DEFINE by the test owner (founder). The post-abort soak is not a J-F1b state (C9 (4)). DECIDED *Lead decision, per mechanical.json v2.4* T4829 Prescale TO-VERIFY, J-F1b method TO-DEFINE, LOCTITE 243 range TO-VERIFY (founder / test owner); TO-VERIFY T4830
LOCTITE 243 service range (TO-VERIFY)	LOCTITE 243 tap zone at TB_a150: 128.0 C GVLO / 116.0 C GVHI (148.7 / 146.9 C with the whole history shifted, ASSUMED), ACCEPTED as below 150 C; the TDS text states no service range and its hot-strength charts could not be read here (HTTP 403), so the range and the hot-strength fractions are TO-VERIFY (founder action). DECIDED *Lead decision, line 248* T4831 Prescale TO-VERIFY, J-F1b method TO-DEFINE, LOCTITE 243 range TO-VERIFY (founder / test owner); TO-VERIFY T4832
Single-use J-FWD screws and springs; removed-screw inspection	J-FWD screws are new at every assembly, never reused or re-torqued; removed screws are kept per hole and inspected at 10x plus dye penetrant at the thread root AND the head-to-shank fillet, as data (lead 16:53Z R3 + LEAD ADDITION 18:36:37Z) DECIDED *Lead decision, per mechanical.json v2.4* T4833 Founder rule until C6 closes: the J-FWD screws are single-use (new at every assembly, never reused or re-torqued); removed screws are kept per hole and inspected at 10x with dye penetrant at the thread root and the under-head fillet, as data. DECIDED *Lead decision, line 243* T4834
Post-abort joint replacement rule	(4) POST-ABORT RULE (founder rule, lead 248 (7)): after any case-TC or TB-state abort the J-FWD joint counts as used: the ten ENG-HW-101-IN718 screws and their HW-170 springs are replaced (new, single-use) and the TC-01 holes (pins GO 5.50 / NO-GO 5.62) and the TC-02 taps (M5-6H GO / NO-GO) are gauged before the next run. TB-state aborts (CONFIRMED by Mechanical on the lead's provisional set of 253 (8), lc_tb70_envelope_v2.json F4_abort_set): HW_OVERPRESSURE, PC_OVERPRESSURE, HARD_START and PC_SENSOR on the HIGH side (transducer output >= 4.75 V = 146.5 bar abs), with the case-TC OVER_TEMP abort of F-14. Reason: the post-abort soak (ASSUMED shifted history, J-FWD band up to 201.4 C GVLO / 164.0 C GVHI, jf1b_loctite_v352_v1.json) is not a rows or J-F1b state. BESIDE IT, for every run whatever its end (ridden-through FA-1 pulses included), the J-FWD REPLACE RULE of the LC-TB70 envelope v2 (lc_tb70_envelope_v2.json 7ebd2fd63dc878ef; it supersedes v1's 'any head excursion above 60.0 bar abs'): when Q_JFWD exceeds 60.0 bar abs at any time in a run (Q_JFWD = the larger of the head-end peak and the head-end undamped SDOF ESP at the J-FWD periods 0.2-0.5 ms, on the light window and on each excursion window; no minimum duration) the joint counts as used: before the next run the line-231 J-FWD inspection (C6 (2) list) and the TC-02 taps gauged (M5-6H GO / NO-GO). Basis: at 60.0 bar abs every J-FWD row meets the nominal factors at the LC2 state 78.6 C (minimum MS_y +0.328 TC-01 ligament (LC2U), MS_u +0.312 TC-01 ligament trend 78.6 C) [SIM / DERIVED]; above it the J-FWD has only the SF 1.0 FA-1 rows (line 231). Logged signal: SIL, the plant head pressure (exact); stand, B-DPC-1 DC-restored with C1 (after a no-event noise check, TO-VERIFY); flight, the C1 LIGHT_SPIKE raw peak, a lower bound, against Electronics' chain-model threshold (TO-VERIFY). NOT meant for normal-light transients (R4-10): a normal light stays at or below MEOP 45 bar abs and reaches Q_JFWD 60 only with a 10-90 % rise below 0.288 ms [DERIVED]; 298 of the 896 FA-1 hard-start bracket cases exceed it, 185 by the head peak alone [SIM grid counts on an ASSUMED construct, not rates of normal lights]. OPEN (lead): in flight a flagged first light leaves the relight on a used joint, and the line-231 one-off basis covers one Z3 event per J-FWD assembly. DECIDED *Lead decision, per mechanical.json v2.4* T4835 FOUNDER RULE: after any case-TC or TB-state abort the J-FWD joint counts as used (HW-101 and HW-170 replaced, TC-01 holes and TC-02 taps gauged before the next run), beside the 60 bar abs rule of the LC-TB70 envelope. DECIDED *Lead decision, line 248* T4836
HW-170 orientation	drawing_note_ENG-HW-101-IN718_and_HW-170: HW-170 DISC SPRING ORIENTATION (route (i); Mechanical, DERIVED from the rows v8 load path; lead 248 (7)): one spring under each head, never stacked; concave (hollow) side DOWN, its outer edge (De 10.0, 9.85 min) bearing on the TC-01 spot-face flat; convex (crowned) side UP, its inner edge (Di 5.2, 5.32 max) toward the screw head; centred on the 5.15 h9 pilot of the F3 head; NEVER inverted (an inverted spring is removed and scrapped, C9 (2)).; assembly_step: C4 step 7 (the spring ORIENTED per C9 (1) as each flight screw is fitted; checked by eye before the screw engages); derivation: DERIVED; the DIN EN 16983 / DIN 2093 installation convention (load through the upper inner edge and the lower outer edge) is AS RECALLED, TO-VERIFY on the maker's sheet (Schnorr 001 500 A / Lesjofors 7909). The rows take the spring flat (solid) at every J-F1 preload (derated F_flat 360 N spring A / 206.5 N spring B against the 2057.6 N minimum, jfwd_spring_v352_v5.json); in the cool frames after embedding the spring force falls below F_flat (spring A F_S min 335.2 / 333.1 N at k_b lo / hi, R 0, the spring active in those frames), where the spring carries the load at its edges. Standard orientation: the outer edge (r 4.925-5.00) bears on the flat (the d11.2 spot face, with the disclosed tangential overhang, P(i)-2) and the head meets the crown at its bearing-face inner corner (da 5.55-5.80, r 2.775-2.90; F3); the T-head Kt at that corner is NOT SOURCED (disclosed with the head-fillet rows). Inverted: the inner edge (r 2.60-2.66) would sit inside the 5.82 hole-plus-break edge (r 2.91) with no support, and the outer edge (r 4.925 min) lies outside the ISO 7380-1 head (dk 9.5 max, r 4.75), so the head would ride on the spring's bowl: the load path the rows assume would not exist. DERIVED T4837
Stand T0 <= 24 C rule and its checklist row	STAND T0 RULE (founder action): B1-length stand burns only with the engine soaked at a stand ambient <= 24 C (facility thermometer) [DERIVED: 20 + 4.48 / 0.96 = 24.67 C less 0.5 K thermometer, rounded down; dT_station / dT0 about 0.96, DERIVED]; in stand mode v2.11 refuses ARM if TC1 or TC2 reads > 24 C (297.15 K, own refuse code, test_core case) as a backstop for gross cases; the founder pre-fire checklist row is the control; no hot-day FE run; re-derive after B-TC-1 as T0_max = 20 + (12.26 - e_hi,cal) / 0.96; the A1 flight has no T0 rule (> 45 K margin). DECIDED *Lead decision, line 249 (2)* T4838 STAND T0 RULE (founder action, lead CHANGELOG 249 (2); cross-reference only, not a Mechanical rule): B1-length stand burns only with the engine soaked at a stand ambient <= 24 C (facility thermometer) [DERIVED]; in stand mode v2.11 refuses ARM if TC1 or TC2 reads > 24 C (backstop); the founder pre-fire checklist row is the control; re-derive after B-TC-1 as T0_max = 20 + (12.26 - e_hi,cal) / 0.96; the A1 flight has no T0 rule. The J-FWD rows use the B1 thermal fields as solved (T0 20 C; no hot-day FE run). INFORMATION (jfwd_pq_band_v352_v1.json, ASSUMED uniform +4 K on the seat-line temperature, FE P+Q unchanged): at T0 24 C the band edge 0.0686 mm would read MS -0.041 (SOAK) and MS 0 would need g <= 0.0638 mm; at the ruled R2-B edge g 0.0631 mm it would read MS +0.007 (SOAK, k_b hi, 3750 N) [CALC, ASSUMED +4 K]; not a criterion. DECIDED *Lead decision, per mechanical.json v2.4* T4839

Founder action	Decision and text (quoted in full)
RA-03, J-FWD hole edges (TC-01 rev 5, C1): edge breaks, deburring and the 10x inspection before assembly (240 (5); 241 (8) C1)	TC-01 drawing, 10x d5.5 J-FWD holes (rev 5): (a) bore-side edge: break R0.1-R0.2, blended and polished, burr-free, no chamfer or step; R0.2 is a HARD MAXIMUM: a hole whose break exceeds R0.2 after rework is rejected or re-assessed, never accepted on sight. (b) flat-side (spot-face) edge of the hole: break 0.1 max. (c) spot-face flat-to-OD edge: break 0.1 max. (d) deburr after match-drilling with TC-02 and after any rework, before anodise. (e) 100 % inspection at 10x before assembly: no burr, nick or raised metal at the hole edges or on the bore within 3 mm of a hole. (f) drawing note: bearing length = wall at the hole minus both breaks (2.665 - 0.2 - 0.1 = 2.365 mm at the drawing-limit wall; the FA1S hole bearing is quoted at this length, +0.005 / +0.239, CHANGELOG 240 (5)). (g) hole size (ADOPTED, lead 14:34Z, Option A): d5.5 +0.12/0 (H12). ENG-HW-101-IN718 in-service gauge pins GO 5.50 / NO-GO 5.62; the line-231 post-FA-1 hole check uses the 5.62 pin. Route (i) (lead 16:53Z R2): the drawn maximum 5.62 plus the 0.1 flat-side break each side (5.82) is the inner boundary of the HW-170 disc-spring footprint on the flat (rows v8 route_i_spring_footprint, unchanged from rows v7: spring De 9.85 / Di 5.32 at its limits, 46.06 mm2 at the registered stack 0.4865 (lead 17:55:55Z F2), 45.352 mm2 at full float 0.557); the head bears on the spring, with da 5.55-5.80 stated on the ENG-HW-101-IN718 drawing (F3; it replaces the v2 'da 5.7 TO-VERIFY'); the v2 basis (head on the flat, float 0.397, rows v5) is superseded. (h) hole position (ADOPTED, lead 14:34Z; datums as checked in the rev-5 CAD and ACCEPTED 17:55:55Z): datum A = the axis of the forward seal-land bore 57.25 H8; datum B = the TC-01 forward end face, from which the hole row is dimensioned (12.000 BASIC from B); each hole axis 0.10 dia to A|B; the d11.2 spot face (rev-5 cutter, P(i)-1) coaxial with its hole within 0.10 dia. The rows v8 flat-bearing row takes the registered stack (tangential 0.4865, F2; axial 0.3705, F4) with the OD 63.5 h9 and its 0.04 runout to A. The v2 drawing-consistency FLAG (A had been the end face) is CLOSED by these datums: r_flat 31.313 +0.020/0 from A and the OD runout 0.04 to A read as radial quantities from the bore axis. DECIDED *Design value, item_c_tf1_texts_v352_v5.json v5* T4840 HOLE EDGE BREAK (item C amended 10:08Z; callout ruled 10:15Z) [DERIVED]: the bore-side edge of the 10x d5.5 J-FWD holes is broken R0.1-R0.2, replacing the R0.2-R0.4 of 239 (13)(i): blended and polished, burr-free, no chamfer or step; R0.2 is a hard maximum (a hole over it after rework is rejected or re-assessed); spot-face side 0.1 max; deburr after match-drilling and any rework, before anodise; 100 % 10x inspection (no burr, nick or raised metal at the hole edges or on the bore within 3 mm of a hole); drawing note: bearing length = wall at the hole minus both breaks. A larger break shortens the bearing length and lengthens the screw bending arm (R0.4 + 0.1 gives -0.070 nominal / -0.080 at the limit wall). FA1S hole bearing (SF 1.0, 45.6 C, Fbry 330.5 / Fbru 407.4 MPa, CALC V): drawing-limit wall 2.665, L 2.365: +0.005 / +0.239, GOVERNING and quoted in the datasheet (nominal wall 2.688: +0.015 / +0.251). FEA3D's (b1) FE bearing with the break replaces the FE-V row when it lands; the CALC row governs unless that is worse. The 231 post-FA-1 inspection names hole elongation and bearing set at the 10 holes. O1 is re-issued on the spot-face wall (jfwd_v352_v2.json, replacing the CAD-wall +0.338 / +0.649). The fitting plugs of 239 (13)(ii) carry the O-ring protection. DECIDED *Lead decision, line 240 (5)* T4841 C1, TC-01 rev 5 at the 10 x d5.5 J-FWD holes: (a) bore-side break R0.1-R0.2 blended and polished, no chamfer or step, R0.2 a HARD MAXIMUM (over it after rework: rejected or re-assessed, never accepted on sight); (b) flat-side edge 0.1 max; (c) spot-face flat-to-OD edge 0.1 max; (d) deburr after match-drilling with TC-02 and after any rework, before anodise; (e) 100 % 10x inspection before assembly; (f) bearing length = wall minus both breaks, 2.365 at the 2.665 limit wall (FA1S hole bearing +0.005 / +0.239); (g) and (h) as item 6 (c) and (d). DECIDED *Lead decision, line 241 (8)* T4842
B-TC-1 case-TC channel: acceptance on the DESIGN install (tau and the static terms, else the tolerance is re-derived; 249 (1)); CJTO trim and mean-shift EMI pass (253 (11))	B-TC-1 (founder: tau and the static terms); DECIDED *Lead decision, line 249* T4843 low-reading tolerance tol_lo = 9.05 + 0.630 r K worst case (r = station heating rate at the trip, K/s) on the DESIGN install (ungrounded junction spot-welded to a 0.05 mm SS shim, 25 um polyimide, leads along the hoop isotherm, continuous Class 1 wire, no connector; tau 0.25 s DESIGN target); a B-TC-1 bench row verifies tau <= 0.25 s and the static terms, else the tolerance is re-derived. DECIDED *Lead decision, line 249 (1)* T4844 Founder items added: F-10 also asks Kulite for the thermal-zero maximum and for a 100 F compensated band from the coldest ARM to the hottest relight seat (R3-05); B-PC-1 measures the zero coefficient per serial; B-TC-1 gets the CJTO trim step and a mean-shift EMI pass (R3-03); F-13 gets the soak condition and the +/-0.5 K thermometer (R3-06). DECIDED *Lead decision, line 253 (11)* T4845
T-F1, short-time exposure allowables of TC-01 / TC-02: a GATE before the first static fire; routes (i) and (ii) and the PASS criteria in full (241 (8), 249 (3), 254 (2))	T-F1. Short-time exposure allowables for the 6061-T6 parts TC-01 (case tube) and TC-02 (injector head / forward closure). GATE: before the first static fire. WHY: criterion 183 (1)(d) / 188 (5) needs each part's temperature within a SOURCED time-at-temperature limit. The usable limit on file for 6061-T6 is 150 C (long time). The strength cells use the engmat 10,000 h allowables (conservative for strength), so (d) is reported OPEN, never as passed, for TC-01 and TC-02; it gates nominal operation as well as the abort value. PREDICTED EXPOSURE [SIM, v3.5.2 rev-4 thermal, B1 p99 tg711, gap-conductance bounds GVLO | GVHI]: TC-02 part maximum 188.4 | 181.9 C at the end of burn, above 150 C for 12.3 | 9.9 s per firing. TC-01 part maximum 191.7 | 187.0 C at the chamber bore about 14 mm aft of the injector face, in the soak (51.65 | 43.65 s), above 150 C for 488.8 | 479.5 s per firing; at the J-FWD screw row 159.3 | 129.6 C, 48.5 | 0.0 s above 150 C; while pressurised (burn) TC-01 stays at or below 109.6 | 118.6 C. Over the 50-cycle design life (183 (1)(c)) the cumulative time above 150 C is TC-02 10.2 | 8.2 min and TC-01 6.8 | 6.7 h (part maxima). EXPOSURE BASIS, TB STATE (RESTATED, lead 2026-10-21T05:50:45Z R9; CHANGELOG 249 (3)) [ASSUMED surrogate, sax.py: the end-of-burn field shifted uniformly so that the TC-01 maximum equals A; the shifted-history times apply the same shift to the whole B1 history and are for sizing only]: the basis is the A 175 rows (TB; FEA3D's tb352g GVLO / tb352h GVHI; both gap-conductance bounds; they cover a TC-01 rise rate r <= about 22.8 K/s): A 175 C: TC-02 248.4 | 233.0 C, shifted-history time above 150 C TC-02 1192.0 | 929.7 s, TC-01 1435.8 | 1266.0 s (TC-01 shifted-history maximum 257.1 | 243.4 C), margin to the nominal end-of-burn case maximum +65.4 | +56.4 K; BESIDE them the reading-equals-truth case at the adopted abort value (A stays 150 C, 249 (1)): A 150 C: TC-02 223.4 | 208.0 C, shifted-history time above 150 C TC-02 35.6 | 29.7 s, TC-01 999.2 | 866.4 s (TC-01 shifted-history maximum 232.1 | 218.4 C), margin to the nominal end-of-burn case maximum +40.4 | +31.4 K. On the A 175 rows criteria (a)-(c) PASS on both bounds (lowest TC-02 TBS +0.409 / +0.745, lead 249 (3)); (d) is OPEN (this action). Re-basing T-F1 to the A 165 rows is possible only after B-TC-1 calibrates the channel (tol_lo 10.28 K, TC-01 maximum 161.82 C), by the lead, before T-F1 is run; the A 165 / 135 / 120 / 105 rows stay in the json table (T-F1_TB_table) as the record of the closed abort search. ROUTE (i), sourced allowables, no test: obtain short-time exposure data for 6061-T6 from a public, citable document (for example MIL-HDBK-5J chapter 3, the 6061 elevated-temperature strength curves with exposure time, or EN 1999-1-2 Table 1a, the 6061-T6 strength ratios for short exposure) and record document, edition, figure or table, page, URL and access date. Read, at each part maximum (nominal as above; TB on the A 175 basis rows, both bounds) and for an exposure no shorter than the cumulative time above 150 C of the planned firings, (1) the strength at temperature and (2) the room-temperature strength after the exposure. PASS when (1) is at least the value the cells used (engmat 10,000 h, e.g. Fty / Ftu 116.9 / 134.4 MPa at 188.4 C and 111.2 / 129.2 MPa at 191.7 C; the full table is in this json) and (2) meets the T6 minimums of the stock's own specification (engmat basis ASTM B221 / B241: Fty 240 / Ftu 260 MPa); otherwise T-F3 (the line-212 design change). ROUTE (ii), coupon test on the same stock (the heat / lot of the TC-01 tube and of the TC-02 bar): expose coupons in a calibrated oven (+/-3 K) at the TB part maximum of the A 175 basis rows and no lower than the nominal part maximum (GVLO, the hotter bound: TC-02 at 248.4 C, its A 175 TB state; TC-01 at 191.7 C, its nominal soak maximum, above its 175 C TB state), for the cumulative hot time of the planned firings (at least the 50-cycle values above, plus every planned TB / abort test event at its A 175 shifted-history time above 150 C, TC-02 1192.0 s and TC-01 1435.8 s GVLO); then room-temperature tensile tests (EN ISO 6892-1 or ASTM B557) on at least 3 exposed and 3 unexposed control coupons per stock, same orientation, and hardness (plus eddy-current conductivity if available) on every coupon. PASS when every exposed coupon meets the T6 minimums of the stock specification (Rp0.2, Rm and elongation); hardness and conductivity are recorded as the screening trend, not the pass rule. Optional: one hot tensile at the part maximum after the exposure, at least the engmat value at that temperature. If any exposed coupon fails, (d) stays OPEN and the design goes to T-F3. No recalled or estimated value may replace either route. Until one closes, every quoted TC-01 / TC-02 row carries (d) OPEN; (d) is reported OPEN on the A 175 basis rows, with the A 150 rows beside them. CALC T1212 a_to_c: PASS on both bounds (lowest TC-02 TBS +0.409 / +0.745; lead, CHANGELOG 249 (3)); d: OPEN (T-F1): no sourced short-time exposure allowable; the cells use the engmat 10,000 h basis CALC T1214 [CALC on SIM; accepted, 249 (3) T1215] T-F1 (founder action, a GATE before the first static fire): 183 (1)(d) / 188 (5) needs a SOURCED time-at-temperature limit, OPEN for TC-01 and TC-02. Exposure [SIM, GVLO | GVHI]: TC-02 188.4 | 181.9 C at end of burn, 12.3 | 9.9 s above 150 C per firing; TC-01 191.7 | 187.0 C at the chamber bore in the soak, 488.8 | 479.5 s above 150 C per firing; over 50 cycles TC-02 10.2 | 8.2 min and TC-01 6.8 | 6.7 h above 150 C. Route (i): sourced handbook exposure data (MIL-HDBK-5J ch. 3 or EN 1999-1-2 Table 1a, full citation) read at each part maximum for the cumulative time; route (ii): coupons from the same heat / lot (oven at the TB part maximum of the adopted abort value for the cumulative hot time; RT tensile to EN ISO 6892-1 or ASTM B557; >= 3 exposed and 3 control coupons per stock); PASS criteria as written; any failure keeps (d) OPEN (T-F3); no recalled or estimated value replaces either route. DECIDED *Lead decision, line 241 (8)* T4846 T-F1 EXPOSURE BASIS (21:51:07Z, SUPERSEDING the 21:46:49Z 'A 165 rows' wording): the A 175 rows (TB, tb352g / h, both bounds; they cover r <= about 22.8 K/s), because the GVHI rev-4 B1 end-of-burn station rate is 7.03 K/s [SIM; GVLO 6.74] above the 7.00 K/s limit of the A 165 rows (one method lands 0.004 K inside 165 C, below its resolution); the surrogate sets the TC-01 maximum to A and the station reads A - 1.43 / 1.54 K (GVLO / GVHI). A 175 rows [SIM / ASSUMED surrogate]: GVLO TC-02 248.4 C, above 150 C for 1192.0 s, TC-01 257.1 C (1435.8 s above 150 C); GVHI TC-02 233.0 C, 929.7 s, TC-01 243.4 C; (a)-(c) PASS on both bounds (lowest TC-02 TBS +0.409 / +0.745), no leak; (d) OPEN (T-F1). The TB rows quoted for the abort are the A 175 rows, with the A 150 rows beside them as the reading-equals-truth case; A stays 150 C (a lower A would trip a nominal merged-bound burn with the uncalibrated channel: 137.74 + 7.78 = 145.52 C); re-basing route: if B-TC-1 calibrates the channel (Electronics: tol_lo 10.28 K at 7.03 K/s, TC-01 maximum 161.82 C [DERIVED]), the lead may move T-F1 to the A 165 rows before T-F1 is run; Mechanical's re-map restates tf1_exposure_v352 (d23e72f089cca33a) and the T-F1 founder text on the A 175 rows. DECIDED *Lead decision, line 249 (3)* T1215 T-F1 restated per R9 (A 175 rows on both bounds, A 150 beside; (a)-(c) PASS, (d) OPEN) T4847 Mechanical's re-map restates tf1_exposure_v352 (d23e72f089cca33a) and the T-F1 founder text on the A 175 rows. DECIDED *Lead decision, line 249* T4848
Shoulder-screw registration (F2) and the index pin (F5)	F2 ADOPTED into C4: two GSE shoulder screws selected to the pin-gauged holes at two opposite holes first, flight screws one at a time and recorded, shoulder screws out last; tangential basis = the registered stack 0.4865 unless the method shows the drilling registration is reproduced. F5: clocking +/-0.25 deg at match-drilling by an index pin in the drill fixture (torch-tap wall net about 1.64; +/-0.5 deg 1.54 information) plus a CALC tap-wall row. DECIDED *Lead decision, line 245* T4849
RA-06, make or buy: ENG-TC-12 J-FWD hole-plug set (GSE; TO-DRAW / TO-QUOTE, delivered to Sønderborg DK-6400; 241 (8) C3)	**BOM GSE line (§5.13):** ENG-TC-12: J-FWD hole plug set, 10 plugs + band, O-ring passage aid; drawing TO-DRAW; price TO-QUOTE (delivery Sonderborg, Denmark, DK-6400); gse (not flight) DECIDED *Design value, BOM_v3.5.2.json v3.5.2* T269 ENG-TC-12 J-FWD hole plug set (GSE, not flight mass; TO-DRAW; TO-QUOTE, or made from POM rod): 10 plugs and 1 retaining band. Plug: POM, shank d5.5 -0.02/-0.05; a head shoulder seats on the spot face and sets the depth positively; the inner end is contoured R28.6 (the 57.25 bore; contour depth 0.132 mm over the d5.5 face) with the contour axis parallel to the bore axis, keyed by a tab on the head in a slot of the band (+/-15 deg is enough: the contour error at the plug edge is 0.009 mm at 15 deg [CALC]). Length from the shoulder to the contour centre = the wall measured at that hole (flat to bore along the hole axis) minus 0.025 +/- 0.010, so the end sits 0.015-0.035 below the bore surface (inside the ruled 0 to 0.05; never proud); plugs numbered to their holes. Reason: the drawn wall at the hole spans 2.665-2.708 mm, wider than the 0.05 window, so one fixed length cannot meet it at every stack [CALC]. Band: a split ring over the ten heads on the TC-01 OD that holds each plug against the passing ring's outward push. The plugs come out only after both rings have passed the hole row. BOM line (rev 5): 'ENG-TC-12 | J-FWD hole plug set, 10 plugs + band, O-ring passage aid | GSE (not flight) | POM | qty 1 set | TO-DRAW / TO-QUOTE | used at every TC-02 insertion'. DECIDED *Design value, item_c_tf1_texts_v352_v5.json v5* T4850 C3: ENG-TC-12 J-FWD hole-plug set (GSE, not flight; 10 POM plugs and a band; shank d5.5 -0.02/-0.05; contoured R28.6 end, keyed; length = wall measured at that hole - 0.025 +/- 0.010, numbered; TO-DRAW / TO-QUOTE). DECIDED *Lead decision, line 241 (8)* T4851
RA-06, make or buy: ENG-TC-13 J-FWD registration shoulder-screw pair (GSE; TO-DRAW / TO-QUOTE, delivered to Sønderborg DK-6400; 245 (4) F2, 246 (2) C3b)	**BOM GSE line (§5.13):** ENG-TC-13: J-FWD registration shoulder screw pair (C4 registration); drawing TO-DRAW; price TO-QUOTE (delivery Sonderborg, Denmark, DK-6400); gse (not flight) DECIDED *Design value, BOM_v3.5.2.json v3.5.2* T271 ENG-TC-13 J-FWD registration shoulder screw pair (GSE, not flight mass; TO-DRAW / TO-QUOTE; lead 17:55:55Z F2, ADOPTED into C4): two shoulder screws, the shoulder selected to the pin-gauged sizes of their TC-01 holes (d5.5 H12; pins GO 5.50 / NO-GO 5.62), M5 into the TC-02 taps; they go into two opposite holes before any flight screw and come out last, each replaced by a flight screw (C4 steps 6 and 8); never left in the engine; shoulder length, material and the shoulder size steps TO-DRAW. BOM line (v3.5.2 BOM): 'ENG-TC-13 | J-FWD registration shoulder screw pair (C4 registration) | gse (not flight) | TO-SELECT | qty 1 | TO-DRAW / TO-QUOTE'. DECIDED *Design value, item_c_tf1_texts_v352_v5.json v5* T4852 F2 ADOPTED into C4: two GSE shoulder screws selected to the pin-gauged holes at two opposite holes first, flight screws one at a time and recorded, shoulder screws out last DECIDED *Lead decision, line 245 (4)* T4853 new C3b (ENG-TC-13 GSE) T4854

Founder action	Decision and text (quoted in full)
RA-06, make: the match-drill fixture with the index pin and the guide bushings (shop tooling, TO-DRAW; not a BOM line; 245 (4) F5)	match-drill fixture with the index pin and the guide bushings: shop tooling TO-DRAW (not BOM lines) DECIDED *Design value, item_c_tf1_texts_v352_v5.json v5* T4855 J-FWD taps 10x M5-6H x 8.0 full thread, drill 4.2 × 9.0, radial (rev 5, lead 16:53Z R2): position 42.000 BASIC below A (head top face), pattern az 18 + 36 k deg BASIC from the torch-seat axis (0 deg); MATCH-DRILLED and tapped at a dry first fit (no O-rings, before finishing) through the ENG-TC-01 d5.5 H12 holes: TC-02 clocked to the TC-01 pattern within +/-0.25 deg by an index pin in the match-drill fixture (shop tooling, TO-DRAW; lead 17:55:55Z F5: tap-to-torch-passage wall 1.6355 net at +/-0.25 deg, 04-fea/v352/jfwd_tapwall_v352_v1.json; +/-0.5 deg (1.5415 net) is information only), head shoulder set 2.00 +/-0.02 from the TC-01 fwd end face with a gap gauge (TO-CONFIRM; the same gauge at every later assembly); at each hole a hardened guide bushing (GSE, TO-DRAW / TO-QUOTE: OD g6 in 0.01 mm steps selected to the hole size gauged with pins, bore 4.2 F7, bore concentric to the OD within 0.005 TIR) seated in the TC-01 hole guides the 4.2 h8 drill, then a tap guide of the same OD guides the M5-6H tap; tap axis coaxial with its TC-01 hole within dia 0.10 at match-drilling; separate, deburr (TC-01 C1 (d)), inspect, finish, oxygen-clean, assemble per C4 DECIDED *Design value, item_c_tf1_texts_v352_v5.json v5* T4856 F5: clocking +/-0.25 deg at match-drilling by an index pin in the drill fixture (torch-tap wall net about 1.64; +/-0.5 deg 1.54 information) plus a CALC tap-wall row. DECIDED *Lead decision, line 245 (4)* T4857
Preload-retention audit	PRELOAD-RETENTION AUDIT ADOPTED (11:25Z) as a founder procedure and the physical arbiter between the seat-line and ring-line bounds: J-F1 on the IN718 flight lot calibrates the tightening-direction crack-on torque against clamp (load washer, LOCTITE 243 cured); after the first hot fire or a TB soak test, before disassembly, each screw's crack-on torque is read and converted to residual clamp; breakaway torque is logged; 10x inspection for fretting and set (231 list). DECIDED *Lead decision, line 240* T4858
TC-02 spigot select fit (R2-B; U TO-DEFINE)	LEAD RULING R2-B (2026-10-21T07:37:34Z): the ENG-TC-02 plug OD is a SELECT FIT to its own measured TC-01 bore, flag-free band: radial gap g in [0.040, 0.0631] mm, i.e. D_meas - 0.1262 <= OD <= D_meas - 0.080 mm (window 0.0462 mm on the diameter, less 2U, U the expanded (k=2) uncertainty of the bore and OD measurements, TO-DEFINE); bore and OD measured at 20 +/- 1 C and recorded with the paired TC-01 / TC-02 serials; the TC-02 OD is finished last, from a blank with finishing stock, to the measured bore. DECIDED *Lead ruling R2-B (2026-10-21T07:37:34Z), quoted verbatim in Mechanical line 254 (1)* T525 ENG-TC-02 PLUG OD, J-FWD SEAT-LINE BAND (lead 2026-10-21T05:50:45Z R2, restated to the lead ruling R2-B; jfwd_pq_band_v352_v1.json on relax v8, the final anchors s6 + C1; the TC-01 bore 57.25 H8, 57.250-57.296): LEAD RULING R2-B (2026-10-21T07:37:34Z): the ENG-TC-02 plug OD is a SELECT FIT to its own measured TC-01 bore, flag-free band: radial gap g in [0.040, 0.0631] mm, i.e. D_meas - 0.1262 <= OD <= D_meas - 0.080 mm (window 0.0462 mm on the diameter, less 2U, U the expanded (k=2) uncertainty of the bore and OD measurements, TO-DEFINE); bore and OD measured at 20 +/- 1 C and recorded with the paired TC-01 / TC-02 serials; the TC-02 OD is finished last, from a blank with finishing stock, to the measured bore. Basis: jfwd_pq_band_v352_v1.json 1ba3147b42edfeed [CALC on SIM]: seat-line MS >= +0.05 at 3750 N at the band edge, and positive in the ASSUMED +4 K stand T0 24 C information case (MS 0 there at g 0.0638). Option A (drawn 57.20 -0.030/-0.040, MS +0.005, FLAG) is not adopted (information only). The parts / BOM / hardware data strings stay unedited in v3.5.2 and are superseded by this drawing note. The seat-line P+Q stays NOT shown mesh-converged (R4 MF note); this is CALC on SIM, not physical approval. At the ruled edge g 0.0631 mm [CALC on SIM, jfwd_pq_band_v352_v1.json]: seat-line MS +0.0500 (k_b hi) / +0.0534 (k_b lo) at 3750 N (SOAK); 4096 N information +0.0479 / +0.0513; ASSUMED +4 K information (stand T0 24 C, k_b hi, 3750 N) +0.0067 (SOAK). INFORMATION: MS 0 at g 0.06862 mm (k_b hi) / 0.06892 mm (k_b lo) at 3750 N. The drawn 57.20 f7 (radial 0.040-0.078) leaves the seat line a FOUND NEGATIVE (rows v8 at 3750 N: SOAK -0.075 (b2) rep, SOAK70 -0.059, SOAK52 -0.033 (b1) under C1); the line-240 band [0.040, 0.0761] does not close it at its upper edge (MS -0.061). The edge is set by SOAK ((b2) rep 366.8 MPa at 158.0 C), not SOAK52 (370.3 MPa under C1 at 152.7 C). MS >= +0.05 would need g <= 0.0631 (select fit only). Lead CALC 'about 0.011 mm' CONFIRMED: 0.0108 mm at 0.0684 (the 4096 N edge), 0.0112 mm at 0.0686 (the 3750 N edge). MF NOTE (R4): the seat-line P+Q is NOT shown mesh-converged (s9 → s6 +12 to +18 %, s6 → cD -1.5 to -7.3 %, not monotonic); the FE are at g 0.078, outside the band, so the band MS are the model's scaling of the 078 FE [DERIVED], quoted as found. The band acts on the seat line only; the screw rows take no credit. No CAD change: the TC-02 STEP is unchanged. CALC T2279
S01 / S02 and the screws replaced after any FA-1-class event	GLAND-1 FA1S CLOSED (10:10Z; gland_fwd_v352_v3.json) [DATASHEET as cited; DERIVED]: the clearances of all three sources are DIAMETRAL as worded and nothing centres the plug, so the one-side gap is the full 0.2772 mm (0.156 at rest + 0.1212 FA1S growth). The charts are entered with the fluid pressure, their own axis: the FA-1 fluid bound at the head is 93.64 bar abs = 1343 psig (combustion fa1_spectra_v2p4b); FA1S at 151.04 bar abs is the structural DLF 2.0 case (line 194). FNST one-side at its 1500 psi row (0.30 mm) gives +0.082 (governing); Apple Rubber halved at 1343 psi +0.16; the 2176 psi rows and the FNST-scaled -0.093 are information; Parker PTD5705 has no readable extrusion chart. Gland-1 closes with NO BACK-UP RING (the 239 (11) condition is met); the datasheet states the residual (the maker curves differ in shape; no printed basis). NEW RULE: S01 and S02 are replaced after any FA-1-class event, together with the 231 J-FWD inspection (founder action list). DECIDED *Lead decision, line 240 (6)* T4859 (1) S01 and S02 are replaced after any FA-1-class event, together with the line-231 J-FWD inspection (lead 10:10Z rule). (2) Line-231 post-FA-1 inspection list: the ten J-FWD screws and their HW-170 disc springs replaced (new, single-use, (3)); the ten holes gauged for elongation and bearing set with the HW-101 pins GO 5.50 / NO-GO 5.62 (C1 (g)); a hole that takes the 5.62 pin is rejected; hole edges and the bore within 3 mm of a hole at 10x for nicks or raised metal; S01 / S02 replaced; the removed screws inspected per (3). (3) J-FWD screw life: SINGLE-USE founder rule until C6 closes (lead 16:53Z R3): the J-FWD screws are new at every assembly, never reused or re-torqued; removed screws are kept per hole and inspected at 10x plus dye penetrant at the thread root AND the head-to-shank fillet (LEAD ADDITION 18:36:37Z), as data. C6 is OPEN: the screw LCF at the thread root is a FOUND NEGATIVE (183 (1)(c); jfwd_screw_lcf_v352_v3.json on rows v8, SOAK (b1) as solved, Kf 3.0: per firing N_allow 5.6 (MS -0.888), assembly from zero N_allow 0.6 (MS -0.988); the Kf 2.2 sensitivity row per firing N_allow 26.9 (MS -0.462; v2 +0.360: SIGN CHANGE in v3); with (b1)'s moment x 1.4764 (R5, information) per firing N_allow 1.3 (MS -0.974)); the head-to-shank fillet from zero is a FOUND NEGATIVE that does not govern (jfwd_head_fillet_v352_v2.json: N_allow 29.3 (MS -0.414); per firing 1092.0 (MS +20.840); Kt LITERATURE, secondary via AMESweb citing Pilkey 2005, the Pilkey page and the T-head Kt TO-SOURCE; the no-spring head-end resultants carried over ASSUMED); the thread root governs. Both are quoted as negative, never as passing. Closure paths 2 (FE levers: (b1) is solved and entered, rows v8 under C1 / R5, and the row stays negative; the P+Q select-fit band, C9 (6), is sized for the seat line and takes no screw credit) and 3 (qualification test, TO-DEFINE, founder) remain; path 1 (MJ thread) closed NOT SOURCED. HW-170 disc spring SINGLE-USE (lead 19:53:27Z, CHANGELOG 248 (7)): a new spring with each new screw at every J-FWD assembly; removed springs are scrapped (C9 (2)). (4) POST-ABORT RULE (founder rule, lead 248 (7)): after any case-TC or TB-state abort the J-FWD joint counts as used: the ten ENG-HW-101-IN718 screws and their HW-170 springs are replaced (new, single-use) and the TC-01 holes (pins GO 5.50 / NO-GO 5.62) and the TC-02 taps (M5-6H GO / NO-GO) are gauged before the next run. TB-state aborts (CONFIRMED by Mechanical on the lead's provisional set of 253 (8), lc_tb70_envelope_v2.json F4_abort_set): HW_OVERPRESSURE, PC_OVERPRESSURE, HARD_START and PC_SENSOR on the HIGH side (transducer output >= 4.75 V = 146.5 bar abs), with the case-TC OVER_TEMP abort of F-14. Reason: the post-abort soak (ASSUMED shifted history, J-FWD band up to 201.4 C GVLO / 164.0 C GVHI, jf1b_loctite_v352_v1.json) is not a rows or J-F1b state. BESIDE IT, for every run whatever its end (ridden-through FA-1 pulses included), the J-FWD REPLACE RULE of the LC-TB70 envelope v2 (lc_tb70_envelope_v2.json 7ebd2fd63dc878ef; it supersedes v1's 'any head excursion above 60.0 bar abs'): when Q_JFWD exceeds 60.0 bar abs at any time in a run (Q_JFWD = the larger of the head-end peak and the head-end undamped SDOF ESP at the J-FWD periods 0.2-0.5 ms, on the light window and on each excursion window; no minimum duration) the joint counts as used: before the next run the line-231 J-FWD inspection (C6 (2) list) and the TC-02 taps gauged (M5-6H GO / NO-GO). Basis: at 60.0 bar abs every J-FWD row meets the nominal factors at the LC2 state 78.6 C (minimum MS_y +0.328 TC-01 ligament (LC2U), MS_u +0.312 TC-01 ligament trend 78.6 C) [SIM / DERIVED]; above it the J-FWD has only the SF 1.0 FA-1 rows (line 231). Logged signal: SIL, the plant head pressure (exact); stand, B-DPC-1 DC-restored with C1 (after a no-event noise check, TO-VERIFY); flight, the C1 LIGHT_SPIKE raw peak, a lower bound, against Electronics' chain-model threshold (TO-VERIFY). NOT meant for normal-light transients (R4-10): a normal light stays at or below MEOP 45 bar abs and reaches Q_JFWD 60 only with a 10-90 % rise below 0.288 ms [DERIVED]; 298 of the 896 FA-1 hard-start bracket cases exceed it, 185 by the head peak alone [SIM grid counts on an ASSUMED construct, not rates of normal lights]. OPEN (lead): in flight a flagged first light leaves the relight on a used joint, and the line-231 one-off basis covers one Z3 event per J-FWD assembly. DECIDED *Lead decision, per mechanical.json v2.4* T4860
RA-15, C6 closure path 3: the screw qualification test (TO-DEFINE, founder; 243 (5), 254 (6))	Closure paths: (1) an MJ-form external thread (ISO 5855-2 root radius) with a sourced Kf for the root geometry (no AS RECALLED value, and no credit for rolling residual stress, which the plastic root cycling relaxes) and the mating with the M5-6H TC-02 checked; on LCF v2's tables a Kf <= 2.2 with single-use screws gives from-zero N_allow 2.4 / 3.1 >= 1, which would close C6 for single use (reuse needs a Miner row, the first cycle from zero plus per-firing cycles), so LCF v3 runs if a Kf is sourced; (2) the bending levers through FE instances ((b1) mu 0.2 in chain E's tail, the select-fit band); (3) a qualification test (TO-DEFINE, founder). DECIDED *Lead decision, line 243 (5)* T4861 C6 closure path 1 is negative, paths 2 ((b1) FE levers) and 3 (qualification test) remain; C6 OPEN DECIDED *Lead decision, line 245 (4)* T4862 screw thread-root LCF paths 2 / 3 (Mechanical / founder test TO-DEFINE) T4863
J-FWD build procedure C4 (every assembly and re-assembly)	**Printed in full, step by step, in Annex C4** (after this table) DECIDED *Lead decision, per mechanical.json v2.4* T4824
Kulite F-10, B-PC-1 zero coefficient per serial, F-13 soak and thermometer (253 (11))	Founder items added: F-10 also asks Kulite for the thermal-zero maximum and for a 100 F compensated band from the coldest ARM to the hottest relight seat (R3-05); B-PC-1 measures the zero coefficient per serial; B-TC-1 gets the CJTO trim step and a mean-shift EMI pass (R3-03); F-13 gets the soak condition and the +/-0.5 K thermometer (R3-06). DECIDED *Lead decision, line 253 (11)* T4845
B-PC-1: the Pc-zero coefficient per serial and T_cal, with the bench plan v5 acceptance (264 (2))	with the per-serial coefficient and T_cal from B-PC-1 (0-85 C) and the bench plan v5 acceptance (coefficient within 5 % at k = 3) DECIDED *Lead decision, line 264 (2)* T4864 Residuals stay OPEN with owners: the per-serial coefficient and T_cal (founders, B-PC-1), the thermal-zero maximum (F-10), the seat-TC mount at the C1 boss and the P_mid / U570 temperature (Mechanical, MISSING), and the ASSUMED error-budget terms (coefficient 10 %, gradient, 0.25 s lag, 0.3 K). DECIDED *Lead decision, line 264 (2)* T4642 **Corrected by 266 (7), E4-02:** E4-02 (the coefficient-within-5-percent-at-k-3 acceptance, also in the 264 (2) wording, is replaced by the k-dependent limit of its Q3 table, eps_max 11.2 / 8.1 / 6.0 / 3.4 / 1.8 / 0.8 % at k = 1 / 1.25 / 1.5 / 2 / 2.5 / 3 with the 0.1012 bar adders [DERIVED]; beyond it the stand relight after B1 is planned REFUSED NOT_VENTED, fail-safe, with no relight objective claimed) DECIDED *Lead decision, line 266 (7)* T4241
F-10: the thermal-zero maximum from Kulite (264 (2))	Residuals stay OPEN with owners: the per-serial coefficient and T_cal (founders, B-PC-1), the thermal-zero maximum (F-10), the seat-TC mount at the C1 boss and the P_mid / U570 temperature (Mechanical, MISSING), and the ASSUMED error-budget terms (coefficient 10 %, gradient, 0.25 s lag, 0.3 K). DECIDED *Lead decision, line 264 (2)* T4642
RA-32, stand fallback (coefficient 0): allowed on the stand only within the 264 (2) limits, by F-10; never a flight configuration	The fallback (coefficient 0, the flagged v2.11 behaviour) is NOT a flight configuration: it loses the A1 relight at k = 3 (default build) and from k = 1.5 on the minus sign (B-PC-1-shown build) [SIM S13T]; it is allowed on the stand only with k <= 2 (default build) or k <= 1 (B-PC-1-shown build) by F-10 and k x 0.0248 bar/K x (T0 - T_cal) clearly inside 0.5 bar at ARM [DERIVED]. DECIDED *Lead decision, line 264 (2)* T4865 Option (b) (seat coefficient loaded, setPcSeatCoef != 0) is not used for any start with ignition (pad first start, stand starts, relights) until a v2.13 delta with the light-window hold and the TC4 step / rate checks is reviewed. The coefficient may be loaded for B-PC-1 / B-TC-1 bench work without ignition. DECIDED *Lead decision, line 266 (6)* T024 Consequences under the rulings in force: A1 FLIGHT IS NO-GO until v2.13 is reviewed (264 (2): the A1 pad is no-go unless the coefficient is loaded and TC4 is valid at ARM); stand first starts may run on the fallback only within 264 (2) (k <= 2 default build, k <= 1 B-PC-1-shown build, with k measured at B-PC-1 first; F-10 MISSING); no stand relight (the fallback refuses it NOT_VENTED [SIM]), so the T4 relight waits for v2.13. Option (b) stays the design: it is the only configuration that keeps the A1 relight at k = 3 [SIM S13T]. DECIDED *Lead decision, line 266 (6)* T025
Confirm the A1 flight T0 design range against the launch-site climate (264 (6))	A1 FLIGHT T0 0-40 C: ADOPTED as the ASSUMED design range (DECIDED) of the A1 relight analyses; the founders confirm it against the launch-site climate for 1 Sept 2027 before flight; the stand keeps T0 <= 24 C. DECIDED *Lead decision, line 264 (6)* T4487
Decide OI-JFWD-FLIGHT at flight acceptance, with T-F1 (264 (7))	OI-JFWD-FLIGHT: PROVISIONAL lead rule until flight acceptance: a flagged first light (f3 JFWD_LS) inhibits the A1 relight and the vehicle takes the lost-relight outcome of gnc.json v3.2 (a relight on a joint that has used its one analysed Z3 event, 231 / 258 (4), is not flown); the founders confirm or change it at flight acceptance, with T-F1. DECIDED *Lead decision, line 264 (7)* T4490
Buy the P_torch unit, a C1-family Kulite (264 (4)); TO-QUOTE delivered to Sønderborg DK-6400	the BOM gains one C1-family unit, TO-QUOTE delivered to Sonderborg DK-6400. DECIDED *Lead decision, line 264 (4)* T4866

Founder action	Decision and text (quoted in full)
RA-26, CF-4 / B-BLEED-1: purge acceptance (measured Cd x A) and the re-set rule of the purge flow and the both-open cap (256 (2), 257 (2), 258 (2))	with three conditions: (a) wherever 1.9 s is quoted as the needed flow, 2.2 s at the 283.15 K cold corner is stated with it (REPORT 2.13, CF-4 line 35, Electronics); (b) CF-4 / B-BLEED-1 shows a measured Cd x A of the two-tap path >= 0.91 x the Cd 0.70 value (>= 0.95 x with the FD-01 element bound), else the flow is re-sized; (c) the line stays unlined (lined, 283.15 K needs 2.5 s). DECIDED *Lead decision, line 256 (2)* T4867 DECIDED: after CF-4 / B-BLEED-1, kPurgeFlow_s and the both-open cap are re-set from the measured Cd x A by the measured-time form of purge_basis_line216_v2p4b, with the mass flow taken at 297.15 K (overlap <= 76 g / mdot(297.15 K) - (t_close - t_open)), never above 2.43 / 2.39 s. The 297.15 K vent then stays <= 76 g at any measured Cd x A (<= 75.99 g at the cap), and the 283.15 K margin stays positive because need and limit both scale with 1 / (Cd x A): 0.23 s at a Cd x A ratio of 1, 0.21 s at 1.08 and 1.13 (Cd 0.80), 0.19 s at 1.2 on the 10 ms steps, 0.10-0.18 s with the FD-01 element bounds [DERIVED by the lead; the scaling ASSUMED as in (b)]. Without the re-set, the measured Cd x A must also be <= 1.08 x the Cd 0.70 value (vent <= 82 g, REV-GNC-7's bound), or GNC re-runs the reserve test at the measured vent. DECIDED *Lead decision, line 257 (2)* T4868 DECIDED: the two REV-GNC-7 additions are adopted as (d1) the re-set uses the upper 95 % bound of the measured N2O Cd x A (at r 1.1326 a ratio measured 7.9 % low gives a true vent of 81.99 g, just under the 82.1 g first fail) and (d2) CF-4 shows the 283.15 K need at the re-set flow; below r 1 condition (b) governs. DECIDED *Lead decision, line 258 (2)* T4869 The same conditions as published in combustion.json /purge_decision_v2p4c: §7.5. The test list (§12.6) points here.
Q_JFWD > 60.0 bar abs: inspection and TC-02 tap gauging before the next run	ADOPTED by LEAD 258 (3): Q_JFWD = max(head peak over the record, max of S(T) over T 0.2-0.5 ms on the light window and each excursion window), not the peak alone; Q_JFWD > 60.0 bar abs at any time in a run, no minimum duration, means the line-231 J-FWD inspection (C6 (2) list) and the TC-02 taps gauged (M5-6H GO / NO-GO) before the next run DECIDED *Lead decision, per combustion.json v2.4d* T4870 Q_JFWD RULING: the J-FWD replace rule uses Q_JFWD = max(head peak over the record, max of S(T) over T 0.2-0.5 ms on the light window and each excursion window), not the peak alone: Q_JFWD > 60.0 bar abs at any time in a run, no minimum duration, means the line-231 J-FWD inspection (C6 (2) list) and the TC-02 taps gauged (M5-6H GO / NO-GO) before the next run; grid 298 / 896 against 185 for the peak alone [SIM grid counts, not rates]. Why: Q_JFWD >= the peak by construction, so it keeps every flag of the 256 (4) peak rule and adds the 113 cases whose J-FWD band load exceeds 60 bar abs below a 60 bar peak (Q / peak up to 1.2989); a false trigger costs an inspection, a missed one is an unanalysed joint load. LOW side not added), the inside_v2 re-classification of the S6R runs (mechanical requests_v352[4]) and Q_JFWD logging go to Electronics v2.12; the combustion.json fa1_p_torch_v2p4c mechanical_raw wording (governing row SOAK52 +0.670 / +0.785; +0.275 at SOAK70 is the screw-temperature sensitivity, same numbers) is corrected in combustion v2.4d together with the Z3 ruling and (d1) / (d2). DECIDED *Lead decision, line 258* T4871
Stand no-event B-DPC-1 record for a noise-floor rule (TO-VERIFY)	on measured stand traces a noise-driven trigger still means the inspection until a noise-floor rule is derived from a no-event B-DPC-1 record and reviewed (TO-VERIFY) TO-VERIFY T4872
Electronics v2.13 delta (the E4-01 fix) and its review: QUEUED, NOT STARTED; the founder says when (266 (8)); until it is reviewed, option (b) is not used for any start with ignition (266 (6))	QUEUED, NOT STARTED (the founder says when): the Electronics v2.13 delta and its review: the light-window hold (latched at IGNITION entry until E2 + 0.45 s, bumpless resume), the TC4 step check (1.5 K per conversion) and rate check (5 K/s slew, 3 K trip) with a latched f3 SEAT_STEP, the E4-03 fix, unit tests UT-1 to UT-4, the S13T TC-fault set, the FMEA row, the B-TC-1 TC4 row with the spark on, the v4.2 folds, and notes E4-N01 (T_cal 10-30 C or kPcSeatArmDevMax_K 40 K), E4-N02 (the 264 (2) pad rule and the 264 (7) inhibit implemented), E4-N03 (47.5 against 47.75 bar abs, TO-VERIFY), E4-N04 (the P_torch re-base needs compensation or a photodiode-guarded gate; torch local heat MISSING) and E4-N11 (a SIL MANIFEST_v2). DECIDED *Lead decision, line 266 (8)* T030 Option (b) (seat coefficient loaded, setPcSeatCoef != 0) is not used for any start with ignition (pad first start, stand starts, relights) until a v2.13 delta with the light-window hold and the TC4 step / rate checks is reviewed. The coefficient may be loaded for B-PC-1 / B-TC-1 bench work without ignition. DECIDED *Lead decision, line 266 (6)* T024
Electronics founder block F-1 to F-14 (bench plan v5 section 0; electronics.json v4.1): action | content | closes or feeds | done when	**Status:** Bench plan v5 does not go to the founders before these corrections. DECIDED *Lead decision, line 266 (7)* T4929 REV-ELEC-4 found F-13 and F-14 executable as written (§12.6); the plan as a whole waits for the corrections. **F-1** RFQ BOM 237 DECIDED *Design value, bench_plan_v5.md v5* T4873 | PCB Piezotronics 113B22 × 3 (two in use, one spare) with calibration certificates; flush seal rings 065A02, recess seal sleeves 065A05, clamp nuts 060A03 5/16-24; mounting adapter TO-SELECT; delivered DK-6400 DECIDED *Design value, bench_plan_v5.md v5* T4874 | B-DPC-1, IGN-1, T series DECIDED *Design value, bench_plan_v5.md v5* T4875 | quote with lead time on file DECIDED *Design value, bench_plan_v5.md v5* T4876 **F-2** RFQ BOM 238 DECIDED *Design value, bench_plan_v5.md v5* T4877 | NI 9234 + CompactDAQ chassis + 10-32 low-noise coax (length TO-DEFINE); delivered DK-6400 DECIDED *Design value, bench_plan_v5.md v5* T4878 | B-DPC-1 DECIDED *Design value, bench_plan_v5.md v5* T4879 | quote on file DECIDED *Design value, bench_plan_v5.md v5* T4880 **F-3** RFQ BOM 244 DECIDED *Design value, bench_plan_v5.md v5* T4881 | Coriolis mass-flow meter, liquid N2O service, 0-200 g/s class, response ≤ 100 ms, O2-clean (RE-22 note on the row); Endress+Hauser DK first, Emerson / Bronkhorst as alternates DECIDED *Design value, bench_plan_v5.md v5* T4882 | CF-3, B-BLEED-1 (mdot for the measured-time purge cap, LEAD 216 (5)) DECIDED *Design value, bench_plan_v5.md v5* T4883 | quote on file DECIDED *Design value, bench_plan_v5.md v5* T4884 **F-4** RFQ BOM 245 DECIDED *Design value, bench_plan_v5.md v5* T4885 | reference pressure gauge / calibrator 0-100 bar, uncertainty ≤ 0.05 bar, traceable certificate; hydraulic hand test pump to ≥ 100 bar g RESERVED for units that never see N2O (R2-14); a clean dry-N2 path (regulator + manifold) for N2O-wetted units DECIDED *Design value, bench_plan_v5.md v5* T4886 | B-PT-1, B-PC-1, F-12 DECIDED *Design value, bench_plan_v5.md v5* T4887 | quote on file DECIDED *Design value, bench_plan_v5.md v5* T4888 **F-5** RFQ BOM 250 DECIDED *Design value, bench_plan_v5.md v5* T4889 | anemometer at 3 m (ultrasonic or cup) with ≥ 1 Hz output, 3 m mast, logger with a UTC time base DECIDED *Design value, bench_plan_v5.md v5* T4890 | GSE wind rule (LEAD 181 (6) HARD) DECIDED *Design value, bench_plan_v5.md v5* T4891 | quote on file DECIDED *Design value, bench_plan_v5.md v5* T4892 **F-6** RFQ BOM 251 (Electronics addition) DECIDED *Design value, bench_plan_v5.md v5* T4893 | Kulite XTL-190SM-2000A × 3 + 1 spare: absolute mode, M5 × 0.8 short body, O2-clean processing, FKM or metal seal instead of the supplied silicone O-ring. Compensated band (R3-05; the v4 "20-120 °C" is WITHDRAWN: the sheet offers "any 100 °F range", 55.6 K): order the 100 °F band 0-55.6 °C (32-132 °F) from the coldest ARM (flight ambient 0 °C, stand ≥ 0 °C ASSUMED) through the hottest flight A1 relight seat (40 + 14.7 = 54.7 °C at the light [DERIVED, Mechanical SIM rise from ARM]); the stand relight after a B1-length burn reaches about 78 °C at E1 (24 + 54.4 K [DERIVED]), outside any 100 °F band that starts at 0 °C: F-10 (b) DECIDED *Design value, bench_plan_v5.md v5* T4894 | flight Pc / P_feed / P_mid (LEAD 205, 206 (3)); option (b) seat compensation DECIDED *Design value, bench_plan_v5.md v5* T4895 | quote and the inquiry F-10 answered DECIDED *Design value, bench_plan_v5.md v5* T4896 **F-7** TE inquiry (M3234-000005-070BG) DECIDED *Design value, bench_plan_v5.md v5* T4897 | (a) output above full scale: the saturation voltage, (b) recovery time after an over-range, (c) zero shift after an over-range of 70-100 bar g (below the 140 bar g proof), (d) the rise time of the 0.5-4.5 V output (the datasheet rows are for mV / mA outputs, R2-17); (e) whether TE has a unit for the line-205 requirement as a second source to C1: seat-mounted, body within Ø12 × 30 mm above a named thread, front dead volume ≤ 20 mm³, proof ≥ 166 bar abs (lead ruling 2026-10-20: 1.1 × FA1S 151.04), range covering the 45 bar MEOP with the 30 bar abort resolvable, operating ≥ 120 °C, N2O / O2 compatible DECIDED *Design value, bench_plan_v5.md v5* T4898 | 070BG behaviour (stand reference, P_torch); second source DECIDED *Design value, bench_plan_v5.md v5* T4899 | written answer on file DECIDED *Design value, bench_plan_v5.md v5* T4900 **F-8** PCB inquiry (113B22) DECIDED *Design value, bench_plan_v5.md v5* T4901 | (a) operation from the NI 9234 IEPE source (2.0 mA, 19 V compliance) with the 8-14 V bias and 1.45 V at 100 bar: does the full 344.75 bar range hold? (b) coating (RTV / ablative) for a hybrid N2O chamber; (c) the conditioner PCB recommends if 19 V is not enough (482C series model; BOM 265 fallback) DECIDED *Design value, bench_plan_v5.md v5* T4902 | B-DPC-1 excitation (LEAD 203 (7) TO-VERIFY) DECIDED *Design value, bench_plan_v5.md v5* T4903 | written answer on file DECIDED *Design value, bench_plan_v5.md v5* T4904 **F-9** NI inquiry (NI 9234) DECIDED *Design value, bench_plan_v5.md v5* T4905 | IEPE compliance at 2.0 mA with the 113B22; AC coupling with IEPE OFF behind an external conditioner (BOM 265); external sync / trigger with the stand DAQ DECIDED *Design value, bench_plan_v5.md v5* T4906 | B-DPC-1 DECIDED *Design value, bench_plan_v5.md v5* T4907 | written answer on file DECIDED *Design value, bench_plan_v5.md v5* T4908 **F-10** Kulite inquiry (XTL-190SM-2000A) DECIDED *Design value, bench_plan_v5.md v5* T4909 | ordering code in absolute mode at 2000 psi; compensated band (R3-05): (a) the 100 °F band 0-55.6 °C of F-6 and (b) a wider compensated band 0-85 °C (coldest ARM to the hottest stand relight seat + margin) if offered; for each, the thermal zero shift MAXIMUM (the sheet gives ±1 % FS / 100 °F typical only = 0.0248 bar/K; the firmware residual budget uses k = 3 and the SIL S13T sweeps k 1-3, ASSUMED) and the zero non-linearity over the band; seal options; O2-clean processing; cable mass per metre (the mass delta uses 10 g/m ASSUMED); shock rating (MISSING in the sheet); current datasheet revision; delivery to DK-6400 DECIDED *Design value, bench_plan_v5.md v5* T4910 | BOM 251, the mass statement, B-PC-1 zero-TC row, kPcRezeroMax_Pa basis DECIDED *Design value, bench_plan_v5.md v5* T4911 | written answer on file DECIDED *Design value, bench_plan_v5.md v5* T4912 **F-11** anemometer mast and logger (BOM 250) DECIDED *Design value, bench_plan_v5.md v5* T4913 | set up at the launch site: 3 m mast, ≥ 1 Hz logging with UTC time; compute W10 (10-min mean) and sigma_u(3 m) (10-min standard deviation of the along-wind component); dry run of the launch-commit rule over one afternoon with the 10-min windows logged DECIDED *Design value, bench_plan_v5.md v5* T4914 | W10 < 6 m/s AND sigma_u ≤ 1.0 m/s is HARD for A1 (LEAD 181 (6)); the R2 0.8 s window is confirmed only inside it (S17 v2.11) DECIDED *Design value, bench_plan_v5.md v5* T4915 | one logged day; the rule evaluated on it DECIDED *Design value, bench_plan_v5.md v5* T4916 **F-12** spare-070BG over-range row (LEAD 203 (2), 216 (6); moved from B-PC-1) DECIDED *Design value, bench_plan_v5.md v5* T4917 | on the SPARE 070BG (label it "HYDRAULIC ONLY - NEVER N2O"; it never returns to N2O service): bench supply 5.00 V (inside 4.75-5.25 V), bench DAQ ≥ 10 kS/s (no ECU), hydraulic pump + reference gauge (BOM 245). Steps 70, 80, 90, 100 bar g (all below the 140 bar g proof, DATASHEET), 3 cycles. Record the output at each step (saturation level: MISSING in the datasheet), the zero before and after (shift within the TEB ±1.5 % FS, DATASHEET) and the output after a fast release at ≥ 10 kS/s (recovery: MISSING; the release time bounds what can be shown) DECIDED *Design value, bench_plan_v5.md v5* T4918 | after line 205 this characterises the 070BG kept for P_torch (J901) and as the B-PC-1 static reference, NOT the flight Pc (C1) DECIDED *Design value, bench_plan_v5.md v5* T4919 | data and a one-page note on file DECIDED *Design value, bench_plan_v5.md v5* T4920 **F-13** stand T0 rule (LEAD ruling 2026-10-20; basis LEAD 249 (2): 20 + 4.48 / 0.96 = 24.67 °C less 0.5 K for the facility thermometer) DECIDED *Design value, bench_plan_v5.md v5* T4921 | founder pre-fire checklist row (R3-06): the engine SOAKED at the stand ambient: both case TCs TC1 / TC2 within 1 K of the facility thermometer for at least 30 min [DESIGN], with the facility thermometer calibrated to ±0.5 K (certificate on file), and that reading ≤ 24 °C (297.15 K) before any stand ARM; the ECU refuses a stand ARM above 24 °C on TC1 / TC2 (Abort CASE_WARM_STAND 31, a backstop); the checklist is the control) DECIDED *Design value, bench_plan_v5.md v5* T4922 | stand start temperature basis of the case-TC rows DECIDED *Design value, bench_plan_v5.md v5* T4923 | checklist row in use DECIDED *Design value, bench_plan_v5.md v5* T4924 **F-14** J-FWD replace rule and TB-state (LC-TB70 v2; LEAD 248, 258 (3); Mechanical F4, line 259) DECIDED *Design value, bench_plan_v5.md v5* T4925 | (1) Quoted verbatim from mechanical.json v2.4 lc_tb70_envelope_v352.jfwd_replace_rule_v2.rule: "J-FWD REPLACE RULE (LC-TB70 v2): when Q_JFWD of a run's head-end pressure exceeds 60.0 bar abs at any time, the joint counts as used: before the next run the line-231 J-FWD inspection (C6 (2) list: the ten ENG-HW-101-IN718 screws and their HW-170 springs replaced, new, single-use; the TC-01 holes gauged with the pins GO 5.50 / NO-GO 5.62, a hole that takes the 5.62 pin rejected; hole edges and the bore within 3 mm of a hole at 10x; S01 / S02 replaced; the removed screws inspected per C6 (3)) and the TC-02 taps gauged (M5-6H GO / NO-GO)". (2) The quantity, verbatim (.quantity): "Q_JFWD = max(the head peak over the record, max over T in 0.2-0.5 ms of S(T) on the light window (record start before the light to 150 ms after the first crossing of 8 bar abs, from rest, held after it) and on each excursion window (50 ms before to 150 ms after an upward crossing of 45 bar abs later than the light window, from equilibrium at its start)); S(T) the undamped SDOF ESP of Z3 v2; sample step <= 0.05 ms in the windows, else Q_JFWD is a lower bound (the peak)"; level 60.0 bar abs, duration: none (one exceedance at any time in the run). (3) Envelope rules, verbatim (.rules[1], [2]): "an event that leaves the envelope is outside the structural basis: no margin claim, engine quarantined for inspection (v1, kept)"; "one Z3 event per J-FWD assembly: the line-231 basis is a one-off abnormal event; a second Z3 excursion before the replacement is outside the envelope (stated in v2; OPEN for the lead where it bears on the flight relight)". (4) Founder rule (LEAD 248), beside (1): after ANY case-TC abort (Abort OVER_TEMP 6 from TC1 / TC2 at 423.15 K) or a TB-state abort the J-FWD joint counts as used: the same inspection as (1) before the next firing. TB-STATE ABORT = the ECU abort cause HW_OVERPRESSURE, PC_OVERPRESSURE, HARD_START, OVER_TEMP, or PC_SENSOR on the HIGH side (transducer output ≥ 4.75 V = 146.5 bar abs); the LOW side (below 0.25 V: open wire, short) is NOT one; OVER_THRUST is information only (Mechanical F4 verdict, verbatim: "CONFIRMED, with one addition: PC_SENSOR raised on the HIGH side (transducer output >= 4.75 V = 146.5 bar abs: the stand latches and aborts; in flight the non-sticky over-range bit, log v12 ovr_flags b0)"). Firmware v2.12 latches it as f3 TB_STATE (log v13 f3_flags b0; b1 TB_PCS_HIGH for the HIGH cause), cleared at the next ARM. (5) Signal for Q_JFWD: stand = B-DPC-1 DC-restored with C1, sample step ≤ 0.05 ms in the windows, Q_JFWD computed post-run with inside_v2 (04-fea/v352/lc_tb70_envelope.py d14489808338006b), usable only after a no-event B-DPC-1 noise record has been run through inside_v2 (B-DPC-1 row; TO-VERIFY: until a noise-floor rule is derived and reviewed, a noise-driven trigger still means the inspection); flight = the C1 LIGHT_SPIKE raw peak ≥ 47.5 bar abs sets f3 JFWD_LS (log v13 b5) [DERIVED from the C1 chain model on the 896-case grid: 0 misses of the 298 Q_JFWD > 60 cases on each of the four C1 chains (lowest raw peak among them 47.750 bar abs), and 128-152 of the 598 other cases flagged as well; TO-VERIFY on hardware; the ideal sampler gives 40.381 bar abs, information]; the 1 kHz flight record is a LOWER BOUND and Z3 v2 is NOT EVALUATED in flight. In flight the joint cannot be replaced before the relight (OPEN, lead) DECIDED *Design value, bench_plan_v5.md v5* T4926 | joint life (Mechanical) DECIDED *Design value, bench_plan_v5.md v5* T4927 | rule in the stand procedure; the post-run Q_JFWD script on the stand PC DECIDED *Design value, bench_plan_v5.md v5* T4928

Annex C4. J-FWD build procedure C4 (every assembly and re-assembly), in full

mechanical.json /jfwd_v352/founder_rules/C4_procedure (the same text as item_c_tf1_texts_v352_v5.json /item_C/C4: the generator checks that they are identical, row "Check-T5"), split at its step numbers; nothing is left out. TO-SELECT, TO-CONFIRM and TO-VERIFY items stay as tagged in the text.

Step	Text (quoted in full)
Scope	J-FWD build and every re-assembly (v3: route (i), lead 16:53Z R2 / R3, 17:55:55Z F1 / F2, 18:36:37Z; v4: the 248 (7) rulings and the lead 2026-10-21T05:50:45Z re-map R2 / R10): DECIDED *Lead decision, per mechanical.json v2.4* T4930
Step 0	0. GATE: J-F1 complete on the flight lots (ENG-HW-101-IN718 screws, HW-170 springs, LOCTITE 243): it sets the lot torque T_lot and carries the spring items (K with the spring under the head, the spring flattening, the crack-on torque) and the Fuji Prescale footprint item (TO-VERIFY) (hardware_v352 v1.7 HW-101 test). DECIDED *Lead decision, per mechanical.json v2.4* T4931

Step	Text (quoted in full)
Step 1	1. inspect the hole edges and the bore per C1 (e); gauge the ten TC-01 holes with the pins GO 5.50 / NO-GO 5.62 and record each hole's size (ENG-TC-13 is selected to it; a hole that takes the 5.62 pin is rejected, C1 (g)); gauge the ten TC-02 taps M5-6H GO / NO-GO (cured locker removed by a thread chaser only, no re-tap); check the TC-02 plug OD against its select-fit pairing record to this TC-01 bore (C9 (6): bore and OD at 20 +/- 1 C, D_meas - 0.1262 <= OD <= D_meas - 0.080, less 2U); clean. DECIDED *Lead decision, per mechanical.json v2.4* T4932
Step 2	2. fit the ten ENG-TC-12 plugs in their numbered holes from outside, contour aligned, band on. DECIDED *Lead decision, per mechanical.json v2.4* T4933
Step 3	3. lubricate S01 and S02 with the selected lubricant (FKM- and N2O-compatible, TO-SELECT with a datasheet; oxygen-clean per the wetted-part rule); fit them into the TC-02 grooves with an even stretch over a cone, no twist (check the mould line). DECIDED *Lead decision, per mechanical.json v2.4* T4934
Step 4	4. clock TC-02 to the match marks BEFORE the rings enter the lead-in, then push it in axially without rotation until the head shoulder stands 2.00 +/-0.02 from the TC-01 forward end face on the gap gauge (rev-5 TC-02 tap note, TO-CONFIRM; TC-02 does not seat axially on TC-01, lead 16:53Z R4; v2 read 'until seated'). DECIDED *Lead decision, per mechanical.json v2.4* T4935
Step 5	5. remove the band and the plugs; look through each hole (10x loupe or borescope) for rubber debris or a cut ring; ANY debris: withdraw TC-02, replace both rings and repeat from step 1. DECIDED *Lead decision, per mechanical.json v2.4* T4936
Step 6	6. REGISTRATION (lead 17:55:55Z F2): fit the two GSE shoulder screws ENG-TC-13 (not flight), selected to the pin-gauged sizes of their holes, into two opposite holes before any flight screw. DECIDED *Lead decision, per mechanical.json v2.4* T4937
Step 7	7. fit the flight screws ONE AT A TIME into the other eight holes: a NEW ENG-HW-101-IN718 (M5 × 10.7 with spring A; M5 × 10.6 with spring B; never mixed in one joint) with a NEW HW-170 disc spring under the head, ORIENTED per C9 (1): concave (hollow) side DOWN on the TC-01 flat (outer edge on the flat), convex side UP to the head (inner edge toward the head), centred on the 5.15 h9 pilot, NEVER inverted; LOCTITE 243 (flight lot) on the screw thread only, never on the spring or the flat; tighten to the J-F1 LOT TORQUE by method M-T1J (T_lot = 3750 N x K_min(lot) x 5 mm, so that the lot's K_min gives a maximum preload of 3750 N or less, lead 17:55:55Z F1; today's K window 0.1465-0.267: 2.747 N m, minimum 2057.6 N); record per hole: hole number and its pin size, screw and spring lot, torque, wrench (ID and calibration date). DECIDED *Lead decision, per mechanical.json v2.4* T4938
Step 8	8. remove the two shoulder screws LAST and replace each with a flight screw per step 7, recorded. DECIDED *Lead decision, per mechanical.json v2.4* T4939
Step 9	9. never re-torque a fitted screw; a screw removed for any reason is replaced by a new screw and a new spring (single-use, C6 (3)); removed springs are scrapped (HW-170 SINGLE-USE, C9 (2)); after any case-TC or TB-state abort the whole joint counts as used (C6 (4)); removed screws are bagged and labelled with their hole number for the C6 (3) inspection; then the proof / leak test. DECIDED *Lead decision, per mechanical.json v2.4* T4940
Check-T5	texts v5 C4 DECIDED *Lead decision, per item_c_tf1_texts_v352_v5.json v5* T4941 and the text above: identical DERIVED T4942
Source, 241 (8) C4	C4: assembly order (inspect; plugs and band in; lubricate S01/S02 with an oxygen-clean FKM- and N2O-compatible lubricant TO-SELECT; clock TC-02 and push axially; plugs out and check each hole for debris, any debris: withdraw, replace both rings, repeat; screws per J-F1 with LOCTITE 243 at 3.0 N m; proof / leak test). DECIDED *Lead decision, line 241 (8)* T4943 [246 (2) rewrote C4: the torque of 241 (8) is replaced by the J-F1 lot torque by method M-T1J (step 7)]
Source, 246 (2) C4	C4 rewritten: J-F1 gate (lot torque, spring items, Prescale footprint item TO-VERIFY), hole pins GO 5.50 / NO-GO 5.62 and tap gauging, head shoulder 2.00 +/-0.02 on the gap gauge (the v2 'until seated' corrected: TC-02 does not seat axially, R4), registration with two GSE shoulder screws ENG-TC-13 in two opposite holes first (F2), flight screws one at a time each with a new HW-170 spring to the J-F1 lot torque by M-T1J (maximum preload capped at 3750 N; today 2.747 N m, minimum 2057.6 N), recorded, shoulder screws out last, never re-torqued; TO-VERIFY T4944
Source, 254 (2)	C4 step 1 checks the select-fit pairing record T4945

12.4 Open items with owners

Item	Status (quoted)	Owner
A1 flight: NO-GO until v2.13 is reviewed; stand first starts on the fallback only within 264 (2); no stand relight (266 (6))	Option (b) (seat coefficient loaded, setPcSeatCoef != 0) is not used for any start with ignition (pad first start, stand starts, relights) until a v2.13 delta with the light-window hold and the TC4 step / rate checks is reviewed. The coefficient may be loaded for B-PC-1 / B-TC-1 bench work without ignition. DECIDED *Lead decision, line 266 (6)* T024 Consequences under the rulings in force: A1 FLIGHT IS NO-GO until v2.13 is reviewed (264 (2): the A1 pad is no-go unless the coefficient is loaded and TC4 is valid at ARM); stand first starts may run on the fallback only within 264 (2) (k <= 2 default build, k <= 1 B-PC-1-shown build, with k measured at B-PC-1 first; F-10 MISSING); no stand relight (the fallback refuses it NOT_VENTED [SIM]), so the T4 relight waits for v2.13. Option (b) stays the design: it is the only configuration that keeps the A1 relight at k = 3 [SIM S13T]. DECIDED *Lead decision, line 266 (6)* T025	lead; founders; Electronics (v2.13)
Electronics v2.13 delta (E4-01 fix: light-window hold, TC4 step / rate checks, E4-03, UT-1 to UT-4, S13T TC-fault set, FMEA row, B-TC-1 TC4 row) and its review	QUEUED, NOT STARTED (the founder says when): the Electronics v2.13 delta and its review: the light-window hold (latched at IGNITION entry until E2 + 0.45 s, bumpless resume), the TC4 step check (1.5 K per conversion) and rate check (5 K/s slew, 3 K trip) with a latched f3 SEAT_STEP, the E4-03 fix, unit tests UT-1 to UT-4, the S13T TC-fault set, the FMEA row, the B-TC-1 TC4 row with the spark on, the v4.2 folds, and notes E4-N01 (T_cal 10-30 C or kPcSeatArmDevMax_K 40 K), E4-N02 (the 264 (2) pad rule and the 264 (7) inhibit implemented), E4-N03 (47.5 against 47.75 bar abs, TO-VERIFY), E4-N04 (the P_torch re-base needs compensation or a photodiode-guarded gate; torch local heat MISSING) and E4-N11 (a SIL MANIFEST_v2). DECIDED *Lead decision, line 266 (8)* T030	QUEUED, NOT STARTED; the founder says when (266 (8))
Routed with line 266 (8): E4-N08 (the both-open backstop up to 2.741 s against the 2.39 s never-above value); the FC rule for SEAT_STEP and the compensated-Pc release predicate	Routed: E4-N08 (the hardware both-open backstop allows up to 2.741 s against Combustion's 2.39 s never-above value) to Combustion and the lead at B-BLEED-1; the FC rule for SEAT_STEP before a relight and the compensated-Pc release predicate are GNC and lead items with v2.13. DECIDED *Lead decision, line 266 (8)* T031	Combustion and the lead at B-BLEED-1 (E4-N08); GNC and the lead with v2.13
Before flight acceptance (LEAD 258 (3))	the flight consequence of a flagged first light (relight on a used joint; line 231 covers one Z3 event per assembly) (LEAD) TO-VERIFY T1496 the damped ESP variant (structural damping MISSING) TO-VERIFY T1497 the 0.2-0.5 ms J-FWD band (ASSUMED; TO-VERIFY on a v3.5.2 modal) TO-VERIFY T1498 a shutdown from a case hotter than the rows' soak basis (F4 observation; Mechanical, no limit derived) TO-VERIFY T1499	LEAD / Mechanical
Noise-floor rule for the J-FWD trigger on measured traces	on measured stand traces a noise-driven trigger still means the inspection until a noise-floor rule is derived from a no-event B-DPC-1 record and reviewed (TO-VERIFY) TO-VERIFY T4872	founders (B-DPC-1) / review
OI-T4-RESIZE: the T4 re-run with the shorter relight	T4 RE-SIZED (DECIDED): GNC's N-07 answer in gnc.json v3.2 (duty_cycles.note_n07_rev_gnc_7) confirms that no GNC objective needs the DC-D duty as one firing and that GNC's DC-D segments fly as their own segments or firings inside the A = 150 C case limit (REV-ELEC-3 N-07: the DC-D first burn would cross it at about 23.5 s, merged bound [DERIVED]). T4 is re-sized to REV-COMB-4's shortest firing (B1 p99 burn 21.76 s, a coast of about 19 s, one relight of about 4.6 s at about 0.47 throttle): planning value about 2.0 kg of N2O [ESTIMATE] against 7.02 kg. The T4 rows of T1-T4 lines 61, 143-145 and 171 stay the DC-D computation [SIM] until Combustion re-runs them (OPEN, Combustion; not in the finish run). Why: a T4 that trips the case abort at about 23.5 s of its first burn cannot be flown as written, and REV-COMB-4, REV-ELEC-3 and GNC all point to the shortest firing. DECIDED *Lead decision, line 261 (2)* T1554 Combustion sets the length in the T4 re-run of 261 (2) with Mechanical's soak-back. DECIDED *Lead decision, line 264 (5)* T494 T4 then demonstrates relight latency from a soaked state, not a 4.59 s landing-length relight from the B1-soaked state; bench plan v5 and the datasheet say so. DECIDED *Lead decision, line 264 (5)* T1556	Combustion / Mechanical
OI-C1-SEAT-DRIFT residuals (B-PC-1; F-10; the seat-TC mount; the P_mid temperature, MISSING)	Residuals stay OPEN with owners: the per-serial coefficient and T_cal (founders, B-PC-1), the thermal-zero maximum (F-10), the seat-TC mount at the C1 boss and the P_mid / U570 temperature (Mechanical, MISSING), and the ASSUMED error-budget terms (coefficient 10 %, gradient, 0.25 s lag, 0.3 K). DECIDED *Lead decision, line 264 (2)* T4642	founders / Mechanical
A1 pad no-go rule (NEW RULE, not implemented yet)	NEW RULE (lead; not implemented yet): A1 pad no-go unless the B-PC-1 coefficient is loaded (non-zero) and the seat TC is valid at ARM; DECIDED *Lead decision, line 264 (2)* T4528	Electronics v2.13 (E4-N02, 266 (8))
OI-PTORCH-PROOF: re-base of the torch thresholds on the C1 zero	OPEN for the next Electronics round after REV-ELEC-4: re-base kTorchArmZeroMax_Pa and kPtorchRezeroMax_Pa (0.60 bar now, on the 070BG basis) on the C1 zero including the torch-body thermal zero (the local torch heat is MISSING: Combustion / Mechanical); the output saturates at the INA333 rail above 146.5 bar abs during FA-1 and its recovery is TO-VERIFY at B-PC-1 / F-12; the BOM gains one C1-family unit, TO-QUOTE delivered to Sonderborg DK-6400. DECIDED *Lead decision, line 264 (4)* T4946	Electronics; Combustion / Mechanical (torch heat)
Purge: the measured-time form of the flow and the cap after CF-4 / B-BLEED-1	PURGE CONDITION (d) ADDED to 256 (2), on REV-GNC-7 interim 2 (MAJOR, 09:25:51Z): the 80.5 g of 256 (2) is the 289.5 K figure; at 297.15 K the fixed 2.43 / 2.39 s schedule at Cd 0.80 vents about 85.5 g (85.8 g at the cap) [ESTIMATE], and GNC's line-208 A1 reserve test against the v3d reference passes 76 g (UB95 0.854 pp) and 80.5 g (0.972 pp, margin 0.028 pp) and first fails at 82.1 g [DERIVED from SIM; ESTIMATE above 80 g]. DECIDED: after CF-4 / B-BLEED-1, kPurgeFlow_s and the both-open cap are re-set from the measured Cd x A by the measured-time form of purge_basis_line216_v2p4b, with the mass flow taken at 297.15 K (overlap <= 76 g / mdot(297.15 K) - (t_close - t_open)), never above 2.43 / 2.39 s. The 297.15 K vent then stays <= 76 g at any measured Cd x A (<= 75.99 g at the cap), and the 283.15 K margin stays positive because need and limit both scale with 1 / (Cd x A): 0.23 s at a Cd x A ratio of 1, 0.21 s at 1.08 and 1.13 (Cd 0.80), 0.19 s at 1.2 on the 10 ms steps, 0.10-0.18 s with the FD-01 element bounds [DERIVED by the lead; the scaling ASSUMED as in (b)]. Without the re-set, the measured Cd x A must also be <= 1.08 x the Cd 0.70 value (vent <= 82 g, REV-GNC-7's bound), or GNC re-runs the reserve test at the measured vent. A temperature-scheduled flow (1.8 s needed at 297.15 K) is NOT adopted. Folded into v2.4c (purge_decision_v2p4c.condition_d_LEAD_257_2, CF-4 P5 and line 66, the DATASHEET-INPUTS purge row, REPORT 17.1); routed to Electronics v2.12 (both values become calibration constants) and sent to REV-GNC-7 to check in its final. DECIDED *Lead decision, line 257 (2)* T4448	founders (CF-4 / B-BLEED-1); Combustion / Electronics (re-set)
Electronics v2.12 work NOT DONE: the v2.12 SIL MANIFEST; the S1-S11 / S14-S17 re-runs	The author did NOT produce the v2.12 SIL MANIFEST or re-run S1-S11 / S14-S17 on v2.12: REV-ELEC-4 judges which of them the v2.12 changes can affect. DECIDED *Lead decision, line 264 (8)* T4947	Electronics: v2.13 (a SIL MANIFEST_v2, 266 (8))
Items routed to Electronics v2.12 (in v4.1, promoted, line 266)	ROUTING: the F4 PC_SENSOR HIGH abort (>= 4.75 V = 146.5 bar abs; LOW side not added), the inside_v2 re-classification of the S6R runs (mechanical requests_v352[4]) and Q_JFWD logging go to Electronics v2.12; the combustion.json fa1_p_torch_v2p4c mechanical_row wording (governing row SOAK52 +0.670 / +0.785; +0.275 at SOAK70 is the screw-temperature sensitivity, same numbers) is corrected in combustion v2.4d together with the Z3 ruling and (d1) / (d2). DECIDED *Lead decision, line 258 (4)* T4482	Electronics
ENG-11 spread element (alpha / beta spread across poses): OPEN under 188 (7)	RM1-08 ENG-11: GNC assesses TVC loop robustness over the V5c stiffness range (3133-7406 N m/rad); if robust, the lead amends the spread requirement, otherwise Mechanical stiffens. Mechanical publishes V5c in mechanical.json v2.2 with a mesh study and compliant mount boundaries. DECIDED *Lead decision, line 188 (7)* T945 **Spread element:** not met on the v3.5.2 R3 FE beta values (±**28.7** % DERIVED T944 half-range over mid-range, DERIVED from §3.3; the requirement is ±**20** % DERIVED T941); OPEN under 188 (7) T945: GNC robustness assessment, then the lead amends the requirement or Mechanical stiffens; see §3.3 and open item 11b (§11.2). Owners (188 (7)): GNC (assessment), lead (amendment), Mechanical (stiffening). The hard minimum ≥ 3,100 N·m/rad per axis DERIVED T942 holds on the FE values (beta minimum **3971.84** N m/rad SIM T928).	GNC (robustness assessment); lead (amendment); Mechanical (stiffening)

Item	Status (quoted)	Owner
Found negatives of the margin matrix	**33** DERIVED T697	per class, §9.3; owners §9.10
FEA3D open items	13. Open items and owners (2026-10-21) DERIVED T4948	per item, §9.10
GNC open items 11a-11q	Open item: 210 m gate-delay arm TO-VERIFY T4949	per item, §11.2
Electronics open issues (electronics.json), open	**31** DERIVED T4666	per item, §10.9
Electronics open issues closed by a lead line (shown for the record in §10.9; only a residual a lead line names stays open)	**3** DERIVED T4667	§10.9
Airframe items kept for when the airframe is tackled (ICD-AIRFRAME out of the finish scope)	The items that were queued for it are kept as engine data or open items in the engine datasheet and this log, for when the airframe is tackled: bay bore >= 140 mm down to the nozzle and the v3.5.1 envelope; mounting loads at the 16 M5 holes and ring resultants (FEA3D lines 217/220); engine-mount modes (line 221); DECIDED *Lead decision, line 222* T4950	airframe work, deferred by the founder (222)

12.5 Requirement verification methods

00-requirements/REQUIREMENTS.md §2 and its amendments (REQ rev 2). No requirement has been verified physically. ENG-11 (REQ) is the gimbal axis stiffness; the GNC servo output stiffness K_s that the GNC inputs file under ENG-11 is in §3.2.

ID	Requirement (rev 2 text, start)	Amendments after rev 2 (sentences naming the ID)	Verification method
ENG-01	Max thrust 500 N at sea level, with N2O at ~~15-25 °C~~ ARM 15-24 C; T_tank <= 24 C at ARM and at ignition **OVR-04**. Normal Pc is capped at 25 bar in software. DECIDED *Requirement, ENG-01* T970	ENG-01 at the cap (CHANGELOG 170): the 25 bar software Pc cap stays; if T1 measures the low end of the eta_cf band (0.95), rated thrust is ~487 N at 25 bar (DERIVED) and the measured thrust coefficient k feeds the GNC cap rule. DECIDED *Requirement, ENG-01* T4952	Analysis, then static fire DECIDED *Requirement, ENG-01* T4951
ENG-02	Throttle floor ≤ 20 % of max (5:1, 100–500 N) with stable combustion. Stretch: 12.5 % (8:1). Throttle by a servo-driven variable-area pintle, with a separate NC isolation valve for shutoff. DERIVED T4953	none	Analysis, cold flow, static fire T3 with 1 kHz Pc DECIDED *Requirement, ENG-02* T4954
ENG-03	Throttle response: t10–90 ≤ max(0.15 s, 0.8·ΔF / 1000 N/s). Ramp (amended 2026-10-13, lead on GNC eng03_rate_v1.1): closing ≥ 2500 N/s in every state; opening ≥ the HV69 corner rate map ... DERIVED T4955	none	SIL, B-PNT-1, static fire DECIDED *Requirement, ENG-03* T4956
ENG-04	Restartable: ≥ 3 starts per flight. Command to 90 % thrust ≤ 0.5 s (target 0.3 s), repeatability ±50 ms; flight software plans relight on 0.81 s (amended 2026-10-13; the 0.6 s rule is withdrawn). DEVIATION recorded 2026-10-13: ... DERIVED T4957	ENG-04 / relight: deterministic relight (RELIGHT at planned T_fire; DECIDED *Requirement, ENG-04* T4959	Analysis, static fire DECIDED *Requirement, ENG-04* T4958
ENG-05	Shutdown: command to < 10 % thrust ≤ 0.10 s. Met with a mandatory pre-MECO throttle-down to ≤ 75 % (a 100 % MECO at 3.5 MPa tank fails at 0.106–0.124 s). No flashback or decomposition in the feed. NC main valve closed during ... DERIVED T4960	ENG-05 / hard start (stand): unignited-inventory allowable set by the two-volume relief analysis; stand rule R1-R4; stand relief per the 2026-10-13 decision, fallback (b) ADOPTED: R1 = R2 = 0.35 s windows, R3 deleted, rounded grain port entrance r >= 4 mm, ... DECIDED *Requirement, ENG-05* T4962	Analysis, B-PNT-1 closing time at 25 bar, static fire DECIDED *Requirement, ENG-05* T4961
ENG-06	Accumulated burn per flight sized by the duty cycles in gnc.json v3. Grain replaced each flight; web ≥ 1 mm over the whole regression-law envelope (a ×0.6–1.5, n 0.35–0.68). Throat insert: plan on one insert per flight until ... ASSUMED T4963	none	Analysis, static fire bore measurement DECIDED *Requirement, ENG-06* T4964
ENG-07	TVC by a two-axis gimbal of the whole engine, pivot at the engine-group mid-burn CG. Range ±7° on the vector norm, 8° conical mechanical stop. Slew ≥ 100°/s at 7° in the worst vector direction at the 7.4 V nominal servo plug ... DECIDED *Lead decision, per REQUIREMENTS.md rev 2* T967	none	SIL, B-TVC-1 DECIDED *Requirement, ENG-07* T4965
ENG-07b	No actuator saturation episode longer than 0.1 s during the landing burn. Logged in flight as SAT_LONG. DERIVED T4966	none	MC, flight logs DECIDED *Requirement, ENG-07b* T4967
ENG-07c	Reflected servo rotor inertia J_int ≤ 1e-3 kg m² at the output (provisional), unless a saturation-aware controller meets ENG-07b up to 2.7e-3. Estimated value for the SB-2292SG: 1.6e-3 (0.76–2.7e-3). DERIVED T4968	none	B-TVC-1 DECIDED *Requirement, ENG-07c* T4969
ENG-07d	Flex gain margin ≥ 6 dB at the gimbal rocking mode. ECU notch required, with separate alpha and beta centre frequencies set from B-TVC-1. DERIVED T4970	none	CSI analysis, B-TVC-1 DECIDED *Requirement, ENG-07d* T4971
ENG-08	Gimbal sweep fits the airframe ICD: bay ID ≥ 140 mm, zone [−0.238, +0.252] m. Feed across the gimbal (amended 2026-10-13; Option A/O3b supersedes the 0.40 m radial line): coil on the case, fixed 3/8 wall line at αz 219.5° clamped ... DERIVED *Requirement, ENG-08* T4972	none	CAD, B-FD-1 DECIDED *Requirement, ENG-08* T4973
ENG-09	Pressure parts: SF ≥ 1.5 on yield and ≥ 2.0 on ultimate, at MEOP and at temperature. Chamber MEOP 45 bar (~~stuck: pintle case, 30 °C tank~~ ignition interlock T_tank <= 24 C; basis 44.33 bar (meop_basis_v2p4) **OVR-06**), proof 67.5 bar, which covers the blocked-nozzle bound of about 63 bar below yield. Normal Pc ≤ 25 bar; ECU over-pressure abort at about 30 bar. Tank MEOP 7.0 MPa, with a passive relief device (relief valve and/or burst disc, set below proof, T-vent). ~~Trapped-liquid relief ≤ 7.5 MPa.~~ tank relief 6.9 MPa, MAWP >= 8.42 MPa, proof >= 10.5 MPa, back-relief check (Cv >= 0.1), disc 9.5 MPa +/- 5 % **OVR-05** Chamber Pc transducer range ≥ 70 bar. DECIDED *Lead decision, per REQUIREMENTS.md rev 2* T4974	ENG-09: MEOP 45 bar kept; ignition interlock T_tank <= 24 C at ARM and at ignition. DECIDED *Requirement, ENG-09* T4976 ENG-09 trapped-liquid relief text superseded: tank relief 6.9 MPa, MAWP >= 8.42 MPa, proof >= 10.5 MPa, back-relief check Cv >= 0.1, secondary disc 9.5 MPa +/- 5 %. DECIDED *Requirement, ENG-09* T4977	FEA, hydro proof DECIDED *Requirement, ENG-09* T4975
ENG-10	Structure (gimbal, mounts, fasteners): SF ≥ 1.5 on yield and ≥ 2.0 on ultimate under design limit loads. Gimbal design loads: engine moment 35.3 N·m (both servos at 8.4 V stall); rod force 441 N at centre, 552 N at 7°, 609 N for ... DERIVED *Lead decision, per REQUIREMENTS.md rev 2* T4978	REV-MECH-1 decisions (CHANGELOG 188): safety factors: nominal states (LC1-LC4, LC6, TS1, TS2, SEGP, TB70) ENG-10 1.5/2.0; abnormal states (LC2F, BN, TBS) 1.0/1.5; DECIDED *Requirement, ENG-10* T4980	FEA, hand calculations DECIDED *Requirement, ENG-10* T4979
ENG-11 (REQ): gimbal axis stiffness	Gimbal axis stiffness (structure plus rods) ≥ 3,100 N·m/rad per axis at every pose (hard minimum); goal 5,200 N·m/rad. Alpha/beta spread across poses ≤ ±20 %. Rocking mode as required by ENG-07d. DERIVED T940 **Spread element:** not met on the v3.5.2 R3 FE beta values (**±28.7** % DERIVED T944 half-range over mid-range, DERIVED from §3.3; the requirement is **±20** % DERIVED T941); OPEN under 188 (7) T945: GNC robustness assessment, then the lead amends the requirement or Mechanical stiffens; see §3.3 and open item 11b (§11.2). Owners (188 (7)): GNC (assessment), lead (amendment), Mechanical (stiffening). The hard minimum ≥ 3,100 N·m/rad per axis DERIVED T942 holds on the FE values (beta minimum **3971.84** N m/rad SIM T920).	ENG-11 gimbal stiffness spread (3133-7406 N m/rad) is assessed by GNC for loop robustness. DECIDED *Requirement, ENG-11* T4982 [the spread range in this sentence is the V5c plant, CAD v3.3 (188 (7) T945, 227 T4983; published in mechanical.json v2.2 per 188 (7)), not the v3.5.2 model; the v3.5.2 R3 FE values per axis (§3.3): alpha **7294** T916–**9337** N m/rad SIM T917; beta **3972** T920–**7172** N m/rad SIM T921]	Modal and stiffness FEA, B-TVC-1 DECIDED *Requirement, ENG-11* T4981
ENG-12	Propellants: self-pressurised N2O, tank 2.00 kg / 3.17 L conditioned to ~~15-25 °C~~ ARM 15-24 C; T_tank <= 24 C at ARM and at ignition **OVR-04**, with heater hard-disabled at ARM; N2 supercharge not baselined. 3D-printed ABS grain (PETG backup): straight centred 25 mm port, 53.5 OD × 230 mm, 0.404 kg. Nozzle ε 2.0 with a fine-grain graphite insert. DECIDED *Requirement, ENG-12* T4984	ENG-12: tank temperature range at ARM 15-24 C (was 15-25 C); heater hard-disabled at ARM. DECIDED *Requirement, ENG-12* T4986	Trade studies, static fire DECIDED *Requirement, ENG-12* T4985
ENG-13	Engine controller (ECU) works with a Teensy-class flight computer. It provides closed-loop thrust and gimbal control, telemetry, arming interlocks and logging. ARM requires predicted both-servo-stall plug voltage ≥ 6.6 V and ... DERIVED T4987	none	Bench, SIL, HIL DECIDED *Requirement, ENG-13* T4988
ENG-14	Cost: cheapest viable. Every BOM line has a source and price, or is marked TO-QUOTE with a named vendor (delivery: Sønderborg, DK-6400). No zero or "from" prices. DECIDED *Requirement, ENG-14* T4989	none	BOM review DECIDED *Requirement, ENG-14* T4990
ENG-15	Manufacturability: printed parts on a hobby FDM printer; metal parts from stock tube and bar by online CNC or a simple lathe; a PETG fit-check version of every part. DECIDED *Requirement, ENG-15* T4991	none	CAD review DECIDED *Requirement, ENG-15* T4992
ENG-16	Safety: N2O-compatible, oxygen-clean wetted paths (PFPE lubricant only); passive relief plus remote vent; arming plug; remote firing; written hazard list. Cold flows only into a back-pressured catch volume or with N2 at ≤ 3.5 ... DECIDED *Requirement, ENG-16* T4993	none	Review, bench DECIDED *Requirement, ENG-16* T4994

12.6 Founder test list

Bench plan v5 (the plan of electronics.json v4.1), for information, not for execution. Bench plan v5 does not go to the founders before these corrections. DECIDED *Lead decision, line 266 (7)* T4929 **REV-ELEC-4:** ACCEPT WITH COMMENTS (text corrections E4-02, E4-06 before founder use; the TC4 rows of v2.13) DERIVED T5043 F-14, F-13, B-TC-1 (case TCs), B-BLEED-1 (d) / (d1) / (d2), B-MV-1 and T1-T4 inside A = 150 C are executable as written. The B-PC-1 stand sentence, the A1 pad row and the T4 row are not DERIVED T5044 **The corrections (266 (7)):** E4-06 (bench plan v5: A1 pad NO-GO until v2.13; T4 restated as a relight-latency demonstration from a soaked state, its length from Combustion's T4 re-run, waiting for v2.13; BOM row 187 relabelled; the C7 stand-DAQ TC added as a stand-only row) DECIDED *Lead decision, line 266 (7)* T5042
Corrected by 266 (7), E4-02: E4-02 (the coefficient-within-5-percent-at-k-3 acceptance, also in the 264 (2) wording, is replaced by the k-dependent limit of its Q3 table, eps_max 11.2 / 8.1 / 6.0 / 3.4 / 1.8 / 0.8 % at k = 1 / 1.25 / 1.5 / 2 / 2.5 / 3 with the 0.1012 bar adders [DERIVED]; beyond it the stand relight after B1 is planned REFUSED NOT_VENTED, fail-safe, with no relight objective claimed) DECIDED *Lead decision, line 266 (7)* T4241
Under the carve-out the A1 pad is NO-GO and the stand relight (T4 among it) waits for v2.13 (§10.0). The CF-4 / B-BLEED-1 purge acceptance and re-set rule: founder row in §12.3 (and §7.5).

Bench plan v5 section 1 (BENCH v5): for information, not for execution: it does not go to the founders before the corrections of CHANGELOG 266 (7). REQ §4: early purchases. CF-4 and T1-T4 criteria: §7.9; the CF-4 / B-BLEED-1 purge acceptance and re-set rule: §12.3. None has been run. Cost: TO-QUOTE, delivered to Sønderborg, DK-6400.

Kind	Test	Gate or measurement	Needs / source
Bench step F	section 0 (founder block) DECIDED *Design value, bench_plan_v5.md v5* T4995	RFQs, inquiries, anemometer, spare-070BG over-range, stand rules DECIDED *Design value, bench_plan_v5.md v5* T4996	~~nothing executable now~~ DECIDED *Design value, bench_plan_v5.md v5* T4997 [not for execution: bench plan v5 does not go to the founders before the corrections of 266 (7) T4929]
Bench step 0	RE-15 board checks, HAL-1 DECIDED *Design value, bench_plan_v5.md v5* T4998	board fit for B-PC-1; HAL in place DECIDED *Design value, bench_plan_v5.md v5* T4999	schematic v2.6 (the v2.11 rows of BOM v2.3; NOT drawn) + layout (not started) DECIDED *Design value, bench_plan_v5.md v5* T5000
Bench step 1	B-PWR-1, B-PWR-2 DECIDED *Design value, bench_plan_v5.md v5* T5001	rails, e-fuse, ARM interlock, overlaps at 6.6 / 8.1 V (RE-09) DECIDED *Design value, bench_plan_v5.md v5* T5002	bare ECU, bench PSU DECIDED *Design value, bench_plan_v5.md v5* T5003
Bench step 2	B-ECU-1 DECIDED *Design value, bench_plan_v5.md v5* T5004	I/O, CAN, watchdog, log v13, ladder decode with decade box (RE-19) DECIDED *Design value, bench_plan_v5.md v5* T5005	step 1 DECIDED *Design value, bench_plan_v5.md v5* T5006
Bench step 3	B-PT-1, B-PC-1, B-TC-1 DECIDED *Design value, bench_plan_v5.md v5* T5007	P_tank calibrated budget (guard band); C1 Pc trims, post-zero residual (`kBpc1Shown`), zero TC per serial (seat coefficient and T_cal, option (b)), seat-TC channel, 20 s drift, comparator; case-TC CJTO trim, offset and tau DECIDED *Design value, bench_plan_v5.md v5* T5008	step 2, clean dry N2 0-70 bar g, reference gauge (BOM 245), TC reference (BOM 266), temperature chamber 0-85 °C (TO-DEFINE) DECIDED *Design value, bench_plan_v5.md v5* T5009 **Corrected by 266 (7), E4-02:** E4-02 (the coefficient-within-5-percent-at-k-3 acceptance, also in the 264 (2) wording, is replaced by the k-dependent limit of its Q3 table, eps_max 11.2 / 8.1 / 6.0 / 3.4 / 1.8 / 0.8 % at k = 1 / 1.25 / 1.5 / 2 / 2.5 / 3 with the 0.1012 bar adders [DERIVED]; beyond it the stand relight after B1 is planned REFUSED NOT_VENTED, fail-safe, with no relight objective claimed) DECIDED *Lead decision, line 266 (7)* T4241 SUSPENDED UNTIL v2.13 (266 (6)) the TC4 rows of B-PC-1 / B-TC-1 T026
Bench step 4	B-MV-1, B-PNT-1, B-TVC-1 DECIDED *Design value, bench_plan_v5.md v5* T5010	actuator timing, ENG-03, MV ladder, TVC jam threshold (RE-24) DECIDED *Design value, bench_plan_v5.md v5* T5011	step 2 DECIDED *Design value, bench_plan_v5.md v5* T5012
Bench step 5	CHK-PW, CHK-BT, B-BLEED-1 DECIDED *Design value, bench_plan_v5.md v5* T5013	purge window, both-open timer, valve, closure timing, purge mass, comparator, pad BIT DECIDED *Design value, bench_plan_v5.md v5* T5014	steps 1-4, selected valves (none yet) DECIDED *Design value, bench_plan_v5.md v5* T5015
Bench step 6	B-IGN-1, B-FC-1 DECIDED *Design value, bench_plan_v5.md v5* T5016	spark EMI on Pc, FC link, TLM_FAST bits, ABORT path, PC_VALID, encoder TLM DECIDED *Design value, bench_plan_v5.md v5* T5017	step 2, FC DECIDED *Design value, bench_plan_v5.md v5* T5018
Bench step 7	HIL-OT-1, HIL-TR-1, HIL-RST-1 DECIDED *Design value, bench_plan_v5.md v5* T5019	over-thrust and transient replay; reset behaviour (RE-28) DECIDED *Design value, bench_plan_v5.md v5* T5020	steps 2, 6, HAL DECIDED *Design value, bench_plan_v5.md v5* T5021
Bench step 8	LEAK-1, CF-3, CF-4 DECIDED *Design value, bench_plan_v5.md v5* T5022	feed leak, ignition-lift table, segment void DECIDED *Design value, bench_plan_v5.md v5* T5023	Mechanical / Combustion rigs DECIDED *Design value, bench_plan_v5.md v5* T5024
Bench step 9	B-SAFE-1 (procedure dry run), B-DPC-1, IGN-1, T1-T4 DECIDED *Design value, bench_plan_v5.md v5* T5025	relative light rule, start latency, LIGHT_SPIKE, F_acc DECIDED *Design value, bench_plan_v5.md v5* T5026	hot-fire stand DECIDED *Design value, bench_plan_v5.md v5* T5027 [266 (7), E4-06 (2): T4 restated as a relight-latency demonstration from a soaked state, its length from Combustion's T4 re-run, waiting for v2.13 T5028]
Early purchase (REQ §4)	**B-PNT-1**	Commanded travel ≥ 103°; unpowered spring closing at the MEOP-equivalent load; guide μ ≤ 0.20; break-out after a dwell; closing time at 25 bar; HV69 torque-speed under load (0.42-0.55 N·m at 7.335 and 7.49 V) against the rate-map model speed, and closed-loop t10-90 ≤ 100 ms under load including the drive (ENG-03; if the drive lag adds to the MC tau, B1 loses ~1.2 %, all in 6-8 m/s wind); N2 leak test at 5 and 70 bar (≤ 1 scc/min, TO-AGREE). If travel falls short, fall back to the jackshaft layout. DECIDED *Requirement, REQUIREMENTS.md rev 2* T5030	1× MKS HV69 (EUR 62.99 at Lindinger, catalogue price incl. AT VAT; TO-QUOTE to DK-6400) DECIDED *Requirement, REQUIREMENTS.md rev 2* T5031
Early purchase (REQ §4)	**B-TVC-1**	Measure servo output stiffness K_s at 6.6 and 7.4 V, damping, J_int and the rotor/structure split; stall-hold heating. If K_s < ~100 N·m/rad, switch to the actuator contingency (maxon ECX 22 M class). Reduced landing bandwidth is a degraded mode only, not a fallback (P 0.676 / 0.968; amended 2026-10-13, REV-GNC-1 RG-12). DECIDED *Requirement, REQUIREMENTS.md rev 2* T5033	2× SB-2292SG (TO-QUOTE) with the n 4.8 linkage on a 0.0545 kg·m² inertia dummy DECIDED *Requirement, REQUIREMENTS.md rev 2* T5034
Early purchase (REQ §4)	**B-PWR-1/2**	Source and branch resistance against the 15 / 20 mΩ assumptions. DECIDED *Requirement, REQUIREMENTS.md rev 2* T5036	Flight battery pack and harness DECIDED *Requirement, REQUIREMENTS.md rev 2* T5037
Early purchase (REQ §4)	**B-FD-1**	Hinge stiffness while pressurised ≤ 20 N·m/rad per axis. DECIDED *Requirement, REQUIREMENTS.md rev 2* T5039	Gimbal hose DECIDED *Requirement, REQUIREMENTS.md rev 2* T5040

13 Decision overrides and the contradiction register

13.1 Decision overrides

Table gen/decision_overrides.json (ovr-7, sha256/16 6746adf53921b1ef). Each decision is quoted from its source; the generator checks that the cited text contains the key phrases and every number of the decision, and applies a target only while the stale wording is still in it. An entry with no active target is reported REDUNDANT and can be dropped.

Retired in ovr-7, each REDUNDANT at the rev1 draft build (the table's drop rule), kept verbatim in the file under "retired" and in gen/history/: OVR-02, OVR-03, OVR-07, OVR-08.

Override	Decision shown	Cited sources (quoted)	Targets: key, mode, state, displays
OVR-01 30 bar over-pressure abort: channel and persistence basis: REV-COMB-2 R2-01	30 bar firmware abort on the FILTERED channel with 5 ms persistence (amends 'raw' in CHANGELOG 160 (2)) DECIDED *Lead decision, per decision_overrides.json ovr-7* T004 **ACTIVE** T5067	CHANGELOG.md L166(4) RE-20: the 30 bar firmware abort runs on the FILTERED channel with 5 ms persistence (probe F: comparator stays first; Combustion: filtered light-spike peak <= 21.4 bar); amends the word "raw" in l.160 (2). DECIDED *Lead decision, line 166 (4)* T001 REQUIREMENTS.md §Amendments after rev 2 bullet "Electronics review decisions (CHANGELOG 166)"[sentence containing "filtered channel, 5 ms"] Electronics review decisions (CHANGELOG 166): light-rule baseline never substituted by ambient on the high side (pad/stand refuse with PC_SENSOR, A1 relight proceeds on measured P0); nominal-profile plant Pc <= 28 bar abs at p99.9 with zero PC_OVERPRESSURE in S5 (30 bar abort unchanged, filtered channel, 5 ms); DECIDED *Lead decision, per REQUIREMENTS.md rev 2* T003	combustion.json /abort_matrix_v2p4/v/A4: annotate, REDUNDANT, shown 0× CHANGELOG.md L160(2): annotate, ACTIVE, shown 0× T5066
OVR-04 Tank temperature range at ARM and ignition interlock basis: REV-COMB-2 R2-05, R2-07	tank temperature range at ARM 15-24 C (was 15-25 C); T_tank <= 24 C at ARM and at ignition; thresholds sit below 24 C / 55.28 bar abs by the channel uncertainty DECIDED *Lead decision, per decision_overrides.json ovr-7* T008 **ACTIVE** T5069	REQUIREMENTS.md §Amendments after rev 2 bullet "ENG-12:" ENG-12: tank temperature range at ARM 15-24 C (was 15-25 C); heater hard-disabled at ARM. DECIDED *Requirement, ENG-12* T005 REQUIREMENTS.md §Amendments after rev 2 bullet "ENG-09:" ENG-09: MEOP 45 bar kept; ignition interlock T_tank <= 24 C at ARM and at ignition. DECIDED *Requirement, ENG-09* T006 CHANGELOG.md L172(4) R2-07 interlock guard band: T_tank and P_tank thresholds sit below 24 C / 55.28 bar abs by the total channel uncertainty (Electronics states it); CF-1/CF-3 acceptance includes full-stroke Cd <= 0.80 and line loss >= model at the failure flow, otherwise MEOP is restated. DECIDED *Lead decision, line 172 (4)* T007	combustion.json /decisions/tank_pressurisation: annotate, REDUNDANT, shown 0× combustion.json /igniter/sequence/v/arm_preconditions/0: annotate, REDUNDANT, shown 0× REQUIREMENTS.md §2. Top-level row "ENG-12" col 1: annotate, ACTIVE, shown 1× REQUIREMENTS.md §2. Top-level row "ENG-01" col 1: annotate, ACTIVE, shown 1× T5068
OVR-05 Relief set points (feed and tank) basis: REV-COMB-2 R2-05; CHANGELOG 172 (6)	tank relief 6.9 MPa, MAWP >= 8.42 MPa, proof >= 10.5 MPa, back-relief check MV segment to tank (Cv >= 0.1), secondary disc 9.5 MPa +/- 5 %; the trapped-liquid relief <= 7.5 MPa text is superseded DECIDED *Lead decision, per decision_overrides.json ovr-7* T013 **ACTIVE** T5071	REQUIREMENTS.md §Amendments after rev 2 bullet "Combustion re-review decisions (CHANGELOG 172)"[sentence containing "trapped-liquid relief"] ENG-09 trapped-liquid relief text superseded: tank relief 6.9 MPa, MAWP >= 8.42 MPa, proof >= 10.5 MPa, back-relief check Cv >= 0.1, secondary disc 9.5 MPa +/- 5 %. DECIDED *Lead decision, per REQUIREMENTS.md rev 2* T010 CHANGELOG.md L172(6) R2-01, R2-05, R2-06 (incl. CF-4 P4 per tap and phase), R2-08, R2-09 go into a Combustion v2.4a text/interface patch (relief_v2p4 block added); the ENG-09 "trapped-liquid relief <= 7.5 MPa" text is superseded by the tank relief 6.9 MPa / MAWP 8.42 MPa / back-relief check / 9.5 MPa disc. DECIDED *Lead decision, line 172 (6)* T011 REQUIREMENTS.md §Amendments after rev 2 bullet "Relief:" Relief: back-relief check MV segment → tank (Cv >= 0.1); secondary rupture disc 9.5 MPa +/- 5 % on FS-01, discharge overboard. DECIDED *Requirement, REQUIREMENTS.md rev 2* T012	combustion.json /valves_and_actuators/isolation_valve/trapped_liquid_relief: replace, REDUNDANT, shown 0× combustion.json /valves_and_actuators/priming_rule/v/2: annotate, REDUNDANT, shown 0× REQUIREMENTS.md §2. Top-level row "ENG-09" col 1: annotate, ACTIVE, shown 1× T5070
OVR-06 MEOP basis: ignition interlock instead of the v2.3 tank temperature basis: REV-COMB-2 R2-05 (RC-08 closed); CHANGELOG 172 (4)	MEOP 45 bar kept with the ignition interlock T_tank <= 24 C at ARM and at ignition; basis 44.33 bar at the interlock (meop_basis_v2p4), margin 0.67 bar DECIDED *Lead decision, per decision_overrides.json ovr-7* T015 **ACTIVE** T5073	REQUIREMENTS.md §Amendments after rev 2 bullet "ENG-09:" ENG-09: MEOP 45 bar kept; ignition interlock T_tank <= 24 C at ARM and at ignition. DECIDED *Requirement, ENG-09* T006 combustion.json /meop_basis_v2p4/v {"basis_bar": 44.33, "margin_to_MEOP_bar": 0.67, "proof_margin_min_bar": 21.9, "supersedes": "v2.3 basis [withheld: superseded value, SUP-01] bar (RC-08)"} SIM T014	combustion.json /decisions/chamber_MEOP_Pa/tag: replace, ACTIVE, shown 0× REQUIREMENTS.md §2. Top-level row "ENG-09" col 1: annotate, ACTIVE, shown 1× ICD-AIRFRAME.md §3. Loads row "Physical worst-case thrust" col 1: annotate, ACTIVE, shown 0× T5072
OVR-09 Drogue release with Pc invalid: accelerometer-confirmed thrust instead of the time-only term basis: GNC rev2 amendment c	The time-only 0xFFFF release term (line 168 (1)) is REPLACED in design by R2: with Pc invalid, release iff ECU RUN AND accelerometer thrust >= 37.16 N for 3 frames at 100 Hz; otherwise hold the drogue and command ECU ABORT at t_E2 + pcf_tmax, pcf_tmax 0.8 s (PROPOSAL to be flown) DECIDED *Lead decision, per decision_overrides.json ovr-7* T019 **ACTIVE** T5075	CHANGELOG.md L175(5) Amendment c: the time-only 0xFFFF release term (line 168 (1)) is REPLACED in design by R2: with Pc invalid, release iff ECU RUN AND accelerometer thrust >= 37.16 N for 3 frames at 100 Hz; otherwise hold the drogue and command ECU ABORT at t_E2 + pcf_tmax, pcf_tmax 0.8 s (PROPOSAL to be flown; margins 33.0 vs 37.16 N ... DECIDED *Lead decision, line 175 (5)* T016 REQUIREMENTS.md §Amendments after rev 2 bullet "GNC rev2 decisions (CHANGELOG 175)"[sentence containing "accelerometer-confirmed thrust"] With Pc invalid after E2 the drogue is released only on accelerometer-confirmed thrust (>= 37.16 N, 3 frames at 100 Hz) with the ECU in RUN; otherwise the drogue is held and the ECU aborted at E2 + 0.8 s (proposal, to be flown). DECIDED *Lead decision, per REQUIREMENTS.md rev 2* T018	CHANGELOG.md L168(1): annotate, ACTIVE, shown 0× T5074
OVR-10 electronics.json v4.1 status text: STAGED vs promoted basis: CHANGELOG 266 (7)	interfaces/electronics.json is v4.1, PROMOTED 14:45:39Z by the lead; its STAGED status text is superseded until v4.2 DECIDED *Lead decision, per decision_overrides.json ovr-7* T021 **ACTIVE** T5077	CHANGELOG.md L266(7) PROMOTED 14:45:39Z: interfaces/electronics.json = staging_v2.12 results/electronics_v4.1.json c862375163393adc, byte for byte (v4 11f7896ad0d0613b stays as staging_v2.11/results/electronics_v4.json). Firmware v2.12 and its SIL are the design baseline under the carve-out of (6). Until v4.2 this line governs v4.1 where ... DECIDED *Lead decision, line 266 (7)* T020	electronics.json /status: annotate, ACTIVE, shown 1× T5076

13.2 Superseded values withheld

Values the Combustion author withdrew in REPORT v2.4 §13 (their list: combustion.json /superseded_v2p4), wherever combustion.json v2.4d still carries them. They are not shown in this datasheet; masked text reads "[withheld: superseded value, SUP-xx]".

ID	Shown instead	Keys withheld	Text masks	In the author's withdrawal list
SUP-01 v2.3 MEOP basis (stuck pintle, tank above the ignition interlock)	v2.4 basis /meop_basis_v2p4 (OVR-06) basis: REV-COMB-2 R2-05; combustion.json /superseded_v2p4	combustion.json /operating_points/pressure_limits/full_stroke_failure_S230_max_bar: WITHHELD; combustion.json /operating_points/pressure_limits/full_stroke_failure_S230/v/11/Pc_bar: WITHHELD T5078	3 displayed fields masked T5079	yes T5080
SUP-02 v2.3 throat-insert life in A1 flights	one insert per flight (ENG-06 as amended) basis: REV-COMB-2 R2-05; combustion.json /superseded_v2p4	combustion.json /thermal_and_loads/flights_per_insert_A1: WITHHELD T5081	0 displayed fields masked T5082	yes T5083
SUP-03 v2.3 coast seat-leak ceiling	coast leak ceiling from the Combustion report v2.4 (§7.3 of this datasheet) basis: REV-COMB-2 R2-05; combustion.json /superseded_v2p4	combustion.json /valves_and_actuators/pintle_actuator/seat_and_seal_review_v2p3/v/b_seat/leak_tightness: WITHHELD T5084	0 displayed fields masked T5085	yes T5086
SUP-04 ICD-AIRFRAME quotes the v2.3 MEOP-basis pressure rounded	v2.4 basis (OVR-06); worst-case thrust at that basis not published basis: same physical case as SUP-01 (ICD tag: SIM, Combustion v2.3)	no key listed T5087	0 displayed fields masked T5088	yes T5089

13.3 Contradiction register

Each entry quotes every side, then the source that governs and the status of the disagreement. An entry is generated when the current sources disagree; a governing source already quoted under the sides is referenced, not repeated. An ID is never reused: an entry whose sides agree at this build is kept as RETIRED, with the reason (1 at this build). "See also" links related entries.

ID	Sides (quoted)	Governing source	Status
C-01 HW-101 J-FWD u cell: FEA3D-2 report text vs the rows-v8 matrix of record	**FEA3D-2 report §13 item 2:** Eq. 22 inputs V 4079.2 N (FE, (b1)) and M 4354.7 N mm (CALC) (lead 01:30:18Z (6)); the cell quotes +0.301 at V 3888.1 N until the re-map; Mechanical's soak fibre rows negative (SOAK -0.257 at 3750 N; 10A.8); (b1) moments as solved, M x 1.404 / x 1.476 as information (lead 02:01:34Z (B)) DERIVED T5045 **rows-v8 matrix delta:** part: HW-101 (J-FWD); column: J-FWD; old_MS_y_MS_u: [0.65, 0.301]; new_MS_y_MS_u: [0.65, 0.278] DERIVED T5046 **rows-v8 why:** Mechanical's single J-FWD re-map on the FINAL envelope v6 was published under the lead go 2026-10-21T07:37:34Z (CHANGELOG line 254, sha16 c7d52385af9ce594): rows v8 jfwd_in718_rows_v352_v8.json af072cc2e70711ca, with HW-101 5020A Eq. 22 on the R6 basis (V 4079.2 N FE (b1), M 4354.6 N mm CALC). DERIVED T5047	**rows-v8 matrix of record:** v352 · sha256/16 6c694fa7147fb511 DERIVED T720 **CHANGELOG 255 (3):** It supersedes only assembly3d_v352/tables/margin_matrix_v352.md / .json and that script; assembly3d_v352 (MANIFEST a4b920eb96f50053) stands as published. DECIDED *Lead decision, line 255 (3)* T5048	The matrix of record governs; the FEA3D-2 report text is stale. The matrix value is shown (§9.5.3). OPEN: Mechanical / FEA3D align the report wording. DERIVED T5090
C-02 electronics.json v4.1 status: "STAGED" vs promoted (line 266)	**electronics.json v4.1 /status:** DESIGN + SIL + schematic, ~~STAGED for the lead (06-electronics/staging_v2.12/; interfaces/electronics.json v4 stays the published file until the lead promotes this one)~~ PROMOTED 14:45:39Z (v4.1 is the published file); the STAGED status text is superseded until v4.2 **OVR-10**. Firmware v2.12 runs only in SIL (no Teensy HAL): the REV-ELEC-3 R3-01 option (b) seat compensation of the C1 Pc reading on the working basis given at 11:25Z (LEAD 261 (3); the lead confirms or changes it, the fallback and decision (3) at this handback), the TB-state of the LC-TB70 v2 F4 set with PC_SENSOR HIGH, the flight J-FWD indication, the purge FINAL as calibration constants with rule (d), log v13. The ECU v2.5 schematic does NOT show the v2.11 / v2.12 board items (schematic v2.6 NOT drawn); no board has been built and nothing here is hardware-validated or physical approval. Values carry tags; open items are in open_issues. DECIDED *Design value, electronics.json v4.1* T4126 **CHANGELOG 266 (7):** PROMOTED 14:45:39Z: interfaces/electronics.json = staging_v2.12 results/electronics_v4.1.json c862375163393adc, byte for byte (v4 11f7896ad0d0613b stays as staging_v2.11/results/electronics_v4.json). Firmware v2.12 and its SIL are the design baseline under the carve-out of (6). DECIDED *Lead decision, line 266 (7)* T023	**CHANGELOG 266 (7)** (quoted under the sides) DECIDED *Lead decision, line 266 (7)* T023 **CHANGELOG 266 (7), the governing clause:** Until v4.2 this line governs v4.1 where they differ: its STAGED status text is superseded; DECIDED *Lead decision, line 266 (7)* T4137	Line 266 (7) governs: electronics.json v4.1 is the source of record and its internal STAGED status string is stale; it is annotated (OVR-10). OPEN: Electronics restates it at v4.2. DERIVED T5091
C-03 A1 mission-burn records: "300 m comparators until gnc.json v3.2" vs v3.2 published without a 210 m record	**combustion datasheet inputs note:** A1 apogee target 210 m is decided (lines 214, 227 (1)); the A1 records above are the 300 m comparators until gnc.json v3.2 (OI-49). DECIDED *Design value, DATASHEET-INPUTS-COMBUSTION.md v2.4d* T5049 **GNC datasheet inputs 11h:** The coverage set at 210 m (e19, L400, wind sweep) and a 210 m A1 duty-cycle record. B1 governs thermal and erosion sizing. TO-VERIFY T5050 **CHANGELOG 260 (1):** (1) GNC.JSON v3.2 PUBLISHED by the lead (built and stamped 10:56:48Z, promoted 10:57:09Z): interfaces/gnc.json v3.2 bde317782574cbd9 (1,415,700 B) = the candidate 01-systems-gnc/staging_v3.1/rev4b/v32/gnc_v3.2_rev4b.json 43d58c4aff164b86 with the 34 REV-GNC-7 operations of reviews/gnc/rev4b/data/b/rv7_v32_corrections.json f7c0c6051efd5a08, applied independently by the lead and byte-equal to the reviewer trial fea33eacdd8a7d56 without its trial marker (236107a89ca4a240), plus lead folds L1-L4: L1 the status header (published) and the status item (4) sentences replaced by the texts of ops 6 and ... DECIDED *Lead decision, line 260* T5051	**GNC datasheet inputs 11h** (quoted under the sides) TO-VERIFY T5050 **CHANGELOG 213:** Consequences: gnc.json v3.2 carries the trim; the A1 duty cycle shortens (B1 still governs thermal and erosion sizing, no new structural case); the datasheet and ICD-AIRFRAME v2.1 state about 210 m; tank stays 2.00 kg; lighter bleed valves remain the lever if a higher apogee is wanted later. DECIDED *Lead decision, line 213* T5052	OPEN (GNC): the 210 m A1 mission-burn and duty-cycle record is not flown (row 11h). The 300 m records stay, labelled as 300 m comparators; B1 governs the thermal and erosion sizing. No 210 m value is invented here. DERIVED T5092
C-04 TC-02 plug fit: BOM notes "57.20 f7" vs the R2-B select fit	**BOM ENG-TC-02 notes:** plug 57.20 f7 with 2 AS568-226 FKM 90A glands (51.65 h9 groove, 4.85 wide) DECIDED *Design value, BOM_v3.5.2.json v3.5.2* T5053 **CHANGELOG 254 (1):** LEAD RULING R2-B (2026-10-21T07:37:34Z): the ENG-TC-02 plug OD is a SELECT FIT to its own measured TC-01 bore, flag-free band: radial gap g in [0.040, 0.0631] mm, i.e. D_meas - 0.1262 <= OD <= D_meas - 0.080 mm (window 0.0462 mm on the diameter, less 2U, U the expanded (k=2) uncertainty of the bore and OD measurements, TO-DEFINE); bore and OD measured at 20 +/- 1 C and recorded with the paired TC-01 / TC-02 serials; the TC-02 OD is finished last, from a blank with finishing stock, to the measured bore. DECIDED *Lead ruling R2-B (2026-10-21T07:37:34Z), quoted verbatim in Mechanical line 254 (1)* T525 **matrix TC-01 J-FWD cell (drawn fit):** -0.075 / +0.239 CALC T5054	**CHANGELOG 254 (1), R2-B** (quoted under the sides) DECIDED *Lead ruling R2-B (2026-10-21T07:37:34Z), quoted verbatim in Mechanical line 254 (1)* T525	The R2-B select fit governs (line 254); the BOM fit note is superseded and the drawn fit is FOUND NEGATIVE. OPEN: Mechanical restates the BOM note (generated from the part script; no geometry change). See also C-11. DERIVED T5093
C-05 HW-101 action: "after any Pc above 60 bar abs" vs the Q_JFWD replace rule	**BOM ENG-HW-101 notes:** after any Pc above 60 bar abs the line-231 J-FWD inspection (C6 (2) list) DECIDED *Design value, BOM_v3.5.2.json v3.5.2* T5055 **combustion.json z3_jfwd_rulings_v2p4d:** the v1 'any head excursion above 60.0 bar abs: replace all J-FWD screws' rule and the HW-101 '60 bar' action (drawing texts v2 58683aa9a66a65dd; no hardware-data, BOM or geometry edit) DECIDED *Lead decision, per combustion.json v2.4d* T5056	**combustion.json z3_jfwd_rulings_v2p4d (LEAD 258 (3))** (quoted under the sides) DECIDED *Lead decision, per combustion.json v2.4d* T5056 **CHANGELOG 259 (6):** no CAD, hardware-data or BOM edit: the HW-101 'after any Pc above 60 bar abs' and tap-wall v1 strings are superseded by the drawing texts v2. T5057	The Q_JFWD replace rule (258 (3)) governs; line 259 (6) states that the HW-101 string is superseded by the drawing texts v2. OPEN: Mechanical restates the hardware string at the next hardware-data revision. DERIVED T5094
C-06 T4: the T1-T4 rows (DC-D, 7.02 kg) vs the re-sized shortest firing (about 2.0 kg)	**T1-T4 spec T4 row:** | T4, DC-D duty cycle (S230 grain, 3 starts; v3.2: re-sized, line 294) | life and relight data; law check only | 1.023-1.077 | 0.46-1.00 | SIM T5058 **combustion.json t4_duration_note_v2p4c:** CONFIRMED by GNC in gnc.json v3.2 (bde317782574cbd9; duty_cycles.note_n07_rev_gnc_7; published in CHANGELOG 260): no GNC objective needs the DC-D duty as one firing; GNC's static-fire objectives need DC-D segments, each flown as its own segment or firing inside the A = 150 C case limit (REV-ELEC-3 N-07: the DC-D first burn would cross it at about 23.5 s, merged bound [DERIVED]); the A1-like relight objective sits with the shortest T4 firing. T4 is re-sized to that firing (LEAD 261): planning value about 2.0 kg of N2O [ESTIMATE]; the T4 rows of T1-T4 v3.2 lines 61, 143-145 and 171 stay the ... DECIDED *Lead decision, per combustion.json v2.4d* T1553 **CHANGELOG 261 (2):** T4 RE-SIZED (DECIDED): GNC's N-07 answer in gnc.json v3.2 (duty_cycles.note_n07_rev_gnc_7) confirms that no GNC objective needs the DC-D duty as one firing and that GNC's DC-D segments fly as their own segments or firings inside the A = 150 C case limit (REV-ELEC-3 N-07: the DC-D first burn would cross it at about 23.5 s, merged bound [DERIVED]). T4 is re-sized to REV-COMB-4's shortest firing (B1 p99 burn 21.76 s, a coast of about 19 s, one relight of about 4.6 s at about 0.47 throttle): planning value about 2.0 kg of N2O [ESTIMATE] against 7.02 kg. The T4 rows of T1-T4 lines 61, 143-145 and ... DECIDED *Lead decision, line 261 (2)* T1554	**CHANGELOG 261 (2)** (quoted under the sides) DECIDED *Lead decision, line 261 (2)* T1554 **CHANGELOG 264 (5):** N-07 (OI-T4-RESIZE): SHORTER RELIGHT ADOPTED: the T4 relight after the B1 p99 burn (case about 137.8 C at its end [DERIVED]) is cut so that no planned segment crosses A = 150 C, keeping the light, the latency measurement and a steady-Pc segment; Combustion sets the length in the T4 re-run of 261 (2) with Mechanical's soak-back. The coast is not lengthened to make margin (soak-back can raise the case after shutdown and is not modelled). T4 then demonstrates relight latency from a soaked state, not a 4.59 s landing-length relight from the B1-soaked state; bench plan v5 and the datasheet say so. DECIDED *Lead decision, line 264 (5)* T1555	The re-size governs (261 (2), 264 (5)); the published T4 rows (DC-D) are superseded. OPEN (OI-T4-RESIZE): the T4 re-run sets the length. See also C-15. DERIVED T5095
C-07 Bench plan v5 location: electronics.json path vs the hub tree	**electronics.json /bench_plan/file:** 06-electronics/docs/bench_plan_v5.md 360c1d6695cc305c DECIDED *Design value, electronics.json v4.1* T5059 **file in the hub:** 06-electronics/staging_v2.12/docs/bench_plan_v5.md DERIVED T5060	**file in the hub** (quoted under the sides) DERIVED T5060	The hub tree governs; the path is printed as it is in the hub (§15.2). OPEN: Electronics states the hub path. DERIVED T5096
C-08 CF-4 baseline: "mechanical.json v2.2 not yet published" vs mechanical.json v2.4 published	**combustion.json /cf4_gate/baseline_v2p4b:** re-baselined on the frozen v3.5.1 bleed path (LEAD 194 (8)); bleed-path part numbers and lengths are taken from Mechanical's v3.5.1 freeze (mechanical.json v2.2 not yet published: MISSING in the hub) DECIDED *Design value, combustion.json v2.4d* T5061 **mechanical.json version:** v2.4 DECIDED *Design value, mechanical.json v2.4* T5062	**mechanical.json version** (quoted under the sides) DECIDED *Design value, mechanical.json v2.4* T5062	mechanical.json v2.4 is current; the CF-4 text predates it (the bleed-path basis stays the frozen v3.5.1 path as written). OPEN: Combustion restates it at the next text revision. DERIVED T5097

ID	Sides (quoted)	Governing source	Status
C-09 RETIRED BOM mass_sum_kg vs the sum of its lines	**BOM /mass_sum_kg: 5.0466** kg DERIVED T412 **generator sum of the BOM lines: 5.047** kg DERIVED T409	–	RETIRED: the two agree at this build (difference below 0.0005 kg; §4.4), so there is no contradiction. The ID is kept, and not reused, so that the IDs of the rev1-draft stay valid (REV-DS-1 D1-04). DERIVED T5098
C-10 P_torch transducer: electronics.json v4.1 (TE M3234-000005-070BG) vs lead 264 (4) (C1-family Kulite XTL-190SM-2000A)	**electronics.json v4.1 /sensors/pressure_transducers/selection:** PC: Kulite XTL-190SM-2000A (C1, seat-mounted on the head, J601); P_FEED: Kulite XTL-190SM-2000A (C1, J602); P_MID: Kulite XTL-190SM-2000A (C1, J511); P_TORCH: TE M3234-000005-070BG (J901); P_TANK: TE M3234-000005-170BG (J603); stand_static_reference: TE M3234-000005-070BG (B-PC-1, BOM 199) DECIDED *Design value, electronics.json v4.1* T4156 **CHANGELOG 264 (4):** P_torch (J901, on the torch body, not in the head; torch_port_bl08_v352_v1.json 20220647d32cbbbc) moves from the TE M3234-000005-070BG (proof 141.0 bar abs) to a C1-family Kulite XTL-190SM-2000A (proof 206.8 bar abs [DATASHEET]) against the 166.14 bar abs requirement (1.1 x FA1S 151.04 bar abs) [DERIVED]; it is the same part as P_mid (ENG-BL-08) and C1. DECIDED *Lead decision, line 264 (4)* T802	**CHANGELOG 264 (4)** (quoted under the sides) DECIDED *Lead decision, line 264 (4)* T802 **CHANGELOG 266 (7), E4-05 (7):** the 264 (1)-(8) folds such as P_TORCH XTL-190SM-2000A DECIDED *Lead decision, line 266 (7)* T4157	Line 264 (4) governs: the C1-family Kulite XTL-190SM-2000A, its proof [DATASHEET] against the required proof [DERIVED] (§2.3). The electronics.json v4.1 P_torch entry is superseded for the design (line 266 (7) applies the 264 fold, E4-05 (7); the file keeps the old entry until v4.2). OPEN (OI-PTORCH-PROOF): the threshold re-base (E4-N04, queued with v2.13, 266 (8)). See also C-14. DERIVED T5099
C-11 TC-01 J-FWD select-fit band: the line-240 band [0.040, 0.0761] mm quoted in the matrix and the FEA3D-2 report vs R2-B	**margin matrix negative-cell table, TC-01 J-FWD closure (the line-240 band):** flags (0 to +0.05): FA1S (ii) hole bearing, CALC V 3888.1 N, L 2.365 (governing, 240 (5)) +0.005/+0.239 [CALC]; SOAK52 IN718 under-head band Neuber (local strain) [ASSUMED (sigma_el x A_ANN 47.12 / A_supp; edge loading at the tangential overhang not represented)] at 3750 N +0.039/n/a [CALC / DERIVED; ASSUMED scaling]. R0.2 bore-side break adopted (240 (5)); route (i) adopted (HW-170 disc spring under the head, lead 16:53Z R2; rev-5 CAD check accepted 17:55:55Z); J-F1 maximum capped at 3750 N (lead 17:56:12Z). (b1) at C1 is negative (SOAK52 -0.030), so the 240 (10) seat-line P+Q select-fit ... T524 **FEA3D-2 report §13 item 1, closure (the line-240 band):** the band inside [0.040, 0.0761] mm that closes it, or the seat-line P+Q stays a found negative in the datasheet; the HW-170 disc-spring route (i) and the preload-retention audit (240 (10)) DERIVED T4042 **CHANGELOG 254 (1), LEAD RULING R2-B:** LEAD RULING R2-B (2026-10-21T07:37:34Z): the ENG-TC-02 plug OD is a SELECT FIT to its own measured TC-01 bore, flag-free band: radial gap g in [0.040, 0.0631] mm, i.e. D_meas - 0.1262 <= OD <= D_meas - 0.080 mm on the diameter, less 2U, U the expanded (k=2) uncertainty of the bore and OD measurements, TO-DEFINE); bore and OD measured at 20 +/- 1 C and recorded with the paired TC-01 / TC-02 serials; the TC-02 OD is finished last, from a blank with finishing stock, to the measured bore. DECIDED *Lead ruling R2-B (2026-10-21T07:37:34Z), quoted verbatim in Mechanical line 254 (1)* T525	**CHANGELOG 254 (1), R2-B** (quoted under the sides) DECIDED *Lead ruling R2-B (2026-10-21T07:37:34Z), quoted verbatim in Mechanical line 254 (1)* T525 **band file of record:** v1 · sha256/16 1ba3147b42edfeed DERIVED T528	R2-B governs (254 (1)): the select-fit band is the R2-B band. Band re-size done (band v1 1ba3147b42edfeed); the seat-line P+Q stays a found negative at the drawn fit (§9.3, §9.10 item 1). See also C-04. DERIVED T5100
C-12 Purge both-open cap: 2.55 s (v2.11, electronics.json v4) vs the 256 (2) FINAL 2.43 s flow / 2.39 s cap	**electronics.json v4 /limits_v2p11/4 (v2.11 cap) [kept v4 file]:** 2.55 s DECIDED *Lead decision, line 203 (3)* T5063 **CHANGELOG 203 (3):** OI-PURGE-CAP: the ECU both-open cap is set to <= 2.55 s (Combustion basis 75.9 g <= 76 g). Electronics shows the 2.5 s scheduled flow is not truncated under the valve timing tolerances; if it would be, the lead re-opens the 76 g allowance instead. The flow-basis difference (SIL 32.0 vs Combustion 29.8 g/s, +7 %) goes to Combustion v2.4b; until CF-4/B-BLEED-1 measure it, GNC carries the purge at the higher basis (about 80 g for the 2.5 s schedule) as a sensitivity. gnc.json purge_bound wording is corrected in v3.2. DECIDED *Lead decision, line 203 (3)* T4450 **CHANGELOG 256 (2):** PURGE, 253 (7)(ii), now FINAL: the 2.43 s scheduled flow with the 2.39 s cap is ADOPTED [DERIVED by REV-COMB-4 on the author's SIM model: needed flow 2.2 / 1.9 / 1.8 s at 283.15 / 289.5 / 297.15 K unlined; vented N2O <= 66.7 / 70.8 / 75.1 g at the ASSUMED Cd 0.70], with three conditions: (a) wherever 1.9 s is quoted as the needed flow, 2.2 s at the 283.15 K cold corner is stated with it (REPORT 2.13, CF-4 line 35, Electronics); (b) CF-4 / B-BLEED-1 shows a measured Cd x A of the two-tap path >= 0.91 x the Cd 0.70 value (>= 0.95 x with the FD-01 element bound), else the flow is re-sized; (c) ... DECIDED *Lead decision, line 256 (2)* T4445	**CHANGELOG 256 (2)** (quoted under the sides) DECIDED *Lead decision, line 256 (2)* T4445 **CHANGELOG 257 (2):** PURGE CONDITION (d) ADDED to 256 (2), on REV-GNC-7 interim 2 (MAJOR, 09:25:51Z): the 80.5 g of 256 (2) is the 289.5 K figure; at 297.15 K the fixed 2.43 / 2.39 s schedule at Cd 0.80 vents about 85.5 g (85.8 g at the cap) [ESTIMATE], and GNC's line-208 A1 reserve test against the v3d reference passes 76 g (UB95 0.854 pp) and 80.5 g (0.972 pp, margin 0.028 pp) and first fails at 82.1 g [DERIVED from SIM; ESTIMATE above 80 g]. DECIDED: after CF-4 / B-BLEED-1, kPurgeFlow_s and the both-open cap are re-set from the measured Cd x A by the measured-time form of purge_basis_line216_v2p4b, with the ... DECIDED *Lead decision, line 257 (2)* T4448 **CHANGELOG 258 (2):** PURGE: condition (d) of 257 (2) holds (RV7-04, review section 5 (ii): vent at the cap <= 75.99 g for Cd x A ratio r 1.0-2.0; 283.15 K margin 0.178-0.23 s for r 1.0-1.3, 0.10-0.18 s with FD-01 [DERIVED; the 1/(Cd x A) need scaling ASSUMED]). DECIDED: the two REV-GNC-7 additions are adopted as (d1) the re-set uses the upper 95 % bound of the measured N2O Cd x A (at r 1.1326 a ratio measured 7.9 % low gives a true vent of 81.99 g, just under the 82.1 g first fail) and (d2) CF-4 shows the 283.15 K need at the re-set flow; below r 1 condition (b) governs. Routed to combustion v2.4d and Electronics ... DECIDED *Lead decision, line 258 (2)* T4449 **implementation record:** PROMOTED 14:45:39Z: interfaces/electronics.json = staging_v2.12 results/electronics_v4.1.json c862375163393adc, byte for byte (v4 11f7896ad0d0613b stays as staging_v2.11/results/electronics_v4.json). Firmware v2.12 and its SIL are the design baseline under the carve-out of (6). DECIDED *Lead decision, line 266 (7)* T023 **implementation record:** Accepted now: purge FINAL 2.43 / 2.39 s with rule (d), (d1), (d2); TB-state; JFWD_LS; kPtorchRezeroMax_Pa 0.60 bar (re-base OPEN, 264 (4)); log v13; kIgnBaseCheck_Pa 0.76 / 0.46 bar; the U330 band rows; R4-02, R4-09, R4-13; F-13, F-14, B-MV-1; the T1-T4 planning rule. DECIDED *Lead decision, line 266 (6)* T027	256 (2) governs (2.43 s flow, 2.39 s cap, conditions (a)-(d), (d1), (d2) of 256 (2) / 257 (2) / 258 (2)); implemented in v4.1: ACCEPTED (REV-ELEC-4, line 266); the measured-time form is open after CF-4 / B-BLEED-1. DERIVED T5101

ID	Sides (quoted)	Governing source	Status
C-13 C1 re-zero and BASE constants: v2.11 klgnBaseCheck_Pa 70000 Pa / kPcRezeroMax_Pa 30000 Pa (not accepted, 253 (5)) vs 264 (2)	**electronics.json v4 /light_rule/klgnBaseCheck_Pa (v2.11) [kept v4 file]:** 70000 Pa DECIDED *Design value, Electronics v2.11 (electronics.json v4)* T5064 **electronics.json v4 coast re-zero max, kPcRezeroMax_Pa (v2.11) [kept v4 file]:** 30000 Pa DECIDED *Design value, Electronics v2.11 (electronics.json v4)* T5065 **CHANGELOG 264 (2):** kPcRezeroMax_Pa stays 0.30 bar on the compensated residual; klgnBaseCheck_Pa 0.76 / 0.46 bar (default / B-PC-1-shown build) DECIDED *Lead decision, line 264 (2)* T4329	**CHANGELOG 253 (5):** EXCEPT the C1 re-zero and BASE constants (kPcRezeroMax_Pa, klgnBaseCheck_Pa), which are not accepted (R3-01, open MAJOR) DECIDED *Lead decision, line 253 (5)* T4330 **CHANGELOG 264 (2)** (quoted under the sides) DECIDED *Lead decision, line 264 (2)* T4329 **implementation record:** PROMOTED 14:45:39Z: interfaces/electronics.json = staging_v2.12 results/electronics_v4.1.json c862375163393adc, byte for byte (v4 11f7896ad0d0613b stays as staging_v2.11/results/electronics_v4.json). Firmware v2.12 and its SIL are the design baseline under the carve-out of (6). DECIDED *Lead decision, line 266 (7)* T023 **implementation record:** Accepted now: purge FINAL 2.43 / 2.39 s with rule (d), (d1), (d2); TB-state; JFWD_LS; kPtorchRezeroMax_Pa 0.60 bar (re-base OPEN, 264 (4)); log v13; klgnBaseCheck_Pa 0.76 / 0.46 bar; the U330 band rows; R4-02, R4-09, R4-13; F-13, F-14, B-MV-1; the T1-T4 planning rule. DECIDED *Lead decision, line 266 (6)* T027 **implementation record:** Suspended until v2.13, and marked so wherever shown: the option (b) firmware rows, residuals and S13T outcomes; the klgnBaseDrift 0.06 bar basis for any relight; the evidence (not the decision) of 264 (3), to be re-shown on v2.13; the A1 pad seat row; the stand relight after B1; the B-PC-1 / B-TC-1 TC4 rows. DECIDED *Lead decision, line 266 (6)* T026	253 (5) did not accept the v2.11 values; 264 (2) governs (kPcRezeroMax_Pa 0.30 bar on the compensated residual, klgnBaseCheck_Pa 0.76 / 0.46 bar default / B-PC-1-shown build); v4.1 implementation: klgnBaseCheck_Pa 0.76 / 0.46 bar ACCEPTED (REV-ELEC-4, line 266); the option (b) rows and residuals SUSPENDED UNTIL v2.13 (266 (6)). DERIVED T5102
C-14 P_torch on the combustion side: the 070BG / "reselected" transducer texts vs lead 264 (4)	**combustion datasheet inputs, row "Pc transducer proof" (condition / basis):** path period > 1 ms not acceptable for the dynamic channel; the P_torch 070BG over-ranges during FA-1 (its proof and recovery are Electronics v2.12 items) DATASHEET *Lead decision, line 249 (4)* T801 **combustion.json /igniter/sequence ARM precondition 4 (P_torch clause):** Pc transducer: flight unit RESELECTED by Electronics (LEAD 205): miniature seat-mounted, body within 12 mm x 30 mm, front dead volume <= 20 mm3, proof >= 166.14 bar abs (1.1 x FA1S 151.04; LEAD 249 (4)); C1 Kulite XTL-190SM-2000A 206.8 bar abs (DATASHEET 1.5x rule) meets it; OI-41 DECIDED by LEAD 230 (2), range covering the 45 bar MEOP with the 30 bar abort resolvable, >= 120 C; over-range bit and above-full-scale flag (LEAD 203 (1)(a)); P_torch 0-70 bar class (ENG-09; it sees the FA-1 passage pressure, design case FA1S 151.04 bar abs: its proof (>= 166.14 bar abs, or isolation) and its ... DERIVED *Lead decision, line 153* T1436 **CHANGELOG 264 (4):** P_torch (J901, on the torch body, not in the head; torch_port_bl08_v352_v1.json 20220647d32cbbbc) moves from the TE M3234-000005-070BG (proof 141.0 bar abs) to a C1-family Kulite XTL-190SM-2000A (proof 206.8 bar abs [DATASHEET]) against the 166.14 bar abs requirement (1.1 x FA1S 151.04 bar abs) [DERIVED]; it is the same part as P_mid (ENG-BL-08) and C1. DECIDED *Lead decision, line 264 (4)* T802	**CHANGELOG 264 (4)** (quoted under the sides) DECIDED *Lead decision, line 264 (4)* T802	264 (4) governs: P_torch is the C1-family Kulite XTL-190SM-2000A. The re-base of the torch thresholds is OPEN (OI-PTORCH-PROOF; §12.4); it needs compensation or a photodiode-guarded gate and the torch local heat first (E4-N04, 266 (8)). See also C-10. DERIVED T5103
C-15 T4 relight: about 4.6 s after the B1 burn (261 (2)) vs the shorter relight of 264 (5)	**CHANGELOG 261 (2):** T4 RE-SIZED (DECIDED): GNC's N-07 answer in gnc.json v3.2 (duty_cycles.note_n07_rev_gnc_7) confirms that no GNC objective needs the DC-D duty as one firing and that GNC's DC-D segments fly as their own segments or firings inside the A = 150 C case limit (REV-ELEC-3 N-07: the DC-D first burn would cross it at about 23.5 s, merged bound [DERIVED]). T4 is re-sized to REV-COMB-4's shortest firing (B1 p99 burn 21.76 s, a coast of about 19 s, one relight of about 4.6 s at about 0.47 throttle): planning value about 2.0 kg of N2O [ESTIMATE] against 7.02 kg. The T4 rows of T1-T4 lines 61, 143-145 and ... DECIDED *Lead decision, line 261 (2)* T1554 **CHANGELOG 264 (5):** N-07 (OI-T4-RESIZE): SHORTER RELIGHT ADOPTED: the T4 relight after the B1 p99 burn (case about 137.8 C at its end [DERIVED]) is cut so that no planned segment crosses A = 150 C, keeping the light, the latency measurement and a steady-Pc segment; Combustion sets the length in the T4 re-run of 261 (2) with Mechanical's soak-back. The coast is not lengthened to make margin (soak-back can raise the case after shutdown and is not modelled). T4 then demonstrates relight latency from a soaked state, not a 4.59 s landing-length relight from the B1-soaked state; bench plan v5 and the datasheet say so. DECIDED *Lead decision, line 264 (5)* T1555 **CHANGELOG 264 (5), what T4 demonstrates:** T4 then demonstrates relight latency from a soaked state, not a 4.59 s landing-length relight from the B1-soaked state; bench plan v5 and the datasheet say so. DECIDED *Lead decision, line 264 (5)* T1556	**CHANGELOG 264 (5)** (quoted under the sides) DECIDED *Lead decision, line 264 (5)* T1555 **CHANGELOG 266 (7), E4-06 (2):** T4 restated as a relight-latency demonstration from a soaked state, its length from Combustion's T4 re-run, waiting for v2.13 DECIDED *Lead decision, line 266 (7)* T5028	264 (5) governs: the shorter relight is adopted and T4 demonstrates relight latency from a soaked state. The length is OPEN: Combustion sets it in the T4 re-run with Mechanical's soak-back (OI-T4-RESIZE); T4 is a stand relight and waits for v2.13 (266 (6)). See also C-06. DERIVED T5104
C-16 OI-A1-BASE-AMB: listed open in electronics.json vs closed by 264 (3)	**electronics.json /open_issues OI-A1-BASE-AMB, issue:** S13T with the drift model: option (b) 0 BASE_AMB in the A1 sweep (both builds); fallback BASE_AMB → PC_SENSOR at k = 3 (default build) and at k >= 1.5 on the - sign (B-PC-1-shown build) (r3_01_c1_drift.decision_3); LEAD decision (3) at this handback TO-VERIFY T4525 **electronics.json OI-A1-BASE-AMB, closure:** accept the outcome or rule the alternative (keep the measured P0 below ambient; not implemented) DECIDED *Design value, electronics.json v4.1* T4524 **CHANGELOG 264 (3):** DECISION (3), A1 BASE_AMB: ADOPTED as built: a BASE_AMB at the A1 relight aborts PC_SENSOR (fail-safe, no false E2, relight lost: the lost-relight outcome of gnc.json v3.2, drogue descent and vehicle loss); with option (b) it occurs in 0 of the S13T A1 runs (T0 0 / 20 / 40 C, k 1-3, both signs, coefficient +/-10 %, TC +/-0.3 K) [SIM]. DECIDED *Lead decision, line 264 (3)* T4485	**CHANGELOG 264 (3)** (quoted under the sides) DECIDED *Lead decision, line 264 (3)* T4485 **CHANGELOG 266 (6):** Suspended until v2.13, and marked so wherever shown: the option (b) firmware rows, residuals and S13T outcomes; the klgnBaseDrift 0.06 bar basis for any relight; the evidence (not the decision) of 264 (3), to be re-shown on v2.13; the A1 pad seat row; the stand relight after B1; the B-PC-1 / B-TC-1 TC4 rows. DECIDED *Lead decision, line 266 (6)* T026	Closed: 264 (3) adopts the A1 BASE_AMB outcome as built; the item is listed under the closed group in §10.9. Only the residual that a lead line names stays open: the A1 pad no-go rule of 264 (2) (NEW RULE, §12.4). The evidence (not the decision) of 264 (3) is SUSPENDED UNTIL v2.13 and is re-shown on v2.13 (266 (6)). DERIVED T5105

14 Unknowns register

Every value the sources mark MISSING, TO-DEFINE, PENDING or TO-VERIFY, plus every key the generator looked for and did not find. They are listed so that none is mistaken for a design value.

MISSING 23 · PENDING 0 · TO-DEFINE 0 · TO-VERIFY 59

Trace	Class	Source key	Value or text
T488	MISSING	–: mass of the items listed as not modelled in mechanical.json (v2.1 record)	MISSING
T264	MISSING	BOM: /provision_lines/0/unit_mass_kg	MISSING
T266	MISSING	BOM: /provision_lines/1/unit_mass_kg	MISSING
T268	MISSING	BOM: /contingency_lines/0/unit_mass_kg	TO-DEFINE
T270	MISSING	BOM: /gse_tool_lines/0/unit_mass_kg	MISSING
T272	MISSING	BOM: /gse_tool_lines/1/unit_mass_kg	MISSING
T322	MISSING	BOM: /hardware/lines/24/unit_mass_kg	MISSING
T362	MISSING	BOM: /hardware/lines/44/unit_mass_kg	MISSING
T364	MISSING	BOM: /hardware/lines/45/unit_mass_kg	MISSING
T366	MISSING	BOM: /hardware/lines/46/unit_mass_kg	MISSING
T368	MISSING	BOM: /hardware/lines/47/unit_mass_kg	MISSING
T372	MISSING	BOM: /hardware/lines/49/unit_mass_kg	MISSING
T374	MISSING	BOM: /hardware/lines/50/unit_mass_kg	MISSING
T376	MISSING	BOM: /hardware/lines/51/unit_mass_kg	MISSING
T378	MISSING	BOM: /hardware/lines/52/unit_mass_kg	MISSING
T382	MISSING	BOM: /hardware/lines/54/unit_mass_kg	MISSING
T413	MISSING	BOM: /lines/*/unit_price	TO-QUOTE (not summed)
T1259	MISSING	CMB: /interlock_guard_band/v/thresholds	T_tank <= 24 C and P_tank <= 55.28 bar abs, each lowered by the total channel uncertainty (Electronics states it)
T1327	MISSING	CMB: /igniter/sequence/v/start/6/t_s	MISSING
T4698	MISSING	DSIG: line 36 row "11e" col 2	A notch on the measured bending mode (RESEARCH PROPOSAL).
T4704	MISSING	DSIG: line 39 row "11h" col 2	The coverage set at 210 m (e19, L400, wind sweep) and a 210 m A1 duty-cycle record. B1 governs thermal and erosion sizing.
T487	MISSING	MEC: /fixed_module/not_modelled	thrust-ring radial screws (16 x M5); isolation valve + actuator (mass UNKNOWN: Electronics/Combustion); tee, filter (<= 15 um, OI-7), relief, fill/vent; hose and lines fixed shares (~0.07 kg ASSUMED); tank
T1028	MISSING	MP: /hardware_carried/lines_without_mass	HW-128; HW-158; HW-159; HW-160; HW-161; HW-163; HW-164; HW-165; HW-166; HW-168
T094	TO-VERIFY	BOM: /lines/22/unit_mass_kg	0.0379
T216	TO-VERIFY	BOM: /lines/83/unit_mass_kg	0.03
T1211	TO-VERIFY	BOM: /provision_mass_estimate_kg/valves	TO-VERIFY (datasheet)
T4830	TO-VERIFY	CL: L254[sentence containing "J-F1b method TO-DEFINE"]	Prescale TO-VERIFY, J-F1b method TO-DEFINE, LOCTITE 243 range TO-VERIFY (founder / test owner);
T4832	TO-VERIFY	CL: L254[sentence containing "LOCTITE 243 range TO-VERIFY"]	Prescale TO-VERIFY, J-F1b method TO-DEFINE, LOCTITE 243 range TO-VERIFY (founder / test owner);
T4944	TO-VERIFY	CL: L246(2)[from "C4 rewritten:" to "never re-torqued;"]	C4 rewritten: J-F1 gate (lot torque, spring items, Prescale footprint item TO-VERIFY), hole pins GO 5.50 / NO-GO 5.62 and tap gauging, head shoulder 2.00 +/-0.02 on the gap gauge (the v2 'until seated' corrected: TC-02 does not seat axially, R4) ...
T1496	TO-VERIFY	CMB: /z3_jfwd_rulings_v2p4d/v/open_before_flight_acceptance/0	the flight consequence of a flagged first light (relight on a used joint; line 231 covers one Z3 event per assembly) (LEAD)
T1497	TO-VERIFY	CMB: /z3_jfwd_rulings_v2p4d/v/open_before_flight_acceptance/1	the damped ESP variant (structural damping MISSING)
T1498	TO-VERIFY	CMB: /z3_jfwd_rulings_v2p4d/v/open_before_flight_acceptance/2	the 0.2-0.5 ms J-FWD band (ASSUMED; TO-VERIFY on a v3.5.2 modal)
T1499	TO-VERIFY	CMB: /z3_jfwd_rulings_v2p4d/v/open_before_flight_acceptance/3	a shutdown from a case hotter than the rows' soak basis (F4 observation; Mechanical, no limit derived)
T1541	TO-VERIFY	CMB: /stability/hybrid_LF_note	classical hybrid low-frequency (thermal-lag/boundary-layer) instability 0.3-2 Hz band is identified but its amplitude is NOT predicted; static fire T2/T3 with the 1 kHz control channel for the 0.3-2 Hz band and the stand dynamic channel (>= 10-20 kHz ...
T4872	TO-VERIFY	CMB: /z3_jfwd_rulings_v2p4d/v/jfwd_replace_rule/measured_traces	on measured stand traces a noise-driven trigger still means the inspection until a noise-floor rule is derived from a no-event B-DPC-1 record and reviewed (TO-VERIFY)
T4690	TO-VERIFY	DSIG: line 32 row "11a" col 2	D = 0.25 / 0.5 / 0.75 / 1.0 s, both states, N = 1000 per main base (about 16,000 flights), plus a bound on the drogue deploy-to-inflation delay. Needed before the 210 m relight P is used for flight acceptance; not needed for v3.2 or the datasheet.
T4694	TO-VERIFY	DSIG: line 34 row "11c" col 2	Measure K_s with its uncertainty, J at the servo output shaft, the reflected-inertia convention, and the rocking modes at the poses. K_s is frozen only on an ENG-07d PASS.
T4696	TO-VERIFY	DSIG: line 35 row "11d" col 2	The full and landing modes, measured or from the airframe FE. airframe_first_bending_min_Hz 10 Hz is NOT CONSISTENT with ENG-07d at the window with one fixed notch.
T4700	TO-VERIFY	DSIG: line 37 row "11f" col 2	Electronics v2.11 S17 CONFIRMED pcf_tmax 0.8 s for W10 < 6 m/s with conditions: the ECU R2 hold, the mode-3 flush, the photodiode main-chamber view (or the reference re-taken), the FC Pc branch off or moved with margin (OI-FC-PC-BRANCH), FC frame noise <= ...
T4702	TO-VERIFY	DSIG: line 38 row "11g" col 2	Its B1, thermal and erosion check (A1 engine-on p99 8.92 / 8.71 s, max 9.24 s, > 8.258 s).
T4710	TO-VERIFY	DSIG: line 42 row "11k" col 2	Recompute with the exact linkage kinematics (GNC extrapolation 97.8-111.2 deg/s).
T4712	TO-VERIFY	DSIG: line 43 row "11l" col 2	The final cap after T1-T4 (regression law measured). B1 is not re-flown on massprops v3.5.1 (row 2).
T4949	TO-VERIFY	DSIG: §* row c0="11a" col 1 (line 32)	Open item: 210 m gate-delay arm
T5050	TO-VERIFY	DSIG: §* row c0="11h" col 2 (line 39)	The coverage set at 210 m (e19, L400, wind sweep) and a 210 m A1 duty-cycle record. B1 governs thermal and erosion sizing.
T4132	TO-VERIFY	ELE: /firmware/target	PJRC Teensy 4.1 (i.MX RT1062). HAL NOT written (RE-06; LEAD 166 (9): Electronics open item, required before B-PC-1). The core runs in SIL and test_core only
T4237	TO-VERIFY	ELE: /start_monitors/A7_flight_blank/conditions/2	n: 3; condition: the FC owner confirms no decision uses the A7 high flag (d6 b7 via PC_SENSOR) before E2 + 0.45 s; status: OPEN: FC owner (requests)
T4404	TO-VERIFY	ELE: /abort_matrix/causes/25/thresholds/kBayArmMax_K	318.15
T4428	TO-VERIFY	ELE: /abort_matrix/causes/29/thresholds/kTankColdPrepressDrop_K	0.23
T4495	TO-VERIFY	ELE: /open_issues/0/issue	Teensy HAL not written (RE-06); TLM_SLOW undefined; 96 ARM -Wdouble-promotion warnings (v2.10: 93); objects only, not linked
T4500	TO-VERIFY	ELE: /open_issues/1/issue	RE-09: +12V_ARM and the ARM bus over rating in the purge and E0 overlaps; the E0 row CANNOT PASS as designed (OI-E0-SUPPLY, bench plan v4 B-PWR-2); time-resolved budget, sequencing and peak-and-hold not done; v2.11 adds about 15 mA of C1 bridge current on ...
T4505	TO-VERIFY	ELE: /open_issues/2/issue	bleed valve not selected (LEAD 206 (4) search continues); peak-and-hold not drawn; valve times ASSUMED (30 / 80 ms) in the cap, the masses and the timer margins
T4510	TO-VERIFY	ELE: /open_issues/3/issue	FINAL: 2.43 s flow / 2.39 s cap as calibration constants, rule (d) (d1) (d2) (purge_final_v2p12); residual: CF-4 / B-BLEED-1 re-set
T4515	TO-VERIFY	ELE: /open_issues/4/issue	stuck-high purge-window chain and the both-open timer are invisible to the ECU (no free pin); a pre-flight check with PURGE_EN low is not in firmware v2.11
T4520	TO-VERIFY	ELE: /open_issues/5/issue	S12 (SIM): both bleed valves stuck open (the 203 (4) label is corrected in head_end_bleed_provision.oi_bleed_detect) are not detected on the stand and are flagged 5.05 s after injection in the flight first burn (P_mid below the U570 band on the decided ...
T4525	TO-VERIFY	ELE: /open_issues/6/issue	S13T with the drift model: option (b) 0 BASE_AMB in the A1 sweep (both builds); fallback BASE_AMB → PC_SENSOR at k = 3 (default build) and at k >= 1.5 on the - sign (B-PC-1-shown build) (r3_01_c1_drift.decision_3); LEAD decision (3) at this handback
T4532	TO-VERIFY	ELE: /open_issues/7/issue	S9 +/-1.034 bar TEB cases fail as in v2.9 / v2.10 (the v2.11 run uses the C1 chain with the same injected biases)
T4537	TO-VERIFY	ELE: /open_issues/8/issue	cold side CLOSED by LEAD 216 (7) (ARM margin 0.42 K, PREPRESS safe exit; S5: TANK_COLD 26, safe exits 0). Open: the uncalibrated P_tank hot interlock needs the tank <= about 19.8 C; wall-vs-bulk offset MISSING; countdown margin above 15 C MISSING
T4542	TO-VERIFY	ELE: /open_issues/9/issue	E2 - light (SIL): p50 12 / max 13 ms in S5, 12 / 13 ms in S15 (flight), 12 / 13 ms (stand); A1 relight 13-20 ms over the S13 B / C / D lit cases, 18 ms at the offset corners within the 0.3 bar B-PC-1 residual, 20 ms at the -0.6 bar corners: the IGN-1 ...
T4547	TO-VERIFY	ELE: /open_issues/10/issue	no layout; pin-function check, footprints, derating table not done
T4552	TO-VERIFY	ELE: /open_issues/11/issue	schematic v2.6 NOT drawn: three C1 bridge channels, U330 / U570 values on C1, the ride-through C330B variant, the both-open timer U576-U579, the C1 harness breaks; ERC and the netlist cross-check cover v2.5 only (ecu.schematic.v2p11_board_items_not_drawn)
T4557	TO-VERIFY	ELE: /open_issues/12/issue	S6 gimbal_jam_moving_cmd: small-amplitude jam undetected (RE-24 test-based, LEAD 203 (6)); the abort is not inhibited near touchdown (LEAD 227 (7))
T4562	TO-VERIFY	ELE: /open_issues/13/issue	S6 throat block: plant Pc 53.42 bar abs, 33 ms above MEOP (R2-15: not 72 ms); inside LC-TB70 (S6C)
T4567	TO-VERIFY	ELE: /open_issues/14/issue	FC: PURGE_EN discrete; R2 v2 at W 0.8 s with the mode-3 flush at the Pc-invalid onset (227 (6)), the reference choice and the frame-noise budget (<= 2.51 N rms iid at 0.8 s); DROGUE reports with the re-send rule (<= 10 re-sends, 0.2 s) and the DROGUE_ECHO ...
T4572	TO-VERIFY	ELE: /open_issues/15/issue	ECU side in SIL: lift-commanded hold at 0.21776 mm, backstop E2 + 850 ms, per-tick Pc validity (S17 v2.11: 0 dead releases with the hold and the flush at either window, W10 < 6 m/s, photodiode view). Open: the FC side; GNC R2 v2 re-run with the flush / ...
T4577	TO-VERIFY	ELE: /open_issues/16/issue	C1 chain (v2.11): 0 of 896 FA-1 grid cases reach the over-range or the above-FS flag; as built the A1 relight aborts 374-379 of 896 (filtered 30 bar / 5 ms and the comparator); the ride-through option (default OFF) aborts 0 and S6R aborts 105 of 105 required ...

Trace	Class	Source key	Value or text
T4582	TO-VERIFY	ELE: /open_issues/17/issue	A1: no over-thrust detector after an in-flight Pc loss (fmea_rows_a1)
T4587	TO-VERIFY	ELE: /open_issues/18/issue	S15 v2.11 stand: 4 of 230 lit plants end NO_IGNITION (light later than the stand W); flight 0 of 230; fail-safe by design (R2-19)
T4592	TO-VERIFY	ELE: /open_issues/19/issue	the ECU guard is DONE (LEAD 203 (7): flight build with the bleed fitted refuses a RELIGHT in COAST, UNPURGED 16). Remaining: (a) with an UNPRIMED / UNKNOWN line in flight with the bleed fitted there is no relight path (PURGE needs the direct path: refused ...
T4597	TO-VERIFY	ELE: /open_issues/20/issue	gen_tables.py v3.3 cannot parse the combustion.json v2.4b igniter sequence; engine_params.h / plant_tables.h / gnc_tables.h / valve_map_table.h stay generated from combustion.json v2.3 (HISTORICAL); v2.3 → v2.4b leaf diff of the read keys: 0 numeric changes ...
T4602	TO-VERIFY	ELE: /open_issues/21/issue	lit F_min 33.14 N (3 of 72 Combustion corners never reach 37.16 N at 0.21776 mm, LEAD 230 (5)): S17 v2.11 with the lit thrust capped there: no lit run released at either window with the flush (relight lost, drogue kept); without the flush, reference none ...
T4607	TO-VERIFY	ELE: /open_issues/22/issue	the dead-side result with reference none needs the photodiode main-chamber view (FLAMEOUT_OPT armed): photodiode-blind families with the flush release 20 / 42 (normal_noopt) and 42 / 98 (starved040_noopt) of 1000 dead runs at 0.5 / 0.8 s; with the reference ...
T4612	TO-VERIFY	ELE: /open_issues/23/issue	with the FC Pc branch (release on Pc valid and filtered Pc >= 3.0 bar abs) every normal-family run is released at the E2 frame: Pc_f at E2 3.04 bar abs, 0.04 bar above the branch threshold, so R2 never decides those runs ...
T4617	TO-VERIFY	ELE: /open_issues/24/issue	a HIGH over-range glitch later in the burn (E2 + 350 ms) is also seen by the filtered channel and the U330 tap: as built HW_OVERPRESSURE at every command, the ride-through build PC_OVERPRESSURE / HW_OVERPRESSURE except at 200 N (fail-safe, relight lost); the ...
T4622	TO-VERIFY	ELE: /open_issues/25/issue	C1 seat-face peaks 104.5-125.1 bar abs (SIM on the ASSUMED 1-D lumped passage model) against the 102 bar abs T <= 1 ms ultimate of line 205; LEAD 249 (4) restated the proof requirement (>= 166 bar abs; C1 206.8 bar abs) and the peaks lie within FA1S 151.04
T4627	TO-VERIFY	ELE: /open_issues/26/issue	10 S6R runs have their INPUT outside LC-TB70: the Combustion v2.4a FA-1 replay max-face case peaks 88.02 bar abs > Z3 75.519 bar abs; 46 of 896 grid cases peak above 75.519 (141 above 63.1); no v2.11 claim depends on them
T4632	TO-VERIFY	ELE: /open_issues/27/issue	ideal 25 us steps ring the C1 seat passage in the SIL model (laminar-only damping ASSUMED): the T45 guard does not fire on them (record: the ride-build ideal-step aborts come at the window end or the comparator, ride_through.seat_ringing_finding); the 1 kHz ...
T4637	TO-VERIFY	ELE: /open_issues/28/issue	config.h comments corrected in v2.12 (rev_elec_3_author_status.N-09)
T4643	TO-VERIFY	ELE: /open_issues/29/issue	R3-01: option (b) implemented in SIL; needs the lead ruling, the per-serial coefficient and T_cal (B-PC-1 over 0-85 C), the thermal-zero maximum (F-10), the seat TC mounting at the C1 boss and the P_mid / U570 temperature (MISSING)
T4648	TO-VERIFY	ELE: /open_issues/30/issue	S13T: a stand relight after a B1-length burn is refused NOT_VENTED when the seat coefficient is 10 % low at k >= 2 (0.27-0.40 bar residual on the 0.30 bar limit): B-PC-1 must hold the coefficient to about 5 % at k = 3 (safe refusal)
T4653	TO-VERIFY	ELE: /open_issues/31/issue	R4-12: 070BG proof 141.0 bar abs < 166.14 bar abs required at FA1S; isolate or move P_torch to a C1-family unit (rev_comb_4_items_v2p12.R4-12)
T4658	TO-VERIFY	ELE: /open_issues/32/issue	N-07: at the B1 end-of-burn rate the re-sized T4 relight (4.59 s) needs the case TCs <= 112.7 C at the relight; the B1 p99 burn ends at about 137.8 C; a coast longer than about 19 s or a shorter relight (bench plan v5 T1-T4)
T4663	TO-VERIFY	ELE: /open_issues/33/issue	a flagged first light (f3 JFWD_LS) leaves the relight on a used joint; line 231 covers one Z3 event per assembly (LEAD 258 (4) OPEN)

15 Revision history and sources

15.1 Revision history

Revision	Generated (UTC)	CAD	Generator	Note
rev0-draft	2026-10-13T13:25:31Z	v3.4.1	build_datasheet.py 0.2.0	first draft generated from the hub state; FEA, RG-01 re-fly, Electronics v2.10 and the Combustion v2.4a interface patch pending (FA-1 envelope data folded in)
rev1-draft	2026-10-21T13:10:25Z	v3.5.2	build_datasheet.py 0.3.0	rev1 draft for the FINAL CAD v3.5.2; inputs as they stand (PENDING where an input is pending)
rev1	2026-10-21T16:02:27Z	v3.5.2	build_datasheet.py 0.3.1	rev1 final for the FINAL CAD v3.5.2 (PRELIMINARY): the REV-DS-1 corrections applied; electronics.json v4.1 promoted (CHANGELOG 266) with the option (b) carve-out of 266 (6); under REV-DS-1's check of this build; not physical approval

15.2 Sources

ID	Role	Path (hub-relative)	Version	sha256/16	Modified (UTC)	Reads
CMB	Combustion interface v2.4d (design point, grain, injector, igniter, sequences, FA-1 rulings)	interfaces/combustion.json	v2.4d	7dbf425c12300658	2026-10-21T11:24:52Z	666
MEC	Mechanical interface (its _v352 sections govern for CAD v3.5.2)	interfaces/mechanical.json	v2.4	66bc53d88d444679	2026-10-21T10:52:44Z	39
ELE	Electronics interface v4.1 (ECU, sensors, firmware v2.12 SIL, abort causes): promoted, CHANGELOG 266 (7); option (b) is carved out of every start with ignition until a reviewed v2.13 delta (266 (6)); line 266 governs v4.1 where they differ	interfaces/electronics.json	v4.1	c862375163393adc	2026-10-21T14:45:39Z	661
ELE4	Electronics interface v4 (firmware v2.11), the file published before line 266 (kept as staging_v2.11/results/electronics_v4.json, 266 (7)): used only for the superseded sides of the contradiction register	06-electronics/staging_v2.11/results/electronics_v4.json	v4	11f7896ad0d0613b	2026-10-21T05:38:06Z	3
GNC	Space/GNC interface v3.2 (reference vehicle, mission, ENG-11)	interfaces/gnc.json	v3.2	bde317782574cbd9	2026-10-21T10:57:09Z	11
CL	Decision log (line-numbered)	interfaces/CHANGELOG.md	266 lines	4cde8acb297d195c	2026-10-21T14:46:29Z	0
REQ	Engine requirements	00-requirements/REQUIREMENTS.md	rev 2	012b14cbaaddfdc6	2026-10-13T19:00:41Z	0
ICD	ICD-AIRFRAME draft 0.2 (2026-10-13; out of the finish scope since 2026-10-19, CHANGELOG 222; not updated): used only where no current source covers a value	00-requirements/ICD-AIRFRAME.md	draft 0.2	706455abf4098182	2026-10-13T11:13:06Z	0
SUP	Independent review register	reviews/SUPERVISION.json	18 reviews	076172727cddd074	2026-10-21T10:40:35Z	93
OVR	Decision override table (datasheet generator input)	07-datasheet/gen/decision_overrides.json	ovr-7	6746adf53921b1ef	2026-10-21T15:59:35Z	0
REV0S	Datasheet rev0-draft sources record (kept; CAD v3.4.1)	07-datasheet/out/sources.json	v3.4.1	fd2f3566f2dc345d	2026-10-13T13:26:04Z	6
RCMB	Combustion report	02-combustion/REPORT.md	2.4d	0f2af843861176c8	2026-10-21T11:24:52Z	0
DSIC	Combustion datasheet inputs	02-combustion/DATASHEET-INPUTS-COMBUSTION.md	v2.4d	8dc583578748b06f	2026-10-21T11:24:52Z	0
CF4	CF-4 relight-purge cold-flow specification	02-combustion/CF-4-SPEC.md	2.3	6f0d4afd7fb3a798	2026-10-21T11:24:52Z	0
T14	T1-T4 regression measurement specification	02-combustion/T1-T4-REGRESSION-SPEC.md	v3.2	edde58cf7e1aadf7	2026-10-21T11:24:52Z	0
FA1SP	FA-1 equivalent-static-pressure spectra, 406 periods (Combustion v2.4b data, SIM histories, DERIVED spectra)	02-combustion/data/v2p4b/fa1_pressure_spectra_published_v2p4b.csv	406 periods	a2c380515cada638	2026-10-19T22:21:00Z	0
RCMB4	Independent review REV-COMB-4 (text)	reviews/REVIEW-COMBUSTION-R4.md	REVIEW-COMBUSTION-R4 (REV-COMB-4)	2908321d7f75d439	2026-10-21T08:38:49Z	0
RPL	Plume CFD report	05-plume-cfd/REPORT.md	v2.2b	140b3c59f8323bba	2026-10-13T11:24:48Z	0
PVAL	Plume CFD validation results	05-plume-cfd/validation/results/validation_v2.json	v2.1a	cbfbce89c41689f3	2026-10-12T21:31:13Z	34
PF-MACH	Plume figure	05-plume-cfd/ownruns/f_2s_100/fig/f2s_mach_near.png	figure	39cfd050d53b9f68	2026-10-12T21:31:30Z	0
PF-SCHL	Plume figure	05-plume-cfd/ownruns/f_2s_100/fig/f2s_schlieren_near.png	figure	5a559717ba7c3a76	2026-10-12T21:31:35Z	0
PF-TEMP	Plume figure	05-plume-cfd/ownruns/f_2s_100/fig/f2s_T_full.png	figure	b7fe094ff216873c	2026-10-12T21:31:32Z	0
PF-AXIS	Plume figure	05-plume-cfd/ownruns/f_2s_100/fig/f2s_centreline.png	figure	de88abf544299723	2026-10-12T21:31:36Z	0
PF-V2MACH	Plume figure	05-plume-cfd/ownruns/c_2s_100_v2op/fig/c2s_v2op_mach_near.png	figure	034d7f18ff3075f3	2026-10-12T21:29:26Z	0
PF-ART	Plume figure	05-plume-cfd/blender/final/argos_plume_artistic_side.png	figure	7c722b7f001b2ae4	2026-10-12T20:43:12Z	0
DSIG	GNC datasheet inputs (gnc.json v3.2)	01-systems-gnc/DATASHEET-INPUTS-GNC.md	v3.2	b8b0dcaf45a5d915	2026-10-21T10:57:09Z	0
RGNC5	Independent review REV-GNC-7 (text)	reviews/REVIEW-GNC-R5.md	REVIEW-GNC-R5 (REV-GNC-7)	6b86fcd326568646	2026-10-21T10:39:53Z	0
RELE	Electronics v2.11 handback report	06-electronics/REPORT-v2.11-HANDBACK.md	v2.11	44166deaa23609e0	2026-10-21T05:49:57Z	0
BENCH	Electronics bench and test plan v5, the plan of electronics.json v4.1: ACCEPT WITH COMMENTS (REV-ELEC-4); it does not go to the founders before the corrections of CHANGELOG 266 (7) (E4-02, E4-06)	06-electronics/staging_v2.12/docs/bench_plan_v5.md	v5	360c1d6695cc305c	2026-10-21T12:38:44Z	0
RELE3	Independent review REV-ELEC-3 (text)	reviews/REVIEW-ELECTRONICS-R3.md	REVIEW-ELECTRONICS-R3 (REV-ELEC-3)	fcf05f8bc7382cd1	2026-10-21T07:05:48Z	0
RELE4	Independent review REV-ELEC-4 of Electronics v2.12 (text; the corrections that line 266 (7) applies)	reviews/REVIEW-ELECTRONICS-R4.md	REVIEW-ELECTRONICS-R4 (REV-ELEC-4)	70b7103472eebab4	2026-10-21T14:43:16Z	0
RDS1	Independent review REV-DS-1 of this datasheet's rev1-draft (text)	reviews/REVIEW-DATASHEET-R1.md	REVIEW-DATASHEET-R1 (REV-DS-1)	1ef95f92ae8aef27	2026-10-21T14:30:45Z	0
F3D	FEA3D v3.5.2 report (FEA3D-2, final)	04-fea/assembly3d_v352/REPORT-FEA3D-v352.md	FINAL handback of the v3.5.2 re-check	dc53f4cd45d75f4e	2026-10-21T05:43:45Z	0
F3DMAN	FEA3D v3.5.2 manifest	04-fea/assembly3d_v352/MANIFEST.json	—	09eb820dfa384cd0	2026-10-21T05:43:45Z	0
MOUNT	Engine mounting data v3.5.2 (hole loads)	04-fea/assembly3d_v352/results/v352/engine_mount_data_v352.json	—	e5f41bde43a00ba4	2026-10-20T23:07:03Z	57
MOUNTMD	Engine mounting data v3.5.2, datasheet tables	04-fea/assembly3d_v352/results/v352/tables/engine_mount_data_v352.md	v3.5.2 re-check with RM2-11 folded	ec986ff940cfed33	2026-10-20T23:07:03Z	0
KTAB	Axis stiffness K table at R3 (v3.5.2)	04-fea/assembly3d_v352/results/v352/k_table_r3_v352.json	—	f3c7cfbcfb98bcd7	2026-10-20T18:48:19Z	145
ROCK	Rocking modes vs servo J (v3.5.2)	04-fea/assembly3d_v352/results/v352/rocking_servo_J_v352.json	—	e655a92b3818bf32	2026-10-20T05:58:57Z	18
MODAL	Modal ca2c vs ca2b (v3.5.2)	04-fea/assembly3d_v352/results/v352/modal_ca2c_vs_ca2b_v352.json	—	1bdaf41b4611094c	2026-10-20T17:02:03Z	0
F4J	F4 joints J-FD01 / J-FD09 (v3.5.2)	04-fea/assembly3d_v352/results/v352/f4_v352.json	v1	fd5a06671fd4456a	2026-10-20T03:00:39Z	24
CDJ	Mesh-convergence check cD (J-FWD seat)	04-fea/assembly3d_v352/results/v352/jfwd_cD_result_v352.json	—	8209e80f09f28af7	2026-10-21T05:16:00Z	0
ENV6	J-FWD soak envelope v6 (final)	04-fea/assembly3d_v352/results/v352/jfwd_soak_envelope_v352_v6.json	v6	43a5d78b67c615fc	2026-10-21T05:19:18Z	0
ICDMF	Moment-free hole-load estimate (v3.5.2)	04-fea/assembly3d_v352/results/v352/icd_momentfree_v352.json	—	785b4e48f407ddfb	2026-10-20T23:04:00Z	0
MMX	Margin matrix v3.5.2 on J-FWD rows v8 (every part x every case)	04-fea/assembly3d_v352_rows_v8/tables/margin_matrix_v352.json	v352	6c694fa7147fb511	2026-10-21T07:51:43Z	2566
MMXMD	Margin matrix v3.5.2 on rows v8 (published table)	04-fea/assembly3d_v352_rows_v8/tables/margin_matrix_v352.md	FEA3D margin matrix v3.5.2	a0917e000f3a8375	2026-10-21T07:51:43Z	0
MMXD	Margin matrix delta, rows v8 vs the FEA3D-2 matrix	04-fea/assembly3d_v352_rows_v8/delta_vs_assembly3d_v352.json	v352	e4006ef6b9464e6c	2026-10-21T07:51:43Z	8
ROWS8	J-FWD IN718 rows v8 (Mechanical)	04-fea/v352/jfwd_in718_rows_v352_v8.json	v8	af072cc2e70711ca	2026-10-21T07:47:40Z	0

ID	Role	Path (hub-relative)	Version	sha256/16	Modified (UTC)	Reads
BAND	Seat-line P+Q select-fit band v1 (Mechanical, R2-B)	04-fea/v352/jfwd_pq_band_v352_v1.json	v1	1ba3147b42edfeed	2026-10-21T07:47:40Z	0
TAPW	TC-02 torch-tap wall at FA1S, v3 (Mechanical, R4-01)	04-fea/v352/jfwd_tapwall_v352_v3.json	v3	ef70572ec92b05b0	2026-10-21T10:52:26Z	7
TB70	LC-TB70 envelope v2: Z3 v2 and the J-FWD replace rule (Mechanical)	04-fea/v352/lc_tb70_envelope_v2.json	v2	7ebd2fd63dc878ef	2026-10-21T10:52:26Z	4
TXT5	Item C / T-F1 texts v5 (drawing, founder and datasheet texts)	04-fea/v352/texts/item_c_tf1_texts_v352_v5.json	v5	c4ea791b7aa68e2e	2026-10-21T10:52:26Z	41
LCF	J-FWD screw LCF v3 (Mechanical)	04-fea/v352/jfwd_screw_lcf_v352_v3.json	v3	b6ff0ea0f731c7dc	2026-10-21T07:47:40Z	5
FIL	J-FWD screw head fillet v2 (Mechanical)	04-fea/v352/jfwd_head_fillet_v352_v2.json	v2	704f55ecb7a18dea	2026-10-21T07:47:40Z	2
SPR	HW-170 disc spring v5 (Mechanical)	04-fea/v352/jfwd_spring_v352_v5.json	v5	7d2e27d74f202418	2026-10-21T07:47:40Z	0
C1S	C1 seat (Mechanical F1)	04-fea/v352/c1seat/c1_seat_v352_v1.json	—	435f5d71f6213160	2026-10-21T10:52:26Z	0
TPB	Torch-port rise F2 (Mechanical; not a bound)	04-fea/v352/c1seat/torch_port_bl08_v352_v1.json	v1	20220647d32cbbbc	2026-10-21T10:52:26Z	0
MMAN	Mechanical 04-fea/v352 manifest	04-fea/v352/MANIFEST_v352.sha256	226 rows	4f7e7655ccc25f70	2026-10-21T10:52:27Z	0
BOM	Bill of materials (CAD v3.5.2)	03-cad/v3.5.2/BOM_v3.5.2.json	v3.5.2	9234c953c5b6beb8	2026-10-20T19:16:52Z	22
MP	Mass properties (CAD v3.5.2)	03-cad/v3.5.2/massprops_v3.5.2.json	v3.5.2	c1ff170be0ac8cc5	2026-10-20T19:16:53Z	31
MDEL	Mass delta v3.5.2 vs v3.5.1	03-cad/v3.5.2/massdelta_v3.5.2_vs_v3.5.1.json	v3.5.2	6f39368668acafd1	2026-10-20T19:18:21Z	11
SET	CAD set record (v3.5.2)	03-cad/v3.5.2/set_v3.5.2.json	v3.5.2 final	621481ddeccc228b	2026-10-20T20:46:49Z	8
MAN	Assembly manifest, instance bounds (CAD v3.5.2)	03-cad/v3.5.2/assembly/assembly_manifest_v3.5.2.json	v3.5.2	b2ffac1bdefd288c	2026-10-20T06:41:35Z	8
GLB	Assembly GLB, visualisation geometry (CAD v3.5.2)	03-cad/v3.5.2/assembly/ENGINE_v3.5.2_neutral.glb	v3.5.2	4bcc264f683e128f	2026-10-20T19:16:52Z	0
HWD	Hardware data v3.5.2 (fasteners, springs)	03-cad/v3.5.2/data/hardware_v352.json	v1.7	c2c7b13d3a0086e3	2026-10-20T18:51:59Z	1
CHK	Final-set check (round trips, changed parts, BOM delta)	03-cad/v3.5.2/checks/test_final352_v1.json	v1	aa064886ee64ac94	2026-10-20T19:24:36Z	2
CLR	Interference and clearance check at 18 poses (geometry unchanged in the final)	03-cad/v3.5.2/checks/clear_interim352_v2.json	v2	846275cf5620e56d	2026-10-20T00:43:41Z	75
GLBR	GLB read-back check	03-cad/v3.5.2/checks/glb_readback_v3.5.2.json	—	e95fe38bffac9c44	2026-10-20T19:16:52Z	3
DRW	J-FWD drawing texts v2 (TC-01 / TC-02 / HW-101 notes)	03-cad/v3.5.2/texts/jfwd_drawing_texts_v352_v2.json	v2	58683aa9a66a65dd	2026-10-21T10:52:43Z	0
CADMAN	CAD v3.5.2 manifest	03-cad/v3.5.2/MANIFEST_v3.5.2.sha256	456 rows	a4713d45b3b24a0f	2026-10-21T10:52:44Z	0
RMAN	Render manifest (this generator)	07-datasheet/build/renders/v3.5.2/render_manifest.json	v3.5.2	9e3d1d0e8156ad4a	2026-10-21T11:23:02Z	0
RSCR	Render script (Blender, this generator)	07-datasheet/gen/render_views_blender.py	—	6604b8cfa8cff3de	2026-10-21T11:17:36Z	0
R-ISO	Render metadata: iso (camera, anchors)	07-datasheet/build/renders/v3.5.2/iso_meta.json	—	d9985e8966a7f30f	2026-10-21T11:19:06Z	1
R-FRONT	Render metadata: front (camera, anchors)	07-datasheet/build/renders/v3.5.2/front_meta.json	—	fea086e8fbd19db8	2026-10-21T11:19:37Z	0
R-SIDE	Render metadata: side (camera, anchors)	07-datasheet/build/renders/v3.5.2/side_meta.json	—	849d19ffa80378ec	2026-10-21T11:20:07Z	0
R-SECTION	Render metadata: section (camera, anchors)	07-datasheet/build/renders/v3.5.2/section_meta.json	—	ed3009f55d6ac69e	2026-10-21T11:20:37Z	0
R-LAYOUT	Render metadata: layout (camera, anchors)	07-datasheet/build/renders/v3.5.2/layout_meta.json	—	3405446a58446390	2026-10-21T11:20:58Z	0
R-FAM-BL	Render metadata: family_BL (camera, anchors)	07-datasheet/build/renders/v3.5.2/family_BL_meta.json	—	6db6ac3b107b56b3	2026-10-21T11:21:06Z	0
R-FAM-EL	Render metadata: family_EL (camera, anchors)	07-datasheet/build/renders/v3.5.2/family_EL_meta.json	—	a992b63ec2bd820b	2026-10-21T11:21:09Z	0
R-FAM-FD	Render metadata: family_FD (camera, anchors)	07-datasheet/build/renders/v3.5.2/family_FD_meta.json	—	a28d36cd71eee5e7	2026-10-21T11:21:16Z	0
R-FAM-FS	Render metadata: family_FS (camera, anchors)	07-datasheet/build/renders/v3.5.2/family_FS_meta.json	—	c79ecfe0b10b9f8d	2026-10-21T11:21:35Z	0
R-FAM-GM	Render metadata: family_GM (camera, anchors)	07-datasheet/build/renders/v3.5.2/family_GM_meta.json	—	3287ced23fe37915	2026-10-21T11:22:04Z	0
R-FAM-GR	Render metadata: family_GR (camera, anchors)	07-datasheet/build/renders/v3.5.2/family_GR_meta.json	—	b668337919726512	2026-10-21T11:22:12Z	0
R-FAM-HW	Render metadata: family_HW (camera, anchors)	07-datasheet/build/renders/v3.5.2/family_HW_meta.json	—	9014a4bd5011ccfd	2026-10-21T11:22:19Z	0
R-FAM-IG	Render metadata: family_IG (camera, anchors)	07-datasheet/build/renders/v3.5.2/family_IG_meta.json	—	7312734deafef981	2026-10-21T11:22:27Z	0
R-FAM-NZ	Render metadata: family_NZ (camera, anchors)	07-datasheet/build/renders/v3.5.2/family_NZ_meta.json	—	47f90eede451eede	2026-10-21T11:22:39Z	0
R-FAM-PD	Render metadata: family_PD (camera, anchors)	07-datasheet/build/renders/v3.5.2/family_PD_meta.json	—	c8af44f69581630c	2026-10-21T11:22:56Z	0
R-FAM-TC	Render metadata: family_TC (camera, anchors)	07-datasheet/build/renders/v3.5.2/family_TC_meta.json	—	2c9336f24b28f442	2026-10-21T11:23:02Z	0

Paths are relative to the engine programme hub (argos-engine). The same list is written to out/rev1/sources.json. "Reads" counts JSON key reads; text sources are read by line, row or sentence (Appendix A).

Appendix A Value trace

Every traced value used in this document: trace id, value (a value read from a source at the precision of that source; a value the generator computes, to 6 significant digits; the body may show it rounded), class, source and version, key (JSON pointer, markdown row, CHANGELOG line and item, or generator formula) and the source tag text. A DECIDED value also shows its source class (design value, lead decision or requirement). Identical source keys share one id. Display rules of the body and of this appendix (trace.json keeps the source strings): the review status written in Spanish in the FEA3D-2 report prints as "to be reviewed" (REV-DS-1 D1-12); a bare "§n" in a combustion.json text prints as "combustion REPORT §n" (D1-11); a number read from a source prints at the precision of that source, and a number read from a text keeps the digits of the text (D1-08).

ID	Value	Class	Source	Key	Source tag / formula
T001	RE-20: the 30 bar firmware abort runs on the FILTERED channel with 5 ms persistence ...	DECIDED *Lead decision, line 166 (4)*	CL 266 lines	L166(4)	lead (decision) 2026-10-13 (line sha 3fde4047cba94dbc)
T002	Electronics review decisions (CHANGELOG 166): light-rule baseline never substituted by ...	DECIDED *Requirement, REQUIREMENTS.md rev 2*	REQ rev 2	§Amendments after rev 2 bullet "Electronics review decisions (CHANGELOG 166)"	tag in REQUIREMENTS.md rev 2: REQ bullet
T003	Electronics review decisions (CHANGELOG 166): light-rule baseline never substituted by ...	DECIDED *Lead decision, per REQUIREMENTS.md rev 2*	REQ rev 2	§Amendments after rev 2 bullet "Electronics review decisions (CHANGELOG 166)"[sentence containing "filtered channel, 5 ms"]	tag in REQUIREMENTS.md rev 2: REQ bullet
T004	30 bar firmware abort on the FILTERED channel with 5 ms persistence (amends 'raw' in ...	DECIDED *Lead decision, per decision_overrides.json ovr-7*	OVR ovr-7	/overrides/0/value	tag in decision_overrides.json ovr-7: override OVR-01 (ovr-7); numbers and key phrases checked against the cited sources
T005	ENG-12: tank temperature range at ARM 15-24 C (was 15-25 C); heater hard-disabled at ARM.	DECIDED *Requirement, ENG-12*	REQ rev 2	§Amendments after rev 2 bullet "ENG-12:"	tag in REQUIREMENTS.md rev 2: REQ bullet
T006	ENG-09: MEOP 45 bar kept; ignition interlock T_tank <= 24 C at ARM and at ignition.	DECIDED *Requirement, ENG-09*	REQ rev 2	§Amendments after rev 2 bullet "ENG-09:"	tag in REQUIREMENTS.md rev 2: REQ bullet
T007	R2-07 interlock guard band: T_tank and P_tank thresholds sit below 24 C / 55.28 bar abs ...	DECIDED *Lead decision, line 172 (4)*	CL 266 lines	L172(4)	lead (decision) 2026-10-13 (line sha 6d0e30766faf3c84)
T008	tank temperature range at ARM 15-24 C (was 15-25 C); T_tank <= 24 C at ARM and at ...	DECIDED *Lead decision, per decision_overrides.json ovr-7*	OVR ovr-7	/overrides/1/value	tag in decision_overrides.json ovr-7: override OVR-04 (ovr-7); numbers and key phrases checked against the cited sources
T009	Combustion re-review decisions (CHANGELOG 172): stand safing after a no-light = N2 purge ...	DECIDED *Requirement, REQUIREMENTS.md rev 2*	REQ rev 2	§Amendments after rev 2 bullet "Combustion re-review decisions (CHANGELOG 172)"	tag in REQUIREMENTS.md rev 2: REQ bullet
T010	ENG-09 trapped-liquid relief text superseded: tank relief 6.9 MPa, MAWP >= 8.42 MPa ...	DECIDED *Lead decision, per REQUIREMENTS.md rev 2*	REQ rev 2	§Amendments after rev 2 bullet "Combustion re-review decisions (CHANGELOG 172)"[sentence containing "trapped-liquid relief"]	tag in REQUIREMENTS.md rev 2: REQ bullet
T011	R2-01, R2-05, R2-06 (incl. CF-4 P4 per tap and phase), R2-08, R2-09 go into a Combustion ...	DECIDED *Lead decision, line 172 (6)*	CL 266 lines	L172(6)	lead (decision) 2026-10-13 (line sha 6d0e30766faf3c84)
T012	Relief: back-relief check MV segment → tank (Cv >= 0.1); secondary rupture disc 9.5 MPa ...	DECIDED *Requirement, REQUIREMENTS.md rev 2*	REQ rev 2	§Amendments after rev 2 bullet "Relief:"	tag in REQUIREMENTS.md rev 2: REQ bullet
T013	tank relief 6.9 MPa, MAWP >= 8.42 MPa, proof >= 10.5 MPa, back-relief check MV segment ...	DECIDED *Lead decision, per decision_overrides.json ovr-7*	OVR ovr-7	/overrides/2/value	tag in decision_overrides.json ovr-7: override OVR-05 (ovr-7); numbers and key phrases checked against the cited sources
T014	{"basis_bar": 44.33, "margin_to_MEOP_bar": 0.67, "proof_margin_min_bar": 21.9 ...	SIM	CMB v2.4d	/meop_basis_v2p4/v	SIM meop_basis_v2p4.json
T015	MEOP 45 bar kept with the ignition interlock T_tank <= 24 C at ARM and at ignition ...	DECIDED *Lead decision, per decision_overrides.json ovr-7*	OVR ovr-7	/overrides/3/value	tag in decision_overrides.json ovr-7: override OVR-06 (ovr-7); numbers and key phrases checked against the cited sources
T016	Amendment c: the time-only 0xFFFF release term (line 168 (1)) is REPLACED in design by ...	DECIDED *Lead decision, line 175 (5)*	CL 266 lines	L175(5)	lead (decision) 2026-10-13 (line sha 779493c2be25e6d0)
T017	GNC rev2 decisions (CHANGELOG 175): A1 apogee target 285 m. PURGE commits no later than ...	DECIDED *Requirement, REQUIREMENTS.md rev 2*	REQ rev 2	§Amendments after rev 2 bullet "GNC rev2 decisions (CHANGELOG 175)"	tag in REQUIREMENTS.md rev 2: REQ bullet
T018	With Pc invalid after E2 the drogue is released only on accelerometer-confirmed thrust ...	DECIDED *Lead decision, per REQUIREMENTS.md rev 2*	REQ rev 2	§Amendments after rev 2 bullet "GNC rev2 decisions (CHANGELOG 175)"[sentence containing "accelerometer-confirmed thrust"]	tag in REQUIREMENTS.md rev 2: REQ bullet
T019	The time-only 0xFFFF release term (line 168 (1)) is REPLACED in design by R2: with Pc ...	DECIDED *Lead decision, per decision_overrides.json ovr-7*	OVR ovr-7	/overrides/4/value	tag in decision_overrides.json ovr-7: override OVR-09 (ovr-7); numbers and key phrases checked against the cited sources
T020	PROMOTED 14:45:39Z: interfaces/electronics.json = staging_v2.12 ...	DECIDED *Lead decision, line 266 (7)*	CL 266 lines	L266(7)	lead (decision) 2026-10-21 (line sha 6f3181b859f02937)
T021	interfaces/electronics.json is v4.1, PROMOTED 14:45:39Z by the lead; its STAGED status ...	DECIDED *Lead decision, per decision_overrides.json ovr-7*	OVR ovr-7	/overrides/5/value	tag in decision_overrides.json ovr-7: override OVR-10 (ovr-7); numbers and key phrases checked against the cited sources
T022	interfaces/electronics.json stays v4 11f7896ad0d0613b; results/electronics_v4.1.json ...	DECIDED *Lead decision, line 264 (1)*	CL 266 lines	L264(1)[from "interfaces/electronics.json stays v4" to "as PENDING REVIEW."]	lead (decision) 2026-10-21 (line sha 7cdbe51041ae6b8b)
T023	PROMOTED 14:45:39Z: interfaces/electronics.json = staging_v2.12 ...	DECIDED *Lead decision, line 266 (7)*	CL 266 lines	L266(7)[from "PROMOTED 14:45:39Z" to "under the carve-out of (6)."]	lead (decision) 2026-10-21 (line sha 6f3181b859f02937)
T024	Option (b) (seat coefficient loaded, setPcSeatCoef != 0) is not used for any start with ...	DECIDED *Lead decision, line 266 (6)*	CL 266 lines	L266(6)[from "Option (b) (seat coefficient loaded" to "bench work without ignition."]	lead (decision) 2026-10-21 (line sha 6f3181b859f02937)
T025	Consequences under the rulings in force: A1 FLIGHT IS NO-GO until v2.13 is reviewed (264 ...	DECIDED *Lead decision, line 266 (6)*	CL 266 lines	L266(6)[from "Consequences under the rulings in force:" to "keeps the A1 relight at k = 3 [SIM S13T]."]	lead (decision) 2026-10-21 (line sha 6f3181b859f02937)
T026	Suspended until v2.13, and marked so wherever shown: the option (b) firmware rows ...	DECIDED *Lead decision, line 266 (6)*	CL 266 lines	L266(6)[from "Suspended until v2.13, and marked so wherever shown:" to "the B-PC-1 / B-TC-1 TC4 rows."]	lead (decision) 2026-10-21 (line sha 6f3181b859f02937)
T027	Accepted now: purge FINAL 2.43 / 2.39 s with rule (d), (d1), (d2); TB-state; JFWD_LS ...	DECIDED *Lead decision, line 266 (6)*	CL 266 lines	L266(6)[from "Accepted now:" to "the T1-T4 planning rule."]	lead (decision) 2026-10-21 (line sha 6f3181b859f02937)
T028	Until v4.2 this line governs v4.1 where they differ: its STAGED status text is ...	DECIDED *Lead decision, line 266 (7)*	CL 266 lines	L266(7)[from "Until v4.2 this line governs v4.1 where they differ:" to "Bench plan v5 does not go to the founders before these corrections."]	lead (decision) 2026-10-21 (line sha 6f3181b859f02937)
T029	REV-ELEC-4 (independent delta review of Electronics v2.12; interim E4-01 / E4-02 13:57Z ...	DECIDED *Lead decision, line 266 (4)*	CL 266 lines	L266(4)[from "REV-ELEC-4 (independent delta review of Electronics v2.12" to "11 NOTES"]	lead (decision) 2026-10-21 (line sha 6f3181b859f02937)
T030	QUEUED, NOT STARTED (the founder says when): the Electronics v2.13 delta and its review ...	DECIDED *Lead decision, line 266 (8)*	CL 266 lines	L266(8)[from "QUEUED, NOT STARTED (the founder says when)" to "a SIL MANIFEST_v2)."]	lead (decision) 2026-10-21 (line sha 6f3181b859f02937)
T031	Routed: E4-N08 (the hardware both-open backstop allows up to 2.741 s against ...	DECIDED *Lead decision, line 266 (8)*	CL 266 lines	L266(8)[from "Routed: E4-N08" to "are GNC and lead items with v2.13."]	lead (decision) 2026-10-21 (line sha 6f3181b859f02937)
T032	E4-01 (MAJOR) [SIM reviewer SIL variant; CALC probes]: with option (b) the seat ...	DECIDED *Lead decision, line 266 (5)*	CL 266 lines	L266(5)	lead (decision) 2026-10-21 (line sha 6f3181b859f02937)
T033	500 N	SIM	CMB v2.4d	/operating_points/throttle_points_20C_initial_port/0/F_N	20 C saturated tank (P_tank 50 bar row of the map for lift and dP/Pc), initial port 25 mm, nominal law. SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* ...
T034	193.1 s	SIM	CMB v2.4d	/operating_points/throttle_points_20C_initial_port/0/Isp_SL_s	20 C saturated tank (P_tank 50 bar row of the map for lift and dP/Pc), initial port 25 mm, nominal law. SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* ...
T035	10.435	SIM	CMB v2.4d	/operating_points/throttle_points_20C_initial_port/0/OF	20 C saturated tank (P_tank 50 bar row of the map for lift and dP/Pc), initial port 25 mm, nominal law. SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* ...
T036	2501007 Pa	SIM	CMB v2.4d	/operating_points/throttle_points_20C_initial_port/0/Pc_Pa	20 C saturated tank (P_tank 50 bar row of the map for lift and dP/Pc), initial port 25 mm, nominal law. SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* ...
T037	100 N	SIM	CMB v2.4d	/operating_points/throttle_points_20C_initial_port/5/F_N	20 C saturated tank (P_tank 50 bar row of the map for lift and dP/Pc), initial port 25 mm, nominal law. SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* ...
T038	5.34303 kg	DERIVED	MP v3.5.2	/states/wet/all/m_kg	(class set at the call site: DERIVED) [CAD model total, wet (full grain); hardware CALC carried]
T039	2.73802 kg	DERIVED	MEC v2.4	/gimballed_mass_properties_v352/states/mid/engine/m_kg	(class set at the call site: DERIVED)
T040	593.3 mm	DERIVED	MAN v3.5.2	/instances/*/bounds_mm	DERIVED by build_datasheet.py: overall z extent over assembly-manifest instance bounds
T041	355.2 mm	DERIVED	MAN v3.5.2	/instances/*/bounds_mm	DERIVED by build_datasheet.py: z of the forward end (pivot at z 0) over assembly-manifest instance bounds
T042	238.1 mm	DERIVED	MAN v3.5.2	/instances/*/bounds_mm	DERIVED by build_datasheet.py: distance pivot to aft end over assembly-manifest instance bounds
T043	139.5 mm	DERIVED	MAN v3.5.2	/instances/*/bounds_mm	DERIVED by build_datasheet.py: x extent over assembly-manifest instance bounds
T044	140.23000000000002 mm	DERIVED	MAN v3.5.2	/instances/*/bounds_mm	DERIVED by build_datasheet.py: y extent over assembly-manifest instance bounds

ID	Value	Class	Source	Key	Source tag / formula
T045	488.75 mm	DERIVED	MAN v3.5.2	/instances/*/bounds_mm	DERIVED by build_datasheet.py: z extent over assembly-manifest instance bounds (group = engine)
T046	−238.1–250.65 mm	DERIVED	MAN v3.5.2	/instances/*/bounds_mm	DERIVED by build_datasheet.py: z range over assembly-manifest instance bounds (group = engine)
T047	455 mm	DERIVED	MAN v3.5.2	/instances/*/bounds_mm	DERIVED by build_datasheet.py: z extent over assembly-manifest instance bounds (group = fixed)
T048	−99.8–355.2 mm	DERIVED	MAN v3.5.2	/instances/*/bounds_mm	DERIVED by build_datasheet.py: z range over assembly-manifest instance bounds (group = fixed)
T049	ENG-BL-01: Bleed tap block T4 crown (316L) with R1 0.71 mm restrictor insert, O-ring ...	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/lines/0	tag in BOM_v3.5.2.json v3.5.2: BOM row (CAD parts/ENG-BL-01.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T050	0.00832 kg	DERIVED	BOM v3.5.2	/lines/0/unit_mass_kg	CAD volume x rho(SS316L)
T051	ENG-BL-02: Bleed tap adapter on FD-01 (316L): M8×1 port, <= 15 um sintered 316L disc, R2 ...	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/lines/1	tag in BOM_v3.5.2.json v3.5.2: BOM row (CAD parts/ENG-BL-02.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T052	0.00614 kg	DERIVED	BOM v3.5.2	/lines/1/unit_mass_kg	CAD volume x rho(SS316L)
T053	ENG-BL-03A: Bleed line T4 leg BL-01 → BL-02 tee: 1/8 × 0.028 in 316L, 49 mm developed ...	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/lines/2	tag in BOM_v3.5.2.json v3.5.2: BOM row (CAD parts/ENG-BL-03A.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T054	0.00214 kg	DERIVED	BOM v3.5.2	/lines/2/unit_mass_kg	COTS mass (DERIVED annulus 44.0 g/m x 49 mm); CG and inertia of a uniform bounding box
T055	ENG-BL-03E: Bleed line common line BL-02 → BL-04A: 1/8 × 0.028 in 316L, 53 mm developed ...	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/lines/3	tag in BOM_v3.5.2.json v3.5.2: BOM row (CAD parts/ENG-BL-03E.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T056	0.00233 kg	DERIVED	BOM v3.5.2	/lines/3/unit_mass_kg	COTS mass (DERIVED annulus 44.0 g/m x 53 mm); CG and inertia of a uniform bounding box
T057	ENG-BL-03F: Bleed line fixed side: 1/8 × 0.028 in 316L, 266 mm developed (wall clamps to ...	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/lines/4	tag in BOM_v3.5.2.json v3.5.2: BOM row (CAD parts/ENG-BL-03F.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T058	0.01169 kg	DERIVED	BOM v3.5.2	/lines/4/unit_mass_kg	COTS mass (DERIVED annulus x 266 mm); CG and inertia of a uniform bounding box
T059	ENG-BL-04: Bleed crossing hose -2 class PTFE / 316 braid, bundled 23 mm above the FD-07 ...	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/lines/5	tag in BOM_v3.5.2.json v3.5.2: BOM row (CAD parts/ENG-BL-04.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T060	0.0126 kg	ASSUMED	BOM v3.5.2	/lines/5/unit_mass_kg	COTS mass (ESTIMATE 70 g/m x 180 mm); CG and inertia of a uniform bounding box
T061	ENG-BL-05: Bleed valve 2/2 NC solenoid, 12 V, seat >= 2 mm, sub-base mount (ENVELOPE Ø36 ...	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/lines/6	tag in BOM_v3.5.2.json v3.5.2: BOM row (CAD parts/ENG-BL-05.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T062	0.3 kg	ASSUMED	BOM v3.5.2	/lines/6/unit_mass_kg	COTS mass (ASSUMED 0.30 kg (MASS RISK; target <= 0.15 kg)); CG and inertia of a uniform bounding box
T063	ENG-BL-06: V2 vent bulkhead fitting (316L) through the bay wall at az 112; outside ...	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/lines/7	tag in BOM_v3.5.2.json v3.5.2: BOM row (CAD parts/ENG-BL-06.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T064	0.01 kg	ASSUMED	BOM v3.5.2	/lines/7/unit_mass_kg	COTS mass (ESTIMATE 10 g); CG and inertia of a uniform bounding box
T065	ENG-BL-07: Bleed valve manifold (6061-T6): V1/V2 pads, Ø1.5 mid passage with P_mid port ...	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/lines/8	tag in BOM_v3.5.2.json v3.5.2: BOM row (CAD parts/ENG-BL-07.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T066	0.07494 kg	DERIVED	BOM v3.5.2	/lines/8/unit_mass_kg	CAD volume x rho(AL6061-T6)
T067	ENG-BL-08: P_mid pressure transducer (ENVELOPE Ø22 × 55, Electronics selects) on the ...	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/lines/9	tag in BOM_v3.5.2.json v3.5.2: BOM row (CAD parts/ENG-BL-08.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T068	0.05 kg	ASSUMED	BOM v3.5.2	/lines/9/unit_mass_kg	COTS mass (ASSUMED 50 g); CG and inertia of a uniform bounding box
T069	ENG-BL-10: V1 inlet fitting (316L): fixed line BL-03F into the manifold	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/lines/10	tag in BOM_v3.5.2.json v3.5.2: BOM row (CAD parts/ENG-BL-10.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T070	0.005 kg	ASSUMED	BOM v3.5.2	/lines/10/unit_mass_kg	COTS mass (ESTIMATE 5 g); CG and inertia of a uniform bounding box
T071	ENG-EL-03: Gimbal encoder, inner axis (beta): AS5048A on a round PCB 20 × 2.8 (REQUEST ...	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/lines/11	tag in BOM_v3.5.2.json v3.5.2: BOM row (CAD parts/ENG-EL-03.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T072	0.003 kg	ASSUMED	BOM v3.5.2	/lines/11/unit_mass_kg	COTS mass (ASSUMED 2-5 g incl. holder); CG and inertia of a uniform bounding box
T073	ENG-EL-04: Gimbal encoder, outer axis (alpha): AS5048A on a round PCB 20 × 2.8 recessed ...	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/lines/12	tag in BOM_v3.5.2.json v3.5.2: BOM row (CAD parts/ENG-EL-04.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T074	0.003 kg	ASSUMED	BOM v3.5.2	/lines/12/unit_mass_kg	COTS mass (ASSUMED 2-5 g); CG and inertia of a uniform bounding box
T075	ENG-EL-05: Pintle-position encoder AS5048A on a 14 mm round PCB (chip + PCB envelope 14 ...	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/lines/13	tag in BOM_v3.5.2.json v3.5.2: BOM row (CAD parts/ENG-EL-05.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T076	0.002 kg	ASSUMED	BOM v3.5.2	/lines/13/unit_mass_kg	COTS mass (ASSUMED 1-3 g); CG and inertia of a uniform bounding box
T077	ENG-EL-06: Diametral NdFeB magnet 6 × 2.5 (AS5048A), bonded on the pin end	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/lines/14	tag in BOM_v3.5.2.json v3.5.2: BOM row (CAD parts/ENG-EL-06.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T078	0.0005 kg	DERIVED	BOM v3.5.2	/lines/14/unit_mass_kg	COTS mass (DERIVED 6 × 2.5 mm x 7.5 g/cm3); CG and inertia of a uniform bounding box
T079	ENG-EL-11: P_feed transducer J602 TE M3234-000005 class, 1/4-18 NPT male, vertical on ...	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/lines/15	tag in BOM_v3.5.2.json v3.5.2: BOM row (CAD parts/ENG-EL-11.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T080	0.05 kg	ASSUMED	BOM v3.5.2	/lines/15/unit_mass_kg	COTS mass (ASSUMED 50 g (electronics.json)); CG and inertia of a uniform bounding box
T081	ENG-EL-12: MSS-2 Type T MI thermocouple 1/16 in in a 1/16 × 1/8-27 NPT male connector ...	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/lines/16	tag in BOM_v3.5.2.json v3.5.2: BOM row (CAD parts/ENG-EL-12.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T082	0.015 kg	ASSUMED	BOM v3.5.2	/lines/16/unit_mass_kg	COTS mass (ESTIMATE 15 g incl. connector); CG and inertia of a uniform bounding box
T083	ENG-FD-01: Oxidiser inlet block E1-M (316L): riser socket, 8 mm passages, P_feed tap ...	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/lines/17	tag in BOM_v3.5.2.json v3.5.2: BOM row (CAD parts/ENG-FD-01.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T084	0.04344 kg	DERIVED	BOM v3.5.2	/lines/17/unit_mass_kg	CAD volume x rho(SS316L)
T085	ENG-FD-02: Filter cartridge 15 um absolute, pleated 316L Dutch-twill mesh, area >= 1450 ...	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/lines/18	tag in BOM_v3.5.2.json v3.5.2: BOM row (CAD parts/ENG-FD-02.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T086	0.004 kg	ASSUMED	BOM v3.5.2	/lines/18/unit_mass_kg	COTS mass (ESTIMATE 3-6 g (envelope 13.9 × 19, 316L)); CG and inertia of a uniform bounding box
T087	ENG-FD-03: Oxidiser riser 3/8 × 0.035 in 316L seamless, 71 mm developed, 2 S-bends (R >= ...	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/lines/19	tag in BOM_v3.5.2.json v3.5.2: BOM row (CAD parts/ENG-FD-03.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T088	0.0137 kg	DERIVED	BOM v3.5.2	/lines/19/unit_mass_kg	CAD volume x rho(SS316L)
T089	ENG-FD-04: Riser radiation shield: 0.2 mm polished 316 foil + 2 band-clip stand-offs ...	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/lines/20	tag in BOM_v3.5.2.json v3.5.2: BOM row (CAD parts/ENG-FD-04.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T090	0.00538 kg	DERIVED	BOM v3.5.2	/lines/20/unit_mass_kg	CAD volume x rho(SS316L)
T091	ENG-FD-05: Riser-foot TEE (316L): riser socket (top), hose socket (side, 25.9 deg) ...	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/lines/21	tag in BOM_v3.5.2.json v3.5.2: BOM row (CAD parts/ENG-FD-05.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T092	0.04066 kg	DERIVED	BOM v3.5.2	/lines/21/unit_mass_kg	CAD volume x rho(SS316L)
T093	ENG-FD-07: Feed hose 3/8 in convoluted PTFE / 316 braid, rising coil 203 deg, flexible ...	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/lines/22	tag in BOM_v3.5.2.json v3.5.2: BOM row (CAD parts/ENG-FD-07.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]

ID	Value	Class	Source	Key	Source tag / formula
T094	0.0379 kg	TO-VERIFY	BOM v3.5.2	/lines/22/unit_mass_kg	COTS mass (ESTIMATE 0.21 kg/m x 0.180 m (datasheet TO-VERIFY)); CG and inertia of a uniform bounding box
T095	ENG-FD-08: Fixed feed line 3/8 × 0.035 in 316L, 279 mm developed: 3D bend, vertical on ...	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/lines/23	tag in BOM_v3.5.2.json v3.5.2: BOM row (CAD parts/ENG-FD-08.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T096	0.05379 kg	DERIVED	BOM v3.5.2	/lines/23/unit_mass_kg	CAD volume x rho(SS316L)
T097	ENG-FD-08F: Line flange (316L): socket for FD-08, AS568-012 FKM face seal to FS-01A, 2x M4	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/lines/24	tag in BOM_v3.5.2.json v3.5.2: BOM row (CAD parts/ENG-FD-08F.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T098	0.01346 kg	DERIVED	BOM v3.5.2	/lines/24/unit_mass_kg	CAD volume x rho(SS316L)
T099	ENG-FD-09: Fixed hose-end clamp (6061-T6, base + cap, 2x M4 into GM-09 top face, bare ...	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/lines/25	tag in BOM_v3.5.2.json v3.5.2: BOM row (CAD parts/ENG-FD-09.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T100	0.03803 kg	DERIVED	BOM v3.5.2	/lines/25/unit_mass_kg	CAD volume x rho(AL6061-T6)
T101	ENG-FD-11: P-clip 3/8 in, 316 band + silicone cushion, M4 to the airframe wall ...	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/lines/26	tag in BOM_v3.5.2.json v3.5.2: BOM row (CAD parts/ENG-FD-11.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T102	0.004 kg	ASSUMED	BOM v3.5.2	/lines/26/unit_mass_kg	COTS mass (ESTIMATE 3-5 g); CG and inertia of a uniform bounding box
T103	ENG-FD-12F: Tank-line flange (316L) on FS-01B: socket for the tank line (route TO-DEFINE ...	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/lines/27	tag in BOM_v3.5.2.json v3.5.2: BOM row (CAD parts/ENG-FD-12F.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T104	0.0135 kg	DERIVED	BOM v3.5.2	/lines/27/unit_mass_kg	CAD volume x rho(SS316L)
T105	ENG-FS-01A: MV module lower body (6061-T6): outlet chamber 18 × 12.3 (segment side) ...	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/lines/28	tag in BOM_v3.5.2.json v3.5.2: BOM row (CAD parts/ENG-FS-01A.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T106	0.0521 kg	DERIVED	BOM v3.5.2	/lines/28/unit_mass_kg	CAD volume x rho(AL6061-T6)
T107	ENG-FS-01B: MV module upper body (6061-T6): seat web + PCTFE seat counterbore, inlet ...	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/lines/29	tag in BOM_v3.5.2.json v3.5.2: BOM row (CAD parts/ENG-FS-01B.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T108	0.04435 kg	DERIVED	BOM v3.5.2	/lines/29/unit_mass_kg	CAD volume x rho(AL6061-T6)
T109	ENG-FS-02: MV-P poppet: disc 14 × 3 + stem 3 h7, hollow 1.2 (balance hole), M3 into FS-04	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/lines/30	tag in BOM_v3.5.2.json v3.5.2: BOM row (CAD parts/ENG-FS-02.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T110	0.00464 kg	DERIVED	BOM v3.5.2	/lines/30/unit_mass_kg	CAD volume x rho(SS17-4PH-H1150)
T111	ENG-FS-03: MV-P soft seat insert PCTFE: land 11.5-12.5 × 0.3, OD 16 × 2.3 (retention lip ...	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/lines/31	tag in BOM_v3.5.2.json v3.5.2: BOM row (CAD parts/ENG-FS-03.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T112	0.00023 kg	DERIVED	BOM v3.5.2	/lines/31/unit_mass_kg	CAD volume x rho(PCTFE)
T113	ENG-FS-04: MV-P balance piston 11.9 f7 + rod 4 h7 (one piece), seal groove, cross hole ...	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/lines/32	tag in BOM_v3.5.2.json v3.5.2: BOM row (CAD parts/ENG-FS-04.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T114	0.00772 kg	DERIVED	BOM v3.5.2	/lines/32/unit_mass_kg	CAD volume x rho(SS17-4PH-H1150)
T115	ENG-FS-06: MV actuator housing (dry side, 6061-T6): spring pocket, cam slot, 2x MR85 ...	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/lines/33	tag in BOM_v3.5.2.json v3.5.2: BOM row (CAD parts/ENG-FS-06.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T116	0.02918 kg	DERIVED	BOM v3.5.2	/lines/33/unit_mass_kg	CAD volume x rho(AL6061-T6)
T117	ENG-FS-07: MV closing spring EN 10270-3 1.4310, d 1.8 / Dm 12 / n 6.6 active, L0 31.3 ...	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/lines/34	tag in BOM_v3.5.2.json v3.5.2: BOM row (CAD parts/ENG-FS-07.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T118	0.008 kg	ASSUMED	BOM v3.5.2	/lines/34/unit_mass_kg	COTS mass (ESTIMATE wire volume 8 g); CG and inertia of a uniform bounding box
T119	ENG-FS-08: MV follower (17-4PH H900, flat hardened face) on the rod top, M3	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/lines/35	tag in BOM_v3.5.2.json v3.5.2: BOM row (CAD parts/ENG-FS-08.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T120	0.00448 kg	DERIVED	BOM v3.5.2	/lines/35/unit_mass_kg	CAD volume x rho(SS17-4PH-H900)
T121	ENG-FS-09: MV cam (r 6.0 → 8.8 over 45 deg, 15 deg dwell, width 5; closed 25 deg before ...	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/lines/36	tag in BOM_v3.5.2.json v3.5.2: BOM row (CAD parts/ENG-FS-09.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T122	0.00813 kg	DERIVED	BOM v3.5.2	/lines/36/unit_mass_kg	CAD volume x rho(SS17-4PH-H900)
T123	ENG-FS-10: MV opener servo Savox SB-2292SG (Electronics, commonality) on the housing +u ...	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/lines/37	tag in BOM_v3.5.2.json v3.5.2: BOM row (CAD parts/ENG-FS-10.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T124	0.081 kg	DATASHEET	BOM v3.5.2	/lines/37/unit_mass_kg	COTS mass (DATASHEET 81 g (electronics.json)); CG and inertia of a uniform bounding box
T125	ENG-FS-11: MV latch + release + dashpot + OPEN/CLOSED microswitches: RESERVED ENVELOPE ...	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/lines/38	tag in BOM_v3.5.2.json v3.5.2: BOM row (CAD parts/ENG-FS-11.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T126	0.025 kg	ASSUMED	BOM v3.5.2	/lines/38/unit_mass_kg	COTS mass (ESTIMATE reserve 25 g (no design yet)); CG and inertia of a uniform bounding box
T127	ENG-FS-12: S1 back-relief check, insert cartridge 5.5 mm class, crack 0.3-0.7 bar ...	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/lines/39	tag in BOM_v3.5.2.json v3.5.2: BOM row (CAD parts/ENG-FS-12.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T128	0.002 kg	ASSUMED	BOM v3.5.2	/lines/39/unit_mass_kg	COTS mass (ESTIMATE 1-3 g); CG and inertia of a uniform bounding box
T129	ENG-FS-13: Secondary rupture disc 9.5 MPa +/-5 %, reverse-acting, 1/4-18 NPT holder with ...	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/lines/40	tag in BOM_v3.5.2.json v3.5.2: BOM row (CAD parts/ENG-FS-13.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T130	0.045 kg	ASSUMED	BOM v3.5.2	/lines/40/unit_mass_kg	COTS mass (ESTIMATE 35-60 g holder + disc); CG and inertia of a uniform bounding box
T131	ENG-FS-14: Drain port plug 1/8-27 NPT flush socket plug 316 (ground drain/purge; segment ...	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/lines/41	tag in BOM_v3.5.2.json v3.5.2: BOM row (CAD parts/ENG-FS-14.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T132	0.004 kg	ASSUMED	BOM v3.5.2	/lines/41/unit_mass_kg	COTS mass (ESTIMATE 4 g); CG and inertia of a uniform bounding box
T133	ENG-FS-15: MV module mounting bracket (6061-T6): 2x M4 to FS-01A/B, 2x M5 radial into ...	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/lines/42	tag in BOM_v3.5.2.json v3.5.2: BOM row (CAD parts/ENG-FS-15.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T134	0.0206 kg	DERIVED	BOM v3.5.2	/lines/42/unit_mass_kg	CAD volume x rho(AL6061-T6)
T135	ENG-GM-01: Engine collar half +Y (servo 1 lever, 135 deg)	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/lines/43	tag in BOM_v3.5.2.json v3.5.2: BOM row (CAD parts/ENG-GM-01.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T136	0.06779 kg	DERIVED	BOM v3.5.2	/lines/43/unit_mass_kg	CAD volume x rho(AL7075-T6)
T137	ENG-GM-02: Engine collar half -Y (servo 2 lever, 225 deg), mirror of GM-01	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/lines/44	tag in BOM_v3.5.2.json v3.5.2: BOM row (CAD parts/ENG-GM-02.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T138	0.06907 kg	DERIVED	BOM v3.5.2	/lines/44/unit_mass_kg	CAD volume x rho(AL7075-T6)
T139	ENG-GM-03: Gimbal ring (Cardan)	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/lines/45	tag in BOM_v3.5.2.json v3.5.2: BOM row (CAD parts/ENG-GM-03.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T140	0.18988 kg	DERIVED	BOM v3.5.2	/lines/45/unit_mass_kg	CAD volume x rho(AL6061-T6)
T141	ENG-GM-04: Inner pivot pin 8 h5 × 18, needle raceway (AISI 440C, 58-62 HRC)	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/lines/46	tag in BOM_v3.5.2.json v3.5.2: BOM row (CAD parts/ENG-GM-04.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T142	0.00676 kg	DERIVED	BOM v3.5.2	/lines/46/unit_mass_kg	CAD volume x rho(SS440C-HT)
T143	ENG-GM-05: Outer pivot pin 8 h5 × 18.5, needle raceway (AISI 440C, 58-62 HRC)	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/lines/47	tag in BOM_v3.5.2.json v3.5.2: BOM row (CAD parts/ENG-GM-05.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T144	0.00695 kg	DERIVED	BOM v3.5.2	/lines/47/unit_mass_kg	CAD volume x rho(SS440C-HT)
T145	ENG-GM-06: Inner-axis thrust washer 8.1/14 x 1.5 (PTFE composite)	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/lines/48	tag in BOM_v3.5.2.json v3.5.2: BOM row (CAD parts/ENG-GM-06.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]

ID	Value	Class	Source	Key	Source tag / formula
T146	0.0012 kg	ASSUMED	BOM v3.5.2	/lines/48/unit_mass_kg	COTS mass (ESTIMATE annulus x 7.5 g/cm3); CG and inertia of a uniform bounding box
T147	ENG-GM-07: Outer-axis thrust washer 8.1/14 x 1.4 (PTFE composite)	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/lines/49	tag in BOM_v3.5.2.json v3.5.2: BOM row (CAD parts/ENG-GM-07.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T148	0.0011 kg	ASSUMED	BOM v3.5.2	/lines/49/unit_mass_kg	COTS mass (ESTIMATE annulus x 7.5 g/cm3); CG and inertia of a uniform bounding box
T149	ENG-GM-08: Yoke post (outer-axis bearing housing + frame post), x2	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/lines/50	tag in BOM_v3.5.2.json v3.5.2: BOM row (CAD parts/ENG-GM-08.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T150	0.07737 kg	DERIVED	BOM v3.5.2	/lines/50/unit_mass_kg	CAD volume x rho(AL6061-T6)
T151	ENG-GM-09: Upper thrust ring (engine module to airframe, thrust path)	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/lines/51	tag in BOM_v3.5.2.json v3.5.2: BOM row (CAD parts/ENG-GM-09.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T152	0.15203 kg	DERIVED	BOM v3.5.2	/lines/51/unit_mass_kg	CAD volume x rho(AL6061-T6)
T153	ENG-GM-10: Lower ring (stop ring, TVC servo base, frame tie)	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/lines/52	tag in BOM_v3.5.2.json v3.5.2: BOM row (CAD parts/ENG-GM-10.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T154	0.19269 kg	DERIVED	BOM v3.5.2	/lines/52/unit_mass_kg	CAD volume x rho(AL6061-T6)
T155	ENG-GM-11: Conical stop bumper, PU 90A, 8 deg line contact	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/lines/53	tag in BOM_v3.5.2.json v3.5.2: BOM row (CAD parts/ENG-GM-11.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T156	0.0092 kg	DERIVED	BOM v3.5.2	/lines/53/unit_mass_kg	CAD volume x rho(ELASTOMER-PU)
T157	ENG-GM-12: Transport / neutral lock, 2 half rings (REMOVE BEFORE FLIGHT)	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/lines/54	tag in BOM_v3.5.2.json v3.5.2: BOM row (CAD parts/ENG-GM-12.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T158	0.06225 kg	DERIVED	BOM v3.5.2	/lines/54/unit_mass_kg	CAD volume x rho(PETG)
T159	ENG-GM-13: TVC servo 1 bracket (C-frame: top/bottom ear blocks + back rail)	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/lines/55	tag in BOM_v3.5.2.json v3.5.2: BOM row (CAD parts/ENG-GM-13.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T160	0.04893 kg	DERIVED	BOM v3.5.2	/lines/55/unit_mass_kg	CAD volume x rho(AL6061-T6)
T161	ENG-GM-14: TVC servo 2 bracket (mirror of GM-13)	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/lines/56	tag in BOM_v3.5.2.json v3.5.2: BOM row (CAD parts/ENG-GM-14.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T162	0.04893 kg	DERIVED	BOM v3.5.2	/lines/56/unit_mass_kg	CAD volume x rho(AL6061-T6)
T163	ENG-GM-15: TVC servo Savox SB-2292SG (electronics.json)	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/lines/57	tag in BOM_v3.5.2.json v3.5.2: BOM row (CAD parts/ENG-GM-15.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T164	0.081 kg	DATASHEET	BOM v3.5.2	/lines/57/unit_mass_kg	COTS mass (INTERFACE electronics.json v3 (actuators.tvc_A.mass_kg)); CG and inertia of a uniform bounding box
T165	ENG-GM-16: TVC horn r 11.1 mm, 25T clamp (7075-T6)	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/lines/58	tag in BOM_v3.5.2.json v3.5.2: BOM row (CAD parts/ENG-GM-16.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T166	0.00218 kg	DERIVED	BOM v3.5.2	/lines/58/unit_mass_kg	CAD volume x rho(AL7075-T6)
T167	ENG-GM-17: TVC pushrod dog-bone 42.3 mm between eyes (7075-T6), fixed length	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/lines/59	tag in BOM_v3.5.2.json v3.5.2: BOM row (CAD parts/ENG-GM-17.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T168	0.00421 kg	DERIVED	BOM v3.5.2	/lines/59/unit_mass_kg	CAD volume x rho(AL7075-T6)
T169	ENG-GM-18: Lever pivot stud: flange 7.5 × 1.5 (= washer), 5 h7 shoulder x 7.5, M4 × 4.5 ...	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/lines/60	tag in BOM_v3.5.2.json v3.5.2: BOM row (CAD parts/ENG-GM-18.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T170	0.002 kg	DERIVED	BOM v3.5.2	/lines/60/unit_mass_kg	CAD volume x rho(SS17-4PH-H1150)
T171	ENG-GM-19: Joint spacer washer 5.1 × 7.5 x 1.5 (17-4PH H1150), ground faces	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/lines/61	tag in BOM_v3.5.2.json v3.5.2: BOM row (CAD parts/ENG-GM-19.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T172	0.00028 kg	DERIVED	BOM v3.5.2	/lines/61/unit_mass_kg	CAD volume x rho(SS17-4PH-H1150)
T173	ENG-GM-20: Horn-joint shoulder screw: head 8.5 × 4, 5 h7 × 10 shoulder, M4 × 3.5 (17-4PH)	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/lines/62	tag in BOM_v3.5.2.json v3.5.2: BOM row (CAD parts/ENG-GM-20.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T174	0.00352 kg	DERIVED	BOM v3.5.2	/lines/62/unit_mass_kg	CAD volume x rho(SS17-4PH-H1150)
T175	ENG-GR-01: Fuel grain 1/2: ABS 53.5 OD x 115.0, straight centred port 25 mm (grain v2 ...	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/lines/63	tag in BOM_v3.5.2.json v3.5.2: BOM row (CAD parts/ENG-GR-01.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T176	0.21219 kg	DERIVED	BOM v3.5.2	/lines/63/unit_mass_kg	CAD volume x rho(ABS)
T177	ENG-GR-02: Fuel grain 2/2: ABS 53.5 OD x 115.0, straight centred port 25 mm (grain v2 ...	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/lines/64	tag in BOM_v3.5.2.json v3.5.2: BOM row (CAD parts/ENG-GR-02.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T178	0.21217 kg	DERIVED	BOM v3.5.2	/lines/64/unit_mass_kg	CAD volume x rho(ABS)
T179	ENG-HW-20: Drawn-cup needle roller bearing HK0808 (Fw 8 x D 12 x C 8), open ends	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/lines/65	tag in BOM_v3.5.2.json v3.5.2: BOM row (CAD parts/ENG-HW-20.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T180	0.003 kg	DATASHEET	BOM v3.5.2	/lines/65/unit_mass_kg	COTS mass (DATASHEET Koyo HK0808 0.003 kg (koyo.jtekt.co.jp product page, accessed 2026-10-13)); CG and inertia of a uniform bounding box
T181	ENG-HW-21: Retaining washer DIN 6799-6 (shaft 7-9 mm), spring steel	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/lines/66	tag in BOM_v3.5.2.json v3.5.2: BOM row (CAD parts/ENG-HW-21.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T182	0.0003 kg	ASSUMED	BOM v3.5.2	/lines/66/unit_mass_kg	COTS mass (ASSUMED 0.2-0.4 g); CG and inertia of a uniform bounding box
T183	ENG-HW-22: Spherical plain bearing GE 5 E (steel/steel) - outer ring (pressed in the ...	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/lines/67	tag in BOM_v3.5.2.json v3.5.2: BOM row (CAD parts/ENG-HW-22.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T184	0.0025 kg	DERIVED	BOM v3.5.2	/lines/67/unit_mass_kg	CAD volume x rho(STEEL)
T185	ENG-HW-25: Horn axial screw ISO 10642 M3 x 6 (countersunk, 10.9), Loctite 243	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/lines/68	tag in BOM_v3.5.2.json v3.5.2: BOM row (CAD parts/ENG-HW-25.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T186	0.0005 kg	ASSUMED	BOM v3.5.2	/lines/68/unit_mass_kg	COTS mass (ASSUMED 0.4-0.6 g); CG and inertia of a uniform bounding box
T187	ENG-IG-01: Torch igniter (N2O/propane, spark) - ENVELOPE 20 × 45 mm + M10×1 nose + 2 ...	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/lines/69	tag in BOM_v3.5.2.json v3.5.2: BOM row (CAD parts/ENG-IG-01.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T188	0.07 kg	ASSUMED	BOM v3.5.2	/lines/69/unit_mass_kg	COTS mass (ASSUMED body 50 g + plug 10 g (ignition_v2.json) + 2 fittings 10 g); CG and inertia of a uniform bounding box
T189	ENG-IG-02: Torch inlet check valve, cartridge insert in the torch inlet bore (N2O and ...	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/lines/70	tag in BOM_v3.5.2.json v3.5.2: BOM row (CAD parts/ENG-IG-02.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T190	0.002 kg	ASSUMED	BOM v3.5.2	/lines/70/unit_mass_kg	COTS mass (ASSUMED 1-3 g cartridge); CG and inertia of a uniform bounding box
T191	ENG-NZ-01: Graphite throat insert (Dt 13.72, De 19.41, OD 44, aft spigot 30)	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/lines/71	tag in BOM_v3.5.2.json v3.5.2: BOM row (CAD parts/ENG-NZ-01.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T192	0.04685 kg	DERIVED	BOM v3.5.2	/lines/71/unit_mass_kg	CAD volume x rho(GRAPHITE-ISO)
T193	ENG-NZ-02: Insert insulating sleeve (carbon-phenolic 50/44, 3 mm)	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/lines/72	tag in BOM_v3.5.2.json v3.5.2: BOM row (CAD parts/ENG-NZ-02.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T194	0.01196 kg	DERIVED	BOM v3.5.2	/lines/72/unit_mass_kg	CAD volume x rho(CARBON-PHENOLIC)
T195	ENG-NZ-03: Nozzle-joint hot layer J1: carbon-phenolic washer 30.6/49.8 × 2.0	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/lines/73	tag in BOM_v3.5.2.json v3.5.2: BOM row (CAD parts/ENG-NZ-03.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T196	0.00352 kg	DERIVED	BOM v3.5.2	/lines/73/unit_mass_kg	CAD volume x rho(CARBON-PHENOLIC)

ID	Value	Class	Source	Key	Source tag / formula
T197	ENG-NZ-04: Nozzle-joint lip-face seal J1: flexible graphite ring 30.6/49.8 × 0.5 ...	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/lines/74	tag in BOM_v3.5.2.json v3.5.2: BOM row (CAD parts/ENG-NZ-04.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T198	0.00068 kg	DERIVED	BOM v3.5.2	/lines/74/unit_mass_kg	CAD volume x rho(FLEX-GRAPHITE-OI)
T199	ENG-NZ-06: Nozzle-joint load-spreading foil (v3.5.2 N-F2): flexible graphite ring ...	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/lines/75	tag in BOM_v3.5.2.json v3.5.2: BOM row (CAD parts/ENG-NZ-06.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T200	0.00019 kg	DERIVED	BOM v3.5.2	/lines/75/unit_mass_kg	CAD volume x rho(FLEX-GRAPHITE-TH)
T201	ENG-PD-01: Pintle drive frame, 2 pieces: A (-Y foot + bridge + leg + servo plate + cap) ...	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/lines/76	tag in BOM_v3.5.2.json v3.5.2: BOM row (CAD parts/ENG-PD-01.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T202	0.03886 kg	DERIVED	BOM v3.5.2	/lines/76/unit_mass_kg	CAD volume x rho(AL6061-T6)
T203	ENG-PD-02: Pintle crank housing (bearing seats A 18 and B 19, crank cavity, spring ...	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/lines/77	tag in BOM_v3.5.2.json v3.5.2: BOM row (CAD parts/ENG-PD-02.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T204	0.01983 kg	DERIVED	BOM v3.5.2	/lines/77/unit_mass_kg	CAD volume x rho(AL6061-T6)
T205	ENG-PD-03: Crank-housing cover with encoder PCB pocket	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/lines/78	tag in BOM_v3.5.2.json v3.5.2: BOM row (CAD parts/ENG-PD-03.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T206	0.00261 kg	DERIVED	BOM v3.5.2	/lines/78/unit_mass_kg	CAD volume x rho(AL6061-T6)
T207	ENG-PD-04: Pintle crankshaft (throw r 3.0, pin 3, journals 12 / 10, dog end, spring hub ...	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/lines/79	tag in BOM_v3.5.2.json v3.5.2: BOM row (CAD parts/ENG-PD-04.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T208	0.01249 kg	DERIVED	BOM v3.5.2	/lines/79/unit_mass_kg	CAD volume x rho(SS17-4PH-H900)
T209	ENG-PD-05: Dog hub on the HV69 6 mm 25T spline (one-sided drive tooth 60 deg)	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/lines/80	tag in BOM_v3.5.2.json v3.5.2: BOM row (CAD parts/ENG-PD-05.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T210	0.00061 kg	DERIVED	BOM v3.5.2	/lines/80/unit_mass_kg	CAD volume x rho(AL7075-T6)
T211	ENG-PD-06: Pintle connecting link 28 mm centres, eyes OD 8 × 3.8 with 4.5 bores, shank 4 ...	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/lines/81	tag in BOM_v3.5.2.json v3.5.2: BOM row (CAD parts/ENG-PD-06.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T212	0.00394 kg	DERIVED	BOM v3.5.2	/lines/81/unit_mass_kg	CAD volume x rho(SS17-4PH-H1150)
T213	ENG-PD-07: Pintle carriage: needle clamp arm (M3 stem thread) + link clevis + plate on ...	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/lines/82	tag in BOM_v3.5.2.json v3.5.2: BOM row (CAD parts/ENG-PD-07.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T214	0.01003 kg	DERIVED	BOM v3.5.2	/lines/82/unit_mass_kg	CAD volume x rho(AL6061-T6)
T215	ENG-PD-08: Miniature linear guide block HIWIN MGN09H (C0 4020 N, M0Y 18.6 N m)	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/lines/83	tag in BOM_v3.5.2.json v3.5.2: BOM row (CAD parts/ENG-PD-08.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T216	0.03 kg	TO-VERIFY	BOM v3.5.2	/lines/83/unit_mass_kg	COTS mass (DATASHEET HIWIN MG catalogue 2023 (copy) 0.03 kg incl. rail share? - block only TO-VERIFY); CG and inertia of a uniform bounding box
T217	ENG-PD-09: Spiral (clock) return spring, 0.25 N m at closed + 0.12 N m/rad (+/-10 %) ...	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/lines/84	tag in BOM_v3.5.2.json v3.5.2: BOM row (CAD parts/ENG-PD-09.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T218	0.0055 kg	ASSUMED	BOM v3.5.2	/lines/84/unit_mass_kg	COTS mass (ESTIMATE 5-6 g (strip 4.5 x ~0.71, ~206 mm active, 1.4310 +C1900 class: calcs/pd09_spring_fit_v1.json; v3.4 erratum: the earlier 4.5 × 0.25 strip note gave 9.7 GPa at 0.455 N m)); CG and ...
T219	ENG-PD-10: Radiation shield under the HV69: 0.5 mm polished aluminium L-sheet on 2 G11 ...	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/lines/85	tag in BOM_v3.5.2.json v3.5.2: BOM row (CAD parts/ENG-PD-10.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T220	0.0007 kg	DERIVED	BOM v3.5.2	/lines/85/unit_mass_kg	CAD volume x rho(AL6061-T6)
T221	ENG-PD-11: Isolator washer G11 7/3.4 × 2 (frame feet to head top)	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/lines/86	tag in BOM_v3.5.2.json v3.5.2: BOM row (CAD parts/ENG-PD-11.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T222	0.00011 kg	DERIVED	BOM v3.5.2	/lines/86/unit_mass_kg	CAD volume x rho(G11)
T223	ENG-PD-12: G11 pad 1.5 mm under the HV69 ears (window for the case)	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/lines/87	tag in BOM_v3.5.2.json v3.5.2: BOM row (CAD parts/ENG-PD-12.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T224	0.00086 kg	DERIVED	BOM v3.5.2	/lines/87/unit_mass_kg	CAD volume x rho(G11)
T225	ENG-PD-13: Pintle servo MKS HV69 (29 × 13 x 28.7, 21 g, 6 mm 25T spline), case below the ...	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/lines/88	tag in BOM_v3.5.2.json v3.5.2: BOM row (CAD parts/ENG-PD-13.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T226	0.021 kg	DATASHEET	BOM v3.5.2	/lines/88/unit_mass_kg	COTS mass (DATASHEET 21 g (Electronics pintle_mini_servo_v1, retailer copies of MKS data)); CG and inertia of a uniform bounding box
T227	ENG-PD-14: Thin-section ball bearing 12×18×4 (NSK/NMB 6701 class, 'NB701' in ...	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/lines/89	tag in BOM_v3.5.2.json v3.5.2: BOM row (CAD parts/ENG-PD-14.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T228	0.0031 kg	ASSUMED	BOM v3.5.2	/lines/89/unit_mass_kg	COTS mass (ASSUMED catalogue class 3 g (6701)); CG and inertia of a uniform bounding box
T229	ENG-PD-15: Deep-groove ball bearing 6800-2Z 10×19×5, C0r 840 N	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/lines/90	tag in BOM_v3.5.2.json v3.5.2: BOM row (CAD parts/ENG-PD-15.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T230	0.0055 kg	ASSUMED	BOM v3.5.2	/lines/90/unit_mass_kg	COTS mass (ASSUMED catalogue class 5.5 g (61800)); CG and inertia of a uniform bounding box
T231	ENG-PD-16: Pin 3 g6 × 12 (17-4PH H900) with e-clip grooves (crank pin bush / carriage ...	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/lines/91	tag in BOM_v3.5.2.json v3.5.2: BOM row (CAD parts/ENG-PD-16.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T232	0.00066 kg	DERIVED	BOM v3.5.2	/lines/91/unit_mass_kg	CAD volume x rho(SS17-4PH-H900)
T233	ENG-PD-17: PTFE-lined steel-backed bush 3 x 4.5 × 3.8 (DU class)	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/lines/92	tag in BOM_v3.5.2.json v3.5.2: BOM row (CAD parts/ENG-PD-17.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T234	0.0003 kg	ASSUMED	BOM v3.5.2	/lines/92/unit_mass_kg	COTS mass (ESTIMATE 0.3 g); CG and inertia of a uniform bounding box
T235	ENG-PD-18: Encoder magnet NdFeB N35 diametral 6 × 1	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/lines/93	tag in BOM_v3.5.2.json v3.5.2: BOM row (CAD parts/ENG-PD-18.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T236	0.00021 kg	DERIVED	BOM v3.5.2	/lines/93/unit_mass_kg	CAD volume x rho(NDFEB)
T237	ENG-PD-19: Miniature rail HIWIN MGNR09R, length 46.1 mm (cut to length)	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/lines/94	tag in BOM_v3.5.2.json v3.5.2: BOM row (CAD parts/ENG-PD-19.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T238	0.0175 kg	DATASHEET	BOM v3.5.2	/lines/94/unit_mass_kg	COTS mass (DATASHEET 0.38 kg/m (HIWIN MG catalogue copy)); CG and inertia of a uniform bounding box
T239	ENG-PD-20: G11 stand-off 5/2.2 × 2 (shield to servo plate)	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/lines/95	tag in BOM_v3.5.2.json v3.5.2: BOM row (CAD parts/ENG-PD-20.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T240	6e-05 kg	DERIVED	BOM v3.5.2	/lines/95/unit_mass_kg	CAD volume x rho(G11)
T241	ENG-TC-01: Combustion case tube	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/lines/96	tag in BOM_v3.5.2.json v3.5.2: BOM row (CAD parts/ENG-TC-01.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T242	0.46134 kg	DERIVED	BOM v3.5.2	/lines/96/unit_mass_kg	CAD volume x rho(AL6061-T6)
T243	ENG-TC-02: Injector head / forward closure (pintle seat bore, manifold, Pc and torch ...	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/lines/97	tag in BOM_v3.5.2.json v3.5.2: BOM row (CAD parts/ENG-TC-02.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T244	0.43723 kg	DERIVED	BOM v3.5.2	/lines/97/unit_mass_kg	CAD volume x rho(AL6061-T6)
T245	ENG-TC-03: Pintle seat insert (5.5 mm sharp-edged seat, 37.5 deg diffuser)	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/lines/98	tag in BOM_v3.5.2.json v3.5.2: BOM row (CAD parts/ENG-TC-03.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T246	0.01488 kg	DERIVED	BOM v3.5.2	/lines/98/unit_mass_kg	CAD volume x rho(GLIDCOP-AL15)

ID	Value	Class	Source	Key	Source tag / formula
T247	ENG-TC-04: Bonnet: stem seal + guide housing, E1-M inlet boss (8 mm transfer ...	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/lines/99	tag in BOM_v3.5.2.json v3.5.2: BOM row (CAD parts/ENG-TC-04.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T248	0.05354 kg	DERIVED	BOM v3.5.2	/lines/99/unit_mass_kg	CAD volume x rho(AL6061-T6)
T249	ENG-TC-05: Pintle needle (15 deg metering cone, 6 mm body, 3 mm stem)	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/lines/100	tag in BOM_v3.5.2.json v3.5.2: BOM row (CAD parts/ENG-TC-05.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T250	0.01663 kg	DERIVED	BOM v3.5.2	/lines/100/unit_mass_kg	CAD volume x rho(SS17-4PH-H1150)
T251	ENG-TC-06: Pre-combustion chamber insulator (phenolic, bore 49.9, wall 3.60 mm, 30 mm ...	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/lines/101	tag in BOM_v3.5.2.json v3.5.2: BOM row (CAD parts/ENG-TC-06.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T252	0.02445 kg	DERIVED	BOM v3.5.2	/lines/101/unit_mass_kg	CAD volume x rho(PHENOLIC)
T253	ENG-TC-07: Grain liner (phenolic tube 57.10/53.95)	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/lines/102	tag in BOM_v3.5.2.json v3.5.2: BOM row (CAD parts/ENG-TC-07.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T254	0.08531 kg	DERIVED	BOM v3.5.2	/lines/102/unit_mass_kg	CAD volume x rho(PHENOLIC)
T255	ENG-TC-08: Post-chamber + convergent insulator (phenolic, 3.50 mm over the post-chamber ...	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/lines/103	tag in BOM_v3.5.2.json v3.5.2: BOM row (CAD parts/ENG-TC-08.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T256	0.0515 kg	DERIVED	BOM v3.5.2	/lines/103/unit_mass_kg	CAD volume x rho(PHENOLIC)
T257	ENG-TC-09: Aft section v3.2: post-chamber barrel + nozzle holder + retaining lip (IN718 ...	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/lines/104	tag in BOM_v3.5.2.json v3.5.2: BOM row (CAD parts/ENG-TC-09.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T258	0.51434 kg	DERIVED	BOM v3.5.2	/lines/104/unit_mass_kg	CAD volume x rho(IN718)
T259	ENG-TC-10: Injector-face insulator disc, phenolic 2 mm	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/lines/105	tag in BOM_v3.5.2.json v3.5.2: BOM row (CAD parts/ENG-TC-10.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T260	0.00468 kg	DERIVED	BOM v3.5.2	/lines/105/unit_mass_kg	CAD volume x rho(PHENOLIC)
T261	ENG-TC-11: Stem guide bushing 5/3 × 4 (PTFE + 25 % glass)	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/lines/106	tag in BOM_v3.5.2.json v3.5.2: BOM row (CAD parts/ENG-TC-11.step) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T262	0.00011 kg	DERIVED	BOM v3.5.2	/lines/106/unit_mass_kg	CAD volume x rho(PTFE)
T263	ENG-BL-06N: bleed vent non-propulsive opposed tee outside the airframe wall (on ENG-BL-06)	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/provision_lines/0	tag in BOM_v3.5.2.json v3.5.2: BOM row (CAD) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T264	MISSING	MISSING	BOM v3.5.2	/provision_lines/0/unit_mass_kg	ESTIMATE 10-20 g (no datasheet)
T265	ENG-TK-OUT: N2O tank outlet on the aft bulkhead (OI-32): position, bore, sump / dip tube ...	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/provision_lines/1	tag in BOM_v3.5.2.json v3.5.2: BOM row (CAD) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T266	MISSING	MISSING	BOM v3.5.2	/provision_lines/1/unit_mass_kg	MISSING (tank not in CAD)
T267	ENG-NZ-05: J1b contingency: insulating layer 0.5 mm phlogopite mica between the NZ-03 ...	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/contingency_lines/0	tag in BOM_v3.5.2.json v3.5.2: BOM row (CAD) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T268	TO-DEFINE	MISSING	BOM v3.5.2	/contingency_lines/0/unit_mass_kg	not fitted
T269	ENG-TC-12: J-FWD hole plug set, 10 plugs + band, O-ring passage aid	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/gse_tool_lines/0	tag in BOM_v3.5.2.json v3.5.2: BOM row (CAD) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T270	MISSING	MISSING	BOM v3.5.2	/gse_tool_lines/0/unit_mass_kg	GSE: not flight mass
T271	ENG-TC-13: J-FWD registration shoulder screw pair (C4 registration)	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/gse_tool_lines/1	tag in BOM_v3.5.2.json v3.5.2: BOM row (CAD) [row trace: name, qty, groups, material_name, process, unit_mass_kg, mass_source, unit_price, supplier]
T272	MISSING	MISSING	BOM v3.5.2	/gse_tool_lines/1/unit_mass_kg	GSE: not flight mass
T273	ENG-HW-101: J-FWD forward-closure radial screw ENG-HW-101-IN718 (case TC-01 to head plug ...	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/hardware/lines/0	tag in BOM_v3.5.2.json v3.5.2: BOM hardware row (standard part, not modelled) [row trace: name, qty, host, size, grade, torque, tightening method, unit_mass_kg, mass_source, unit_price]
T274	0.002179 kg	CALC	BOM v3.5.2	/hardware/lines/0/unit_mass_kg	CALC (hardware_v351 ISO 7380-1 M5 rule: nominal head AS RECALLED + d2 shank x L 10.7, plus the pilot ring 5.15 × 0.30; x IN718 8190 kg/m3 LITERATURE, engmat 'Special Metals 718'); the ...
T275	ENG-HW-102: J-BON bonnet screw (TC-04 on TC-02)	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/hardware/lines/1	tag in BOM_v3.5.2.json v3.5.2: BOM hardware row (standard part, not modelled) [row trace: name, qty, host, size, grade, torque, tightening method, unit_mass_kg, mass_source, unit_price]
T276	0.002055 kg	CALC	BOM v3.5.2	/hardware/lines/1/unit_mass_kg	CALC (standard nominal geometry AS RECALLED x engmat density); DATASHEET TO-FETCH
T277	ENG-HW-103: FD-01 inlet block screw, lower pair (FD-01 ears into the TC-02 flat r 29.5)	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/hardware/lines/2	tag in BOM_v3.5.2.json v3.5.2: BOM hardware row (standard part, not modelled) [row trace: name, qty, host, size, grade, torque, tightening method, unit_mass_kg, mass_source, unit_price]
T278	0.002372 kg	CALC	BOM v3.5.2	/hardware/lines/2/unit_mass_kg	CALC (standard nominal geometry AS RECALLED x engmat density); DATASHEET TO-FETCH
T279	ENG-HW-103B: FD-01 inlet block screw, upper (FD-01 body into the TC-02 flat r 29.5 ...	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/hardware/lines/3	tag in BOM_v3.5.2.json v3.5.2: BOM hardware row (standard part, not modelled) [row trace: name, qty, host, size, grade, torque, tightening method, unit_mass_kg, mass_source, unit_price]
T280	0.002056 kg	CALC	BOM v3.5.2	/hardware/lines/3/unit_mass_kg	CALC (standard nominal geometry AS RECALLED x engmat density); DATASHEET TO-FETCH
T281	ENG-HW-104: BL-01 bleed tap block screw (into the TC-04 boss)	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/hardware/lines/4	tag in BOM_v3.5.2.json v3.5.2: BOM hardware row (standard part, not modelled) [row trace: name, qty, host, size, grade, torque, tightening method, unit_mass_kg, mass_source, unit_price]
T282	0.000655 kg	CALC	BOM v3.5.2	/hardware/lines/4/unit_mass_kg	CALC (standard nominal geometry AS RECALLED x engmat density); DATASHEET TO-FETCH
T283	ENG-HW-105: PD-01 frame foot screw through the G11 isolator washer PD-11 (into TC-02)	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/hardware/lines/5	tag in BOM_v3.5.2.json v3.5.2: BOM hardware row (standard part, not modelled) [row trace: name, qty, host, size, grade, torque, tightening method, unit_mass_kg, mass_source, unit_price]
T284	0.0010400000000000001 kg	CALC	BOM v3.5.2	/hardware/lines/5/unit_mass_kg	CALC (standard nominal geometry AS RECALLED x engmat density); DATASHEET TO-FETCH
T285	ENG-HW-106: PD-02 crank housing to the PD-01 shelf and cap	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/hardware/lines/6	tag in BOM_v3.5.2.json v3.5.2: BOM hardware row (standard part, not modelled) [row trace: name, qty, host, size, grade, torque, tightening method, unit_mass_kg, mass_source, unit_price]
T286	0.0008629999999999999 kg	CALC	BOM v3.5.2	/hardware/lines/6/unit_mass_kg	CALC (standard nominal geometry AS RECALLED x engmat density); DATASHEET TO-FETCH
T287	ENG-HW-107: HV69 pintle servo ear screw through the G11 pad PD-12 (into PD-01)	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/hardware/lines/7	tag in BOM_v3.5.2.json v3.5.2: BOM hardware row (standard part, not modelled) [row trace: name, qty, host, size, grade, torque, tightening method, unit_mass_kg, mass_source, unit_price]
T288	0.000314 kg	CALC	BOM v3.5.2	/hardware/lines/7/unit_mass_kg	CALC (standard nominal geometry AS RECALLED x engmat density); DATASHEET TO-FETCH
T289	ENG-HW-108: MGNR09R rail screw (into the PD-01 rail plate)	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/hardware/lines/8	tag in BOM_v3.5.2.json v3.5.2: BOM hardware row (standard part, not modelled) [row trace: name, qty, host, size, grade, torque, tightening method, unit_mass_kg, mass_source, unit_price]
T290	0.0008629999999999999 kg	CALC	BOM v3.5.2	/hardware/lines/8/unit_mass_kg	CALC (standard nominal geometry AS RECALLED x engmat density); DATASHEET TO-FETCH
T291	ENG-HW-109: PD-07 carriage plate to the MGN09H block	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/hardware/lines/9	tag in BOM_v3.5.2.json v3.5.2: BOM hardware row (standard part, not modelled) [row trace: name, qty, host, size, grade, torque, tightening method, unit_mass_kg, mass_source, unit_price]
T292	0.0007740000000000001 kg	CALC	BOM v3.5.2	/hardware/lines/9/unit_mass_kg	CALC (standard nominal geometry AS RECALLED x engmat density); DATASHEET TO-FETCH
T293	ENG-HW-110: PD-03 cover screw (into PD-02 end face)	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/hardware/lines/10	tag in BOM_v3.5.2.json v3.5.2: BOM hardware row (standard part, not modelled) [row trace: name, qty, host, size, grade, torque, tightening method, unit_mass_kg, mass_source, unit_price]
T294	0.000277 kg	CALC	BOM v3.5.2	/hardware/lines/10/unit_mass_kg	CALC (standard nominal geometry AS RECALLED x engmat density); DATASHEET TO-FETCH
T295	ENG-HW-111: PD-05 dog-hub clamp screw	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/hardware/lines/11	tag in BOM_v3.5.2.json v3.5.2: BOM hardware row (standard part, not modelled) [row trace: name, qty, host, size, grade, torque, tightening method, unit_mass_kg, mass_source, unit_price]

ID	Value	Class	Source	Key	Source tag / formula
T296	0.000277 kg	CALC	BOM v3.5.2	/hardware/lines/11/unit_mass_kg	CALC (standard nominal geometry AS RECALLED x engmat density); DATASHEET TO-FETCH
T297	ENG-HW-112: PD-10 radiation shield screw through the G11 stand-offs	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/hardware/lines/12	tag in BOM_v3.5.2.json v3.5.2: BOM hardware row (standard part, not modelled) [row trace: name, qty, host, size, grade, torque, tightening method, unit_mass_kg, mass_source, unit_price]
T298	0.000314 kg	CALC	BOM v3.5.2	/hardware/lines/12/unit_mass_kg	CALC (standard nominal geometry AS RECALLED x engmat density); DATASHEET TO-FETCH
T299	ENG-HW-113: Collar clamp screw (GM-01 clearance + counterbore into GM-02 tapped)	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/hardware/lines/13	tag in BOM_v3.5.2.json v3.5.2: BOM hardware row (standard part, not modelled) [row trace: name, qty, host, size, grade, torque, tightening method, unit_mass_kg, mass_source, unit_price]
T300	0.0007970000000000001 kg	CALC	BOM v3.5.2	/hardware/lines/13/unit_mass_kg	CALC (standard nominal geometry AS RECALLED x engmat density); DATASHEET TO-FETCH
T301	ENG-HW-114: Horn radial pinch-clamp screw	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/hardware/lines/14	tag in BOM_v3.5.2.json v3.5.2: BOM hardware row (standard part, not modelled) [row trace: name, qty, host, size, grade, torque, tightening method, unit_mass_kg, mass_source, unit_price]
T302	0.0009459999999999999 kg	CALC	BOM v3.5.2	/hardware/lines/14/unit_mass_kg	CALC (standard nominal geometry AS RECALLED x engmat density); DATASHEET TO-FETCH
T303	ENG-HW-115: EL-03 encoder holder screw (into the GM-03 ring)	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/hardware/lines/15	tag in BOM_v3.5.2.json v3.5.2: BOM hardware row (standard part, not modelled) [row trace: name, qty, host, size, grade, torque, tightening method, unit_mass_kg, mass_source, unit_price]
T304	0.000277 kg	CALC	BOM v3.5.2	/hardware/lines/15/unit_mass_kg	CALC (standard nominal geometry AS RECALLED x engmat density); DATASHEET TO-FETCH
T305	ENG-HW-120: Yoke post to the upper thrust ring GM-09 (counterbored from the top)	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/hardware/lines/16	tag in BOM_v3.5.2.json v3.5.2: BOM hardware row (standard part, not modelled) [row trace: name, qty, host, size, grade, torque, tightening method, unit_mass_kg, mass_source, unit_price]
T306	0.0035059999999999996 kg	CALC	BOM v3.5.2	/hardware/lines/16/unit_mass_kg	CALC (standard nominal geometry AS RECALLED x engmat density); DATASHEET TO-FETCH
T307	ENG-HW-121: Yoke post to the lower ring GM-10 (counterbored from below)	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/hardware/lines/17	tag in BOM_v3.5.2.json v3.5.2: BOM hardware row (standard part, not modelled) [row trace: name, qty, host, size, grade, torque, tightening method, unit_mass_kg, mass_source, unit_price]
T308	0.001941 kg	CALC	BOM v3.5.2	/hardware/lines/17/unit_mass_kg	CALC (A4-70 CALC unit mass x 4430/8000); DATASHEET TO-FETCH
T309	ENG-HW-122: Dowel pin 4 m6 × 16, hardened (posts to rings)	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/hardware/lines/18	tag in BOM_v3.5.2.json v3.5.2: BOM hardware row (standard part, not modelled) [row trace: name, qty, host, size, grade, torque, tightening method, unit_mass_kg, mass_source, unit_price]
T310	0.001578 kg	CALC	BOM v3.5.2	/hardware/lines/18/unit_mass_kg	CALC (cylinder x 7850)
T311	ENG-HW-123: TVC servo bracket to the lower ring GM-10 (counterbored from below)	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/hardware/lines/19	tag in BOM_v3.5.2.json v3.5.2: BOM hardware row (standard part, not modelled) [row trace: name, qty, host, size, grade, torque, tightening method, unit_mass_kg, mass_source, unit_price]
T312	0.001138 kg	CALC	BOM v3.5.2	/hardware/lines/19/unit_mass_kg	CALC (A4-70 CALC unit mass x 4430/8000); DATASHEET TO-FETCH
T313	ENG-HW-124: TVC servo ear screw (servo to bracket GM-13/14)	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/hardware/lines/20	tag in BOM_v3.5.2.json v3.5.2: BOM hardware row (standard part, not modelled) [row trace: name, qty, host, size, grade, torque, tightening method, unit_mass_kg, mass_source, unit_price]
T314	0.001897 kg	CALC	BOM v3.5.2	/hardware/lines/20/unit_mass_kg	CALC (standard nominal geometry AS RECALLED x engmat density); DATASHEET TO-FETCH
T315	ENG-HW-125: Airframe screw into the upper thrust ring GM-09 (radial, 8x) - ICD-AIRFRAME	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/hardware/lines/21	tag in BOM_v3.5.2.json v3.5.2: BOM hardware row (standard part, not modelled) [row trace: name, qty, host, size, grade, torque, tightening method, unit_mass_kg, mass_source, unit_price]
T316	0.002092 kg	CALC	BOM v3.5.2	/hardware/lines/21/unit_mass_kg	CALC (standard nominal geometry AS RECALLED x engmat density); DATASHEET TO-FETCH
T317	ENG-HW-126: Airframe screw into the lower ring GM-10 (radial, 8x) - ICD-AIRFRAME	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/hardware/lines/22	tag in BOM_v3.5.2.json v3.5.2: BOM hardware row (standard part, not modelled) [row trace: name, qty, host, size, grade, torque, tightening method, unit_mass_kg, mass_source, unit_price]
T318	0.002092 kg	CALC	BOM v3.5.2	/hardware/lines/22/unit_mass_kg	CALC (standard nominal geometry AS RECALLED x engmat density); DATASHEET TO-FETCH
T319	ENG-HW-127: FD-09 hose-end clamp foot screw into GM-09 (with helicoil)	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/hardware/lines/23	tag in BOM_v3.5.2.json v3.5.2: BOM hardware row (standard part, not modelled) [row trace: name, qty, host, size, grade, torque, tightening method, unit_mass_kg, mass_source, unit_price]
T320	0.002055 kg	CALC	BOM v3.5.2	/hardware/lines/23/unit_mass_kg	CALC (standard nominal geometry AS RECALLED x engmat density); DATASHEET TO-FETCH
T321	ENG-HW-128: Wire thread insert M4 × 1.5 D free-running, 304 (Helicoil class)	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/hardware/lines/24	tag in BOM_v3.5.2.json v3.5.2: BOM hardware row (standard part, not modelled) [row trace: name, qty, host, size, grade, torque, tightening method, unit_mass_kg, mass_source, unit_price]
T322	MISSING	MISSING	BOM v3.5.2	/hardware/lines/24/unit_mass_kg	TO-FETCH (supplier sheet); inside the allowance
T323	ENG-HW-129: FD-09 clamp cap screw	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/hardware/lines/25	tag in BOM_v3.5.2.json v3.5.2: BOM hardware row (standard part, not modelled) [row trace: name, qty, host, size, grade, torque, tightening method, unit_mass_kg, mass_source, unit_price]
T324	0.001413 kg	CALC	BOM v3.5.2	/hardware/lines/25/unit_mass_kg	CALC (standard nominal geometry AS RECALLED x engmat density); DATASHEET TO-FETCH
T325	ENG-HW-130: P-clip FD-11 screw into the airframe insert / rivet nut (ICD)	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/hardware/lines/26	tag in BOM_v3.5.2.json v3.5.2: BOM hardware row (standard part, not modelled) [row trace: name, qty, host, size, grade, torque, tightening method, unit_mass_kg, mass_source, unit_price]
T326	0.001019 kg	CALC	BOM v3.5.2	/hardware/lines/26/unit_mass_kg	CALC (standard nominal geometry AS RECALLED x engmat density); DATASHEET TO-FETCH
T327	ENG-HW-131: FD-08F line flange screw (into FS-01A)	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/hardware/lines/27	tag in BOM_v3.5.2.json v3.5.2: BOM hardware row (standard part, not modelled) [row trace: name, qty, host, size, grade, torque, tightening method, unit_mass_kg, mass_source, unit_price]
T328	0.002055 kg	CALC	BOM v3.5.2	/hardware/lines/27/unit_mass_kg	CALC (standard nominal geometry AS RECALLED x engmat density); DATASHEET TO-FETCH
T329	ENG-HW-132: FD-12F tank-line flange screw (into FS-01B)	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/hardware/lines/28	tag in BOM_v3.5.2.json v3.5.2: BOM hardware row (standard part, not modelled) [row trace: name, qty, host, size, grade, torque, tightening method, unit_mass_kg, mass_source, unit_price]
T330	0.002055 kg	CALC	BOM v3.5.2	/hardware/lines/28/unit_mass_kg	CALC (standard nominal geometry AS RECALLED x engmat density); DATASHEET TO-FETCH
T331	ENG-HW-133: MV-P body screw FS-01A to FS-01B	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/hardware/lines/29	tag in BOM_v3.5.2.json v3.5.2: BOM hardware row (standard part, not modelled) [row trace: name, qty, host, size, grade, torque, tightening method, unit_mass_kg, mass_source, unit_price]
T332	0.003081 kg	CALC	BOM v3.5.2	/hardware/lines/29/unit_mass_kg	CALC (standard nominal geometry AS RECALLED x engmat density); DATASHEET TO-FETCH
T333	ENG-HW-134: MV-P cover FS-06 screw (into FS-01B)	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/hardware/lines/30	tag in BOM_v3.5.2.json v3.5.2: BOM hardware row (standard part, not modelled) [row trace: name, qty, host, size, grade, torque, tightening method, unit_mass_kg, mass_source, unit_price]
T334	0.001054 kg	CALC	BOM v3.5.2	/hardware/lines/30/unit_mass_kg	CALC (standard nominal geometry AS RECALLED x engmat density); DATASHEET TO-FETCH
T335	ENG-HW-135: MV follower FS-08 screw on the FS-04 rod top	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/hardware/lines/31	tag in BOM_v3.5.2.json v3.5.2: BOM hardware row (standard part, not modelled) [row trace: name, qty, host, size, grade, torque, tightening method, unit_mass_kg, mass_source, unit_price]
T336	0.000535 kg	CALC	BOM v3.5.2	/hardware/lines/31/unit_mass_kg	CALC (standard nominal geometry AS RECALLED x engmat density); DATASHEET TO-FETCH
T337	ENG-HW-136: FS-15 bracket to FS-01A/B	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/hardware/lines/32	tag in BOM_v3.5.2.json v3.5.2: BOM hardware row (standard part, not modelled) [row trace: name, qty, host, size, grade, torque, tightening method, unit_mass_kg, mass_source, unit_price]
T338	0.002055 kg	CALC	BOM v3.5.2	/hardware/lines/32/unit_mass_kg	CALC (standard nominal geometry AS RECALLED x engmat density); DATASHEET TO-FETCH
T339	ENG-HW-137: FS-15 bracket to the airframe wall (radial, ICD-2)	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/hardware/lines/33	tag in BOM_v3.5.2.json v3.5.2: BOM hardware row (standard part, not modelled) [row trace: name, qty, host, size, grade, torque, tightening method, unit_mass_kg, mass_source, unit_price]
T340	0.003254 kg	CALC	BOM v3.5.2	/hardware/lines/33/unit_mass_kg	CALC (standard nominal geometry AS RECALLED x engmat density); DATASHEET TO-FETCH
T341	ENG-HW-138: Bleed valve to manifold BL-07 (sub-base screws, per valve)	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/hardware/lines/34	tag in BOM_v3.5.2.json v3.5.2: BOM hardware row (standard part, not modelled) [row trace: name, qty, host, size, grade, torque, tightening method, unit_mass_kg, mass_source, unit_price]

ID	Value	Class	Source	Key	Source tag / formula
T342	0.001638 kg	CALC	BOM v3.5.2	/hardware/lines/34/unit_mass_kg	CALC (standard nominal geometry AS RECALLED x engmat density); DATASHEET TO-FETCH
T343	ENG-HW-139: Bleed manifold BL-07 to the airframe (ICD-14)	DECIDED Design value, BOM_v3.5.2.json v3.5.2	BOM v3.5.2	/hardware/lines/35	tag in BOM_v3.5.2.json v3.5.2: BOM hardware row (standard part, not modelled) [row trace: name, qty, host, size, grade, torque, tightening method, unit_mass_kg, mass_source, unit_price]
T344	0.002055 kg	CALC	BOM v3.5.2	/hardware/lines/35/unit_mass_kg	CALC (standard nominal geometry AS RECALLED x engmat density); DATASHEET TO-FETCH
T345	ENG-HW-150: O-ring AS568-226 FKM	DECIDED Design value, BOM_v3.5.2.json v3.5.2	BOM v3.5.2	/hardware/lines/36	tag in BOM_v3.5.2.json v3.5.2: BOM hardware row (standard part, not modelled) [row trace: name, qty, host, size, grade, torque, tightening method, unit_mass_kg, mass_source, unit_price]
T346	0.00315 kg	CALC	BOM v3.5.2	/hardware/lines/36/unit_mass_kg	CALC (torus at AS568 nominal dims x FKM density 1900 ASSUMED); DATASHEET TO-FETCH | v1.5: HELD on the superseded ID 47.22 (sourced-ID torus 3.150 g, not carried) | v1.6: CARRIED (lead 08:40Z item ...
T347	ENG-HW-151: O-ring AS568-016 FFKM	DECIDED Design value, BOM_v3.5.2.json v3.5.2	BOM v3.5.2	/hardware/lines/37	tag in BOM_v3.5.2.json v3.5.2: BOM hardware row (standard part, not modelled) [row trace: name, qty, host, size, grade, torque, tightening method, unit_mass_kg, mass_source, unit_price]
T348	0.000272 kg	CALC	BOM v3.5.2	/hardware/lines/37/unit_mass_kg	CALC (torus at AS568 nominal dims x FFKM density 2000 ASSUMED); DATASHEET TO-FETCH
T349	ENG-HW-152: O-ring AS568-013 FKM	DECIDED Design value, BOM_v3.5.2.json v3.5.2	BOM v3.5.2	/hardware/lines/38	tag in BOM_v3.5.2.json v3.5.2: BOM hardware row (standard part, not modelled) [row trace: name, qty, host, size, grade, torque, tightening method, unit_mass_kg, mass_source, unit_price]
T350	0.000187 kg	CALC	BOM v3.5.2	/hardware/lines/38/unit_mass_kg	CALC (torus at AS568 nominal dims x FKM density 1900 ASSUMED); DATASHEET TO-FETCH
T351	ENG-HW-153: O-ring AS568-012 FKM	DECIDED Design value, BOM_v3.5.2.json v3.5.2	BOM v3.5.2	/hardware/lines/39	tag in BOM_v3.5.2.json v3.5.2: BOM hardware row (standard part, not modelled) [row trace: name, qty, host, size, grade, torque, tightening method, unit_mass_kg, mass_source, unit_price]
T352	0.000164 kg	CALC	BOM v3.5.2	/hardware/lines/39/unit_mass_kg	CALC (torus at AS568 nominal dims x FKM density 1900 ASSUMED); DATASHEET TO-FETCH
T353	ENG-HW-154: O-ring AS568-010 FKM	DECIDED Design value, BOM_v3.5.2.json v3.5.2	BOM v3.5.2	/hardware/lines/40	tag in BOM_v3.5.2.json v3.5.2: BOM hardware row (standard part, not modelled) [row trace: name, qty, host, size, grade, torque, tightening method, unit_mass_kg, mass_source, unit_price]
T354	0.00011700000000000001 kg	CALC	BOM v3.5.2	/hardware/lines/40/unit_mass_kg	CALC (torus at AS568 nominal dims x FKM density 1900 ASSUMED); DATASHEET TO-FETCH
T355	ENG-HW-155: O-ring AS568-013 FKM	DECIDED Design value, BOM_v3.5.2.json v3.5.2	BOM v3.5.2	/hardware/lines/41	tag in BOM_v3.5.2.json v3.5.2: BOM hardware row (standard part, not modelled) [row trace: name, qty, host, size, grade, torque, tightening method, unit_mass_kg, mass_source, unit_price]
T356	0.000187 kg	CALC	BOM v3.5.2	/hardware/lines/41/unit_mass_kg	CALC (torus at AS568 nominal dims x FKM density 1900 ASSUMED); DATASHEET TO-FETCH
T357	ENG-HW-156: O-ring AS568-229 FKM	DECIDED Design value, BOM_v3.5.2.json v3.5.2	BOM v3.5.2	/hardware/lines/42	tag in BOM_v3.5.2.json v3.5.2: BOM hardware row (standard part, not modelled) [row trace: name, qty, host, size, grade, torque, tightening method, unit_mass_kg, mass_source, unit_price]
T358	0.003521 kg	CALC	BOM v3.5.2	/hardware/lines/42/unit_mass_kg	CALC (torus at AS568 nominal dims x FKM density 1900 ASSUMED); DATASHEET TO-FETCH
T359	ENG-HW-157: O-ring AS568-012 FKM	DECIDED Design value, BOM_v3.5.2.json v3.5.2	BOM v3.5.2	/hardware/lines/43	tag in BOM_v3.5.2.json v3.5.2: BOM hardware row (standard part, not modelled) [row trace: name, qty, host, size, grade, torque, tightening method, unit_mass_kg, mass_source, unit_price]
T360	0.000164 kg	CALC	BOM v3.5.2	/hardware/lines/43/unit_mass_kg	CALC (torus at AS568 nominal dims x FKM density 1900 ASSUMED); DATASHEET TO-FETCH
T361	ENG-HW-158: Spring-energised PTFE rod seal 4 mm (pintle stem, TC-04)	DECIDED Design value, BOM_v3.5.2.json v3.5.2	BOM v3.5.2	/hardware/lines/44	tag in BOM_v3.5.2.json v3.5.2: BOM hardware row (standard part, not modelled) [row trace: name, qty, host, size, grade, torque, tightening method, unit_mass_kg, mass_source, unit_price]
T362	MISSING	MISSING	BOM v3.5.2	/hardware/lines/44/unit_mass_kg	TO-FETCH; inside the allowance
T363	ENG-HW-159: PTFE / 25 % glass guide bushing (pintle stem, TC-04)	DECIDED Design value, BOM_v3.5.2.json v3.5.2	BOM v3.5.2	/hardware/lines/45	tag in BOM_v3.5.2.json v3.5.2: BOM hardware row (standard part, not modelled) [row trace: name, qty, host, size, grade, torque, tightening method, unit_mass_kg, mass_source, unit_price]
T364	MISSING	MISSING	BOM v3.5.2	/hardware/lines/45/unit_mass_kg	TO-FETCH; inside the allowance
T365	ENG-HW-160: BL-01 face O-ring (FFKM/FKM, N2O)	DECIDED Design value, BOM_v3.5.2.json v3.5.2	BOM v3.5.2	/hardware/lines/46	tag in BOM_v3.5.2.json v3.5.2: BOM hardware row (standard part, not modelled) [row trace: name, qty, host, size, grade, torque, tightening method, unit_mass_kg, mass_source, unit_price]
T366	MISSING	MISSING	BOM v3.5.2	/hardware/lines/46/unit_mass_kg	TO-FETCH; inside the allowance
T367	ENG-HW-161: BL-02 M8×1 port seal (bonded seal or O-ring boss)	DECIDED Design value, BOM_v3.5.2.json v3.5.2	BOM v3.5.2	/hardware/lines/47	tag in BOM_v3.5.2.json v3.5.2: BOM hardware row (standard part, not modelled) [row trace: name, qty, host, size, grade, torque, tightening method, unit_mass_kg, mass_source, unit_price]
T368	MISSING	MISSING	BOM v3.5.2	/hardware/lines/47/unit_mass_kg	TO-FETCH; inside the allowance
T369	ENG-HW-162: Crush washer M10 copper (torch IG-01 nose)	DECIDED Design value, BOM_v3.5.2.json v3.5.2	BOM v3.5.2	/hardware/lines/48	tag in BOM_v3.5.2.json v3.5.2: BOM hardware row (standard part, not modelled) [row trace: name, qty, host, size, grade, torque, tightening method, unit_mass_kg, mass_source, unit_price]
T370	0.000671 kg	CALC	BOM v3.5.2	/hardware/lines/48/unit_mass_kg	CALC (annulus x 8900)
T371	ENG-HW-163: Port seals of the M10×1 transducer ports (P_feed FD-01, P_manifold TC-04) ...	DECIDED Design value, BOM_v3.5.2.json v3.5.2	BOM v3.5.2	/hardware/lines/49	tag in BOM_v3.5.2.json v3.5.2: BOM hardware row (standard part, not modelled) [row trace: name, qty, host, size, grade, torque, tightening method, unit_mass_kg, mass_source, unit_price]
T372	MISSING	MISSING	BOM v3.5.2	/hardware/lines/49/unit_mass_kg	TO-FETCH; inside the allowance
T373	ENG-HW-164: MSS-1 thermocouple compression fitting 1/16 tube x 1/8-27 NPT male (Swagelok ...	DECIDED Design value, BOM_v3.5.2.json v3.5.2	BOM v3.5.2	/hardware/lines/50	tag in BOM_v3.5.2.json v3.5.2: BOM hardware row (standard part, not modelled) [row trace: name, qty, host, size, grade, torque, tightening method, unit_mass_kg, mass_source, unit_price]
T374	MISSING	MISSING	BOM v3.5.2	/hardware/lines/50/unit_mass_kg	TO-FETCH; inside the allowance
T375	ENG-HW-165: MV-P balance-piston seal (FS-04 groove) and poppet seat seal	DECIDED Design value, BOM_v3.5.2.json v3.5.2	BOM v3.5.2	/hardware/lines/51	tag in BOM_v3.5.2.json v3.5.2: BOM hardware row (standard part, not modelled) [row trace: name, qty, host, size, grade, torque, tightening method, unit_mass_kg, mass_source, unit_price]
T376	MISSING	MISSING	BOM v3.5.2	/hardware/lines/51/unit_mass_kg	TO-FETCH; inside the allowance
T377	ENG-HW-166: Bleed valve sub-base O-rings (2 per valve)	DECIDED Design value, BOM_v3.5.2.json v3.5.2	BOM v3.5.2	/hardware/lines/52	tag in BOM_v3.5.2.json v3.5.2: BOM hardware row (standard part, not modelled) [row trace: name, qty, host, size, grade, torque, tightening method, unit_mass_kg, mass_source, unit_price]
T378	MISSING	MISSING	BOM v3.5.2	/hardware/lines/52/unit_mass_kg	TO-FETCH; inside the allowance
T379	ENG-HW-167: G11 isolator washers under the HV69 ears (not modelled)	DECIDED Design value, BOM_v3.5.2.json v3.5.2	BOM v3.5.2	/hardware/lines/53	tag in BOM_v3.5.2.json v3.5.2: BOM hardware row (standard part, not modelled) [row trace: name, qty, host, size, grade, torque, tightening method, unit_mass_kg, mass_source, unit_price]
T380	2.9999999999999997e-05 kg	CALC	BOM v3.5.2	/hardware/lines/53/unit_mass_kg	CALC (5/2.2 × 1 ASSUMED x 1900)
T381	ENG-HW-168: Bundle clip BL-04 bleed hose to the FD-07 coil	DECIDED Design value, BOM_v3.5.2.json v3.5.2	BOM v3.5.2	/hardware/lines/54	tag in BOM_v3.5.2.json v3.5.2: BOM hardware row (standard part, not modelled) [row trace: name, qty, host, size, grade, torque, tightening method, unit_mass_kg, mass_source, unit_price]
T382	MISSING	MISSING	BOM v3.5.2	/hardware/lines/54/unit_mass_kg	TO-FETCH; inside the allowance
T383	ENG-HW-169: Thread locker / retaining compounds: Loctite 243 (GM-18, GM-20, HW-25, TC-05 ...	DECIDED Design value, BOM_v3.5.2.json v3.5.2	BOM v3.5.2	/hardware/lines/55	tag in BOM_v3.5.2.json v3.5.2: BOM hardware row (standard part, not modelled) [row trace: name, qty, host, size, grade, torque, tightening method, unit_mass_kg, mass_source, unit_price]
T384	0 kg	ASSUMED	BOM v3.5.2	/hardware/lines/55/unit_mass_kg	negligible (g-level), inside the allowance
T385	ENG-HW-170: J-FWD disc spring under each ENG-HW-101-IN718 head (route (i), lead 16:53Z ...	DECIDED Design value, BOM_v3.5.2.json v3.5.2	BOM v3.5.2	/hardware/lines/56	tag in BOM_v3.5.2.json v3.5.2: BOM hardware row (standard part, not modelled) [row trace: name, qty, host, size, grade, torque, tightening method, unit_mass_kg, mass_source, unit_price]
T386	0.00022600000000000002 kg	CALC	BOM v3.5.2	/hardware/lines/56/unit_mass_kg	CALC (flat annulus De 10 / Di 5.2 x t 0.5 x SS-SPRING 7900 kg/m3 LITERATURE, engmat); spring B (t 0.4) 0.181 g; DATASHEET TO-FETCH
T387	11	DERIVED	generator	formula: count of BOM lines with part-number family BL	DERIVED by build_datasheet.py: count of BOM lines with part-number family BL ← T049, T051, T053, T055, T057, T059, T061, T063 ...
T388	0.78316 kg	DERIVED	generator	formula: Σ qty × unit mass over the BL lines	DERIVED by build_datasheet.py: Σ qty × unit mass over the BL lines ← T050, T052, T054, T056, T058, T060, T062, T064 ...
T389	6	DERIVED	generator	formula: count of BOM lines with part-number family EL	DERIVED by build_datasheet.py: count of BOM lines with part-number family EL ← T071, T073, T075, T077, T079, T081
T390	0.074 kg	DERIVED	generator	formula: Σ qty × unit mass over the EL lines	DERIVED by build_datasheet.py: Σ qty × unit mass over the EL lines ← T072, T074, T076, T078, T080, T082

ID	Value	Class	Source	Key	Source tag / formula
T391	11	DERIVED	generator	formula: count of BOM lines with part-number family FD	DERIVED by build_datasheet.py: count of BOM lines with part-number family FD ← T083, T085, T087, T089, T091, T093, T095, T097 …
T392	0.27586 kg	DERIVED	generator	formula: Σ qty × unit mass over the FD lines	DERIVED by build_datasheet.py: Σ qty × unit mass over the FD lines ← T084, T086, T088, T090, T092, T094, T096, T098 …
T393	15	DERIVED	generator	formula: count of BOM lines with part-number family FS	DERIVED by build_datasheet.py: count of BOM lines with part-number family FS ← T105, T107, T109, T111, T113, T115, T117, T119 …
T394	0.33643 kg	DERIVED	generator	formula: Σ qty × unit mass over the FS lines	DERIVED by build_datasheet.py: Σ qty × unit mass over the FS lines ← T106, T108, T110, T112, T114, T116, T118, T120 …
T395	19	DERIVED	generator	formula: count of BOM lines with part-number family GM	DERIVED by build_datasheet.py: count of BOM lines with part-number family GM ← T135, T137, T139, T141, T143, T145, T147, T149 …
T396	1.15278 kg	DERIVED	generator	formula: Σ qty × unit mass over the GM lines	DERIVED by build_datasheet.py: Σ qty × unit mass over the GM lines ← T136, T138, T140, T142, T144, T146, T148, T150 …
T397	2	DERIVED	generator	formula: count of BOM lines with part-number family GR	DERIVED by build_datasheet.py: count of BOM lines with part-number family GR ← T175, T177
T398	0.42436 kg	DERIVED	generator	formula: Σ qty × unit mass over the GR lines	DERIVED by build_datasheet.py: Σ qty × unit mass over the GR lines ← T176, T178
T399	4	DERIVED	generator	formula: count of BOM lines with part-number family HW	DERIVED by build_datasheet.py: count of BOM lines with part-number family HW ← T179, T181, T183, T185
T400	0.0242 kg	DERIVED	generator	formula: Σ qty × unit mass over the HW lines	DERIVED by build_datasheet.py: Σ qty × unit mass over the HW lines ← T180, T182, T184, T186
T401	2	DERIVED	generator	formula: count of BOM lines with part-number family IG	DERIVED by build_datasheet.py: count of BOM lines with part-number family IG ← T187, T189
T402	0.074 kg	DERIVED	generator	formula: Σ qty × unit mass over the IG lines	DERIVED by build_datasheet.py: Σ qty × unit mass over the IG lines ← T188, T190
T403	5	DERIVED	generator	formula: count of BOM lines with part-number family NZ	DERIVED by build_datasheet.py: count of BOM lines with part-number family NZ ← T191, T193, T195, T197, T199
T404	0.0632 kg	DERIVED	generator	formula: Σ qty × unit mass over the NZ lines	DERIVED by build_datasheet.py: Σ qty × unit mass over the NZ lines ← T192, T194, T196, T198, T200
T405	20	DERIVED	generator	formula: count of BOM lines with part-number family PD	DERIVED by build_datasheet.py: count of BOM lines with part-number family PD ← T201, T203, T205, T207, T209, T211, T213, T215 …
T406	0.17456 kg	DERIVED	generator	formula: Σ qty × unit mass over the PD lines	DERIVED by build_datasheet.py: Σ qty × unit mass over the PD lines ← T202, T204, T206, T208, T210, T212, T214, T216 …
T407	11	DERIVED	generator	formula: count of BOM lines with part-number family TC	DERIVED by build_datasheet.py: count of BOM lines with part-number family TC ← T241, T243, T245, T247, T249, T251, T253, T255 …
T408	1.66401 kg	DERIVED	generator	formula: Σ qty × unit mass over the TC lines	DERIVED by build_datasheet.py: Σ qty × unit mass over the TC lines ← T242, T244, T246, T248, T250, T252, T254, T256 …
T409	5.04656 kg	DERIVED	generator	formula: Σ qty × unit mass over all BOM lines (transport lock excluded)	DERIVED by build_datasheet.py: Σ qty × unit mass over all BOM lines (transport lock excluded) ← T388, T390, T392, T394, T396, T398, T400, T402 …
T410	0.06225 kg	DERIVED	generator	formula: Σ qty × unit mass over the transport-lock lines (group lock)	DERIVED by build_datasheet.py: Σ qty × unit mass over the transport-lock lines (group lock) ← T158
T411	0.223142 kg	CALC	generator	formula: Σ qty × unit mass over the BOM hardware lines (CALC)	DERIVED by build_datasheet.py: Σ qty × unit mass over the BOM hardware lines (CALC) ← T274, T276, T278, T280, T282, T284, T286, T288 … [lines without a mass: ENG-HW-128, ENG-HW-158, ENG-HW-159 …
T412	5.0466 kg	DERIVED	BOM v3.5.2	/mass_sum_kg	sum of unit masses x qty (CAD volume x density, or tagged COTS masses); includes the transport-lock items only if they are placed (they are not: group 'lock' is excluded from the instance list) …
T413	TO-QUOTE (not summed)	MISSING	BOM v3.5.2	/lines/*/unit_price	BOM unit_price: every line TO-QUOTE (delivery Sonderborg, Denmark, DK-6400) [total not summed: no line carries a DK-6400 quotation]
T414	16	DERIVED	generator	formula: count of BOM lines with material_name "Stainless steel 1.4404 / 316L, annealed bar (EN 10088-3), oxidiser-wetted parts"	DERIVED by build_datasheet.py: count of BOM lines with material_name "Stainless steel 1.4404 / 316L, annealed bar (EN 10088-3), oxidiser-wetted parts" ← T049, T051, T053, T055, T057, T061, T063 …
T415	0.87955 kg	DERIVED	generator	formula: Σ qty × unit mass, material "Stainless steel 1.4404 / 316L, annealed bar (EN 10088-3), oxidiser-wetted parts"	DERIVED by build_datasheet.py: Σ qty × unit mass, material "Stainless steel 1.4404 / 316L, annealed bar (EN 10088-3), oxidiser-wetted parts" ← T050, T052, T054, T056, T058, T062, T064, T068 …
T416	15	DERIVED	generator	formula: count of oxygen-clean lines, material "Stainless steel 1.4404 / 316L, annealed bar (EN 10088-3), oxidiser-wetted parts"	DERIVED by build_datasheet.py: count of oxygen-clean lines, material "Stainless steel 1.4404 / 316L, annealed bar (EN 10088-3), oxidiser-wetted parts" ← T049, T051, T053, T055, T057, T061, T063 …
T417	2	DERIVED	generator	formula: count of BOM lines with material_name "PTFE"	DERIVED by build_datasheet.py: count of BOM lines with material_name "PTFE" ← T059, T261
T418	0.01271 kg	DERIVED	generator	formula: Σ qty × unit mass, material "PTFE"	DERIVED by build_datasheet.py: Σ qty × unit mass, material "PTFE" ← T060, T262
T419	2	DERIVED	generator	formula: count of oxygen-clean lines, material "PTFE"	DERIVED by build_datasheet.py: count of oxygen-clean lines, material "PTFE" ← T059, T261
T420	20	DERIVED	generator	formula: count of BOM lines with material_name "Aluminium alloy 6061-T6 (EN AW-6061 T6), bar/tube"	DERIVED by build_datasheet.py: count of BOM lines with material_name "Aluminium alloy 6061-T6 (EN AW-6061 T6), bar/tube" ← T065, T099, T105, T107, T115, T133, T139, T149 …
T421	2.07054 kg	DERIVED	generator	formula: Σ qty × unit mass, material "Aluminium alloy 6061-T6 (EN AW-6061 T6), bar/tube"	DERIVED by build_datasheet.py: Σ qty × unit mass, material "Aluminium alloy 6061-T6 (EN AW-6061 T6), bar/tube" ← T066, T100, T106, T108, T116, T134, T140, T150 …
T422	5	DERIVED	generator	formula: count of oxygen-clean lines, material "Aluminium alloy 6061-T6 (EN AW-6061 T6), bar/tube"	DERIVED by build_datasheet.py: count of oxygen-clean lines, material "Aluminium alloy 6061-T6 (EN AW-6061 T6), bar/tube" ← T065, T099, T105, T107, T115, T133, T139, T149 …
T423	17	DERIVED	generator	formula: count of BOM lines with material_name "COTS assembly (mixed materials), mass from datasheet or assumed"	DERIVED by build_datasheet.py: count of BOM lines with material_name "COTS assembly (mixed materials), mass from datasheet or assumed" ← T071, T073, T075, T079, T081, T085, T093, T101 …
T424	0.5409 kg	DERIVED	generator	formula: Σ qty × unit mass, material "COTS assembly (mixed materials), mass from datasheet or assumed"	DERIVED by build_datasheet.py: Σ qty × unit mass, material "COTS assembly (mixed materials), mass from datasheet or assumed" ← T072, T074, T076, T080, T082, T086, T094, T102 …
T425	8	DERIVED	generator	formula: count of oxygen-clean lines, material "COTS assembly (mixed materials), mass from datasheet or assumed"	DERIVED by build_datasheet.py: count of oxygen-clean lines, material "COTS assembly (mixed materials), mass from datasheet or assumed" ← T071, T073, T075, T079, T081, T085, T093, T101 …
T426	2	DERIVED	generator	formula: count of BOM lines with material_name "NdFeB magnet N35 (diametral, Ni-plated)"	DERIVED by build_datasheet.py: count of BOM lines with material_name "NdFeB magnet N35 (diametral, Ni-plated)" ← T077, T235
T427	0.00121 kg	DERIVED	generator	formula: Σ qty × unit mass, material "NdFeB magnet N35 (diametral, Ni-plated)"	DERIVED by build_datasheet.py: Σ qty × unit mass, material "NdFeB magnet N35 (diametral, Ni-plated)" ← T078, T236
T428	0	DERIVED	generator	formula: count of oxygen-clean lines, material "NdFeB magnet N35 (diametral, Ni-plated)"	DERIVED by build_datasheet.py: count of oxygen-clean lines, material "NdFeB magnet N35 (diametral, Ni-plated)" ← T077, T235
T429	7	DERIVED	generator	formula: count of BOM lines with material_name "17-4PH (1.4542) condition H1150, pins"	DERIVED by build_datasheet.py: count of BOM lines with material_name "17-4PH (1.4542) condition H1150, pins" ← T109, T113, T169, T171, T173, T211, T249
T430	0.04565 kg	DERIVED	generator	formula: Σ qty × unit mass, material "17-4PH (1.4542) condition H1150, pins"	DERIVED by build_datasheet.py: Σ qty × unit mass, material "17-4PH (1.4542) condition H1150, pins" ← T110, T114, T170, T172, T174, T212, T250
T431	3	DERIVED	generator	formula: count of oxygen-clean lines, material "17-4PH (1.4542) condition H1150, pins"	DERIVED by build_datasheet.py: count of oxygen-clean lines, material "17-4PH (1.4542) condition H1150, pins" ← T109, T113, T169, T171, T173, T211, T249
T432	1	DERIVED	generator	formula: count of BOM lines with material_name "PCTFE (Kel-F 81 / Neoflon M-300 class) soft seat, oxygen-compatible"	DERIVED by build_datasheet.py: count of BOM lines with material_name "PCTFE (Kel-F 81 / Neoflon M-300 class) soft seat, oxygen-compatible" ← T111
T433	0.00023 kg	DERIVED	generator	formula: Σ qty × unit mass, material "PCTFE (Kel-F 81 / Neoflon M-300 class) soft seat, oxygen-compatible"	DERIVED by build_datasheet.py: Σ qty × unit mass, material "PCTFE (Kel-F 81 / Neoflon M-300 class) soft seat, oxygen-compatible" ← T112

ID	Value	Class	Source	Key	Source tag / formula
T434	1	DERIVED	generator	formula: count of oxygen-clean lines, material "PCTFE (Kel-F 81 / Neoflon M-300 class) soft seat, oxygen-compatible"	DERIVED by build_datasheet.py: count of oxygen-clean lines, material "PCTFE (Kel-F 81 / Neoflon M-300 class) soft seat, oxygen-compatible" ← T111
T435	2	DERIVED	generator	formula: count of BOM lines with material_name "Stainless spring wire 1.4310 (302/301) EN 10270-3 or 17-7PH (springs: MV-P closing, pintle spiral spring)"	DERIVED by build_datasheet.py: count of BOM lines with material_name "Stainless spring wire 1.4310 (302/301) EN 10270-3 or 17-7PH (springs: MV-P closing, pintle spiral spring)" ← T117, T217
T436	0.0135 kg	DERIVED	generator	formula: Σ qty × unit mass, material "Stainless spring wire 1.4310 (302/301) EN 10270-3 or 17-7PH (springs: MV-P closing, pintle spiral spring)"	DERIVED by build_datasheet.py: Σ qty × unit mass, material "Stainless spring wire 1.4310 (302/301) EN 10270-3 or 17-7PH (springs: MV-P closing, pintle spiral spring)" ← T118, T218
T437	0	DERIVED	generator	formula: count of oxygen-clean lines, material "Stainless spring wire 1.4310 (302/301) EN 10270-3 or 17-7PH (springs: MV-P closing, pintle spiral spring)"	DERIVED by build_datasheet.py: count of oxygen-clean lines, material "Stainless spring wire 1.4310 (302/301) EN 10270-3 or 17-7PH (springs: MV-P closing, pintle spiral spring)" ← T117, T217
T438	4	DERIVED	generator	formula: count of BOM lines with material_name "17-4PH (1.4542) condition H900: pintle crankshaft, crank/link pins, MV-P poppet"	DERIVED by build_datasheet.py: count of BOM lines with material_name "17-4PH (1.4542) condition H900: pintle crankshaft, crank/link pins, MV-P poppet" ← T119, T121, T207, T231
T439	0.02576 kg	DERIVED	generator	formula: Σ qty × unit mass, material "17-4PH (1.4542) condition H900: pintle crankshaft, crank/link pins, MV-P poppet"	DERIVED by build_datasheet.py: Σ qty × unit mass, material "17-4PH (1.4542) condition H900: pintle crankshaft, crank/link pins, MV-P poppet" ← T120, T122, T208, T232
T440	0	DERIVED	generator	formula: count of oxygen-clean lines, material "17-4PH (1.4542) condition H900: pintle crankshaft, crank/link pins, MV-P poppet"	DERIVED by build_datasheet.py: count of oxygen-clean lines, material "17-4PH (1.4542) condition H900: pintle crankshaft, crank/link pins, MV-P poppet" ← T119, T121, T207, T231
T441	5	DERIVED	generator	formula: count of BOM lines with material_name "Aluminium alloy 7075-T6 (EN AW-7075 T6), bar"	DERIVED by build_datasheet.py: count of BOM lines with material_name "Aluminium alloy 7075-T6 (EN AW-7075 T6), bar" ← T135, T137, T165, T167, T209
T442	0.15025 kg	DERIVED	generator	formula: Σ qty × unit mass, material "Aluminium alloy 7075-T6 (EN AW-7075 T6), bar"	DERIVED by build_datasheet.py: Σ qty × unit mass, material "Aluminium alloy 7075-T6 (EN AW-7075 T6), bar" ← T136, T138, T166, T168, T210
T443	0	DERIVED	generator	formula: count of oxygen-clean lines, material "Aluminium alloy 7075-T6 (EN AW-7075 T6), bar"	DERIVED by build_datasheet.py: count of oxygen-clean lines, material "Aluminium alloy 7075-T6 (EN AW-7075 T6), bar" ← T135, T137, T165, T167, T209
T444	2	DERIVED	generator	formula: count of BOM lines with material_name "Martensitic stainless 440C (UNS S44004, 1.4125), hardened + low temper (raceway pins for drawn-cup needle …	DERIVED by build_datasheet.py: count of BOM lines with material_name "Martensitic stainless 440C (UNS S44004, 1.4125), hardened + low temper (raceway pins for drawn-cup needle bearings, v3.5.1)" ← …
T445	0.02742 kg	DERIVED	generator	formula: Σ qty × unit mass, material "Martensitic stainless 440C (UNS S44004, 1.4125), hardened + low temper (raceway pins for drawn-cup needle bearings …	DERIVED by build_datasheet.py: Σ qty × unit mass, material "Martensitic stainless 440C (UNS S44004, 1.4125), hardened + low temper (raceway pins for drawn-cup needle bearings, v3.5.1)" ← T142, T144
T446	0	DERIVED	generator	formula: count of oxygen-clean lines, material "Martensitic stainless 440C (UNS S44004, 1.4125), hardened + low temper (raceway pins for drawn-cup needle …	DERIVED by build_datasheet.py: count of oxygen-clean lines, material "Martensitic stainless 440C (UNS S44004, 1.4125), hardened + low temper (raceway pins for drawn-cup needle bearings, v3.5.1)" ← …
T447	2	DERIVED	generator	formula: count of BOM lines with material_name "Stainless steel 1.4301 / AISI 304, annealed bar (EN 10088-3)"	DERIVED by build_datasheet.py: count of BOM lines with material_name "Stainless steel 1.4301 / AISI 304, annealed bar (EN 10088-3)" ← T145, T147
T448	0.0046 kg	DERIVED	generator	formula: Σ qty × unit mass, material "Stainless steel 1.4301 / AISI 304, annealed bar (EN 10088-3)"	DERIVED by build_datasheet.py: Σ qty × unit mass, material "Stainless steel 1.4301 / AISI 304, annealed bar (EN 10088-3)" ← T146, T148
T449	0	DERIVED	generator	formula: count of oxygen-clean lines, material "Stainless steel 1.4301 / AISI 304, annealed bar (EN 10088-3)"	DERIVED by build_datasheet.py: count of oxygen-clean lines, material "Stainless steel 1.4301 / AISI 304, annealed bar (EN 10088-3)" ← T145, T147
T450	1	DERIVED	generator	formula: count of BOM lines with material_name "Polyurethane bumper 90A"	DERIVED by build_datasheet.py: count of BOM lines with material_name "Polyurethane bumper 90A" ← T155
T451	0.0092 kg	DERIVED	generator	formula: Σ qty × unit mass, material "Polyurethane bumper 90A"	DERIVED by build_datasheet.py: Σ qty × unit mass, material "Polyurethane bumper 90A" ← T156
T452	0	DERIVED	generator	formula: count of oxygen-clean lines, material "Polyurethane bumper 90A"	DERIVED by build_datasheet.py: count of oxygen-clean lines, material "Polyurethane bumper 90A" ← T155
T453	2	DERIVED	generator	formula: count of BOM lines with material_name "ABS, FDM-printed fuel grain (100 % infill)"	DERIVED by build_datasheet.py: count of BOM lines with material_name "ABS, FDM-printed fuel grain (100 % infill)" ← T175, T177
T454	0.42436 kg	DERIVED	generator	formula: Σ qty × unit mass, material "ABS, FDM-printed fuel grain (100 % infill)"	DERIVED by build_datasheet.py: Σ qty × unit mass, material "ABS, FDM-printed fuel grain (100 % infill)" ← T176, T178
T455	0	DERIVED	generator	formula: count of oxygen-clean lines, material "ABS, FDM-printed fuel grain (100 % infill)"	DERIVED by build_datasheet.py: count of oxygen-clean lines, material "ABS, FDM-printed fuel grain (100 % infill)" ← T175, T177
T456	9	DERIVED	generator	formula: count of BOM lines with material_name "Carbon/bearing steel (generic COTS)"	DERIVED by build_datasheet.py: count of BOM lines with material_name "Carbon/bearing steel (generic COTS)" ← T179, T181, T183, T185, T215, T227, T229, T233 …
T457	0.0809 kg	DERIVED	generator	formula: Σ qty × unit mass, material "Carbon/bearing steel (generic COTS)"	DERIVED by build_datasheet.py: Σ qty × unit mass, material "Carbon/bearing steel (generic COTS)" ← T180, T182, T184, T186, T216, T228, T230, T234 …
T458	0	DERIVED	generator	formula: count of oxygen-clean lines, material "Carbon/bearing steel (generic COTS)"	DERIVED by build_datasheet.py: count of oxygen-clean lines, material "Carbon/bearing steel (generic COTS)" ← T179, T181, T183, T185, T215, T227, T229, T233 …
T459	1	DERIVED	generator	formula: count of BOM lines with material_name "Isomolded fine-grain graphite (reference grade Toyo Tanso IG-110 or equivalent)"	DERIVED by build_datasheet.py: count of BOM lines with material_name "Isomolded fine-grain graphite (reference grade Toyo Tanso IG-110 or equivalent)" ← T191
T460	0.04685 kg	DERIVED	generator	formula: Σ qty × unit mass, material "Isomolded fine-grain graphite (reference grade Toyo Tanso IG-110 or equivalent)"	DERIVED by build_datasheet.py: Σ qty × unit mass, material "Isomolded fine-grain graphite (reference grade Toyo Tanso IG-110 or equivalent)" ← T192
T461	0	DERIVED	generator	formula: count of oxygen-clean lines, material "Isomolded fine-grain graphite (reference grade Toyo Tanso IG-110 or equivalent)"	DERIVED by build_datasheet.py: count of oxygen-clean lines, material "Isomolded fine-grain graphite (reference grade Toyo Tanso IG-110 or equivalent)" ← T191
T462	2	DERIVED	generator	formula: count of BOM lines with material_name "Carbon-cloth phenolic laminate (MX-4926N / FM-5055 class), tube or plate: NZ-02 sleeve, J1 hot-layer washer …	DERIVED by build_datasheet.py: count of BOM lines with material_name "Carbon-cloth phenolic laminate (MX-4926N / FM-5055 class), tube or plate: NZ-02 sleeve, J1 hot-layer washer NZ-03" ← T193, T195
T463	0.01548 kg	DERIVED	generator	formula: Σ qty × unit mass, material "Carbon-cloth phenolic laminate (MX-4926N / FM-5055 class), tube or plate: NZ-02 sleeve, J1 hot-layer washer NZ-03"	DERIVED by build_datasheet.py: Σ qty × unit mass, material "Carbon-cloth phenolic laminate (MX-4926N / FM-5055 class), tube or plate: NZ-02 sleeve, J1 hot-layer washer NZ-03" ← T194, T196
T464	0	DERIVED	generator	formula: count of oxygen-clean lines, material "Carbon-cloth phenolic laminate (MX-4926N / FM-5055 class), tube or plate: NZ-02 sleeve, J1 hot-layer washer …	DERIVED by build_datasheet.py: count of oxygen-clean lines, material "Carbon-cloth phenolic laminate (MX-4926N / FM-5055 class), tube or plate: NZ-02 sleeve, J1 hot-layer washer NZ-03" ← T193, T195
T465	1	DERIVED	generator	formula: count of BOM lines with material_name "Flexible graphite sheet, oxidation-inhibited grade (GrafTech Grafoil GTJ / SGL Sigraflex inhibited class), 0.5 …	DERIVED by build_datasheet.py: count of BOM lines with material_name "Flexible graphite sheet, oxidation-inhibited grade (GrafTech Grafoil GTJ / SGL Sigraflex inhibited class), 0.5 mm: J1 lip-face …
T466	0.00068 kg	DERIVED	generator	formula: Σ qty × unit mass, material "Flexible graphite sheet, oxidation-inhibited grade (GrafTech Grafoil GTJ / SGL Sigraflex inhibited class), 0.5 mm: J1 …	DERIVED by build_datasheet.py: Σ qty × unit mass, material "Flexible graphite sheet, oxidation-inhibited grade (GrafTech Grafoil GTJ / SGL Sigraflex inhibited class), 0.5 mm: J1 lip-face seal NZ-04" …
T467	0	DERIVED	generator	formula: count of oxygen-clean lines, material "Flexible graphite sheet, oxidation-inhibited grade (GrafTech Grafoil GTJ / SGL Sigraflex inhibited class), 0.5 …	DERIVED by build_datasheet.py: count of oxygen-clean lines, material "Flexible graphite sheet, oxidation-inhibited grade (GrafTech Grafoil GTJ / SGL Sigraflex inhibited class), 0.5 mm: J1 lip-face …
T468	1	DERIVED	generator	formula: count of BOM lines with material_name "Flexible graphite roll foil 0.25 mm, >= 99.85 % C, 1.0 g/cm3 (SGL SIGRAFLEX TH class): v3.5.2 N-F2 …	DERIVED by build_datasheet.py: count of BOM lines with material_name "Flexible graphite roll foil 0.25 mm, >= 99.85 % C, 1.0 g/cm3 (SGL SIGRAFLEX TH class): v3.5.2 N-F2 load-spreading foil NZ-06" ← …
T469	0.00019 kg	DERIVED	generator	formula: Σ qty × unit mass, material "Flexible graphite roll foil 0.25 mm, >= 99.85 % C, 1.0 g/cm3 (SGL SIGRAFLEX TH class): v3.5.2 N-F2 load-spreading foil …	DERIVED by build_datasheet.py: Σ qty × unit mass, material "Flexible graphite roll foil 0.25 mm, >= 99.85 % C, 1.0 g/cm3 (SGL SIGRAFLEX TH class): v3.5.2 N-F2 load-spreading foil NZ-06" ← T200

ID	Value	Class	Source	Key	Source tag / formula
T470	0	DERIVED	generator	formula: count of oxygen-clean lines, material "Flexible graphite roll foil 0.25 mm, >= 99.85 % C, 1.0 g/cm3 (SGL SIGRAFLEX TH class): v3.5.2 N-F2 …	DERIVED by build_datasheet.py: count of oxygen-clean lines, material "Flexible graphite roll foil 0.25 mm, >= 99.85 % C, 1.0 g/cm3 (SGL SIGRAFLEX TH class): v3.5.2 N-F2 load-spreading foil NZ-06" ← …
T471	3	DERIVED	generator	formula: count of BOM lines with material_name "Glass-epoxy laminate NEMA LI-1 G11 (EN 60893-3-2 EP GC 203 class), high-temperature epoxy"	DERIVED by build_datasheet.py: count of BOM lines with material_name "Glass-epoxy laminate NEMA LI-1 G11 (EN 60893-3-2 EP GC 203 class), high-temperature epoxy" ← T221, T223, T239
T472	0.00142 kg	DERIVED	generator	formula: Σ qty × unit mass, material "Glass-epoxy laminate NEMA LI-1 G11 (EN 60893-3-2 EP GC 203 class), high-temperature epoxy"	DERIVED by build_datasheet.py: Σ qty × unit mass, material "Glass-epoxy laminate NEMA LI-1 G11 (EN 60893-3-2 EP GC 203 class), high-temperature epoxy" ← T222, T224, T240
T473	0	DERIVED	generator	formula: count of oxygen-clean lines, material "Glass-epoxy laminate NEMA LI-1 G11 (EN 60893-3-2 EP GC 203 class), high-temperature epoxy"	DERIVED by build_datasheet.py: count of oxygen-clean lines, material "Glass-epoxy laminate NEMA LI-1 G11 (EN 60893-3-2 EP GC 203 class), high-temperature epoxy" ← T221, T223, T239
T474	1	DERIVED	generator	formula: count of BOM lines with material_name "Dispersion-strengthened copper GlidCop AL-15 (UNS C15715), as-consolidated rod (v3.5.1 TC-03 pintle seat …	DERIVED by build_datasheet.py: count of BOM lines with material_name "Dispersion-strengthened copper GlidCop AL-15 (UNS C15715), as-consolidated rod (v3.5.1 TC-03 pintle seat insert)" ← T245
T475	0.01488 kg	DERIVED	generator	formula: Σ qty × unit mass, material "Dispersion-strengthened copper GlidCop AL-15 (UNS C15715), as-consolidated rod (v3.5.1 TC-03 pintle seat insert)"	DERIVED by build_datasheet.py: Σ qty × unit mass, material "Dispersion-strengthened copper GlidCop AL-15 (UNS C15715), as-consolidated rod (v3.5.1 TC-03 pintle seat insert)" ← T246
T476	1	DERIVED	generator	formula: count of oxygen-clean lines, material "Dispersion-strengthened copper GlidCop AL-15 (UNS C15715), as-consolidated rod (v3.5.1 TC-03 pintle seat …	DERIVED by build_datasheet.py: count of oxygen-clean lines, material "Dispersion-strengthened copper GlidCop AL-15 (UNS C15715), as-consolidated rod (v3.5.1 TC-03 pintle seat insert)" ← T245
T477	4	DERIVED	generator	formula: count of BOM lines with material_name "Cotton/paper phenolic laminate tube (NEMA LI-1 grade CE/XX class)"	DERIVED by build_datasheet.py: count of BOM lines with material_name "Cotton/paper phenolic laminate tube (NEMA LI-1 grade CE/XX class)" ← T251, T253, T255, T259
T478	0.16594 kg	DERIVED	generator	formula: Σ qty × unit mass, material "Cotton/paper phenolic laminate tube (NEMA LI-1 grade CE/XX class)"	DERIVED by build_datasheet.py: Σ qty × unit mass, material "Cotton/paper phenolic laminate tube (NEMA LI-1 grade CE/XX class)" ← T252, T254, T256, T260
T479	0	DERIVED	generator	formula: count of oxygen-clean lines, material "Cotton/paper phenolic laminate tube (NEMA LI-1 grade CE/XX class)"	DERIVED by build_datasheet.py: count of oxygen-clean lines, material "Cotton/paper phenolic laminate tube (NEMA LI-1 grade CE/XX class)" ← T251, T253, T255, T259
T480	1	DERIVED	generator	formula: count of BOM lines with material_name "Inconel 718 (UNS N07718, AMS 5662 solution + AMS 5663 aged bar): TC-03 seat-insert alternative"	DERIVED by build_datasheet.py: count of BOM lines with material_name "Inconel 718 (UNS N07718, AMS 5662 solution + AMS 5663 aged bar): TC-03 seat-insert alternative" ← T257
T481	0.51434 kg	DERIVED	generator	formula: Σ qty × unit mass, material "Inconel 718 (UNS N07718, AMS 5662 solution + AMS 5663 aged bar): TC-03 seat-insert alternative"	DERIVED by build_datasheet.py: Σ qty × unit mass, material "Inconel 718 (UNS N07718, AMS 5662 solution + AMS 5663 aged bar): TC-03 seat-insert alternative" ← T258
T482	0	DERIVED	generator	formula: count of oxygen-clean lines, material "Inconel 718 (UNS N07718, AMS 5662 solution + AMS 5663 aged bar): TC-03 seat-insert alternative"	DERIVED by build_datasheet.py: count of oxygen-clean lines, material "Inconel 718 (UNS N07718, AMS 5662 solution + AMS 5663 aged bar): TC-03 seat-insert alternative" ← T257
T483	1.179 kg	DERIVED	MEC v2.4	/fixed_module/tank/tube_mass_kg	DERIVED [mechanical.json v2.1 record (not re-derived for v3.5.2: not_re_derived_v352)]
T484	6061-T6 tube 101.6 × 3.175 + 2 bolted bulkheads (radial M6, 2 FKM O-rings each)	DERIVED	MEC v2.4	/fixed_module/tank/concept	DERIVED [mechanical.json v2.1 record (not re-derived for v3.5.2: not_re_derived_v352)]
T485	0.01–0.02 kg	ASSUMED	BOM v3.5.2	/provision_mass_estimate_kg/estimate_range_kg	(class set at the call site: ASSUMED)
T486	v3.5: bleed MODELLED (ENG-BL-01..10 in the BOM lines); provision = opposed vent tee …	ASSUMED	BOM v3.5.2	/provision_mass_estimate_kg/note	(class set at the call site: ASSUMED)
T487	thrust-ring radial screws (16 x M5); isolation valve + actuator (mass UNKNOWN …	MISSING	MEC v2.4	/fixed_module/not_modelled	OPEN [mechanical.json v2.1 record (not re-derived for v3.5.2: not_re_derived_v352)]
T488	MISSING	MISSING	—		[mass of the items listed as not modelled in mechanical.json (v2.1 record)]
T489	0.16 kg	ASSUMED	ELE v4.1	/ecu/mass_kg	ASSUMED (PCB+Teensy+connectors 0.11, enclosure 0.05)
T490	0.12–0.22 kg	ASSUMED	ELE v4.1	/ecu/mass_kg/range	(class set at the call site: ASSUMED)
T491	separate engine ECU (Teensy 4.1) on the fixed thrust structure, CAN link to flight …	DERIVED	ELE v4.1	/ecu/architecture	DERIVED trade (REPORT Architecture)
T492	2.0018 kg	DECIDED *Lead decision of 2026-10-12*	CMB v2.4d	/decisions/N2O_load_and_tank/v/m_N2O_kg	tag in combustion.json v2.4d: LEAD decision 2026-10-12 (CHANGELOG): 2.00 kg / 3.17 L (85 % fill)
T493	LEAD decision 2026-10-12 (CHANGELOG): 2.00 kg / 3.17 L (85 % fill)	DECIDED *Lead decision, per combustion.json v2.4d*	CMB v2.4d	/decisions/N2O_load_and_tank/tag	tag in combustion.json v2.4d:
T494	Combustion sets the length in the T4 re-run of 261 (2) with Mechanical's soak-back.	DECIDED *Lead decision, line 264 (5)*	CL 266 lines	L264(5)[from "Combustion sets the length" to "soak-back."]	lead (decision) 2026-10-21 (line sha 7cdbe51041ae6b8b)
T495	FA1U	SIM	MMXMD FEA3D margin matrix v3.5.2	line 425 row "TC-01" col 1	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T496	-0.083 / -0.012	SIM	MMXMD FEA3D margin matrix v3.5.2	line 425 row "TC-01" col 2	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T497	n9090, A1_22.2, regular , None, 45.7, -0.058, +0.833	SIM	MMXMD FEA3D margin matrix v3.5.2	line 425 row "TC-01" col 3	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T498	MF	DERIVED	MMXMD FEA3D margin matrix v3.5.2	line 425 row "TC-01" col 4	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T499	MF (n9090 regular, 1.0451 mm from the J-FWD RBE3 in the J-FWD region; the all-node …	DERIVED	MMXMD FEA3D margin matrix v3.5.2	line 425 row "TC-01" col 5	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T500	lead 21:15:29Z (3) under 234 (4); basis as the 239 TB70 ruling (lines 231 (3), 239)		MMXMD FEA3D margin matrix v3.5.2	line 425 row "TC-01" col 6	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T501	FA1S	SIM	MMXMD FEA3D margin matrix v3.5.2	line 426 row "TC-01" col 1	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T502	-0.107 / -0.037	SIM	MMXMD FEA3D margin matrix v3.5.2	line 426 row "TC-01" col 2	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T503	n9090, A1_22.2, regular , None, 45.7, -0.083, +0.785	SIM	MMXMD FEA3D margin matrix v3.5.2	line 426 row "TC-01" col 3	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T504	MF	DERIVED	MMXMD FEA3D margin matrix v3.5.2	line 426 row "TC-01" col 4	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T505	MF (n9090 regular, 1.0451 mm from the J-FWD RBE3 in the J-FWD region; the all-node …	DERIVED	MMXMD FEA3D margin matrix v3.5.2	line 426 row "TC-01" col 5	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T506	lead 21:15:29Z (3) under 234 (4); basis as the 239 TB70 ruling (lines 231 (3), 239)		MMXMD FEA3D margin matrix v3.5.2	line 426 row "TC-01" col 6	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T507	LC6 soak	SIM	MMXMD FEA3D margin matrix v3.5.2	line 427 row "TC-01" col 1	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T508	-0.356 / -0.482	SIM	MMXMD FEA3D margin matrix v3.5.2	line 427 row "TC-01" col 2	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T509	n13829, B1_70.7, regular , None, 159.2, +42.139, +0.932	SIM	MMXMD FEA3D margin matrix v3.5.2	line 427 row "TC-01" col 3	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T510	TS-183	DERIVED	MMXMD FEA3D margin matrix v3.5.2	line 427 row "TC-01" col 4	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T511	thermal-secondary (provisional, lead 08:04Z); in the J-FWD region (z >= 97.5) the J-FWD …	DERIVED	MMXMD FEA3D margin matrix v3.5.2	line 427 row "TC-01" col 5	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T512	lead 07:51Z (d), 08:04Z, 08:40Z		MMXMD FEA3D margin matrix v3.5.2	line 427 row "TC-01" col 6	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T513	TB70	SIM	MMXMD FEA3D margin matrix v3.5.2	line 428 row "TC-01" col 1	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T514	+0.120 / -0.110	SIM	MMXMD FEA3D margin matrix v3.5.2	line 428 row "TC-01" col 2	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T515	n9090, TB, regular , None, 132.0, +0.353, +2.360	SIM	MMXMD FEA3D margin matrix v3.5.2	line 428 row "TC-01" col 3	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T516	MF	DERIVED	MMXMD FEA3D margin matrix v3.5.2	line 428 row "TC-01" col 4	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T517	n9090 1.05 mm from the J-FWD RBE3: connector artefact, closed by the J-FWD sub-model …	DERIVED	MMXMD FEA3D margin matrix v3.5.2	line 428 row "TC-01" col 5	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table

ID	Value	Class	Source	Key	Source tag / formula
T518	lead 08:58Z		MMXMD FEA3D margin matrix v3.5.2	`line 428 row "TC-01" col 6`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T519	J-FWD	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 429 row "TC-01" col 1`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); J-FWD cell classes (margin_matrix_v352.json): governing[0] SOAK (MS_y -0.075): CALC / ...
T520	-0.075 / +0.239	CALC	MMXMD FEA3D margin matrix v3.5.2	`line 429 row "TC-01" col 2`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); J-FWD cell classes (margin_matrix_v352.json): governing[0] SOAK (MS_y -0.075): CALC / ...
T521		CALC	MMXMD FEA3D margin matrix v3.5.2	`line 429 row "TC-01" col 3`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); J-FWD cell classes (margin_matrix_v352.json): governing[0] SOAK (MS_y -0.075): CALC / ...
T522	DC	DERIVED	MMXMD FEA3D margin matrix v3.5.2	`line 429 row "TC-01" col 4`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); J-FWD cell classes (margin_matrix_v352.json): governing[0] SOAK (MS_y -0.075): CALC / ...
T523	TC-01 seat / ring line P+Q (secondary, shakedown, P+Q <= 2 Fty(T)) FOUND NEGATIVE on the ...	DERIVED	MMXMD FEA3D margin matrix v3.5.2	`line 429 row "TC-01" col 5`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); J-FWD cell classes (margin_matrix_v352.json): governing[0] SOAK (MS_y -0.075): CALC / ...
T524	flags (0 to +0.05): FA1S (ii) hole bearing, CALC V 3888.1 N, L 2.365 (governing, 240 ...		MMXMD FEA3D margin matrix v3.5.2	`line 429 row "TC-01" col 6`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); J-FWD cell classes (margin_matrix_v352.json): governing[0] SOAK (MS_y -0.075): CALC / ...
T525	LEAD RULING R2-B (2026-10-21T07:37:34Z): the ENG-TC-02 plug OD is a SELECT FIT to its ...	DECIDED *Lead ruling R2-B (2026-10-21T07:37:34Z), quoted verbatim in Mechanical line 254 (1)*	CL 266 lines	`L254(1)[from "LEAD RULING R2-B" to "to the measured bore."]`	mechanical (result) 2026-10-21 (line sha c7d52385af9ce594); class set at the call site: the text quotes LEAD RULING R2-B verbatim
T526	0.04 mm	DECIDED *Lead ruling R2-B (2026-10-21T07:37:34Z), quoted verbatim in Mechanical line 254 (1)*	CL 266 lines	`L254(1)[from "LEAD RULING R2-B" to "to the measured bore."] [regex radial gap g in \ [([\d.]+), [\d.]+\] mm]`	mechanical (result) 2026-10-21 (line sha c7d52385af9ce594); class set at the call site: the text quotes LEAD RULING R2-B verbatim
T527	0.0631 mm	DECIDED *Lead ruling R2-B (2026-10-21T07:37:34Z), quoted verbatim in Mechanical line 254 (1)*	CL 266 lines	`L254(1)[from "LEAD RULING R2-B" to "to the measured bore."] [regex radial gap g in \ [[\d.]+, ([\d.]+)\] mm]`	mechanical (result) 2026-10-21 (line sha c7d52385af9ce594); class set at the call site: the text quotes LEAD RULING R2-B verbatim
T528	v1 · sha256/16 1ba3147b42edfeed	DERIVED	BAND v1	`(file)`	generator: version and sha256/16 of source BAND at this build [jfwd_pq_band_v352_v1.json]
T529	TB70	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 430 row "TC-02" col 1`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T530	-0.236 / -0.271	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 430 row "TC-02" col 2`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T531	n30266, TB, regular , None, 228.7, +1.267, +1.292	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 430 row "TC-02" col 3`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T532	TS-212	DERIVED	MMXMD FEA3D margin matrix v3.5.2	`line 430 row "TC-02" col 4`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T533	TC-02 under 212 (1): (a)-(c) pass; (d) time at temperature open with T-F1 (Mechanical's ...	DERIVED	MMXMD FEA3D margin matrix v3.5.2	`line 430 row "TC-02" col 5`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T534	REV-MECH-2 rows 2-4; lead 08:58Z, 09:15Z (3)		MMXMD FEA3D margin matrix v3.5.2	`line 430 row "TC-02" col 6`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T535	TBS	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 431 row "TC-02" col 1`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T536	+0.138 / -0.035	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 431 row "TC-02" col 2`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T537	n30266, TB, regular , None, 228.7, +2.131, +1.276	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 431 row "TC-02" col 3`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T538	TS-212	DERIVED	MMXMD FEA3D margin matrix v3.5.2	`line 431 row "TC-02" col 4`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T539	TC-02 under 212 (1): (a)-(c) pass; (d) time at temperature open with T-F1 (Mechanical's ...	DERIVED	MMXMD FEA3D margin matrix v3.5.2	`line 431 row "TC-02" col 5`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T540	REV-MECH-2 rows 2-4; lead 08:58Z, 09:15Z (3)		MMXMD FEA3D margin matrix v3.5.2	`line 431 row "TC-02" col 6`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T541	LC1	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 432 row "TC-03" col 1`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T542	-0.088 / -0.170	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 432 row "TC-03" col 2`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T543	n46040, B1_1, regular , None, 89.6, +8.043, +1.737	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 432 row "TC-03" col 3`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T544	TS-183	DERIVED	MMXMD FEA3D margin matrix v3.5.2	`line 432 row "TC-03" col 4`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T545	thermal-secondary 183 (1): (a) primary and (b) P+Q pass; (c) LCF and (d) GlidCop time at ...	DERIVED	MMXMD FEA3D margin matrix v3.5.2	`line 432 row "TC-03" col 5`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T546	REV-MECH-2 rows 5-7; lead 07:51Z (a), 08:58Z		MMXMD FEA3D margin matrix v3.5.2	`line 432 row "TC-03" col 6`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T547	LC2	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 433 row "TC-03" col 1`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T548	-0.092 / -0.175	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 433 row "TC-03" col 2`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T549	n46040, B1_1, regular , None, 89.6, +7.684, +1.723	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 433 row "TC-03" col 3`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T550	TS-183	DERIVED	MMXMD FEA3D margin matrix v3.5.2	`line 433 row "TC-03" col 4`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T551	thermal-secondary 183 (1): (a) primary and (b) P+Q pass; (c) LCF and (d) GlidCop time at ...	DERIVED	MMXMD FEA3D margin matrix v3.5.2	`line 433 row "TC-03" col 5`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T552	REV-MECH-2 rows 5-7; lead 07:51Z (a), 08:58Z		MMXMD FEA3D margin matrix v3.5.2	`line 433 row "TC-03" col 6`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T553	LC3 hot bound	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 434 row "TC-03" col 1`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T554	-0.107 / -0.188	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 434 row "TC-03" col 2`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T555	n38065, B1_1, regular , None, 89.6, n/a, n/a	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 434 row "TC-03" col 3`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T556	TS-183	DERIVED	MMXMD FEA3D margin matrix v3.5.2	`line 434 row "TC-03" col 4`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T557	thermal-secondary 183 (1): (a) primary and (b) P+Q pass; (c) LCF and (d) GlidCop time at ...	DERIVED	MMXMD FEA3D margin matrix v3.5.2	`line 434 row "TC-03" col 5`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T558	REV-MECH-2 rows 5-7; lead 07:51Z (a), 08:58Z		MMXMD FEA3D margin matrix v3.5.2	`line 434 row "TC-03" col 6`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T559	F2 hot bound	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 435 row "TC-03" col 1`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T560	-0.107 / -0.188	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 435 row "TC-03" col 2`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T561	n38065, B1_1, regular , None, 89.6, n/a, n/a	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 435 row "TC-03" col 3`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T562	TS-183	DERIVED	MMXMD FEA3D margin matrix v3.5.2	`line 435 row "TC-03" col 4`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T563	thermal-secondary 183 (1): (a) primary and (b) P+Q pass; (c) LCF and (d) GlidCop time at ...	DERIVED	MMXMD FEA3D margin matrix v3.5.2	`line 435 row "TC-03" col 5`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T564	REV-MECH-2 rows 5-7; lead 07:51Z (a), 08:58Z		MMXMD FEA3D margin matrix v3.5.2	`line 435 row "TC-03" col 6`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T565	TB70	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 436 row "TC-03" col 1`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T566	-0.121 / -0.201	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 436 row "TC-03" col 2`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table

ID	Value	Class	Source	Key	Source tag / formula
T567	n48254, TB, regular , None, 181.3, +6.297, +1.638	SIM	MMXMD FEA3D margin matrix v3.5.2	line 436 row "TC-03" col 3	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T568	TS-183	DERIVED	MMXMD FEA3D margin matrix v3.5.2	line 436 row "TC-03" col 4	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T569	thermal-secondary 183 (1): (a) primary and (b) P+Q pass; (c) LCF and (d) GlidCop time at ...	DERIVED	MMXMD FEA3D margin matrix v3.5.2	line 436 row "TC-03" col 5	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T570	REV-MECH-2 rows 5-7; lead 07:51Z (a), 08:58Z		MMXMD FEA3D margin matrix v3.5.2	line 436 row "TC-03" col 6	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T571	LC3 hot bound	SIM	MMXMD FEA3D margin matrix v3.5.2	line 437 row "TC-04" col 1	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T572	+0.228 / -0.004	SIM	MMXMD FEA3D margin matrix v3.5.2	line 437 row "TC-04" col 2	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T573	n42993, B1_19, BT bad_tets, 0.0, 30.0, n/a, n/a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 437 row "TC-04" col 3	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T574	BT	DERIVED	MMXMD FEA3D margin matrix v3.5.2	line 437 row "TC-04" col 4	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T575	found negative: DERIVED lower bound at a mesh-defect node (not an exception under 234 ...	DERIVED	MMXMD FEA3D margin matrix v3.5.2	line 437 row "TC-04" col 5	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T576	lead 21:15:29Z (4)		MMXMD FEA3D margin matrix v3.5.2	line 437 row "TC-04" col 6	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T577	F2 hot bound	SIM	MMXMD FEA3D margin matrix v3.5.2	line 438 row "TC-04" col 1	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T578	+0.223 / -0.008	SIM	MMXMD FEA3D margin matrix v3.5.2	line 438 row "TC-04" col 2	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T579	n42993, B1_19, BT bad_tets, 0.0, 30.0, n/a, n/a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 438 row "TC-04" col 3	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T580	BT	DERIVED	MMXMD FEA3D margin matrix v3.5.2	line 438 row "TC-04" col 4	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T581	found negative: DERIVED lower bound at a mesh-defect node (not an exception under 234 ...	DERIVED	MMXMD FEA3D margin matrix v3.5.2	line 438 row "TC-04" col 5	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T582	lead 21:15:29Z (4)		MMXMD FEA3D margin matrix v3.5.2	line 438 row "TC-04" col 6	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T583	LC1	SIM	MMXMD FEA3D margin matrix v3.5.2	line 439 row "NZ-01" col 1	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T584	-0.520 / n/a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 439 row "NZ-01" col 2	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T585	n73561, B1_19, RE reentrant_edges, 0.5, 1311.6, n/a, n/a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 439 row "NZ-01" col 3	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T586	BR	DERIVED	MMXMD FEA3D margin matrix v3.5.2	line 439 row "NZ-01" col 4	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T587	brittle and thermal: 183 (2) capture / containment shown + test basis, else design change	DERIVED	MMXMD FEA3D margin matrix v3.5.2	line 439 row "NZ-01" col 5	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T588	REV-MECH-2 rows 8-10		MMXMD FEA3D margin matrix v3.5.2	line 439 row "NZ-01" col 6	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T589	LC2	SIM	MMXMD FEA3D margin matrix v3.5.2	line 440 row "NZ-01" col 1	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T590	-0.489 / n/a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 440 row "NZ-01" col 2	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T591	n74175, B1_1, BT bad_tets, 0.5389, 1332.4, n/a, n/a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 440 row "NZ-01" col 3	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T592	BR	DERIVED	MMXMD FEA3D margin matrix v3.5.2	line 440 row "NZ-01" col 4	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T593	brittle and thermal: 183 (2) capture / containment shown + test basis, else design change	DERIVED	MMXMD FEA3D margin matrix v3.5.2	line 440 row "NZ-01" col 5	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T594	REV-MECH-2 rows 8-10		MMXMD FEA3D margin matrix v3.5.2	line 440 row "NZ-01" col 6	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T595	LC3 hot bound	SIM	MMXMD FEA3D margin matrix v3.5.2	line 441 row "NZ-01" col 1	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T596	-0.403 / n/a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 441 row "NZ-01" col 2	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T597	part-wide bound (every node; hb_partwide_v352), stored minimum at: n51953, B1_1, regular ...	SIM	MMXMD FEA3D margin matrix v3.5.2	line 441 row "NZ-01" col 3	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T598	BR	DERIVED	MMXMD FEA3D margin matrix v3.5.2	line 441 row "NZ-01" col 4	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T599	brittle and thermal: 183 (2) capture / containment shown + test basis, else design change	DERIVED	MMXMD FEA3D margin matrix v3.5.2	line 441 row "NZ-01" col 5	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T600	REV-MECH-2 rows 8-10		MMXMD FEA3D margin matrix v3.5.2	line 441 row "NZ-01" col 6	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T601	F2 hot bound	SIM	MMXMD FEA3D margin matrix v3.5.2	line 442 row "NZ-01" col 1	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T602	-0.401 / n/a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 442 row "NZ-01" col 2	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T603	n51953, B1_1, regular , None, 610.6, n/a, n/a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 442 row "NZ-01" col 3	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T604	BR	DERIVED	MMXMD FEA3D margin matrix v3.5.2	line 442 row "NZ-01" col 4	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T605	brittle and thermal: 183 (2) capture / containment shown + test basis, else design change	DERIVED	MMXMD FEA3D margin matrix v3.5.2	line 442 row "NZ-01" col 5	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T606	REV-MECH-2 rows 8-10		MMXMD FEA3D margin matrix v3.5.2	line 442 row "NZ-01" col 6	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T607	LC6 soak	SIM	MMXMD FEA3D margin matrix v3.5.2	line 443 row "NZ-01" col 1	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T608	-0.426 / n/a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 443 row "NZ-01" col 2	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T609	n75257, B1_70.7, regular , None, 911.8, n/a, n/a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 443 row "NZ-01" col 3	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T610	BR	DERIVED	MMXMD FEA3D margin matrix v3.5.2	line 443 row "NZ-01" col 4	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T611	brittle and thermal: 183 (2) capture / containment shown + test basis, else design change	DERIVED	MMXMD FEA3D margin matrix v3.5.2	line 443 row "NZ-01" col 5	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T612	REV-MECH-2 rows 8-10		MMXMD FEA3D margin matrix v3.5.2	line 443 row "NZ-01" col 6	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T613	TB70	SIM	MMXMD FEA3D margin matrix v3.5.2	line 444 row "NZ-01" col 1	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T614	-0.315 / n/a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 444 row "NZ-01" col 2	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T615	n75213, TB, RE reentrant_edges, 0.2789, 1374.0, n/a, n/a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 444 row "NZ-01" col 3	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T616	BR	DERIVED	MMXMD FEA3D margin matrix v3.5.2	line 444 row "NZ-01" col 4	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T617	brittle and thermal: 183 (2) capture / containment shown + test basis, else design change	DERIVED	MMXMD FEA3D margin matrix v3.5.2	line 444 row "NZ-01" col 5	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T618	REV-MECH-2 rows 8-10		MMXMD FEA3D margin matrix v3.5.2	line 444 row "NZ-01" col 6	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table

ID	Value	Class	Source	Key	Source tag / formula
T619	TBS	SIM	MMXMD FEA3D margin matrix v3.5.2	line 445 row "NZ-01" col 1	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T620	-0.054 / n/a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 445 row "NZ-01" col 2	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T621	n75213, TB, RE reentrant_edges, 0.2789, 1374.0, n/a, n/a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 445 row "NZ-01" col 3	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T622	BR	DERIVED	MMXMD FEA3D margin matrix v3.5.2	line 445 row "NZ-01" col 4	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T623	brittle and thermal: 183 (2) capture / containment shown + test basis, else design change	DERIVED	MMXMD FEA3D margin matrix v3.5.2	line 445 row "NZ-01" col 5	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T624	REV-MECH-2 rows 8-10		MMXMD FEA3D margin matrix v3.5.2	line 445 row "NZ-01" col 6	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T625	LC2	SIM	MMXMD FEA3D margin matrix v3.5.2	line 446 row "NZ-03" col 1	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T626	-0.088 / n/a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 446 row "NZ-03" col 2	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T627	n81357, B1_21.65, regular , None, 1294.1, n/a, n/a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 446 row "NZ-03" col 3	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T628	CH	DERIVED	MMXMD FEA3D margin matrix v3.5.2	line 446 row "NZ-03" col 4	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T629	char primary vs the ASSUMED 20 MPa: N-F1 (>= 21.96 MPa inside the 23.7 MPa acceptance) ...	DERIVED	MMXMD FEA3D margin matrix v3.5.2	line 446 row "NZ-03" col 5	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T630	REV-MECH-2 LC2 table		MMXMD FEA3D margin matrix v3.5.2	line 446 row "NZ-03" col 6	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T631	TB70	SIM	MMXMD FEA3D margin matrix v3.5.2	line 447 row "NZ-03" col 1	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T632	-0.392 / n/a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 447 row "NZ-03" col 2	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T633	n81357, TB, regular , None, 1359.5, n/a, n/a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 447 row "NZ-03" col 3	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T634	CH	DERIVED	MMXMD FEA3D margin matrix v3.5.2	line 447 row "NZ-03" col 4	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T635	char primary vs the ASSUMED 20 MPa: N-F1 (>= 21.96 MPa inside the 23.7 MPa acceptance) ...	DERIVED	MMXMD FEA3D margin matrix v3.5.2	line 447 row "NZ-03" col 5	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T636	REV-MECH-2 LC2 table		MMXMD FEA3D margin matrix v3.5.2	line 447 row "NZ-03" col 6	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T637	TBS	SIM	MMXMD FEA3D margin matrix v3.5.2	line 448 row "NZ-03" col 1	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T638	-0.253 / n/a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 448 row "NZ-03" col 2	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T639	n81357, TB, regular , None, 1359.5, n/a, n/a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 448 row "NZ-03" col 3	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T640	CH	DERIVED	MMXMD FEA3D margin matrix v3.5.2	line 448 row "NZ-03" col 4	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T641	char primary vs the ASSUMED 20 MPa: N-F1 (>= 21.96 MPa inside the 23.7 MPa acceptance) ...	DERIVED	MMXMD FEA3D margin matrix v3.5.2	line 448 row "NZ-03" col 5	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T642	REV-MECH-2 LC2 table		MMXMD FEA3D margin matrix v3.5.2	line 448 row "NZ-03" col 6	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T643	LC6 soak	SIM	MMXMD FEA3D margin matrix v3.5.2	line 449 row "FD-01" col 1	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T644	-0.187 / +0.780	SIM	MMXMD FEA3D margin matrix v3.5.2	line 449 row "FD-01" col 2	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T645	n124943, B1_70.7, ID tie_patch_edges, 0.0, 99.2, +80.663, +1.440	SIM	MMXMD FEA3D margin matrix v3.5.2	line 449 row "FD-01" col 3	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T646	TS-183	DERIVED	MMXMD FEA3D margin matrix v3.5.2	line 449 row "FD-01" col 4	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T647	thermal-secondary 183 (1), all-node value at the FD01_TC02_screws bonded-patch edge kept ...	DERIVED	MMXMD FEA3D margin matrix v3.5.2	line 449 row "FD-01" col 5	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T648	lead 08:58Z, 09:47Z (6)		MMXMD FEA3D margin matrix v3.5.2	line 449 row "FD-01" col 6	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T649	TB70	SIM	MMXMD FEA3D margin matrix v3.5.2	line 450 row "FD-01" col 1	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T650	-0.336 / +0.484	SIM	MMXMD FEA3D margin matrix v3.5.2	line 450 row "FD-01" col 2	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T651	n124901, TB, ID tie_patch_edges, 0.0, 113.8, +1.939, +0.991	SIM	MMXMD FEA3D margin matrix v3.5.2	line 450 row "FD-01" col 3	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T652	TS-183	DERIVED	MMXMD FEA3D margin matrix v3.5.2	line 450 row "FD-01" col 4	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T653	thermal-secondary 183 (1), all-node value at the FD01_TC02_screws bonded-patch edge kept ...	DERIVED	MMXMD FEA3D margin matrix v3.5.2	line 450 row "FD-01" col 5	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T654	lead 08:58Z, 09:47Z (6)		MMXMD FEA3D margin matrix v3.5.2	line 450 row "FD-01" col 6	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T655	TBS	SIM	MMXMD FEA3D margin matrix v3.5.2	line 451 row "FD-01" col 1	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T656	-0.005 / +0.976	SIM	MMXMD FEA3D margin matrix v3.5.2	line 451 row "FD-01" col 2	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T657	n124901, TB, ID tie_patch_edges, 0.0, 113.8, +3.427, +0.989	SIM	MMXMD FEA3D margin matrix v3.5.2	line 451 row "FD-01" col 3	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T658	TS-183	DERIVED	MMXMD FEA3D margin matrix v3.5.2	line 451 row "FD-01" col 4	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T659	thermal-secondary 183 (1), all-node value at the FD01_TC02_screws bonded-patch edge kept ...	DERIVED	MMXMD FEA3D margin matrix v3.5.2	line 451 row "FD-01" col 5	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T660	lead 08:58Z, 09:47Z (6)		MMXMD FEA3D margin matrix v3.5.2	line 451 row "FD-01" col 6	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T661	LC6 soak	SIM	MMXMD FEA3D margin matrix v3.5.2	line 452 row "PD-01R" col 1	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T662	-0.315 / -0.451	SIM	MMXMD FEA3D margin matrix v3.5.2	line 452 row "PD-01R" col 2	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T663	n115356, B1_70.7, regular , None, 96.4, +24.533, +1.055	SIM	MMXMD FEA3D margin matrix v3.5.2	line 452 row "PD-01R" col 3	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T664	MF	DERIVED	MMXMD FEA3D margin matrix v3.5.2	line 452 row "PD-01R" col 4	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T665	PD-02 rigid in the model (same alloy): the PD-01 / PD-02 mismatch is a model artefact ...	DERIVED	MMXMD FEA3D margin matrix v3.5.2	line 452 row "PD-01R" col 5	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T666	lead 07:51Z (e), 08:04Z		MMXMD FEA3D margin matrix v3.5.2	line 452 row "PD-01R" col 6	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T667	TB70	SIM	MMXMD FEA3D margin matrix v3.5.2	line 453 row "PD-01R" col 1	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T668	-0.387 / -0.509	SIM	MMXMD FEA3D margin matrix v3.5.2	line 453 row "PD-01R" col 2	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T669	n115356, TB, regular , None, 104.1, +22.563, +0.840	SIM	MMXMD FEA3D margin matrix v3.5.2	line 453 row "PD-01R" col 3	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T670	MF	DERIVED	MMXMD FEA3D margin matrix v3.5.2	line 453 row "PD-01R" col 4	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T671	PD-02 rigid in the model (same alloy): the PD-01 / PD-02 mismatch is a model artefact ...	DERIVED	MMXMD FEA3D margin matrix v3.5.2	line 453 row "PD-01R" col 5	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table

ID	Value	Class	Source	Key	Source tag / formula
T672	lead 07:51Z (e), 08:04Z		MMXMD FEA3D margin matrix v3.5.2	`line 453 row "PD-01R" col 6`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T673	TBS	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 454 row "PD-01R" col 1`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T674	-0.081 / -0.346	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 454 row "PD-01R" col 2`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T675	n115356, TB, regular , None, 104.1, +22.266, +0.839	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 454 row "PD-01R" col 3`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T676	MF	DERIVED	MMXMD FEA3D margin matrix v3.5.2	`line 454 row "PD-01R" col 4`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T677	PD-02 rigid in the model (same alloy): the PD-01 / PD-02 mismatch is a model artefact ...	DERIVED	MMXMD FEA3D margin matrix v3.5.2	`line 454 row "PD-01R" col 5`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T678	lead 07:51Z (e), 08:04Z		MMXMD FEA3D margin matrix v3.5.2	`line 454 row "PD-01R" col 6`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T679	LC6 soak	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 455 row "PD-01S" col 1`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T680	-0.378 / -0.502	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 455 row "PD-01S" col 2`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T681	n118875, B1_70.7, regular , None, 96.4, +26.014, +0.865	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 455 row "PD-01S" col 3`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T682	MF	DERIVED	MMXMD FEA3D margin matrix v3.5.2	`line 455 row "PD-01S" col 4`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T683	PD-02 rigid in the model (same alloy): the PD-01 / PD-02 mismatch is a model artefact ...	DERIVED	MMXMD FEA3D margin matrix v3.5.2	`line 455 row "PD-01S" col 5`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T684	lead 07:51Z (e), 08:04Z		MMXMD FEA3D margin matrix v3.5.2	`line 455 row "PD-01S" col 6`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T685	TB70	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 456 row "PD-01S" col 1`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T686	-0.444 / -0.555	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 456 row "PD-01S" col 2`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T687	n118875, TB, regular , None, 104.1, +23.533, +0.669	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 456 row "PD-01S" col 3`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T688	MF	DERIVED	MMXMD FEA3D margin matrix v3.5.2	`line 456 row "PD-01S" col 4`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T689	PD-02 rigid in the model (same alloy): the PD-01 / PD-02 mismatch is a model artefact ...	DERIVED	MMXMD FEA3D margin matrix v3.5.2	`line 456 row "PD-01S" col 5`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T690	lead 07:51Z (e), 08:04Z		MMXMD FEA3D margin matrix v3.5.2	`line 456 row "PD-01S" col 6`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T691	TBS	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 457 row "PD-01S" col 1`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T692	-0.166 / -0.407	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 457 row "PD-01S" col 2`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T693	n118875, TB, regular , None, 104.1, +23.225, +0.669	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 457 row "PD-01S" col 3`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T694	MF	DERIVED	MMXMD FEA3D margin matrix v3.5.2	`line 457 row "PD-01S" col 4`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T695	PD-02 rigid in the model (same alloy): the PD-01 / PD-02 mismatch is a model artefact ...	DERIVED	MMXMD FEA3D margin matrix v3.5.2	`line 457 row "PD-01S" col 5`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T696	lead 07:51Z (e), 08:04Z		MMXMD FEA3D margin matrix v3.5.2	`line 457 row "PD-01S" col 6`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T697	33	DERIVED	generator	`formula: count of rows of the matrix table "Negative cells and their classes"`	DERIVED by build_datasheet.py: count of rows of the matrix table "Negative cells and their classes"
T698	7	DERIVED	generator	`formula: count of negative cells with class BR`	DERIVED by build_datasheet.py: count of negative cells with class BR ← T586, T592, T598, T604, T610, T616, T622
T699	2	DERIVED	generator	`formula: count of negative cells with class BT`	DERIVED by build_datasheet.py: count of negative cells with class BT ← T574, T580
T700	3	DERIVED	generator	`formula: count of negative cells with class CH`	DERIVED by build_datasheet.py: count of negative cells with class CH ← T628, T634, T640
T701	1	DERIVED	generator	`formula: count of negative cells with class DC`	DERIVED by build_datasheet.py: count of negative cells with class DC ← T522
T702	9	DERIVED	generator	`formula: count of negative cells with class MF`	DERIVED by build_datasheet.py: count of negative cells with class MF ← T498, T504, T516, T664, T670, T676, T682, T688 ...
T703	9	DERIVED	generator	`formula: count of negative cells with class TS-183`	DERIVED by build_datasheet.py: count of negative cells with class TS-183 ← T510, T544, T550, T556, T562, T568, T646, T652 ...
T704	2	DERIVED	generator	`formula: count of negative cells with class TS-212`	DERIVED by build_datasheet.py: count of negative cells with class TS-212 ← T532, T538
T705	2	DERIVED PENDING	generator	`formula: count of inputs pending at this build (§1.4)`	DERIVED by build_datasheet.py: count of inputs pending at this build (§1.4) ← T030, T494
T706	ovr-7: 6 active, 0 redundant, 0 in error; 4 superseded values withheld (4 interface ...	DERIVED	generator	`formula: count of override entries by status on this build`	DERIVED by build_datasheet.py: count of override entries by status on this build ← T004, T008, T013, T015, T019, T021
T707	v3.5.2 final · sha256/16 621481ddeccc228b	DERIVED	SET v3.5.2 final	`(file)`	generator: version and sha256/16 of source SET at this build [set_v3.5.2.json]
T708	456 rows · sha256/16 a4713d45b3b24a0f	DERIVED	CADMAN 456 rows	`(file)`	generator: version and sha256/16 of source CADMAN at this build [MANIFEST_v3.5.2.sha256]
T709	v3.5.2 · sha256/16 4bcc264f683e128f	DERIVED	GLB v3.5.2	`(file)`	generator: version and sha256/16 of source GLB at this build [ENGINE_v3.5.2_neutral.glb]
T710	v3.5.2 · sha256/16 9234c953c5b6beb8	DERIVED	BOM v3.5.2	`(file)`	generator: version and sha256/16 of source BOM at this build [BOM_v3.5.2.json]
T711	v3.5.2 · sha256/16 c1ff170be0ac8cc5	DERIVED	MP v3.5.2	`(file)`	generator: version and sha256/16 of source MP at this build [massprops_v3.5.2.json]
T712	v2.4d · sha256/16 7dbf425c12300658	DERIVED	CMB v2.4d	`(file)`	generator: version and sha256/16 of source CMB at this build [combustion.json]
T713	v2.4d · sha256/16 8dc583578748b06f	DERIVED	DSIC v2.4d	`(file)`	generator: version and sha256/16 of source DSIC at this build [DATASHEET-INPUTS-COMBUSTION.md]
T714	v2.4 · sha256/16 66bc53d88d444679	DERIVED	MEC v2.4	`(file)`	generator: version and sha256/16 of source MEC at this build [mechanical.json]
T715	226 rows · sha256/16 4f7e7655ccc25f70	DERIVED	MMAN 226 rows	`(file)`	generator: version and sha256/16 of source MMAN at this build [MANIFEST_v352.sha256]
T716	v3.2 · sha256/16 bde317782574cbd9	DERIVED	GNC v3.2	`(file)`	generator: version and sha256/16 of source GNC at this build [gnc.json]
T717	v3.2 · sha256/16 b8b0dcaf45a5d915	DERIVED	DSIG v3.2	`(file)`	generator: version and sha256/16 of source DSIG at this build [DATASHEET-INPUTS-GNC.md]
T718	v4.1 · sha256/16 c862375163393adc	DERIVED	ELE v4.1	`(file)`	generator: version and sha256/16 of source ELE at this build [electronics.json]
T719	FINAL handback of the v3.5.2 re-check · sha256/16 dc53f4cd45d75f4e	DERIVED	F3D FINAL handback of the v3.5.2 re-check	`(file)`	generator: version and sha256/16 of source F3D at this build [REPORT-FEA3D-v352.md]
T720	v352 · sha256/16 6c694fa7147fb511	DERIVED	MMX v352	`(file)`	generator: version and sha256/16 of source MMX at this build [margin_matrix_v352.json]
T721	266 lines · sha256/16 4cde8acb297d195c	DERIVED	CL 266 lines	`(file)`	generator: version and sha256/16 of source CL at this build [CHANGELOG.md]
T722	18 reviews · sha256/16 076172727cddd074	DERIVED	SUP 18 reviews	`(file)`	generator: version and sha256/16 of source SUP at this build [SUPERVISION.json]
T723	rev 2 · sha256/16 012b14cbaaddfdc6	DERIVED	REQ rev 2	`(file)`	generator: version and sha256/16 of source REQ at this build [REQUIREMENTS.md]
T724	draft 0.2 · sha256/16 706455abf4098182	DERIVED	ICD draft 0.2	`(file)`	generator: version and sha256/16 of source ICD at this build [ICD-AIRFRAME.md]
T725	FINAL v3.5.2 CAD set (lead release 19:09:50Z, chain E ended 19:06:26Z; publication ...	DECIDED *Design value, set_v3.5.2.json v3.5.2 final*	SET v3.5.2 final	`/status`	set_v3.5.2.json v3.5.2 final
T726	v3.4.1	DERIVED	REV0S v3.4.1	`/cad_version`	(class set at the call site: DERIVED) [rev0 sources.json]
T727	v3.5.2	DECIDED *Design value, render_manifest.json v3.5.2*	RMAN v3.5.2	`/cad_version`	tag in render_manifest.json v3.5.2: CAD version of this build (render manifest and CAD folder)

ID	Value	Class	Source	Key	Source tag / formula
T728	v2.4	DERIVED	REV0S v3.4.1	/sources/CMB/version	(class set at the call site: DERIVED) [rev0 sources.json]
T729	v2.4d	DERIVED	CMB v2.4d	(file version)	version of source CMB at this build
T730	v2	DERIVED	REV0S v3.4.1	/sources/MEC/version	(class set at the call site: DERIVED) [rev0 sources.json]
T731	v2.4	DERIVED	MEC v2.4	(file version)	version of source MEC at this build
T732	v3	DERIVED	REV0S v3.4.1	/sources/GNC/version	(class set at the call site: DERIVED) [rev0 sources.json]
T733	v3.2	DERIVED	GNC v3.2	(file version)	version of source GNC at this build
T734	v2	DERIVED	REV0S v3.4.1	/sources/ELE/version	(class set at the call site: DERIVED) [rev0 sources.json]
T735	v4.1	DERIVED	ELE v4.1	(file version)	version of source ELE at this build
T736	176 lines	DERIVED	REV0S v3.4.1	/sources/CL/version	(class set at the call site: DERIVED) [rev0 sources.json]
T737	266 lines	DERIVED	CL 266 lines	(file version)	version of source CL at this build
T738	ENG-FD-01; ENG-FD-02; ENG-FD-04; ENG-GM-01; ENG-GM-02; ENG-NZ-01; ENG-NZ-06; ENG-TC-01 ...	DECIDED *Design value, set_v3.5.2.json v3.5.2 final*	SET v3.5.2 final	/changed_parts_vs_v3.5.1	set_v3.5.2.json v3.5.2 final
T739	BELOW the trigger with and without the conditional TC-09 thinning: no re-fly by mass	DERIVED	MEC v2.4	/gimballed_mass_properties_v352/delta_vs_v3.5.1/gnc_trigger/verdict	(class set at the call site: DERIVED)
T740	REV-GNC-7 ACCEPTED: its final (reviews/REVIEW-GNC-R5.md 6b86fcd326568646, saved verbatim ...	DECIDED *Lead decision, line 258 (1)*	CL 266 lines	L258(1)	lead decision 2026-10-21 (line sha cdd58cc9fdc95079)
T741	120 N m/rad	DERIVED	CL 266 lines	L258(1) [regex rev0 datasheet row 8 window (\d+)-(\d+) is withdrawn]	lead decision 2026-10-21 (line sha cdd58cc9fdc95079); (class set at the call site: DERIVED) [upper end of the rev0 window (DERIVED on CAD v3.5.1, withdrawn by line 258 (1))]
T742	Design-to 100-110 N m/rad at the nominal servo J 1.6e-3 kg m2 (cons): the V5c window ...	DERIVED	DSIG v3.2	line 29 row "ENG-11 servo output stiffness K_s" col 2	DERIVED (linear ENG-07d, one fixed ECU notch, K_s grid 90-160 step 5, edges +-5 N m/rad); CAD v3.5.2 R3 plant and band ends checked by REV-GNC-7 (the 100-120 of CAD v3.5.1 does not hold on CAD ...
T743	100 N m/rad	DERIVED	DSIG v3.2	line 29 row "ENG-11 servo output stiffness K_s" col 2 [regex Design-to (\d+)-(\d+) N m/rad]	DERIVED (linear ENG-07d, one fixed ECU notch, K_s grid 90-160 step 5, edges +-5 N m/rad); CAD v3.5.2 R3 plant and band ends checked by REV-GNC-7 (the 100-120 of CAD v3.5.1 does not hold on CAD ...
T744	110 N m/rad	DERIVED	DSIG v3.2	line 29 row "ENG-11 servo output stiffness K_s" col 2 [regex Design-to (\d+)-(\d+) N m/rad]	DERIVED (linear ENG-07d, one fixed ECU notch, K_s grid 90-160 step 5, edges +-5 N m/rad); CAD v3.5.2 R3 plant and band ends checked by REV-GNC-7 (the 100-120 of CAD v3.5.1 does not hold on CAD ...
T745	0.0016 kg m²	ASSUMED	DSIG v3.2	line 29 row "ENG-11 servo output stiffness K_s" col 2 [regex servo J ([\d.]+e-\d+) kg m2]	DERIVED (linear ENG-07d, one fixed ECU notch, K_s grid 90-160 step 5, edges +-5 N m/rad); CAD v3.5.2 R3 plant and band ends checked by REV-GNC-7 (the 100-120 of CAD v3.5.1 does not hold on CAD ...
T746	It is not robust: it moves with J (R3 pose table, cons: 0.76e-3 130→= 160; 2.7e-3 ...	DERIVED	DSIG v3.2	line 29 row "ENG-11 servo output stiffness K_s" col 2 [sentence containing "not robust"]	DERIVED (linear ENG-07d, one fixed ECU notch, K_s grid 90-160 step 5, edges +-5 N m/rad); CAD v3.5.2 R3 plant and band ends checked by REV-GNC-7 (the 100-120 of CAD v3.5.1 does not hold on CAD ...
T747	Lead consistency fold found while applying: the case-TC in-burn abort 423.15 K (243 (3) ...	DECIDED *Lead decision, line 257*	CL 266 lines	L257[sentence containing "423.15"]	lead decision 2026-10-21 (line sha cf364643652fceb0)
T748	2026-10-21, FINAL handback of the v3.5.2 re-check (seat FEA3D-2 from 2026-10-21T00:41Z ...	DERIVED	F3D FINAL handback of the v3.5.2 re-check	§REPORT FEA3D v3.5.2 row c0="Status" col 1 (line 5)	FEA3D-2 report header
T749	Lead decisions and verifications after line 252 (2026-10-21 05:24Z to 07:09:15Z, times ...	DECIDED *Lead decision, line 253*	CL 266 lines	L253	lead (decision) 2026-10-21 (line sha 3a7f695ca59d621e)
T750	Lead line covering 07:13Z-07:52Z (times from date -u; SIM / CALC / DERIVED as tagged ...	DECIDED *Lead decision, line 255*	CL 266 lines	L255	lead (decision) 2026-10-21 (line sha c541950ce85dcf2b)
T751	Lead line covering 07:53Z-08:45Z (times from date -u; SIM / CALC / DERIVED as tagged ...	DECIDED *Lead decision, line 256*	CL 266 lines	L256	lead (decision) 2026-10-21 (line sha cf51938a477079b6)
T752	08:46-09:39Z (date -u). (1) COMBUSTION v2.4c PUBLISHED by the lead at 09:38:09Z (text ...	DECIDED *Lead decision, line 257*	CL 266 lines	L257	lead decision 2026-10-21 (line sha cf364643652fceb0)
T753	09:40-10:47Z (date -u). (1) REV-GNC-7 ACCEPTED: its final (reviews/REVIEW-GNC-R5.md ...	DECIDED *Lead decision, line 258*	CL 266 lines	L258	lead decision 2026-10-21 (line sha cdd58cc9fdc95079)
T754	10:48-10:57Z (date -u). (1) GNC.JSON v3.2 PUBLISHED by the lead (built and stamped ...	DECIDED *Lead decision, line 260*	CL 266 lines	L260	lead decision 2026-10-21 (line sha 33d3167adecb2242)
T755	10:58-11:26Z (date -u). (1) COMBUSTION v2.4d PUBLISHED by the lead at 11:24:52Z (text ...	DECIDED *Lead decision, line 261*	CL 266 lines	L261	lead decision 2026-10-21 (line sha d510d8077319e234)
T756	Lead decisions and verifications after line 261 (2026-10-21 11:29Z to 12:14Z, times from ...	DECIDED *Lead decision, line 262*	CL 266 lines	L262	lead (decision) 2026-10-21 (line sha 4d0bf4e937506f42)
T757	Lead rulings on the Electronics v2.12 handback (line 263), 2026-10-21 12:43:15Z to ...	DECIDED *Lead decision, line 264*	CL 266 lines	L264	lead (decision) 2026-10-21 (line sha 7cdbe51041ae6b8b) [1 working-path mention(s) replaced by a marker]
T758	Lead decisions and verifications after line 264 (2026-10-21 12:50Z to 14:46Z, times from ...	DECIDED *Lead decision, line 266*	CL 266 lines	L266	lead (decision) 2026-10-21 (line sha 6f3181b859f02937) [1 working-path mention(s) replaced by a marker]
T759	20 C saturated tank, initial port 25 mm, nominal regression law, eta_c* 0.92, eta_cf ...	SIM	DSIC v2.4d	line 7 row "Rated thrust (maximum commanded)" col 2	SIM (grain_q1d v2 on ASSUMED law, Cd and efficiencies; CEA)
T760	500 N; Pc 25.01 bar abs; O/F 10.44; Isp (sea level) 193.1 s; c* 1401 m/s delivered ...	SIM	DSIC v2.4d	line 7 row "Rated thrust (maximum commanded)" col 1	SIM (grain_q1d v2 on ASSUMED law, Cd and efficiencies; CEA)
T761	`combustion.json` operating_points.max_thrust		DSIC v2.4d	line 7 row "Rated thrust (maximum commanded)" col 4	source column
T762	MEOP basis: late light at the 24 C interlock with the designed line loss and the ...	SIM *Lead decision, per DATASHEET-INPUTS-COMBUSTION.md v2.4d*	DSIC v2.4d	line 8 row "Chamber pressure limits" col 2	LEAD (limits); SIM (basis)
T763	software cap 25 bar; ECU abort 30 bar; MEOP 45 bar; proof 67.5 bar; MEOP basis 44.33 bar ...	SIM *Lead decision, per DATASHEET-INPUTS-COMBUSTION.md v2.4d*	DSIC v2.4d	line 8 row "Chamber pressure limits" col 1	LEAD (limits); SIM (basis)
T764	`combustion.json` operating_points.pressure_limits, meop_basis_v2p4		DSIC v2.4d	line 8 row "Chamber pressure limits" col 4	source column
T765	ceiling = F at min(full stroke 4 mm, injector dP/Pc >= 0.25)	SIM	DSIC v2.4d	line 9 row "Thrust ceiling (available, not commanded)" col 2	SIM (injector models; Cd ASSUMED 0.5-0.8)
T766	ceiling 603-747 N for Cd 0.5-0.8 at 20 C (Dyer); worst model case HEM at 15 C, Cd 0.6 ...	SIM	DSIC v2.4d	line 9 row "Thrust ceiling (available, not commanded)" col 1	SIM (injector models; Cd ASSUMED 0.5-0.8)
T767	`combustion.json` operating_points.ceiling_vs_Cd_20C_and_models		DSIC v2.4d	line 9 row "Thrust ceiling (available, not commanded)" col 4	source column
T768	initial port, steady; the ECU abort (30 bar) acts on it; the hard-start transient is the ...	SIM	DSIC v2.4d	line 10 row "Worst-case steady thrust and Pc (pintle full-stroke failure)" col 2	SIM (grain_q1d v2 steady)
T769	619-822 N, Pc 30.7-40.3 bar abs at 15-25 C (Cd 0.6-0.8); 876 N, [withheld: superseded ...	SIM	DSIC v2.4d	line 10 row "Worst-case steady thrust and Pc (pintle full-stroke failure)" col 1	SIM (grain_q1d v2 steady)
T770	`combustion.json` operating_points.pressure_limits.full_stroke_failure_S230		DSIC v2.4d	line 10 row "Worst-case steady thrust and Pc (pintle full-stroke failure)" col 4	source column
T771	25 % floor (125 N): met (dP/Pc 5.8); 20 % target (100 N): met (dP/Pc 7.1); 12.5 % ...	SIM	DSIC v2.4d	line 11 row "Throttle range" col 2	SIM (floors REQ)
T772	62.5-500 N (12.5-100 %): Pc 4.37-25.01 bar abs; O/F 4.06-10.44; Isp (sea level) ...	SIM	DSIC v2.4d	line 11 row "Throttle range" col 1	SIM (floors REQ)
T773	`combustion.json` operating_points.throttle_points_20C_initial_port, floor_*		DSIC v2.4d	line 11 row "Throttle range" col 4	source column
T774	15-25 C tank (the 25 C end is a bound: ARM 15-24 C), equilibrium and non-equilibrium tank	SIM	DSIC v2.4d	line 12 row "Burn time and total impulse at a 500 N hold (2.0 kg tank)" col 2	SIM
T775	500 N held 4.06-7.01 s; liquid end 6.77-7.50 s; impulse to liquid end 3387-3616 N s; N2O ...	SIM	DSIC v2.4d	line 12 row "Burn time and total impulse at a 500 N hold (2.0 kg tank)" col 1	SIM
T776	`combustion.json` operating_points.max_thrust_hold_baseline_tank		DSIC v2.4d	line 12 row "Burn time and total impulse at a 500 N hold (2.0 kg tank)" col 4	source column
T777	gnc.json v3 records flown on the S230 grain (non-equilibrium tank); the A1 rows are the ...	SIM	DSIC v2.4d	line 13 row "Mission burns (2.0 kg tank, 3.167 L)" col 2	SIM

ID	Value	Class	Source	Key	Source tag / formula
T778	A1 dispersed p99: 8.26 s, 2968.2 N s, O/F 9.05, Isp 190.7 s (T0 294.12 K); A1 p99 ...	SIM	DSIC v2.4d	line 13 row "Mission burns (2.0 kg tank, 3.167 L)" col 1	SIM
T779	`02-combustion/data/v2/dc_gncv3/summary_v2p4.json` (`7f4f181b80d55331`); CHANGELOG 227 (9)		DSIC v2.4d	line 13 row "Mission burns (2.0 kg tank, 3.167 L)" col 4	source column
T780	2.20 kg: B1, thermal and erosion TO-VERIFY (line 227 (9)); no B1 record with 2.20 kg ...	DERIVED *Lead decision, per DATASHEET-INPUTS-COMBUSTION.md v2.4d*	DSIC v2.4d	line 14 row "N2O tank" col 2	LEAD (2.00 kg); DERIVED (first-order scaling)
T781	2.00 kg (founder, line 219). Option 2.20 kg: A1 burn p99 8.92 / 8.71 s (max 9.24 s) over ...	DERIVED *Lead decision, per DATASHEET-INPUTS-COMBUSTION.md v2.4d*	DSIC v2.4d	line 14 row "N2O tank" col 1	LEAD (2.00 kg); DERIVED (first-order scaling)
T782	`data/v2p4b/tank_2p20kg_option_v2p4b.json` (`59995244124f7c9a`)		DSIC v2.4d	line 14 row "N2O tank" col 4	source column
T783	graphite grade TO-QUOTE (Sønderborg DK-6400)	DECIDED *Design value, DATASHEET-INPUTS-COMBUSTION.md v2.4d*	DSIC v2.4d	line 15 row "Nozzle and grain" col 2	tag in DATASHEET-INPUTS-COMBUSTION.md v2.4d: DESIGN
T784	throat 13.72 mm, exit 19.41 mm, expansion ratio 2.0 (graphite insert, one per flight) ...	DECIDED *Design value, DATASHEET-INPUTS-COMBUSTION.md v2.4d*	DSIC v2.4d	line 15 row "Nozzle and grain" col 1	tag in DATASHEET-INPUTS-COMBUSTION.md v2.4d: DESIGN
T785	`combustion.json` gas_path		DSIC v2.4d	line 15 row "Nozzle and grain" col 4	source column
T786	lit thrust: grain_q1d over 72 corners of the ASSUMED bands; lift: ECU table (Electronics)	SIM / DERIVED *Design value, DATASHEET-INPUTS-COMBUSTION.md v2.4d*	DSIC v2.4d	line 16 row "Start level (relight)" col 2	DESIGN (lift); SIM (lit); DERIVED bound (cold flow)
T787	start-level lift 0.21776 mm at the A1 relight state (286.70 K), 0.2039 mm at 20 C (the ...	SIM / DERIVED *Design value, DATASHEET-INPUTS-COMBUSTION.md v2.4d*	DSIC v2.4d	line 16 row "Start level (relight)" col 1	DESIGN (lift); SIM (lit); DERIVED bound (cold flow)
T788	`data/v2p4b/p4c_lit_side_v2p4b.json` (`12f7699ddbeeecda`) ...		DSIC v2.4d	line 16 row "Start level (relight)" col 4	source column
T789	conditions (256 (2)): (a) 2.2 s at 283.15 K stated wherever 1.9 s is quoted; (b) ...	DERIVED / ASSUMED *Lead decision, per DATASHEET-INPUTS-COMBUSTION.md v2.4d; basis SIM*	DSIC v2.4d	line 17 row "Relight purge flow (decided)" col 2	LEAD (flow, cap, re-set rule); DERIVED (REV-COMB-4 on the SIM model; ASSUMED Cd); A1 reserve test DERIVED from SIM (REV-GNC-7)
T790	2.43 s scheduled flow (both-open overlap 2.38 s + 0.05 s) with the 2.39 s both-open cap ...	DERIVED / ASSUMED *Lead decision, per DATASHEET-INPUTS-COMBUSTION.md v2.4d; basis SIM*	DSIC v2.4d	line 17 row "Relight purge flow (decided)" col 1	LEAD (flow, cap, re-set rule); DERIVED (REV-COMB-4 on the SIM model; ASSUMED Cd); A1 reserve test DERIVED from SIM (REV-GNC-7)
T791	`data/v2p4b/purge_basis_line216_v2p4b.json` (`ca2ee1f0ea0f0194`) ...		DSIC v2.4d	line 17 row "Relight purge flow (decided)" col 4	source column
T792	band values 94 / 121 / 147 / 94 bar abs DECIDED for structural SDOF paths (LEAD 230 (2) ...	SIM	DSIC v2.4d	line 18 row "FA-1 hard start: structural load" col 2	SIM histories (ASSUMED inputs); DERIVED spectra; LEAD (230 (2))
T793	head equivalent static pressure, all 896 cases, undamped: 93.6 bar abs (T <= 0.1 ms) ...	SIM	DSIC v2.4d	line 18 row "FA-1 hard start: structural load" col 1	SIM histories (ASSUMED inputs); DERIVED spectra; LEAD (230 (2))
T794	`data/v2p4b/fa1_spectra_damped_v2p4b.json` (`3a84a905052d5364`)		DSIC v2.4d	line 18 row "FA-1 hard start: structural load" col 4	source column
T795	linear lossless duct (ASSUMED); not modelled: wave steepening and shock reflection ...	DERIVED	DSIC v2.4d	line 19 row "FA-1 in gas-filled passages (P_torch passage, Pc port)" col 2	DERIVED (REV-COMB-4 on SIM histories); LEAD design case (230 (1))
T796	design case FA1S 151.04 bar abs uniform over the passage; linear lossless passage peaks ...	DERIVED	DSIC v2.4d	line 19 row "FA-1 in gas-filled passages (P_torch passage, Pc port)" col 1	DERIVED (REV-COMB-4 on SIM histories); LEAD design case (230 (1))
T797	`reviews/combustion/r4/r4_torch_passage.json` (`8cb4b0c80173b88d`) ...		DSIC v2.4d	line 19 row "FA-1 in gas-filled passages (P_torch passage, Pc port)" col 4	source column
T798	grid counts on an ASSUMED construct, not rates; Z3 v2 NOT EVALUATED in flight (1 kHz ...	DERIVED *Lead decision, line 258 (3); basis SIM*	DSIC v2.4d	line 20 row "FA-1 class Z3 v2 and the J-FWD replace rule" col 2	LEAD (258 (3)); DERIVED on SIM (Mechanical; REV-COMB-4)
T799	Z3 v2: 63.1 < head <= 88.1 bar abs inside the light window, with the undamped SDOF ESP ...	DERIVED *Lead decision, line 258 (3); basis SIM*	DSIC v2.4d	line 20 row "FA-1 class Z3 v2 and the J-FWD replace rule" col 1	LEAD (258 (3)); DERIVED on SIM (Mechanical; REV-COMB-4)
T800	`interfaces/mechanical.json` v2.4 (`66bc53d88d444679`) `lc_tb70_envelope_v352` ...		DSIC v2.4d	line 20 row "FA-1 class Z3 v2 and the J-FWD replace rule" col 4	source column
T801	path period > 1 ms not acceptable for the dynamic channel; the P_torch 070BG over-ranges ...	DATASHEET *Lead decision, line 249 (4)*	DSIC v2.4d	line 21 row "Pc transducer proof" col 2	LEAD (249 (4), 230 (2)); DATASHEET (C1 rating, Electronics)
T802	P_torch (J901, on the torch body, not in the head; torch_port_bl08_v352_v1.json ...	DECIDED *Lead decision, line 264 (4)*	CL 266 lines	L264(4)[from "P_torch (J901" to "same part as P_mid (ENG-BL-08) and C1."]	lead (decision) 2026-10-21 (line sha 7cdbe51041ae6b8b)
T803	flight unit >= 166.14 bar abs (1.1 x FA1S 151.04); the C1 Kulite XTL-190SM-2000A pick ...	DATASHEET *Lead decision, line 249 (4)*	DSIC v2.4d	line 21 row "Pc transducer proof" col 1	LEAD (249 (4), 230 (2)); DATASHEET (C1 rating, Electronics)
T804	`combustion.json` v2.4c pc_dynamic_channel		DSIC v2.4d	line 21 row "Pc transducer proof" col 4	source column
T805	worst of the FE frequency factors x0.85 / 0.90 / 1.00; intervals because FEA3D publishes ...	DERIVED	DSIC v2.4d	line 22 row "FA-1 at TC-01 J-FWD (per-mode)" col 2	DERIVED on SIM shares (FEA3D) and SIM histories; damping ASSUMED; analysis path only (LEAD 230 (2)): TC-01 J-FWD was not met conservatively and the design change is sized to FA1S (230 (1)); shares ...
T806	time-domain modal bound 114.6 bar abs undamped (108.9 at zeta 0.02), E rule 127.0 ...	DERIVED	DSIC v2.4d	line 22 row "FA-1 at TC-01 J-FWD (per-mode)" col 1	DERIVED on SIM shares (FEA3D) and SIM histories; damping ASSUMED; analysis path only (LEAD 230 (2)): TC-01 J-FWD was not met conservatively and the design change is sized to FA1S (230 (1)); shares ...
T807	`data/v2p4b/fa1_modal_th_bound_v2p4b.json` (`639f5fbf3db77ee8`) ...		DSIC v2.4d	line 22 row "FA-1 at TC-01 J-FWD (per-mode)" col 4	source column
T808	A1 apogee target 210 m is decided (lines 214, 227 (1)); the A1 records above are the 300 ...	DECIDED *Design value, DATASHEET-INPUTS-COMBUSTION.md v2.4d*	DSIC v2.4d	notes line 25	tag in DATASHEET-INPUTS-COMBUSTION.md v2.4d: datasheet inputs note
T809	Prices: none quoted here; any quote is TO-QUOTE, delivered to Sønderborg DK-6400.	DECIDED *Design value, DATASHEET-INPUTS-COMBUSTION.md v2.4d*	DSIC v2.4d	notes line 26	tag in DATASHEET-INPUTS-COMBUSTION.md v2.4d: datasheet inputs note
T810	OI-41 is DECIDED (LEAD 230 (2) band values; 249 (4) flight Pc proof); the per-mode row ...	DECIDED *Design value, DATASHEET-INPUTS-COMBUSTION.md v2.4d*	DSIC v2.4d	notes line 27	tag in DATASHEET-INPUTS-COMBUSTION.md v2.4d: datasheet inputs note
T811	pcf_tmax 0.8 s (LEAD 227 (5)) holds on the dead side only with the photodiode ...	DECIDED *Design value, DATASHEET-INPUTS-COMBUSTION.md v2.4d*	DSIC v2.4d	notes line 28	tag in DATASHEET-INPUTS-COMBUSTION.md v2.4d: datasheet inputs note
T812	25.010070000000002 bar	SIM	CMB v2.4d	/operating_points/throttle_points_20C_initial_port/0/Pc_Pa	20 C saturated tank (P_tank 50 bar row of the map for lift and dP/Pc), initial port 25 mm, nominal law. SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* ...
T813	240.99 g/s	SIM	CMB v2.4d	/operating_points/throttle_points_20C_initial_port/0/mdot_ox_kg_s	20 C saturated tank (P_tank 50 bar row of the map for lift and dP/Pc), initial port 25 mm, nominal law. SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* ...
T814	23.09 g/s	SIM	CMB v2.4d	/operating_points/throttle_points_20C_initial_port/0/mdot_fuel_kg_s	20 C saturated tank (P_tank 50 bar row of the map for lift and dP/Pc), initial port 25 mm, nominal law. SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* ...
T815	1.916 mm	SIM	CMB v2.4d	/operating_points/throttle_points_20C_initial_port/0/pintle_lift_m	20 C saturated tank (P_tank 50 bar row of the map for lift and dP/Pc), initial port 25 mm, nominal law. SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* ...
T816	0.907	SIM	CMB v2.4d	/operating_points/throttle_points_20C_initial_port/0/dP_inj_over_Pc	20 C saturated tank (P_tank 50 bar row of the map for lift and dP/Pc), initial port 25 mm, nominal law. SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* ...
T817	1	SIM	CMB v2.4d	/operating_points/throttle_points_20C_initial_port/0/fraction	20 C saturated tank (P_tank 50 bar row of the map for lift and dP/Pc), initial port 25 mm, nominal law. SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* ...
T818	375 N	SIM	CMB v2.4d	/operating_points/throttle_points_20C_initial_port/1/F_N	20 C saturated tank (P_tank 50 bar row of the map for lift and dP/Pc), initial port 25 mm, nominal law. SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* ...
T819	19.09896 bar	SIM	CMB v2.4d	/operating_points/throttle_points_20C_initial_port/1/Pc_Pa	20 C saturated tank (P_tank 50 bar row of the map for lift and dP/Pc), initial port 25 mm, nominal law. SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* ...
T820	8.994	SIM	CMB v2.4d	/operating_points/throttle_points_20C_initial_port/1/OF	20 C saturated tank (P_tank 50 bar row of the map for lift and dP/Pc), initial port 25 mm, nominal law. SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* ...

ID	Value	Class	Source	Key	Source tag / formula
T821	192.2 s	SIM	CMB v2.4d	/operating_points/throttle_points_20C_initial_port/1/Isp_SL_s	20 C saturated tank (P_tank 50 bar row of the map for lift and dP/Pc), initial port 25 mm, nominal law. SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* ...
T822	179.01 g/s	SIM	CMB v2.4d	/operating_points/throttle_points_20C_initial_port/1/mdot_ox_kg_s	20 C saturated tank (P_tank 50 bar row of the map for lift and dP/Pc), initial port 25 mm, nominal law. SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* ...
T823	19.900000000000002 g/s	SIM	CMB v2.4d	/operating_points/throttle_points_20C_initial_port/1/mdot_fuel_kg_s	20 C saturated tank (P_tank 50 bar row of the map for lift and dP/Pc), initial port 25 mm, nominal law. SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* ...
T824	1.252 mm	SIM	CMB v2.4d	/operating_points/throttle_points_20C_initial_port/1/pintle_lift_m	20 C saturated tank (P_tank 50 bar row of the map for lift and dP/Pc), initial port 25 mm, nominal law. SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* ...
T825	1.551	SIM	CMB v2.4d	/operating_points/throttle_points_20C_initial_port/1/dP_inj_over_Pc	20 C saturated tank (P_tank 50 bar row of the map for lift and dP/Pc), initial port 25 mm, nominal law. SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* ...
T826	0.75	SIM	CMB v2.4d	/operating_points/throttle_points_20C_initial_port/1/fraction	20 C saturated tank (P_tank 50 bar row of the map for lift and dP/Pc), initial port 25 mm, nominal law. SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* ...
T827	250 N	SIM	CMB v2.4d	/operating_points/throttle_points_20C_initial_port/2/F_N	20 C saturated tank (P_tank 50 bar row of the map for lift and dP/Pc), initial port 25 mm, nominal law. SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* ...
T828	13.202140000000002 bar	SIM	CMB v2.4d	/operating_points/throttle_points_20C_initial_port/2/Pc_Pa	20 C saturated tank (P_tank 50 bar row of the map for lift and dP/Pc), initial port 25 mm, nominal law. SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* ...
T829	7.336	SIM	CMB v2.4d	/operating_points/throttle_points_20C_initial_port/2/OF	20 C saturated tank (P_tank 50 bar row of the map for lift and dP/Pc), initial port 25 mm, nominal law. SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* ...
T830	188.4 s	SIM	CMB v2.4d	/operating_points/throttle_points_20C_initial_port/2/Isp_SL_s	20 C saturated tank (P_tank 50 bar row of the map for lift and dP/Pc), initial port 25 mm, nominal law. SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* ...
T831	119.1 g/s	SIM	CMB v2.4d	/operating_points/throttle_points_20C_initial_port/2/mdot_ox_kg_s	20 C saturated tank (P_tank 50 bar row of the map for lift and dP/Pc), initial port 25 mm, nominal law. SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* ...
T832	16.240000000000002 g/s	SIM	CMB v2.4d	/operating_points/throttle_points_20C_initial_port/2/mdot_fuel_kg_s	20 C saturated tank (P_tank 50 bar row of the map for lift and dP/Pc), initial port 25 mm, nominal law. SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* ...
T833	0.755 mm	SIM	CMB v2.4d	/operating_points/throttle_points_20C_initial_port/2/pintle_lift_m	20 C saturated tank (P_tank 50 bar row of the map for lift and dP/Pc), initial port 25 mm, nominal law. SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* ...
T834	2.744	SIM	CMB v2.4d	/operating_points/throttle_points_20C_initial_port/2/dP_inj_over_Pc	20 C saturated tank (P_tank 50 bar row of the map for lift and dP/Pc), initial port 25 mm, nominal law. SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* ...
T835	0.5	SIM	CMB v2.4d	/operating_points/throttle_points_20C_initial_port/2/fraction	20 C saturated tank (P_tank 50 bar row of the map for lift and dP/Pc), initial port 25 mm, nominal law. SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* ...
T836	200 N	SIM	CMB v2.4d	/operating_points/throttle_points_20C_initial_port/3/F_N	20 C saturated tank (P_tank 50 bar row of the map for lift and dP/Pc), initial port 25 mm, nominal law. SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* ...
T837	10.848400000000002 bar	SIM	CMB v2.4d	/operating_points/throttle_points_20C_initial_port/3/Pc_Pa	20 C saturated tank (P_tank 50 bar row of the map for lift and dP/Pc), initial port 25 mm, nominal law. SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* ...
T838	6.583	SIM	CMB v2.4d	/operating_points/throttle_points_20C_initial_port/3/OF	20 C saturated tank (P_tank 50 bar row of the map for lift and dP/Pc), initial port 25 mm, nominal law. SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* ...
T839	184.6 s	SIM	CMB v2.4d	/operating_points/throttle_points_20C_initial_port/3/Isp_SL_s	20 C saturated tank (P_tank 50 bar row of the map for lift and dP/Pc), initial port 25 mm, nominal law. SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* ...
T840	95.92 g/s	SIM	CMB v2.4d	/operating_points/throttle_points_20C_initial_port/3/mdot_ox_kg_s	20 C saturated tank (P_tank 50 bar row of the map for lift and dP/Pc), initial port 25 mm, nominal law. SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* ...
T841	14.57 g/s	SIM	CMB v2.4d	/operating_points/throttle_points_20C_initial_port/3/mdot_fuel_kg_s	20 C saturated tank (P_tank 50 bar row of the map for lift and dP/Pc), initial port 25 mm, nominal law. SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* ...
T842	0.588 mm	SIM	CMB v2.4d	/operating_points/throttle_points_20C_initial_port/3/pintle_lift_m	20 C saturated tank (P_tank 50 bar row of the map for lift and dP/Pc), initial port 25 mm, nominal law. SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* ...
T843	3.575	SIM	CMB v2.4d	/operating_points/throttle_points_20C_initial_port/3/dP_inj_over_Pc	20 C saturated tank (P_tank 50 bar row of the map for lift and dP/Pc), initial port 25 mm, nominal law. SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* ...
T844	0.4	SIM	CMB v2.4d	/operating_points/throttle_points_20C_initial_port/3/fraction	20 C saturated tank (P_tank 50 bar row of the map for lift and dP/Pc), initial port 25 mm, nominal law. SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* ...
T845	125 N	SIM	CMB v2.4d	/operating_points/throttle_points_20C_initial_port/4/F_N	20 C saturated tank (P_tank 50 bar row of the map for lift and dP/Pc), initial port 25 mm, nominal law. SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* ...
T846	7.333900000000001 bar	SIM	CMB v2.4d	/operating_points/throttle_points_20C_initial_port/4/Pc_Pa	20 C saturated tank (P_tank 50 bar row of the map for lift and dP/Pc), initial port 25 mm, nominal law. SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* ...
T847	5.319	SIM	CMB v2.4d	/operating_points/throttle_points_20C_initial_port/4/OF	20 C saturated tank (P_tank 50 bar row of the map for lift and dP/Pc), initial port 25 mm, nominal law. SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* ...
T848	171.4 s	SIM	CMB v2.4d	/operating_points/throttle_points_20C_initial_port/4/Isp_SL_s	20 C saturated tank (P_tank 50 bar row of the map for lift and dP/Pc), initial port 25 mm, nominal law. SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* ...
T849	62.61 g/s	SIM	CMB v2.4d	/operating_points/throttle_points_20C_initial_port/4/mdot_ox_kg_s	20 C saturated tank (P_tank 50 bar row of the map for lift and dP/Pc), initial port 25 mm, nominal law. SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* ...
T850	11.77 g/s	SIM	CMB v2.4d	/operating_points/throttle_points_20C_initial_port/4/mdot_fuel_kg_s	20 C saturated tank (P_tank 50 bar row of the map for lift and dP/Pc), initial port 25 mm, nominal law. SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* ...
T851	0.367 mm	SIM	CMB v2.4d	/operating_points/throttle_points_20C_initial_port/4/pintle_lift_m	20 C saturated tank (P_tank 50 bar row of the map for lift and dP/Pc), initial port 25 mm, nominal law. SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* ...
T852	5.796	SIM	CMB v2.4d	/operating_points/throttle_points_20C_initial_port/4/dP_inj_over_Pc	20 C saturated tank (P_tank 50 bar row of the map for lift and dP/Pc), initial port 25 mm, nominal law. SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* ...
T853	0.25	SIM	CMB v2.4d	/operating_points/throttle_points_20C_initial_port/4/fraction	20 C saturated tank (P_tank 50 bar row of the map for lift and dP/Pc), initial port 25 mm, nominal law. SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* ...
T854	6.16171 bar	SIM	CMB v2.4d	/operating_points/throttle_points_20C_initial_port/5/Pc_Pa	20 C saturated tank (P_tank 50 bar row of the map for lift and dP/Pc), initial port 25 mm, nominal law. SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* ...
T855	4.848	SIM	CMB v2.4d	/operating_points/throttle_points_20C_initial_port/5/OF	20 C saturated tank (P_tank 50 bar row of the map for lift and dP/Pc), initial port 25 mm, nominal law. SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* ...
T856	162.5 s	SIM	CMB v2.4d	/operating_points/throttle_points_20C_initial_port/5/Isp_SL_s	20 C saturated tank (P_tank 50 bar row of the map for lift and dP/Pc), initial port 25 mm, nominal law. SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* ...
T857	52.019999999999996 g/s	SIM	CMB v2.4d	/operating_points/throttle_points_20C_initial_port/5/mdot_ox_kg_s	20 C saturated tank (P_tank 50 bar row of the map for lift and dP/Pc), initial port 25 mm, nominal law. SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* ...
T858	10.73 g/s	SIM	CMB v2.4d	/operating_points/throttle_points_20C_initial_port/5/mdot_fuel_kg_s	20 C saturated tank (P_tank 50 bar row of the map for lift and dP/Pc), initial port 25 mm, nominal law. SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* ...
T859	0.301 mm	SIM	CMB v2.4d	/operating_points/throttle_points_20C_initial_port/5/pintle_lift_m	20 C saturated tank (P_tank 50 bar row of the map for lift and dP/Pc), initial port 25 mm, nominal law. SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* ...

ID	Value	Class	Source	Key	Source tag / formula
T860	7.097	SIM	CMB v2.4d	/operating_points/throttle_points_20C_initial_port/5/dP_inj_over_Pc	20 C saturated tank (P_tank 50 bar row of the map for lift and dP/Pc), initial port 25 mm, nominal law. SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* ...
T861	0.2	SIM	CMB v2.4d	/operating_points/throttle_points_20C_initial_port/5/fraction	20 C saturated tank (P_tank 50 bar row of the map for lift and dP/Pc), initial port 25 mm, nominal law. SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* ...
T862	75 N	SIM	CMB v2.4d	/operating_points/throttle_points_20C_initial_port/6/F_N	20 C saturated tank (P_tank 50 bar row of the map for lift and dP/Pc), initial port 25 mm, nominal law. SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* ...
T863	4.98407 bar	SIM	CMB v2.4d	/operating_points/throttle_points_20C_initial_port/6/Pc_Pa	20 C saturated tank (P_tank 50 bar row of the map for lift and dP/Pc), initial port 25 mm, nominal law. SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* ...
T864	4.342	SIM	CMB v2.4d	/operating_points/throttle_points_20C_initial_port/6/OF	20 C saturated tank (P_tank 50 bar row of the map for lift and dP/Pc), initial port 25 mm, nominal law. SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* ...
T865	149 s	SIM	CMB v2.4d	/operating_points/throttle_points_20C_initial_port/6/Isp_SL_s	20 C saturated tank (P_tank 50 bar row of the map for lift and dP/Pc), initial port 25 mm, nominal law. SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* ...
T866	41.730000000000004 g/s	SIM	CMB v2.4d	/operating_points/throttle_points_20C_initial_port/6/mdot_ox_kg_s	20 C saturated tank (P_tank 50 bar row of the map for lift and dP/Pc), initial port 25 mm, nominal law. SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* ...
T867	9.610000000000001 g/s	SIM	CMB v2.4d	/operating_points/throttle_points_20C_initial_port/6/mdot_fuel_kg_s	20 C saturated tank (P_tank 50 bar row of the map for lift and dP/Pc), initial port 25 mm, nominal law. SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* ...
T868	0.23800000000000002 mm	SIM	CMB v2.4d	/operating_points/throttle_points_20C_initial_port/6/pintle_lift_m	20 C saturated tank (P_tank 50 bar row of the map for lift and dP/Pc), initial port 25 mm, nominal law. SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* ...
T869	9.018	SIM	CMB v2.4d	/operating_points/throttle_points_20C_initial_port/6/dP_inj_over_Pc	20 C saturated tank (P_tank 50 bar row of the map for lift and dP/Pc), initial port 25 mm, nominal law. SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* ...
T870	0.15	SIM	CMB v2.4d	/operating_points/throttle_points_20C_initial_port/6/fraction	20 C saturated tank (P_tank 50 bar row of the map for lift and dP/Pc), initial port 25 mm, nominal law. SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* ...
T871	62.5 N	SIM	CMB v2.4d	/operating_points/throttle_points_20C_initial_port/7/F_N	20 C saturated tank (P_tank 50 bar row of the map for lift and dP/Pc), initial port 25 mm, nominal law. SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* ...
T872	4.3698500000000005 bar	SIM	CMB v2.4d	/operating_points/throttle_points_20C_initial_port/7/Pc_Pa	20 C saturated tank (P_tank 50 bar row of the map for lift and dP/Pc), initial port 25 mm, nominal law. SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* ...
T873	4.059	SIM	CMB v2.4d	/operating_points/throttle_points_20C_initial_port/7/OF	20 C saturated tank (P_tank 50 bar row of the map for lift and dP/Pc), initial port 25 mm, nominal law. SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* ...
T874	140.2 s	SIM	CMB v2.4d	/operating_points/throttle_points_20C_initial_port/7/Isp_SL_s	20 C saturated tank (P_tank 50 bar row of the map for lift and dP/Pc), initial port 25 mm, nominal law. SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* ...
T875	36.46 g/s	SIM	CMB v2.4d	/operating_points/throttle_points_20C_initial_port/7/mdot_ox_kg_s	20 C saturated tank (P_tank 50 bar row of the map for lift and dP/Pc), initial port 25 mm, nominal law. SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* ...
T876	8.98 g/s	SIM	CMB v2.4d	/operating_points/throttle_points_20C_initial_port/7/mdot_fuel_kg_s	20 C saturated tank (P_tank 50 bar row of the map for lift and dP/Pc), initial port 25 mm, nominal law. SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* ...
T877	0.207 mm	SIM	CMB v2.4d	/operating_points/throttle_points_20C_initial_port/7/pintle_lift_m	20 C saturated tank (P_tank 50 bar row of the map for lift and dP/Pc), initial port 25 mm, nominal law. SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* ...
T878	10.43	SIM	CMB v2.4d	/operating_points/throttle_points_20C_initial_port/7/dP_inj_over_Pc	20 C saturated tank (P_tank 50 bar row of the map for lift and dP/Pc), initial port 25 mm, nominal law. SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* ...
T879	0.125	SIM	CMB v2.4d	/operating_points/throttle_points_20C_initial_port/7/fraction	20 C saturated tank (P_tank 50 bar row of the map for lift and dP/Pc), initial port 25 mm, nominal law. SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* ...
T880	25.000000000000004 bar	DECIDED *Lead decision of 2026-10-12*	CMB v2.4d	/operating_points/pressure_limits/normal_cap_Pa/v	tag in combustion.json v2.4d: LEAD 2026-10-12 [display = raw x 1e-05 (Pa to bar)]
T881	30.000000000000004 bar	DECIDED *Lead decision of 2026-10-12*	CMB v2.4d	/operating_points/pressure_limits/abort_Pa/v	tag in combustion.json v2.4d: LEAD 2026-10-12 (~30 bar) [display = raw x 1e-05 (Pa to bar)]
T882	45.00000000000001 bar	DECIDED *Lead decision of 2026-10-12*	CMB v2.4d	/operating_points/pressure_limits/chamber_MEOP_Pa/v	tag in combustion.json v2.4d: LEAD 2026-10-12 [display = raw x 1e-05 (Pa to bar)]
T883	67.5 bar	DECIDED *Lead decision of 2026-10-12*	CMB v2.4d	/operating_points/pressure_limits/proof_Pa/v	tag in combustion.json v2.4d: LEAD 2026-10-12 [display = raw x 1e-05 (Pa to bar)]
T884	44.33 bar	SIM	CMB v2.4d	/meop_basis_v2p4/v/basis_bar	SIM meop_basis_v2p4.json
T885	93.641 bar abs	SIM	CMB v2.4d	/fa1_spectra_damped_v2p4b/v/head_all_band_maxima_bar_abs/T_le_0p1ms/zeta0	SIM histories (v2.4a construct, ASSUMED inputs); DERIVED spectra; damping ASSUMED (0.02, 0.05; 0 = bound); proposal for the LEAD [undamped (zeta 0) band maximum, head, all cases]
T886	120.302 bar abs	SIM	CMB v2.4d	/fa1_spectra_damped_v2p4b/v/head_all_band_maxima_bar_abs/T_0p1_to_1ms/zeta0	SIM histories (v2.4a construct, ASSUMED inputs); DERIVED spectra; damping ASSUMED (0.02, 0.05; 0 = bound); proposal for the LEAD [undamped (zeta 0) band maximum, head, all cases]
T887	146.881 bar abs	SIM	CMB v2.4d	/fa1_spectra_damped_v2p4b/v/head_all_band_maxima_bar_abs/T_1_to_20ms/zeta0	SIM histories (v2.4a construct, ASSUMED inputs); DERIVED spectra; damping ASSUMED (0.02, 0.05; 0 = bound); proposal for the LEAD [undamped (zeta 0) band maximum, head, all cases]
T888	93.641 bar abs	SIM	CMB v2.4d	/fa1_spectra_damped_v2p4b/v/head_all_band_maxima_bar_abs/T_20_to_50ms/zeta0	SIM histories (v2.4a construct, ASSUMED inputs); DERIVED spectra; damping ASSUMED (0.02, 0.05; 0 = bound); proposal for the LEAD [undamped (zeta 0) band maximum, head, all cases]
T889	P_torch during FA-1: design case FA1S 151.04 bar abs uniform over the passage ...	DERIVED	CMB v2.4d	/fa1_p_torch_v2p4c/v/design_case	DERIVED by the reviewer (linear lossless two-segment duct, ASSUMED) on SIM FA-1 head histories; design case LEAD 230 (1); Mechanical row CALC on SIM
T890	151.04 bar abs	DERIVED	CMB v2.4d	/fa1_p_torch_v2p4c/v/design_case [regex FA1S ([\d.]+) bar abs]	DERIVED by the reviewer (linear lossless two-segment duct, ASSUMED) on SIM FA-1 head histories; design case LEAD 230 (1); Mechanical row CALC on SIM
T891	flight unit >= 166.14 bar abs (1.1 x FA1S 151.04); the C1 Kulite XTL-190SM-2000A pick ...	DECIDED *Lead decision, line 249 (4)*	DSIC v2.4d	§* row c0="Pc transducer proof" col 1 (line 21)	tag in DATASHEET-INPUTS-COMBUSTION.md v2.4d: LEAD (249 (4), 230 (2))
T892	166.14 bar abs	DERIVED	DSIC v2.4d	§* row c0="Pc transducer proof" col 1 (line 21) [regex >= ([\d.]+) bar abs]	LEAD (249 (4), 230 (2)); (class set at the call site: DERIVED) [1.1 x FA1S; the rule is the lead's (249 (4)), the number is DERIVED]
T893	1.1	DECIDED *Lead decision, line 249 (4)*	DSIC v2.4d	§* row c0="Pc transducer proof" col 1 (line 21) [regex \(([\d.]+) x FA1S [\d.]+\)]	tag in DATASHEET-INPUTS-COMBUSTION.md v2.4d: LEAD (249 (4), 230 (2)) [factor of the rule (249 (4))]
T894	R4-12 (070BG proof rating against FA1S 151.04 bar abs: >= 166.14 bar abs or isolate the ...	DECIDED *Lead decision, line 256 (5)*	CL 266 lines	L256(5)[from "R4-12 (070BG proof rating" to "state saturation and recovery)"]	lead (decision) 2026-10-21 (line sha cf51938a477079b6)
T895	flight unit >= 166.14 bar abs (1.1 x FA1S 151.04); the C1 Kulite XTL-190SM-2000A pick ...	DATASHEET	DSIC v2.4d	§* row c0="Pc transducer proof" col 1 (line 21)	DATASHEET (C1 rating)
T896	206.8 bar abs	DATASHEET	DSIC v2.4d	§* row c0="Pc transducer proof" col 1 (line 21) [regex XTL-190SM-2000A pick, ([\d.]+) bar abs]	DATASHEET (C1 rating)
T897	406 periods, columns head|all|zeta0, zeta0.02, zeta0.05, head|baseline|zeta0	SIM	FA1SP 406 periods	(csv columns period_ms, head|all|zeta*, head|baseline|zeta0)	SIM histories (ASSUMED inputs); DERIVED spectra [plotted as published]
T898	band values 94 / 121 / 147 / 94 bar abs DECIDED for structural SDOF paths (LEAD 230 (2) ...	DECIDED *Lead decision, line 230 (2)*	DSIC v2.4d	§* row c0="FA-1 hard start: structural load" col 2 (line 18)	tag in DATASHEET-INPUTS-COMBUSTION.md v2.4d: LEAD (230 (2))
T899	94 bar	DECIDED *Lead decision, line 230 (2)*	DSIC v2.4d	§* row c0="FA-1 hard start: structural load" col 2 (line 18) [regex band values ([\d]+) / ([\d]+) / ([\d]+) / ([\d]+) bar abs DECIDED]	tag in DATASHEET-INPUTS-COMBUSTION.md v2.4d: LEAD (230 (2))
T900	a2c380515cada638		CMB v2.4d	/fa1_spectra_damped_v2p4b/v/files/02-combustion~1data~1v2p4b~1fa1_pressure_spectra_published_v2p4b.csv	(class set at the call site: REF)
T901	same file: combustion.json v2.4d names 02-combustion/data/v2p4b/fa1_pressure_spectra_publ ...	DERIVED	generator	formula: generator check: sha named by combustion.json vs the file read	DERIVED by build_datasheet.py: generator check: sha named by combustion.json vs the file read ← T900

ID	Value	Class	Source	Key	Source tag / formula
T902	Applied by the lead with `src/apply_v2p4c_rev_comb_4.py` (sha256/16 `063195b1b3a12bed` ...		RCMB 2.4d	§17. v2.4c line 3165 [sentence containing "Text only"]	combustion REPORT.md §17 (v2.4c) header paragraph
T903	Applied by the lead with `src/apply_v2p4d_lead.py` (sha256/16 `6c43216fc7b22306`, run at ...		RCMB 2.4d	§18. v2.4d line 3318 [sentence containing "Text only"]	combustion REPORT.md §18 (v2.4d) header paragraph
T904	- No Combustion number changed except the T4 planning value; the FA-1 grid histories are ...		RCMB 2.4d	§18.3 Not changed line 3381 [sentence containing "FA-1 grid histories"]	combustion REPORT.md §18.3
T905	F_N: v: 125.0; tag: REQ; Pc_Pa: v: 733390.0; tag: SIM grain_q1d.py v2 (quasi-1D port ...	SIM	CMB v2.4d	/operating_points/floor_requirement_25pct	(class set at the call site: SIM)
T906	F_N: v: 100.0; tag: REQ; Pc_Pa: v: 616171.0; tag: SIM grain_q1d.py v2 (quasi-1D port ...	SIM	CMB v2.4d	/operating_points/floor_target_20pct	(class set at the call site: SIM)
T907	F_N: v: 62.5; tag: REQ stretch; Pc_Pa: v: 436985.0; tag: SIM grain_q1d.py v2 (quasi-1D ...	SIM	CMB v2.4d	/operating_points/floor_stretch_12p5pct	(class set at the call site: SIM)
T908	F_ceiling = F at min(stroke 4 mm, injector dP/Pc >= 0.25)	SIM	CMB v2.4d	/operating_points/thrust_ceiling_rule	DERIVED stiffness rule (stability section)
T909	0.0034	SIM	CMB v2.4d	/operating_points/chamber_fill_time_constant_s	DERIVED L*/(c*_del Gamma^2); step-response fit 0.004 s (1 ms sampling)
T910	Servo-driven variable-area pintle (needle-in-seat) at the head end + separate NC main ...	DECIDED *Lead decision of 2026-10-11*	CMB v2.4d	/decisions/throttle_architecture	tag in combustion.json v2.4d: LEAD decision 2026-10-11; v2 trade (data/v2/throttle_trade_v2.json) finds no showstopper
T911	±7.0 deg commanded limit, on the vector norm; 8.0 deg conical mechanical stop. Flown ...	SIM	DSIG v3.2	line 27 row "Gimbal range" col 2	SIM (m2_mc6 v2 limit cases) + mechanical.json v1 (stop)
T912	7 deg gives P 0.984. 6 deg gives P 0.982 (3 lost / 2 gained, p 1). 5 deg gives P 0.924 ...	SIM	DSIG v3.2	line 27 row "Gimbal range" col 4	SIM (m2_mc6 v2 limit cases) + mechanical.json v1 (stop)
T913	Slew >= 100 deg/s in the worst direction at 7 deg, at the 7.4 V nominal plug (ENG-07). ...	DECIDED *Requirement, DATASHEET-INPUTS-GNC.md v3.2; basis SIM*	DSIG v3.2	line 28 row "Gimbal slew and dynamics" col 2	tag in DATASHEET-INPUTS-GNC.md v3.2: REQ (lead 2026-10-12) from SIM
T914	Comparators: 80 deg/s loses (11 lost / 2 gained, p 0.022); 60 deg/s gives P 0.716. ...	SIM	DSIG v3.2	line 28 row "Gimbal slew and dynamics" col 4	REQ (lead 2026-10-12) from SIM (class set at the call site: SIM: the basis cell states the GNC MC results; REV-DS-1 D1-07)
T915	24 / 24 variants (RV6-14 (a)-(e)); check run reproduces REV-GNC-6 v05 part B on V5c. ...	DERIVED	DSIG v3.2	line 29 row "ENG-11 servo output stiffness K_s" col 4	DERIVED (linear ENG-07d, one fixed ECU notch, K_s grid 90-160 step 5, edges +-5 N m/rad); CAD v3.5.2 R3 plant and band ends checked by REV-GNC-7 (the 100-120 of CAD v3.5.1 does not hold on CAD ...
T916	7293.95 N m/rad	SIM	KTAB	/range_over_posted_R3_poses/FE_coupled_alpha/min	(class set at the call site: SIM)
T917	9337.08 N m/rad	SIM	KTAB	/range_over_posted_R3_poses/FE_coupled_alpha/max	(class set at the call site: SIM)
T918	stop000	DERIVED	KTAB	/range_over_posted_R3_poses/FE_coupled_alpha/min_pose	(class set at the call site: DERIVED)
T919	stop180	DERIVED	KTAB	/range_over_posted_R3_poses/FE_coupled_alpha/max_pose	(class set at the call site: DERIVED)
T920	3971.84 N m/rad	SIM	KTAB	/range_over_posted_R3_poses/FE_coupled_beta/min	(class set at the call site: SIM)
T921	7172.12 N m/rad	SIM	KTAB	/range_over_posted_R3_poses/FE_coupled_beta/max	(class set at the call site: SIM)
T922	stop180	DERIVED	KTAB	/range_over_posted_R3_poses/FE_coupled_beta/min_pose	(class set at the call site: DERIVED)
T923	stop000	DERIVED	KTAB	/range_over_posted_R3_poses/FE_coupled_beta/max_pose	(class set at the call site: DERIVED)
T924	1149.93 N m/rad	DERIVED	KTAB	/range_over_posted_R3_poses/k_s58_coupled_alpha/min	(class set at the call site: DERIVED)
T925	1426.19 N m/rad	DERIVED	KTAB	/range_over_posted_R3_poses/k_s58_coupled_alpha/max	(class set at the call site: DERIVED)
T926	neutral	DERIVED	KTAB	/range_over_posted_R3_poses/k_s58_coupled_alpha/min_pose	(class set at the call site: DERIVED)
T927	stop000	DERIVED	KTAB	/range_over_posted_R3_poses/k_s58_coupled_alpha/max_pose	(class set at the call site: DERIVED)
T928	790.77 N m/rad	DERIVED	KTAB	/range_over_posted_R3_poses/k_s58_coupled_beta/min	(class set at the call site: DERIVED)
T929	1799.09 N m/rad	DERIVED	KTAB	/range_over_posted_R3_poses/k_s58_coupled_beta/max	(class set at the call site: DERIVED)
T930	stop180	DERIVED	KTAB	/range_over_posted_R3_poses/k_s58_coupled_beta/min_pose	(class set at the call site: DERIVED)
T931	stop000	DERIVED	KTAB	/range_over_posted_R3_poses/k_s58_coupled_beta/max_pose	(class set at the call site: DERIVED)
T932	2697.48 N m/rad	DERIVED	KTAB	/range_over_posted_R3_poses/k_s174_coupled_alpha/min	(class set at the call site: DERIVED)
T933	3075.77 N m/rad	DERIVED	KTAB	/range_over_posted_R3_poses/k_s174_coupled_alpha/max	(class set at the call site: DERIVED)
T934	neutral	DERIVED	KTAB	/range_over_posted_R3_poses/k_s174_coupled_alpha/min_pose	(class set at the call site: DERIVED)
T935	stop000	DERIVED	KTAB	/range_over_posted_R3_poses/k_s174_coupled_alpha/max_pose	(class set at the call site: DERIVED)
T936	1696.7 N m/rad	DERIVED	KTAB	/range_over_posted_R3_poses/k_s174_coupled_beta/min	(class set at the call site: DERIVED)
T937	3594.13 N m/rad	DERIVED	KTAB	/range_over_posted_R3_poses/k_s174_coupled_beta/max	(class set at the call site: DERIVED)
T938	stop180	DERIVED	KTAB	/range_over_posted_R3_poses/k_s174_coupled_beta/min_pose	(class set at the call site: DERIVED)
T939	stop000	DERIVED	KTAB	/range_over_posted_R3_poses/k_s174_coupled_beta/max_pose	(class set at the call site: DERIVED)
T940	Gimbal axis stiffness (structure plus rods) ≥ 3,100 N·m/rad per axis at every pose (hard ...	DERIVED	REQ rev 2	§2. Top-level row "ENG-11" col 1	DERIVED (GNC csi_v3)
T941	20 %	DERIVED	REQ rev 2	§2. Top-level row "ENG-11" col 1 [regex spread across poses ≤ ±(\d+) %]	DERIVED (GNC csi_v3)
T942	3,100 N·m/rad per axis	DERIVED	REQ rev 2	§2. Top-level row "ENG-11" col 1 [sentence containing "3,100 N·m/rad per axis"]	DERIVED (GNC csi_v3)
T943	12.285 %	DERIVED	generator	formula: half-range over mid-range of the FE-coupled alpha axis stiffness over the 17 R3 poses: 100 x (max - min) / (max + min) (§3.3)	DERIVED by build_datasheet.py: half-range over mid-range of the FE-coupled alpha axis stiffness over the 17 R3 poses: 100 x (max - min) / (max + min) (§3.3) ← T916, T917
T944	28.7176 %	DERIVED	generator	formula: half-range over mid-range of the FE-coupled beta axis stiffness over the 17 R3 poses: 100 x (max - min) / (max + min) (§3.3)	DERIVED by build_datasheet.py: half-range over mid-range of the FE-coupled beta axis stiffness over the 17 R3 poses: 100 x (max - min) / (max + min) (§3.3) ← T920, T921
T945	RM1-08 ENG-11: GNC assesses TVC loop robustness over the V5c stiffness range (3133-7406 ...	DECIDED *Lead decision, line 188 (7)*	CL 266 lines	L188(7)	lead (decision) 2026-10-13 (line sha 45df43a2aeab0326)
T946	2 x Savox SB-2292SG on the fixed lower ring GM-10 below the gimbal ring, shafts ...	DERIVED *Design value, mechanical.json v2.4*	MEC v2.4	/tvc_actuation/concept	DERIVED [v2.1 record; not_re_derived_v352]
T947	4.8	DECIDED *Lead decision, per mechanical.json v2.4*	MEC v2.4	/tvc_actuation/gear_ratio_n	tag in mechanical.json v2.4: LEAD-DECISION [v2.1 record; not_re_derived_v352]
T948	cmd_7deg: amp_deg: 7.0; psi_min_deg: [-32.37, -32.37]; psi_max_deg: [37.39, 37.39] ...	DERIVED *Design value, mechanical.json v2.4*	MEC v2.4	/tvc_actuation/servo_angle_range_deg	DERIVED [v2.1 record; not_re_derived_v352]
T949	6.6: worst_dps: 95.3; worst_dir_deg: 315.0; alpha_dps: 153.3; beta_dps: 152.2 ...	DERIVED *Design value, mechanical.json v2.4*	MEC v2.4	/tvc_actuation/rate_engine_dps_under_3.19Nm	DERIVED [v2.1 record; not_re_derived_v352]
T950	as built: worst-direction slew at 7 deg 111.2 deg/s at 7.4 V, 118.3 at 7.6 V (>= 100) ...	CALC	MEC v2.4	/tvc_actuation/ENG-07_rate_status	(class set at the call site: CALC / DERIVED, with SIM for the P value it quotes; REV-DS-1 D1-07) [v2.1 record; not_re_derived_v352]
T951	37.95	DERIVED *Design value, mechanical.json v2.4*	MEC v2.4	/tvc_actuation/hinge_moment_design_Nm	DERIVED [v2.1 record; not_re_derived_v352]
T952	first rocking mode 16.0-27.8 Hz (excl. J_int) vs the 25 Hz minimum: below it whenever ...	SIM	MEC v2.4	/tvc_actuation/ENG-11_status	(class set at the call site: SIM / DERIVED; REV-DS-1 D1-07) [v2.1 record; not_re_derived_v352]
T953	two-axis Cardan gimbal of the whole thrust chamber; split engine collar (GM-01/02) ...	DERIVED *Design value, mechanical.json v2.4*	MEC v2.4	/gimbal/architecture	DERIVED [v2.1 record; not_re_derived_v352]
T954	7	DECIDED *Requirement, mechanical.json v2.4*	MEC v2.4	/gimbal/range_deg	tag in mechanical.json v2.4: REQ [v2.1 record; not_re_derived_v352]

ID	Value	Class	Source	Key	Source tag / formula
T955	8	DERIVED *Design value, mechanical.json v2.4*	MEC v2.4	/gimbal/mechanical_stop_deg	DERIVED [v2.1 record; not_re_derived_v352]
T956	4 × 608-2RS (8×22×7); pins 8 h6 17-4PH H1150	DECIDED *Design value, mechanical.json v2.4*	MEC v2.4	/gimbal/bearings	tag in mechanical.json v2.4: DATASHEET-PENDING [v2.1 record; not_re_derived_v352]
T957	149	DERIVED	GLBR	/readback_meshes	(class set at the call site: DERIVED)
T958	0.0867064403129586 mm	DERIVED	GLBR	/bounds_err_m	(class set at the call site: DERIVED) [display = raw x 1000 (m to mm)]
T959	yes	DERIVED	GLBR	/passed	(class set at the call site: DERIVED)
T960	0	DERIVED	CLR v2	/summary/n_not_whitelisted	(class set at the call site: DERIVED)
T961	0.525 mm	DERIVED	CLR v2	/summary/min_moving_clearance_mm	(class set at the call site: DERIVED)
T962	≥ 140 mm	DECIDED *Requirement, REQUIREMENTS.md rev 2*	REQ rev 2	§1. Mission row "Engine-bay inner diameter" col 1	tag in REQUIREMENTS.md rev 2: REQ (founder approved 2026-10-12)
T963	140 mm	DECIDED *Requirement, REQUIREMENTS.md rev 2*	REQ rev 2	§1. Mission row "Engine-bay inner diameter" col 1 [regex ≥\s*([\d.]+)\s*mm]	tag in REQUIREMENTS.md rev 2: REQ (founder approved 2026-10-12)
T964	139.5 mm	DERIVED	MAN v3.5.2	/instances/*/bounds_mm	DERIVED by build_datasheet.py: largest x or y extent of the thrust rings over assembly-manifest instance bounds (instance prefix ENG-GM-09 / ENG-GM-10)
T965	0.25 mm	DERIVED	generator	formula: (bay inner diameter − thrust-ring outer diameter) / 2	DERIVED by build_datasheet.py: (bay inner diameter − thrust-ring outer diameter) / 2 ← T963, T964
T966	ENG-BL-06#0: 70.48 mm from the axis (V2 vent bulkhead fitting (316L) through the bay ...	DERIVED	generator	formula: instances whose bounding box reaches beyond the thrust-ring radius (largest |x| or |y| of the bounds)	DERIVED by build_datasheet.py: instances whose bounding box reaches beyond the thrust-ring radius (largest |x| or |y| of the bounds) ← T964, T063
T967	TVC by a two-axis gimbal of the whole engine, pivot at the engine-group mid-burn CG. ...	DECIDED *Lead decision, per REQUIREMENTS.md rev 2*	REQ rev 2	§2. Top-level row "ENG-07" col 1	tag in REQUIREMENTS.md rev 2: LEAD (GNC MC: 80°/s worse, p 0.022)
T968	7 °	DECIDED *Requirement, ENG-07*	REQ rev 2	§2. Top-level row "ENG-07" col 1 [regex Range ± (\d+)°]	tag in REQUIREMENTS.md rev 2: LEAD (GNC MC: 80°/s worse, p 0.022)
T969	8 °	DECIDED *Requirement, ENG-07*	REQ rev 2	§2. Top-level row "ENG-07" col 1 [regex (\d+)° conical mechanical stop]	tag in REQUIREMENTS.md rev 2: LEAD (GNC MC: 80°/s worse, p 0.022)
T970	Max thrust 500 N at sea level, with N2O at 15–25 °C. Normal Pc is capped at 25 bar in ...	DECIDED *Requirement, ENG-01*	REQ rev 2	§2. Top-level row "ENG-01" col 1	tag in REQUIREMENTS.md rev 2: REQ
T971	500 N	DECIDED *Requirement, ENG-01*	REQ rev 2	§2. Top-level row "ENG-01" col 1 [regex Max thrust (\d+) N]	tag in REQUIREMENTS.md rev 2: REQ
T972	60.9347 N	DERIVED	generator	formula: max thrust × sin(gimbal range)	DERIVED by build_datasheet.py: max thrust × sin(gimbal range) ← T971, T968
T973	ENG-BL-01; ENG-BL-02; ENG-BL-03A; ENG-BL-03E; ENG-BL-03F; ENG-BL-04; ENG-BL-04A ...	DERIVED	R-ISO	/pns	(class set at the call site: DERIVED) [part numbers of the GLB objects in the render (render script, from the GLB)]
T974	Not in the GLB display mesh, so shown in no render: all 57 of the 57 standard-hardware ...	DERIVED	generator	formula: BOM /lines and BOM /hardware/lines whose part number is not among the render metadata /pns (isometric view)	DERIVED by build_datasheet.py: BOM /lines and BOM /hardware/lines whose part number is not among the render metadata /pns (isometric view) ← T973, T157, T273, T275, T277, T279, T281, T283 ...
T975	2.82102 kg	DERIVED	MP v3.5.2	/states/wet/engine/m_kg	(class set at the call site: DERIVED) [CAD volume × density, COTS masses and the hardware CALC (massprops hardware_rule)]
T976	2.73802 kg	DERIVED	MP v3.5.2	/states/mid/engine/m_kg	(class set at the call site: DERIVED) [CAD volume × density, COTS masses and the hardware CALC (massprops hardware_rule)]
T977	2.65502 kg	DERIVED	MP v3.5.2	/states/burnout/engine/m_kg	(class set at the call site: DERIVED) [CAD volume × density, COTS masses and the hardware CALC (massprops hardware_rule)]
T978	2.39666 kg	DERIVED	MP v3.5.2	/states/nograin/engine/m_kg	(class set at the call site: DERIVED) [CAD volume × density, COTS masses and the hardware CALC (massprops hardware_rule)]
T979	0.21664 kg	DERIVED	MP v3.5.2	/states/wet/ring/m_kg	(class set at the call site: DERIVED) [CAD volume × density, COTS masses and the hardware CALC (massprops hardware_rule)]
T980	0.21664 kg	DERIVED	MP v3.5.2	/states/mid/ring/m_kg	(class set at the call site: DERIVED) [CAD volume × density, COTS masses and the hardware CALC (massprops hardware_rule)]
T981	0.21664 kg	DERIVED	MP v3.5.2	/states/burnout/ring/m_kg	(class set at the call site: DERIVED) [CAD volume × density, COTS masses and the hardware CALC (massprops hardware_rule)]
T982	0.21664 kg	DERIVED	MP v3.5.2	/states/nograin/ring/m_kg	(class set at the call site: DERIVED) [CAD volume × density, COTS masses and the hardware CALC (massprops hardware_rule)]
T983	0.02005 kg	DERIVED	MP v3.5.2	/states/wet/horn/m_kg	(class set at the call site: DERIVED) [CAD volume × density, COTS masses and the hardware CALC (massprops hardware_rule)]
T984	0.02005 kg	DERIVED	MP v3.5.2	/states/mid/horn/m_kg	(class set at the call site: DERIVED) [CAD volume × density, COTS masses and the hardware CALC (massprops hardware_rule)]
T985	0.02005 kg	DERIVED	MP v3.5.2	/states/burnout/horn/m_kg	(class set at the call site: DERIVED) [CAD volume × density, COTS masses and the hardware CALC (massprops hardware_rule)]
T986	0.02005 kg	DERIVED	MP v3.5.2	/states/nograin/horn/m_kg	(class set at the call site: DERIVED) [CAD volume × density, COTS masses and the hardware CALC (massprops hardware_rule)]
T987	0.01841 kg	DERIVED	MP v3.5.2	/states/wet/pushrod/m_kg	(class set at the call site: DERIVED) [CAD volume × density, COTS masses and the hardware CALC (massprops hardware_rule)]
T988	0.01841 kg	DERIVED	MP v3.5.2	/states/mid/pushrod/m_kg	(class set at the call site: DERIVED) [CAD volume × density, COTS masses and the hardware CALC (massprops hardware_rule)]
T989	0.01841 kg	DERIVED	MP v3.5.2	/states/burnout/pushrod/m_kg	(class set at the call site: DERIVED) [CAD volume × density, COTS masses and the hardware CALC (massprops hardware_rule)]
T990	0.01841 kg	DERIVED	MP v3.5.2	/states/nograin/pushrod/m_kg	(class set at the call site: DERIVED) [CAD volume × density, COTS masses and the hardware CALC (massprops hardware_rule)]
T991	2.26691 kg	DERIVED	MP v3.5.2	/states/wet/fixed/m_kg	(class set at the call site: DERIVED) [CAD volume × density, COTS masses and the hardware CALC (massprops hardware_rule)]
T992	2.26691 kg	DERIVED	MP v3.5.2	/states/mid/fixed/m_kg	(class set at the call site: DERIVED) [CAD volume × density, COTS masses and the hardware CALC (massprops hardware_rule)]
T993	2.26691 kg	DERIVED	MP v3.5.2	/states/burnout/fixed/m_kg	(class set at the call site: DERIVED) [CAD volume × density, COTS masses and the hardware CALC (massprops hardware_rule)]
T994	2.26691 kg	DERIVED	MP v3.5.2	/states/nograin/fixed/m_kg	(class set at the call site: DERIVED) [CAD volume × density, COTS masses and the hardware CALC (massprops hardware_rule)]
T995	5.34303 kg	DERIVED	MP v3.5.2	/states/wet/all/m_kg	(class set at the call site: DERIVED) [CAD volume × density, COTS masses and the hardware CALC (massprops hardware_rule)]
T996	5.26003 kg	DERIVED	MP v3.5.2	/states/mid/all/m_kg	(class set at the call site: DERIVED) [CAD volume × density, COTS masses and the hardware CALC (massprops hardware_rule)]
T997	5.17703 kg	DERIVED	MP v3.5.2	/states/burnout/all/m_kg	(class set at the call site: DERIVED) [CAD volume × density, COTS masses and the hardware CALC (massprops hardware_rule)]
T998	4.91867 kg	DERIVED	MP v3.5.2	/states/nograin/all/m_kg	(class set at the call site: DERIVED) [CAD volume × density, COTS masses and the hardware CALC (massprops hardware_rule)]
T999	2.82102 kg	DERIVED	MEC v2.4	/gimballed_mass_properties_v352/states/wet/engine/m_kg	(class set at the call site: DERIVED) [mechanical.json _v352 (FINAL set massprops)]
T1000	−2.543, 0.892, 4.539 mm	DERIVED	MEC v2.4	/gimballed_mass_properties_v352/states/wet/engine/cg_mm	(class set at the call site: DERIVED)
T1001	[[0.057518, 2.22e-05, 0.0002685], [2.22e-05, 0.0577508, -0.0004486], [0.0002685 ...	DERIVED	MEC v2.4	/gimballed_mass_properties_v352/states/wet/engine/I_pivot_kgm2	(class set at the call site: DERIVED)
T1002	0.057518, 0.0577508, 0.0021804 kg·m²	DERIVED	generator	formula: diagonal of I_pivot (Ixx, Iyy, Izz)	DERIVED by build_datasheet.py: diagonal of I_pivot (Ixx, Iyy, Izz) ← T1001
T1003	2.73802 kg	DERIVED	MEC v2.4	/gimballed_mass_properties_v352/states/mid/engine/m_kg	(class set at the call site: DERIVED) [mechanical.json _v352 (FINAL set massprops)]
T1004	−2.62, 0.919, 5.934 mm	DERIVED	MEC v2.4	/gimballed_mass_properties_v352/states/mid/engine/cg_mm	(class set at the call site: DERIVED)
T1005	[[0.0569911, 2.22e-05, 0.0002685], [2.22e-05, 0.0572239, -0.0004486], [0.0002685 ...	DERIVED	MEC v2.4	/gimballed_mass_properties_v352/states/mid/engine/I_pivot_kgm2	(class set at the call site: DERIVED)
T1006	0.0569911, 0.0572239, 0.0021442 kg·m²	DERIVED	generator	formula: diagonal of I_pivot (Ixx, Iyy, Izz)	DERIVED by build_datasheet.py: diagonal of I_pivot (Ixx, Iyy, Izz) ← T1005
T1007	2.65502 kg	DERIVED	MEC v2.4	/gimballed_mass_properties_v352/states/burnout/engine/m_kg	(class set at the call site: DERIVED) [mechanical.json _v352 (FINAL set massprops)]
T1008	−2.702, 0.948, 7.417 mm	DERIVED	MEC v2.4	/gimballed_mass_properties_v352/states/burnout/engine/cg_mm	(class set at the call site: DERIVED)
T1009	[[0.0564643, 2.22e-05, 0.0002685], [2.22e-05, 0.0566971, -0.0004486], [0.0002685 ...	DERIVED	MEC v2.4	/gimballed_mass_properties_v352/states/burnout/engine/I_pivot_kgm2	(class set at the call site: DERIVED)

ID	Value	Class	Source	Key	Source tag / formula
T1010	0.0564643, 0.0566971, 0.002108 kg·m²	DERIVED	generator	formula: diagonal of I_pivot (Ixx, Iyy, Izz)	DERIVED by build_datasheet.py: diagonal of I_pivot (Ixx, Iyy, Izz) ← T1009
T1011	2.39666 kg	DERIVED	MEC v2.4	/gimballed_mass_properties_v352/states/nograin/engine/m_kg	(class set at the call site: DERIVED) [mechanical.json _v352 (FINAL set massprops)]
T1012	−2.993, 1.05, 12.688 mm	DERIVED	MEC v2.4	/gimballed_mass_properties_v352/states/nograin/engine/cg_mm	(class set at the call site: DERIVED)
T1013	[[0.0548243, 2.22e-05, 0.0002685], [2.22e-05, 0.0550572, -0.0004486], [0.0002685 ...	DERIVED	MEC v2.4	/gimballed_mass_properties_v352/states/nograin/engine/I_pivot_kgm2	(class set at the call site: DERIVED)
T1014	0.0548243, 0.0550572, 0.0019954 kg·m²	DERIVED	generator	formula: diagonal of I_pivot (Ixx, Iyy, Izz)	DERIVED by build_datasheet.py: diagonal of I_pivot (Ixx, Iyy, Izz) ← T1013
T1015	Rated 500 N; liftoff mass 17.44 kg; nominal deterministic A1 flight; 0 kg payload; 2.00 ...	DERIVED	DSIG v3.2	§* row c0="4" col 4 (line 25)	DERIVED
T1016	17.44 kg	DERIVED	DSIG v3.2	§* row c0="4" col 4 (line 25) [regex liftoff mass ([\d.]+) kg]	DERIVED [GNC basis: massprops v3.5.1 (row 4 basis text)]
T1017	2.92	DERIVED	DSIG v3.2	§* row c0="4" col 2 (line 25)	DERIVED
T1018	2.92	DERIVED	DSIG v3.2	§* row c0="4" col 2 (line 25) [regex ([\d.]+)]	DERIVED
T1019	−0.0662 kg	CALC	MDEL v3.5.2	/gnc_trigger/dm_landing_kg	CALC (CAD volume x engmat density + hardware CALC); not weighed, so not a 'confirmed' mass; COTS masses unchanged
T1020	0.0118 kg	CALC	MDEL v3.5.2	/gnc_trigger/dm_landing_without_TC09_thinning_kg	CALC (CAD volume x engmat density + hardware CALC); not weighed, so not a 'confirmed' mass; COTS masses unchanged
T1021	−0.078 kg	CALC	MDEL v3.5.2	/gnc_trigger/TC09_thinning_kg	CALC (CAD volume x engmat density + hardware CALC); not weighed, so not a 'confirmed' mass; COTS masses unchanged
T1022	0.1 kg	DECIDED *Lead decision, per massdelta_v3.5.2_vs_v3.5.1.json v3.5.2*	MDEL v3.5.2	/gnc_trigger/rule [regex >= ([\d.]+) kg]	tag in massdelta_v3.5.2_vs_v3.5.1.json v3.5.2: [GNC's re-fly trigger as written in the rule (lead / GNC trigger)]
T1023	BELOW the trigger with and without the conditional TC-09 thinning: no re-fly by mass	CALC	MDEL v3.5.2	/gnc_trigger/verdict	CALC (CAD volume x engmat density + hardware CALC); not weighed, so not a 'confirmed' mass; COTS masses unchanged
T1024	CALC (CAD volume x engmat density + hardware CALC); not weighed, so not a 'confirmed' ...	CALC	MDEL v3.5.2	/gnc_trigger/evidence	CALC (CAD volume x engmat density + hardware CALC); not weighed, so not a 'confirmed' mass; COTS masses unchanged
T1025	0.2 kg	CALC *Lead decision, line 188 (11)*	MP v3.5.2	/hardware_carried/allowance_kg	v3.5.1 (lead 188 (11)): 'states' INCLUDE the standard hardware (data/hardware_v352.json) as point masses at the host parts, carried as max(0.20 kg allowance, CALC sum) until DATASHEET masses are in ...
T1026	0.22314 kg	CALC	MP v3.5.2	/hardware_carried/calc_sum_kg	(class set at the call site: CALC)
T1027	0.22314 kg	CALC	MP v3.5.2	/hardware_carried/carried_kg	(class set at the call site: CALC)
T1028	HW-128; HW-158; HW-159; HW-160; HW-161; HW-163; HW-164; HW-165; HW-166; HW-168	MISSING	MP v3.5.2	/hardware_carried/lines_without_mass	(class set at the call site: MISSING)
T1029	massprops_v3.5.2.json (engine group mid 2.73802 kg, CG [-2.62, 0.919, 5.934] mm ...	CALC	SET v3.5.2 final	/mass	(class set at the call site: CALC)
T1030	Electronics net -0.061 kg; worst combination: Mechanical v3.5.2 +0.0118 kg (without the ...	DERIVED	ELE v4.1	/mass_statement/against_trigger	DERIVED + Mechanical CALC (mechanical.json gimballed_mass_properties_v352)
T1031	16	DERIVED	generator	formula: Σ qty of the airframe screw lines ENG-HW-125 + ENG-HW-126	DERIVED by build_datasheet.py: Σ qty of the airframe screw lines ENG-HW-125 + ENG-HW-126 ← T315, T317
T1032	-6 PTFE-lined stainless-braided hose (oxygen-cleaned, PFPE only), JIC-37 ends: fixed end ...	DERIVED	MEC v2.4	/feed_across_gimbal/hose	DERIVED [mechanical.json /not_re_derived_v352 lists feed_across_gimbal]
T1033	650 N	DECIDED *Lead decision of 2026-10-13*	ELE v4.1	/abort_matrix/causes/23/thresholds/kOverThrust_N	tag in electronics.json v4.1: LEAD 2026-10-13 ADOPTED
T1034	4500000 Pa	DECIDED *Lead decision of 2026-10-12*	CMB v2.4d	/operating_points/pressure_limits/chamber_MEOP_Pa	tag in combustion.json v2.4d: LEAD 2026-10-12
T1035	6750000 Pa	DECIDED *Lead decision of 2026-10-12*	CMB v2.4d	/operating_points/pressure_limits/proof_Pa	tag in combustion.json v2.4d: LEAD 2026-10-12
T1036	4.04 g	SIM	GNC v3.2	/design_limit_loads/axial_accel_flight_g	SIM m2_mc6 v2 (sim6 v2.4, 500 dispersed runs, brief C.7 dispersions, relay engine model) max (flight, before touchdown)
T1037	3.77 g	SIM	GNC v3.2	/design_limit_loads/axial_accel_flight_g/note [regex p99 ([\d.]+) g]	(class set at the call site: SIM) [p99 of the same dispersed set, from the record note]
T1038	5.43 g	SIM	GNC v3.2	/design_limit_loads/axial_accel_touchdown_g	SIM m2_mc6 v2 (sim6 v2.4, 500 dispersed runs, brief C.7 dispersions, relay engine model) max (after first contact, rigid body at the CG)
T1039	5 g	SIM	GNC v3.2	/design_limit_loads/axial_accel_touchdown_g/note [regex p99 ([\d.]+) g]	(class set at the call site: SIM) [p99 of the same dispersed set, from the record note]
T1040	6.29 g	SIM	GNC v3.2	/design_limit_loads/lateral_accel_at_pivot_touchdown_g	SIM m2_mc6 v2 (sim6 v2.4, 500 dispersed runs, brief C.7 dispersions, relay engine model) max
T1041	5.22 g	SIM	GNC v3.2	/design_limit_loads/lateral_accel_at_pivot_touchdown_g/note [regex p99 ([\d.]+) g]	(class set at the call site: SIM) [p99 of the same dispersed set, from the record note]
T1042	10 g	DERIVED	GNC v3.2	/design_limit_loads/axial_accel_landing_shock_envelope_g	DERIVED v1: 3 m/s touchdown, 0.10 m stroke, 50 % efficiency
T1043	ICD-AIRFRAME v2.1 is REMOVED from the finish scope and is not written now ...	DECIDED *Lead decision, line 222*	CL 266 lines	L222[from "ICD-AIRFRAME v2.1 is REMOVED" to "(superseded in places, not updated)."]	lead (founder decision) 2026-10-19 (line sha 1c5073790c99185c)
T1044	The items that were queued for it are kept as engine data or open items in the engine ...	DECIDED *Lead decision, line 222*	CL 266 lines	L222[from "The items that were queued for it" to "the 2.20 kg tank option (line 219)."]	lead (founder decision) 2026-10-19 (line sha 1c5073790c99185c)
T1045	checks/make_all_log_v3.5.2.json, checks/build_log_v3.5.2.json ...	DERIVED	SET v3.5.2 final	/build_evidence	(class set at the call site: DERIVED)
T1046	112	DERIVED	CHK v1	/roundtrip_parts/n_passed	(class set at the call site: DERIVED)
T1047	112	DERIVED	CHK v1	/roundtrip_parts/n	(class set at the call site: DERIVED)
T1048	checks/test_final352_v1.json: of the 112 part numbers only ENG-TC-01 differs from the ...	DERIVED	SET v3.5.2 final	/final_vs_interim	(class set at the call site: DERIVED)
T1049	checks/test_final352_v1.json parts_changed_vs_v351: STL sha256 of every final part ...	DERIVED	SET v3.5.2 final	/how_changed_was_found	(class set at the call site: DERIVED)
T1050	interim checks (unchanged geometry): checks/clear_interim352_v2.json ...	DERIVED	SET v3.5.2 final	/interference	(class set at the call site: DERIVED)
T1051	the v3.5.2 set rebuilds from the hub copy of the archive with the five named inputs ...	DERIVED	SET v3.5.2 final	/rebuild_proof/claim	(class set at the call site: DERIVED)
T1052	neutral	DECIDED *Design value, clear_interim352_v2.json v2*	CLR v2	/poses/0/pose	clear_interim352_v2.json v2
T1053	[{"a": "ENG-GM-01#0", "b": "ENG-GM-18#0", "common_mm3": 18.05, "whitelisted": "M4 thread ...	DERIVED	CLR v2	/poses/0/interference	(class set at the call site: DERIVED)
T1054		DERIVED	CLR v2	/poses/0/not_whitelisted	(class set at the call site: DERIVED)
T1055	[{"a": "ENG-GM-01#0", "b": "ENG-GM-03#0", "dist_mm": 1.188, "at": [8.0, 39.0, 0.0] ...	DERIVED	CLR v2	/poses/0/close_lt_3mm	(class set at the call site: DERIVED)
T1056	4	DERIVED	generator	formula: count of interference records at the pose (whitelisted thread overlaps included)	DERIVED by build_datasheet.py: count of interference records at the pose (whitelisted thread overlaps included) ← T1053
T1057	0	DERIVED	generator	formula: count of not-whitelisted interferences at the pose	DERIVED by build_datasheet.py: count of not-whitelisted interferences at the pose ← T1054
T1058	8	DERIVED	generator	formula: count of pairs closer than 3 mm at the pose	DERIVED by build_datasheet.py: count of pairs closer than 3 mm at the pose ← T1055
T1059	1.188 mm	DERIVED	generator	formula: smallest pair distance among the pairs closer than 3 mm	DERIVED by build_datasheet.py: smallest pair distance among the pairs closer than 3 mm ← T1055
T1060	alpha+	DECIDED *Design value, clear_interim352_v2.json v2*	CLR v2	/poses/1/pose	clear_interim352_v2.json v2
T1061		DERIVED	CLR v2	/poses/1/interference	(class set at the call site: DERIVED)
T1062		DERIVED	CLR v2	/poses/1/not_whitelisted	(class set at the call site: DERIVED)
T1063	[{"a": "ENG-GM-01#0", "b": "ENG-GM-03#0", "dist_mm": 1.188, "at": [8.0, 38.7, 4.8] ...	DERIVED	CLR v2	/poses/1/close_lt_3mm	(class set at the call site: DERIVED)
T1064	0	DERIVED	generator	formula: count of interference records at the pose (whitelisted thread overlaps included)	DERIVED by build_datasheet.py: count of interference records at the pose (whitelisted thread overlaps included) ← T1061
T1065	0	DERIVED	generator	formula: count of not-whitelisted interferences at the pose	DERIVED by build_datasheet.py: count of not-whitelisted interferences at the pose ← T1062

ID	Value	Class	Source	Key	Source tag / formula
T1066	10	DERIVED	generator	formula: count of pairs closer than 3 mm at the pose	DERIVED by build_datasheet.py: count of pairs closer than 3 mm at the pose ← T1063
T1067	1.188 mm	DERIVED	generator	formula: smallest pair distance among the pairs closer than 3 mm	DERIVED by build_datasheet.py: smallest pair distance among the pairs closer than 3 mm ← T1063
T1068	alpha-	DECIDED *Design value, clear_interim352_v2.json v2*	CLR v2	/poses/2/pose	clear_interim352_v2.json v2
T1069		DERIVED	CLR v2	/poses/2/interference	(class set at the call site: DERIVED)
T1070		DERIVED	CLR v2	/poses/2/not_whitelisted	(class set at the call site: DERIVED)
T1071	[{"a": "ENG-GM-01#0", "b": "ENG-GM-03#0", "dist_mm": 1.188, "at": [8.0, 38.7, -4.8] ...	DERIVED	CLR v2	/poses/2/close_lt_3mm	(class set at the call site: DERIVED)
T1072	0	DERIVED	generator	formula: count of interference records at the pose (whitelisted thread overlaps included)	DERIVED by build_datasheet.py: count of interference records at the pose (whitelisted thread overlaps included) ← T1069
T1073	0	DERIVED	generator	formula: count of not-whitelisted interferences at the pose	DERIVED by build_datasheet.py: count of not-whitelisted interferences at the pose ← T1070
T1074	9	DERIVED	generator	formula: count of pairs closer than 3 mm at the pose	DERIVED by build_datasheet.py: count of pairs closer than 3 mm at the pose ← T1071
T1075	1.188 mm	DERIVED	generator	formula: smallest pair distance among the pairs closer than 3 mm	DERIVED by build_datasheet.py: smallest pair distance among the pairs closer than 3 mm ← T1071
T1076	beta+	DECIDED *Design value, clear_interim352_v2.json v2*	CLR v2	/poses/3/pose	clear_interim352_v2.json v2
T1077		DERIVED	CLR v2	/poses/3/interference	(class set at the call site: DERIVED)
T1078		DERIVED	CLR v2	/poses/3/not_whitelisted	(class set at the call site: DERIVED)
T1079	[{"a": "ENG-GM-01#0", "b": "ENG-GM-17#0", "dist_mm": 0.816, "at": [-44.3, 37.0, -14.3] ...	DERIVED	CLR v2	/poses/3/close_lt_3mm	(class set at the call site: DERIVED)
T1080	0	DERIVED	generator	formula: count of interference records at the pose (whitelisted thread overlaps included)	DERIVED by build_datasheet.py: count of interference records at the pose (whitelisted thread overlaps included) ← T1077
T1081	0	DERIVED	generator	formula: count of not-whitelisted interferences at the pose	DERIVED by build_datasheet.py: count of not-whitelisted interferences at the pose ← T1078
T1082	13	DERIVED	generator	formula: count of pairs closer than 3 mm at the pose	DERIVED by build_datasheet.py: count of pairs closer than 3 mm at the pose ← T1079
T1083	0.816 mm	DERIVED	generator	formula: smallest pair distance among the pairs closer than 3 mm	DERIVED by build_datasheet.py: smallest pair distance among the pairs closer than 3 mm ← T1079
T1084	beta-	DECIDED *Design value, clear_interim352_v2.json v2*	CLR v2	/poses/4/pose	clear_interim352_v2.json v2
T1085		DERIVED	CLR v2	/poses/4/interference	(class set at the call site: DERIVED)
T1086		DERIVED	CLR v2	/poses/4/not_whitelisted	(class set at the call site: DERIVED)
T1087	[{"a": "ENG-GM-01#0", "b": "ENG-GM-17#0", "dist_mm": 0.948, "at": [-38.5, 41.4, -40.0] ...	DERIVED	CLR v2	/poses/4/close_lt_3mm	(class set at the call site: DERIVED)
T1088	0	DERIVED	generator	formula: count of interference records at the pose (whitelisted thread overlaps included)	DERIVED by build_datasheet.py: count of interference records at the pose (whitelisted thread overlaps included) ← T1085
T1089	0	DERIVED	generator	formula: count of not-whitelisted interferences at the pose	DERIVED by build_datasheet.py: count of not-whitelisted interferences at the pose ← T1086
T1090	13	DERIVED	generator	formula: count of pairs closer than 3 mm at the pose	DERIVED by build_datasheet.py: count of pairs closer than 3 mm at the pose ← T1087
T1091	0.948 mm	DERIVED	generator	formula: smallest pair distance among the pairs closer than 3 mm	DERIVED by build_datasheet.py: smallest pair distance among the pairs closer than 3 mm ← T1087
T1092	combined045	DECIDED *Design value, clear_interim352_v2.json v2*	CLR v2	/poses/5/pose	clear_interim352_v2.json v2
T1093		DERIVED	CLR v2	/poses/5/interference	(class set at the call site: DERIVED)
T1094		DERIVED	CLR v2	/poses/5/not_whitelisted	(class set at the call site: DERIVED)
T1095	[{"a": "ENG-GM-01#0", "b": "ENG-GM-17#0", "dist_mm": 1.052, "at": [-43.7, 35.1, -18.9] ...	DERIVED	CLR v2	/poses/5/close_lt_3mm	(class set at the call site: DERIVED)
T1096	0	DERIVED	generator	formula: count of interference records at the pose (whitelisted thread overlaps included)	DERIVED by build_datasheet.py: count of interference records at the pose (whitelisted thread overlaps included) ← T1093
T1097	0	DERIVED	generator	formula: count of not-whitelisted interferences at the pose	DERIVED by build_datasheet.py: count of not-whitelisted interferences at the pose ← T1094
T1098	13	DERIVED	generator	formula: count of pairs closer than 3 mm at the pose	DERIVED by build_datasheet.py: count of pairs closer than 3 mm at the pose ← T1095
T1099	1.052 mm	DERIVED	generator	formula: smallest pair distance among the pairs closer than 3 mm	DERIVED by build_datasheet.py: smallest pair distance among the pairs closer than 3 mm ← T1095
T1100	combined135	DECIDED *Design value, clear_interim352_v2.json v2*	CLR v2	/poses/6/pose	clear_interim352_v2.json v2
T1101		DERIVED	CLR v2	/poses/6/interference	(class set at the call site: DERIVED)
T1102		DERIVED	CLR v2	/poses/6/not_whitelisted	(class set at the call site: DERIVED)
T1103	[{"a": "ENG-GM-01#0", "b": "ENG-GM-03#0", "dist_mm": 1.188, "at": [8.0, 38.9, -3.4] ...	DERIVED	CLR v2	/poses/6/close_lt_3mm	(class set at the call site: DERIVED)
T1104	0	DERIVED	generator	formula: count of interference records at the pose (whitelisted thread overlaps included)	DERIVED by build_datasheet.py: count of interference records at the pose (whitelisted thread overlaps included) ← T1101
T1105	0	DERIVED	generator	formula: count of not-whitelisted interferences at the pose	DERIVED by build_datasheet.py: count of not-whitelisted interferences at the pose ← T1102
T1106	13	DERIVED	generator	formula: count of pairs closer than 3 mm at the pose	DERIVED by build_datasheet.py: count of pairs closer than 3 mm at the pose ← T1103
T1107	1.188 mm	DERIVED	generator	formula: smallest pair distance among the pairs closer than 3 mm	DERIVED by build_datasheet.py: smallest pair distance among the pairs closer than 3 mm ← T1103
T1108	combined225	DECIDED *Design value, clear_interim352_v2.json v2*	CLR v2	/poses/7/pose	clear_interim352_v2.json v2
T1109		DERIVED	CLR v2	/poses/7/interference	(class set at the call site: DERIVED)
T1110		DERIVED	CLR v2	/poses/7/not_whitelisted	(class set at the call site: DERIVED)
T1111	[{"a": "ENG-GM-01#0", "b": "ENG-GM-03#0", "dist_mm": 1.188, "at": [8.0, 38.9, 3.4] ...	DERIVED	CLR v2	/poses/7/close_lt_3mm	(class set at the call site: DERIVED)
T1112	0	DERIVED	generator	formula: count of interference records at the pose (whitelisted thread overlaps included)	DERIVED by build_datasheet.py: count of interference records at the pose (whitelisted thread overlaps included) ← T1109
T1113	0	DERIVED	generator	formula: count of not-whitelisted interferences at the pose	DERIVED by build_datasheet.py: count of not-whitelisted interferences at the pose ← T1110
T1114	13	DERIVED	generator	formula: count of pairs closer than 3 mm at the pose	DERIVED by build_datasheet.py: count of pairs closer than 3 mm at the pose ← T1111
T1115	1.188 mm	DERIVED	generator	formula: smallest pair distance among the pairs closer than 3 mm	DERIVED by build_datasheet.py: smallest pair distance among the pairs closer than 3 mm ← T1111
T1116	combined315	DECIDED *Design value, clear_interim352_v2.json v2*	CLR v2	/poses/8/pose	clear_interim352_v2.json v2
T1117		DERIVED	CLR v2	/poses/8/interference	(class set at the call site: DERIVED)
T1118		DERIVED	CLR v2	/poses/8/not_whitelisted	(class set at the call site: DERIVED)
T1119	[{"a": "ENG-GM-02#0", "b": "ENG-GM-17#1", "dist_mm": 1.052, "at": [-43.7, -35.1, -18.9] ...	DERIVED	CLR v2	/poses/8/close_lt_3mm	(class set at the call site: DERIVED)
T1120	0	DERIVED	generator	formula: count of interference records at the pose (whitelisted thread overlaps included)	DERIVED by build_datasheet.py: count of interference records at the pose (whitelisted thread overlaps included) ← T1117
T1121	0	DERIVED	generator	formula: count of not-whitelisted interferences at the pose	DERIVED by build_datasheet.py: count of not-whitelisted interferences at the pose ← T1118
T1122	15	DERIVED	generator	formula: count of pairs closer than 3 mm at the pose	DERIVED by build_datasheet.py: count of pairs closer than 3 mm at the pose ← T1119
T1123	1.052 mm	DERIVED	generator	formula: smallest pair distance among the pairs closer than 3 mm	DERIVED by build_datasheet.py: smallest pair distance among the pairs closer than 3 mm ← T1119
T1124	stop000	DECIDED *Design value, clear_interim352_v2.json v2*	CLR v2	/poses/9/pose	clear_interim352_v2.json v2
T1125		DERIVED	CLR v2	/poses/9/interference	(class set at the call site: DERIVED)
T1126		DERIVED	CLR v2	/poses/9/not_whitelisted	(class set at the call site: DERIVED)
T1127	[{"a": "ENG-GM-11#0", "b": "ENG-TC-01#0", "dist_mm": 0.0, "at": [-44.7, 0.0, -89.8] ...	DERIVED	CLR v2	/poses/9/close_lt_3mm	(class set at the call site: DERIVED)
T1128	0	DERIVED	generator	formula: count of interference records at the pose (whitelisted thread overlaps included)	DERIVED by build_datasheet.py: count of interference records at the pose (whitelisted thread overlaps included) ← T1125

ID	Value	Class	Source	Key	Source tag / formula
T1129	0	DERIVED	generator	formula: count of not-whitelisted interferences at the pose	DERIVED by build_datasheet.py: count of not-whitelisted interferences at the pose ← T1126
T1130	14	DERIVED	generator	formula: count of pairs closer than 3 mm at the pose	DERIVED by build_datasheet.py: count of pairs closer than 3 mm at the pose ← T1127
T1131	0 mm	DERIVED	generator	formula: smallest pair distance among the pairs closer than 3 mm	DERIVED by build_datasheet.py: smallest pair distance among the pairs closer than 3 mm ← T1127
T1132	stop045	DECIDED *Design value, clear_interim352_v2.json v2*	CLR v2	/poses/10/pose	clear_interim352_v2.json v2
T1133		DERIVED	CLR v2	/poses/10/interference	(class set at the call site: DERIVED)
T1134		DERIVED	CLR v2	/poses/10/not_whitelisted	(class set at the call site: DERIVED)
T1135	[{"a": "ENG-GM-11#0", "b": "ENG-TC-01#0", "dist_mm": 0.0, "at": [-31.6, -31.6, -89.8] ...	DERIVED	CLR v2	/poses/10/close_lt_3mm	(class set at the call site: DERIVED)
T1136	0	DERIVED	generator	formula: count of interference records at the pose (whitelisted thread overlaps included)	DERIVED by build_datasheet.py: count of interference records at the pose (whitelisted thread overlaps included) ← T1133
T1137	0	DERIVED	generator	formula: count of not-whitelisted interferences at the pose	DERIVED by build_datasheet.py: count of not-whitelisted interferences at the pose ← T1134
T1138	14	DERIVED	generator	formula: count of pairs closer than 3 mm at the pose	DERIVED by build_datasheet.py: count of pairs closer than 3 mm at the pose ← T1135
T1139	0 mm	DERIVED	generator	formula: smallest pair distance among the pairs closer than 3 mm	DERIVED by build_datasheet.py: smallest pair distance among the pairs closer than 3 mm ← T1135
T1140	stop090	DECIDED *Design value, clear_interim352_v2.json v2*	CLR v2	/poses/11/pose	clear_interim352_v2.json v2
T1141		DERIVED	CLR v2	/poses/11/interference	(class set at the call site: DERIVED)
T1142		DERIVED	CLR v2	/poses/11/not_whitelisted	(class set at the call site: DERIVED)
T1143	[{"a": "ENG-GM-11#0", "b": "ENG-TC-01#0", "dist_mm": 0.0, "at": [-0.0, -44.7, -89.8] ...	DERIVED	CLR v2	/poses/11/close_lt_3mm	(class set at the call site: DERIVED)
T1144	0	DERIVED	generator	formula: count of interference records at the pose (whitelisted thread overlaps included)	DERIVED by build_datasheet.py: count of interference records at the pose (whitelisted thread overlaps included) ← T1141
T1145	0	DERIVED	generator	formula: count of not-whitelisted interferences at the pose	DERIVED by build_datasheet.py: count of not-whitelisted interferences at the pose ← T1142
T1146	12	DERIVED	generator	formula: count of pairs closer than 3 mm at the pose	DERIVED by build_datasheet.py: count of pairs closer than 3 mm at the pose ← T1143
T1147	0 mm	DERIVED	generator	formula: smallest pair distance among the pairs closer than 3 mm	DERIVED by build_datasheet.py: smallest pair distance among the pairs closer than 3 mm ← T1143
T1148	stop135	DECIDED *Design value, clear_interim352_v2.json v2*	CLR v2	/poses/12/pose	clear_interim352_v2.json v2
T1149		DERIVED	CLR v2	/poses/12/interference	(class set at the call site: DERIVED)
T1150		DERIVED	CLR v2	/poses/12/not_whitelisted	(class set at the call site: DERIVED)
T1151	[{"a": "ENG-GM-11#0", "b": "ENG-TC-01#0", "dist_mm": 0.0, "at": [31.6, -31.6, -89.8] ...	DERIVED	CLR v2	/poses/12/close_lt_3mm	(class set at the call site: DERIVED)
T1152	0	DERIVED	generator	formula: count of interference records at the pose (whitelisted thread overlaps included)	DERIVED by build_datasheet.py: count of interference records at the pose (whitelisted thread overlaps included) ← T1149
T1153	0	DERIVED	generator	formula: count of not-whitelisted interferences at the pose	DERIVED by build_datasheet.py: count of not-whitelisted interferences at the pose ← T1150
T1154	14	DERIVED	generator	formula: count of pairs closer than 3 mm at the pose	DERIVED by build_datasheet.py: count of pairs closer than 3 mm at the pose ← T1151
T1155	0 mm	DERIVED	generator	formula: smallest pair distance among the pairs closer than 3 mm	DERIVED by build_datasheet.py: smallest pair distance among the pairs closer than 3 mm ← T1151
T1156	stop180	DECIDED *Design value, clear_interim352_v2.json v2*	CLR v2	/poses/13/pose	clear_interim352_v2.json v2
T1157		DERIVED	CLR v2	/poses/13/interference	(class set at the call site: DERIVED)
T1158		DERIVED	CLR v2	/poses/13/not_whitelisted	(class set at the call site: DERIVED)
T1159	[{"a": "ENG-GM-11#0", "b": "ENG-TC-01#0", "dist_mm": 0.0, "at": [44.7, 0.0, -89.8] ...	DERIVED	CLR v2	/poses/13/close_lt_3mm	(class set at the call site: DERIVED)
T1160	0	DERIVED	generator	formula: count of interference records at the pose (whitelisted thread overlaps included)	DERIVED by build_datasheet.py: count of interference records at the pose (whitelisted thread overlaps included) ← T1157
T1161	0	DERIVED	generator	formula: count of not-whitelisted interferences at the pose	DERIVED by build_datasheet.py: count of not-whitelisted interferences at the pose ← T1158
T1162	14	DERIVED	generator	formula: count of pairs closer than 3 mm at the pose	DERIVED by build_datasheet.py: count of pairs closer than 3 mm at the pose ← T1159
T1163	0 mm	DERIVED	generator	formula: smallest pair distance among the pairs closer than 3 mm	DERIVED by build_datasheet.py: smallest pair distance among the pairs closer than 3 mm ← T1159
T1164	stop225	DECIDED *Design value, clear_interim352_v2.json v2*	CLR v2	/poses/14/pose	clear_interim352_v2.json v2
T1165		DERIVED	CLR v2	/poses/14/interference	(class set at the call site: DERIVED)
T1166		DERIVED	CLR v2	/poses/14/not_whitelisted	(class set at the call site: DERIVED)
T1167	[{"a": "ENG-GM-11#0", "b": "ENG-TC-01#0", "dist_mm": 0.0, "at": [31.6, 31.6, -89.8] ...	DERIVED	CLR v2	/poses/14/close_lt_3mm	(class set at the call site: DERIVED)
T1168	0	DERIVED	generator	formula: count of interference records at the pose (whitelisted thread overlaps included)	DERIVED by build_datasheet.py: count of interference records at the pose (whitelisted thread overlaps included) ← T1165
T1169	0	DERIVED	generator	formula: count of not-whitelisted interferences at the pose	DERIVED by build_datasheet.py: count of not-whitelisted interferences at the pose ← T1166
T1170	14	DERIVED	generator	formula: count of pairs closer than 3 mm at the pose	DERIVED by build_datasheet.py: count of pairs closer than 3 mm at the pose ← T1167
T1171	0 mm	DERIVED	generator	formula: smallest pair distance among the pairs closer than 3 mm	DERIVED by build_datasheet.py: smallest pair distance among the pairs closer than 3 mm ← T1167
T1172	stop270	DECIDED *Design value, clear_interim352_v2.json v2*	CLR v2	/poses/15/pose	clear_interim352_v2.json v2
T1173		DERIVED	CLR v2	/poses/15/interference	(class set at the call site: DERIVED)
T1174		DERIVED	CLR v2	/poses/15/not_whitelisted	(class set at the call site: DERIVED)
T1175	[{"a": "ENG-GM-11#0", "b": "ENG-TC-01#0", "dist_mm": 0.0, "at": [0.0, 44.7, -89.8] ...	DERIVED	CLR v2	/poses/15/close_lt_3mm	(class set at the call site: DERIVED)
T1176	0	DERIVED	generator	formula: count of interference records at the pose (whitelisted thread overlaps included)	DERIVED by build_datasheet.py: count of interference records at the pose (whitelisted thread overlaps included) ← T1173
T1177	0	DERIVED	generator	formula: count of not-whitelisted interferences at the pose	DERIVED by build_datasheet.py: count of not-whitelisted interferences at the pose ← T1174
T1178	14	DERIVED	generator	formula: count of pairs closer than 3 mm at the pose	DERIVED by build_datasheet.py: count of pairs closer than 3 mm at the pose ← T1175
T1179	0 mm	DERIVED	generator	formula: smallest pair distance among the pairs closer than 3 mm	DERIVED by build_datasheet.py: smallest pair distance among the pairs closer than 3 mm ← T1175
T1180	stop315	DECIDED *Design value, clear_interim352_v2.json v2*	CLR v2	/poses/16/pose	clear_interim352_v2.json v2
T1181		DERIVED	CLR v2	/poses/16/interference	(class set at the call site: DERIVED)
T1182		DERIVED	CLR v2	/poses/16/not_whitelisted	(class set at the call site: DERIVED)
T1183	[{"a": "ENG-GM-11#0", "b": "ENG-TC-01#0", "dist_mm": 0.0, "at": [-31.6, 31.6, -89.8] ...	DERIVED	CLR v2	/poses/16/close_lt_3mm	(class set at the call site: DERIVED)
T1184	0	DERIVED	generator	formula: count of interference records at the pose (whitelisted thread overlaps included)	DERIVED by build_datasheet.py: count of interference records at the pose (whitelisted thread overlaps included) ← T1181
T1185	0	DERIVED	generator	formula: count of not-whitelisted interferences at the pose	DERIVED by build_datasheet.py: count of not-whitelisted interferences at the pose ← T1182
T1186	16	DERIVED	generator	formula: count of pairs closer than 3 mm at the pose	DERIVED by build_datasheet.py: count of pairs closer than 3 mm at the pose ← T1183
T1187	0 mm	DERIVED	generator	formula: smallest pair distance among the pairs closer than 3 mm	DERIVED by build_datasheet.py: smallest pair distance among the pairs closer than 3 mm ← T1183
T1188	stopFD07B	DECIDED *Design value, clear_interim352_v2.json v2*	CLR v2	/poses/17/pose	clear_interim352_v2.json v2
T1189		DERIVED	CLR v2	/poses/17/interference	(class set at the call site: DERIVED)
T1190		DERIVED	CLR v2	/poses/17/not_whitelisted	(class set at the call site: DERIVED)
T1191	[{"a": "ENG-GM-11#0", "b": "ENG-TC-01#0", "dist_mm": 0.0, "at": [-17.5, 41.1, -89.8] ...	DERIVED	CLR v2	/poses/17/close_lt_3mm	(class set at the call site: DERIVED)

ID	Value	Class	Source	Key	Source tag / formula
T1192	0	DERIVED	generator	formula: count of interference records at the pose (whitelisted thread overlaps included)	DERIVED by build_datasheet.py: count of interference records at the pose (whitelisted thread overlaps included) ← T1189
T1193	0	DERIVED	generator	formula: count of not-whitelisted interferences at the pose	DERIVED by build_datasheet.py: count of not-whitelisted interferences at the pose ← T1190
T1194	19	DERIVED	generator	formula: count of pairs closer than 3 mm at the pose	DERIVED by build_datasheet.py: count of pairs closer than 3 mm at the pose ← T1191
T1195	0 mm	DERIVED	generator	formula: smallest pair distance among the pairs closer than 3 mm	DERIVED by build_datasheet.py: smallest pair distance among the pairs closer than 3 mm ← T1191
T1196	Sonderborg, Denmark, DK-6400	DECIDED *Requirement, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/delivery	tag in BOM_v3.5.2.json v3.5.2:
T1197	prices not given: every line TO-QUOTE unless dated	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/currency	BOM_v3.5.2.json v3.5.2
T1198	107	DERIVED	BOM v3.5.2	/n_lines	(class set at the call site: DERIVED)
T1199	142	DERIVED	BOM v3.5.2	/n_instances	(class set at the call site: DERIVED)
T1200	57	DERIVED	BOM v3.5.2	/hardware/n_lines	(class set at the call site: DERIVED)
T1201	174	DERIVED	BOM v3.5.2	/hardware/n_items	(class set at the call site: DERIVED)
T1202	CALC masses (standard nominal geometry), NOT datasheet: the hardware is carried in ...	CALC	BOM v3.5.2	/hardware/mass_rule	(class set at the call site: CALC)
T1203	torque-controlled: calibrated setting torque wrench (ISO 6789-1 Type II; tool tolerance ...	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/tightening_methods/M-T1	BOM_v3.5.2.json v3.5.2
T1204	as M-T1 with a calibrated setting torque screwdriver (ISO 6789-1 Type II) for final ...	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/tightening_methods/M-T2	BOM_v3.5.2.json v3.5.2
T1205	as M-T1 (M-T2 tool below 1 N m) on threads coated with the thread locker of the line ...	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/tightening_methods/M-T1L	BOM_v3.5.2.json v3.5.2
T1206	as M-T1; torque from a torque-tension test of the anodised (AMS 2488 Type 2) + ...	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/tightening_methods/M-T1Ti	BOM_v3.5.2.json v3.5.2
T1207	not a torque joint (dowel, insert, seal, washer, clip, consumable)	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/tightening_methods/M-NONE	BOM_v3.5.2.json v3.5.2
T1208	J-FWD (HW-101, lead 17:55:55Z F1 / F2, 16:53Z R3): as M-T1L (wet thread locker of the ...	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/tightening_methods/M-T1J	BOM_v3.5.2.json v3.5.2
T1209	+/-25 % about F = T / (K d), K 0.20 (both ASSUMED; the FEA3D F4 basis, alpha_A = 1.25 / ...	ASSUMED	BOM v3.5.2	/preload_scatter_rule	(class set at the call site: ASSUMED)
T1210	0.01–0.02 kg	ASSUMED	BOM v3.5.2	/provision_mass_estimate_kg/estimate_range_kg	v3.5: bleed MODELLED (ENG-BL-01..10 in the BOM lines); provision = opposed vent tee outside the wall only
T1211	TO-VERIFY (datasheet)	TO-VERIFY	BOM v3.5.2	/provision_mass_estimate_kg/valves	(class set at the call site: TO-VERIFY)
T1212	T-F1. Short-time exposure allowables for the 6061-T6 parts TC-01 (case tube) and TC-02 ...	CALC	TXT5 v5	/T-F1	(class set at the call site: CALC)
T1213	A = 150 C (CHANGELOG 249 (1) ACCEPTED)	DECIDED *Lead decision, per item_c_tf1_texts_v352_v5.json v5*	TXT5 v5	/T-F1_exposure_basis_R9/abort_value	tag in item_c_tf1_texts_v352_v5.json v5:
T1214	a_to_c: PASS on both bounds (lowest TC-02 TBS +0.409 / +0.745; lead, CHANGELOG 249 (3)) ...	CALC	TXT5 v5	/T-F1_exposure_basis_R9/criteria_status	(class set at the call site: CALC on SIM; the criteria results were accepted by the lead in 249 (3); REV-DS-1 D1-07)
T1215	T-F1 EXPOSURE BASIS (21:51:07Z, SUPERSEDING the 21:46:49Z 'A 165 rows' wording): the A ...	DECIDED *Lead decision, line 249 (3)*	CL 266 lines	L249(3)	lead (decision) 2026-10-21 (line sha 21373a357d2dd391)
T1216	184.3 MPa	LITERATURE	TXT5 v5	/T-F1_engmat_10000h_values_used_by_the_cells/150/Fty_MPa	(class set at the call site: LITERATURE)
T1217	194.7 MPa	LITERATURE	TXT5 v5	/T-F1_engmat_10000h_values_used_by_the_cells/150/Ftu_MPa	(class set at the call site: LITERATURE)
T1218	140.5 MPa	LITERATURE	TXT5 v5	/T-F1_engmat_10000h_values_used_by_the_cells/175/Fty_MPa	(class set at the call site: LITERATURE)
T1219	155.4 MPa	LITERATURE	TXT5 v5	/T-F1_engmat_10000h_values_used_by_the_cells/175/Ftu_MPa	(class set at the call site: LITERATURE)
T1220	134.5 MPa	LITERATURE	TXT5 v5	/T-F1_engmat_10000h_values_used_by_the_cells/178.4/Fty_MPa	(class set at the call site: LITERATURE)
T1221	150.1 MPa	LITERATURE	TXT5 v5	/T-F1_engmat_10000h_values_used_by_the_cells/178.4/Ftu_MPa	(class set at the call site: LITERATURE)
T1222	116.9 MPa	LITERATURE	TXT5 v5	/T-F1_engmat_10000h_values_used_by_the_cells/188.4/Fty_MPa	(class set at the call site: LITERATURE)
T1223	134.4 MPa	LITERATURE	TXT5 v5	/T-F1_engmat_10000h_values_used_by_the_cells/188.4/Ftu_MPa	(class set at the call site: LITERATURE)
T1224	111.2 MPa	LITERATURE	TXT5 v5	/T-F1_engmat_10000h_values_used_by_the_cells/191.7/Fty_MPa	(class set at the call site: LITERATURE)
T1225	129.2 MPa	LITERATURE	TXT5 v5	/T-F1_engmat_10000h_values_used_by_the_cells/191.7/Ftu_MPa	(class set at the call site: LITERATURE)
T1226	108.2 MPa	LITERATURE	TXT5 v5	/T-F1_engmat_10000h_values_used_by_the_cells/193.4/Fty_MPa	(class set at the call site: LITERATURE)
T1227	126.5 MPa	LITERATURE	TXT5 v5	/T-F1_engmat_10000h_values_used_by_the_cells/193.4/Ftu_MPa	(class set at the call site: LITERATURE)
T1228	96.6 MPa	LITERATURE	TXT5 v5	/T-F1_engmat_10000h_values_used_by_the_cells/200/Fty_MPa	(class set at the call site: LITERATURE)
T1229	116.2 MPa	LITERATURE	TXT5 v5	/T-F1_engmat_10000h_values_used_by_the_cells/200/Ftu_MPa	(class set at the call site: LITERATURE)
T1230	84.9 MPa	LITERATURE	TXT5 v5	/T-F1_engmat_10000h_values_used_by_the_cells/208.4/Fty_MPa	(class set at the call site: LITERATURE)
T1231	104.6 MPa	LITERATURE	TXT5 v5	/T-F1_engmat_10000h_values_used_by_the_cells/208.4/Ftu_MPa	(class set at the call site: LITERATURE)
T1232	68.8 MPa	LITERATURE	TXT5 v5	/T-F1_engmat_10000h_values_used_by_the_cells/223.4/Fty_MPa	(class set at the call site: LITERATURE)
T1233	86.6 MPa	LITERATURE	TXT5 v5	/T-F1_engmat_10000h_values_used_by_the_cells/223.4/Ftu_MPa	(class set at the call site: LITERATURE)
T1234	67.1 MPa	LITERATURE	TXT5 v5	/T-F1_engmat_10000h_values_used_by_the_cells/225/Fty_MPa	(class set at the call site: LITERATURE)
T1235	84.7 MPa	LITERATURE	TXT5 v5	/T-F1_engmat_10000h_values_used_by_the_cells/225/Ftu_MPa	(class set at the call site: LITERATURE)
T1236	52.7 MPa	LITERATURE	TXT5 v5	/T-F1_engmat_10000h_values_used_by_the_cells/238.4/Fty_MPa	(class set at the call site: LITERATURE)
T1237	68.7 MPa	LITERATURE	TXT5 v5	/T-F1_engmat_10000h_values_used_by_the_cells/238.4/Ftu_MPa	(class set at the call site: LITERATURE)

ID	Value	Class	Source	Key	Source tag / formula
T1238	42 MPa	LITERATURE	TXT5 v5	/T-F1_engmat_10000h_values_used_by_the_cells/248.4/Fty_MPa	(class set at the call site: LITERATURE)
T1239	56.7 MPa	LITERATURE	TXT5 v5	/T-F1_engmat_10000h_values_used_by_the_cells/248.4/Ftu_MPa	(class set at the call site: LITERATURE)
T1240	40.3 MPa	LITERATURE	TXT5 v5	/T-F1_engmat_10000h_values_used_by_the_cells/250/Fty_MPa	(class set at the call site: LITERATURE)
T1241	54.8 MPa	LITERATURE	TXT5 v5	/T-F1_engmat_10000h_values_used_by_the_cells/250/Ftu_MPa	(class set at the call site: LITERATURE)
T1242	21.919697 MW/m²	SIM	CMB v2.4d	/thermal_and_loads/peak_convective_heat_flux_W_m2	SIM Bartz at the throat, wall 600 K, max thrust, S230 (thermal_bc_summary_v2.json) [display = raw x 1e-06 (W/m² to MW/m²)]
T1243	10258.7 W/(m²·K)	SIM	CMB v2.4d	/thermal_and_loads/peak_h_g_W_m2K	SIM
T1244	2736.7 K	SIM	CMB v2.4d	/thermal_and_loads/T_aw_throat_K	SIM
T1245	27.1199999999999997–31.5499999999999997 MW/m²	SIM	CMB v2.4d	/thermal_and_loads/peak_incident_flux_cold_wall_transient_W_m2	SIM coupled insert model at start (wall near ambient), nominal/conservative (erosion_S230/erosion_dc_v2.json) [display = raw x 1e-06 (W/m² to MW/m²)]
T1246	423.15 K	DECIDED *Lead decision, line 243 (3)*	ELE v4.1	/abort_matrix/causes/5/thresholds/kCaseOverTemp_K	tag in electronics.json v4.1: LEAD CHANGELOG 243 (3)
T1247	403.15 K	DECIDED *Lead decision of 2026-10-20*	ELE v4.1	/abort_matrix/causes/5/thresholds/kCaseRelightMax_K	tag in electronics.json v4.1: LEAD 2026-10-20T21:46:49Z item 5, Electronics basis
T1248	523.15 K	ASSUMED	ELE v4.1	/abort_matrix/causes/5/thresholds/kNozzleHolderOverTemp_K	ASSUMED
T1249	1.6 mm	DECIDED *Design value, combustion.json v2.4d*	CMB v2.4d	/gas_path/liner_grain_zone/thickness_m	tag in combustion.json v2.4d: v1 (grain zone only: the residual web >= 1 mm shields the case) [display = raw x 1000 (m to mm)]
T1250	phenolic, thickness set by thermal FEA; 1-D bound needs >= 5 mm and still exceeds the ...	DERIVED	CMB v2.4d	/gas_path/post_chamber_insulator	DERIVED data/v2/soakback_wall_v2.json
T1251	replace the insert when dA_t/A_t reaches 5 % (measured pin gauge / bore scan after each ...	DERIVED	CMB v2.4d	/thermal_and_loads/throat_replacement_rule	DERIVED
T1252	LEAD 186 (3) as worded by 194 (5): refutation is any of: a throat-area change more than ...	SIM *Lead decision, per combustion.json v2.4d*	CMB v2.4d	/nz01_criterion_v2p4b/v/criterion	LEAD criterion; DESIGN procedure; SIM erosion model
T1253	mica/vermiculite 900 C rating vs 1595 C FE: fails by 695 K	DERIVED	CMB v2.4d	/nozzle_joint/v/NZ03_gasket	DERIVED nozzle_joint_v2p4.json, nozzle_lip_melt_v2p4.json; FEA Mechanical; materials as recalled TO-VERIFY
T1254	2.0 mm carbon-phenolic flange + 0.5 mm flexible-graphite seal at the lip face	DERIVED	CMB v2.4d	/nozzle_joint/v/proposal_J1	DERIVED nozzle_joint_v2p4.json, nozzle_lip_melt_v2p4.json; FEA Mechanical; materials as recalled TO-VERIFY
T1255	Lead consistency fold found while applying: the case-TC in-burn abort 423.15 K (243 (3) ...	DECIDED *Lead decision, line 257*	CL 266 lines	L257[sentence containing "423.15" | "403.15"]	lead decision 2026-10-21 (line sha cf364643652fceb0)
T1256	ABS (FDM printed, 100 % infill); PETG backup	DECIDED *Requirement, combustion.json v2.4d*	CMB v2.4d	/decisions/fuel	tag in combustion.json v2.4d: REQ founder/LEAD; ABS card c* within +/-0.3 % across compositions/hf sources (ceq_cards.py)
T1257	3.17 L	DECIDED *Lead decision of 2026-10-12*	CMB v2.4d	/decisions/N2O_load_and_tank/v/V_tank_m3	tag in combustion.json v2.4d: LEAD decision 2026-10-12 (CHANGELOG): 2.00 kg / 3.17 L (85 % fill) [display = raw x 1000 (m³ to L)]
T1258	Self-pressurised N2O conditioned 15-24 C at ARM (ENG-12 as amended; was 15-25 C) with ...	DERIVED	CMB v2.4d	/decisions/tank_pressurisation	DERIVED data/v2/supercharge_trade_v2.json: N2 pad 60 bar gives +3.4 bar at MECO vs heating to 25 C +4.3 bar; regulated N2 (0.21 L at 300 bar) adds a pressure system for +13 bar; ENG-12 amended ...
T1259	T_tank <= 24 C and P_tank <= 55.28 bar abs, each lowered by the total channel ...	MISSING *Lead decision, line 172 (4)*	CMB v2.4d	/interlock_guard_band/v/thresholds	LEAD line 172 (4); REV-COMB-2 R2-07; values MISSING until Electronics states the channel uncertainty
T1260	20 C saturated tank (P_tank 50 bar row of the map for lift and dP/Pc), initial port 25 ...	SIM	CMB v2.4d	/operating_points/basis	(class set at the call site: SIM)
T1261	0.92	ASSUMED	CMB v2.4d	/operating_points/basis [regex eta_c* ([\d.]+)]	(class set at the call site: ASSUMED)
T1262	0.975	ASSUMED	CMB v2.4d	/operating_points/basis [regex eta_cf ([\d.]+)]	(class set at the call site: ASSUMED)
T1263	491 kg/(m²·s)	SIM	CMB v2.4d	/operating_points/max_thrust/G_ox_initial_kg_m2s	SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* 0.92, eta_cf 0.975)
T1264	r = a * G_ox^n (oxidiser-flux law, local G_ox along the port)	ASSUMED	CMB v2.4d	/gas_path/grain/regression_law/form	(class set at the call site: ASSUMED)
T1265	5.77e-05 m/s·(kg/m²s)^-n	ASSUMED	CMB v2.4d	/gas_path/grain/regression_law/a	ASSUMED band x0.6-1.5 [m/s (kg/m2/s)^-n]
T1266	0.5	ASSUMED	CMB v2.4d	/gas_path/grain/regression_law/n	ASSUMED band 0.35-0.68
T1267	No verified N2O/ABS law in accessible literature; every result is checked at x0.6 and ...	ASSUMED	CMB v2.4d	/gas_path/grain/regression_law/note	(class set at the call site: ASSUMED)
T1268	{"basis_bar": 44.33, "margin_to_MEOP_bar": 0.67, "proof_margin_min_bar": 21.9 ...	SIM	CMB v2.4d	/meop_basis_v2p4	SIM meop_basis_v2p4.json
T1269	decomposition_TNT_kg: 0.7737; R_6p9kPa_m: 16.4	DERIVED	CMB v2.4d	/stored_energy/v/flight_tank_20C	DERIVED safing_relief_v2p4.json, stored_energy_sup_v2p4.json; scaled distance 17.9 m/kg^(1/3) LITERATURE as recalled, TO-VERIFY
T1270	116.80000000000001 mm	DERIVED	MAN v3.5.2	/instances/*/bounds_mm	DERIVED by build_datasheet.py: z extent of ENG-GR-01 over assembly-manifest instance bounds (instance prefix ENG-GR-01#)
T1271	115 mm	DERIVED	MAN v3.5.2	/instances/*/bounds_mm	DERIVED by build_datasheet.py: z extent of ENG-GR-02 over assembly-manifest instance bounds (instance prefix ENG-GR-02#)
T1272	53.5 mm	DECIDED *Design value, combustion.json v2.4d*	CMB v2.4d	/gas_path/grain/outer_diameter_m	tag in combustion.json v2.4d: combustion.json v1 [m] [display = raw x 1000 (m to mm)]
T1273	230 mm	DERIVED	CMB v2.4d	/gas_path/grain/length_m	DERIVED v2 baseline (grain trade 2026-10-12; v1 0.200) [m] [display = raw x 1000 (m to mm)]
T1274	single centred straight round port	DECIDED *Lead decision of 2026-10-12*	CMB v2.4d	/gas_path/grain/port/type	tag in combustion.json v2.4d: LEAD decision 2026-10-12
T1275	25 mm	DECIDED *Design value, combustion.json v2.4d*	CMB v2.4d	/gas_path/grain/port/initial_diameter_m	tag in combustion.json v2.4d: combustion.json v1 [m] [display = raw x 1000 (m to mm)]
T1276	14.25 mm	DERIVED	CMB v2.4d	/gas_path/grain/port/initial_web_m	DERIVED (OD - port)/2 [display = raw x 1000 (m to mm)]
T1277	Rounded port entrance r >= 4 mm is now REQUIRED on every grain, flight and stand ...	DECIDED *Lead decision, line 160 (4)*	CL 266 lines	L160(4)	lead (decision) 2026-10-13 (line sha c3c13e6adb771e7a)
T1278	1000 kg/m³	ASSUMED	CMB v2.4d	/gas_path/grain/density_kg_m3	ASSUMED printed ABS [kg/m3]; filament 1040
T1279	0.404 kg	DERIVED	CMB v2.4d	/gas_path/grain/mass_kg	DERIVED geometry x 1000 kg/m3
T1280	0.42436 kg	DERIVED	MP v3.5.2	/grain_mass_wet_kg	(class set at the call site: DERIVED)
T1281	web >= 1 mm at the x1.5 regression bound for every flight duty cycle	DERIVED	CMB v2.4d	/gas_path/grain/residual_web_rule	DERIVED: ABS thermal layer kappa/r 0.1-0.3 mm (soakback_v2.json grain_zone); >= 1 mm shields the case
T1282	forward face on the pre-chamber insulator shoulder, aft face on the post-chamber ...	DECIDED *Design value, combustion.json v2.4d*	CMB v2.4d	/gas_path/grain/retention	combustion.json v2.4d
T1283	0.2–0.3 kg	ASSUMED	CMB v2.4d	/igniter/mass_kg	ASSUMED: body 50 g, plug 10 g, two micro solenoids 60-120 g, propane cylinder 50 g, lines/fittings 50 g
T1284	tank → NC main isolation valve (off-gimbal) → -6 PTFE-lined braided hose across the ...	DECIDED *Design value, combustion.json v2.4d*	CMB v2.4d	/valves_and_actuators/architecture	combustion.json v2.4d
T1285	Dyer et al. AIAA 2007-5702 (SPI/HEM blend); HEM and SPI bound it (G ratio 0.55-0.68 and ...	LITERATURE	CMB v2.4d	/gas_path/injector/flow_model	(class set at the call site: LITERATURE)
T1286	sintered <= 15 um, Cv >= 2.2	DERIVED	CMB v2.4d	/gas_path/injector/filter	DERIVED: seat gap at the 12.5 % floor = lift 0.207 mm x sin 15 deg = 54 um; filter <= 1/3 of the smallest gap; Cv from the line budget (feed_line_checks_v2.json); v1 said 40 um
T1287	3	DERIVED	CMB v2.4d	/valves_and_actuators/isolation_valve/Cv_min	DERIVED feed budget (feed_line_checks_v2.json): valve Cv 3 + filter Cv 2.2 give 1.17 bar total at 0.26 kg/s vs the 2.70 bar budget; Cv 1.5 + Cv 1.0 give 2.85 bar
T1288	7000000 Pa	DECIDED *Lead decision, per combustion.json v2.4d*	CMB v2.4d	/valves_and_actuators/isolation_valve/MEOP_Pa	tag in combustion.json v2.4d: LEAD
T1289	0.15 s	DERIVED	CMB v2.4d	/valves_and_actuators/isolation_valve/close_time_max_s	DERIVED
T1290	SUPERSEDED (v2.4a): v1 "required (<= 7.5 MPa) between the isolation valve and the ...	DECIDED *Lead decision, line 172 (6)*	CMB v2.4d	/valves_and_actuators/isolation_valve/trapped_liquid_relief	tag in combustion.json v2.4d: LEAD line 172 (6); REQUIREMENTS amendments (ENG-09 trapped-liquid relief text superseded)
T1291	tank_relief_set_MPa: 6.9; tank_relief_basis: below the 7.0 MPa tank MEOP (ENG-09); fire ...	DERIVED *Lead decision, line 172 (6)*	CMB v2.4d	/relief_v2p4a	LEAD line 172 (6); REPORT v2.4 §2.7 (DERIVED sizing); REQUIREMENTS amendments. Verification CF-3 / proof test (founders)

ID	Value	Class	Source	Key	Source tag / formula
T1292	-6 accepted: 6.3 m/s at 0.241 kg/s nominal, 6.5-7.1 m/s at 0.26 kg/s (15-25 C computed ...	DERIVED	CMB v2.4d	/valves_and_actuators/hose	DERIVED feed_line_checks_v2.json (answers Mechanical REQUEST 2026-10-12)
T1293	head top (Mechanical 2026-10-12) via sense line to the fixed side	DECIDED *Design value, combustion.json v2.4d*	CMB v2.4d	/gas_path/pc_tap/location	tag in combustion.json v2.4d: mechanical CHANGELOG
T1294	1/8 in OD (1.6 mm ID) SS, length <= 0.3 m, transducer dead volume <= 0.3 cm3	DERIVED	CMB v2.4d	/gas_path/pc_tap/sense_line	DERIVED data/v2/feed_line_checks_v2.json: lowest f_n per length (dead volume 0.1-0.5 cm3) {"L0.20m": [164.0, 325.0], "L0.30m": [125.0, 230.0], "L0.45m": [94.0, 160.0], "L0.60m": [75.0, 123.0]} Hz ...
T1295	spark-ignited N2O/propane torch, radial into the pre-chamber	DECIDED *Design value, combustion.json v2.4d*	CMB v2.4d	/igniter/type	tag in combustion.json v2.4d: ignition_v2.json
T1296	3 bar	DERIVED	CMB v2.4d	/igniter/torch_sizing/v/Pc_torch_bar	DERIVED/CEA ignition_v2.json
T1297	5.5	DERIVED	CMB v2.4d	/igniter/torch_sizing/v/OF	DERIVED/CEA ignition_v2.json
T1298	2 mm	DERIVED	CMB v2.4d	/igniter/torch_sizing/v/throat_d_m	DERIVED/CEA ignition_v2.json [display = raw x 1000 (m to mm)]
T1299	0.503 g/s	DERIVED	CMB v2.4d	/igniter/torch_sizing/v/mdot_n2o_kg_s	DERIVED/CEA ignition_v2.json [display = raw x 1000 (kg/s to g/s)]
T1300	0.091 g/s	DERIVED	CMB v2.4d	/igniter/torch_sizing/v/mdot_propane_kg_s	DERIVED/CEA ignition_v2.json [display = raw x 1000 (kg/s to g/s)]
T1301	12 V	DERIVED	CMB v2.4d	/igniter/electrical/v/spark_supply_V	DERIVED; electronics.json v1 IGN_HV coil mode + SOL1/SOL2
T1302	0.05 J	DERIVED	CMB v2.4d	/igniter/electrical/v/spark_energy_J	DERIVED; electronics.json v1 IGN_HV coil mode + SOL1/SOL2
T1303	50 Hz	DERIVED	CMB v2.4d	/igniter/electrical/v/rate_Hz	DERIVED; electronics.json v1 IGN_HV coil mode + SOL1/SOL2
T1304	>= 3; limited by propane (0.055 g/start) and N2O torch flow (0.30 g/start)	DERIVED	CMB v2.4d	/igniter/starts_per_flight	DERIVED
T1305	0.4 s	DECIDED *Lead decision, per combustion.json v2.4d*	CMB v2.4d	/light_windows/v/flight_first_start	tag in combustion.json v2.4d: LEAD; REPORT v2.4 §7.3
T1306	0.65 s	DECIDED *Lead decision, per combustion.json v2.4d*	CMB v2.4d	/light_windows/v/flight_relight	tag in combustion.json v2.4d: LEAD; REPORT v2.4 §7.3
T1307	0.35 s	DECIDED *Lead decision, per combustion.json v2.4d*	CMB v2.4d	/light_windows/v/stand	tag in combustion.json v2.4d: LEAD; REPORT v2.4 §7.3
T1308	PURGE (A1 relight only)	DERIVED *Lead decision, line 153*	CMB v2.4d	/igniter/sequence/v/start/0/epoch	LEAD (CHANGELOG 153-177) + DERIVED (REPORT v2.4a §7.3-7.7, §2.8); firmware v2.8 (Electronics). Verify at IGN-1, CF-4, T1-T4
T1309	−2.9 s	DERIVED *Lead decision, line 153*	CMB v2.4d	/igniter/sequence/v/start/0/t_s	LEAD (CHANGELOG 153-177) + DERIVED (REPORT v2.4a §7.3-7.7, §2.8); firmware v2.8 (Electronics). Verify at IGN-1, CF-4, T1-T4
T1310	PURGE commits iff the lead to the plain relight trigger is >= L* = 1.3 s (LEAD 186 (1) ...	DERIVED *Lead decision, line 153*	CMB v2.4d	/igniter/sequence/v/start/0/action	LEAD (CHANGELOG 153-177) + DERIVED (REPORT v2.4a §7.3-7.7, §2.8); firmware v2.8 (Electronics). Verify at IGN-1, CF-4, T1-T4
T1311	E0	DERIVED *Lead decision, line 153*	CMB v2.4d	/igniter/sequence/v/start/1/epoch	LEAD (CHANGELOG 153-177) + DERIVED (REPORT v2.4a §7.3-7.7, §2.8); firmware v2.8 (Electronics). Verify at IGN-1, CF-4, T1-T4
T1312	0 s	DERIVED *Lead decision, line 153*	CMB v2.4d	/igniter/sequence/v/start/1/t_s	LEAD (CHANGELOG 153-177) + DERIVED (REPORT v2.4a §7.3-7.7, §2.8); firmware v2.8 (Electronics). Verify at IGN-1, CF-4, T1-T4
T1313	START or RELIGHT at T_fire: MV opens with the pintle seated (already open after PURGE) ...	DERIVED *Lead decision, line 153*	CMB v2.4d	/igniter/sequence/v/start/1/action	LEAD (CHANGELOG 153-177) + DERIVED (REPORT v2.4a §7.3-7.7, §2.8); firmware v2.8 (Electronics). Verify at IGN-1, CF-4, T1-T4
T1314	torch gate	DERIVED *Lead decision, line 153*	CMB v2.4d	/igniter/sequence/v/start/2/epoch	LEAD (CHANGELOG 153-177) + DERIVED (REPORT v2.4a §7.3-7.7, §2.8); firmware v2.8 (Electronics). Verify at IGN-1, CF-4, T1-T4
T1315	0.16 s	DERIVED *Lead decision, line 153*	CMB v2.4d	/igniter/sequence/v/start/2/t_s	LEAD (CHANGELOG 153-177) + DERIVED (REPORT v2.4a §7.3-7.7, §2.8); firmware v2.8 (Electronics). Verify at IGN-1, CF-4, T1-T4
T1316	no E1 by E0 + 0.16 s → TORCH_FAULT (A1); no main N2O admitted	DERIVED *Lead decision, line 153*	CMB v2.4d	/igniter/sequence/v/start/2/action	LEAD (CHANGELOG 153-177) + DERIVED (REPORT v2.4a §7.3-7.7, §2.8); firmware v2.8 (Electronics). Verify at IGN-1, CF-4, T1-T4
T1317	E1	DERIVED *Lead decision, line 153*	CMB v2.4d	/igniter/sequence/v/start/3/epoch	LEAD (CHANGELOG 153-177) + DERIVED (REPORT v2.4a §7.3-7.7, §2.8); firmware v2.8 (Electronics). Verify at IGN-1, CF-4, T1-T4
T1318	0.09 s	DERIVED *Lead decision, line 153*	CMB v2.4d	/igniter/sequence/v/start/3/t_s	LEAD (CHANGELOG 153-177) + DERIVED (REPORT v2.4a §7.3-7.7, §2.8); firmware v2.8 (Electronics). Verify at IGN-1, CF-4, T1-T4
T1319	E1 = torch flame confirmed on P_torch AND photodiode (2oo2, FA-1 (a)), 10 ms ...	DERIVED *Lead decision, line 153*	CMB v2.4d	/igniter/sequence/v/start/3/action	LEAD (CHANGELOG 153-177) + DERIVED (REPORT v2.4a §7.3-7.7, §2.8); firmware v2.8 (Electronics). Verify at IGN-1, CF-4, T1-T4
T1320	W	DERIVED *Lead decision, line 153*	CMB v2.4d	/igniter/sequence/v/start/4/epoch	LEAD (CHANGELOG 153-177) + DERIVED (REPORT v2.4a §7.3-7.7, §2.8); firmware v2.8 (Electronics). Verify at IGN-1, CF-4, T1-T4
T1321	0.81 s	DERIVED *Lead decision, line 153*	CMB v2.4d	/igniter/sequence/v/start/4/t_s	LEAD (CHANGELOG 153-177) + DERIVED (REPORT v2.4a §7.3-7.7, §2.8); firmware v2.8 (Electronics). Verify at IGN-1, CF-4, T1-T4
T1322	abort deadline W from the lift command: 0.40 s flight first start, 0.65 s flight ...	DERIVED *Lead decision, line 153*	CMB v2.4d	/igniter/sequence/v/start/4/action	LEAD (CHANGELOG 153-177) + DERIVED (REPORT v2.4a §7.3-7.7, §2.8); firmware v2.8 (Electronics). Verify at IGN-1, CF-4, T1-T4
T1323	E2	DERIVED *Lead decision, line 153*	CMB v2.4d	/igniter/sequence/v/start/5/epoch	LEAD (CHANGELOG 153-177) + DERIVED (REPORT v2.4a §7.3-7.7, §2.8); firmware v2.8 (Electronics). Verify at IGN-1, CF-4, T1-T4
T1324	0.442 s	DERIVED *Lead decision, line 153*	CMB v2.4d	/igniter/sequence/v/start/5/t_s	LEAD (CHANGELOG 153-177) + DERIVED (REPORT v2.4a §7.3-7.7, §2.8); firmware v2.8 (Electronics). Verify at IGN-1, CF-4, T1-T4
T1325	light: Pc >= 0.8 bar above the pre-lift baseline on filtered AND raw Pc for 10 ms, above ...	DERIVED *Lead decision, line 153*	CMB v2.4d	/igniter/sequence/v/start/5/action	LEAD (CHANGELOG 153-177) + DERIVED (REPORT v2.4a §7.3-7.7, §2.8); firmware v2.8 (Electronics). Verify at IGN-1, CF-4, T1-T4
T1326	E3	DERIVED *Lead decision, line 153*	CMB v2.4d	/igniter/sequence/v/start/6/epoch	LEAD (CHANGELOG 153-177) + DERIVED (REPORT v2.4a §7.3-7.7, §2.8); firmware v2.8 (Electronics). Verify at IGN-1, CF-4, T1-T4
T1327	MISSING	MISSING *Lead decision, line 153*	CMB v2.4d	/igniter/sequence/v/start/6/t_s	LEAD (CHANGELOG 153-177) + DERIVED (REPORT v2.4a §7.3-7.7, §2.8); firmware v2.8 (Electronics). Verify at IGN-1, CF-4, T1-T4
T1328	thrust >= 90 % of the command (ENG-04 latency, combustion REPORT §7.7)	DERIVED *Lead decision, line 153*	CMB v2.4d	/igniter/sequence/v/start/6/action	LEAD (CHANGELOG 153-177) + DERIVED (REPORT v2.4a §7.3-7.7, §2.8); firmware v2.8 (Electronics). Verify at IGN-1, CF-4, T1-T4
T1329	1.3 s	DECIDED *Lead decision, per combustion.json v2.4d*	CMB v2.4d	/igniter_sequence_fields_v2p4c/v/fields/0	tag in combustion.json v2.4d: LEAD
T1330	igniter.sequence.v.start[0]; LEAD 186 (1)	DECIDED *Design value, combustion.json v2.4d*	CMB v2.4d	/igniter_sequence_fields_v2p4c/v/fields/0/src	combustion.json v2.4d
T1331	−2.9 s	DECIDED *Design value, combustion.json v2.4d*	CMB v2.4d	/igniter_sequence_fields_v2p4c/v/fields/1	tag in combustion.json v2.4d: DESIGN (analysed)
T1332	relight_purge.v.timing_s; Electronics v2.12 states the flight value	DECIDED *Design value, combustion.json v2.4d*	CMB v2.4d	/igniter_sequence_fields_v2p4c/v/fields/1/src	combustion.json v2.4d
T1333	0.35 s	DECIDED *Design value, combustion.json v2.4d*	CMB v2.4d	/igniter_sequence_fields_v2p4c/v/fields/2	tag in combustion.json v2.4d: DESIGN
T1334	relight_purge.v.timing_s	DECIDED *Design value, combustion.json v2.4d*	CMB v2.4d	/igniter_sequence_fields_v2p4c/v/fields/2/src	combustion.json v2.4d
T1335	2.43 s	DECIDED *Lead decision, per combustion.json v2.4d*	CMB v2.4d	/igniter_sequence_fields_v2p4c/v/fields/3	tag in combustion.json v2.4d: LEAD
T1336	purge_decision_v2p4c; LEAD 253 (7)(ii), 256 (2)	DECIDED *Design value, combustion.json v2.4d*	CMB v2.4d	/igniter_sequence_fields_v2p4c/v/fields/3/src	combustion.json v2.4d
T1337	−0.05 s	DECIDED *Lead decision, per combustion.json v2.4d*	CMB v2.4d	/igniter_sequence_fields_v2p4c/v/fields/4	tag in combustion.json v2.4d: LEAD
T1338	LEAD line 164	DECIDED *Design value, combustion.json v2.4d*	CMB v2.4d	/igniter_sequence_fields_v2p4c/v/fields/4/src	combustion.json v2.4d
T1339	2.39 s	DECIDED *Lead decision, per combustion.json v2.4d*	CMB v2.4d	/igniter_sequence_fields_v2p4c/v/fields/5	tag in combustion.json v2.4d: LEAD
T1340	purge_decision_v2p4c; LEAD 256 (2)	DECIDED *Design value, combustion.json v2.4d*	CMB v2.4d	/igniter_sequence_fields_v2p4c/v/fields/5/src	combustion.json v2.4d
T1341	2.2 s	DERIVED	CMB v2.4d	/igniter_sequence_fields_v2p4c/v/fields/6	DERIVED
T1342	purge_decision_v2p4c; REV-COMB-4 R4-07	DECIDED *Design value, combustion.json v2.4d*	CMB v2.4d	/igniter_sequence_fields_v2p4c/v/fields/6/src	combustion.json v2.4d
T1343	1.9 s	SIM	CMB v2.4d	/igniter_sequence_fields_v2p4c/v/fields/7	SIM
T1344	relight_purge.v.timing_s.needed_flow_unlined	DECIDED *Design value, combustion.json v2.4d*	CMB v2.4d	/igniter_sequence_fields_v2p4c/v/fields/7/src	combustion.json v2.4d
T1345	0.03–0.08 s	ASSUMED	CMB v2.4d	/igniter_sequence_fields_v2p4c/v/fields/8	ASSUMED
T1346	purge_basis_line216_v2p4b.v.cap_rule	DECIDED *Design value, combustion.json v2.4d*	CMB v2.4d	/igniter_sequence_fields_v2p4c/v/fields/8/src	combustion.json v2.4d
T1347	0.16 s	DECIDED *Design value, combustion.json v2.4d*	CMB v2.4d	/igniter_sequence_fields_v2p4c/v/fields/9	tag in combustion.json v2.4d: DESIGN
T1348	igniter.sequence.v.start[3]; Electronics v4 kTorchGate_s (= the torch gate)	DECIDED *Design value, combustion.json v2.4d*	CMB v2.4d	/igniter_sequence_fields_v2p4c/v/fields/9/src	combustion.json v2.4d
T1349	50 Hz	DECIDED *Design value, combustion.json v2.4d*	CMB v2.4d	/igniter_sequence_fields_v2p4c/v/fields/10	tag in combustion.json v2.4d: DESIGN
T1350	igniter.sequence.v.start[1]	DECIDED *Design value, combustion.json v2.4d*	CMB v2.4d	/igniter_sequence_fields_v2p4c/v/fields/10/src	combustion.json v2.4d

ID	Value	Class	Source	Key	Source tag / formula
T1351	0.05 J	DECIDED *Design value, combustion.json v2.4d*	CMB v2.4d	/igniter_sequence_fields_v2p4c/v/fields/11	tag in combustion.json v2.4d: DESIGN
T1352	igniter.sequence.v.start[1] (50 mJ)	DECIDED *Design value, combustion.json v2.4d*	CMB v2.4d	/igniter_sequence_fields_v2p4c/v/fields/11/src	combustion.json v2.4d
T1353	0.02 s	DECIDED *Design value, combustion.json v2.4d*	CMB v2.4d	/igniter_sequence_fields_v2p4c/v/fields/12	tag in combustion.json v2.4d: DESIGN
T1354	igniter.sequence.v.start[1]	DECIDED *Design value, combustion.json v2.4d*	CMB v2.4d	/igniter_sequence_fields_v2p4c/v/fields/12/src	combustion.json v2.4d
T1355	0.5 s	DECIDED *Design value, combustion.json v2.4d*	CMB v2.4d	/igniter_sequence_fields_v2p4c/v/fields/13	tag in combustion.json v2.4d: DESIGN
T1356	firmware_constant_sources_v2p4b kSparkTrain_s	DECIDED *Design value, combustion.json v2.4d*	CMB v2.4d	/igniter_sequence_fields_v2p4c/v/fields/13/src	combustion.json v2.4d
T1357	0.16 s	DECIDED *Design value, combustion.json v2.4d*	CMB v2.4d	/igniter_sequence_fields_v2p4c/v/fields/14	tag in combustion.json v2.4d: DESIGN
T1358	igniter.sequence.v.start[2]	DECIDED *Design value, combustion.json v2.4d*	CMB v2.4d	/igniter_sequence_fields_v2p4c/v/fields/14/src	combustion.json v2.4d
T1359	0.09 s	SIM	CMB v2.4d	/igniter_sequence_fields_v2p4c/v/fields/15	SIM
T1360	igniter.sequence.v.start[3] (torch_gate_v2p4)	DECIDED *Design value, combustion.json v2.4d*	CMB v2.4d	/igniter_sequence_fields_v2p4c/v/fields/15/src	combustion.json v2.4d
T1361	0.01 s	DECIDED *Design value, combustion.json v2.4d*	CMB v2.4d	/igniter_sequence_fields_v2p4c/v/fields/16	tag in combustion.json v2.4d: DESIGN
T1362	igniter.sequence.v.start[3]	DECIDED *Design value, combustion.json v2.4d*	CMB v2.4d	/igniter_sequence_fields_v2p4c/v/fields/16/src	combustion.json v2.4d
T1363	0.005 s	DECIDED *Design value, combustion.json v2.4d*	CMB v2.4d	/igniter_sequence_fields_v2p4c/v/fields/17	tag in combustion.json v2.4d: DESIGN
T1364	igniter.sequence.v.start[3] (FA-1 (b), A16)	DECIDED *Design value, combustion.json v2.4d*	CMB v2.4d	/igniter_sequence_fields_v2p4c/v/fields/17/src	combustion.json v2.4d
T1365	0.2039 mm	DECIDED *Design value, combustion.json v2.4d*	CMB v2.4d	/igniter_sequence_fields_v2p4c/v/fields/18	tag in combustion.json v2.4d: DESIGN
T1366	igniter.sequence.v.start[3]; start_level_lift_v2p4b	DECIDED *Design value, combustion.json v2.4d*	CMB v2.4d	/igniter_sequence_fields_v2p4c/v/fields/18/src	combustion.json v2.4d
T1367	0.21776 mm	DECIDED *Design value, combustion.json v2.4d*	CMB v2.4d	/igniter_sequence_fields_v2p4c/v/fields/19	tag in combustion.json v2.4d: DESIGN
T1368	start_level_lift_v2p4b.v.line216_9 (ECU row)	DECIDED *Design value, combustion.json v2.4d*	CMB v2.4d	/igniter_sequence_fields_v2p4c/v/fields/19/src	combustion.json v2.4d
T1369	0.0287–0.0295 s	DERIVED	CMB v2.4d	/igniter_sequence_fields_v2p4c/v/fields/20	DERIVED
T1370	igniter.sequence.v.start[3]	DECIDED *Design value, combustion.json v2.4d*	CMB v2.4d	/igniter_sequence_fields_v2p4c/v/fields/20/src	combustion.json v2.4d
T1371	37.1 g/s	DECIDED *Lead decision, per combustion.json v2.4d*	CMB v2.4d	/igniter_sequence_fields_v2p4c/v/fields/21	tag in combustion.json v2.4d: LEAD (cap)
T1372	igniter.sequence.v.start[3]; CF-3	DECIDED *Design value, combustion.json v2.4d*	CMB v2.4d	/igniter_sequence_fields_v2p4c/v/fields/21/src	combustion.json v2.4d
T1373	37.4 g/s	DECIDED *Lead decision, per combustion.json v2.4d*	CMB v2.4d	/igniter_sequence_fields_v2p4c/v/fields/22	tag in combustion.json v2.4d: LEAD (cap)
T1374	igniter.sequence.v.start[3]; CF-3	DECIDED *Design value, combustion.json v2.4d*	CMB v2.4d	/igniter_sequence_fields_v2p4c/v/fields/22/src	combustion.json v2.4d
T1375	0.4 s	DECIDED *Lead decision, per combustion.json v2.4d*	CMB v2.4d	/igniter_sequence_fields_v2p4c/v/fields/23	tag in combustion.json v2.4d: LEAD
T1376	igniter.sequence.v.light_windows_W_s	DECIDED *Design value, combustion.json v2.4d*	CMB v2.4d	/igniter_sequence_fields_v2p4c/v/fields/23/src	combustion.json v2.4d
T1377	0.65 s	DECIDED *Lead decision, per combustion.json v2.4d*	CMB v2.4d	/igniter_sequence_fields_v2p4c/v/fields/24	tag in combustion.json v2.4d: LEAD
T1378	igniter.sequence.v.light_windows_W_s	DECIDED *Design value, combustion.json v2.4d*	CMB v2.4d	/igniter_sequence_fields_v2p4c/v/fields/24/src	combustion.json v2.4d
T1379	0.35 s	DECIDED *Lead decision, per combustion.json v2.4d*	CMB v2.4d	/igniter_sequence_fields_v2p4c/v/fields/25	tag in combustion.json v2.4d: LEAD
T1380	igniter.sequence.v.light_windows_W_s	DECIDED *Design value, combustion.json v2.4d*	CMB v2.4d	/igniter_sequence_fields_v2p4c/v/fields/25/src	combustion.json v2.4d
T1381	0.81 s	DECIDED *Lead decision, per combustion.json v2.4d*	CMB v2.4d	/igniter_sequence_fields_v2p4c/v/fields/26	tag in combustion.json v2.4d: LEAD
T1382	igniter.sequence.v.start[4] (torch gate 0.16 + W 0.65)	DECIDED *Design value, combustion.json v2.4d*	CMB v2.4d	/igniter_sequence_fields_v2p4c/v/fields/26/src	combustion.json v2.4d
T1383	0.8 bar	DECIDED *Design value, combustion.json v2.4d*	CMB v2.4d	/igniter_sequence_fields_v2p4c/v/fields/27	tag in combustion.json v2.4d: DESIGN
T1384	igniter.sequence.v.start[5] (filtered AND raw Pc)	DECIDED *Design value, combustion.json v2.4d*	CMB v2.4d	/igniter_sequence_fields_v2p4c/v/fields/27/src	combustion.json v2.4d
T1385	0.01 s	DECIDED *Design value, combustion.json v2.4d*	CMB v2.4d	/igniter_sequence_fields_v2p4c/v/fields/28	tag in combustion.json v2.4d: DESIGN
T1386	igniter.sequence.v.start[5]	DECIDED *Design value, combustion.json v2.4d*	CMB v2.4d	/igniter_sequence_fields_v2p4c/v/fields/28/src	combustion.json v2.4d
T1387	1.8 bar abs	DECIDED *Design value, combustion.json v2.4d*	CMB v2.4d	/igniter_sequence_fields_v2p4c/v/fields/29	tag in combustion.json v2.4d: DESIGN
T1388	igniter.sequence.v.start[5]	DECIDED *Design value, combustion.json v2.4d*	CMB v2.4d	/igniter_sequence_fields_v2p4c/v/fields/29/src	combustion.json v2.4d
T1389	0.02 s	DECIDED *Design value, combustion.json v2.4d*	CMB v2.4d	/igniter_sequence_fields_v2p4c/v/fields/30	tag in combustion.json v2.4d: DESIGN
T1390	igniter.sequence.v.start[5]	DECIDED *Design value, combustion.json v2.4d*	CMB v2.4d	/igniter_sequence_fields_v2p4c/v/fields/30/src	combustion.json v2.4d
T1391	0.0147 s	DERIVED	CMB v2.4d	/igniter_sequence_fields_v2p4c/v/fields/31	DERIVED
T1392	E2_rule_v2p4b.v.shift_bound_s	DECIDED *Design value, combustion.json v2.4d*	CMB v2.4d	/igniter_sequence_fields_v2p4c/v/fields/31/src	combustion.json v2.4d
T1393	0.442, 0.589, 0.626 s	SIM	CMB v2.4d	/igniter_sequence_fields_v2p4c/v/fields/32	SIM
T1394	igniter.sequence.v.start[5] (rg01_line_states_v2p4)	DECIDED *Design value, combustion.json v2.4d*	CMB v2.4d	/igniter_sequence_fields_v2p4c/v/fields/32/src	combustion.json v2.4d
T1395	0.9 -	DECIDED *Requirement, combustion.json v2.4d*	CMB v2.4d	/igniter_sequence_fields_v2p4c/v/fields/33	tag in combustion.json v2.4d: REQ
T1396	igniter.sequence.v.start[6] (ENG-04)	DECIDED *Design value, combustion.json v2.4d*	CMB v2.4d	/igniter_sequence_fields_v2p4c/v/fields/33/src	combustion.json v2.4d
T1397	0.4 s	DECIDED *Design value, combustion.json v2.4d*	CMB v2.4d	/igniter_sequence_fields_v2p4c/v/fields/34	tag in combustion.json v2.4d: DESIGN
T1398	firmware_constant_sources_v2p4b kPropaneOffPcAbs_Pa	DECIDED *Design value, combustion.json v2.4d*	CMB v2.4d	/igniter_sequence_fields_v2p4c/v/fields/34/src	combustion.json v2.4d
T1399	5 bar abs	DERIVED	CMB v2.4d	/igniter_sequence_fields_v2p4c/v/fields/35	DERIVED
T1400	firmware_constant_sources_v2p4b kPropaneOffPcAbs_Pa	DECIDED *Design value, combustion.json v2.4d*	CMB v2.4d	/igniter_sequence_fields_v2p4c/v/fields/35/src	combustion.json v2.4d
T1401	2500 N/s	DECIDED *Design value, combustion.json v2.4d*	CMB v2.4d	/igniter_sequence_fields_v2p4c/v/fields/36	tag in combustion.json v2.4d: DESIGN
T1402	firmware_constant_sources_v2p4b kStartRamp_Nps	DECIDED *Design value, combustion.json v2.4d*	CMB v2.4d	/igniter_sequence_fields_v2p4c/v/fields/36/src	combustion.json v2.4d
T1403	0.12 s	DECIDED *Design value, combustion.json v2.4d*	CMB v2.4d	/igniter_sequence_fields_v2p4c/v/fields/37	tag in combustion.json v2.4d: DESIGN (legacy)
T1404	firmware_constant_sources_v2p4b kPintleToIgnitionLift_s	DECIDED *Design value, combustion.json v2.4d*	CMB v2.4d	/igniter_sequence_fields_v2p4c/v/fields/37/src	combustion.json v2.4d
T1405	0.75 -	DECIDED *Lead decision, per combustion.json v2.4d*	CMB v2.4d	/igniter_sequence_fields_v2p4c/v/fields/38	tag in combustion.json v2.4d: LEAD
T1406	igniter.sequence.v.shutdown[0] (ENG-05)	DECIDED *Design value, combustion.json v2.4d*	CMB v2.4d	/igniter_sequence_fields_v2p4c/v/fields/38/src	combustion.json v2.4d
T1407	2 bar abs	DECIDED *Design value, combustion.json v2.4d*	CMB v2.4d	/igniter_sequence_fields_v2p4c/v/fields/39	tag in combustion.json v2.4d: DESIGN

ID	Value	Class	Source	Key	Source tag / formula
T1408	igniter.sequence.v.shutdown[2]	DECIDED *Design value, combustion.json v2.4d*	CMB v2.4d	/igniter_sequence_fields_v2p4c/v/fields/39/src	combustion.json v2.4d
T1409	0.3 s	DECIDED *Design value, combustion.json v2.4d*	CMB v2.4d	/igniter_sequence_fields_v2p4c/v/fields/40	tag in combustion.json v2.4d: DESIGN
T1410	igniter.sequence.v.shutdown[2]	DECIDED *Design value, combustion.json v2.4d*	CMB v2.4d	/igniter_sequence_fields_v2p4c/v/fields/40/src	combustion.json v2.4d
T1411	30 s	DECIDED *Design value, combustion.json v2.4d*	CMB v2.4d	/igniter_sequence_fields_v2p4c/v/fields/41	tag in combustion.json v2.4d: DESIGN
T1412	igniter.sequence.v.shutdown[3]	DECIDED *Design value, combustion.json v2.4d*	CMB v2.4d	/igniter_sequence_fields_v2p4c/v/fields/41/src	combustion.json v2.4d
T1413	15–24 C	DECIDED *Requirement, combustion.json v2.4d*	CMB v2.4d	/igniter_sequence_fields_v2p4c/v/fields/42	tag in combustion.json v2.4d: REQ
T1414	igniter.sequence.v.arm_preconditions[0] (ENG-12; interlock_guard_band)	DECIDED *Design value, combustion.json v2.4d*	CMB v2.4d	/igniter_sequence_fields_v2p4c/v/fields/42/src	combustion.json v2.4d
T1415	1.05 bar abs	DECIDED *Design value, combustion.json v2.4d*	CMB v2.4d	/igniter_sequence_fields_v2p4c/v/fields/43	tag in combustion.json v2.4d: DESIGN
T1416	igniter.sequence.v.arm_preconditions[2]	DECIDED *Design value, combustion.json v2.4d*	CMB v2.4d	/igniter_sequence_fields_v2p4c/v/fields/43/src	combustion.json v2.4d
T1417	403.15 K	DECIDED *Lead decision, per combustion.json v2.4d*	CMB v2.4d	/igniter_sequence_fields_v2p4c/v/fields/44	tag in combustion.json v2.4d: LEAD
T1418	igniter.sequence.v.arm_preconditions[5]; LEAD 249 (4) kCaseRelightMax_K	DECIDED *Design value, combustion.json v2.4d*	CMB v2.4d	/igniter_sequence_fields_v2p4c/v/fields/44/src	combustion.json v2.4d
T1419	37.16 N	DECIDED *Lead decision, per combustion.json v2.4d*	CMB v2.4d	/igniter_sequence_fields_v2p4c/v/fields/45	tag in combustion.json v2.4d: LEAD
T1420	drogue_release_hold.v.line175_5_note_v2p4a (LEAD 175 (5))	DECIDED *Design value, combustion.json v2.4d*	CMB v2.4d	/igniter_sequence_fields_v2p4c/v/fields/45/src	combustion.json v2.4d
T1421	3 -	DECIDED *Lead decision, per combustion.json v2.4d*	CMB v2.4d	/igniter_sequence_fields_v2p4c/v/fields/46	tag in combustion.json v2.4d: LEAD
T1422	drogue_release_hold.v.line175_5_note_v2p4a	DECIDED *Design value, combustion.json v2.4d*	CMB v2.4d	/igniter_sequence_fields_v2p4c/v/fields/46/src	combustion.json v2.4d
T1423	100 Hz	DECIDED *Lead decision, per combustion.json v2.4d*	CMB v2.4d	/igniter_sequence_fields_v2p4c/v/fields/47	tag in combustion.json v2.4d: LEAD
T1424	drogue_release_hold.v.line175_5_note_v2p4a	DECIDED *Design value, combustion.json v2.4d*	CMB v2.4d	/igniter_sequence_fields_v2p4c/v/fields/47/src	combustion.json v2.4d
T1425	0.8 s	DECIDED *Lead decision, per combustion.json v2.4d*	CMB v2.4d	/igniter_sequence_fields_v2p4c/v/fields/48	tag in combustion.json v2.4d: LEAD
T1426	drogue_release_hold.v.line175_5_note_v2p4a; LEAD 227 (5)	DECIDED *Design value, combustion.json v2.4d*	CMB v2.4d	/igniter_sequence_fields_v2p4c/v/fields/48/src	combustion.json v2.4d
T1427	60 s	DECIDED *Lead decision, per combustion.json v2.4d*	CMB v2.4d	/igniter_sequence_fields_v2p4c/v/fields/49	tag in combustion.json v2.4d: LEAD
T1428	igniter.sequence.v.abort_if_no_ignition[1]	DECIDED *Design value, combustion.json v2.4d*	CMB v2.4d	/igniter_sequence_fields_v2p4c/v/fields/49/src	combustion.json v2.4d
T1429	10 g/s	DECIDED *Lead decision, per combustion.json v2.4d*	CMB v2.4d	/igniter_sequence_fields_v2p4c/v/fields/50	tag in combustion.json v2.4d: LEAD
T1430	igniter.sequence.v.abort_if_no_ignition[1]	DECIDED *Design value, combustion.json v2.4d*	CMB v2.4d	/igniter_sequence_fields_v2p4c/v/fields/50/src	combustion.json v2.4d
T1431	15 min	DECIDED *Lead decision, per combustion.json v2.4d*	CMB v2.4d	/igniter_sequence_fields_v2p4c/v/fields/51	tag in combustion.json v2.4d: LEAD
T1432	igniter.sequence.v.abort_if_no_ignition[1]	DECIDED *Design value, combustion.json v2.4d*	CMB v2.4d	/igniter_sequence_fields_v2p4c/v/fields/51/src	combustion.json v2.4d
T1433	tank 15-24 C at ARM (ENG-12 as amended); T_tank <= 24 C at ARM and at ignition ...	DERIVED *Lead decision, line 153*	CMB v2.4d	/igniter/sequence/v/arm_preconditions/0	LEAD (CHANGELOG 153-177) + DERIVED (REPORT v2.4a §7.3-7.7, §2.8); firmware v2.8 (Electronics). Verify at IGN-1, CF-4, T1-T4
T1434	pintle seated, encoder healthy; MV-P closed; bleed valves V1 and V2 closed, drivers ...	DERIVED *Lead decision, line 153*	CMB v2.4d	/igniter/sequence/v/arm_preconditions/1	LEAD (CHANGELOG 153-177) + DERIVED (REPORT v2.4a §7.3-7.7, §2.8); firmware v2.8 (Electronics). Verify at IGN-1, CF-4, T1-T4
T1435	P_torch auto-zero; ARM refused if P_torch > 1.05 bar abs	DERIVED *Lead decision, line 153*	CMB v2.4d	/igniter/sequence/v/arm_preconditions/2	LEAD (CHANGELOG 153-177) + DERIVED (REPORT v2.4a §7.3-7.7, §2.8); firmware v2.8 (Electronics). Verify at IGN-1, CF-4, T1-T4
T1436	Pc transducer: flight unit RESELECTED by Electronics (LEAD 205): miniature seat-mounted ...	DERIVED *Lead decision, line 153*	CMB v2.4d	/igniter/sequence/v/arm_preconditions/3	LEAD (CHANGELOG 153-177) + DERIVED (REPORT v2.4a §7.3-7.7, §2.8); firmware v2.8 (Electronics). Verify at IGN-1, CF-4, T1-T4
T1437	igniter continuity (spark coil, SOL1, SOL2); photodiode healthy (needed for the 2oo2 E1)	DERIVED *Lead decision, line 153*	CMB v2.4d	/igniter/sequence/v/arm_preconditions/4	LEAD (CHANGELOG 153-177) + DERIVED (REPORT v2.4a §7.3-7.7, §2.8); firmware v2.8 (Electronics). Verify at IGN-1, CF-4, T1-T4
T1438	relight only: chamber vented; RELIGHT refused if a case thermocouple reads > 403.15 K ...	DERIVED *Lead decision, line 153*	CMB v2.4d	/igniter/sequence/v/arm_preconditions/5	LEAD (CHANGELOG 153-177) + DERIVED (REPORT v2.4a §7.3-7.7, §2.8); firmware v2.8 (Electronics). Verify at IGN-1, CF-4, T1-T4
T1439	MV closed during coasts (LEAD); the v2.3 text "the isolation valve may stay open between ...	DERIVED *Lead decision, line 153*	CMB v2.4d	/igniter/sequence/v/relight/0	LEAD (CHANGELOG 153-177) + DERIVED (REPORT v2.4a §7.3-7.7, §2.8); firmware v2.8 (Electronics). Verify at IGN-1, CF-4, T1-T4
T1440	A1 relight: PURGE at E0 - 2.90 s (2.5 s of flow as analysed in the v2.4b records; the ...	DERIVED *Lead decision, line 153*	CMB v2.4d	/igniter/sequence/v/relight/1	LEAD (CHANGELOG 153-177) + DERIVED (REPORT v2.4a §7.3-7.7, §2.8); firmware v2.8 (Electronics). Verify at IGN-1, CF-4, T1-T4
T1441	settling: liquid at the tank outlet from E0 - 2.90 s - t_settle (combustion REPORT §2.4 ...	DERIVED *Lead decision, line 153*	CMB v2.4d	/igniter/sequence/v/relight/2	LEAD (CHANGELOG 153-177) + DERIVED (REPORT v2.4a §7.3-7.7, §2.8); firmware v2.8 (Electronics). Verify at IGN-1, CF-4, T1-T4
T1442	light window W 0.65 s; E0 → E2 max 0.626 s on the worst purged line state ...	DERIVED *Lead decision, line 153*	CMB v2.4d	/igniter/sequence/v/relight/3	LEAD (CHANGELOG 153-177) + DERIVED (REPORT v2.4a §7.3-7.7, §2.8); firmware v2.8 (Electronics). Verify at IGN-1, CF-4, T1-T4
T1443	pre-MECO throttle-down to <= 75 % (ENG-05)	DERIVED *Lead decision, line 153*	CMB v2.4d	/igniter/sequence/v/shutdown/0	LEAD (CHANGELOG 153-177) + DERIVED (REPORT v2.4a §7.3-7.7, §2.8); firmware v2.8 (Electronics). Verify at IGN-1, CF-4, T1-T4
T1444	pintle closes at full drive speed	DERIVED *Lead decision, line 153*	CMB v2.4d	/igniter/sequence/v/shutdown/1	LEAD (CHANGELOG 153-177) + DERIVED (REPORT v2.4a §7.3-7.7, §2.8); firmware v2.8 (Electronics). Verify at IGN-1, CF-4, T1-T4
T1445	MV de-energised on seat confirmation (encoder at seat and Pc < 2 bar abs) or 0.30 s ...	DERIVED *Lead decision, line 153*	CMB v2.4d	/igniter/sequence/v/shutdown/2	LEAD (CHANGELOG 153-177) + DERIVED (REPORT v2.4a §7.3-7.7, §2.8); firmware v2.8 (Electronics). Verify at IGN-1, CF-4, T1-T4
T1446	trapped liquid relieves to the tank through S1 (combustion REPORT §2.7); ground: 30 s ...	DERIVED *Lead decision, line 153*	CMB v2.4d	/igniter/sequence/v/shutdown/3	LEAD (CHANGELOG 153-177) + DERIVED (REPORT v2.4a §7.3-7.7, §2.8); firmware v2.8 (Electronics). Verify at IGN-1, CF-4, T1-T4
T1447	pintle closed; torch solenoids closed (SOL1 not after SOL2); MV closed (abort is MV-first)	DERIVED *Lead decision, line 153*	CMB v2.4d	/igniter/sequence/v/abort_if_no_ignition/0	LEAD (CHANGELOG 153-177) + DERIVED (REPORT v2.4a §7.3-7.7, §2.8); firmware v2.8 (Electronics). Verify at IGN-1, CF-4, T1-T4
T1448	no retry into a pooled chamber; stand: N2 purge >= 60 s at >= 10 g/s, then a hold of >= ...	DERIVED *Lead decision, line 153*	CMB v2.4d	/igniter/sequence/v/abort_if_no_ignition/1	LEAD (CHANGELOG 153-177) + DERIVED (REPORT v2.4a §7.3-7.7, §2.8); firmware v2.8 (Electronics). Verify at IGN-1, CF-4, T1-T4
T1449	flight: no retry; vehicle-level response per GNC	DERIVED *Lead decision, line 153*	CMB v2.4d	/igniter/sequence/v/abort_if_no_ignition/2	LEAD (CHANGELOG 153-177) + DERIVED (REPORT v2.4a §7.3-7.7, §2.8); firmware v2.8 (Electronics). Verify at IGN-1, CF-4, T1-T4
T1450	A5 (Pc > 8 bar for 5 ms) is DELETED (LEAD line 160): it tripped on 98 % of the ...	DERIVED *Lead decision, line 153*	CMB v2.4d	/igniter/sequence/v/abort_if_no_ignition/3	LEAD (CHANGELOG 153-177) + DERIVED (REPORT v2.4a §7.3-7.7, §2.8); firmware v2.8 (Electronics). Verify at IGN-1, CF-4, T1-T4
T1451	no retry into a pooled chamber; N2 purge of the chamber >= 60 s at >= 10 g/s; hold >= 15 ...	DECIDED *Lead decision, line 172 (1)*	CMB v2.4d	/stand_safing_v2p4a	tag in combustion.json v2.4d: LEAD line 172 (1); supersedes "wait >= 5 s" everywhere (H9, §10.2, igniter.sequence.abort_if_no_ignition)
T1452	PURGE commits iff the lead to the plain relight trigger is >= L* = 1.3 s; below L* no ...	SIM / DERIVED *Lead decision, per combustion.json v2.4d*	CMB v2.4d	/purge_commit_rule_v2p4b/v/rule	LEAD decisions; Combustion SIM/DERIVED basis
T1453	2.43 s scheduled purge flow (both-open overlap 2.38 s + 0.05 s) with the 2.39 s ...	DERIVED / ASSUMED / ESTIMATE *Lead decision, per combustion.json v2.4d; basis SIM*	CMB v2.4d	/purge_decision_v2p4c/v/decision	LEAD (decision and conditions); DERIVED by REV-COMB-4 on the author's SIM model (Cd 0.70 ASSUMED); ESTIMATE (Cd 0.80 mass); A1 reserve test DERIVED from SIM by REV-GNC-7 (ESTIMATE above 80 g)
T1454	2.43 s	DERIVED *Lead decision, per combustion.json v2.4d; basis SIM*	CMB v2.4d	/purge_decision_v2p4c/v/flow_s	LEAD (decision and conditions); DERIVED by REV-COMB-4 on the author's SIM model (Cd 0.70 ASSUMED); ESTIMATE (Cd 0.80 mass); A1 reserve test DERIVED from SIM by REV-GNC-7 (ESTIMATE above 80 g)
T1455	2.38 s	DERIVED *Lead decision, per combustion.json v2.4d; basis SIM*	CMB v2.4d	/purge_decision_v2p4c/v/both_open_overlap_s	LEAD (decision and conditions); DERIVED by REV-COMB-4 on the author's SIM model (Cd 0.70 ASSUMED); ESTIMATE (Cd 0.80 mass); A1 reserve test DERIVED from SIM by REV-GNC-7 (ESTIMATE above 80 g)
T1456	2.39 s	DERIVED *Lead decision, per combustion.json v2.4d; basis SIM*	CMB v2.4d	/purge_decision_v2p4c/v/cap_both_open_s	LEAD (decision and conditions); DERIVED by REV-COMB-4 on the author's SIM model (Cd 0.70 ASSUMED); ESTIMATE (Cd 0.80 mass); A1 reserve test DERIVED from SIM by REV-GNC-7 (ESTIMATE above 80 g)
T1457	2.44 s	DERIVED *Lead decision, per combustion.json v2.4d; basis SIM*	CMB v2.4d	/purge_decision_v2p4c/v/flow_at_cap_s	LEAD (decision and conditions); DERIVED by REV-COMB-4 on the author's SIM model (Cd 0.70 ASSUMED); ESTIMATE (Cd 0.80 mass); A1 reserve test DERIVED from SIM by REV-GNC-7 (ESTIMATE above 80 g)
T1458	283.15 K: 2.2; 289.5 K: 1.9; 297.15 K: 1.8	DERIVED	CMB v2.4d	/purge_decision_v2p4c/v/needed_flow_unlined_s	(class set at the call site: DERIVED)

ID	Value	Class	Source	Key	Source tag / formula
T1459	283.15 K: 0.23; 289.5 K: 0.53; 297.15 K: 0.63	DERIVED	CMB v2.4d	/purge_decision_v2p4c/v/margins_at_2p43s_unlined_s	(class set at the call site: DERIVED)
T1460	283.15 K: 66.4; 289.5 K: 70.7; 297.15 K: 75.4; at_cap_2p44s_of_flow: 75.75	DERIVED	CMB v2.4d	/purge_decision_v2p4c/v/vented_N2O_bound_law_g	(class set at the call site: DERIVED)
T1461	289.5 K: 80.5; 297.15 K: 85.5; 297.15 K, 2.44 s of flow at the cap: 85.8	ESTIMATE	CMB v2.4d	/purge_decision_v2p4c/v/vented_N2O_Cd0p80_fixed_schedule_g	(class set at the call site: ESTIMATE)
T1462	ESTIMATE: 2.43 s x 33.13 g/s at 289.5 K (average rate); at 297.15 K the Cd 0.80 / 0.70 ...	ESTIMATE	CMB v2.4d	/purge_decision_v2p4c/v/vented_N2O_Cd0p80_tag	(class set at the call site: ESTIMATE)
T1463	test: CHANGELOG line 208 against the v3d reference (A1 210 m, v351, N = 500 per base ...	DERIVED	CMB v2.4d	/purge_decision_v2p4c/v/a1_reserve_test_REV_GNC_7	(class set at the call site: DERIVED)
T1464	wherever 1.9 s is quoted as the needed flow, 2.2 s at the 283.15 K cold corner is stated ...	DECIDED *Lead decision, per combustion.json v2.4d*	CMB v2.4d	/purge_decision_v2p4c/v/conditions_LEAD_256_2/a	tag in combustion.json v2.4d:
T1465	CF-4 / B-BLEED-1 shows a measured Cd x A of the two-tap path >= 0.91 x the Cd 0.70 value ...	DECIDED *Lead decision, per combustion.json v2.4d*	CMB v2.4d	/purge_decision_v2p4c/v/conditions_LEAD_256_2/b	tag in combustion.json v2.4d:
T1466	the line stays unlined (lined, 283.15 K needs 2.5 s)	DECIDED *Lead decision, per combustion.json v2.4d*	CMB v2.4d	/purge_decision_v2p4c/v/conditions_LEAD_256_2/c	tag in combustion.json v2.4d:
T1467	after CF-4 / B-BLEED-1, kPurgeFlow_s and the both-open cap are re-set from the measured ...	DECIDED *Lead decision, per combustion.json v2.4d*	CMB v2.4d	/purge_decision_v2p4c/v/condition_d_LEAD_257_2	tag in combustion.json v2.4d:
T1468	the (d) re-set takes the upper 95 % bound of the measured N2O Cd x A of the two-tap ...	DECIDED *Lead decision, per combustion.json v2.4d*	CMB v2.4d	/purge_decision_v2p4c/v/conditions_LEAD_258_2/d1	tag in combustion.json v2.4d:
T1469	CF-4 shows the 283.15 K need at the re-set flow (need and limit both scale with 1 / (Cd ...	DECIDED *Lead decision, per combustion.json v2.4d*	CMB v2.4d	/purge_decision_v2p4c/v/conditions_LEAD_258_2/d2	tag in combustion.json v2.4d:
T1470	open: Electronics v2.12 states E0 - 2.90 s (the flow ends at E0 - 0.12 s) or E0 - 2.83 s ...	DERIVED	CMB v2.4d	/purge_decision_v2p4c/v/PURGE_command_time	(class set at the call site: DERIVED)
T1471	CF-4 v2.3 (v2.4d) runs the decided flow, adds a run at or below 283.15 K or states the ...	DECIDED *Lead decision, per combustion.json v2.4d*	CMB v2.4d	/purge_decision_v2p4c/v/CF4	tag in combustion.json v2.4d:
T1472	kPurgeFlow_s 2.43 s; command span 2.38 s inside the 2.39 s cap; flow at the cap 2.44 s ...	DECIDED *Design value, combustion.json v2.4d*	CMB v2.4d	/purge_decision_v2p4c/v/electronics	combustion.json v2.4d
T1473	vented N2O is judged against the 76 g basis (G73 decided, G80 sensitivity) and the A1 ...	DECIDED *Lead decision, per combustion.json v2.4d*	CMB v2.4d	/purge_decision_v2p4c/v/reserve	tag in combustion.json v2.4d:
T1474	no E2 within W of the lift command (0.40 / 0.65 / 0.35 s)	DECIDED *Lead decision, line 160*	CMB v2.4d	/abort_matrix_v2p4/v/A2	tag in combustion.json v2.4d: LEAD lines 160, 164, 166, 168, 175; REPORT v2.4a §7.4
T1475	Pc > 30 bar abs on the FILTERED channel for 5 ms (LEAD line 166 (4); amends line 160 ...	DECIDED *Lead decision, line 160*	CMB v2.4d	/abort_matrix_v2p4/v/A4	tag in combustion.json v2.4d: LEAD lines 160, 164, 166, 168, 175; REPORT v2.4a §7.4
T1476	DELETED → LIGHT_SPIKE log	DECIDED *Lead decision, line 160*	CMB v2.4d	/abort_matrix_v2p4/v/A5	tag in combustion.json v2.4d: LEAD lines 160, 164, 166, 168, 175; REPORT v2.4a §7.4
T1477	Pc loss or implausibility, bands and the flight START high-side rule (Electronics sets ...	DECIDED *Lead decision, line 160*	CMB v2.4d	/abort_matrix_v2p4/v/A7	tag in combustion.json v2.4d: LEAD lines 160, 164, 166, 168, 175; REPORT v2.4a §7.4
T1478	bleed valve fault, or closure not confirmed by the closure deadline (closure deadline = ...	DECIDED *Lead decision, line 160*	CMB v2.4d	/abort_matrix_v2p4/v/A14	tag in combustion.json v2.4d: LEAD lines 160, 164, 166, 168, 175; REPORT v2.4a §7.4
T1479	pintle or chamber anomaly during PURGE: inhibit torch, close MV	DECIDED *Lead decision, line 160*	CMB v2.4d	/abort_matrix_v2p4/v/A15	tag in combustion.json v2.4d: LEAD lines 160, 164, 166, 168, 175; REPORT v2.4a §7.4
T1480	torch loss on BOTH sensors >= 5 ms between E1 and E2: close pintle and MV	DECIDED *Lead decision, line 160*	CMB v2.4d	/abort_matrix_v2p4/v/A16	tag in combustion.json v2.4d: LEAD lines 160, 164, 166, 168, 175; REPORT v2.4a §7.4
T1481	A3 trips on the post-light vapour gap of an unpurged start (ratio <= 0.24 against 0.40 ...	DECIDED *Lead decision, line 160*	CMB v2.4d	/abort_matrix_v2p4/v/A3_unpurged_note_v2p4a	tag in combustion.json v2.4d: LEAD lines 160, 164, 166, 168, 175; REPORT v2.4a §7.4
T1482	every abort de-energises the NC valve at once and commands the pintle closed in ...	DECIDED *Lead decision, per electronics.json v4.1*	ELE v4.1	/abort_matrix/actions	tag in electronics.json v4.1:
T1483	ADOPTED by LEAD 258 (3) at the go 2026-10-21T10:47:17Z: REV-COMB-4 Z3' as written (head ...	DERIVED / CALC *Lead decision, line 258 (3); basis SIM / ASSUMED*	CMB v2.4d	/z3_jfwd_rulings_v2p4d/v/Z3_v2/decision	LEAD (rulings, CHANGELOG 258 (3)); DERIVED on SIM by Mechanical and REV-COMB-4 (grid counts on an ASSUMED construct); CALC on SIM (basis rows); nothing here is physical approval
T1484	896 / 896 grid cases inside (850 inside v1); ESP max 146.881 bar abs at 3.5 ms, margin ...	DERIVED / CALC *Lead decision, line 258 (3); basis SIM / ASSUMED*	CMB v2.4d	/z3_jfwd_rulings_v2p4d/v/Z3_v2/grid	LEAD (rulings, CHANGELOG 258 (3)); DERIVED on SIM by Mechanical and REV-COMB-4 (grid counts on an ASSUMED construct); CALC on SIM (basis rows); nothing here is physical approval
T1485	NOT EVALUATED in flight (1 kHz record; the LIGHT_SPIKE raw peak is a lower bound)	DERIVED / CALC *Lead decision, line 258 (3); basis SIM / ASSUMED*	CMB v2.4d	/z3_jfwd_rulings_v2p4d/v/Z3_v2/flight	LEAD (rulings, CHANGELOG 258 (3)); DERIVED on SIM by Mechanical and REV-COMB-4 (grid counts on an ASSUMED construct); CALC on SIM (basis rows); nothing here is physical approval
T1486	shape limits (option 2: the smallest box that admits every grid case admits a trapezoid ...	DERIVED / CALC *Lead decision, line 258 (3); basis SIM / ASSUMED*	CMB v2.4d	/z3_jfwd_rulings_v2p4d/v/Z3_v2/not_adopted	LEAD (rulings, CHANGELOG 258 (3)); DERIVED on SIM by Mechanical and REV-COMB-4 (grid counts on an ASSUMED construct); CALC on SIM (basis rows); nothing here is physical approval
T1487	ADOPTED by LEAD 258 (3): Q_JFWD = max(head peak over the record, max of S(T) over T ...	DERIVED / CALC *Lead decision, line 258 (3); basis SIM / ASSUMED*	CMB v2.4d	/z3_jfwd_rulings_v2p4d/v/jfwd_replace_rule/decision	LEAD (rulings, CHANGELOG 258 (3)); DERIVED on SIM by Mechanical and REV-COMB-4 (grid counts on an ASSUMED construct); CALC on SIM (basis rows); nothing here is physical approval
T1488	298 / 896 grid cases against 185 for the peak alone; the 113 added cases have a head ...	DERIVED / CALC *Lead decision, line 258 (3); basis SIM / ASSUMED*	CMB v2.4d	/z3_jfwd_rulings_v2p4d/v/jfwd_replace_rule/grid	LEAD (rulings, CHANGELOG 258 (3)); DERIVED on SIM by Mechanical and REV-COMB-4 (grid counts on an ASSUMED construct); CALC on SIM (basis rows); nothing here is physical approval
T1489	Q_JFWD >= the peak by construction, so it keeps every flag of the 256 (4) peak rule; a ...	DERIVED / CALC *Lead decision, line 258 (3); basis SIM / ASSUMED*	CMB v2.4d	/z3_jfwd_rulings_v2p4d/v/jfwd_replace_rule/why	LEAD (rulings, CHANGELOG 258 (3)); DERIVED on SIM by Mechanical and REV-COMB-4 (grid counts on an ASSUMED construct); CALC on SIM (basis rows); nothing here is physical approval
T1490	not meant for normal lights (REV-COMB-4 R4-10): a light to MEOP 45 bar abs trips it only ...	DERIVED / CALC *Lead decision, line 258 (3); basis SIM / ASSUMED*	CMB v2.4d	/z3_jfwd_rulings_v2p4d/v/jfwd_replace_rule/normal_lights	LEAD (rulings, CHANGELOG 258 (3)); DERIVED on SIM by Mechanical and REV-COMB-4 (grid counts on an ASSUMED construct); CALC on SIM (basis rows); nothing here is physical approval
T1491	on measured stand traces a noise-driven trigger still means the inspection until a ...	DERIVED / CALC *Lead decision, line 258 (3); basis SIM / ASSUMED*	CMB v2.4d	/z3_jfwd_rulings_v2p4d/v/jfwd_replace_rule/measured_traces	LEAD (rulings, CHANGELOG 258 (3)); DERIVED on SIM by Mechanical and REV-COMB-4 (grid counts on an ASSUMED construct); CALC on SIM (basis rows); nothing here is physical approval
T1492	Mechanical's rows at 60 bar abs, nominal factors: MS_y +0.328 (TC-01 ligament, LC2U) / ...	DERIVED / CALC *Lead decision, line 258 (3); basis SIM / ASSUMED*	CMB v2.4d	/z3_jfwd_rulings_v2p4d/v/jfwd_replace_rule/basis	LEAD (rulings, CHANGELOG 258 (3)); DERIVED on SIM by Mechanical and REV-COMB-4 (grid counts on an ASSUMED construct); CALC on SIM (basis rows); nothing here is physical approval
T1493	0.2-0.5 ms J-FWD band ASSUMED (no v3.5.2 J-FWD modal in this package: TO-VERIFY)	DERIVED / CALC *Lead decision, line 258 (3); basis SIM / ASSUMED*	CMB v2.4d	/z3_jfwd_rulings_v2p4d/v/jfwd_replace_rule/band	LEAD (rulings, CHANGELOG 258 (3)); DERIVED on SIM by Mechanical and REV-COMB-4 (grid counts on an ASSUMED construct); CALC on SIM (basis rows); nothing here is physical approval
T1494	the v1 'any head excursion above 60.0 bar abs: replace all J-FWD screws' rule and the ...	DERIVED / CALC *Lead decision, line 258 (3); basis SIM / ASSUMED*	CMB v2.4d	/z3_jfwd_rulings_v2p4d/v/jfwd_replace_rule/replaces	LEAD (rulings, CHANGELOG 258 (3)); DERIVED on SIM by Mechanical and REV-COMB-4 (grid counts on an ASSUMED construct); CALC on SIM (basis rows); nothing here is physical approval
T1495	Q_JFWD logging and the inside_v2 re-classification of the S6R runs and the N-04 burn-on ...	DERIVED / CALC *Lead decision, line 258 (3); basis SIM / ASSUMED*	CMB v2.4d	/z3_jfwd_rulings_v2p4d/v/jfwd_replace_rule/electronics	LEAD (rulings, CHANGELOG 258 (3)); DERIVED on SIM by Mechanical and REV-COMB-4 (grid counts on an ASSUMED construct); CALC on SIM (basis rows); nothing here is physical approval
T1496	the flight consequence of a flagged first light (relight on a used joint; line 231 ...	TO-VERIFY	CMB v2.4d	/z3_jfwd_rulings_v2p4d/v/open_before_flight_acceptance/0	(class set at the call site: TO-VERIFY)
T1497	the damped ESP variant (structural damping MISSING)	TO-VERIFY	CMB v2.4d	/z3_jfwd_rulings_v2p4d/v/open_before_flight_acceptance/1	(class set at the call site: TO-VERIFY)
T1498	the 0.2-0.5 ms J-FWD band (ASSUMED; TO-VERIFY on a v3.5.2 modal)	TO-VERIFY	CMB v2.4d	/z3_jfwd_rulings_v2p4d/v/open_before_flight_acceptance/2	(class set at the call site: TO-VERIFY)
T1499	a shutdown from a case hotter than the rows' soak basis (F4 observation; Mechanical, no ...	TO-VERIFY	CMB v2.4d	/z3_jfwd_rulings_v2p4d/v/open_before_flight_acceptance/3	(class set at the call site: TO-VERIFY)
T1500	P_torch during FA-1: design case FA1S 151.04 bar abs uniform over the passage ...	DECIDED *Lead decision, per combustion.json v2.4d*	CMB v2.4d	/fa1_p_torch_v2p4c/v/design_case	tag in combustion.json v2.4d:
T1501	tap-wall v3, published (04-fea/v352/jfwd_tapwall_v352_v3.json ef70572ec92b05b0 ...	CALC	CMB v2.4d	/fa1_p_torch_v2p4c/v/mechanical_row	(class set at the call site: CALC)

ID	Value	Class	Source	Key	Source tag / formula
T1502	CLOSED (LEAD 258 (3), go 2026-10-21T10:47:17Z): Z3 v2 ADOPTED = Z3' above as written ...	DECIDED *Lead decision, per combustion.json v2.4d*	CMB v2.4d	/fa1_grid_statistics_v2p4c/v/status	tag in combustion.json v2.4d:
T1503	no Combustion number changes: the FA-1 grid histories are the v2.4b set and ...	DERIVED / CALC *Lead decision, line 258 (3); basis SIM / ASSUMED*	CMB v2.4d	/z3_jfwd_rulings_v2p4d/v/combustion_side	LEAD (rulings, CHANGELOG 258 (3)); DERIVED on SIM by Mechanical and REV-COMB-4 (grid counts on an ASSUMED construct); CALC on SIM (basis rows); nothing here is physical approval
T1504	0.93	SIM	CMB v2.4d	/stability/points_initial_port/v/1.0/dP_over_Pc	SIM stability.py v2 (stability_v2.json)
T1505	6.5	SIM	CMB v2.4d	/stability/points_initial_port/v/1.0/GM_tau5ms	SIM stability.py v2 (stability_v2.json)
T1506	4.7	SIM	CMB v2.4d	/stability/points_initial_port/v/1.0/GM_tau10ms	SIM stability.py v2 (stability_v2.json)
T1507	1460 Hz	SIM	CMB v2.4d	/stability/points_initial_port/v/1.0/f_1L_Hz	SIM stability.py v2 (stability_v2.json)
T1508	1125 Hz	SIM	CMB v2.4d	/stability/points_initial_port/v/1.0/f_Helmholtz_Hz	SIM stability.py v2 (stability_v2.json)
T1509	2.17 Hz	SIM	CMB v2.4d	/stability/points_initial_port/v/1.0/f_thermal_lag_Hz	SIM stability.py v2 (stability_v2.json)
T1510	1.58	SIM	CMB v2.4d	/stability/points_initial_port/v/0.75/dP_over_Pc	SIM stability.py v2 (stability_v2.json)
T1511	10.2	SIM	CMB v2.4d	/stability/points_initial_port/v/0.75/GM_tau5ms	SIM stability.py v2 (stability_v2.json)
T1512	7.5	SIM	CMB v2.4d	/stability/points_initial_port/v/0.75/GM_tau10ms	SIM stability.py v2 (stability_v2.json)
T1513	1477 Hz	SIM	CMB v2.4d	/stability/points_initial_port/v/0.75/f_1L_Hz	SIM stability.py v2 (stability_v2.json)
T1514	1138 Hz	SIM	CMB v2.4d	/stability/points_initial_port/v/0.75/f_Helmholtz_Hz	SIM stability.py v2 (stability_v2.json)
T1515	1.61 Hz	SIM	CMB v2.4d	/stability/points_initial_port/v/0.75/f_thermal_lag_Hz	SIM stability.py v2 (stability_v2.json)
T1516	2.78	SIM	CMB v2.4d	/stability/points_initial_port/v/0.5/dP_over_Pc	SIM stability.py v2 (stability_v2.json)
T1517	17	SIM	CMB v2.4d	/stability/points_initial_port/v/0.5/GM_tau5ms	SIM stability.py v2 (stability_v2.json)
T1518	12.5	SIM	CMB v2.4d	/stability/points_initial_port/v/0.5/GM_tau10ms	SIM stability.py v2 (stability_v2.json)
T1519	1499 Hz	SIM	CMB v2.4d	/stability/points_initial_port/v/0.5/f_1L_Hz	SIM stability.py v2 (stability_v2.json)
T1520	1155 Hz	SIM	CMB v2.4d	/stability/points_initial_port/v/0.5/f_Helmholtz_Hz	SIM stability.py v2 (stability_v2.json)
T1521	1.07 Hz	SIM	CMB v2.4d	/stability/points_initial_port/v/0.5/f_thermal_lag_Hz	SIM stability.py v2 (stability_v2.json)
T1522	5.86	SIM	CMB v2.4d	/stability/points_initial_port/v/0.25/dP_over_Pc	SIM stability.py v2 (stability_v2.json)
T1523	32.1	SIM	CMB v2.4d	/stability/points_initial_port/v/0.25/GM_tau5ms	SIM stability.py v2 (stability_v2.json)
T1524	23.7	SIM	CMB v2.4d	/stability/points_initial_port/v/0.25/GM_tau10ms	SIM stability.py v2 (stability_v2.json)
T1525	1532 Hz	SIM	CMB v2.4d	/stability/points_initial_port/v/0.25/f_1L_Hz	SIM stability.py v2 (stability_v2.json)
T1526	1180 Hz	SIM	CMB v2.4d	/stability/points_initial_port/v/0.25/f_Helmholtz_Hz	SIM stability.py v2 (stability_v2.json)
T1527	0.56 Hz	SIM	CMB v2.4d	/stability/points_initial_port/v/0.25/f_thermal_lag_Hz	SIM stability.py v2 (stability_v2.json)
T1528	7.18	SIM	CMB v2.4d	/stability/points_initial_port/v/0.2/dP_over_Pc	SIM stability.py v2 (stability_v2.json)
T1529	36.8	SIM	CMB v2.4d	/stability/points_initial_port/v/0.2/GM_tau5ms	SIM stability.py v2 (stability_v2.json)
T1530	27.2	SIM	CMB v2.4d	/stability/points_initial_port/v/0.2/GM_tau10ms	SIM stability.py v2 (stability_v2.json)
T1531	1539 Hz	SIM	CMB v2.4d	/stability/points_initial_port/v/0.2/f_1L_Hz	SIM stability.py v2 (stability_v2.json)
T1532	1186 Hz	SIM	CMB v2.4d	/stability/points_initial_port/v/0.2/f_Helmholtz_Hz	SIM stability.py v2 (stability_v2.json)
T1533	0.47 Hz	SIM	CMB v2.4d	/stability/points_initial_port/v/0.2/f_thermal_lag_Hz	SIM stability.py v2 (stability_v2.json)
T1534	10.54	SIM	CMB v2.4d	/stability/points_initial_port/v/0.125/dP_over_Pc	SIM stability.py v2 (stability_v2.json)
T1535	52.2	SIM	CMB v2.4d	/stability/points_initial_port/v/0.125/GM_tau5ms	SIM stability.py v2 (stability_v2.json)
T1536	38.6	SIM	CMB v2.4d	/stability/points_initial_port/v/0.125/GM_tau10ms	SIM stability.py v2 (stability_v2.json)
T1537	1544 Hz	SIM	CMB v2.4d	/stability/points_initial_port/v/0.125/f_1L_Hz	SIM stability.py v2 (stability_v2.json)
T1538	1190 Hz	SIM	CMB v2.4d	/stability/points_initial_port/v/0.125/f_Helmholtz_Hz	SIM stability.py v2 (stability_v2.json)
T1539	0.33 Hz	SIM	CMB v2.4d	/stability/points_initial_port/v/0.125/f_thermal_lag_Hz	SIM stability.py v2 (stability_v2.json)
T1540	injector dP/Pc >= 0.25 everywhere (ceiling rule); chug gain margin from the lumped ...	DERIVED	CMB v2.4d	/stability/criteria	(class set at the call site: DERIVED)
T1541	classical hybrid low-frequency (thermal-lag/boundary-layer) instability 0.3-2 Hz band is ...	TO-VERIFY	CMB v2.4d	/stability/hybrid_LF_note	(class set at the call site: TO-VERIFY)
T1542	A1 relight lights within W 0.65 s with P_abort <= 1e-3, venting <= 76 g, theta <= 2 deg	DECIDED *Lead decision, line 164*	CMB v2.4d	/cf4_gate/v/claim	tag in combustion.json v2.4d: LEAD line 164 (accepted as the gate), 166 (3); CF-4-SPEC v2.0; physical execution by the founders; CF-4-SPEC v2.1 (v2.4b): red lines, abort criteria, IS-1/IS-2 ...
T1543	CF-4-SPEC v2.3 (v2.4d, 6f0d4afd7fb3a798)	DECIDED *Design value, combustion.json v2.4d*	CMB v2.4d	/cf4_gate/spec_version	combustion.json v2.4d
T1544	P1 liquid at the pintle inlet within 18.86 cm3 of lift-flow volume after reaching the ...	DECIDED *Lead decision, line 164*	CMB v2.4d	/cf4_gate/v/pass_criteria/0	tag in combustion.json v2.4d: LEAD line 164 (accepted as the gate), 166 (3); CF-4-SPEC v2.0; physical execution by the founders; CF-4-SPEC v2.1 (v2.4b): red lines, abort criteria, IS-1/IS-2 ...
T1545	P2 >= 0.35 s continuous liquid or inside the void budget	DECIDED *Lead decision, line 164*	CMB v2.4d	/cf4_gate/v/pass_criteria/1	tag in combustion.json v2.4d: LEAD line 164 (accepted as the gate), 166 (3); CF-4-SPEC v2.0; physical execution by the founders; CF-4-SPEC v2.1 (v2.4b): red lines, abort criteria, IS-1/IS-2 ...
T1546	P3 cap clears FD-01 crown <= 2.3 s after flow start	DECIDED *Lead decision, line 164*	CMB v2.4d	/cf4_gate/v/pass_criteria/2	tag in combustion.json v2.4d: LEAD line 164 (accepted as the gate), 166 (3); CF-4-SPEC v2.0; physical execution by the founders; CF-4-SPEC v2.1 (v2.4b): red lines, abort criteria, IS-1/IS-2 ...
T1547	P4 per tap and per phase: vapour >= 3.4 g/s per tap (80 % of HEM 4.25), liquid >= 12.5 ...	DECIDED *Lead decision, line 164*	CMB v2.4d	/cf4_gate/v/pass_criteria/3	tag in combustion.json v2.4d: LEAD line 164 (accepted as the gate), 166 (3); CF-4-SPEC v2.0; physical execution by the founders; CF-4-SPEC v2.1 (v2.4b): red lines, abort criteria, IS-1/IS-2 ...
T1548	P5 <= 76 g vented (IS-1, IS-2)	DECIDED *Lead decision, line 164*	CMB v2.4d	/cf4_gate/v/pass_criteria/4	tag in combustion.json v2.4d: LEAD line 164 (accepted as the gate), 166 (3); CF-4-SPEC v2.0; physical execution by the founders; CF-4-SPEC v2.1 (v2.4b): red lines, abort criteria, IS-1/IS-2 ...
T1549	P6 flow stopped by E0 - 0.05 s (scheduled); closure confirmed (V1 first, P_mid) by the ...	DECIDED *Lead decision, line 164*	CMB v2.4d	/cf4_gate/v/pass_criteria/5	tag in combustion.json v2.4d: LEAD line 164 (accepted as the gate), 166 (3); CF-4-SPEC v2.0; physical execution by the founders; CF-4-SPEC v2.1 (v2.4b): red lines, abort criteria, IS-1/IS-2 ...
T1550	P7 no unfiltered path to the seat	DECIDED *Lead decision, line 164*	CMB v2.4d	/cf4_gate/v/pass_criteria/6	tag in combustion.json v2.4d: LEAD line 164 (accepted as the gate), 166 (3); CF-4-SPEC v2.0; physical execution by the founders; CF-4-SPEC v2.1 (v2.4b): red lines, abort criteria, IS-1/IS-2 ...
T1551	default gate "current rule OR a_meas >= 1.25 → change grain" unless the oxidiser-flux ...	DERIVED	CMB v2.4d	/t1t4_law_v2/v/rule	DERIVED from SIM grain_q1d histories (78 B1 records on gnc.json v3 and v3.1 cycles; O/F is not a gnc.json quantity); ASSUMED durations, s1, ss
T1552	T1-T4 regression-rate measurement and law form (RG-03), v3.2	DECIDED *Design value, T1-T4-REGRESSION-SPEC.md v3.2*	T14 v3.2	heading line 1	tag in T1-T4-REGRESSION-SPEC.md v3.2: document heading
T1553	CONFIRMED by GNC in gnc.json v3.2 (bde317782574cbd9; duty_cycles.note_n07_rev_gnc_7 ...	DECIDED *Lead decision, per combustion.json v2.4d*	CMB v2.4d	/t4_duration_note_v2p4c/v/status	tag in combustion.json v2.4d:
T1554	T4 RE-SIZED (DECIDED): GNC's N-07 answer in gnc.json v3.2 ...	DECIDED *Lead decision, line 261 (2)*	CL 266 lines	L261(2)	lead decision 2026-10-21 (line sha d510d8077319e234)
T1555	N-07 (OI-T4-RESIZE): SHORTER RELIGHT ADOPTED: the T4 relight after the B1 p99 burn (case ...	DECIDED *Lead decision, line 264 (5)*	CL 266 lines	L264(5)	lead (decision) 2026-10-21 (line sha 7cdbe51041ae6b8b)
T1556	T4 then demonstrates relight latency from a soaked state, not a 4.59 s landing-length ...	DECIDED *Lead decision, line 264 (5)*	CL 266 lines	L264(5)[from "T4 then demonstrates" to "say so."]	lead (decision) 2026-10-21 (line sha 7cdbe51041ae6b8b)
T1557	T1_100pct_2t_8s_kg: 1.928; share_of_flight_load: 0.964; T1_max_2t_s_on_flight_tank ...	DERIVED	CMB v2.4d	/test_budgets_v2p4b	DERIVED (SIM map flows, ASSUMED durations/repeats, DESIGN loads); stand tank MISSING
T1558	0.019409 m	DECIDED *Design value, combustion.json v2.4d*	CMB v2.4d	/gas_path/nozzle/exit_diameter_m	tag in combustion.json v2.4d: v1 geometry
T1559	Version v2.1a for validation and v2.2 for the time-matched afterburn, 2026-10-12. Owner ...	SIM	RPL v2.2b	line 3	RPL line 3

ID	Value	Class	Source	Key	Source tag / formula
T1560	v1 (combustion.json v1): Pc 24.99 bar, O/F 7.76, T0 = eta_c**2 Tc = 2844.9 K. All grid ...	SIM	RPL v2.2b	§0. Checkpoint bullet "v1 (combustion.json v1)"	RPL bullet
T1561	v2 (combustion.json v2.1 S230 point): Pc 25.01 bar, O/F 10.435, T0 2757 K.	SIM	RPL v2.2b	§0. Checkpoint bullet "v2 (combustion.json v2.1 S230 point)"	RPL bullet
T1562	7.76	SIM	RPL v2.2b	§0. Checkpoint bullet "v1 (combustion.json v1)" [regex O/F ([\d.]+)]	RPL bullet
T1563	24.99 bar	SIM	RPL v2.2b	§0. Checkpoint bullet "v1 (combustion.json v1)" [regex Pc ([\d.]+) bar]	RPL bullet
T1564	10.435	SIM	PVAL v2.1a	/v2op/gas/OF	(class set at the call site: SIM) [validation_v2.json carries no tags: the v2-point inputs of the CFD run]
T1565	25.010070000000002 bar	SIM	PVAL v2.1a	/v2op/gas/p0	(class set at the call site: SIM) [display = raw x 1e-05 (Pa to bar)]
T1566	500 N	SIM	CMB v2.4d	/operating_points/max_thrust/F_N	SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* 0.92, eta_cf 0.975)
T1567	25.010070000000002 bar	SIM	CMB v2.4d	/operating_points/max_thrust/Pc_Pa	SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* 0.92, eta_cf 0.975) [display = raw x 1e-05 (Pa to bar)]
T1568	2757 K	SIM	CMB v2.4d	/operating_points/throttle_points_20C_initial_port/0/T0_delivered_K	20 C saturated tank (P_tank 50 bar row of the map for lift and dP/Pc), initial port 25 mm, nominal law. SIM grain_q1d.py v2 (quasi-1D port, S230, nominal ASSUMED law, Dyer pintle Cd 0.70, eta_c* ...
T1569	0.014923027687774315 %	SIM	PVAL v2.1a	/v2op_vs_v1/thrust_exit_N/diff_pct	(class set at the call site: SIM)
T1570	2.4648496078546245 %	SIM	PVAL v2.1a	/v2op_vs_v1/x_mach_disk_De/diff_pct	(class set at the call site: SIM)
T1571	3.224512018732506 %	SIM	PVAL v2.1a	/v2op_vs_v1/mdot_throat_kg_s/diff_pct	(class set at the call site: SIM)
T1572	−5.758596911322145 %	SIM	PVAL v2.1a	/v2op_vs_v1/T_axis_10De_K/diff_pct	(class set at the call site: SIM)
T1573	The current max-thrust point (combustion.json v2.4d: 500 N, Pc 25.01 bar, O/F 10.435) is ...	DERIVED	generator	formula: statement built from the traced operating points and the v2-vs-v1 coarse-grid differences	DERIVED by build_datasheet.py: statement built from the traced operating points and the v2-vs-v1 coarse-grid differences ← T1560, T1562, T1563, T1564, T1565, T1566, T1567, T035 ...
T1574	RPL (05-plume-cfd/REPORT.md, text revision v2.2b, 140b3c59f8323bba) and PVAL ...	DERIVED	generator	formula: generator check on this build: newest modification time under 05-plume-cfd/ and the REPORT.md / REPORT_v2p2b.txt sha	DERIVED by build_datasheet.py: generator check on this build: newest modification time under 05-plume-cfd/ and the REPORT.md / REPORT_v2p2b.txt sha
T1575	2.6759252764405907 De	SIM	PVAL v2.1a	/correlations/own_f_2s_rans/x_md_cfd	(class set at the call site: SIM) [validation_v2.json carries no tags: CFD outputs SIM, 1-D and correlation values DERIVED]
T1576	2.662555623319815 De	DERIVED	PVAL v2.1a	/correlations/own_f_2s_rans/x_md_lewis_carlson	(class set at the call site: DERIVED) [validation_v2.json carries no tags: CFD outputs SIM, 1-D and correlation values DERIVED]
T1577	3.959272515274379 De	SIM	PVAL v2.1a	/correlations/own_f_2s_rans/spacing_first_mach	(class set at the call site: SIM) [validation_v2.json carries no tags: CFD outputs SIM, 1-D and correlation values DERIVED]
T1578	4.101274690948743 De	SIM	PVAL v2.1a	/correlations/own_f_2s_rans/spacing_mean3_mach	(class set at the call site: SIM) [validation_v2.json carries no tags: CFD outputs SIM, 1-D and correlation values DERIVED]
T1579	4.816460953753267 De	DERIVED	PVAL v2.1a	/correlations/own_f_2s_rans/spacing_pp_tam	(class set at the call site: DERIVED) [validation_v2.json carries no tags: CFD outputs SIM, 1-D and correlation values DERIVED]
T1580	14.90004230068159 De	SIM	PVAL v2.1a	/gci/core_length_Z095_De/fine	(class set at the call site: SIM) [validation_v2.json carries no tags: CFD outputs SIM, 1-D and correlation values DERIVED]
T1581	2222.2512513813263 K	SIM	PVAL v2.1a	/gci/T_axis_10De_K/fine	(class set at the call site: SIM) [validation_v2.json carries no tags: CFD outputs SIM, 1-D and correlation values DERIVED]
T1582	2057.669651883192 K	SIM	PVAL v2.1a	/gci/T_axis_20De_K/fine	(class set at the call site: SIM) [validation_v2.json carries no tags: CFD outputs SIM, 1-D and correlation values DERIVED]
T1583	500.8162021154685 N	SIM	PVAL v2.1a	/thrust_vs_1d/own_f_2s_rans/F_cfd	(class set at the call site: SIM) [validation_v2.json carries no tags: CFD outputs SIM, 1-D and correlation values DERIVED]
T1584	505.78770230207766 N	DERIVED	PVAL v2.1a	/one_d/F_ideal_lambda_N	(class set at the call site: DERIVED) [validation_v2.json carries no tags: CFD outputs SIM, 1-D and correlation values DERIVED]
T1585	−0.9829223138446253 %	SIM	PVAL v2.1a	/thrust_vs_1d/own_f_2s_rans/dev_vs_ideal_lambda_pct	(class set at the call site: SIM) [validation_v2.json carries no tags: CFD outputs SIM, 1-D and correlation values DERIVED]
T1586	0.25415365293759107 kg/s	SIM	PVAL v2.1a	/thrust_vs_1d/own_f_2s_rans/mdot_cfd	(class set at the call site: SIM) [validation_v2.json carries no tags: CFD outputs SIM, 1-D and correlation values DERIVED]
T1587	0.2558231798839737 kg/s	DERIVED	PVAL v2.1a	/one_d/mdot	(class set at the call site: DERIVED) [validation_v2.json carries no tags: CFD outputs SIM, 1-D and correlation values DERIVED]
T1588	−0.6526097233017778 %	SIM	PVAL v2.1a	/thrust_vs_1d/own_f_2s_rans/mdot_dev_pct	(class set at the call site: SIM) [validation_v2.json carries no tags: CFD outputs SIM, 1-D and correlation values DERIVED]
T1589	2.0119829447987048	DERIVED	PVAL v2.1a	/one_d/Me	(class set at the call site: DERIVED) [validation_v2.json carries no tags: CFD outputs SIM, 1-D and correlation values DERIVED]
T1590	3.2357942188232363	DERIVED	PVAL v2.1a	/one_d/pe_over_pa	(class set at the call site: DERIVED) [validation_v2.json carries no tags: CFD outputs SIM, 1-D and correlation values DERIVED]
T1591	2.6232413790801554	DERIVED	PVAL v2.1a	/one_d/Mj	(class set at the call site: DERIVED) [validation_v2.json carries no tags: CFD outputs SIM, 1-D and correlation values DERIVED]
T1592	51.937 mm	DERIVED	generator	formula: x/De × De (exit diameter from combustion.json, tagged v1 geometry)	DERIVED by build_datasheet.py: x/De × De (exit diameter from combustion.json, tagged v1 geometry) ← T1575, T1558
T1593	grid differences within 2 sigma of the temporal scatter: GCI not meaningful; report the ...	SIM	PVAL v2.1a	/gci_resolvability/x_mach_disk_De/verdict	(class set at the call site: SIM) [validation_v2.json carries no tags: CFD outputs SIM, 1-D and correlation values DERIVED]
T1594	76.8455 mm	DERIVED	generator	formula: x/De × De (exit diameter from combustion.json, tagged v1 geometry)	DERIVED by build_datasheet.py: x/De × De (exit diameter from combustion.json, tagged v1 geometry) ← T1577, T1558
T1595	79.6016 mm	DERIVED	generator	formula: x/De × De (exit diameter from combustion.json, tagged v1 geometry)	DERIVED by build_datasheet.py: x/De × De (exit diameter from combustion.json, tagged v1 geometry) ← T1578, T1558
T1596	6.542174825858145 %	SIM	PVAL v2.1a	/gci/spacing_mean_3_De/gci_fine_pct	(class set at the call site: SIM) [validation_v2.json carries no tags: CFD outputs SIM, 1-D and correlation values DERIVED]
T1597	oscillatory: 3 x max-deviation bound	SIM	PVAL v2.1a	/gci/spacing_mean_3_De/note	(class set at the call site: SIM) [validation_v2.json carries no tags: CFD outputs SIM, 1-D and correlation values DERIVED]
T1598	3.6875368031844005	SIM	PVAL v2.1a	/gci/M_max_before_disk/fine	(class set at the call site: SIM) [validation_v2.json carries no tags: CFD outputs SIM, 1-D and correlation values DERIVED]
T1599	289.195 mm	DERIVED	generator	formula: x/De × De (exit diameter from combustion.json, tagged v1 geometry)	DERIVED by build_datasheet.py: x/De × De (exit diameter from combustion.json, tagged v1 geometry) ← T1580, T1558
T1600	0.15219464936381047 %	SIM	PVAL v2.1a	/gci/thrust_exit_N/gci_fine_pct	(class set at the call site: SIM) [validation_v2.json carries no tags: CFD outputs SIM, 1-D and correlation values DERIVED]
T1601	monotone without observed convergence (p < 0.5): 3 x max-deviation bound	SIM	PVAL v2.1a	/gci/thrust_exit_N/note	(class set at the call site: SIM) [validation_v2.json carries no tags: CFD outputs SIM, 1-D and correlation values DERIVED]
T1602	44.9 De	SIM	PVAL v2.1a	/jet_front_x_De/own_f_2s_rans	(class set at the call site: SIM) [validation_v2.json carries no tags: CFD outputs SIM, 1-D and correlation values DERIVED]
T1603	2.708786126767669 De	SIM	PVAL v2.1a	/v2op/x_md_cfd	(class set at the call site: SIM) [validation_v2.json carries no tags: CFD outputs SIM, 1-D and correlation values DERIVED]
T1604	2.6662630489711634 De	DERIVED	PVAL v2.1a	/v2op/x_md_lewis_carlson	(class set at the call site: DERIVED) [validation_v2.json carries no tags: CFD outputs SIM, 1-D and correlation values DERIVED]
T1605	−1.0866730201599784 %	SIM	PVAL v2.1a	/v2op/thrust_dev_vs_ideal_lambda_pct	(class set at the call site: SIM) [validation_v2.json carries no tags: CFD outputs SIM, 1-D and correlation values DERIVED]
T1606	same coarse grid and models as own_c_2s_rans (v1 point O/F 7.76); v2 - v1 differences in ...	SIM	PVAL v2.1a	/v2op/note	(class set at the call site: SIM) [validation_v2.json carries no tags: CFD outputs SIM, 1-D and correlation values DERIVED]
T1607	adds 172-182 K at O/F 7.76 (temporal band 172-178 K; time-matched fine maximum 179.2 K ...	SIM	RPL v2.2b	§0. Checkpoint bullet "adds 172"	RPL bullet
T1608	adds nothing at the v2 point for an equilibrium exhaust (O/F 10.4, oxidiser-rich; 0.44 ...	SIM	RPL v2.2b	§0. Checkpoint bullet "adds nothing"	RPL bullet
T1609	adds up to 1476 K at the fuel-rich 12.5 % floor.	SIM	RPL v2.2b	§0. Checkpoint bullet "adds up to"	RPL bullet
T1610	**Soot and radiation.** There is no soot model and no radiative transfer. The synthetic ...	SIM	RPL v2.2b	§10. What the model cannot show bullet "**Soot and radiation"	RPL bullet
T1611	**Finite-rate chemistry and turbulence-chemistry interaction.** Afterburning is a ...	SIM	RPL v2.2b	§10. What the model cannot show bullet "**Finite-rate chemistry"	RPL bullet
T1612	**Exhaust gas model.** The exhaust is a calorically perfect gas with frozen calibrated ...	SIM	RPL v2.2b	§10. What the model cannot show bullet "**Exhaust gas model"	RPL bullet
T1613	**Geometry and setting.** The model is axisymmetric only (no gimbal angle, no 3-D ...	SIM	RPL v2.2b	§10. What the model cannot show bullet "**Geometry and setting"	RPL bullet
T1614	**Duration.** Simulated time is 3-4 ms. The far field (> 15 De) is not statistically ...	SIM	RPL v2.2b	§10. What the model cannot show bullet "**Duration"	RPL bullet
T1615	**No acoustics or plume loads on the vehicle.**	SIM	RPL v2.2b	§10. What the model cannot show bullet "**No acoustics"	RPL bullet

ID	Value	Class	Source	Key	Source tag / formula
T1616	0 unchanged, 1 changed since review (05-plume-cfd/REPORT.md), 0 not found	DERIVED	generator	formula: sha256/16 of each reviewed hub file vs the register	DERIVED by build_datasheet.py: sha256/16 of each reviewed hub file vs the register
T1617	REV-PLUME-1		SUP 18 reviews	/reviews/1/id	(class set at the call site: REF)
T1618	ACCEPT WITH COMMENTS	DERIVED	SUP 18 reviews	/reviews/1/verdict	(class set at the call site: DERIVED)
T1619	RC-10 thrust GCI R=1.09 diverges: reword; RC-11 correlation claim wording; RC-19 ...	DERIVED	SUP 18 reviews	/reviews/1/comments	(class set at the call site: DERIVED)
T1620	1 unchanged, 0 changed since review (none), 0 not found	DERIVED	generator	formula: sha256/16 of each reviewed hub file vs the register	DERIVED by build_datasheet.py: sha256/16 of each reviewed hub file vs the register
T1621	REV-PLUME-2		SUP 18 reviews	/reviews/6/id	(class set at the call site: REF)
T1622	ACCEPT	DERIVED	SUP 18 reviews	/reviews/6/verdict	(class set at the call site: DERIVED)
T1623	R2-12 summary wording (lead file) corrected: HANDBACK-SUMMARY d12c8862bf5524a4	DERIVED	SUP 18 reviews	/reviews/6/comments	(class set at the call site: DERIVED)
T1624	Delivered by / reviewed by / date: fea3d (FEA3D-2) / not reviewed: status to be reviewed ...	DERIVED	F3D FINAL handback of the v3.5.2 re-check	line 16 (starts "Delivered by")	FEA3D-2 report §0 checkpoint
T1625	2026-10-21, FINAL handback of the v3.5.2 re-check (seat FEA3D-2 from 2026-10-21T00:41Z ...	DERIVED	F3D FINAL handback of the v3.5.2 re-check	§REPORT FEA3D v3.5.2 row c0="Status" col 1 (line 5)	FEA3D-2 report header table
T1626	fea3d (FEA3D-1, then FEA3D-2), lead briefs and rulings as cited per section (CHANGELOG ...	DERIVED	F3D FINAL handback of the v3.5.2 re-check	§REPORT FEA3D v3.5.2 row c0="Author" col 1 (line 6)	FEA3D-2 report header table
T1627	CAD v3.5.2 INTERIM set as relayed (manifests in 2.1; rev 1 `05a48a8efb89ea8f`	DERIVED	F3D FINAL handback of the v3.5.2 re-check	§REPORT FEA3D v3.5.2 row c0="Configuration" col 1 (line 7)	FEA3D-2 report header table
T1628	456	DERIVED	generator	formula: count of the non-empty rows of 03-cad/v3.5.2/MANIFEST_v3.5.2.sha256	DERIVED by build_datasheet.py: count of the non-empty rows of 03-cad/v3.5.2/MANIFEST_v3.5.2.sha256 ← T708
T1629	CalculiX 2.21 (SPOOLES direct), C3D10 tets (gmsh 4.15.2), own Python pre / post ...	DERIVED	F3D FINAL handback of the v3.5.2 re-check	§REPORT FEA3D v3.5.2 row c0="Tool" col 1 (line 8)	FEA3D-2 report header table
T1630	mm, N, MPa, degC. MS = allowable / (SF x stress) - 1	DERIVED	F3D FINAL handback of the v3.5.2 re-check	§REPORT FEA3D v3.5.2 row c0="Units" col 1 (line 9)	FEA3D-2 report header table
T1631	SIM (FE output), DERIVED (computed from SIM / data), CALC (Mechanical's hand ...	DERIVED	F3D FINAL handback of the v3.5.2 re-check	§REPORT FEA3D v3.5.2 row c0="Tags" col 1 (line 10)	FEA3D-2 report header table
T1632	DECIDED: no independent review of the FEA3D v3.5.2 run (04-fea/assembly3d_v352/, FINAL ...	DECIDED *Lead decision, line 262 (3)*	CL 266 lines	L262(3)	lead (decision) 2026-10-21 (line sha 4d0bf4e937506f42)
T1633	Cell = MS_y / MS_u after the 234 (4) all-node rule. Flags: r = regular value stands (the ...	DERIVED	MMXMD FEA3D margin matrix v3.5.2	line 3 (starts "Cell = MS_y")	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1634	Mechanical's single J-FWD re-map on the FINAL envelope v6 was published under the lead ...	DERIVED	MMXD v352	/why	CALC / SIM / DERIVED as in the table; a table re-run, not physical approval
T1635	MF note (lead 01:30:18Z (5b)). The seat-line P+Q is not shown converged: s9 → s6 raises ...	SIM	F3D FINAL handback of the v3.5.2 re-check	line 319 (starts "MF note")	FEA3D-2 report §5.3 MF note
T1636	cD END [SIM]: seat_sDL12_mu0 (s6 + two Ball fields h 0.04 mm at the d+6.051 z'0 ends ...	SIM	CL 266 lines	L252(1)	fea3d (result) 2026-10-21 (line sha 38e099299d2c1bed)
T1637	cD RULING (05:17:36Z): s6 + C1 stand in every state; band anchors unchanged (PRELOAD ...	DECIDED *Lead decision, line 250 (10)*	CL 266 lines	L250(10)	lead (decision) 2026-10-21 (line sha 155022d655ee3cc3)
T1638	+0.955 / +0.580 r	SIM	MMXMD FEA3D margin matrix v3.5.2	line 7 row "TC-01" col 1	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1639	+0.440 / +0.164 r	SIM	MMXMD FEA3D margin matrix v3.5.2	line 7 row "TC-01" col 2	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1640	-	SIM	MMXMD FEA3D margin matrix v3.5.2	line 7 row "TC-01" col 3	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1641	+0.412 / +0.522 r	SIM	MMXMD FEA3D margin matrix v3.5.2	line 7 row "TC-01" col 4	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1642	+0.107 / +0.194 r	SIM	MMXMD FEA3D margin matrix v3.5.2	line 7 row "TC-01" col 5	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1643	-0.083 / -0.012 r MF (gov. sub-model +0.005 / +0.239)	SIM	MMXMD FEA3D margin matrix v3.5.2	line 7 row "TC-01" col 6	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1644	-0.107 / -0.037 r MF (gov. sub-model +0.005 / +0.239)	SIM	MMXMD FEA3D margin matrix v3.5.2	line 7 row "TC-01" col 7	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1645	-	SIM	MMXMD FEA3D margin matrix v3.5.2	line 7 row "TC-01" col 8	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1646	+0.496 / +0.223 r	SIM	MMXMD FEA3D margin matrix v3.5.2	line 8 row "TC-02" col 1	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1647	+0.469 / +0.202 r	SIM	MMXMD FEA3D margin matrix v3.5.2	line 8 row "TC-02" col 2	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1648	-	SIM	MMXMD FEA3D margin matrix v3.5.2	line 8 row "TC-02" col 3	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1649	+1.705 / +1.918 r	SIM	MMXMD FEA3D margin matrix v3.5.2	line 8 row "TC-02" col 4	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1650	+1.098 / +1.263 r	SIM	MMXMD FEA3D margin matrix v3.5.2	line 8 row "TC-02" col 5	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1651	+0.725 / +0.861 r	SIM	MMXMD FEA3D margin matrix v3.5.2	line 8 row "TC-02" col 6	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1652	+0.679 / +0.811 r	SIM	MMXMD FEA3D margin matrix v3.5.2	line 8 row "TC-02" col 7	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1653	-	SIM	MMXMD FEA3D margin matrix v3.5.2	line 8 row "TC-02" col 8	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1654	-0.088 / -0.170	SIM	MMXMD FEA3D margin matrix v3.5.2	line 9 row "TC-03" col 1	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1655	-0.092 / -0.175	SIM	MMXMD FEA3D margin matrix v3.5.2	line 9 row "TC-03" col 2	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1656	-	SIM	MMXMD FEA3D margin matrix v3.5.2	line 9 row "TC-03" col 3	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1657	+9.382 / +11.587	SIM	MMXMD FEA3D margin matrix v3.5.2	line 9 row "TC-03" col 4	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1658	+8.834 / +10.922	SIM	MMXMD FEA3D margin matrix v3.5.2	line 9 row "TC-03" col 5	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1659	+8.247 / +10.211	SIM	MMXMD FEA3D margin matrix v3.5.2	line 9 row "TC-03" col 6	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1660	+8.153 / +10.097	SIM	MMXMD FEA3D margin matrix v3.5.2	line 9 row "TC-03" col 7	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1661	-	SIM	MMXMD FEA3D margin matrix v3.5.2	line 9 row "TC-03" col 8	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1662	+0.459 / +0.184 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 10 row "TC-04" col 1	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1663	+0.459 / +0.183 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 10 row "TC-04" col 2	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1664	-	SIM	MMXMD FEA3D margin matrix v3.5.2	line 10 row "TC-04" col 3	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1665	+1.803 / +2.030 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 10 row "TC-04" col 4	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1666	+1.802 / +2.029 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 10 row "TC-04" col 5	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1667	+1.800 / +2.027 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 10 row "TC-04" col 6	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1668	+1.800 / +2.027 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 10 row "TC-04" col 7	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1669	-	SIM	MMXMD FEA3D margin matrix v3.5.2	line 10 row "TC-04" col 8	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1670	+1.062 / +0.796 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 11 row "TC-09" col 1	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1671	+1.003 / +0.744 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 11 row "TC-09" col 2	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)

ID	Value	Class	Source	Key	Source tag / formula
T1672	-	SIM	MMXMD FEA3D margin matrix v3.5.2	line 11 row "TC-09" col 3	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1673	+2.053 / +2.636	SIM	MMXMD FEA3D margin matrix v3.5.2	line 11 row "TC-09" col 4	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1674	+1.548 / +2.034	SIM	MMXMD FEA3D margin matrix v3.5.2	line 11 row "TC-09" col 5	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1675	+1.197 / +1.617	SIM	MMXMD FEA3D margin matrix v3.5.2	line 11 row "TC-09" col 6	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1676	+1.151 / +1.563	SIM	MMXMD FEA3D margin matrix v3.5.2	line 11 row "TC-09" col 7	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1677	-	SIM	MMXMD FEA3D margin matrix v3.5.2	line 11 row "TC-09" col 8	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1678	-0.520 / n/a a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 12 row "NZ-01" col 1	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1679	-0.489 / n/a a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 12 row "NZ-01" col 2	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1680	-	SIM	MMXMD FEA3D margin matrix v3.5.2	line 12 row "NZ-01" col 3	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1681	+3.234 / n/a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 12 row "NZ-01" col 4	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1682	+2.134 / n/a a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 12 row "NZ-01" col 5	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1683	+1.504 / n/a a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 12 row "NZ-01" col 6	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1684	+1.476 / n/a r	SIM	MMXMD FEA3D margin matrix v3.5.2	line 12 row "NZ-01" col 7	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1685	-	SIM	MMXMD FEA3D margin matrix v3.5.2	line 12 row "NZ-01" col 8	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1686	+0.595 / n/a a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 13 row "NZ-02" col 1	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1687	+0.596 / n/a a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 13 row "NZ-02" col 2	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1688	-	SIM	MMXMD FEA3D margin matrix v3.5.2	line 13 row "NZ-02" col 3	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1689	+4.107 / n/a a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 13 row "NZ-02" col 4	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1690	+2.967 / n/a a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 13 row "NZ-02" col 5	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1691	+2.265 / n/a a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 13 row "NZ-02" col 6	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1692	+2.178 / n/a a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 13 row "NZ-02" col 7	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1693	-	SIM	MMXMD FEA3D margin matrix v3.5.2	line 13 row "NZ-02" col 8	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1694	+0.445 / n/a r	SIM	MMXMD FEA3D margin matrix v3.5.2	line 14 row "NZ-03" col 1	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1695	-0.088 / n/a r	SIM	MMXMD FEA3D margin matrix v3.5.2	line 14 row "NZ-03" col 2	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1696	-	SIM	MMXMD FEA3D margin matrix v3.5.2	line 14 row "NZ-03" col 3	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1697	+1.615 / n/a r	SIM	MMXMD FEA3D margin matrix v3.5.2	line 14 row "NZ-03" col 4	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1698	+1.032 / n/a r	SIM	MMXMD FEA3D margin matrix v3.5.2	line 14 row "NZ-03" col 5	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1699	+0.672 / n/a r	SIM	MMXMD FEA3D margin matrix v3.5.2	line 14 row "NZ-03" col 6	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1700	+0.628 / n/a r	SIM	MMXMD FEA3D margin matrix v3.5.2	line 14 row "NZ-03" col 7	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1701	-	SIM	MMXMD FEA3D margin matrix v3.5.2	line 14 row "NZ-03" col 8	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1702	+3.584 / n/a a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 15 row "NZ-04" col 1	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1703	+2.897 / n/a a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 15 row "NZ-04" col 2	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1704	-	SIM	MMXMD FEA3D margin matrix v3.5.2	line 15 row "NZ-04" col 3	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1705	+6.534 / n/a a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 15 row "NZ-04" col 4	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1706	+5.282 / n/a a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 15 row "NZ-04" col 5	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1707	+4.415 / n/a a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 15 row "NZ-04" col 6	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1708	+4.301 / n/a a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 15 row "NZ-04" col 7	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1709	-	SIM	MMXMD FEA3D margin matrix v3.5.2	line 15 row "NZ-04" col 8	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1710	+1.227 / +3.422 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 16 row "FD-01" col 1	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1711	+1.197 / +3.362 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 16 row "FD-01" col 2	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1712	-	SIM	MMXMD FEA3D margin matrix v3.5.2	line 16 row "FD-01" col 3	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1713	+3.170 / +9.743 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 16 row "FD-01" col 4	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1714	+3.088 / +9.531 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 16 row "FD-01" col 5	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1715	+3.011 / +9.334 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 16 row "FD-01" col 6	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1716	+3.000 / +9.304 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 16 row "FD-01" col 7	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1717	-	SIM	MMXMD FEA3D margin matrix v3.5.2	line 16 row "FD-01" col 8	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1718	+3.352 / +2.750 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 17 row "GM-01" col 1	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1719	+1.439 / +1.102 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 17 row "GM-01" col 2	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1720	+2.551 / +1.720 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 17 row "GM-01" col 3	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1721	+0.767 / +1.030 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 17 row "GM-01" col 4	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1722	+0.373 / +0.578 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 17 row "GM-01" col 5	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1723	+0.130 / +0.299 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 17 row "GM-01" col 6	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1724	+0.100 / +0.264 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 17 row "GM-01" col 7	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1725	+0.777 / +1.042 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 17 row "GM-01" col 8	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1726	+6.556 / +5.511 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 18 row "GM-02" col 1	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1727	+3.255 / +2.667 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 18 row "GM-02" col 2	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)

ID	Value	Class	Source	Key	Source tag / formula
T1728	+3.585 / +2.512 α	SIM	MMXMD FEA3D margin matrix v3.5.2	line 18 row "GM-02" col 3	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1729	+2.091 / +2.551 α	SIM	MMXMD FEA3D margin matrix v3.5.2	line 18 row "GM-02" col 4	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1730	+1.404 / +1.762 α	SIM	MMXMD FEA3D margin matrix v3.5.2	line 18 row "GM-02" col 5	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1731	+0.980 / +1.275 α	SIM	MMXMD FEA3D margin matrix v3.5.2	line 18 row "GM-02" col 6	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1732	+0.927 / +1.215 α	SIM	MMXMD FEA3D margin matrix v3.5.2	line 18 row "GM-02" col 7	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1733	+2.105 / +2.568 α	SIM	MMXMD FEA3D margin matrix v3.5.2	line 18 row "GM-02" col 8	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1734	+13.611 / +10.787 α	SIM	MMXMD FEA3D margin matrix v3.5.2	line 19 row "GM-03" col 1	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1735	+6.254 / +4.852 α	SIM	MMXMD FEA3D margin matrix v3.5.2	line 19 row "GM-03" col 2	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1736	+8.795 / +6.024 α	SIM	MMXMD FEA3D margin matrix v3.5.2	line 19 row "GM-03" col 3	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1737	+3.585 / +3.932 α	SIM	MMXMD FEA3D margin matrix v3.5.2	line 19 row "GM-03" col 4	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1738	+2.564 / +2.833 α	SIM	MMXMD FEA3D margin matrix v3.5.2	line 19 row "GM-03" col 5	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1739	+1.934 / +2.156 α	SIM	MMXMD FEA3D margin matrix v3.5.2	line 19 row "GM-03" col 6	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1740	+1.856 / +2.072 α	SIM	MMXMD FEA3D margin matrix v3.5.2	line 19 row "GM-03" col 7	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1741	+1.353 / +1.531 α	SIM	MMXMD FEA3D margin matrix v3.5.2	line 19 row "GM-03" col 8	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1742	+61.651 / +49.544 α	SIM	MMXMD FEA3D margin matrix v3.5.2	line 20 row "GM-08a" col 1	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1743	+32.135 / +25.732 α	SIM	MMXMD FEA3D margin matrix v3.5.2	line 20 row "GM-08a" col 2	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1744	+32.080 / +22.722 α	SIM	MMXMD FEA3D margin matrix v3.5.2	line 20 row "GM-08a" col 3	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1745	+21.291 / +22.978 α	SIM	MMXMD FEA3D margin matrix v3.5.2	line 20 row "GM-08a" col 4	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1746	+16.337 / +17.649 α	SIM	MMXMD FEA3D margin matrix v3.5.2	line 20 row "GM-08a" col 5	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1747	+13.281 / +14.362 α	SIM	MMXMD FEA3D margin matrix v3.5.2	line 20 row "GM-08a" col 6	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1748	+12.900 / +13.952 α	SIM	MMXMD FEA3D margin matrix v3.5.2	line 20 row "GM-08a" col 7	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1749	+10.455 / +11.322 α	SIM	MMXMD FEA3D margin matrix v3.5.2	line 20 row "GM-08a" col 8	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1750	+56.638 / +45.500 α	SIM	MMXMD FEA3D margin matrix v3.5.2	line 21 row "GM-08b" col 1	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1751	+28.473 / +22.777 α	SIM	MMXMD FEA3D margin matrix v3.5.2	line 21 row "GM-08b" col 2	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1752	+35.973 / +25.514 α	SIM	MMXMD FEA3D margin matrix v3.5.2	line 21 row "GM-08b" col 3	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1753	+18.126 / +19.573 α	SIM	MMXMD FEA3D margin matrix v3.5.2	line 21 row "GM-08b" col 4	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1754	+13.877 / +15.003 α	SIM	MMXMD FEA3D margin matrix v3.5.2	line 21 row "GM-08b" col 5	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1755	+11.256 / +12.183 α	SIM	MMXMD FEA3D margin matrix v3.5.2	line 21 row "GM-08b" col 6	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1756	+10.929 / +11.832 α	SIM	MMXMD FEA3D margin matrix v3.5.2	line 21 row "GM-08b" col 7	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1757	+8.839 / +9.584 α	SIM	MMXMD FEA3D margin matrix v3.5.2	line 21 row "GM-08b" col 8	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1758	+64.003 / +51.441 α	SIM	MMXMD FEA3D margin matrix v3.5.2	line 22 row "GM-09" col 1	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1759	+31.350 / +25.099 α	SIM	MMXMD FEA3D margin matrix v3.5.2	line 22 row "GM-09" col 2	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1760	+35.853 / +25.428 α	SIM	MMXMD FEA3D margin matrix v3.5.2	line 22 row "GM-09" col 3	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1761	+19.350 / +20.890 α	SIM	MMXMD FEA3D margin matrix v3.5.2	line 22 row "GM-09" col 4	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1762	+14.799 / +15.994 α	SIM	MMXMD FEA3D margin matrix v3.5.2	line 22 row "GM-09" col 5	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1763	+11.999 / +12.983 α	SIM	MMXMD FEA3D margin matrix v3.5.2	line 22 row "GM-09" col 6	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1764	+11.651 / +12.608 α	SIM	MMXMD FEA3D margin matrix v3.5.2	line 22 row "GM-09" col 7	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1765	+9.410 / +10.198 α	SIM	MMXMD FEA3D margin matrix v3.5.2	line 22 row "GM-09" col 8	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1766	+89.997 / +72.412	SIM	MMXMD FEA3D margin matrix v3.5.2	line 23 row "GM-10" col 1	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1767	+46.871 / +37.620	SIM	MMXMD FEA3D margin matrix v3.5.2	line 23 row "GM-10" col 2	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1768	+22.208 / +15.643	SIM	MMXMD FEA3D margin matrix v3.5.2	line 23 row "GM-10" col 3	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1769	+30.505 / +32.889	SIM	MMXMD FEA3D margin matrix v3.5.2	line 23 row "GM-10" col 4	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1770	+23.597 / +25.458	SIM	MMXMD FEA3D margin matrix v3.5.2	line 23 row "GM-10" col 5	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1771	+19.309 / +20.846	SIM	MMXMD FEA3D margin matrix v3.5.2	line 23 row "GM-10" col 6	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1772	+18.773 / +20.270	SIM	MMXMD FEA3D margin matrix v3.5.2	line 23 row "GM-10" col 7	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1773	+19.238 / +20.769 α	SIM	MMXMD FEA3D margin matrix v3.5.2	line 23 row "GM-10" col 8	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1774	+67.053 / +53.901 α	SIM	MMXMD FEA3D margin matrix v3.5.2	line 24 row "GM-13" col 1	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1775	+34.896 / +27.959 α	SIM	MMXMD FEA3D margin matrix v3.5.2	line 24 row "GM-13" col 2	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1776	+15.399 / +10.760 α	SIM	MMXMD FEA3D margin matrix v3.5.2	line 24 row "GM-13" col 3	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1777	+22.696 / +24.490 α	SIM	MMXMD FEA3D margin matrix v3.5.2	line 24 row "GM-13" col 4	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1778	+17.500 / +18.900 α	SIM	MMXMD FEA3D margin matrix v3.5.2	line 24 row "GM-13" col 5	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1779	+14.274 / +15.430 α	SIM	MMXMD FEA3D margin matrix v3.5.2	line 24 row "GM-13" col 6	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1780	+13.872 / +14.997 α	SIM	MMXMD FEA3D margin matrix v3.5.2	line 24 row "GM-13" col 7	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1781	+22.434 / +24.208 α	SIM	MMXMD FEA3D margin matrix v3.5.2	line 24 row "GM-13" col 8	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1782	>+99 / >+99 α	SIM	MMXMD FEA3D margin matrix v3.5.2	line 25 row "GM-14" col 1	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1783	>+99 / >+99 α	SIM	MMXMD FEA3D margin matrix v3.5.2	line 25 row "GM-14" col 2	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)

ID	Value	Class	Source	Key	Source tag / formula
T1784	+18.968 / +13.319 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 25 row "GM-14" col 3	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1785	>+99 / >+99 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 25 row "GM-14" col 4	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1786	>+99 / >+99 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 25 row "GM-14" col 5	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1787	>+99 / >+99 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 25 row "GM-14" col 6	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1788	>+99 / >+99 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 25 row "GM-14" col 7	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1789	>+99 / >+99 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 25 row "GM-14" col 8	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1790	+29.165 / +24.993 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 26 row "GM-16a" col 1	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1791	+15.354 / +13.092 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 26 row "GM-16a" col 2	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1792	+5.523 / +3.996 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 26 row "GM-16a" col 3	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1793	+10.140 / +11.799 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 26 row "GM-16a" col 4	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1794	+7.697 / +8.993 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 26 row "GM-16a" col 5	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1795	+6.181 / +7.251 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 26 row "GM-16a" col 6	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1796	+5.992 / +7.033 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 26 row "GM-16a" col 7	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1797	+10.140 / +11.799 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 26 row "GM-16a" col 8	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1798	>+99 / +89.126 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 27 row "GM-16b" col 1	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1799	+89.402 / +76.900 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 27 row "GM-16b" col 2	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1800	+8.123 / +5.988 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 27 row "GM-16b" col 3	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1801	>+99 / >+99 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 27 row "GM-16b" col 4	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1802	>+99 / >+99 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 27 row "GM-16b" col 5	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1803	>+99 / >+99 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 27 row "GM-16b" col 6	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1804	>+99 / >+99 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 27 row "GM-16b" col 7	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1805	>+99 / >+99 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 27 row "GM-16b" col 8	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1806	+1.886 / +1.338 r	SIM	MMXMD FEA3D margin matrix v3.5.2	line 28 row "PD-01R" col 1	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1807	+1.862 / +1.318 r	SIM	MMXMD FEA3D margin matrix v3.5.2	line 28 row "PD-01R" col 2	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1808	-	SIM	MMXMD FEA3D margin matrix v3.5.2	line 28 row "PD-01R" col 3	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1809	+4.444 / +4.883 r	SIM	MMXMD FEA3D margin matrix v3.5.2	line 28 row "PD-01R" col 4	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1810	+4.207 / +4.628 r	SIM	MMXMD FEA3D margin matrix v3.5.2	line 28 row "PD-01R" col 5	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1811	+2.827 / +3.136	SIM	MMXMD FEA3D margin matrix v3.5.2	line 28 row "PD-01R" col 6	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1812	+2.670 / +2.966	SIM	MMXMD FEA3D margin matrix v3.5.2	line 28 row "PD-01R" col 7	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1813	-	SIM	MMXMD FEA3D margin matrix v3.5.2	line 28 row "PD-01R" col 8	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1814	+1.590 / +1.097 r	SIM	MMXMD FEA3D margin matrix v3.5.2	line 29 row "PD-01S" col 1	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1815	+1.590 / +1.097 r	SIM	MMXMD FEA3D margin matrix v3.5.2	line 29 row "PD-01S" col 2	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1816	-	SIM	MMXMD FEA3D margin matrix v3.5.2	line 29 row "PD-01S" col 3	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1817	+4.273 / +4.699 r	SIM	MMXMD FEA3D margin matrix v3.5.2	line 29 row "PD-01S" col 4	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1818	+4.008 / +4.412 r	SIM	MMXMD FEA3D margin matrix v3.5.2	line 29 row "PD-01S" col 5	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1819	+2.850 / +3.161 r	SIM	MMXMD FEA3D margin matrix v3.5.2	line 29 row "PD-01S" col 6	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1820	+2.698 / +2.997 r	SIM	MMXMD FEA3D margin matrix v3.5.2	line 29 row "PD-01S" col 7	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1821	-	SIM	MMXMD FEA3D margin matrix v3.5.2	line 29 row "PD-01S" col 8	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1822	-	SIM	MMXMD FEA3D margin matrix v3.5.2	line 30 row "HW-101 (J-FWD)" col 1	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1823	-	SIM	MMXMD FEA3D margin matrix v3.5.2	line 30 row "HW-101 (J-FWD)" col 2	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1824	-	SIM	MMXMD FEA3D margin matrix v3.5.2	line 30 row "HW-101 (J-FWD)" col 3	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1825	-	SIM	MMXMD FEA3D margin matrix v3.5.2	line 30 row "HW-101 (J-FWD)" col 4	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1826	-	SIM	MMXMD FEA3D margin matrix v3.5.2	line 30 row "HW-101 (J-FWD)" col 5	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1827	-	SIM	MMXMD FEA3D margin matrix v3.5.2	line 30 row "HW-101 (J-FWD)" col 6	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1828	-	SIM	MMXMD FEA3D margin matrix v3.5.2	line 30 row "HW-101 (J-FWD)" col 7	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1829	-	SIM	MMXMD FEA3D margin matrix v3.5.2	line 30 row "HW-101 (J-FWD)" col 8	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1830	-	SIM	MMXMD FEA3D margin matrix v3.5.2	line 31 row "HW-103 (J-FD01)" col 1	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1831	-	SIM	MMXMD FEA3D margin matrix v3.5.2	line 31 row "HW-103 (J-FD01)" col 2	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1832	-	SIM	MMXMD FEA3D margin matrix v3.5.2	line 31 row "HW-103 (J-FD01)" col 3	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1833	-	SIM	MMXMD FEA3D margin matrix v3.5.2	line 31 row "HW-103 (J-FD01)" col 4	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1834	-	SIM	MMXMD FEA3D margin matrix v3.5.2	line 31 row "HW-103 (J-FD01)" col 5	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1835	-	SIM	MMXMD FEA3D margin matrix v3.5.2	line 31 row "HW-103 (J-FD01)" col 6	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1836	-	SIM	MMXMD FEA3D margin matrix v3.5.2	line 31 row "HW-103 (J-FD01)" col 7	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1837	-	SIM	MMXMD FEA3D margin matrix v3.5.2	line 31 row "HW-103 (J-FD01)" col 8	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1838	-	SIM	MMXMD FEA3D margin matrix v3.5.2	line 32 row "HW-127 (J-FD09)" col 1	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1839	-	SIM	MMXMD FEA3D margin matrix v3.5.2	line 32 row "HW-127 (J-FD09)" col 2	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)

ID	Value	Class	Source	Key	Source tag / formula
T1840	-	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 32 row "HW-127 (J-FD09)" col 3`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1841	-	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 32 row "HW-127 (J-FD09)" col 4`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1842	-	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 32 row "HW-127 (J-FD09)" col 5`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1843	-	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 32 row "HW-127 (J-FD09)" col 6`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1844	-	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 32 row "HW-127 (J-FD09)" col 7`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1845	-	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 32 row "HW-127 (J-FD09)" col 8`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1846	+1.044 / +0.657 a	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 7 row "TC-01" col 9`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1847	+0.890 / +0.528 r	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 7 row "TC-01" col 10`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1848	+1.080 / +0.686 a	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 7 row "TC-01" col 11`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1849	+1.118 / +0.717 a	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 7 row "TC-01" col 12`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1850	+30.584 / +24.649 a	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 7 row "TC-01" col 13`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1851	+0.877 / +0.518 r	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 7 row "TC-01" col 14`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1852	-0.356 / -0.482 r	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 7 row "TC-01" col 15`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1853	+7.561 / +5.883 r	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 8 row "TC-02" col 9`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1854	+0.504 / +0.190 r	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 8 row "TC-02" col 10`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1855	+7.254 / +5.637 r	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 8 row "TC-02" col 11`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1856	+7.689 / +5.986 r	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 8 row "TC-02" col 12`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1857	>+99 / >+99 r	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 8 row "TC-02" col 13`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1858	+0.493 / +0.181 r	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 8 row "TC-02" col 14`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1859	+1.068 / +0.656 a	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 8 row "TC-02" col 15`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1860	+25.346 / +22.957 a	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 9 row "TC-03" col 9`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1861	-0.107 / -0.188 r	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 9 row "TC-03" col 10`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1862	+25.274 / +22.892 a	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 9 row "TC-03" col 11`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1863	+25.529 / +23.124 a	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 9 row "TC-03" col 12`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1864	>+99 / >+99 a	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 9 row "TC-03" col 13`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1865	-0.107 / -0.188 r	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 9 row "TC-03" col 14`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1866	+0.992 / +0.811 r	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 9 row "TC-03" col 15`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1867	>+99 / >+99 a	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 10 row "TC-04" col 9`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1868	+0.228 / -0.004 as BT	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 10 row "TC-04" col 10`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1869	>+99 / +92.532 a	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 10 row "TC-04" col 11`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1870	>+99 / >+99 a	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 10 row "TC-04" col 12`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1871	>+99 / >+99 a	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 10 row "TC-04" col 13`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1872	+0.223 / -0.008 as BT	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 10 row "TC-04" col 14`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1873	+6.477 / +5.017 a	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 10 row "TC-04" col 15`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1874	+10.348 / +9.106 a	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 11 row "TC-09" col 9`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1875	+1.031 / +0.774 a	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 11 row "TC-09" col 10`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1876	+10.456 / +9.202 a	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 11 row "TC-09" col 11`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1877	+10.545 / +9.281 a	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 11 row "TC-09" col 12`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1878	>+99 / >+99 a	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 11 row "TC-09" col 13`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1879	+1.091 / +0.826 aw (part-wide; stored +1.092 / +0.827)	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 11 row "TC-09" col 14`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1880	+1.075 / +0.705	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 11 row "TC-09" col 15`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1881	+4.804 / n/a r	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 12 row "NZ-01" col 9`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1882	-0.403 / n/a aw (part-wide; stored -0.402 / n/a)	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 12 row "NZ-01" col 10`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1883	+4.860 / n/a r	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 12 row "NZ-01" col 11`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1884	+4.866 / n/a r	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 12 row "NZ-01" col 12`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1885	>+99 / n/a r	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 12 row "NZ-01" col 13`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1886	-0.401 / n/a	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 12 row "NZ-01" col 14`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1887	-0.426 / n/a r	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 12 row "NZ-01" col 15`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1888	+1.559 / n/a a	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 13 row "NZ-02" col 9`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1889	+0.781 / n/a a	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 13 row "NZ-02" col 10`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1890	+1.581 / n/a a	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 13 row "NZ-02" col 11`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1891	+1.581 / n/a a	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 13 row "NZ-02" col 12`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1892	>+99 / n/a a	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 13 row "NZ-02" col 13`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1893	+0.784 / n/a as	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 13 row "NZ-02" col 14`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1894	>+99 / n/a a	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 13 row "NZ-02" col 15`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1895	+0.179 / n/a r	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 14 row "NZ-03" col 9`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)

ID	Value	Class	Source	Key	Source tag / formula
T1896	+0.243 / n/a r	SIM	MMXMD FEA3D margin matrix v3.5.2	line 14 row "NZ-03" col 10	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1897	+0.187 / n/a r	SIM	MMXMD FEA3D margin matrix v3.5.2	line 14 row "NZ-03" col 11	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1898	+0.190 / n/a r	SIM	MMXMD FEA3D margin matrix v3.5.2	line 14 row "NZ-03" col 12	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1899	>+99 / n/a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 14 row "NZ-03" col 13	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1900	+0.252 / n/a r	SIM	MMXMD FEA3D margin matrix v3.5.2	line 14 row "NZ-03" col 14	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1901	>+99 / n/a r	SIM	MMXMD FEA3D margin matrix v3.5.2	line 14 row "NZ-03" col 15	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1902	+13.606 / n/a a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 15 row "NZ-04" col 9	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1903	+3.004 / n/a a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 15 row "NZ-04" col 10	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1904	+13.685 / n/a a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 15 row "NZ-04" col 11	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1905	+13.702 / n/a a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 15 row "NZ-04" col 12	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1906	>+99 / n/a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 15 row "NZ-04" col 13	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1907	+3.007 / n/a as	SIM	MMXMD FEA3D margin matrix v3.5.2	line 15 row "NZ-04" col 14	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1908	+6.093 / n/a a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 15 row "NZ-04" col 15	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1909	+33.700 / +66.045 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 16 row "FD-01" col 9	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1910	+0.837 / +2.620 aw (part-wide; stored +0.848 / +2.640)	SIM	MMXMD FEA3D margin matrix v3.5.2	line 16 row "FD-01" col 10	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1911	+35.220 / +68.982 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 16 row "FD-01" col 11	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1912	+32.090 / +62.934 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 16 row "FD-01" col 12	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1913	+85.170 / >+99 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 16 row "FD-01" col 13	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1914	+0.830 / +2.606 aw (part-wide; stored +0.840 / +2.625)	SIM	MMXMD FEA3D margin matrix v3.5.2	line 16 row "FD-01" col 14	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1915	-0.187 / +0.780 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 16 row "FD-01" col 15	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1916	+0.534 / +0.322	SIM	MMXMD FEA3D margin matrix v3.5.2	line 17 row "GM-01" col 9	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1917	+0.220 / +0.052	SIM	MMXMD FEA3D margin matrix v3.5.2	line 17 row "GM-01" col 10	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1918	+0.648 / +0.420	SIM	MMXMD FEA3D margin matrix v3.5.2	line 17 row "GM-01" col 11	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1919	+2.660 / +2.154 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 17 row "GM-01" col 12	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1920	+42.346 / +36.351	SIM	MMXMD FEA3D margin matrix v3.5.2	line 17 row "GM-01" col 13	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1921	+0.289 / +0.111	SIM	MMXMD FEA3D margin matrix v3.5.2	line 17 row "GM-01" col 14	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1922	+0.166 / +0.005 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 17 row "GM-01" col 15	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1923	+0.400 / +0.206	SIM	MMXMD FEA3D margin matrix v3.5.2	line 18 row "GM-02" col 9	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1924	+0.202 / +0.036	SIM	MMXMD FEA3D margin matrix v3.5.2	line 18 row "GM-02" col 10	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1925	+0.591 / +0.371	SIM	MMXMD FEA3D margin matrix v3.5.2	line 18 row "GM-02" col 11	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1926	+3.643 / +3.001 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 18 row "GM-02" col 12	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1927	+57.337 / +49.269	SIM	MMXMD FEA3D margin matrix v3.5.2	line 18 row "GM-02" col 13	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1928	+0.337 / +0.152	SIM	MMXMD FEA3D margin matrix v3.5.2	line 18 row "GM-02" col 14	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1929	+0.200 / +0.034 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 18 row "GM-02" col 15	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1930	+1.227 / +0.796 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 19 row "GM-03" col 9	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1931	+1.225 / +0.795 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 19 row "GM-03" col 10	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1932	+1.893 / +1.334 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 19 row "GM-03" col 11	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1933	+4.793 / +3.674 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 19 row "GM-03" col 12	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1934	+11.646 / +9.202 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 19 row "GM-03" col 13	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1935	+1.881 / +1.324 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 19 row "GM-03" col 14	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1936	+13.300 / +10.536 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 19 row "GM-03" col 15	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1937	+5.528 / +4.266 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 20 row "GM-08a" col 9	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1938	+5.523 / +4.262 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 20 row "GM-08a" col 10	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1939	+12.101 / +9.569 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 20 row "GM-08a" col 11	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1940	+15.110 / +11.997 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 20 row "GM-08a" col 12	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1941	+37.240 / +29.850 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 20 row "GM-08a" col 13	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1942	+12.079 / +9.551 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 20 row "GM-08a" col 14	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1943	+37.360 / +29.947 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 20 row "GM-08a" col 15	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1944	+10.694 / +8.434 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 21 row "GM-08b" col 9	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1945	+10.667 / +8.412 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 21 row "GM-08b" col 10	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1946	+10.525 / +8.298 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 21 row "GM-08b" col 11	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1947	+22.858 / +18.247 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 21 row "GM-08b" col 12	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1948	+49.715 / +39.914 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 21 row "GM-08b" col 13	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1949	+10.499 / +8.277 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 21 row "GM-08b" col 14	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1950	+49.877 / +40.045 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 21 row "GM-08b" col 15	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1951	+5.134 / +3.948 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 22 row "GM-09" col 9	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)

ID	Value	Class	Source	Key	Source tag / formula
T1952	+5.131 / +3.946 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 22 row "GM-09" col 10	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1953	+12.056 / +9.533 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 22 row "GM-09" col 11	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1954	+14.319 / +11.359 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 22 row "GM-09" col 12	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1955	+42.664 / +34.226 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 22 row "GM-09" col 13	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1956	+11.963 / +9.458 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 22 row "GM-09" col 14	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1957	+43.439 / +34.851 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 22 row "GM-09" col 15	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1958	+3.708 / +2.798	SIM	MMXMD FEA3D margin matrix v3.5.2	line 23 row "GM-10" col 9	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1959	+3.646 / +2.748	SIM	MMXMD FEA3D margin matrix v3.5.2	line 23 row "GM-10" col 10	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1960	+3.618 / +2.726	SIM	MMXMD FEA3D margin matrix v3.5.2	line 23 row "GM-10" col 11	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1961	+14.128 / +11.205	SIM	MMXMD FEA3D margin matrix v3.5.2	line 23 row "GM-10" col 12	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1962	+76.758 / +61.732 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 23 row "GM-10" col 13	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1963	+3.596 / +2.708	SIM	MMXMD FEA3D margin matrix v3.5.2	line 23 row "GM-10" col 14	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1964	+77.024 / +61.946 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 23 row "GM-10" col 15	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1965	+2.418 / +1.757 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 24 row "GM-13" col 9	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1966	+2.375 / +1.723 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 24 row "GM-13" col 10	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1967	+2.578 / +1.887 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 24 row "GM-13" col 11	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1968	+9.510 / +7.479 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 24 row "GM-13" col 12	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1969	+61.967 / +49.799 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 24 row "GM-13" col 13	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1970	+2.531 / +1.849 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 24 row "GM-13" col 14	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1971	+76.436 / +61.471 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 24 row "GM-13" col 15	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1972	+2.434 / +1.771 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 25 row "GM-14" col 9	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1973	+2.418 / +1.758 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 25 row "GM-14" col 10	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1974	+2.509 / +1.831 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 25 row "GM-14" col 11	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1975	+10.565 / +8.330 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 25 row "GM-14" col 12	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1976	+77.419 / +62.264 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 25 row "GM-14" col 13	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1977	+2.492 / +1.817 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 25 row "GM-14" col 14	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1978	>+99 / +81.872 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 25 row "GM-14" col 15	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1979	+0.372 / +0.183 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 26 row "GM-16a" col 9	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1980	+0.369 / +0.179 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 26 row "GM-16a" col 10	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1981	+0.568 / +0.351 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 26 row "GM-16a" col 11	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1982	+3.197 / +2.616 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 26 row "GM-16a" col 12	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1983	+29.491 / +25.274 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 26 row "GM-16a" col 13	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1984	+0.548 / +0.334 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 26 row "GM-16a" col 14	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1985	+47.292 / +40.613 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 26 row "GM-16a" col 15	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1986	+0.450 / +0.250 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 27 row "GM-16b" col 9	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1987	+0.449 / +0.248 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 27 row "GM-16b" col 10	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1988	+0.531 / +0.319 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 27 row "GM-16b" col 11	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1989	+4.299 / +3.566 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 27 row "GM-16b" col 12	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1990	+45.191 / +38.802 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 27 row "GM-16b" col 13	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1991	+0.524 / +0.313 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 27 row "GM-16b" col 14	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1992	+71.675 / +61.624 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 27 row "GM-16b" col 15	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1993	+4.237 / +3.243	SIM	MMXMD FEA3D margin matrix v3.5.2	line 28 row "PD-01R" col 9	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1994	+1.477 / +1.006 r	SIM	MMXMD FEA3D margin matrix v3.5.2	line 28 row "PD-01R" col 10	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1995	+2.484 / +1.823	SIM	MMXMD FEA3D margin matrix v3.5.2	line 28 row "PD-01R" col 11	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1996	+3.956 / +3.015	SIM	MMXMD FEA3D margin matrix v3.5.2	line 28 row "PD-01R" col 12	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1997	+20.413 / +16.349	SIM	MMXMD FEA3D margin matrix v3.5.2	line 28 row "PD-01R" col 13	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1998	+0.981 / +0.604 r	SIM	MMXMD FEA3D margin matrix v3.5.2	line 28 row "PD-01R" col 14	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T1999	-0.315 / -0.451 r	SIM	MMXMD FEA3D margin matrix v3.5.2	line 28 row "PD-01R" col 15	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2000	+5.864 / +4.562	SIM	MMXMD FEA3D margin matrix v3.5.2	line 29 row "PD-01S" col 9	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2001	+1.323 / +0.881 r	SIM	MMXMD FEA3D margin matrix v3.5.2	line 29 row "PD-01S" col 10	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2002	+3.513 / +2.657	SIM	MMXMD FEA3D margin matrix v3.5.2	line 29 row "PD-01S" col 11	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2003	+5.515 / +4.279	SIM	MMXMD FEA3D margin matrix v3.5.2	line 29 row "PD-01S" col 12	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2004	+21.487 / +17.219	SIM	MMXMD FEA3D margin matrix v3.5.2	line 29 row "PD-01S" col 13	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2005	+0.971 / +0.596 r	SIM	MMXMD FEA3D margin matrix v3.5.2	line 29 row "PD-01S" col 14	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2006	-0.378 / -0.502 r	SIM	MMXMD FEA3D margin matrix v3.5.2	line 29 row "PD-01S" col 15	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2007	-	SIM	MMXMD FEA3D margin matrix v3.5.2	line 30 row "HW-101 (J-FWD)" col 9	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)

ID	Value	Class	Source	Key	Source tag / formula
T2008	-	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 30 row "HW-101 (J-FWD)" col 10`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2009	-	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 30 row "HW-101 (J-FWD)" col 11`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2010	-	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 30 row "HW-101 (J-FWD)" col 12`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2011	-	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 30 row "HW-101 (J-FWD)" col 13`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2012	-	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 30 row "HW-101 (J-FWD)" col 14`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2013	-	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 30 row "HW-101 (J-FWD)" col 15`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2014	-	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 31 row "HW-103 (J-FD01)" col 9`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2015	-	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 31 row "HW-103 (J-FD01)" col 10`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2016	-	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 31 row "HW-103 (J-FD01)" col 11`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2017	-	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 31 row "HW-103 (J-FD01)" col 12`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2018	-	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 31 row "HW-103 (J-FD01)" col 13`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2019	-	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 31 row "HW-103 (J-FD01)" col 14`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2020	-	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 31 row "HW-103 (J-FD01)" col 15`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2021	-	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 32 row "HW-127 (J-FD09)" col 9`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2022	-	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 32 row "HW-127 (J-FD09)" col 10`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2023	-	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 32 row "HW-127 (J-FD09)" col 11`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2024	-	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 32 row "HW-127 (J-FD09)" col 12`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2025	-	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 32 row "HW-127 (J-FD09)" col 13`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2026	-	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 32 row "HW-127 (J-FD09)" col 14`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2027	-	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 32 row "HW-127 (J-FD09)" col 15`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2028	+0.120 / -0.110 r	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 7 row "TC-01" col 16`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2029	+0.570 / +0.109 r	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 7 row "TC-01" col 17`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2030	-0.075 / +0.239	CALC	MMXMD FEA3D margin matrix v3.5.2	`line 7 row "TC-01" col 18`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); J-FWD cell classes (margin_matrix_v352.json): governing[0] SOAK (MS_y -0.075): CALC / ...
T2031	-	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 7 row "TC-01" col 19`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2032	-	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 7 row "TC-01" col 20`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2033	-0.482 (LC6 soak u)	DERIVED	MMXMD FEA3D margin matrix v3.5.2	`line 7 row "TC-01" col 21`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2034	-0.236 / -0.271 r	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 8 row "TC-02" col 16`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2035	+0.138 / -0.035 r	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 8 row "TC-02" col 17`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2036	+0.045 / +0.162	CALC	MMXMD FEA3D margin matrix v3.5.2	`line 8 row "TC-02" col 18`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); J-FWD cell classes (margin_matrix_v352.json): governing[0] SOAK52 (MS_y +0.045; MS_u ...
T2037	+0.404 / n/a	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 8 row "TC-02" col 19`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2038	-	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 8 row "TC-02" col 20`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2039	-0.271 (TB70 u)	DERIVED	MMXMD FEA3D margin matrix v3.5.2	`line 8 row "TC-02" col 21`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2040	-0.121 / -0.201 r	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 9 row "TC-03" col 16`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2041	+0.325 / +0.071 r	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 9 row "TC-03" col 17`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2042	-	CALC	MMXMD FEA3D margin matrix v3.5.2	`line 9 row "TC-03" col 18`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2043	-	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 9 row "TC-03" col 19`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2044	-	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 9 row "TC-03" col 20`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2045	-0.201 (TB70 u)	DERIVED	MMXMD FEA3D margin matrix v3.5.2	`line 9 row "TC-03" col 21`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2046	+0.401 / +0.122 a	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 10 row "TC-04" col 16`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2047	+1.102 / +0.496 a	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 10 row "TC-04" col 17`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2048	-	CALC	MMXMD FEA3D margin matrix v3.5.2	`line 10 row "TC-04" col 18`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2049	-	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 10 row "TC-04" col 19`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2050	-	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 10 row "TC-04" col 20`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2051	-0.008 (F2 hot bound u)	DERIVED	MMXMD FEA3D margin matrix v3.5.2	`line 10 row "TC-04" col 21`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2052	+0.861 / +0.604 a	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 11 row "TC-09" col 16`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2053	+1.763 / +1.117 a	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 11 row "TC-09" col 17`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2054	-	CALC	MMXMD FEA3D margin matrix v3.5.2	`line 11 row "TC-09" col 18`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2055	-	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 11 row "TC-09" col 19`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2056	-	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 11 row "TC-09" col 20`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2057	+0.604 (TB70 u)	DERIVED	MMXMD FEA3D margin matrix v3.5.2	`line 11 row "TC-09" col 21`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2058	-0.315 / n/a a	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 12 row "NZ-01" col 16`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2059	-0.054 / n/a a	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 12 row "NZ-01" col 17`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2060	-	CALC	MMXMD FEA3D margin matrix v3.5.2	`line 12 row "NZ-01" col 18`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2061	-	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 12 row "NZ-01" col 19`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)

ID	Value	Class	Source	Key	Source tag / formula
T2062	-	SIM	MMXMD FEA3D margin matrix v3.5.2	line 12 row "NZ-01" col 20	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2063	-0.520 (LC1 y)	DERIVED	MMXMD FEA3D margin matrix v3.5.2	line 12 row "NZ-01" col 21	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2064	+0.356 / n/a a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 13 row "NZ-02" col 16	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2065	+0.665 / n/a a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 13 row "NZ-02" col 17	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2066	-	CALC	MMXMD FEA3D margin matrix v3.5.2	line 13 row "NZ-02" col 18	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2067	-	SIM	MMXMD FEA3D margin matrix v3.5.2	line 13 row "NZ-02" col 19	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2068	-	SIM	MMXMD FEA3D margin matrix v3.5.2	line 13 row "NZ-02" col 20	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2069	+0.356 (TB70 y)	DERIVED	MMXMD FEA3D margin matrix v3.5.2	line 13 row "NZ-02" col 21	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2070	-0.392 / n/a r	SIM	MMXMD FEA3D margin matrix v3.5.2	line 14 row "NZ-03" col 16	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2071	-0.253 / n/a r	SIM	MMXMD FEA3D margin matrix v3.5.2	line 14 row "NZ-03" col 17	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2072	-	CALC	MMXMD FEA3D margin matrix v3.5.2	line 14 row "NZ-03" col 18	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2073	-	SIM	MMXMD FEA3D margin matrix v3.5.2	line 14 row "NZ-03" col 19	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2074	-	SIM	MMXMD FEA3D margin matrix v3.5.2	line 14 row "NZ-03" col 20	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2075	-0.392 (TB70 y)	DERIVED	MMXMD FEA3D margin matrix v3.5.2	line 14 row "NZ-03" col 21	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2076	+2.265 / n/a a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 15 row "NZ-04" col 16	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2077	+3.195 / n/a a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 15 row "NZ-04" col 17	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2078	-	CALC	MMXMD FEA3D margin matrix v3.5.2	line 15 row "NZ-04" col 18	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2079	-	SIM	MMXMD FEA3D margin matrix v3.5.2	line 15 row "NZ-04" col 19	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2080	-	SIM	MMXMD FEA3D margin matrix v3.5.2	line 15 row "NZ-04" col 20	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2081	+2.265 (TB70 y)	DERIVED	MMXMD FEA3D margin matrix v3.5.2	line 15 row "NZ-04" col 21	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2082	-0.336 / +0.484 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 16 row "FD-01" col 16	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2083	-0.005 / +0.976 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 16 row "FD-01" col 17	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2084	-	CALC	MMXMD FEA3D margin matrix v3.5.2	line 16 row "FD-01" col 18	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2085	-	SIM	MMXMD FEA3D margin matrix v3.5.2	line 16 row "FD-01" col 19	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2086	-	SIM	MMXMD FEA3D margin matrix v3.5.2	line 16 row "FD-01" col 20	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2087	-0.336 (TB70 y)	DERIVED	MMXMD FEA3D margin matrix v3.5.2	line 16 row "FD-01" col 21	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2088	+0.291 / +0.113 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 17 row "GM-01" col 16	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2089	+0.936 / +0.483 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 17 row "GM-01" col 17	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2090	-	CALC	MMXMD FEA3D margin matrix v3.5.2	line 17 row "GM-01" col 18	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2091	-	SIM	MMXMD FEA3D margin matrix v3.5.2	line 17 row "GM-01" col 19	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2092	-	SIM	MMXMD FEA3D margin matrix v3.5.2	line 17 row "GM-01" col 20	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2093	+0.005 (LC6 soak u)	DERIVED	MMXMD FEA3D margin matrix v3.5.2	line 17 row "GM-01" col 21	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2094	+0.241 / +0.069 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 18 row "GM-02" col 16	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2095	+0.861 / +0.425 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 18 row "GM-02" col 17	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2096	-	CALC	MMXMD FEA3D margin matrix v3.5.2	line 18 row "GM-02" col 18	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2097	-	SIM	MMXMD FEA3D margin matrix v3.5.2	line 18 row "GM-02" col 19	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2098	-	SIM	MMXMD FEA3D margin matrix v3.5.2	line 18 row "GM-02" col 20	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2099	+0.034 (LC6 soak u)	DERIVED	MMXMD FEA3D margin matrix v3.5.2	line 18 row "GM-02" col 21	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2100	+72.008 / +57.900 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 19 row "GM-03" col 16	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2101	>+99 / +71.545 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 19 row "GM-03" col 17	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2102	-	CALC	MMXMD FEA3D margin matrix v3.5.2	line 19 row "GM-03" col 18	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2103	-	SIM	MMXMD FEA3D margin matrix v3.5.2	line 19 row "GM-03" col 19	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2104	-	SIM	MMXMD FEA3D margin matrix v3.5.2	line 19 row "GM-03" col 20	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2105	+0.795 (LC3 hot bound u)	DERIVED	MMXMD FEA3D margin matrix v3.5.2	line 19 row "GM-03" col 21	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2106	>+99 / >+99 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 20 row "GM-08a" col 16	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2107	>+99 / >+99 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 20 row "GM-08a" col 17	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2108	-	CALC	MMXMD FEA3D margin matrix v3.5.2	line 20 row "GM-08a" col 18	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2109	-	SIM	MMXMD FEA3D margin matrix v3.5.2	line 20 row "GM-08a" col 19	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2110	-	SIM	MMXMD FEA3D margin matrix v3.5.2	line 20 row "GM-08a" col 20	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2111	+4.262 (LC3 hot bound u)	DERIVED	MMXMD FEA3D margin matrix v3.5.2	line 20 row "GM-08a" col 21	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2112	>+99 / >+99 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 21 row "GM-08b" col 16	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2113	>+99 / >+99 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 21 row "GM-08b" col 17	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2114	-	CALC	MMXMD FEA3D margin matrix v3.5.2	line 21 row "GM-08b" col 18	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2115	-	SIM	MMXMD FEA3D margin matrix v3.5.2	line 21 row "GM-08b" col 19	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2116	-	SIM	MMXMD FEA3D margin matrix v3.5.2	line 21 row "GM-08b" col 20	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2117	+8.277 (F2 hot bound u)	DERIVED	MMXMD FEA3D margin matrix v3.5.2	line 21 row "GM-08b" col 21	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)

ID	Value	Class	Source	Key	Source tag / formula
T2118	>+99 / >+99 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 22 row "GM-09" col 16	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2119	>+99 / >+99 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 22 row "GM-09" col 17	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2120	-	CALC	MMXMD FEA3D margin matrix v3.5.2	line 22 row "GM-09" col 18	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2121	-	SIM	MMXMD FEA3D margin matrix v3.5.2	line 22 row "GM-09" col 19	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2122	+0.626 / n/a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 22 row "GM-09" col 20	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2123	+0.626 (J-FD09 y)	DERIVED	MMXMD FEA3D margin matrix v3.5.2	line 22 row "GM-09" col 21	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2124	+32.616 / +26.119	SIM	MMXMD FEA3D margin matrix v3.5.2	line 23 row "GM-10" col 16	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2125	+45.559 / +32.388	SIM	MMXMD FEA3D margin matrix v3.5.2	line 23 row "GM-10" col 17	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2126	-	CALC	MMXMD FEA3D margin matrix v3.5.2	line 23 row "GM-10" col 18	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2127	-	SIM	MMXMD FEA3D margin matrix v3.5.2	line 23 row "GM-10" col 19	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2128	-	SIM	MMXMD FEA3D margin matrix v3.5.2	line 23 row "GM-10" col 20	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2129	+2.708 (F2 hot bound u)	DERIVED	MMXMD FEA3D margin matrix v3.5.2	line 23 row "GM-10" col 21	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2130	+24.422 / +19.509 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 24 row "GM-13" col 16	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2131	+34.211 / +24.250 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 24 row "GM-13" col 17	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2132	-	CALC	MMXMD FEA3D margin matrix v3.5.2	line 24 row "GM-13" col 18	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2133	-	SIM	MMXMD FEA3D margin matrix v3.5.2	line 24 row "GM-13" col 19	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2134	-	SIM	MMXMD FEA3D margin matrix v3.5.2	line 24 row "GM-13" col 20	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2135	+1.723 (LC3 hot bound u)	DERIVED	MMXMD FEA3D margin matrix v3.5.2	line 24 row "GM-13" col 21	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2136	>+99 / >+99 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 25 row "GM-14" col 16	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2137	>+99 / >+99 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 25 row "GM-14" col 17	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2138	-	CALC	MMXMD FEA3D margin matrix v3.5.2	line 25 row "GM-14" col 18	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2139	-	SIM	MMXMD FEA3D margin matrix v3.5.2	line 25 row "GM-14" col 19	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2140	-	SIM	MMXMD FEA3D margin matrix v3.5.2	line 25 row "GM-14" col 20	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2141	+1.758 (LC3 hot bound u)	DERIVED	MMXMD FEA3D margin matrix v3.5.2	line 25 row "GM-14" col 21	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2142	+10.802 / +9.170 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 26 row "GM-16a" col 16	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2143	+15.346 / +11.521 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 26 row "GM-16a" col 17	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2144	-	CALC	MMXMD FEA3D margin matrix v3.5.2	line 26 row "GM-16a" col 18	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2145	-	SIM	MMXMD FEA3D margin matrix v3.5.2	line 26 row "GM-16a" col 19	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2146	-	SIM	MMXMD FEA3D margin matrix v3.5.2	line 26 row "GM-16a" col 20	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2147	+0.179 (LC3 hot bound u)	DERIVED	MMXMD FEA3D margin matrix v3.5.2	line 26 row "GM-16a" col 21	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2148	>+99 / >+99 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 27 row "GM-16b" col 16	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2149	>+99 / >+99 a	SIM	MMXMD FEA3D margin matrix v3.5.2	line 27 row "GM-16b" col 17	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2150	-	CALC	MMXMD FEA3D margin matrix v3.5.2	line 27 row "GM-16b" col 18	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2151	-	SIM	MMXMD FEA3D margin matrix v3.5.2	line 27 row "GM-16b" col 19	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2152	-	SIM	MMXMD FEA3D margin matrix v3.5.2	line 27 row "GM-16b" col 20	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2153	+0.248 (LC3 hot bound u)	DERIVED	MMXMD FEA3D margin matrix v3.5.2	line 27 row "GM-16b" col 21	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2154	-0.387 / -0.509 r	SIM	MMXMD FEA3D margin matrix v3.5.2	line 28 row "PD-01R" col 16	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2155	-0.081 / -0.346 r	SIM	MMXMD FEA3D margin matrix v3.5.2	line 28 row "PD-01R" col 17	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2156	-	CALC	MMXMD FEA3D margin matrix v3.5.2	line 28 row "PD-01R" col 18	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2157	-	SIM	MMXMD FEA3D margin matrix v3.5.2	line 28 row "PD-01R" col 19	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2158	-	SIM	MMXMD FEA3D margin matrix v3.5.2	line 28 row "PD-01R" col 20	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2159	-0.509 (TB70 u)	DERIVED	MMXMD FEA3D margin matrix v3.5.2	line 28 row "PD-01R" col 21	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2160	-0.444 / -0.555 r	SIM	MMXMD FEA3D margin matrix v3.5.2	line 29 row "PD-01S" col 16	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2161	-0.166 / -0.407 r	SIM	MMXMD FEA3D margin matrix v3.5.2	line 29 row "PD-01S" col 17	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2162	-	CALC	MMXMD FEA3D margin matrix v3.5.2	line 29 row "PD-01S" col 18	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2163	-	SIM	MMXMD FEA3D margin matrix v3.5.2	line 29 row "PD-01S" col 19	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2164	-	SIM	MMXMD FEA3D margin matrix v3.5.2	line 29 row "PD-01S" col 20	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2165	-0.555 (TB70 u)	DERIVED	MMXMD FEA3D margin matrix v3.5.2	line 29 row "PD-01S" col 21	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2166	-	SIM	MMXMD FEA3D margin matrix v3.5.2	line 30 row "HW-101 (J-FWD)" col 16	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2167	-	SIM	MMXMD FEA3D margin matrix v3.5.2	line 30 row "HW-101 (J-FWD)" col 17	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2168	+0.650 / +0.278	CALC	MMXMD FEA3D margin matrix v3.5.2	line 30 row "HW-101 (J-FWD)" col 18	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); J-FWD cell classes (margin_matrix_v352.json): governing[0] SOAK52 (MS_y +0.650; MS_u ...
T2169	-	SIM	MMXMD FEA3D margin matrix v3.5.2	line 30 row "HW-101 (J-FWD)" col 19	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2170	-	SIM	MMXMD FEA3D margin matrix v3.5.2	line 30 row "HW-101 (J-FWD)" col 20	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2171	+0.278 (J-FWD u)	DERIVED	MMXMD FEA3D margin matrix v3.5.2	line 30 row "HW-101 (J-FWD)" col 21	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2172	-	SIM	MMXMD FEA3D margin matrix v3.5.2	line 31 row "HW-103 (J-FD01)" col 16	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)

ID	Value	Class	Source	Key	Source tag / formula
T2173	-	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 31 row "HW-103 (J-FD01)" col 17`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2174	-	CALC	MMXMD FEA3D margin matrix v3.5.2	`line 31 row "HW-103 (J-FD01)" col 18`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2175	+0.121 / +0.379	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 31 row "HW-103 (J-FD01)" col 19`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2176	-	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 31 row "HW-103 (J-FD01)" col 20`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2177	+0.121 (J-FD01 y)	DERIVED	MMXMD FEA3D margin matrix v3.5.2	`line 31 row "HW-103 (J-FD01)" col 21`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2178	-	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 32 row "HW-127 (J-FD09)" col 16`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2179	-	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 32 row "HW-127 (J-FD09)" col 17`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2180	-	CALC	MMXMD FEA3D margin matrix v3.5.2	`line 32 row "HW-127 (J-FD09)" col 18`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2181	-	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 32 row "HW-127 (J-FD09)" col 19`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2182	+0.221 / +0.485	SIM	MMXMD FEA3D margin matrix v3.5.2	`line 32 row "HW-127 (J-FD09)" col 20`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2183	+0.221 (J-FD09 y)	DERIVED	MMXMD FEA3D margin matrix v3.5.2	`line 32 row "HW-127 (J-FD09)" col 21`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2184	−0.482	DERIVED	MMX v352	`/rows/0/min/0`	(class set at the call site: DERIVED) [lowest cell of the part (matrix json min)]
T2185	LC6 soak	DERIVED	MMX v352	`/rows/0/min/1`	(class set at the call site: DERIVED)
T2186	−0.271	DERIVED	MMX v352	`/rows/1/min/0`	(class set at the call site: DERIVED) [lowest cell of the part (matrix json min)]
T2187	TB70	DERIVED	MMX v352	`/rows/1/min/1`	(class set at the call site: DERIVED)
T2188	−0.201	DERIVED	MMX v352	`/rows/2/min/0`	(class set at the call site: DERIVED) [lowest cell of the part (matrix json min)]
T2189	TB70	DERIVED	MMX v352	`/rows/2/min/1`	(class set at the call site: DERIVED)
T2190	−0.008	DERIVED	MMX v352	`/rows/3/min/0`	(class set at the call site: DERIVED) [lowest cell of the part (matrix json min)]
T2191	F2 hot bound	DERIVED	MMX v352	`/rows/3/min/1`	(class set at the call site: DERIVED)
T2192	0.604	DERIVED	MMX v352	`/rows/4/min/0`	(class set at the call site: DERIVED) [lowest cell of the part (matrix json min)]
T2193	TB70	DERIVED	MMX v352	`/rows/4/min/1`	(class set at the call site: DERIVED)
T2194	−0.52	DERIVED	MMX v352	`/rows/5/min/0`	(class set at the call site: DERIVED) [lowest cell of the part (matrix json min)]
T2195	LC1	DERIVED	MMX v352	`/rows/5/min/1`	(class set at the call site: DERIVED)
T2196	0.356	DERIVED	MMX v352	`/rows/6/min/0`	(class set at the call site: DERIVED) [lowest cell of the part (matrix json min)]
T2197	TB70	DERIVED	MMX v352	`/rows/6/min/1`	(class set at the call site: DERIVED)
T2198	−0.392	DERIVED	MMX v352	`/rows/7/min/0`	(class set at the call site: DERIVED) [lowest cell of the part (matrix json min)]
T2199	TB70	DERIVED	MMX v352	`/rows/7/min/1`	(class set at the call site: DERIVED)
T2200	2.265	DERIVED	MMX v352	`/rows/8/min/0`	(class set at the call site: DERIVED) [lowest cell of the part (matrix json min)]
T2201	TB70	DERIVED	MMX v352	`/rows/8/min/1`	(class set at the call site: DERIVED)
T2202	−0.336	DERIVED	MMX v352	`/rows/9/min/0`	(class set at the call site: DERIVED) [lowest cell of the part (matrix json min)]
T2203	TB70	DERIVED	MMX v352	`/rows/9/min/1`	(class set at the call site: DERIVED)
T2204	0.005	DERIVED	MMX v352	`/rows/10/min/0`	(class set at the call site: DERIVED) [lowest cell of the part (matrix json min)]
T2205	LC6 soak	DERIVED	MMX v352	`/rows/10/min/1`	(class set at the call site: DERIVED)
T2206	0.034	DERIVED	MMX v352	`/rows/11/min/0`	(class set at the call site: DERIVED) [lowest cell of the part (matrix json min)]
T2207	LC6 soak	DERIVED	MMX v352	`/rows/11/min/1`	(class set at the call site: DERIVED)
T2208	0.795	DERIVED	MMX v352	`/rows/12/min/0`	(class set at the call site: DERIVED) [lowest cell of the part (matrix json min)]
T2209	LC3 hot bound	DERIVED	MMX v352	`/rows/12/min/1`	(class set at the call site: DERIVED)
T2210	4.262	DERIVED	MMX v352	`/rows/13/min/0`	(class set at the call site: DERIVED) [lowest cell of the part (matrix json min)]
T2211	LC3 hot bound	DERIVED	MMX v352	`/rows/13/min/1`	(class set at the call site: DERIVED)
T2212	8.277	DERIVED	MMX v352	`/rows/14/min/0`	(class set at the call site: DERIVED) [lowest cell of the part (matrix json min)]
T2213	F2 hot bound	DERIVED	MMX v352	`/rows/14/min/1`	(class set at the call site: DERIVED)
T2214	0.626	DERIVED	MMX v352	`/rows/15/min/0`	(class set at the call site: DERIVED) [lowest cell of the part (matrix json min)]
T2215	J-FD09	DERIVED	MMX v352	`/rows/15/min/1`	(class set at the call site: DERIVED)
T2216	2.708	DERIVED	MMX v352	`/rows/16/min/0`	(class set at the call site: DERIVED) [lowest cell of the part (matrix json min)]
T2217	F2 hot bound	DERIVED	MMX v352	`/rows/16/min/1`	(class set at the call site: DERIVED)
T2218	1.723	DERIVED	MMX v352	`/rows/17/min/0`	(class set at the call site: DERIVED) [lowest cell of the part (matrix json min)]
T2219	LC3 hot bound	DERIVED	MMX v352	`/rows/17/min/1`	(class set at the call site: DERIVED)
T2220	1.758	DERIVED	MMX v352	`/rows/18/min/0`	(class set at the call site: DERIVED) [lowest cell of the part (matrix json min)]
T2221	LC3 hot bound	DERIVED	MMX v352	`/rows/18/min/1`	(class set at the call site: DERIVED)
T2222	0.179	DERIVED	MMX v352	`/rows/19/min/0`	(class set at the call site: DERIVED) [lowest cell of the part (matrix json min)]
T2223	LC3 hot bound	DERIVED	MMX v352	`/rows/19/min/1`	(class set at the call site: DERIVED)
T2224	0.248	DERIVED	MMX v352	`/rows/20/min/0`	(class set at the call site: DERIVED) [lowest cell of the part (matrix json min)]
T2225	LC3 hot bound	DERIVED	MMX v352	`/rows/20/min/1`	(class set at the call site: DERIVED)
T2226	−0.509	DERIVED	MMX v352	`/rows/21/min/0`	(class set at the call site: DERIVED) [lowest cell of the part (matrix json min)]
T2227	TB70	DERIVED	MMX v352	`/rows/21/min/1`	(class set at the call site: DERIVED)
T2228	−0.555	DERIVED	MMX v352	`/rows/22/min/0`	(class set at the call site: DERIVED) [lowest cell of the part (matrix json min)]
T2229	TB70	DERIVED	MMX v352	`/rows/22/min/1`	(class set at the call site: DERIVED)
T2230	0.278	DERIVED	MMX v352	`/rows/23/min/0`	(class set at the call site: DERIVED) [lowest cell of the part (matrix json min)]
T2231	J-FWD	DERIVED	MMX v352	`/rows/23/min/1`	(class set at the call site: DERIVED)
T2232	0.121	DERIVED	MMX v352	`/rows/24/min/0`	(class set at the call site: DERIVED) [lowest cell of the part (matrix json min)]
T2233	J-FD01	DERIVED	MMX v352	`/rows/24/min/1`	(class set at the call site: DERIVED)
T2234	0.221	DERIVED	MMX v352	`/rows/25/min/0`	(class set at the call site: DERIVED) [lowest cell of the part (matrix json min)]
T2235	J-FD09	DERIVED	MMX v352	`/rows/25/min/1`	(class set at the call site: DERIVED)
T2236	**FA1T**: evaluated on the gimbal / mount parts only (post_f1 GIMBAL_PARTS: GM-01 ...	DERIVED	MMXMD FEA3D margin matrix v3.5.2	`§Cells shown "-" and why line 482`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2237	**LC2F**: evaluated on the gimbal / mount parts only (post_s3d GIMBAL_PARTS; the ...	DERIVED	MMXMD FEA3D margin matrix v3.5.2	`§Cells shown "-" and why line 483`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2238	**J-FWD**: only TC-01, TC-02 and the HW-101 screw are in the J-FWD sub-model; the global ...	DERIVED	MMXMD FEA3D margin matrix v3.5.2	`§Cells shown "-" and why line 484`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2239	**J-FD01**: F4 joint: the HW-103 / 103B screws and the TC-02 host thread only	DERIVED	MMXMD FEA3D margin matrix v3.5.2	`§Cells shown "-" and why line 485`	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)

ID	Value	Class	Source	Key	Source tag / formula
T2240	**J-FD09**: F4 joint: the HW-127 screws and the GM-09 host thread only	DERIVED	MMXMD FEA3D margin matrix v3.5.2	§Cells shown "-" and why line 486	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2241	**joint rows**: HW-101 / HW-103 / HW-127 are connectors in the global model (no FE ...	DERIVED	MMXMD FEA3D margin matrix v3.5.2	§Cells shown "-" and why line 487	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2242	−0.034	SIM	MMX v352	/rows/0/cells/TS2 (info)/value/0	(class set at the call site: SIM) [TC-01 TS2 (info): y]
T2243	UNCLASSIFIED	DERIVED	MMX v352	/rows/0/cells/TS2 (info)/class	(class set at the call site: DERIVED)
T2244	to the lead	DERIVED	MMX v352	/rows/0/cells/TS2 (info)/closure	(class set at the call site: DERIVED)
T2245	−0.332	SIM	MMX v352	/rows/2/cells/LC1 merged v351 map (info)/value/0	(class set at the call site: SIM) [TC-03 LC1 merged v351 map (info): y]
T2246	−0.393	SIM	MMX v352	/rows/2/cells/LC1 merged v351 map (info)/value/1	(class set at the call site: SIM) [TC-03 LC1 merged v351 map (info): u]
T2247	TS-183	DERIVED	MMX v352	/rows/2/cells/LC1 merged v351 map (info)/class	(class set at the call site: DERIVED)
T2248	thermal-secondary 183 (1): (a) primary and (b) P+Q pass; (c) LCF and (d) GlidCop time at ...	DERIVED	MMX v352	/rows/2/cells/LC1 merged v351 map (info)/closure	(class set at the call site: DERIVED)
T2249	−0.257	SIM	MMX v352	/rows/5/cells/TS2 (info)/value/0	(class set at the call site: SIM) [NZ-01 TS2 (info): y]
T2250	BR	DERIVED	MMX v352	/rows/5/cells/TS2 (info)/class	(class set at the call site: DERIVED)
T2251	brittle and thermal: 183 (2) capture / containment shown + test basis, else design change	DERIVED	MMX v352	/rows/5/cells/TS2 (info)/closure	(class set at the call site: DERIVED)
T2252	−0.522	SIM	MMX v352	/rows/5/cells/LC1 merged v351 map (info)/value/0	(class set at the call site: SIM) [NZ-01 LC1 merged v351 map (info): y]
T2253	BR	DERIVED	MMX v352	/rows/5/cells/LC1 merged v351 map (info)/class	(class set at the call site: DERIVED)
T2254	brittle and thermal: 183 (2) capture / containment shown + test basis, else design change	DERIVED	MMX v352	/rows/5/cells/LC1 merged v351 map (info)/closure	(class set at the call site: DERIVED)
T2255	−0.087	SIM	MMX v352	/rows/7/cells/TB70 SF1.0 (info)/value/0	(class set at the call site: SIM) [NZ-03 TB70 SF1.0 (info): y]
T2256	CH	DERIVED	MMX v352	/rows/7/cells/TB70 SF1.0 (info)/class	(class set at the call site: DERIVED)
T2257	char primary vs the ASSUMED 20 MPa: N-F1 (>= 21.96 MPa inside the 23.7 MPa acceptance) ...	DERIVED	MMX v352	/rows/7/cells/TB70 SF1.0 (info)/closure	(class set at the call site: DERIVED)
T2258	−0.022	SIM	MMX v352	/rows/9/cells/TS2 (info)/value/0	(class set at the call site: SIM) [FD-01 TS2 (info): y]
T2259	TS-183	DERIVED	MMX v352	/rows/9/cells/TS2 (info)/class	(class set at the call site: DERIVED)
T2260	thermal-secondary 183 (1), all-node value at the FD01_TC02_screws bonded-patch edge kept ...	DERIVED	MMX v352	/rows/9/cells/TS2 (info)/closure	(class set at the call site: DERIVED)
T2261	−0.004	SIM	MMX v352	/rows/9/cells/TB70 SF1.0 (info)/value/0	(class set at the call site: SIM) [FD-01 TB70 SF1.0 (info): y]
T2262	TS-183	DERIVED	MMX v352	/rows/9/cells/TB70 SF1.0 (info)/class	(class set at the call site: DERIVED)
T2263	thermal-secondary 183 (1), all-node value at the FD01_TC02_screws bonded-patch edge kept ...	DERIVED	MMX v352	/rows/9/cells/TB70 SF1.0 (info)/closure	(class set at the call site: DERIVED)
T2264	−0.204	SIM	MMX v352	/rows/21/cells/TS2 (info)/value/0	(class set at the call site: SIM) [PD-01R TS2 (info): y]
T2265	MF	DERIVED	MMX v352	/rows/21/cells/TS2 (info)/class	(class set at the call site: DERIVED)
T2266	PD-02 rigid in the model (same alloy): the PD-01 / PD-02 mismatch is a model artefact ...	DERIVED	MMX v352	/rows/21/cells/TS2 (info)/closure	(class set at the call site: DERIVED)
T2267	−0.08	SIM	MMX v352	/rows/21/cells/TB70 SF1.0 (info)/value/0	(class set at the call site: SIM) [PD-01R TB70 SF1.0 (info): y]
T2268	MF	DERIVED	MMX v352	/rows/21/cells/TB70 SF1.0 (info)/class	(class set at the call site: DERIVED)
T2269	PD-02 rigid in the model (same alloy): the PD-01 / PD-02 mismatch is a model artefact ...	DERIVED	MMX v352	/rows/21/cells/TB70 SF1.0 (info)/closure	(class set at the call site: DERIVED)
T2270	−0.279	SIM	MMX v352	/rows/22/cells/TS2 (info)/value/0	(class set at the call site: SIM) [PD-01S TS2 (info): y]
T2271	MF	DERIVED	MMX v352	/rows/22/cells/TS2 (info)/class	(class set at the call site: DERIVED)
T2272	PD-02 rigid in the model (same alloy): the PD-01 / PD-02 mismatch is a model artefact ...	DERIVED	MMX v352	/rows/22/cells/TS2 (info)/closure	(class set at the call site: DERIVED)
T2273	−0.166	SIM	MMX v352	/rows/22/cells/TB70 SF1.0 (info)/value/0	(class set at the call site: SIM) [PD-01S TB70 SF1.0 (info): y]
T2274	MF	DERIVED	MMX v352	/rows/22/cells/TB70 SF1.0 (info)/class	(class set at the call site: DERIVED)
T2275	PD-02 rigid in the model (same alloy): the PD-01 / PD-02 mismatch is a model artefact ...	DERIVED	MMX v352	/rows/22/cells/TB70 SF1.0 (info)/closure	(class set at the call site: DERIVED)
T2276	11	DERIVED	generator	formula: count of negative cells in the information columns of margin_matrix_v352.json	DERIVED by build_datasheet.py: count of negative cells in the information columns of margin_matrix_v352.json
T2277	LC3 hot bound +0.228 / -0.004, F2 hot bound +0.223 / -0.008 [DERIVED node-wise lower ...	DERIVED	F3D FINAL handback of the v3.5.2 re-check	§13. Open items and owners row c0="10" col 2 (line 1705)	FEA3D-2 report §13 item 10
T2278	lead (ruled 2026-10-21T03:30:46Z (2): the settling route NOT RUN, final, for disk)	DECIDED *Lead decision, per REPORT-FEA3D-v352.md FINAL handback of the v3.5.2 re-check*	F3D FINAL handback of the v3.5.2 re-check	§13. Open items and owners row c0="10" col 3 (line 1705)	tag in REPORT-FEA3D-v352.md FINAL handback of the v3.5.2 re-check: FEA3D-2 report §13 item 10 owner
T2279	ENG-TC-02 PLUG OD, J-FWD SEAT-LINE BAND (lead 2026-10-21T05:50:45Z R2, restated to the ...	CALC	TXT5 v5	/item_C/C9_founder_texts_remap_v352/(6)_TC-02_select_fit_band_R2	(class set at the call site: CALC)
T2280	(1) S01 and S02 are replaced after any FA-1-class event, together with the line-231 ...	CALC	TXT5 v5	/item_C/C6_founder_action_list_related	(class set at the call site: CALC)
T2281	At the J-F1 cap 3750 N the fillet root (d 5.120, A 20.59 mm2, Z 13.18 mm3; head-end FE ...	CALC	FIL v2	/verdict	CALC; Kt basis WEB-SOURCED, BOOK PAGE TO-SOURCE; bolt-head (T-head) loading NOT SOURCED; negatives SHOWN AS FOUND
T2282	FOUND NEGATIVE (lead 14:14Z): quoted as negative, never as passing	DECIDED *Lead decision, per jfwd_screw_lcf_v352_v3.json v3*	LCF v3	/status_183_1c	tag in jfwd_screw_lcf_v352_v3.json v3:
T2283	−0.888	CALC	LCF v3	/governing/i_firing|Kf3.0|worst_of_histories|3750/MS_cycles_vs_50	(class set at the call site: CALC)
T2284	5.6	CALC	LCF v3	/governing/i_firing|Kf3.0|worst_of_histories|3750/N_allow	(class set at the call site: CALC)
T2285	−0.988	CALC	LCF v3	/governing/ii_assembly_from_zero|Kf3.0|worst_of_histories|3750/MS_cycles_vs_50	(class set at the call site: CALC)
T2286	0.6	CALC	LCF v3	/governing/ii_assembly_from_zero|Kf3.0|worst_of_histories|3750/N_allow	(class set at the call site: CALC)
T2287	J-FWD column rows (cell = minimum over the criterion rows; information rows listed after ...	DERIVED	MMXMD FEA3D margin matrix v3.5.2	heading line 489	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1)
T2288	−0.075–0.239	CALC	MMX v352	/rows/0/cells/J-FWD/value	(class set at the call site: CALC) [J-FWD cell = minimum over the criterion rows]
T2289	FA1S	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/0/state	(class set at the call site: DERIVED)
T2290	(i) ligament Pm+Pb, half-ligament average	SIM	MMX v352	/rows/0/cells/J-FWD/rows/0/row	SIM / DERIVED
T2291	0.171	SIM	MMX v352	/rows/0/cells/J-FWD/rows/0/MS_y	SIM / DERIVED
T2292	0.263	SIM	MMX v352	/rows/0/cells/J-FWD/rows/0/MS_u	SIM / DERIVED
T2293	231 (1)(i)	DECIDED *Lead decision, per margin_matrix_v352.json v352*	MMX v352	/rows/0/cells/J-FWD/rows/0/criterion	tag in margin_matrix_v352.json v352:
T2294	FA1S	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/1/state	(class set at the call site: DERIVED)
T2295	(i) ligament Pm, half-ligament average	SIM	MMX v352	/rows/0/cells/J-FWD/rows/1/row	SIM / DERIVED
T2296	1.268	SIM	MMX v352	/rows/0/cells/J-FWD/rows/1/MS_y	SIM / DERIVED
T2297	1.446	SIM	MMX v352	/rows/0/cells/J-FWD/rows/1/MS_u	SIM / DERIVED
T2298	231 (1)(i)	DECIDED *Lead decision, per margin_matrix_v352.json v352*	MMX v352	/rows/0/cells/J-FWD/rows/1/criterion	tag in margin_matrix_v352.json v352:
T2299	FA1S	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/2/state	(class set at the call site: DERIVED)
T2300	(i) shear-out flank+ Pm	SIM	MMX v352	/rows/0/cells/J-FWD/rows/2/row	SIM / DERIVED
T2301	0.991	SIM	MMX v352	/rows/0/cells/J-FWD/rows/2/MS_y	SIM / DERIVED
T2302	1.147	SIM	MMX v352	/rows/0/cells/J-FWD/rows/2/MS_u	SIM / DERIVED
T2303	231 (1)(i)	DECIDED *Lead decision, per margin_matrix_v352.json v352*	MMX v352	/rows/0/cells/J-FWD/rows/2/criterion	tag in margin_matrix_v352.json v352:
T2304	FA1S	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/3/state	(class set at the call site: DERIVED)
T2305	(i) shear-out flank+ Pm+Pb	SIM	MMX v352	/rows/0/cells/J-FWD/rows/3/row	SIM / DERIVED
T2306	0.809	SIM	MMX v352	/rows/0/cells/J-FWD/rows/3/MS_y	SIM / DERIVED
T2307	0.951	SIM	MMX v352	/rows/0/cells/J-FWD/rows/3/MS_u	SIM / DERIVED

ID	Value	Class	Source	Key	Source tag / formula
T2308	231 (1)(i)	DECIDED *Lead decision, per margin_matrix_v352.json v352*	MMX v352	/rows/0/cells/J-FWD/rows/3/criterion	tag in margin_matrix_v352.json v352:
T2309	FA1S	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/4/state	(class set at the call site: DERIVED)
T2310	(i) shear-out flank- Pm	SIM	MMX v352	/rows/0/cells/J-FWD/rows/4/row	SIM / DERIVED
T2311	0.952	SIM	MMX v352	/rows/0/cells/J-FWD/rows/4/MS_y	SIM / DERIVED
T2312	1.105	SIM	MMX v352	/rows/0/cells/J-FWD/rows/4/MS_u	SIM / DERIVED
T2313	231 (1)(i)	DECIDED *Lead decision, per margin_matrix_v352.json v352*	MMX v352	/rows/0/cells/J-FWD/rows/4/criterion	tag in margin_matrix_v352.json v352:
T2314	FA1S	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/5/state	(class set at the call site: DERIVED)
T2315	(i) shear-out flank- Pm+Pb	SIM	MMX v352	/rows/0/cells/J-FWD/rows/5/row	SIM / DERIVED
T2316	0.769	SIM	MMX v352	/rows/0/cells/J-FWD/rows/5/MS_y	SIM / DERIVED
T2317	0.907	SIM	MMX v352	/rows/0/cells/J-FWD/rows/5/MS_u	SIM / DERIVED
T2318	231 (1)(i)	DECIDED *Lead decision, per margin_matrix_v352.json v352*	MMX v352	/rows/0/cells/J-FWD/rows/5/criterion	tag in margin_matrix_v352.json v352:
T2319	FA1S	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/6/state	(class set at the call site: DERIVED)
T2320	(ii) hole bearing, exact V 3603.6 N on t 2.688 (sharp)	SIM	MMX v352	/rows/0/cells/J-FWD/rows/6/row	SIM / DERIVED
T2321	0.233	SIM	MMX v352	/rows/0/cells/J-FWD/rows/6/MS_y	SIM / DERIVED
T2322	0.519	SIM	MMX v352	/rows/0/cells/J-FWD/rows/6/MS_u	SIM / DERIVED
T2323	231 (1)(ii)	DECIDED *Lead decision, per margin_matrix_v352.json v352*	MMX v352	/rows/0/cells/J-FWD/rows/6/criterion	tag in margin_matrix_v352.json v352:
T2324	FA1S	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/7/state	(class set at the call site: DERIVED)
T2325	(iii) band Neuber n30 (local strain)	SIM	MMX v352	/rows/0/cells/J-FWD/rows/7/row	SIM / DERIVED
T2326	0.518	SIM	MMX v352	/rows/0/cells/J-FWD/rows/7/MS_y	SIM / DERIVED
T2327	MISSING	SIM	MMX v352	/rows/0/cells/J-FWD/rows/7/MS_u	SIM / DERIVED
T2328	231 (1)(iii)	DECIDED *Lead decision, per margin_matrix_v352.json v352*	MMX v352	/rows/0/cells/J-FWD/rows/7/criterion	tag in margin_matrix_v352.json v352:
T2329	FA1S	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/10/state	(class set at the call site: DERIVED)
T2330	(ii) hole bearing, CALC V 3888.1 N, L 2.365 (governing, 240 (5))	CALC	MMX v352	/rows/0/cells/J-FWD/rows/10/row	CALC
T2331	0.005	CALC	MMX v352	/rows/0/cells/J-FWD/rows/10/MS_y	CALC
T2332	0.239	CALC	MMX v352	/rows/0/cells/J-FWD/rows/10/MS_u	CALC
T2333	231 (1)(ii)	DECIDED *Lead decision, per margin_matrix_v352.json v352*	MMX v352	/rows/0/cells/J-FWD/rows/10/criterion	tag in margin_matrix_v352.json v352:
T2334	FA1S	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/11/state	(class set at the call site: DERIVED)
T2335	(i) shear-out	CALC	MMX v352	/rows/0/cells/J-FWD/rows/11/row	CALC
T2336	0.727	CALC	MMX v352	/rows/0/cells/J-FWD/rows/11/MS_y	CALC
T2337	0.937	CALC	MMX v352	/rows/0/cells/J-FWD/rows/11/MS_u	CALC
T2338	231 (1)(i)	DECIDED *Lead decision, per margin_matrix_v352.json v352*	MMX v352	/rows/0/cells/J-FWD/rows/11/criterion	tag in margin_matrix_v352.json v352:
T2339	FA1S	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/12/state	(class set at the call site: DERIVED)
T2340	(i) net section Pm	CALC	MMX v352	/rows/0/cells/J-FWD/rows/12/row	CALC
T2341	1.483	CALC	MMX v352	/rows/0/cells/J-FWD/rows/12/MS_y	CALC
T2342	1.678	CALC	MMX v352	/rows/0/cells/J-FWD/rows/12/MS_u	CALC
T2343	231 (1)(i)	DECIDED *Lead decision, per margin_matrix_v352.json v352*	MMX v352	/rows/0/cells/J-FWD/rows/12/criterion	tag in margin_matrix_v352.json v352:
T2344	FA1S	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/13/state	(class set at the call site: DERIVED)
T2345	(i) ligament Pm+Pb	CALC	MMX v352	/rows/0/cells/J-FWD/rows/13/row	CALC
T2346	0.161	CALC	MMX v352	/rows/0/cells/J-FWD/rows/13/MS_y	CALC
T2347	0.252	CALC	MMX v352	/rows/0/cells/J-FWD/rows/13/MS_u	CALC
T2348	231 (1)(i)	DECIDED *Lead decision, per margin_matrix_v352.json v352*	MMX v352	/rows/0/cells/J-FWD/rows/13/criterion	tag in margin_matrix_v352.json v352:
T2349	SOAK70	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/20/state	(class set at the call site: DERIVED)
T2350	band Neuber n30 (under-head band, local strain)	SIM	MMX v352	/rows/0/cells/J-FWD/rows/20/row	SIM / DERIVED
T2351	0.788	SIM	MMX v352	/rows/0/cells/J-FWD/rows/20/MS_y	SIM / DERIVED
T2352	MISSING	SIM	MMX v352	/rows/0/cells/J-FWD/rows/20/MS_u	SIM / DERIVED
T2353	lead 06:36Z (DC head seat)	DECIDED *Lead decision, per margin_matrix_v352.json v352*	MMX v352	/rows/0/cells/J-FWD/rows/20/criterion	tag in margin_matrix_v352.json v352:
T2354	SOAK70	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/24/state	(class set at the call site: DERIVED)
T2355	54 seat lines P+Q <= 2 Fty(T), envelope minimum (line d+6.051 z'+0.0, (b2); A4-80 FE)	SIM	MMX v352	/rows/0/cells/J-FWD/rows/24/row	SIM / DERIVED
T2356	−0.055	SIM	MMX v352	/rows/0/cells/J-FWD/rows/24/MS_y	SIM / DERIVED
T2357	MISSING	SIM	MMX v352	/rows/0/cells/J-FWD/rows/24/MS_u	SIM / DERIVED
T2358	240 (9)(i)	DECIDED *Lead decision, per margin_matrix_v352.json v352*	MMX v352	/rows/0/cells/J-FWD/rows/24/criterion	tag in margin_matrix_v352.json v352:
T2359	SOAK	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/25/state	(class set at the call site: DERIVED)
T2360	band Neuber n30 (under-head band, local strain)	SIM	MMX v352	/rows/0/cells/J-FWD/rows/25/row	SIM / DERIVED
T2361	0.832	SIM	MMX v352	/rows/0/cells/J-FWD/rows/25/MS_y	SIM / DERIVED
T2362	MISSING	SIM	MMX v352	/rows/0/cells/J-FWD/rows/25/MS_u	SIM / DERIVED
T2363	lead 06:36Z (DC head seat)	DECIDED *Lead decision, per margin_matrix_v352.json v352*	MMX v352	/rows/0/cells/J-FWD/rows/25/criterion	tag in margin_matrix_v352.json v352:
T2364	SOAK	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/29/state	(class set at the call site: DERIVED)
T2365	54 seat lines P+Q <= 2 Fty(T), envelope minimum (line d+6.051 z'+0.0, (b2) rep; A4-80 FE)	SIM	MMX v352	/rows/0/cells/J-FWD/rows/29/row	SIM / DERIVED
T2366	−0.071	SIM	MMX v352	/rows/0/cells/J-FWD/rows/29/MS_y	SIM / DERIVED
T2367	MISSING	SIM	MMX v352	/rows/0/cells/J-FWD/rows/29/MS_u	SIM / DERIVED
T2368	240 (9)(i)	DECIDED *Lead decision, per margin_matrix_v352.json v352*	MMX v352	/rows/0/cells/J-FWD/rows/29/criterion	tag in margin_matrix_v352.json v352:
T2369	SOAK52	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/30/state	(class set at the call site: DERIVED)
T2370	band Neuber n30 (under-head band, local strain)	SIM	MMX v352	/rows/0/cells/J-FWD/rows/30/row	SIM / DERIVED
T2371	0.479	SIM	MMX v352	/rows/0/cells/J-FWD/rows/30/MS_y	SIM / DERIVED
T2372	MISSING	SIM	MMX v352	/rows/0/cells/J-FWD/rows/30/MS_u	SIM / DERIVED
T2373	lead 06:36Z (DC head seat)	DECIDED *Lead decision, per margin_matrix_v352.json v352*	MMX v352	/rows/0/cells/J-FWD/rows/30/criterion	tag in margin_matrix_v352.json v352:
T2374	SOAK52	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/34/state	(class set at the call site: DERIVED)
T2375	54 seat lines P+Q <= 2 Fty(T), envelope minimum (line d+6.051 z'+0.0, (b1); A4-80 FE)	SIM	MMX v352	/rows/0/cells/J-FWD/rows/34/row	SIM / DERIVED
T2376	−0.03	SIM	MMX v352	/rows/0/cells/J-FWD/rows/34/MS_y	SIM / DERIVED
T2377	MISSING	SIM	MMX v352	/rows/0/cells/J-FWD/rows/34/MS_u	SIM / DERIVED
T2378	240 (9)(i)	DECIDED *Lead decision, per margin_matrix_v352.json v352*	MMX v352	/rows/0/cells/J-FWD/rows/34/criterion	tag in margin_matrix_v352.json v352:
T2379	FA1U	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/35/state	(class set at the call site: DERIVED)
T2380	js5b FA1U (FA1, SF [1.0, 1.0]): (i) ligament Pm+Pb, half-ligament average	SIM	MMX v352	/rows/0/cells/J-FWD/rows/35/row	SIM / DERIVED
T2381	0.222	SIM	MMX v352	/rows/0/cells/J-FWD/rows/35/MS_y	SIM / DERIVED
T2382	0.318	SIM	MMX v352	/rows/0/cells/J-FWD/rows/35/MS_u	SIM / DERIVED
T2383	231 (1)(i)	DECIDED *Lead decision, per margin_matrix_v352.json v352*	MMX v352	/rows/0/cells/J-FWD/rows/35/criterion	tag in margin_matrix_v352.json v352:
T2384	FA1U	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/36/state	(class set at the call site: DERIVED)
T2385	js5b FA1U (FA1, SF [1.0, 1.0]): (i) ligament Pm, half-ligament average	SIM	MMX v352	/rows/0/cells/J-FWD/rows/36/row	SIM / DERIVED

ID	Value	Class	Source	Key	Source tag / formula
T2386	1.319	SIM	MMX v352	/rows/0/cells/J-FWD/rows/36/MS_y	SIM / DERIVED
T2387	1.501	SIM	MMX v352	/rows/0/cells/J-FWD/rows/36/MS_u	SIM / DERIVED
T2388	231 (1)(i)	DECIDED *Lead decision, per margin_matrix_v352.json v352*	MMX v352	/rows/0/cells/J-FWD/rows/36/criterion	tag in margin_matrix_v352.json v352:
T2389	FA1U	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/37/state	(class set at the call site: DERIVED)
T2390	js5b FA1U (FA1, SF [1.0, 1.0]): (i) shear-out flank+ Pm+Pb	SIM	MMX v352	/rows/0/cells/J-FWD/rows/37/row	SIM / DERIVED
T2391	0.769	SIM	MMX v352	/rows/0/cells/J-FWD/rows/37/MS_y	SIM / DERIVED
T2392	0.908	SIM	MMX v352	/rows/0/cells/J-FWD/rows/37/MS_u	SIM / DERIVED
T2393	231 (1)(i)	DECIDED *Lead decision, per margin_matrix_v352.json v352*	MMX v352	/rows/0/cells/J-FWD/rows/37/criterion	tag in margin_matrix_v352.json v352:
T2394	FA1U	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/38/state	(class set at the call site: DERIVED)
T2395	js5b FA1U (FA1, SF [1.0, 1.0]): (i) shear-out flank- Pm+Pb	SIM	MMX v352	/rows/0/cells/J-FWD/rows/38/row	SIM / DERIVED
T2396	0.733	SIM	MMX v352	/rows/0/cells/J-FWD/rows/38/MS_y	SIM / DERIVED
T2397	0.869	SIM	MMX v352	/rows/0/cells/J-FWD/rows/38/MS_u	SIM / DERIVED
T2398	231 (1)(i)	DECIDED *Lead decision, per margin_matrix_v352.json v352*	MMX v352	/rows/0/cells/J-FWD/rows/38/criterion	tag in margin_matrix_v352.json v352:
T2399	FA1U	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/39/state	(class set at the call site: DERIVED)
T2400	js5b FA1U (FA1, SF [1.0, 1.0]): (ii) hole bearing, exact V 3562.7 N on t 2.688	SIM	MMX v352	/rows/0/cells/J-FWD/rows/39/row	SIM / DERIVED
T2401	0.247	SIM	MMX v352	/rows/0/cells/J-FWD/rows/39/MS_y	SIM / DERIVED
T2402	0.537	SIM	MMX v352	/rows/0/cells/J-FWD/rows/39/MS_u	SIM / DERIVED
T2403	231 (1)(ii)	DECIDED *Lead decision, per margin_matrix_v352.json v352*	MMX v352	/rows/0/cells/J-FWD/rows/39/criterion	tag in margin_matrix_v352.json v352:
T2404	FA1U	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/40/state	(class set at the call site: DERIVED)
T2405	js5b FA1U (FA1, SF [1.0, 1.0]): (iii) band Neuber n30 (local strain)	SIM	MMX v352	/rows/0/cells/J-FWD/rows/40/row	SIM / DERIVED
T2406	0.403	SIM	MMX v352	/rows/0/cells/J-FWD/rows/40/MS_y	SIM / DERIVED
T2407	MISSING	SIM	MMX v352	/rows/0/cells/J-FWD/rows/40/MS_u	SIM / DERIVED
T2408	231 (1)(iii)	DECIDED *Lead decision, per margin_matrix_v352.json v352*	MMX v352	/rows/0/cells/J-FWD/rows/40/criterion	tag in margin_matrix_v352.json v352:
T2409	FA1D	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/41/state	(class set at the call site: DERIVED)
T2410	js5b FA1D (FA1, SF [1.0, 1.0]): (i) ligament Pm+Pb, half-ligament average	SIM	MMX v352	/rows/0/cells/J-FWD/rows/41/row	SIM / DERIVED
T2411	0.257	SIM	MMX v352	/rows/0/cells/J-FWD/rows/41/MS_y	SIM / DERIVED
T2412	0.355	SIM	MMX v352	/rows/0/cells/J-FWD/rows/41/MS_u	SIM / DERIVED
T2413	231 (1)(i)	DECIDED *Lead decision, per margin_matrix_v352.json v352*	MMX v352	/rows/0/cells/J-FWD/rows/41/criterion	tag in margin_matrix_v352.json v352:
T2414	FA1D	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/42/state	(class set at the call site: DERIVED)
T2415	js5b FA1D (FA1, SF [1.0, 1.0]): (i) ligament Pm, half-ligament average	SIM	MMX v352	/rows/0/cells/J-FWD/rows/42/row	SIM / DERIVED
T2416	1.641	SIM	MMX v352	/rows/0/cells/J-FWD/rows/42/MS_y	SIM / DERIVED
T2417	1.848	SIM	MMX v352	/rows/0/cells/J-FWD/rows/42/MS_u	SIM / DERIVED
T2418	231 (1)(i)	DECIDED *Lead decision, per margin_matrix_v352.json v352*	MMX v352	/rows/0/cells/J-FWD/rows/42/criterion	tag in margin_matrix_v352.json v352:
T2419	FA1D	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/43/state	(class set at the call site: DERIVED)
T2420	js5b FA1D (FA1, SF [1.0, 1.0]): (i) shear-out flank+ Pm+Pb	SIM	MMX v352	/rows/0/cells/J-FWD/rows/43/row	SIM / DERIVED
T2421	1.121	SIM	MMX v352	/rows/0/cells/J-FWD/rows/43/MS_y	SIM / DERIVED
T2422	1.288	SIM	MMX v352	/rows/0/cells/J-FWD/rows/43/MS_u	SIM / DERIVED
T2423	231 (1)(i)	DECIDED *Lead decision, per margin_matrix_v352.json v352*	MMX v352	/rows/0/cells/J-FWD/rows/43/criterion	tag in margin_matrix_v352.json v352:
T2424	FA1D	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/44/state	(class set at the call site: DERIVED)
T2425	js5b FA1D (FA1, SF [1.0, 1.0]): (i) shear-out flank- Pm+Pb	SIM	MMX v352	/rows/0/cells/J-FWD/rows/44/row	SIM / DERIVED
T2426	1.08	SIM	MMX v352	/rows/0/cells/J-FWD/rows/44/MS_y	SIM / DERIVED
T2427	1.244	SIM	MMX v352	/rows/0/cells/J-FWD/rows/44/MS_u	SIM / DERIVED
T2428	231 (1)(i)	DECIDED *Lead decision, per margin_matrix_v352.json v352*	MMX v352	/rows/0/cells/J-FWD/rows/44/criterion	tag in margin_matrix_v352.json v352:
T2429	FA1D	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/45/state	(class set at the call site: DERIVED)
T2430	js5b FA1D (FA1, SF [1.0, 1.0]): (ii) hole bearing, exact V 2825.1 N on t 2.688	SIM	MMX v352	/rows/0/cells/J-FWD/rows/45/row	SIM / DERIVED
T2431	0.573	SIM	MMX v352	/rows/0/cells/J-FWD/rows/45/MS_y	SIM / DERIVED
T2432	0.938	SIM	MMX v352	/rows/0/cells/J-FWD/rows/45/MS_u	SIM / DERIVED
T2433	231 (1)(ii)	DECIDED *Lead decision, per margin_matrix_v352.json v352*	MMX v352	/rows/0/cells/J-FWD/rows/45/criterion	tag in margin_matrix_v352.json v352:
T2434	FA1D	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/46/state	(class set at the call site: DERIVED)
T2435	js5b FA1D (FA1, SF [1.0, 1.0]): (iii) band Neuber n30 (local strain)	SIM	MMX v352	/rows/0/cells/J-FWD/rows/46/row	SIM / DERIVED
T2436	1.231	SIM	MMX v352	/rows/0/cells/J-FWD/rows/46/MS_y	SIM / DERIVED
T2437	MISSING	SIM	MMX v352	/rows/0/cells/J-FWD/rows/46/MS_u	SIM / DERIVED
T2438	231 (1)(iii)	DECIDED *Lead decision, per margin_matrix_v352.json v352*	MMX v352	/rows/0/cells/J-FWD/rows/46/criterion	tag in margin_matrix_v352.json v352:
T2439	LC2U	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/47/state	(class set at the call site: DERIVED)
T2440	js5b LC2U (FE-factored, SF [1.0, 1.0]): (i) ligament Pm+Pb, half-ligament average	SIM	MMX v352	/rows/0/cells/J-FWD/rows/47/row	SIM / DERIVED
T2441	MISSING	SIM	MMX v352	/rows/0/cells/J-FWD/rows/47/MS_y	SIM / DERIVED
T2442	0.424	SIM	MMX v352	/rows/0/cells/J-FWD/rows/47/MS_u	SIM / DERIVED
T2443	231 (1)(i)	DECIDED *Lead decision, per margin_matrix_v352.json v352*	MMX v352	/rows/0/cells/J-FWD/rows/47/criterion	tag in margin_matrix_v352.json v352:
T2444	FE-factored: the x1.5 / x2.0 state solved with contact at the 3750 N maximum preload ...	SIM	MMX v352	/rows/0/cells/J-FWD/rows/47/note	SIM / DERIVED
T2445	LC2U	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/48/state	(class set at the call site: DERIVED)
T2446	js5b LC2U (FE-factored, SF [1.0, 1.0]): (i) ligament Pm, half-ligament average	SIM	MMX v352	/rows/0/cells/J-FWD/rows/48/row	SIM / DERIVED
T2447	MISSING	SIM	MMX v352	/rows/0/cells/J-FWD/rows/48/MS_y	SIM / DERIVED
T2448	2.277	SIM	MMX v352	/rows/0/cells/J-FWD/rows/48/MS_u	SIM / DERIVED
T2449	231 (1)(i)	DECIDED *Lead decision, per margin_matrix_v352.json v352*	MMX v352	/rows/0/cells/J-FWD/rows/48/criterion	tag in margin_matrix_v352.json v352:
T2450	FE-factored: the x1.5 / x2.0 state solved with contact at the 3750 N maximum preload ...	SIM	MMX v352	/rows/0/cells/J-FWD/rows/48/note	SIM / DERIVED
T2451	LC2U	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/49/state	(class set at the call site: DERIVED)
T2452	js5b LC2U (FE-factored, SF [1.0, 1.0]): (i) shear-out flank+ Pm+Pb	SIM	MMX v352	/rows/0/cells/J-FWD/rows/49/row	SIM / DERIVED
T2453	MISSING	SIM	MMX v352	/rows/0/cells/J-FWD/rows/49/MS_y	SIM / DERIVED
T2454	1.544	SIM	MMX v352	/rows/0/cells/J-FWD/rows/49/MS_u	SIM / DERIVED
T2455	231 (1)(i)	DECIDED *Lead decision, per margin_matrix_v352.json v352*	MMX v352	/rows/0/cells/J-FWD/rows/49/criterion	tag in margin_matrix_v352.json v352:
T2456	FE-factored: the x1.5 / x2.0 state solved with contact at the 3750 N maximum preload ...	SIM	MMX v352	/rows/0/cells/J-FWD/rows/49/note	SIM / DERIVED
T2457	LC2U	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/50/state	(class set at the call site: DERIVED)
T2458	js5b LC2U (FE-factored, SF [1.0, 1.0]): (i) shear-out flank- Pm+Pb	SIM	MMX v352	/rows/0/cells/J-FWD/rows/50/row	SIM / DERIVED
T2459	MISSING	SIM	MMX v352	/rows/0/cells/J-FWD/rows/50/MS_y	SIM / DERIVED
T2460	1.512	SIM	MMX v352	/rows/0/cells/J-FWD/rows/50/MS_u	SIM / DERIVED

ID	Value	Class	Source	Key	Source tag / formula
T2461	231 (1)(i)	DECIDED *Lead decision, per margin_matrix_v352.json v352*	MMX v352	/rows/0/cells/J-FWD/rows/50/criterion	tag in margin_matrix_v352.json v352:
T2462	FE-factored: the x1.5 / x2.0 state solved with contact at the 3750 N maximum preload ...	SIM	MMX v352	/rows/0/cells/J-FWD/rows/50/note	SIM / DERIVED
T2463	LC2U	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/51/state	(class set at the call site: DERIVED)
T2464	js5b LC2U (FE-factored, SF [1.0, 1.0]): (ii) hole bearing, exact V 2044.7 N on t 2.688	SIM	MMX v352	/rows/0/cells/J-FWD/rows/51/row	SIM / DERIVED
T2465	MISSING	SIM	MMX v352	/rows/0/cells/J-FWD/rows/51/MS_y	SIM / DERIVED
T2466	1.605	SIM	MMX v352	/rows/0/cells/J-FWD/rows/51/MS_u	SIM / DERIVED
T2467	231 (1)(ii)	DECIDED *Lead decision, per margin_matrix_v352.json v352*	MMX v352	/rows/0/cells/J-FWD/rows/51/criterion	tag in margin_matrix_v352.json v352:
T2468	FE-factored: the x1.5 / x2.0 state solved with contact at the 3750 N maximum preload ...	SIM	MMX v352	/rows/0/cells/J-FWD/rows/51/note	SIM / DERIVED
T2469	LC2Y	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/52/state	(class set at the call site: DERIVED)
T2470	js5b LC2Y (FE-factored, SF [1.0, 1.0]): (i) ligament Pm+Pb, half-ligament average	SIM	MMX v352	/rows/0/cells/J-FWD/rows/52/row	SIM / DERIVED
T2471	0.421	SIM	MMX v352	/rows/0/cells/J-FWD/rows/52/MS_y	SIM / DERIVED
T2472	MISSING	SIM	MMX v352	/rows/0/cells/J-FWD/rows/52/MS_u	SIM / DERIVED
T2473	231 (1)(i)	DECIDED *Lead decision, per margin_matrix_v352.json v352*	MMX v352	/rows/0/cells/J-FWD/rows/52/criterion	tag in margin_matrix_v352.json v352:
T2474	FE-factored: the x1.5 / x2.0 state solved with contact at the 3750 N maximum preload ...	SIM	MMX v352	/rows/0/cells/J-FWD/rows/52/note	SIM / DERIVED
T2475	LC2Y	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/53/state	(class set at the call site: DERIVED)
T2476	js5b LC2Y (FE-factored, SF [1.0, 1.0]): (i) ligament Pm, half-ligament average	SIM	MMX v352	/rows/0/cells/J-FWD/rows/53/row	SIM / DERIVED
T2477	2.539	SIM	MMX v352	/rows/0/cells/J-FWD/rows/53/MS_y	SIM / DERIVED
T2478	MISSING	SIM	MMX v352	/rows/0/cells/J-FWD/rows/53/MS_u	SIM / DERIVED
T2479	231 (1)(i)	DECIDED *Lead decision, per margin_matrix_v352.json v352*	MMX v352	/rows/0/cells/J-FWD/rows/53/criterion	tag in margin_matrix_v352.json v352:
T2480	FE-factored: the x1.5 / x2.0 state solved with contact at the 3750 N maximum preload ...	SIM	MMX v352	/rows/0/cells/J-FWD/rows/53/note	SIM / DERIVED
T2481	LC2Y	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/54/state	(class set at the call site: DERIVED)
T2482	js5b LC2Y (FE-factored, SF [1.0, 1.0]): (i) shear-out flank+ Pm+Pb	SIM	MMX v352	/rows/0/cells/J-FWD/rows/54/row	SIM / DERIVED
T2483	1.631	SIM	MMX v352	/rows/0/cells/J-FWD/rows/54/MS_y	SIM / DERIVED
T2484	MISSING	SIM	MMX v352	/rows/0/cells/J-FWD/rows/54/MS_u	SIM / DERIVED
T2485	231 (1)(i)	DECIDED *Lead decision, per margin_matrix_v352.json v352*	MMX v352	/rows/0/cells/J-FWD/rows/54/criterion	tag in margin_matrix_v352.json v352:
T2486	FE-factored: the x1.5 / x2.0 state solved with contact at the 3750 N maximum preload ...	SIM	MMX v352	/rows/0/cells/J-FWD/rows/54/note	SIM / DERIVED
T2487	LC2Y	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/55/state	(class set at the call site: DERIVED)
T2488	js5b LC2Y (FE-factored, SF [1.0, 1.0]): (i) shear-out flank- Pm+Pb	SIM	MMX v352	/rows/0/cells/J-FWD/rows/55/row	SIM / DERIVED
T2489	1.615	SIM	MMX v352	/rows/0/cells/J-FWD/rows/55/MS_y	SIM / DERIVED
T2490	MISSING	SIM	MMX v352	/rows/0/cells/J-FWD/rows/55/MS_u	SIM / DERIVED
T2491	231 (1)(i)	DECIDED *Lead decision, per margin_matrix_v352.json v352*	MMX v352	/rows/0/cells/J-FWD/rows/55/criterion	tag in margin_matrix_v352.json v352:
T2492	FE-factored: the x1.5 / x2.0 state solved with contact at the 3750 N maximum preload ...	SIM	MMX v352	/rows/0/cells/J-FWD/rows/55/note	SIM / DERIVED
T2493	LC2Y	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/56/state	(class set at the call site: DERIVED)
T2494	js5b LC2Y (FE-factored, SF [1.0, 1.0]): (ii) hole bearing, exact V 1514.6 N on t 2.688	SIM	MMX v352	/rows/0/cells/J-FWD/rows/56/row	SIM / DERIVED
T2495	1.871	SIM	MMX v352	/rows/0/cells/J-FWD/rows/56/MS_y	SIM / DERIVED
T2496	MISSING	SIM	MMX v352	/rows/0/cells/J-FWD/rows/56/MS_u	SIM / DERIVED
T2497	231 (1)(ii)	DECIDED *Lead decision, per margin_matrix_v352.json v352*	MMX v352	/rows/0/cells/J-FWD/rows/56/criterion	tag in margin_matrix_v352.json v352:
T2498	FE-factored: the x1.5 / x2.0 state solved with contact at the 3750 N maximum preload ...	SIM	MMX v352	/rows/0/cells/J-FWD/rows/56/note	SIM / DERIVED
T2499	LC2Y	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/57/state	(class set at the call site: DERIVED)
T2500	js5b LC2Y (FE-factored, SF [1.0, 1.0]): (iii) band Neuber n30 (local strain)	SIM	MMX v352	/rows/0/cells/J-FWD/rows/57/row	SIM / DERIVED
T2501	3.07	SIM	MMX v352	/rows/0/cells/J-FWD/rows/57/MS_y	SIM / DERIVED
T2502	MISSING	SIM	MMX v352	/rows/0/cells/J-FWD/rows/57/MS_u	SIM / DERIVED
T2503	231 (1)(iii)	DECIDED *Lead decision, per margin_matrix_v352.json v352*	MMX v352	/rows/0/cells/J-FWD/rows/57/criterion	tag in margin_matrix_v352.json v352:
T2504	FE-factored: the x1.5 / x2.0 state solved with contact at the 3750 N maximum preload ...	SIM	MMX v352	/rows/0/cells/J-FWD/rows/57/note	SIM / DERIVED
T2505	SOAK70	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/84/state	(class set at the call site: DERIVED)
T2506	IN718 flat bearing vs Fbry on the route (i) spring footprint (A 46.06 mm2, registered ...	CALC	MMX v352	/rows/0/cells/J-FWD/rows/84/row	CALC / DERIVED
T2507	0.355	CALC	MMX v352	/rows/0/cells/J-FWD/rows/84/MS_y	CALC / DERIVED
T2508	MISSING	CALC	MMX v352	/rows/0/cells/J-FWD/rows/84/MS_u	CALC / DERIVED
T2509	3750	CALC	MMX v352	/rows/0/cells/J-FWD/rows/84/preload_N	CALC / DERIVED
T2510	no-spring records (information): union 36.401 mm2 0.071, A_ROW 38.171 mm2 0.123	CALC	MMX v352	/rows/0/cells/J-FWD/rows/84/note	CALC / DERIVED
T2511	SOAK70	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/85/state	(class set at the call site: DERIVED)
T2512	IN718 under-head band Neuber (local strain) [ASSUMED (sigma_el x A_ANN 47.12 / A_supp ...	CALC	MMX v352	/rows/0/cells/J-FWD/rows/85/row	CALC / DERIVED; ASSUMED scaling
T2513	0.092	CALC	MMX v352	/rows/0/cells/J-FWD/rows/85/MS_y	CALC / DERIVED; ASSUMED scaling
T2514	MISSING	CALC	MMX v352	/rows/0/cells/J-FWD/rows/85/MS_u	CALC / DERIVED; ASSUMED scaling
T2515	3750	CALC	MMX v352	/rows/0/cells/J-FWD/rows/85/preload_N	CALC / DERIVED; ASSUMED scaling
T2516	(b2); full_float_0.557 (lead 17:55:55Z F1), mean-pressure scaling ASSUMED	CALC	MMX v352	/rows/0/cells/J-FWD/rows/85/note	CALC / DERIVED; ASSUMED scaling
T2517	SOAK70	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/87/state	(class set at the call site: DERIVED)
T2518	IN718 seat line d6.051 z'0 P+Q <= 2 Fty(T), F_sh form	CALC	MMX v352	/rows/0/cells/J-FWD/rows/87/row	CALC / DERIVED
T2519	−0.059	CALC	MMX v352	/rows/0/cells/J-FWD/rows/87/MS_y	CALC / DERIVED
T2520	MISSING	CALC	MMX v352	/rows/0/cells/J-FWD/rows/87/MS_u	CALC / DERIVED
T2521	3750	CALC	MMX v352	/rows/0/cells/J-FWD/rows/87/preload_N	CALC / DERIVED
T2522	240 (9)(i), (10)	DECIDED *Lead decision, per margin_matrix_v352.json v352*	MMX v352	/rows/0/cells/J-FWD/rows/87/criterion	tag in margin_matrix_v352.json v352:
T2523	(b2)	CALC	MMX v352	/rows/0/cells/J-FWD/rows/87/note	CALC / DERIVED
T2524	SOAK70	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/89/state	(class set at the call site: DERIVED)
T2525	IN718 ring line P+Q <= 2 Fty(T), F_sh form	CALC	MMX v352	/rows/0/cells/J-FWD/rows/89/row	CALC / DERIVED
T2526	−0.059	CALC	MMX v352	/rows/0/cells/J-FWD/rows/89/MS_y	CALC / DERIVED
T2527	MISSING	CALC	MMX v352	/rows/0/cells/J-FWD/rows/89/MS_u	CALC / DERIVED
T2528	3750	CALC	MMX v352	/rows/0/cells/J-FWD/rows/89/preload_N	CALC / DERIVED
T2529	240 (9)(i), (10)	DECIDED *Lead decision, per margin_matrix_v352.json v352*	MMX v352	/rows/0/cells/J-FWD/rows/89/criterion	tag in margin_matrix_v352.json v352:
T2530	(b2)	CALC	MMX v352	/rows/0/cells/J-FWD/rows/89/note	CALC / DERIVED
T2531	SOAK	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/100/state	(class set at the call site: DERIVED)
T2532	IN718 flat bearing vs Fbry on the route (i) spring footprint (A 46.06 mm2, registered ...	CALC	MMX v352	/rows/0/cells/J-FWD/rows/100/row	CALC / DERIVED
T2533	0.354	CALC	MMX v352	/rows/0/cells/J-FWD/rows/100/MS_y	CALC / DERIVED
T2534	MISSING	CALC	MMX v352	/rows/0/cells/J-FWD/rows/100/MS_u	CALC / DERIVED
T2535	3750	CALC	MMX v352	/rows/0/cells/J-FWD/rows/100/preload_N	CALC / DERIVED

ID	Value	Class	Source	Key	Source tag / formula
T2536	no-spring records (information): union 36.401 mm2 0.07, A_ROW 38.171 mm2 0.122	CALC	MMX v352	/rows/0/cells/J-FWD/rows/100/note	CALC / DERIVED
T2537	SOAK	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/101/state	(class set at the call site: DERIVED)
T2538	IN718 under-head band Neuber (local strain) [ASSUMED (sigma_el x A_ANN 47.12 / A_supp ...	CALC	MMX v352	/rows/0/cells/J-FWD/rows/101/row	CALC / DERIVED; ASSUMED scaling
T2539	0.191	CALC	MMX v352	/rows/0/cells/J-FWD/rows/101/MS_y	CALC / DERIVED; ASSUMED scaling
T2540	MISSING	CALC	MMX v352	/rows/0/cells/J-FWD/rows/101/MS_u	CALC / DERIVED; ASSUMED scaling
T2541	3750	CALC	MMX v352	/rows/0/cells/J-FWD/rows/101/preload_N	CALC / DERIVED; ASSUMED scaling
T2542	(b1); full_float_0.557 (lead 17:55:55Z F1), mean-pressure scaling ASSUMED	CALC	MMX v352	/rows/0/cells/J-FWD/rows/101/note	CALC / DERIVED; ASSUMED scaling
T2543	SOAK	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/103/state	(class set at the call site: DERIVED)
T2544	IN718 seat line d6.051 z'0 P+Q <= 2 Fty(T), F_sh form	CALC	MMX v352	/rows/0/cells/J-FWD/rows/103/row	CALC / DERIVED
T2545	−0.075	CALC	MMX v352	/rows/0/cells/J-FWD/rows/103/MS_y	CALC / DERIVED
T2546	MISSING	CALC	MMX v352	/rows/0/cells/J-FWD/rows/103/MS_u	CALC / DERIVED
T2547	3750	CALC	MMX v352	/rows/0/cells/J-FWD/rows/103/preload_N	CALC / DERIVED
T2548	240 (9)(i), (10)	DECIDED *Lead decision, per margin_matrix_v352.json v352*	MMX v352	/rows/0/cells/J-FWD/rows/103/criterion	tag in margin_matrix_v352.json v352:
T2549	(b2) rep	CALC	MMX v352	/rows/0/cells/J-FWD/rows/103/note	CALC / DERIVED
T2550	SOAK	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/105/state	(class set at the call site: DERIVED)
T2551	IN718 ring line P+Q <= 2 Fty(T), F_sh form	CALC	MMX v352	/rows/0/cells/J-FWD/rows/105/row	CALC / DERIVED
T2552	−0.075	CALC	MMX v352	/rows/0/cells/J-FWD/rows/105/MS_y	CALC / DERIVED
T2553	MISSING	CALC	MMX v352	/rows/0/cells/J-FWD/rows/105/MS_u	CALC / DERIVED
T2554	3750	CALC	MMX v352	/rows/0/cells/J-FWD/rows/105/preload_N	CALC / DERIVED
T2555	240 (9)(i), (10)	DECIDED *Lead decision, per margin_matrix_v352.json v352*	MMX v352	/rows/0/cells/J-FWD/rows/105/criterion	tag in margin_matrix_v352.json v352:
T2556	(b2) rep	CALC	MMX v352	/rows/0/cells/J-FWD/rows/105/note	CALC / DERIVED
T2557	SOAK52	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/116/state	(class set at the call site: DERIVED)
T2558	IN718 flat bearing vs Fbry on the route (i) spring footprint (A 46.06 mm2, registered ...	CALC	MMX v352	/rows/0/cells/J-FWD/rows/116/row	CALC / DERIVED
T2559	0.373	CALC	MMX v352	/rows/0/cells/J-FWD/rows/116/MS_y	CALC / DERIVED
T2560	MISSING	CALC	MMX v352	/rows/0/cells/J-FWD/rows/116/MS_u	CALC / DERIVED
T2561	3750	CALC	MMX v352	/rows/0/cells/J-FWD/rows/116/preload_N	CALC / DERIVED
T2562	no-spring records (information): union 36.401 mm2 0.085, A_ROW 38.171 mm2 0.138	CALC	MMX v352	/rows/0/cells/J-FWD/rows/116/note	CALC / DERIVED
T2563	SOAK52	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/117/state	(class set at the call site: DERIVED)
T2564	IN718 under-head band Neuber (local strain) [ASSUMED (sigma_el x A_ANN 47.12 / A_supp ...	CALC	MMX v352	/rows/0/cells/J-FWD/rows/117/row	CALC / DERIVED; ASSUMED scaling
T2565	0.039	CALC	MMX v352	/rows/0/cells/J-FWD/rows/117/MS_y	CALC / DERIVED; ASSUMED scaling
T2566	MISSING	CALC	MMX v352	/rows/0/cells/J-FWD/rows/117/MS_u	CALC / DERIVED; ASSUMED scaling
T2567	3750	CALC	MMX v352	/rows/0/cells/J-FWD/rows/117/preload_N	CALC / DERIVED; ASSUMED scaling
T2568	(b2); full_float_0.557 (lead 17:55:55Z F1), mean-pressure scaling ASSUMED	CALC	MMX v352	/rows/0/cells/J-FWD/rows/117/note	CALC / DERIVED; ASSUMED scaling
T2569	SOAK52	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/119/state	(class set at the call site: DERIVED)
T2570	IN718 seat line d6.051 z'0 P+Q <= 2 Fty(T), F_sh form	CALC	MMX v352	/rows/0/cells/J-FWD/rows/119/row	CALC / DERIVED
T2571	−0.033	CALC	MMX v352	/rows/0/cells/J-FWD/rows/119/MS_y	CALC / DERIVED
T2572	MISSING	CALC	MMX v352	/rows/0/cells/J-FWD/rows/119/MS_u	CALC / DERIVED
T2573	3750	CALC	MMX v352	/rows/0/cells/J-FWD/rows/119/preload_N	CALC / DERIVED
T2574	240 (9)(i), (10)	DECIDED *Lead decision, per margin_matrix_v352.json v352*	MMX v352	/rows/0/cells/J-FWD/rows/119/criterion	tag in margin_matrix_v352.json v352:
T2575	(b1)	CALC	MMX v352	/rows/0/cells/J-FWD/rows/119/note	CALC / DERIVED
T2576	SOAK52	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/121/state	(class set at the call site: DERIVED)
T2577	IN718 ring line P+Q <= 2 Fty(T), F_sh form	CALC	MMX v352	/rows/0/cells/J-FWD/rows/121/row	CALC / DERIVED
T2578	−0.033	CALC	MMX v352	/rows/0/cells/J-FWD/rows/121/MS_y	CALC / DERIVED
T2579	MISSING	CALC	MMX v352	/rows/0/cells/J-FWD/rows/121/MS_u	CALC / DERIVED
T2580	3750	CALC	MMX v352	/rows/0/cells/J-FWD/rows/121/preload_N	CALC / DERIVED
T2581	240 (9)(i), (10)	DECIDED *Lead decision, per margin_matrix_v352.json v352*	MMX v352	/rows/0/cells/J-FWD/rows/121/criterion	tag in margin_matrix_v352.json v352:
T2582	(b1)	CALC	MMX v352	/rows/0/cells/J-FWD/rows/121/note	CALC / DERIVED
T2583	FA1S	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/132/state	(class set at the call site: DERIVED)
T2584	IN718 under-head band Neuber (local strain) [ASSUMED (sigma_el x A_ANN 47.12 / A_supp ...	CALC	MMX v352	/rows/0/cells/J-FWD/rows/132/row	CALC / DERIVED; ASSUMED scaling
T2585	0.259	CALC	MMX v352	/rows/0/cells/J-FWD/rows/132/MS_y	CALC / DERIVED; ASSUMED scaling
T2586	MISSING	CALC	MMX v352	/rows/0/cells/J-FWD/rows/132/MS_u	CALC / DERIVED; ASSUMED scaling
T2587	3750	CALC	MMX v352	/rows/0/cells/J-FWD/rows/132/preload_N	CALC / DERIVED; ASSUMED scaling
T2588	(b2); full_float_0.557 (lead 17:55:55Z F1), mean-pressure scaling ASSUMED	CALC	MMX v352	/rows/0/cells/J-FWD/rows/132/note	CALC / DERIVED; ASSUMED scaling
T2589	LC2N	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/58/state	(class set at the call site: DERIVED)
T2590	js5b LC2N (LC2, SF [1.5, 2.0]): (i) ligament Pm+Pb, half-ligament average	SIM	MMX v352	/rows/0/cells/J-FWD/rows/58/row	SIM / DERIVED
T2591	0.016	SIM	MMX v352	/rows/0/cells/J-FWD/rows/58/MS_y	SIM / DERIVED
T2592	−0.183	SIM	MMX v352	/rows/0/cells/J-FWD/rows/58/MS_u	SIM / DERIVED
T2593	231 (1)(i)	DECIDED *Lead decision, per margin_matrix_v352.json v352*	MMX v352	/rows/0/cells/J-FWD/rows/58/criterion	tag in margin_matrix_v352.json v352:
T2594	hand form: SF on the total stress including preload, linear superposition ASSUMED ...	SIM	MMX v352	/rows/0/cells/J-FWD/rows/58/note	SIM / DERIVED
T2595	LC2N	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/59/state	(class set at the call site: DERIVED)
T2596	js5b LC2N (LC2, SF [1.5, 2.0]): (i) ligament Pm, half-ligament average	SIM	MMX v352	/rows/0/cells/J-FWD/rows/59/row	SIM / DERIVED
T2597	1.751	SIM	MMX v352	/rows/0/cells/J-FWD/rows/59/MS_y	SIM / DERIVED
T2598	1.212	SIM	MMX v352	/rows/0/cells/J-FWD/rows/59/MS_u	SIM / DERIVED
T2599	231 (1)(i)	DECIDED *Lead decision, per margin_matrix_v352.json v352*	MMX v352	/rows/0/cells/J-FWD/rows/59/criterion	tag in margin_matrix_v352.json v352:
T2600	hand form: SF on the total stress including preload, linear superposition ASSUMED ...	SIM	MMX v352	/rows/0/cells/J-FWD/rows/59/note	SIM / DERIVED
T2601	LC2N	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/60/state	(class set at the call site: DERIVED)
T2602	js5b LC2N (LC2, SF [1.5, 2.0]): (i) shear-out flank+ Pm+Pb	SIM	MMX v352	/rows/0/cells/J-FWD/rows/60/row	SIM / DERIVED
T2603	0.944	SIM	MMX v352	/rows/0/cells/J-FWD/rows/60/MS_y	SIM / DERIVED
T2604	0.563	SIM	MMX v352	/rows/0/cells/J-FWD/rows/60/MS_u	SIM / DERIVED
T2605	231 (1)(i)	DECIDED *Lead decision, per margin_matrix_v352.json v352*	MMX v352	/rows/0/cells/J-FWD/rows/60/criterion	tag in margin_matrix_v352.json v352:
T2606	hand form: SF on the total stress including preload, linear superposition ASSUMED ...	SIM	MMX v352	/rows/0/cells/J-FWD/rows/60/note	SIM / DERIVED
T2607	LC2N	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/61/state	(class set at the call site: DERIVED)
T2608	js5b LC2N (LC2, SF [1.5, 2.0]): (i) shear-out flank- Pm+Pb	SIM	MMX v352	/rows/0/cells/J-FWD/rows/61/row	SIM / DERIVED
T2609	0.948	SIM	MMX v352	/rows/0/cells/J-FWD/rows/61/MS_y	SIM / DERIVED
T2610	0.566	SIM	MMX v352	/rows/0/cells/J-FWD/rows/61/MS_u	SIM / DERIVED
T2611	231 (1)(i)	DECIDED *Lead decision, per margin_matrix_v352.json v352*	MMX v352	/rows/0/cells/J-FWD/rows/61/criterion	tag in margin_matrix_v352.json v352:
T2612	hand form: SF on the total stress including preload, linear superposition ASSUMED ...	SIM	MMX v352	/rows/0/cells/J-FWD/rows/61/note	SIM / DERIVED
T2613	LC2N	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/62/state	(class set at the call site: DERIVED)

ID	Value	Class	Source	Key	Source tag / formula
T2614	js5b LC2N (LC2, SF [1.5, 2.0]): (ii) hole bearing, exact V 988.2 N on t 2.688	SIM	MMX v352	/rows/0/cells/J-FWD/rows/62/row	SIM / DERIVED
T2615	1.933	SIM	MMX v352	/rows/0/cells/J-FWD/rows/62/MS_y	SIM / DERIVED
T2616	1.695	SIM	MMX v352	/rows/0/cells/J-FWD/rows/62/MS_u	SIM / DERIVED
T2617	231 (1)(ii)	DECIDED Lead decision, per margin_matrix_v352.json v352	MMX v352	/rows/0/cells/J-FWD/rows/62/criterion	tag in margin_matrix_v352.json v352:
T2618	hand form: SF on the total stress including preload, linear superposition ASSUMED ...	SIM	MMX v352	/rows/0/cells/J-FWD/rows/62/note	SIM / DERIVED
T2619	LC2N	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/63/state	(class set at the call site: DERIVED)
T2620	js5b LC2N (LC2, SF [1.5, 2.0]): (iii) band Neuber n30 (local strain)	SIM	MMX v352	/rows/0/cells/J-FWD/rows/63/row	SIM / DERIVED
T2621	5.146	SIM	MMX v352	/rows/0/cells/J-FWD/rows/63/MS_y	SIM / DERIVED
T2622	MISSING	SIM	MMX v352	/rows/0/cells/J-FWD/rows/63/MS_u	SIM / DERIVED
T2623	231 (1)(iii)	DECIDED Lead decision, per margin_matrix_v352.json v352	MMX v352	/rows/0/cells/J-FWD/rows/63/criterion	tag in margin_matrix_v352.json v352:
T2624	hand form: SF on the total stress including preload, linear superposition ASSUMED ...	SIM	MMX v352	/rows/0/cells/J-FWD/rows/63/note	SIM / DERIVED
T2625	LC2M	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/64/state	(class set at the call site: DERIVED)
T2626	js5b LC2M (LC2, SF [1.5, 2.0]): (i) ligament Pm+Pb, half-ligament average	SIM	MMX v352	/rows/0/cells/J-FWD/rows/64/row	SIM / DERIVED
T2627	0.266	SIM	MMX v352	/rows/0/cells/J-FWD/rows/64/MS_y	SIM / DERIVED
T2628	0.021	SIM	MMX v352	/rows/0/cells/J-FWD/rows/64/MS_u	SIM / DERIVED
T2629	231 (1)(i)	DECIDED Lead decision, per margin_matrix_v352.json v352	MMX v352	/rows/0/cells/J-FWD/rows/64/criterion	tag in margin_matrix_v352.json v352:
T2630	hand form: SF on the total stress including preload, linear superposition ASSUMED ...	SIM	MMX v352	/rows/0/cells/J-FWD/rows/64/note	SIM / DERIVED
T2631	LC2M	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/65/state	(class set at the call site: DERIVED)
T2632	js5b LC2M (LC2, SF [1.5, 2.0]): (i) ligament Pm, half-ligament average	SIM	MMX v352	/rows/0/cells/J-FWD/rows/65/row	SIM / DERIVED
T2633	2.22	SIM	MMX v352	/rows/0/cells/J-FWD/rows/65/MS_y	SIM / DERIVED
T2634	1.599	SIM	MMX v352	/rows/0/cells/J-FWD/rows/65/MS_u	SIM / DERIVED
T2635	231 (1)(i)	DECIDED Lead decision, per margin_matrix_v352.json v352	MMX v352	/rows/0/cells/J-FWD/rows/65/criterion	tag in margin_matrix_v352.json v352:
T2636	hand form: SF on the total stress including preload, linear superposition ASSUMED ...	SIM	MMX v352	/rows/0/cells/J-FWD/rows/65/note	SIM / DERIVED
T2637	LC2M	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/66/state	(class set at the call site: DERIVED)
T2638	js5b LC2M (LC2, SF [1.5, 2.0]): (i) shear-out flank+ Pm+Pb	SIM	MMX v352	/rows/0/cells/J-FWD/rows/66/row	SIM / DERIVED
T2639	1.279	SIM	MMX v352	/rows/0/cells/J-FWD/rows/66/MS_y	SIM / DERIVED
T2640	0.84	SIM	MMX v352	/rows/0/cells/J-FWD/rows/66/MS_u	SIM / DERIVED
T2641	231 (1)(i)	DECIDED Lead decision, per margin_matrix_v352.json v352	MMX v352	/rows/0/cells/J-FWD/rows/66/criterion	tag in margin_matrix_v352.json v352:
T2642	hand form: SF on the total stress including preload, linear superposition ASSUMED ...	SIM	MMX v352	/rows/0/cells/J-FWD/rows/66/note	SIM / DERIVED
T2643	LC2M	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/67/state	(class set at the call site: DERIVED)
T2644	js5b LC2M (LC2, SF [1.5, 2.0]): (i) shear-out flank- Pm+Pb	SIM	MMX v352	/rows/0/cells/J-FWD/rows/67/row	SIM / DERIVED
T2645	1.276	SIM	MMX v352	/rows/0/cells/J-FWD/rows/67/MS_y	SIM / DERIVED
T2646	0.837	SIM	MMX v352	/rows/0/cells/J-FWD/rows/67/MS_u	SIM / DERIVED
T2647	231 (1)(i)	DECIDED Lead decision, per margin_matrix_v352.json v352	MMX v352	/rows/0/cells/J-FWD/rows/67/criterion	tag in margin_matrix_v352.json v352:
T2648	hand form: SF on the total stress including preload, linear superposition ASSUMED ...	SIM	MMX v352	/rows/0/cells/J-FWD/rows/67/note	SIM / DERIVED
T2649	LC2M	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/68/state	(class set at the call site: DERIVED)
T2650	js5b LC2M (LC2, SF [1.5, 2.0]): (ii) hole bearing, exact V 995.4 N on t 2.688	SIM	MMX v352	/rows/0/cells/J-FWD/rows/68/row	SIM / DERIVED
T2651	1.952	SIM	MMX v352	/rows/0/cells/J-FWD/rows/68/MS_y	SIM / DERIVED
T2652	1.723	SIM	MMX v352	/rows/0/cells/J-FWD/rows/68/MS_u	SIM / DERIVED
T2653	231 (1)(ii)	DECIDED Lead decision, per margin_matrix_v352.json v352	MMX v352	/rows/0/cells/J-FWD/rows/68/criterion	tag in margin_matrix_v352.json v352:
T2654	hand form: SF on the total stress including preload, linear superposition ASSUMED ...	SIM	MMX v352	/rows/0/cells/J-FWD/rows/68/note	SIM / DERIVED
T2655	LC2M	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/69/state	(class set at the call site: DERIVED)
T2656	js5b LC2M (LC2, SF [1.5, 2.0]): (iii) band Neuber n30 (local strain)	SIM	MMX v352	/rows/0/cells/J-FWD/rows/69/row	SIM / DERIVED
T2657	6.441	SIM	MMX v352	/rows/0/cells/J-FWD/rows/69/MS_y	SIM / DERIVED
T2658	MISSING	SIM	MMX v352	/rows/0/cells/J-FWD/rows/69/MS_u	SIM / DERIVED
T2659	231 (1)(iii)	DECIDED Lead decision, per margin_matrix_v352.json v352	MMX v352	/rows/0/cells/J-FWD/rows/69/criterion	tag in margin_matrix_v352.json v352:
T2660	hand form: SF on the total stress including preload, linear superposition ASSUMED ...	SIM	MMX v352	/rows/0/cells/J-FWD/rows/69/note	SIM / DERIVED
T2661	FA1S	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/8/state	(class set at the call site: DERIVED)
T2662	elastic band (contact-edge field)	SIM	MMX v352	/rows/0/cells/J-FWD/rows/8/row	SIM / DERIVED
T2663	−0.576	SIM	MMX v352	/rows/0/cells/J-FWD/rows/8/MS_y	SIM / DERIVED
T2664	−0.543	SIM	MMX v352	/rows/0/cells/J-FWD/rows/8/MS_u	SIM / DERIVED
T2665	FA1S	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/9/state	(class set at the call site: DERIVED)
T2666	all-node (hole / head contact edge)	SIM	MMX v352	/rows/0/cells/J-FWD/rows/9/row	SIM / DERIVED
T2667	−0.946	SIM	MMX v352	/rows/0/cells/J-FWD/rows/9/MS_y	SIM / DERIVED
T2668	−0.942	SIM	MMX v352	/rows/0/cells/J-FWD/rows/9/MS_u	SIM / DERIVED
T2669	n388 4390.4 MPa	SIM	MMX v352	/rows/0/cells/J-FWD/rows/9/note	SIM / DERIVED
T2670	FA1S	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/14/state	(class set at the call site: DERIVED)
T2671	(iii) hole-edge Neuber (Mechanical's information row)	CALC	MMX v352	/rows/0/cells/J-FWD/rows/14/row	CALC
T2672	1.384	CALC	MMX v352	/rows/0/cells/J-FWD/rows/14/MS_y	CALC
T2673	MISSING	CALC	MMX v352	/rows/0/cells/J-FWD/rows/14/MS_u	CALC
T2674	PRELOAD	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/15/state	(class set at the call site: DERIVED)
T2675	band Neuber n30 (under-head band, local strain)	SIM	MMX v352	/rows/0/cells/J-FWD/rows/15/row	SIM / DERIVED
T2676	9.3	SIM	MMX v352	/rows/0/cells/J-FWD/rows/15/MS_y	SIM / DERIVED
T2677	MISSING	SIM	MMX v352	/rows/0/cells/J-FWD/rows/15/MS_u	SIM / DERIVED
T2678	lead 06:36Z (DC head seat)	DECIDED Lead decision, per margin_matrix_v352.json v352	MMX v352	/rows/0/cells/J-FWD/rows/15/criterion	tag in margin_matrix_v352.json v352:
T2679	PRELOAD	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/16/state	(class set at the call site: DERIVED)
T2680	flat-seat head bearing vs Fbry (A4-80 FE head force)	SIM	MMX v352	/rows/0/cells/J-FWD/rows/16/row	SIM / DERIVED
T2681	3.222	SIM	MMX v352	/rows/0/cells/J-FWD/rows/16/MS_y	SIM / DERIVED
T2682	MISSING	SIM	MMX v352	/rows/0/cells/J-FWD/rows/16/MS_u	SIM / DERIVED
T2683	flight row = IN718 below	SIM	MMX v352	/rows/0/cells/J-FWD/rows/16/note	SIM / DERIVED
T2684	PRELOAD	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/17/state	(class set at the call site: DERIVED)
T2685	ligament station P+Q (0.087 mm from the hole wall; contact-edge field)	SIM	MMX v352	/rows/0/cells/J-FWD/rows/17/row	SIM / DERIVED
T2686	−0.096	SIM	MMX v352	/rows/0/cells/J-FWD/rows/17/MS_y	SIM / DERIVED
T2687	MISSING	SIM	MMX v352	/rows/0/cells/J-FWD/rows/17/MS_u	SIM / DERIVED
T2688	240 (9): information	DECIDED Lead decision, per margin_matrix_v352.json v352	MMX v352	/rows/0/cells/J-FWD/rows/17/criterion	tag in margin_matrix_v352.json v352:
T2689	PRELOAD	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/18/state	(class set at the call site: DERIVED)

ID	Value	Class	Source	Key	Source tag / formula
T2690	all-node (hole / head contact edge)	SIM	MMX v352	/rows/0/cells/J-FWD/rows/18/row	SIM / DERIVED
T2691	−0.941	SIM	MMX v352	/rows/0/cells/J-FWD/rows/18/MS_y	SIM / DERIVED
T2692	−0.936	SIM	MMX v352	/rows/0/cells/J-FWD/rows/18/MS_u	SIM / DERIVED
T2693	n388 4073.3 MPa	SIM	MMX v352	/rows/0/cells/J-FWD/rows/18/note	SIM / DERIVED
T2694	PRELOAD	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/19/state	(class set at the call site: DERIVED)
T2695	54 seat lines P+Q <= 2 Fty(T), envelope minimum (line d+6.051 z'+0.0, (b2); A4-80 FE)	SIM	MMX v352	/rows/0/cells/J-FWD/rows/19/row	SIM / DERIVED
T2696	1.038	SIM	MMX v352	/rows/0/cells/J-FWD/rows/19/MS_y	SIM / DERIVED
T2697	MISSING	SIM	MMX v352	/rows/0/cells/J-FWD/rows/19/MS_u	SIM / DERIVED
T2698	240 (9)(i)	DECIDED *Lead decision, per margin_matrix_v352.json v352*	MMX v352	/rows/0/cells/J-FWD/rows/19/criterion	tag in margin_matrix_v352.json v352:
T2699	SOAK70	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/21/state	(class set at the call site: DERIVED)
T2700	flat-seat head bearing vs Fbry (A4-80 FE head force)	SIM	MMX v352	/rows/0/cells/J-FWD/rows/21/row	SIM / DERIVED
T2701	1.003	SIM	MMX v352	/rows/0/cells/J-FWD/rows/21/MS_y	SIM / DERIVED
T2702	MISSING	SIM	MMX v352	/rows/0/cells/J-FWD/rows/21/MS_u	SIM / DERIVED
T2703	flight row = IN718 below	SIM	MMX v352	/rows/0/cells/J-FWD/rows/21/note	SIM / DERIVED
T2704	SOAK70	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/22/state	(class set at the call site: DERIVED)
T2705	ligament station P+Q (0.087 mm from the hole wall; contact-edge field)	SIM	MMX v352	/rows/0/cells/J-FWD/rows/22/row	SIM / DERIVED
T2706	−0.395	SIM	MMX v352	/rows/0/cells/J-FWD/rows/22/MS_y	SIM / DERIVED
T2707	MISSING	SIM	MMX v352	/rows/0/cells/J-FWD/rows/22/MS_u	SIM / DERIVED
T2708	240 (9): information	DECIDED *Lead decision, per margin_matrix_v352.json v352*	MMX v352	/rows/0/cells/J-FWD/rows/22/criterion	tag in margin_matrix_v352.json v352:
T2709	SOAK70	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/23/state	(class set at the call site: DERIVED)
T2710	all-node (hole / head contact edge)	SIM	MMX v352	/rows/0/cells/J-FWD/rows/23/row	SIM / DERIVED
T2711	−0.95	SIM	MMX v352	/rows/0/cells/J-FWD/rows/23/MS_y	SIM / DERIVED
T2712	−0.946	SIM	MMX v352	/rows/0/cells/J-FWD/rows/23/MS_u	SIM / DERIVED
T2713	n398 3371.4 MPa	SIM	MMX v352	/rows/0/cells/J-FWD/rows/23/note	SIM / DERIVED
T2714	SOAK	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/26/state	(class set at the call site: DERIVED)
T2715	flat-seat head bearing vs Fbry (A4-80 FE head force)	SIM	MMX v352	/rows/0/cells/J-FWD/rows/26/row	SIM / DERIVED
T2716	0.522	SIM	MMX v352	/rows/0/cells/J-FWD/rows/26/MS_y	SIM / DERIVED
T2717	MISSING	SIM	MMX v352	/rows/0/cells/J-FWD/rows/26/MS_u	SIM / DERIVED
T2718	flight row = IN718 below	SIM	MMX v352	/rows/0/cells/J-FWD/rows/26/note	SIM / DERIVED
T2719	SOAK	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/27/state	(class set at the call site: DERIVED)
T2720	ligament station P+Q (0.087 mm from the hole wall; contact-edge field)	SIM	MMX v352	/rows/0/cells/J-FWD/rows/27/row	SIM / DERIVED
T2721	−0.2	SIM	MMX v352	/rows/0/cells/J-FWD/rows/27/MS_y	SIM / DERIVED
T2722	MISSING	SIM	MMX v352	/rows/0/cells/J-FWD/rows/27/MS_u	SIM / DERIVED
T2723	240 (9): information	DECIDED *Lead decision, per margin_matrix_v352.json v352*	MMX v352	/rows/0/cells/J-FWD/rows/27/criterion	tag in margin_matrix_v352.json v352:
T2724	SOAK	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/28/state	(class set at the call site: DERIVED)
T2725	all-node (hole / head contact edge)	SIM	MMX v352	/rows/0/cells/J-FWD/rows/28/row	SIM / DERIVED
T2726	−0.956	SIM	MMX v352	/rows/0/cells/J-FWD/rows/28/MS_y	SIM / DERIVED
T2727	−0.953	SIM	MMX v352	/rows/0/cells/J-FWD/rows/28/MS_u	SIM / DERIVED
T2728	n338 3842.7 MPa	SIM	MMX v352	/rows/0/cells/J-FWD/rows/28/note	SIM / DERIVED
T2729	SOAK52	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/31/state	(class set at the call site: DERIVED)
T2730	flat-seat head bearing vs Fbry (A4-80 FE head force)	SIM	MMX v352	/rows/0/cells/J-FWD/rows/31/row	SIM / DERIVED
T2731	0.836	SIM	MMX v352	/rows/0/cells/J-FWD/rows/31/MS_y	SIM / DERIVED
T2732	MISSING	SIM	MMX v352	/rows/0/cells/J-FWD/rows/31/MS_u	SIM / DERIVED
T2733	flight row = IN718 below	SIM	MMX v352	/rows/0/cells/J-FWD/rows/31/note	SIM / DERIVED
T2734	SOAK52	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/32/state	(class set at the call site: DERIVED)
T2735	ligament station P+Q (0.087 mm from the hole wall; contact-edge field)	SIM	MMX v352	/rows/0/cells/J-FWD/rows/32/row	SIM / DERIVED
T2736	−0.208	SIM	MMX v352	/rows/0/cells/J-FWD/rows/32/MS_y	SIM / DERIVED
T2737	MISSING	SIM	MMX v352	/rows/0/cells/J-FWD/rows/32/MS_u	SIM / DERIVED
T2738	240 (9): information	DECIDED *Lead decision, per margin_matrix_v352.json v352*	MMX v352	/rows/0/cells/J-FWD/rows/32/criterion	tag in margin_matrix_v352.json v352:
T2739	SOAK52	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/33/state	(class set at the call site: DERIVED)
T2740	all-node (hole / head contact edge)	SIM	MMX v352	/rows/0/cells/J-FWD/rows/33/row	SIM / DERIVED
T2741	−0.964	SIM	MMX v352	/rows/0/cells/J-FWD/rows/33/MS_y	SIM / DERIVED
T2742	−0.962	SIM	MMX v352	/rows/0/cells/J-FWD/rows/33/MS_u	SIM / DERIVED
T2743	n395 5034.6 MPa	SIM	MMX v352	/rows/0/cells/J-FWD/rows/33/note	SIM / DERIVED
T2744	PRELOAD	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/70/state	(class set at the call site: DERIVED)
T2745	IN718 flat bearing vs Fbry on the route (i) spring footprint (A 46.06 mm2, registered ...	CALC	MMX v352	/rows/0/cells/J-FWD/rows/70/row	CALC / DERIVED
T2746	3.127	CALC	MMX v352	/rows/0/cells/J-FWD/rows/70/MS_y	CALC / DERIVED
T2747	MISSING	CALC	MMX v352	/rows/0/cells/J-FWD/rows/70/MS_u	CALC / DERIVED
T2748	3750	CALC	MMX v352	/rows/0/cells/J-FWD/rows/70/preload_N	CALC / DERIVED
T2749	no-spring records (information): union 36.401 mm2 2.261, A_ROW 38.171 mm2 2.42	CALC	MMX v352	/rows/0/cells/J-FWD/rows/70/note	CALC / DERIVED
T2750	PRELOAD	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/71/state	(class set at the call site: DERIVED)
T2751	IN718 under-head band Neuber (local strain) [ASSUMED (sigma_el x A_ANN 47.12 / A_supp ...	CALC	MMX v352	/rows/0/cells/J-FWD/rows/71/row	CALC / DERIVED; ASSUMED scaling
T2752	5.91	CALC	MMX v352	/rows/0/cells/J-FWD/rows/71/MS_y	CALC / DERIVED; ASSUMED scaling
T2753	MISSING	CALC	MMX v352	/rows/0/cells/J-FWD/rows/71/MS_u	CALC / DERIVED; ASSUMED scaling
T2754	3750	CALC	MMX v352	/rows/0/cells/J-FWD/rows/71/preload_N	CALC / DERIVED; ASSUMED scaling
T2755	(b1); full_float_0.557 (lead 17:55:55Z F1), mean-pressure scaling ASSUMED	CALC	MMX v352	/rows/0/cells/J-FWD/rows/71/note	CALC / DERIVED; ASSUMED scaling
T2756	PRELOAD	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/72/state	(class set at the call site: DERIVED)
T2757	IN718 ligament station P+Q (0.08 mm from the hole wall; contact-edge field)	CALC	MMX v352	/rows/0/cells/J-FWD/rows/72/row	CALC / DERIVED
T2758	−0.096	CALC	MMX v352	/rows/0/cells/J-FWD/rows/72/MS_y	CALC / DERIVED
T2759	MISSING	CALC	MMX v352	/rows/0/cells/J-FWD/rows/72/MS_u	CALC / DERIVED
T2760	3750	CALC	MMX v352	/rows/0/cells/J-FWD/rows/72/preload_N	CALC / DERIVED
T2761	240 (9): information	DECIDED *Lead decision, per margin_matrix_v352.json v352*	MMX v352	/rows/0/cells/J-FWD/rows/72/criterion	tag in margin_matrix_v352.json v352:
T2762	PRELOAD	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/73/state	(class set at the call site: DERIVED)
T2763	IN718 seat line d6.051 z'0 P+Q <= 2 Fty(T), F_sh form	CALC	MMX v352	/rows/0/cells/J-FWD/rows/73/row	CALC / DERIVED
T2764	1.038	CALC	MMX v352	/rows/0/cells/J-FWD/rows/73/MS_y	CALC / DERIVED
T2765	MISSING	CALC	MMX v352	/rows/0/cells/J-FWD/rows/73/MS_u	CALC / DERIVED
T2766	3750	CALC	MMX v352	/rows/0/cells/J-FWD/rows/73/preload_N	CALC / DERIVED
T2767	240 (9)(i), (10)	DECIDED *Lead decision, per margin_matrix_v352.json v352*	MMX v352	/rows/0/cells/J-FWD/rows/73/criterion	tag in margin_matrix_v352.json v352:
T2768	(b2)	CALC	MMX v352	/rows/0/cells/J-FWD/rows/73/note	CALC / DERIVED
T2769	PRELOAD	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/74/state	(class set at the call site: DERIVED)
T2770	IN718 seat line d6.051 z'0 P+Q, head-force bound	CALC	MMX v352	/rows/0/cells/J-FWD/rows/74/row	CALC / DERIVED
T2771	1.038	CALC	MMX v352	/rows/0/cells/J-FWD/rows/74/MS_y	CALC / DERIVED
T2772	MISSING	CALC	MMX v352	/rows/0/cells/J-FWD/rows/74/MS_u	CALC / DERIVED
T2773	3750	CALC	MMX v352	/rows/0/cells/J-FWD/rows/74/preload_N	CALC / DERIVED

ID	Value	Class	Source	Key	Source tag / formula
T2774	(b2)	CALC	MMX v352	/rows/0/cells/J-FWD/rows/74/note	CALC / DERIVED
T2775	PRELOAD	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/75/state	(class set at the call site: DERIVED)
T2776	IN718 ring line P+Q <= 2 Fty(T), F_sh form	CALC	MMX v352	/rows/0/cells/J-FWD/rows/75/row	CALC / DERIVED
T2777	1.038	CALC	MMX v352	/rows/0/cells/J-FWD/rows/75/MS_y	CALC / DERIVED
T2778	MISSING	CALC	MMX v352	/rows/0/cells/J-FWD/rows/75/MS_u	CALC / DERIVED
T2779	3750	CALC	MMX v352	/rows/0/cells/J-FWD/rows/75/preload_N	CALC / DERIVED
T2780	240 (9)(i), (10)	DECIDED *Lead decision, per margin_matrix_v352.json v352*	MMX v352	/rows/0/cells/J-FWD/rows/75/criterion	tag in margin_matrix_v352.json v352:
T2781	(b2)	CALC	MMX v352	/rows/0/cells/J-FWD/rows/75/note	CALC / DERIVED
T2782	PRELOAD	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/76/state	(class set at the call site: DERIVED)
T2783	IN718 ring line P+Q, head-force bound	CALC	MMX v352	/rows/0/cells/J-FWD/rows/76/row	CALC / DERIVED
T2784	1.038	CALC	MMX v352	/rows/0/cells/J-FWD/rows/76/MS_y	CALC / DERIVED
T2785	MISSING	CALC	MMX v352	/rows/0/cells/J-FWD/rows/76/MS_u	CALC / DERIVED
T2786	3750	CALC	MMX v352	/rows/0/cells/J-FWD/rows/76/preload_N	CALC / DERIVED
T2787	(b2)	CALC	MMX v352	/rows/0/cells/J-FWD/rows/76/note	CALC / DERIVED
T2788	PRELOAD	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/77/state	(class set at the call site: DERIVED)
T2789	IN718 flat bearing vs Fbry on the route (i) spring footprint (A 46.06 mm2, registered ...	CALC	MMX v352	/rows/0/cells/J-FWD/rows/77/row	CALC / DERIVED
T2790	2.778	CALC	MMX v352	/rows/0/cells/J-FWD/rows/77/MS_y	CALC / DERIVED
T2791	MISSING	CALC	MMX v352	/rows/0/cells/J-FWD/rows/77/MS_u	CALC / DERIVED
T2792	4096	CALC	MMX v352	/rows/0/cells/J-FWD/rows/77/preload_N	CALC / DERIVED
T2793	no-spring records (information): union 36.401 mm2 1.986, A_ROW 38.171 mm2 2.131; 4096 N ...	CALC	MMX v352	/rows/0/cells/J-FWD/rows/77/note	CALC / DERIVED
T2794	PRELOAD	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/78/state	(class set at the call site: DERIVED)
T2795	IN718 under-head band Neuber (local strain) [ASSUMED (sigma_el x A_ANN 47.12 / A_supp ...	CALC	MMX v352	/rows/0/cells/J-FWD/rows/78/row	CALC / DERIVED; ASSUMED scaling
T2796	4.973	CALC	MMX v352	/rows/0/cells/J-FWD/rows/78/MS_y	CALC / DERIVED; ASSUMED scaling
T2797	MISSING	CALC	MMX v352	/rows/0/cells/J-FWD/rows/78/MS_u	CALC / DERIVED; ASSUMED scaling
T2798	4096	CALC	MMX v352	/rows/0/cells/J-FWD/rows/78/preload_N	CALC / DERIVED; ASSUMED scaling
T2799	(b1); full_float_0.557 (lead 17:55:55Z F1), mean-pressure scaling ASSUMED; 4096 N column ...	CALC	MMX v352	/rows/0/cells/J-FWD/rows/78/note	CALC / DERIVED; ASSUMED scaling
T2800	PRELOAD	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/79/state	(class set at the call site: DERIVED)
T2801	IN718 ligament station P+Q (0.08 mm from the hole wall; contact-edge field)	CALC	MMX v352	/rows/0/cells/J-FWD/rows/79/row	CALC / DERIVED
T2802	−0.173	CALC	MMX v352	/rows/0/cells/J-FWD/rows/79/MS_y	CALC / DERIVED
T2803	MISSING	CALC	MMX v352	/rows/0/cells/J-FWD/rows/79/MS_u	CALC / DERIVED
T2804	4096	CALC	MMX v352	/rows/0/cells/J-FWD/rows/79/preload_N	CALC / DERIVED
T2805	240 (9): information	DECIDED *Lead decision, per margin_matrix_v352.json v352*	MMX v352	/rows/0/cells/J-FWD/rows/79/criterion	tag in margin_matrix_v352.json v352:
T2806	4096 N column = information (J-F1 maximum capped at 3750 N, lead 17:56:12Z)	CALC	MMX v352	/rows/0/cells/J-FWD/rows/79/note	CALC / DERIVED
T2807	PRELOAD	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/80/state	(class set at the call site: DERIVED)
T2808	IN718 seat line d6.051 z'0 P+Q <= 2 Fty(T), F_sh form	CALC	MMX v352	/rows/0/cells/J-FWD/rows/80/row	CALC / DERIVED
T2809	1.033	CALC	MMX v352	/rows/0/cells/J-FWD/rows/80/MS_y	CALC / DERIVED
T2810	MISSING	CALC	MMX v352	/rows/0/cells/J-FWD/rows/80/MS_u	CALC / DERIVED
T2811	4096	CALC	MMX v352	/rows/0/cells/J-FWD/rows/80/preload_N	CALC / DERIVED
T2812	240 (9)(i), (10)	DECIDED *Lead decision, per margin_matrix_v352.json v352*	MMX v352	/rows/0/cells/J-FWD/rows/80/criterion	tag in margin_matrix_v352.json v352:
T2813	(b2); 4096 N column = information (J-F1 maximum capped at 3750 N, lead 17:56:12Z)	CALC	MMX v352	/rows/0/cells/J-FWD/rows/80/note	CALC / DERIVED
T2814	PRELOAD	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/81/state	(class set at the call site: DERIVED)
T2815	IN718 seat line d6.051 z'0 P+Q, head-force bound	CALC	MMX v352	/rows/0/cells/J-FWD/rows/81/row	CALC / DERIVED
T2816	0.866	CALC	MMX v352	/rows/0/cells/J-FWD/rows/81/MS_y	CALC / DERIVED
T2817	MISSING	CALC	MMX v352	/rows/0/cells/J-FWD/rows/81/MS_u	CALC / DERIVED
T2818	4096	CALC	MMX v352	/rows/0/cells/J-FWD/rows/81/preload_N	CALC / DERIVED
T2819	(b2); 4096 N column = information (J-F1 maximum capped at 3750 N, lead 17:56:12Z)	CALC	MMX v352	/rows/0/cells/J-FWD/rows/81/note	CALC / DERIVED
T2820	PRELOAD	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/82/state	(class set at the call site: DERIVED)
T2821	IN718 ring line P+Q <= 2 Fty(T), F_sh form	CALC	MMX v352	/rows/0/cells/J-FWD/rows/82/row	CALC / DERIVED
T2822	1.033	CALC	MMX v352	/rows/0/cells/J-FWD/rows/82/MS_y	CALC / DERIVED
T2823	MISSING	CALC	MMX v352	/rows/0/cells/J-FWD/rows/82/MS_u	CALC / DERIVED
T2824	4096	CALC	MMX v352	/rows/0/cells/J-FWD/rows/82/preload_N	CALC / DERIVED
T2825	240 (9)(i), (10)	DECIDED *Lead decision, per margin_matrix_v352.json v352*	MMX v352	/rows/0/cells/J-FWD/rows/82/criterion	tag in margin_matrix_v352.json v352:
T2826	(b2); 4096 N column = information (J-F1 maximum capped at 3750 N, lead 17:56:12Z)	CALC	MMX v352	/rows/0/cells/J-FWD/rows/82/note	CALC / DERIVED
T2827	PRELOAD	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/83/state	(class set at the call site: DERIVED)
T2828	IN718 ring line P+Q, head-force bound	CALC	MMX v352	/rows/0/cells/J-FWD/rows/83/row	CALC / DERIVED
T2829	0.866	CALC	MMX v352	/rows/0/cells/J-FWD/rows/83/MS_y	CALC / DERIVED
T2830	MISSING	CALC	MMX v352	/rows/0/cells/J-FWD/rows/83/MS_u	CALC / DERIVED
T2831	4096	CALC	MMX v352	/rows/0/cells/J-FWD/rows/83/preload_N	CALC / DERIVED
T2832	(b2); 4096 N column = information (J-F1 maximum capped at 3750 N, lead 17:56:12Z)	CALC	MMX v352	/rows/0/cells/J-FWD/rows/83/note	CALC / DERIVED
T2833	SOAK70	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/86/state	(class set at the call site: DERIVED)
T2834	IN718 ligament station P+Q (0.08 mm from the hole wall; contact-edge field)	CALC	MMX v352	/rows/0/cells/J-FWD/rows/86/row	CALC / DERIVED
T2835	−0.44	CALC	MMX v352	/rows/0/cells/J-FWD/rows/86/MS_y	CALC / DERIVED
T2836	MISSING	CALC	MMX v352	/rows/0/cells/J-FWD/rows/86/MS_u	CALC / DERIVED
T2837	3750	CALC	MMX v352	/rows/0/cells/J-FWD/rows/86/preload_N	CALC / DERIVED
T2838	240 (9): information	DECIDED *Lead decision, per margin_matrix_v352.json v352*	MMX v352	/rows/0/cells/J-FWD/rows/86/criterion	tag in margin_matrix_v352.json v352:
T2839	SOAK70	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/88/state	(class set at the call site: DERIVED)
T2840	IN718 seat line d6.051 z'0 P+Q, head-force bound	CALC	MMX v352	/rows/0/cells/J-FWD/rows/88/row	CALC / DERIVED
T2841	−0.142	CALC	MMX v352	/rows/0/cells/J-FWD/rows/88/MS_y	CALC / DERIVED
T2842	MISSING	CALC	MMX v352	/rows/0/cells/J-FWD/rows/88/MS_u	CALC / DERIVED
T2843	3750	CALC	MMX v352	/rows/0/cells/J-FWD/rows/88/preload_N	CALC / DERIVED
T2844	(b2)	CALC	MMX v352	/rows/0/cells/J-FWD/rows/88/note	CALC / DERIVED
T2845	SOAK70	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/90/state	(class set at the call site: DERIVED)
T2846	IN718 ring line P+Q, head-force bound	CALC	MMX v352	/rows/0/cells/J-FWD/rows/90/row	CALC / DERIVED
T2847	−0.142	CALC	MMX v352	/rows/0/cells/J-FWD/rows/90/MS_y	CALC / DERIVED
T2848	MISSING	CALC	MMX v352	/rows/0/cells/J-FWD/rows/90/MS_u	CALC / DERIVED
T2849	3750	CALC	MMX v352	/rows/0/cells/J-FWD/rows/90/preload_N	CALC / DERIVED
T2850	(b2)	CALC	MMX v352	/rows/0/cells/J-FWD/rows/90/note	CALC / DERIVED
T2851	SOAK70	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/91/state	(class set at the call site: DERIVED)
T2852	IN718 TC-01 lines with Pm > Fty (Mechanical)	CALC	MMX v352	/rows/0/cells/J-FWD/rows/91/row	CALC / DERIVED
T2853	MISSING	CALC	MMX v352	/rows/0/cells/J-FWD/rows/91/MS_y	CALC / DERIVED
T2854	MISSING	CALC	MMX v352	/rows/0/cells/J-FWD/rows/91/MS_u	CALC / DERIVED
T2855	3750	CALC	MMX v352	/rows/0/cells/J-FWD/rows/91/preload_N	CALC / DERIVED

ID	Value	Class	Source	Key	Source tag / formula
T2856	{"per_history": {"js5": {"seat": {"A4-80": 12, "IN718_Fsh": 13, "IN718_headforce_bound" ...	CALC	MMX v352	/rows/0/cells/J-FWD/rows/91/note	CALC / DERIVED
T2857	SOAK70	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/92/state	(class set at the call site: DERIVED)
T2858	IN718 flat bearing vs Fbry on the route (i) spring footprint (A 46.06 mm2, registered ...	CALC	MMX v352	/rows/0/cells/J-FWD/rows/92/row	CALC / DERIVED
T2859	0.3	CALC	MMX v352	/rows/0/cells/J-FWD/rows/92/MS_y	CALC / DERIVED
T2860	MISSING	CALC	MMX v352	/rows/0/cells/J-FWD/rows/92/MS_u	CALC / DERIVED
T2861	4096	CALC	MMX v352	/rows/0/cells/J-FWD/rows/92/preload_N	CALC / DERIVED
T2862	no-spring records (information): union 36.401 mm2 0.028, A_ROW 38.171 mm2 0.078; 4096 N ...	CALC	MMX v352	/rows/0/cells/J-FWD/rows/92/note	CALC / DERIVED
T2863	SOAK70	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/93/state	(class set at the call site: DERIVED)
T2864	IN718 under-head band Neuber (local strain) [ASSUMED (sigma_el x A_ANN 47.12 / A_supp ...	CALC	MMX v352	/rows/0/cells/J-FWD/rows/93/row	CALC / DERIVED; ASSUMED scaling
T2865	0.009	CALC	MMX v352	/rows/0/cells/J-FWD/rows/93/MS_y	CALC / DERIVED; ASSUMED scaling
T2866	MISSING	CALC	MMX v352	/rows/0/cells/J-FWD/rows/93/MS_u	CALC / DERIVED; ASSUMED scaling
T2867	4096	CALC	MMX v352	/rows/0/cells/J-FWD/rows/93/preload_N	CALC / DERIVED; ASSUMED scaling
T2868	(b2); full_float_0.557 (lead 17:55:55Z F1), mean-pressure scaling ASSUMED; 4096 N column ...	CALC	MMX v352	/rows/0/cells/J-FWD/rows/93/note	CALC / DERIVED; ASSUMED scaling
T2869	SOAK70	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/94/state	(class set at the call site: DERIVED)
T2870	IN718 ligament station P+Q (0.08 mm from the hole wall; contact-edge field)	CALC	MMX v352	/rows/0/cells/J-FWD/rows/94/row	CALC / DERIVED
T2871	−0.471	CALC	MMX v352	/rows/0/cells/J-FWD/rows/94/MS_y	CALC / DERIVED
T2872	MISSING	CALC	MMX v352	/rows/0/cells/J-FWD/rows/94/MS_u	CALC / DERIVED
T2873	4096	CALC	MMX v352	/rows/0/cells/J-FWD/rows/94/preload_N	CALC / DERIVED
T2874	240 (9): information	DECIDED *Lead decision, per margin_matrix_v352.json v352*	MMX v352	/rows/0/cells/J-FWD/rows/94/criterion	tag in margin_matrix_v352.json v352:
T2875	4096 N column = information (J-F1 maximum capped at 3750 N, lead 17:56:12Z)	CALC	MMX v352	/rows/0/cells/J-FWD/rows/94/note	CALC / DERIVED
T2876	SOAK70	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/95/state	(class set at the call site: DERIVED)
T2877	IN718 seat line d6.051 z'0 P+Q <= 2 Fty(T), F_sh form	CALC	MMX v352	/rows/0/cells/J-FWD/rows/95/row	CALC / DERIVED
T2878	−0.061	CALC	MMX v352	/rows/0/cells/J-FWD/rows/95/MS_y	CALC / DERIVED
T2879	MISSING	CALC	MMX v352	/rows/0/cells/J-FWD/rows/95/MS_u	CALC / DERIVED
T2880	4096	CALC	MMX v352	/rows/0/cells/J-FWD/rows/95/preload_N	CALC / DERIVED
T2881	240 (9)(i), (10)	DECIDED *Lead decision, per margin_matrix_v352.json v352*	MMX v352	/rows/0/cells/J-FWD/rows/95/criterion	tag in margin_matrix_v352.json v352:
T2882	(b2); 4096 N column = information (J-F1 maximum capped at 3750 N, lead 17:56:12Z)	CALC	MMX v352	/rows/0/cells/J-FWD/rows/95/note	CALC / DERIVED
T2883	SOAK70	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/96/state	(class set at the call site: DERIVED)
T2884	IN718 seat line d6.051 z'0 P+Q, head-force bound	CALC	MMX v352	/rows/0/cells/J-FWD/rows/96/row	CALC / DERIVED
T2885	−0.177	CALC	MMX v352	/rows/0/cells/J-FWD/rows/96/MS_y	CALC / DERIVED
T2886	MISSING	CALC	MMX v352	/rows/0/cells/J-FWD/rows/96/MS_u	CALC / DERIVED
T2887	4096	CALC	MMX v352	/rows/0/cells/J-FWD/rows/96/preload_N	CALC / DERIVED
T2888	(b2); 4096 N column = information (J-F1 maximum capped at 3750 N, lead 17:56:12Z)	CALC	MMX v352	/rows/0/cells/J-FWD/rows/96/note	CALC / DERIVED
T2889	SOAK70	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/97/state	(class set at the call site: DERIVED)
T2890	IN718 ring line P+Q <= 2 Fty(T), F_sh form	CALC	MMX v352	/rows/0/cells/J-FWD/rows/97/row	CALC / DERIVED
T2891	−0.061	CALC	MMX v352	/rows/0/cells/J-FWD/rows/97/MS_y	CALC / DERIVED
T2892	MISSING	CALC	MMX v352	/rows/0/cells/J-FWD/rows/97/MS_u	CALC / DERIVED
T2893	4096	CALC	MMX v352	/rows/0/cells/J-FWD/rows/97/preload_N	CALC / DERIVED
T2894	240 (9)(i), (10)	DECIDED *Lead decision, per margin_matrix_v352.json v352*	MMX v352	/rows/0/cells/J-FWD/rows/97/criterion	tag in margin_matrix_v352.json v352:
T2895	(b2); 4096 N column = information (J-F1 maximum capped at 3750 N, lead 17:56:12Z)	CALC	MMX v352	/rows/0/cells/J-FWD/rows/97/note	CALC / DERIVED
T2896	SOAK70	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/98/state	(class set at the call site: DERIVED)
T2897	IN718 ring line P+Q, head-force bound	CALC	MMX v352	/rows/0/cells/J-FWD/rows/98/row	CALC / DERIVED
T2898	−0.177	CALC	MMX v352	/rows/0/cells/J-FWD/rows/98/MS_y	CALC / DERIVED
T2899	MISSING	CALC	MMX v352	/rows/0/cells/J-FWD/rows/98/MS_u	CALC / DERIVED
T2900	4096	CALC	MMX v352	/rows/0/cells/J-FWD/rows/98/preload_N	CALC / DERIVED
T2901	(b2); 4096 N column = information (J-F1 maximum capped at 3750 N, lead 17:56:12Z)	CALC	MMX v352	/rows/0/cells/J-FWD/rows/98/note	CALC / DERIVED
T2902	SOAK70	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/99/state	(class set at the call site: DERIVED)
T2903	IN718 TC-01 lines with Pm > Fty (Mechanical)	CALC	MMX v352	/rows/0/cells/J-FWD/rows/99/row	CALC / DERIVED
T2904	MISSING	CALC	MMX v352	/rows/0/cells/J-FWD/rows/99/MS_y	CALC / DERIVED
T2905	MISSING	CALC	MMX v352	/rows/0/cells/J-FWD/rows/99/MS_u	CALC / DERIVED
T2906	4096	CALC	MMX v352	/rows/0/cells/J-FWD/rows/99/preload_N	CALC / DERIVED
T2907	{"per_history": {"js5": {"seat": {"A4-80": 12, "IN718_Fsh": 13, "IN718_headforce_bound" ...	CALC	MMX v352	/rows/0/cells/J-FWD/rows/99/note	CALC / DERIVED
T2908	SOAK	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/102/state	(class set at the call site: DERIVED)
T2909	IN718 ligament station P+Q (0.08 mm from the hole wall; contact-edge field)	CALC	MMX v352	/rows/0/cells/J-FWD/rows/102/row	CALC / DERIVED
T2910	−0.272	CALC	MMX v352	/rows/0/cells/J-FWD/rows/102/MS_y	CALC / DERIVED
T2911	MISSING	CALC	MMX v352	/rows/0/cells/J-FWD/rows/102/MS_u	CALC / DERIVED
T2912	3750	CALC	MMX v352	/rows/0/cells/J-FWD/rows/102/preload_N	CALC / DERIVED
T2913	240 (9): information	DECIDED *Lead decision, per margin_matrix_v352.json v352*	MMX v352	/rows/0/cells/J-FWD/rows/102/criterion	tag in margin_matrix_v352.json v352:
T2914	SOAK	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/104/state	(class set at the call site: DERIVED)
T2915	IN718 seat line d6.051 z'0 P+Q, head-force bound	CALC	MMX v352	/rows/0/cells/J-FWD/rows/104/row	CALC / DERIVED
T2916	−0.142	CALC	MMX v352	/rows/0/cells/J-FWD/rows/104/MS_y	CALC / DERIVED
T2917	MISSING	CALC	MMX v352	/rows/0/cells/J-FWD/rows/104/MS_u	CALC / DERIVED
T2918	3750	CALC	MMX v352	/rows/0/cells/J-FWD/rows/104/preload_N	CALC / DERIVED
T2919	(b2) rep	CALC	MMX v352	/rows/0/cells/J-FWD/rows/104/note	CALC / DERIVED
T2920	SOAK	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/106/state	(class set at the call site: DERIVED)
T2921	IN718 ring line P+Q, head-force bound	CALC	MMX v352	/rows/0/cells/J-FWD/rows/106/row	CALC / DERIVED
T2922	−0.142	CALC	MMX v352	/rows/0/cells/J-FWD/rows/106/MS_y	CALC / DERIVED
T2923	MISSING	CALC	MMX v352	/rows/0/cells/J-FWD/rows/106/MS_u	CALC / DERIVED
T2924	3750	CALC	MMX v352	/rows/0/cells/J-FWD/rows/106/preload_N	CALC / DERIVED
T2925	(b2) rep	CALC	MMX v352	/rows/0/cells/J-FWD/rows/106/note	CALC / DERIVED
T2926	SOAK	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/107/state	(class set at the call site: DERIVED)
T2927	IN718 TC-01 lines with Pm > Fty (Mechanical)	CALC	MMX v352	/rows/0/cells/J-FWD/rows/107/row	CALC / DERIVED
T2928	MISSING	CALC	MMX v352	/rows/0/cells/J-FWD/rows/107/MS_y	CALC / DERIVED
T2929	MISSING	CALC	MMX v352	/rows/0/cells/J-FWD/rows/107/MS_u	CALC / DERIVED
T2930	3750	CALC	MMX v352	/rows/0/cells/J-FWD/rows/107/preload_N	CALC / DERIVED
T2931	{"per_history": {"js5": {"seat": {"A4-80": 12, "IN718_Fsh": 13, "IN718_headforce_bound" ...	CALC	MMX v352	/rows/0/cells/J-FWD/rows/107/note	CALC / DERIVED
T2932	SOAK	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/108/state	(class set at the call site: DERIVED)
T2933	IN718 flat bearing vs Fbry on the route (i) spring footprint (A 46.06 mm2, registered ...	CALC	MMX v352	/rows/0/cells/J-FWD/rows/108/row	CALC / DERIVED
T2934	0.299	CALC	MMX v352	/rows/0/cells/J-FWD/rows/108/MS_y	CALC / DERIVED
T2935	MISSING	CALC	MMX v352	/rows/0/cells/J-FWD/rows/108/MS_u	CALC / DERIVED
T2936	4096	CALC	MMX v352	/rows/0/cells/J-FWD/rows/108/preload_N	CALC / DERIVED

ID	Value	Class	Source	Key	Source tag / formula
T2937	no-spring records (information): union 36.401 mm2 0.026, A_ROW 38.171 mm2 0.076; 4096 N ...	CALC	MMX v352	/rows/0/cells/J-FWD/rows/108/note	CALC / DERIVED
T2938	SOAK	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/109/state	(class set at the call site: DERIVED)
T2939	IN718 under-head band Neuber (local strain) [ASSUMED (sigma_el x A_ANN 47.12 / A_supp ...	CALC	MMX v352	/rows/0/cells/J-FWD/rows/109/row	CALC / DERIVED; ASSUMED scaling
T2940	0.087	CALC	MMX v352	/rows/0/cells/J-FWD/rows/109/MS_y	CALC / DERIVED; ASSUMED scaling
T2941	MISSING	CALC	MMX v352	/rows/0/cells/J-FWD/rows/109/MS_u	CALC / DERIVED; ASSUMED scaling
T2942	4096	CALC	MMX v352	/rows/0/cells/J-FWD/rows/109/preload_N	CALC / DERIVED; ASSUMED scaling
T2943	(b1); full_float_0.557 (lead 17:55:55Z F1), mean-pressure scaling ASSUMED; 4096 N column ...	CALC	MMX v352	/rows/0/cells/J-FWD/rows/109/note	CALC / DERIVED; ASSUMED scaling
T2944	SOAK	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/110/state	(class set at the call site: DERIVED)
T2945	IN718 ligament station P+Q (0.08 mm from the hole wall; contact-edge field)	CALC	MMX v352	/rows/0/cells/J-FWD/rows/110/row	CALC / DERIVED
T2946	−0.302	CALC	MMX v352	/rows/0/cells/J-FWD/rows/110/MS_y	CALC / DERIVED
T2947	MISSING	CALC	MMX v352	/rows/0/cells/J-FWD/rows/110/MS_u	CALC / DERIVED
T2948	4096	CALC	MMX v352	/rows/0/cells/J-FWD/rows/110/preload_N	CALC / DERIVED
T2949	240 (9): information	DECIDED *Lead decision, per margin_matrix_v352.json v352*	MMX v352	/rows/0/cells/J-FWD/rows/110/criterion	tag in margin_matrix_v352.json v352:
T2950	4096 N column = information (J-F1 maximum capped at 3750 N, lead 17:56:12Z)	CALC	MMX v352	/rows/0/cells/J-FWD/rows/110/note	CALC / DERIVED
T2951	SOAK	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/111/state	(class set at the call site: DERIVED)
T2952	IN718 seat line d6.051 z'0 P+Q <= 2 Fty(T), F_sh form	CALC	MMX v352	/rows/0/cells/J-FWD/rows/111/row	CALC / DERIVED
T2953	−0.076	CALC	MMX v352	/rows/0/cells/J-FWD/rows/111/MS_y	CALC / DERIVED
T2954	MISSING	CALC	MMX v352	/rows/0/cells/J-FWD/rows/111/MS_u	CALC / DERIVED
T2955	4096	CALC	MMX v352	/rows/0/cells/J-FWD/rows/111/preload_N	CALC / DERIVED
T2956	240 (9)(i), (10)	DECIDED *Lead decision, per margin_matrix_v352.json v352*	MMX v352	/rows/0/cells/J-FWD/rows/111/criterion	tag in margin_matrix_v352.json v352:
T2957	(b2) rep; 4096 N column = information (J-F1 maximum capped at 3750 N, lead 17:56:12Z)	CALC	MMX v352	/rows/0/cells/J-FWD/rows/111/note	CALC / DERIVED
T2958	SOAK	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/112/state	(class set at the call site: DERIVED)
T2959	IN718 seat line d6.051 z'0 P+Q, head-force bound	CALC	MMX v352	/rows/0/cells/J-FWD/rows/112/row	CALC / DERIVED
T2960	−0.183	CALC	MMX v352	/rows/0/cells/J-FWD/rows/112/MS_y	CALC / DERIVED
T2961	MISSING	CALC	MMX v352	/rows/0/cells/J-FWD/rows/112/MS_u	CALC / DERIVED
T2962	4096	CALC	MMX v352	/rows/0/cells/J-FWD/rows/112/preload_N	CALC / DERIVED
T2963	(b2) rep; 4096 N column = information (J-F1 maximum capped at 3750 N, lead 17:56:12Z)	CALC	MMX v352	/rows/0/cells/J-FWD/rows/112/note	CALC / DERIVED
T2964	SOAK	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/113/state	(class set at the call site: DERIVED)
T2965	IN718 ring line P+Q <= 2 Fty(T), F_sh form	CALC	MMX v352	/rows/0/cells/J-FWD/rows/113/row	CALC / DERIVED
T2966	−0.076	CALC	MMX v352	/rows/0/cells/J-FWD/rows/113/MS_y	CALC / DERIVED
T2967	MISSING	CALC	MMX v352	/rows/0/cells/J-FWD/rows/113/MS_u	CALC / DERIVED
T2968	4096	CALC	MMX v352	/rows/0/cells/J-FWD/rows/113/preload_N	CALC / DERIVED
T2969	240 (9)(i), (10)	DECIDED *Lead decision, per margin_matrix_v352.json v352*	MMX v352	/rows/0/cells/J-FWD/rows/113/criterion	tag in margin_matrix_v352.json v352:
T2970	(b2) rep; 4096 N column = information (J-F1 maximum capped at 3750 N, lead 17:56:12Z)	CALC	MMX v352	/rows/0/cells/J-FWD/rows/113/note	CALC / DERIVED
T2971	SOAK	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/114/state	(class set at the call site: DERIVED)
T2972	IN718 ring line P+Q, head-force bound	CALC	MMX v352	/rows/0/cells/J-FWD/rows/114/row	CALC / DERIVED
T2973	−0.183	CALC	MMX v352	/rows/0/cells/J-FWD/rows/114/MS_y	CALC / DERIVED
T2974	MISSING	CALC	MMX v352	/rows/0/cells/J-FWD/rows/114/MS_u	CALC / DERIVED
T2975	4096	CALC	MMX v352	/rows/0/cells/J-FWD/rows/114/preload_N	CALC / DERIVED
T2976	(b2) rep; 4096 N column = information (J-F1 maximum capped at 3750 N, lead 17:56:12Z)	CALC	MMX v352	/rows/0/cells/J-FWD/rows/114/note	CALC / DERIVED
T2977	SOAK	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/115/state	(class set at the call site: DERIVED)
T2978	IN718 TC-01 lines with Pm > Fty (Mechanical)	CALC	MMX v352	/rows/0/cells/J-FWD/rows/115/row	CALC / DERIVED
T2979	MISSING	CALC	MMX v352	/rows/0/cells/J-FWD/rows/115/MS_y	CALC / DERIVED
T2980	MISSING	CALC	MMX v352	/rows/0/cells/J-FWD/rows/115/MS_u	CALC / DERIVED
T2981	4096	CALC	MMX v352	/rows/0/cells/J-FWD/rows/115/preload_N	CALC / DERIVED
T2982	{"per_history": {"js5": {"seat": {"A4-80": 12, "IN718_Fsh": 13, "IN718_headforce_bound" ...	CALC	MMX v352	/rows/0/cells/J-FWD/rows/115/note	CALC / DERIVED
T2983	SOAK52	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/118/state	(class set at the call site: DERIVED)
T2984	IN718 ligament station P+Q (0.08 mm from the hole wall; contact-edge field)	CALC	MMX v352	/rows/0/cells/J-FWD/rows/118/row	CALC / DERIVED
T2985	−0.283	CALC	MMX v352	/rows/0/cells/J-FWD/rows/118/MS_y	CALC / DERIVED
T2986	MISSING	CALC	MMX v352	/rows/0/cells/J-FWD/rows/118/MS_u	CALC / DERIVED
T2987	3750	CALC	MMX v352	/rows/0/cells/J-FWD/rows/118/preload_N	CALC / DERIVED
T2988	240 (9): information	DECIDED *Lead decision, per margin_matrix_v352.json v352*	MMX v352	/rows/0/cells/J-FWD/rows/118/criterion	tag in margin_matrix_v352.json v352:
T2989	SOAK52	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/120/state	(class set at the call site: DERIVED)
T2990	IN718 seat line d6.051 z'0 P+Q, head-force bound	CALC	MMX v352	/rows/0/cells/J-FWD/rows/120/row	CALC / DERIVED
T2991	−0.085	CALC	MMX v352	/rows/0/cells/J-FWD/rows/120/MS_y	CALC / DERIVED
T2992	MISSING	CALC	MMX v352	/rows/0/cells/J-FWD/rows/120/MS_u	CALC / DERIVED
T2993	3750	CALC	MMX v352	/rows/0/cells/J-FWD/rows/120/preload_N	CALC / DERIVED
T2994	(b2)	CALC	MMX v352	/rows/0/cells/J-FWD/rows/120/note	CALC / DERIVED
T2995	SOAK52	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/122/state	(class set at the call site: DERIVED)
T2996	IN718 ring line P+Q, head-force bound	CALC	MMX v352	/rows/0/cells/J-FWD/rows/122/row	CALC / DERIVED
T2997	−0.085	CALC	MMX v352	/rows/0/cells/J-FWD/rows/122/MS_y	CALC / DERIVED
T2998	MISSING	CALC	MMX v352	/rows/0/cells/J-FWD/rows/122/MS_u	CALC / DERIVED
T2999	3750	CALC	MMX v352	/rows/0/cells/J-FWD/rows/122/preload_N	CALC / DERIVED
T3000	(b2)	CALC	MMX v352	/rows/0/cells/J-FWD/rows/122/note	CALC / DERIVED
T3001	SOAK52	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/123/state	(class set at the call site: DERIVED)
T3002	IN718 TC-01 lines with Pm > Fty (Mechanical)	CALC	MMX v352	/rows/0/cells/J-FWD/rows/123/row	CALC / DERIVED
T3003	MISSING	CALC	MMX v352	/rows/0/cells/J-FWD/rows/123/MS_y	CALC / DERIVED
T3004	MISSING	CALC	MMX v352	/rows/0/cells/J-FWD/rows/123/MS_u	CALC / DERIVED
T3005	3750	CALC	MMX v352	/rows/0/cells/J-FWD/rows/123/preload_N	CALC / DERIVED
T3006	{"per_history": {"js5": {"seat": {"A4-80": 0, "IN718_Fsh": 0, "IN718_headforce_bound" ...	CALC	MMX v352	/rows/0/cells/J-FWD/rows/123/note	CALC / DERIVED
T3007	SOAK52	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/124/state	(class set at the call site: DERIVED)
T3008	IN718 flat bearing vs Fbry on the route (i) spring footprint (A 46.06 mm2, registered ...	CALC	MMX v352	/rows/0/cells/J-FWD/rows/124/row	CALC / DERIVED
T3009	0.32	CALC	MMX v352	/rows/0/cells/J-FWD/rows/124/MS_y	CALC / DERIVED
T3010	MISSING	CALC	MMX v352	/rows/0/cells/J-FWD/rows/124/MS_u	CALC / DERIVED
T3011	4096	CALC	MMX v352	/rows/0/cells/J-FWD/rows/124/preload_N	CALC / DERIVED
T3012	no-spring records (information): union 36.401 mm2 0.043, A_ROW 38.171 mm2 0.094; 4096 N ...	CALC	MMX v352	/rows/0/cells/J-FWD/rows/124/note	CALC / DERIVED
T3013	SOAK52	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/125/state	(class set at the call site: DERIVED)
T3014	IN718 under-head band Neuber (local strain) [ASSUMED (sigma_el x A_ANN 47.12 / A_supp ...	CALC	MMX v352	/rows/0/cells/J-FWD/rows/125/row	CALC / DERIVED; ASSUMED scaling
T3015	−0.057	CALC	MMX v352	/rows/0/cells/J-FWD/rows/125/MS_y	CALC / DERIVED; ASSUMED scaling
T3016	MISSING	CALC	MMX v352	/rows/0/cells/J-FWD/rows/125/MS_u	CALC / DERIVED; ASSUMED scaling

ID	Value	Class	Source	Key	Source tag / formula
T3017	4096	CALC	MMX v352	/rows/0/cells/J-FWD/rows/125/preload_N	CALC / DERIVED; ASSUMED scaling
T3018	(b2); full_float_0.557 (lead 17:55:55Z F1), mean-pressure scaling ASSUMED; 4096 N column ...	CALC	MMX v352	/rows/0/cells/J-FWD/rows/125/note	CALC / DERIVED; ASSUMED scaling
T3019	SOAK52	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/126/state	(class set at the call site: DERIVED)
T3020	IN718 ligament station P+Q (0.08 mm from the hole wall; contact-edge field)	CALC	MMX v352	/rows/0/cells/J-FWD/rows/126/row	CALC / DERIVED
T3021	−0.316	CALC	MMX v352	/rows/0/cells/J-FWD/rows/126/MS_y	CALC / DERIVED
T3022	MISSING	CALC	MMX v352	/rows/0/cells/J-FWD/rows/126/MS_u	CALC / DERIVED
T3023	4096	CALC	MMX v352	/rows/0/cells/J-FWD/rows/126/preload_N	CALC / DERIVED
T3024	240 (9): information	DECIDED *Lead decision, per margin_matrix_v352.json v352*	MMX v352	/rows/0/cells/J-FWD/rows/126/criterion	tag in margin_matrix_v352.json v352:
T3025	4096 N column = information (J-F1 maximum capped at 3750 N, lead 17:56:12Z)	CALC	MMX v352	/rows/0/cells/J-FWD/rows/126/note	CALC / DERIVED
T3026	SOAK52	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/127/state	(class set at the call site: DERIVED)
T3027	IN718 seat line d6.051 z'0 P+Q <= 2 Fty(T), F_sh form	CALC	MMX v352	/rows/0/cells/J-FWD/rows/127/row	CALC / DERIVED
T3028	−0.035	CALC	MMX v352	/rows/0/cells/J-FWD/rows/127/MS_y	CALC / DERIVED
T3029	MISSING	CALC	MMX v352	/rows/0/cells/J-FWD/rows/127/MS_u	CALC / DERIVED
T3030	4096	CALC	MMX v352	/rows/0/cells/J-FWD/rows/127/preload_N	CALC / DERIVED
T3031	240 (9)(i), (10)	DECIDED *Lead decision, per margin_matrix_v352.json v352*	MMX v352	/rows/0/cells/J-FWD/rows/127/criterion	tag in margin_matrix_v352.json v352:
T3032	(b1); 4096 N column = information (J-F1 maximum capped at 3750 N, lead 17:56:12Z)	CALC	MMX v352	/rows/0/cells/J-FWD/rows/127/note	CALC / DERIVED
T3033	SOAK52	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/128/state	(class set at the call site: DERIVED)
T3034	IN718 seat line d6.051 z'0 P+Q, head-force bound	CALC	MMX v352	/rows/0/cells/J-FWD/rows/128/row	CALC / DERIVED
T3035	−0.131	CALC	MMX v352	/rows/0/cells/J-FWD/rows/128/MS_y	CALC / DERIVED
T3036	MISSING	CALC	MMX v352	/rows/0/cells/J-FWD/rows/128/MS_u	CALC / DERIVED
T3037	4096	CALC	MMX v352	/rows/0/cells/J-FWD/rows/128/preload_N	CALC / DERIVED
T3038	(b1); 4096 N column = information (J-F1 maximum capped at 3750 N, lead 17:56:12Z)	CALC	MMX v352	/rows/0/cells/J-FWD/rows/128/note	CALC / DERIVED
T3039	SOAK52	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/129/state	(class set at the call site: DERIVED)
T3040	IN718 ring line P+Q <= 2 Fty(T), F_sh form	CALC	MMX v352	/rows/0/cells/J-FWD/rows/129/row	CALC / DERIVED
T3041	−0.035	CALC	MMX v352	/rows/0/cells/J-FWD/rows/129/MS_y	CALC / DERIVED
T3042	MISSING	CALC	MMX v352	/rows/0/cells/J-FWD/rows/129/MS_u	CALC / DERIVED
T3043	4096	CALC	MMX v352	/rows/0/cells/J-FWD/rows/129/preload_N	CALC / DERIVED
T3044	240 (9)(i), (10)	DECIDED *Lead decision, per margin_matrix_v352.json v352*	MMX v352	/rows/0/cells/J-FWD/rows/129/criterion	tag in margin_matrix_v352.json v352:
T3045	(b1); 4096 N column = information (J-F1 maximum capped at 3750 N, lead 17:56:12Z)	CALC	MMX v352	/rows/0/cells/J-FWD/rows/129/note	CALC / DERIVED
T3046	SOAK52	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/130/state	(class set at the call site: DERIVED)
T3047	IN718 ring line P+Q, head-force bound	CALC	MMX v352	/rows/0/cells/J-FWD/rows/130/row	CALC / DERIVED
T3048	−0.131	CALC	MMX v352	/rows/0/cells/J-FWD/rows/130/MS_y	CALC / DERIVED
T3049	MISSING	CALC	MMX v352	/rows/0/cells/J-FWD/rows/130/MS_u	CALC / DERIVED
T3050	4096	CALC	MMX v352	/rows/0/cells/J-FWD/rows/130/preload_N	CALC / DERIVED
T3051	(b1); 4096 N column = information (J-F1 maximum capped at 3750 N, lead 17:56:12Z)	CALC	MMX v352	/rows/0/cells/J-FWD/rows/130/note	CALC / DERIVED
T3052	SOAK52	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/131/state	(class set at the call site: DERIVED)
T3053	IN718 TC-01 lines with Pm > Fty (Mechanical)	CALC	MMX v352	/rows/0/cells/J-FWD/rows/131/row	CALC / DERIVED
T3054	MISSING	CALC	MMX v352	/rows/0/cells/J-FWD/rows/131/MS_y	CALC / DERIVED
T3055	MISSING	CALC	MMX v352	/rows/0/cells/J-FWD/rows/131/MS_u	CALC / DERIVED
T3056	4096	CALC	MMX v352	/rows/0/cells/J-FWD/rows/131/preload_N	CALC / DERIVED
T3057	{"per_history": {"js5": {"seat": {"A4-80": 0, "IN718_Fsh": 0, "IN718_headforce_bound" ...	CALC	MMX v352	/rows/0/cells/J-FWD/rows/131/note	CALC / DERIVED
T3058	FA1S	DERIVED	MMX v352	/rows/0/cells/J-FWD/rows/133/state	(class set at the call site: DERIVED)
T3059	IN718 under-head band Neuber (local strain) [ASSUMED (sigma_el x A_ANN 47.12 / A_supp ...	CALC	MMX v352	/rows/0/cells/J-FWD/rows/133/row	CALC / DERIVED; ASSUMED scaling
T3060	0.012	CALC	MMX v352	/rows/0/cells/J-FWD/rows/133/MS_y	CALC / DERIVED; ASSUMED scaling
T3061	MISSING	CALC	MMX v352	/rows/0/cells/J-FWD/rows/133/MS_u	CALC / DERIVED; ASSUMED scaling
T3062	4096	CALC	MMX v352	/rows/0/cells/J-FWD/rows/133/preload_N	CALC / DERIVED; ASSUMED scaling
T3063	(b2); full_float_0.557 (lead 17:55:55Z F1), mean-pressure scaling ASSUMED; 4096 N column ...	CALC	MMX v352	/rows/0/cells/J-FWD/rows/133/note	CALC / DERIVED; ASSUMED scaling
T3064	0.045–0.162	CALC	MMX v352	/rows/1/cells/J-FWD/value	(class set at the call site: CALC) [J-FWD cell = minimum over the criterion rows]
T3065	FA1S	DERIVED	MMX v352	/rows/1/cells/J-FWD/rows/0/state	(class set at the call site: DERIVED)
T3066	(i) shear-out to g2 at x' 28.0 Pm+Pb	SIM	MMX v352	/rows/1/cells/J-FWD/rows/0/row	SIM / DERIVED
T3067	0.981	SIM	MMX v352	/rows/1/cells/J-FWD/rows/0/MS_y	SIM / DERIVED
T3068	1.136	SIM	MMX v352	/rows/1/cells/J-FWD/rows/0/MS_u	SIM / DERIVED
T3069	231 (1)(i)	DECIDED *Lead decision, per margin_matrix_v352.json v352*	MMX v352	/rows/1/cells/J-FWD/rows/0/criterion	tag in margin_matrix_v352.json v352:
T3070	FA1S	DERIVED	MMX v352	/rows/1/cells/J-FWD/rows/1/state	(class set at the call site: DERIVED)
T3071	(i) shear-out to g2 at x' 28.0 Pm	SIM	MMX v352	/rows/1/cells/J-FWD/rows/1/row	SIM / DERIVED
T3072	3.593	SIM	MMX v352	/rows/1/cells/J-FWD/rows/1/MS_y	SIM / DERIVED
T3073	3.954	SIM	MMX v352	/rows/1/cells/J-FWD/rows/1/MS_u	SIM / DERIVED
T3074	231 (1)(i)	DECIDED *Lead decision, per margin_matrix_v352.json v352*	MMX v352	/rows/1/cells/J-FWD/rows/1/criterion	tag in margin_matrix_v352.json v352:
T3075	FA1S	DERIVED	MMX v352	/rows/1/cells/J-FWD/rows/2/state	(class set at the call site: DERIVED)
T3076	(ii) thread shear on A_n 68.4	SIM	MMX v352	/rows/1/cells/J-FWD/rows/2/row	SIM / DERIVED
T3077	1.969	SIM	MMX v352	/rows/1/cells/J-FWD/rows/2/MS_y	SIM / DERIVED
T3078	2.33	SIM	MMX v352	/rows/1/cells/J-FWD/rows/2/MS_u	SIM / DERIVED
T3079	231 (1)(ii)	DECIDED *Lead decision, per margin_matrix_v352.json v352*	MMX v352	/rows/1/cells/J-FWD/rows/2/criterion	tag in margin_matrix_v352.json v352:
T3080	FA1S	DERIVED	MMX v352	/rows/1/cells/J-FWD/rows/3/state	(class set at the call site: DERIVED)
T3081	(ii) tap bearing on L_e 6.84	SIM	MMX v352	/rows/1/cells/J-FWD/rows/3/row	SIM / DERIVED
T3082	2.115	SIM	MMX v352	/rows/1/cells/J-FWD/rows/3/MS_y	SIM / DERIVED
T3083	2.839	SIM	MMX v352	/rows/1/cells/J-FWD/rows/3/MS_u	SIM / DERIVED
T3084	231 (1)(ii)	DECIDED *Lead decision, per margin_matrix_v352.json v352*	MMX v352	/rows/1/cells/J-FWD/rows/3/criterion	tag in margin_matrix_v352.json v352:
T3085	FA1S	DERIVED	MMX v352	/rows/1/cells/J-FWD/rows/4/state	(class set at the call site: DERIVED)
T3086	(iii) band Neuber n15	SIM	MMX v352	/rows/1/cells/J-FWD/rows/4/row	SIM / DERIVED
T3087	7.893	SIM	MMX v352	/rows/1/cells/J-FWD/rows/4/MS_y	SIM / DERIVED
T3088	MISSING	SIM	MMX v352	/rows/1/cells/J-FWD/rows/4/MS_u	SIM / DERIVED
T3089	231 (1)(iii)	DECIDED *Lead decision, per margin_matrix_v352.json v352*	MMX v352	/rows/1/cells/J-FWD/rows/4/criterion	tag in margin_matrix_v352.json v352:
T3090	FA1S	DERIVED	MMX v352	/rows/1/cells/J-FWD/rows/8/state	(class set at the call site: DERIVED)
T3091	(ii) tapped-thread bearing	CALC	MMX v352	/rows/1/cells/J-FWD/rows/8/row	CALC
T3092	1.385	CALC	MMX v352	/rows/1/cells/J-FWD/rows/8/MS_y	CALC
T3093	1.94	CALC	MMX v352	/rows/1/cells/J-FWD/rows/8/MS_u	CALC
T3094	231 (1)(ii)	DECIDED *Lead decision, per margin_matrix_v352.json v352*	MMX v352	/rows/1/cells/J-FWD/rows/8/criterion	tag in margin_matrix_v352.json v352:
T3095	FA1S	DERIVED	MMX v352	/rows/1/cells/J-FWD/rows/9/state	(class set at the call site: DERIVED)
T3096	(ii) internal thread shear at 3965.0 N on A_n 67.9	CALC	MMX v352	/rows/1/cells/J-FWD/rows/9/row	CALC

ID	Value	Class	Source	Key	Source tag / formula
T3097	1.332	CALC	MMX v352	/rows/1/cells/J-FWD/rows/9/MS_y	CALC
T3098	1.615	CALC	MMX v352	/rows/1/cells/J-FWD/rows/9/MS_u	CALC
T3099	231 (1)(ii)	DECIDED *Lead decision, per margin_matrix_v352.json v352*	MMX v352	/rows/1/cells/J-FWD/rows/9/criterion	tag in margin_matrix_v352.json v352:
T3100	FA1U	DERIVED	MMX v352	/rows/1/cells/J-FWD/rows/21/state	(class set at the call site: DERIVED)
T3101	js5b FA1U (FA1, SF [1.0, 1.0]): (i) shear-out to g2 at x' 28.0 Pm+Pb	SIM	MMX v352	/rows/1/cells/J-FWD/rows/21/row	SIM / DERIVED
T3102	1.449	SIM	MMX v352	/rows/1/cells/J-FWD/rows/21/MS_y	SIM / DERIVED
T3103	1.641	SIM	MMX v352	/rows/1/cells/J-FWD/rows/21/MS_u	SIM / DERIVED
T3104	231 (1)(i)	DECIDED *Lead decision, per margin_matrix_v352.json v352*	MMX v352	/rows/1/cells/J-FWD/rows/21/criterion	tag in margin_matrix_v352.json v352:
T3105	FA1U	DERIVED	MMX v352	/rows/1/cells/J-FWD/rows/22/state	(class set at the call site: DERIVED)
T3106	js5b FA1U (FA1, SF [1.0, 1.0]): (ii) thread shear on A_n 68.4	SIM	MMX v352	/rows/1/cells/J-FWD/rows/22/row	SIM / DERIVED
T3107	1.65	SIM	MMX v352	/rows/1/cells/J-FWD/rows/22/MS_y	SIM / DERIVED
T3108	1.971	SIM	MMX v352	/rows/1/cells/J-FWD/rows/22/MS_u	SIM / DERIVED
T3109	231 (1)(ii)	DECIDED *Lead decision, per margin_matrix_v352.json v352*	MMX v352	/rows/1/cells/J-FWD/rows/22/criterion	tag in margin_matrix_v352.json v352:
T3110	FA1U	DERIVED	MMX v352	/rows/1/cells/J-FWD/rows/23/state	(class set at the call site: DERIVED)
T3111	js5b FA1U (FA1, SF [1.0, 1.0]): (ii) tap bearing on L_e 6.84	SIM	MMX v352	/rows/1/cells/J-FWD/rows/23/row	SIM / DERIVED
T3112	2.112	SIM	MMX v352	/rows/1/cells/J-FWD/rows/23/MS_y	SIM / DERIVED
T3113	2.835	SIM	MMX v352	/rows/1/cells/J-FWD/rows/23/MS_u	SIM / DERIVED
T3114	231 (1)(ii)	DECIDED *Lead decision, per margin_matrix_v352.json v352*	MMX v352	/rows/1/cells/J-FWD/rows/23/criterion	tag in margin_matrix_v352.json v352:
T3115	FA1D	DERIVED	MMX v352	/rows/1/cells/J-FWD/rows/24/state	(class set at the call site: DERIVED)
T3116	js5b FA1D (FA1, SF [1.0, 1.0]): (i) shear-out to g2 at x' 28.0 Pm+Pb	SIM	MMX v352	/rows/1/cells/J-FWD/rows/24/row	SIM / DERIVED
T3117	1.645	SIM	MMX v352	/rows/1/cells/J-FWD/rows/24/MS_y	SIM / DERIVED
T3118	1.853	SIM	MMX v352	/rows/1/cells/J-FWD/rows/24/MS_u	SIM / DERIVED
T3119	231 (1)(i)	DECIDED *Lead decision, per margin_matrix_v352.json v352*	MMX v352	/rows/1/cells/J-FWD/rows/24/criterion	tag in margin_matrix_v352.json v352:
T3120	FA1D	DERIVED	MMX v352	/rows/1/cells/J-FWD/rows/25/state	(class set at the call site: DERIVED)
T3121	js5b FA1D (FA1, SF [1.0, 1.0]): (ii) thread shear on A_n 68.4	SIM	MMX v352	/rows/1/cells/J-FWD/rows/25/row	SIM / DERIVED
T3122	1.597	SIM	MMX v352	/rows/1/cells/J-FWD/rows/25/MS_y	SIM / DERIVED
T3123	1.913	SIM	MMX v352	/rows/1/cells/J-FWD/rows/25/MS_u	SIM / DERIVED
T3124	231 (1)(ii)	DECIDED *Lead decision, per margin_matrix_v352.json v352*	MMX v352	/rows/1/cells/J-FWD/rows/25/criterion	tag in margin_matrix_v352.json v352:
T3125	FA1D	DERIVED	MMX v352	/rows/1/cells/J-FWD/rows/26/state	(class set at the call site: DERIVED)
T3126	js5b FA1D (FA1, SF [1.0, 1.0]): (ii) tap bearing on L_e 6.84	SIM	MMX v352	/rows/1/cells/J-FWD/rows/26/row	SIM / DERIVED
T3127	2.97	SIM	MMX v352	/rows/1/cells/J-FWD/rows/26/MS_y	SIM / DERIVED
T3128	3.894	SIM	MMX v352	/rows/1/cells/J-FWD/rows/26/MS_u	SIM / DERIVED
T3129	231 (1)(ii)	DECIDED *Lead decision, per margin_matrix_v352.json v352*	MMX v352	/rows/1/cells/J-FWD/rows/26/criterion	tag in margin_matrix_v352.json v352:
T3130	LC2U	DERIVED	MMX v352	/rows/1/cells/J-FWD/rows/27/state	(class set at the call site: DERIVED)
T3131	js5b LC2U (FE-factored, SF [1.0, 1.0]): (i) shear-out to g2 at x' 28.0 Pm+Pb	SIM	MMX v352	/rows/1/cells/J-FWD/rows/27/row	SIM / DERIVED
T3132	MISSING	SIM	MMX v352	/rows/1/cells/J-FWD/rows/27/MS_y	SIM / DERIVED
T3133	3.239	SIM	MMX v352	/rows/1/cells/J-FWD/rows/27/MS_u	SIM / DERIVED
T3134	231 (1)(i)	DECIDED *Lead decision, per margin_matrix_v352.json v352*	MMX v352	/rows/1/cells/J-FWD/rows/27/criterion	tag in margin_matrix_v352.json v352:
T3135	FE-factored: the x1.5 / x2.0 state solved with contact at the 3750 N maximum preload ...	SIM	MMX v352	/rows/1/cells/J-FWD/rows/27/note	SIM / DERIVED
T3136	LC2U	DERIVED	MMX v352	/rows/1/cells/J-FWD/rows/28/state	(class set at the call site: DERIVED)
T3137	js5b LC2U (FE-factored, SF [1.0, 1.0]): (ii) thread shear on A_n 68.4	SIM	MMX v352	/rows/1/cells/J-FWD/rows/28/row	SIM / DERIVED
T3138	MISSING	SIM	MMX v352	/rows/1/cells/J-FWD/rows/28/MS_y	SIM / DERIVED
T3139	1.633	SIM	MMX v352	/rows/1/cells/J-FWD/rows/28/MS_u	SIM / DERIVED
T3140	231 (1)(ii)	DECIDED *Lead decision, per margin_matrix_v352.json v352*	MMX v352	/rows/1/cells/J-FWD/rows/28/criterion	tag in margin_matrix_v352.json v352:
T3141	FE-factored: the x1.5 / x2.0 state solved with contact at the 3750 N maximum preload ...	SIM	MMX v352	/rows/1/cells/J-FWD/rows/28/note	SIM / DERIVED
T3142	LC2U	DERIVED	MMX v352	/rows/1/cells/J-FWD/rows/29/state	(class set at the call site: DERIVED)
T3143	js5b LC2U (FE-factored, SF [1.0, 1.0]): (ii) tap bearing on L_e 6.84	SIM	MMX v352	/rows/1/cells/J-FWD/rows/29/row	SIM / DERIVED
T3144	MISSING	SIM	MMX v352	/rows/1/cells/J-FWD/rows/29/MS_y	SIM / DERIVED
T3145	5.493	SIM	MMX v352	/rows/1/cells/J-FWD/rows/29/MS_u	SIM / DERIVED
T3146	231 (1)(ii)	DECIDED *Lead decision, per margin_matrix_v352.json v352*	MMX v352	/rows/1/cells/J-FWD/rows/29/criterion	tag in margin_matrix_v352.json v352:
T3147	FE-factored: the x1.5 / x2.0 state solved with contact at the 3750 N maximum preload ...	SIM	MMX v352	/rows/1/cells/J-FWD/rows/29/note	SIM / DERIVED
T3148	LC2Y	DERIVED	MMX v352	/rows/1/cells/J-FWD/rows/30/state	(class set at the call site: DERIVED)
T3149	js5b LC2Y (FE-factored, SF [1.0, 1.0]): (i) shear-out to g2 at x' 28.0 Pm+Pb	SIM	MMX v352	/rows/1/cells/J-FWD/rows/30/row	SIM / DERIVED
T3150	5.228	SIM	MMX v352	/rows/1/cells/J-FWD/rows/30/MS_y	SIM / DERIVED
T3151	MISSING	SIM	MMX v352	/rows/1/cells/J-FWD/rows/30/MS_u	SIM / DERIVED
T3152	231 (1)(i)	DECIDED *Lead decision, per margin_matrix_v352.json v352*	MMX v352	/rows/1/cells/J-FWD/rows/30/criterion	tag in margin_matrix_v352.json v352:
T3153	FE-factored: the x1.5 / x2.0 state solved with contact at the 3750 N maximum preload ...	SIM	MMX v352	/rows/1/cells/J-FWD/rows/30/note	SIM / DERIVED
T3154	LC2Y	DERIVED	MMX v352	/rows/1/cells/J-FWD/rows/31/state	(class set at the call site: DERIVED)
T3155	js5b LC2Y (FE-factored, SF [1.0, 1.0]): (ii) thread shear on A_n 68.4	SIM	MMX v352	/rows/1/cells/J-FWD/rows/31/row	SIM / DERIVED
T3156	1.33	SIM	MMX v352	/rows/1/cells/J-FWD/rows/31/MS_y	SIM / DERIVED
T3157	MISSING	SIM	MMX v352	/rows/1/cells/J-FWD/rows/31/MS_u	SIM / DERIVED
T3158	231 (1)(ii)	DECIDED *Lead decision, per margin_matrix_v352.json v352*	MMX v352	/rows/1/cells/J-FWD/rows/31/criterion	tag in margin_matrix_v352.json v352:
T3159	FE-factored: the x1.5 / x2.0 state solved with contact at the 3750 N maximum preload ...	SIM	MMX v352	/rows/1/cells/J-FWD/rows/31/note	SIM / DERIVED
T3160	LC2Y	DERIVED	MMX v352	/rows/1/cells/J-FWD/rows/32/state	(class set at the call site: DERIVED)
T3161	js5b LC2Y (FE-factored, SF [1.0, 1.0]): (ii) tap bearing on L_e 6.84	SIM	MMX v352	/rows/1/cells/J-FWD/rows/32/row	SIM / DERIVED
T3162	6.085	SIM	MMX v352	/rows/1/cells/J-FWD/rows/32/MS_y	SIM / DERIVED
T3163	MISSING	SIM	MMX v352	/rows/1/cells/J-FWD/rows/32/MS_u	SIM / DERIVED
T3164	231 (1)(ii)	DECIDED *Lead decision, per margin_matrix_v352.json v352*	MMX v352	/rows/1/cells/J-FWD/rows/32/criterion	tag in margin_matrix_v352.json v352:
T3165	FE-factored: the x1.5 / x2.0 state solved with contact at the 3750 N maximum preload ...	SIM	MMX v352	/rows/1/cells/J-FWD/rows/32/note	SIM / DERIVED
T3166	SOAK70	DERIVED	MMX v352	/rows/1/cells/J-FWD/rows/43/state	(class set at the call site: DERIVED)
T3167	IN718 thread shear on THE ROW A_n 67.93 (route (i): M5 × 10.7 + HW-170 spring, L_e min ...	CALC	MMX v352	/rows/1/cells/J-FWD/rows/43/row	CALC / DERIVED
T3168	0.087	CALC	MMX v352	/rows/1/cells/J-FWD/rows/43/MS_y	CALC / DERIVED
T3169	0.207	CALC	MMX v352	/rows/1/cells/J-FWD/rows/43/MS_u	CALC / DERIVED
T3170	3750	CALC	MMX v352	/rows/1/cells/J-FWD/rows/43/preload_N	CALC / DERIVED

ID	Value	Class	Source	Key	Source tag / formula
T3171	234 (1)(b)	DECIDED *Lead decision, per margin_matrix_v352.json v352*	MMX v352	/rows/1/cells/J-FWD/rows/43/criterion	tag in margin_matrix_v352.json v352:
T3172	N 8227.4 ((b2)), T 97.7 C	CALC	MMX v352	/rows/1/cells/J-FWD/rows/43/note	CALC / DERIVED
T3173	SOAK	DERIVED	MMX v352	/rows/1/cells/J-FWD/rows/47/state	(class set at the call site: DERIVED)
T3174	IN718 thread shear on THE ROW A_n 67.93 (route (i): M5 × 10.7 + HW-170 spring, L_e min ...	CALC	MMX v352	/rows/1/cells/J-FWD/rows/47/row	CALC / DERIVED
T3175	0.104	CALC	MMX v352	/rows/1/cells/J-FWD/rows/47/MS_y	CALC / DERIVED
T3176	0.227	CALC	MMX v352	/rows/1/cells/J-FWD/rows/47/MS_u	CALC / DERIVED
T3177	3750	CALC	MMX v352	/rows/1/cells/J-FWD/rows/47/preload_N	CALC / DERIVED
T3178	234 (1)(b)	DECIDED *Lead decision, per margin_matrix_v352.json v352*	MMX v352	/rows/1/cells/J-FWD/rows/47/criterion	tag in margin_matrix_v352.json v352:
T3179	N 8110.3 (js5), T 96.0 C	CALC	MMX v352	/rows/1/cells/J-FWD/rows/47/note	CALC / DERIVED
T3180	SOAK52	DERIVED	MMX v352	/rows/1/cells/J-FWD/rows/51/state	(class set at the call site: DERIVED)
T3181	IN718 thread shear on THE ROW A_n 67.93 (route (i): M5 × 10.7 + HW-170 spring, L_e min ...	CALC	MMX v352	/rows/1/cells/J-FWD/rows/51/row	CALC / DERIVED
T3182	0.045	CALC	MMX v352	/rows/1/cells/J-FWD/rows/51/MS_y	CALC / DERIVED
T3183	0.162	CALC	MMX v352	/rows/1/cells/J-FWD/rows/51/MS_u	CALC / DERIVED
T3184	3750	CALC	MMX v352	/rows/1/cells/J-FWD/rows/51/preload_N	CALC / DERIVED
T3185	234 (1)(b)	DECIDED *Lead decision, per margin_matrix_v352.json v352*	MMX v352	/rows/1/cells/J-FWD/rows/51/criterion	tag in margin_matrix_v352.json v352:
T3186	N 8572.4 ((b2) rep), T 94.9 C; within 0.05: FE confirmation	CALC	MMX v352	/rows/1/cells/J-FWD/rows/51/note	CALC / DERIVED
T3187	FA1S	DERIVED	MMX v352	/rows/1/cells/J-FWD/rows/55/state	(class set at the call site: DERIVED)
T3188	IN718 thread shear on THE ROW A_n 67.93 (route (i): M5 × 10.7 + HW-170 spring, L_e min ...	CALC	MMX v352	/rows/1/cells/J-FWD/rows/55/row	CALC / DERIVED
T3189	1.802	CALC	MMX v352	/rows/1/cells/J-FWD/rows/55/MS_y	CALC / DERIVED
T3190	2.143	CALC	MMX v352	/rows/1/cells/J-FWD/rows/55/MS_u	CALC / DERIVED
T3191	3750	CALC	MMX v352	/rows/1/cells/J-FWD/rows/55/preload_N	CALC / DERIVED
T3192	234 (1)(b)	DECIDED *Lead decision, per margin_matrix_v352.json v352*	MMX v352	/rows/1/cells/J-FWD/rows/55/criterion	tag in margin_matrix_v352.json v352:
T3193	N 3302.3 (js5), T 45.6 C	CALC	MMX v352	/rows/1/cells/J-FWD/rows/55/note	CALC / DERIVED
T3194	FA1S	DERIVED	MMX v352	/rows/1/cells/J-FWD/rows/59/state	(class set at the call site: DERIVED)
T3195	(ii) tap bearing, CALC V 3888.1 N on FEA3D's d x L_e basis (Mechanical rows)	CALC	MMX v352	/rows/1/cells/J-FWD/rows/59/row	CALC
T3196	1.891	CALC	MMX v352	/rows/1/cells/J-FWD/rows/59/MS_y	CALC
T3197	2.564	CALC	MMX v352	/rows/1/cells/J-FWD/rows/59/MS_u	CALC
T3198	231 (1)(ii)	DECIDED *Lead decision, per margin_matrix_v352.json v352*	MMX v352	/rows/1/cells/J-FWD/rows/59/criterion	tag in margin_matrix_v352.json v352:
T3199	FEA3D tap_bearing_derived sigma 106.0 at V 3603.6 (d x L_e 34.00 mm2, L_e 6.85) ...	CALC	MMX v352	/rows/1/cells/J-FWD/rows/59/note	CALC
T3200	LC2N	DERIVED	MMX v352	/rows/1/cells/J-FWD/rows/33/state	(class set at the call site: DERIVED)
T3201	js5b LC2N (LC2, SF [1.5, 2.0]): (i) shear-out to g2 at x' 28.0 Pm+Pb	SIM	MMX v352	/rows/1/cells/J-FWD/rows/33/row	SIM / DERIVED
T3202	6.251	SIM	MMX v352	/rows/1/cells/J-FWD/rows/33/MS_y	SIM / DERIVED
T3203	4.829	SIM	MMX v352	/rows/1/cells/J-FWD/rows/33/MS_u	SIM / DERIVED
T3204	231 (1)(i)	DECIDED *Lead decision, per margin_matrix_v352.json v352*	MMX v352	/rows/1/cells/J-FWD/rows/33/criterion	tag in margin_matrix_v352.json v352:
T3205	hand form: SF on the total stress including preload, linear superposition ASSUMED ...	SIM	MMX v352	/rows/1/cells/J-FWD/rows/33/note	SIM / DERIVED
T3206	LC2N	DERIVED	MMX v352	/rows/1/cells/J-FWD/rows/34/state	(class set at the call site: DERIVED)
T3207	js5b LC2N (LC2, SF [1.5, 2.0]): (ii) thread shear on A_n 68.4	SIM	MMX v352	/rows/1/cells/J-FWD/rows/34/row	SIM / DERIVED
T3208	0.526	SIM	MMX v352	/rows/1/cells/J-FWD/rows/34/MS_y	SIM / DERIVED
T3209	0.276	SIM	MMX v352	/rows/1/cells/J-FWD/rows/34/MS_u	SIM / DERIVED
T3210	231 (1)(ii)	DECIDED *Lead decision, per margin_matrix_v352.json v352*	MMX v352	/rows/1/cells/J-FWD/rows/34/criterion	tag in margin_matrix_v352.json v352:
T3211	hand form: SF on the total stress including preload, linear superposition ASSUMED ...	SIM	MMX v352	/rows/1/cells/J-FWD/rows/34/note	SIM / DERIVED
T3212	LC2N	DERIVED	MMX v352	/rows/1/cells/J-FWD/rows/35/state	(class set at the call site: DERIVED)
T3213	js5b LC2N (LC2, SF [1.5, 2.0]): (ii) tap bearing on L_e 6.84	SIM	MMX v352	/rows/1/cells/J-FWD/rows/35/row	SIM / DERIVED
T3214	6.137	SIM	MMX v352	/rows/1/cells/J-FWD/rows/35/MS_y	SIM / DERIVED
T3215	5.558	SIM	MMX v352	/rows/1/cells/J-FWD/rows/35/MS_u	SIM / DERIVED
T3216	231 (1)(ii)	DECIDED *Lead decision, per margin_matrix_v352.json v352*	MMX v352	/rows/1/cells/J-FWD/rows/35/criterion	tag in margin_matrix_v352.json v352:
T3217	hand form: SF on the total stress including preload, linear superposition ASSUMED ...	SIM	MMX v352	/rows/1/cells/J-FWD/rows/35/note	SIM / DERIVED
T3218	LC2M	DERIVED	MMX v352	/rows/1/cells/J-FWD/rows/36/state	(class set at the call site: DERIVED)
T3219	js5b LC2M (LC2, SF [1.5, 2.0]): (i) shear-out to g2 at x' 28.0 Pm+Pb	SIM	MMX v352	/rows/1/cells/J-FWD/rows/36/row	SIM / DERIVED
T3220	2.851	SIM	MMX v352	/rows/1/cells/J-FWD/rows/36/MS_y	SIM / DERIVED
T3221	2.089	SIM	MMX v352	/rows/1/cells/J-FWD/rows/36/MS_u	SIM / DERIVED
T3222	231 (1)(i)	DECIDED *Lead decision, per margin_matrix_v352.json v352*	MMX v352	/rows/1/cells/J-FWD/rows/36/criterion	tag in margin_matrix_v352.json v352:
T3223	hand form: SF on the total stress including preload, linear superposition ASSUMED ...	SIM	MMX v352	/rows/1/cells/J-FWD/rows/36/note	SIM / DERIVED
T3224	LC2M	DERIVED	MMX v352	/rows/1/cells/J-FWD/rows/37/state	(class set at the call site: DERIVED)
T3225	js5b LC2M (LC2, SF [1.5, 2.0]): (ii) thread shear on A_n 68.4	SIM	MMX v352	/rows/1/cells/J-FWD/rows/37/row	SIM / DERIVED
T3226	0.793	SIM	MMX v352	/rows/1/cells/J-FWD/rows/37/MS_y	SIM / DERIVED
T3227	0.496	SIM	MMX v352	/rows/1/cells/J-FWD/rows/37/MS_u	SIM / DERIVED
T3228	231 (1)(ii)	DECIDED *Lead decision, per margin_matrix_v352.json v352*	MMX v352	/rows/1/cells/J-FWD/rows/37/criterion	tag in margin_matrix_v352.json v352:
T3229	hand form: SF on the total stress including preload, linear superposition ASSUMED ...	SIM	MMX v352	/rows/1/cells/J-FWD/rows/37/note	SIM / DERIVED
T3230	LC2M	DERIVED	MMX v352	/rows/1/cells/J-FWD/rows/38/state	(class set at the call site: DERIVED)
T3231	js5b LC2M (LC2, SF [1.5, 2.0]): (ii) tap bearing on L_e 6.84	SIM	MMX v352	/rows/1/cells/J-FWD/rows/38/row	SIM / DERIVED
T3232	6.096	SIM	MMX v352	/rows/1/cells/J-FWD/rows/38/MS_y	SIM / DERIVED
T3233	5.506	SIM	MMX v352	/rows/1/cells/J-FWD/rows/38/MS_u	SIM / DERIVED
T3234	231 (1)(ii)	DECIDED *Lead decision, per margin_matrix_v352.json v352*	MMX v352	/rows/1/cells/J-FWD/rows/38/criterion	tag in margin_matrix_v352.json v352:
T3235	hand form: SF on the total stress including preload, linear superposition ASSUMED ...	SIM	MMX v352	/rows/1/cells/J-FWD/rows/38/note	SIM / DERIVED
T3236	FA1S	DERIVED	MMX v352	/rows/1/cells/J-FWD/rows/5/state	(class set at the call site: DERIVED)
T3237	condition: through-wall SCL at the band node (must pass)	SIM	MMX v352	/rows/1/cells/J-FWD/rows/5/row	SIM / DERIVED
T3238	0.346	SIM	MMX v352	/rows/1/cells/J-FWD/rows/5/MS_y	SIM / DERIVED
T3239	0.452	SIM	MMX v352	/rows/1/cells/J-FWD/rows/5/MS_u	SIM / DERIVED
T3240	lead 07:21Z (4) condition	DECIDED *Lead decision, per margin_matrix_v352.json v352*	MMX v352	/rows/1/cells/J-FWD/rows/5/criterion	tag in margin_matrix_v352.json v352:
T3241	passes	SIM	MMX v352	/rows/1/cells/J-FWD/rows/5/note	SIM / DERIVED
T3242	FA1S	DERIVED	MMX v352	/rows/1/cells/J-FWD/rows/6/state	(class set at the call site: DERIVED)

ID	Value	Class	Source	Key	Source tag / formula
T3243	elastic band (contact-edge field, 0.99 mm from the TC-01 hole-edge footprint)	SIM	MMX v352	/rows/1/cells/J-FWD/rows/6/row	SIM / DERIVED
T3244	0.155	SIM	MMX v352	/rows/1/cells/J-FWD/rows/6/MS_y	SIM / DERIVED
T3245	0.245	SIM	MMX v352	/rows/1/cells/J-FWD/rows/6/MS_u	SIM / DERIVED
T3246	FA1S	DERIVED	MMX v352	/rows/1/cells/J-FWD/rows/7/state	(class set at the call site: DERIVED)
T3247	all-node (bonded-thread tie edge)	SIM	MMX v352	/rows/1/cells/J-FWD/rows/7/row	SIM / DERIVED
T3248	−0.89	SIM	MMX v352	/rows/1/cells/J-FWD/rows/7/MS_y	SIM / DERIVED
T3249	−0.881	SIM	MMX v352	/rows/1/cells/J-FWD/rows/7/MS_u	SIM / DERIVED
T3250	n1660 2148.3 MPa	SIM	MMX v352	/rows/1/cells/J-FWD/rows/7/note	SIM / DERIVED
T3251	PRELOAD	DERIVED	MMX v352	/rows/1/cells/J-FWD/rows/10/state	(class set at the call site: DERIVED)
T3252	thread shear on A_n 68.4, js5 N (A4-80 FE)	SIM	MMX v352	/rows/1/cells/J-FWD/rows/10/row	SIM / DERIVED
T3253	1.526	SIM	MMX v352	/rows/1/cells/J-FWD/rows/10/MS_y	SIM / DERIVED
T3254	1.846	SIM	MMX v352	/rows/1/cells/J-FWD/rows/10/MS_u	SIM / DERIVED
T3255	A4-80 FE force (A4-80 withdrawn, 240 (4)): the flight row is Mechanical's IN718 THE ROW	SIM	MMX v352	/rows/1/cells/J-FWD/rows/10/note	SIM / DERIVED
T3256	PRELOAD	DERIVED	MMX v352	/rows/1/cells/J-FWD/rows/11/state	(class set at the call site: DERIVED)
T3257	all-node (bonded-thread tie edge)	SIM	MMX v352	/rows/1/cells/J-FWD/rows/11/row	SIM / DERIVED
T3258	−0.865	SIM	MMX v352	/rows/1/cells/J-FWD/rows/11/MS_y	SIM / DERIVED
T3259	−0.854	SIM	MMX v352	/rows/1/cells/J-FWD/rows/11/MS_u	SIM / DERIVED
T3260	n1660 1779.6 MPa	SIM	MMX v352	/rows/1/cells/J-FWD/rows/11/note	SIM / DERIVED
T3261	SOAK70	DERIVED	MMX v352	/rows/1/cells/J-FWD/rows/12/state	(class set at the call site: DERIVED)
T3262	thread shear on A_n 68.4, js5 N (A4-80 FE)	SIM	MMX v352	/rows/1/cells/J-FWD/rows/12/row	SIM / DERIVED
T3263	0.581	SIM	MMX v352	/rows/1/cells/J-FWD/rows/12/MS_y	SIM / DERIVED
T3264	0.756	SIM	MMX v352	/rows/1/cells/J-FWD/rows/12/MS_u	SIM / DERIVED
T3265	A4-80 FE force (A4-80 withdrawn, 240 (4)): the flight row is Mechanical's IN718 THE ROW	SIM	MMX v352	/rows/1/cells/J-FWD/rows/12/note	SIM / DERIVED
T3266	SOAK70	DERIVED	MMX v352	/rows/1/cells/J-FWD/rows/13/state	(class set at the call site: DERIVED)
T3267	thread shear on A_n 68.4 at the envelope N 7471.8 ((b2); A4-80 FE)	DERIVED	MMX v352	/rows/1/cells/J-FWD/rows/13/row	DERIVED
T3268	0.206	DERIVED	MMX v352	/rows/1/cells/J-FWD/rows/13/MS_y	DERIVED
T3269	0.339	DERIVED	MMX v352	/rows/1/cells/J-FWD/rows/13/MS_u	DERIVED
T3270	A4-80 FE force (A4-80 withdrawn, 240 (4)): the flight row is Mechanical's IN718 THE ROW	DERIVED	MMX v352	/rows/1/cells/J-FWD/rows/13/note	DERIVED
T3271	SOAK70	DERIVED	MMX v352	/rows/1/cells/J-FWD/rows/14/state	(class set at the call site: DERIVED)
T3272	all-node (bonded-thread tie edge)	SIM	MMX v352	/rows/1/cells/J-FWD/rows/14/row	SIM / DERIVED
T3273	−0.779	SIM	MMX v352	/rows/1/cells/J-FWD/rows/14/MS_y	SIM / DERIVED
T3274	−0.764	SIM	MMX v352	/rows/1/cells/J-FWD/rows/14/MS_u	SIM / DERIVED
T3275	n1660 1034.8 MPa	SIM	MMX v352	/rows/1/cells/J-FWD/rows/14/note	SIM / DERIVED
T3276	SOAK	DERIVED	MMX v352	/rows/1/cells/J-FWD/rows/15/state	(class set at the call site: DERIVED)
T3277	thread shear on A_n 68.4, js5 N (A4-80 FE)	SIM	MMX v352	/rows/1/cells/J-FWD/rows/15/row	SIM / DERIVED
T3278	0.221	SIM	MMX v352	/rows/1/cells/J-FWD/rows/15/MS_y	SIM / DERIVED
T3279	0.357	SIM	MMX v352	/rows/1/cells/J-FWD/rows/15/MS_u	SIM / DERIVED
T3280	A4-80 FE force (A4-80 withdrawn, 240 (4)): the flight row is Mechanical's IN718 THE ROW	SIM	MMX v352	/rows/1/cells/J-FWD/rows/15/note	SIM / DERIVED
T3281	SOAK	DERIVED	MMX v352	/rows/1/cells/J-FWD/rows/16/state	(class set at the call site: DERIVED)
T3282	thread shear on A_n 68.4 at the envelope N 7382.9 (js5; A4-80 FE)	DERIVED	MMX v352	/rows/1/cells/J-FWD/rows/16/row	DERIVED
T3283	0.221	DERIVED	MMX v352	/rows/1/cells/J-FWD/rows/16/MS_y	DERIVED
T3284	0.357	DERIVED	MMX v352	/rows/1/cells/J-FWD/rows/16/MS_u	DERIVED
T3285	A4-80 FE force (A4-80 withdrawn, 240 (4)): the flight row is Mechanical's IN718 THE ROW	DERIVED	MMX v352	/rows/1/cells/J-FWD/rows/16/note	DERIVED
T3286	SOAK	DERIVED	MMX v352	/rows/1/cells/J-FWD/rows/17/state	(class set at the call site: DERIVED)
T3287	all-node (bonded-thread tie edge)	SIM	MMX v352	/rows/1/cells/J-FWD/rows/17/row	SIM / DERIVED
T3288	−0.891	SIM	MMX v352	/rows/1/cells/J-FWD/rows/17/MS_y	SIM / DERIVED
T3289	−0.884	SIM	MMX v352	/rows/1/cells/J-FWD/rows/17/MS_u	SIM / DERIVED
T3290	n1642 2100.4 MPa	SIM	MMX v352	/rows/1/cells/J-FWD/rows/17/note	SIM / DERIVED
T3291	SOAK52	DERIVED	MMX v352	/rows/1/cells/J-FWD/rows/18/state	(class set at the call site: DERIVED)
T3292	thread shear on A_n 68.4, js5 N (A4-80 FE)	SIM	MMX v352	/rows/1/cells/J-FWD/rows/18/row	SIM / DERIVED
T3293	0.375	SIM	MMX v352	/rows/1/cells/J-FWD/rows/18/MS_y	SIM / DERIVED
T3294	0.528	SIM	MMX v352	/rows/1/cells/J-FWD/rows/18/MS_u	SIM / DERIVED
T3295	A4-80 FE force (A4-80 withdrawn, 240 (4)): the flight row is Mechanical's IN718 THE ROW	SIM	MMX v352	/rows/1/cells/J-FWD/rows/18/note	SIM / DERIVED
T3296	SOAK52	DERIVED	MMX v352	/rows/1/cells/J-FWD/rows/19/state	(class set at the call site: DERIVED)
T3297	thread shear on A_n 68.4 at the envelope N 7764.0 ((b2) rep; A4-80 FE)	DERIVED	MMX v352	/rows/1/cells/J-FWD/rows/19/row	DERIVED
T3298	0.162	DERIVED	MMX v352	/rows/1/cells/J-FWD/rows/19/MS_y	DERIVED
T3299	0.291	DERIVED	MMX v352	/rows/1/cells/J-FWD/rows/19/MS_u	DERIVED
T3300	A4-80 FE force (A4-80 withdrawn, 240 (4)): the flight row is Mechanical's IN718 THE ROW	DERIVED	MMX v352	/rows/1/cells/J-FWD/rows/19/note	DERIVED
T3301	SOAK52	DERIVED	MMX v352	/rows/1/cells/J-FWD/rows/20/state	(class set at the call site: DERIVED)
T3302	all-node (bonded-thread tie edge)	SIM	MMX v352	/rows/1/cells/J-FWD/rows/20/row	SIM / DERIVED
T3303	−0.824	SIM	MMX v352	/rows/1/cells/J-FWD/rows/20/MS_y	SIM / DERIVED
T3304	−0.812	SIM	MMX v352	/rows/1/cells/J-FWD/rows/20/MS_u	SIM / DERIVED
T3305	n17845 1298.7 MPa	SIM	MMX v352	/rows/1/cells/J-FWD/rows/20/note	SIM / DERIVED
T3306	PRELOAD	DERIVED	MMX v352	/rows/1/cells/J-FWD/rows/39/state	(class set at the call site: DERIVED)
T3307	IN718 thread shear on THE ROW A_n 67.93 (route (i): M5 × 10.7 + HW-170 spring, L_e min …	CALC	MMX v352	/rows/1/cells/J-FWD/rows/39/row	CALC / DERIVED
T3308	1.509	CALC	MMX v352	/rows/1/cells/J-FWD/rows/39/MS_y	CALC / DERIVED
T3309	1.826	CALC	MMX v352	/rows/1/cells/J-FWD/rows/39/MS_u	CALC / DERIVED
T3310	3750	CALC	MMX v352	/rows/1/cells/J-FWD/rows/39/preload_N	CALC / DERIVED
T3311	234 (1)(b)	DECIDED *Lead decision, per margin_matrix_v352.json v352*	MMX v352	/rows/1/cells/J-FWD/rows/39/criterion	tag in margin_matrix_v352.json v352:
T3312	N 3750.0 (js5), T 20.0 C	CALC	MMX v352	/rows/1/cells/J-FWD/rows/39/note	CALC / DERIVED
T3313	PRELOAD	DERIVED	MMX v352	/rows/1/cells/J-FWD/rows/40/state	(class set at the call site: DERIVED)
T3314	IN718 thread shear on FEA3D's drawn-basis A_n 68.4 (M5 × 10.2, lead 13:14Z; superseded …	CALC	MMX v352	/rows/1/cells/J-FWD/rows/40/row	CALC / DERIVED
T3315	1.526	CALC	MMX v352	/rows/1/cells/J-FWD/rows/40/MS_y	CALC / DERIVED
T3316	1.845	CALC	MMX v352	/rows/1/cells/J-FWD/rows/40/MS_u	CALC / DERIVED
T3317	3750	CALC	MMX v352	/rows/1/cells/J-FWD/rows/40/preload_N	CALC / DERIVED
T3318	N 3750.0 (js5), T 20.0 C	CALC	MMX v352	/rows/1/cells/J-FWD/rows/40/note	CALC / DERIVED
T3319	PRELOAD	DERIVED	MMX v352	/rows/1/cells/J-FWD/rows/41/state	(class set at the call site: DERIVED)
T3320	IN718 thread shear on THE ROW A_n 67.93 (route (i): M5 × 10.7 + HW-170 spring, L_e min …	CALC	MMX v352	/rows/1/cells/J-FWD/rows/41/row	CALC / DERIVED
T3321	1.297	CALC	MMX v352	/rows/1/cells/J-FWD/rows/41/MS_y	CALC / DERIVED
T3322	1.587	CALC	MMX v352	/rows/1/cells/J-FWD/rows/41/MS_u	CALC / DERIVED
T3323	4096	CALC	MMX v352	/rows/1/cells/J-FWD/rows/41/preload_N	CALC / DERIVED
T3324	234 (1)(b)	DECIDED *Lead decision, per margin_matrix_v352.json v352*	MMX v352	/rows/1/cells/J-FWD/rows/41/criterion	tag in margin_matrix_v352.json v352:
T3325	N 4096.0 (js5), T 20.0 C; 4096 N column = information (J-F1 maximum capped at 3750 N …	CALC	MMX v352	/rows/1/cells/J-FWD/rows/41/note	CALC / DERIVED

ID	Value	Class	Source	Key	Source tag / formula
T3326	PRELOAD	DERIVED	MMX v352	/rows/1/cells/J-FWD/rows/42/state	(class set at the call site: DERIVED)
T3327	IN718 thread shear on FEA3D's drawn-basis A_n 68.4 (M5 × 10.2, lead 13:14Z; superseded ...	CALC	MMX v352	/rows/1/cells/J-FWD/rows/42/row	CALC / DERIVED
T3328	1.313	CALC	MMX v352	/rows/1/cells/J-FWD/rows/42/MS_y	CALC / DERIVED
T3329	1.605	CALC	MMX v352	/rows/1/cells/J-FWD/rows/42/MS_u	CALC / DERIVED
T3330	4096	CALC	MMX v352	/rows/1/cells/J-FWD/rows/42/preload_N	CALC / DERIVED
T3331	N 4096.0 (js5), T 20.0 C; 4096 N column = information (J-F1 maximum capped at 3750 N ...	CALC	MMX v352	/rows/1/cells/J-FWD/rows/42/note	CALC / DERIVED
T3332	SOAK70	DERIVED	MMX v352	/rows/1/cells/J-FWD/rows/44/state	(class set at the call site: DERIVED)
T3333	IN718 thread shear on FEA3D's drawn-basis A_n 68.4 (M5 × 10.2, lead 13:14Z; superseded ...	CALC	MMX v352	/rows/1/cells/J-FWD/rows/44/row	CALC / DERIVED
T3334	0.095	CALC	MMX v352	/rows/1/cells/J-FWD/rows/44/MS_y	CALC / DERIVED
T3335	0.215	CALC	MMX v352	/rows/1/cells/J-FWD/rows/44/MS_u	CALC / DERIVED
T3336	3750	CALC	MMX v352	/rows/1/cells/J-FWD/rows/44/preload_N	CALC / DERIVED
T3337	N 8227.4 ((b2)), T 97.7 C	CALC	MMX v352	/rows/1/cells/J-FWD/rows/44/note	CALC / DERIVED
T3338	SOAK70	DERIVED	MMX v352	/rows/1/cells/J-FWD/rows/45/state	(class set at the call site: DERIVED)
T3339	IN718 thread shear on THE ROW A_n 67.93 (route (i): M5 × 10.7 + HW-170 spring, L_e min ...	CALC	MMX v352	/rows/1/cells/J-FWD/rows/45/row	CALC / DERIVED
T3340	0.043	CALC	MMX v352	/rows/1/cells/J-FWD/rows/45/MS_y	CALC / DERIVED
T3341	0.159	CALC	MMX v352	/rows/1/cells/J-FWD/rows/45/MS_u	CALC / DERIVED
T3342	4096	CALC	MMX v352	/rows/1/cells/J-FWD/rows/45/preload_N	CALC / DERIVED
T3343	234 (1)(b)	DECIDED *Lead decision, per margin_matrix_v352.json v352*	MMX v352	/rows/1/cells/J-FWD/rows/45/criterion	tag in margin_matrix_v352.json v352:
T3344	N 8573.4 ((b2)), T 97.7 C; within 0.05: FE confirmation; 4096 N column = information ...	CALC	MMX v352	/rows/1/cells/J-FWD/rows/45/note	CALC / DERIVED
T3345	SOAK70	DERIVED	MMX v352	/rows/1/cells/J-FWD/rows/46/state	(class set at the call site: DERIVED)
T3346	IN718 thread shear on FEA3D's drawn-basis A_n 68.4 (M5 × 10.2, lead 13:14Z; superseded ...	CALC	MMX v352	/rows/1/cells/J-FWD/rows/46/row	CALC / DERIVED
T3347	0.051	CALC	MMX v352	/rows/1/cells/J-FWD/rows/46/MS_y	CALC / DERIVED
T3348	0.166	CALC	MMX v352	/rows/1/cells/J-FWD/rows/46/MS_u	CALC / DERIVED
T3349	4096	CALC	MMX v352	/rows/1/cells/J-FWD/rows/46/preload_N	CALC / DERIVED
T3350	N 8573.4 ((b2)), T 97.7 C; 4096 N column = information (J-F1 maximum capped at 3750 N ...	CALC	MMX v352	/rows/1/cells/J-FWD/rows/46/note	CALC / DERIVED
T3351	SOAK	DERIVED	MMX v352	/rows/1/cells/J-FWD/rows/48/state	(class set at the call site: DERIVED)
T3352	IN718 thread shear on FEA3D's drawn-basis A_n 68.4 (M5 × 10.2, lead 13:14Z; superseded ...	CALC	MMX v352	/rows/1/cells/J-FWD/rows/48/row	CALC / DERIVED
T3353	0.112	CALC	MMX v352	/rows/1/cells/J-FWD/rows/48/MS_y	CALC / DERIVED
T3354	0.235	CALC	MMX v352	/rows/1/cells/J-FWD/rows/48/MS_u	CALC / DERIVED
T3355	3750	CALC	MMX v352	/rows/1/cells/J-FWD/rows/48/preload_N	CALC / DERIVED
T3356	N 8110.3 (js5), T 96.0 C	CALC	MMX v352	/rows/1/cells/J-FWD/rows/48/note	CALC / DERIVED
T3357	SOAK	DERIVED	MMX v352	/rows/1/cells/J-FWD/rows/49/state	(class set at the call site: DERIVED)
T3358	IN718 thread shear on THE ROW A_n 67.93 (route (i): M5 × 10.7 + HW-170 spring, L_e min ...	CALC	MMX v352	/rows/1/cells/J-FWD/rows/49/row	CALC / DERIVED
T3359	0.059	CALC	MMX v352	/rows/1/cells/J-FWD/rows/49/MS_y	CALC / DERIVED
T3360	0.176	CALC	MMX v352	/rows/1/cells/J-FWD/rows/49/MS_u	CALC / DERIVED
T3361	4096	CALC	MMX v352	/rows/1/cells/J-FWD/rows/49/preload_N	CALC / DERIVED
T3362	234 (1)(b)	DECIDED *Lead decision, per margin_matrix_v352.json v352*	MMX v352	/rows/1/cells/J-FWD/rows/49/criterion	tag in margin_matrix_v352.json v352:
T3363	N 8456.3 (js5), T 96.0 C; 4096 N column = information (J-F1 maximum capped at 3750 N ...	CALC	MMX v352	/rows/1/cells/J-FWD/rows/49/note	CALC / DERIVED
T3364	SOAK	DERIVED	MMX v352	/rows/1/cells/J-FWD/rows/50/state	(class set at the call site: DERIVED)
T3365	IN718 thread shear on FEA3D's drawn-basis A_n 68.4 (M5 × 10.2, lead 13:14Z; superseded ...	CALC	MMX v352	/rows/1/cells/J-FWD/rows/50/row	CALC / DERIVED
T3366	0.066	CALC	MMX v352	/rows/1/cells/J-FWD/rows/50/MS_y	CALC / DERIVED
T3367	0.184	CALC	MMX v352	/rows/1/cells/J-FWD/rows/50/MS_u	CALC / DERIVED
T3368	4096	CALC	MMX v352	/rows/1/cells/J-FWD/rows/50/preload_N	CALC / DERIVED
T3369	N 8456.3 (js5), T 96.0 C; 4096 N column = information (J-F1 maximum capped at 3750 N ...	CALC	MMX v352	/rows/1/cells/J-FWD/rows/50/note	CALC / DERIVED
T3370	SOAK52	DERIVED	MMX v352	/rows/1/cells/J-FWD/rows/52/state	(class set at the call site: DERIVED)
T3371	IN718 thread shear on FEA3D's drawn-basis A_n 68.4 (M5 × 10.2, lead 13:14Z; superseded ...	CALC	MMX v352	/rows/1/cells/J-FWD/rows/52/row	CALC / DERIVED
T3372	0.052	CALC	MMX v352	/rows/1/cells/J-FWD/rows/52/MS_y	CALC / DERIVED
T3373	0.17	CALC	MMX v352	/rows/1/cells/J-FWD/rows/52/MS_u	CALC / DERIVED
T3374	3750	CALC	MMX v352	/rows/1/cells/J-FWD/rows/52/preload_N	CALC / DERIVED
T3375	N 8572.4 ((b2) rep), T 94.9 C	CALC	MMX v352	/rows/1/cells/J-FWD/rows/52/note	CALC / DERIVED
T3376	SOAK52	DERIVED	MMX v352	/rows/1/cells/J-FWD/rows/53/state	(class set at the call site: DERIVED)
T3377	IN718 thread shear on THE ROW A_n 67.93 (route (i): M5 × 10.7 + HW-170 spring, L_e min ...	CALC	MMX v352	/rows/1/cells/J-FWD/rows/53/row	CALC / DERIVED
T3378	0.005	CALC	MMX v352	/rows/1/cells/J-FWD/rows/53/MS_y	CALC / DERIVED
T3379	0.116	CALC	MMX v352	/rows/1/cells/J-FWD/rows/53/MS_u	CALC / DERIVED
T3380	4096	CALC	MMX v352	/rows/1/cells/J-FWD/rows/53/preload_N	CALC / DERIVED
T3381	234 (1)(b)	DECIDED *Lead decision, per margin_matrix_v352.json v352*	MMX v352	/rows/1/cells/J-FWD/rows/53/criterion	tag in margin_matrix_v352.json v352:
T3382	N 8918.4 ((b2) rep), T 94.9 C; within 0.05: FE confirmation; 4096 N column = information ...	CALC	MMX v352	/rows/1/cells/J-FWD/rows/53/note	CALC / DERIVED
T3383	SOAK52	DERIVED	MMX v352	/rows/1/cells/J-FWD/rows/54/state	(class set at the call site: DERIVED)
T3384	IN718 thread shear on FEA3D's drawn-basis A_n 68.4 (M5 × 10.2, lead 13:14Z; superseded ...	CALC	MMX v352	/rows/1/cells/J-FWD/rows/54/row	CALC / DERIVED
T3385	0.012	CALC	MMX v352	/rows/1/cells/J-FWD/rows/54/MS_y	CALC / DERIVED
T3386	0.124	CALC	MMX v352	/rows/1/cells/J-FWD/rows/54/MS_u	CALC / DERIVED
T3387	4096	CALC	MMX v352	/rows/1/cells/J-FWD/rows/54/preload_N	CALC / DERIVED
T3388	N 8918.4 ((b2) rep), T 94.9 C; 4096 N column = information (J-F1 maximum capped at 3750 ...	CALC	MMX v352	/rows/1/cells/J-FWD/rows/54/note	CALC / DERIVED
T3389	FA1S	DERIVED	MMX v352	/rows/1/cells/J-FWD/rows/56/state	(class set at the call site: DERIVED)
T3390	IN718 thread shear on FEA3D's drawn-basis A_n 68.4 (M5 × 10.2, lead 13:14Z; superseded ...	CALC	MMX v352	/rows/1/cells/J-FWD/rows/56/row	CALC / DERIVED
T3391	1.821	CALC	MMX v352	/rows/1/cells/J-FWD/rows/56/MS_y	CALC / DERIVED
T3392	2.165	CALC	MMX v352	/rows/1/cells/J-FWD/rows/56/MS_u	CALC / DERIVED
T3393	3750	CALC	MMX v352	/rows/1/cells/J-FWD/rows/56/preload_N	CALC / DERIVED
T3394	N 3302.3 (js5), T 45.6 C	CALC	MMX v352	/rows/1/cells/J-FWD/rows/56/note	CALC / DERIVED
T3395	FA1S	DERIVED	MMX v352	/rows/1/cells/J-FWD/rows/57/state	(class set at the call site: DERIVED)

ID	Value	Class	Source	Key	Source tag / formula
T3396	IN718 thread shear on THE ROW A_n 67.93 (route (i): M5 × 10.7 + HW-170 spring, L_e min ...	CALC	MMX v352	/rows/1/cells/J-FWD/rows/57/row	CALC / DERIVED
T3397	1.537	CALC	MMX v352	/rows/1/cells/J-FWD/rows/57/MS_y	CALC / DERIVED
T3398	1.845	CALC	MMX v352	/rows/1/cells/J-FWD/rows/57/MS_u	CALC / DERIVED
T3399	4096	CALC	MMX v352	/rows/1/cells/J-FWD/rows/57/preload_N	CALC / DERIVED
T3400	234 (1)(b)	DECIDED *Lead decision, per margin_matrix_v352.json v352*	MMX v352	/rows/1/cells/J-FWD/rows/57/criterion	tag in margin_matrix_v352.json v352:
T3401	N 3648.3 (js5), T 45.6 C; 4096 N column = information (J-F1 maximum capped at 3750 N ...	CALC	MMX v352	/rows/1/cells/J-FWD/rows/57/note	CALC / DERIVED
T3402	FA1S	DERIVED	MMX v352	/rows/1/cells/J-FWD/rows/58/state	(class set at the call site: DERIVED)
T3403	IN718 thread shear on FEA3D's drawn-basis A_n 68.4 (M5 × 10.2, lead 13:14Z; superseded ...	CALC	MMX v352	/rows/1/cells/J-FWD/rows/58/row	CALC / DERIVED
T3404	1.554	CALC	MMX v352	/rows/1/cells/J-FWD/rows/58/MS_y	CALC / DERIVED
T3405	1.865	CALC	MMX v352	/rows/1/cells/J-FWD/rows/58/MS_u	CALC / DERIVED
T3406	4096	CALC	MMX v352	/rows/1/cells/J-FWD/rows/58/preload_N	CALC / DERIVED
T3407	N 3648.3 (js5), T 45.6 C; 4096 N column = information (J-F1 maximum capped at 3750 N ...	CALC	MMX v352	/rows/1/cells/J-FWD/rows/58/note	CALC / DERIVED
T3408	0.65–0.278	CALC	MMX v352	/rows/23/cells/J-FWD/value	(class set at the call site: CALC) [J-FWD cell = minimum over the criterion rows]
T3409	FA1S	DERIVED	MMX v352	/rows/23/cells/J-FWD/rows/1/state	(class set at the call site: DERIVED)
T3410	IN718 5020A Eq. 22, preload omitted (criterion, 240 (3)); R6 basis: V 4079.2 N FE (b1) ...	CALC	MMX v352	/rows/23/cells/J-FWD/rows/1/row	CALC
T3411	MISSING	CALC	MMX v352	/rows/23/cells/J-FWD/rows/1/MS_y	CALC
T3412	0.278	CALC	MMX v352	/rows/23/cells/J-FWD/rows/1/MS_u	CALC
T3413	NASA-STD-5020A Eq. 22	DECIDED *Lead decision, per margin_matrix_v352.json v352*	MMX v352	/rows/23/cells/J-FWD/rows/1/criterion	tag in margin_matrix_v352.json v352:
T3414	preload included 0.065 (information); fix v2 CALC record (V 3888.1 N) 0.301 / included ...	CALC	MMX v352	/rows/23/cells/J-FWD/rows/1/note	CALC
T3415	FA1S	DERIVED	MMX v352	/rows/23/cells/J-FWD/rows/2/state	(class set at the call site: DERIVED)
T3416	IN718 direct shear on the minor area (231 (ii))	CALC	MMX v352	/rows/23/cells/J-FWD/rows/2/row	CALC
T3417	0.933	CALC	MMX v352	/rows/23/cells/J-FWD/rows/2/MS_y	CALC
T3418	1.408	CALC	MMX v352	/rows/23/cells/J-FWD/rows/2/MS_u	CALC
T3419	231 (ii)	DECIDED *Lead decision, per margin_matrix_v352.json v352*	MMX v352	/rows/23/cells/J-FWD/rows/2/criterion	tag in margin_matrix_v352.json v352:
T3420	LC2N	DERIVED	MMX v352	/rows/23/cells/J-FWD/rows/11/state	(class set at the call site: DERIVED)
T3421	js5b LC2N (LC2, SF [1.5, 2.0]): IN718 FE screw, LC2: combined vM (preload on A_s + ...	SIM	MMX v352	/rows/23/cells/J-FWD/rows/11/row	SIM / DERIVED
T3422	1.894	SIM	MMX v352	/rows/23/cells/J-FWD/rows/11/MS_y	SIM / DERIVED
T3423	2.11	SIM	MMX v352	/rows/23/cells/J-FWD/rows/11/MS_u	SIM / DERIVED
T3424	234 (1)(b)	DECIDED *Lead decision, per margin_matrix_v352.json v352*	MMX v352	/rows/23/cells/J-FWD/rows/11/criterion	tag in margin_matrix_v352.json v352:
T3425	LC2M	DERIVED	MMX v352	/rows/23/cells/J-FWD/rows/12/state	(class set at the call site: DERIVED)
T3426	js5b LC2M (LC2, SF [1.5, 2.0]): IN718 FE screw, LC2: combined vM (preload on A_s + ...	SIM	MMX v352	/rows/23/cells/J-FWD/rows/12/row	SIM / DERIVED
T3427	2.218	SIM	MMX v352	/rows/23/cells/J-FWD/rows/12/MS_y	SIM / DERIVED
T3428	2.394	SIM	MMX v352	/rows/23/cells/J-FWD/rows/12/MS_u	SIM / DERIVED
T3429	234 (1)(b)	DECIDED *Lead decision, per margin_matrix_v352.json v352*	MMX v352	/rows/23/cells/J-FWD/rows/12/criterion	tag in margin_matrix_v352.json v352:
T3430	SOAK70	DERIVED	MMX v352	/rows/23/cells/J-FWD/rows/17/state	(class set at the call site: DERIVED)
T3431	IN718 tension on A_s (234 (1)(b))	CALC	MMX v352	/rows/23/cells/J-FWD/rows/17/row	CALC / DERIVED
T3432	0.717	CALC	MMX v352	/rows/23/cells/J-FWD/rows/17/MS_y	CALC / DERIVED
T3433	1.034	CALC	MMX v352	/rows/23/cells/J-FWD/rows/17/MS_u	CALC / DERIVED
T3434	3750	CALC	MMX v352	/rows/23/cells/J-FWD/rows/17/preload_N	CALC / DERIVED
T3435	N 8227.4 ((b2)), T 158.6 C	CALC	MMX v352	/rows/23/cells/J-FWD/rows/17/note	CALC / DERIVED
T3436	SOAK70	DERIVED	MMX v352	/rows/23/cells/J-FWD/rows/18/state	(class set at the call site: DERIVED)
T3437	IN718 head on the HW-170 spring, A 39.19 mm2 (IN718 Fty / Ftu, no bearing credit)	CALC	MMX v352	/rows/23/cells/J-FWD/rows/18/row	CALC / DERIVED
T3438	3.745	CALC	MMX v352	/rows/23/cells/J-FWD/rows/18/MS_y	CALC / DERIVED
T3439	4.622	CALC	MMX v352	/rows/23/cells/J-FWD/rows/18/MS_u	CALC / DERIVED
T3440	3750	CALC	MMX v352	/rows/23/cells/J-FWD/rows/18/preload_N	CALC / DERIVED
T3441	N 8227.4 ((b2)), T 158.6 C; spring side TO-SOURCE (Mechanical)	CALC	MMX v352	/rows/23/cells/J-FWD/rows/18/note	CALC / DERIVED
T3442	SOAK	DERIVED	MMX v352	/rows/23/cells/J-FWD/rows/23/state	(class set at the call site: DERIVED)
T3443	IN718 tension on A_s (234 (1)(b))	CALC	MMX v352	/rows/23/cells/J-FWD/rows/23/row	CALC / DERIVED
T3444	0.757	CALC	MMX v352	/rows/23/cells/J-FWD/rows/23/MS_y	CALC / DERIVED
T3445	1.088	CALC	MMX v352	/rows/23/cells/J-FWD/rows/23/MS_u	CALC / DERIVED
T3446	3750	CALC	MMX v352	/rows/23/cells/J-FWD/rows/23/preload_N	CALC / DERIVED
T3447	N 8110.3 (js5), T 127.0 C	CALC	MMX v352	/rows/23/cells/J-FWD/rows/23/note	CALC / DERIVED
T3448	SOAK	DERIVED	MMX v352	/rows/23/cells/J-FWD/rows/24/state	(class set at the call site: DERIVED)
T3449	IN718 head on the HW-170 spring, A 39.19 mm2 (IN718 Fty / Ftu, no bearing credit)	CALC	MMX v352	/rows/23/cells/J-FWD/rows/24/row	CALC / DERIVED
T3450	3.855	CALC	MMX v352	/rows/23/cells/J-FWD/rows/24/MS_y	CALC / DERIVED
T3451	4.77	CALC	MMX v352	/rows/23/cells/J-FWD/rows/24/MS_u	CALC / DERIVED
T3452	3750	CALC	MMX v352	/rows/23/cells/J-FWD/rows/24/preload_N	CALC / DERIVED
T3453	N 8110.3 (js5), T 127.0 C; spring side TO-SOURCE (Mechanical)	CALC	MMX v352	/rows/23/cells/J-FWD/rows/24/note	CALC / DERIVED
T3454	SOAK52	DERIVED	MMX v352	/rows/23/cells/J-FWD/rows/29/state	(class set at the call site: DERIVED)
T3455	IN718 tension on A_s (234 (1)(b))	CALC	MMX v352	/rows/23/cells/J-FWD/rows/29/row	CALC / DERIVED
T3456	0.65	CALC	MMX v352	/rows/23/cells/J-FWD/rows/29/MS_y	CALC / DERIVED
T3457	0.955	CALC	MMX v352	/rows/23/cells/J-FWD/rows/29/MS_u	CALC / DERIVED
T3458	3750	CALC	MMX v352	/rows/23/cells/J-FWD/rows/29/preload_N	CALC / DERIVED
T3459	N 8572.4 ((b2) rep), T 154.1 C	CALC	MMX v352	/rows/23/cells/J-FWD/rows/29/note	CALC / DERIVED
T3460	SOAK52	DERIVED	MMX v352	/rows/23/cells/J-FWD/rows/30/state	(class set at the call site: DERIVED)
T3461	IN718 head on the HW-170 spring, A 39.19 mm2 (IN718 Fty / Ftu, no bearing credit)	CALC	MMX v352	/rows/23/cells/J-FWD/rows/30/row	CALC / DERIVED
T3462	3.56	CALC	MMX v352	/rows/23/cells/J-FWD/rows/30/MS_y	CALC / DERIVED
T3463	4.405	CALC	MMX v352	/rows/23/cells/J-FWD/rows/30/MS_u	CALC / DERIVED
T3464	3750	CALC	MMX v352	/rows/23/cells/J-FWD/rows/30/preload_N	CALC / DERIVED
T3465	N 8572.4 ((b2) rep), T 154.1 C; spring side TO-SOURCE (Mechanical)	CALC	MMX v352	/rows/23/cells/J-FWD/rows/30/note	CALC / DERIVED
T3466	FA1S	DERIVED	MMX v352	/rows/23/cells/J-FWD/rows/36/state	(class set at the call site: DERIVED)
T3467	IN718 head on the HW-170 spring, A 39.19 mm2 (IN718 Fty / Ftu, no bearing credit)	CALC	MMX v352	/rows/23/cells/J-FWD/rows/36/row	CALC / DERIVED
T3468	11.188	CALC	MMX v352	/rows/23/cells/J-FWD/rows/36/MS_y	CALC / DERIVED
T3469	13.595	CALC	MMX v352	/rows/23/cells/J-FWD/rows/36/MS_u	CALC / DERIVED
T3470	3750	CALC	MMX v352	/rows/23/cells/J-FWD/rows/36/preload_N	CALC / DERIVED
T3471	N 3302.3 (js5), T 45.6 C; spring side TO-SOURCE (Mechanical)	CALC	MMX v352	/rows/23/cells/J-FWD/rows/36/note	CALC / DERIVED
T3472	FA1S	DERIVED	MMX v352	/rows/23/cells/J-FWD/rows/0/state	(class set at the call site: DERIVED)
T3473	A4-80 FE: interaction (234 (1)(b)) / shear-only yield on A_3	SIM	MMX v352	/rows/23/cells/J-FWD/rows/0/row	SIM / DERIVED

ID	Value	Class	Source	Key	Source tag / formula
T3474	0.145	SIM	MMX v352	/rows/23/cells/J-FWD/rows/0/MS_y	SIM / DERIVED
T3475	0.508	SIM	MMX v352	/rows/23/cells/J-FWD/rows/0/MS_u	SIM / DERIVED
T3476	A4-80 withdrawn (240 (4))	SIM	MMX v352	/rows/23/cells/J-FWD/rows/0/note	SIM / DERIVED
T3477	PRELOAD	DERIVED	MMX v352	/rows/23/cells/J-FWD/rows/3/state	(class set at the call site: DERIVED)
T3478	A4-80 FE: tension on A_s (234 (1)(b))	SIM	MMX v352	/rows/23/cells/J-FWD/rows/3/row	SIM / DERIVED
T3479	1.269	SIM	MMX v352	/rows/23/cells/J-FWD/rows/3/MS_y	SIM / DERIVED
T3480	2.025	SIM	MMX v352	/rows/23/cells/J-FWD/rows/3/MS_u	SIM / DERIVED
T3481	A4-80 withdrawn (240 (4))	SIM	MMX v352	/rows/23/cells/J-FWD/rows/3/note	SIM / DERIVED
T3482	SOAK70	DERIVED	MMX v352	/rows/23/cells/J-FWD/rows/4/state	(class set at the call site: DERIVED)
T3483	A4-80 FE: tension on A_s (234 (1)(b))	SIM	MMX v352	/rows/23/cells/J-FWD/rows/4/row	SIM / DERIVED
T3484	0.113	SIM	MMX v352	/rows/23/cells/J-FWD/rows/4/MS_y	SIM / DERIVED
T3485	0.89	SIM	MMX v352	/rows/23/cells/J-FWD/rows/4/MS_u	SIM / DERIVED
T3486	A4-80 withdrawn (240 (4))	SIM	MMX v352	/rows/23/cells/J-FWD/rows/4/note	SIM / DERIVED
T3487	SOAK	DERIVED	MMX v352	/rows/23/cells/J-FWD/rows/5/state	(class set at the call site: DERIVED)
T3488	A4-80 FE: tension on A_s (234 (1)(b))	SIM	MMX v352	/rows/23/cells/J-FWD/rows/5/row	SIM / DERIVED
T3489	−0.089	SIM	MMX v352	/rows/23/cells/J-FWD/rows/5/MS_y	SIM / DERIVED
T3490	0.476	SIM	MMX v352	/rows/23/cells/J-FWD/rows/5/MS_u	SIM / DERIVED
T3491	A4-80 withdrawn (240 (4))	SIM	MMX v352	/rows/23/cells/J-FWD/rows/5/note	SIM / DERIVED
T3492	SOAK52	DERIVED	MMX v352	/rows/23/cells/J-FWD/rows/6/state	(class set at the call site: DERIVED)
T3493	A4-80 FE: tension on A_s (234 (1)(b))	SIM	MMX v352	/rows/23/cells/J-FWD/rows/6/row	SIM / DERIVED
T3494	−0.025	SIM	MMX v352	/rows/23/cells/J-FWD/rows/6/MS_y	SIM / DERIVED
T3495	0.644	SIM	MMX v352	/rows/23/cells/J-FWD/rows/6/MS_u	SIM / DERIVED
T3496	A4-80 withdrawn (240 (4))	SIM	MMX v352	/rows/23/cells/J-FWD/rows/6/note	SIM / DERIVED
T3497	FA1U	DERIVED	MMX v352	/rows/23/cells/J-FWD/rows/7/state	(class set at the call site: DERIVED)
T3498	js5b FA1U (FA1, SF [1.0, 1.0]): IN718 FE screw, FA1 (234 (1)(b)): ultimate interaction = ...	SIM	MMX v352	/rows/23/cells/J-FWD/rows/7/row	SIM / DERIVED
T3499	1.07	SIM	MMX v352	/rows/23/cells/J-FWD/rows/7/MS_y	SIM / DERIVED
T3500	1.287	SIM	MMX v352	/rows/23/cells/J-FWD/rows/7/MS_u	SIM / DERIVED
T3501	234 (1)(b)	DECIDED *Lead decision, per margin_matrix_v352.json v352*	MMX v352	/rows/23/cells/J-FWD/rows/7/criterion	tag in margin_matrix_v352.json v352:
T3502	FA1D	DERIVED	MMX v352	/rows/23/cells/J-FWD/rows/8/state	(class set at the call site: DERIVED)
T3503	js5b FA1D (FA1, SF [1.0, 1.0]): IN718 FE screw, FA1 (234 (1)(b)): ultimate interaction = ...	SIM	MMX v352	/rows/23/cells/J-FWD/rows/8/row	SIM / DERIVED
T3504	1.641	SIM	MMX v352	/rows/23/cells/J-FWD/rows/8/MS_y	SIM / DERIVED
T3505	1.724	SIM	MMX v352	/rows/23/cells/J-FWD/rows/8/MS_u	SIM / DERIVED
T3506	234 (1)(b)	DECIDED *Lead decision, per margin_matrix_v352.json v352*	MMX v352	/rows/23/cells/J-FWD/rows/8/criterion	tag in margin_matrix_v352.json v352:
T3507	LC2U	DERIVED	MMX v352	/rows/23/cells/J-FWD/rows/9/state	(class set at the call site: DERIVED)
T3508	js5b LC2U (FE-factored, SF [1.0, 1.0]): IN718 FE screw, FE-factored: combined vM ...	SIM	MMX v352	/rows/23/cells/J-FWD/rows/9/row	SIM / DERIVED
T3509	MISSING	SIM	MMX v352	/rows/23/cells/J-FWD/rows/9/MS_y	SIM / DERIVED
T3510	2.145	SIM	MMX v352	/rows/23/cells/J-FWD/rows/9/MS_u	SIM / DERIVED
T3511	234 (1)(b)	DECIDED *Lead decision, per margin_matrix_v352.json v352*	MMX v352	/rows/23/cells/J-FWD/rows/9/criterion	tag in margin_matrix_v352.json v352:
T3512	LC2Y	DERIVED	MMX v352	/rows/23/cells/J-FWD/rows/10/state	(class set at the call site: DERIVED)
T3513	js5b LC2Y (FE-factored, SF [1.0, 1.0]): IN718 FE screw, FE-factored: combined vM ...	SIM	MMX v352	/rows/23/cells/J-FWD/rows/10/row	SIM / DERIVED
T3514	1.919	SIM	MMX v352	/rows/23/cells/J-FWD/rows/10/MS_y	SIM / DERIVED
T3515	MISSING	SIM	MMX v352	/rows/23/cells/J-FWD/rows/10/MS_u	SIM / DERIVED
T3516	234 (1)(b)	DECIDED *Lead decision, per margin_matrix_v352.json v352*	MMX v352	/rows/23/cells/J-FWD/rows/10/criterion	tag in margin_matrix_v352.json v352:
T3517	PRELOAD	DERIVED	MMX v352	/rows/23/cells/J-FWD/rows/13/state	(class set at the call site: DERIVED)
T3518	IN718 tension on A_s (234 (1)(b))	CALC	MMX v352	/rows/23/cells/J-FWD/rows/13/row	CALC / DERIVED
T3519	2.91	CALC	MMX v352	/rows/23/cells/J-FWD/rows/13/MS_y	CALC / DERIVED
T3520	3.693	CALC	MMX v352	/rows/23/cells/J-FWD/rows/13/MS_u	CALC / DERIVED
T3521	3750	CALC	MMX v352	/rows/23/cells/J-FWD/rows/13/preload_N	CALC / DERIVED
T3522	N 3750.0 (js5), T 20.0 C	CALC	MMX v352	/rows/23/cells/J-FWD/rows/13/note	CALC / DERIVED
T3523	PRELOAD	DERIVED	MMX v352	/rows/23/cells/J-FWD/rows/14/state	(class set at the call site: DERIVED)
T3524	IN718 head on the HW-170 spring, A 39.19 mm2 (IN718 Fty / Ftu, no bearing credit)	CALC	MMX v352	/rows/23/cells/J-FWD/rows/14/row	CALC / DERIVED
T3525	9.806	CALC	MMX v352	/rows/23/cells/J-FWD/rows/14/MS_y	CALC / DERIVED
T3526	11.97	CALC	MMX v352	/rows/23/cells/J-FWD/rows/14/MS_u	CALC / DERIVED
T3527	3750	CALC	MMX v352	/rows/23/cells/J-FWD/rows/14/preload_N	CALC / DERIVED
T3528	N 3750.0 (js5), T 20.0 C; spring side TO-SOURCE (Mechanical)	CALC	MMX v352	/rows/23/cells/J-FWD/rows/14/note	CALC / DERIVED
T3529	PRELOAD	DERIVED	MMX v352	/rows/23/cells/J-FWD/rows/15/state	(class set at the call site: DERIVED)
T3530	IN718 tension on A_s (234 (1)(b))	CALC	MMX v352	/rows/23/cells/J-FWD/rows/15/row	CALC / DERIVED
T3531	2.58	CALC	MMX v352	/rows/23/cells/J-FWD/rows/15/MS_y	CALC / DERIVED
T3532	3.296	CALC	MMX v352	/rows/23/cells/J-FWD/rows/15/MS_u	CALC / DERIVED
T3533	4096	CALC	MMX v352	/rows/23/cells/J-FWD/rows/15/preload_N	CALC / DERIVED
T3534	N 4096.0 (js5), T 20.0 C; 4096 N column = information (J-F1 maximum capped at 3750 N ...	CALC	MMX v352	/rows/23/cells/J-FWD/rows/15/note	CALC / DERIVED
T3535	PRELOAD	DERIVED	MMX v352	/rows/23/cells/J-FWD/rows/16/state	(class set at the call site: DERIVED)
T3536	IN718 head on the HW-170 spring, A 39.19 mm2 (IN718 Fty / Ftu, no bearing credit)	CALC	MMX v352	/rows/23/cells/J-FWD/rows/16/row	CALC / DERIVED
T3537	8.893	CALC	MMX v352	/rows/23/cells/J-FWD/rows/16/MS_y	CALC / DERIVED
T3538	10.874	CALC	MMX v352	/rows/23/cells/J-FWD/rows/16/MS_u	CALC / DERIVED
T3539	4096	CALC	MMX v352	/rows/23/cells/J-FWD/rows/16/preload_N	CALC / DERIVED
T3540	N 4096.0 (js5), T 20.0 C; spring side TO-SOURCE (Mechanical); 4096 N column = ...	CALC	MMX v352	/rows/23/cells/J-FWD/rows/16/note	CALC / DERIVED
T3541	SOAK70	DERIVED	MMX v352	/rows/23/cells/J-FWD/rows/19/state	(class set at the call site: DERIVED)
T3542	IN718 soak fibre yield (secondary: relaxation / shakedown / LCF, lead 10:14Z)	CALC	MMX v352	/rows/23/cells/J-FWD/rows/19/row	CALC / DERIVED
T3543	−0.219	CALC	MMX v352	/rows/23/cells/J-FWD/rows/19/MS_y	CALC / DERIVED
T3544	MISSING	CALC	MMX v352	/rows/23/cells/J-FWD/rows/19/MS_u	CALC / DERIVED
T3545	3750	CALC	MMX v352	/rows/23/cells/J-FWD/rows/19/preload_N	CALC / DERIVED
T3546	(cD); shakedown 1.078, range 958.8 MPa	CALC	MMX v352	/rows/23/cells/J-FWD/rows/19/note	CALC / DERIVED
T3547	SOAK70	DERIVED	MMX v352	/rows/23/cells/J-FWD/rows/20/state	(class set at the call site: DERIVED)
T3548	IN718 tension on A_s (234 (1)(b))	CALC	MMX v352	/rows/23/cells/J-FWD/rows/20/row	CALC / DERIVED
T3549	0.648	CALC	MMX v352	/rows/23/cells/J-FWD/rows/20/MS_y	CALC / DERIVED
T3550	0.952	CALC	MMX v352	/rows/23/cells/J-FWD/rows/20/MS_u	CALC / DERIVED
T3551	4096	CALC	MMX v352	/rows/23/cells/J-FWD/rows/20/preload_N	CALC / DERIVED
T3552	N 8573.4 ((b2)), T 158.6 C; 4096 N column = information (J-F1 maximum capped at 3750 N ...	CALC	MMX v352	/rows/23/cells/J-FWD/rows/20/note	CALC / DERIVED
T3553	SOAK70	DERIVED	MMX v352	/rows/23/cells/J-FWD/rows/21/state	(class set at the call site: DERIVED)
T3554	IN718 head on the HW-170 spring, A 39.19 mm2 (IN718 Fty / Ftu, no bearing credit)	CALC	MMX v352	/rows/23/cells/J-FWD/rows/21/row	CALC / DERIVED
T3555	3.553	CALC	MMX v352	/rows/23/cells/J-FWD/rows/21/MS_y	CALC / DERIVED
T3556	4.395	CALC	MMX v352	/rows/23/cells/J-FWD/rows/21/MS_u	CALC / DERIVED
T3557	4096	CALC	MMX v352	/rows/23/cells/J-FWD/rows/21/preload_N	CALC / DERIVED
T3558	N 8573.4 ((b2)), T 158.6 C; spring side TO-SOURCE (Mechanical); 4096 N column = ...	CALC	MMX v352	/rows/23/cells/J-FWD/rows/21/note	CALC / DERIVED

ID	Value	Class	Source	Key	Source tag / formula
T3559	SOAK70	DERIVED	MMX v352	/rows/23/cells/J-FWD/rows/22/state	(class set at the call site: DERIVED)
T3560	IN718 soak fibre yield (secondary: relaxation / shakedown / LCF, lead 10:14Z)	CALC	MMX v352	/rows/23/cells/J-FWD/rows/22/row	CALC / DERIVED
T3561	−0.267	CALC	MMX v352	/rows/23/cells/J-FWD/rows/22/MS_y	CALC / DERIVED
T3562	MISSING	CALC	MMX v352	/rows/23/cells/J-FWD/rows/22/MS_u	CALC / DERIVED
T3563	4096	CALC	MMX v352	/rows/23/cells/J-FWD/rows/22/preload_N	CALC / DERIVED
T3564	(cD); shakedown 0.964, range 1014.4 MPa; 4096 N column = information (J-F1 maximum ...	CALC	MMX v352	/rows/23/cells/J-FWD/rows/22/note	CALC / DERIVED
T3565	SOAK	DERIVED	MMX v352	/rows/23/cells/J-FWD/rows/25/state	(class set at the call site: DERIVED)
T3566	IN718 soak fibre yield (secondary: relaxation / shakedown / LCF, lead 10:14Z)	CALC	MMX v352	/rows/23/cells/J-FWD/rows/25/row	CALC / DERIVED
T3567	−0.343	CALC	MMX v352	/rows/23/cells/J-FWD/rows/25/MS_y	CALC / DERIVED
T3568	MISSING	CALC	MMX v352	/rows/23/cells/J-FWD/rows/25/MS_u	CALC / DERIVED
T3569	3750	CALC	MMX v352	/rows/23/cells/J-FWD/rows/25/preload_N	CALC / DERIVED
T3570	(b1); shakedown 0.657, range 1212.5 MPa	CALC	MMX v352	/rows/23/cells/J-FWD/rows/25/note	CALC / DERIVED
T3571	SOAK	DERIVED	MMX v352	/rows/23/cells/J-FWD/rows/26/state	(class set at the call site: DERIVED)
T3572	IN718 tension on A_s (234 (1)(b))	CALC	MMX v352	/rows/23/cells/J-FWD/rows/26/row	CALC / DERIVED
T3573	0.685	CALC	MMX v352	/rows/23/cells/J-FWD/rows/26/MS_y	CALC / DERIVED
T3574	1.002	CALC	MMX v352	/rows/23/cells/J-FWD/rows/26/MS_u	CALC / DERIVED
T3575	4096	CALC	MMX v352	/rows/23/cells/J-FWD/rows/26/preload_N	CALC / DERIVED
T3576	N 8456.3 (js5), T 127.0 C; 4096 N column = information (J-F1 maximum capped at 3750 N ...	CALC	MMX v352	/rows/23/cells/J-FWD/rows/26/note	CALC / DERIVED
T3577	SOAK	DERIVED	MMX v352	/rows/23/cells/J-FWD/rows/27/state	(class set at the call site: DERIVED)
T3578	IN718 head on the HW-170 spring, A 39.19 mm2 (IN718 Fty / Ftu, no bearing credit)	CALC	MMX v352	/rows/23/cells/J-FWD/rows/27/row	CALC / DERIVED
T3579	3.657	CALC	MMX v352	/rows/23/cells/J-FWD/rows/27/MS_y	CALC / DERIVED
T3580	4.534	CALC	MMX v352	/rows/23/cells/J-FWD/rows/27/MS_u	CALC / DERIVED
T3581	4096	CALC	MMX v352	/rows/23/cells/J-FWD/rows/27/preload_N	CALC / DERIVED
T3582	N 8456.3 (js5), T 127.0 C; spring side TO-SOURCE (Mechanical); 4096 N column = ...	CALC	MMX v352	/rows/23/cells/J-FWD/rows/27/note	CALC / DERIVED
T3583	SOAK	DERIVED	MMX v352	/rows/23/cells/J-FWD/rows/28/state	(class set at the call site: DERIVED)
T3584	IN718 soak fibre yield (secondary: relaxation / shakedown / LCF, lead 10:14Z)	CALC	MMX v352	/rows/23/cells/J-FWD/rows/28/row	CALC / DERIVED
T3585	−0.398	CALC	MMX v352	/rows/23/cells/J-FWD/rows/28/MS_y	CALC / DERIVED
T3586	MISSING	CALC	MMX v352	/rows/23/cells/J-FWD/rows/28/MS_u	CALC / DERIVED
T3587	4096	CALC	MMX v352	/rows/23/cells/J-FWD/rows/28/preload_N	CALC / DERIVED
T3588	(b1); shakedown 0.518, range 1323.9 MPa; 4096 N column = information (J-F1 maximum ...	CALC	MMX v352	/rows/23/cells/J-FWD/rows/28/note	CALC / DERIVED
T3589	SOAK52	DERIVED	MMX v352	/rows/23/cells/J-FWD/rows/31/state	(class set at the call site: DERIVED)
T3590	IN718 soak fibre yield (secondary: relaxation / shakedown / LCF, lead 10:14Z)	CALC	MMX v352	/rows/23/cells/J-FWD/rows/31/row	CALC / DERIVED
T3591	−0.201	CALC	MMX v352	/rows/23/cells/J-FWD/rows/31/MS_y	CALC / DERIVED
T3592	MISSING	CALC	MMX v352	/rows/23/cells/J-FWD/rows/31/MS_u	CALC / DERIVED
T3593	3750	CALC	MMX v352	/rows/23/cells/J-FWD/rows/31/preload_N	CALC / DERIVED
T3594	(b2) rep; shakedown 1.14, range 932.1 MPa	CALC	MMX v352	/rows/23/cells/J-FWD/rows/31/note	CALC / DERIVED
T3595	SOAK52	DERIVED	MMX v352	/rows/23/cells/J-FWD/rows/32/state	(class set at the call site: DERIVED)
T3596	IN718 tension on A_s (234 (1)(b))	CALC	MMX v352	/rows/23/cells/J-FWD/rows/32/row	CALC / DERIVED
T3597	0.586	CALC	MMX v352	/rows/23/cells/J-FWD/rows/32/MS_y	CALC / DERIVED
T3598	0.88	CALC	MMX v352	/rows/23/cells/J-FWD/rows/32/MS_u	CALC / DERIVED
T3599	4096	CALC	MMX v352	/rows/23/cells/J-FWD/rows/32/preload_N	CALC / DERIVED
T3600	N 8918.4 ((b2) rep), T 154.1 C; 4096 N column = information (J-F1 maximum capped at 3750 ...	CALC	MMX v352	/rows/23/cells/J-FWD/rows/32/note	CALC / DERIVED
T3601	SOAK52	DERIVED	MMX v352	/rows/23/cells/J-FWD/rows/33/state	(class set at the call site: DERIVED)
T3602	IN718 head on the HW-170 spring, A 39.19 mm2 (IN718 Fty / Ftu, no bearing credit)	CALC	MMX v352	/rows/23/cells/J-FWD/rows/33/row	CALC / DERIVED
T3603	3.383	CALC	MMX v352	/rows/23/cells/J-FWD/rows/33/MS_y	CALC / DERIVED
T3604	4.195	CALC	MMX v352	/rows/23/cells/J-FWD/rows/33/MS_u	CALC / DERIVED
T3605	4096	CALC	MMX v352	/rows/23/cells/J-FWD/rows/33/preload_N	CALC / DERIVED
T3606	N 8918.4 ((b2) rep), T 154.1 C; spring side TO-SOURCE (Mechanical); 4096 N column = ...	CALC	MMX v352	/rows/23/cells/J-FWD/rows/33/note	CALC / DERIVED
T3607	SOAK52	DERIVED	MMX v352	/rows/23/cells/J-FWD/rows/34/state	(class set at the call site: DERIVED)
T3608	IN718 soak fibre yield (secondary: relaxation / shakedown / LCF, lead 10:14Z)	CALC	MMX v352	/rows/23/cells/J-FWD/rows/34/row	CALC / DERIVED
T3609	−0.247	CALC	MMX v352	/rows/23/cells/J-FWD/rows/34/MS_y	CALC / DERIVED
T3610	MISSING	CALC	MMX v352	/rows/23/cells/J-FWD/rows/34/MS_u	CALC / DERIVED
T3611	4096	CALC	MMX v352	/rows/23/cells/J-FWD/rows/34/preload_N	CALC / DERIVED
T3612	(b2) rep; shakedown 1.037, range 979.1 MPa; 4096 N column = information (J-F1 maximum ...	CALC	MMX v352	/rows/23/cells/J-FWD/rows/34/note	CALC / DERIVED
T3613	FA1S	DERIVED	MMX v352	/rows/23/cells/J-FWD/rows/35/state	(class set at the call site: DERIVED)
T3614	IN718 tension on A_s at FA1S (information beside Eq. 22)	CALC	MMX v352	/rows/23/cells/J-FWD/rows/35/row	CALC / DERIVED
T3615	3.41	CALC	MMX v352	/rows/23/cells/J-FWD/rows/35/MS_y	CALC / DERIVED
T3616	4.281	CALC	MMX v352	/rows/23/cells/J-FWD/rows/35/MS_u	CALC / DERIVED
T3617	3750	CALC	MMX v352	/rows/23/cells/J-FWD/rows/35/preload_N	CALC / DERIVED
T3618	N 3302.3 (js5), T 45.6 C	CALC	MMX v352	/rows/23/cells/J-FWD/rows/35/note	CALC / DERIVED
T3619	FA1S	DERIVED	MMX v352	/rows/23/cells/J-FWD/rows/37/state	(class set at the call site: DERIVED)
T3620	IN718 tension on A_s at FA1S (information beside Eq. 22)	CALC	MMX v352	/rows/23/cells/J-FWD/rows/37/row	CALC / DERIVED
T3621	2.992	CALC	MMX v352	/rows/23/cells/J-FWD/rows/37/MS_y	CALC / DERIVED
T3622	3.78	CALC	MMX v352	/rows/23/cells/J-FWD/rows/37/MS_u	CALC / DERIVED
T3623	4096	CALC	MMX v352	/rows/23/cells/J-FWD/rows/37/preload_N	CALC / DERIVED
T3624	N 3648.3 (js5), T 45.6 C; 4096 N column = information (J-F1 maximum capped at 3750 N ...	CALC	MMX v352	/rows/23/cells/J-FWD/rows/37/note	CALC / DERIVED
T3625	FA1S	DERIVED	MMX v352	/rows/23/cells/J-FWD/rows/38/state	(class set at the call site: DERIVED)
T3626	IN718 head on the HW-170 spring, A 39.19 mm2 (IN718 Fty / Ftu, no bearing credit)	CALC	MMX v352	/rows/23/cells/J-FWD/rows/38/row	CALC / DERIVED
T3627	10.032	CALC	MMX v352	/rows/23/cells/J-FWD/rows/38/MS_y	CALC / DERIVED
T3628	12.211	CALC	MMX v352	/rows/23/cells/J-FWD/rows/38/MS_u	CALC / DERIVED
T3629	4096	CALC	MMX v352	/rows/23/cells/J-FWD/rows/38/preload_N	CALC / DERIVED
T3630	N 3648.3 (js5), T 45.6 C; spring side TO-SOURCE (Mechanical); 4096 N column = ...	CALC	MMX v352	/rows/23/cells/J-FWD/rows/38/note	CALC / DERIVED
T3631	FA1S	DERIVED	MMX v352	/rows/23/cells/J-FWD/rows/39/state	(class set at the call site: DERIVED)
T3632	IN718 5020A Eq. 22 at arm +0.4 mm (sensitivity), preload omitted	CALC	MMX v352	/rows/23/cells/J-FWD/rows/39/row	CALC
T3633	MISSING	CALC	MMX v352	/rows/23/cells/J-FWD/rows/39/MS_y	CALC
T3634	0.165	CALC	MMX v352	/rows/23/cells/J-FWD/rows/39/MS_u	CALC
T3635	NASA-STD-5020A Eq. 22	DECIDED *Lead decision, per margin_matrix_v352.json v352*	MMX v352	/rows/23/cells/J-FWD/rows/39/criterion	tag in margin_matrix_v352.json v352:
T3636	preload included -0.035 (information)	CALC	MMX v352	/rows/23/cells/J-FWD/rows/39/note	CALC
T3637	lead 22:54:03Z: governing = the larger of the as-found value and the moment-free ...	DERIVED	MOUNT	/datasheet_v352_governing/rule	(class set at the call site: DERIVED)

ID	Value	Class	Source	Key	Source tag / formula
T3638	Generated 2026-10-20T23:06:59Z by scripts/icd_export_v352.py from the f1.dat reaction ...	SIM	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 3 (starts "Generated")	FEA3D mounting data (SIM unless tagged)
T3639	OP-MOMENT ARTEFACT AND HOLE LOADS (22:24:09Z, 22:54:03Z, 23:12:52Z): FOUND (FEA3D): the ...	DECIDED *Lead decision, line 249 (6)*	CL 266 lines	L249(6)	lead (decision) 2026-10-21 (line sha 21373a357d2dd391)
T3640	Cell = max shear / max |axial| / max pull-out, N; ! = an estimate governs that value by ...	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 16 (starts "Cell = max shear")	FEA3D mounting data legend
T3641	76! / 68! / 20!	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 20 row "AF_GM09_0 (59.4, 24.6, 27.0)" col 1	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3642	150! / 136! / 33!	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 20 row "AF_GM09_0 (59.4, 24.6, 27.0)" col 2	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3643	348! / 320! / 64!	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 20 row "AF_GM09_0 (59.4, 24.6, 27.0)" col 3	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3644	448! / 412! / 82!	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 20 row "AF_GM09_0 (59.4, 24.6, 27.0)" col 4	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3645	544! / 500! / 99!	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 20 row "AF_GM09_0 (59.4, 24.6, 27.0)" col 5	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3646	559! / 514! / 102!	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 20 row "AF_GM09_0 (59.4, 24.6, 27.0)" col 6	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3647	657! / 604! / 120!	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 20 row "AF_GM09_0 (59.4, 24.6, 27.0)" col 7	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3648	5 / 3 / 0 *	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 21 row "AF_GM09_1 (24.6, 59.4, 27.0)" col 1	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3649	5 / 5 / 0	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 21 row "AF_GM09_1 (24.6, 59.4, 27.0)" col 2	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3650	10! / 9! / 0	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 21 row "AF_GM09_1 (24.6, 59.4, 27.0)" col 3	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3651	13! / 12! / 0	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 21 row "AF_GM09_1 (24.6, 59.4, 27.0)" col 4	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3652	16! / 14! / 0	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 21 row "AF_GM09_1 (24.6, 59.4, 27.0)" col 5	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3653	16! / 14! / 0	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 21 row "AF_GM09_1 (24.6, 59.4, 27.0)" col 6	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3654	19! / 17! / 0	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 21 row "AF_GM09_1 (24.6, 59.4, 27.0)" col 7	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3655	3 / 3 / 0	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 22 row "AF_GM09_2 (-24.6, 59.4, 27.0)" col 1	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3656	5 / 5 / 0	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 22 row "AF_GM09_2 (-24.6, 59.4, 27.0)" col 2	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3657	10! / 9! / 0	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 22 row "AF_GM09_2 (-24.6, 59.4, 27.0)" col 3	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3658	14! / 12! / 0	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 22 row "AF_GM09_2 (-24.6, 59.4, 27.0)" col 4	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3659	16! / 14! / 0	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 22 row "AF_GM09_2 (-24.6, 59.4, 27.0)" col 5	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3660	17! / 15! / 0	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 22 row "AF_GM09_2 (-24.6, 59.4, 27.0)" col 6	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3661	20! / 18! / 0	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 22 row "AF_GM09_2 (-24.6, 59.4, 27.0)" col 7	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3662	71! / 65! / 14!	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 23 row "AF_GM09_3 (-59.4, 24.6, 27.0)" col 1	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3663	143! / 131! / 27!	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 23 row "AF_GM09_3 (-59.4, 24.6, 27.0)" col 2	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3664	336! / 310! / 61!	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 23 row "AF_GM09_3 (-59.4, 24.6, 27.0)" col 3	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3665	434! / 400! / 78!	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 23 row "AF_GM09_3 (-59.4, 24.6, 27.0)" col 4	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3666	527! / 486! / 95!	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 23 row "AF_GM09_3 (-59.4, 24.6, 27.0)" col 5	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3667	542! / 499! / 98!	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 23 row "AF_GM09_3 (-59.4, 24.6, 27.0)" col 6	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3668	638! / 588! / 115!	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 23 row "AF_GM09_3 (-59.4, 24.6, 27.0)" col 7	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3669	76! / 69! / 15!	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 24 row "AF_GM09_4 (-59.4, -24.6, 27.0)" col 1	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3670	149! / 136! / 28!	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 24 row "AF_GM09_4 (-59.4, -24.6, 27.0)" col 2	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3671	346! / 320! / 62!	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 24 row "AF_GM09_4 (-59.4, -24.6, 27.0)" col 3	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3672	446! / 411! / 79!	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 24 row "AF_GM09_4 (-59.4, -24.6, 27.0)" col 4	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3673	541! / 499! / 96!	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 24 row "AF_GM09_4 (-59.4, -24.6, 27.0)" col 5	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3674	556! / 513! / 99!	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 24 row "AF_GM09_4 (-59.4, -24.6, 27.0)" col 6	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3675	654! / 603! / 116!	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 24 row "AF_GM09_4 (-59.4, -24.6, 27.0)" col 7	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3676	3 / 3 / 1 *	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 25 row "AF_GM09_5 (-24.6, -59.4, 27.0)" col 1	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)

ID	Value	Class	Source	Key	Source tag / formula
T3677	5 / 5 / 0	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 25 row "AF_GM09_5 (-24.6, -59.4, 27.0)" col 2	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3678	10! / 9! / 0	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 25 row "AF_GM09_5 (-24.6, -59.4, 27.0)" col 3	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3679	13! / 12! / 0	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 25 row "AF_GM09_5 (-24.6, -59.4, 27.0)" col 4	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3680	16! / 14! / 0	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 25 row "AF_GM09_5 (-24.6, -59.4, 27.0)" col 5	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3681	17! / 15! / 0	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 25 row "AF_GM09_5 (-24.6, -59.4, 27.0)" col 6	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3682	20! / 18! / 0	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 25 row "AF_GM09_5 (-24.6, -59.4, 27.0)" col 7	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3683	6 / 6 / 0 *	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 26 row "AF_GM09_6 (24.6, -59.4, 27.0)" col 1	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3684	8 / 8 / 0	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 26 row "AF_GM09_6 (24.6, -59.4, 27.0)" col 2	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3685	10! / 9! / 0	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 26 row "AF_GM09_6 (24.6, -59.4, 27.0)" col 3	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3686	13! / 11! / 0	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 26 row "AF_GM09_6 (24.6, -59.4, 27.0)" col 4	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3687	15! / 14! / 0	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 26 row "AF_GM09_6 (24.6, -59.4, 27.0)" col 5	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3688	16! / 14! / 0	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 26 row "AF_GM09_6 (24.6, -59.4, 27.0)" col 6	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3689	19! / 17! / 0	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 26 row "AF_GM09_6 (24.6, -59.4, 27.0)" col 7	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3690	72! / 66! / 18!	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 27 row "AF_GM09_7 (59.4, -24.6, 27.0)" col 1	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3691	145! / 133! / 32!	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 27 row "AF_GM09_7 (59.4, -24.6, 27.0)" col 2	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3692	343! / 316! / 62!	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 27 row "AF_GM09_7 (59.4, -24.6, 27.0)" col 3	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3693	442! / 408! / 80!	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 27 row "AF_GM09_7 (59.4, -24.6, 27.0)" col 4	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3694	538! / 495! / 97!	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 27 row "AF_GM09_7 (59.4, -24.6, 27.0)" col 5	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3695	553! / 509! / 100!	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 27 row "AF_GM09_7 (59.4, -24.6, 27.0)" col 6	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3696	651! / 600! / 118!	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 27 row "AF_GM09_7 (59.4, -24.6, 27.0)" col 7	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3697	47 / 47 / 0 *	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 28 row "AF_GM10_0 (53.1, 22.0, -94.8)" col 1	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3698	76! / 72! / 0	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 28 row "AF_GM10_0 (53.1, 22.0, -94.8)" col 2	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3699	183! / 174! / 0	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 28 row "AF_GM10_0 (53.1, 22.0, -94.8)" col 3	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3700	236! / 224! / 0	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 28 row "AF_GM10_0 (53.1, 22.0, -94.8)" col 4	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3701	286! / 272! / 0	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 28 row "AF_GM10_0 (53.1, 22.0, -94.8)" col 5	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3702	294! / 279! / 0	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 28 row "AF_GM10_0 (53.1, 22.0, -94.8)" col 6	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3703	347! / 329! / 0	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 28 row "AF_GM10_0 (53.1, 22.0, -94.8)" col 7	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3704	19 / 19 / 4	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 29 row "AF_GM10_1 (22.0, 53.1, -94.8)" col 1	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3705	33 / 32 / 6	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 29 row "AF_GM10_1 (22.0, 53.1, -94.8)" col 2	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3706	64 / 63 / 12	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 29 row "AF_GM10_1 (22.0, 53.1, -94.8)" col 3	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3707	81 / 80 / 16	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 29 row "AF_GM10_1 (22.0, 53.1, -94.8)" col 4	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3708	99 / 97 / 19	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 29 row "AF_GM10_1 (22.0, 53.1, -94.8)" col 5	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3709	101 / 100 / 19	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 29 row "AF_GM10_1 (22.0, 53.1, -94.8)" col 6	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3710	68 / 68 / 13	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 29 row "AF_GM10_1 (22.0, 53.1, -94.8)" col 7	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3711	39 / 39 / 1	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 30 row "AF_GM10_2 (-22.0, 53.1, -94.8)" col 1	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3712	76 / 75 / 0	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 30 row "AF_GM10_2 (-22.0, 53.1, -94.8)" col 2	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3713	176 / 175 / 0	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 30 row "AF_GM10_2 (-22.0, 53.1, -94.8)" col 3	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3714	225 / 224 / 0	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 30 row "AF_GM10_2 (-22.0, 53.1, -94.8)" col 4	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3715	273 / 272 / 0	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 30 row "AF_GM10_2 (-22.0, 53.1, -94.8)" col 5	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)

ID	Value	Class	Source	Key	Source tag / formula
T3716	280 / 279 / 0	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 30 row "AF_GM10_2 (-22.0, 53.1, -94.8)" col 6	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3717	170 / 169 / 0	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 30 row "AF_GM10_2 (-22.0, 53.1, -94.8)" col 7	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3718	48 / 48 / 0 *	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 31 row "AF_GM10_3 (-53.1, 22.0, -94.8)" col 1	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3719	76! / 73! / 0	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 31 row "AF_GM10_3 (-53.1, 22.0, -94.8)" col 2	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3720	183! / 176! / 0	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 31 row "AF_GM10_3 (-53.1, 22.0, -94.8)" col 3	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3721	236! / 227! / 0	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 31 row "AF_GM10_3 (-53.1, 22.0, -94.8)" col 4	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3722	286! / 276! / 0	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 31 row "AF_GM10_3 (-53.1, 22.0, -94.8)" col 5	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3723	294! / 283! / 0	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 31 row "AF_GM10_3 (-53.1, 22.0, -94.8)" col 6	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3724	346! / 333! / 0	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 31 row "AF_GM10_3 (-53.1, 22.0, -94.8)" col 7	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3725	45 / 45 / 0 *	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 32 row "AF_GM10_4 (-53.1, -22.0, -94.8)" col 1	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3726	76! / 73! / 0	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 32 row "AF_GM10_4 (-53.1, -22.0, -94.8)" col 2	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3727	184! / 177! / 0	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 32 row "AF_GM10_4 (-53.1, -22.0, -94.8)" col 3	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3728	237! / 228! / 0	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 32 row "AF_GM10_4 (-53.1, -22.0, -94.8)" col 4	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3729	288! / 277! / 0	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 32 row "AF_GM10_4 (-53.1, -22.0, -94.8)" col 5	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3730	296! / 285! / 0	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 32 row "AF_GM10_4 (-53.1, -22.0, -94.8)" col 6	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3731	349! / 336! / 0	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 32 row "AF_GM10_4 (-53.1, -22.0, -94.8)" col 7	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3732	20 / 20 / 1	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 33 row "AF_GM10_5 (-22.0, -53.1, -94.8)" col 1	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3733	23 / 23 / 0	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 33 row "AF_GM10_5 (-22.0, -53.1, -94.8)" col 2	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3734	21 / 21 / 0	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 33 row "AF_GM10_5 (-22.0, -53.1, -94.8)" col 3	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3735	25 / 25 / 0	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 33 row "AF_GM10_5 (-22.0, -53.1, -94.8)" col 4	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3736	30 / 30 / 0	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 33 row "AF_GM10_5 (-22.0, -53.1, -94.8)" col 5	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3737	30 / 30 / 0	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 33 row "AF_GM10_5 (-22.0, -53.1, -94.8)" col 6	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3738	27 / 27 / 0	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 33 row "AF_GM10_5 (-22.0, -53.1, -94.8)" col 7	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3739	3 / 3 / 1 *	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 34 row "AF_GM10_6 (22.0, -53.1, -94.8)" col 1	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3740	4! / 4! / 1	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 34 row "AF_GM10_6 (22.0, -53.1, -94.8)" col 2	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3741	6! / 4! / 1!	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 34 row "AF_GM10_6 (22.0, -53.1, -94.8)" col 3	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3742	8! / 6! / 1!	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 34 row "AF_GM10_6 (22.0, -53.1, -94.8)" col 4	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3743	10! / 8! / 2!	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 34 row "AF_GM10_6 (22.0, -53.1, -94.8)" col 5	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3744	10! / 8! / 2!	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 34 row "AF_GM10_6 (22.0, -53.1, -94.8)" col 6	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3745	12! / 10! / 2!	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 34 row "AF_GM10_6 (22.0, -53.1, -94.8)" col 7	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3746	47 / 47 / 0 *	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 35 row "AF_GM10_7 (53.1, -22.0, -94.8)" col 1	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3747	75! / 71! / 0	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 35 row "AF_GM10_7 (53.1, -22.0, -94.8)" col 2	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3748	180! / 171! / 0	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 35 row "AF_GM10_7 (53.1, -22.0, -94.8)" col 3	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3749	232! / 220! / 0	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 35 row "AF_GM10_7 (53.1, -22.0, -94.8)" col 4	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3750	282! / 267! / 0	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 35 row "AF_GM10_7 (53.1, -22.0, -94.8)" col 5	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3751	290! / 274! / 0	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 35 row "AF_GM10_7 (53.1, -22.0, -94.8)" col 6	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3752	341! / 323! / 0	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 35 row "AF_GM10_7 (53.1, -22.0, -94.8)" col 7	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3753	306! / 303 / 295	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 20 row "AF_GM09_0 (59.4, 24.6, 27.0)" col 8	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3754	333 / 298 / 86	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 20 row "AF_GM09_0 (59.4, 24.6, 27.0)" col 9	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)

ID	Value	Class	Source	Key	Source tag / formula
T3755	113 / 91 / 88	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 20 row "AF_GM09_0 (59.4, 24.6, 27.0)" col 10	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3756	186! / 157! / 120!	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 20 row "AF_GM09_0 (59.4, 24.6, 27.0)" col 11	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3757	79 / 69 / 44	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 20 row "AF_GM09_0 (59.4, 24.6, 27.0)" col 12	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3758	197! / 175! / 75!	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 20 row "AF_GM09_0 (59.4, 24.6, 27.0)" col 13	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3759	102 / 16 / 20! *	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 21 row "AF_GM09_1 (24.6, 59.4, 27.0)" col 8	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3760	10 / 9 / 5	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 21 row "AF_GM09_1 (24.6, 59.4, 27.0)" col 9	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3761	6 / 4 / 6	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 21 row "AF_GM09_1 (24.6, 59.4, 27.0)" col 10	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3762	9 / 6! / 7	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 21 row "AF_GM09_1 (24.6, 59.4, 27.0)" col 11	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3763	3 / 2 / 2	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 21 row "AF_GM09_1 (24.6, 59.4, 27.0)" col 12	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3764	7! / 6! / 4	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 21 row "AF_GM09_1 (24.6, 59.4, 27.0)" col 13	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3765	72 / 7 / 5 *	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 22 row "AF_GM09_2 (-24.6, 59.4, 27.0)" col 8	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3766	10 / 9 / 1	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 22 row "AF_GM09_2 (-24.6, 59.4, 27.0)" col 9	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3767	2 / 2 / 2	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 22 row "AF_GM09_2 (-24.6, 59.4, 27.0)" col 10	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3768	5! / 4! / 2	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 22 row "AF_GM09_2 (-24.6, 59.4, 27.0)" col 11	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3769	1 / 1 / 1	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 22 row "AF_GM09_2 (-24.6, 59.4, 27.0)" col 12	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3770	5! / 5! / 0	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 22 row "AF_GM09_2 (-24.6, 59.4, 27.0)" col 13	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3771	323! / 269 / 94!	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 23 row "AF_GM09_3 (-59.4, 24.6, 27.0)" col 8	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3772	338 / 301 / 74	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 23 row "AF_GM09_3 (-59.4, 24.6, 27.0)" col 9	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3773	118 / 95 / 31	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 23 row "AF_GM09_3 (-59.4, 24.6, 27.0)" col 10	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3774	176! / 149! / 48!	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 23 row "AF_GM09_3 (-59.4, 24.6, 27.0)" col 11	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3775	78 / 68 / 20	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 23 row "AF_GM09_3 (-59.4, 24.6, 27.0)" col 12	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3776	191! / 170! / 42!	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 23 row "AF_GM09_3 (-59.4, 24.6, 27.0)" col 13	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3777	332! / 276! / 97!	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 24 row "AF_GM09_4 (-59.4, -24.6, 27.0)" col 8	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3778	349 / 311 / 76	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 24 row "AF_GM09_4 (-59.4, -24.6, 27.0)" col 9	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3779	121 / 97 / 32	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 24 row "AF_GM09_4 (-59.4, -24.6, 27.0)" col 10	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3780	181! / 153! / 48!	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 24 row "AF_GM09_4 (-59.4, -24.6, 27.0)" col 11	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3781	79 / 69 / 20	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 24 row "AF_GM09_4 (-59.4, -24.6, 27.0)" col 12	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3782	196! / 175! / 43!	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 24 row "AF_GM09_4 (-59.4, -24.6, 27.0)" col 13	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3783	36 / 9 / 4 *	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 25 row "AF_GM09_5 (-24.6, -59.4, 27.0)" col 8	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3784	10 / 9 / 1	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 25 row "AF_GM09_5 (-24.6, -59.4, 27.0)" col 9	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3785	4 / 4 / 2	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 25 row "AF_GM09_5 (-24.6, -59.4, 27.0)" col 10	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3786	6! / 6! / 2	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 25 row "AF_GM09_5 (-24.6, -59.4, 27.0)" col 11	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3787	2 / 2 / 1	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 25 row "AF_GM09_5 (-24.6, -59.4, 27.0)" col 12	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3788	5! / 5! / 0	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 25 row "AF_GM09_5 (-24.6, -59.4, 27.0)" col 13	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3789	140 / 20 / 22! *	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 26 row "AF_GM09_6 (24.6, -59.4, 27.0)" col 8	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3790	14 / 12 / 6	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 26 row "AF_GM09_6 (24.6, -59.4, 27.0)" col 9	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3791	14 / 8 / 8	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 26 row "AF_GM09_6 (24.6, -59.4, 27.0)" col 10	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3792	17 / 11 / 10	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 26 row "AF_GM09_6 (24.6, -59.4, 27.0)" col 11	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3793	10 / 9 / 4	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 26 row "AF_GM09_6 (24.6, -59.4, 27.0)" col 12	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)

ID	Value	Class	Source	Key	Source tag / formula
T3794	10! / 9! / 4	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 26 row "AF_GM09_6 (24.6, -59.4, 27.0)" col 13	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3795	304! / 302 / 296	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 27 row "AF_GM09_7 (59.4, -24.6, 27.0)" col 8	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3796	329 / 295 / 85	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 27 row "AF_GM09_7 (59.4, -24.6, 27.0)" col 9	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3797	113 / 92 / 89	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 27 row "AF_GM09_7 (59.4, -24.6, 27.0)" col 10	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3798	185! / 157! / 121	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 27 row "AF_GM09_7 (59.4, -24.6, 27.0)" col 11	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3799	80 / 70 / 44	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 27 row "AF_GM09_7 (59.4, -24.6, 27.0)" col 12	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3800	195! / 174! / 75!	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 27 row "AF_GM09_7 (59.4, -24.6, 27.0)" col 13	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3801	130! / 116! / 162!	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 28 row "AF_GM10_0 (53.1, 22.0, -94.8)" col 8	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3802	161 / 148 / 23	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 28 row "AF_GM10_0 (53.1, 22.0, -94.8)" col 9	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3803	50 / 48 / 27	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 28 row "AF_GM10_0 (53.1, 22.0, -94.8)" col 10	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3804	87! / 83! / 26	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 28 row "AF_GM10_0 (53.1, 22.0, -94.8)" col 11	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3805	41 / 40 / 15	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 28 row "AF_GM10_0 (53.1, 22.0, -94.8)" col 12	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3806	98! / 94! / 2	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 28 row "AF_GM10_0 (53.1, 22.0, -94.8)" col 13	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3807	329! / 311 / 164!	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 29 row "AF_GM10_1 (22.0, 53.1, -94.8)" col 8	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3808	284 / 274 / 83	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 29 row "AF_GM10_1 (22.0, 53.1, -94.8)" col 9	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3809	56 / 52 / 17	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 29 row "AF_GM10_1 (22.0, 53.1, -94.8)" col 10	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3810	109 / 104 / 34	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 29 row "AF_GM10_1 (22.0, 53.1, -94.8)" col 11	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3811	10 / 8 / 4	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 29 row "AF_GM10_1 (22.0, 53.1, -94.8)" col 12	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3812	100 / 98 / 31	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 29 row "AF_GM10_1 (22.0, 53.1, -94.8)" col 13	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3813	890 / 889 / 214	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 30 row "AF_GM10_2 (-22.0, 53.1, -94.8)" col 8	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3814	808 / 806 / 98	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 30 row "AF_GM10_2 (-22.0, 53.1, -94.8)" col 9	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3815	169 / 169 / 28	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 30 row "AF_GM10_2 (-22.0, 53.1, -94.8)" col 10	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3816	306 / 304 / 28!	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 30 row "AF_GM10_2 (-22.0, 53.1, -94.8)" col 11	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3817	55 / 55 / 10	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 30 row "AF_GM10_2 (-22.0, 53.1, -94.8)" col 12	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3818	288 / 287 / 6	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 30 row "AF_GM10_2 (-22.0, 53.1, -94.8)" col 13	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3819	195 / 160 / 217	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 31 row "AF_GM10_3 (-53.1, 22.0, -94.8)" col 8	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3820	162 / 156 / 29	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 31 row "AF_GM10_3 (-53.1, 22.0, -94.8)" col 9	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3821	48 / 44 / 13	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 31 row "AF_GM10_3 (-53.1, 22.0, -94.8)" col 10	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3822	78! / 73! / 11	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 31 row "AF_GM10_3 (-53.1, 22.0, -94.8)" col 11	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3823	39 / 37 / 8	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 31 row "AF_GM10_3 (-53.1, 22.0, -94.8)" col 12	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3824	96! / 92! / 0	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 31 row "AF_GM10_3 (-53.1, 22.0, -94.8)" col 13	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3825	196 / 161 / 211	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 32 row "AF_GM10_4 (-53.1, -22.0, -94.8)" col 8	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3826	162 / 156 / 34	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 32 row "AF_GM10_4 (-53.1, -22.0, -94.8)" col 9	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3827	49 / 45 / 14	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 32 row "AF_GM10_4 (-53.1, -22.0, -94.8)" col 10	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3828	80! / 75! / 12	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 32 row "AF_GM10_4 (-53.1, -22.0, -94.8)" col 11	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3829	40 / 38 / 8	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 32 row "AF_GM10_4 (-53.1, -22.0, -94.8)" col 12	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3830	97! / 93! / 0	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 32 row "AF_GM10_4 (-53.1, -22.0, -94.8)" col 13	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3831	889! / 889! / 221	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 33 row "AF_GM10_5 (-22.0, -53.1, -94.8)" col 8	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3832	807 / 807 / 108	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 33 row "AF_GM10_5 (-22.0, -53.1, -94.8)" col 9	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)

ID	Value	Class	Source	Key	Source tag / formula
T3833	163 / 163 / 30	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 33 row "AF_GM10_5 (-22.0, -53.1, -94.8)" col 10	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3834	275 / 275 / 29!	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 33 row "AF_GM10_5 (-22.0, -53.1, -94.8)" col 11	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3835	45 / 45 / 9	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 33 row "AF_GM10_5 (-22.0, -53.1, -94.8)" col 12	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3836	233 / 233 / 2	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 33 row "AF_GM10_5 (-22.0, -53.1, -94.8)" col 13	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3837	322! / 311! / 164!	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 34 row "AF_GM10_6 (22.0, -53.1, -94.8)" col 8	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3838	281 / 275 / 81	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 34 row "AF_GM10_6 (22.0, -53.1, -94.8)" col 9	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3839	54 / 50 / 16	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 34 row "AF_GM10_6 (22.0, -53.1, -94.8)" col 10	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3840	98 / 95 / 30	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 34 row "AF_GM10_6 (22.0, -53.1, -94.8)" col 11	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3841	12 / 10 / 3	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 34 row "AF_GM10_6 (22.0, -53.1, -94.8)" col 12	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3842	82 / 81 / 25	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 34 row "AF_GM10_6 (22.0, -53.1, -94.8)" col 13	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3843	162 / 160 / 163! *	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 35 row "AF_GM10_7 (53.1, -22.0, -94.8)" col 8	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3844	160 / 146 / 22	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 35 row "AF_GM10_7 (53.1, -22.0, -94.8)" col 9	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3845	52 / 49 / 27	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 35 row "AF_GM10_7 (53.1, -22.0, -94.8)" col 10	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3846	88! / 84! / 25	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 35 row "AF_GM10_7 (53.1, -22.0, -94.8)" col 11	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3847	42 / 40 / 15	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 35 row "AF_GM10_7 (53.1, -22.0, -94.8)" col 12	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3848	98! / 94! / 2	DERIVED	MOUNTMD v3.5.2 re-check with RM2-11 folded	line 35 row "AF_GM10_7 (53.1, -22.0, -94.8)" col 13	FEA3D mounting rows: SIM as found, moment-free estimate DERIVED (! = estimate governs)
T3849	24.376840145847613, 24.448554584674905, 346.75108128364144 N	DERIVED	MOUNT	/datasheet_v352_governing/groups/LC1/total_on_airframe_rigid/max_abs_F_N	DERIVED
T3850	2431.558737226811, 995.3076369140049, 496.3842782604187 N mm	DERIVED	MOUNT	/datasheet_v352_governing/groups/LC1/total_on_airframe_rigid/max_abs_M_Nmm	DERIVED
T3851	24.585419160185076, 24.714478438730474, 751.5510623409074 N	DERIVED	MOUNT	/datasheet_v352_governing/groups/LC2/total_on_airframe_rigid/max_abs_F_N	DERIVED
T3852	3915.4671450804167, 2056.769119742903, 567.6814897774219 N mm	DERIVED	MOUNT	/datasheet_v352_governing/groups/LC2/total_on_airframe_rigid/max_abs_M_Nmm	DERIVED
T3853	0.980851520202501, 1.2504649162674042, 1903.5181915870955 N	DERIVED	MOUNT	/datasheet_v352_governing/groups/FA1Q/total_on_airframe_rigid/max_abs_F_N	DERIVED
T3854	7294.623451005375, 4978.554113813162, 696.9849024049167 N mm	DERIVED	MOUNT	/datasheet_v352_governing/groups/FA1Q/total_on_airframe_rigid/max_abs_M_Nmm	DERIVED
T3855	1.262575063924979, 1.6096402835321584, 2450.2734284585476 N	DERIVED	MOUNT	/datasheet_v352_governing/groups/FA1D/total_on_airframe_rigid/max_abs_F_N	DERIVED
T3856	9298.908237507385, 6412.248822540962, 793.2845720734908 N mm	DERIVED	MOUNT	/datasheet_v352_governing/groups/FA1D/total_on_airframe_rigid/max_abs_M_Nmm	DERIVED
T3857	1.5338644023244044, 1.9555128594167428, 2976.778471371797 N	DERIVED	MOUNT	/datasheet_v352_governing/groups/FA1U/total_on_airframe_rigid/max_abs_F_N	DERIVED
T3858	11228.960254138958, 7792.843727241783, 886.0175873098956 N mm	DERIVED	MOUNT	/datasheet_v352_governing/groups/FA1U/total_on_airframe_rigid/max_abs_M_Nmm	DERIVED
T3859	1.5759977234189009, 2.009229531009936, 3058.5487545750166 N	DERIVED	MOUNT	/datasheet_v352_governing/groups/FA1S/total_on_airframe_rigid/max_abs_F_N	DERIVED
T3860	11528.712178875809, 8007.260735902622, 900.4197379069947 N mm	DERIVED	MOUNT	/datasheet_v352_governing/groups/FA1S/total_on_airframe_rigid/max_abs_M_Nmm	DERIVED
T3861	0.9808488800092334, 1.2504537565312148, 3599.9995321181636 N	DERIVED	MOUNT	/datasheet_v352_governing/groups/FA1T/total_on_airframe_rigid/max_abs_F_N	DERIVED
T3862	7341.009654505095, 5065.5368631222045, 723.0757700057138 N mm	DERIVED	MOUNT	/datasheet_v352_governing/groups/FA1T/total_on_airframe_rigid/max_abs_M_Nmm	DERIVED
T3863	321.5755704204915, 320.7239059436448, 506.4604877539808 N	DERIVED	MOUNT	/datasheet_v352_governing/groups/LC3/total_on_airframe_rigid/max_abs_F_N	DERIVED
T3864	22897.12374112958, 11990.928344766022, 2577.2652269762248 N mm	DERIVED	MOUNT	/datasheet_v352_governing/groups/LC3/total_on_airframe_rigid/max_abs_M_Nmm	DERIVED
T3865	140.2557447265061, 140.49866409325574, 514.4110701506388 N	DERIVED	MOUNT	/datasheet_v352_governing/groups/GS/total_on_airframe_rigid/max_abs_F_N	DERIVED
T3866	56100.49230970387, 50983.24744547273, 7415.355414331895 N mm	DERIVED	MOUNT	/datasheet_v352_governing/groups/GS/total_on_airframe_rigid/max_abs_M_Nmm	DERIVED
T3867	293.98765523988965, 294.1281319612113, 215.24810774919953 N	DERIVED	MOUNT	/datasheet_v352_governing/groups/TD0/total_on_airframe_rigid/max_abs_F_N	DERIVED
T3868	8898.5574781837, 6023.886704964216, 1513.7648590601264 N mm	DERIVED	MOUNT	/datasheet_v352_governing/groups/TD0/total_on_airframe_rigid/max_abs_M_Nmm	DERIVED
T3869	357.2153574945862, 356.35564294013545, 511.3025168711481 N	DERIVED	MOUNT	/datasheet_v352_governing/groups/TDP/total_on_airframe_rigid/max_abs_F_N	DERIVED
T3870	10100.284946647773, 11757.04536574755, 1967.2312956213136 N mm	DERIVED	MOUNT	/datasheet_v352_governing/groups/TDP/total_on_airframe_rigid/max_abs_M_Nmm	DERIVED
T3871	118.52759406, 118.52757839999998, 395.09190572999995 N	DERIVED	MOUNT	/datasheet_v352_governing/groups/LC6/total_on_airframe_rigid/max_abs_F_N	DERIVED
T3872	1435.8829371283646, 1311.8368964338097, 312.028106946702 N mm	DERIVED	MOUNT	/datasheet_v352_governing/groups/LC6/total_on_airframe_rigid/max_abs_M_Nmm	DERIVED
T3873	137.5850598957507, 135.71146880863972, 907.7000011861716 N	DERIVED	MOUNT	/datasheet_v352_governing/groups/LC2F/total_on_airframe_rigid/max_abs_F_N	DERIVED
T3874	2669.2046560969948, 11644.72429281058, 1426.7201851874327 N mm	DERIVED	MOUNT	/datasheet_v352_governing/groups/LC2F/total_on_airframe_rigid/max_abs_M_Nmm	DERIVED
T3875	4.03255 kg	DERIVED	MOUNT	/mass/v352/model_kg	(class set at the call site: DERIVED)
T3876	2.82932 kg	DERIVED	MOUNT	/mass/v352/engine	(class set at the call site: DERIVED)
T3877	1.30993 kg	DERIVED	MOUNT	/mass/v352/excluded_kg	(class set at the call site: DERIVED)
T3878	SIM (FE unit steps on the v3.5.2 R3 pose runs) + DERIVED (servo in series; k_s 58 / 174 ...	SIM	KTAB	/evidence	(class set at the call site: SIM)
T3879	7293.95 N m/rad	SIM	KTAB	/rows/0/values/FE_coupled_alpha/R3	(class set at the call site: SIM)
T3880	7172.12 N m/rad	SIM	KTAB	/rows/0/values/FE_coupled_beta/R3	(class set at the call site: SIM)
T3881	1426.19 N m/rad	DERIVED	KTAB	/rows/0/values/k_s58_coupled_alpha/R3	(class set at the call site: DERIVED)

ID	Value	Class	Source	Key	Source tag / formula
T3882	1799.09 N m/rad	DERIVED	KTAB	/rows/0/values/k_s58_coupled_beta/R3	(class set at the call site: DERIVED)
T3883	3075.77 N m/rad	DERIVED	KTAB	/rows/0/values/k_s174_coupled_alpha/R3	(class set at the call site: DERIVED)
T3884	3594.13 N m/rad	DERIVED	KTAB	/rows/0/values/k_s174_coupled_beta/R3	(class set at the call site: DERIVED)
T3885	1.2085	DERIVED	KTAB	/rows/0/fraction_max	(class set at the call site: DERIVED)
T3886	7403.23 N m/rad	SIM	KTAB	/rows/1/values/FE_coupled_alpha/R3	(class set at the call site: SIM)
T3887	6880.6 N m/rad	SIM	KTAB	/rows/1/values/FE_coupled_beta/R3	(class set at the call site: SIM)
T3888	1362.07 N m/rad	DERIVED	KTAB	/rows/1/values/k_s58_coupled_alpha/R3	(class set at the call site: DERIVED)
T3889	1559.2 N m/rad	DERIVED	KTAB	/rows/1/values/k_s58_coupled_beta/R3	(class set at the call site: DERIVED)
T3890	2987.06 N m/rad	DERIVED	KTAB	/rows/1/values/k_s174_coupled_alpha/R3	(class set at the call site: DERIVED)
T3891	3218.78 N m/rad	DERIVED	KTAB	/rows/1/values/k_s174_coupled_beta/R3	(class set at the call site: DERIVED)
T3892	1	DERIVED	KTAB	/rows/1/fraction_max	(class set at the call site: DERIVED)
T3893	7756.13 N m/rad	SIM	KTAB	/rows/2/values/FE_coupled_alpha/R3	(class set at the call site: SIM)
T3894	5808.44 N m/rad	SIM	KTAB	/rows/2/values/FE_coupled_beta/R3	(class set at the call site: SIM)
T3895	1190.24 N m/rad	DERIVED	KTAB	/rows/2/values/k_s58_coupled_alpha/R3	(class set at the call site: DERIVED)
T3896	1123.86 N m/rad	DERIVED	KTAB	/rows/2/values/k_s58_coupled_beta/R3	(class set at the call site: DERIVED)
T3897	2732.18 N m/rad	DERIVED	KTAB	/rows/2/values/k_s174_coupled_alpha/R3	(class set at the call site: DERIVED)
T3898	2430.89 N m/rad	DERIVED	KTAB	/rows/2/values/k_s174_coupled_beta/R3	(class set at the call site: DERIVED)
T3899	0.8722	DERIVED	KTAB	/rows/2/fraction_max	(class set at the call site: DERIVED)
T3900	8594.86 N m/rad	SIM	KTAB	/rows/3/values/FE_coupled_alpha/R3	(class set at the call site: SIM)
T3901	4568.72 N m/rad	SIM	KTAB	/rows/3/values/FE_coupled_beta/R3	(class set at the call site: SIM)
T3902	1184.01 N m/rad	DERIVED	KTAB	/rows/3/values/k_s58_coupled_alpha/R3	(class set at the call site: DERIVED)
T3903	871.44 N m/rad	DERIVED	KTAB	/rows/3/values/k_s58_coupled_beta/R3	(class set at the call site: DERIVED)
T3904	2784.78 N m/rad	DERIVED	KTAB	/rows/3/values/k_s174_coupled_alpha/R3	(class set at the call site: DERIVED)
T3905	1892.4 N m/rad	DERIVED	KTAB	/rows/3/values/k_s174_coupled_beta/R3	(class set at the call site: DERIVED)
T3906	0.7247	DERIVED	KTAB	/rows/3/fraction_max	(class set at the call site: DERIVED)
T3907	9337.08 N m/rad	SIM	KTAB	/rows/4/values/FE_coupled_alpha/R3	(class set at the call site: SIM)
T3908	3971.84 N m/rad	SIM	KTAB	/rows/4/values/FE_coupled_beta/R3	(class set at the call site: SIM)
T3909	1290.85 N m/rad	DERIVED	KTAB	/rows/4/values/k_s58_coupled_alpha/R3	(class set at the call site: DERIVED)
T3910	790.77 N m/rad	DERIVED	KTAB	/rows/4/values/k_s58_coupled_beta/R3	(class set at the call site: DERIVED)
T3911	3033.72 N m/rad	DERIVED	KTAB	/rows/4/values/k_s174_coupled_alpha/R3	(class set at the call site: DERIVED)
T3912	1696.7 N m/rad	DERIVED	KTAB	/rows/4/values/k_s174_coupled_beta/R3	(class set at the call site: DERIVED)
T3913	0.6636	DERIVED	KTAB	/rows/4/fraction_max	(class set at the call site: DERIVED)
T3914	8605.84 N m/rad	SIM	KTAB	/rows/5/values/FE_coupled_alpha/R3	(class set at the call site: SIM)
T3915	4267.94 N m/rad	SIM	KTAB	/rows/5/values/FE_coupled_beta/R3	(class set at the call site: SIM)
T3916	1184.22 N m/rad	DERIVED	KTAB	/rows/5/values/k_s58_coupled_alpha/R3	(class set at the call site: DERIVED)
T3917	859.88 N m/rad	DERIVED	KTAB	/rows/5/values/k_s58_coupled_beta/R3	(class set at the call site: DERIVED)
T3918	2785.93 N m/rad	DERIVED	KTAB	/rows/5/values/k_s174_coupled_alpha/R3	(class set at the call site: DERIVED)
T3919	1838.73 N m/rad	DERIVED	KTAB	/rows/5/values/k_s174_coupled_beta/R3	(class set at the call site: DERIVED)
T3920	0.7247	DERIVED	KTAB	/rows/5/fraction_max	(class set at the call site: DERIVED)
T3921	7763.92 N m/rad	SIM	KTAB	/rows/6/values/FE_coupled_alpha/R3	(class set at the call site: SIM)
T3922	5442.59 N m/rad	SIM	KTAB	/rows/6/values/FE_coupled_beta/R3	(class set at the call site: SIM)
T3923	1190.43 N m/rad	DERIVED	KTAB	/rows/6/values/k_s58_coupled_alpha/R3	(class set at the call site: DERIVED)
T3924	1109.43 N m/rad	DERIVED	KTAB	/rows/6/values/k_s58_coupled_beta/R3	(class set at the call site: DERIVED)
T3925	2733.14 N m/rad	DERIVED	KTAB	/rows/6/values/k_s174_coupled_alpha/R3	(class set at the call site: DERIVED)
T3926	2364.37 N m/rad	DERIVED	KTAB	/rows/6/values/k_s174_coupled_beta/R3	(class set at the call site: DERIVED)
T3927	0.871	DERIVED	KTAB	/rows/6/fraction_max	(class set at the call site: DERIVED)
T3928	7402.04 N m/rad	SIM	KTAB	/rows/7/values/FE_coupled_alpha/R3	(class set at the call site: SIM)
T3929	6677.77 N m/rad	SIM	KTAB	/rows/7/values/FE_coupled_beta/R3	(class set at the call site: SIM)
T3930	1362.03 N m/rad	DERIVED	KTAB	/rows/7/values/k_s58_coupled_alpha/R3	(class set at the call site: DERIVED)
T3931	1548.54 N m/rad	DERIVED	KTAB	/rows/7/values/k_s58_coupled_beta/R3	(class set at the call site: DERIVED)
T3932	2986.87 N m/rad	DERIVED	KTAB	/rows/7/values/k_s174_coupled_alpha/R3	(class set at the call site: DERIVED)
T3933	3173.69 N m/rad	DERIVED	KTAB	/rows/7/values/k_s174_coupled_beta/R3	(class set at the call site: DERIVED)
T3934	0.9976	DERIVED	KTAB	/rows/7/fraction_max	(class set at the call site: DERIVED)
T3935	8246.32 N m/rad	SIM	KTAB	/rows/8/values/FE_coupled_alpha/R3	(class set at the call site: SIM)
T3936	5763.5 N m/rad	SIM	KTAB	/rows/8/values/FE_coupled_beta/R3	(class set at the call site: SIM)
T3937	1149.93 N m/rad	DERIVED	KTAB	/rows/8/values/k_s58_coupled_alpha/R3	(class set at the call site: DERIVED)
T3938	1084.77 N m/rad	DERIVED	KTAB	/rows/8/values/k_s58_coupled_beta/R3	(class set at the call site: DERIVED)
T3939	2697.48 N m/rad	DERIVED	KTAB	/rows/8/values/k_s174_coupled_alpha/R3	(class set at the call site: DERIVED)
T3940	2364.31 N m/rad	DERIVED	KTAB	/rows/8/values/k_s174_coupled_beta/R3	(class set at the call site: DERIVED)
T3941	0.9896	DERIVED	KTAB	/rows/8/fraction_max	(class set at the call site: DERIVED)
T3942	7879.67 N m/rad	SIM	KTAB	/rows/9/values/FE_coupled_alpha/R3	(class set at the call site: SIM)
T3943	5501.24 N m/rad	SIM	KTAB	/rows/9/values/FE_coupled_beta/R3	(class set at the call site: SIM)
T3944	1178.6 N m/rad	DERIVED	KTAB	/rows/9/values/k_s58_coupled_alpha/R3	(class set at the call site: DERIVED)
T3945	1105.53 N m/rad	DERIVED	KTAB	/rows/9/values/k_s58_coupled_beta/R3	(class set at the call site: DERIVED)
T3946	2721.63 N m/rad	DERIVED	KTAB	/rows/9/values/k_s174_coupled_alpha/R3	(class set at the call site: DERIVED)
T3947	2365.75 N m/rad	DERIVED	KTAB	/rows/9/values/k_s174_coupled_beta/R3	(class set at the call site: DERIVED)
T3948	0.9575	DERIVED	KTAB	/rows/9/fraction_max	(class set at the call site: DERIVED)
T3949	7873.34 N m/rad	SIM	KTAB	/rows/10/values/FE_coupled_alpha/R3	(class set at the call site: SIM)
T3950	5823.16 N m/rad	SIM	KTAB	/rows/10/values/FE_coupled_beta/R3	(class set at the call site: SIM)
T3951	1178.46 N m/rad	DERIVED	KTAB	/rows/10/values/k_s58_coupled_alpha/R3	(class set at the call site: DERIVED)
T3952	1117.95 N m/rad	DERIVED	KTAB	/rows/10/values/k_s58_coupled_beta/R3	(class set at the call site: DERIVED)
T3953	2720.88 N m/rad	DERIVED	KTAB	/rows/10/values/k_s174_coupled_alpha/R3	(class set at the call site: DERIVED)
T3954	2423.36 N m/rad	DERIVED	KTAB	/rows/10/values/k_s174_coupled_beta/R3	(class set at the call site: DERIVED)
T3955	0.9578	DERIVED	KTAB	/rows/10/fraction_max	(class set at the call site: DERIVED)
T3956	7393.15 N m/rad	SIM	KTAB	/rows/11/values/FE_coupled_alpha/R3	(class set at the call site: SIM)
T3957	7004.96 N m/rad	SIM	KTAB	/rows/11/values/FE_coupled_beta/R3	(class set at the call site: SIM)
T3958	1334.39 N m/rad	DERIVED	KTAB	/rows/11/values/k_s58_coupled_alpha/R3	(class set at the call site: DERIVED)
T3959	1633.47 N m/rad	DERIVED	KTAB	/rows/11/values/k_s58_coupled_beta/R3	(class set at the call site: DERIVED)
T3960	2941.38 N m/rad	DERIVED	KTAB	/rows/11/values/k_s174_coupled_alpha/R3	(class set at the call site: DERIVED)
T3961	3341.85 N m/rad	DERIVED	KTAB	/rows/11/values/k_s174_coupled_beta/R3	(class set at the call site: DERIVED)
T3962	1.2013	DERIVED	KTAB	/rows/11/fraction_max	(class set at the call site: DERIVED)
T3963	9216.31 N m/rad	SIM	KTAB	/rows/12/values/FE_coupled_alpha/R3	(class set at the call site: SIM)
T3964	4222.88 N m/rad	SIM	KTAB	/rows/12/values/FE_coupled_beta/R3	(class set at the call site: SIM)
T3965	1263.3 N m/rad	DERIVED	KTAB	/rows/12/values/k_s58_coupled_alpha/R3	(class set at the call site: DERIVED)
T3966	826.23 N m/rad	DERIVED	KTAB	/rows/12/values/k_s58_coupled_beta/R3	(class set at the call site: DERIVED)
T3967	2974.46 N m/rad	DERIVED	KTAB	/rows/12/values/k_s174_coupled_alpha/R3	(class set at the call site: DERIVED)
T3968	1781.55 N m/rad	DERIVED	KTAB	/rows/12/values/k_s174_coupled_beta/R3	(class set at the call site: DERIVED)
T3969	0.7637	DERIVED	KTAB	/rows/12/fraction_max	(class set at the call site: DERIVED)
T3970	7507.87 N m/rad	SIM	KTAB	/rows/13/values/FE_coupled_alpha/R3	(class set at the call site: SIM)
T3971	6744.26 N m/rad	SIM	KTAB	/rows/13/values/FE_coupled_beta/R3	(class set at the call site: SIM)
T3972	1290.09 N m/rad	DERIVED	KTAB	/rows/13/values/k_s58_coupled_alpha/R3	(class set at the call site: DERIVED)
T3973	1456.37 N m/rad	DERIVED	KTAB	/rows/13/values/k_s58_coupled_beta/R3	(class set at the call site: DERIVED)
T3974	2880.39 N m/rad	DERIVED	KTAB	/rows/13/values/k_s174_coupled_alpha/R3	(class set at the call site: DERIVED)
T3975	3051.3 N m/rad	DERIVED	KTAB	/rows/13/values/k_s174_coupled_beta/R3	(class set at the call site: DERIVED)
T3976	1.081	DERIVED	KTAB	/rows/13/fraction_max	(class set at the call site: DERIVED)
T3977	8626.86 N m/rad	SIM	KTAB	/rows/14/values/FE_coupled_alpha/R3	(class set at the call site: SIM)
T3978	4754.14 N m/rad	SIM	KTAB	/rows/14/values/FE_coupled_beta/R3	(class set at the call site: SIM)
T3979	1184.73 N m/rad	DERIVED	KTAB	/rows/14/values/k_s58_coupled_alpha/R3	(class set at the call site: DERIVED)
T3980	899.42 N m/rad	DERIVED	KTAB	/rows/14/values/k_s58_coupled_beta/R3	(class set at the call site: DERIVED)
T3981	2788.35 N m/rad	DERIVED	KTAB	/rows/14/values/k_s174_coupled_alpha/R3	(class set at the call site: DERIVED)

ID	Value	Class	Source	Key	Source tag / formula
T3982	1957.57 N m/rad	DERIVED	KTAB	/rows/14/values/k_s174_coupled_beta/R3	(class set at the call site: DERIVED)
T3983	0.8202	DERIVED	KTAB	/rows/14/fraction_max	(class set at the call site: DERIVED)
T3984	8635.82 N m/rad	SIM	KTAB	/rows/15/values/FE_coupled_alpha/R3	(class set at the call site: SIM)
T3985	4491.88 N m/rad	SIM	KTAB	/rows/15/values/FE_coupled_beta/R3	(class set at the call site: SIM)
T3986	1184.9 N m/rad	DERIVED	KTAB	/rows/15/values/k_s58_coupled_alpha/R3	(class set at the call site: DERIVED)
T3987	889.59 N m/rad	DERIVED	KTAB	/rows/15/values/k_s58_coupled_beta/R3	(class set at the call site: DERIVED)
T3988	2789.29 N m/rad	DERIVED	KTAB	/rows/15/values/k_s174_coupled_alpha/R3	(class set at the call site: DERIVED)
T3989	1911.61 N m/rad	DERIVED	KTAB	/rows/15/values/k_s174_coupled_beta/R3	(class set at the call site: DERIVED)
T3990	0.8207	DERIVED	KTAB	/rows/15/fraction_max	(class set at the call site: DERIVED)
T3991	7507.82 N m/rad	SIM	KTAB	/rows/16/values/FE_coupled_alpha/R3	(class set at the call site: SIM)
T3992	6559.77 N m/rad	SIM	KTAB	/rows/16/values/FE_coupled_beta/R3	(class set at the call site: SIM)
T3993	1290.09 N m/rad	DERIVED	KTAB	/rows/16/values/k_s58_coupled_alpha/R3	(class set at the call site: DERIVED)
T3994	1447.58 N m/rad	DERIVED	KTAB	/rows/16/values/k_s58_coupled_beta/R3	(class set at the call site: DERIVED)
T3995	2880.38 N m/rad	DERIVED	KTAB	/rows/16/values/k_s174_coupled_alpha/R3	(class set at the call site: DERIVED)
T3996	3012.96 N m/rad	DERIVED	KTAB	/rows/16/values/k_s174_coupled_beta/R3	(class set at the call site: DERIVED)
T3997	1.077	DERIVED	KTAB	/rows/16/fraction_max	(class set at the call site: DERIVED)
T3998	18.311763993365044–29.835854443322933 Hz	DERIVED	ROCK	/pose_range_Hz/58/J_free	DERIVED (SIM modal + SIM K; k_s ASSUMED 58 / 174 N m/rad TO-VERIFY B-TVC; J ASSUMED GNC SB-2292SG datasheet-derived, TO-VERIFY B-TVC-1)
T3999	16.595579248366466–23.026246331622346 Hz	DERIVED	ROCK	/pose_range_Hz/58/J_low	DERIVED (SIM modal + SIM K; k_s ASSUMED 58 / 174 N m/rad TO-VERIFY B-TVC; J ASSUMED GNC SB-2292SG datasheet-derived, TO-VERIFY B-TVC-1)
T4000	15.165659595157251–19.142535236265772 Hz	DERIVED	ROCK	/pose_range_Hz/58/J_nom	DERIVED (SIM modal + SIM K; k_s ASSUMED 58 / 174 N m/rad TO-VERIFY B-TVC; J ASSUMED GNC SB-2292SG datasheet-derived, TO-VERIFY B-TVC-1)
T4001	13.753232959209369–16.152725989204292 Hz	DERIVED	ROCK	/pose_range_Hz/58/J_high	DERIVED (SIM modal + SIM K; k_s ASSUMED 58 / 174 N m/rad TO-VERIFY B-TVC; J ASSUMED GNC SB-2292SG datasheet-derived, TO-VERIFY B-TVC-1)
T4002	26.757201563980637–40.31177340631051 Hz	DERIVED	ROCK	/pose_range_Hz/174/J_free	DERIVED (SIM modal + SIM K; k_s ASSUMED 58 / 174 N m/rad TO-VERIFY B-TVC; J ASSUMED GNC SB-2292SG datasheet-derived, TO-VERIFY B-TVC-1)
T4003	24.249494797847266–31.113081344495665 Hz	DERIVED	ROCK	/pose_range_Hz/174/J_low	DERIVED (SIM modal + SIM K; k_s ASSUMED 58 / 174 N m/rad TO-VERIFY B-TVC; J ASSUMED GNC SB-2292SG datasheet-derived, TO-VERIFY B-TVC-1)
T4004	22.160082286964617–27.16812912838974 Hz	DERIVED	ROCK	/pose_range_Hz/174/J_nom	DERIVED (SIM modal + SIM K; k_s ASSUMED 58 / 174 N m/rad TO-VERIFY B-TVC; J ASSUMED GNC SB-2292SG datasheet-derived, TO-VERIFY B-TVC-1)
T4005	20.096231752000943–24.006692209070543 Hz	DERIVED	ROCK	/pose_range_Hz/174/J_high	DERIVED (SIM modal + SIM K; k_s ASSUMED 58 / 174 N m/rad TO-VERIFY B-TVC; J ASSUMED GNC SB-2292SG datasheet-derived, TO-VERIFY B-TVC-1)
T4006	J_low: 0.00076; J_nom: 0.0016; J_high: 0.0027	DERIVED	ROCK	/J_kgm2	DERIVED (SIM modal + SIM K; k_s ASSUMED 58 / 174 N m/rad TO-VERIFY B-TVC; J ASSUMED GNC SB-2292SG datasheet-derived, TO-VERIFY B-TVC-1)
T4007	HW-103 2x M4×20 DIN 7984 A4-80 (lower, y +/-8.5, z 146.51, engagement 8.5) + HW-103B 1x ...	DECIDED *Design value, f4_v352.json v1*	F4J v1	/joints/J-FD01/hardware	f4_v352.json v1
T4008	1.8 N m	DECIDED *Design value, f4_v352.json v1*	F4J v1	/joints/J-FD01/preload/T_Nm	f4_v352.json v1
T4009	2250 N	DERIVED	F4J v1	/joints/J-FD01/preload/F_nom_N	(class set at the call site: DERIVED)
T4010	0.121	SIM	F4J v1	/joints/J-FD01/worst/a_F4_bounding/bounding/MS	(class set at the call site: SIM)
T4011	LC3 hard-landing envelope (10 g)	SIM	F4J v1	/joints/J-FD01/worst/a_F4_bounding/bounding/case	(class set at the call site: SIM)
T4012	R7 bounding	SIM	F4J v1	/joints/J-FD01/worst/a_F4_bounding/bounding/area	(class set at the call site: SIM)
T4013	0.403	SIM	F4J v1	/joints/J-FD01/worst/a_F4_bounding/phi02/MS	(class set at the call site: SIM)
T4014	0.128	SIM	F4J v1	/joints/J-FD01/worst/b_bolt_group_vector/bounding/MS	(class set at the call site: SIM)
T4015	LC3 touchdown (5.43 g SIM)	SIM	F4J v1	/joints/J-FD01/worst/b_bolt_group_vector/bounding/case	(class set at the call site: SIM)
T4016	R7 bounding	SIM	F4J v1	/joints/J-FD01/worst/b_bolt_group_vector/bounding/area	(class set at the call site: SIM)
T4017	0.405	SIM	F4J v1	/joints/J-FD01/worst/b_bolt_group_vector/phi02/MS	(class set at the call site: SIM)
T4018	HW-127 2x M4×12 ISO 4762 A4-80, 1.8 N m M-T1, into GM-09 with helicoil 1.5 D (thread ...	DECIDED *Design value, f4_v352.json v1*	F4J v1	/joints/J-FD09/hardware	f4_v352.json v1
T4019	1.8 N m	DECIDED *Design value, f4_v352.json v1*	F4J v1	/joints/J-FD09/preload/T_Nm	f4_v352.json v1
T4020	2250 N	DERIVED	F4J v1	/joints/J-FD09/preload/F_nom_N	(class set at the call site: DERIVED)
T4021	0.476	SIM	F4J v1	/joints/J-FD09/worst/0.184/bounding/MS	(class set at the call site: SIM)
T4022	LC3 hard-landing envelope (10 g)	SIM	F4J v1	/joints/J-FD09/worst/0.184/bounding/case	(class set at the call site: SIM)
T4023	EI 0.184	SIM	F4J v1	/joints/J-FD09/worst/0.184/bounding/area	(class set at the call site: SIM)
T4024	0.66	SIM	F4J v1	/joints/J-FD09/worst/0.184/phi02/MS	(class set at the call site: SIM)
T4025	0.221	SIM	F4J v1	/joints/J-FD09/worst/0.5/bounding/MS	(class set at the call site: SIM)
T4026	LC3 hard-landing envelope (10 g)	SIM	F4J v1	/joints/J-FD09/worst/0.5/bounding/case	(class set at the call site: SIM)
T4027	EI 0.5	SIM	F4J v1	/joints/J-FD09/worst/0.5/bounding/area	(class set at the call site: SIM)
T4028	0.586	SIM	F4J v1	/joints/J-FD09/worst/0.5/phi02/MS	(class set at the call site: SIM)
T4029	CALC on SIM; ASSUMED conservative ligament model; negatives SHOWN AS FOUND; not physical ...	CALC	TAPW v3	/status	(class set at the call site: CALC)
T4030	Tap wall to the torch passage (az 18 / 342), wall 1.6355 mm net at +/-0.25 deg, CALC on ...	CALC	TAPW v3	/verdict	CALC on SIM; ASSUMED conservative ligament model; negatives SHOWN AS FOUND; not physical approval
T4031	0.67	CALC	TAPW v3	/rows/3750/governing/TC02_thread_T/MS_y	(class set at the call site: CALC)
T4032	0.785	CALC	TAPW v3	/rows/3750/governing/TC02_thread_T/MS_u	(class set at the call site: CALC)
T4033	0.275	CALC	TAPW v3	/rows/3750/governing/worst_with_screw_T_sensitivity/MS_y	(class set at the call site: CALC)
T4034	0.365	CALC	TAPW v3	/rows/3750/governing/worst_with_screw_T_sensitivity/MS_u	(class set at the call site: CALC)
T4035	Z3 (LC-TB70 v2): 63.1 < P_head <= 88.1 bar abs inside the light window (record start ...	DECIDED *Lead decision, per lc_tb70_envelope_v2.json v2*	TB70 v2	/Z3_v2_text	tag in lc_tb70_envelope_v2.json v2:
T4036	J-FWD replace rule (LC-TB70 v2): Q_JFWD > 60.0 bar abs at any time in a run (Q_JFWD = ...	DECIDED *Lead decision, per lc_tb70_envelope_v2.json v2*	TB70 v2	/jfwd_rule_v2_text	tag in lc_tb70_envelope_v2.json v2:
T4037	CONFIRMED, with one addition: PC_SENSOR raised on the HIGH side (transducer output >= ...	CALC	TB70 v2	/F4_abort_set/verdict	(class set at the call site: CALC)
T4038	DERIVED from the analysed structural cases and the Combustion v2.4b FA-1 grid (SIM on an ...	CALC	TB70 v2	/status	(class set at the call site: CALC)
T4039	TC-01 J-FWD seat / ring line P+Q (secondary, shakedown; class DC)	DERIVED	F3D FINAL handback of the v3.5.2 re-check	line 1696 row "1" col 1	FEA3D-2 report §13 open items
T4040	found negative on the final envelope v6 `43a5d78b67c615fc` (lead 05:17:36Z (3); the seat ...	SIM	F3D FINAL handback of the v3.5.2 re-check	line 1696 row "1" col 2	FEA3D-2 report §13 open items
T4041	Mechanical (IN718 re-map once on the final envelope v6 with ...	DECIDED *Lead decision, per REPORT-FEA3D-v352.md FINAL handback of the v3.5.2 re-check*	F3D FINAL handback of the v3.5.2 re-check	line 1696 row "1" col 3	tag in REPORT-FEA3D-v352.md FINAL handback of the v3.5.2 re-check: FEA3D-2 report §13 open items
T4042	the band inside [0.040, 0.0761] mm that closes it, or the seat-line P+Q stays a found ...	DERIVED	F3D FINAL handback of the v3.5.2 re-check	line 1696 row "1" col 4	FEA3D-2 report §13 open items
T4043	HW-101 IN718 screw rows on the final envelope	DERIVED	F3D FINAL handback of the v3.5.2 re-check	line 1697 row "2" col 1	FEA3D-2 report §13 open items
T4044	Eq. 22 inputs V 4079.2 N (FE, (b1)) and M 4354.7 N mm (CALC) (lead 01:30:18Z (6)); the ...	SIM	F3D FINAL handback of the v3.5.2 re-check	line 1697 row "2" col 2	FEA3D-2 report §13 open items

ID	Value	Class	Source	Key	Source tag / formula
T4045	Mechanical	DECIDED *Lead decision, per REPORT-FEA3D-v352.md FINAL handback of the v3.5.2 re-check*	F3D FINAL handback of the v3.5.2 re-check	line 1697 row "2" col 3	tag in REPORT-FEA3D-v352.md FINAL handback of the v3.5.2 re-check: FEA3D-2 report §13 open items
T4046	the re-map on the final envelope v6; the relaxation / shakedown / LCF rows	DERIVED	F3D FINAL handback of the v3.5.2 re-check	line 1697 row "2" col 4	FEA3D-2 report §13 open items
T4047	TC-02 TB70 / TBS (n30266, 228.7 C; TS-212)	DERIVED	F3D FINAL handback of the v3.5.2 re-check	line 1698 row "3" col 1	FEA3D-2 report §13 open items
T4048	-0.236 / -0.271; +0.138 / -0.035 [SIM]	SIM	F3D FINAL handback of the v3.5.2 re-check	line 1698 row "3" col 2	FEA3D-2 report §13 open items
T4049	Mechanical / founder	DECIDED *Lead decision, per REPORT-FEA3D-v352.md FINAL handback of the v3.5.2 re-check*	F3D FINAL handback of the v3.5.2 re-check	line 1698 row "3" col 3	tag in REPORT-FEA3D-v352.md FINAL handback of the v3.5.2 re-check: FEA3D-2 report §13 open items
T4050	212 (1) (d) time at temperature with T-F1; Mechanical's (b) package	DERIVED	F3D FINAL handback of the v3.5.2 re-check	line 1698 row "3" col 4	FEA3D-2 report §13 open items
T4051	The seat region at TB (4.3)	DERIVED	F3D FINAL handback of the v3.5.2 re-check	line 1699 row "4" col 1	FEA3D-2 report §13 open items
T4052	Mechanical's S-AX v352r4 fails line 106 at SF 1.0 at TB70 (TC-02 -0.326 / -0.371; TC-03 ...	SIM	F3D FINAL handback of the v3.5.2 re-check	line 1699 row "4" col 2	FEA3D-2 report §13 open items
T4053	Mechanical	DECIDED *Lead decision, per REPORT-FEA3D-v352.md FINAL handback of the v3.5.2 re-check*	F3D FINAL handback of the v3.5.2 re-check	line 1699 row "4" col 3	tag in REPORT-FEA3D-v352.md FINAL handback of the v3.5.2 re-check: FEA3D-2 report §13 open items
T4054	T-F2 contact runs and the abort search	DERIVED	F3D FINAL handback of the v3.5.2 re-check	line 1699 row "4" col 4	FEA3D-2 report §13 open items
T4055	TC-03 LC1 / LC2 / LC3 / F2 hot bound / TB70 (TS-183)	DERIVED	F3D FINAL handback of the v3.5.2 re-check	line 1700 row "5" col 1	FEA3D-2 report §13 open items
T4056	-0.088 / -0.170 ... -0.121 / -0.201 [SIM / DERIVED]; S-AX governs +2.564 / +2.241 (lead ...	SIM	F3D FINAL handback of the v3.5.2 re-check	line 1700 row "5" col 2	FEA3D-2 report §13 open items
T4057	Mechanical	DECIDED *Lead decision, per REPORT-FEA3D-v352.md FINAL handback of the v3.5.2 re-check*	F3D FINAL handback of the v3.5.2 re-check	line 1700 row "5" col 3	tag in REPORT-FEA3D-v352.md FINAL handback of the v3.5.2 re-check: FEA3D-2 report §13 open items
T4058	183 (1) (c) LCF and (d) GlidCop time at temperature	DERIVED	F3D FINAL handback of the v3.5.2 re-check	line 1700 row "5" col 4	FEA3D-2 report §13 open items
T4059	FD-01 bonded-patch edge (TS-183)	DERIVED	F3D FINAL handback of the v3.5.2 re-check	line 1701 row "6" col 1	FEA3D-2 report §13 open items
T4060	LC6 soak -0.187 / +0.780, TB70 -0.336 / +0.484, TBS -0.005 / +0.976 (A 175 C, hotter ...	SIM	F3D FINAL handback of the v3.5.2 re-check	line 1701 row "6" col 2	FEA3D-2 report §13 open items
T4061	Mechanical	DECIDED *Lead decision, per REPORT-FEA3D-v352.md FINAL handback of the v3.5.2 re-check*	F3D FINAL handback of the v3.5.2 re-check	line 1701 row "6" col 3	tag in REPORT-FEA3D-v352.md FINAL handback of the v3.5.2 re-check: FEA3D-2 report §13 open items
T4062	183 (1) (c) LCF; (d) on the sourced long-time allowable where T <= 150 C, else T-F1	DERIVED	F3D FINAL handback of the v3.5.2 re-check	line 1701 row "6" col 4	FEA3D-2 report §13 open items
T4063	NZ-01 graphite (BR)	DERIVED	F3D FINAL handback of the v3.5.2 re-check	line 1702 row "7" col 1	FEA3D-2 report §13 open items
T4064	-0.520 ... -0.054 (yield only, brittle, thermal) [SIM]; S-AX v352_NF2b is the named ...	SIM	F3D FINAL handback of the v3.5.2 re-check	line 1702 row "7" col 2	FEA3D-2 report §13 open items
T4065	Mechanical / Combustion	DECIDED *Lead decision, per REPORT-FEA3D-v352.md FINAL handback of the v3.5.2 re-check*	F3D FINAL handback of the v3.5.2 re-check	line 1702 row "7" col 3	tag in REPORT-FEA3D-v352.md FINAL handback of the v3.5.2 re-check: FEA3D-2 report §13 open items
T4066	183 (2) capture / containment plus a test basis, else a design change	DERIVED	F3D FINAL handback of the v3.5.2 re-check	line 1702 row "7" col 4	FEA3D-2 report §13 open items
T4067	NZ-03 char (CH)	DERIVED	F3D FINAL handback of the v3.5.2 re-check	line 1703 row "8" col 1	FEA3D-2 report §13 open items
T4068	LC2 -0.088, TB70 -0.392, TBS -0.253 (n81357) against the ASSUMED 20 MPa	SIM	F3D FINAL handback of the v3.5.2 re-check	line 1703 row "8" col 2	FEA3D-2 report §13 open items
T4069	founder / Mechanical	DECIDED *Lead decision, per REPORT-FEA3D-v352.md FINAL handback of the v3.5.2 re-check*	F3D FINAL handback of the v3.5.2 re-check	line 1703 row "8" col 3	tag in REPORT-FEA3D-v352.md FINAL handback of the v3.5.2 re-check: FEA3D-2 report §13 open items
T4070	N-F1 (>= 21.96 MPa inside the 23.7 MPa acceptance); 188 (3) contact FE at the tie edge	DERIVED	F3D FINAL handback of the v3.5.2 re-check	line 1703 row "8" col 4	FEA3D-2 report §13 open items
T4071	PD-01R / PD-01S (MF)	DERIVED	F3D FINAL handback of the v3.5.2 re-check	line 1704 row "9" col 1	FEA3D-2 report §13 open items
T4072	-0.315 / -0.451 ... -0.444 / -0.555; primary-only above +20	SIM	F3D FINAL handback of the v3.5.2 re-check	line 1704 row "9" col 2	FEA3D-2 report §13 open items
T4073	fea3d (model)	DECIDED *Lead decision, per REPORT-FEA3D-v352.md FINAL handback of the v3.5.2 re-check*	F3D FINAL handback of the v3.5.2 re-check	line 1704 row "9" col 3	tag in REPORT-FEA3D-v352.md FINAL handback of the v3.5.2 re-check: FEA3D-2 report §13 open items
T4074	PD-02 is rigid in the model; a deformable PD-02 if the lead asks for it (lead 07:51Z ...	DERIVED	F3D FINAL handback of the v3.5.2 re-check	line 1704 row "9" col 4	FEA3D-2 report §13 open items
T4075	TC-04 BT node n42993 (q 0.040 / 0.027)	DERIVED	F3D FINAL handback of the v3.5.2 re-check	line 1705 row "10" col 1	FEA3D-2 report §13 open items
T4076	LC3 hot bound +0.228 / -0.004, F2 hot bound +0.223 / -0.008 [DERIVED node-wise lower ...	SIM	F3D FINAL handback of the v3.5.2 re-check	line 1705 row "10" col 2	FEA3D-2 report §13 open items
T4077	lead (ruled 2026-10-21T03:30:46Z (2): the settling route NOT RUN, final, for disk)	DECIDED *Lead decision, per REPORT-FEA3D-v352.md FINAL handback of the v3.5.2 re-check*	F3D FINAL handback of the v3.5.2 re-check	line 1705 row "10" col 3	tag in REPORT-FEA3D-v352.md FINAL handback of the v3.5.2 re-check: FEA3D-2 report §13 open items
T4078	the settling route (remesh TC-04 to q >= 0.1, re-solve the hot-bound bases and the 17 R3 ...	DERIVED	F3D FINAL handback of the v3.5.2 re-check	line 1705 row "10" col 4	FEA3D-2 report §13 open items
T4079	OP-moment residue (MF observation, 4.5, 9B.3)	DERIVED	F3D FINAL handback of the v3.5.2 re-check	line 1706 row "11" col 1	FEA3D-2 report §13 open items
T4080	L1 Q [539, 484] N mm/MPa; R3 poses [304-307, 2298]; cells stand as found with the ...	SIM	F3D FINAL handback of the v3.5.2 re-check	line 1706 row "11" col 2	FEA3D-2 report §13 open items
T4081	lead (ruled 03:30:46Z (2): NOT RUN, final, for disk)	DECIDED *Lead decision, per REPORT-FEA3D-v352.md FINAL handback of the v3.5.2 re-check*	F3D FINAL handback of the v3.5.2 re-check	line 1706 row "11" col 3	tag in REPORT-FEA3D-v352.md FINAL handback of the v3.5.2 re-check: FEA3D-2 report §13 open items
T4082	a moment-free OP step on the shared route of item 10, if disk is ever granted	DERIVED	F3D FINAL handback of the v3.5.2 re-check	line 1706 row "11" col 4	FEA3D-2 report §13 open items
T4083	J-FWD contact history dependence (10A.4, 10A.5)	DERIVED	F3D FINAL handback of the v3.5.2 re-check	line 1707 row "12" col 1	FEA3D-2 report §13 open items
T4084	the soak states depend on the predecessor state at mu 0 and mu 0.2; carried as the ...	SIM	F3D FINAL handback of the v3.5.2 re-check	line 1707 row "12" col 2	FEA3D-2 report §13 open items
T4085	fea3d / lead	DECIDED *Lead decision, per REPORT-FEA3D-v352.md FINAL handback of the v3.5.2 re-check*	F3D FINAL handback of the v3.5.2 re-check	line 1707 row "12" col 3	tag in REPORT-FEA3D-v352.md FINAL handback of the v3.5.2 re-check: FEA3D-2 report §13 open items
T4086	a history-robust contact model (mortar, or a softened penalty at the ring) is not run ...	DERIVED	F3D FINAL handback of the v3.5.2 re-check	line 1707 row "12" col 4	FEA3D-2 report §13 open items
T4087	J-FWD on the FINAL TC-01 (2.1, 10A.2)	DERIVED	F3D FINAL handback of the v3.5.2 re-check	line 1708 row "13" col 1	FEA3D-2 report §13 open items
T4088	all J-FWD FE except cD on the interim TC-01 (d10.5 flats; the final d11.2 flats are ...	SIM	F3D FINAL handback of the v3.5.2 re-check	line 1708 row "13" col 2	FEA3D-2 report §13 open items
T4089	fea3d / Mechanical	DECIDED *Lead decision, per REPORT-FEA3D-v352.md FINAL handback of the v3.5.2 re-check*	F3D FINAL handback of the v3.5.2 re-check	line 1708 row "13" col 3	tag in REPORT-FEA3D-v352.md FINAL handback of the v3.5.2 re-check: FEA3D-2 report §13 open items
T4090	effect on the other J-FWD rows ASSUMED small; Mechanical cites the same basis in its ...	DERIVED	F3D FINAL handback of the v3.5.2 re-check	line 1708 row "13" col 4	FEA3D-2 report §13 open items
T4091	LC2 hand form, TC-01 ligament (information, 10A.13)	DERIVED	F3D FINAL handback of the v3.5.2 re-check	line 1709 row "13a" col 1	FEA3D-2 report §13 open items
T4092	LC2N Pm+Pb +0.016 / -0.183, LC2M +0.266 / +0.021 with SF 1.5 / 2.0 on the total stress ...	SIM	F3D FINAL handback of the v3.5.2 re-check	line 1709 row "13a" col 2	FEA3D-2 report §13 open items
T4093	lead (ruled) / datasheet owner	DECIDED *Lead decision, per REPORT-FEA3D-v352.md FINAL handback of the v3.5.2 re-check*	F3D FINAL handback of the v3.5.2 re-check	line 1709 row "13a" col 3	tag in REPORT-FEA3D-v352.md FINAL handback of the v3.5.2 re-check: FEA3D-2 report §13 open items

ID	Value	Class	Source	Key	Source tag / formula
T4094	shown as information beside the cells (10A, the datasheet J-FWD table); no closure ...	DERIVED	F3D FINAL handback of the v3.5.2 re-check	line 1709 row "13a" col 4	FEA3D-2 report §13 open items
T4095	Bearing, washer and screw datasheets	DERIVED	F3D FINAL handback of the v3.5.2 re-check	line 1710 row "14" col 1	FEA3D-2 report §13 open items
T4096	HK0808 C0 / C, PTFE washers: TO-VERIFY; loads in 6.1 and 8	SIM	F3D FINAL handback of the v3.5.2 re-check	line 1710 row "14" col 2	FEA3D-2 report §13 open items
T4097	Mechanical	DECIDED *Lead decision, per REPORT-FEA3D-v352.md FINAL handback of the v3.5.2 re-check*	F3D FINAL handback of the v3.5.2 re-check	line 1710 row "14" col 3	tag in REPORT-FEA3D-v352.md FINAL handback of the v3.5.2 re-check: FEA3D-2 report §13 open items
T4098	datasheet values into the hub (none invented here)	DERIVED	F3D FINAL handback of the v3.5.2 re-check	line 1710 row "14" col 4	FEA3D-2 report §13 open items
T4099	7075-T6 temperature table (RM2-10, RM1-06)	DERIVED	F3D FINAL handback of the v3.5.2 re-check	line 1711 row "15" col 1	FEA3D-2 report §13 open items
T4100	allowables temperature-independent 470 / 540 MPa [LITERATURE]	SIM	F3D FINAL handback of the v3.5.2 re-check	line 1711 row "15" col 2	FEA3D-2 report §13 open items
T4101	Mechanical	DECIDED *Lead decision, per REPORT-FEA3D-v352.md FINAL handback of the v3.5.2 re-check*	F3D FINAL handback of the v3.5.2 re-check	line 1711 row "15" col 3	tag in REPORT-FEA3D-v352.md FINAL handback of the v3.5.2 re-check: FEA3D-2 report §13 open items
T4102	the 188 (5) table and the mill certificate	DERIVED	F3D FINAL handback of the v3.5.2 re-check	line 1711 row "15" col 4	FEA3D-2 report §13 open items
T4103	Servo output stiffness and inertia	DERIVED	F3D FINAL handback of the v3.5.2 re-check	line 1712 row "16" col 1	FEA3D-2 report §13 open items
T4104	k_s 58-174 N m/rad and J 0.76-2.7e-3 kg m2 ASSUMED; rocking 14.9-29.0 Hz at neutral (9A)	SIM	F3D FINAL handback of the v3.5.2 re-check	line 1712 row "16" col 2	FEA3D-2 report §13 open items
T4105	Electronics / bench B-TVC-1; REV-GNC	DECIDED *Lead decision, per REPORT-FEA3D-v352.md FINAL handback of the v3.5.2 re-check*	F3D FINAL handback of the v3.5.2 re-check	line 1712 row "16" col 3	tag in REPORT-FEA3D-v352.md FINAL handback of the v3.5.2 re-check: FEA3D-2 report §13 open items
T4106	bench measurement	DERIVED	F3D FINAL handback of the v3.5.2 re-check	line 1712 row "16" col 4	FEA3D-2 report §13 open items
T4107	Runs not made (section 11)	DERIVED	F3D FINAL handback of the v3.5.2 re-check	line 1713 row "17" col 1	FEA3D-2 report §13 open items
T4108	c71x tie-impact runs and the J-FWD spring-footprint run (NOT RUN: no space after js5b ...	SIM	F3D FINAL handback of the v3.5.2 re-check	line 1713 row "17" col 2	FEA3D-2 report §13 open items
T4109	fea3d (if disk space or memory is granted)	DECIDED *Lead decision, per REPORT-FEA3D-v352.md FINAL handback of the v3.5.2 re-check*	F3D FINAL handback of the v3.5.2 re-check	line 1713 row "17" col 3	tag in REPORT-FEA3D-v352.md FINAL handback of the v3.5.2 re-check: FEA3D-2 report §13 open items
T4110	as section 11	DERIVED	F3D FINAL handback of the v3.5.2 re-check	line 1713 row "17" col 4	FEA3D-2 report §13 open items
T4111	0.301	CALC	MMXD v352	/cell_value_changes/0/old_MS_y_MS_u/1	CALC / SIM / DERIVED as in the table; a table re-run, not physical approval
T4112	0.278	CALC	MMXD v352	/cell_value_changes/0/new_MS_y_MS_u/1	CALC / SIM / DERIVED as in the table; a table re-run, not physical approval
T4113	**c71x** (the tie-impact runs `pose_stop045_c71x`, `pose_neutral_c71x`, `lc1v351_c71x` ...	DERIVED	F3D FINAL handback of the v3.5.2 re-check	§11 line 1550	FEA3D-2 report §11
T4114	**The J-FWD spring-footprint run** (route (i) for the flat bearing under the screw ...	DERIVED	F3D FINAL handback of the v3.5.2 re-check	§11 line 1551	FEA3D-2 report §11
T4115	**The shared re-solve route** for the TC-04 BT cells (hot bounds -0.004 / -0.008 u at ...	DERIVED	F3D FINAL handback of the v3.5.2 re-check	§11 line 1552	FEA3D-2 report §11
T4116	**js5c** (the TC-01 / TC-02 fit contact): dropped. The fit closure is CLOSED on the Lame ...	DERIVED	F3D FINAL handback of the v3.5.2 re-check	§11 line 1553	FEA3D-2 report §11
T4117	No solved 21.5 s thermal state: the 21.5 s TC-02 rows are constant-stress [DERIVED] ...	DERIVED	F3D FINAL handback of the v3.5.2 re-check	§11 line 1556	FEA3D-2 report §11
T4118	No LC1 on the v3.5.2 NF2b merged field: the v3.5.1 merged map stands for TC-02 / TC-03 ...	DERIVED	F3D FINAL handback of the v3.5.2 re-check	§11 line 1557	FEA3D-2 report §11
T4119	No R4 lever-root mesh: where R2 → R3 changes by more than 5 %, the cell quotes the p = ...	DERIVED	F3D FINAL handback of the v3.5.2 re-check	§11 line 1558	FEA3D-2 report §11
T4120	No compliant-airframe v3.5.2 run: the datasheet rows carry the v3.5.1 compliant rows ...	DERIVED	F3D FINAL handback of the v3.5.2 re-check	§11 line 1559	FEA3D-2 report §11
T4121	The runs that would isolate the stop045 K change (`pose_stop045_v352L1`, `v352L1x` ...	DERIVED	F3D FINAL handback of the v3.5.2 re-check	§11 line 1560	FEA3D-2 report §11
T4122	**lc1v352hb** (rev-1 lever, R3 hot-bound base, default S-AX map; DONE 04:01:54Z) ...	DERIVED	F3D FINAL handback of the v3.5.2 re-check	§11 line 1563	FEA3D-2 report §11
T4123	The R2 / R3 stop045 runs without a check_fbd JSON are ACCEPTED as found (lead 21:47:29Z ...	DERIVED	F3D FINAL handback of the v3.5.2 re-check	§11 line 1564	FEA3D-2 report §11
T4124	`modal_v351ca2b_c71` is superseded by `modal_v351ca2c_c71` for RM2-06 only. The 9A mode ...	DERIVED	F3D FINAL handback of the v3.5.2 re-check	§11 line 1565	FEA3D-2 report §11
T4125	The v3.5.1 K files (`k_stop_c71`, `k_cone_c71`, `k_range_v351`) are superseded for every ...	DERIVED	F3D FINAL handback of the v3.5.2 re-check	§11 line 1566	FEA3D-2 report §11
T4126	DESIGN + SIL + schematic, STAGED for the lead (06-electronics/staging_v2.12/ ...	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/status	electronics.json v4.1
T4127	PJRC Teensy 4.1 (i.MX RT1062 600 MHz, native SDIO microSD, 3 CAN incl. 1 CAN-FD)	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/ecu/mcu	electronics.json v4.1
T4128	120, 60, 1.6 mm	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/ecu/pcb_outline_mm	tag in electronics.json v4.1: DESIGN v1 (4-layer, hand-solderable)
T4129	128, 68, 32 mm	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/ecu/enclosure_envelope_mm	tag in electronics.json v4.1: DESIGN v1 reservation (PETG or 1 mm Al); +25 mm connector/cable clearance on both long sides
T4130	version: v2.5 (KiCad 7.0.11, generated from hardware/kicad_gen/ecu_design.py); erc: v: 0 ...	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/ecu/schematic	electronics.json v4.1
T4131	v2.12	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/firmware/version	electronics.json v4.1
T4132	PJRC Teensy 4.1 (i.MX RT1062). HAL NOT written (RE-06; LEAD 166 (9): Electronics open ...	TO-VERIFY	ELE v4.1	/firmware/target	(class set at the call site: TO-VERIFY)
T4133	hooks_build: 326 passed, 0 failed (results/sil_v2p11/test_core_log.txt ...	SIM	ELE v4.1	/firmware/unit_tests	(class set at the call site: SIM)
T4134	comparison: results/sil_v2p11/compare_v2p10_v2p11.json 7bb97d757dbdcc46 (+ .txt) ...	SIM	ELE v4.1	/firmware/sil_record_v2p11	(class set at the call site: SIM)
T4135	ISO 11898-2 CAN, classic 2.0B 1 Mbit/s, 11-bit IDs; CAN-FD-capable transceivers so FD is ...	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/can_interface/physical	tag in electronics.json v4.1: DESIGN
T4136	v3 (firmware v2.10) → v4 (firmware v2.11, SIL record 8c1a534c3af1b0ba, MANIFEST ...	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/change_summary_v3_to_v4/0	electronics.json v4.1
T4137	Until v4.2 this line governs v4.1 where they differ: its STAGED status text is superseded;	DECIDED *Lead decision, line 266 (7)*	CL 266 lines	L266(7)[from "Until v4.2 this line governs v4.1 where they differ: its STAGED status text is superseded;" to "Until v4.2 this line governs v4.1 where they …	lead (decision) 2026-10-21 (line sha 6f3181b859f02937)
T4138	Reproduction [CALC / SIM, a repeat]: rebuild byte-identical (4 binaries); test_core 332 ...	DECIDED *Lead decision, line 266 (4)*	CL 266 lines	L266(4)[from "Reproduction [CALC / SIM, a repeat]" to "as predicted."]	lead (decision) 2026-10-21 (line sha 6f3181b859f02937)
T4139	332 passed, 0 failed	SIM	ELE v4.1	/firmware/v2p12/tests/test_core	(class set at the call site: SIM)
T4140	331 passed, 0 failed	SIM	ELE v4.1	/firmware/v2p12/tests/test_core_nohooks	(class set at the call site: SIM)
T4141	seat compensation (R3-01 option (b)): Pc_comp = Pc_in - coef x (T_seat - T_seat,ARM) ...	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/firmware/v2p12/changes/0	electronics.json v4.1
T4142	ARM sanity static term: |gauge - coef x (T_seat - T_cal)| <= kPcArmZeroMax_Pa (T_cal per ...	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/firmware/v2p12/changes/1	electronics.json v4.1
T4143	E4-03 (canArm subtracts the static term twice after an earlier ARM; fixed in v2.13)	DECIDED *Lead decision, line 266 (7)*	CL 266 lines	L266(7)[from "E4-03 (canArm subtracts the static term twice after an earlier ARM" to "fixed in v2.13)"]	lead (decision) 2026-10-21 (line sha 6f3181b859f02937)

ID	Value	Class	Source	Key	Source tag / formula
T4144	TB-state (f3 b0/b1): HW_OVERPRESSURE, PC_OVERPRESSURE, HARD_START, OVER_TEMP, PC_SENSOR ...	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/firmware/v2p12/changes/2	electronics.json v4.1
T4145	flight J-FWD indication (f3 b5): LIGHT_SPIKE raw >= kJfwdLsRawAbs_Pa 47.5 bar abs or raw ...	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/firmware/v2p12/changes/3	electronics.json v4.1
T4146	E4-N03 (47.5 against 47.75 bar abs, TO-VERIFY)	DECIDED *Lead decision, line 266 (8)*	CL 266 lines	L266(8)[from "E4-N03 (47.5 against 47.75 bar abs, TO-VERIFY)" to "E4-N03 (47.5 against 47.75 bar abs, TO-VERIFY)"]	lead (decision) 2026-10-21 (line sha 6f3181b859f02937)
T4147	torch re-zero limit kPtorchRezeroMax_Pa 0.60 bar (070BG basis; f3 b6 on a rejection)	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/firmware/v2p12/changes/4	electronics.json v4.1
T4148	E4-N04 (the P_torch re-base needs compensation or a photodiode-guarded gate; torch local ...	DECIDED *Lead decision, line 266 (8)*	CL 266 lines	L266(8)[from "E4-N04 (the P_torch re-base needs compensation or a photodiode-guarded gate; torch local heat MISSING)" to "E4-N04 (the P_torch re-base needs …	lead (decision) 2026-10-21 (line sha 6f3181b859f02937)
T4149	kIgnBaseCheck_Pa = kIgnResidual_Pa + kIgnBaseDrift_Pa 0.06 bar + 0.10 bar	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/firmware/v2p12/changes/5	electronics.json v4.1
T4150	purge FINAL: kPurgeFlow_s 2.43 s, kHbleedCap_s 2.39 s, both calibration constants ...	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/firmware/v2p12/changes/6	electronics.json v4.1
T4151	log v13: + seat_comp_hPa, seat_ref_dK, f3_flags (185 B)	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/firmware/v2p12/changes/7	electronics.json v4.1
T4152	5000 Pa	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/sensors/pressure_transducers/coast_rezero/span	tag in electronics.json v4.1: DESIGN
T4153	30000 Pa	DECIDED *Lead decision, line 264 (2) (kPcRezeroMax_Pa stays 0.30 bar on the compensated residual); v4.1 is the source of record (266 (7))*	ELE v4.1	/sensors/pressure_transducers/coast_rezero/max	tag in electronics.json v4.1: LEAD 2026-10-13 B-PC-1 target
T4154	0.5 s	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/sensors/pressure_transducers/coast_rezero/second_rezero_max_age	tag in electronics.json v4.1: DESIGN
T4155	second re-zero ending before PURGE/RELIGHT (RE-07); flight NOT_VENTED = flag, stand = ...	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/sensors/pressure_transducers/coast_rezero/rule	electronics.json v4.1
T4156	PC: Kulite XTL-190SM-2000A (C1, seat-mounted on the head, J601); P_FEED: Kulite ...	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/sensors/pressure_transducers/selection	tag in electronics.json v4.1: DESIGN pc_transducer_v3 (LEAD 205, 206 (3)); BOM v2.3 lines 199, 200, 251
T4157	the 264 (1)-(8) folds such as P_TORCH XTL-190SM-2000A	DECIDED *Lead decision, line 266 (7)*	CL 266 lines	L266(7)[from "the 264 (1)-(8) folds such as P_TORCH XTL-190SM-2000A" to "the 264 (1)-(8) folds such as P_TORCH XTL-190SM-2000A"]	lead (decision) 2026-10-21 (line sha 6f3181b859f02937)
T4158	206.8 bar abs	DATASHEET	CL 266 lines	L264(4)[from "P_torch (J901" to "same part as P_mid (ENG-BL-08) and C1."] [regex proof ([\d.]+) bar abs \[DATASHEET\]]	lead (decision) 2026-10-21 (line sha 7cdbe51041ae6b8b); (class set at the call site: DATASHEET)
T4159	166.14 bar abs	DERIVED	CL 266 lines	L264(4)[from "P_torch (J901" to "same part as P_mid (ENG-BL-08) and C1."] [regex against the ([\d.]+) bar abs requirement]	lead (decision) 2026-10-21 (line sha 7cdbe51041ae6b8b); (class set at the call site: DERIVED)
T4160	U330 on the C1 chain: trip 32.59 bar abs nominal, band 31.08-34.11 bar abs; firmware ...	DERIVED	ELE v4.1	/sensors/pressure_transducers/hw_ovp	DERIVED pc_transducer_v3 F
T4161	TC1 type K case az 90 (U604); TC2 type K case az 270 (U605); TC3 type K nozzle holder ...	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/sensors/thermocouples/channels	electronics.json v4.1
T4162	TC4 is the C1 seat TC (MAX31856 U904, J904, BOM row 205), not bore top (the 264 (2) ...	DECIDED *Lead decision, line 266 (7)*	CL 266 lines	L266(7)[from "TC4 is the C1 seat TC (MAX31856 U904, J904, BOM row 205), not bore top" to "the file keeps the old label until v4.2)"]	lead (decision) 2026-10-21 (line sha 6f3181b859f02937)
T4163	423.15 K	DECIDED *Lead decision, line 243 (3)*	ELE v4.1	/sensors/thermocouples/case_abort	tag in electronics.json v4.1: LEAD CHANGELOG 243 (3)
T4164	403.15 K	DECIDED *Lead decision of 2026-10-20*	ELE v4.1	/sensors/thermocouples/relight_refusal	tag in electronics.json v4.1: LEAD 2026-10-20T21:46:49Z item 5, Electronics basis
T4165	in-burn abort on TC1 / TC2 above the case abort for the persistence (abort_matrix ...	DECIDED *Lead decision, per electronics.json v4.1*	ELE v4.1	/sensors/thermocouples/case_abort_rule	tag in electronics.json v4.1:
T4166	0.2 s	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/sensors/thermocouples/persistence	tag in electronics.json v4.1: MECH 2026-10-13: 0.2 s hold
T4167	403.15 K	DECIDED *Lead decision of 2026-10-20*	ELE v4.1	/sensors/thermocouples/start_and_arm_limit	tag in electronics.json v4.1: LEAD 2026-10-20T21:46:49Z item 5, Electronics basis
T4168	297.15 K	DECIDED *Lead decision, per electronics.json v4.1*	ELE v4.1	/sensors/thermocouples/stand_arm_max	tag in electronics.json v4.1: LEAD ADOPTED
T4169	523.15 K	ASSUMED	ELE v4.1	/sensors/thermocouples/nozzle_holder_limit	ASSUMED
T4170	FA-1 head on J902 (Si photodiode + log TIA, 0-5 V), part TO-SELECT (BOM row); valid 0.02 ...	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/sensors/photodiode	electronics.json v4.1
T4171	AS5048A x3 (gimbal A/B J610/J611, pintle crank J612), SPI1 1 MHz	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/sensors/encoders	electronics.json v4.1
T4172	B-DPC-1: PCB 113B22 (5 kpsi = 344.75 bar at +/-5 V; maximum pressure 15 kpsi = 1034 bar) ...	DATASHEET *Lead decision, line 172 (5)*	ELE v4.1	/sensors/stand_dynamic_pc	LEAD CHANGELOG 172 (5), 194 (7), 203 (7); DATASHEET pcb.com 113B22, NI 377849B-01
T4173	423.15 K	DECIDED *Lead decision, line 243 (3)*	ELE v4.1	/limits_v2p11/0/value	tag in electronics.json v4.1: LEAD CHANGELOG 243 (3)
T4174	A = 150 C (LEAD 243 (3)); false-trip side 7.78 K worst case / 3.55 K RSS inside +12.3 K ...	DERIVED	ELE v4.1	/limits_v2p11/0/basis	(class set at the call site: DERIVED)
T4175	403.15 K	DECIDED *Lead decision of 2026-10-20*	ELE v4.1	/limits_v2p11/1/value	tag in electronics.json v4.1: LEAD 2026-10-20T21:46:49Z item 5, Electronics basis
T4176	A1 purge-to-end-of-relight rise 11.9 K + 1 s longer relight 3.0 K + 1 K relative channel ...	DERIVED	ELE v4.1	/limits_v2p11/1/basis	(class set at the call site: DERIVED)
T4177	523.15 K	ASSUMED	ELE v4.1	/limits_v2p11/2/value	ASSUMED
T4178	250 C ASSUMED (unchanged); the ARM refusal on it is accepted	DERIVED	ELE v4.1	/limits_v2p11/2/basis	(class set at the call site: DERIVED)
T4179	297.15 K	DECIDED *Lead decision, per electronics.json v4.1*	ELE v4.1	/limits_v2p11/3/value	tag in electronics.json v4.1: LEAD ADOPTED
T4180	24 C: the 12.26 K false-trip margin is a 20 C-start margin (dT_station / dT0 about ...	DERIVED	ELE v4.1	/limits_v2p11/3/basis	(class set at the call site: DERIVED)
T4181	2.39 s	DECIDED *Lead decision, line 256 (2)*	ELE v4.1	/limits_v2p11/4/value	tag in electronics.json v4.1: LEAD 256 (2) / 257 (2)
T4182	LEAD 203 (3) <= 2.55 s; 230 (3): the scheduled command span 2.45 s does not fit a 2.39 s ...	DERIVED	ELE v4.1	/limits_v2p11/4/basis	(class set at the call site: DERIVED)
T4183	1. limits_v2p11[4]: the basis text is stale (2.55 s / 76.05 g / 80.76 g). Replace it ...	DERIVED	RELE4 REVIEW-ELECTRONICS-R4 (REV-ELEC-4)	line 210 (starts "1. limits_v2p11[4]: the basis text is st")	REV-ELEC-4 review text (section 2), the correction that CHANGELOG line 266 (7) applies to v4.1 as written in the review
T4184	2.659 s	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/limits_v2p11/5/value	tag in electronics.json v4.1: DESIGN
T4185	LTC6993-2 RSET 63.4k, DIVCODE 8 (AS-RECALLED, TO-VERIFY): 2.577-2.741 s with the 3.1 % ...	DERIVED	ELE v4.1	/limits_v2p11/5/basis	(class set at the call site: DERIVED)
T4186	0.076 kg	DECIDED *Lead decision, line 216 (5)*	ELE v4.1	/limits_v2p11/6/value	tag in electronics.json v4.1: LEAD 216 (5)
T4187	LEAD 216 (5): the limit is the vented mass; one flow basis 29.25 g/s (LEAD 230 (3)) ...	DERIVED	ELE v4.1	/limits_v2p11/6/basis	(class set at the call site: DERIVED)
T4188	0.015 s	DERIVED	ELE v4.1	/limits_v2p11/7/value/kRideOvpTime_s	DERIVED fa1_chain_v2p11: longest continuous filtered > 30 bar abs 12.0 ms; LEAD 216 (3) >= 13 ms + margin: 13 + 2 ms; grid: first zero-trip persistence 15 ms on every C1 chain
T4189	0.015 s	DERIVED	ELE v4.1	/limits_v2p11/7/value/kRideWinPostE2_s	DERIVED: latest end of a filtered > 30 bar episode E2 + 7.3 ms (C1 chains; reviewer 070BG 8.0 ms); LEAD 216 (3) >= E2 + 8 ms + margin: 8 + 7 ms
T4190	0.007 s	DERIVED	ELE v4.1	/limits_v2p11/7/value/kRideT45_s	DERIVED: longest continuous raw > 45 bar abs 5.0 ms + 2 ms (head history 4.40 ms)
T4191	2	DECIDED *Lead decision, line 216 (3)*	ELE v4.1	/limits_v2p11/7/value/kRideOverrangeN	tag in electronics.json v4.1: LEAD 216 (3)

ID	Value	Class	Source	Key	Source tag / formula
T4192	0.012825 s	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/limits_v2p11/7/value/kRideCmpTauNom_s	tag in electronics.json v4.1: DESIGN ride-through BOM variant
T4193	0.0121 s	DERIVED	ELE v4.1	/limits_v2p11/7/value/kRideCmpTauMin_s	DERIVED
T4194	0.013552 s	DERIVED	ELE v4.1	/limits_v2p11/7/value/kRideCmpTauMax_s	DERIVED
T4195	see ride_through.constants (each with its basis)	DERIVED	ELE v4.1	/limits_v2p11/7/basis	(class set at the call site: DERIVED)
T4196	4. limits_v2p11[7] still says "(CANDIDATE, default OFF)". Restate it as ADOPTED on the ...	DERIVED	RELE4 REVIEW-ELECTRONICS-R4 (REV-ELEC-4)	line 218 (starts "4. limits_v2p11[7] still says")	REV-ELEC-4 review text (section 2), the correction that CHANGELOG line 266 (7) applies to v4.1 as written in the review
T4197	C1 206.8 bar abs	DATASHEET	ELE v4.1	/limits_v2p11/8/value	DATASHEET (1.5 × 2000 psi)
T4198	requirement >= 166.1 bar abs (1.1 x FA1S 151.04, 230 (2)), replacing 112 bar abs	DERIVED	ELE v4.1	/limits_v2p11/8/basis	(class set at the call site: DERIVED)
T4199	4.75 V	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/limits_v2p11/9/value/kPcSensorMax_V	tag in electronics.json v4.1: DESIGN
T4200	4.56 V	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/limits_v2p11/9/value/kPcAboveFs_V	tag in electronics.json v4.1: DESIGN v2.11, LEAD 203 (1)(a)
T4201	146.5 / 139.96 bar abs on the C1 transfer (pc_transducer_v3 E)	DERIVED	ELE v4.1	/limits_v2p11/9/basis	(class set at the call site: DERIVED)
T4202	3000000 Pa	DECIDED *Lead decision, per electronics.json v4.1*	ELE v4.1	/limits_v2p11/10/value/kPcOverPressureAbs_Pa	tag in electronics.json v4.1: LEAD via COMB
T4203	0.005 s	DECIDED *Lead decision of 2026-10-13*	ELE v4.1	/limits_v2p11/10/value/kPcOverPressureTime_s	tag in electronics.json v4.1: LEAD REV-ELEC-1 RE-20, 2026-10-13: FILTERED channel, 5 ms
T4204	filtered absolute Pc (RE-20); hardware U330 on C1 32.59 bar abs (31.08-34.11)	DERIVED	ELE v4.1	/limits_v2p11/10/basis	(class set at the call site: DERIVED)
T4205	0.8 s	DECIDED *Lead decision, line 227 (5)*	ELE v4.1	/limits_v2p11/11/value/kR2WindowMax_s	tag in electronics.json v4.1: LEAD CHANGELOG 227 (5)
T4206	0.85 s	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/limits_v2p11/11/value/kR2HoldBackstop_s	tag in electronics.json v4.1: DESIGN, LEAD 216 (2)(b)
T4207	0.00021776 m	DECIDED *Lead decision, line 230 (4)*	ELE v4.1	/limits_v2p11/11/value/kR2HoldLift_m	tag in electronics.json v4.1: COMB v2.4b start-level lift at the A1 relight state, ECU hold-lift row, LEAD 230 (4)
T4208	10	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/limits_v2p11/11/value/kFcDrogueResendMax	tag in electronics.json v4.1: DESIGN, FC side
T4209	0.8 s for W10 < 6 m/s (227 (5), confirmed with conditions by S17 v2.11); backstop = ...	DERIVED	ELE v4.1	/limits_v2p11/11/basis	(class set at the call site: DERIVED)
T4210	288.15 K	DECIDED *Requirement, ENG-12*	ELE v4.1	/limits_v2p11/12/value/kTankIgnMin_K	tag in electronics.json v4.1: REQ ENG-12 l.106, REV-ELEC-1 RE-17
T4211	0.42 K	DERIVED	ELE v4.1	/limits_v2p11/12/value/kTankColdArmMargin_K	DERIVED
T4212	ARM margin = 1.5 x the PREPRESS Tsat drop 0.23 K + repeatability 0.07 K, rounded up (SIL ...	DERIVED	ELE v4.1	/limits_v2p11/12/basis	(class set at the call site: DERIVED)
T4213	0.45 s	DECIDED *Lead decision, line 166 (7)*	ELE v4.1	/limits_v2p11/13/value	tag in electronics.json v4.1: LEAD CHANGELOG 166 (7); phase / flight extension DESIGN
T4214	E2 .. E2 + 0.45 s on the stand and in flight (four conditions ...	DERIVED	ELE v4.1	/limits_v2p11/13/basis	(class set at the call site: DERIVED)
T4215	76000 Pa	DECIDED *Lead decision, line 264 (2) (klgnBaseCheck_Pa 0.76 / 0.46 bar, default / B-PC-1-shown build)*	ELE v4.1	/limits_v2p11/14/value	tag in electronics.json v4.1: LEAD form + R3-01 item 4
T4216	residual + 0.1 bar; per location (pad / stand refuse, A1 keeps / zeroes P0)	DERIVED	ELE v4.1	/limits_v2p11/14/basis	(class set at the call site: DERIVED)
T4217	2. limits_v2p11[14]: the basis "residual + 0.1 bar" lacks the drift term. Write ...	DERIVED	RELE4 REVIEW-ELECTRONICS-R4 (REV-ELEC-4)	line 211 (starts "2. limits_v2p11[14]: the basis")	REV-ELEC-4 review text (section 2), the correction that CHANGELOG line 266 (7) applies to v4.1 as written in the review
T4218	3. Missing limit rows, each with its value, tag and source:	DERIVED	RELE4 REVIEW-ELECTRONICS-R4 (REV-ELEC-4)	line 212 (starts "3. Missing limit rows, each with its val")	REV-ELEC-4 review text (section 2), the correction that CHANGELOG line 266 (7) applies to v4.1 as written in the review
T4219	the seat constants (kPcSeatCoefMax_PaPerK 1.0e4, kPcSeatDevMax_K 150, kPcSeatTcMin_K / ...	DERIVED	RELE4 REVIEW-ELECTRONICS-R4 (REV-ELEC-4)	line 213 (starts "the seat constants (kPcSeatCoefMax_PaPer")	REV-ELEC-4 review text (section 2), the correction that CHANGELOG line 266 (7) applies to v4.1 as written in the review
T4220	klgnBaseDrift 0.06 bar;	DERIVED	RELE4 REVIEW-ELECTRONICS-R4 (REV-ELEC-4)	line 214 (starts "kIgnBaseDrift 0.06 bar;")	REV-ELEC-4 review text (section 2), the correction that CHANGELOG line 266 (7) applies to v4.1 as written in the review
T4221	kPtorchRezeroMax_Pa 0.60 bar;	DERIVED	RELE4 REVIEW-ELECTRONICS-R4 (REV-ELEC-4)	line 215 (starts "kPtorchRezeroMax_Pa 0.60 bar;")	REV-ELEC-4 review text (section 2), the correction that CHANGELOG line 266 (7) applies to v4.1 as written in the review
T4222	kJfwdLsRawAbs 47.5 bar abs.	DERIVED	RELE4 REVIEW-ELECTRONICS-R4 (REV-ELEC-4)	line 216 (starts "kJfwdLsRawAbs 47.5 bar abs.")	REV-ELEC-4 review text (section 2), the correction that CHANGELOG line 266 (7) applies to v4.1 as written in the review
T4223	P0 = mean of filtered / raw gauge Pc over [E1 - 20 ms, E1]. Check against the auto-zero ...	DECIDED *Lead decision, line 216 (4)*	ELE v4.1	/light_rule/rule	tag in electronics.json v4.1: LEAD/COMB; test_core "216 (4) BASE_AMB per location"
T4224	80000 Pa	DECIDED *Lead decision, per electronics.json v4.1*	ELE v4.1	/light_rule/kIgnRise_Pa	tag in electronics.json v4.1: LEAD/COMB
T4225	10	DECIDED *Lead decision, per electronics.json v4.1*	ELE v4.1	/light_rule/kIgnConfirmRelN	tag in electronics.json v4.1: LEAD/COMB
T4226	20	DECIDED *Lead decision, per electronics.json v4.1*	ELE v4.1	/light_rule/kIgnPreN	tag in electronics.json v4.1: LEAD/COMB
T4227	60000 Pa	DECIDED *Lead decision, per electronics.json v4.1*	ELE v4.1	/light_rule/kIgnResidual_Pa	tag in electronics.json v4.1: LEAD B-PC-1 target / COMB present band
T4228	76000 Pa	DECIDED *Lead decision, line 264 (2) (klgnBaseCheck_Pa 0.76 / 0.46 bar, default / B-PC-1-shown build)*	ELE v4.1	/light_rule/kIgnBaseCheck_Pa	tag in electronics.json v4.1: LEAD form + R3-01 item 4
T4229	180000 Pa	DECIDED *Lead decision, per electronics.json v4.1*	ELE v4.1	/light_rule/kIgnFloorAbs_Pa	tag in electronics.json v4.1: LEAD/COMB
T4230	(Pc - P0)/(Pc_map_lo - P0) < 0.4 for 20 ms in RUN; denominator >= 0.5 bar	DECIDED *Lead decision, per electronics.json v4.1*	ELE v4.1	/start_monitors/A3	tag in electronics.json v4.1: LEAD/COMB + DESIGN
T4231	HOLD band from E1 (low edge P0 - 0.3 bar), START band E2 → E3, RUN_MAP after E3; 50 ms ...	DECIDED *Lead decision, per electronics.json v4.1*	ELE v4.1	/start_monitors/A7	tag in electronics.json v4.1: LEAD/COMB + DESIGN
T4232	first RUN tick with the ramp done and Pc >= P0 + 0.9 (setpoint - P0), or E2 + 600 ms	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/start_monitors/E3	tag in electronics.json v4.1: DESIGN
T4233	CHANGELOG line 216 (lead, 2026-10-13): ...ed time, never a wait with the MV open. (8) ...	DECIDED *Lead decision, line 216 (8), as quoted in electronics.json v4.1*	ELE v4.1	/start_monitors/A7_flight_blank/decision	tag in electronics.json v4.1:
T4234	E2 .. E2 + 450 ms, A7 HIGH side only (stand and flight). Still ARMED in it: the filtered ...	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/start_monitors/A7_flight_blank/blind_window	tag in electronics.json v4.1: DESIGN config.h l.413-416
T4235	n: 1; condition: config.h comment corrected (REV-ELEC-2 cited v2.10 l.369-372); status ...	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/start_monitors/A7_flight_blank/conditions/0	(class set at the call site: DESIGN: a condition of the 216 (8) acceptance that the Electronics author reports DONE in electronics.json v4.1; REV-DS-1 D1-07)
T4236	n: 2; condition: electronics.json states the blind window and what stays armed; status ...	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/start_monitors/A7_flight_blank/conditions/1	(class set at the call site: DESIGN: a condition of the 216 (8) acceptance that the Electronics author reports DONE in electronics.json v4.1; REV-DS-1 D1-07)
T4237	n: 3; condition: the FC owner confirms no decision uses the A7 high flag (d6 b7 via ...	TO-VERIFY	ELE v4.1	/start_monitors/A7_flight_blank/conditions/2	(class set at the call site: TO-VERIFY: an open condition of the 216 (8) acceptance, status as stated in electronics.json v4.1; REV-DS-1 D1-07)
T4238	n: 4; condition: test v210-2 extended on d6 b7; status: DONE: PASS v2.11 LEAD 216 (8) ...	SIM	ELE v4.1	/start_monitors/A7_flight_blank/conditions/3	(class set at the call site: SIM: the status reports a passing SIL unit test of firmware v2.11, as recorded in electronics.json v4.1; REV-DS-1 D1-07)
T4239	no late-light abort (the v2.9 pressure-hold abort is deleted, RE-01, LEAD CHANGELOG ...	DECIDED *Lead decision, per electronics.json v4.1*	ELE v4.1	/start_monitors/light_spike_record	tag in electronics.json v4.1: LEAD; test_core "above-FS flag"
T4240	option (b): RUN (all comp / comp_over runs; comp_under refused NOT_VENTED at k >= 2-3: a ...	SIM	ELE v4.1	/r3_01_c1_drift/stand_relight_after_B1	SIM (S13T)
T4241	E4-02 (the coefficient-within-5-percent-at-k-3 acceptance, also in the 264 (2) wording ...	DECIDED *Lead decision, line 266 (7)*	CL 266 lines	L266(7)[from "E4-02 (the coefficient-within-5-percent-at-k-3 acceptance" to "with no relight objective claimed)"]	lead (decision) 2026-10-21 (line sha 6f3181b859f02937)
T4242	by_k: 1.0: A1_relight_burn_end_T0_0C_shift_bar: -0.114 ...	DERIVED	ELE v4.1	/r3_01_c1_drift/A8_U330	(no source tag; class set at the call site: DERIVED)

ID	Value	Class	Source	Key	Source tag / formula
T4243	E4-04 (the U330 ordering gains the stand relight light: the U330 low edge falls below ...	DECIDED *Lead decision, line 266 (7)*	CL 266 lines	`L266(7)[from "E4-04 (the U330 ordering gains the stand relight light" to "[DERIVED])"]`	lead (decision) 2026-10-21 (line sha 6f3181b859f02937)
T4244	1	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	`/abort_matrix/causes/0/code`	electronics.json v4.1
T4245	CMD	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	`/abort_matrix/causes/0/cause`	electronics.json v4.1
T4246	FC ABORT frame 0x010 (magic 0xA5 0x5A, code, seq, CRC-8; no sequence check) or CMD_STATE ...	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	`/abort_matrix/causes/0/trigger`	electronics.json v4.1
T4247	2	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	`/abort_matrix/causes/1/code`	electronics.json v4.1
T4248	HARD_START	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	`/abort_matrix/causes/1/cause`	electronics.json v4.1
T4249	IGNITION, or RUN within 300 ms of entry: filtered absolute Pc above 30 bar abs for 5 ms ...	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	`/abort_matrix/causes/1/trigger`	electronics.json v4.1
T4250	0.3 s	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	`/abort_matrix/causes/1/thresholds/kRunSettle_s`	tag in electronics.json v4.1: DESIGN
T4251	3000000 Pa	DECIDED *Lead decision, per electronics.json v4.1*	ELE v4.1	`/abort_matrix/causes/1/thresholds/kPcOverPressureAbs_Pa`	tag in electronics.json v4.1: LEAD via COMB
T4252	0.005 s	DECIDED *Lead decision of 2026-10-13*	ELE v4.1	`/abort_matrix/causes/1/thresholds/kPcOverPressureTime_s`	tag in electronics.json v4.1: LEAD REV-ELEC-1 RE-20, 2026-10-13: FILTERED channel, 5 ms
T4253	4.75 V	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	`/abort_matrix/causes/1/thresholds/kPcSensorMax_V`	tag in electronics.json v4.1: DESIGN
T4254	0	DECIDED *Lead decision, line 203 (1)*	ELE v4.1	`/abort_matrix/causes/1/thresholds/kRideThroughBuild`	tag in electronics.json v4.1: LEAD 203 (1)(b) / 216 (3) CANDIDATE
T4255	0.015 s	DERIVED	ELE v4.1	`/abort_matrix/causes/1/thresholds/kRideWinPostE2_s`	DERIVED: latest end of a filtered > 30 bar episode E2 + 7.3 ms (C1 chains; reviewer 070BG 8.0 ms); LEAD 216 (3) >= E2 + 8 ms + margin: 8 + 7 ms
T4256	0.015 s	DERIVED	ELE v4.1	`/abort_matrix/causes/1/thresholds/kRideOvpTime_s`	DERIVED fa1_chain_v2p11: longest continuous filtered > 30 bar abs 12.0 ms; LEAD 216 (3) >= 13 ms + margin: 13 + 2 ms; grid: first zero-trip persistence 15 ms on every C1 chain
T4257	2	DECIDED *Lead decision, line 216 (3)*	ELE v4.1	`/abort_matrix/causes/1/thresholds/kRideOverrangeN`	tag in electronics.json v4.1: LEAD 216 (3)
T4258	4500000 Pa	DECIDED *Lead decision, per electronics.json v4.1*	ELE v4.1	`/abort_matrix/causes/1/thresholds/kPcMeopLeadAbs_Pa`	tag in electronics.json v4.1: LEAD via COMB
T4259	0.007 s	DERIVED	ELE v4.1	`/abort_matrix/causes/1/thresholds/kRideT45_s`	DERIVED: longest continuous raw > 45 bar abs 5.0 ms + 2 ms (head history 4.40 ms)
T4260	0.012825 s	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	`/abort_matrix/causes/1/thresholds/kRideCmpTauNom_s`	tag in electronics.json v4.1: DESIGN ride-through BOM variant
T4261	800000 Pa	DECIDED *Lead decision, line 160*	ELE v4.1	`/abort_matrix/causes/1/thresholds/kLightSpikeLoAbs_Pa`	tag in electronics.json v4.1: LEAD CHANGELOG 160
T4262	4.56 V	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	`/abort_matrix/causes/1/thresholds/kPcAboveFs_V`	tag in electronics.json v4.1: DESIGN v2.11, LEAD 203 (1)(a)
T4263	3	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	`/abort_matrix/causes/2/code`	electronics.json v4.1
T4264	NO_IGNITION	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	`/abort_matrix/causes/2/cause`	electronics.json v4.1
T4265	no light within W from the lift by the relative rule: filtered AND raw rise over P0 at ...	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	`/abort_matrix/causes/2/trigger`	electronics.json v4.1
T4266	80000 Pa	DECIDED *Lead decision, per electronics.json v4.1*	ELE v4.1	`/abort_matrix/causes/2/thresholds/kIgnRise_Pa`	tag in electronics.json v4.1: LEAD/COMB
T4267	180000 Pa	DECIDED *Lead decision, per electronics.json v4.1*	ELE v4.1	`/abort_matrix/causes/2/thresholds/kIgnFloorAbs_Pa`	tag in electronics.json v4.1: LEAD/COMB
T4268	10	DECIDED *Lead decision, per electronics.json v4.1*	ELE v4.1	`/abort_matrix/causes/2/thresholds/kIgnConfirmRelN`	tag in electronics.json v4.1: LEAD/COMB
T4269	0.4 s	DECIDED *Lead decision of 2026-10-13*	ELE v4.1	`/abort_matrix/causes/2/thresholds/kIgnWindowFirst_s`	tag in electronics.json v4.1: LEAD 2026-10-13
T4270	0.65 s	DECIDED *Lead decision of 2026-10-13*	ELE v4.1	`/abort_matrix/causes/2/thresholds/kIgnWindowRelightLong_s`	tag in electronics.json v4.1: LEAD 2026-10-13
T4271	0.35 s	DECIDED *Lead decision of 2026-10-13*	ELE v4.1	`/abort_matrix/causes/2/thresholds/kIgnWindowFirstStand_s`	tag in electronics.json v4.1: LEAD 2026-10-13 fallback (b)
T4272	4	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	`/abort_matrix/causes/3/code`	electronics.json v4.1
T4273	FLAMEOUT	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	`/abort_matrix/causes/3/cause`	electronics.json v4.1
T4274	RUN, Pc valid: A3 ratio (Pc - P0)/(Pc_map_lo - P0) below 0.4 for 20 ms (Pc_map lagged by ...	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	`/abort_matrix/causes/3/trigger`	electronics.json v4.1
T4275	0.4	DECIDED *Lead decision of 2026-10-13*	ELE v4.1	`/abort_matrix/causes/3/thresholds/kFlameoutRel`	tag in electronics.json v4.1: LEAD/COMB 2026-10-13
T4276	0.02 s	DECIDED *Lead decision of 2026-10-13*	ELE v4.1	`/abort_matrix/causes/3/thresholds/kFlameoutTime_s`	tag in electronics.json v4.1: LEAD/COMB 2026-10-13
T4277	0.04 s	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	`/abort_matrix/causes/3/thresholds/kFlameoutMapTau_s`	tag in electronics.json v4.1: DESIGN
T4278	0.02 s	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	`/abort_matrix/causes/3/thresholds/kPhotoFlameoutTime_s`	tag in electronics.json v4.1: DESIGN, = A3 persistence
T4279	20	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	`/abort_matrix/causes/3/thresholds/kPhotoViewProveN`	tag in electronics.json v4.1: DESIGN
T4280	150000 Pa	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	`/abort_matrix/causes/3/thresholds/kFlameoutPc_Pa`	tag in electronics.json v4.1: DESIGN v2.2-v2.8.3, legacy rule only
T4281	0.35	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	`/abort_matrix/causes/3/thresholds/kFlameoutFrac`	tag in electronics.json v4.1: DESIGN v2.2-v2.8.3, legacy rule only
T4282	5	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	`/abort_matrix/causes/4/code`	electronics.json v4.1
T4283	PC_BAND	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	`/abort_matrix/causes/4/cause`	electronics.json v4.1
T4284	RUN after 300 ms: |Pc - Pc_ref| above 0.3 of Pc_ref and above 2 bar for 150 ms	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	`/abort_matrix/causes/4/trigger`	electronics.json v4.1
T4285	0.3 s	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	`/abort_matrix/causes/4/thresholds/kRunSettle_s`	tag in electronics.json v4.1: DESIGN
T4286	0.3	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	`/abort_matrix/causes/4/thresholds/kPcBandFrac`	tag in electronics.json v4.1: DESIGN
T4287	200000 Pa	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	`/abort_matrix/causes/4/thresholds/kPcBandMin_Pa`	tag in electronics.json v4.1: DESIGN
T4288	0.15 s	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	`/abort_matrix/causes/4/thresholds/kPcBandTime_s`	tag in electronics.json v4.1: DESIGN
T4289	6	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	`/abort_matrix/causes/5/code`	electronics.json v4.1
T4290	OVER_TEMP	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	`/abort_matrix/causes/5/cause`	electronics.json v4.1
T4291	case TC (TC1 / TC2) above 423.15 K (150.0 C) or nozzle holder (TC3) above 523.15 K ...	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	`/abort_matrix/causes/5/trigger`	electronics.json v4.1
T4292	0.2 s	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	`/abort_matrix/causes/5/thresholds/kOverTempTime_s`	tag in electronics.json v4.1: MECH 2026-10-13: 0.2 s hold
T4293	333.15 K	DERIVED	ELE v4.1	`/abort_matrix/causes/5/thresholds/kServoBracketOverTemp_K`	DS-derived
T4294	7	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	`/abort_matrix/causes/6/code`	electronics.json v4.1
T4295	TANK_OVERPRESSURE	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	`/abort_matrix/causes/6/cause`	electronics.json v4.1
T4296	P_tank above 65 bar for 200 ms in ARMED or PREPRESS (never aborts a burn)	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	`/abort_matrix/causes/6/trigger`	electronics.json v4.1

ID	Value	Class	Source	Key	Source tag / formula
T4297	6500000 Pa	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/abort_matrix/causes/6/thresholds/kTankOverPressure_Pa	tag in electronics.json v4.1: DESIGN
T4298	0.2 s	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/abort_matrix/causes/6/thresholds/kTankOverPressureTime_s	tag in electronics.json v4.1: DESIGN
T4299	8	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/abort_matrix/causes/7/code	electronics.json v4.1
T4300	GIMBAL_FAULT	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/abort_matrix/causes/7/cause	electronics.json v4.1
T4301	command-model vs encoder mismatch above 1.5 deg for 50 ms while burning; the allowance ...	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/abort_matrix/causes/7/trigger	electronics.json v4.1
T4302	0.02618 rad	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/abort_matrix/causes/7/thresholds/kGimbalFaultErr_rad	tag in electronics.json v4.1: DESIGN
T4303	0.05 s	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/abort_matrix/causes/7/thresholds/kGimbalFaultTime_s	tag in electronics.json v4.1: DESIGN
T4304	0.04 s	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/abort_matrix/causes/7/thresholds/kGimbalFaultRateWin_s	tag in electronics.json v4.1: DESIGN v2.2
T4305	9	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/abort_matrix/causes/8/code	electronics.json v4.1
T4306	PINTLE_FAULT	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/abort_matrix/causes/8/cause	electronics.json v4.1
T4307	pintle rail below 5.5 V for 2 ms (LTC4365 trip or regulator failure); jam: no progress ...	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/abort_matrix/causes/8/trigger	electronics.json v4.1
T4308	5.5 V	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/abort_matrix/causes/8/thresholds/kPrailLost_V	tag in electronics.json v4.1: DESIGN
T4309	0.002 s	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/abort_matrix/causes/8/thresholds/kPrailLostTime_s	tag in electronics.json v4.1: DESIGN
T4310	0.1 s	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/abort_matrix/causes/8/thresholds/kPintleStallTime_s	tag in electronics.json v4.1: DESIGN, LEAD approved
T4311	0.05 s	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/abort_matrix/causes/8/thresholds/kPintleStallTimeI_s	tag in electronics.json v4.1: DESIGN, LEAD approved
T4312	10	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/abort_matrix/causes/9/code	electronics.json v4.1
T4313	LINK_LOST	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/abort_matrix/causes/9/cause	electronics.json v4.1
T4314	no valid CMD_FAST: degraded after 20 ms, lost after 100 ms; SAFE/ARMED/PREPRESS → SAFE ...	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/abort_matrix/causes/9/trigger	electronics.json v4.1
T4315	0.02 s	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/abort_matrix/causes/9/thresholds/kLinkDegraded_s	tag in electronics.json v4.1: DESIGN
T4316	0.1 s	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/abort_matrix/causes/9/thresholds/kLinkLost_s	tag in electronics.json v4.1: DESIGN
T4317	11	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/abort_matrix/causes/10/code	electronics.json v4.1
T4318	PREPRESS_FAIL	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/abort_matrix/causes/10/cause	electronics.json v4.1
T4319	feed not primed (P_feed above 0.8 of P_tank) within 600 ms, or the NC valve OPEN switch ...	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/abort_matrix/causes/10/trigger	electronics.json v4.1
T4320	0.8	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/abort_matrix/causes/10/thresholds/kFeedPrimedFrac	tag in electronics.json v4.1: DESIGN
T4321	0.6 s	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/abort_matrix/causes/10/thresholds/kPrepressTimeout_s	tag in electronics.json v4.1: DESIGN
T4322	12	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/abort_matrix/causes/11/code	electronics.json v4.1
T4323	PC_SENSOR	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/abort_matrix/causes/11/cause	electronics.json v4.1
T4324	transducer output outside 0.25 V to 4.75 V: IGNITION abort (low side), stand RUN abort ...	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/abort_matrix/causes/11/trigger	electronics.json v4.1
T4325	R3-01 (OI-C1-SEAT-DRIFT): option (b) ADOPTED for flight and stand: Pc_comp = Pc - coef x ...	DECIDED *Lead decision, line 264 (2)*	CL 266 lines	L264(2)	lead (decision) 2026-10-21 (line sha 7cdbe51041ae6b8b)
T4326	0.3 bar	DECIDED *Lead decision, line 264 (2)*	CL 266 lines	L264(2) [regex kPcRezeroMax_Pa stays ([\d.]+) bar]	lead (decision) 2026-10-21 (line sha 7cdbe51041ae6b8b)
T4327	0.76 bar	DECIDED *Lead decision, line 264 (2)*	CL 266 lines	L264(2) [regex kIgnBaseCheck_Pa ([\d.]+) / [\d.]+ bar]	lead (decision) 2026-10-21 (line sha 7cdbe51041ae6b8b)
T4328	0.46 bar	DECIDED *Lead decision, line 264 (2)*	CL 266 lines	L264(2) [regex kIgnBaseCheck_Pa [\d.]+ / ([\d.]+) bar]	lead (decision) 2026-10-21 (line sha 7cdbe51041ae6b8b)
T4329	kPcRezeroMax_Pa stays 0.30 bar on the compensated residual; kIgnBaseCheck_Pa 0.76 / 0.46 ...	DECIDED *Lead decision, line 264 (2)*	CL 266 lines	L264(2)[from "kPcRezeroMax_Pa stays" to "(default / B-PC-1-shown build)"]	lead (decision) 2026-10-21 (line sha 7cdbe51041ae6b8b)
T4330	EXCEPT the C1 re-zero and BASE constants (kPcRezeroMax_Pa, kIgnBaseCheck_Pa), which are ...	DECIDED *Lead decision, line 253 (5)*	CL 266 lines	L253(5)[from "EXCEPT the C1 re-zero and BASE constants" to "open MAJOR)"]	lead (decision) 2026-10-21 (line sha 3a7f695ca59d621e)
T4331	0.25 V	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/abort_matrix/causes/11/thresholds/kPcSensorMin_V	tag in electronics.json v4.1: DESIGN
T4332	4.75 V	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/abort_matrix/causes/11/thresholds/kPcSensorMax_V	tag in electronics.json v4.1: DESIGN
T4333	0.005 s	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/abort_matrix/causes/11/thresholds/kPcSensorLossStand_s	tag in electronics.json v4.1: DESIGN
T4334	yes	DECIDED *Lead decision, line 203 (1)*	ELE v4.1	/abort_matrix/causes/11/thresholds/kPcOverrangeNonSticky	tag in electronics.json v4.1: LEAD 203 (1)(a)
T4335	0.05 s	ASSUMED	ELE v4.1	/abort_matrix/causes/11/thresholds/kA7Time_s	COMB A7 ASSUMED, Electronics sets
T4336	30000 Pa	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/abort_matrix/causes/11/thresholds/kA7HoldLow_Pa	tag in electronics.json v4.1: DESIGN
T4337	0.45 s	DECIDED *Lead decision, line 166 (7)*	ELE v4.1	/abort_matrix/causes/11/thresholds/kA7StartHiBlank_s	tag in electronics.json v4.1: LEAD CHANGELOG 166 (7); phase / flight extension DESIGN
T4338	76000 Pa	DECIDED *Lead decision, line 264 (2) (kIgnBaseCheck_Pa 0.76 / 0.46 bar, default / B-PC-1-shown build)*	ELE v4.1	/abort_matrix/causes/11/thresholds/kIgnBaseCheck_Pa	tag in electronics.json v4.1: LEAD form + R3-01 item 4
T4339	50000 Pa	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/abort_matrix/causes/11/thresholds/kPcArmZeroMax_Pa	tag in electronics.json v4.1: DESIGN
T4340	13	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/abort_matrix/causes/12/code	electronics.json v4.1
T4341	BATT_LOW	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/abort_matrix/causes/12/cause	electronics.json v4.1
T4342	VBAT below 6.6 V for 200 ms in pad states before the first start (flag in flight); ARM ...	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/abort_matrix/causes/12/trigger	electronics.json v4.1
T4343	6.6 V	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/abort_matrix/causes/12/thresholds/kBattLow_V	tag in electronics.json v4.1: DESIGN
T4344	0.2 s	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/abort_matrix/causes/12/thresholds/kBattLowTime_s	tag in electronics.json v4.1: DESIGN
T4345	8.1 V	DECIDED *Lead decision of 2026-10-12*	ELE v4.1	/abort_matrix/causes/12/thresholds/kBattArmMin_V	tag in electronics.json v4.1: LEAD 2026-10-12 ARM rule, rest voltage
T4346	14	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/abort_matrix/causes/13/code	electronics.json v4.1
T4347	MV_FAULT	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/abort_matrix/causes/13/cause	electronics.json v4.1
T4348	NC valve CLOSED switch missing at ARM, ladder OPEN+CLOSED or short (RE-19); direct ...	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/abort_matrix/causes/13/trigger	electronics.json v4.1
T4349	0.95 s	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/abort_matrix/causes/13/thresholds/kMvOpenBy_s	tag in electronics.json v4.1: DESIGN

ID	Value	Class	Source	Key	Source tag / formula
T4350	0.5 s	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/abort_matrix/causes/13/thresholds/kMvCloseConfirm_s	tag in electronics.json v4.1: DESIGN
T4351	15	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/abort_matrix/causes/14/code	electronics.json v4.1
T4352	ARM_LOST	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/abort_matrix/causes/14/cause	electronics.json v4.1
T4353	arming plug removed, or ARM bus not on after the request	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/abort_matrix/causes/14/trigger	electronics.json v4.1
T4354	16	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/abort_matrix/causes/15/code	electronics.json v4.1
T4355	WATCHDOG	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/abort_matrix/causes/15/cause	electronics.json v4.1
T4356	hardware watchdog timeout 50 ms: reboot lands in SAFE, MV and bleed closed, RUN never ...	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/abort_matrix/causes/15/trigger	electronics.json v4.1
T4357	0.05 s	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/abort_matrix/causes/15/thresholds/kHwWatchdog_s	tag in electronics.json v4.1: DESIGN
T4358	17	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/abort_matrix/causes/16/code	electronics.json v4.1
T4359	PC_OVERPRESSURE	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/abort_matrix/causes/16/cause	electronics.json v4.1
T4360	outside the ignition phase: filtered absolute Pc above 30 bar for 5 ms	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/abort_matrix/causes/16/trigger	electronics.json v4.1
T4361	3000000 Pa	DECIDED *Lead decision, per electronics.json v4.1*	ELE v4.1	/abort_matrix/causes/16/thresholds/kPcOverPressureAbs_Pa	tag in electronics.json v4.1: LEAD via COMB
T4362	0.005 s	DECIDED *Lead decision of 2026-10-13*	ELE v4.1	/abort_matrix/causes/16/thresholds/kPcOverPressureTime_s	tag in electronics.json v4.1: LEAD REV-ELEC-1 RE-20, 2026-10-13: FILTERED channel, 5 ms
T4363	18	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/abort_matrix/causes/17/code	electronics.json v4.1
T4364	HW_OVERPRESSURE	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/abort_matrix/causes/17/cause	electronics.json v4.1
T4365	hardware comparator U330 on the C1 Pc chain (INA333 output before the ADC; tap RC, trip ...	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/abort_matrix/causes/17/trigger	electronics.json v4.1
T4366	0.002 s	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/abort_matrix/causes/17/thresholds/kHwOvpTime_s	tag in electronics.json v4.1: DESIGN
T4367	0.005 s	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/abort_matrix/causes/17/thresholds/kHwOvpArmSettle_s	tag in electronics.json v4.1: DESIGN
T4368	19	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/abort_matrix/causes/18/code	electronics.json v4.1
T4369	TVC_SUPPLY	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/abort_matrix/causes/18/cause	electronics.json v4.1
T4370	pad only: predicted both-stall TVC plug voltage below 6.6 V for 100 ms	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/abort_matrix/causes/18/trigger	electronics.json v4.1
T4371	6.6 V	DECIDED *Lead decision of 2026-10-12*	ELE v4.1	/abort_matrix/causes/18/thresholds/kTvcSupplyMin_V	tag in electronics.json v4.1: LEAD 2026-10-12, V_val
T4372	0.1 s	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/abort_matrix/causes/18/thresholds/kTvcSupplyTime_s	tag in electronics.json v4.1: DESIGN
T4373	20	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/abort_matrix/causes/19/code	electronics.json v4.1
T4374	OVERCURRENT	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/abort_matrix/causes/19/cause	electronics.json v4.1
T4375	firmware e-fuse before the first start: battery current above 22 A for 1 s	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/abort_matrix/causes/19/trigger	electronics.json v4.1
T4376	22 A	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/abort_matrix/causes/19/thresholds/kEfuse_A	tag in electronics.json v4.1: DESIGN
T4377	1 s	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/abort_matrix/causes/19/thresholds/kEfuseTime_s	tag in electronics.json v4.1: DESIGN
T4378	21	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/abort_matrix/causes/20/code	electronics.json v4.1
T4379	MAINT_JAM	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/abort_matrix/causes/20/cause	electronics.json v4.1
T4380	ARM refused: a pintle jam recorded in NV memory and not cleared (CLEAR_JAM, bearing A ...	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/abort_matrix/causes/20/trigger	electronics.json v4.1
T4381	22	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/abort_matrix/causes/21/code	electronics.json v4.1
T4382	TORCH_FAULT	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/abort_matrix/causes/21/cause	electronics.json v4.1
T4383	torch not confirmed by 160 ms from E0: P_torch above 2 bar abs AND photodiode lit (2oo2 ...	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/abort_matrix/causes/21/trigger	electronics.json v4.1
T4384	0.16 s	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/abort_matrix/causes/21/thresholds/kTorchGate_s	tag in electronics.json v4.1: DESIGN; COMB v2.4 CONFIRMED 2026-10-13: E0→E1 max 0.1494 s over 1e6 samples, conditional on IGN-1 p99.9 <= 0.15 s
T4385	200000 Pa	ESTIMATE *Design value, electronics.json v4.1*	ELE v4.1	/abort_matrix/causes/21/thresholds/kTorchLitAbs_Pa	DESIGN, ESTIMATE
T4386	3	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/abort_matrix/causes/21/thresholds/kTorchConfirmN	tag in electronics.json v4.1: DESIGN
T4387	100000 Pa	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/abort_matrix/causes/21/thresholds/kTorchGateMaxPcG_Pa	tag in electronics.json v4.1: DESIGN
T4388	23	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/abort_matrix/causes/22/code	electronics.json v4.1
T4389	RELIGHT_LOCKOUT	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/abort_matrix/causes/22/cause	electronics.json v4.1
T4390	ARM, IGNITE and RELIGHT refused after an abort following a lit burn or on a relight ...	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/abort_matrix/causes/22/trigger	electronics.json v4.1
T4391	960 s	DERIVED	ELE v4.1	/abort_matrix/causes/22/thresholds/kNoLightSafingMin_s	DERIVED
T4392	24	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/abort_matrix/causes/23/code	electronics.json v4.1
T4393	OVER_THRUST	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/abort_matrix/causes/23/cause	electronics.json v4.1
T4394	flight RUN: FC_THRUST above 650 N on 3 consecutive fresh valid frames (rate 100 Hz ...	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/abort_matrix/causes/23/trigger	electronics.json v4.1
T4395	3	DECIDED *Lead decision of 2026-10-13*	ELE v4.1	/abort_matrix/causes/23/thresholds/kOverThrustFrames	tag in electronics.json v4.1: LEAD 2026-10-13 ADOPTED
T4396	100 Hz	DECIDED *Lead decision, per electronics.json v4.1*	ELE v4.1	/abort_matrix/causes/23/thresholds/kFcThrustRate_Hz	tag in electronics.json v4.1: GNC m2_fc_thrust v1.1 / LEAD
T4397	0.05 s	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/abort_matrix/causes/23/thresholds/kFcThrustStale_s	tag in electronics.json v4.1: GNC decision 2026-10-13
T4398	25	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/abort_matrix/causes/24/code	electronics.json v4.1
T4399	MODE_MISMATCH	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/abort_matrix/causes/24/cause	electronics.json v4.1
T4400	ARM refused: mode strap invalid, or strap and load-cell reading disagree	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/abort_matrix/causes/24/trigger	electronics.json v4.1
T4401	26	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/abort_matrix/causes/25/code	electronics.json v4.1
T4402	BAY_HOT	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/abort_matrix/causes/25/cause	electronics.json v4.1
T4403	ARM refused: bay temperature above 318.15 K (45.0 C) or no valid reading	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/abort_matrix/causes/25/trigger	electronics.json v4.1

ID	Value	Class	Source	Key	Source tag / formula
T4404	318.15 K	TO-VERIFY TO-DEFINE	ELE v4.1	/abort_matrix/causes/25/thresholds/kBayArmMax_K	LEAD approx 45 C, TO-DEFINE (class set at the call site: TO-VERIFY, status TO-DEFINE: the source tag says the value is still to be defined; REV-DS-1 D1-07)
T4405	27	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/abort_matrix/causes/26/code	electronics.json v4.1
T4406	TANK_HOT	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/abort_matrix/causes/26/cause	electronics.json v4.1
T4407	ARM refused: tank wall above 296.59 K (23.4 C) (limit 297.15 K (24.0 C) minus the ...	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/abort_matrix/causes/26/trigger	electronics.json v4.1
T4408	296.587 K	DERIVED	ELE v4.1	/abort_matrix/causes/26/thresholds/kTankIgnMaxWall_K	296.59 K = 23.44 C
T4409	297.15 K	DECIDED *Lead decision of 2026-10-13*	ELE v4.1	/abort_matrix/causes/26/thresholds/kTankIgnMax_K	tag in electronics.json v4.1: LEAD 2026-10-13
T4410	0.5625 K	DERIVED	ELE v4.1	/abort_matrix/causes/26/thresholds/kTankWallUnc_K	DERIVED from DATASHEET; mounting and drift MISSING, see above
T4411	5033700 Pa	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/abort_matrix/causes/26/thresholds/kTankIgnMaxP_Pa	tag in electronics.json v4.1: abs 50.34 bar: uncalibrated channel (Tsat 19.8 C)
T4412	5429700 Pa	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/abort_matrix/causes/26/thresholds/kTankIgnMaxPCal_Pa	tag in electronics.json v4.1: abs 54.30 bar: after B-PT-1 (Tsat 23.2 C)
T4413	28	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/abort_matrix/causes/27/code	electronics.json v4.1
T4414	BLEED_FAULT	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/abort_matrix/causes/27/cause	electronics.json v4.1
T4415	head-end bleed: pad BIT step failed (any mode); STAND: closure not confirmed, purge path ...	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/abort_matrix/causes/27/trigger	electronics.json v4.1
T4416	0.2 s	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/abort_matrix/causes/27/thresholds/kHbleedLeakTime_s	tag in electronics.json v4.1: DESIGN
T4417	0.21 s	DERIVED	ELE v4.1	/abort_matrix/causes/27/thresholds/kHbleedCloseConfirm_s	DERIVED
T4418	0.16 s	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/abort_matrix/causes/27/thresholds/kTorchGate_s	tag in electronics.json v4.1: DESIGN; COMB v2.4 CONFIRMED 2026-10-13: E0→E1 max 0.1494 s over 1e6 samples, conditional on IGN-1 p99.9 <= 0.15 s
T4419	29	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/abort_matrix/causes/28/code	electronics.json v4.1
T4420	TORCH_LOST	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/abort_matrix/causes/28/cause	electronics.json v4.1
T4421	torch confirmed (E1, 2oo2) but BOTH flame sensors dark for 5 ms before E2: pintle and NC ...	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/abort_matrix/causes/28/trigger	electronics.json v4.1
T4422	0.005 s	DECIDED *Lead decision, line 164 (8b)*	ELE v4.1	/abort_matrix/causes/28/thresholds/kTorchLossTime_s	tag in electronics.json v4.1: LEAD CHANGELOG 164 (8b)
T4423	30	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/abort_matrix/causes/29/code	electronics.json v4.1
T4424	TANK_COLD	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/abort_matrix/causes/29/cause	electronics.json v4.1
T4425	ARM refused when no valid tank temperature exists or the coldest tank estimate (lower of ...	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/abort_matrix/causes/29/trigger	electronics.json v4.1
T4426	288.15 K	DECIDED *Requirement, ENG-12*	ELE v4.1	/abort_matrix/causes/29/thresholds/kTankIgnMin_K	tag in electronics.json v4.1: REQ ENG-12 l.106, REV-ELEC-1 RE-17
T4427	0.42 K	DERIVED	ELE v4.1	/abort_matrix/causes/29/thresholds/kTankColdArmMargin_K	DERIVED
T4428	0.23 K	TO-VERIFY	ELE v4.1	/abort_matrix/causes/29/thresholds/kTankColdPrepressDrop_K	SIL S10v211 cold_rule, v2.11 pre-record 2026-10-21: drop 0.143-0.168 K at the nominal fill (REV-ELEC-2 0.154-0.156 K), 0.204-0.222 K at 60 % fill; max 0.2219 K rounded up. Fills below 60 % not run ...
T4429	0.07 K	DATASHEET	ELE v4.1	/abort_matrix/causes/29/thresholds/kTankColdRepeat_K	DATASHEET DS18B20 12-bit 0.0625 K + DERIVED Tsat noise < 0.01 K
T4430	0.5 s	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/abort_matrix/causes/29/thresholds/kMvCloseConfirm_s	tag in electronics.json v4.1: DESIGN
T4431	31	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/abort_matrix/causes/30/code	electronics.json v4.1
T4432	CASE_WARM_STAND	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/abort_matrix/causes/30/cause	electronics.json v4.1
T4433	stand mode only: ARM refused when TC1 or TC2 reads above 297.15 K (24.0 C) (stand T0 ...	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/abort_matrix/causes/30/trigger	electronics.json v4.1
T4434	297.15 K	DECIDED *Lead decision, per electronics.json v4.1*	ELE v4.1	/abort_matrix/causes/30/thresholds/kCaseStandArmMax_K	tag in electronics.json v4.1: LEAD ADOPTED
T4435	LiPo 2S 2.2 Ah (16.3 Wh), usable 13 Wh	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/power_budget/battery	electronics.json v4.1
T4436	1.25 W	DERIVED	ELE v4.1	/power_budget/logic_W/v/typ	DERIVED
T4437	2.6 W	DERIVED	ELE v4.1	/power_budget/logic_W/v/peak	DERIVED
T4438	1.73 Wh	DERIVED	ELE v4.1	/power_budget/flight_day_Wh	DERIVED (30 min pad hold + 60 s flight + 3 starts)
T4439	2.79 Wh	DERIVED	ELE v4.1	/power_budget/static_fire_session_Wh	DERIVED
T4440	35.7 A	DERIVED	ELE v4.1	/power_budget/peak_current_A	DERIVED worst case: 3 servos stalled
T4441	6.11 V	DERIVED	ELE v4.1	/power_budget/min_voltage_at_ECU_V	DERIVED at 3.5 V/cell OCV, ASSUMED 10 mOhm pack
T4442	F101: 25 A ATM main; F102: 2 A 1206 logic; F301: 15 A ATM ARM bus (feeds the 12 V ...	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/power_budget/fusing	electronics.json v4.1
T4443	2 NC valves in series (V1, V2), restrictors 2 x 0.707 mm upstream of V1 (1.0 mm ...	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/head_end_bleed_provision/architecture	tag in electronics.json v4.1: DESIGN / LEAD 164, 166 (3), 206 (3), 216 (5); schematic v2.6 NOT drawn for the C1 P_mid and the timer
T4444	purge_to_E0: v: 2.9; unit: s; tag: LEAD CHANGELOG 164 (2); src: config.h ...	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/head_end_bleed_provision/schedule	electronics.json v4.1
T4445	PURGE, 253 (7)(ii), now FINAL: the 2.43 s scheduled flow with the 2.39 s cap is ADOPTED ...	DECIDED *Lead decision, line 256 (2)*	CL 266 lines	L256(2)	lead (decision) 2026-10-21 (line sha cf51938a477079b6)
T4446	2.43 s	DECIDED *Lead decision, line 256 (2)*	CL 266 lines	L256(2) [regex the ([\d.]+) s scheduled flow]	lead (decision) 2026-10-21 (line sha cf51938a477079b6)
T4447	2.39 s	DECIDED *Lead decision, line 256 (2)*	CL 266 lines	L256(2) [regex with the ([\d.]+) s cap]	lead (decision) 2026-10-21 (line sha cf51938a477079b6)
T4448	PURGE CONDITION (d) ADDED to 256 (2), on REV-GNC-7 interim 2 (MAJOR, 09:25:51Z): the ...	DECIDED *Lead decision, line 257 (2)*	CL 266 lines	L257(2)	lead decision 2026-10-21 (line sha cf364643652fceb0)
T4449	PURGE: condition (d) of 257 (2) holds (RV7-04, review section 5 (ii): vent at the cap <= ...	DECIDED *Lead decision, line 258 (2)*	CL 266 lines	L258(2)	lead decision 2026-10-21 (line sha cdd58cc9fdc95079)
T4450	OI-PURGE-CAP: the ECU both-open cap is set to <= 2.55 s (Combustion basis 75.9 g <= 76 ...	DECIDED *Lead decision, line 203 (3)*	CL 266 lines	L203(3)	lead (decision) 2026-10-13 (line sha 46370e2c65162336)
T4451	Purge FINAL 2.43 / 2.39 s as calibration constants with rule (d) (d1) (d2)		CL 266 lines	L263(3)[from "Purge FINAL 2.43 / 2.39 s" to "(d) (d1) (d2)"]	electronics (result) 2026-10-21 (line sha 5f9263d5cf3262ab); class set at the call site: the Electronics author's record of the v2.12 implementation (v4.1, promoted, line 266 (7)); no value is taken ...
T4452	E1 = 2oo2 torch AND [closure confirmed OR deadline 160 ms from E0]; flight: not ...	DECIDED *Lead decision, line 166 (3)*	ELE v4.1	/head_end_bleed_provision/closure_rule	tag in electronics.json v4.1: LEAD CHANGELOG 166 (3)
T4453	question: CHANGELOG line 230 (lead, 2026-10-19): ...ve on FA1S, so no other margin ...	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/head_end_bleed_provision/cap_decision_230_3	electronics.json v4.1
T4454	decision: CHANGELOG line 216 (lead, 2026-10-13): ...TORCH_FAULT; all unlit runs 0 false ...	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/head_end_bleed_provision/hardware_both_open_timer	electronics.json v4.1
T4455	PREPRESS steps S0-S4 (valve pattern vs P_mid); any failure → BLEED_FAULT; a missing ...	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/head_end_bleed_provision/pad_bit	electronics.json v4.1
T4456	2.43 s	DECIDED *Lead decision, line 256 (2)*	ELE v4.1	/purge_final_v2p12/kPurgeFlow_s	tag in electronics.json v4.1: LEAD 256 (2) / 257 (2)
T4457	2.39 s	DECIDED *Lead decision, line 256 (2)*	ELE v4.1	/purge_final_v2p12/kHbleedCap_s	tag in electronics.json v4.1: LEAD 256 (2) / 257 (2)
T4458	both constants are calibration values: the config values are the DEFAULTS and the ...	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/purge_final_v2p12/calibration	tag in electronics.json v4.1: DESIGN; test_core v212-3

ID	Value	Class	Source	Key	Source tag / formula
T4459	after CF-4 / B-BLEED-1: cap <= 76 g / mdot(297.15 K) - (t_close - t_open) with mdot from ...	DECIDED *Lead decision, line 257 (2)*	ELE v4.1	/purge_final_v2p12/rule_d	tag in electronics.json v4.1: LEAD 257 (2), 258 (2); config.h
T4460	PURGE at E0 - 2.90 s; flow start PURGE + 0.42 s; flow end E0 - 0.05 s; scheduled ...	DERIVED	ELE v4.1	/purge_final_v2p12/purge_command_time	DERIVED config.h
T4461	scheduled 2.43 s: 71.1 g at 29.25 g/s (289.5 K), 75.5 g at 31.06 g/s (297.15 K); at the ...	DERIVED	ELE v4.1	/purge_final_v2p12/vented_mass_DERIVED	DERIVED
T4462	9. The stale "<= 73.0 g at 2.5 s" at v4.1 l.9974, sil_main.cpp l.2505 / 2668 and ...	DERIVED	RELE4 REVIEW-ELECTRONICS-R4 (REV-ELEC-4)	line 231 (starts "9. The stale")	REV-ELEC-4 review text (section 2), the correction that CHANGELOG line 266 (7) applies to v4.1 as written in the review
T4463	MV latch drops → MV-P spring + pressure bias close it; the pintle spring closes the ...	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/fail_safe_states/ECU power loss	electronics.json v4.1
T4464	hardware watchdog (50 ms) resets MCU and drops ARM bus → same as above; reboot lands in ...	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/fail_safe_states/MCU hang	electronics.json v4.1
T4465	see link_loss	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/fail_safe_states/link loss	electronics.json v4.1
T4466	RUN: open-loop feed-forward map on P_feed + fault flag; IGNITION: abort; flight HIGH ...	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/fail_safe_states/Pc sensor fault	electronics.json v4.1
T4467	flag; RUN: continue on servo commands (default) unless mismatch abort criterion met	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/fail_safe_states/gimbal encoder fault	electronics.json v4.1
T4468	ARM bus dead: igniter, pintle and main valve power removed regardless of firmware	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/fail_safe_states/arming plug removed	electronics.json v4.1
T4469	reboot to SAFE, MV and bleed closed, RUN never reported after the reset, the drogue is ...	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/fail_safe_states/ECU reset in flight	electronics.json v4.1
T4470	LTC4365 OV disconnect 8.06-8.34 V: servo unpowered, spring closes; PRAIL_OK low drops ...	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/fail_safe_states/pintle rail fail-high	electronics.json v4.1
T4471	hardware comparator U330 on the C1 chain (trip 32.59 bar abs nominal, band 31.08-34.11) ...	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/fail_safe_states/Pc over-pressure	electronics.json v4.1
T4472	FC PURGE_EN low or the one-shot expired: both bleed drives forced off in hardware; both ...	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/fail_safe_states/purge window	electronics.json v4.1
T4473	both case TCs lost: stand treated hot (abort), flight burn continues and the relight is ...	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/fail_safe_states/case TC	electronics.json v4.1
T4474	CHANGELOG line 249 (lead, 2026-10-21): Lead decisions and verifications after line 248 ...	DECIDED *Lead decision, per electronics.json v4.1*	ELE v4.1	/lead_rulings_v2p11/case_tc_abort_channel	tag in electronics.json v4.1:
T4475	CHANGELOG line 249 (lead, 2026-10-21): ...erms, else the tolerance is re-derived. (2) ...	DECIDED *Lead decision, per electronics.json v4.1*	ELE v4.1	/lead_rulings_v2p11/stand_T0_rule	tag in electronics.json v4.1:
T4476	CHANGELOG line 249 (lead, 2026-10-21): ... flight has no T0 rule (> 45 K margin). (3) ...	DECIDED *Lead decision, per electronics.json v4.1*	ELE v4.1	/lead_rulings_v2p11/T_F1_basis	tag in electronics.json v4.1:
T4477	CHANGELOG line 249 (lead, 2026-10-21): ...he T-F1 founder text on the A 175 rows. (4) ...	DECIDED *Lead decision, per electronics.json v4.1*	ELE v4.1	/lead_rulings_v2p11/relight_and_transducer	tag in electronics.json v4.1:
T4478	CHANGELOG line 249 (lead, 2026-10-21): ...nges an envelope quantity, v6 is final. (12) ...	DECIDED *Lead decision, per electronics.json v4.1*	ELE v4.1	/lead_rulings_v2p11/disk	tag in electronics.json v4.1:
T4479	CHANGELOG line 249 (lead, 2026-10-21): ...tarted 03:18:46Z (start checks passed). ...	DECIDED *Lead decision, per electronics.json v4.1*	ELE v4.1	/lead_rulings_v2p11/runs	tag in electronics.json v4.1:
T4480	the MANIFEST cited in 249 (13) is superseded by two re-records with byte-identical ...	DECIDED *Lead decision, per electronics.json v4.1*	ELE v4.1	/lead_rulings_v2p11/runs_note	tag in electronics.json v4.1:
T4481	CHANGELOG line 249 (lead, 2026-10-21): ..._torch, the face-peak inputs of (4) and ...	DECIDED *Lead decision, per electronics.json v4.1*	ELE v4.1	/lead_rulings_v2p11/open	tag in electronics.json v4.1:
T4482	ROUTING: the F4 PC_SENSOR HIGH abort (>= 4.75 V = 146.5 bar abs; LOW side not added) ...	DECIDED *Lead decision, line 258 (4)*	CL 266 lines	L258(4)	lead decision 2026-10-21 (line sha cdd58cc9fdc95079)
T4483	ELECTRONICS v2.12: the Part A interim (R3-01) arrived at 11:19Z and is saved verbatim in ...	DECIDED *Lead decision, line 261 (3)*	CL 266 lines	L261(3)	lead decision 2026-10-21 (line sha d510d8077319e234) [1 working-path mention(s) replaced by a marker]
T4484	Hub 06-electronics/staging_v2.12/: 84 files + SHA256SUMS-v2.12.txt f6fd3a63817dfb34 ...		CL 266 lines	L263[sentence containing "candidate results/electronics_v4.1.json"]	electronics (result) 2026-10-21 (line sha 5f9263d5cf3262ab); class set at the call site: provenance record of the staged candidate
T4485	DECISION (3), A1 BASE_AMB: ADOPTED as built: a BASE_AMB at the A1 relight aborts ...	DECIDED *Lead decision, line 264 (3)*	CL 266 lines	L264(3)	lead (decision) 2026-10-21 (line sha 7cdbe51041ae6b8b)
T4486	R4-12 (OI-PTORCH-PROOF): option 2 ADOPTED: P_torch (J901, on the torch body, not in the ...	DECIDED *Lead decision, line 264 (4)*	CL 266 lines	L264(4)	lead (decision) 2026-10-21 (line sha 7cdbe51041ae6b8b)
T4487	A1 FLIGHT T0 0-40 C: ADOPTED as the ASSUMED design range (DECIDED) of the A1 relight ...	DECIDED *Lead decision, line 264 (6)*	CL 266 lines	L264(6)	lead (decision) 2026-10-21 (line sha 7cdbe51041ae6b8b)
T4488	0 °C	DECIDED *Lead decision, line 264 (6); an ASSUMED design range (264 (6))*	CL 266 lines	L264(6) [regex T0 (\d+)-\d+ C]	lead (decision) 2026-10-21 (line sha 7cdbe51041ae6b8b)
T4489	40 °C	DECIDED *Lead decision, line 264 (6); an ASSUMED design range (264 (6))*	CL 266 lines	L264(6) [regex T0 \d+-(\d+) C]	lead (decision) 2026-10-21 (line sha 7cdbe51041ae6b8b)
T4490	OI-JFWD-FLIGHT: PROVISIONAL lead rule until flight acceptance: a flagged first light (f3 ...	DECIDED *Lead decision, line 264 (7)*	CL 266 lines	L264(7)	lead (decision) 2026-10-21 (line sha 7cdbe51041ae6b8b)
T4491	E4-N02 (the 264 (2) pad rule and the 264 (7) inhibit implemented)	DECIDED *Lead decision, line 266 (8)*	CL 266 lines	L266(8)[from "E4-N02 (the 264 (2) pad rule and the 264 (7) inhibit implemented)" to "E4-N02 (the 264 (2) pad rule and the 264 (7) inhibit implemented)"]	lead (decision) 2026-10-21 (line sha 6f3181b859f02937)
T4492	v4.1 is promoted only after its verdict. The author did NOT produce the v2.12 SIL ...	DECIDED *Lead decision, line 264 (8)*	CL 266 lines	L264(8)[from "v4.1 is promoted only after its verdict" to "can affect."]	lead (decision) 2026-10-21 (line sha 7cdbe51041ae6b8b)
T4493	OI-HAL	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/0/id	electronics.json v4.1
T4494	HAL-1, B-ECU-1	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/0/closure	electronics.json v4.1
T4495	Teensy HAL not written (RE-06); TLM_SLOW undefined; 96 ARM -Wdouble-promotion warnings ...	TO-VERIFY	ELE v4.1	/open_issues/0/issue	(class set at the call site: TO-VERIFY)
T4496	HIGH	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/0/severity	electronics.json v4.1
T4497	Electronics	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/0/owner	electronics.json v4.1
T4498	OI-POWER	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/1/id	electronics.json v4.1
T4499	power sequencing / capacity decided before the layout; B-PWR-2	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/1/closure	electronics.json v4.1
T4500	RE-09: +12V_ARM and the ARM bus over rating in the purge and E0 overlaps; the E0 row ...	TO-VERIFY	ELE v4.1	/open_issues/1/issue	(class set at the call site: TO-VERIFY)
T4501	HIGH	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/1/severity	electronics.json v4.1
T4502	Electronics	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/1/owner	electronics.json v4.1
T4503	OI-BLEED-VALVE	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/2/id	electronics.json v4.1
T4504	selection; B-BLEED-1	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/2/closure	electronics.json v4.1
T4505	bleed valve not selected (LEAD 206 (4) search continues); peak-and-hold not drawn; valve ...	TO-VERIFY	ELE v4.1	/open_issues/2/issue	(class set at the call site: TO-VERIFY)
T4506	HIGH	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/2/severity	electronics.json v4.1
T4507	Electronics / Mechanical	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/2/owner	electronics.json v4.1
T4508	OI-PURGE-CAP	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/3/id	electronics.json v4.1

ID	Value	Class	Source	Key	Source tag / formula
T4509	lead decision on the cap; measured-time form after CF-4 / B-BLEED-1	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/3/closure	electronics.json v4.1
T4510	FINAL: 2.43 s flow / 2.39 s cap as calibration constants, rule (d) (d1) (d2) ...	TO-VERIFY	ELE v4.1	/open_issues/3/issue	(class set at the call site: TO-VERIFY)
T4511	CLOSED (LEAD 256 (2), 257 (2), 258 (2))	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/3/severity	electronics.json v4.1
T4512	LEAD	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/3/owner	electronics.json v4.1
T4513	OI-WINDOW-LATENT	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/4/id	electronics.json v4.1
T4514	CHK-PW, CHK-BT; firmware BIT step	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/4/closure	electronics.json v4.1
T4515	stuck-high purge-window chain and the both-open timer are invisible to the ECU (no free ...	TO-VERIFY	ELE v4.1	/open_issues/4/issue	(class set at the call site: TO-VERIFY)
T4516	MEDIUM	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/4/severity	electronics.json v4.1
T4517	Electronics / FC owner	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/4/owner	electronics.json v4.1
T4518	OI-BLEED-DETECT	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/5/id	electronics.json v4.1
T4519	LEAD: accept the FMEA residual (FMEA-A1-BLEED-BOTH-OPEN) or ask for a detection design ...	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/5/closure	electronics.json v4.1
T4520	S12 (SIM): both bleed valves stuck open (the 203 (4) label is corrected in ...	TO-VERIFY	ELE v4.1	/open_issues/5/issue	(class set at the call site: TO-VERIFY)
T4521	MEDIUM	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/5/severity	electronics.json v4.1
T4522	Electronics / LEAD	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/5/owner	electronics.json v4.1
T4523	OI-A1-BASE-AMB	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/6/id	electronics.json v4.1
T4524	accept the outcome or rule the alternative (keep the measured P0 below ambient; not ...	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/6/closure	electronics.json v4.1
T4525	S13T with the drift model: option (b) 0 BASE_AMB in the A1 sweep (both builds); fallback ...	TO-VERIFY	ELE v4.1	/open_issues/6/issue	(class set at the call site: TO-VERIFY)
T4526	MEDIUM	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/6/severity	electronics.json v4.1
T4527	LEAD	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/6/owner	electronics.json v4.1
T4528	NEW RULE (lead; not implemented yet): A1 pad no-go unless the B-PC-1 coefficient is ...	DECIDED *Lead decision, line 264 (2)*	CL 266 lines	L264(2)[sentence containing "NEW RULE"]	lead (decision) 2026-10-21 (line sha 7cdbe51041ae6b8b)
T4529	the '0 of 500' evidence of OI-A1-BASE-AMB is structural (the SIL Pc model has no thermal ...	DECIDED *Lead decision, line 253 (5)*	CL 266 lines	L253(5)[from "the '0 of 500' evidence of OI-A1-BASE-AMB" to "not an occurrence rate"]	lead (decision) 2026-10-21 (line sha 3a7f695ca59d621e)
T4530	OI-TEB-FALSE-LIGHT	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/7/id	electronics.json v4.1
T4531	B-PC-1 residual <= 0.3 bar, then kBpc1Shown	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/7/closure	electronics.json v4.1
T4532	S9 +/-1.034 bar TEB cases fail as in v2.9 / v2.10 (the v2.11 run uses the C1 chain with ...	TO-VERIFY	ELE v4.1	/open_issues/7/issue	(class set at the call site: TO-VERIFY)
T4533	MEDIUM	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/7/severity	electronics.json v4.1
T4534	LEAD / Electronics	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/7/owner	electronics.json v4.1
T4535	OI-GUARD-BAND	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/8/id	electronics.json v4.1
T4536	B-PT-1	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/8/closure	electronics.json v4.1
T4537	cold side CLOSED by LEAD 216 (7) (ARM margin 0.42 K, PREPRESS safe exit; S5: TANK_COLD ...	TO-VERIFY	ELE v4.1	/open_issues/8/issue	(class set at the call site: TO-VERIFY)
T4538	MEDIUM	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/8/severity	electronics.json v4.1
T4539	Electronics / Mechanical	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/8/owner	electronics.json v4.1
T4540	OI-E2-LATENCY	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/9/id	electronics.json v4.1
T4541	IGN-1	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/9/closure	electronics.json v4.1
T4542	E2 - light (SIL): p50 12 / max 13 ms in S5, 12 / 13 ms in S15 (flight), 12 / 13 ms ...	TO-VERIFY	ELE v4.1	/open_issues/9/issue	(class set at the call site: TO-VERIFY)
T4543	LOW	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/9/severity	electronics.json v4.1
T4544	Combustion / Electronics / GNC	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/9/owner	electronics.json v4.1
T4545	OI-RE15	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/10/id	electronics.json v4.1
T4546	RE-15 checks before fabrication	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/10/closure	electronics.json v4.1
T4547	no layout; pin-function check, footprints, derating table not done	TO-VERIFY	ELE v4.1	/open_issues/10/issue	(class set at the call site: TO-VERIFY)
T4548	HIGH	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/10/severity	electronics.json v4.1
T4549	Electronics	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/10/owner	electronics.json v4.1
T4550	OI-SCHEMATIC-V2P6	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/11/id	electronics.json v4.1
T4551	schematic v2.6 + ERC + netlist cross-check before RE-15	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/11/closure	electronics.json v4.1
T4552	schematic v2.6 NOT drawn: three C1 bridge channels, U330 / U570 values on C1, the ...	TO-VERIFY	ELE v4.1	/open_issues/11/issue	(class set at the call site: TO-VERIFY)
T4553	HIGH	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/11/severity	electronics.json v4.1
T4554	Electronics	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/11/owner	electronics.json v4.1
T4555	OI-GIMBAL-JAM	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/12/id	electronics.json v4.1
T4556	B-TVC-1	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/12/closure	electronics.json v4.1
T4557	S6 gimbal_jam_moving_cmd: small-amplitude jam undetected (RE-24 test-based, LEAD 203 ...	TO-VERIFY	ELE v4.1	/open_issues/12/issue	(class set at the call site: TO-VERIFY)
T4558	MEDIUM	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/12/severity	electronics.json v4.1
T4559	Electronics + GNC	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/12/owner	electronics.json v4.1
T4560	OI-TB-MEOP	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/13/id	electronics.json v4.1
T4561	LC-TB70 (04-fea/v352/lc_tb70_envelope_v1.json)	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/13/closure	electronics.json v4.1
T4562	S6 throat block: plant Pc 53.42 bar abs, 33 ms above MEOP (R2-15: not 72 ms); inside ...	TO-VERIFY	ELE v4.1	/open_issues/13/issue	(class set at the call site: TO-VERIFY)

ID	Value	Class	Source	Key	Source tag / formula
T4563	ACCEPTED	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/13/severity	electronics.json v4.1
T4564	Mechanical FEA	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/13/owner	electronics.json v4.1
T4565	OI-FC	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/14/id	electronics.json v4.1
T4566	B-FC-1, HIL-OT-1	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/14/closure	electronics.json v4.1
T4567	FC: PURGE_EN discrete; R2 v2 at W 0.8 s with the mode-3 flush at the Pc-invalid onset ...	TO-VERIFY	ELE v4.1	/open_issues/14/issue	(class set at the call site: TO-VERIFY)
T4568	HIGH	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/14/severity	electronics.json v4.1
T4569	FC owner + GNC	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/14/owner	electronics.json v4.1
T4570	OI-38-R2-COMMAND-HOLD	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/15/id	electronics.json v4.1
T4571	GNC R2 v2 re-run; B-FC-1; CF-3 start-lift point	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/15/closure	electronics.json v4.1
T4572	ECU side in SIL: lift-commanded hold at 0.21776 mm, backstop E2 + 850 ms, per-tick Pc ...	TO-VERIFY	ELE v4.1	/open_issues/15/issue	(class set at the call site: TO-VERIFY)
T4573	HIGH	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/15/severity	electronics.json v4.1
T4574	GNC (FC) / Combustion / Electronics	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/15/owner	electronics.json v4.1
T4575	OI-FA1-PC-OVERRANGE	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/16/id	electronics.json v4.1
T4576	LEAD decision; B-PC-1; B-DPC-1 at IGN-1 / T series; HIL-TR-1	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/16/closure	electronics.json v4.1
T4577	C1 chain (v2.11): 0 of 896 FA-1 grid cases reach the over-range or the above-FS flag; as ...	TO-VERIFY	ELE v4.1	/open_issues/16/issue	(class set at the call site: TO-VERIFY)
T4578	HIGH	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/16/severity	electronics.json v4.1
T4579	LEAD (adopt the ride-through or not) / Electronics / founders (F-10, B-PC-1, B-DPC-1)	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/16/owner	electronics.json v4.1
T4580	OI-A1-PCLOSS-OVERTHRUST	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/17/id	electronics.json v4.1
T4581	none on A1; RG-16 budget before FC_THRUST is armed on any mission	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/17/closure	electronics.json v4.1
T4582	A1: no over-thrust detector after an in-flight Pc loss (fmea_rows_a1)	TO-VERIFY	ELE v4.1	/open_issues/17/issue	(class set at the call site: TO-VERIFY)
T4583	ACCEPTED (FMEA record)	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/17/severity	electronics.json v4.1
T4584	LEAD / GNC	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/17/owner	electronics.json v4.1
T4585	OI-STAND-LATE-LIGHT	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/18/id	electronics.json v4.1
T4586	IGN-1 (record E1 → light on every stand start)	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/18/closure	electronics.json v4.1
T4587	S15 v2.11 stand: 4 of 230 lit plants end NO_IGNITION (light later than the stand W) ...	TO-VERIFY	ELE v4.1	/open_issues/18/issue	(class set at the call site: TO-VERIFY)
T4588	LOW	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/18/severity	electronics.json v4.1
T4589	Combustion / LEAD	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/18/owner	electronics.json v4.1
T4590	OI-UNPURGED-RELIGHT	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/19/id	electronics.json v4.1
T4591	LEAD on (a); FC rule in the FSW record (B-FC-1); CF-4 O-3, IGN-1	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/19/closure	electronics.json v4.1
T4592	the ECU guard is DONE (LEAD 203 (7): flight build with the bleed fitted refuses a ...	TO-VERIFY	ELE v4.1	/open_issues/19/issue	(class set at the call site: TO-VERIFY)
T4593	MEDIUM	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/19/severity	electronics.json v4.1
T4594	LEAD / Combustion / GNC (FC)	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/19/owner	electronics.json v4.1
T4595	OI-GEN-TABLES	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/20/id	electronics.json v4.1
T4596	gen_tables v3.4, regenerate, diff, re-run the SIL record	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/20/closure	electronics.json v4.1
T4597	gen_tables.py v3.3 cannot parse the combustion.json v2.4b igniter sequence ...	TO-VERIFY	ELE v4.1	/open_issues/20/issue	(class set at the call site: TO-VERIFY)
T4598	MEDIUM	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/20/severity	electronics.json v4.1
T4599	Electronics (parser) + Combustion (machine-readable sequence)	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/20/owner	electronics.json v4.1
T4600	OI-FMIN-LIT	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/21/id	electronics.json v4.1
T4601	230 (5) open item	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/21/closure	electronics.json v4.1
T4602	lit F_min 33.14 N (3 of 72 Combustion corners never reach 37.16 N at 0.21776 mm, LEAD ...	TO-VERIFY	ELE v4.1	/open_issues/21/issue	(class set at the call site: TO-VERIFY)
T4603	MEDIUM	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/21/severity	electronics.json v4.1
T4604	GNC (R2 v2 re-run) / LEAD	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/21/owner	electronics.json v4.1
T4605	OI-PHOTODIODE-VIEW	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/22/id	electronics.json v4.1
T4606	IGN-1 photodiode row (view proven with the torch closed) or the GNC reference choice	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/22/closure	electronics.json v4.1
T4607	the dead-side result with reference none needs the photodiode main-chamber view ...	TO-VERIFY	ELE v4.1	/open_issues/22/issue	(class set at the call site: TO-VERIFY)
T4608	HIGH	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/22/severity	electronics.json v4.1
T4609	Combustion / Mechanical (view) / GNC (reference)	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/22/owner	electronics.json v4.1
T4610	OI-FC-PC-BRANCH	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/23/id	electronics.json v4.1
T4611	state the branch margin or move the threshold; GNC R2 v2 re-run	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/23/closure	electronics.json v4.1
T4612	with the FC Pc branch (release on Pc valid and filtered Pc >= 3.0 bar abs) every ...	TO-VERIFY	ELE v4.1	/open_issues/23/issue	(class set at the call site: TO-VERIFY)
T4613	MEDIUM	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/23/severity	electronics.json v4.1
T4614	GNC / FC owner	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/23/owner	electronics.json v4.1
T4615	OI-R2-23-GLITCH	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/24/id	electronics.json v4.1
T4616	B-PC-1, HIL-TR-1	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/24/closure	electronics.json v4.1

ID	Value	Class	Source	Key	Source tag / formula
T4617	a HIGH over-range glitch later in the burn (E2 + 350 ms) is also seen by the filtered ...	TO-VERIFY	ELE v4.1	/open_issues/24/issue	(class set at the call site: TO-VERIFY)
T4618	LOW	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/24/severity	electronics.json v4.1
T4619	Electronics / founders	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/24/owner	electronics.json v4.1
T4620	OI-FACE-PEAK	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/25/id	electronics.json v4.1
T4621	REV-COMB-4	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/25/closure	electronics.json v4.1
T4622	C1 seat-face peaks 104.5-125.1 bar abs (SIM on the ASSUMED 1-D lumped passage model) ...	TO-VERIFY	ELE v4.1	/open_issues/25/issue	(class set at the call site: TO-VERIFY)
T4623	ROUTED	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/25/severity	electronics.json v4.1
T4624	REV-COMB-4 / Combustion / Mechanical (CHANGELOG 249 (15))	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/25/owner	electronics.json v4.1
T4625	OI-S6R-INPUT	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/26/id	electronics.json v4.1
T4626	REV-COMB-4; LC-TB70	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/26/closure	electronics.json v4.1
T4627	10 S6R runs have their INPUT outside LC-TB70: the Combustion v2.4a FA-1 replay max-face ...	TO-VERIFY	ELE v4.1	/open_issues/26/issue	(class set at the call site: TO-VERIFY)
T4628	ROUTED	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/26/severity	electronics.json v4.1
T4629	REV-COMB-4 / Mechanical (CHANGELOG 249 (15))	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/26/owner	electronics.json v4.1
T4630	OI-SEAT-RINGING	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/27/id	electronics.json v4.1
T4631	B-DPC-1, HIL-TR-1	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/27/closure	electronics.json v4.1
T4632	ideal 25 us steps ring the C1 seat passage in the SIL model (laminar-only damping ...	TO-VERIFY	ELE v4.1	/open_issues/27/issue	(class set at the call site: TO-VERIFY)
T4633	MEDIUM	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/27/severity	electronics.json v4.1
T4634	Electronics / Combustion / founders	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/27/owner	electronics.json v4.1
T4635	OI-STALE-LABELS	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/28/id	electronics.json v4.1
T4636	v2.12 text pass (comments, labels)	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/28/closure	electronics.json v4.1
T4637	config.h comments corrected in v2.12 (rev_elec_3_author_status.N-09)	TO-VERIFY	ELE v4.1	/open_issues/28/issue	(class set at the call site: TO-VERIFY)
T4638	CLOSED (N-09, v2.12)	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/28/severity	electronics.json v4.1
T4639	Electronics	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/28/owner	electronics.json v4.1
T4640	OI-C1-SEAT-DRIFT	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/29/id	electronics.json v4.1
T4641	lead ruling; B-PC-1 zero-TC row; F-10 answer	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/29/closure	electronics.json v4.1
T4642	Residuals stay OPEN with owners: the per-serial coefficient and T_cal (founders ...	DECIDED *Lead decision, line 264 (2)*	CL 266 lines	L264(2)[sentence containing "Residuals stay OPEN"]	lead (decision) 2026-10-21 (line sha 7cdbe51041ae6b8b)
T4643	R3-01: option (b) implemented in SIL; needs the lead ruling, the per-serial coefficient ...	TO-VERIFY	ELE v4.1	/open_issues/29/issue	(class set at the call site: TO-VERIFY)
T4644	HIGH	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/29/severity	electronics.json v4.1
T4645	LEAD (ruling) / Electronics / founders (B-PC-1, F-10) / Mechanical (seat TC mount)	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/29/owner	electronics.json v4.1
T4646	OI-STAND-RELIGHT-COEF	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/30/id	electronics.json v4.1
T4647	B-PC-1 per-serial residual over the 50.4 K span	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/30/closure	electronics.json v4.1
T4648	S13T: a stand relight after a B1-length burn is refused NOT_VENTED when the seat ...	TO-VERIFY	ELE v4.1	/open_issues/30/issue	(class set at the call site: TO-VERIFY)
T4649	MEDIUM	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/30/severity	electronics.json v4.1
T4650	Electronics / founders	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/30/owner	electronics.json v4.1
T4651	OI-PTORCH-PROOF	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/31/id	electronics.json v4.1
T4652	lead choice; F-7 / F-12 data	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/31/closure	electronics.json v4.1
T4653	R4-12: 070BG proof 141.0 bar abs < 166.14 bar abs required at FA1S; isolate or move ...	TO-VERIFY	ELE v4.1	/open_issues/31/issue	(class set at the call site: TO-VERIFY)
T4654	HIGH	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/31/severity	electronics.json v4.1
T4655	LEAD / Electronics / Combustion / Mechanical	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/31/owner	electronics.json v4.1
T4656	OI-T4-RESIZE	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/32/id	electronics.json v4.1
T4657	Combustion re-run of the T4 rows on the re-sized profile; Mechanical soak-back	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/32/closure	electronics.json v4.1
T4658	N-07: at the B1 end-of-burn rate the re-sized T4 relight (4.59 s) needs the case TCs <= ...	TO-VERIFY	ELE v4.1	/open_issues/32/issue	(class set at the call site: TO-VERIFY)
T4659	MEDIUM	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/32/severity	electronics.json v4.1
T4660	Combustion / Mechanical / LEAD	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/32/owner	electronics.json v4.1
T4661	OI-JFWD-FLIGHT	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/33/id	electronics.json v4.1
T4662	lead ruling before flight acceptance	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/33/closure	electronics.json v4.1
T4663	a flagged first light (f3 JFWD_LS) leaves the relight on a used joint; line 231 covers ...	TO-VERIFY	ELE v4.1	/open_issues/33/issue	(class set at the call site: TO-VERIFY)
T4664	MEDIUM	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/33/severity	electronics.json v4.1
T4665	LEAD	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/open_issues/33/owner	electronics.json v4.1
T4666	31	DERIVED	generator	formula: count of electronics.json /open_issues not closed (by a lead line or in the file)	DERIVED by build_datasheet.py: count of electronics.json /open_issues not closed (by a lead line or in the file) ← T4493, T4498, T4503, T4513, T4518, T4530, T4535, T4540 ...
T4667	3	DERIVED	generator	formula: count of electronics.json /open_issues closed by a lead line or marked CLOSED in the file (shown for the record)	DERIVED by build_datasheet.py: count of electronics.json /open_issues closed by a lead line or marked CLOSED in the file (shown for the record) ← T4508, T4523, T4635
T4668	The basis is the published gnc.json v3.2, `interfaces/gnc.json` (bde317782574cbd9 ...	DECIDED *Design value, DATASHEET-INPUTS-GNC.md v3.2*	DSIG v3.2	line 5 (starts "The basis is the published gnc.json v3.2")	tag in DATASHEET-INPUTS-GNC.md v3.2: GNC datasheet inputs header
T4669	210 m. Profile: TVC ascent, nose drogue at apogee, relight, throttled TVC landing.	DECIDED *Requirement, DATASHEET-INPUTS-GNC.md v3.2; basis SIM*	DSIG v3.2	line 22 row "Mission set: A1 apogee target (baseline flight)" col 2	tag in DATASHEET-INPUTS-GNC.md v3.2: REQ (lead 213 / 214, ACCEPTED in 227 (1)) on SIM

ID	Value	Class	Source	Key	Source tag / formula
T4670	The line-208 band rule, judged against the v3d 285 m reference with the dip onset at 80 ...	DECIDED *Requirement, DATASHEET-INPUTS-GNC.md v3.2; basis SIM*	DSIG v3.2	`line 22 row "Mission set: A1 apogee target (baseline flight)" col 4`	tag in DATASHEET-INPUTS-GNC.md v3.2: REQ (lead 213 / 214, ACCEPTED in 227 (1)) on SIM
T4671	v3.2 `v32_items_rev4b[3]`; `01-systems-gnc/staging_v3.1/rev4/out_rev4/p2/p2_rev4.json` ...		DSIG v3.2	`line 22 row "Mission set: A1 apogee target (baseline flight)" col 5`	source column (reference)
T4672	62 m, PROVISIONAL until the T1-T4 regression data. The 85 m dispersed cycles stay the ...	ASSUMED *Requirement, DATASHEET-INPUTS-GNC.md v3.2; basis SIM*	DSIG v3.2	`line 23 row "Mission set: B1 development hop cap" col 2`	REQ (lead 2026-10-13) on SIM; regression band ASSUMED
T4673	At 62 m (regression band plus 7-11 % throat growth): P 0.996 (SIM, N = 500, ledger v3d / ...	ASSUMED *Requirement, DATASHEET-INPUTS-GNC.md v3.2; basis SIM*	DSIG v3.2	`line 23 row "Mission set: B1 development hop cap" col 4`	REQ (lead 2026-10-13) on SIM; regression band ASSUMED
T4674	v3.2 `b1_regression_band.lead_cap`, `flight_profile.mission_plan_v3.v.B1` ...		DSIG v3.2	`line 23 row "Mission set: B1 development hop cap" col 5`	source column (reference)
T4675	P = 0.9835. This is P(sim6 landing AND >= 80 g liquid at touchdown), pooled A1_none ...	SIM	DSIG v3.2	`line 24 row "A1 landing P at 210 m" col 2`	SIM
T4676	N = 1000 per main base (2000 flights), state liq|G73, v3.5.1, sim6 v2.9-dev. This P is ...	SIM	DSIG v3.2	`line 24 row "A1 landing P at 210 m" col 4`	SIM
T4677	`p2_rev4.json` (68cf1f643effd5b9) `variants.v351.states.liq.bands.210`		DSIG v3.2	`line 24 row "A1 landing P at 210 m" col 5`	source column (reference)
T4678	2.92	DERIVED	DSIG v3.2	`line 25 row "Liftoff T/W" col 2`	DERIVED
T4679	Rated 500 N; liftoff mass 17.44 kg; nominal deterministic A1 flight; 0 kg payload; 2.00 ...	DERIVED	DSIG v3.2	`line 25 row "Liftoff T/W" col 4`	DERIVED
T4680	v3.2 `v32_items_rev4b[15]` (c); `01-systems-gnc/staging_v3.1/rev4/out_rev4/payload_n500/p` ...		DSIG v3.2	`line 25 row "Liftoff T/W" col 5`	source column (reference)
T4681	3.16; 0.633. The floor ratio is the 20 % floor thrust over the touchdown weight.	DERIVED	DSIG v3.2	`line 26 row "Touchdown T/W; landing floor ratio" col 2`	DERIVED
T4682	Nominal touchdown mass 16.11 kg (as row 4). The landing ceiling is >= 237.6 N, from 1.5 ...	DERIVED	DSIG v3.2	`line 26 row "Touchdown T/W; landing floor ratio" col 4`	DERIVED
T4683	as row 4; v3.2 `requirements.throttle`		DSIG v3.2	`line 26 row "Touchdown T/W; landing floor ratio" col 5`	source column (reference)
T4684	Nominal planning touchdown mass <= 16.15 kg. A planning mass above 16.15 kg needs its ...	DERIVED *Requirement, DATASHEET-INPUTS-GNC.md v3.2; basis SIM*	DSIG v3.2	`line 30 row "Landing-mass basis" col 2`	REQ (lead 227 (4)) on SIM; lower bounds DERIVED (hover rule m_land >= F_floor / (0.9 g0))
T4685	16.15 kg is the v3.5.1 estimate covered by the 210 m decision MC (SIM, N = 1000 per main ...	DERIVED *Requirement, DATASHEET-INPUTS-GNC.md v3.2; basis SIM*	DSIG v3.2	`line 30 row "Landing-mass basis" col 4`	REQ (lead 227 (4)) on SIM; lower bounds DERIVED (hover rule m_land >= F_floor / (0.9 g0))
T4686	v3.2 `icd_inputs_for_airframe.landing_mass_window_kg` ...		DSIG v3.2	`line 30 row "Landing-mass basis" col 5`	source column (reference)
T4687	Baseline: no payload, 2.00 kg N2O tank (founder, line 219). +0.5 kg: 190 m band. +1 kg ...	SIM	DSIG v3.2	`line 31 row "Payload record (informational)" col 2`	SIM (flights) / DERIVED (iso-margin, T/W); payload as a point mass at 1.15 m and the tank layout ASSUMED
T4688	Criterion: the line-208 rule with the sweep floor at 180 m, N = 500 per base. +0.5 kg ...	SIM	DSIG v3.2	`line 31 row "Payload record (informational)" col 4`	SIM (flights) / DERIVED (iso-margin, T/W); payload as a point mass at 1.15 m and the tank layout ASSUMED
T4689	v3.2 `v32_items_rev4b[15]`; `payload_rev4.json` (6d62104dac3f4fcd)		DSIG v3.2	`line 31 row "Payload record (informational)" col 5`	source column (reference)
T4690	D = 0.25 / 0.5 / 0.75 / 1.0 s, both states, N = 1000 per main base (about 16,000 ...	TO-VERIFY	DSIG v3.2	`line 32 row "11a" col 2`	TO-VERIFY
T4691	Owner: GNC (arm); recovery / airframe with GNC (canopy bound); v3.2 `open_items[42]`		DSIG v3.2	`line 32 row "11a" col 5`	owner / source column (reference)
T4692	The CAD v3.5.1 re-run (`rev4b/eng11_fe/`) and REV-GNC-7's CAD v3.5.2 check are done (row ...	DERIVED	DSIG v3.2	`line 33 row "11b" col 2`	DERIVED
T4693	Owner: REV-GNC reviewer (check); lead (datasheet sentence); v3.2 `open_items[43]`		DSIG v3.2	`line 33 row "11b" col 5`	owner / source column (reference)
T4694	Measure K_s with its uncertainty, J at the servo output shaft, the reflected-inertia ...	TO-VERIFY	DSIG v3.2	`line 34 row "11c" col 2`	TO-VERIFY
T4695	Owner: founders / Electronics (bench); GNC (re-run); v3.2 `open_items[46]`		DSIG v3.2	`line 34 row "11c" col 5`	owner / source column (reference)
T4696	The full and landing modes, measured or from the airframe FE. ...	TO-VERIFY	DSIG v3.2	`line 35 row "11d" col 2`	TO-VERIFY
T4697	Owner: airframe work (deferred by the founder, line 222); GNC re-runs ENG-07d; v3.2 ...		DSIG v3.2	`line 35 row "11d" col 5`	owner / source column (reference)
T4698	A notch on the measured bending mode (RESEARCH PROPOSAL).	MISSING	DSIG v3.2	`line 36 row "11e" col 2`	not flown (class set at the call site: MISSING: the source tag says no flight or simulation exists for the item)
T4699	Owner: GNC; v3.2 `open_items[45]`		DSIG v3.2	`line 36 row "11e" col 5`	owner / source column (reference)
T4700	Electronics v2.11 S17 CONFIRMED pcf_tmax 0.8 s for W10 < 6 m/s with conditions: the ECU ...	TO-VERIFY	DSIG v3.2	`line 37 row "11f" col 2`	TO-VERIFY
T4701	Owner: Electronics v2.11 (S17, budget); Combustion v2.4b (lit side); then GNC re-runs R2 ...		DSIG v3.2	`line 37 row "11f" col 5`	owner / source column (reference)
T4702	Its B1, thermal and erosion check (A1 engine-on p99 8.92 / 8.71 s, max 9.24 s, > 8.258 s).	TO-VERIFY	DSIG v3.2	`line 38 row "11g" col 2`	TO-VERIFY
T4703	Owner: Combustion / Mechanical; v3.2 `open_items[49]`		DSIG v3.2	`line 38 row "11g" col 5`	owner / source column (reference)
T4704	The coverage set at 210 m (e19, L400, wind sweep) and a 210 m A1 duty-cycle record. B1 ...	MISSING	DSIG v3.2	`line 39 row "11h" col 2`	not flown (class set at the call site: MISSING: the source tag says no flight or simulation exists for the item)
T4705	Owner: GNC; v3.2 `open_items[50]`		DSIG v3.2	`line 39 row "11h" col 5`	owner / source column (reference)
T4706	100 N in the sim vs Combustion's 62.5 N start lift (open model item).	SIM	DSIG v3.2	`line 40 row "11i" col 2`	SIM model item
T4707	Owner: GNC; v3.2 `open_items[51]`		DSIG v3.2	`line 40 row "11i" col 5`	owner / source column (reference)
T4708	The abort is NOT inhibited. Revisit it if B-TVC-1 gives a jam rate. Encoder noise ...	DECIDED *Requirement, DATASHEET-INPUTS-GNC.md v3.2; basis SIM*	DSIG v3.2	`line 41 row "11j" col 2`	tag in DATASHEET-INPUTS-GNC.md v3.2: REQ (lead 227 (7)) on SIM
T4709	Owner: lead (revisit); Electronics / founders (jam rate); GNC (re-run); v3.2 ...		DSIG v3.2	`line 41 row "11j" col 5`	owner / source column (reference)
T4710	Recompute with the exact linkage kinematics (GNC extrapolation 97.8-111.2 deg/s).	TO-VERIFY	DSIG v3.2	`line 42 row "11k" col 2`	TO-VERIFY
T4711	Owner: Mechanical / Electronics		DSIG v3.2	`line 42 row "11k" col 5`	owner / source column (reference)
T4712	The final cap after T1-T4 (regression law measured). B1 is not re-flown on massprops ...	TO-VERIFY	DSIG v3.2	`line 43 row "11l" col 2`	TO-VERIFY
T4713	Owner: lead with the Combustion data (cap); GNC (re-fly); v3.2 `open_items[25]`		DSIG v3.2	`line 43 row "11l" col 5`	owner / source column (reference)
T4714	The FC Pc branch of the release predicate (Pc valid AND Pc >= 300 kPa abs) releases ...	SIM	DSIG v3.2	`line 44 row "11m" col 2`	SIM (Electronics v2.11 S17)
T4715	Owner: GNC / FC owner; v3.2 `open_items[53]`		DSIG v3.2	`line 44 row "11m" col 5`	owner / source column (reference)
T4716	pcf_tmax 0.8 s (W10 < 6 m/s, 227 (5)) holds on the dead side only with the photodiode ...	SIM	DSIG v3.2	`line 45 row "11n" col 2`	SIM (S17)
T4717	Owner: Combustion / Mechanical (view); GNC (reference option); v3.2 `open_items[54]`		DSIG v3.2	`line 45 row "11n" col 5`	owner / source column (reference)
T4718	At the 0.21776 mm start-level lift 3 of 72 P4c corners never reach 37.16 N within 0.8 s ...	SIM	DSIG v3.2	`line 46 row "11o" col 2`	SIM; TO-VERIFY (CF-3)
T4719	Owner: GNC / lead; Combustion (CF-3); v3.2 `open_items[55]`		DSIG v3.2	`line 46 row "11o" col 5`	owner / source column (reference)
T4720	The C1 transducer thermal zero is not in the re-zero / BASE limits: a negative drift ...	DERIVED	DSIG v3.2	`line 47 row "11p" col 2`	DERIVED (REVIEWER); MISSING (F-10)
T4721	Owner: Electronics v2.12; Mechanical (seat node); founders (F-10); GNC (re-fly); v3.2 ...		DSIG v3.2	`line 47 row "11p" col 5`	owner / source column (reference)
T4722	kPurgeFlow_s and the both-open cap are re-set from the CF-4 / B-BLEED-1 measured N2O Cd ...	DECIDED *Requirement, DATASHEET-INPUTS-GNC.md v3.2; basis DERIVED*	DSIG v3.2	`line 48 row "11q" col 2`	tag in DATASHEET-INPUTS-GNC.md v3.2: REQ (lead 257 (2), 258 (2)) on DERIVED

ID	Value	Class	Source	Key	Source tag / formula
T4723	Owner: Combustion / Electronics (CF-4, constants); GNC (reserve test if needed); v3.2 ...		DSIG v3.2	line 48 row "11q" col 5	owner / source column (reference)
T4724	Independent review verdicts for the engine team's digital deliverables. A verdict is not ...	DECIDED *Design value, SUPERVISION.json 18 reviews*	SUP 18 reviews	/note	SUPERVISION.json 18 reviews
T4725	ACCEPT WITH CORRECTIONS	DERIVED	SUP 18 reviews	/reviews/16/verdict	(class set at the call site: DERIVED)
T4726	(1) REV-COMB-4, the independent delta re-review of Combustion v2.4b: interim 07:52:49Z ...	DECIDED *Lead decision, line 256*	CL 266 lines	L256[sentence containing "REV-COMB-4"]	lead (decision) 2026-10-21 (line sha cf51938a477079b6)
T4727	ACCEPT WITH COMMENTS	DERIVED	SUP 18 reviews	/reviews/15/verdict	(class set at the call site: DERIVED)
T4728	(4) REV-ELEC-3 (independent delta review of Electronics v2.11; started 05:45:38Z ...	DECIDED *Lead decision, line 253*	CL 266 lines	L253[sentence containing "REV-ELEC-3"]	lead (decision) 2026-10-21 (line sha 3a7f695ca59d621e)
T4729	PUBLISHABLE WITH CORRECTIONS	DERIVED	SUP 18 reviews	/reviews/17/verdict	(class set at the call site: DERIVED)
T4730	(1) REV-GNC-7 ACCEPTED: its final (reviews/REVIEW-GNC-R5.md 6b86fcd326568646, saved ...	DECIDED *Lead decision, line 258*	CL 266 lines	L258[sentence containing "REV-GNC-7"]	lead decision 2026-10-21 (line sha cdd58cc9fdc95079)
T4731	REVISE	DERIVED	SUP 18 reviews	/reviews/14/verdict	(class set at the call site: DERIVED)
T4732	CHANGELOG line 234	DECIDED *Lead decision, per SUPERVISION.json 18 reviews*	SUP 18 reviews	/reviews/14/lead_disposition	tag in SUPERVISION.json 18 reviews:
T4733	Overall: ACCEPT WITH COMMENTS, with option (b) carved out of every start with ignition ...	DERIVED	RELE4 REVIEW-ELECTRONICS-R4 (REV-ELEC-4)	line 23 (starts "Overall: ACCEPT WITH COMMENTS")	REV-ELEC-4 review text, section 1
T4734	REVIEW-ELECTRONICS-R4 (REV-ELEC-4) · sha256/16 70b7103472eebab4	DERIVED	RELE4 REVIEW-ELECTRONICS-R4 (REV-ELEC-4)	(file)	generator: version and sha256/16 of source RELE4 at this build [REVIEW-ELECTRONICS-R4.md]
T4735	1. VERDICT: ACCEPT WITH CORRECTIONS	DERIVED	RDS1 REVIEW-DATASHEET-R1 (REV-DS-1)	line 23 (starts "1. VERDICT: ACCEPT WITH CORRECTIONS")	REV-DS-1 review text, section 1
T4736	No finding is BLOCKING.	DERIVED	RDS1 REVIEW-DATASHEET-R1 (REV-DS-1)	line 31 (starts "No finding is BLOCKING.")	REV-DS-1 review text, section 1
T4737	There are 5 MAJOR findings (D1-01, -02, -03, -04, -06) and 8 MINOR (D1-05, D1-07 to ...	DERIVED	RDS1 REVIEW-DATASHEET-R1 (REV-DS-1)	line 32 (starts "There are 5 MAJOR findings")	REV-DS-1 review text, section 1
T4738	REV-DS-1 (independent review of the rev1-draft): ACCEPT WITH CORRECTIONS, 0 BLOCKING, 5 ...	DERIVED	generator	REV-DS-1 status in this build (text set by the lead 2026-10-21 14:30:54Z)	status text set by the lead for this build; the counts are those of the review file (section 1)
T4739	REV-DS-1 (independent review of the rev1-draft; phase B from 13:23Z; batches 13:41:04Z ...	DECIDED *Lead decision, line 266 (3)*	CL 266 lines	L266(3)[from "REV-DS-1 (independent review of the rev1-draft" to "Every finding ACCEPTED and sent to the author (13:47:38Z, 14:05:35Z, 14:30:54Z)."]	lead (decision) 2026-10-21 (line sha 6f3181b859f02937)
T4740	REV-COMB-1	DERIVED	SUP 18 reviews	/reviews/0/id	(class set at the call site: DERIVED)
T4741	2026-10-12/13	DERIVED	SUP 18 reviews	/reviews/0/date	(class set at the call site: DERIVED)
T4742	REVISE	DERIVED	SUP 18 reviews	/reviews/0/verdict	(class set at the call site: DERIVED)
T4743	02-combustion (REPORT.md v2.3, combustion.json v2.3, HANDBACK-SUMMARY.md)	DERIVED	SUP 18 reviews	/reviews/0/deliverable	(class set at the call site: DERIVED)
T4744	0 of 3 reviewed files unchanged; changed since review: 02-combustion/REPORT.md ...	DERIVED	generator	formula: sha256/16 of each reviewed hub file vs the register entry	DERIVED by build_datasheet.py: sha256/16 of each reviewed hub file vs the register entry
T4745	REV-PLUME-1	DERIVED	SUP 18 reviews	/reviews/1/id	(class set at the call site: DERIVED)
T4746	2026-10-12/13	DERIVED	SUP 18 reviews	/reviews/1/date	(class set at the call site: DERIVED)
T4747	05-plume-cfd (REPORT.md)	DERIVED	SUP 18 reviews	/reviews/1/deliverable	(class set at the call site: DERIVED)
T4748	0 of 1 reviewed files unchanged; changed since review: 05-plume-cfd/REPORT.md	DERIVED	generator	formula: sha256/16 of each reviewed hub file vs the register entry	DERIVED by build_datasheet.py: sha256/16 of each reviewed hub file vs the register entry
T4749	REV-GNC-1	DERIVED	SUP 18 reviews	/reviews/2/id	(class set at the call site: DERIVED)
T4750	2026-10-13	DERIVED	SUP 18 reviews	/reviews/2/date	(class set at the call site: DERIVED)
T4751	REVISE	DERIVED	SUP 18 reviews	/reviews/2/verdict	(class set at the call site: DERIVED)
T4752	01-systems-gnc v3 (REPORT-v3.md, gnc.json v3) with addendum on v3.1 staging (FC_THRUST)	DERIVED	SUP 18 reviews	/reviews/2/deliverable	(class set at the call site: DERIVED)
T4753	4 of 5 reviewed files unchanged; changed since review: interfaces/gnc.json	DERIVED	generator	formula: sha256/16 of each reviewed hub file vs the register entry	DERIVED by build_datasheet.py: sha256/16 of each reviewed hub file vs the register entry
T4754	REV-ELEC-1	DERIVED	SUP 18 reviews	/reviews/3/id	(class set at the call site: DERIVED)
T4755	2026-10-13	DERIVED	SUP 18 reviews	/reviews/3/date	(class set at the call site: DERIVED)
T4756	REVISE	DERIVED	SUP 18 reviews	/reviews/3/verdict	(class set at the call site: DERIVED)
T4757	06-electronics v2.9 (firmware v2.9, SIL sil_v2p9, KiCad ECU v2.4, BOM v2.1 ...	DERIVED	SUP 18 reviews	/reviews/3/deliverable	(class set at the call site: DERIVED)
T4758	3 of 10 reviewed files unchanged; changed since review ...	DERIVED	generator	formula: sha256/16 of each reviewed hub file vs the register entry	DERIVED by build_datasheet.py: sha256/16 of each reviewed hub file vs the register entry
T4759	REV-GNC-2	DERIVED	SUP 18 reviews	/reviews/4/id	(class set at the call site: DERIVED)
T4760	2026-10-13	DERIVED	SUP 18 reviews	/reviews/4/date	(class set at the call site: DERIVED)
T4761	CONDICIONADO	DERIVED	SUP 18 reviews	/reviews/4/verdict	(class set at the call site: DERIVED)
T4762	01-systems-gnc/staging_v3.1/rev1 (REPORT-GNC-rev1.md, rev1_items.json, FSW record v1.4 ...	DERIVED	SUP 18 reviews	/reviews/4/deliverable	(class set at the call site: DERIVED)
T4763	6 of 6 reviewed files unchanged	DERIVED	generator	formula: sha256/16 of each reviewed hub file vs the register entry	DERIVED by build_datasheet.py: sha256/16 of each reviewed hub file vs the register entry
T4764	REV-COMB-2	DERIVED	SUP 18 reviews	/reviews/5/id	(class set at the call site: DERIVED)
T4765	2026-10-13	DERIVED	SUP 18 reviews	/reviews/5/date	(class set at the call site: DERIVED)
T4766	REVISE (targeted)	DERIVED	SUP 18 reviews	/reviews/5/verdict	(class set at the call site: DERIVED)
T4767	02-combustion v2.4 (REPORT.md, REVISION-RESPONSE.md, CF-4-SPEC.md ...	DERIVED	SUP 18 reviews	/reviews/5/deliverable	(class set at the call site: DERIVED)
T4768	0 of 5 reviewed files unchanged; changed since review: 02-combustion/REPORT.md ...	DERIVED	generator	formula: sha256/16 of each reviewed hub file vs the register entry	DERIVED by build_datasheet.py: sha256/16 of each reviewed hub file vs the register entry
T4769	REV-PLUME-2	DERIVED	SUP 18 reviews	/reviews/6/id	(class set at the call site: DERIVED)
T4770	2026-10-13	DERIVED	SUP 18 reviews	/reviews/6/date	(class set at the call site: DERIVED)
T4771	05-plume-cfd v2.2b (REPORT.md)	DERIVED	SUP 18 reviews	/reviews/6/deliverable	(class set at the call site: DERIVED)
T4772	1 of 1 reviewed files unchanged	DERIVED	generator	formula: sha256/16 of each reviewed hub file vs the register entry	DERIVED by build_datasheet.py: sha256/16 of each reviewed hub file vs the register entry
T4773	REV-GNC-3	DERIVED	SUP 18 reviews	/reviews/7/id	(class set at the call site: DERIVED)
T4774	2026-10-13	DERIVED	SUP 18 reviews	/reviews/7/date	(class set at the call site: DERIVED)
T4775	CONDICIONADO	DERIVED	SUP 18 reviews	/reviews/7/verdict	(class set at the call site: DERIVED)
T4776	01-systems-gnc/staging_v3.1/rev2 (REPORT-GNC-rev2.md, staging JSON, sim6 v2.9-dev ...	DERIVED	SUP 18 reviews	/reviews/7/deliverable	(class set at the call site: DERIVED)
T4777	6 of 7 reviewed files unchanged; not in the hub: interfaces/CHANGELOG.md (lines 155-175 ...	DERIVED	generator	formula: sha256/16 of each reviewed hub file vs the register entry	DERIVED by build_datasheet.py: sha256/16 of each reviewed hub file vs the register entry
T4778	REV-MECH-1	DERIVED	SUP 18 reviews	/reviews/8/id	(class set at the call site: DERIVED)
T4779	2026-10-13	DERIVED	SUP 18 reviews	/reviews/8/date	(class set at the call site: DERIVED)
T4780	REVISE (FEA); REVISE, not freezable (CAD v3.5)	DERIVED	SUP 18 reviews	/reviews/8/verdict	(class set at the call site: DERIVED)
T4781	04-fea (S-AX L1_TC09IN718_thJ1revB_TC09IN, S-3D v1.1, joints v1p1, stiffness V5c) and ...	DERIVED	SUP 18 reviews	/reviews/8/deliverable	(class set at the call site: DERIVED)
T4782	5 of 12 reviewed files unchanged; not in the hub: interfaces/mechanical.json v2.1 ...	DERIVED	generator	formula: sha256/16 of each reviewed hub file vs the register entry	DERIVED by build_datasheet.py: sha256/16 of each reviewed hub file vs the register entry
T4783	REV-GNC-4	DERIVED	SUP 18 reviews	/reviews/9/id	(class set at the call site: DERIVED)
T4784	2026-10-13	DERIVED	SUP 18 reviews	/reviews/9/date	(class set at the call site: DERIVED)
T4785	PUBLISHABLE WITH CORRECTIONS (not publishable as staged)	DERIVED	SUP 18 reviews	/reviews/9/verdict	(class set at the call site: DERIVED)
T4786	01-systems-gnc/staging_v3.1/rev2p1 (gnc.json v3.1 publication candidate)	DERIVED	SUP 18 reviews	/reviews/9/deliverable	(class set at the call site: DERIVED)

ID	Value	Class	Source	Key	Source tag / formula
T4787	1 of 5 reviewed files unchanged; not in the hub ...	DERIVED	generator	formula: sha256/16 of each reviewed hub file vs the register entry	DERIVED by build_datasheet.py: sha256/16 of each reviewed hub file vs the register entry
T4788	REV-COMB-3	DERIVED	SUP 18 reviews	/reviews/10/id	(class set at the call site: DERIVED)
T4789	2026-10-13	DERIVED	SUP 18 reviews	/reviews/10/date	(class set at the call site: DERIVED)
T4790	REVISE (targeted)	DERIVED	SUP 18 reviews	/reviews/10/verdict	(class set at the call site: DERIVED)
T4791	02-combustion v2.4a (REPORT.md, REVISION-RESPONSE-R2.md, CF-4-SPEC.md ...	DERIVED	SUP 18 reviews	/reviews/10/deliverable	(class set at the call site: DERIVED)
T4792	2 of 6 reviewed files unchanged; changed since review: 02-combustion/REPORT.md ...	DERIVED	generator	formula: sha256/16 of each reviewed hub file vs the register entry	DERIVED by build_datasheet.py: sha256/16 of each reviewed hub file vs the register entry
T4793	REV-GNC-5	DERIVED	SUP 18 reviews	/reviews/11/id	(class set at the call site: DERIVED)
T4794	2026-10-13	DERIVED	SUP 18 reviews	/reviews/11/date	(class set at the call site: DERIVED)
T4795	PUBLISHABLE WITH CORRECTIONS	DERIVED	SUP 18 reviews	/reviews/11/verdict	(class set at the call site: DERIVED)
T4796	01-systems-gnc/staging_v3.1/rev3 (GNC rev3: P2-P6, P8-P9; 11 v3.2 proposals) and lead ...	DERIVED	SUP 18 reviews	/reviews/11/deliverable	(class set at the call site: DERIVED)
T4797	3 of 4 reviewed files unchanged; changed since review: interfaces/gnc.json	DERIVED	generator	formula: sha256/16 of each reviewed hub file vs the register entry	DERIVED by build_datasheet.py: sha256/16 of each reviewed hub file vs the register entry
T4798	REV-ELEC-2	DERIVED	SUP 18 reviews	/reviews/12/id	(class set at the call site: DERIVED)
T4799	2026-10-13	DERIVED	SUP 18 reviews	/reviews/12/date	(class set at the call site: DERIVED)
T4800	REVISE	DERIVED	SUP 18 reviews	/reviews/12/verdict	(class set at the call site: DERIVED)
T4801	06-electronics v2.10 hub copy (firmware, SIL sil_v2p10, FA-1 harness, bench_plan_v3, BOM ...	DERIVED	SUP 18 reviews	/reviews/12/deliverable	(class set at the call site: DERIVED)
T4802	10 of 11 reviewed files unchanged; changed since review: interfaces/electronics.json	DERIVED	generator	formula: sha256/16 of each reviewed hub file vs the register entry	DERIVED by build_datasheet.py: sha256/16 of each reviewed hub file vs the register entry
T4803	REV-GNC-6	DERIVED	SUP 18 reviews	/reviews/13/id	(class set at the call site: DERIVED)
T4804	2026-10-19	DERIVED	SUP 18 reviews	/reviews/13/date	(class set at the call site: DERIVED)
T4805	PUBLISHABLE WITH CORRECTIONS	DERIVED	SUP 18 reviews	/reviews/13/verdict	(class set at the call site: DERIVED)
T4806	01-systems-gnc/staging_v3.1/rev4 (GNC rev4: A1 band on v3.5.1, landing mass, ENG-11 ...	DERIVED	SUP 18 reviews	/reviews/13/deliverable	(class set at the call site: DERIVED)
T4807	4 of 6 reviewed files unchanged; changed since review: interfaces/gnc.json; not in the ...	DERIVED	generator	formula: sha256/16 of each reviewed hub file vs the register entry	DERIVED by build_datasheet.py: sha256/16 of each reviewed hub file vs the register entry
T4808	REV-MECH-2	DERIVED	SUP 18 reviews	/reviews/14/id	(class set at the call site: DERIVED)
T4809	2026-10-20	DERIVED	SUP 18 reviews	/reviews/14/date	(class set at the call site: DERIVED)
T4810	04-fea/assembly3d (FEA3D final on CAD v3.5.1, asm v1.2, CalculiX 2.21 C3D10, L1 171,123 ...	DERIVED	SUP 18 reviews	/reviews/14/deliverable	(class set at the call site: DERIVED)
T4811	5 of 6 reviewed files unchanged; not in the hub: interfaces/CHANGELOG.md (230 lines at ...	DERIVED	generator	formula: sha256/16 of each reviewed hub file vs the register entry	DERIVED by build_datasheet.py: sha256/16 of each reviewed hub file vs the register entry
T4812	REV-ELEC-3	DERIVED	SUP 18 reviews	/reviews/15/id	(class set at the call site: DERIVED)
T4813	2026-10-21	DERIVED	SUP 18 reviews	/reviews/15/date	(class set at the call site: DERIVED)
T4814	06-electronics/staging_v2.11 (Electronics v2.11: firmware v2.11 core, SIL record ...	DERIVED	SUP 18 reviews	/reviews/15/deliverable	(class set at the call site: DERIVED)
T4815	4 of 6 reviewed files unchanged; not in the hub: interfaces/combustion.json (v2.4b ...	DERIVED	generator	formula: sha256/16 of each reviewed hub file vs the register entry	DERIVED by build_datasheet.py: sha256/16 of each reviewed hub file vs the register entry
T4816	REV-COMB-4	DERIVED	SUP 18 reviews	/reviews/16/id	(class set at the call site: DERIVED)
T4817	2026-10-21	DERIVED	SUP 18 reviews	/reviews/16/date	(class set at the call site: DERIVED)
T4818	02-combustion v2.4b (REPORT.md, REVISION-RESPONSE-R3.md, DATASHEET-INPUTS-COMBUSTION.md ...	DERIVED	SUP 18 reviews	/reviews/16/deliverable	(class set at the call site: DERIVED)
T4819	1 of 8 reviewed files unchanged; changed since review: 02-combustion/REPORT.md ...	DERIVED	generator	formula: sha256/16 of each reviewed hub file vs the register entry	DERIVED by build_datasheet.py: sha256/16 of each reviewed hub file vs the register entry
T4820	REV-GNC-7	DERIVED	SUP 18 reviews	/reviews/17/id	(class set at the call site: DERIVED)
T4821	2026-10-21	DERIVED	SUP 18 reviews	/reviews/17/date	(class set at the call site: DERIVED)
T4822	01-systems-gnc/staging_v3.1/rev4b (GNC rev4b: gnc.json v3.2 candidate, ENG-11 FE-plant ...	DERIVED	SUP 18 reviews	/reviews/17/deliverable	(class set at the call site: DERIVED)
T4823	3 of 6 reviewed files unchanged; not in the hub: interfaces/gnc.json (v3.1 at the ...	DERIVED	generator	formula: sha256/16 of each reviewed hub file vs the register entry	DERIVED by build_datasheet.py: sha256/16 of each reviewed hub file vs the register entry
T4824	J-FWD build and every re-assembly (v3: route (i), lead 16:53Z R2 / R3, 17:55:55Z F1 / ...	DECIDED *Lead decision, per mechanical.json v2.4*	MEC v2.4	/jfwd_v352/founder_rules/C4_procedure	tag in mechanical.json v2.4:
T4825	RULED: the J-FWD screw is IN718 M5 × 10, ENG-HW-101-IN718: AMS 5662 bar aged to AMS ...	DECIDED *Lead decision, line 240*	CL 266 lines	L240[sentence containing "RULED: the J-FWD screw is IN718 M5" | "FALLBACK only if IN718"]	lead (decision) 2026-10-20 (line sha 6ced8f178b20ce0f)
T4826	Spring A (Schnorr 001 500 A in 1.4568) is the baseline and spring B (Lesjofors 7909) the ...	DECIDED *Lead decision, line 243*	CL 266 lines	L243[sentence containing "Spring A (Schnorr 001 500 A in 1.4568)"]	lead (decision) 2026-10-20 (line sha 71288a1cb10b1ffd)
T4827	HW-170 disc spring SINGLE-USE (lead 19:53:27Z, CHANGELOG 248 (7)): a new spring with ...	DECIDED *Lead decision, per mechanical.json v2.4*	MEC v2.4	/jfwd_v352/founder_rules/HW-170_single_use	tag in mechanical.json v2.4:
T4828	F1: the J-F1 MAXIMUM PRELOAD IS CAPPED AT 3750 N (J-F1 sets the flight-lot torque so ...	DECIDED *Lead decision, line 245*	CL 266 lines	L245[sentence containing "J-F1 MAXIMUM PRELOAD IS CAPPED AT 3750 N" | "Prescale"]	lead (decision) 2026-10-20 (line sha 890c52f613f00c14)
T4829	J-F1b (the optional J-F1 extension of hardware v1.7 HW-101; rule lead 19:53:27Z, PEAK ...	DECIDED *Lead decision, per mechanical.json v2.4*	MEC v2.4	/jfwd_v352/founder_rules/J-F1b	tag in mechanical.json v2.4:
T4830	Prescale TO-VERIFY, J-F1b method TO-DEFINE, LOCTITE 243 range TO-VERIFY (founder / test ...	TO-VERIFY	CL 266 lines	L254[sentence containing "J-F1b method TO-DEFINE"]	mechanical (result) 2026-10-21 (line sha c7d52385af9ce594)
T4831	LOCTITE 243 tap zone at TB_a150: 128.0 C GVLO / 116.0 C GVHI (148.7 / 146.9 C with the ...	DECIDED *Lead decision, line 248*	CL 266 lines	L248[sentence containing "LOCTITE 243 tap zone"]	lead (decision) 2026-10-20 (line sha 93f48a7284090206)
T4832	Prescale TO-VERIFY, J-F1b method TO-DEFINE, LOCTITE 243 range TO-VERIFY (founder / test ...	TO-VERIFY	CL 266 lines	L254[sentence containing "LOCTITE 243 range TO-VERIFY"]	mechanical (result) 2026-10-21 (line sha c7d52385af9ce594)
T4833	J-FWD screws are new at every assembly, never reused or re-torqued; removed screws are ...	DECIDED *Lead decision, per mechanical.json v2.4*	MEC v2.4	/jfwd_v352/founder_rules/single_use	tag in mechanical.json v2.4:
T4834	Founder rule until C6 closes: the J-FWD screws are single-use (new at every assembly ...	DECIDED *Lead decision, line 243*	CL 266 lines	L243[sentence containing "Founder rule until C6 closes"]	lead (decision) 2026-10-20 (line sha 71288a1cb10b1ffd)
T4835	(4) POST-ABORT RULE (founder rule, lead 248 (7)): after any case-TC or TB-state abort ...	DECIDED *Lead decision, per mechanical.json v2.4*	MEC v2.4	/jfwd_v352/founder_rules/post_abort_joint_replacement	tag in mechanical.json v2.4:
T4836	FOUNDER RULE: after any case-TC or TB-state abort the J-FWD joint counts as used (HW-101 ...	DECIDED *Lead decision, line 248*	CL 266 lines	L248[sentence containing "FOUNDER RULE: after any case-TC"]	lead (decision) 2026-10-20 (line sha 93f48a7284090206)
T4837	drawing_note_ENG-HW-101-IN718_and_HW-170: HW-170 DISC SPRING ORIENTATION (route (i) ...	DERIVED	MEC v2.4	/jfwd_v352/founder_rules/HW-170_orientation	(class set at the call site: DERIVED)
T4838	STAND T0 RULE (founder action): B1-length stand burns only with the engine soaked at a ...	DECIDED *Lead decision, line 249 (2)*	CL 266 lines	L249(2)	lead (decision) 2026-10-21 (line sha 21373a357d2dd391)
T4839	STAND T0 RULE (founder action, lead CHANGELOG 249 (2); cross-reference only, not a ...	DECIDED *Lead decision, per mechanical.json v2.4*	MEC v2.4	/jfwd_v352/founder_rules/stand_T0_cross_reference	tag in mechanical.json v2.4:
T4840	TC-01 drawing, 10x d5.5 J-FWD holes (rev 5): (a) bore-side edge: break R0.1-R0.2 ...	DECIDED *Design value, item_c_tf1_texts_v352_v5.json v5*	TXT5 v5	/item_C/C1_TC01_drawing_callout_rev5	item_c_tf1_texts_v352_v5.json v5
T4841	HOLE EDGE BREAK (item C amended 10:08Z; callout ruled 10:15Z) [DERIVED]: the bore-side ...	DECIDED *Lead decision, line 240 (5)*	CL 266 lines	L240(5)	lead (decision) 2026-10-20 (line sha 6ced8f178b20ce0f)
T4842	C1, TC-01 rev 5 at the 10 x d5.5 J-FWD holes: (a) bore-side break R0.1-R0.2 blended and ...	DECIDED *Lead decision, line 241 (8)*	CL 266 lines	L241(8)[from "C1, TC-01 rev 5 at the 10 x d5.5 J-FWD holes" to "(g) and (h) as item 6 (c) and (d)."]	lead (decision) 2026-10-20 (line sha 0e301646c032c306)
T4843	B-TC-1 (founder: tau and the static terms);	DECIDED *Lead decision, line 249*	CL 266 lines	L249[sentence containing "B-TC-1 (founder"]	lead (decision) 2026-10-21 (line sha 21373a357d2dd391)
T4844	low-reading tolerance tol_lo = 9.05 + 0.630 r K worst case (r = station heating rate at ...	DECIDED *Lead decision, line 249 (1)*	CL 266 lines	L249(1)[from "low-reading tolerance tol_lo" to "else the tolerance is re-derived."]	lead (decision) 2026-10-21 (line sha 21373a357d2dd391)
T4845	Founder items added: F-10 also asks Kulite for the thermal-zero maximum and for a 100 F ...	DECIDED *Lead decision, line 253 (11)*	CL 266 lines	L253(11)	lead (decision) 2026-10-21 (line sha 3a7f695ca59d621e)

ID	Value	Class	Source	Key	Source tag / formula
T4846	T-F1 (founder action, a GATE before the first static fire): 183 (1)(d) / 188 (5) needs a ...	DECIDED *Lead decision, line 241 (8)*	CL 266 lines	L241(8)[from "T-F1 (founder action, a GATE before the first static fire)" to "no recalled or estimated value replaces either route."]	lead (decision) 2026-10-20 (line sha 0e301646c032c306)
T4847	T-F1 restated per R9 (A 175 rows on both bounds, A 150 beside; (a)-(c) PASS, (d) OPEN)		CL 266 lines	L254(2)[from "T-F1 restated per R9" to "(d) OPEN)"]	mechanical (result) 2026-10-21 (line sha c7d52385af9ce594); class set at the call site: a source line of the founder action (the specialist's record line, quoted as its reference)
T4848	Mechanical's re-map restates tf1_exposure_v352 (d23e72f089cca33a) and the T-F1 founder ...	DECIDED *Lead decision, line 249*	CL 266 lines	L249[sentence containing "T-F1 founder text on the A 175 rows"]	lead (decision) 2026-10-21 (line sha 21373a357d2dd391)
T4849	F2 ADOPTED into C4: two GSE shoulder screws selected to the pin-gauged holes at two ...	DECIDED *Lead decision, line 245*	CL 266 lines	L245[sentence containing "F2 ADOPTED into C4" | "F5: clocking"]	lead (decision) 2026-10-20 (line sha 890c52f613f00c14)
T4850	ENG-TC-12 J-FWD hole plug set (GSE, not flight mass; TO-DRAW; TO-QUOTE, or made from POM ...	DECIDED *Design value, item_c_tf1_texts_v352_v5.json v5*	TXT5 v5	/item_C/C3_fitting_aid_GSE	item_c_tf1_texts_v352_v5.json v5
T4851	C3: ENG-TC-12 J-FWD hole-plug set (GSE, not flight; 10 POM plugs and a band; shank d5.5 ...	DECIDED *Lead decision, line 241 (8)*	CL 266 lines	L241(8)[from "C3: ENG-TC-12 J-FWD hole-plug set" to "TO-DRAW / TO-QUOTE)."]	lead (decision) 2026-10-20 (line sha 0e301646c032c306)
T4852	ENG-TC-13 J-FWD registration shoulder screw pair (GSE, not flight mass; TO-DRAW / ...	DECIDED *Design value, item_c_tf1_texts_v352_v5.json v5*	TXT5 v5	/item_C/C3b_registration_shoulder_screws_GSE	item_c_tf1_texts_v352_v5.json v5
T4853	F2 ADOPTED into C4: two GSE shoulder screws selected to the pin-gauged holes at two ...	DECIDED *Lead decision, line 245 (4)*	CL 266 lines	L245(4)[from "F2 ADOPTED into C4" to "shoulder screws out last"]	lead (decision) 2026-10-20 (line sha 890c52f613f00c14)
T4854	new C3b (ENG-TC-13 GSE)		CL 266 lines	L246(2)[from "new C3b (ENG-TC-13 GSE)" to "new C3b (ENG-TC-13 GSE)"]	mechanical (result: hub publication 1 of 2 after the release 19:09:50Z = 04-fea/v352 J-FWD route (i) rows v7 + four companions as ACCEPTED 18:36:37Z, the drawing texts, the rev-5 CAD check they ...
T4855	match-drill fixture with the index pin and the guide bushings: shop tooling TO-DRAW (not ...	DECIDED *Design value, item_c_tf1_texts_v352_v5.json v5*	TXT5 v5	/item_C/C8_jfwd_drawing_texts_rev5/open_items/2	item_c_tf1_texts_v352_v5.json v5
T4856	J-FWD taps 10x M5-6H x 8.0 full thread, drill 4.2 × 9.0, radial (rev 5, lead 16:53Z R2) ...	DECIDED *Design value, item_c_tf1_texts_v352_v5.json v5*	TXT5 v5	/item_C/C8_jfwd_drawing_texts_rev5/ENG-TC-02_rev5/jfwd_fits_items/0	item_c_tf1_texts_v352_v5.json v5
T4857	F5: clocking +/-0.25 deg at match-drilling by an index pin in the drill fixture ...	DECIDED *Lead decision, line 245 (4)*	CL 266 lines	L245(4)[from "F5: clocking +/-0.25 deg at match-drilling" to "tap-wall row."]	lead (decision) 2026-10-20 (line sha 890c52f613f00c14)
T4858	PRELOAD-RETENTION AUDIT ADOPTED (11:25Z) as a founder procedure and the physical arbiter ...	DECIDED *Lead decision, line 240*	CL 266 lines	L240[sentence containing "PRELOAD-RETENTION AUDIT ADOPTED"]	lead (decision) 2026-10-20 (line sha 6ced8f178b20ce0f)
T4859	GLAND-1 FA1S CLOSED (10:10Z; gland_fwd_v352_v3.json) [DATASHEET as cited; DERIVED]: the ...	DECIDED *Lead decision, line 240 (6)*	CL 266 lines	L240(6)	lead (decision) 2026-10-20 (line sha 6ced8f178b20ce0f)
T4860	(1) S01 and S02 are replaced after any FA-1-class event, together with the line-231 ...	DECIDED *Lead decision, per mechanical.json v2.4*	MEC v2.4	/jfwd_v352/founder_rules/C6	tag in mechanical.json v2.4:
T4861	Closure paths: (1) an MJ-form external thread (ISO 5855-2 root radius) with a sourced Kf ...	DECIDED *Lead decision, line 243 (5)*	CL 266 lines	L243(5)[from "Closure paths:" to "(3) a qualification test (TO-DEFINE, founder)."]	lead (decision) 2026-10-20 (line sha 71288a1cb10b1ffd)
T4862	C6 closure path 1 is negative, paths 2 ((b1) FE levers) and 3 (qualification test) ...	DECIDED *Lead decision, line 245 (4)*	CL 266 lines	L245(4)[from "C6 closure path 1 is negative" to "C6 OPEN"]	lead (decision) 2026-10-20 (line sha 890c52f613f00c14)
T4863	screw thread-root LCF paths 2 / 3 (Mechanical / founder test TO-DEFINE)		CL 266 lines	L254(6)[from "screw thread-root LCF paths 2 / 3" to "(Mechanical / founder test TO-DEFINE)"]	mechanical (result) 2026-10-21 (line sha c7d52385af9ce594); class set at the call site: a source line of the founder action (the specialist's record line, quoted as its reference)
T4864	with the per-serial coefficient and T_cal from B-PC-1 (0-85 C) and the bench plan v5 ...	DECIDED *Lead decision, line 264 (2)*	CL 266 lines	L264(2)[from "with the per-serial coefficient and T_cal from B-PC-1" to "at k = 3)"]	lead (decision) 2026-10-21 (line sha 7cdbe51041ae6b8b)
T4865	The fallback (coefficient 0, the flagged v2.11 behaviour) is NOT a flight configuration ...	DECIDED *Lead decision, line 264 (2)*	CL 266 lines	L264(2)[from "The fallback (coefficient 0" to "at ARM [DERIVED]."]	lead (decision) 2026-10-21 (line sha 7cdbe51041ae6b8b)
T4866	the BOM gains one C1-family unit, TO-QUOTE delivered to Sonderborg DK-6400.	DECIDED *Lead decision, line 264 (4)*	CL 266 lines	L264(4)[from "the BOM gains one C1-family unit" to "DK-6400."]	lead (decision) 2026-10-21 (line sha 7cdbe51041ae6b8b)
T4867	with three conditions: (a) wherever 1.9 s is quoted as the needed flow, 2.2 s at the ...	DECIDED *Lead decision, line 256 (2)*	CL 266 lines	L256(2)[from "with three conditions:" to "needs 2.5 s)."]	lead (decision) 2026-10-21 (line sha cf51938a477079b6)
T4868	DECIDED: after CF-4 / B-BLEED-1, kPurgeFlow_s and the both-open cap are re-set from the ...	DECIDED *Lead decision, line 257 (2)*	CL 266 lines	L257(2)[from "DECIDED: after CF-4 / B-BLEED-1" to "at the measured vent."]	lead decision 2026-10-21 (line sha cf364643652fceb0)
T4869	DECIDED: the two REV-GNC-7 additions are adopted as (d1) the re-set uses the upper 95 % ...	DECIDED *Lead decision, line 258 (2)*	CL 266 lines	L258(2)[from "DECIDED: the two REV-GNC-7 additions" to "condition (b) governs."]	lead decision 2026-10-21 (line sha cdd58cc9fdc95079)
T4870	ADOPTED by LEAD 258 (3): Q_JFWD = max(head peak over the record, max of S(T) over T ...	DECIDED *Lead decision, per combustion.json v2.4d*	CMB v2.4d	/z3_jfwd_rulings_v2p4d/v/jfwd_replace_rule/decision	tag in combustion.json v2.4d:
T4871	Q_JFWD RULING: the J-FWD replace rule uses Q_JFWD = max(head peak over the record, max ...	DECIDED *Lead decision, line 258*	CL 266 lines	L258[sentence containing "Q_JFWD"]	lead decision 2026-10-21 (line sha cdd58cc9fdc95079)
T4872	on measured stand traces a noise-driven trigger still means the inspection until a ...	TO-VERIFY	CMB v2.4d	/z3_jfwd_rulings_v2p4d/v/jfwd_replace_rule/measured_traces	(class set at the call site: TO-VERIFY)
T4873	RFQ BOM 237	DECIDED *Design value, bench_plan_v5.md v5*	BENCH v5	line 53 row "F-1" col 1	tag in bench_plan_v5.md v5: bench plan v5 section 0 (founder block)
T4874	PCB Piezotronics 113B22 × 3 (two in use, one spare) with calibration certificates; flush ...	DECIDED *Design value, bench_plan_v5.md v5*	BENCH v5	line 53 row "F-1" col 2	tag in bench_plan_v5.md v5: bench plan v5 section 0 (founder block)
T4875	B-DPC-1, IGN-1, T series	DECIDED *Design value, bench_plan_v5.md v5*	BENCH v5	line 53 row "F-1" col 3	tag in bench_plan_v5.md v5: bench plan v5 section 0 (founder block)
T4876	quote with lead time on file	DECIDED *Design value, bench_plan_v5.md v5*	BENCH v5	line 53 row "F-1" col 4	tag in bench_plan_v5.md v5: bench plan v5 section 0 (founder block)
T4877	RFQ BOM 238	DECIDED *Design value, bench_plan_v5.md v5*	BENCH v5	line 54 row "F-2" col 1	tag in bench_plan_v5.md v5: bench plan v5 section 0 (founder block)
T4878	NI 9234 + CompactDAQ chassis + 10-32 low-noise coax (length TO-DEFINE); delivered DK-6400	DECIDED *Design value, bench_plan_v5.md v5*	BENCH v5	line 54 row "F-2" col 2	tag in bench_plan_v5.md v5: bench plan v5 section 0 (founder block)
T4879	B-DPC-1	DECIDED *Design value, bench_plan_v5.md v5*	BENCH v5	line 54 row "F-2" col 3	tag in bench_plan_v5.md v5: bench plan v5 section 0 (founder block)
T4880	quote on file	DECIDED *Design value, bench_plan_v5.md v5*	BENCH v5	line 54 row "F-2" col 4	tag in bench_plan_v5.md v5: bench plan v5 section 0 (founder block)
T4881	RFQ BOM 244	DECIDED *Design value, bench_plan_v5.md v5*	BENCH v5	line 55 row "F-3" col 1	tag in bench_plan_v5.md v5: bench plan v5 section 0 (founder block)
T4882	Coriolis mass-flow meter, liquid N2O service, 0-200 g/s class, response ≤ 100 ms ...	DECIDED *Design value, bench_plan_v5.md v5*	BENCH v5	line 55 row "F-3" col 2	tag in bench_plan_v5.md v5: bench plan v5 section 0 (founder block)
T4883	CF-3, B-BLEED-1 (mdot for the measured-time purge cap, LEAD 216 (5))	DECIDED *Design value, bench_plan_v5.md v5*	BENCH v5	line 55 row "F-3" col 3	tag in bench_plan_v5.md v5: bench plan v5 section 0 (founder block)
T4884	quote on file	DECIDED *Design value, bench_plan_v5.md v5*	BENCH v5	line 55 row "F-3" col 4	tag in bench_plan_v5.md v5: bench plan v5 section 0 (founder block)
T4885	RFQ BOM 245	DECIDED *Design value, bench_plan_v5.md v5*	BENCH v5	line 56 row "F-4" col 1	tag in bench_plan_v5.md v5: bench plan v5 section 0 (founder block)
T4886	reference pressure gauge / calibrator 0-100 bar, uncertainty ≤ 0.05 bar, traceable ...	DECIDED *Design value, bench_plan_v5.md v5*	BENCH v5	line 56 row "F-4" col 2	tag in bench_plan_v5.md v5: bench plan v5 section 0 (founder block)
T4887	B-PT-1, B-PC-1, F-12	DECIDED *Design value, bench_plan_v5.md v5*	BENCH v5	line 56 row "F-4" col 3	tag in bench_plan_v5.md v5: bench plan v5 section 0 (founder block)
T4888	quote on file	DECIDED *Design value, bench_plan_v5.md v5*	BENCH v5	line 56 row "F-4" col 4	tag in bench_plan_v5.md v5: bench plan v5 section 0 (founder block)
T4889	RFQ BOM 250	DECIDED *Design value, bench_plan_v5.md v5*	BENCH v5	line 57 row "F-5" col 1	tag in bench_plan_v5.md v5: bench plan v5 section 0 (founder block)
T4890	anemometer at 3 m (ultrasonic or cup) with ≥ 1 Hz output, 3 m mast, logger with a UTC ...	DECIDED *Design value, bench_plan_v5.md v5*	BENCH v5	line 57 row "F-5" col 2	tag in bench_plan_v5.md v5: bench plan v5 section 0 (founder block)
T4891	GSE wind rule (LEAD 181 (6) HARD)	DECIDED *Design value, bench_plan_v5.md v5*	BENCH v5	line 57 row "F-5" col 3	tag in bench_plan_v5.md v5: bench plan v5 section 0 (founder block)
T4892	quote on file	DECIDED *Design value, bench_plan_v5.md v5*	BENCH v5	line 57 row "F-5" col 4	tag in bench_plan_v5.md v5: bench plan v5 section 0 (founder block)
T4893	RFQ BOM 251 (Electronics addition)	DECIDED *Design value, bench_plan_v5.md v5*	BENCH v5	line 58 row "F-6" col 1	tag in bench_plan_v5.md v5: bench plan v5 section 0 (founder block)
T4894	Kulite XTL-190SM-2000A × 3 + 1 spare: absolute mode, M5 × 0.8 short body, O2-clean ...	DECIDED *Design value, bench_plan_v5.md v5*	BENCH v5	line 58 row "F-6" col 2	tag in bench_plan_v5.md v5: bench plan v5 section 0 (founder block)

ID	Value	Class	Source	Key	Source tag / formula
T4895	flight Pc / P_feed / P_mid (LEAD 205, 206 (3)); option (b) seat compensation	DECIDED *Design value, bench_plan_v5.md v5*	BENCH v5	line 58 row "F-6" col 3	tag in bench_plan_v5.md v5: bench plan v5 section 0 (founder block)
T4896	quote and the inquiry F-10 answered	DECIDED *Design value, bench_plan_v5.md v5*	BENCH v5	line 58 row "F-6" col 4	tag in bench_plan_v5.md v5: bench plan v5 section 0 (founder block)
T4897	TE inquiry (M3234-000005-070BG)	DECIDED *Design value, bench_plan_v5.md v5*	BENCH v5	line 59 row "F-7" col 1	tag in bench_plan_v5.md v5: bench plan v5 section 0 (founder block)
T4898	(a) output above full scale: the saturation voltage, (b) recovery time after an ...	DECIDED *Design value, bench_plan_v5.md v5*	BENCH v5	line 59 row "F-7" col 2	tag in bench_plan_v5.md v5: bench plan v5 section 0 (founder block)
T4899	070BG behaviour (stand reference, P_torch); second source	DECIDED *Design value, bench_plan_v5.md v5*	BENCH v5	line 59 row "F-7" col 3	tag in bench_plan_v5.md v5: bench plan v5 section 0 (founder block)
T4900	written answer on file	DECIDED *Design value, bench_plan_v5.md v5*	BENCH v5	line 59 row "F-7" col 4	tag in bench_plan_v5.md v5: bench plan v5 section 0 (founder block)
T4901	PCB inquiry (113B22)	DECIDED *Design value, bench_plan_v5.md v5*	BENCH v5	line 60 row "F-8" col 1	tag in bench_plan_v5.md v5: bench plan v5 section 0 (founder block)
T4902	(a) operation from the NI 9234 IEPE source (2.0 mA, 19 V compliance) with the 8-14 V ...	DECIDED *Design value, bench_plan_v5.md v5*	BENCH v5	line 60 row "F-8" col 2	tag in bench_plan_v5.md v5: bench plan v5 section 0 (founder block)
T4903	B-DPC-1 excitation (LEAD 203 (7) TO-VERIFY)	DECIDED *Design value, bench_plan_v5.md v5*	BENCH v5	line 60 row "F-8" col 3	tag in bench_plan_v5.md v5: bench plan v5 section 0 (founder block)
T4904	written answer on file	DECIDED *Design value, bench_plan_v5.md v5*	BENCH v5	line 60 row "F-8" col 4	tag in bench_plan_v5.md v5: bench plan v5 section 0 (founder block)
T4905	NI inquiry (NI 9234)	DECIDED *Design value, bench_plan_v5.md v5*	BENCH v5	line 61 row "F-9" col 1	tag in bench_plan_v5.md v5: bench plan v5 section 0 (founder block)
T4906	IEPE compliance at 2.0 mA with the 113B22; AC coupling with IEPE OFF behind an external ...	DECIDED *Design value, bench_plan_v5.md v5*	BENCH v5	line 61 row "F-9" col 2	tag in bench_plan_v5.md v5: bench plan v5 section 0 (founder block)
T4907	B-DPC-1	DECIDED *Design value, bench_plan_v5.md v5*	BENCH v5	line 61 row "F-9" col 3	tag in bench_plan_v5.md v5: bench plan v5 section 0 (founder block)
T4908	written answer on file	DECIDED *Design value, bench_plan_v5.md v5*	BENCH v5	line 61 row "F-9" col 4	tag in bench_plan_v5.md v5: bench plan v5 section 0 (founder block)
T4909	Kulite inquiry (XTL-190SM-2000A)	DECIDED *Design value, bench_plan_v5.md v5*	BENCH v5	line 62 row "F-10" col 1	tag in bench_plan_v5.md v5: bench plan v5 section 0 (founder block)
T4910	ordering code in absolute mode at 2000 psi; compensated band (R3-05): (a) the 100 °F ...	DECIDED *Design value, bench_plan_v5.md v5*	BENCH v5	line 62 row "F-10" col 2	tag in bench_plan_v5.md v5: bench plan v5 section 0 (founder block)
T4911	BOM 251, the mass statement, B-PC-1 zero-TC row, kPcRezeroMax_Pa basis	DECIDED *Design value, bench_plan_v5.md v5*	BENCH v5	line 62 row "F-10" col 3	tag in bench_plan_v5.md v5: bench plan v5 section 0 (founder block)
T4912	written answer on file	DECIDED *Design value, bench_plan_v5.md v5*	BENCH v5	line 62 row "F-10" col 4	tag in bench_plan_v5.md v5: bench plan v5 section 0 (founder block)
T4913	anemometer mast and logger (BOM 250)	DECIDED *Design value, bench_plan_v5.md v5*	BENCH v5	line 63 row "F-11" col 1	tag in bench_plan_v5.md v5: bench plan v5 section 0 (founder block)
T4914	set up at the launch site: 3 m mast, ≥ 1 Hz logging with UTC time; compute W10 (10-min ...	DECIDED *Design value, bench_plan_v5.md v5*	BENCH v5	line 63 row "F-11" col 2	tag in bench_plan_v5.md v5: bench plan v5 section 0 (founder block)
T4915	W10 < 6 m/s AND sigma_u ≤ 1.0 m/s is HARD for A1 (LEAD 181 (6)); the R2 0.8 s window is ...	DECIDED *Design value, bench_plan_v5.md v5*	BENCH v5	line 63 row "F-11" col 3	tag in bench_plan_v5.md v5: bench plan v5 section 0 (founder block)
T4916	one logged day; the rule evaluated on it	DECIDED *Design value, bench_plan_v5.md v5*	BENCH v5	line 63 row "F-11" col 4	tag in bench_plan_v5.md v5: bench plan v5 section 0 (founder block)
T4917	spare-070BG over-range row (LEAD 203 (2), 216 (6); moved from B-PC-1)	DECIDED *Design value, bench_plan_v5.md v5*	BENCH v5	line 64 row "F-12" col 1	tag in bench_plan_v5.md v5: bench plan v5 section 0 (founder block)
T4918	on the SPARE 070BG (label it "HYDRAULIC ONLY - NEVER N2O"; it never returns to N2O ...	DECIDED *Design value, bench_plan_v5.md v5*	BENCH v5	line 64 row "F-12" col 2	tag in bench_plan_v5.md v5: bench plan v5 section 0 (founder block)
T4919	after line 205 this characterises the 070BG kept for P_torch (J901) and as the B-PC-1 ...	DECIDED *Design value, bench_plan_v5.md v5*	BENCH v5	line 64 row "F-12" col 3	tag in bench_plan_v5.md v5: bench plan v5 section 0 (founder block)
T4920	data and a one-page note on file	DECIDED *Design value, bench_plan_v5.md v5*	BENCH v5	line 64 row "F-12" col 4	tag in bench_plan_v5.md v5: bench plan v5 section 0 (founder block)
T4921	stand T0 rule (LEAD ruling 2026-10-20; basis LEAD 249 (2): 20 + 4.48 / 0.96 = 24.67 °C ...	DECIDED *Design value, bench_plan_v5.md v5*	BENCH v5	line 65 row "F-13" col 1	tag in bench_plan_v5.md v5: bench plan v5 section 0 (founder block)
T4922	founder pre-fire checklist row (R3-06): the engine SOAKED at the stand ambient: both ...	DECIDED *Design value, bench_plan_v5.md v5*	BENCH v5	line 65 row "F-13" col 2	tag in bench_plan_v5.md v5: bench plan v5 section 0 (founder block)
T4923	stand start temperature basis of the case-TC rows	DECIDED *Design value, bench_plan_v5.md v5*	BENCH v5	line 65 row "F-13" col 3	tag in bench_plan_v5.md v5: bench plan v5 section 0 (founder block)
T4924	checklist row in use	DECIDED *Design value, bench_plan_v5.md v5*	BENCH v5	line 65 row "F-13" col 4	tag in bench_plan_v5.md v5: bench plan v5 section 0 (founder block)
T4925	J-FWD replace rule and TB-state (LC-TB70 v2; LEAD 248, 258 (3); Mechanical F4, line 259)	DECIDED *Design value, bench_plan_v5.md v5*	BENCH v5	line 66 row "F-14" col 1	tag in bench_plan_v5.md v5: bench plan v5 section 0 (founder block)
T4926	(1) Quoted verbatim from mechanical.json v2.4 lc_tb70_envelope_v352.jfwd_replace_rule_v2. ...	DECIDED *Design value, bench_plan_v5.md v5*	BENCH v5	line 66 row "F-14" col 2	tag in bench_plan_v5.md v5: bench plan v5 section 0 (founder block)
T4927	joint life (Mechanical)	DECIDED *Design value, bench_plan_v5.md v5*	BENCH v5	line 66 row "F-14" col 3	tag in bench_plan_v5.md v5: bench plan v5 section 0 (founder block)
T4928	rule in the stand procedure; the post-run Q_JFWD script on the stand PC	DECIDED *Design value, bench_plan_v5.md v5*	BENCH v5	line 66 row "F-14" col 4	tag in bench_plan_v5.md v5: bench plan v5 section 0 (founder block)
T4929	Bench plan v5 does not go to the founders before these corrections.	DECIDED *Lead decision, line 266 (7)*	CL 266 lines	L266(7)[from "Bench plan v5 does not go to the founders before these corrections." to "Bench plan v5 does not go to the founders before these corrections."]	lead (decision) 2026-10-21 (line sha 6f3181b859f02937)
T4930	J-FWD build and every re-assembly (v3: route (i), lead 16:53Z R2 / R3, 17:55:55Z F1 / ...	DECIDED *Lead decision, per mechanical.json v2.4*	MEC v2.4	/jfwd_v352/founder_rules/C4_procedure	tag in mechanical.json v2.4: [scope of the text at this pointer, quoted verbatim]
T4931	0. GATE: J-F1 complete on the flight lots (ENG-HW-101-IN718 screws, HW-170 springs ...	DECIDED *Lead decision, per mechanical.json v2.4*	MEC v2.4	/jfwd_v352/founder_rules/C4_procedure	tag in mechanical.json v2.4: [step 0 of the text at this pointer, quoted verbatim]
T4932	1. inspect the hole edges and the bore per C1 (e); gauge the ten TC-01 holes with the ...	DECIDED *Lead decision, per mechanical.json v2.4*	MEC v2.4	/jfwd_v352/founder_rules/C4_procedure	tag in mechanical.json v2.4: [step 1 of the text at this pointer, quoted verbatim]
T4933	2. fit the ten ENG-TC-12 plugs in their numbered holes from outside, contour aligned ...	DECIDED *Lead decision, per mechanical.json v2.4*	MEC v2.4	/jfwd_v352/founder_rules/C4_procedure	tag in mechanical.json v2.4: [step 2 of the text at this pointer, quoted verbatim]
T4934	3. lubricate S01 and S02 with the selected lubricant (FKM- and N2O-compatible, TO-SELECT ...	DECIDED *Lead decision, per mechanical.json v2.4*	MEC v2.4	/jfwd_v352/founder_rules/C4_procedure	tag in mechanical.json v2.4: [step 3 of the text at this pointer, quoted verbatim]
T4935	4. clock TC-02 to the match marks BEFORE the rings enter the lead-in, then push it in ...	DECIDED *Lead decision, per mechanical.json v2.4*	MEC v2.4	/jfwd_v352/founder_rules/C4_procedure	tag in mechanical.json v2.4: [step 4 of the text at this pointer, quoted verbatim]
T4936	5. remove the band and the plugs; look through each hole (10x loupe or borescope) for ...	DECIDED *Lead decision, per mechanical.json v2.4*	MEC v2.4	/jfwd_v352/founder_rules/C4_procedure	tag in mechanical.json v2.4: [step 5 of the text at this pointer, quoted verbatim]
T4937	6. REGISTRATION (lead 17:55:55Z F2): fit the two GSE shoulder screws ENG-TC-13 (not ...	DECIDED *Lead decision, per mechanical.json v2.4*	MEC v2.4	/jfwd_v352/founder_rules/C4_procedure	tag in mechanical.json v2.4: [step 6 of the text at this pointer, quoted verbatim]
T4938	7. fit the flight screws ONE AT A TIME into the other eight holes: a NEW ...	DECIDED *Lead decision, per mechanical.json v2.4*	MEC v2.4	/jfwd_v352/founder_rules/C4_procedure	tag in mechanical.json v2.4: [step 7 of the text at this pointer, quoted verbatim]
T4939	8. remove the two shoulder screws LAST and replace each with a flight screw per step 7 ...	DECIDED *Lead decision, per mechanical.json v2.4*	MEC v2.4	/jfwd_v352/founder_rules/C4_procedure	tag in mechanical.json v2.4: [step 8 of the text at this pointer, quoted verbatim]
T4940	9. never re-torque a fitted screw; a screw removed for any reason is replaced by a new ...	DECIDED *Lead decision, per mechanical.json v2.4*	MEC v2.4	/jfwd_v352/founder_rules/C4_procedure	tag in mechanical.json v2.4: [step 9 of the text at this pointer, quoted verbatim]
T4941	J-FWD build and every re-assembly (v3: route (i), lead 16:53Z R2 / R3, 17:55:55Z F1 / ...	DECIDED *Lead decision, per item_c_tf1_texts_v352_v5.json v5*	TXT5 v5	/item_C/C4_assembly_step_founder_procedure	tag in item_c_tf1_texts_v352_v5.json v5:
T4942	identical	DERIVED	generator	formula: string comparison of mechanical.json /jfwd_v352/founder_rules/C4_procedure with texts v5 /item_C/C4_assembly_step_founder_procedure	DERIVED by build_datasheet.py: string comparison of mechanical.json /jfwd_v352/founder_rules/C4_procedure with texts v5 /item_C/C4_assembly_step_founder_procedure ← T4824, T4941
T4943	C4: assembly order (inspect; plugs and band in; lubricate S01/S02 with an oxygen-clean ...	DECIDED *Lead decision, line 241 (8)*	CL 266 lines	L241(8)[from "C4: assembly order" to "proof / leak test)."]	lead (decision) 2026-10-20 (line sha 0e301646c032c306)
T4944	C4 rewritten: J-F1 gate (lot torque, spring items, Prescale footprint item TO-VERIFY) ...	TO-VERIFY	CL 266 lines	L246(2)[from "C4 rewritten:" to "never re-torqued;"]	mechanical (result: hub publication 1 of 2 after the release 19:09:50Z = 04-fea/v352 J-FWD route (i) rows v7 + four companions as ACCEPTED 18:36:37Z, the drawing texts, the rev-5 CAD check they ...
T4945	C4 step 1 checks the select-fit pairing record		CL 266 lines	L254(2)[from "C4 step 1 checks the select-fit pairing record" to "select-fit pairing record"]	mechanical (result) 2026-10-21 (line sha c7d52385af9ce594); class set at the call site: a source line of the founder action (the specialist's record line, quoted as its reference)
T4946	OPEN for the next Electronics round after REV-ELEC-4: re-base kTorchArmZeroMax_Pa and ...	DECIDED *Lead decision, line 264 (4)*	CL 266 lines	L264(4)[from "OPEN for the next Electronics round" to "DK-6400."]	lead (decision) 2026-10-21 (line sha 7cdbe51041ae6b8b)

ID	Value	Class	Source	Key	Source tag / formula
T4947	The author did NOT produce the v2.12 SIL MANIFEST or re-run S1-S11 / S14-S17 on v2.12 ...	DECIDED *Lead decision, line 264 (8)*	CL 266 lines	L264(8)[from "The author did NOT produce" to "can affect."]	lead (decision) 2026-10-21 (line sha 7cdbe51041ae6b8b)
T4948	13. Open items and owners (2026-10-21)	DERIVED	F3D FINAL handback of the v3.5.2 re-check	heading line 1690	FEA3D-2 report heading
T4949	Open item: 210 m gate-delay arm	TO-VERIFY	DSIG v3.2	§* row c0="11a" col 1 (line 32)	GNC datasheet inputs row 11a
T4950	The items that were queued for it are kept as engine data or open items in the engine ...	DECIDED *Lead decision, line 222*	CL 266 lines	L222[sentence containing "The items that were queued for it"]	lead (founder decision) 2026-10-19 (line sha 1c5073790c99185c)
T4951	Analysis, then static fire	DECIDED *Requirement, ENG-01*	REQ rev 2	§2. Top-level row "ENG-01" col 3	tag in REQUIREMENTS.md rev 2:
T4952	ENG-01 at the cap (CHANGELOG 170): the 25 bar software Pc cap stays; if T1 measures the ...	DECIDED *Requirement, ENG-01*	REQ rev 2	§Amendments after rev 2 bullet "ENG-01 at the cap (CHANGELOG 170): the 25 bar so" [sentence containing "ENG-01"]	tag in REQUIREMENTS.md rev 2: REQ amendment bullet
T4953	Throttle floor ≤ 20 % of max (5:1, 100–500 N) with stable combustion. Stretch: 12.5 % ...	DERIVED	REQ rev 2	§2. Top-level row "ENG-02" col 1	DERIVED (GNC MC) / LEAD
T4954	Analysis, cold flow, static fire T3 with 1 kHz Pc	DECIDED *Requirement, ENG-02*	REQ rev 2	§2. Top-level row "ENG-02" col 3	tag in REQUIREMENTS.md rev 2:
T4955	Throttle response: t10–90 ≤ max(0.15 s, 0.8·ΔF / 1000 N/s). Ramp (amended 2026-10-13 ...	DERIVED	REQ rev 2	§2. Top-level row "ENG-03" col 1	DERIVED (GNC)
T4956	SIL, B-PNT-1, static fire	DECIDED *Requirement, ENG-03*	REQ rev 2	§2. Top-level row "ENG-03" col 3	tag in REQUIREMENTS.md rev 2:
T4957	Restartable: ≥ 3 starts per flight. Command to 90 % thrust ≤ 0.5 s (target 0.3 s) ...	DERIVED	REQ rev 2	§2. Top-level row "ENG-04" col 1	DERIVED / LEAD
T4958	Analysis, static fire	DECIDED *Requirement, ENG-04*	REQ rev 2	§2. Top-level row "ENG-04" col 3	tag in REQUIREMENTS.md rev 2:
T4959	ENG-04 / relight: deterministic relight (RELIGHT at planned T_fire;	DECIDED *Requirement, ENG-04*	REQ rev 2	§Amendments after rev 2 bullet "ENG-04 / relight: deterministic relight (RELIGHT" [sentence containing "ENG-04"]	tag in REQUIREMENTS.md rev 2: REQ amendment bullet
T4960	Shutdown: command to < 10 % thrust ≤ 0.10 s. Met with a mandatory pre-MECO throttle-down ...	DERIVED	REQ rev 2	§2. Top-level row "ENG-05" col 1	DERIVED / LEAD
T4961	Analysis, B-PNT-1 closing time at 25 bar, static fire	DECIDED *Requirement, ENG-05*	REQ rev 2	§2. Top-level row "ENG-05" col 3	tag in REQUIREMENTS.md rev 2:
T4962	ENG-05 / hard start (stand): unignited-inventory allowable set by the two-volume relief ...	DECIDED *Requirement, ENG-05*	REQ rev 2	§Amendments after rev 2 bullet "ENG-05 / hard start (stand): unignited-inventory" [sentence containing "ENG-05"]	tag in REQUIREMENTS.md rev 2: REQ amendment bullet
T4963	Accumulated burn per flight sized by the duty cycles in gnc.json v3. Grain replaced each ...	ASSUMED	REQ rev 2	§2. Top-level row "ENG-06" col 1	ASSUMED / SIM
T4964	Analysis, static fire bore measurement	DECIDED *Requirement, ENG-06*	REQ rev 2	§2. Top-level row "ENG-06" col 3	tag in REQUIREMENTS.md rev 2:
T4965	SIL, B-TVC-1	DECIDED *Requirement, ENG-07*	REQ rev 2	§2. Top-level row "ENG-07" col 3	tag in REQUIREMENTS.md rev 2:
T4966	No actuator saturation episode longer than 0.1 s during the landing burn. Logged in ...	DERIVED	REQ rev 2	§2. Top-level row "ENG-07b" col 1	DERIVED (GNC)
T4967	MC, flight logs	DECIDED *Requirement, ENG-07b*	REQ rev 2	§2. Top-level row "ENG-07b" col 3	tag in REQUIREMENTS.md rev 2:
T4968	Reflected servo rotor inertia J_int ≤ 1e-3 kg m² at the output (provisional), unless a ...	DERIVED	REQ rev 2	§2. Top-level row "ENG-07c" col 1	DERIVED (GNC) / ASSUMED
T4969	B-TVC-1	DECIDED *Requirement, ENG-07c*	REQ rev 2	§2. Top-level row "ENG-07c" col 3	tag in REQUIREMENTS.md rev 2:
T4970	Flex gain margin ≥ 6 dB at the gimbal rocking mode. ECU notch required, with separate ...	DERIVED	REQ rev 2	§2. Top-level row "ENG-07d" col 1	DERIVED (GNC csi_v3)
T4971	CSI analysis, B-TVC-1	DECIDED *Requirement, ENG-07d*	REQ rev 2	§2. Top-level row "ENG-07d" col 3	tag in REQUIREMENTS.md rev 2:
T4972	Gimbal sweep fits the airframe ICD: bay ID ≥ 140 mm, zone [−0.238, +0.252] m. Feed ...	DERIVED *Requirement, ENG-08*	REQ rev 2	§2. Top-level row "ENG-08" col 1	REQ / DERIVED
T4973	CAD, B-FD-1	DECIDED *Requirement, ENG-08*	REQ rev 2	§2. Top-level row "ENG-08" col 3	tag in REQUIREMENTS.md rev 2:
T4974	Pressure parts: SF ≥ 1.5 on yield and ≥ 2.0 on ultimate, at MEOP and at temperature. ...	DECIDED *Lead decision, per REQUIREMENTS.md rev 2*	REQ rev 2	§2. Top-level row "ENG-09" col 1	tag in REQUIREMENTS.md rev 2: LEAD
T4975	FEA, hydro proof	DECIDED *Requirement, ENG-09*	REQ rev 2	§2. Top-level row "ENG-09" col 3	tag in REQUIREMENTS.md rev 2:
T4976	ENG-09: MEOP 45 bar kept; ignition interlock T_tank <= 24 C at ARM and at ignition.	DECIDED *Requirement, ENG-09*	REQ rev 2	§Amendments after rev 2 bullet "ENG-09: MEOP 45 bar kept; ignition interlock T_t" [sentence containing "ENG-09"]	tag in REQUIREMENTS.md rev 2: REQ amendment bullet
T4977	ENG-09 trapped-liquid relief text superseded: tank relief 6.9 MPa, MAWP >= 8.42 MPa ...	DECIDED *Requirement, ENG-09*	REQ rev 2	§Amendments after rev 2 bullet "Combustion re-review decisions (CHANGELOG 172): " [sentence containing "ENG-09"]	tag in REQUIREMENTS.md rev 2: REQ amendment bullet
T4978	Structure (gimbal, mounts, fasteners): SF ≥ 1.5 on yield and ≥ 2.0 on ultimate under ...	DERIVED *Lead decision, per REQUIREMENTS.md rev 2*	REQ rev 2	§2. Top-level row "ENG-10" col 1	LEAD / DERIVED
T4979	FEA, hand calculations	DECIDED *Requirement, ENG-10*	REQ rev 2	§2. Top-level row "ENG-10" col 3	tag in REQUIREMENTS.md rev 2:
T4980	REV-MECH-1 decisions (CHANGELOG 188): safety factors: nominal states (LC1-LC4, LC6, TS1 ...	DECIDED *Requirement, ENG-10*	REQ rev 2	§Amendments after rev 2 bullet "REV-MECH-1 decisions (CHANGELOG 188): safety fac" [sentence containing "ENG-10"]	tag in REQUIREMENTS.md rev 2: REQ amendment bullet
T4981	Modal and stiffness FEA, B-TVC-1	DECIDED *Requirement, ENG-11*	REQ rev 2	§2. Top-level row "ENG-11" col 3	tag in REQUIREMENTS.md rev 2:
T4982	ENG-11 gimbal stiffness spread (3133-7406 N m/rad) is assessed by GNC for loop robustness.	DECIDED *Requirement, ENG-11*	REQ rev 2	§Amendments after rev 2 bullet "REV-MECH-1 decisions (CHANGELOG 188): safety fac" [sentence containing "ENG-11"]	tag in REQUIREMENTS.md rev 2: REQ amendment bullet
T4983	Design-to window 100 <= K_s <= 130 N m/rad (two-sided) on the V5c plant (CAD v3.3) at ...	DECIDED *Lead decision, line 227*	CL 266 lines	L227[sentence containing "V5c plant (CAD v3.3)"]	lead (decision) 2026-10-19 (line sha 07d96566579945c1)
T4984	Propellants: self-pressurised N2O, tank 2.00 kg / 3.17 L conditioned to 15–25 °C, with ...	DECIDED *Requirement, ENG-12*	REQ rev 2	§2. Top-level row "ENG-12" col 1	tag in REQUIREMENTS.md rev 2: REQ / LEAD
T4985	Trade studies, static fire	DECIDED *Requirement, ENG-12*	REQ rev 2	§2. Top-level row "ENG-12" col 3	tag in REQUIREMENTS.md rev 2:
T4986	ENG-12: tank temperature range at ARM 15-24 C (was 15-25 C); heater hard-disabled at ARM.	DECIDED *Requirement, ENG-12*	REQ rev 2	§Amendments after rev 2 bullet "ENG-12: tank temperature range at ARM 15-24 C (w" [sentence containing "ENG-12"]	tag in REQUIREMENTS.md rev 2: REQ amendment bullet
T4987	Engine controller (ECU) works with a Teensy-class flight computer. It provides ...	DERIVED	REQ rev 2	§2. Top-level row "ENG-13" col 1	DERIVED / LEAD
T4988	Bench, SIL, HIL	DECIDED *Requirement, ENG-13*	REQ rev 2	§2. Top-level row "ENG-13" col 3	tag in REQUIREMENTS.md rev 2:
T4989	Cost: cheapest viable. Every BOM line has a source and price, or is marked TO-QUOTE with ...	DECIDED *Requirement, ENG-14*	REQ rev 2	§2. Top-level row "ENG-14" col 1	tag in REQUIREMENTS.md rev 2: REQ
T4990	BOM review	DECIDED *Requirement, ENG-14*	REQ rev 2	§2. Top-level row "ENG-14" col 3	tag in REQUIREMENTS.md rev 2:
T4991	Manufacturability: printed parts on a hobby FDM printer; metal parts from stock tube and ...	DECIDED *Requirement, ENG-15*	REQ rev 2	§2. Top-level row "ENG-15" col 1	tag in REQUIREMENTS.md rev 2: REQ
T4992	CAD review	DECIDED *Requirement, ENG-15*	REQ rev 2	§2. Top-level row "ENG-15" col 3	tag in REQUIREMENTS.md rev 2:
T4993	Safety: N2O-compatible, oxygen-clean wetted paths (PFPE lubricant only); passive relief ...	DECIDED *Requirement, ENG-16*	REQ rev 2	§2. Top-level row "ENG-16" col 1	tag in REQUIREMENTS.md rev 2: REQ / LEAD
T4994	Review, bench	DECIDED *Requirement, ENG-16*	REQ rev 2	§2. Top-level row "ENG-16" col 3	tag in REQUIREMENTS.md rev 2:
T4995	section 0 (founder block)	DECIDED *Design value, bench_plan_v5.md v5*	BENCH v5	§1. Order and gates row "F" col 1	bench_plan_v5.md v5
T4996	RFQs, inquiries, anemometer, spare-070BG over-range, stand rules	DECIDED *Design value, bench_plan_v5.md v5*	BENCH v5	§1. Order and gates row "F" col 2	bench_plan_v5.md v5
T4997	nothing: executable now	DECIDED *Design value, bench_plan_v5.md v5*	BENCH v5	§1. Order and gates row "F" col 3	bench_plan_v5.md v5
T4998	RE-15 board checks, HAL-1	DECIDED *Design value, bench_plan_v5.md v5*	BENCH v5	§1. Order and gates row "0" col 1	bench_plan_v5.md v5
T4999	board fit for B-PC-1; HAL in place	DECIDED *Design value, bench_plan_v5.md v5*	BENCH v5	§1. Order and gates row "0" col 2	bench_plan_v5.md v5
T5000	schematic v2.6 (the v2.11 rows of BOM v2.3; NOT drawn) + layout (not started)	DECIDED *Design value, bench_plan_v5.md v5*	BENCH v5	§1. Order and gates row "0" col 3	bench_plan_v5.md v5
T5001	B-PWR-1, B-PWR-2	DECIDED *Design value, bench_plan_v5.md v5*	BENCH v5	§1. Order and gates row "1" col 1	bench_plan_v5.md v5
T5002	rails, e-fuse, ARM interlock, overlaps at 6.6 / 8.1 V (RE-09)	DECIDED *Design value, bench_plan_v5.md v5*	BENCH v5	§1. Order and gates row "1" col 2	bench_plan_v5.md v5
T5003	bare ECU, bench PSU	DECIDED *Design value, bench_plan_v5.md v5*	BENCH v5	§1. Order and gates row "1" col 3	bench_plan_v5.md v5
T5004	B-ECU-1	DECIDED *Design value, bench_plan_v5.md v5*	BENCH v5	§1. Order and gates row "2" col 1	bench_plan_v5.md v5
T5005	I/O, CAN, watchdog, log v13, ladder decode with decade box (RE-19)	DECIDED *Design value, bench_plan_v5.md v5*	BENCH v5	§1. Order and gates row "2" col 2	bench_plan_v5.md v5

ID	Value	Class	Source	Key	Source tag / formula
T5006	step 1	DECIDED *Design value, bench_plan_v5.md v5*	BENCH v5	§1. Order and gates row "2" col 3	bench_plan_v5.md v5
T5007	B-PT-1, B-PC-1, B-TC-1	DECIDED *Design value, bench_plan_v5.md v5*	BENCH v5	§1. Order and gates row "3" col 1	bench_plan_v5.md v5
T5008	P_tank calibrated budget (guard band); C1 Pc trims, post-zero residual (`kBpc1Shown`) ...	DECIDED *Design value, bench_plan_v5.md v5*	BENCH v5	§1. Order and gates row "3" col 2	bench_plan_v5.md v5
T5009	step 2, clean dry N2 0-70 bar g, reference gauge (BOM 245), TC reference (BOM 266) ...	DECIDED *Design value, bench_plan_v5.md v5*	BENCH v5	§1. Order and gates row "3" col 3	bench_plan_v5.md v5
T5010	B-MV-1, B-PNT-1, B-TVC-1	DECIDED *Design value, bench_plan_v5.md v5*	BENCH v5	§1. Order and gates row "4" col 1	bench_plan_v5.md v5
T5011	actuator timing, ENG-03, MV ladder, TVC jam threshold (RE-24)	DECIDED *Design value, bench_plan_v5.md v5*	BENCH v5	§1. Order and gates row "4" col 2	bench_plan_v5.md v5
T5012	step 2	DECIDED *Design value, bench_plan_v5.md v5*	BENCH v5	§1. Order and gates row "4" col 3	bench_plan_v5.md v5
T5013	CHK-PW, CHK-BT, B-BLEED-1	DECIDED *Design value, bench_plan_v5.md v5*	BENCH v5	§1. Order and gates row "5" col 1	bench_plan_v5.md v5
T5014	purge window, both-open timer, valve, closure timing, purge mass, comparator, pad BIT	DECIDED *Design value, bench_plan_v5.md v5*	BENCH v5	§1. Order and gates row "5" col 2	bench_plan_v5.md v5
T5015	steps 1-4, selected valves (none yet)	DECIDED *Design value, bench_plan_v5.md v5*	BENCH v5	§1. Order and gates row "5" col 3	bench_plan_v5.md v5
T5016	B-IGN-1, B-FC-1	DECIDED *Design value, bench_plan_v5.md v5*	BENCH v5	§1. Order and gates row "6" col 1	bench_plan_v5.md v5
T5017	spark EMI on Pc, FC link, TLM_FAST bits, ABORT path, PC_VALID, encoder TLM	DECIDED *Design value, bench_plan_v5.md v5*	BENCH v5	§1. Order and gates row "6" col 2	bench_plan_v5.md v5
T5018	step 2, FC	DECIDED *Design value, bench_plan_v5.md v5*	BENCH v5	§1. Order and gates row "6" col 3	bench_plan_v5.md v5
T5019	HIL-OT-1, HIL-TR-1, HIL-RST-1	DECIDED *Design value, bench_plan_v5.md v5*	BENCH v5	§1. Order and gates row "7" col 1	bench_plan_v5.md v5
T5020	over-thrust and transient replay; reset behaviour (RE-28)	DECIDED *Design value, bench_plan_v5.md v5*	BENCH v5	§1. Order and gates row "7" col 2	bench_plan_v5.md v5
T5021	steps 2, 6, HAL	DECIDED *Design value, bench_plan_v5.md v5*	BENCH v5	§1. Order and gates row "7" col 3	bench_plan_v5.md v5
T5022	LEAK-1, CF-3, CF-4	DECIDED *Design value, bench_plan_v5.md v5*	BENCH v5	§1. Order and gates row "8" col 1	bench_plan_v5.md v5
T5023	feed leak, ignition-lift table, segment void	DECIDED *Design value, bench_plan_v5.md v5*	BENCH v5	§1. Order and gates row "8" col 2	bench_plan_v5.md v5
T5024	Mechanical / Combustion rigs	DECIDED *Design value, bench_plan_v5.md v5*	BENCH v5	§1. Order and gates row "8" col 3	bench_plan_v5.md v5
T5025	B-SAFE-1 (procedure dry run), B-DPC-1, IGN-1, T1-T4	DECIDED *Design value, bench_plan_v5.md v5*	BENCH v5	§1. Order and gates row "9" col 1	bench_plan_v5.md v5
T5026	relative light rule, start latency, LIGHT_SPIKE, F_acc	DECIDED *Design value, bench_plan_v5.md v5*	BENCH v5	§1. Order and gates row "9" col 2	bench_plan_v5.md v5
T5027	hot-fire stand	DECIDED *Design value, bench_plan_v5.md v5*	BENCH v5	§1. Order and gates row "9" col 3	bench_plan_v5.md v5
T5028	T4 restated as a relight-latency demonstration from a soaked state, its length from ...	DECIDED *Lead decision, line 266 (7)*	CL 266 lines	L266(7)[from "T4 restated as a relight-latency demonstration" to "waiting for v2.13"]	lead (decision) 2026-10-21 (line sha 6f3181b859f02937)
T5029	B-PNT-1	DECIDED *Requirement, REQUIREMENTS.md rev 2*	REQ rev 2	§4. Gating early bench tests row "B-PNT-1" col 0	tag in REQUIREMENTS.md rev 2:
T5030	Commanded travel ≥ 103°; unpowered spring closing at the MEOP-equivalent load; guide μ ≤ ...	DECIDED *Requirement, REQUIREMENTS.md rev 2*	REQ rev 2	§4. Gating early bench tests row "B-PNT-1" col 2	tag in REQUIREMENTS.md rev 2:
T5031	1× MKS HV69 (EUR 62.99 at Lindinger, catalogue price incl. AT VAT; TO-QUOTE to DK-6400)	DECIDED *Requirement, REQUIREMENTS.md rev 2*	REQ rev 2	§4. Gating early bench tests row "B-PNT-1" col 1	tag in REQUIREMENTS.md rev 2:
T5032	B-TVC-1	DECIDED *Requirement, REQUIREMENTS.md rev 2*	REQ rev 2	§4. Gating early bench tests row "B-TVC-1" col 0	tag in REQUIREMENTS.md rev 2:
T5033	Measure servo output stiffness K_s at 6.6 and 7.4 V, damping, J_int and the ...	DECIDED *Requirement, REQUIREMENTS.md rev 2*	REQ rev 2	§4. Gating early bench tests row "B-TVC-1" col 2	tag in REQUIREMENTS.md rev 2:
T5034	2× SB-2292SG (TO-QUOTE) with the n 4.8 linkage on a 0.0545 kg·m² inertia dummy	DECIDED *Requirement, REQUIREMENTS.md rev 2*	REQ rev 2	§4. Gating early bench tests row "B-TVC-1" col 1	tag in REQUIREMENTS.md rev 2:
T5035	B-PWR-1/2	DECIDED *Requirement, REQUIREMENTS.md rev 2*	REQ rev 2	§4. Gating early bench tests row "B-PWR-1/2" col 0	tag in REQUIREMENTS.md rev 2:
T5036	Source and branch resistance against the 15 / 20 mΩ assumptions.	DECIDED *Requirement, REQUIREMENTS.md rev 2*	REQ rev 2	§4. Gating early bench tests row "B-PWR-1/2" col 2	tag in REQUIREMENTS.md rev 2:
T5037	Flight battery pack and harness	DECIDED *Requirement, REQUIREMENTS.md rev 2*	REQ rev 2	§4. Gating early bench tests row "B-PWR-1/2" col 1	tag in REQUIREMENTS.md rev 2:
T5038	B-FD-1	DECIDED *Requirement, REQUIREMENTS.md rev 2*	REQ rev 2	§4. Gating early bench tests row "B-FD-1" col 0	tag in REQUIREMENTS.md rev 2:
T5039	Hinge stiffness while pressurised ≤ 20 N·m/rad per axis.	DECIDED *Requirement, REQUIREMENTS.md rev 2*	REQ rev 2	§4. Gating early bench tests row "B-FD-1" col 2	tag in REQUIREMENTS.md rev 2:
T5040	Gimbal hose	DECIDED *Requirement, REQUIREMENTS.md rev 2*	REQ rev 2	§4. Gating early bench tests row "B-FD-1" col 1	tag in REQUIREMENTS.md rev 2:
T5041	bench plan v4 is REVISE (text) and does not go to the founders before v5.	DECIDED *Lead decision, line 253 (5)*	CL 266 lines	L253(5)[from "bench plan v4 is REVISE" to "before v5."]	lead (decision) 2026-10-21 (line sha 3a7f695ca59d621e)
T5042	E4-06 (bench plan v5: A1 pad NO-GO until v2.13; T4 restated as a relight-latency ...	DECIDED *Lead decision, line 266 (7)*	CL 266 lines	L266(7)[from "E4-06 (bench plan v5:" to "the C7 stand-DAQ TC added as a stand-only row)"]	lead (decision) 2026-10-21 (line sha 6f3181b859f02937)
T5043	ACCEPT WITH COMMENTS (text corrections E4-02, E4-06 before founder use; the TC4 rows of ...	DERIVED	RELE4 REVIEW-ELECTRONICS-R4 (REV-ELEC-4)	§1. VERDICT row c0="Bench plan v5" col 1 (line 31)	REV-ELEC-4 verdict table
T5044	F-14, F-13, B-TC-1 (case TCs), B-BLEED-1 (d) / (d1) / (d2), B-MV-1 and T1-T4 inside A = ...	DERIVED	RELE4 REVIEW-ELECTRONICS-R4 (REV-ELEC-4)	§1. VERDICT row c0="Bench plan v5" col 2 (line 31)	REV-ELEC-4 verdict table
T5045	Eq. 22 inputs V 4079.2 N (FE, (b1)) and M 4354.7 N mm (CALC) (lead 01:30:18Z (6)); the ...	DERIVED	F3D FINAL handback of the v3.5.2 re-check	§13. Open items and owners row c0="2" col 2 (line 1697)	FEA3D-2 report §13 item 2
T5046	part: HW-101 (J-FWD); column: J-FWD; old_MS_y_MS_u: [0.65, 0.301]; new_MS_y_MS_u: [0.65 ...	DERIVED	MMXD v352	/cell_value_changes/0	(class set at the call site: DERIVED)
T5047	Mechanical's single J-FWD re-map on the FINAL envelope v6 was published under the lead ...	DERIVED	MMXD v352	/why	(class set at the call site: DERIVED)
T5048	It supersedes only assembly3d_v352/tables/margin_matrix_v352.md / .json and that script ...	DECIDED *Lead decision, line 255 (3)*	CL 266 lines	L255(3)[from "It supersedes only" to "stands as published."]	lead (decision) 2026-10-21 (line sha c541950ce85dcf2b)
T5049	A1 apogee target 210 m is decided (lines 214, 227 (1)); the A1 records above are the 300 ...	DECIDED *Design value, DATASHEET-INPUTS-COMBUSTION.md v2.4d*	DSIC v2.4d	line 25 (starts "A1 apogee target 210 m is decided")	tag in DATASHEET-INPUTS-COMBUSTION.md v2.4d: combustion datasheet inputs note
T5050	The coverage set at 210 m (e19, L400, wind sweep) and a 210 m A1 duty-cycle record. B1 ...	TO-VERIFY	DSIG v3.2	§* row c0="11h" col 2 (line 39)	GNC datasheet inputs row 11h
T5051	(1) GNC.JSON v3.2 PUBLISHED by the lead (built and stamped 10:56:48Z, promoted ...	DECIDED *Lead decision, line 260*	CL 266 lines	L260[sentence containing "GNC.JSON v3.2 PUBLISHED"]	lead decision 2026-10-21 (line sha 33d3167adecb2242)
T5052	Consequences: gnc.json v3.2 carries the trim; the A1 duty cycle shortens (B1 still ...	DECIDED *Lead decision, line 213*	CL 266 lines	L213[sentence containing "B1 still governs thermal and erosion sizing"]	lead (decision, founder) 2026-10-13 (line sha b5683ac804871b39)
T5053	plug 57.20 f7 with 2 AS568-226 FKM 90A glands (51.65 h9 groove, 4.85 wide)	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/lines/97/notes	tag in BOM_v3.5.2.json v3.5.2: BOM line notes (clause quoted) [the clause containing "57.20 f7" of the notes string at this pointer, quoted verbatim]
T5054	-0.075 / +0.239	CALC	MMXMD FEA3D margin matrix v3.5.2	§Negative cells and their classes row c0="TC-01" & c1="J-FWD" col 2 (line 429)	margin matrix v3.5.2 on rows v8 (lead re-run; 234 (4) all-node rule; MS = allowable / (SF x stress) - 1); negative-cell table
T5055	after any Pc above 60 bar abs the line-231 J-FWD inspection (C6 (2) list)	DECIDED *Design value, BOM_v3.5.2.json v3.5.2*	BOM v3.5.2	/hardware/lines/0/notes	tag in BOM_v3.5.2.json v3.5.2: BOM hardware notes (clause quoted) [the clause containing "60 bar abs" of the notes string at this pointer, quoted verbatim]

ID	Value	Class	Source	Key	Source tag / formula
T5056	the v1 'any head excursion above 60.0 bar abs: replace all J-FWD screws' rule and the ...	DECIDED *Lead decision, per combustion.json v2.4d*	CMB v2.4d	/z3_jfwd_rulings_v2p4d/v/jfwd_replace_rule/replaces	tag in combustion.json v2.4d:
T5057	no CAD, hardware-data or BOM edit: the HW-101 'after any Pc above 60 bar abs' and ...		CL 266 lines	L259(6)[from "no CAD, hardware-data or BOM edit" to "superseded by the drawing texts v2."]	mechanical (result) 2026-10-21 (line sha be16c854fa798851); class set at the call site: Mechanical record that the string is superseded; it states no value
T5058	| T4, DC-D duty cycle (S230 grain, 3 starts; v3.2: re-sized, line 294) | life and ...	SIM	T14 v3.2	line 61 (starts "| T4, DC-D duty cycle")	T1-T4 spec §3 T4 row
T5059	06-electronics/docs/bench_plan_v5.md 360c1d6695cc305c	DECIDED *Design value, electronics.json v4.1*	ELE v4.1	/bench_plan/file	electronics.json v4.1
T5060	06-electronics/staging_v2.12/docs/bench_plan_v5.md	DERIVED	BENCH v5	(file)	generator: file found at this path (sha256/16 360c1d6695cc305c)
T5061	re-baselined on the frozen v3.5.1 bleed path (LEAD 194 (8)); bleed-path part numbers and ...	DECIDED *Design value, combustion.json v2.4d*	CMB v2.4d	/cf4_gate/v/baseline_v2p4b	combustion.json v2.4d
T5062	v2.4	DECIDED *Design value, mechanical.json v2.4*	MEC v2.4	/version	mechanical.json v2.4
T5063	2.55 s	DECIDED *Lead decision, line 203 (3)*	ELE4 v4	/limits_v2p11/4/value	tag in electronics_v4.json v4: LEAD CHANGELOG 203 (3), PROVISIONAL 216 (5) / 230 (3)
T5064	70000 Pa	DECIDED *Design value, Electronics v2.11 (electronics.json v4)*	ELE4 v4	/light_rule/kIgnBaseCheck_Pa	source tag: LEAD; class set at the call site: DESIGN (not accepted by the lead: 253 (5), R3-01 (lead 2026-10-21 13:47Z, REV-DS-1 D1-03))
T5065	30000 Pa	DECIDED *Design value, Electronics v2.11 (electronics.json v4)*	ELE4 v4	/sensors/pressure_transducers/coast_rezero/max	source tag: LEAD 2026-10-13 B-PC-1 target; class set at the call site: DESIGN (not accepted by the lead: 253 (5), R3-01 (lead 2026-10-21 13:47Z, REV-DS-1 D1-03))
T5066	combustion.json /abort_matrix_v2p4/v/A4: annotate, REDUNDANT, shown 0×; CHANGELOG.md ...	DERIVED	generator	override OVR-01 targets (this build)	generator target check on this build
T5067	ACTIVE	DERIVED	generator	override OVR-01 status (this build)	generator check on this build
T5068	combustion.json /decisions/tank_pressurisation: annotate, REDUNDANT, shown 0× ...	DERIVED	generator	override OVR-04 targets (this build)	generator target check on this build
T5069	ACTIVE	DERIVED	generator	override OVR-04 status (this build)	generator check on this build
T5070	combustion.json /valves_and_actuators/isolation_valve/trapped_liquid_relief: replace ...	DERIVED	generator	override OVR-05 targets (this build)	generator target check on this build
T5071	ACTIVE	DERIVED	generator	override OVR-05 status (this build)	generator check on this build
T5072	combustion.json /decisions/chamber_MEOP_Pa/tag: replace, ACTIVE, shown 0× ...	DERIVED	generator	override OVR-06 targets (this build)	generator target check on this build
T5073	ACTIVE	DERIVED	generator	override OVR-06 status (this build)	generator check on this build
T5074	CHANGELOG.md L168(1): annotate, ACTIVE, shown 0×	DERIVED	generator	override OVR-09 targets (this build)	generator target check on this build
T5075	ACTIVE	DERIVED	generator	override OVR-09 status (this build)	generator check on this build
T5076	electronics.json /status: annotate, ACTIVE, shown 1×	DERIVED	generator	override OVR-10 targets (this build)	generator target check on this build
T5077	ACTIVE	DERIVED	generator	override OVR-10 status (this build)	generator check on this build
T5078	combustion.json /operating_points/pressure_limits/full_stroke_failure_S230_max_bar ...	DERIVED	generator	withheld SUP-01 keys (this build)	generator check on this build
T5079	3 displayed fields masked	DERIVED	generator	withheld SUP-01 masks (this build)	generator count on this build
T5080	yes	DERIVED	generator	withheld SUP-01 corroboration	key present in combustion.json /superseded_v2p4 (the author's withdrawal list)
T5081	combustion.json /thermal_and_loads/flights_per_insert_A1: WITHHELD	DERIVED	generator	withheld SUP-02 keys (this build)	generator check on this build
T5082	0 displayed fields masked	DERIVED	generator	withheld SUP-02 masks (this build)	generator count on this build
T5083	yes	DERIVED	generator	withheld SUP-02 corroboration	key present in combustion.json /superseded_v2p4 (the author's withdrawal list)
T5084	combustion.json /valves_and_actuators/pintle_actuator/seat_and_seal_review_v2p3/v/b_seat/ ...	DERIVED	generator	withheld SUP-03 keys (this build)	generator check on this build
T5085	0 displayed fields masked	DERIVED	generator	withheld SUP-03 masks (this build)	generator count on this build
T5086	yes	DERIVED	generator	withheld SUP-03 corroboration	key present in combustion.json /superseded_v2p4 (the author's withdrawal list)
T5087	no key listed	DERIVED	generator	withheld SUP-04 keys (this build)	generator check on this build
T5088	0 displayed fields masked	DERIVED	generator	withheld SUP-04 masks (this build)	generator count on this build
T5089	yes	DERIVED	generator	withheld SUP-04 corroboration	key present in combustion.json /superseded_v2p4 (the author's withdrawal list)
T5090	The matrix of record governs; the FEA3D-2 report text is stale. The matrix value is ...	DERIVED	generator	contradiction C-01 status	generator summary of the cited sides and the governing source
T5091	Line 266 (7) governs: electronics.json v4.1 is the source of record and its internal ...	DERIVED	generator	contradiction C-02 status	generator summary of the cited sides and the governing source
T5092	OPEN (GNC): the 210 m A1 mission-burn and duty-cycle record is not flown (row 11h). The ...	DERIVED	generator	contradiction C-03 status	generator summary of the cited sides and the governing source
T5093	The R2-B select fit governs (line 254); the BOM fit note is superseded and the drawn fit ...	DERIVED	generator	contradiction C-04 status	generator summary of the cited sides and the governing source
T5094	The Q_JFWD replace rule (258 (3)) governs; line 259 (6) states that the HW-101 string is ...	DERIVED	generator	contradiction C-05 status	generator summary of the cited sides and the governing source
T5095	The re-size governs (261 (2), 264 (5)); the published T4 rows (DC-D) are superseded. ...	DERIVED	generator	contradiction C-06 status	generator summary of the cited sides and the governing source
T5096	The hub tree governs; the path is printed as it is in the hub (§15.2). OPEN: Electronics ...	DERIVED	generator	contradiction C-07 status	generator summary of the cited sides and the governing source
T5097	mechanical.json v2.4 is current; the CF-4 text predates it (the bleed-path basis stays ...	DERIVED	generator	contradiction C-08 status	generator summary of the cited sides and the governing source
T5098	RETIRED: the two agree at this build (difference below 0.0005 kg; §4.4), so there is no ...	DERIVED	generator	contradiction C-09 status	generator summary of the cited sides and the governing source
T5099	Line 264 (4) governs: the C1-family Kulite XTL-190SM-2000A, its proof [DATASHEET] ...	DERIVED	generator	contradiction C-10 status	generator summary of the cited sides and the governing source
T5100	R2-B governs (254 (1)): the select-fit band is the R2-B band. Band re-size done (band v1 ...	DERIVED	generator	contradiction C-11 status	generator summary of the cited sides and the governing source
T5101	256 (2) governs (2.43 s flow, 2.39 s cap, conditions (a)-(d), (d1), (d2) of 256 (2) / ...	DERIVED	generator	contradiction C-12 status	generator summary of the cited sides and the governing source
T5102	253 (5) did not accept the v2.11 values; 264 (2) governs (kPcRezeroMax_Pa 0.30 bar on ...	DERIVED	generator	contradiction C-13 status	generator summary of the cited sides and the governing source
T5103	264 (4) governs: P_torch is the C1-family Kulite XTL-190SM-2000A. The re-base of the ...	DERIVED	generator	contradiction C-14 status	generator summary of the cited sides and the governing source
T5104	264 (5) governs: the shorter relight is adopted and T4 demonstrates relight latency from ...	DERIVED	generator	contradiction C-15 status	generator summary of the cited sides and the governing source
T5105	Closed: 264 (3) adopts the A1 BASE_AMB outcome as built; the item is listed under the ...	DERIVED	generator	contradiction C-16 status	generator summary of the cited sides and the governing source